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Preface

Regression is taught under at least five different descriptions, and each one is correct. In one, regression is least squares together with the distribution theory of quadratic forms. In another, it is orthogonal projection onto a subspace, and everything else follows as a corollary. A third starts from likelihood and treats the normal linear model as the Gaussian member of the generalized linear family. A fourth defines regression as estimating a conditional distribution, with linearity one structural assumption among many. A fifth treats it as a data-analytic procedure whose assumptions must be checked, one at a time, against the data.

Most textbooks pick one of these descriptions and stay with it. This book uses all of them, in order, as rungs of a single ladder. Parts I–III set up the classical linear model and its exact theory. After that, *each part relaxes exactly one assumption*, and asks what survives, what must be rebuilt, and what new tools the relaxation requires.

Part	What changes	What we study
I	—	The apparatus: matrices, random vectors, the normal distribution, quadratic forms
II	—	$\mathbf{Y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}$, $E(\boldsymbol{\varepsilon}) = \mathbf{0}$, $\text{Cov}(\boldsymbol{\varepsilon}) = \sigma^2\mathbf{I}$, as a projection problem
III	adds $\boldsymbol{\varepsilon} \sim N(\mathbf{0}, \sigma^2\mathbf{I})$	Exact finite-sample inference
IV	drops $\text{rank}(\mathbf{X}) = p$	Estimability, analysis of variance, designed structure
V	audits every assumption	What breaks, how to see it, what inference survives
VI	drops unbiasedness	Shrinkage, selection, regularization, $p > n$
VII	drops $\text{Cov}(\boldsymbol{\varepsilon}) = \sigma^2\mathbf{I}$	Generalized least squares, general Gauss–Markov, mixed models
VIII	drops normality of $\mathbf{Y}$	Exponential families and generalized linear models
IX	drops the linear predictor	Polynomials, splines, smoothing, additive models
X	drops modeling only the mean	Quantile regression and distributional models

Choices this book makes

Geometry before distribution theory. Least squares is introduced as a nearest-point problem ([Chapter 6](#)) before any sampling theory. Fitted values, residuals and sums of squares are projections. Coefficients are coordinates, and coordinates depend on the basis you choose. This is the only framing in which rank-deficient models need no special treatment.

Estimability is an estimation idea. It appears in Part II, not as a preliminary to the analysis of variance. Putting it there is what allows [Part IV](#) to treat ANOVA as ordinary subspace comparison.

Computation is part of the theory. Conditioning, QR and the singular value decomposition come immediately after least squares ([Chapter 10](#)), because numerical stability and statistical stability are closely linked.

The Bayesian treatment is a thread, not a quarantined chapter. Boxed “Bayesian thread” sections close Parts II, III, VII and VIII. Each develops the Bayesian counterpart of the result just proved.

Departures have a theory. Part V opens by deriving what omitted variables, wrong covariance, non-normality, outliers and collinearity actually do to the estimator. Diagnostics and remedies come after that.

Material most regression texts omit. This book adds chapters on bootstrap and permutation inference ([Chapter 23](#)), errors in variables ([Chapter 24](#)), the causal interpretation of coefficients ([Chapter 25](#)), high-dimensional regression ([Chapter 28](#)) and missing data ([Chapter 41](#)).

Code

Every figure and every number obtained by computation is produced by a Python script in the book’s `code/` directory, and the scripts check their own claims with assertions. The code in the text is taken from those scripts. On this website each code block is a live cell: press **Run** (or Shift+Enter) to execute it in your browser, edit it, and run it again. Nothing needs to be installed; the first run downloads a Python environment with numpy, scipy, pandas and statsmodels. Cells on a page share their variables, and running a cell first runs the cells above it that have not run yet.

Sources

The coverage of this book was planned against six standard texts: Rencher and Schaalje (2008), Seber and Lee (2003), Christensen (2020), Sen and Srivastava (1990), Agresti (2015) and Fahrmeir et al. (2021). Their chapters were mapped onto a single prerequisite graph, and the order of this book respects that graph. Each chapter’s *Notes and sources* section names the treatments it drew on and the primary literature. The exposition, proofs, examples and exercises are original to this book.

How to Use This Book

Parts and chapters

Parts I–VI (Chapters 1–30) are the classical linear model, complete in itself: estimation, exact inference, designed experiments, diagnostics, and the move from unbiased estimation to regularization. Parts VII–X (Chapters 31–45) climb the rest of the ladder: correlated errors, non-normal responses, nonlinear predictors, and models for entire conditional distributions. Chapters are published here as they are written; the ones not yet on the site are named in plain text.

Reading paths

- **A first course on linear models.** Chapters 5–9, 11–12, 15, 19–20 and 29. Consult Part I as needed.
- **The theory spine.** Part I, then Chapters 6, 7, 8, 9, 11 and 31. This is the shortest route to a full structural understanding of the linear model.
- **For implementers.** Chapters 5, 6, 10 and 20, Part VI and Chapter 34.
- **A course on generalized models.** Chapters 5, 6 and 11, then Parts VII–X.

Prerequisites

Each chapter opens with a list of the earlier chapters it uses, taken from the prerequisite graph the book was planned against. Part I is written as reference material. Read it quickly the first time and come back to it when a later chapter sends you there.

Exercises

Exercises come at the end of each section in three groups: **A. Check your understanding** (routine: a definition or computation has been understood), **B. Practice** (core: the result is used later in the book) and **C. Going deeper** (hard: may need an idea not in the text). Many have a worked solution, folded under the exercise; try the exercise before opening it.

Notation

Symbol	Meaning
$\mathbf{Y}, \mathbf{y}$	response vector (random, observed), $n \times 1$
$\mathbf{X}$	model matrix, $n \times p$; columns $\mathbf{x}_1, \dots, \mathbf{x}_p$
$\boldsymbol{\beta}, \hat{\boldsymbol{\beta}}$	coefficient vector and a least squares estimate
$\boldsymbol{\varepsilon}$	error vector, $\boldsymbol{\varepsilon} = \mathbf{Y} - \mathbf{X}\hat{\boldsymbol{\beta}}$
$\hat{\mathbf{Y}}, \hat{\boldsymbol{\varepsilon}}$	fitted values $\mathbf{X}\hat{\boldsymbol{\beta}}$ and residuals $\mathbf{Y} - \hat{\mathbf{Y}}$
σ^2	error variance
$\mathbb{R}^n$	n -dimensional real space with $\langle \mathbf{u}, \mathbf{v} \rangle = \mathbf{u}^\top \mathbf{v}$
$\mathcal{C}(\mathbf{A}), \mathcal{N}(\mathbf{A})$	column space and null space of $\mathbf{A}$
$\mathcal{S}^\perp$	orthogonal complement of a subspace $\mathcal{S}$
$\mathbf{A}^\top, \mathbf{A}^-$	transpose; a generalized inverse ($\mathbf{A}\mathbf{A}^-\mathbf{A} = \mathbf{A}$)
$\mathbf{M}$	orthogonal projection onto $\mathcal{C}(\mathbf{X})$
$\mathbf{M}_0$	orthogonal projection onto the column space of a reduced model

Symbol	Meaning
rank, tr, diag	rank, trace, diagonal matrix
1 , 0 , I	vector of ones, zero vector, identity
$\ \cdot\ $	Euclidean norm unless stated otherwise
E, Var, Cov	expectation, variance, covariance (matrix)
$N_n(\boldsymbol{\mu}, \boldsymbol{\Sigma})$	n -variate normal distribution
$\chi^2(r, \gamma)$, $F(r, s, \gamma)$	noncentral distributions, noncentrality γ

Vectors are columns. Matrices and vectors are set in bold upright type and scalars in italic. A bare “projection” always means an orthogonal projection with respect to the Euclidean inner product. [Section 6.9](#) covers the other kinds.

Chapter 1

Matrix Algebra for Linear Models

A linear model is a statement about vectors and matrices, and nearly every argument in this book is an argument about them. This chapter collects the matrix algebra the book relies on: rank and the fundamental subspaces, inverses and generalized inverses, consistent systems of equations, orthogonal matrices, determinants and partitioned matrices, the spectral theorem and positive definiteness, the singular value decomposition, idempotent matrices, Kronecker products and the calculus of quadratic forms. It is a reference chapter, meant to be skimmed first and consulted later, but every central result is proved.

What you need. None beyond the Notation chapter. A first course in linear algebra helps, but is not assumed.

1.0.1 Roadmap

- [Vectors, matrices and the operations we use](#)
- [Rank, column space and null space](#)
- [Inverses and generalized inverses](#)
- [Systems of equations and consistency](#)
- [Orthogonality and orthogonal matrices](#)
- [Trace, determinants and partitioned matrices](#)
- [Eigenvalues, the spectral theorem and positive definiteness](#)
- [The singular value decomposition and conditioning](#)
- [Idempotent matrices](#)
- [Kronecker products and the vec operator](#)
- [Vector and matrix calculus](#)
- [Summary and notes](#)

1.1 Vectors, matrices and the operations we use

This chapter works in $\mathbb{R}^n$ and with real matrices only. It fixes the conventions, states the results later chapters cite, and proves them. A reader who already knows linear algebra can go straight

to the results that are specific to statistics: generalized inverses (Section 1.3 and Section 1.4), Schur complements and the Woodbury identity (Section 1.6), nonnegative definite matrices (Section 1.7), conditioning (Section 1.8) and idempotent matrices (Section 1.9). Orthogonal projections, the geometric heart of least squares, get only the minimum here. Chapter 6 develops them.

1.1.1 Conventions

An $m \times n$ matrix $\mathbf{A} = (a_{ij})$ has m rows and n columns. We write $\mathbf{A} = [\mathbf{a}_1, \dots, \mathbf{a}_n]$ for its columns and $\mathbf{a}_{(i)}^\top$ for its i th row. A **vector** is always a column, so $\mathbf{x} \in \mathbb{R}^n$ is an $n \times 1$ matrix and $\mathbf{x}^\top$ is a row. Bold upright letters denote vectors and matrices, and italic letters denote scalars. The special matrices we use constantly are the identity $\mathbf{I}_n$, the zero matrix $\mathbf{0}$, the vector of ones $\mathbf{1}_n$, the matrix of ones $\mathbf{J}_n = \mathbf{1}_n \mathbf{1}_n^\top$, and the coordinate vectors $\mathbf{e}_1, \dots, \mathbf{e}_n$, the columns of $\mathbf{I}_n$. Subscripts giving sizes are dropped when the size is clear. $\text{diag}(d_1, \dots, d_n)$ is the diagonal matrix with the given diagonal entries. A square matrix is **symmetric** if $\mathbf{A}^\top = \mathbf{A}$, *upper triangular* if $a_{ij} = 0$ for $i > j$, and *lower triangular* if $\mathbf{A}^\top$ is upper triangular.

The **transpose** satisfies $(\mathbf{A}^\top)^\top = \mathbf{A}$, $(\mathbf{A} + \mathbf{B})^\top = \mathbf{A}^\top + \mathbf{B}^\top$ and, most usefully,

$$(\mathbf{AB})^\top = \mathbf{B}^\top \mathbf{A}^\top. \quad (1.1.1)$$

1.1.2 Products as linear combinations

If $\mathbf{A}$ is $m \times n$ and $\mathbf{B}$ is $n \times q$, the (i, j) entry of $\mathbf{AB}$ is $\sum_k a_{ik} b_{kj}$. Three equivalent readings of the product carry most of the arguments in this book:

$$\mathbf{Ab} = \sum_{j=1}^n b_j \mathbf{a}_j, \quad \mathbf{AB} = [\mathbf{Ab}_1, \dots, \mathbf{Ab}_q], \quad \mathbf{AB} = \sum_{k=1}^n \mathbf{a}_k \mathbf{b}_{(k)}^\top. \quad (1.1.2)$$

The first says that $\mathbf{Ab}$ combines the columns of $\mathbf{A}$, with weights taken from $\mathbf{b}$. The second says that each column of $\mathbf{AB}$ is such a combination. The third writes a product as a sum of n rank-one matrices (outer products), and is the form in which spectral and singular value decompositions are usually used. Transposing the first reading shows that each row of $\mathbf{AB}$ is a combination of the rows of $\mathbf{B}$.

Matrices partitioned into conformable blocks multiply as if the blocks were scalars, as long as the order of factors is kept:

$$\begin{pmatrix} \mathbf{A}_{11} & \mathbf{A}_{12} \\ \mathbf{A}_{21} & \mathbf{A}_{22} \end{pmatrix} \begin{pmatrix} \mathbf{B}_{11} & \mathbf{B}_{12} \\ \mathbf{B}_{21} & \mathbf{B}_{22} \end{pmatrix} = \begin{pmatrix} \mathbf{A}_{11}\mathbf{B}_{11} + \mathbf{A}_{12}\mathbf{B}_{21} & \mathbf{A}_{11}\mathbf{B}_{12} + \mathbf{A}_{12}\mathbf{B}_{22} \\ \mathbf{A}_{21}\mathbf{B}_{11} + \mathbf{A}_{22}\mathbf{B}_{21} & \mathbf{A}_{21}\mathbf{B}_{12} + \mathbf{A}_{22}\mathbf{B}_{22} \end{pmatrix}.$$

Multiplication is associative and distributive but not commutative, and $\mathbf{AB} = \mathbf{0}$ does not force $\mathbf{A} = \mathbf{0}$ or $\mathbf{B} = \mathbf{0}$. The **Hadamard product** $\mathbf{A} \circ \mathbf{B} = (a_{ij} b_{ij})$ multiplies entrywise. It appears only occasionally.

1.1.3 Inner products, lengths and the Cauchy–Schwarz inequality

The **inner product** of $\mathbf{u}, \mathbf{v} \in \mathbb{R}^n$ is $\langle \mathbf{u}, \mathbf{v} \rangle = \mathbf{u}^\top \mathbf{v} = \sum_i u_i v_i$, and the **Euclidean norm** is $\|\mathbf{u}\| = (\mathbf{u}^\top \mathbf{u})^{1/2}$. The entries of $\mathbf{A}^\top \mathbf{A}$ are the inner products of the columns of $\mathbf{A}$, and those of $\mathbf{A}\mathbf{A}^\top$ are the inner products of its rows. In particular

$$\sum_{i,j} a_{ij}^2 = \sum_j \|\mathbf{a}_j\|^2, \quad (1.1.3)$$

so $\mathbf{A}^\top \mathbf{A} = \mathbf{0}$ forces every column to be zero, that is, $\mathbf{A} = \mathbf{0}$. The square root of Eq. 1.1.3 is the **Frobenius norm** $\|\mathbf{A}\|_F$.

Proposition 1.1.1 — Cauchy–Schwarz

For $\mathbf{u}, \mathbf{v} \in \mathbb{R}^n$, $(\mathbf{u}^\top \mathbf{v})^2 \leq (\mathbf{u}^\top \mathbf{u})(\mathbf{v}^\top \mathbf{v})$, with equality iff one vector is a multiple of the other. Consequently $\|\mathbf{u} + \mathbf{v}\| \leq \|\mathbf{u}\| + \|\mathbf{v}\|$.

Proof. If $\mathbf{v} = \mathbf{0}$ there is nothing to prove. Otherwise, the quadratic $q(t) = \|\mathbf{u} - t\mathbf{v}\|^2 = \mathbf{u}^\top \mathbf{u} - 2t \mathbf{u}^\top \mathbf{v} + t^2 \mathbf{v}^\top \mathbf{v}$ is nonnegative for all t , so its discriminant $4(\mathbf{u}^\top \mathbf{v})^2 - 4(\mathbf{u}^\top \mathbf{u})(\mathbf{v}^\top \mathbf{v})$ is at most zero. Equality holds iff q has a real root, that is, iff $\mathbf{u} = t\mathbf{v}$ for some t . The triangle inequality follows by expanding $\|\mathbf{u} + \mathbf{v}\|^2$ and bounding the cross term. ■

1.1.4 Quadratic forms

A **quadratic form** in $\mathbf{x} \in \mathbb{R}^n$ is a function $\mathbf{x}^\top \mathbf{A} \mathbf{x} = \sum_{i,j} a_{ij} x_i x_j$. Since $\mathbf{x}^\top \mathbf{A} \mathbf{x}$ is a scalar it equals its own transpose $\mathbf{x}^\top \mathbf{A}^\top \mathbf{x}$, so

$$\mathbf{x}^\top \mathbf{A} \mathbf{x} = \mathbf{x}^\top \left(\frac{1}{2} (\mathbf{A} + \mathbf{A}^\top) \right) \mathbf{x}. \quad (1.1.4)$$

Every quadratic form therefore has a symmetric matrix, and that matrix is unique: if $\mathbf{x}^\top \mathbf{S} \mathbf{x} = 0$ for all $\mathbf{x}$ with $\mathbf{S}$ symmetric, then taking $\mathbf{x} = \mathbf{e}_i$ gives $s_{ii} = 0$, and $\mathbf{x} = \mathbf{e}_i + \mathbf{e}_j$ gives $2s_{ij} = 0$. From now on we always take the matrix in a quadratic form to be symmetric.

Example. (Sums of squares as quadratic forms)

The sum of squared deviations of $\mathbf{y} \in \mathbb{R}^n$ from its mean is

$$\sum_{i=1}^n (y_i - \bar{y})^2 = \mathbf{y}^\top \mathbf{y} - n\bar{y}^2 = \mathbf{y}^\top \left(\mathbf{I}_n - \frac{1}{n} \mathbf{J}_n \right) \mathbf{y},$$

because $n\bar{y}^2 = (\mathbf{1}^\top \mathbf{y})^2 / n = \mathbf{y}^\top \mathbf{J}_n \mathbf{y} / n$. The **centring matrix** $\mathbf{I}_n - n^{-1} \mathbf{J}_n$ is symmetric and satisfies $(\mathbf{I}_n - n^{-1} \mathbf{J}_n)^2 = \mathbf{I}_n - n^{-1} \mathbf{J}_n$, because $\mathbf{J}_n^2 = n \mathbf{J}_n$. Every sum of squares

in an analysis of variance is a quadratic form whose matrix has these two properties (Section 1.9).

1.2 Rank, column space and null space

1.2.1 Subspaces and dimension

A nonempty set $\mathcal{S} \subseteq \mathbb{R}^n$ is a **subspace** if $\mathbf{u}, \mathbf{v} \in \mathcal{S}$ and $a, b \in \mathbb{R}$ imply $a\mathbf{u} + b\mathbf{v} \in \mathcal{S}$. Vectors $\mathbf{v}_1, \dots, \mathbf{v}_k$ are **linearly independent** if $\sum_j c_j \mathbf{v}_j = \mathbf{0}$ only for $c_1 = \dots = c_k = 0$, that is, if the matrix $[\mathbf{v}_1, \dots, \mathbf{v}_k]$ sends only the zero vector to zero. Their **span** is the set of all their linear combinations. A **basis** of $\mathcal{S}$ is an independent set that spans it.

We use the following facts of elementary linear algebra without proof (Halmos 1958, chapter 1). Every subspace of $\mathbb{R}^n$ has a basis. All bases of a subspace have the same number of elements, its **dimension** $\dim \mathcal{S}$. In an r -dimensional subspace, any r independent vectors form a basis, and any independent set can be extended to a basis. If $\mathcal{S} \subseteq \mathcal{T}$ and $\dim \mathcal{S} = \dim \mathcal{T}$, then $\mathcal{S} = \mathcal{T}$. Finally, if $\mathcal{S} \cap \mathcal{T} = \{\mathbf{0}\}$, the union of a basis of $\mathcal{S}$ and a basis of $\mathcal{T}$ is independent, so the sum $\mathcal{S} + \mathcal{T} = \{\mathbf{u} + \mathbf{v} : \mathbf{u} \in \mathcal{S}, \mathbf{v} \in \mathcal{T}\}$ has dimension $\dim \mathcal{S} + \dim \mathcal{T}$. Such a sum is called **direct** and written $\mathcal{S} \oplus \mathcal{T}$.

1.2.2 The four subspaces of a matrix

An $m \times n$ matrix $\mathbf{A}$ carries two subspaces of $\mathbb{R}^m$ and two of $\mathbb{R}^n$:

$$\begin{aligned} \mathcal{C}(\mathbf{A}) &= \{\mathbf{A}\mathbf{b} : \mathbf{b} \in \mathbb{R}^n\} \subseteq \mathbb{R}^m, & \mathcal{N}(\mathbf{A}) &= \{\mathbf{b} \in \mathbb{R}^n : \mathbf{A}\mathbf{b} = \mathbf{0}\} \subseteq \mathbb{R}^n, \\ \mathcal{C}(\mathbf{A}^\top) &\subseteq \mathbb{R}^n, & \mathcal{N}(\mathbf{A}^\top) &\subseteq \mathbb{R}^m. \end{aligned}$$

By Eq. 1.1.2, $\mathcal{C}(\mathbf{A})$ is the span of the columns, the **column space**, and $\mathcal{C}(\mathbf{A}^\top)$ is the span of the rows, the **row space**. $\mathcal{N}(\mathbf{A})$ is the **null space**. The **rank** of $\mathbf{A}$ is $\text{rank}(\mathbf{A}) = \dim \mathcal{C}(\mathbf{A})$, the largest number of independent columns. The number of independent rows, $\dim \mathcal{C}(\mathbf{A}^\top)$, looks like a different quantity. It is not.

Theorem 1.2.1 — Row rank equals column rank

For every matrix $\mathbf{A}$, $\dim \mathcal{C}(\mathbf{A}^\top) = \dim \mathcal{C}(\mathbf{A})$. That is, $\text{rank}(\mathbf{A}^\top) = \text{rank}(\mathbf{A})$. Moreover, if $\text{rank}(\mathbf{A}) = r \geq 1$ there are an $m \times r$ matrix $\mathbf{B}$ and an $r \times n$ matrix $\mathbf{F}$, both of rank r , with $\mathbf{A} = \mathbf{B}\mathbf{F}$.

Proof. Let $r = \text{rank}(\mathbf{A})$ and let the columns of the $m \times r$ matrix $\mathbf{B}$ be a basis of $\mathcal{C}(\mathbf{A})$. Each column $\mathbf{a}_j$ is a combination $\mathbf{B}\mathbf{f}_j$ of these, so $\mathbf{A} = \mathbf{B}\mathbf{F}$ with $\mathbf{F} = [\mathbf{f}_1, \dots, \mathbf{f}_n]$. By the transposed reading of Eq. 1.1.2, every row of $\mathbf{A} = \mathbf{B}\mathbf{F}$ is a combination of the r rows of $\mathbf{F}$, so $\dim \mathcal{C}(\mathbf{A}^\top) \leq r = \dim \mathcal{C}(\mathbf{A})$. Applying this to $\mathbf{A}^\top$ gives the reverse inequality. Finally $\text{rank}(\mathbf{F}) \leq r$ because $\mathbf{F}$ has r rows, and $r = \dim \mathcal{C}(\mathbf{A}^\top) \leq \dim \mathcal{C}(\mathbf{F}^\top) = \text{rank}(\mathbf{F})$, so $\text{rank}(\mathbf{F}) = r$. ■

The factorization $\mathbf{A} = \mathbf{B}\mathbf{F}$ is a **rank factorization**: $\mathbf{B}$ has **full column rank** (its columns are independent) and $\mathbf{F}$ has **full row rank**. It is a convenient tool for proofs, as [Proposition 1.9.1](#) shows.

Theorem 1.2.2 — Rank-nullity

For an $m \times n$ matrix $\mathbf{A}$, $\text{rank}(\mathbf{A}) + \dim \mathcal{N}(\mathbf{A}) = n$.

Proof. Let $\mathbf{v}_1, \dots, \mathbf{v}_k$ be a basis of $\mathcal{N}(\mathbf{A})$ and extend it to a basis $\mathbf{v}_1, \dots, \mathbf{v}_k, \mathbf{w}_1, \dots, \mathbf{w}_{n-k}$ of $\mathbb{R}^n$. Every $\mathbf{A}\mathbf{b}$ is a combination of $\mathbf{A}\mathbf{w}_1, \dots, \mathbf{A}\mathbf{w}_{n-k}$, since $\mathbf{A}\mathbf{v}_i = \mathbf{0}$. These vectors are independent: if $\sum_j c_j \mathbf{A}\mathbf{w}_j = \mathbf{0}$, then $\sum_j c_j \mathbf{w}_j \in \mathcal{N}(\mathbf{A})$ is a combination of the $\mathbf{v}_i$, and independence of the full basis forces every $c_j = 0$. So $\text{rank}(\mathbf{A}) = n - k$. ■

In a linear model with $n \times p$ model matrix $\mathbf{X}$, $\dim \mathcal{N}(\mathbf{X}) = p - \text{rank}(\mathbf{X})$ counts the directions in which the coefficient vector can move without changing the mean $\mathbf{X}\boldsymbol{\beta}$. $\mathbf{X}$ has full column rank iff $\mathcal{N}(\mathbf{X}) = \{\mathbf{0}\}$.

1.2.3 Rank of products and of $\mathbf{A}^\top \mathbf{A}$

Proposition 1.2.3

Let $\mathbf{A}$ be $m \times n$ and $\mathbf{B}$ be $n \times q$.

- (a) $\mathcal{C}(\mathbf{AB}) \subseteq \mathcal{C}(\mathbf{A})$ and $\mathcal{N}(\mathbf{B}) \subseteq \mathcal{N}(\mathbf{AB})$, so $\text{rank}(\mathbf{AB}) \leq \min\{\text{rank}(\mathbf{A}), \text{rank}(\mathbf{B})\}$.
- (b) If $\mathbf{B}$ has full row rank, then $\mathcal{C}(\mathbf{AB}) = \mathcal{C}(\mathbf{A})$. If $\mathbf{A}$ has full column rank, then $\mathcal{N}(\mathbf{AB}) = \mathcal{N}(\mathbf{B})$. So $\text{rank}(\mathbf{AB}) = \text{rank}(\mathbf{A})$ in the first case and $\text{rank}(\mathbf{AB}) = \text{rank}(\mathbf{B})$ in the second. In particular, multiplying by a nonsingular matrix does not change rank.
- (c) $\mathcal{N}(\mathbf{A}^\top \mathbf{A}) = \mathcal{N}(\mathbf{A})$ and $\mathcal{C}(\mathbf{A}^\top \mathbf{A}) = \mathcal{C}(\mathbf{A}^\top)$, so $\text{rank}(\mathbf{A}^\top \mathbf{A}) = \text{rank}(\mathbf{AA}^\top) = \text{rank}(\mathbf{A})$.
- (d) $\text{rank}(\mathbf{A} + \mathbf{C}) \leq \text{rank}(\mathbf{A}) + \text{rank}(\mathbf{C})$ for $\mathbf{C}$ of the same size as $\mathbf{A}$.

Proof.

- (a) is [Eq. 1.1.2](#), together with $\text{rank}(\mathbf{AB}) = \text{rank}(\mathbf{B}^\top \mathbf{A}^\top) \leq \text{rank}(\mathbf{B}^\top)$ by the first inclusion and [Theorem 1.2.1](#). (b) If $\mathbf{B}$ has full row rank, $\mathcal{C}(\mathbf{B}) = \mathbb{R}^n$ because it is an n -dimensional subspace of $\mathbb{R}^n$, so every $\mathbf{A}\mathbf{u}$ equals $\mathbf{A}\mathbf{B}\mathbf{v}$ for some $\mathbf{v}$. If $\mathbf{A}$ has full column rank, $\mathbf{A}\mathbf{B}\mathbf{v} = \mathbf{0}$ forces $\mathbf{B}\mathbf{v} = \mathbf{0}$. The rank statements follow from rank-nullity. (c) If $\mathbf{A}^\top \mathbf{A}\mathbf{v} = \mathbf{0}$ then $\|\mathbf{A}\mathbf{v}\|^2 = \mathbf{v}^\top \mathbf{A}^\top \mathbf{A}\mathbf{v} = 0$, so $\mathbf{A}\mathbf{v} = \mathbf{0}$. Thus the null spaces agree, and by [Theorem 1.2.2](#) $\text{rank}(\mathbf{A}^\top \mathbf{A}) = \text{rank}(\mathbf{A})$. Since $\mathcal{C}(\mathbf{A}^\top \mathbf{A}) \subseteq \mathcal{C}(\mathbf{A}^\top)$ and both have dimension $\text{rank}(\mathbf{A})$, they are equal. Applying this to $\mathbf{A}^\top$ gives $\text{rank}(\mathbf{AA}^\top) = \text{rank}(\mathbf{A})$. (d) $\mathcal{C}(\mathbf{A} + \mathbf{C}) \subseteq \mathcal{C}(\mathbf{A}) + \mathcal{C}(\mathbf{C})$, and the dimension of a sum is at most the sum of the dimensions. ■

Part (c) is the reason the normal equations $\mathbf{X}^\top \mathbf{X} \mathbf{b} = \mathbf{X}^\top \mathbf{y}$ always have a solution: the right-hand side lies in $\mathcal{C}(\mathbf{X}^\top) = \mathcal{C}(\mathbf{X}^\top \mathbf{X})$ (Section 1.4).

1.2.4 Nonsingular submatrices

Proposition 1.2.4

Let $\mathbf{A}$ have rank $r \geq 1$. Choose any r linearly independent rows, indexed by I , and any r linearly independent columns, indexed by J . Then the $r \times r$ submatrix $\mathbf{A}_{IJ}$ at their intersection is nonsingular, and

$$\mathbf{A} = \mathbf{A}_{\cdot J} \mathbf{A}_{IJ}^{-1} \mathbf{A}_{I\cdot}, \quad (1.2.1)$$

where $\mathbf{A}_{\cdot J}$ holds the columns in J and $\mathbf{A}_{I\cdot}$ the rows in I .

Proof. The columns in J are a basis of $\mathcal{C}(\mathbf{A})$, so $\mathbf{A} = \mathbf{A}_{\cdot J} \mathbf{F}$ for some $r \times n$ matrix $\mathbf{F}$. Keeping only the rows in I gives $\mathbf{A}_{I\cdot} = \mathbf{A}_{IJ} \mathbf{F}$. The left side has rank r , so by Proposition 1.2.3(a) the $r \times r$ matrix $\mathbf{A}_{IJ}$ has rank r and is nonsingular. Then $\mathbf{F} = \mathbf{A}_{IJ}^{-1} \mathbf{A}_{I\cdot}$, which gives Eq. 1.2.1. ■

Conversely, a nonsingular $k \times k$ submatrix makes the k columns of $\mathbf{A}$ that contain it linearly independent, so $\text{rank}(\mathbf{A}) \geq k$. In particular, a matrix has rank r iff it has a nonsingular $r \times r$ submatrix and no nonsingular larger one. This proposition gives an explicit generalized inverse in Theorem 1.3.2.

1.2.5 Exercises

1.2.5.1 A. Check your understanding

Exercise A1.

Show that $\sum_{i < j} (y_i - y_j)^2 = n \sum_i (y_i - \bar{y})^2$, and write both sides as $\mathbf{y}^\top \mathbf{A} \mathbf{y}$ with $\mathbf{A}$ symmetric. Find the symmetric matrix of the quadratic form $2x_1^2 - x_2^2 + 6x_1x_2 - 4x_2x_3$.

Exercise A2.

Show that $\mathbf{X}^\top \mathbf{X} \mathbf{B} = \mathbf{X}^\top \mathbf{X} \mathbf{C}$ implies $\mathbf{X} \mathbf{B} = \mathbf{X} \mathbf{C}$. Deduce that $\mathbf{X}^\top \mathbf{X} \mathbf{b} = \mathbf{0}$ iff $\mathbf{X} \mathbf{b} = \mathbf{0}$.

Solution. Put $\mathbf{D} = \mathbf{B} - \mathbf{C}$, so that $\mathbf{X}^\top \mathbf{X} \mathbf{D} = \mathbf{0}$. Then $(\mathbf{X} \mathbf{D})^\top (\mathbf{X} \mathbf{D}) = \mathbf{D}^\top \mathbf{X}^\top \mathbf{X} \mathbf{D} = \mathbf{0}$, and by Eq. 1.1.3 a matrix $\mathbf{Z}$ with $\mathbf{Z}^\top \mathbf{Z} = \mathbf{0}$ is zero. So $\mathbf{X} \mathbf{D} = \mathbf{0}$. The second statement is the case $\mathbf{C} = \mathbf{0}$, with the converse obvious.

Exercise A3.

For the additive two-way layout with two rows, three columns and one observation per cell, write the 6×6 model matrix $\mathbf{X} = [\mathbf{1}, \mathbf{R}, \mathbf{C}]$, where $\mathbf{R}$ and $\mathbf{C}$ hold the row and column indicators. Find $\text{rank}(\mathbf{X})$ and a basis of $\mathcal{N}(\mathbf{X})$. Compare $[\mathbf{R}, \mathbf{C}]^\top$ with the matrix $\mathbf{A}$ of [Example \(Filling a table from its margins\)](#).

1.2.5.2 B. Practice**Exercise B1.**

Let $\mathbf{A}$ be $m \times n$ and $\mathbf{B}$ be $n \times q$. Prove $\text{rank}(\mathbf{AB}) \geq \text{rank}(\mathbf{A}) + \text{rank}(\mathbf{B}) - n$. *Hint:* apply rank-nullity to $\mathbf{AB}_0$, where the columns of $\mathbf{B}_0$ are a basis of $\mathcal{C}(\mathbf{B})$.

Solution. Let the $n \times s$ matrix $\mathbf{B}_0$, $s = \text{rank}(\mathbf{B})$, have columns forming a basis of $\mathcal{C}(\mathbf{B})$. Then $\mathcal{C}(\mathbf{AB}) = \mathcal{C}(\mathbf{AB}_0)$, so by rank-nullity $\text{rank}(\mathbf{AB}) = s - \dim \mathcal{N}(\mathbf{AB}_0)$. The map $\mathbf{c} \mapsto \mathbf{B}_0 \mathbf{c}$ is one-to-one and sends $\mathcal{N}(\mathbf{AB}_0)$ into $\mathcal{N}(\mathbf{A})$, so $\dim \mathcal{N}(\mathbf{AB}_0) \leq \dim \mathcal{N}(\mathbf{A}) = n - \text{rank}(\mathbf{A})$. Hence $\text{rank}(\mathbf{AB}) \geq \text{rank}(\mathbf{B}) - n + \text{rank}(\mathbf{A})$.

Exercise B2.

Show that if $\mathbf{A} = \mathbf{B}_1 \mathbf{F}_1 = \mathbf{B}_2 \mathbf{F}_2$ are two rank factorizations, then $\mathbf{B}_2 = \mathbf{B}_1 \mathbf{T}$ and $\mathbf{F}_2 = \mathbf{T}^{-1} \mathbf{F}_1$ for a nonsingular $\mathbf{T}$.

1.3 Inverses and generalized inverses

1.3.1 The inverse of a square matrix

An $n \times n$ matrix $\mathbf{A}$ is **nonsingular** if there is a matrix $\mathbf{A}^{-1}$ with $\mathbf{AA}^{-1} = \mathbf{A}^{-1}\mathbf{A} = \mathbf{I}_n$. The inverse is unique: if $\mathbf{B}$ and $\mathbf{C}$ both qualify, then $\mathbf{B} = \mathbf{B}(\mathbf{AC}) = (\mathbf{BA})\mathbf{C} = \mathbf{C}$.

Proposition 1.3.1

For an $n \times n$ matrix $\mathbf{A}$ the following are equivalent: (a) $\mathbf{A}$ is nonsingular; (b) $\text{rank}(\mathbf{A}) = n$; (c) $\mathcal{N}(\mathbf{A}) = \{\mathbf{0}\}$; (d) $\mathbf{BA} = \mathbf{I}$ for some $\mathbf{B}$; (e) $\mathbf{AC} = \mathbf{I}$ for some $\mathbf{C}$. When they hold, $\mathbf{B} = \mathbf{C} = \mathbf{A}^{-1}$, and

$$(\mathbf{A}^\top)^{-1} = (\mathbf{A}^{-1})^\top, \quad (\mathbf{AB})^{-1} = \mathbf{B}^{-1}\mathbf{A}^{-1}$$

for nonsingular $\mathbf{A}, \mathbf{B}$ of the same size. A nonsingular symmetric matrix has a symmetric inverse.

Proof. (b) $\Leftrightarrow$ (c) is rank-nullity. (a) $\Rightarrow$ (d) and (a) $\Rightarrow$ (e) are trivial. (d) $\Rightarrow$ (c): $\mathbf{Ax} = \mathbf{0}$ gives $\mathbf{x} = \mathbf{BAx} = \mathbf{0}$. (e) $\Rightarrow$ (b): $\mathbf{I} = \mathbf{AC}$ has rank n , and $\text{rank}(\mathbf{AC}) \leq \text{rank}(\mathbf{A})$. (b) $\Rightarrow$ (a): since $\mathcal{C}(\mathbf{A}) = \mathbb{R}^n$, we can solve $\mathbf{Ac}_j = \mathbf{e}_j$ for each j , giving $\mathbf{AC} = \mathbf{I}$. Then $\mathbf{A}(\mathbf{CA} - \mathbf{I}) = \mathbf{0}$,

and because $\mathcal{N}(\mathbf{A}) = \{\mathbf{0}\}$ each column of $\mathbf{CA} - \mathbf{I}$ is zero. So $\mathbf{CA} = \mathbf{I}$ as well. If $\mathbf{BA} = \mathbf{I}$ then $\mathbf{B} = \mathbf{BAA}^{-1} = \mathbf{A}^{-1}$, and similarly for $\mathbf{C}$. The formulas are verified by multiplication: $\mathbf{A}^\top (\mathbf{A}^{-1})^\top = (\mathbf{A}^{-1}\mathbf{A})^\top = \mathbf{I}$ and $(\mathbf{AB})(\mathbf{B}^{-1}\mathbf{A}^{-1}) = \mathbf{I}$. If $\mathbf{A} = \mathbf{A}^\top$, then $\mathbf{A}^{-1} = (\mathbf{A}^\top)^{-1} = (\mathbf{A}^{-1})^\top$.

■

A rectangular matrix has no two-sided inverse. If $\mathbf{X}$ is $n \times p$ with full column rank, then $\mathbf{X}^\top \mathbf{X}$ is nonsingular by Proposition 1.2.3(c), and $(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top$ is a *left inverse*: it multiplies $\mathbf{X}$ on the left to give $\mathbf{I}_p$. This is the matrix that computes least squares coefficients.

1.3.2 Generalized inverses

When $\mathbf{A}$ is singular or rectangular we still want a matrix that inverts $\mathbf{A}$ on the vectors where inversion makes sense, namely those in $\mathcal{C}(\mathbf{A})$. The following weak requirement turns out to be exactly what solving equations needs.

Definition 1.3.1 — Generalized inverse

A **generalized inverse** of an $m \times n$ matrix $\mathbf{A}$ is any $n \times m$ matrix $\mathbf{A}^-$ with

$$\mathbf{AA}^-\mathbf{A} = \mathbf{A}.$$

Read columnwise, the definition says that $\mathbf{AA}^-$ acts as the identity on every column of $\mathbf{A}$, hence on all of $\mathcal{C}(\mathbf{A})$: if $\mathbf{u} = \mathbf{Ab}$ then $\mathbf{A}(\mathbf{A}^-\mathbf{u}) = \mathbf{AA}^-\mathbf{Ab} = \mathbf{u}$. So $\mathbf{A}^-\mathbf{u}$ is a solution of $\mathbf{Ax} = \mathbf{u}$ whenever one exists. If $\mathbf{A}$ is nonsingular, multiplying the definition by $\mathbf{A}^{-1}$ on both sides gives $\mathbf{A}^- = \mathbf{A}^{-1}$. In general there are many generalized inverses. Chapter 6 uses them to write projections, and Chapter 8 uses them to describe estimable functions in models of less than full rank.

Theorem 1.3.2 — Existence

Every matrix has a generalized inverse. Explicitly, if $\text{rank}(\mathbf{A}) = r \geq 1$ and $\mathbf{A}_{IJ}$ is the nonsingular $r \times r$ submatrix of Proposition 1.2.4, the $n \times m$ matrix $\mathbf{G}$ that holds $\mathbf{A}_{IJ}^{-1}$ in the rows indexed by J and the columns indexed by I , and zeros elsewhere, is a generalized inverse of $\mathbf{A}$.

Proof. For $\mathbf{A} = \mathbf{0}$ any matrix of the right size works. Otherwise let $\mathbf{E}_I$ and $\mathbf{E}_J$ be the columns of $\mathbf{I}_m$ and $\mathbf{I}_n$ indexed by I and J . Then $\mathbf{G} = \mathbf{E}_J \mathbf{A}_{IJ}^{-1} \mathbf{E}_I^\top$, $\mathbf{AE}_J = \mathbf{A}_{.J}$ and $\mathbf{E}_I^\top \mathbf{A} = \mathbf{A}_{I.}$. So $\mathbf{AGA} = \mathbf{A}_{.J} \mathbf{A}_{IJ}^{-1} \mathbf{A}_{I.} = \mathbf{A}$ by Eq. 1.2.1.

■

Proposition 1.3.3 — Properties of generalized inverses

Let $\mathbf{A}$ be $m \times n$ of rank r and let $\mathbf{G}$ be a generalized inverse of $\mathbf{A}$.

- (a) $\mathbf{AG}$ and $\mathbf{GA}$ are idempotent. Moreover $\mathcal{C}(\mathbf{AG}) = \mathcal{C}(\mathbf{A})$ and $\mathcal{N}(\mathbf{GA}) = \mathcal{N}(\mathbf{A})$, and $\text{rank}(\mathbf{AG}) = \text{rank}(\mathbf{GA}) = r \leq \text{rank}(\mathbf{G})$.
- (b) $\mathbf{G}^\top$ is a generalized inverse of $\mathbf{A}^\top$.
- (c) If $\mathbf{G}_1, \mathbf{G}_2$ are generalized inverses of $\mathbf{A}$, so is $\mathbf{G}_1\mathbf{AG}_2$. In particular $\mathbf{GAG}$ is a generalized inverse which also satisfies $(\mathbf{GAG})\mathbf{A}(\mathbf{GAG}) = \mathbf{GAG}$.
- (d) If $\mathbf{A}$ is symmetric, then $\mathbf{G}^\top$ and $\frac{1}{2}(\mathbf{G} + \mathbf{G}^\top)$ are generalized inverses of $\mathbf{A}$, so a symmetric generalized inverse always exists.
- (e) $\mathbf{G} + (\mathbf{I} - \mathbf{GA})\mathbf{U} + \mathbf{V}(\mathbf{I} - \mathbf{AG})$ is a generalized inverse of $\mathbf{A}$ for every $\mathbf{U}$ and $\mathbf{V}$ of size $n \times m$.

Proof.

- (a) $(\mathbf{AG})^2 = (\mathbf{AGA})\mathbf{G} = \mathbf{AG}$, and likewise for $\mathbf{GA}$. Next, $\mathcal{C}(\mathbf{A}) = \mathcal{C}(\mathbf{AGA}) \subseteq \mathcal{C}(\mathbf{AG}) \subseteq \mathcal{C}(\mathbf{A})$, and $\mathcal{N}(\mathbf{A}) \subseteq \mathcal{N}(\mathbf{GA}) \subseteq \mathcal{N}(\mathbf{AGA}) = \mathcal{N}(\mathbf{A})$. The ranks follow from these and rank-nullity, and $r = \text{rank}(\mathbf{AG}) \leq \text{rank}(\mathbf{G})$.
- (b) Transpose $\mathbf{AGA} = \mathbf{A}$. (c) $\mathbf{AG}_1\mathbf{AG}_2\mathbf{A} = \mathbf{AG}_2\mathbf{A} = \mathbf{A}$. The second statement is $\mathbf{GAGAGAG} = \mathbf{G}(\mathbf{AGA})\mathbf{GAG} = \mathbf{GAGAG} = \mathbf{GAG}$.
- (c) By (b), $\mathbf{G}^\top$ is a generalized inverse of $\mathbf{A}^\top = \mathbf{A}$. The condition $\mathbf{AGA} = \mathbf{A}$ is linear in $\mathbf{G}$, so it is preserved by averaging.
- (d) Multiply out, using $\mathbf{A}(\mathbf{I} - \mathbf{GA}) = \mathbf{0}$ and $(\mathbf{I} - \mathbf{AG})\mathbf{A} = \mathbf{0}$.

■

A generalized inverse with the extra property in (c), $\mathbf{GAG} = \mathbf{G}$, is called **reflexive**. [Exercise \(B1\)](#) shows that the reflexive ones are exactly those of rank r . Part (e) in fact produces *every* generalized inverse from a single one as $\mathbf{U}$ and $\mathbf{V}$ vary. Idempotence in (a) means that $\mathbf{AG}$ and $\mathbf{GA}$ are projections, generally not orthogonal ones. [Chapter 6](#) shows that the particular combination $\mathbf{X}(\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{X}^\top$ is symmetric, does not depend on the choice of generalized inverse, and is the orthogonal projection onto $\mathcal{C}(\mathbf{X})$.

1.3.3 The Moore–Penrose inverse

One generalized inverse is singled out by symmetry requirements.

Definition 1.3.2 — Moore–Penrose inverse

The **Moore–Penrose inverse** of $\mathbf{A}$ is the matrix $\mathbf{A}^+$ satisfying

$$\mathbf{AA}^+\mathbf{A} = \mathbf{A}, \quad \mathbf{A}^+\mathbf{AA}^+ = \mathbf{A}^+, \quad (\mathbf{AA}^+)^\top = \mathbf{AA}^+, \quad (\mathbf{A}^+\mathbf{A})^\top = \mathbf{A}^+\mathbf{A}.$$

There is at most one such matrix. If $\mathbf{G}$ and $\mathbf{H}$ both satisfy the four conditions, then

$$\mathbf{G} = \mathbf{GAG} = \mathbf{G}(\mathbf{AG})^\top = \mathbf{GG}^\top(\mathbf{AHA})^\top = \mathbf{G}(\mathbf{AG})^\top(\mathbf{AH})^\top = \mathbf{GAGAH} = \mathbf{GAH},$$

where the steps use the second and third conditions for $\mathbf{G}$, the first for $\mathbf{H}$, and the third for both. The same argument with the fourth condition in place of the third gives

$$\mathbf{H} = (\mathbf{HA})^\top\mathbf{H} = \mathbf{A}^\top\mathbf{H}^\top\mathbf{H} = (\mathbf{AGA})^\top\mathbf{H}^\top\mathbf{H} = (\mathbf{GA})^\top(\mathbf{HA})^\top\mathbf{H} = \mathbf{GAHAH} = \mathbf{GAH},$$

so $\mathbf{G} = \mathbf{H}$. Existence follows from the singular value decomposition, which gives $\mathbf{A}^+$ explicitly (Theorem 1.8.1).

1.4 Systems of equations and consistency

A system $\mathbf{Ax} = \mathbf{b}$ with $\mathbf{A}$ of size $m \times n$ is **consistent** if it has at least one solution. Least squares leads to such systems in which $\mathbf{A} = \mathbf{X}^\top \mathbf{X}$ may be singular, so we need to know when solutions exist, what all of them look like, and which features are common to all of them.

Theorem 1.4.1 — Consistency and the general solution

Let $\mathbf{G}$ be any generalized inverse of $\mathbf{A}$. The following are equivalent:

- (a) $\mathbf{Ax} = \mathbf{b}$ is consistent;
- (b) $\mathbf{b} \in \mathcal{C}(\mathbf{A})$;
- (c) $\text{rank}[\mathbf{A}, \mathbf{b}] = \text{rank}(\mathbf{A})$;
- (d) $\mathbf{AGb} = \mathbf{b}$.

If the system is consistent, $\mathbf{x}_0 = \mathbf{Gb}$ is a solution, and the set of all solutions is

$$\{\mathbf{Gb} + (\mathbf{I} - \mathbf{GA})\mathbf{z} : \mathbf{z} \in \mathbb{R}^n\} = \mathbf{x}_0 + \mathcal{N}(\mathbf{A}). \quad (1.4.1)$$

Proof. (a) $\Leftrightarrow$ (b) is the first reading in Eq. 1.1.2. (b) $\Leftrightarrow$ (c): appending $\mathbf{b}$ as a column leaves the column space unchanged iff $\mathbf{b}$ is already in it, and a subspace of $\mathcal{C}([\mathbf{A}, \mathbf{b}])$ with the same dimension equals it. (d) $\Rightarrow$ (a): $\mathbf{Gb}$ is a solution. (b) $\Rightarrow$ (d): if $\mathbf{b} = \mathbf{Au}$, then $\mathbf{AGb} = \mathbf{AGA}\mathbf{u} = \mathbf{Au} = \mathbf{b}$.

The solutions of a consistent system are $\mathbf{x}_0 + \mathcal{N}(\mathbf{A})$, since $\mathbf{Ax} = \mathbf{b}$ iff $\mathbf{A}(\mathbf{x} - \mathbf{x}_0) = \mathbf{0}$. It remains to show $\mathcal{N}(\mathbf{A}) = \mathcal{C}(\mathbf{I} - \mathbf{GA})$. The inclusion $\supseteq$ holds because $\mathbf{A}(\mathbf{I} - \mathbf{GA}) = \mathbf{0}$. Conversely, if $\mathbf{Av} = \mathbf{0}$, then $\mathbf{v} = (\mathbf{I} - \mathbf{GA})\mathbf{v}$.

■

Criterion (d) is useful in practice: it tests consistency with a single matrix product, for any generalized inverse. By Proposition 1.3.3(a) and rank-nullity, $\text{rank}(\mathbf{I} - \mathbf{GA}) = n - r$, so the solution set is an affine subspace of dimension $n - r$. The solution is unique iff $\text{rank}(\mathbf{A}) = n$, and then $\mathbf{Gb}$ is the same for every $\mathbf{G}$.

Idea. Any generalized inverse does three jobs for $\mathbf{Ax} = \mathbf{b}$. It tests consistency ($\mathbf{AGb} = \mathbf{b}$), produces one solution ($\mathbf{Gb}$), and parametrizes all of them ($\mathbf{Gb} + (\mathbf{I} - \mathbf{GA})\mathbf{z}$). Which $\mathbf{G}$ is used changes the particular solution but never the solution set, and a linear function $\mathbf{q}^\top \mathbf{x}$ is the same across solutions exactly when $\mathbf{q} \in \mathcal{C}(\mathbf{A}^\top)$.

Corollary 1.4.2 — Invariant linear functions

Let $\mathbf{Ax} = \mathbf{b}$ be consistent and let $\mathbf{q} \in \mathbb{R}^n$. The value $\mathbf{q}^\top \mathbf{x}$ is the same for every solution

$\mathbf{x}$ iff $\mathbf{q} \in \mathcal{C}(\mathbf{A}^\top)$, that is, iff $\mathbf{q}^\top = \mathbf{q}^\top \mathbf{G} \mathbf{A}$ for one (and then every) generalized inverse $\mathbf{G}$.

Proof. By Eq. 1.4.1, $\mathbf{q}^\top \mathbf{x}$ is constant over solutions iff $\mathbf{q}^\top (\mathbf{I} - \mathbf{G} \mathbf{A}) \mathbf{z} = 0$ for all $\mathbf{z}$, that is, iff $\mathbf{q}^\top = \mathbf{q}^\top \mathbf{G} \mathbf{A}$. That equation puts $\mathbf{q}$ in $\mathcal{C}(\mathbf{A}^\top)$. Conversely, if $\mathbf{q} = \mathbf{A}^\top \mathbf{w}$, then $\mathbf{q}^\top \mathbf{G} \mathbf{A} = \mathbf{w}^\top \mathbf{A} \mathbf{G} \mathbf{A} = \mathbf{w}^\top \mathbf{A} = \mathbf{q}^\top$. ■

Applied to the normal equations $\mathbf{X}^\top \mathbf{X} \mathbf{b} = \mathbf{X}^\top \mathbf{y}$, which are consistent by Proposition 1.2.3(c), the corollary says that $\mathbf{q}^\top \hat{\boldsymbol{\beta}}$ is the same for all least squares solutions iff $\mathbf{q} \in \mathcal{C}(\mathbf{X}^\top \mathbf{X}) = \mathcal{C}(\mathbf{X}^\top)$. This is the algebra behind estimable functions (Theorem 8.3.1).

Proposition 1.4.3 — Minimum-norm solution

If $\mathbf{A} \mathbf{x} = \mathbf{b}$ is consistent, then $\mathbf{A}^+ \mathbf{b}$ is its unique solution lying in $\mathcal{C}(\mathbf{A}^\top)$, and $\|\mathbf{A}^+ \mathbf{b}\| < \|\mathbf{x}\|$ for every other solution $\mathbf{x}$.

Proof. $\mathbf{A}^+ \mathbf{b}$ is a solution because $\mathbf{A}^+$ is a generalized inverse. From the second and fourth Penrose conditions, $\mathbf{A}^+ = (\mathbf{A}^+ \mathbf{A}) \mathbf{A}^+ = (\mathbf{A}^+ \mathbf{A})^\top \mathbf{A}^+ = \mathbf{A}^\top (\mathbf{A}^+)^{\top} \mathbf{A}^+$, so $\mathbf{A}^+ \mathbf{b} \in \mathcal{C}(\mathbf{A}^\top)$. Every vector of $\mathcal{C}(\mathbf{A}^\top)$ is orthogonal to every $\mathbf{v} \in \mathcal{N}(\mathbf{A})$, since $(\mathbf{A}^\top \mathbf{w})^\top \mathbf{v} = \mathbf{w}^\top \mathbf{A} \mathbf{v} = 0$. Any other solution is $\mathbf{x} = \mathbf{A}^+ \mathbf{b} + \mathbf{v}$ with $\mathbf{0} \neq \mathbf{v} \in \mathcal{N}(\mathbf{A})$, and then $\|\mathbf{x}\|^2 = \|\mathbf{A}^+ \mathbf{b}\|^2 + \|\mathbf{v}\|^2$. The same computation shows that $\mathbf{x} \notin \mathcal{C}(\mathbf{A}^\top)$, because $\mathbf{x}^\top \mathbf{v} = \|\mathbf{v}\|^2 \neq 0$. ■

Example. (Filling a table from its margins)

Suppose the six entries of a 2×3 table are unknown, but its row totals (9, 6) and column totals (4, 5, 6) are known. With the cells ordered $x_{11}, x_{12}, x_{13}, x_{21}, x_{22}, x_{23}$, the equations are $\mathbf{A} \mathbf{x} = \mathbf{b}$ with

$$\mathbf{A} = \begin{pmatrix} 1 & 1 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 1 & 1 \\ 1 & 0 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 & 0 & 1 \end{pmatrix}, \quad \mathbf{b} = \begin{pmatrix} 9 \\ 6 \\ 4 \\ 5 \\ 6 \end{pmatrix}.$$

The rows of $\mathbf{A}$ satisfy one relation: the two row-total rows and the three column-total rows both add to $\mathbf{1}_6^\top$. So $\text{rank}(\mathbf{A}) = 4$ and there is a $6 - 4 = 2$ -dimensional family of solutions. The same relation must hold for $\mathbf{b}$. By Theorem 1.4.1(c), the system is consistent iff the row totals and the column totals have the same sum, and here both sums are 15.

The listing builds three generalized inverses: $\mathbf{G}_1$ from the nonsingular submatrix in rows 1–4 and columns 1–4 (Theorem 1.3.2), the Moore–Penrose inverse $\mathbf{G}_2 = \mathbf{A}^+$, and $\mathbf{G}_3 = \mathbf{G}_1 + (\mathbf{I} - \mathbf{G}_1 \mathbf{A}) \mathbf{Z}$ for an integer matrix $\mathbf{Z}$ (Proposition 1.3.3(e)). They give the

solutions

$$\begin{aligned}\mathbf{G}_1\mathbf{b} &= (-2, 5, 6, 6, 0, 0)^\top, & \|\mathbf{G}_1\mathbf{b}\| &= 10.050, \\ \mathbf{G}_2\mathbf{b} &= (5/2, 3, 7/2, 3/2, 2, 5/2)^\top, & \|\mathbf{G}_2\mathbf{b}\| &= 6.325, \\ \mathbf{G}_3\mathbf{b} &= (21, -39, 27, -17, 44, -21)^\top, & \|\mathbf{G}_3\mathbf{b}\| &= 73.192.\end{aligned}$$

The individual cell x_{11} takes the values -2 , $5/2$ and 21 : margins do not determine cells. But $x_{12} + x_{13} - x_{21}$ equals 5 for all three, as [Corollary 1.4.2](#) predicts, because its coefficient vector is the first row of $\mathbf{A}$ minus the third. The minimum-norm solution has the additive form $x_{ij} = r_i/3 + c_j/2 - 15/6$ in terms of the row totals r_i and column totals c_j ([Exercise \(A2\)](#)). If the last column total is changed to 7 , then $\mathbf{A}\mathbf{G}\mathbf{b} \neq \mathbf{b}$ for each of the three matrices, and the script confirms that the test detects the inconsistency. The script also checks all 60 submatrices formed by 4 independent rows and 4 independent columns, and finds each one nonsingular, as [Proposition 1.2.4](#) requires.

```
from fractions import Fraction
from itertools import combinations
import numpy as np
m, n = A.shape
# cells ordered (1,1), (1,2), (1,3), (2,1), (2,2), (2,3)
A = np.array([[1, 1, 1, 0, 0, 0],      # row 1 total
              [0, 0, 0, 1, 1, 1],      # row 2 total
              [1, 0, 0, 1, 0, 0],      # column 1 total
              [0, 1, 0, 0, 1, 0],      # column 2 total
              [0, 0, 1, 0, 0, 1]],      # column 3 total
             dtype=float)
b = np.array([9, 6, 4, 5, 6], dtype=float) # both sets of totals add to 15
r = np.linalg.matrix_rank(A)
print("rank(A) =", r, " rank([A, b]) =", np.linalg.matrix_rank(np.column_stack([A, b]
)))

def ginverse_from_submatrix(A, rows, cols):
    """Place the inverse of the nonsingular block A[rows, cols] at [cols, rows]."""
    G = np.zeros(A.T.shape)
    G[np.ix_(cols, rows)] = np.linalg.inv(A[np.ix_(rows, cols)])
    return G

rows, cols = [0, 1, 2, 3], [0, 1, 2, 3] # r independent rows and
columns                                     columns
G1 = ginverse_from_submatrix(A, rows, cols)
G2 = np.linalg.pinv(A) # Moore-Penrose inverse
rng = np.random.default_rng(1)
Z = rng.integers(-2, 3, size=A.T.shape).astype(float)
G3 = G1 + (np.eye(n) - G1 @ A) @ Z # another, non-reflexive
one

for G in (G1, G2, G3):
    assert np.allclose(A @ G @ A, A) # each is a g-inverse
    assert np.allclose(A @ G @ b, b) # consistency test passes
    x = G @ b
    print(np.round(x, 4), " A x - b =", np.abs(A @ x - b).max())
```

1.4.1 Exercises

1.4.1.1 A. Check your understanding

Exercise A1.

Find all generalized inverses of a nonzero vector $\mathbf{a} \in \mathbb{R}^n$, regarded as an $n \times 1$ matrix. Which one is $\mathbf{a}^+$?

Exercise A2.

Generalize [Example \(Filling a table from its margins\)](#) to an $a \times b$ table with row totals r_i , column totals c_j and grand total $T = \sum_i r_i = \sum_j c_j$. Show that $x_{ij} = r_i/b + c_j/a - T/(ab)$ is a solution lying in $\mathcal{C}(\mathbf{A}^\top)$, and hence is the minimum-norm solution.

Solution. Row i of the proposed table sums to $r_i + \sum_j c_j/a - bT/(ab) = r_i$, and column j sums to $\sum_i r_i/b + c_j - T/b = c_j$. The rows of $\mathbf{A}$ are the row indicators and the column indicators, so $\mathcal{C}(\mathbf{A}^\top)$ consists of the tables of the form $x_{ij} = \alpha_i + \beta_j$. The proposed solution has this form. By [Proposition 1.4.3](#), it is $\mathbf{A}^+\mathbf{b}$. For $a = 2$, $b = 3$ and $T = 15$ this is the additive formula quoted in [Example \(Filling a table from its margins\)](#).

Exercise A3.

For which c and d is the system $x_1 + x_2 + x_3 = 2$, $x_1 - x_2 = c$, $3x_1 + x_2 + 2x_3 = d$ consistent? When it is, give its general solution in the form [Eq. 1.4.1](#).

1.4.1.2 B. Practice

Exercise B1.

Let $\mathbf{G}$ be a generalized inverse of $\mathbf{A}$. Show that $\mathbf{GAG} = \mathbf{G}$ iff $\text{rank}(\mathbf{G}) = \text{rank}(\mathbf{A})$.

Solution. If $\mathbf{GAG} = \mathbf{G}$, then $\text{rank}(\mathbf{G}) \leq \text{rank}(\mathbf{A})$ by [Proposition 1.2.3\(a\)](#), and $\text{rank}(\mathbf{A}) \leq \text{rank}(\mathbf{G})$ by [Proposition 1.3.3\(a\)](#). Conversely, suppose $\text{rank}(\mathbf{G}) = \text{rank}(\mathbf{A}) = r$. Since $\mathcal{C}(\mathbf{GA}) \subseteq \mathcal{C}(\mathbf{G})$ and $\text{rank}(\mathbf{GA}) = r$, the two column spaces coincide, so $\mathbf{G} = \mathbf{GAK}$ for some $\mathbf{K}$. Then $\mathbf{GAG} = \mathbf{GAGAK} = \mathbf{GAK} = \mathbf{G}$, using $\mathbf{AGA} = \mathbf{A}$.

Exercise B2.

Let $\mathbf{A} = \mathbf{BF}$ be a rank factorization. Show that $\mathbf{A}^+ = \mathbf{F}^\top(\mathbf{FF}^\top)^{-1}(\mathbf{B}^\top\mathbf{B})^{-1}\mathbf{B}^\top$. Deduce that $\mathbf{A}^+ = (\mathbf{A}^\top\mathbf{A})^{-1}\mathbf{A}^\top$ when $\mathbf{A}$ has full column rank.

Solution. Write $\mathbf{G}$ for the proposed matrix. Then $\mathbf{AG} = \mathbf{B}(\mathbf{B}^\top\mathbf{B})^{-1}\mathbf{B}^\top$ and $\mathbf{GA} = \mathbf{F}^\top(\mathbf{FF}^\top)^{-1}\mathbf{F}$, both symmetric. Also $\mathbf{AGA} = \mathbf{B}(\mathbf{B}^\top\mathbf{B})^{-1}\mathbf{B}^\top\mathbf{BF} = \mathbf{A}$ and $\mathbf{GAG} = \mathbf{F}^\top(\mathbf{FF}^\top)^{-1}\mathbf{FF}^\top(\mathbf{FF}^\top)^{-1}(\mathbf{B}^\top\mathbf{B})^{-1}\mathbf{B}^\top = \mathbf{G}$. By uniqueness $\mathbf{G} = \mathbf{A}^+$. If $\mathbf{A}$ has full column rank, take $\mathbf{B} = \mathbf{A}$ and $\mathbf{F} = \mathbf{I}$.

Exercise B3.

Let $\mathbf{A}$ be $m \times n$ of rank r and let $\mathbf{Ax} = \mathbf{b}$ be consistent with $\mathbf{b} \neq \mathbf{0}$. Show that the system has $n - r + 1$ linearly independent solutions and no more, and that every solution is an affine combination of them.

1.4.1.3 C. Going deeper
Exercise C1.

Show that $(\mathbf{A}^+)^+ = \mathbf{A}$, $(\mathbf{A}^\top)^+ = (\mathbf{A}^+)^^\top$, $(\mathbf{A}^\top \mathbf{A})^+ = \mathbf{A}^+ (\mathbf{A}^+)^^\top$ and $\mathbf{A}^+ = (\mathbf{A}^\top \mathbf{A})^+ \mathbf{A}^\top$. Give an example with $(\mathbf{AB})^+ \neq \mathbf{B}^+ \mathbf{A}^+$.

1.5 Orthogonality and orthogonal matrices

1.5.1 Orthogonal vectors and complements

Vectors $\mathbf{u}, \mathbf{v}$ are **orthogonal**, written $\mathbf{u} \perp \mathbf{v}$, if $\mathbf{u}^\top \mathbf{v} = 0$. For orthogonal vectors the Pythagorean identity $\|\mathbf{u} + \mathbf{v}\|^2 = \|\mathbf{u}\|^2 + \|\mathbf{v}\|^2$ holds. A set of nonzero, mutually orthogonal vectors is independent: if $\sum_j c_j \mathbf{v}_j = \mathbf{0}$, taking the inner product with $\mathbf{v}_k$ leaves $c_k \|\mathbf{v}_k\|^2 = 0$. A set is **orthonormal** if, in addition, each vector has unit length. The columns of an $n \times r$ matrix $\mathbf{Q}$ are orthonormal iff $\mathbf{Q}^\top \mathbf{Q} = \mathbf{I}_r$.

The **orthogonal complement** of a subspace $\mathcal{S} \subseteq \mathbb{R}^n$ is $\mathcal{S}^\perp = \{\mathbf{v} : \mathbf{v}^\top \mathbf{u} = 0 \text{ for all } \mathbf{u} \in \mathcal{S}\}$. The next proposition is all that this chapter needs about complements. [Chapter 6](#) builds the theory of orthogonal projections on it.

Proposition 1.5.1

Let $\mathbf{A}$ be $m \times n$ and let $\mathcal{S} \subseteq \mathbb{R}^n$ be a subspace.

- (a) $\mathcal{N}(\mathbf{A}) = \mathcal{C}(\mathbf{A}^\top)^\perp$ and $\mathcal{N}(\mathbf{A}^\top) = \mathcal{C}(\mathbf{A})^\perp$.
- (b) $\dim \mathcal{S}^\perp = n - \dim \mathcal{S}$, and $\mathbb{R}^n = \mathcal{S} \oplus \mathcal{S}^\perp$. In particular $\mathbb{R}^n = \mathcal{C}(\mathbf{A}^\top) \oplus \mathcal{N}(\mathbf{A})$ and $\mathbb{R}^m = \mathcal{C}(\mathbf{A}) \oplus \mathcal{N}(\mathbf{A}^\top)$.
- (c) $(\mathcal{S}^\perp)^\perp = \mathcal{S}$.

Proof.

- (a) $\mathbf{Av} = \mathbf{0}$ says that $\mathbf{v}$ is orthogonal to every row of $\mathbf{A}$, hence to their span. Apply this to $\mathbf{A}^\top$ for the second identity.
- (b) Put a basis of $\mathcal{S}$ (k vectors) into the columns of $\mathbf{B}$. Then $\mathcal{S}^\perp = \mathcal{N}(\mathbf{B}^\top)$, which has dimension $n - k$ by rank-nullity. If $\mathbf{u} \in \mathcal{S} \cap \mathcal{S}^\perp$ then $\mathbf{u}^\top \mathbf{u} = 0$, so the sum is direct and has dimension n , so it is $\mathbb{R}^n$.
- (c) $\mathcal{S} \subseteq (\mathcal{S}^\perp)^\perp$, and both have dimension $\dim \mathcal{S}$ by (b).

1.5.2 Gram–Schmidt and the QR factorization

Theorem 1.5.2 — Gram–Schmidt; QR factorization

Let $\mathbf{X} = [\mathbf{x}_1, \dots, \mathbf{x}_p]$ be $n \times p$ with full column rank. Define recursively

$$\mathbf{w}_j = \mathbf{x}_j - \sum_{k < j} (\mathbf{u}_k^\top \mathbf{x}_j) \mathbf{u}_k, \quad \mathbf{u}_j = \mathbf{w}_j / \|\mathbf{w}_j\|.$$

Then $\mathbf{u}_1, \dots, \mathbf{u}_p$ are orthonormal and, for each j , span the same subspace as $\mathbf{x}_1, \dots, \mathbf{x}_j$. Equivalently, $\mathbf{X} = \mathbf{Q}\mathbf{R}$ with $\mathbf{Q} = [\mathbf{u}_1, \dots, \mathbf{u}_p]$ satisfying $\mathbf{Q}^\top \mathbf{Q} = \mathbf{I}_p$ and $\mathbf{R}$ upper triangular with positive diagonal entries.

Proof. By induction on j . Suppose $\mathbf{u}_1, \dots, \mathbf{u}_{j-1}$ are orthonormal and span $\text{span}(\mathbf{x}_1, \dots, \mathbf{x}_{j-1})$. Then $\mathbf{w}_j \neq \mathbf{0}$, since otherwise $\mathbf{x}_j$ would lie in that span, contradicting independence. For $i < j$, $\mathbf{u}_i^\top \mathbf{w}_j = \mathbf{u}_i^\top \mathbf{x}_j - \sum_{k < j} (\mathbf{u}_k^\top \mathbf{x}_j) \mathbf{u}_k^\top \mathbf{u}_i = 0$. So $\mathbf{u}_1, \dots, \mathbf{u}_j$ are orthonormal, and they span $\text{span}(\mathbf{x}_1, \dots, \mathbf{x}_j)$ because $\mathbf{x}_j = \|\mathbf{w}_j\| \mathbf{u}_j + \sum_{k < j} (\mathbf{u}_k^\top \mathbf{x}_j) \mathbf{u}_k$. That last identity, read column by column, is $\mathbf{X} = \mathbf{Q}\mathbf{R}$ with $r_{jj} = \|\mathbf{w}_j\| > 0$ and $r_{kj} = \mathbf{u}_k^\top \mathbf{x}_j$ for $k < j$. ■

The factorization is unique ([Exercise \(B2\)](#)). Numerical software computes it by a more stable route, but the object is the same, and [Chapter 6](#) and [Chapter 10](#) rely on it. Two consequences are used repeatedly. First, any orthonormal set $\mathbf{u}_1, \dots, \mathbf{u}_r$ in $\mathbb{R}^n$ extends to an orthonormal basis: apply Gram–Schmidt to $\mathbf{u}_1, \dots, \mathbf{u}_r, \mathbf{e}_1, \dots, \mathbf{e}_n$, discarding each vector that depends on its predecessors. Second, every subspace has an orthonormal basis.

1.5.3 Orthogonal matrices

A square matrix $\mathbf{Q}$ is **orthogonal** if $\mathbf{Q}^\top \mathbf{Q} = \mathbf{I}$. By [Proposition 1.3.1](#), $\mathbf{Q}^\top$ is then the inverse of $\mathbf{Q}$, so also $\mathbf{Q}\mathbf{Q}^\top = \mathbf{I}$: the rows of an orthogonal matrix are orthonormal as well as its columns.

Proposition 1.5.3

Let $\mathbf{Q}$ and $\mathbf{P}$ be $n \times n$ orthogonal matrices.

- (a) $(\mathbf{Q}\mathbf{x})^\top (\mathbf{Q}\mathbf{v}) = \mathbf{x}^\top \mathbf{v}$ and $\|\mathbf{Q}\mathbf{x}\| = \|\mathbf{x}\|$ for all $\mathbf{x}, \mathbf{v}$.
- (b) $\mathbf{Q}^\top$ and $\mathbf{Q}\mathbf{P}$ are orthogonal.
- (c) Every entry satisfies $|q_{ij}| \leq 1$, and $\det \mathbf{Q} = \pm 1$.
- (d) If $[\mathbf{Q}_1, \mathbf{Q}_2]$ is orthogonal with $\mathbf{Q}_1$ of size $n \times r$, then $\mathbf{Q}_1 \mathbf{Q}_1^\top + \mathbf{Q}_2 \mathbf{Q}_2^\top = \mathbf{I}_n$, and $\mathcal{C}(\mathbf{Q}_2) = \mathcal{C}(\mathbf{Q}_1)^\perp$.

Proof.

- (a) $(\mathbf{Q}\mathbf{x})^\top \mathbf{Q}\mathbf{v} = \mathbf{x}^\top \mathbf{Q}^\top \mathbf{Q}\mathbf{v} = \mathbf{x}^\top \mathbf{v}$. (b) $(\mathbf{Q}\mathbf{P})^\top \mathbf{Q}\mathbf{P} = \mathbf{P}^\top \mathbf{P} = \mathbf{I}$.
 (b) Each column has unit length. For the determinant, see [Proposition 1.6.2](#): $1 = \det(\mathbf{Q}^\top \mathbf{Q}) = (\det \mathbf{Q})^2$. (d) The first statement is $\mathbf{Q}\mathbf{Q}^\top = \mathbf{I}$ written in blocks. For the second, $\mathbf{Q}_1^\top \mathbf{Q}_2 = \mathbf{0}$ gives $\mathcal{C}(\mathbf{Q}_2) \subseteq \mathcal{C}(\mathbf{Q}_1)^\perp$, and both have dimension $n - r$.

■

Part (a) says an orthogonal matrix is a rigid motion that fixes the origin: a rotation, a reflection, or a composition of the two. Changing coordinates by an orthogonal matrix changes neither lengths nor angles. This is why the distribution theory of [Chapter 4](#) can rotate a normal vector with independent components into a convenient coordinate system without changing its distribution.

Example. (Helmert matrices)

The $n \times n$ **Helmert matrix** $\mathbf{H}$ has first row $n^{-1/2}\mathbf{1}^\top$ and, for $k = 2, \dots, n$, k th row

$$\mathbf{h}_k^\top = \frac{1}{\sqrt{k(k-1)}} \left(\underbrace{1, \dots, 1}_{k-1}, -(k-1), 0, \dots, 0 \right).$$

Each row has unit length, since $(k-1) + (k-1)^2 = k(k-1)$. Row k is orthogonal to row 1 because its entries sum to zero. For $2 \leq j < k$, the nonzero entries of $\mathbf{h}_j$ sit in positions $1, \dots, j$, where $\mathbf{h}_k$ is constant, so $\mathbf{h}_j^\top \mathbf{h}_k$ is proportional to the sum of the entries of $\mathbf{h}_j$, which is zero. So $\mathbf{H}$ is orthogonal. For $\mathbf{y} \in \mathbb{R}^n$, $(\mathbf{H}\mathbf{y})_1 = \sqrt{n}\bar{y}$ and, by [Proposition 1.5.3\(a\)](#),

$$\sum_{k=2}^n (\mathbf{h}_k^\top \mathbf{y})^2 = \|\mathbf{H}\mathbf{y}\|^2 - n\bar{y}^2 = \sum_{i=1}^n (y_i - \bar{y})^2.$$

The sum of squared deviations is a sum of $n-1$ squares of orthonormal contrasts. With independent normal observations this gives a short proof that the sample mean and variance are independent ([Exercise \(B1\)](#)).

Two other families appear later. **Permutation matrices**, the identity with its rows reordered, are orthogonal. For a unit vector $\mathbf{u}$, the **Householder reflection** $\mathbf{I} - 2\mathbf{u}\mathbf{u}^\top$ is symmetric and orthogonal and reflects $\mathbb{R}^n$ in the hyperplane $\mathbf{u}^\perp$. Householder reflections are how QR factorizations are computed in practice ([Exercise \(B1\)](#)).

1.5.4 Exercises

1.5.4.1 A. Check your understanding

Exercise A1.

Show that every 2×2 orthogonal matrix is either a rotation $\begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix}$ (determinant 1) or a reflection $\begin{pmatrix} \cos \theta & \sin \theta \\ \sin \theta & -\cos \theta \end{pmatrix}$ (determinant -1). Find the line fixed by the reflection.

1.5.4.2 B. Practice

Exercise B1.

Let $\mathbf{u}$ be a unit vector and $\mathbf{H} = \mathbf{I} - 2\mathbf{u}\mathbf{u}^\top$. Show that $\mathbf{H}$ is symmetric and orthogonal, that $\mathbf{H}\mathbf{u} = -\mathbf{u}$, and that $\mathbf{H}\mathbf{v} = \mathbf{v}$ for $\mathbf{v} \perp \mathbf{u}$. Given $\mathbf{x}$ not a nonnegative multiple of $\mathbf{e}_1$, find $\mathbf{u}$ with $\mathbf{H}\mathbf{x} = \|\mathbf{x}\|\mathbf{e}_1$. Explain how repeating this column by column produces a QR factorization.

Solution. $\mathbf{H}^\top = \mathbf{H}$, and $\mathbf{H}^\top\mathbf{H} = \mathbf{I} - 4\mathbf{u}\mathbf{u}^\top + 4\mathbf{u}(\mathbf{u}^\top\mathbf{u})\mathbf{u}^\top = \mathbf{I}$. Also $\mathbf{H}\mathbf{u} = \mathbf{u} - 2\mathbf{u} = -\mathbf{u}$, and $\mathbf{H}\mathbf{v} = \mathbf{v}$ when $\mathbf{u}^\top\mathbf{v} = 0$. Let $\mathbf{w} = \mathbf{x} - \|\mathbf{x}\|\mathbf{e}_1 \neq \mathbf{0}$ and $\mathbf{u} = \mathbf{w}/\|\mathbf{w}\|$. Then $\mathbf{w}^\top\mathbf{w} = 2\|\mathbf{x}\|^2 - 2\|\mathbf{x}\|x_1 = 2\mathbf{w}^\top\mathbf{x}$, so $\mathbf{H}\mathbf{x} = \mathbf{x} - 2\mathbf{w}(\mathbf{w}^\top\mathbf{x})/(\mathbf{w}^\top\mathbf{w}) = \mathbf{x} - \mathbf{w} = \|\mathbf{x}\|\mathbf{e}_1$. Applying such a reflection $\mathbf{H}_1$ to the first column of $\mathbf{X}$, then a reflection acting on coordinates $2, \dots, n$ to the second column of $\mathbf{H}_1\mathbf{X}$, and so on, gives $\mathbf{H}_p \cdots \mathbf{H}_1\mathbf{X} = \begin{pmatrix} \mathbf{R} \\ \mathbf{0} \end{pmatrix}$ with $\mathbf{R}$ upper triangular. Hence $\mathbf{X} = \mathbf{Q}\mathbf{R}$, where $\mathbf{Q}$ holds the first p columns of the orthogonal matrix $\mathbf{H}_1 \cdots \mathbf{H}_p$. In floating point one uses $\mathbf{w} = \mathbf{x} + \text{sgn}(x_1)\|\mathbf{x}\|\mathbf{e}_1$ instead, which avoids cancellation. It gives $\mathbf{H}\mathbf{x} = -\text{sgn}(x_1)\|\mathbf{x}\|\mathbf{e}_1$, so diagonal entries of $\mathbf{R}$ may come out negative; changing the sign of such a row of $\mathbf{R}$ and of the matching column of $\mathbf{Q}$ at the end restores the positive diagonal of [Theorem 1.5.2](#).

Exercise B2.

Show that the QR factorization of a matrix with full column rank, with $\mathbf{R}$ having positive diagonal, is unique, and that $\mathbf{R}^\top$ is the Cholesky factor of $\mathbf{X}^\top\mathbf{X}$.

Solution. If $\mathbf{X} = \mathbf{Q}_1\mathbf{R}_1 = \mathbf{Q}_2\mathbf{R}_2$, then $\mathbf{X}^\top\mathbf{X} = \mathbf{R}_1^\top\mathbf{R}_1 = \mathbf{R}_2^\top\mathbf{R}_2$. Each $\mathbf{R}_k^\top$ is lower triangular with positive diagonal, so both are the Cholesky factor of $\mathbf{X}^\top\mathbf{X}$, which is unique by [Theorem 1.7.4\(d\)](#). So $\mathbf{R}_1 = \mathbf{R}_2$ and $\mathbf{Q}_1 = \mathbf{X}\mathbf{R}_1^{-1} = \mathbf{Q}_2$.

1.6 Trace, determinants and partitioned matrices

1.6.1 Trace

The **trace** of a square matrix is the sum of its diagonal entries, $\text{tr}(\mathbf{A}) = \sum_i a_{ii}$. It is linear, $\text{tr}(a\mathbf{A} + b\mathbf{B}) = a\text{tr}(\mathbf{A}) + b\text{tr}(\mathbf{B})$, and $\text{tr}(\mathbf{A}^\top) = \text{tr}(\mathbf{A})$.

Theorem 1.6.1 — Cyclic property

If $\mathbf{A}$ is $m \times n$ and $\mathbf{B}$ is $n \times m$, then $\text{tr}(\mathbf{AB}) = \text{tr}(\mathbf{BA})$.

Proof. $\text{tr}(\mathbf{AB}) = \sum_{i=1}^m \sum_{k=1}^n a_{ik}b_{ki} = \sum_{k=1}^n \sum_{i=1}^m b_{ki}a_{ik} = \text{tr}(\mathbf{BA})$. ■

The two products need not have the same size. Applied repeatedly, the theorem allows cyclic

permutations such as $\text{tr}(\mathbf{ABC}) = \text{tr}(\mathbf{CAB}) = \text{tr}(\mathbf{BCA})$, but not arbitrary reorderings. Three consequences occur again and again:

$$\text{tr}(\mathbf{A}^\top \mathbf{A}) = \sum_{i,j} a_{ij}^2 = \|\mathbf{A}\|_F^2, \quad \mathbf{x}^\top \mathbf{A} \mathbf{x} = \text{tr}(\mathbf{A} \mathbf{x} \mathbf{x}^\top), \quad \text{tr}(\mathbf{P}^{-1} \mathbf{A} \mathbf{P}) = \text{tr}(\mathbf{A}). \quad (1.6.1)$$

The middle one turns a scalar quadratic form into the trace of a matrix that is linear in $\mathbf{x} \mathbf{x}^\top$. Taking expectations then gives the mean of a random quadratic form ([Chapter 2](#)).

1.6.2 Determinants

The **determinant** of an $n \times n$ matrix is

$$\det \mathbf{A} = \sum_{\pi} \text{sgn}(\pi) a_{1\pi(1)} a_{2\pi(2)} \cdots a_{n\pi(n)},$$

the sum over all permutations π of $\{1, \dots, n\}$, where $\text{sgn}(\pi) = \pm 1$ according to whether π is a product of an even or odd number of transpositions. We collect the standard properties. Parts (a)–(d) are proved in any linear algebra text (Harville 1997, chapter 13), and the others follow from them.

Proposition 1.6.2 — Determinants

Let $\mathbf{A}, \mathbf{B}$ be $n \times n$.

- (a) $\det \mathbf{A}^\top = \det \mathbf{A}$.
- (b) $\det \mathbf{A}$ is a linear function of each column separately. It changes sign when two columns are interchanged, and is unchanged when a multiple of one column is added to another. The same holds for rows.
- (c) A triangular matrix has determinant equal to the product of its diagonal entries.
- (d) $\det(\mathbf{AB}) = \det \mathbf{A} \det \mathbf{B}$.
- (e) $\mathbf{A}$ is nonsingular iff $\det \mathbf{A} \neq 0$, and then $\det \mathbf{A}^{-1} = 1/\det \mathbf{A}$.
- (f) $\det(c\mathbf{A}) = c^n \det \mathbf{A}$, and $\det \mathbf{Q} = \pm 1$ for orthogonal $\mathbf{Q}$.

Proof (Proof of (e) and (f)). If $\mathbf{A}$ is nonsingular, $1 = \det \mathbf{I} = \det \mathbf{A} \det \mathbf{A}^{-1}$ by (d). If $\mathbf{A}$ is singular, some column is a combination of the others. Subtracting that combination leaves a zero column without changing the determinant, by (b), and the determinant of a matrix with a zero column is zero, since every term of the sum contains a factor from that column. For (f), apply (b) to each of the n columns, and use (a) and (d) on $\mathbf{Q}^\top \mathbf{Q} = \mathbf{I}$.

■

1.6.3 Schur complements and block determinants

Partitioned matrices arise whenever the regressors split into two groups, the model matrix $\mathbf{X} = [\mathbf{X}_1, \mathbf{X}_2]$ giving

$$\mathbf{X}^\top \mathbf{X} = \begin{pmatrix} \mathbf{X}_1^\top \mathbf{X}_1 & \mathbf{X}_1^\top \mathbf{X}_2 \\ \mathbf{X}_2^\top \mathbf{X}_1 & \mathbf{X}_2^\top \mathbf{X}_2 \end{pmatrix},$$

and whenever a random vector splits into two blocks (Chapter 3). In what follows

$$\mathbf{A} = \begin{pmatrix} \mathbf{A}_{11} & \mathbf{A}_{12} \\ \mathbf{A}_{21} & \mathbf{A}_{22} \end{pmatrix}, \quad (1.6.2)$$

with $\mathbf{A}_{11}$ of size $k \times k$ and $\mathbf{A}_{22}$ of size $(n - k) \times (n - k)$.

Definition 1.6.1 — Schur complement

If $\mathbf{A}_{11}$ is nonsingular, the **Schur complement** of $\mathbf{A}_{11}$ in $\mathbf{A}$ is $\mathbf{A}_{22 \cdot 1} = \mathbf{A}_{22} - \mathbf{A}_{21} \mathbf{A}_{11}^{-1} \mathbf{A}_{12}$.
If $\mathbf{A}_{22}$ is nonsingular, the Schur complement of $\mathbf{A}_{22}$ is $\mathbf{A}_{11 \cdot 2} = \mathbf{A}_{11} - \mathbf{A}_{12} \mathbf{A}_{22}^{-1} \mathbf{A}_{21}$.

Everything in this subsection comes from one identity, block Gaussian elimination. If $\mathbf{A}_{11}$ is nonsingular, multiplying out shows

$$\mathbf{A} = \begin{pmatrix} \mathbf{I} & \mathbf{0} \\ \mathbf{A}_{21} \mathbf{A}_{11}^{-1} & \mathbf{I} \end{pmatrix} \begin{pmatrix} \mathbf{A}_{11} & \mathbf{0} \\ \mathbf{0} & \mathbf{A}_{22 \cdot 1} \end{pmatrix} \begin{pmatrix} \mathbf{I} & \mathbf{A}_{11}^{-1} \mathbf{A}_{12} \\ \mathbf{0} & \mathbf{I} \end{pmatrix}. \quad (1.6.3)$$

The outer factors are block triangular with identity blocks on the diagonal, and are nonsingular with inverses obtained by changing the sign of the off-diagonal block.

Theorem 1.6.3 — Block determinants

Let $\mathbf{A}$ be partitioned as in Eq. 1.6.2.

- (a) If $\mathbf{A}_{12} = \mathbf{0}$ or $\mathbf{A}_{21} = \mathbf{0}$, then $\det \mathbf{A} = \det \mathbf{A}_{11} \det \mathbf{A}_{22}$.
- (b) If $\mathbf{A}_{11}$ is nonsingular, $\det \mathbf{A} = \det \mathbf{A}_{11} \det \mathbf{A}_{22 \cdot 1}$. If $\mathbf{A}_{22}$ is nonsingular, $\det \mathbf{A} = \det \mathbf{A}_{22} \det \mathbf{A}_{11 \cdot 2}$.
- (c) For $m \times n$ matrices $\mathbf{U}$ and $n \times m$ matrices $\mathbf{V}$, $\det(\mathbf{I}_m + \mathbf{UV}) = \det(\mathbf{I}_n + \mathbf{VU})$. If $\mathbf{B}$ is $m \times m$ and nonsingular, $\det(\mathbf{B} + \mathbf{UV}) = \det \mathbf{B} \det(\mathbf{I}_n + \mathbf{VB}^{-1} \mathbf{U})$.

Proof.

- (a) Suppose $\mathbf{A}_{12} = \mathbf{0}$ (otherwise transpose). In the permutation expansion, a term that is nonzero cannot use an entry of the zero block $\mathbf{A}_{12}$. So each of the first k rows takes its entry from the first k columns, and π maps $\{1, \dots, k\}$ onto itself. Such a permutation splits into a permutation of the first k indices and one of the rest. Its sign is the product of their signs, so the sum factors as $\det \mathbf{A}_{11} \det \mathbf{A}_{22}$.
- (b) Take determinants in Eq. 1.6.3, using (a) and Proposition 1.6.2(d). The outer factors have determinant 1. The second statement follows in the same way from the elimination that starts with $\mathbf{A}_{22}$.

- (c) Apply (b) in both ways to $\begin{pmatrix} \mathbf{I}_m & -\mathbf{U} \\ \mathbf{V} & \mathbf{I}_n \end{pmatrix}$, whose Schur complements are $\mathbf{I}_m + \mathbf{V}\mathbf{U}$ and $\mathbf{I}_n + \mathbf{U}\mathbf{V}$. For the second statement, write $\mathbf{B} + \mathbf{U}\mathbf{V} = \mathbf{B}(\mathbf{I}_m + \mathbf{B}^{-1}\mathbf{U}\mathbf{V})$ and apply the first. ■

Part (c) with $n = 1$ is the **matrix determinant lemma** $\det(\mathbf{B} + \mathbf{u}\mathbf{v}^\top) = \det \mathbf{B} (1 + \mathbf{v}^\top \mathbf{B}^{-1} \mathbf{u})$.

1.6.4 Inverses of partitioned matrices

Theorem 1.6.4 — Partitioned inverse

Let $\mathbf{A}$ be partitioned as in Eq. 1.6.2 with $\mathbf{A}_{11}$ nonsingular. Then $\mathbf{A}$ is nonsingular iff $\mathbf{A}_{22 \cdot 1}$ is, and in that case

$$\mathbf{A}^{-1} = \begin{pmatrix} \mathbf{A}_{11}^{-1} + \mathbf{A}_{11}^{-1} \mathbf{A}_{12} \mathbf{A}_{22 \cdot 1}^{-1} \mathbf{A}_{21} \mathbf{A}_{11}^{-1} & -\mathbf{A}_{11}^{-1} \mathbf{A}_{12} \mathbf{A}_{22 \cdot 1}^{-1} \\ -\mathbf{A}_{22 \cdot 1}^{-1} \mathbf{A}_{21} \mathbf{A}_{11}^{-1} & \mathbf{A}_{22 \cdot 1}^{-1} \end{pmatrix}. \quad (1.6.4)$$

If $\mathbf{A}_{22}$ is also nonsingular, the leading block of $\mathbf{A}^{-1}$ equals $\mathbf{A}_{11 \cdot 2}^{-1}$:

$$(\mathbf{A}_{11} - \mathbf{A}_{12} \mathbf{A}_{22}^{-1} \mathbf{A}_{21})^{-1} = \mathbf{A}_{11}^{-1} + \mathbf{A}_{11}^{-1} \mathbf{A}_{12} \mathbf{A}_{22 \cdot 1}^{-1} \mathbf{A}_{21} \mathbf{A}_{11}^{-1}. \quad (1.6.5)$$

Proof. The nonsingularity statement is Theorem 1.6.3(b). Inverting the three factors of Eq. 1.6.3 and multiplying them in reverse order gives

$$\mathbf{A}^{-1} = \begin{pmatrix} \mathbf{I} & -\mathbf{A}_{11}^{-1} \mathbf{A}_{12} \\ \mathbf{0} & \mathbf{I} \end{pmatrix} \begin{pmatrix} \mathbf{A}_{11}^{-1} & \mathbf{0} \\ \mathbf{0} & \mathbf{A}_{22 \cdot 1}^{-1} \end{pmatrix} \begin{pmatrix} \mathbf{I} & \mathbf{0} \\ -\mathbf{A}_{21} \mathbf{A}_{11}^{-1} & \mathbf{I} \end{pmatrix},$$

which multiplies out to Eq. 1.6.4. Eliminating with $\mathbf{A}_{22}$ first gives a second expression for $\mathbf{A}^{-1}$, whose leading block is $\mathbf{A}_{11 \cdot 2}^{-1}$. The inverse is unique, so the two leading blocks agree. ■

For a symmetric positive definite $\mathbf{X}^\top \mathbf{X}$ (Section 1.7), the theorem says that the block of $(\mathbf{X}^\top \mathbf{X})^{-1}$ belonging to $\mathbf{X}_1$ is $(\mathbf{X}_1^\top \mathbf{X}_1 - \mathbf{X}_1^\top \mathbf{X}_2 (\mathbf{X}_2^\top \mathbf{X}_2)^{-1} \mathbf{X}_2^\top \mathbf{X}_1)^{-1}$. Chapter 6 derives the same formula geometrically, from the Frisch–Waugh–Lovell theorem.

Identity Eq. 1.6.5 is a disguised form of a result that deserves its own name. Rename $\mathbf{A}_{11} = \mathbf{B}$, $\mathbf{A}_{12} = \mathbf{U}$, $\mathbf{A}_{21} = \mathbf{V}^\top$ and $\mathbf{A}_{22} = -\mathbf{C}^{-1}$.

Theorem 1.6.5 — Sherman–Morrison–Woodbury

Let $\mathbf{B}$ ($n \times n$) and $\mathbf{C}$ ($k \times k$) be nonsingular, and let $\mathbf{U}, \mathbf{V}$ be $n \times k$. If $\mathbf{C}^{-1} + \mathbf{V}^\top \mathbf{B}^{-1} \mathbf{U}$ is nonsingular, then so is $\mathbf{B} + \mathbf{U}\mathbf{C}\mathbf{V}^\top$, and

$$(\mathbf{B} + \mathbf{UCV}^\top)^{-1} = \mathbf{B}^{-1} - \mathbf{B}^{-1}\mathbf{U}(\mathbf{C}^{-1} + \mathbf{V}^\top\mathbf{B}^{-1}\mathbf{U})^{-1}\mathbf{V}^\top\mathbf{B}^{-1}. \quad (1.6.6)$$

In particular, for vectors $\mathbf{u}, \mathbf{v}$ with $1 + \mathbf{v}^\top\mathbf{B}^{-1}\mathbf{u} \neq 0$,

$$(\mathbf{B} + \mathbf{uv}^\top)^{-1} = \mathbf{B}^{-1} - \frac{\mathbf{B}^{-1}\mathbf{uv}^\top\mathbf{B}^{-1}}{1 + \mathbf{v}^\top\mathbf{B}^{-1}\mathbf{u}}. \quad (1.6.7)$$

Proof. Apply [Theorem 1.6.4](#) to $\mathbf{N} = \begin{pmatrix} \mathbf{B} & \mathbf{U} \\ \mathbf{V}^\top & \mathbf{C}^{-1} \end{pmatrix}$. Both diagonal blocks are nonsingular. The Schur complement of $\mathbf{B}$ is $-(\mathbf{C}^{-1} + \mathbf{V}^\top\mathbf{B}^{-1}\mathbf{U})$, which is nonsingular by hypothesis, so $\mathbf{N}$ is nonsingular. Then the Schur complement of the lower block, $\mathbf{B} + \mathbf{UCV}^\top$, is nonsingular by the same theorem with the roles of the blocks exchanged, and [Eq. 1.6.5](#) becomes [Eq. 1.6.6](#). For [Eq. 1.6.7](#) take $k = 1$ and $\mathbf{C} = 1$. ■

Idea. Every result in this section comes from one block elimination, [Eq. 1.6.3](#): a partitioned matrix is a unit triangular matrix times $\text{diag}(\mathbf{A}_{11}, \mathbf{A}_{22.1})$ times another. Taking determinants gives [Theorem 1.6.3](#), inverting the factors gives [Theorem 1.6.4](#), and eliminating in the other order and comparing gives the Woodbury identity. When a formula about a block is forgotten, redo the elimination.

The identity replaces the inversion of an $n \times n$ matrix by that of a $k \times k$ matrix. That matters when $\mathbf{B}^{-1}$ is already known or cheap, as in the following two examples.

Example. (Deleting an observation)

Removing the i th row $\mathbf{x}_i^\top$ from a model matrix changes $\mathbf{X}^\top\mathbf{X}$ into $\mathbf{X}^\top\mathbf{X} - \mathbf{x}_i\mathbf{x}_i^\top$. With $\mathbf{u} = -\mathbf{x}_i$, $\mathbf{v} = \mathbf{x}_i$ and $h_i = \mathbf{x}_i^\top(\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{x}_i$, [Eq. 1.6.7](#) gives

$$(\mathbf{X}^\top\mathbf{X} - \mathbf{x}_i\mathbf{x}_i^\top)^{-1} = (\mathbf{X}^\top\mathbf{X})^{-1} + \frac{(\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{x}_i\mathbf{x}_i^\top(\mathbf{X}^\top\mathbf{X})^{-1}}{1 - h_i},$$

valid whenever $h_i < 1$, and the determinant lemma gives $\det(\mathbf{X}^\top\mathbf{X} - \mathbf{x}_i\mathbf{x}_i^\top) = (1 - h_i)\det(\mathbf{X}^\top\mathbf{X})$. The quantity h_i is the leverage of observation i ([Chapter 6](#)), and these formulas are what make the deletion diagnostics of Chapter 20 cheap. For Brownlee's stack-loss data ($n = 21$, $p = 4$), the listing updates the inverse for each of the 21 deletions. The largest relative discrepancy from direct inversion is 1×10^{-13} . Observation 17 has the largest leverage, 0.412, and removing it multiplies $\det(\mathbf{X}^\top\mathbf{X})$ by 0.588.

```
import numpy as np
import statsmodels.api as sm
data = sm.datasets.stackloss.load_pandas().data
X = np.column_stack([np.ones(len(data)), data[["AIRFLOW", "WATERTEMP", "ACIDCONC"]]])
n, p = X.shape

A_inv = np.linalg.inv(X.T @ X)
h = np.einsum("ij,jk,ik->i", X, A_inv, X)           # leverages x_i' (X'X)^{-1} x_i
```

```

worst = 0.0
for i in range(n):
    x = X[i]
    # Sherman-Morrison with u = -x, v = x: (X'X - x x')^{-1}
    B_update = A_inv + np.outer(A_inv @ x, x @ A_inv) / (1 - h[i])
    X_del = np.delete(X, i, axis=0)
    B_direct = np.linalg.inv(X_del.T @ X_del)
    worst = max(worst, np.abs(B_update - B_direct).max() / np.abs(B_direct).max())
print(f"largest relative discrepancy over all {n} deletions: {worst:.1e}")

```

Example. (Diagonal plus low rank)

In factor models and in the linear mixed models of Chapter 32, covariance matrices have the form $\Sigma = D + LL^\top$ with D diagonal and positive and L of size $N \times k$, $k \ll N$. By Eq. 1.6.6 and Theorem 1.6.3(c),

$$\Sigma^{-1} = D^{-1} - D^{-1}L(I_k + L^\top D^{-1}L)^{-1}L^\top D^{-1}, \quad \det \Sigma = \det D \det(I_k + L^\top D^{-1}L).$$

Each needs only a $k \times k$ factorization. The script checks this with $N = 1500$ and $k = 4$. The solution of $\Sigma \mathbf{w} = \mathbf{v}$ agrees with a direct solve to a relative error of 8×10^{-14} .

Example. (Equicorrelation)

The matrix $\mathbf{E} = (1 - \rho)\mathbf{I}_n + \rho\mathbf{J}_n$ has unit diagonal and every off-diagonal entry equal to ρ . It is the correlation matrix of exchangeable observations. Taking $\mathbf{B} = (1 - \rho)\mathbf{I}$ and $\mathbf{u} = \rho\mathbf{1}$, $\mathbf{v} = \mathbf{1}$, for $\rho \neq 1$ and $1 + (n - 1)\rho \neq 0$,

$$\mathbf{E}^{-1} = \frac{1}{1 - \rho} \left(\mathbf{I}_n - \frac{\rho}{1 + (n - 1)\rho} \mathbf{J}_n \right), \quad \det \mathbf{E} = (1 - \rho)^{n-1} (1 + (n - 1)\rho).$$

1.6.5 Exercises

1.6.5.1 A. Check your understanding

Exercise A1.

Show that $\text{tr}(\mathbf{ABC}) = \text{tr}(\mathbf{ACB})$ when $\mathbf{A}, \mathbf{B}, \mathbf{C}$ are symmetric, and find 2×2 matrices for which the two traces differ.

Solution. For symmetric matrices, $\text{tr}(\mathbf{ABC}) = \text{tr}((\mathbf{ABC})^\top) = \text{tr}(\mathbf{CBA}) = \text{tr}(\mathbf{ACB})$, using transposition and then the cyclic property. For a counterexample take $\mathbf{A} = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$, $\mathbf{B} = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}$ and $\mathbf{C} = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$. Then $\mathbf{ABC} = \mathbf{C}$ has trace 1, while $\mathbf{AC} = \mathbf{0}$ gives $\text{tr}(\mathbf{ACB}) = 0$.

Exercise A2.

Let $\mathbf{A}$ be nonsingular, $\mathbf{a}$ a vector and c a scalar with $s = c - \mathbf{a}^\top \mathbf{A}^{-1} \mathbf{a} \neq 0$. Write down

the inverse and the determinant of $\begin{pmatrix} \mathbf{A} & \mathbf{a} \\ \mathbf{a}^\top & c \end{pmatrix}$. Use it to update $(\mathbf{X}^\top \mathbf{X})^{-1}$ when a column is added to $\mathbf{X}$.

1.6.5.2 B. Practice

Exercise B1.

Let $\hat{\boldsymbol{\beta}} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{y}$, $e_i = y_i - \mathbf{x}_i^\top \hat{\boldsymbol{\beta}}$ and $h_i = \mathbf{x}_i^\top (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{x}_i < 1$. Show that the least squares coefficients computed without observation i are

$$\hat{\boldsymbol{\beta}}_{(i)} = \hat{\boldsymbol{\beta}} - \frac{(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{x}_i e_i}{1 - h_i}.$$

Solution. Write $\mathbf{K} = (\mathbf{X}^\top \mathbf{X})^{-1}$ and $\mathbf{x} = \mathbf{x}_i$. Deleting row i turns $\mathbf{X}^\top \mathbf{y}$ into $\mathbf{X}^\top \mathbf{y} - \mathbf{x} y_i$, so by [Example \(Deleting an observation\)](#)

$$\hat{\boldsymbol{\beta}}_{(i)} = \left(\mathbf{K} + \frac{\mathbf{K} \mathbf{x} \mathbf{x}^\top \mathbf{K}}{1 - h_i} \right) (\mathbf{X}^\top \mathbf{y} - \mathbf{x} y_i) = \hat{\boldsymbol{\beta}} - \mathbf{K} \mathbf{x} y_i + \frac{\mathbf{K} \mathbf{x} (\mathbf{x}^\top \hat{\boldsymbol{\beta}} - h_i y_i)}{1 - h_i} = \hat{\boldsymbol{\beta}} - \mathbf{K} \mathbf{x} \frac{y_i - \mathbf{x}^\top \hat{\boldsymbol{\beta}}}{1 - h_i},$$

since $(1 - h_i) y_i - \mathbf{x}^\top \hat{\boldsymbol{\beta}} + h_i y_i = y_i - \mathbf{x}^\top \hat{\boldsymbol{\beta}}$.

Exercise B2.

For an $n \times p$ matrix $\mathbf{X}$ and $\lambda > 0$, show that $(\mathbf{X}^\top \mathbf{X} + \lambda \mathbf{I}_p)^{-1} \mathbf{X}^\top = \mathbf{X}^\top (\mathbf{X} \mathbf{X}^\top + \lambda \mathbf{I}_n)^{-1}$. Explain why the right-hand side is preferable when p is much larger than n , and derive the identity a second way from [Theorem 1.6.5](#).

Solution. Both $\mathbf{X}^\top \mathbf{X} + \lambda \mathbf{I}$ and $\mathbf{X} \mathbf{X}^\top + \lambda \mathbf{I}$ are positive definite. The identity $(\mathbf{X}^\top \mathbf{X} + \lambda \mathbf{I}_p) \mathbf{X}^\top = \mathbf{X}^\top \mathbf{X} \mathbf{X}^\top + \lambda \mathbf{X}^\top = \mathbf{X}^\top (\mathbf{X} \mathbf{X}^\top + \lambda \mathbf{I}_n)$ gives the result after multiplying by the two inverses. When $p \gg n$, the right-hand side inverts an $n \times n$ matrix instead of a $p \times p$ one. For the second derivation, apply [Theorem 1.6.5](#) with $\mathbf{B} = \lambda \mathbf{I}_p$, $\mathbf{U} = \mathbf{X}^\top$, $\mathbf{C} = \mathbf{I}_n$ and $\mathbf{V} = \mathbf{X}^\top$: $(\lambda \mathbf{I}_p + \mathbf{X}^\top \mathbf{X})^{-1} = \lambda^{-1} \mathbf{I}_p - \lambda^{-2} \mathbf{X}^\top \mathbf{W}^{-1} \mathbf{X}$ with $\mathbf{W} = \mathbf{I}_n + \mathbf{X} \mathbf{X}^\top / \lambda$. Multiplying on the right by $\mathbf{X}^\top$ gives $\lambda^{-1} \mathbf{X}^\top [\mathbf{I}_n - \mathbf{W}^{-1} (\mathbf{X} \mathbf{X}^\top / \lambda)] = \lambda^{-1} \mathbf{X}^\top \mathbf{W}^{-1} [\mathbf{W} - \mathbf{X} \mathbf{X}^\top / \lambda] = \lambda^{-1} \mathbf{X}^\top \mathbf{W}^{-1} = \mathbf{X}^\top (\mathbf{X} \mathbf{X}^\top + \lambda \mathbf{I}_n)^{-1}$.

1.7 Eigenvalues, the spectral theorem and positive definiteness

1.7.1 Eigenvalues

A scalar λ is an **eigenvalue** of the $n \times n$ matrix $\mathbf{A}$, with **eigenvector** $\mathbf{x} \neq \mathbf{0}$, if $\mathbf{A} \mathbf{x} = \lambda \mathbf{x}$. Equivalently, $\mathbf{A} - \lambda \mathbf{I}$ is singular, that is, λ is a root of the **characteristic polynomial** $p(\lambda) = \det(\lambda \mathbf{I} - \mathbf{A})$. This polynomial has degree n and leading coefficient one, so over the complex numbers it factors as $\prod_{i=1}^n (\lambda - \lambda_i)$. The eigenvalues $\lambda_1, \dots, \lambda_n$ are listed with multiplicity and may be complex when $\mathbf{A}$ is not symmetric.

Proposition 1.7.1

Let $\mathbf{A}$ be $n \times n$ with eigenvalues $\lambda_1, \dots, \lambda_n$.

- (a) $\det \mathbf{A} = \prod_i \lambda_i$ and $\operatorname{tr} \mathbf{A} = \sum_i \lambda_i$.
- (b) If $\mathbf{A}\mathbf{x} = \lambda\mathbf{x}$, then $f(\mathbf{A})\mathbf{x} = f(\lambda)\mathbf{x}$ for every polynomial f . If $\mathbf{A}$ is nonsingular, $\mathbf{A}^{-1}\mathbf{x} = \lambda^{-1}\mathbf{x}$.
- (c) $\mathbf{A}$ and $\mathbf{P}^{-1}\mathbf{A}\mathbf{P}$ have the same characteristic polynomial for nonsingular $\mathbf{P}$.
- (d) If $\mathbf{C}$ is $m \times n$ and $\mathbf{D}$ is $n \times m$, then $\lambda^n \det(\lambda \mathbf{I}_m - \mathbf{CD}) = \lambda^m \det(\lambda \mathbf{I}_n - \mathbf{DC})$. So $\mathbf{CD}$ and $\mathbf{DC}$ have the same nonzero eigenvalues, with the same multiplicities.

Proof.

- (a) Setting $\lambda = 0$ in $\det(\lambda \mathbf{I} - \mathbf{A}) = \prod(\lambda - \lambda_i)$ gives $(-1)^n \det \mathbf{A} = (-1)^n \prod \lambda_i$. For the trace, compare coefficients of λ^{n-1} . On the right it is $-\sum \lambda_i$. On the left, in the permutation expansion of $\det(\lambda \mathbf{I} - \mathbf{A})$, only the product of diagonal entries $\prod_i (\lambda - a_{ii})$ contains λ^{n-1} , because every other term omits at least two diagonal entries. Its coefficient is $-\sum a_{ii}$.
- (b) $\mathbf{A}^k \mathbf{x} = \lambda^k \mathbf{x}$ by induction, and $\mathbf{x} = \mathbf{A}^{-1} \mathbf{A} \mathbf{x} = \lambda \mathbf{A}^{-1} \mathbf{x}$ with $\lambda \neq 0$. (c) $\det(\lambda \mathbf{I} - \mathbf{P}^{-1} \mathbf{A} \mathbf{P}) = \det(\mathbf{P}^{-1}(\lambda \mathbf{I} - \mathbf{A})\mathbf{P}) = \det(\lambda \mathbf{I} - \mathbf{A})$.
- (d) For $\lambda \neq 0$, Theorem 1.6.3(c) gives $\det(\mathbf{I}_m - \lambda^{-1} \mathbf{CD}) = \det(\mathbf{I}_n - \lambda^{-1} \mathbf{DC})$. Multiplying by λ^{m+n} gives the identity for $\lambda \neq 0$. Both sides are polynomials, so it holds for all λ .

■

1.7.2 The spectral theorem

Symmetric matrices have the best possible eigenstructure, and almost every matrix whose eigenvalues matter in statistics is symmetric.

Theorem 1.7.2 — Spectral theorem

Let $\mathbf{A}$ be an $n \times n$ symmetric matrix. Then all eigenvalues of $\mathbf{A}$ are real, and there is an orthogonal matrix $\mathbf{Q} = [\mathbf{q}_1, \dots, \mathbf{q}_n]$ with

$$\mathbf{A} = \mathbf{Q} \mathbf{\Lambda} \mathbf{Q}^\top = \sum_{i=1}^n \lambda_i \mathbf{q}_i \mathbf{q}_i^\top, \quad \mathbf{\Lambda} = \operatorname{diag}(\lambda_1, \dots, \lambda_n). \quad (1.7.1)$$

The columns of $\mathbf{Q}$ are an orthonormal basis of eigenvectors, $\mathbf{A}\mathbf{q}_i = \lambda_i \mathbf{q}_i$. Eigenvectors belonging to distinct eigenvalues are orthogonal.

Proof. Real eigenvalues. Let $\mathbf{A}\mathbf{z} = \lambda\mathbf{z}$ with $\mathbf{z} \in \mathbb{C}^n$ nonzero, and let $\bar{\mathbf{z}}$ be the complex conjugate. The scalar $s = \bar{\mathbf{z}}^\top \mathbf{A}\mathbf{z}$ is real, since its conjugate is $\mathbf{z}^\top \mathbf{A}\bar{\mathbf{z}} = (\mathbf{z}^\top \mathbf{A}\bar{\mathbf{z}})^\top = \bar{\mathbf{z}}^\top \mathbf{A}\mathbf{z} = s$ by symmetry. Also $s = \lambda \bar{\mathbf{z}}^\top \mathbf{z} = \lambda \sum_i |z_i|^2$, with the sum real and positive. So λ is real. The real singular matrix $\mathbf{A} - \lambda \mathbf{I}$ then has a real nonzero null vector, so there

is a real eigenvector.

Diagonalization. By induction on n , the case $n = 1$ being trivial. Let $\mathbf{q}_1$ be a real unit eigenvector for λ_1 and extend it to an orthogonal matrix $[\mathbf{q}_1, \mathbf{Q}_2]$ (Section 1.5). Since $\mathbf{Q}_2^\top \mathbf{A} \mathbf{q}_1 = \lambda_1 \mathbf{Q}_2^\top \mathbf{q}_1 = \mathbf{0}$ and $\mathbf{A}$ is symmetric,

$$[\mathbf{q}_1, \mathbf{Q}_2]^\top \mathbf{A} [\mathbf{q}_1, \mathbf{Q}_2] = \begin{pmatrix} \lambda_1 & \mathbf{0}^\top \\ \mathbf{0} & \mathbf{Q}_2^\top \mathbf{A} \mathbf{Q}_2 \end{pmatrix}.$$

The $(n-1) \times (n-1)$ block is symmetric, so by induction $\mathbf{Q}_2^\top \mathbf{A} \mathbf{Q}_2 = \mathbf{P} \mathbf{\Lambda}_2 \mathbf{P}^\top$ with $\mathbf{P}$ orthogonal. Then $\mathbf{Q} = [\mathbf{q}_1, \mathbf{Q}_2 \mathbf{P}]$ is orthogonal (Proposition 1.5.3(b)) and $\mathbf{Q}^\top \mathbf{A} \mathbf{Q} = \text{diag}(\lambda_1, \mathbf{\Lambda}_2)$.

Orthogonality. If $\mathbf{A} \mathbf{u} = \lambda \mathbf{u}$ and $\mathbf{A} \mathbf{v} = \mu \mathbf{v}$, then $\lambda \mathbf{u}^\top \mathbf{v} = \mathbf{u}^\top \mathbf{A} \mathbf{v} = \mu \mathbf{u}^\top \mathbf{v}$, so $\mathbf{u}^\top \mathbf{v} = 0$ when $\lambda \neq \mu$. ■

Unless stated otherwise, the eigenvalues of symmetric $\mathbf{A}$ are ordered as $\lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_n$, and we write $\lambda_{\max} = \lambda_1$ and $\lambda_{\min} = \lambda_n$. The following consequences are used without comment later on.

Corollary 1.7.3

Let $\mathbf{A} = \mathbf{Q} \mathbf{\Lambda} \mathbf{Q}^\top$ be symmetric.

- (a) $\text{rank}(\mathbf{A})$ counts the nonzero eigenvalues. $\mathcal{C}(\mathbf{A})$ is spanned by the $\mathbf{q}_i$ with $\lambda_i \neq 0$, and $\mathcal{N}(\mathbf{A})$ by those with $\lambda_i = 0$.
- (b) $\mathbf{A}^k = \mathbf{Q} \mathbf{\Lambda}^k \mathbf{Q}^\top$ for $k = 1, 2, \dots$. If $\mathbf{A}$ is nonsingular, $\mathbf{A}^{-1} = \mathbf{Q} \mathbf{\Lambda}^{-1} \mathbf{Q}^\top$, and $\mathbf{A}^+ = \mathbf{Q} \mathbf{\Lambda}^+ \mathbf{Q}^\top$ in general, where $\mathbf{\Lambda}^+$ inverts the nonzero diagonal entries.
- (c) The eigenspace $\{\mathbf{x} : \mathbf{A} \mathbf{x} = \lambda \mathbf{x}\}$ has dimension equal to the multiplicity of λ as a root of the characteristic polynomial.

Proof. Multiplying by the nonsingular $\mathbf{Q}$ and $\mathbf{Q}^\top$ changes neither rank (Proposition 1.2.3(b)) nor, after the change of variable $\mathbf{x} = \mathbf{Q} \mathbf{c}$, the structure of null spaces and eigenspaces. So each statement reduces to the diagonal matrix $\mathbf{\Lambda}$, where it is immediate. For (b), $\mathbf{Q}^\top \mathbf{Q} = \mathbf{I}$ collapses the middle factors, and the four Penrose conditions for $\mathbf{Q} \mathbf{\Lambda}^+ \mathbf{Q}^\top$ reduce to those for $\mathbf{\Lambda}^+$. ■

Example. (Eigenvalues of the equicorrelation matrix)

For $\mathbf{E} = (1 - \rho) \mathbf{I} + \rho \mathbf{J}_n$ of Example (Equicorrelation), $\mathbf{E} \mathbf{1} = (1 + (n-1)\rho) \mathbf{1}$, and $\mathbf{E} \mathbf{v} = (1 - \rho) \mathbf{v}$ for every $\mathbf{v} \perp \mathbf{1}$. So the eigenvalues are $1 + (n-1)\rho$ once and $1 - \rho$ with multiplicity $n-1$, with eigenvectors $\mathbf{1}/\sqrt{n}$ and the last $n-1$ rows of the Helmert matrix (Example (Helmert matrices)). The determinant and inverse found earlier follow at once from Proposition 1.7.1 and Corollary 1.7.3(b).

1.7.3 Positive definite and nonnegative definite matrices

Definition 1.7.1

A symmetric matrix $\mathbf{A}$ is **positive definite** if $\mathbf{x}^\top \mathbf{A} \mathbf{x} > 0$ for every $\mathbf{x} \neq \mathbf{0}$, and **nonnegative definite** if $\mathbf{x}^\top \mathbf{A} \mathbf{x} \geq 0$ for every $\mathbf{x}$. For symmetric $\mathbf{A}, \mathbf{B}$ of the same size we write $\mathbf{A} \succeq \mathbf{B}$ if $\mathbf{A} - \mathbf{B}$ is nonnegative definite, and $\mathbf{A} \succ \mathbf{B}$ if it is positive definite (the **Loewner order**).

In this book, “nonnegative definite” includes positive definite. Some texts, Rencher and Schaalje among them, reserve “positive semidefinite” for matrices that are nonnegative definite but singular. We use the terms only for symmetric matrices. Covariance matrices are nonnegative definite (Chapter 2), and *Gram matrices* $\mathbf{X}^\top \mathbf{X}$ are too, since $\mathbf{x}^\top \mathbf{X}^\top \mathbf{X} \mathbf{x} = \|\mathbf{X} \mathbf{x}\|^2$.

Theorem 1.7.4 — Characterizations

For a symmetric $n \times n$ matrix $\mathbf{A}$ the following are equivalent:

- (a) $\mathbf{A}$ is positive definite;
- (b) every eigenvalue of $\mathbf{A}$ is positive;
- (c) $\mathbf{A} = \mathbf{B}^\top \mathbf{B}$ for some matrix $\mathbf{B}$ with n columns and full column rank;
- (d) $\mathbf{A} = \mathbf{L} \mathbf{L}^\top$ with $\mathbf{L}$ lower triangular with positive diagonal entries (the **Cholesky factorization**; $\mathbf{L}$ is unique);
- (e) every leading principal minor $\det \mathbf{A}_{[k]}$, $k = 1, \dots, n$, is positive, where $\mathbf{A}_{[k]}$ is the upper left $k \times k$ block.

Likewise, the following are equivalent: (a') $\mathbf{A}$ is nonnegative definite; (b') every eigenvalue is nonnegative; (c') $\mathbf{A} = \mathbf{B}^\top \mathbf{B}$ for some $\mathbf{B}$, which can be taken of size $\text{rank}(\mathbf{A}) \times n$; (e') every principal minor (determinant of a submatrix with the same row and column indices) is nonnegative.

Proof. Write $\mathbf{A} = \mathbf{Q} \mathbf{\Lambda} \mathbf{Q}^\top$. (a) $\Rightarrow$ (b): $\lambda_i = \mathbf{q}_i^\top \mathbf{A} \mathbf{q}_i > 0$. (b) $\Rightarrow$ (c): $\mathbf{B} = \mathbf{A}^{1/2} \mathbf{Q}^\top$ is nonsingular. (c) $\Rightarrow$ (a): $\mathbf{x}^\top \mathbf{B}^\top \mathbf{B} \mathbf{x} = \|\mathbf{B} \mathbf{x}\|^2$, which is positive for $\mathbf{x} \neq \mathbf{0}$ by full column rank. (d) $\Rightarrow$ (c) is immediate, with $\mathbf{B} = \mathbf{L}^\top$.

(a) $\Rightarrow$ (d), by induction on n . Write $\mathbf{A} = \begin{pmatrix} a & \mathbf{b}^\top \\ \mathbf{b} & \mathbf{A}_2 \end{pmatrix}$. Then $a = \mathbf{e}_1^\top \mathbf{A} \mathbf{e}_1 > 0$. The Schur complement $\mathbf{A}_2 - \mathbf{b} \mathbf{b}^\top / a$ is positive definite: for $\mathbf{v} \neq \mathbf{0}$, put $\mathbf{x} = (-\mathbf{b}^\top \mathbf{v} / a, \mathbf{v}^\top)^\top$ and expand to get $\mathbf{x}^\top \mathbf{A} \mathbf{x} = \mathbf{v}^\top (\mathbf{A}_2 - \mathbf{b} \mathbf{b}^\top / a) \mathbf{v} > 0$. By induction $\mathbf{A}_2 - \mathbf{b} \mathbf{b}^\top / a = \mathbf{L}_2 \mathbf{L}_2^\top$, and $\mathbf{L} = \begin{pmatrix} \sqrt{a} & \mathbf{0}^\top \\ \mathbf{b} / \sqrt{a} & \mathbf{L}_2 \end{pmatrix}$ satisfies $\mathbf{L} \mathbf{L}^\top = \mathbf{A}$. For uniqueness, if $\mathbf{L} \mathbf{L}^\top = \mathbf{K} \mathbf{K}^\top$, then $\mathbf{K}^{-1} \mathbf{L} = \mathbf{K}^\top (\mathbf{L}^\top)^{-1}$ is both lower and upper triangular, hence diagonal, say $\mathbf{D}$. From $\mathbf{K}^{-1} \mathbf{L} = \mathbf{D}$ and $\mathbf{L} \mathbf{L}^\top = \mathbf{K} \mathbf{K}^\top$ we get $\mathbf{D} \mathbf{D}^\top = \mathbf{I}$, so $\mathbf{D} = \mathbf{I}$ because its diagonal is positive.

(a) $\Rightarrow$ (e): $\mathbf{A}_{[k]} = \mathbf{E}^\top \mathbf{A} \mathbf{E}$ with $\mathbf{E} = [\mathbf{e}_1, \dots, \mathbf{e}_k]$ is positive definite, since $\mathbf{x}^\top \mathbf{A}_{[k]} \mathbf{x} = (\mathbf{E} \mathbf{x})^\top \mathbf{A} (\mathbf{E} \mathbf{x})$. By (b) and Proposition 1.7.1(a) its determinant is positive. (e) $\Rightarrow$ (a), by induction on n . The block $\mathbf{A}_{[n-1]}$ satisfies (e) and so is positive definite, so by (a) $\Rightarrow$ (d),

which is already proved, $\mathbf{A}_{[n-1]} = \mathbf{L}\mathbf{L}^\top$. Write $\mathbf{A} = \begin{pmatrix} \mathbf{A}_{[n-1]} & \mathbf{c} \\ \mathbf{c}^\top & d \end{pmatrix}$ and $s = d - \mathbf{c}^\top \mathbf{A}_{[n-1]}^{-1} \mathbf{c}$. By Theorem 1.6.3(b), $s = \det \mathbf{A} / \det \mathbf{A}_{[n-1]} > 0$. With $\mathbf{T} = \begin{pmatrix} \mathbf{I} & \mathbf{A}_{[n-1]}^{-1} \mathbf{c} \\ \mathbf{0}^\top & 1 \end{pmatrix}$, a direct check gives $\mathbf{A} = \mathbf{T}^\top \text{diag}(\mathbf{A}_{[n-1]}, s) \mathbf{T} = \mathbf{B}^\top \mathbf{B}$ with $\mathbf{B} = \text{diag}(\mathbf{L}^\top, \sqrt{s}) \mathbf{T}$ nonsingular, so $\mathbf{A}$ is positive definite by (c) $\Rightarrow$ (a).

The primed statements: (a') $\Leftrightarrow$ (b') $\Leftrightarrow$ (c') exactly as above, taking for $\mathbf{B}$ the rows of $\mathbf{A}^{1/2} \mathbf{Q}^\top$ that belong to nonzero eigenvalues. (a') $\Rightarrow$ (e') as above, with an arbitrary index set in place of $\{1, \dots, k\}$. For (e') $\Rightarrow$ (a'), fix $t > 0$. Each column of $\mathbf{A}_{[k]} + t\mathbf{I}$ is a column of $\mathbf{A}_{[k]}$ plus t times a coordinate vector. Expanding the determinant by linearity in each column gives $\det(\mathbf{A}_{[k]} + t\mathbf{I}) = \sum_S t^{k-|S|} \det \mathbf{A}_{SS}$, the sum over all subsets $S \subseteq \{1, \dots, k\}$, with $\det \mathbf{A}_{\emptyset\emptyset} = 1$. Every term is nonnegative and the term for $S = \emptyset$ is $t^k > 0$. So $\mathbf{A} + t\mathbf{I}$ is positive definite by (e) $\Rightarrow$ (a), and $\mathbf{x}^\top \mathbf{A} \mathbf{x} = \lim_{t \rightarrow 0} \mathbf{x}^\top (\mathbf{A} + t\mathbf{I}) \mathbf{x} \geq 0$. ■

Two warnings. Leading principal minors do not suffice for nonnegative definiteness: $\text{diag}(0, -1)$ has leading minors 0 and 0. And (c') holds with many matrices $\mathbf{B}$. The Cholesky factor and the symmetric square root below are two convenient choices.

Proposition 1.7.5

Let $\mathbf{A}$ be $n \times n$ nonnegative definite.

- (a) $a_{ii} \geq 0$, and if $a_{ii} = 0$ then row i and column i of $\mathbf{A}$ are zero.
- (b) $\mathbf{x}^\top \mathbf{A} \mathbf{x} = 0$ iff $\mathbf{A} \mathbf{x} = \mathbf{0}$.
- (c) $\mathbf{B}^\top \mathbf{A} \mathbf{B}$ is nonnegative definite for any $n \times q$ matrix $\mathbf{B}$. If $\mathbf{A}$ is positive definite, $\mathbf{B}^\top \mathbf{A} \mathbf{B}$ is positive definite iff $\mathbf{B}$ has full column rank.
- (d) Every principal submatrix of $\mathbf{A}$ is nonnegative definite, and positive definite if $\mathbf{A}$ is.
- (e) If $\mathbf{A}$ is positive definite, it is nonsingular and $\mathbf{A}^{-1}$ is positive definite. If also $\mathbf{A}$ is partitioned as in Eq. 1.6.2, both Schur complements $\mathbf{A}_{11.2}$ and $\mathbf{A}_{22.1}$ are positive definite.
- (f) If $\mathbf{A} \succeq \mathbf{B} \succ \mathbf{0}$, then $\mathbf{B}^{-1} \succeq \mathbf{A}^{-1}$.

Proof.

- (b) Write $\mathbf{A} = \mathbf{B}^\top \mathbf{B}$ by Theorem 1.7.4(c'). Then $\mathbf{x}^\top \mathbf{A} \mathbf{x} = \|\mathbf{B} \mathbf{x}\|^2$, which vanishes iff $\mathbf{B} \mathbf{x} = \mathbf{0}$, which implies $\mathbf{A} \mathbf{x} = \mathbf{0}$.
- (c) $a_{ii} = \mathbf{e}_i^\top \mathbf{A} \mathbf{e}_i \geq 0$. If $a_{ii} = 0$ then $\mathbf{A} \mathbf{e}_i = \mathbf{0}$ by (b), so column i is zero, and row i by symmetry.
- (d) $\mathbf{x}^\top \mathbf{B}^\top \mathbf{A} \mathbf{B} \mathbf{x} = (\mathbf{B} \mathbf{x})^\top \mathbf{A} (\mathbf{B} \mathbf{x})$ is nonnegative, and when $\mathbf{A}$ is positive definite it is zero only if $\mathbf{B} \mathbf{x} = \mathbf{0}$.
- (e) is (c) with $\mathbf{B}$ a set of columns of $\mathbf{I}$.
- (f) The eigenvalues of $\mathbf{A}^{-1}$ are $1/\lambda_i > 0$. The inverse in Eq. 1.6.4 has $\mathbf{A}_{22.1}^{-1}$ as a principal submatrix, which is positive definite by (d). The other Schur complement follows by symmetry of the argument.
- (g) By Theorem 1.7.6 below, $\mathbf{B} = \mathbf{B}^{1/2} \mathbf{B}^{1/2}$. Then $\mathbf{C} = \mathbf{B}^{-1/2} \mathbf{A} \mathbf{B}^{-1/2} \succeq \mathbf{I}$ by (c), so

every eigenvalue of $\mathbf{C}$ is at least 1 and every eigenvalue of $\mathbf{C}^{-1} = \mathbf{B}^{1/2}\mathbf{A}^{-1}\mathbf{B}^{1/2}$ is at most 1. So $\mathbf{I} \succeq \mathbf{B}^{1/2}\mathbf{A}^{-1}\mathbf{B}^{1/2}$, and multiplying on both sides by $\mathbf{B}^{-1/2}$ gives $\mathbf{B}^{-1} \succeq \mathbf{A}^{-1}$ by (c). ■

Part (f) is used in [Chapter 7](#): an estimator whose covariance matrix is smaller in the Loewner order has smaller variance for every linear combination of its components.

Theorem 1.7.6 — Square root

A nonnegative definite matrix $\mathbf{A} = \mathbf{Q}\mathbf{\Lambda}\mathbf{Q}^\top$ has exactly one nonnegative definite matrix $\mathbf{B}$ with $\mathbf{B}^2 = \mathbf{A}$, namely $\mathbf{A}^{1/2} = \mathbf{Q}\mathbf{\Lambda}^{1/2}\mathbf{Q}^\top$. It satisfies $\mathcal{C}(\mathbf{A}^{1/2}) = \mathcal{C}(\mathbf{A})$. If $\mathbf{A}$ is positive definite, $\mathbf{A}^{1/2}$ is positive definite and its inverse is $\mathbf{A}^{-1/2} = \mathbf{Q}\mathbf{\Lambda}^{-1/2}\mathbf{Q}^\top$.

Proof. $\mathbf{A}^{1/2}$ is symmetric with nonnegative eigenvalues and $(\mathbf{Q}\mathbf{\Lambda}^{1/2}\mathbf{Q}^\top)^2 = \mathbf{Q}\mathbf{\Lambda}\mathbf{Q}^\top$. Its column space is spanned by the $\mathbf{q}_i$ with $\lambda_i > 0$, as is $\mathcal{C}(\mathbf{A})$ ([Corollary 1.7.3\(a\)](#)). For uniqueness, let $\mathbf{B}$ be nonnegative definite with $\mathbf{B}^2 = \mathbf{A}$, and write $\mathbf{B} = \mathbf{P}\mathbf{\Delta}\mathbf{P}^\top$ with $\mathbf{\Delta} = \text{diag}(\delta_i)$, $\delta_i \geq 0$. Then $\mathbf{A} = \mathbf{P}\mathbf{\Delta}^2\mathbf{P}^\top$. Let $\mathbf{A}\mathbf{v} = \lambda\mathbf{v}$ and expand $\mathbf{v} = \sum_i c_i \mathbf{p}_i$. Comparing $\mathbf{A}\mathbf{v} = \sum_i c_i \delta_i^2 \mathbf{p}_i$ with $\lambda\mathbf{v}$ gives $c_i(\delta_i^2 - \lambda) = 0$, so $c_i \neq 0$ only if $\delta_i = \sqrt{\lambda}$. Hence $\mathbf{B}\mathbf{v} = \sum_i c_i \delta_i \mathbf{p}_i = \sqrt{\lambda}\mathbf{v}$. Applying this to each $\mathbf{q}_j$ gives $\mathbf{B}\mathbf{Q} = \mathbf{Q}\mathbf{\Lambda}^{1/2}$, that is, $\mathbf{B} = \mathbf{A}^{1/2}$. The last statement is [Corollary 1.7.3\(b\)](#). ■

1.7.4 Extremal properties of eigenvalues

Theorem 1.7.7 — Rayleigh quotient

Let $\mathbf{A}$ be symmetric with eigenvalues $\lambda_1 \geq \dots \geq \lambda_n$ and orthonormal eigenvectors $\mathbf{q}_1, \dots, \mathbf{q}_n$. For every $\mathbf{x} \neq \mathbf{0}$,

$$\lambda_n \leq \frac{\mathbf{x}^\top \mathbf{A} \mathbf{x}}{\mathbf{x}^\top \mathbf{x}} \leq \lambda_1, \quad (1.7.2)$$

with equality on the right at $\mathbf{x} = \mathbf{q}_1$ and on the left at $\mathbf{x} = \mathbf{q}_n$. More generally, the maximum of $\mathbf{x}^\top \mathbf{A} \mathbf{x} / \mathbf{x}^\top \mathbf{x}$ over nonzero $\mathbf{x}$ orthogonal to $\mathbf{q}_1, \dots, \mathbf{q}_{k-1}$ is λ_k , attained at $\mathbf{q}_k$.

Proof. Put $\mathbf{c} = \mathbf{Q}^\top \mathbf{x}$, so that $\mathbf{x}^\top \mathbf{x} = \mathbf{c}^\top \mathbf{c}$ and

$$\frac{\mathbf{x}^\top \mathbf{A} \mathbf{x}}{\mathbf{x}^\top \mathbf{x}} = \frac{\sum_i \lambda_i c_i^2}{\sum_i c_i^2},$$

a weighted average of the eigenvalues with weights $c_i^2 / \sum_j c_j^2$. It lies between the smallest and largest eigenvalue, and equals λ_1 when all weight is on c_1 , that is, at $\mathbf{x} = \mathbf{q}_1$. The condition $\mathbf{x} \perp \mathbf{q}_1, \dots, \mathbf{q}_{k-1}$ means $c_1 = \dots = c_{k-1} = 0$, so the average is then over

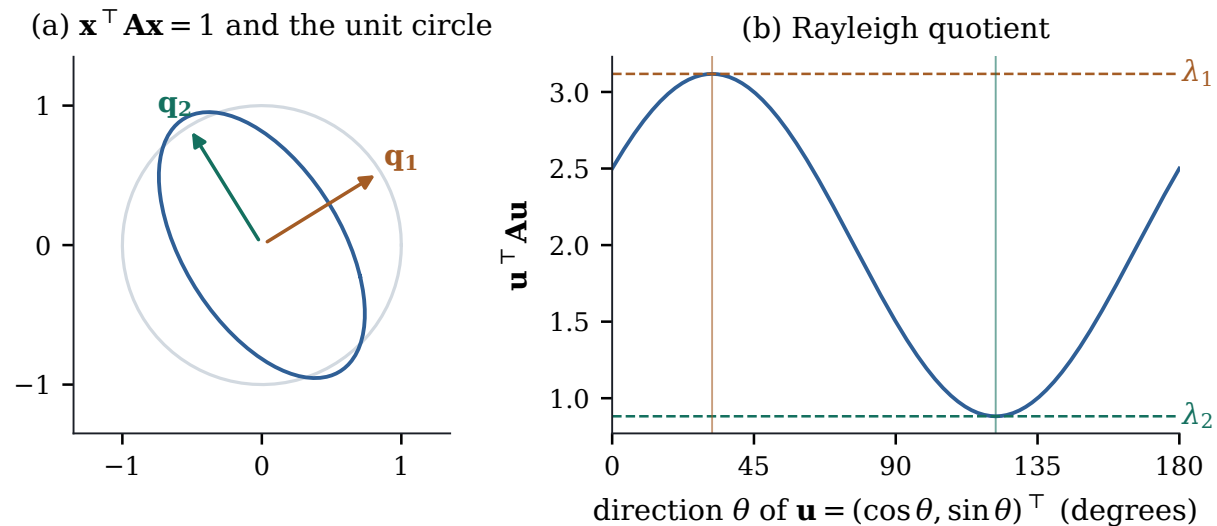


Figure 1.7.1: The Rayleigh quotient of a 2×2 symmetric matrix. (a) The level set $\mathbf{x}^T \mathbf{A} \mathbf{x} = 1$, an ellipse with axes along the orthonormal eigenvectors $\mathbf{q}_1, \mathbf{q}_2$, and the unit circle. (b) The quotient $\mathbf{u}^T \mathbf{A} \mathbf{u}$ along the unit circle. Its maximum λ_1 and minimum λ_2 are attained at the eigenvector directions (Theorem 1.7.7).

$\lambda_k, \dots, \lambda_n$, whose largest member is λ_k .

■

Figure 1.7.1 shows the theorem for $\mathbf{A} = \begin{pmatrix} 2.5 & 1 \\ 1 & 1.5 \end{pmatrix}$. The eigenvalues are 3.118 and 0.882, with eigenvectors at angles 31.7° and 121.7° . As $\mathbf{x}$ turns around the unit circle, the quotient swings between the two eigenvalues. The ellipse $\mathbf{x}^T \mathbf{A} \mathbf{x} = 1$ has its principal axes along the eigenvectors, with semi-axis lengths $\lambda_i^{-1/2}$. Consequently the Rayleigh quotient is largest in the direction in which the ellipse is narrowest.

Corollary 1.7.8 — Relative to a positive definite matrix

Let $\mathbf{A}$ be symmetric and $\mathbf{B}$ positive definite, both $n \times n$.

- (a) $\max_{\mathbf{x} \neq \mathbf{0}} \mathbf{x}^T \mathbf{A} \mathbf{x} / \mathbf{x}^T \mathbf{B} \mathbf{x}$ is the largest eigenvalue of $\mathbf{B}^{-1} \mathbf{A}$, which has the same eigenvalues as the symmetric $\mathbf{B}^{-1/2} \mathbf{A} \mathbf{B}^{-1/2}$.
- (b) There is a nonsingular $\mathbf{P}$ with $\mathbf{P}^T \mathbf{B} \mathbf{P} = \mathbf{I}$ and $\mathbf{P}^T \mathbf{A} \mathbf{P}$ diagonal.
- (c) For $\mathbf{a} \in \mathbb{R}^n$, $\max_{\mathbf{x} \neq \mathbf{0}} (\mathbf{a}^T \mathbf{x})^2 / \mathbf{x}^T \mathbf{B} \mathbf{x} = \mathbf{a}^T \mathbf{B}^{-1} \mathbf{a}$, attained at $\mathbf{x} = \mathbf{B}^{-1} \mathbf{a}$.

Proof. Substitute $\mathbf{x} = \mathbf{B}^{-1/2} \mathbf{z}$. The quotient in (a) becomes $\mathbf{z}^T \mathbf{B}^{-1/2} \mathbf{A} \mathbf{B}^{-1/2} \mathbf{z} / \mathbf{z}^T \mathbf{z}$, to which Theorem 1.7.7 applies, and $\mathbf{B}^{-1} \mathbf{A} = \mathbf{B}^{-1/2} (\mathbf{B}^{-1/2} \mathbf{A} \mathbf{B}^{-1/2}) \mathbf{B}^{1/2}$ is similar to it (Proposition 1.7.1(c)). For (b), if $\mathbf{B}^{-1/2} \mathbf{A} \mathbf{B}^{-1/2} = \mathbf{Q} \mathbf{\Lambda} \mathbf{Q}^T$, take $\mathbf{P} = \mathbf{B}^{-1/2} \mathbf{Q}$. For (c), the matrix $\mathbf{A} = \mathbf{a} \mathbf{a}^T$ has $\mathbf{B}^{-1/2} \mathbf{a} \mathbf{a}^T \mathbf{B}^{-1/2}$ of rank at most one, whose only nonzero eigenvalue is its trace $\mathbf{a}^T \mathbf{B}^{-1} \mathbf{a}$. Substituting $\mathbf{x} = \mathbf{B}^{-1} \mathbf{a}$ attains it.

■

Part (c) is the Cauchy–Schwarz inequality in the inner product $\mathbf{x}^T \mathbf{B} \mathbf{v}$. It is the algebra behind simultaneous confidence intervals (Section 12.2), and part (b) is used for linear discriminants

and for comparing two covariance matrices.

1.7.5 Exercises

1.7.5.1 A. Check your understanding

Exercise A1.

Show that $\begin{pmatrix} a & b \\ b & c \end{pmatrix}$ is positive definite iff $a > 0$ and $ac > b^2$, and nonnegative definite iff $a \geq 0$, $c \geq 0$ and $ac \geq b^2$. Why is $a \geq 0$ and $ac \geq b^2$ not enough?

1.7.5.2 B. Practice

Exercise B1.

Let $\mathbf{A} \succeq \mathbf{B} \succ \mathbf{0}$. Show that $\det \mathbf{A} \geq \det \mathbf{B}$ and $\operatorname{tr} \mathbf{A} \geq \operatorname{tr} \mathbf{B}$.

Exercise B2.

Let $\mathbf{X} = [\mathbf{X}_1, \mathbf{X}_2]$ have full column rank. Show that the leading block of $(\mathbf{X}^\top \mathbf{X})^{-1}$ satisfies $[(\mathbf{X}^\top \mathbf{X})^{-1}]_{11} \succeq (\mathbf{X}_1^\top \mathbf{X}_1)^{-1}$, with equality iff $\mathbf{X}_1^\top \mathbf{X}_2 = \mathbf{0}$. Interpret this for the variances of least squares coefficients when regressors are added to a model.

Solution. Write $\mathbf{A} = \mathbf{X}^\top \mathbf{X}$ in blocks. By [Theorem 1.6.4](#), $[\mathbf{A}^{-1}]_{11} = \mathbf{A}_{11.2}^{-1}$. Now $\mathbf{A}_{11} - \mathbf{A}_{11.2} = \mathbf{A}_{12} \mathbf{A}_{22}^{-1} \mathbf{A}_{21}$ is nonnegative definite by [Proposition 1.7.5\(c,e\)](#), and $\mathbf{A}_{11.2}$ is positive definite. So $\mathbf{A}_{11} \succeq \mathbf{A}_{11.2} \succ \mathbf{0}$, and [Proposition 1.7.5\(f\)](#) gives $\mathbf{A}_{11.2}^{-1} \succeq \mathbf{A}_{11}^{-1}$. Equality forces $\mathbf{A}_{12} \mathbf{A}_{22}^{-1} \mathbf{A}_{21} = \mathbf{0}$, hence $\mathbf{A}_{21} \mathbf{x} = \mathbf{0}$ for all $\mathbf{x}$ by [Proposition 1.7.5\(b\)](#), that is, $\mathbf{X}_2^\top \mathbf{X}_1 = \mathbf{0}$. When $\operatorname{Cov}(\mathbf{Y}) = \sigma^2 \mathbf{I}$, the covariance matrix of the coefficients of $\mathbf{X}_1$ is $\sigma^2 [\mathbf{A}^{-1}]_{11}$ in the larger model and $\sigma^2 \mathbf{A}_{11}^{-1}$ in the smaller one. Adding regressors never decreases the variance of any linear combination of the original coefficients, and leaves it unchanged only when the new regressors are orthogonal to the old ones.

1.7.5.3 C. Going deeper

Exercise C1.

Let $\mathbf{A}$ be symmetric with eigenvalues $\lambda_1 \geq \dots \geq \lambda_n$. Prove the Courant–Fischer theorem

$$\lambda_k = \max_{\dim S=k} \min_{\mathbf{0} \neq \mathbf{x} \in S} \frac{\mathbf{x}^\top \mathbf{A} \mathbf{x}}{\mathbf{x}^\top \mathbf{x}}.$$

Deduce that if $\mathbf{B}$ is obtained from $\mathbf{A}$ by deleting one row and the corresponding column, its eigenvalues $\mu_1 \geq \dots \geq \mu_{n-1}$ interlace: $\lambda_k \geq \mu_k \geq \lambda_{k+1}$.

Exercise C2.

Let $\mathbf{S}$ be positive definite of size $p \times p$. Show that $g(\boldsymbol{\Sigma}) = -\log \det \boldsymbol{\Sigma} - \operatorname{tr}(\boldsymbol{\Sigma}^{-1} \mathbf{S})$, over positive definite $\boldsymbol{\Sigma}$, is maximized uniquely at $\boldsymbol{\Sigma} = \mathbf{S}$. *Hint:* write g in terms of the

eigenvalues of $\mathbf{S}^{-1/2}\mathbf{\Sigma}\mathbf{S}^{-1/2}$.

Solution. Let $\mathbf{M} = \mathbf{S}^{-1/2}\mathbf{\Sigma}\mathbf{S}^{-1/2}$, which is positive definite with eigenvalues $\mu_1, \dots, \mu_p > 0$. Then $\det \mathbf{\Sigma} = \det \mathbf{S} \det \mathbf{M}$ and $\operatorname{tr}(\mathbf{\Sigma}^{-1}\mathbf{S}) = \operatorname{tr}(\mathbf{S}^{1/2}\mathbf{\Sigma}^{-1}\mathbf{S}^{1/2}) = \operatorname{tr}(\mathbf{M}^{-1})$. So

$$g(\mathbf{\Sigma}) = -\log \det \mathbf{S} - \sum_{i=1}^p \left(\log \mu_i + \frac{1}{\mu_i} \right).$$

The function $\mu \mapsto \log \mu + 1/\mu$ has derivative $(\mu - 1)/\mu^2$ and a unique minimum at $\mu = 1$. So g is maximized iff every $\mu_i = 1$, that is, $\mathbf{M} = \mathbf{I}$ and $\mathbf{\Sigma} = \mathbf{S}$. The maximum is $-\log \det \mathbf{S} - p$.

1.8 The singular value decomposition and conditioning

The spectral theorem needs a symmetric matrix. A model matrix is rectangular, and the decomposition that fits it is the singular value decomposition.

Theorem 1.8.1 — Singular value decomposition

Let $\mathbf{A}$ be $m \times n$ of rank $r \geq 1$. There are an $m \times r$ matrix $\mathbf{U}$ and an $n \times r$ matrix $\mathbf{V}$, both with orthonormal columns, and numbers $\sigma_1 \geq \dots \geq \sigma_r > 0$, such that

$$\mathbf{A} = \mathbf{U}\mathbf{D}\mathbf{V}^\top = \sum_{i=1}^r \sigma_i \mathbf{u}_i \mathbf{v}_i^\top, \quad \mathbf{D} = \operatorname{diag}(\sigma_1, \dots, \sigma_r). \quad (1.8.1)$$

The **singular values** σ_i are the positive square roots of the nonzero eigenvalues of $\mathbf{A}^\top \mathbf{A}$ (equivalently of $\mathbf{A}\mathbf{A}^\top$). The columns of $\mathbf{V}$ and $\mathbf{U}$ are eigenvectors of $\mathbf{A}^\top \mathbf{A}$ and $\mathbf{A}\mathbf{A}^\top$, with $\mathcal{C}(\mathbf{V}) = \mathcal{C}(\mathbf{A}^\top)$ and $\mathcal{C}(\mathbf{U}) = \mathcal{C}(\mathbf{A})$. Moreover $\mathbf{A}^+ = \mathbf{V}\mathbf{D}^{-1}\mathbf{U}^\top$ is the Moore–Penrose inverse of $\mathbf{A}$.

Proof. $\mathbf{A}^\top \mathbf{A}$ is nonnegative definite of rank r (Proposition 1.2.3(c)). By the spectral theorem its eigenvalues are $\sigma_1^2 \geq \dots \geq \sigma_r^2 > 0$ and $n-r$ zeros, with orthonormal eigenvectors $\mathbf{V} = [\mathbf{v}_1, \dots, \mathbf{v}_r]$ for the nonzero ones and $\mathbf{V}_0$ for the zeros. Define $\mathbf{U} = \mathbf{A}\mathbf{V}\mathbf{D}^{-1}$. Then $\mathbf{U}^\top \mathbf{U} = \mathbf{D}^{-1}\mathbf{V}^\top \mathbf{A}^\top \mathbf{A}\mathbf{V}\mathbf{D}^{-1} = \mathbf{D}^{-1}\mathbf{D}^2\mathbf{D}^{-1} = \mathbf{I}_r$. For each column $\mathbf{v}$ of $\mathbf{V}_0$, $\|\mathbf{A}\mathbf{v}\|^2 = \mathbf{v}^\top \mathbf{A}^\top \mathbf{A}\mathbf{v} = 0$, so $\mathbf{A}\mathbf{V}_0 = \mathbf{0}$. Since $\mathbf{V}\mathbf{V}^\top + \mathbf{V}_0\mathbf{V}_0^\top = \mathbf{I}$ (Proposition 1.5.3(d)),

$$\mathbf{A} = \mathbf{A}\mathbf{V}\mathbf{V}^\top + \mathbf{A}\mathbf{V}_0\mathbf{V}_0^\top = \mathbf{U}\mathbf{D}\mathbf{V}^\top.$$

Then $\mathbf{A}\mathbf{A}^\top \mathbf{U} = \mathbf{U}\mathbf{D}\mathbf{V}^\top \mathbf{V}\mathbf{D}\mathbf{U}^\top \mathbf{U} = \mathbf{U}\mathbf{D}^2$, so the columns of $\mathbf{U}$ are eigenvectors of $\mathbf{A}\mathbf{A}^\top$. $\mathcal{C}(\mathbf{U}) \subseteq \mathcal{C}(\mathbf{A})$ by definition of $\mathbf{U}$, and both have dimension r . The statement for $\mathbf{V}$ is the same argument applied to $\mathbf{A}^\top = \mathbf{V}\mathbf{D}\mathbf{U}^\top$. The four Penrose conditions for $\mathbf{V}\mathbf{D}^{-1}\mathbf{U}^\top$ follow from $\mathbf{U}^\top \mathbf{U} = \mathbf{V}^\top \mathbf{V} = \mathbf{I}_r$. For instance, $\mathbf{A}\mathbf{A}^+ = \mathbf{U}\mathbf{U}^\top$ and $\mathbf{A}^+ \mathbf{A} = \mathbf{V}\mathbf{V}^\top$ are symmetric. ■

Completing $\mathbf{U}$ and $\mathbf{V}$ to orthogonal matrices gives the *full* SVD $\mathbf{A} = \tilde{\mathbf{U}}\tilde{\mathbf{D}}\tilde{\mathbf{V}}^\top$, with $\tilde{\mathbf{D}}$ of size $m \times n$ carrying $\sigma_1, \dots, \sigma_r$ on its diagonal and zeros elsewhere. By convention, $\sigma_{r+1} = \dots = \sigma_{\min(m,n)} = 0$. Geometrically, $\mathbf{A}$ maps the unit ball of $\mathbb{R}^n$ onto a solid ellipsoid in $\mathcal{C}(\mathbf{A})$ with semi-axes $\sigma_i \mathbf{u}_i$. The input direction $\mathbf{v}_i$ is stretched by the factor σ_i and turned into $\mathbf{u}_i$. The Moore–Penrose inverse undoes the stretching on $\mathcal{C}(\mathbf{A})$ and sends $\mathcal{C}(\mathbf{A})^\perp$ to zero. The matrix $\mathbf{U}\mathbf{U}^\top$ is the orthogonal projection onto $\mathcal{C}(\mathbf{A})$, whose theory is in [Chapter 6](#).

Proposition 1.8.2 — Norms and extremal properties

Let $\mathbf{A}$ be $m \times n$ with singular values $\sigma_1 \geq \sigma_2 \geq \dots$, and let $\|\mathbf{A}\|_2 = \max_{\mathbf{x} \neq \mathbf{0}} \|\mathbf{A}\mathbf{x}\|/\|\mathbf{x}\|$ be the **spectral norm**.

- (a) $\|\mathbf{A}\|_2 = \sigma_1$ and $\|\mathbf{A}\|_F^2 = \sum_i \sigma_i^2$.
- (b) If $\mathbf{A}$ has full column rank n , then $\min_{\mathbf{x} \neq \mathbf{0}} \|\mathbf{A}\mathbf{x}\|/\|\mathbf{x}\| = \sigma_n$, attained at $\mathbf{v}_n$.
- (c) (Eckart–Young.) For $k < r$, let $\mathbf{A}_k = \sum_{i \leq k} \sigma_i \mathbf{u}_i \mathbf{v}_i^\top$. Then $\|\mathbf{A} - \mathbf{B}\|_2 \geq \sigma_{k+1} = \|\mathbf{A} - \mathbf{A}_k\|_2$ for every $\mathbf{B}$ with $\text{rank}(\mathbf{B}) \leq k$.

Proof.

- (a) and (b): $\|\mathbf{A}\mathbf{x}\|^2/\|\mathbf{x}\|^2 = \mathbf{x}^\top \mathbf{A}^\top \mathbf{A} \mathbf{x} / \mathbf{x}^\top \mathbf{x}$, and the eigenvalues of $\mathbf{A}^\top \mathbf{A}$ are the σ_i^2 (padded with zeros), so [Theorem 1.7.7](#) applies. For the Frobenius norm, $\|\mathbf{A}\|_F^2 = \text{tr}(\mathbf{A}^\top \mathbf{A}) = \sum \sigma_i^2$ by [Proposition 1.7.1\(a\)](#).
- (b) $\mathbf{A} - \mathbf{A}_k = \sum_{i > k} \sigma_i \mathbf{u}_i \mathbf{v}_i^\top$ has largest singular value σ_{k+1} . If $\text{rank}(\mathbf{B}) \leq k$, then $\dim \mathcal{N}(\mathbf{B}) \geq n - k$, and since $(n - k) + (k + 1) > n$, a basis of $\mathcal{N}(\mathbf{B})$ together with $\mathbf{v}_1, \dots, \mathbf{v}_{k+1}$ is dependent. So the two subspaces share a unit vector $\mathbf{x} = \sum_{i \leq k+1} c_i \mathbf{v}_i$. Then $\|(\mathbf{A} - \mathbf{B})\mathbf{x}\|^2 = \|\mathbf{A}\mathbf{x}\|^2 = \sum_{i \leq k+1} \sigma_i^2 c_i^2 \geq \sigma_{k+1}^2$.

■

The Frobenius version of (c), $\|\mathbf{A} - \mathbf{B}\|_F^2 \geq \sum_{i > k} \sigma_i^2$, also holds (Eckart and Young 1936). It is the basis of principal components.

1.8.1 The condition number

Definition 1.8.1 — Condition number

The **condition number** of an $m \times n$ matrix $\mathbf{A}$ of full column rank is

$$\kappa(\mathbf{A}) = \frac{\sigma_{\max}(\mathbf{A})}{\sigma_{\min}(\mathbf{A})} = \frac{\sigma_1}{\sigma_n}.$$

For a nonsingular square matrix this equals $\|\mathbf{A}\|_2 \|\mathbf{A}^{-1}\|_2$. A matrix without full column rank has $\kappa(\mathbf{A}) = \infty$ by convention.

By [Proposition 1.8.2](#), $\kappa(\mathbf{A})$ is the ratio of the largest to the smallest stretching that $\mathbf{A}$ applies to a unit vector. It is at least 1, is unchanged by scaling $\mathbf{A}$ or multiplying it on the left by a matrix with orthonormal columns, and equals 1 iff $\mathbf{A}^\top \mathbf{A} = c\mathbf{I}$, that is, iff the columns are orthogonal with equal lengths. Two further facts explain its role in regression. First, the singular values of

$\mathbf{A}^\top \mathbf{A}$ are the σ_i^2 , so

$$\kappa(\mathbf{A}^\top \mathbf{A}) = \kappa(\mathbf{A})^2. \quad (1.8.2)$$

Second, κ bounds how much relative errors in the data can grow when a linear system is solved. If $\mathbf{A}$ is square and nonsingular, $\mathbf{A}\mathbf{x} = \mathbf{b}$ and $\mathbf{A}(\mathbf{x} + \delta\mathbf{x}) = \mathbf{b} + \delta\mathbf{b}$, then $\delta\mathbf{x} = \mathbf{A}^{-1}\delta\mathbf{b}$, and $\|\mathbf{b}\| \leq \sigma_1 \|\mathbf{x}\|$ gives

$$\frac{\|\delta\mathbf{x}\|}{\|\mathbf{x}\|} \leq \frac{\|\mathbf{A}^{-1}\|_2 \|\delta\mathbf{b}\|}{\|\mathbf{b}\|/\sigma_1} = \kappa(\mathbf{A}) \frac{\|\delta\mathbf{b}\|}{\|\mathbf{b}\|}. \quad (1.8.3)$$

In floating point, relative errors of about 10^{-16} in the data can therefore become errors of about $\kappa \times 10^{-16}$ in the solution. Solving the normal equations works with $\mathbf{X}^\top \mathbf{X}$ and so, by [Eq. 1.8.2](#), squares the damage. [Chapter 6](#) and [Chapter 10](#) take this up.

A small $\sigma_{\min}$ has a direct statistical reading. By [Proposition 1.8.2\(b\)](#), $\|\mathbf{X}\mathbf{v}_n\| = \sigma_n$ with $\|\mathbf{v}_n\| = 1$. The coefficients $\mathbf{v}_n$ combine the columns of $\mathbf{X}$ into a vector that is nearly zero, which is a near-dependence among the regressors. This is the basis of the collinearity diagnostics of [Chapter 26](#). The size of κ depends on the units of the columns, so columns are usually rescaled before it is interpreted.

Example. (A macroeconomic design)

Take the 203 quarterly US observations of real GDP, real investment, real government spending, real disposable income, population and unemployment, with an intercept, as a model matrix with $p = 7$ columns. As recorded, $\kappa(\mathbf{X}) = 5.10 \times 10^5$. Most of that is units: GDP is measured in billions and unemployment in percent. Rescaling each column to unit length reduces it to 383, and centring the regressors before rescaling reduces it further to 124. By [Eq. 1.8.2](#) the corresponding cross-product matrix has condition number 15386.

What remains is not an artefact of units. [Figure 1.8.1\(b\)](#) shows the right singular vector for the smallest singular value, $\sigma_{\min} = 0.0175$, of the centred and scaled design. Its weight is concentrated on GDP (0.78) and disposable income (-0.61). The combination of the scaled columns with these coefficients is a vector of length 0.0175, although every column has length one. The two series move almost in lockstep: their correlation is 0.9990. No rescaling can separate their coefficients well. Centring is not a neutral choice, however: it removes near-dependences that involve the intercept, and [Belsley et al. \(1980\)](#) recommend scaling without centring for that reason, so their diagnostics would use the unit-length figure 383. [Chapter 26](#) returns to the choice. The listing computes the singular values, and the script also checks [Eq. 1.8.2](#), the Penrose conditions for $\mathbf{V}\mathbf{D}^{-1}\mathbf{U}^\top$, and the Eckart–Young bound.

```
import numpy as np
import statsmodels.api as sm
data = sm.datasets.macrodta.load_pandas().data
names = ["realgdp", "realinv", "realgovt", "realdpi", "pop", "unemp"]
labels = ["GDP", "investment", "government", "disp. income", "population", "unemployment"]
Z = data[names].to_numpy()
n = len(Z)
```

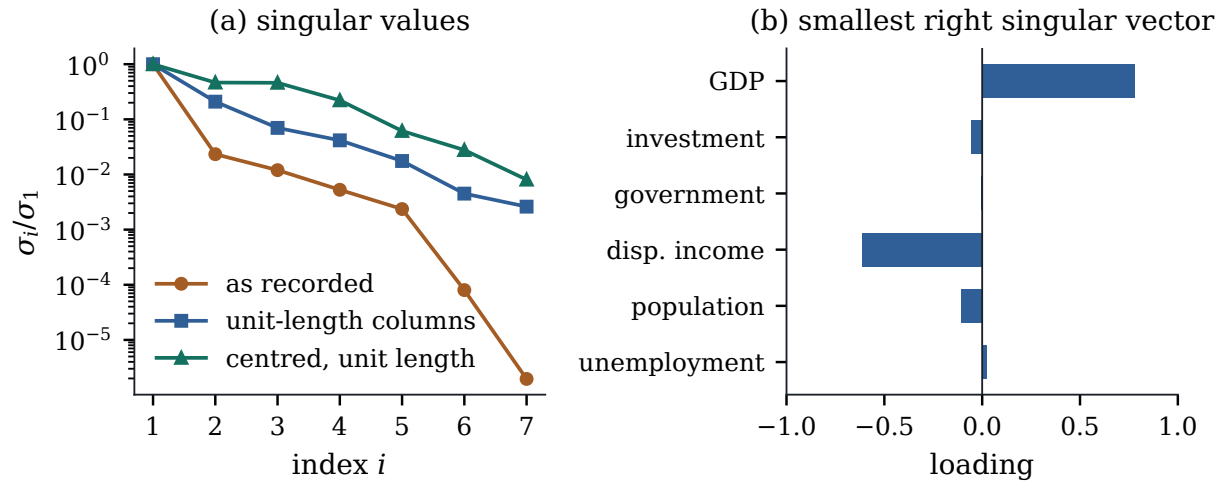


Figure 1.8.1: Singular value decomposition of a macroeconomic model matrix (Example (A macroeconomic design)). (a) Singular values relative to the largest, for the columns as recorded, rescaled to unit length, and centred then rescaled. The ratio of the first to the last is the condition number. (b) The right singular vector for the smallest singular value of the centred design. Its two large loadings reveal a near-dependence between GDP and disposable income.

```

X = np.column_stack([np.ones(n), Z])
X_unit = X / np.linalg.norm(X, axis=0)
Xc = np.column_stack([np.ones(n), Z - Z.mean(axis=0)])
X_std = Xc / np.linalg.norm(Xc, axis=0)

# as recorded
# unit-length columns
# regressors centred
# ... then length 1

for name, M in [("raw", X), ("unit length", X_unit), ("centred", X_std)]:
    U, s, Vt = np.linalg.svd(M, full_matrices=False)
    print(f"{name:12s} kappa = {s[0] / s[-1]:10.1f}    singular values / s_1:",
          np.array2string(s / s[0], precision=4))

```

1.8.2 Exercises

1.8.2.1 B. Practice

Exercise B1.

Let $\mathbf{X} = [\mathbf{x}_1, \mathbf{x}_2]$ with $\|\mathbf{x}_1\| = \|\mathbf{x}_2\| = 1$ and $\mathbf{x}_1^\top \mathbf{x}_2 = r$, $|r| < 1$. Show that $\kappa(\mathbf{X}) = \sqrt{(1 + |r|)/(1 - |r|)}$. How large must $|r|$ be for $\kappa(\mathbf{X}) > 30$?

Solution. $\mathbf{X}^\top \mathbf{X} = \begin{pmatrix} 1 & r \\ r & 1 \end{pmatrix}$ has eigenvalues $1 \pm r$, with eigenvectors $(1, \pm 1)^\top$. The singular values of $\mathbf{X}$ are their square roots, so $\kappa(\mathbf{X})^2 = (1 + |r|)/(1 - |r|)$. For $\kappa > 30$ we need $(1 + |r|)/(1 - |r|) > 900$, that is, $|r| > 899/901 \approx 0.998$. For two columns, even $\kappa(\mathbf{X}) = 100$ requires $|r| > 9999/10001 \approx 0.9998$.

Exercise B2.

Show that $\kappa(\mathbf{X}) \geq \max_j \|\mathbf{x}_j\| / \min_k \|\mathbf{x}_k\|$, so that columns on very different scales force a large condition number. Show by example that rescaling the columns can also increase

κ .

1.9 Idempotent matrices

A square matrix $\mathbf{A}$ is **idempotent** if $\mathbf{A}^2 = \mathbf{A}$. Applying it twice has the same effect as applying it once, which is how a projection behaves. Examples are $\mathbf{I}$, $\mathbf{0}$, the centring matrix of [Example \(Sums of squares as quadratic forms\)](#), the products $\mathbf{AG}$ and $\mathbf{GA}$ for a generalized inverse $\mathbf{G}$ ([Proposition 1.3.3\(a\)](#)), and the non-symmetric $\begin{pmatrix} 1 & 1 \\ 0 & 0 \end{pmatrix}$. In linear models, sums of squares are quadratic forms in *symmetric* idempotent matrices, and their degrees of freedom are ranks.

Proposition 1.9.1 — General idempotent matrices

Let $\mathbf{A}$ be $n \times n$ and idempotent of rank r .

- (a) Every eigenvalue of $\mathbf{A}$ is 0 or 1.
- (b) $\text{rank}(\mathbf{A}) = \text{tr}(\mathbf{A})$.
- (c) $\mathbf{I} - \mathbf{A}$ is idempotent, $\mathbf{A}(\mathbf{I} - \mathbf{A}) = (\mathbf{I} - \mathbf{A})\mathbf{A} = \mathbf{0}$, $\mathcal{C}(\mathbf{I} - \mathbf{A}) = \mathcal{N}(\mathbf{A})$, and $\text{rank}(\mathbf{A}) + \text{rank}(\mathbf{I} - \mathbf{A}) = n$.
- (d) An idempotent matrix that is nonsingular equals $\mathbf{I}$.
- (e) $\mathbf{P}^{-1}\mathbf{A}\mathbf{P}$ is idempotent for nonsingular $\mathbf{P}$, and $\mathbf{A}^\top$ is idempotent.

Proof.

- (a) If $\mathbf{A}\mathbf{x} = \lambda\mathbf{x}$ with $\mathbf{x} \neq \mathbf{0}$, then $\lambda\mathbf{x} = \mathbf{A}\mathbf{x} = \mathbf{A}^2\mathbf{x} = \lambda^2\mathbf{x}$, so $\lambda^2 = \lambda$.
- (b) If $r = 0$ both sides vanish. Otherwise take a rank factorization $\mathbf{A} = \mathbf{BF}$ ([Theorem 1.2.1](#)). Then $\mathbf{BFBF} = \mathbf{BF}$. Since $\mathbf{B}$ has full column rank and $\mathbf{F}$ has full row rank, $\mathbf{B}$ has a left inverse and $\mathbf{F}$ a right inverse ([Section 1.3](#)), and cancelling them gives $\mathbf{FB} = \mathbf{I}_r$. So $\text{tr}(\mathbf{A}) = \text{tr}(\mathbf{BF}) = \text{tr}(\mathbf{FB}) = r$ by [Theorem 1.6.1](#).
- (c) The first two claims are one-line expansions. $\mathcal{C}(\mathbf{I} - \mathbf{A}) \subseteq \mathcal{N}(\mathbf{A})$ because $\mathbf{A}(\mathbf{I} - \mathbf{A}) = \mathbf{0}$, and $\mathbf{v} \in \mathcal{N}(\mathbf{A})$ gives $\mathbf{v} = (\mathbf{I} - \mathbf{A})\mathbf{v}$. The rank statement is then rank-nullity.
- (d) Multiply $\mathbf{A}^2 = \mathbf{A}$ by $\mathbf{A}^{-1}$.
- (e) $(\mathbf{P}^{-1}\mathbf{A}\mathbf{P})^2 = \mathbf{P}^{-1}\mathbf{A}^2\mathbf{P}$ and $(\mathbf{A}^\top)^2 = (\mathbf{A}^2)^\top$.

■

By (c), every $\mathbf{x}$ splits as $\mathbf{A}\mathbf{x} + (\mathbf{I} - \mathbf{A})\mathbf{x}$ with the first part in $\mathcal{C}(\mathbf{A})$ and the second in $\mathcal{N}(\mathbf{A})$. So $\mathbb{R}^n = \mathcal{C}(\mathbf{A}) \oplus \mathcal{N}(\mathbf{A})$, and $\mathbf{A}$ projects onto $\mathcal{C}(\mathbf{A})$ along $\mathcal{N}(\mathbf{A})$. The projection is orthogonal exactly when the two subspaces are orthogonal, which happens iff $\mathbf{A}$ is symmetric ([Exercise \(B1\)](#)).

Theorem 1.9.2 — Symmetric idempotent matrices

Let $\mathbf{A}$ be an $n \times n$ symmetric matrix of rank r . Then $\mathbf{A}$ is idempotent iff all its eigenvalues are 0 or 1. In that case

- (a) $\mathbf{A} = \mathbf{Q}_1 \mathbf{Q}_1^\top$, where the $n \times r$ matrix $\mathbf{Q}_1$ has orthonormal columns spanning $\mathcal{C}(\mathbf{A})$;
- (b) $\text{rank}(\mathbf{A}) = \text{tr}(\mathbf{A}) = r$, the multiplicity of the eigenvalue 1;
- (c) $\mathbf{A}$ and $\mathbf{I} - \mathbf{A}$ are nonnegative definite, and $\mathbf{x}^\top \mathbf{A} \mathbf{x} = \|\mathbf{A} \mathbf{x}\|^2 \leq \|\mathbf{x}\|^2$.

Proof. If $\mathbf{A}$ is idempotent its eigenvalues are 0 or 1 by [Proposition 1.9.1\(a\)](#). Conversely, if $\mathbf{A} = \mathbf{Q} \mathbf{\Lambda} \mathbf{Q}^\top$ with each $\lambda_i \in \{0, 1\}$, then $\mathbf{A}^2 = \mathbf{Q} \mathbf{\Lambda}^2 \mathbf{Q}^\top = \mathbf{Q} \mathbf{\Lambda} \mathbf{Q}^\top = \mathbf{A}$. In that case, order the eigenvalues so that the r ones come first and let $\mathbf{Q}_1$ hold the corresponding eigenvectors. Then $\mathbf{A} = \mathbf{Q} \mathbf{\Lambda} \mathbf{Q}^\top = \mathbf{Q}_1 \mathbf{Q}_1^\top$, and $\mathcal{C}(\mathbf{A}) = \mathcal{C}(\mathbf{Q}_1)$ by [Corollary 1.7.3\(a\)](#). This is (a). The trace is the sum of the eigenvalues, which is (b). For (c), the eigenvalues of $\mathbf{A}$ and of $\mathbf{I} - \mathbf{A}$ are nonnegative, and $\mathbf{x}^\top \mathbf{A} \mathbf{x} = \mathbf{x}^\top \mathbf{A}^\top \mathbf{A} \mathbf{x}$. Also $\|\mathbf{x}\|^2 - \|\mathbf{A} \mathbf{x}\|^2 = \mathbf{x}^\top (\mathbf{I} - \mathbf{A}) \mathbf{x} \geq 0$. ■

[Chapter 6](#) shows that symmetric idempotent matrices are exactly the orthogonal projections, and $\mathbf{Q}_1 \mathbf{Q}_1^\top$ in (a) is the projection onto $\mathcal{C}(\mathbf{A})$. In [Chapter 4](#), part (b) becomes the degrees of freedom of a chi-squared distribution.

Theorem 1.9.3 — Sums of symmetric idempotent matrices

Let $\mathbf{A}_1, \dots, \mathbf{A}_k$ be symmetric $n \times n$ matrices and $\mathbf{A} = \sum_i \mathbf{A}_i$. Consider

- (i) $\mathbf{A}$ is idempotent;
- (ii) every $\mathbf{A}_i$ is idempotent;
- (iii) $\mathbf{A}_i \mathbf{A}_j = \mathbf{0}$ for all $i \neq j$.

Any two of these imply the third. Moreover, if $\mathbf{A} = \mathbf{I}_n$ and $\sum_i \text{rank}(\mathbf{A}_i) = n$, then (ii) and (iii) hold.

Proof.

(ii) and (iii) imply (i): $\mathbf{A}^2 = \sum_i \mathbf{A}_i^2 + \sum_{i \neq j} \mathbf{A}_i \mathbf{A}_j = \sum_i \mathbf{A}_i = \mathbf{A}$.

(iii) and (ii) imply (iii): By [Theorem 1.9.2\(c\)](#), $\|\mathbf{x}\|^2 \geq \mathbf{x}^\top \mathbf{A} \mathbf{x}$ for all $\mathbf{x}$. Fix j and take $\mathbf{x} = \mathbf{A}_j \mathbf{y}$, so that $\mathbf{A}_j \mathbf{x} = \mathbf{x}$ and $\mathbf{x}^\top \mathbf{A}_j \mathbf{x} = \|\mathbf{x}\|^2$. Then

$$\|\mathbf{x}\|^2 \geq \mathbf{x}^\top \mathbf{A} \mathbf{x} = \|\mathbf{x}\|^2 + \sum_{i \neq j} \mathbf{x}^\top \mathbf{A}_i \mathbf{x} = \|\mathbf{x}\|^2 + \sum_{i \neq j} \|\mathbf{A}_i \mathbf{x}\|^2.$$

So $\mathbf{A}_i \mathbf{A}_j \mathbf{y} = \mathbf{A}_i \mathbf{x} = \mathbf{0}$ for every $i \neq j$ and every $\mathbf{y}$.

(iv) and (iii) imply (ii): By (iii), $\mathbf{A} \mathbf{A}_i = \mathbf{A}_i^2$. Using $\mathbf{A}^2 = \mathbf{A}$, $\mathbf{A}_i^2 = \mathbf{A} \mathbf{A}_i = \mathbf{A}^2 \mathbf{A}_i = \mathbf{A} \mathbf{A}_i^2 = \mathbf{A}_i^3$. The eigenvalues of the symmetric $\mathbf{A}_i$ therefore satisfy $\lambda^2 = \lambda^3$, so they lie in $\{0, 1\}$, and $\mathbf{A}_i$ is idempotent by [Theorem 1.9.2](#).

For the last statement, let $r_i = \text{rank}(\mathbf{A}_i)$. For each i , $\mathbf{I} - \mathbf{A}_i = \sum_{j \neq i} \mathbf{A}_j$ has rank at most $\sum_{j \neq i} r_j = n - r_i$ ([Proposition 1.2.3\(d\)](#)). So the eigenvalue 1 of $\mathbf{A}_i$, whose eigenspace is $\mathcal{N}(\mathbf{I} - \mathbf{A}_i)$, has multiplicity at least r_i . But $\mathbf{A}_i$ has exactly r_i nonzero eigenvalues, so all of them equal 1 and $\mathbf{A}_i$ is idempotent. Now (i) holds trivially, and (iii) follows from the

second part of the proof. ■

The last statement is the algebraic core of Cochran's theorem (Chapter 4): if $\mathbf{y}^\top \mathbf{y} = \sum_i \mathbf{y}^\top \mathbf{A}_i \mathbf{y}$ and the ranks add up to n , the quadratic forms are built from mutually orthogonal projections.

1.9.1 Exercises

1.9.1.1 A. Check your understanding

Exercise A1.

Let $\mathbf{A}, \mathbf{B}$ be idempotent of the same size. Show that $\mathbf{AB}$ is idempotent if $\mathbf{AB} = \mathbf{BA}$, and that $\mathbf{A} + \mathbf{B}$ is idempotent iff $\mathbf{AB} + \mathbf{BA} = \mathbf{0}$. Show that the latter forces $\mathbf{AB} = \mathbf{0}$ (symmetry is not needed).

1.9.1.2 B. Practice

Exercise B1.

Let $\mathbf{A}$ be idempotent. Show that $\mathbf{A}$ is symmetric iff $\mathcal{C}(\mathbf{A}) \perp \mathcal{N}(\mathbf{A})$.

Solution. Suppose $\mathbf{A}$ is symmetric. For $\mathbf{u} = \mathbf{Ax} \in \mathcal{C}(\mathbf{A})$ and $\mathbf{v} \in \mathcal{N}(\mathbf{A})$, $\mathbf{u}^\top \mathbf{v} = \mathbf{x}^\top \mathbf{Av} = 0$. Conversely, suppose $\mathcal{C}(\mathbf{A}) \perp \mathcal{N}(\mathbf{A})$. For all $\mathbf{x}, \mathbf{y}$, $(\mathbf{I} - \mathbf{A})\mathbf{y} \in \mathcal{N}(\mathbf{A})$, so $\mathbf{x}^\top \mathbf{A}^\top (\mathbf{I} - \mathbf{A})\mathbf{y} = 0$. Hence $\mathbf{A}^\top (\mathbf{I} - \mathbf{A}) = \mathbf{0}$, that is, $\mathbf{A}^\top = \mathbf{A}^\top \mathbf{A}$. The right side is symmetric, so $\mathbf{A}^\top$ is symmetric, and so is $\mathbf{A}$.

1.10 Kronecker products and the vec operator

Balanced designs have model matrices with a product structure, and multivariate observations have covariance matrices with one. The Kronecker product expresses both.

Definition 1.10.1

The **Kronecker product** of an $m \times n$ matrix $\mathbf{A}$ and a $p \times q$ matrix $\mathbf{B}$ is the $mp \times nq$ block matrix $\mathbf{A} \otimes \mathbf{B} = (a_{ij}\mathbf{B})$. The **vec operator** places the n columns of an $m \times n$ matrix one below another, giving a vector of length mn : $\text{vec}(\mathbf{A}) = (\mathbf{a}_1^\top, \dots, \mathbf{a}_n^\top)^\top$.

Proposition 1.10.1

For matrices of compatible sizes:

- (a) $(\mathbf{A} \otimes \mathbf{B})(\mathbf{C} \otimes \mathbf{D}) = \mathbf{AC} \otimes \mathbf{BD}$ (mixed-product rule);
- (b) $(\mathbf{A} \otimes \mathbf{B})^\top = \mathbf{A}^\top \otimes \mathbf{B}^\top$, and $(\mathbf{A} \otimes \mathbf{B})^{-1} = \mathbf{A}^{-1} \otimes \mathbf{B}^{-1}$ for nonsingular $\mathbf{A}, \mathbf{B}$. More generally $\mathbf{A}^- \otimes \mathbf{B}^-$ is a generalized inverse of $\mathbf{A} \otimes \mathbf{B}$;
- (c) $\text{tr}(\mathbf{A} \otimes \mathbf{B}) = \text{tr}(\mathbf{A}) \text{tr}(\mathbf{B})$ and $\text{rank}(\mathbf{A} \otimes \mathbf{B}) = \text{rank}(\mathbf{A}) \text{rank}(\mathbf{B})$;
- (d) if $\mathbf{A}$ ($m \times m$) and $\mathbf{B}$ ($p \times p$) are symmetric with eigenvalues λ_i and μ_j , the eigenvalues of $\mathbf{A} \otimes \mathbf{B}$ are the products $\lambda_i \mu_j$, and $\det(\mathbf{A} \otimes \mathbf{B}) = (\det \mathbf{A})^p (\det \mathbf{B})^m$;
- (e) Kronecker products of orthogonal, symmetric, idempotent, or nonnegative definite matrices are of the same kind;
- (f) $\text{vec}(\mathbf{ABC}) = (\mathbf{C}^\top \otimes \mathbf{A}) \text{vec}(\mathbf{B})$ and $\text{tr}(\mathbf{A}^\top \mathbf{B}) = \text{vec}(\mathbf{A})^\top \text{vec}(\mathbf{B})$.

Proof.

- (a) The (i, j) block of the left side is $\sum_k a_{ik} c_{kj} \mathbf{B} \mathbf{D}$, which is the (i, j) block of the right side. (b) Transposition is direct, and the rest follows from (a), for example $(\mathbf{A} \otimes \mathbf{B})(\mathbf{A}^- \otimes \mathbf{B}^-)(\mathbf{A} \otimes \mathbf{B}) = \mathbf{AA}^- \otimes \mathbf{BB}^-$.
- (b) With $\mathbf{A} = \mathbf{PAP}^\top$ and $\mathbf{B} = \mathbf{QMQ}^\top$, (a) and (b) give $\mathbf{A} \otimes \mathbf{B} = (\mathbf{P} \otimes \mathbf{Q})(\mathbf{A} \otimes \mathbf{M})(\mathbf{P} \otimes \mathbf{Q})^\top$, a spectral decomposition because $\mathbf{P} \otimes \mathbf{Q}$ is orthogonal and $\mathbf{A} \otimes \mathbf{M}$ is diagonal with entries $\lambda_i \mu_j$. The determinant is the product of these. (c) The trace statement is direct. For the rank, by (a) $(\mathbf{A} \otimes \mathbf{B})(\mathbf{A}^- \otimes \mathbf{B}^-) = \mathbf{AA}^- \otimes \mathbf{BB}^-$ is idempotent with the same rank as $\mathbf{A} \otimes \mathbf{B}$ (Proposition 1.3.3(a)). Its trace is $\text{tr}(\mathbf{AA}^-) \text{tr}(\mathbf{BB}^-) = \text{rank}(\mathbf{A}) \text{rank}(\mathbf{B})$ by Proposition 1.9.1(b).
- (c) follows from (a), (b) and (d). (f) The j th column of $\mathbf{ABC}$ is $\mathbf{ABc}_j = \sum_k c_{kj} \mathbf{Ab}_k$, which is the j th block of $(\mathbf{C}^\top \otimes \mathbf{A}) \text{vec}(\mathbf{B})$. The trace identity is $\sum_{ij} a_{ij} b_{ij}$ written two ways.

■

The determinant formula in (d) holds for all square $\mathbf{A}$ and $\mathbf{B}$, not only symmetric ones. We need only the symmetric case.

Example. (The balanced two-way layout)

Let observations y_{ij} , $i = 1, \dots, a$, $j = 1, \dots, b$, be stacked with j running inside i . The additive model $E(y_{ij}) = \mu + \alpha_i + \beta_j$ has model matrix

$$\mathbf{X} = [\mathbf{1}_a \otimes \mathbf{1}_b, \mathbf{I}_a \otimes \mathbf{1}_b, \mathbf{1}_a \otimes \mathbf{I}_b],$$

because $\mathbf{I}_a \otimes \mathbf{1}_b$ has a 1 in column i exactly for the b observations in row i . With $\bar{\mathbf{J}}_k = k^{-1} \mathbf{J}_k$, a symmetric idempotent averaging matrix, the mixed-product rule shows that $\mathbf{I}_a \otimes \bar{\mathbf{J}}_b$, $\bar{\mathbf{J}}_a \otimes \mathbf{I}_b$ and $\bar{\mathbf{J}}_a \otimes \bar{\mathbf{J}}_b$ replace each observation by its row mean, its column mean and the grand mean. The matrix $\mathbf{M} = \mathbf{I}_a \otimes \bar{\mathbf{J}}_b + \bar{\mathbf{J}}_a \otimes \mathbf{I}_b - \bar{\mathbf{J}}_a \otimes \bar{\mathbf{J}}_b$ is symmetric and idempotent, with trace $a + b - 1 = \text{rank}(\mathbf{X})$. It is the projection onto $\mathcal{C}(\mathbf{X})$ (Theorem 16.1.2 in Chapter 16). The script checks this for $a = 3$ and $b = 4$, together with each part of Proposition 1.10.1.

```
import numpy as np
rng = np.random.default_rng(11)
def vec(M):
    return M.reshape(-1, order="F")
```

stack the columns

```

A, B, C = rng.normal(size=(3, 4)), rng.normal(size=(4, 2)), rng.normal(size=(2, 5))
assert np.allclose(vec(A @ B @ C), np.kron(C.T, A) @ vec(B))           # vec(ABC)

# balanced two-way layout: a = 3 rows, b = 4 columns, one observation per cell
a, b = 3, 4
one = lambda k: np.ones((k, 1))
X = np.hstack([np.kron(one(a), one(b)),           # intercept
               np.kron(np.eye(a), one(b)),       # row indicators
               np.kron(one(a), np.eye(b))])       # column indicators
print("rank(X) =", np.linalg.matrix_rank(X), "= a + b - 1 =", a + b - 1)

```

1.11 Vector and matrix calculus

1.11.1 Gradients of linear and quadratic functions

For a differentiable scalar function f of $\mathbf{x} \in \mathbb{R}^n$, the **gradient** $\partial f / \partial \mathbf{x}$ is the column vector with entries $\partial f / \partial x_i$, and the **Hessian** $\partial^2 f / \partial \mathbf{x} \partial \mathbf{x}^\top$ is the $n \times n$ matrix of second derivatives. For a vector-valued $\mathbf{g}(\mathbf{x}) \in \mathbb{R}^m$, the $m \times n$ *Jacobian* $\partial \mathbf{g} / \partial \mathbf{x}^\top$ has (i, j) entry $\partial g_i / \partial x_j$.

Theorem 1.11.1 — Derivatives of linear and quadratic forms

Let $\mathbf{a} \in \mathbb{R}^n$, let $\mathbf{A}$ be $n \times n$ and $\mathbf{B}$ be $m \times n$. Then

$$\frac{\partial(\mathbf{a}^\top \mathbf{x})}{\partial \mathbf{x}} = \mathbf{a}, \quad \frac{\partial(\mathbf{B}\mathbf{x})}{\partial \mathbf{x}^\top} = \mathbf{B}, \quad \frac{\partial(\mathbf{x}^\top \mathbf{A}\mathbf{x})}{\partial \mathbf{x}} = (\mathbf{A} + \mathbf{A}^\top)\mathbf{x}, \quad \frac{\partial^2(\mathbf{x}^\top \mathbf{A}\mathbf{x})}{\partial \mathbf{x} \partial \mathbf{x}^\top} = \mathbf{A} + \mathbf{A}^\top.$$

In particular $\partial(\mathbf{x}^\top \mathbf{A}\mathbf{x}) / \partial \mathbf{x} = 2\mathbf{A}\mathbf{x}$ when $\mathbf{A}$ is symmetric.

Proof. The first two are immediate from $\mathbf{a}^\top \mathbf{x} = \sum_i a_i x_i$ and $(\mathbf{B}\mathbf{x})_i = \sum_j b_{ij} x_j$. For the quadratic form, expand

$$(\mathbf{x} + \mathbf{h})^\top \mathbf{A}(\mathbf{x} + \mathbf{h}) = \mathbf{x}^\top \mathbf{A}\mathbf{x} + \mathbf{h}^\top (\mathbf{A} + \mathbf{A}^\top)\mathbf{x} + \mathbf{h}^\top \mathbf{A}\mathbf{h}.$$

The middle term is linear in $\mathbf{h}$ and the last is $O(\|\mathbf{h}\|^2)$, so $(\mathbf{A} + \mathbf{A}^\top)\mathbf{x}$ is the gradient. The gradient is linear in $\mathbf{x}$ with matrix $\mathbf{A} + \mathbf{A}^\top$, which is the Hessian by the second formula. ■

Example. (The least squares criterion)

Let $S(\mathbf{b}) = \|\mathbf{y} - \mathbf{X}\mathbf{b}\|^2 = \mathbf{y}^\top \mathbf{y} - 2\mathbf{y}^\top \mathbf{X}\mathbf{b} + \mathbf{b}^\top \mathbf{X}^\top \mathbf{X}\mathbf{b}$. By Theorem 1.11.1,

$$\frac{\partial S}{\partial \mathbf{b}} = -2\mathbf{X}^\top \mathbf{y} + 2\mathbf{X}^\top \mathbf{X}\mathbf{b} = -2\mathbf{X}^\top (\mathbf{y} - \mathbf{X}\mathbf{b}), \quad \frac{\partial^2 S}{\partial \mathbf{b} \partial \mathbf{b}^\top} = 2\mathbf{X}^\top \mathbf{X}.$$

Setting the gradient to zero gives the normal equations $\mathbf{X}^\top \mathbf{X}\mathbf{b} = \mathbf{X}^\top \mathbf{y}$. The Hessian is nonnegative definite, so S is convex and every solution is a global minimizer. The next

result proves this without calculus, and covers the rank-deficient case.

Proposition 1.11.2 — Minimizing a quadratic function

Let $\mathbf{A}$ be $n \times n$ nonnegative definite, $\mathbf{b} \in \mathbb{R}^n$ and $f(\mathbf{x}) = \mathbf{x}^\top \mathbf{A} \mathbf{x} - 2\mathbf{b}^\top \mathbf{x}$. If $\mathbf{b} \in \mathcal{C}(\mathbf{A})$, the minimizers of f are exactly the solutions of $\mathbf{A} \mathbf{x} = \mathbf{b}$, and the minimum value is $-\mathbf{b}^\top \mathbf{A}^- \mathbf{b}$ for any generalized inverse $\mathbf{A}^-$. If $\mathbf{b} \notin \mathcal{C}(\mathbf{A})$, then f is unbounded below.

Proof. Let $\mathbf{b} \in \mathcal{C}(\mathbf{A})$ and let $\mathbf{x}_0$ solve $\mathbf{A} \mathbf{x}_0 = \mathbf{b}$. Completing the square,

$$f(\mathbf{x}) = (\mathbf{x} - \mathbf{x}_0)^\top \mathbf{A} (\mathbf{x} - \mathbf{x}_0) - \mathbf{x}_0^\top \mathbf{A} \mathbf{x}_0.$$

The first term is nonnegative, and by Proposition 1.7.5(b) it is zero iff $\mathbf{A}(\mathbf{x} - \mathbf{x}_0) = \mathbf{0}$, that is, iff $\mathbf{A} \mathbf{x} = \mathbf{b}$. The minimum is $-\mathbf{x}_0^\top \mathbf{A} \mathbf{x}_0 = -\mathbf{x}_0^\top \mathbf{A} \mathbf{A}^- \mathbf{A} \mathbf{x}_0 = -\mathbf{b}^\top \mathbf{A}^- \mathbf{b}$. If $\mathbf{b} \notin \mathcal{C}(\mathbf{A})$, write $\mathbf{b} = \mathbf{b}_1 + \mathbf{b}_2$ with $\mathbf{b}_1 \in \mathcal{C}(\mathbf{A})$ and $\mathbf{0} \neq \mathbf{b}_2 \in \mathcal{C}(\mathbf{A})^\perp = \mathcal{N}(\mathbf{A})$ (Proposition 1.5.1). Then $f(t\mathbf{b}_2) = -2t\|\mathbf{b}_2\|^2 \rightarrow -\infty$ as $t \rightarrow \infty$. ■

With $\mathbf{A} = \mathbf{X}^\top \mathbf{X}$ and $\mathbf{b} = \mathbf{X}^\top \mathbf{y}$, the proposition gives least squares for every model matrix: the minimizers are the solutions of the normal equations, and the minimum of S is $\mathbf{y}^\top \mathbf{y} - \mathbf{y}^\top \mathbf{X}(\mathbf{X}^\top \mathbf{X})^- \mathbf{X}^\top \mathbf{y}$.

1.11.2 Constrained minimization

To minimize a differentiable $f(\mathbf{x})$ subject to linear constraints $\mathbf{H} \mathbf{x} = \mathbf{h}$, with $\mathbf{H}$ of size $q \times n$, the method of **Lagrange multipliers** looks for stationary points of $f(\mathbf{x}) + 2\boldsymbol{\theta}^\top (\mathbf{H} \mathbf{x} - \mathbf{h})$ in $(\mathbf{x}, \boldsymbol{\theta})$. The factor 2 is only for convenience. For quadratic f the stationary point can be written down and checked directly.

Proposition 1.11.3

Let $\mathbf{A}$ be positive definite, $\mathbf{H}$ of size $q \times n$ with full row rank, and $f(\mathbf{x}) = \mathbf{x}^\top \mathbf{A} \mathbf{x} - 2\mathbf{b}^\top \mathbf{x}$. The unique minimizer of f subject to $\mathbf{H} \mathbf{x} = \mathbf{h}$ is

$$\mathbf{x}_* = \mathbf{x}_0 - \mathbf{A}^{-1} \mathbf{H}^\top (\mathbf{H} \mathbf{A}^{-1} \mathbf{H}^\top)^{-1} (\mathbf{H} \mathbf{x}_0 - \mathbf{h}), \quad \mathbf{x}_0 = \mathbf{A}^{-1} \mathbf{b}.$$

Proof. $\mathbf{H} \mathbf{A}^{-1} \mathbf{H}^\top$ is positive definite by Proposition 1.7.5(c,e), so $\mathbf{x}_*$ is well defined, and $\mathbf{H} \mathbf{x}_* = \mathbf{h}$ by direct substitution. Stationarity of the Lagrangian gives $2\mathbf{A} \mathbf{x} - 2\mathbf{b} + 2\mathbf{H}^\top \boldsymbol{\theta} = \mathbf{0}$. Indeed $\mathbf{A} \mathbf{x}_* - \mathbf{b} = -\mathbf{H}^\top \boldsymbol{\theta}_*$ with $\boldsymbol{\theta}_* = (\mathbf{H} \mathbf{A}^{-1} \mathbf{H}^\top)^{-1} (\mathbf{H} \mathbf{x}_0 - \mathbf{h})$. For any feasible $\mathbf{x}$, put $\mathbf{d} = \mathbf{x} - \mathbf{x}_*$, so that $\mathbf{H} \mathbf{d} = \mathbf{0}$. Then

$$f(\mathbf{x}) = f(\mathbf{x}_*) + 2\mathbf{d}^\top (\mathbf{A} \mathbf{x}_* - \mathbf{b}) + \mathbf{d}^\top \mathbf{A} \mathbf{d} = f(\mathbf{x}_*) - 2(\mathbf{H} \mathbf{d})^\top \boldsymbol{\theta}_* + \mathbf{d}^\top \mathbf{A} \mathbf{d} = f(\mathbf{x}_*) + \mathbf{d}^\top \mathbf{A} \mathbf{d},$$

which exceeds $f(\mathbf{x}_*)$ unless $\mathbf{d} = \mathbf{0}$. ■

With $\mathbf{A} = \mathbf{X}^\top \mathbf{X}$ and $\mathbf{b} = \mathbf{X}^\top \mathbf{y}$, this is least squares subject to a linear hypothesis $\mathbf{H}\boldsymbol{\beta} = \mathbf{h}$, the starting point of [Chapter 11](#).

1.11.3 Derivatives with respect to matrices and scalars

Likelihoods for normal models involve $\log \det \boldsymbol{\Sigma}$ and $\text{tr}(\boldsymbol{\Sigma}^{-1} \mathbf{S})$. For a scalar function f of an $m \times n$ matrix $\mathbf{W}$ with functionally independent entries, $\partial f / \partial \mathbf{W}$ is the $m \times n$ matrix with entries $\partial f / \partial w_{ij}$.

Proposition 1.11.4 (a) $\partial \text{tr}(\mathbf{A}\mathbf{W}) / \partial \mathbf{W} = \mathbf{A}^\top$.

(b) If $\det \mathbf{W} > 0$, then $\partial \log \det \mathbf{W} / \partial \mathbf{W} = (\mathbf{W}^{-1})^\top$.

(c) If $\mathbf{A}(t)$ is a nonsingular matrix whose entries are differentiable functions of a scalar t , with derivative $\dot{\mathbf{A}} = \partial \mathbf{A} / \partial t$, then

$$\frac{\partial \mathbf{A}^{-1}}{\partial t} = -\mathbf{A}^{-1} \dot{\mathbf{A}} \mathbf{A}^{-1}, \quad \frac{\partial \log |\det \mathbf{A}|}{\partial t} = \text{tr}(\mathbf{A}^{-1} \dot{\mathbf{A}}).$$

Proof.

(a) $\text{tr}(\mathbf{A}\mathbf{W}) = \sum_{i,j} a_{ji} w_{ij}$. (c) Differentiate $\mathbf{A}\mathbf{A}^{-1} = \mathbf{I}$ by the product rule to get $\dot{\mathbf{A}}\mathbf{A}^{-1} + \mathbf{A}\partial \mathbf{A}^{-1} / \partial t = \mathbf{0}$. For the determinant, $\mathbf{A}(t+h) = \mathbf{A}(t)(\mathbf{I} + h\mathbf{A}^{-1}\dot{\mathbf{A}} + o(h))$. In the permutation expansion of $\det(\mathbf{I} + h\mathbf{C})$ only the diagonal product contributes terms of first order in h , so $\det(\mathbf{I} + h\mathbf{C}) = 1 + h \text{tr} \mathbf{C} + O(h^2)$. Hence $\det \mathbf{A}(t+h) = \det \mathbf{A}(t)(1 + h \text{tr}(\mathbf{A}^{-1}\dot{\mathbf{A}}) + o(h))$, and taking logarithms gives the formula. (b) Apply (c) with $\mathbf{A}(t) = \mathbf{W} + t\mathbf{e}_i\mathbf{e}_j^\top$. Then $\text{tr}(\mathbf{W}^{-1}\mathbf{e}_i\mathbf{e}_j^\top) = \mathbf{e}_j^\top \mathbf{W}^{-1} \mathbf{e}_i$, the (j, i) entry of $\mathbf{W}^{-1}$.

■

When $\mathbf{W}$ is constrained to be symmetric, w_{ij} and w_{ji} are the same variable, so the derivatives with respect to off-diagonal entries double. Some books (Renchner and Schaalje 2008) report formulas of the type $2\mathbf{W}^{-1} - \text{diag}(\mathbf{W}^{-1})$ for this reason. The simplest way to avoid the issue is to differentiate along a symmetric direction $\mathbf{W} + t\mathbf{E}$, as in (c). This is the convenient route for the normal likelihoods of Chapter 32. The listing checks the gradient formulas against central finite differences, and the script also checks part (c) and [Proposition 1.11.2](#). The largest discrepancy in the listing is 3×10^{-9} , which is the size one expects from the finite-difference approximation.

```
import numpy as np
rng = np.random.default_rng(5)
p = 4
def num_grad(f, x, h=1e-6):
    """Central-difference gradient of a scalar function of a vector or matrix."""
    g = np.zeros_like(x)
    for idx in np.ndindex(x.shape):
        e = np.zeros_like(x)
        e[idx] = h
        g[idx] = (f(x + e) - f(x - e)) / (2 * h)
    return g

a = rng.normal(size=p)
```

```

A = rng.normal(size=(p, p))                                # not symmetric
x = rng.normal(size=p)
checks = {
    "a'x": (num_grad(lambda v: a @ v, x), a),
    "x'Ax": (num_grad(lambda v: v @ A @ v, x), (A + A.T) @ x),
}
X = rng.normal(size=(20, p))
y = rng.normal(size=20)
b = rng.normal(size=p)
grad_ls = -2 * X.T @ (y - X @ b)
checks["||y - Xb||^2"] = (num_grad(lambda v: np.sum((y - X @ v) ** 2), b), grad_ls)

S = rng.normal(size=(p, p)); S = S @ S.T + p * np.eye(p)    # positive definite
checks["log det S"] = (num_grad(lambda W: np.linalg.slogdet(W)[1], S), np.linalg.inv(S).T)
checks["tr(AS)"] = (num_grad(lambda W: np.trace(A @ W), S), A.T)
for name, (numeric, formula) in checks.items():
    print(f"{name:14s} max error {np.abs(numeric - formula).max():.1e}")

```

1.11.4 Exercises

1.11.4.1 A. Check your understanding

Exercise A1.

Observations fall in a clusters of size b , and $\Sigma = \sigma^2 \mathbf{I}_a \otimes \mathbf{E}_b$, where $\mathbf{E}_b = (1 - \rho)\mathbf{I}_b + \rho\mathbf{J}_b$. Find the eigenvalues, determinant and inverse of Σ using [Proposition 1.10.1](#), [Example \(Equicorrelation\)](#) and [Example \(Eigenvalues of the equicorrelation matrix\)](#).

Exercise A2.

Use [Proposition 1.11.2](#) to show that $\|\mathbf{y} - \mathbf{X}\mathbf{b}\|^2 + \lambda\|\mathbf{b}\|^2$ with $\lambda > 0$ has the unique minimizer $(\mathbf{X}^\top \mathbf{X} + \lambda \mathbf{I})^{-1} \mathbf{X}^\top \mathbf{y}$, whatever the rank of $\mathbf{X}$.

1.11.4.2 B. Practice

Exercise B1.

Let $\mathbf{X}$ have full column rank, $\hat{\boldsymbol{\beta}} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{y}$, and let $\mathbf{H}$ be $q \times p$ with full row rank. Show that the minimizer of $\|\mathbf{y} - \mathbf{X}\mathbf{b}\|^2$ subject to $\mathbf{H}\mathbf{b} = \mathbf{h}$ is

$$\hat{\boldsymbol{\beta}}_H = \hat{\boldsymbol{\beta}} - (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{H}^\top \mathbf{W}^{-1} (\mathbf{H} \hat{\boldsymbol{\beta}} - \mathbf{h}), \quad \mathbf{W} = \mathbf{H} (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{H}^\top,$$

and that the increase in the residual sum of squares is $(\mathbf{H} \hat{\boldsymbol{\beta}} - \mathbf{h})^\top \mathbf{W}^{-1} (\mathbf{H} \hat{\boldsymbol{\beta}} - \mathbf{h})$.

Solution. Apply [Proposition 1.11.3](#) with $\mathbf{A} = \mathbf{X}^\top \mathbf{X}$ and $\mathbf{b} = \mathbf{X}^\top \mathbf{y}$. Since $\|\mathbf{y} - \mathbf{X}\mathbf{b}\|^2 = \mathbf{y}^\top \mathbf{y} + f(\mathbf{b})$ and $\mathbf{x}_0 = \hat{\boldsymbol{\beta}}$, this gives $\hat{\boldsymbol{\beta}}_H$. By completing the square as in [Proposition 1.11.2](#), $\|\mathbf{y} - \mathbf{X}\mathbf{b}\|^2 = \|\mathbf{y} - \mathbf{X}\hat{\boldsymbol{\beta}}\|^2 + (\mathbf{b} - \hat{\boldsymbol{\beta}})^\top \mathbf{X}^\top \mathbf{X} (\mathbf{b} - \hat{\boldsymbol{\beta}})$ for every $\mathbf{b}$. With

$\mathbf{b} - \hat{\boldsymbol{\beta}} = -(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{H}^\top \mathbf{W}^{-1} (\mathbf{H} \hat{\boldsymbol{\beta}} - \mathbf{h})$, the increase is

$$(\mathbf{H} \hat{\boldsymbol{\beta}} - \mathbf{h})^\top \mathbf{W}^{-1} \mathbf{H} (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{H}^\top \mathbf{W}^{-1} (\mathbf{H} \hat{\boldsymbol{\beta}} - \mathbf{h}) = (\mathbf{H} \hat{\boldsymbol{\beta}} - \mathbf{h})^\top \mathbf{W}^{-1} (\mathbf{H} \hat{\boldsymbol{\beta}} - \mathbf{h}).$$

This quadratic form is the numerator of the F statistic in Chapter 11 (Theorem 11.3.2).

1.12 Summary and Notes

1.12.1 Summary

Idea.

1. Row rank equals column rank, and $\text{rank}(\mathbf{A}) + \dim \mathcal{N}(\mathbf{A})$ is the number of columns. $\mathcal{C}(\mathbf{A}^\top \mathbf{A}) = \mathcal{C}(\mathbf{A}^\top)$, so the normal equations are always consistent (Theorem 1.2.1, Theorem 1.2.2 and Proposition 1.2.3).
2. A generalized inverse satisfies $\mathbf{A} \mathbf{A}^- \mathbf{A} = \mathbf{A}$. Every matrix has one, built from any nonsingular $r \times r$ submatrix. The Moore–Penrose inverse is the unique one that is also reflexive and makes $\mathbf{A} \mathbf{A}^+$ and $\mathbf{A}^+ \mathbf{A}$ symmetric (Definition 1.3.1, Theorem 1.3.2, Proposition 1.2.4 and Definition 1.3.2).
3. $\mathbf{A} \mathbf{x} = \mathbf{b}$ is consistent iff $\mathbf{A} \mathbf{A}^- \mathbf{b} = \mathbf{b}$, and then its solutions are $\mathbf{A}^- \mathbf{b} + (\mathbf{I} - \mathbf{A}^- \mathbf{A}) \mathbf{z}$. A linear function $\mathbf{q}^\top \mathbf{x}$ is the same for all solutions iff $\mathbf{q} \in \mathcal{C}(\mathbf{A}^\top)$ (Theorem 1.4.1 and Corollary 1.4.2).
4. Orthogonal matrices preserve lengths and angles. Gram–Schmidt gives the QR factorization, and $\mathbb{R}^n = \mathcal{C}(\mathbf{A}^\top) \oplus \mathcal{N}(\mathbf{A})$ (Proposition 1.5.3, Theorem 1.5.2 and Proposition 1.5.1).
5. $\text{tr}(\mathbf{A} \mathbf{B}) = \text{tr}(\mathbf{B} \mathbf{A})$. Block elimination gives the determinant and inverse of a partitioned matrix through Schur complements, and the Sherman–Morrison–Woodbury identity as a special case (Theorem 1.6.1, Theorem 1.6.3, Theorem 1.6.4 and Theorem 1.6.5).
6. A symmetric matrix is $\mathbf{Q} \boldsymbol{\Lambda} \mathbf{Q}^\top$ with $\mathbf{Q}$ orthogonal. Positive definiteness is equivalent to positive eigenvalues, to a Cholesky factorization, and to positive leading minors. Nonnegative definite matrices have a unique nonnegative definite square root (Theorem 1.7.2, Theorem 1.7.4 and Theorem 1.7.6).
7. The Rayleigh quotient $\mathbf{x}^\top \mathbf{A} \mathbf{x} / \mathbf{x}^\top \mathbf{x}$ ranges over $[\lambda_{\min}, \lambda_{\max}]$ (Theorem 1.7.7 and Corollary 1.7.8).
8. Every matrix is $\mathbf{U} \mathbf{D} \mathbf{V}^\top$. The condition number $\kappa = \sigma_{\max} / \sigma_{\min}$ bounds error growth, is squared by forming $\mathbf{X}^\top \mathbf{X}$, and its smallest singular vector exhibits near-dependences among the columns (Theorem 1.8.1, Definition 1.8.1 and Proposition 1.8.2).
9. Idempotent matrices have eigenvalues 0 and 1 and rank equal to trace. Symmetric ones are $\mathbf{Q}_1 \mathbf{Q}_1^\top$. If symmetric matrices summing to $\mathbf{I}$ have ranks summing to n , they are idempotent and mutually orthogonal (Proposition 1.9.1, Theorem 1.9.2 and Theorem 1.9.3).
10. Kronecker products factor balanced designs. $\text{vec}(\mathbf{A} \mathbf{B} \mathbf{C}) = (\mathbf{C}^\top \otimes \mathbf{A}) \text{vec}(\mathbf{B})$ (Proposition 1.10.1).
11. $\partial(\mathbf{x}^\top \mathbf{A} \mathbf{x}) / \partial \mathbf{x} = (\mathbf{A} + \mathbf{A}^\top) \mathbf{x}$. Quadratic functions with nonnegative definite ma-

trices are minimized by solving a linear system, with or without linear constraints ([Theorem 1.11.1](#), [Proposition 1.11.2](#) and [Proposition 1.11.3](#)).

1.12.2 Notes and sources

Coverage and framing. The selection of results follows what the rest of the book uses. Rencher and Schaalje (2008, chapter 2) was the checklist for topics: its sections on positive definite matrices, generalized inverses, orthogonal matrices, idempotent matrices and matrix calculus correspond to [Section 1.7](#), [Section 1.3](#), [Section 1.5](#), [Section 1.9](#) and [Section 1.11](#). Several results that Rencher and Schaalje state without proof or with a reference, such as the general solution of a consistent system, the characterizations of positive definiteness and the sums of idempotent matrices, are proved here. The vector-space point of view, in which column spaces, null spaces and orthogonal complements come first and matrices second, is that of Christensen (2020, appendices A and B). Seber and Lee (2003, appendix A) is a compact list of further matrix facts. The Kronecker product results in [Section 1.10](#) are also in Christensen’s appendix B.

Linear algebra for statisticians. Harville (1997) is the most complete treatment written for statisticians, with full proofs of the determinant facts assumed in [Proposition 1.6.2](#) and a long development of generalized inverses. Searle (1982) is more elementary and has many worked examples from linear models. Horn and Johnson (2013) is the standard reference for eigenvalue inequalities, positive definite matrices and norms. Halmos (1958) gives the coordinate-free view of the vector-space facts used in [Section 1.2](#).

Generalized inverses. The four conditions of [Definition 1.3.2](#) are due to Penrose (1955), who proved existence and uniqueness. E.H. Moore had described an equivalent inverse decades earlier. Rao (1973) made the weaker inverse $\mathbf{A}\mathbf{A}^-\mathbf{A} = \mathbf{A}$ the basic tool of linear-model theory, and it is his approach that [Section 1.4](#) and [Chapter 8](#) follow. The construction from a nonsingular submatrix in [Theorem 1.3.2](#) is a matrix-free way of stating an algorithm that is often given as a sequence of steps (Searle 1982).

Schur complements and updating formulas. The name “Schur complement” and the notation for it vary. Zhang (2005) collects the history and many applications. The rank-one formula [Eq. 1.6.7](#) is from Sherman and Morrison (1950), and the general identity [Eq. 1.6.6](#) from Woodbury (1950). Henderson and Searle (1981) trace earlier appearances and compare several derivations. Obtaining the identity by comparing two expressions for one block of a partitioned inverse, as in [Theorem 1.6.5](#), is a classical route.

Singular values and conditioning. Stewart (1993) tells the early history of the singular value decomposition, beginning with Beltrami and Jordan in the 1870s. The low-rank approximation theorem, [Proposition 1.8.2\(c\)](#) and its Frobenius form, is usually credited to Eckart and Young (1936), although Stewart points out that Schmidt had proved it earlier in the setting of integral equations. Golub and Van Loan (2013) is the reference for computing the decomposition and for perturbation bounds such as [Eq. 1.8.3](#), and Higham (2002) for their refinements. The use of scaled condition numbers and singular vectors to diagnose collinearity is due to Belsley et al. (1980). Chapter 26 returns to it.

Kronecker products and calculus. Henderson and Searle (1979) survey the vec operator and its uses in multivariate statistics. Magnus and Neudecker (2019) develop matrix calculus systematically through differentials, which avoid the bookkeeping of the entrywise definitions used in [Section 1.11](#) and handle symmetric matrix arguments cleanly.

1.12.3 References

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Chapter 2

Random Vectors: Expectation and Covariance Operators

A linear model is a statement about the first two moments of a random vector: its mean lies in a subspace and its covariance matrix has a given form. This chapter builds the calculus for such statements. Expectation and covariance become matrix operators that commute with linear maps, covariance matrices turn out to be exactly the nonnegative definite matrices, and a singular covariance means the vector is confined to a flat. The mean of a quadratic form needs only two moments; its variance needs four. Partitioning a vector gives cross-covariances, Schur complements and the best linear predictor, which foreshadow conditional distributions under normality.

What you need. [Chapter 1](#) (trace, spectral theorem, nonnegative definite matrices and their square roots, generalized inverses, partitioned matrices), and probability at the level of expectations, variances and independence of random variables.

2.0.1 Roadmap

- [Mean vectors and covariance matrices](#)
- [Linear and affine transformations](#)
- [Expectation of quadratic forms](#)
- [Correlation matrices](#)
- [Moments of partitioned random vectors](#)
- [Moment generating functions and independence](#)
- [Sample moments and their computation](#)
- [Summary and notes](#)

2.1 Mean vectors and covariance matrices

Random vectors enter regression in two different ways. The first is the response vector. In the model

$$\mathbf{Y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon},$$

$\mathbf{Y} = (Y_1, \dots, Y_n)^\top$ has one entry per observational unit, and so does the unobservable error $\boldsymbol{\varepsilon}$. The assumptions that do the real work in [Chapter 5](#), [Chapter 6](#) and [Chapter 7](#) are not about the shape of any distribution. They are statements about moments: $E(\boldsymbol{\varepsilon}) = \mathbf{0}$, and the errors have equal variances and are uncorrelated. The second way is the measurement vector. When several quantities are recorded on the same unit, such as income, age and years of schooling, the unit contributes a vector whose components are usually correlated and measured in different units. Random regressors, errors-in-variables models and multivariate responses are all built from such vectors.

Both uses need the same toolkit: expectation and covariance as operations on vectors and matrices, their behaviour under linear maps, and the expected values of quadratic forms. Throughout, a random vector *has finite second moments* if every component has finite variance.

2.1.1 Random matrices and their expectation

Definition 2.1.1 — Expectation of a random matrix

A **random matrix** $\mathbf{Z} = (Z_{ij})$ is an $m \times k$ array of random variables. If every Z_{ij} has a finite expectation, the **expectation** of $\mathbf{Z}$ is the $m \times k$ matrix of expectations,

$$E(\mathbf{Z}) = (E(Z_{ij})).$$

A random vector is the case $k = 1$.

The definition works entry by entry, so every rule that is linear in the entries carries over at once.

Proposition 2.1.1 — Linearity of expectation

Let $\mathbf{Z}$ and $\mathbf{Z}_2$ be $m \times k$ random matrices with integrable entries, and let $\mathbf{A}$ ($l \times m$), $\mathbf{B}$ ($k \times s$) and $\mathbf{C}$ ($l \times s$) be constant matrices. Then

- (a) $E(\mathbf{AZB} + \mathbf{C}) = \mathbf{A} E(\mathbf{Z}) \mathbf{B} + \mathbf{C}$;
- (b) $E(\mathbf{Z} + \mathbf{Z}_2) = E(\mathbf{Z}) + E(\mathbf{Z}_2)$ and $E(\mathbf{Z}^\top) = E(\mathbf{Z})^\top$;
- (c) if $m = k$, then $E \operatorname{tr}(\mathbf{Z}) = \operatorname{tr} E(\mathbf{Z})$.

Proof. The (i, j) entry of $\mathbf{AZB} + \mathbf{C}$ is $\sum_r \sum_t a_{ir} Z_{rt} b_{tj} + c_{ij}$, a finite linear combination of integrable random variables with constant coefficients. Its expectation is $\sum_r \sum_t a_{ir} E(Z_{rt}) b_{tj} + c_{ij}$, which is the (i, j) entry of $\mathbf{A} E(\mathbf{Z}) \mathbf{B} + \mathbf{C}$. Parts (b) and (c) are the same argument applied to $Z_{ij} + Z_{2,ij}$, to Z_{ji} , and to $\sum_i Z_{ii}$. ■

Two warnings go with this proposition. The multipliers must be constant: if $\mathbf{A}$ is random, $E(\mathbf{AZ})$ is generally not $E(\mathbf{A})E(\mathbf{Z})$. And the expectation of a product of random matrices is generally not the product of expectations. It is when every entry of the first is independent of every entry of the second, because then $E(Z_{ir}W_{rj}) = E(Z_{ir})E(W_{rj})$ for each term.

2.1.2 Mean vector and covariance matrix

Definition 2.1.2 — Mean vector and covariance matrices

Let $\mathbf{Y}$ be a $p \times 1$ random vector with finite second moments. Its **mean vector** is $\boldsymbol{\mu} = E(\mathbf{Y})$, and its **covariance matrix** is the $p \times p$ matrix

$$\text{Cov}(\mathbf{Y}) = E[(\mathbf{Y} - \boldsymbol{\mu})(\mathbf{Y} - \boldsymbol{\mu})^\top],$$

whose (i, j) entry is $\sigma_{ij} = \text{Cov}(Y_i, Y_j)$ and whose diagonal holds the variances $\sigma_{ii} = \text{Var}(Y_i)$. If $\mathbf{U}$ ($p \times 1$) and $\mathbf{V}$ ($q \times 1$) both have finite second moments, their **cross-covariance matrix** is the $p \times q$ matrix

$$\text{Cov}(\mathbf{U}, \mathbf{V}) = E[(\mathbf{U} - E \mathbf{U})(\mathbf{V} - E \mathbf{V})^\top],$$

with (i, j) entry $\text{Cov}(U_i, V_j)$.

The expectations exist: $|U_i V_j| \leq \frac{1}{2}(U_i^2 + V_j^2)$, so finite second moments make every product integrable. We often write $\boldsymbol{\Sigma}$ for $\text{Cov}(\mathbf{Y})$, and $\text{Cov}(\mathbf{Y}) = \text{Cov}(\mathbf{Y}, \mathbf{Y})$. Some books write $\text{Var}(\mathbf{Y})$ or $D(\mathbf{Y})$ for the covariance matrix. We keep Var for scalars. The symbol Cov then has three uses: $\text{Cov}(Y_i, Y_j)$ is a number, $\text{Cov}(\mathbf{Y})$ is a symmetric square matrix, and $\text{Cov}(\mathbf{U}, \mathbf{V})$ is a rectangular matrix that is usually not symmetric even when $p = q$.

Proposition 2.1.2 — Elementary properties

Let $\mathbf{Y}$, $\mathbf{U}$, $\mathbf{V}$ have finite second moments, with $\boldsymbol{\mu} = E(\mathbf{Y})$ and $\boldsymbol{\Sigma} = \text{Cov}(\mathbf{Y})$, and let $\mathbf{a}$, $\mathbf{b}$ be constant vectors of matching sizes. Then:

- (a) $\boldsymbol{\Sigma}$ is symmetric;
- (b) $\text{Cov}(\mathbf{U}, \mathbf{V}) = E(\mathbf{UV}^\top) - E(\mathbf{U})E(\mathbf{V})^\top$, and in particular $\boldsymbol{\Sigma} = E(\mathbf{YY}^\top) - \boldsymbol{\mu}\boldsymbol{\mu}^\top$;
- (c) $\text{Cov}(\mathbf{V}, \mathbf{U}) = \text{Cov}(\mathbf{U}, \mathbf{V})^\top$;
- (d) $\text{Cov}(\mathbf{U} + \mathbf{a}, \mathbf{V} + \mathbf{b}) = \text{Cov}(\mathbf{U}, \mathbf{V})$;
- (e) $E[(\mathbf{Y} - \mathbf{a})(\mathbf{Y} - \mathbf{a})^\top] = \boldsymbol{\Sigma} + (\boldsymbol{\mu} - \mathbf{a})(\boldsymbol{\mu} - \mathbf{a})^\top$, and hence $E\|\mathbf{Y} - \mathbf{a}\|^2 = \text{tr}(\boldsymbol{\Sigma}) + \|\boldsymbol{\mu} - \mathbf{a}\|^2$.

Proof.

- (a) and (c) hold because $(\mathbf{uv}^\top)^\top = \mathbf{vu}^\top$ and expectation commutes with transposition. For (b), expand $(\mathbf{U} - \boldsymbol{\mu}_U)(\mathbf{V} - \boldsymbol{\mu}_V)^\top = \mathbf{UV}^\top - \mathbf{U}\boldsymbol{\mu}_V^\top - \boldsymbol{\mu}_U\mathbf{V}^\top + \boldsymbol{\mu}_U\boldsymbol{\mu}_V^\top$ and take expectations with Proposition 2.1.1. The two middle terms each give $-\boldsymbol{\mu}_U\boldsymbol{\mu}_V^\top$. (d) holds because adding a constant does not change the centred vectors. For (e), write $\mathbf{Y} - \mathbf{a} = (\mathbf{Y} - \boldsymbol{\mu}) + (\boldsymbol{\mu} - \mathbf{a})$ and expand. The cross terms contain the factor $E(\mathbf{Y} - \boldsymbol{\mu}) = \mathbf{0}$ and vanish. The trace identity follows from $\|\mathbf{v}\|^2 = \text{tr}(\mathbf{vv}^\top)$ and Proposition 2.1.1(c). ■

Part (e) is the vector form of “mean squared error equals variance plus squared bias”. It shows that the mean is the constant vector closest to $\mathbf{Y}$ in expected squared distance, and that the minimal value is $\text{tr}(\boldsymbol{\Sigma}) = \sum_i \text{Var}(Y_i)$. The matrix statement is stronger than the trace statement.

It says that the second-moment matrix about any point exceeds Σ by a nonnegative definite matrix of rank at most one.

2.1.3 Uncorrelated is not independent

Random vectors $\mathbf{U}$ and $\mathbf{V}$ are **uncorrelated** if $\text{Cov}(\mathbf{U}, \mathbf{V}) = \mathbf{0}$. If $\mathbf{U}$ and $\mathbf{V}$ are independent, then $E(U_i V_j) = E(U_i) E(V_j)$ for all i, j , so they are uncorrelated. The converse is false, and it fails in the most innocent-looking cases.

Example. (A point on a circle)

Let Θ be uniform on $[0, 2\pi)$ and $\mathbf{Y} = (\cos \Theta, \sin \Theta)^\top$. Then $E(\cos \Theta) = E(\sin \Theta) = 0$, $E(\cos^2 \Theta) = E(\sin^2 \Theta) = \frac{1}{2}$, and $E(\cos \Theta \sin \Theta) = \frac{1}{2} E(\sin 2\Theta) = 0$. So $\boldsymbol{\mu} = \mathbf{0}$ and $\text{Cov}(\mathbf{Y}) = \frac{1}{2} \mathbf{I}_2$, exactly the moments of two independent coordinates. Yet $Y_1^2 + Y_2^2 = 1$: once Y_1 is known, Y_2 is known up to sign. Covariance measures *linear* association only, and the dependence here is entirely nonlinear.

For jointly normal vectors, zero covariance and independence coincide ([Theorem 3.4.1](#)). That equivalence rests on joint normality, not on normal marginals, and it is one of the main reasons normal theory is so tractable.

Scalar summaries. The **total variance** $\text{tr}(\Sigma)$ ignores correlation; the **generalized variance** $\det(\Sigma)$ does not. If $\lambda_1, \dots, \lambda_p$ are the eigenvalues of Σ ([Theorem 1.7.2](#)), these are $\sum_i \lambda_i$ and $\prod_i \lambda_i$, and both are unchanged by a rotation $\mathbf{Y} \mapsto \mathbf{QY}$. The determinant can be tiny even when every variance is large, if one eigenvalue is near zero. [Theorem 2.2.3](#) shows that a zero eigenvalue means some linear combination of $\mathbf{Y}$ does not vary at all.

2.1.4 Exercises

2.1.4.1 A. Check your understanding

Exercise A1.

Let $\mathbf{Z}$ be an $n \times p$ random matrix whose entries are independent, with mean 0 and variance σ^2 . Show that $E(\mathbf{Z}^\top \mathbf{Z}) = n\sigma^2 \mathbf{I}_p$ and $E(\mathbf{Z}\mathbf{Z}^\top) = p\sigma^2 \mathbf{I}_n$. For a constant $p \times p$ matrix $\mathbf{B}$, find $E(\mathbf{Z}\mathbf{B}\mathbf{Z}^\top)$.

2.1.4.2 B. Practice

Exercise B1.

Let X have a distribution symmetric about 0 with $\text{Var}(X) = 1$ and $E X^4 < \infty$, let S be independent of X with $\Pr(S = 1) = \Pr(S = -1) = \frac{1}{2}$, and put $\mathbf{Y} = (X, SX)^\top$. Show that the two components have the same distribution and are uncorrelated, and that $\text{Cov}(Y_1^2, Y_2^2) = \text{Var}(X^2)$. Deduce that Y_1 and Y_2 are independent iff X is itself a random sign. ([Chapter 3](#) takes X standard normal.)

Solution. Since S is independent of X and X is symmetric, $\Pr(SX \leq y) = \frac{1}{2} \Pr(X \leq y) + \frac{1}{2} \Pr(-X \leq y) = \Pr(X \leq y)$. By independence, $\text{Cov}(X, SX) = E(S)E(X^2) = 0$. Both squares equal X^2 , so $\text{Cov}(Y_1^2, Y_2^2) = \text{Var}(X^2)$. Functions of independent variables are independent, hence uncorrelated, so $\text{Var}(X^2) > 0$ makes Y_1 and Y_2 dependent. If $\text{Var}(X^2) = 0$, then $X^2 = E X^2 = 1$ with probability one, and by symmetry $X = \pm 1$ with probability $\frac{1}{2}$ each. Then $\Pr(Y_1 = a, Y_2 = b) = \Pr(X = a) \Pr(S = ab) = \frac{1}{4}$ for all signs a, b , which is the product of the marginal probabilities, as in [Example \(Pairwise but not mutually independent\)](#).

Exercise B2.

Show that $E[(\mathbf{Y} - \mathbf{a})(\mathbf{Y} - \mathbf{a})^\top] - \text{Cov}(\mathbf{Y})$ is nonnegative definite for every constant $\mathbf{a}$, with equality iff $\mathbf{a} = \boldsymbol{\mu}$. Deduce that $\mathbf{a} = \boldsymbol{\mu}$ minimizes $E[(\mathbf{Y} - \mathbf{a})^\top \mathbf{N}(\mathbf{Y} - \mathbf{a})]$ for every positive definite $\mathbf{N}$.

2.2 Linear and affine transformations

Almost every statistic in a linear model is a linear function of the response: least squares coefficients, fitted values, residuals, contrasts. Their means and covariance matrices follow from one theorem.

2.2.1 The transformation rules

Theorem 2.2.1 — Moments of affine transformations

Let $\mathbf{Y}$ be a $p \times 1$ random vector with finite second moments, $E(\mathbf{Y}) = \boldsymbol{\mu}$ and $\text{Cov}(\mathbf{Y}) = \boldsymbol{\Sigma}$. Let $\mathbf{A}$ be $k \times p$, $\mathbf{B}$ be $m \times p$, and $\mathbf{b}$ be $k \times 1$, all constant. Then

- (a) $E(\mathbf{A}\mathbf{Y} + \mathbf{b}) = \mathbf{A}\boldsymbol{\mu} + \mathbf{b}$;
- (b) $\text{Cov}(\mathbf{A}\mathbf{Y} + \mathbf{b}) = \mathbf{A}\boldsymbol{\Sigma}\mathbf{A}^\top$;
- (c) $\text{Cov}(\mathbf{A}\mathbf{Y}, \mathbf{B}\mathbf{Y}) = \mathbf{A}\boldsymbol{\Sigma}\mathbf{B}^\top$.

More generally, if $\mathbf{U}$ and $\mathbf{V}$ have finite second moments and $\mathbf{A}, \mathbf{B}, \mathbf{b}, \mathbf{c}$ are constants of conformable sizes, then $\text{Cov}(\mathbf{A}\mathbf{U} + \mathbf{b}, \mathbf{B}\mathbf{V} + \mathbf{c}) = \mathbf{A} \text{Cov}(\mathbf{U}, \mathbf{V}) \mathbf{B}^\top$.

Proof.

- (a) is [Proposition 2.1.1\(a\)](#). For the general statement, the centred vectors are $\mathbf{A}\mathbf{U} + \mathbf{b} - E(\mathbf{A}\mathbf{U} + \mathbf{b}) = \mathbf{A}(\mathbf{U} - E\mathbf{U})$ and similarly $\mathbf{B}(\mathbf{V} - E\mathbf{V})$. Hence

$$\text{Cov}(\mathbf{A}\mathbf{U} + \mathbf{b}, \mathbf{B}\mathbf{V} + \mathbf{c}) = E[\mathbf{A}(\mathbf{U} - E\mathbf{U})(\mathbf{V} - E\mathbf{V})^\top \mathbf{B}^\top] = \mathbf{A} E[(\mathbf{U} - E\mathbf{U})(\mathbf{V} - E\mathbf{V})^\top] \mathbf{B}^\top,$$

again by [Proposition 2.1.1\(a\)](#). Parts (c) and (b) are the cases $\mathbf{U} = \mathbf{V} = \mathbf{Y}$, and $\mathbf{B} = \mathbf{A}$ in (b).

For a single linear combination, $\mathbf{A} = \mathbf{a}^\top$ is a row vector, and (b) and (c) give

$$\text{Var}(\mathbf{a}^\top \mathbf{Y}) = \mathbf{a}^\top \boldsymbol{\Sigma} \mathbf{a} = \sum_{i=1}^p \sum_{j=1}^p a_i a_j \sigma_{ij}, \quad \text{Cov}(\mathbf{a}^\top \mathbf{Y}, \mathbf{b}^\top \mathbf{Y}) = \mathbf{a}^\top \boldsymbol{\Sigma} \mathbf{b}. \quad (2.2.1)$$

Every variance and covariance enters $\text{Var}(\mathbf{a}^\top \mathbf{Y})$, weighted by the products of the coefficients. Conversely, $\boldsymbol{\Sigma}$ is determined by the variances of linear combinations, because $\sigma_{ij} = \frac{1}{2}[\text{Var}(Y_i + Y_j) - \text{Var}(Y_i) - \text{Var}(Y_j)]$.

Proposition 2.2.2 — Covariance of sums

Let $\mathbf{U}, \mathbf{U}_2, \mathbf{V}$ have finite second moments, with $\mathbf{U}$ and $\mathbf{U}_2$ of the same size in (a), and $\mathbf{U}$ and $\mathbf{V}$ of the same size in (b). Then

- (a) $\text{Cov}(\mathbf{U} + \mathbf{U}_2, \mathbf{V}) = \text{Cov}(\mathbf{U}, \mathbf{V}) + \text{Cov}(\mathbf{U}_2, \mathbf{V})$;
- (b) $\text{Cov}(\mathbf{U} + \mathbf{V}) = \text{Cov}(\mathbf{U}) + \text{Cov}(\mathbf{U}, \mathbf{V}) + \text{Cov}(\mathbf{V}, \mathbf{U}) + \text{Cov}(\mathbf{V})$;
- (c) if $\mathbf{Y}_1, \dots, \mathbf{Y}_n$ are pairwise uncorrelated $p \times 1$ vectors and $c_1, \dots, c_n$ are constants, then $\text{Cov}(\sum_i c_i \mathbf{Y}_i) = \sum_i c_i^2 \text{Cov}(\mathbf{Y}_i)$. In particular, if every $\text{Cov}(\mathbf{Y}_i) = \boldsymbol{\Sigma}$, then $\text{Cov}(\bar{\mathbf{Y}}) = \boldsymbol{\Sigma}/n$.

Proof.

- (a) Centre each vector and use linearity of the outer product and of E. (b) Apply (a) in each argument; note that the two cross terms are transposes of each other, not equal.
- (b) By (a) and (b), applied repeatedly, $\text{Cov}(\sum_i c_i \mathbf{Y}_i) = \sum_i \sum_j c_i c_j \text{Cov}(\mathbf{Y}_i, \mathbf{Y}_j)$, using $\text{Cov}(c\mathbf{U}, \mathbf{V}) = c \text{Cov}(\mathbf{U}, \mathbf{V})$ from [Theorem 2.2.1](#). The terms with $i \neq j$ vanish.

Example. (Centring a sample)

Let $\mathbf{Y}$ be $n \times 1$ with $\text{Cov}(\mathbf{Y}) = \sigma^2 \mathbf{I}_n$, and write $\mathbf{J} = \mathbf{1}\mathbf{1}^\top$. The mean is $\bar{Y} = n^{-1} \mathbf{1}^\top \mathbf{Y}$, and the vector of deviations is $\mathbf{d} = (\mathbf{I} - n^{-1} \mathbf{J})\mathbf{Y}$. The matrix $\mathbf{I} - n^{-1} \mathbf{J}$ is symmetric and idempotent, so [Theorem 2.2.1](#) gives

$$\text{Cov}(\mathbf{d}) = \sigma^2 (\mathbf{I} - n^{-1} \mathbf{J}), \quad \text{Cov}(\bar{Y}, \mathbf{d}) = \sigma^2 n^{-1} \mathbf{1}^\top (\mathbf{I} - n^{-1} \mathbf{J}) = \mathbf{0}^\top.$$

The deviations have variances $\sigma^2(1 - 1/n)$ and are negatively correlated, with correlation $-1/(n-1)$ between each pair, because they are forced to sum to zero. And each deviation is uncorrelated with the mean. Under normality, uncorrelated becomes independent ([Example \(Sample mean and deviations\)](#)), which is the first half of [Corollary 4.4.5](#).

2.2.2 Covariance matrices are nonnegative definite

A variance cannot be negative, and by Eq. 2.2.1 this simple fact constrains the whole matrix. Recall from Definition 1.7.1 that a symmetric Σ is nonnegative definite if $\mathbf{a}^\top \Sigma \mathbf{a} \geq 0$ for all $\mathbf{a}$, and positive definite if in addition $\mathbf{a}^\top \Sigma \mathbf{a} > 0$ for $\mathbf{a} \neq \mathbf{0}$.

Theorem 2.2.3 — Covariance matrices

Let $\mathbf{Y}$ have finite second moments, mean $\boldsymbol{\mu}$ and covariance matrix Σ .

- (a) Σ is nonnegative definite.
- (b) Conversely, every $p \times p$ nonnegative definite matrix is the covariance matrix of some random vector.
- (c) For a constant $\mathbf{a}$, $\Sigma \mathbf{a} = \mathbf{0}$ iff $\mathbf{a}^\top \mathbf{Y} = \mathbf{a}^\top \boldsymbol{\mu}$ with probability one.
- (d) $\Pr(\mathbf{Y} - \boldsymbol{\mu} \in \mathcal{C}(\Sigma)) = 1$. If $\mathcal{S}$ is any subspace with $\Pr(\mathbf{Y} - \boldsymbol{\mu} \in \mathcal{S}) = 1$, then $\mathcal{C}(\Sigma) \subseteq \mathcal{S}$.

In particular, Σ is singular iff some nontrivial linear combination of $\mathbf{Y}$ is constant with probability one. Then $\mathbf{Y}$ lies, with probability one, in the affine subspace $\boldsymbol{\mu} + \mathcal{C}(\Sigma)$ of dimension $\text{rank}(\Sigma) < p$, and this is the smallest such flat.

Proof.

- (a) Σ is symmetric, and $\mathbf{a}^\top \Sigma \mathbf{a} = \text{Var}(\mathbf{a}^\top \mathbf{Y}) \geq 0$ by Eq. 2.2.1.
- (b) Let Σ be nonnegative definite and let $\Sigma^{1/2}$ be its symmetric nonnegative definite square root (Theorem 1.7.6). Let $\mathbf{Z}$ have independent components with mean 0 and variance 1, for instance independent standard normal variables, so that $\text{Cov}(\mathbf{Z}) = \mathbf{I}$. Then $\text{Cov}(\Sigma^{1/2} \mathbf{Z}) = \Sigma^{1/2} \mathbf{I} \Sigma^{1/2} = \Sigma$.
- (c) If $\Sigma \mathbf{a} = \mathbf{0}$, the random variable $\mathbf{a}^\top (\mathbf{Y} - \boldsymbol{\mu})$ has mean zero and variance $\mathbf{a}^\top \Sigma \mathbf{a} = 0$, so it is zero with probability one. Conversely, if $\mathbf{a}^\top (\mathbf{Y} - \boldsymbol{\mu}) = 0$ with probability one, then $\Sigma \mathbf{a} = \mathbb{E}[(\mathbf{Y} - \boldsymbol{\mu})(\mathbf{Y} - \boldsymbol{\mu})^\top \mathbf{a}] = \mathbf{0}$.
- (d) Let $\mathbf{a}_1, \dots, \mathbf{a}_s$ be a basis of $\mathcal{N}(\Sigma)$. By (c), each event $\{\mathbf{a}_j^\top (\mathbf{Y} - \boldsymbol{\mu}) = 0\}$ has probability one, and so does their intersection. On that intersection $\mathbf{Y} - \boldsymbol{\mu}$ is orthogonal to $\mathcal{N}(\Sigma)$. Because Σ is symmetric, $\mathcal{N}(\Sigma)$ is the orthogonal complement of $\mathcal{C}(\Sigma)$, and so $\mathbf{Y} - \boldsymbol{\mu} \in \mathcal{C}(\Sigma)$. Now let $\mathcal{S}$ be a subspace with $\Pr(\mathbf{Y} - \boldsymbol{\mu} \in \mathcal{S}) = 1$. For every $\mathbf{a} \in \mathcal{S}^\perp$ we have $\mathbf{a}^\top (\mathbf{Y} - \boldsymbol{\mu}) = 0$ with probability one, hence $\mathbf{a} \in \mathcal{N}(\Sigma) = \mathcal{C}(\Sigma)^\perp$ by (c). So $\mathcal{S}^\perp \subseteq \mathcal{C}(\Sigma)^\perp$, and taking orthogonal complements gives $\mathcal{C}(\Sigma) \subseteq \mathcal{S}$. The final statements follow, with $\dim \mathcal{C}(\Sigma) = \text{rank}(\Sigma)$. For minimality among all flats, suppose $\Pr(\mathbf{Y} \in \mathbf{c} + \mathcal{S}) = 1$ for a vector $\mathbf{c}$ and a subspace $\mathcal{S}$. Every $\mathbf{a} \in \mathcal{S}^\perp$ has $\mathbf{a}^\top (\mathbf{Y} - \mathbf{c}) = 0$ with probability one, and taking expectations gives $\mathbf{a}^\top (\boldsymbol{\mu} - \mathbf{c}) = 0$. Hence $\boldsymbol{\mu} - \mathbf{c} \in \mathcal{S}$, so $\mathbf{c} + \mathcal{S} = \boldsymbol{\mu} + \mathcal{S}$, and the subspace case applies.

■

A covariance matrix is therefore positive definite exactly when no linear combination of the components is degenerate. Singular covariance matrices are not pathological. They arise whenever the components satisfy an exact linear constraint, as with the deviations of Example (Centring

a sample), whose covariance matrix has rank $n - 1$ because $\mathbf{1}^\top \mathbf{d} = 0$. The residual vector of a regression is another case: it is orthogonal to every column of $\mathbf{X}$, and when $\text{Cov}(\mathbf{Y}) = \sigma^2 \mathbf{I}$ its covariance matrix has rank $n - \text{rank}(\mathbf{X})$ (Section 2.2, Chapter 6).

Example. (Multinomial counts)

In m independent trials, each outcome falls into one of k categories with probabilities $\boldsymbol{\pi} = (\pi_1, \dots, \pi_k)^\top$, where $\sum_j \pi_j = 1$. Let $\mathbf{Z}_t$ be the indicator vector of trial t , with a single 1 in the position of its category. Then $E(\mathbf{Z}_t) = \boldsymbol{\pi}$, and $\mathbf{Z}_t \mathbf{Z}_t^\top = \text{diag}(\mathbf{Z}_t)$ because the indicator has only one nonzero entry. So $\text{Cov}(\mathbf{Z}_t) = \text{diag}(\boldsymbol{\pi}) - \boldsymbol{\pi} \boldsymbol{\pi}^\top$. The count vector $\mathbf{Y} = \sum_t \mathbf{Z}_t$ is a sum of independent vectors, and Proposition 2.2.2(c) gives

$$\boldsymbol{\Sigma} = \text{Cov}(\mathbf{Y}) = m(\text{diag}(\boldsymbol{\pi}) - \boldsymbol{\pi} \boldsymbol{\pi}^\top).$$

Because $\boldsymbol{\Sigma} \mathbf{1} = m(\boldsymbol{\pi} - \boldsymbol{\pi} \boldsymbol{\pi}^\top \mathbf{1}) = \mathbf{0}$, the matrix is singular, and Theorem 2.2.3(c) recovers the constraint $\mathbf{1}^\top \mathbf{Y} = m$. For $m = 20$ and $\boldsymbol{\pi} = (0.5, 0.3, 0.2)^\top$ the eigenvalues are 7.762, 4.638 and 0. All 5000 simulated count vectors lie in the plane $\boldsymbol{\mu} + \mathcal{C}(\boldsymbol{\Sigma}) = \{\mathbf{y} : \mathbf{1}^\top \mathbf{y} = m\}$.

Linear maps cannot increase the rank of a covariance matrix, since $\text{rank}(\mathbf{A} \boldsymbol{\Sigma} \mathbf{A}^\top) \leq \text{rank}(\boldsymbol{\Sigma})$ (Exercise B2). They can, however, discard degenerate directions. If $\mathbf{Y} = (Z, Z)^\top$ with $\text{Var}(Z) = 1$, then $\boldsymbol{\Sigma}$ has all entries equal to 1 and is singular, while $\mathbf{A} = (1, 0)$ gives $\text{Var}(\mathbf{A} \mathbf{Y}) = 1 > 0$. By Theorem 2.2.1, the covariance matrix of $\mathbf{A} \mathbf{Y}$ is $\mathbf{A} \boldsymbol{\Sigma} \mathbf{A}^\top$. If $\boldsymbol{\Sigma}$ is positive definite and $\mathbf{A}$ has full row rank $k \leq p$, then for $\mathbf{b} \neq \mathbf{0}$ the vector $\mathbf{A}^\top \mathbf{b}$ is nonzero and $\mathbf{b}^\top \mathbf{A} \boldsymbol{\Sigma} \mathbf{A}^\top \mathbf{b} > 0$. So $\mathbf{A} \boldsymbol{\Sigma} \mathbf{A}^\top$ is positive definite. If $k > p$, then $\mathbf{A} \boldsymbol{\Sigma} \mathbf{A}^\top$ has rank at most p and is singular. A vector of more than p linear combinations of p variables always satisfies a linear constraint.

2.2.3 Standardization and Mahalanobis distance

A scalar with mean μ and variance $\sigma^2 > 0$ is standardized by $(Y - \mu)/\sigma$. The vector version must remove correlation as well as scale.

Proposition 2.2.4 — Whitening

Let $\boldsymbol{\Sigma} = \text{Cov}(\mathbf{Y})$ be positive definite, with $E(\mathbf{Y}) = \boldsymbol{\mu}$.

- (a) $\mathbf{Z} = \boldsymbol{\Sigma}^{-1/2}(\mathbf{Y} - \boldsymbol{\mu})$ has $E(\mathbf{Z}) = \mathbf{0}$ and $\text{Cov}(\mathbf{Z}) = \mathbf{I}_p$.
- (b) A $p \times p$ matrix $\mathbf{W}$ satisfies $\mathbf{W} \boldsymbol{\Sigma} \mathbf{W}^\top = \mathbf{I}_p$ iff $\mathbf{W} = \mathbf{Q} \boldsymbol{\Sigma}^{-1/2}$ for some orthogonal $\mathbf{Q}$. In particular, if $\boldsymbol{\Sigma} = \mathbf{L} \mathbf{L}^\top$ is the Cholesky factorization, $\mathbf{W} = \mathbf{L}^{-1}$ is such a matrix.
- (c) For every $\mathbf{W}$ as in (b), $\|\mathbf{W}(\mathbf{Y} - \boldsymbol{\mu})\|^2 = (\mathbf{Y} - \boldsymbol{\mu})^\top \boldsymbol{\Sigma}^{-1}(\mathbf{Y} - \boldsymbol{\mu})$.

Proof.

- (a) By Theorem 2.2.1, $\text{Cov}(\mathbf{Z}) = \boldsymbol{\Sigma}^{-1/2} \boldsymbol{\Sigma} \boldsymbol{\Sigma}^{-1/2} = \mathbf{I}$, since $\boldsymbol{\Sigma}^{-1/2}$ is symmetric and commutes with $\boldsymbol{\Sigma}$. (b) If $\mathbf{W} = \mathbf{Q} \boldsymbol{\Sigma}^{-1/2}$, then $\mathbf{W} \boldsymbol{\Sigma} \mathbf{W}^\top = \mathbf{Q} \mathbf{Q}^\top = \mathbf{I}$. Conversely, if $\mathbf{W} \boldsymbol{\Sigma} \mathbf{W}^\top = \mathbf{I}$, put $\mathbf{Q} = \mathbf{W} \boldsymbol{\Sigma}^{1/2}$. Then $\mathbf{Q} \mathbf{Q}^\top = \mathbf{W} \boldsymbol{\Sigma} \mathbf{W}^\top = \mathbf{I}$, so the square matrix $\mathbf{Q}$ is orthogonal, and $\mathbf{W} = \mathbf{Q} \boldsymbol{\Sigma}^{-1/2}$. For the Cholesky factor, $\mathbf{L}^{-1} \boldsymbol{\Sigma} (\mathbf{L}^\top)^{-1} = \mathbf{L}^{-1} \mathbf{L} \mathbf{L}^\top (\mathbf{L}^\top)^{-1} = \mathbf{I}$. (c) $\mathbf{W}^\top \mathbf{W} = \boldsymbol{\Sigma}^{-1/2} \mathbf{Q}^\top \mathbf{Q} \boldsymbol{\Sigma}^{-1/2} = \boldsymbol{\Sigma}^{-1}$.

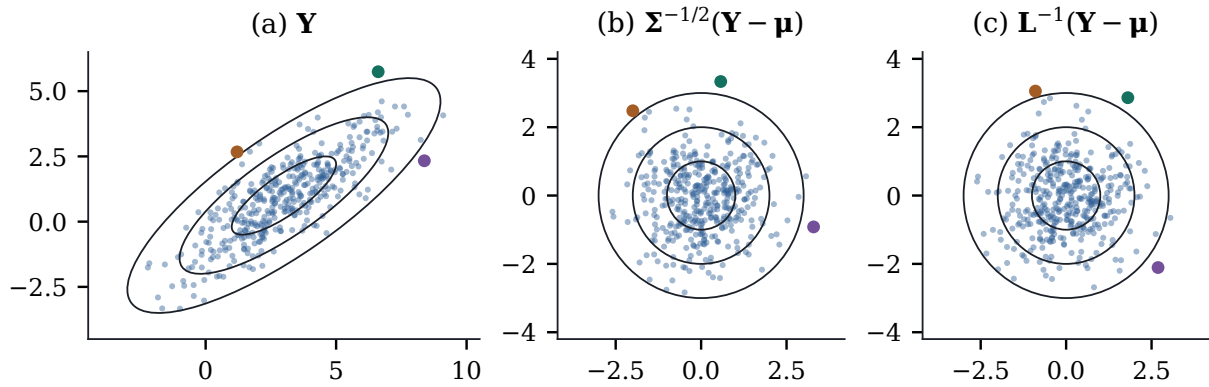


Figure 2.2.1: Whitening a correlated sample. (a) Draws with $\text{Cov}(\mathbf{Y}) = \Sigma$ and the Mahalanobis contours $\Delta = 1, 2, 3$. (b) The symmetric whitening $\Sigma^{-1/2}(\mathbf{Y} - \mu)$. (c) The Cholesky whitening $\mathbf{L}^{-1}(\mathbf{Y} - \mu)$. The contours become circles in both, and the three coloured points keep their distance from the origin. Panels (b) and (c) differ by a rotation ([Proposition 2.2.4](#)).

The quantity in (c),

$$\Delta^2(\mathbf{Y}) = (\mathbf{Y} - \mu)^\top \Sigma^{-1}(\mathbf{Y} - \mu), \quad (2.2.2)$$

is the squared **Mahalanobis distance** of $\mathbf{Y}$ from μ . It measures distance in units of the covariance structure. Its level sets are ellipsoids with axes along the eigenvectors of Σ , with half-lengths proportional to the square roots of the eigenvalues. Whitening maps these ellipsoids to spheres. Part (b) says that whitening is unique only up to a rotation or reflection. Different choices give different coordinates, but the same distances.

Example. (Two whitenings)

Take $\mu = (3, 1)^\top$ and $\Sigma = \begin{pmatrix} 4 & 2.4 \\ 2.4 & 2.25 \end{pmatrix}$, so the standard deviations are 2 and 1.5 and the correlation is 0.8. The eigenvalues are 5.680 and 0.570. The symmetric and Cholesky whitening matrices are

$$\Sigma^{-1/2} = \begin{pmatrix} 0.717 & -0.425 \\ -0.425 & 1.027 \end{pmatrix}, \quad \mathbf{L}^{-1} = \begin{pmatrix} 0.500 & 0 \\ -0.667 & 1.111 \end{pmatrix}.$$

The Cholesky version standardizes Y_1 alone and then removes from Y_2 its linear dependence on Y_1 , in the spirit of Gram–Schmidt. The symmetric version treats both coordinates alike and moves points as little as possible ([Exercise \(B3\)](#)). [Figure 2.2.1](#) applies both to 400 simulated points. The two whitened clouds differ by a rotation through 22.5° , and every point keeps its Mahalanobis distance.

```
import numpy as np
rng = np.random.default_rng(2)

mu = np.array([3.0, 1.0])
Sigma = np.array([[4.0, 2.4],
                  [2.4, 2.25]])

# correlation 2.4 / (2 * 1.5) =
0.8
```

```

lam, U = np.linalg.eigh(Sigma)                # spectral decomposition
W_sym = U @ np.diag(lam ** -0.5) @ U.T        # Sigma^{-1/2}
L = np.linalg.cholesky(Sigma)                # Sigma = L L', L lower triangular
W_chol = np.linalg.inv(L)                    # L^{-1}

n = 400
Y = mu + rng.standard_normal((n, 2)) @ L.T    # any draws with Cov = Sigma will
do
Z_sym = (Y - mu) @ W_sym.T
Z_chol = (Y - mu) @ W_chol.T

Q = W_chol @ np.linalg.inv(W_sym)            # L^{-1} Sigma^{1/2}
print("Q'Q =", np.round(Q.T @ Q, 12))

```

Whitening runs in both directions. If $\text{Cov}(\mathbf{Z}) = \mathbf{I}$, then $\boldsymbol{\mu} + \boldsymbol{\Sigma}^{1/2}\mathbf{Z}$, or $\boldsymbol{\mu} + \mathbf{L}\mathbf{Z}$, has mean $\boldsymbol{\mu}$ and covariance $\boldsymbol{\Sigma}$. This is how correlated vectors are simulated, and with $\mathbf{Z}$ standard normal it is how [Chapter 3](#) constructs $N_p(\boldsymbol{\mu}, \boldsymbol{\Sigma})$. The Mahalanobis distance then appears in the exponent of the density ([Theorem 3.2.2](#)). When $\boldsymbol{\Sigma}$ is singular, no matrix whitens all of $\mathbf{Y}$, since a degenerate combination cannot be scaled to variance one. The best one can do is whiten within the support $\boldsymbol{\mu} + \mathcal{C}(\boldsymbol{\Sigma})$ using the Moore–Penrose inverse square root ([Exercise \(B4\)](#)).

2.2.4 Second moments in the linear model

Here is how the rules of this section enter regression. Suppose $\mathbf{Y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}$ with $\mathbf{X}$ an $n \times p$ constant matrix of rank p , $E(\boldsymbol{\varepsilon}) = \mathbf{0}$ and $\text{Cov}(\boldsymbol{\varepsilon}) = \sigma^2 \mathbf{I}_n$. Adding the constant $\mathbf{X}\boldsymbol{\beta}$ changes the mean but not the covariance, so $E(\mathbf{Y}) = \mathbf{X}\boldsymbol{\beta}$ and $\text{Cov}(\mathbf{Y}) = \sigma^2 \mathbf{I}_n$. The least squares estimator $\hat{\boldsymbol{\beta}} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{Y}$ is $\mathbf{A}\mathbf{Y}$ with $\mathbf{A} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top$, so

$$E(\hat{\boldsymbol{\beta}}) = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{X} \boldsymbol{\beta} = \boldsymbol{\beta}, \quad \text{Cov}(\hat{\boldsymbol{\beta}}) = \sigma^2 (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{X} (\mathbf{X}^\top \mathbf{X})^{-1} = \sigma^2 (\mathbf{X}^\top \mathbf{X})^{-1}.$$

No distributional assumption was used, and none is needed for the Gauss–Markov theorem ([Theorem 7.2.1](#)). Fitted values and residuals are also linear in $\mathbf{Y}$, and their covariance matrices are computed the same way in [Chapter 5](#) and [Chapter 6](#). If instead $\text{Cov}(\boldsymbol{\varepsilon}) = \sigma^2 \mathbf{V}$ with $\mathbf{V}$ positive definite, then $\mathbf{V}^{-1/2} \mathbf{Y} = \mathbf{V}^{-1/2} \mathbf{X} \boldsymbol{\beta} + \mathbf{V}^{-1/2} \boldsymbol{\varepsilon}$ is a model with $\text{Cov}(\mathbf{V}^{-1/2} \boldsymbol{\varepsilon}) = \sigma^2 \mathbf{I}$. Whitening the data turns correlated errors back into the standard case. This is generalized least squares, which [Chapter 6](#) and [Chapter 31](#) develop.

2.2.5 Exercises

2.2.5.1 A. Check your understanding

Exercise A1.

Let $\text{Cov}(\mathbf{Y}) = \sigma^2 \mathbf{I}_3$. Find the covariance and correlation matrices of the successive differences $(Y_2 - Y_1, Y_3 - Y_2)^\top$, and of $(Y_2 - Y_1, Y_3 - Y_2, Y_3 - Y_1)^\top$. Which is singular, and which linear combination is degenerate?

Exercise A2.

For $p = 2$ and $\Sigma = \mathbf{L}\mathbf{L}^\top$, show that the second component of $\mathbf{L}^{-1}(\mathbf{Y} - \boldsymbol{\mu})$ is a multiple of $Y_2 - \mu_2 - (\sigma_{12}/\sigma_{11})(Y_1 - \mu_1)$. Explain the connection with [Theorem 2.5.2](#).

2.2.5.2 B. Practice**Exercise B1.**

Let $\varepsilon_0, \dots, \varepsilon_n$ be uncorrelated with variance σ^2 , and let $Y_t = \varepsilon_t + \theta\varepsilon_{t-1}$ for $t = 1, \dots, n$. Find $\text{Cov}(\mathbf{Y})$, and show that it is positive definite for every real θ , including $\theta = \pm 1$.

Solution. $\mathbf{Y} = \mathbf{A}\boldsymbol{\varepsilon}$, where $\mathbf{A}$ is $n \times (n+1)$ with θ in column t and 1 in column $t+1$ of row t (columns indexed by $\varepsilon_0, \dots, \varepsilon_n$). So $\text{Cov}(\mathbf{Y}) = \sigma^2 \mathbf{A}\mathbf{A}^\top$ is tridiagonal, with $\sigma^2(1 + \theta^2)$ on the diagonal and $\sigma^2\theta$ next to it. The last n columns of $\mathbf{A}$ form a lower bidiagonal matrix with unit diagonal, which is nonsingular. So $\mathbf{A}$ has full row rank n , and $\mathbf{A}\mathbf{A}^\top$ is positive definite ([Section 2.2](#)) whatever the value of θ .

Exercise B2.

Show that $\mathcal{C}(\mathbf{A}\Sigma\mathbf{A}^\top) = \mathcal{C}(\mathbf{A}\Sigma)$ for every nonnegative definite Σ and conformable $\mathbf{A}$. Deduce that $\text{rank Cov}(\mathbf{A}\mathbf{Y}) \leq \text{rank Cov}(\mathbf{Y})$.

Exercise B3.

Let Σ be positive definite. Among all $\mathbf{W}$ with $\mathbf{W}\Sigma\mathbf{W}^\top = \mathbf{I}$, show that $\mathbb{E}\|\mathbf{W}(\mathbf{Y} - \boldsymbol{\mu}) - (\mathbf{Y} - \boldsymbol{\mu})\|^2$ is minimized uniquely by $\mathbf{W} = \Sigma^{-1/2}$. *Hint:* use [Proposition 2.2.4\(b\)](#) and show that $\text{tr}(\mathbf{Q}\Sigma^{1/2}) \leq \text{tr}(\Sigma^{1/2})$ for orthogonal $\mathbf{Q}$.

Solution. Write $\mathbf{W} = \mathbf{Q}\Sigma^{-1/2}$. By [Corollary 2.3.2\(a\)](#) with $\mathbf{A} = (\mathbf{W} - \mathbf{I})^\top(\mathbf{W} - \mathbf{I})$,

$$\mathbb{E}\|(\mathbf{W} - \mathbf{I})(\mathbf{Y} - \boldsymbol{\mu})\|^2 = \text{tr}((\mathbf{W} - \mathbf{I})\Sigma(\mathbf{W} - \mathbf{I})^\top) = \text{tr}(\mathbf{W}\Sigma\mathbf{W}^\top) - 2\text{tr}(\mathbf{W}\Sigma) + \text{tr}(\Sigma) = p + \text{tr}(\Sigma) - 2\text{tr}(\mathbf{Q}\Sigma^{1/2}).$$

Let $\Sigma^{1/2} = \mathbf{\Gamma}\mathbf{\Lambda}^{1/2}\mathbf{\Gamma}^\top$ and $\mathbf{G} = \mathbf{\Gamma}^\top\mathbf{Q}\mathbf{\Gamma}$, which is orthogonal, so each $|g_{ii}| \leq 1$. Then $\text{tr}(\mathbf{Q}\Sigma^{1/2}) = \text{tr}(\mathbf{G}\mathbf{\Lambda}^{1/2}) = \sum_i g_{ii}\lambda_i^{1/2} \leq \sum_i \lambda_i^{1/2}$. Since every $\lambda_i > 0$, equality forces every $g_{ii} = 1$. An orthogonal matrix with unit diagonal is $\mathbf{I}$, because each column has norm one. So $\mathbf{G} = \mathbf{I}$, which means $\mathbf{Q} = \mathbf{I}$.

Exercise B4.

Let $\Sigma = \text{Cov}(\mathbf{Y})$ have rank r , with spectral decomposition $\Sigma = \mathbf{\Gamma}_r\mathbf{\Lambda}_r\mathbf{\Gamma}_r^\top$, where $\mathbf{\Gamma}_r^\top\mathbf{\Gamma}_r = \mathbf{I}_r$ and $\mathbf{\Lambda}_r$ holds the positive eigenvalues. Show that $\mathbf{Z} = \mathbf{\Lambda}_r^{-1/2}\mathbf{\Gamma}_r^\top(\mathbf{Y} - \boldsymbol{\mu})$ has $\text{Cov}(\mathbf{Z}) = \mathbf{I}_r$ and that $\mathbf{Y} = \boldsymbol{\mu} + \mathbf{\Gamma}_r\mathbf{\Lambda}_r^{1/2}\mathbf{Z}$ with probability one. Show that $(\mathbf{Y} - \boldsymbol{\mu})^\top\Sigma^+(\mathbf{Y} - \boldsymbol{\mu}) = \|\mathbf{Z}\|^2$, where Σ^+ is the Moore–Penrose inverse, and that its expectation is r .

Solution. By [Theorem 2.2.1](#), $\text{Cov}(\mathbf{Z}) = \mathbf{\Lambda}_r^{-1/2}\mathbf{\Gamma}_r^\top\mathbf{\Gamma}_r\mathbf{\Lambda}_r\mathbf{\Gamma}_r^\top\mathbf{\Gamma}_r\mathbf{\Lambda}_r^{-1/2} = \mathbf{I}_r$. By [Theorem 2.2.3\(d\)](#), $\mathbf{Y} - \boldsymbol{\mu} \in \mathcal{C}(\Sigma) = \mathcal{C}(\mathbf{\Gamma}_r)$ with probability one, and on that event $\mathbf{Y} -$

$\boldsymbol{\mu} = \boldsymbol{\Gamma}_r \boldsymbol{\Gamma}_r^\top (\mathbf{Y} - \boldsymbol{\mu}) = \boldsymbol{\Gamma}_r \boldsymbol{\Lambda}_r^{1/2} \mathbf{Z}$. The Moore–Penrose inverse is $\boldsymbol{\Sigma}^+ = \boldsymbol{\Gamma}_r \boldsymbol{\Lambda}_r^{-1} \boldsymbol{\Gamma}_r^\top$, so $(\mathbf{Y} - \boldsymbol{\mu})^\top \boldsymbol{\Sigma}^+ (\mathbf{Y} - \boldsymbol{\mu}) = \|\boldsymbol{\Lambda}_r^{-1/2} \boldsymbol{\Gamma}_r^\top (\mathbf{Y} - \boldsymbol{\mu})\|^2 = \|\mathbf{Z}\|^2$, with expectation $\text{tr Cov}(\mathbf{Z}) = r$.

2.2.5.3 C. Going deeper

Exercise C1.

In $\mathbf{Y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}$ with $\mathbf{X}$ of full column rank, let $\text{Cov}(\boldsymbol{\varepsilon}) = \sigma^2 \mathbf{V}$ with $\mathbf{V}$ positive definite. Find $\text{Cov}(\hat{\boldsymbol{\beta}})$ for $\hat{\boldsymbol{\beta}} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{Y}$ and $\text{Cov}(\tilde{\boldsymbol{\beta}})$ for $\tilde{\boldsymbol{\beta}} = (\mathbf{X}^\top \mathbf{V}^{-1} \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{V}^{-1} \mathbf{Y}$. Show that $\text{Cov}(\hat{\boldsymbol{\beta}}) - \text{Cov}(\tilde{\boldsymbol{\beta}})$ is nonnegative definite. *Hint:* show that $\hat{\boldsymbol{\beta}} - \tilde{\boldsymbol{\beta}}$ and $\tilde{\boldsymbol{\beta}}$ are uncorrelated.

Solution. By [Theorem 2.2.1](#),

$$\text{Cov}(\hat{\boldsymbol{\beta}}) = \sigma^2 (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{V} \mathbf{X} (\mathbf{X}^\top \mathbf{X})^{-1}, \quad \text{Cov}(\tilde{\boldsymbol{\beta}}) = \sigma^2 (\mathbf{X}^\top \mathbf{V}^{-1} \mathbf{X})^{-1}.$$

Write $\hat{\boldsymbol{\beta}} - \tilde{\boldsymbol{\beta}} = \mathbf{D}\mathbf{Y}$ with $\mathbf{D} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top - (\mathbf{X}^\top \mathbf{V}^{-1} \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{V}^{-1}$, and note $\mathbf{D}\mathbf{X} = \mathbf{I} - \mathbf{I} = \mathbf{0}$. Then $\text{Cov}(\hat{\boldsymbol{\beta}} - \tilde{\boldsymbol{\beta}}, \tilde{\boldsymbol{\beta}}) = \sigma^2 \mathbf{D} \mathbf{V} \mathbf{V}^{-1} \mathbf{X} (\mathbf{X}^\top \mathbf{V}^{-1} \mathbf{X})^{-1} = \mathbf{0}$, so by [Proposition 2.2.2\(b\)](#), $\text{Cov}(\hat{\boldsymbol{\beta}}) = \text{Cov}(\tilde{\boldsymbol{\beta}}) + \text{Cov}(\hat{\boldsymbol{\beta}} - \tilde{\boldsymbol{\beta}})$, and the last term is nonnegative definite.

2.3 Expectation of quadratic forms

Sums of squares are quadratic forms in the response. The sample variance is $(n-1)^{-1} \mathbf{Y}^\top (\mathbf{I} - n^{-1} \mathbf{J}) \mathbf{Y}$. The residual sum of squares of a regression is $\mathbf{Y}^\top (\mathbf{I} - \mathbf{M}) \mathbf{Y}$, where $\mathbf{M}$ is the projection onto the column space of $\mathbf{X}$ ([Chapter 6](#)). Test statistics are ratios of such forms. Their expectations follow from a short trace argument that needs no distributional assumption.

2.3.1 The basic formula

Theorem 2.3.1 — Mean of a quadratic form

Let $\mathbf{Y}$ be a $p \times 1$ random vector with finite second moments, $E(\mathbf{Y}) = \boldsymbol{\mu}$ and $\text{Cov}(\mathbf{Y}) = \boldsymbol{\Sigma}$, and let $\mathbf{A}$ be a constant $p \times p$ matrix. Then

$$E(\mathbf{Y}^\top \mathbf{A} \mathbf{Y}) = \text{tr}(\mathbf{A} \boldsymbol{\Sigma}) + \boldsymbol{\mu}^\top \mathbf{A} \boldsymbol{\mu}.$$

Proof. A 1×1 matrix equals its trace, and $\text{tr}(\mathbf{a} \mathbf{b}^\top) = \mathbf{b}^\top \mathbf{a}$ ([Theorem 1.6.1](#)), so $\mathbf{Y}^\top \mathbf{A} \mathbf{Y} = \text{tr}(\mathbf{A} \mathbf{Y} \mathbf{Y}^\top)$. By [Proposition 2.1.1](#) and [Proposition 2.1.2\(b\)](#),

$$E(\mathbf{Y}^\top \mathbf{A} \mathbf{Y}) = \text{tr}(\mathbf{A} E(\mathbf{Y} \mathbf{Y}^\top)) = \text{tr}(\mathbf{A} (\boldsymbol{\Sigma} + \boldsymbol{\mu} \boldsymbol{\mu}^\top)) = \text{tr}(\mathbf{A} \boldsymbol{\Sigma}) + \boldsymbol{\mu}^\top \mathbf{A} \boldsymbol{\mu}.$$

■

The formula has two parts. The trace term is what the form picks up from random variation. The second term is the value of the form at the mean, which is what it would be without any

randomness. The theorem does not require independence, normality or symmetry of $\mathbf{A}$. Since $\mathbf{Y}^\top \mathbf{A} \mathbf{Y} = \mathbf{Y}^\top \mathbf{A}_s \mathbf{Y}$ with $\mathbf{A}_s = \frac{1}{2}(\mathbf{A} + \mathbf{A}^\top)$, and $\text{tr}(\mathbf{A} \mathbf{\Sigma}) = \text{tr}(\mathbf{A}_s \mathbf{\Sigma})$ when $\mathbf{\Sigma}$ is symmetric, we may assume $\mathbf{A}$ symmetric whenever that is convenient.

Corollary 2.3.2

Under the conditions of [Theorem 2.3.1](#):

- (a) $E[(\mathbf{Y} - \mathbf{b})^\top \mathbf{A}(\mathbf{Y} - \mathbf{b})] = \text{tr}(\mathbf{A} \mathbf{\Sigma}) + (\boldsymbol{\mu} - \mathbf{b})^\top \mathbf{A}(\boldsymbol{\mu} - \mathbf{b})$ for any constant $\mathbf{b}$;
- (b) if $\mathbf{\Sigma} = \sigma^2 \mathbf{I}$, then $E(\mathbf{Y}^\top \mathbf{A} \mathbf{Y}) = \sigma^2 \text{tr}(\mathbf{A}) + \boldsymbol{\mu}^\top \mathbf{A} \boldsymbol{\mu}$;
- (c) if $\mathbf{\Sigma}$ is positive definite, the squared Mahalanobis distance [Eq. 2.2.2](#) has $E \Delta^2(\mathbf{Y}) = p$;
- (d) for random vectors $\mathbf{U}$ ($p \times 1$) and $\mathbf{V}$ ($q \times 1$) with finite second moments and a constant $p \times q$ matrix $\mathbf{A}$, $E(\mathbf{U}^\top \mathbf{A} \mathbf{V}) = \text{tr}(\mathbf{A} \text{Cov}(\mathbf{V}, \mathbf{U})) + E(\mathbf{U})^\top \mathbf{A} E(\mathbf{V})$.

Proof.

- (a) applies the theorem to $\mathbf{Y} - \mathbf{b}$, which has covariance matrix $\mathbf{\Sigma}$. (b) is immediate.
- (c) is (a) with $\mathbf{b} = \boldsymbol{\mu}$ and $\mathbf{A} = \mathbf{\Sigma}^{-1}$, so the trace is $\text{tr}(\mathbf{I}_p) = p$. For (d), $\mathbf{U}^\top \mathbf{A} \mathbf{V} = \text{tr}(\mathbf{A} \mathbf{V} \mathbf{U}^\top)$ and $E(\mathbf{V} \mathbf{U}^\top) = \text{Cov}(\mathbf{V}, \mathbf{U}) + E(\mathbf{V}) E(\mathbf{U})^\top$.

■

Example. (Sample variance with unequal means and variances)

Let $Y_1, \dots, Y_n$ be uncorrelated with means μ_i and variances σ_i^2 , and let $S^2 = (n - 1)^{-1} \mathbf{Y}^\top \mathbf{C} \mathbf{Y}$ with $\mathbf{C} = \mathbf{I} - n^{-1} \mathbf{J}$. Here $\mathbf{\Sigma} = \text{diag}(\sigma_1^2, \dots, \sigma_n^2)$, so $\text{tr}(\mathbf{C} \mathbf{\Sigma}) = (1 - 1/n) \sum_i \sigma_i^2$, and $\boldsymbol{\mu}^\top \mathbf{C} \boldsymbol{\mu} = \sum_i (\mu_i - \bar{\mu})^2$. Therefore

$$E(S^2) = \overline{\sigma^2} + \frac{1}{n-1} \sum_{i=1}^n (\mu_i - \bar{\mu})^2, \quad \overline{\sigma^2} = \frac{1}{n} \sum_{i=1}^n \sigma_i^2.$$

With a common mean, S^2 is unbiased for the *average* variance, even when the variances differ. Differences in the means inflate it. That is exactly why a residual mean square estimates σ^2 only when the mean structure is right.

Example. (Residual sum of squares under a wrong mean)

Let $\text{Cov}(\mathbf{Y}) = \sigma^2 \mathbf{I}_n$ and $E(\mathbf{Y}) = \boldsymbol{\theta}$, where $\boldsymbol{\theta}$ need not lie in $\mathcal{C}(\mathbf{X})$. The residual sum of squares is $\mathbf{Y}^\top (\mathbf{I} - \mathbf{M}) \mathbf{Y}$, where $\mathbf{M}$ is symmetric and idempotent of rank $r = \text{rank}(\mathbf{X})$, so $\text{tr}(\mathbf{I} - \mathbf{M}) = n - r$ ([Theorem 1.9.2](#)). By [Corollary 2.3.2](#)(b),

$$E(\text{RSS}) = \sigma^2(n - r) + \|(\mathbf{I} - \mathbf{M})\boldsymbol{\theta}\|^2. \quad (2.3.1)$$

If the model is right, $\boldsymbol{\theta} \in \mathcal{C}(\mathbf{X})$, the second term vanishes and $\text{RSS}/(n - r)$ is unbiased for σ^2 . If the model is wrong, the estimate is biased upwards by the squared distance from the true mean to the model space, divided by $n - r$. For instance, fit a straight

line at $x = 1, \dots, 6$ when the true mean is the parabola $\theta_i = \frac{1}{4}(x_i - 3.5)^2$ and $\sigma^2 = 1$. Then $\sigma^2(n - r) = 4$, $\|(\mathbf{I} - \mathbf{M})\boldsymbol{\theta}\|^2 = 2.333$, and $E(\text{RSS}) = 6.333$. Chapter 19 studies such departures systematically.

The theorem uses only two moments, so it holds equally for skewed and heavy-tailed data. The listing checks it for a correlated vector $\mathbf{Y} = \boldsymbol{\mu} + \mathbf{B}\mathbf{U}$, where $\mathbf{U}$ has independent, centred exponential components, which are strongly skewed. With the matrices in the listing, $\text{tr}(\mathbf{A}\boldsymbol{\Sigma}) = 6.9$ and $\boldsymbol{\mu}^\top \mathbf{A} \boldsymbol{\mu} = 17.0$, so $E(\mathbf{Y}^\top \mathbf{A} \mathbf{Y}) = 23.9$. The average over 10^6 draws is 23.940, with Monte Carlo standard error 0.018. The script also verifies the formula exactly, by enumerating a discrete distribution with eight atoms.

```
import itertools
import numpy as np
rng = np.random.default_rng(20260917)

mu = np.array([1.0, -1.0, 2.0])
B = np.array([[1.0, 0.0, 0.0],
              [0.5, 1.0, 0.0],
              [-0.5, 0.8, 0.6]])
Sigma = B @ B.T                                     # Cov(mu + B U) when Cov(U) = I
A = np.array([[2.0, 1.0, 0.0],
              [1.0, 1.0, -1.0],
              [0.0, -1.0, 3.0]])

theory = np.trace(A @ Sigma) + mu @ A @ mu          # Theorem: no normality needed

N = 1_000_000
U = rng.exponential(size=(N, 3)) - 1.0             # skewed, mean 0, variance 1
Y = mu + U @ B.T
Q = np.einsum("ij,jk,ik->i", Y, A, Y)             # Y_i' A Y_i for every draw
print(f"theory {theory:.4f}    simulation {Q.mean():.4f}  {Q.std() / np.sqrt(N):.4f}")
)
```

2.3.2 The variance needs fourth moments

The mean of $\mathbf{Y}^\top \mathbf{A} \mathbf{Y}$ involves only $E(Y_i Y_j)$. Its second moment involves $E(Y_i Y_j Y_k Y_l)$, so no formula for $\text{Var}(\mathbf{Y}^\top \mathbf{A} \mathbf{Y})$ can be written in terms of $\boldsymbol{\mu}$ and $\boldsymbol{\Sigma}$ alone. Two random vectors with the same mean and covariance matrix can give the same quadratic form very different variances. The next theorem handles independent components, the case that matters for residual sums of squares under the classical error assumptions. Chapter 4 gives the normal case with a general $\boldsymbol{\Sigma}$ (Theorem 4.2.3).

Theorem 2.3.3 — Variance of a quadratic form

Let $Y_1, \dots, Y_n$ be independent with means $\theta_1, \dots, \theta_n$ and common central moments $\sigma^2 = E(Y_i - \theta_i)^2$, $\mu_3 = E(Y_i - \theta_i)^3$ and $\mu_4 = E(Y_i - \theta_i)^4 < \infty$. Let $\mathbf{A}$ be a constant symmetric $n \times n$ matrix, and let $\mathbf{a} = (a_{11}, \dots, a_{nn})^\top$ be its diagonal. Then

$$\text{Var}(\mathbf{Y}^\top \mathbf{A} \mathbf{Y}) = (\mu_4 - 3\sigma^4) \mathbf{a}^\top \mathbf{a} + 2\sigma^4 \text{tr}(\mathbf{A}^2) + 4\sigma^2 \boldsymbol{\theta}^\top \mathbf{A}^2 \boldsymbol{\theta} + 4\mu_3 \boldsymbol{\theta}^\top \mathbf{A} \mathbf{a}.$$

Proof. Write $\mathbf{Y} = \boldsymbol{\theta} + \boldsymbol{\varepsilon}$, where the ε_i are independent with mean zero. Then

$$\mathbf{Y}^\top \mathbf{A} \mathbf{Y} = q + \ell + \boldsymbol{\theta}^\top \mathbf{A} \boldsymbol{\theta}, \quad q = \boldsymbol{\varepsilon}^\top \mathbf{A} \boldsymbol{\varepsilon}, \quad \ell = 2\mathbf{b}^\top \boldsymbol{\varepsilon}, \quad \mathbf{b} = \mathbf{A} \boldsymbol{\theta},$$

so $\text{Var}(\mathbf{Y}^\top \mathbf{A} \mathbf{Y}) = \text{Var}(q) + \text{Var}(\ell) + 2 \text{Cov}(q, \ell)$. We compute the three terms.

The linear term. $\text{Cov}(\boldsymbol{\varepsilon}) = \sigma^2 \mathbf{I}$, so $\text{Var}(\ell) = 4\sigma^2 \mathbf{b}^\top \mathbf{b} = 4\sigma^2 \boldsymbol{\theta}^\top \mathbf{A}^2 \boldsymbol{\theta}$.

The cross term. Since $E(\ell) = 0$, $\text{Cov}(q, \ell) = E(q\ell) = 2 \sum_{i,j,k} a_{ij} b_k E(\varepsilon_i \varepsilon_j \varepsilon_k)$. By independence and zero means, $E(\varepsilon_i \varepsilon_j \varepsilon_k)$ vanishes unless $i = j = k$, when it equals μ_3 . So $\text{Cov}(q, \ell) = 2\mu_3 \sum_i a_{ii} b_i = 2\mu_3 \boldsymbol{\theta}^\top \mathbf{A} \mathbf{a}$.

The quadratic term. $E(q^2) = \sum_{i,j,k,l} a_{ij} a_{kl} E(\varepsilon_i \varepsilon_j \varepsilon_k \varepsilon_l)$. The fourth-order moment is μ_4 if all four indices agree. It is σ^4 if the indices form two distinct pairs, which happens in three patterns: $i = j \neq k = l$, $i = k \neq j = l$ and $i = l \neq j = k$. Otherwise some index occurs exactly once, and the moment is zero. Using the symmetry of $\mathbf{A}$,

$$E(q^2) = \mu_4 \sum_i a_{ii}^2 + \sigma^4 \left[\sum_{i \neq k} a_{ii} a_{kk} + 2 \sum_{i \neq j} a_{ij}^2 \right] = \mu_4 \mathbf{a}^\top \mathbf{a} + \sigma^4 \left[(\text{tr } \mathbf{A})^2 - \mathbf{a}^\top \mathbf{a} + 2 \text{tr}(\mathbf{A}^2) - 2\mathbf{a}^\top \mathbf{a} \right],$$

since $\sum_{i,j} a_{ij}^2 = \text{tr}(\mathbf{A}^2)$. By [Theorem 2.3.1](#), $E(q) = \sigma^2 \text{tr } \mathbf{A}$, and subtracting its square gives $\text{Var}(q) = (\mu_4 - 3\sigma^4) \mathbf{a}^\top \mathbf{a} + 2\sigma^4 \text{tr}(\mathbf{A}^2)$. Adding the three terms proves the formula. ■

For normal errors, $\mu_3 = 0$ and $\mu_4 = 3\sigma^4$, and the formula collapses to $2\sigma^4 \text{tr}(\mathbf{A}^2) + 4\sigma^2 \boldsymbol{\theta}^\top \mathbf{A}^2 \boldsymbol{\theta}$. Both extra terms describe how non-normality leaks into second-order behaviour. The kurtosis term depends on $\mathbf{A}$ only through its diagonal. The skewness term needs both a skewed error distribution and a mean that the form does not annihilate. For the residual sum of squares of a correct model, $\mathbf{A} \boldsymbol{\theta} = (\mathbf{I} - \mathbf{M}) \boldsymbol{\theta} = \mathbf{0}$, so the skewness term disappears but the kurtosis term remains.

Example. (Skewed errors and a misspecified line)

Continue [Example \(Residual sum of squares under a wrong mean\)](#), now with independent centred exponential errors, for which $\sigma^2 = 1$, $\mu_3 = 2$ and $\mu_4 = 9$. [Theorem 2.3.3](#) gives $\text{Var}(\text{RSS}) = 29.80$. Of this, the kurtosis term contributes 16.73 and the skewness term -4.27 . Normal errors with the same mean and variance would give 17.33, so the exponential errors multiply the variance by 1.72 even though the mean of RSS is unchanged. Over 10^6 simulated data sets, the variance of RSS is 30.04. The script also confirms the formula exactly for a two-point error distribution.

```
se = Q.std() / np.sqrt(N)
vals, probs = np.array([-1.0, 2.0]), np.array([2 / 3, 1 / 3])
exact = 0.0
for idx in itertools.product(range(2), repeat=3):
    u = vals[list(idx)]
    y = mu + B @ u
    exact += np.prod(probs[list(idx)]) * (y @ A @ y)
x = np.arange(1.0, 7.0) # n = 6 design points
X = np.column_stack([np.ones_like(x), x])
M = X @ np.linalg.solve(X.T @ X, X.T)
A2 = np.eye(6) - M # RSS = Y'(I - M)Y for a straight
```

```

line
theta = 0.25 * (x - 3.5) ** 2          # true mean is curved: the line is
wrong
a = np.diag(A2)

def var_quadform(A, theta, m2, m3, m4):
    """Var(Y'AY) for independent Y_i with means theta_i and common central moments.
    """
    a = np.diag(A)
    return ((m4 - 3 * m2**2) * a @ a + 2 * m2**2 * np.trace(A @ A)
            + 4 * m2 * theta @ A @ A @ theta + 4 * m3 * theta @ A @ a)

# A2 = I - M for a straight line at x = 1..6; theta = true (curved) mean
mean_rss = np.trace(A2) + theta @ A2 @ theta          # sigma^2 = 1
var_expo = var_quadform(A2, theta, 1.0, 2.0, 9.0) # centred exponential errors
var_norm = var_quadform(A2, theta, 1.0, 0.0, 3.0) # what normal errors would give

E = rng.exponential(size=(N, 6)) - 1.0
R = np.einsum("ij,jk,ik->i", theta + E, A2, theta + E)
print(f"mean   {mean_rss:.3f} vs {R.mean():.3f}")
print(f"var     {var_expo:.3f} vs {R.var():.3f}   (normal-theory value {var_norm:.3f})")

```

2.3.3 Exercises

2.3.3.1 A. Check your understanding

Exercise A1.

For random vectors $\mathbf{U}$ and $\mathbf{V}$ of the same size, show that $E\|\mathbf{U} - \mathbf{V}\|^2 = \text{tr Cov}(\mathbf{U}) + \text{tr Cov}(\mathbf{V}) - 2 \text{tr Cov}(\mathbf{U}, \mathbf{V}) + \|E\mathbf{U} - E\mathbf{V}\|^2$.

2.3.3.2 B. Practice

Exercise B1.

Let $E(Y_t) = \mu$ and $\text{Cov}(Y_t, Y_s) = \sigma^2 \phi^{|t-s|}$ for $t, s = 1, \dots, n$, with $|\phi| < 1$. Show that the sample variance satisfies

$$E(S^2) = \sigma^2 \left[1 - \frac{2}{n(n-1)} \sum_{k=1}^{n-1} (n-k) \phi^k \right].$$

Is S^2 biased up or down when $\phi > 0$? What happens as $n \rightarrow \infty$ with ϕ fixed?

Solution. With $\mathbf{C} = \mathbf{I} - n^{-1}\mathbf{1}\mathbf{1}^\top$ and $E(\mathbf{Y}) = \mu\mathbf{1}$, $\mathbf{C}E(\mathbf{Y}) = \mathbf{0}$, so $E(\mathbf{Y}^\top \mathbf{C} \mathbf{Y}) = \text{tr}(\mathbf{C}\Sigma) = \text{tr}(\Sigma) - n^{-1}\mathbf{1}^\top \Sigma \mathbf{1}$. Here $\text{tr}(\Sigma) = n\sigma^2$, and $\mathbf{1}^\top \Sigma \mathbf{1}$ sums all entries. There are n diagonal entries and $2(n-k)$ entries at lag k , so $\mathbf{1}^\top \Sigma \mathbf{1} = \sigma^2 [n + 2 \sum_{k=1}^{n-1} (n-k) \phi^k]$. Dividing $n\sigma^2 - n^{-1}\mathbf{1}^\top \Sigma \mathbf{1}$ by $n-1$ gives the formula. For $\phi > 0$ the sum is positive, so S^2 is biased downwards: positively correlated observations cluster, and the sample looks

less variable than the process is. As $n \rightarrow \infty$ the sum is $O(n)$, so the bias is $O(1/n)$ and vanishes.

Exercise B2.

Suppose $E(\mathbf{Y}) = \mathbf{X}\boldsymbol{\beta}$ and $\text{Cov}(\mathbf{Y}) = \sigma^2 \mathbf{V}$. Show that $E(\text{RSS}) = \sigma^2 \text{tr}((\mathbf{I} - \mathbf{M})\mathbf{V})$. For the equicorrelated $\mathbf{V} = (1 - \rho)\mathbf{I} + \rho\mathbf{1}\mathbf{1}^\top$ and a model with $\mathbf{1} \in \mathcal{C}(\mathbf{X})$, show that $E(\text{RSS})/(n - r) = \sigma^2(1 - \rho)$.

Exercise B3.

Let $Y_1, \dots, Y_n$ be independent and identically distributed with variance σ^2 and finite fourth central moment μ_4 . Use [Theorem 2.3.3](#) to show that $\text{Var}(S^2) = (\mu_4 - 3\sigma^4)/n + 2\sigma^4/(n - 1)$. Compare with the normal case.

Solution. Take $\mathbf{A} = \mathbf{C}/(n - 1)$ and $\boldsymbol{\theta} = \mu\mathbf{1}$. Then $\mathbf{A}\boldsymbol{\theta} = \mathbf{0}$, so the last two terms of [Theorem 2.3.3](#) vanish. Every diagonal entry of $\mathbf{A}$ is $(1 - 1/n)/(n - 1) = 1/n$, so $\mathbf{a}^\top \mathbf{a} = 1/n$, and $\text{tr}(\mathbf{A}^2) = \text{tr}(\mathbf{C})/(n - 1)^2 = 1/(n - 1)$. This gives the formula. Normal data have $\mu_4 = 3\sigma^4$ and $\text{Var}(S^2) = 2\sigma^4/(n - 1)$. Heavy tails ($\mu_4 > 3\sigma^4$) add a term of the same order $1/n$, so the normal-theory variance is wrong even in large samples.

Exercise B4.

Under the conditions of [Theorem 2.3.3](#) with $\boldsymbol{\theta} = \mathbf{0}$, show that $\text{Cov}(\mathbf{Y}^\top \mathbf{A} \mathbf{Y}, \mathbf{Y}^\top \mathbf{B} \mathbf{Y}) = (\mu_4 - 3\sigma^4)\mathbf{a}^\top \mathbf{b} + 2\sigma^4 \text{tr}(\mathbf{A}\mathbf{B})$, where $\mathbf{a}$ and $\mathbf{b}$ are the diagonals of the symmetric matrices $\mathbf{A}$ and $\mathbf{B}$. *Hint:* apply the theorem to $\mathbf{A} + \mathbf{B}$.

Exercise B5.

Let $Y_1, \dots, Y_n$ be independent $N(0, \sigma^2)$. Among symmetric $\mathbf{A}$ with $E(\mathbf{Y}^\top \mathbf{A} \mathbf{Y}) = \sigma^2$, show that $\text{Var}(\mathbf{Y}^\top \mathbf{A} \mathbf{Y})$ is minimized by $\mathbf{A} = n^{-1}\mathbf{I}$. *Hint:* $\text{tr}(\mathbf{A}^2) \geq (\text{tr} \mathbf{A})^2/n$.

2.4 Correlation matrices

Covariances carry units: the covariance of income in dollars with age in years is in dollar-years. To compare the strength of linear association across pairs of variables, each covariance is divided by the two standard deviations.

Definition 2.4.1 — Correlation matrix

Let $\mathbf{Y}$ have covariance matrix $\boldsymbol{\Sigma} = (\sigma_{ij})$ with every $\sigma_{ii} > 0$, and let $\mathbf{D}_\sigma = \text{diag}(\sigma_{11}^{1/2}, \dots, \sigma_{pp}^{1/2})$ be the diagonal matrix of standard deviations. The **correlation matrix** of $\mathbf{Y}$ is

$$\mathbf{R} = \mathbf{D}_\sigma^{-1} \boldsymbol{\Sigma} \mathbf{D}_\sigma^{-1}, \quad \text{with entries} \quad \rho_{ij} = \frac{\sigma_{ij}}{\sqrt{\sigma_{ii}\sigma_{jj}}} = \text{corr}(Y_i, Y_j).$$

Equivalently $\Sigma = \mathbf{D}_\sigma \mathbf{R} \mathbf{D}_\sigma$. For random vectors $\mathbf{U}$ and $\mathbf{V}$ with standard-deviation matrices $\mathbf{D}_U$ and $\mathbf{D}_V$, the cross-correlation matrix is $\mathbf{D}_U^{-1} \text{Cov}(\mathbf{U}, \mathbf{V}) \mathbf{D}_V^{-1}$.

Some books write $\mathbf{P}_\rho$ for the correlation matrix. We use $\mathbf{R}$ in this chapter, and $\hat{\mathbf{R}}$ for its sample counterpart in [Section 2.7](#).

Proposition 2.4.1 — Properties of correlation matrices

Let $\mathbf{R}$ be the correlation matrix of $\mathbf{Y}$.

- (a) $\mathbf{R}$ is the covariance matrix of the standardized vector $\mathbf{Z} = \mathbf{D}_\sigma^{-1}(\mathbf{Y} - \boldsymbol{\mu})$. Hence $\mathbf{R}$ is nonnegative definite with unit diagonal, and $\text{tr}(\mathbf{R}) = p$.
- (b) $|\rho_{ij}| \leq 1$, with $|\rho_{ij}| = 1$ iff $Z_j = \rho_{ij} Z_i$ with probability one.
- (c) $\text{rank}(\mathbf{R}) = \text{rank}(\Sigma)$. In particular, $\mathbf{R}$ is positive definite iff Σ is.
- (d) If $\mathbf{Y}^* = \mathbf{D}\mathbf{Y} + \mathbf{c}$ with $\mathbf{D} = \text{diag}(d_1, \dots, d_p)$ and every $d_i \neq 0$, then $\rho_{ij}^* = \text{sign}(d_i d_j) \rho_{ij}$. Positive rescaling and shifting leave $\mathbf{R}$ unchanged.
- (e) Conversely, every nonnegative definite matrix with unit diagonal is the correlation matrix of some random vector.

Proof.

- (a) [Theorem 2.2.1](#) gives $\text{Cov}(\mathbf{Z}) = \mathbf{D}_\sigma^{-1} \Sigma \mathbf{D}_\sigma^{-1} = \mathbf{R}$, and [Theorem 2.2.3\(a\)](#) applies.
- (b) The 2×2 matrix of rows and columns i, j of $\mathbf{R}$ is the covariance matrix of (Z_i, Z_j) , so it is nonnegative definite and its determinant $1 - \rho_{ij}^2$ is nonnegative. If $\rho_{ij} = \pm 1$, the vector $(\rho_{ij}, -1)^\top$ is in the null space of that 2×2 matrix, so $\rho_{ij} Z_i - Z_j = 0$ with probability one by [Theorem 2.2.3\(c\)](#). Conversely, $Z_j = c Z_i$ with $\text{Var}(Z_i) = \text{Var}(Z_j) = 1$ forces $c = \pm 1$ and $\rho_{ij} = c$.
- (c) $\mathbf{D}_\sigma$ is nonsingular.
- (d) $\text{Cov}(\mathbf{Y}^*) = \mathbf{D} \Sigma \mathbf{D}$ has entries $d_i d_j \sigma_{ij}$ and standard deviations $|d_i| \sigma_{ii}^{1/2}$.
- (e) Such a matrix is a covariance matrix by [Theorem 2.2.3\(b\)](#), and its variances are one.

■

Part (d) is the reason correlation is the natural scale-free summary. Part (e) is less obvious than it looks. A symmetric matrix with unit diagonal and off-diagonal entries in $[-1, 1]$ need not be a correlation matrix, because the correlations must be jointly consistent.

Proposition 2.4.2 — Three correlations

Let $|\rho_{12}| < 1$ and $|\rho_{13}| < 1$. The matrix

$$\mathbf{R} = \begin{pmatrix} 1 & \rho_{12} & \rho_{13} \\ \rho_{12} & 1 & \rho_{23} \\ \rho_{13} & \rho_{23} & 1 \end{pmatrix}$$

is a correlation matrix iff

$$\rho_{12}\rho_{13} - \sqrt{(1 - \rho_{12}^2)(1 - \rho_{13}^2)} \leq \rho_{23} \leq \rho_{12}\rho_{13} + \sqrt{(1 - \rho_{12}^2)(1 - \rho_{13}^2)}.$$

Proof. By [Proposition 2.4.1\(e\)](#) we must decide when $\mathbf{R}$ is nonnegative definite. Write $\mathbf{r} = (\rho_{12}, \rho_{13})^\top$ and let $\mathbf{R}_2$ be the lower right 2×2 block. The nonsingular matrix $\mathbf{T} = \begin{pmatrix} 1 & \mathbf{0}^\top \\ -\mathbf{r} & \mathbf{I}_2 \end{pmatrix}$ gives

$$\mathbf{T}\mathbf{R}\mathbf{T}^\top = \begin{pmatrix} 1 & \mathbf{0}^\top \\ \mathbf{0} & \mathbf{R}_2 - \mathbf{r}\mathbf{r}^\top \end{pmatrix},$$

and $\mathbf{a}^\top \mathbf{R} \mathbf{a} = \mathbf{b}^\top (\mathbf{T}\mathbf{R}\mathbf{T}^\top) \mathbf{b}$ with $\mathbf{b} = (\mathbf{T}^\top)^{-1} \mathbf{a}$ ranges over all vectors as $\mathbf{a}$ does. So $\mathbf{R}$ is nonnegative definite iff the Schur complement

$$\mathbf{R}_2 - \mathbf{r}\mathbf{r}^\top = \begin{pmatrix} 1 - \rho_{12}^2 & \rho_{23} - \rho_{12}\rho_{13} \\ \rho_{23} - \rho_{12}\rho_{13} & 1 - \rho_{13}^2 \end{pmatrix}$$

is. A symmetric 2×2 matrix with positive diagonal is nonnegative definite iff its determinant is nonnegative, because its eigenvalues have positive sum and their product is the determinant. The determinant is $(1 - \rho_{12}^2)(1 - \rho_{13}^2) - (\rho_{23} - \rho_{12}\rho_{13})^2$, and it is nonnegative exactly on the stated interval. ■

If Y_2 and Y_3 both correlate 0.8 with Y_1 , the interval is $[0.28, 1]$. Two variables that are strongly related to a third cannot be negatively correlated with each other, nor even uncorrelated. The quantity $(\rho_{23} - \rho_{12}\rho_{13}) / \sqrt{(1 - \rho_{12}^2)(1 - \rho_{13}^2)}$, which the proof forces into $[-1, 1]$, is the correlation of Y_2 and Y_3 after the linear effect of Y_1 has been removed from both. [Section 2.5](#) develops this residualization, and [Definition 3.6.1](#) names the result the partial correlation.

Correlation matrices do not transform simply under linear maps: the correlation matrix of $\mathbf{A}\mathbf{Y}$ must be recomputed from $\mathbf{A}\Sigma\mathbf{A}^\top$. Theory is therefore done with covariance matrices, and correlation matrices are used for description.

2.4.1 Exercises

2.4.1.1 A. Check your understanding

Exercise A1.

Show that $(1 - \rho)\mathbf{I}_p + \rho\mathbf{1}\mathbf{1}^\top$ is a correlation matrix iff $-1/(p-1) \leq \rho \leq 1$. Explain the lower bound through the variance of $\sum_i Z_i$.

Solution. The matrix sends $\mathbf{1}$ to $(1 + (p-1)\rho)\mathbf{1}$ and every $\mathbf{u} \perp \mathbf{1}$ to $(1 - \rho)\mathbf{u}$. Its eigenvalues are $1 + (p-1)\rho$ (once) and $1 - \rho$ ($p-1$ times). By [Proposition 2.4.1\(e\)](#) it is a correlation matrix iff both are nonnegative. If $\mathbf{Z}$ has this correlation matrix, $\text{Var}(\sum_i Z_i) = \mathbf{1}^\top \mathbf{R} \mathbf{1} = p[1 + (p-1)\rho]$, which must be nonnegative. Many variables cannot all be strongly negatively correlated with one another, because their sum would then have negative variance.

2.4.1.2 B. Practice

Exercise B1.

Show that $\det(\mathbf{\Sigma}) \leq \prod_i \sigma_{ii}$ for every covariance matrix, with equality for positive definite $\mathbf{\Sigma}$ iff $\mathbf{\Sigma}$ is diagonal. *Hint:* factor out $\mathbf{D}_\sigma$ and use the eigenvalues of $\mathbf{R}$.

Solution. If some $\sigma_{ii} = 0$, the i th row of $\mathbf{\Sigma}$ vanishes (apply [Theorem 2.2.3\(c\)](#) with $\mathbf{a}$ the i th coordinate vector) and both sides are zero. Otherwise $\det \mathbf{\Sigma} = \det(\mathbf{D}_\sigma)^2 \det \mathbf{R} = \prod_i \sigma_{ii} \cdot \det \mathbf{R}$. The eigenvalues $\lambda_i \geq 0$ of $\mathbf{R}$ sum to $\text{tr} \mathbf{R} = p$, so by the inequality of arithmetic and geometric means $\det \mathbf{R} = \prod_i \lambda_i \leq (p^{-1} \sum_i \lambda_i)^p = 1$. Equality holds iff every $\lambda_i = 1$, that is, iff $\mathbf{R} = \mathbf{I}$, that is, iff $\mathbf{\Sigma}$ is diagonal.

2.4.1.3 C. Going deeper

Exercise C1.

- (a) Show that a $p \times p$ matrix is a correlation matrix iff it is the Gram matrix $(\mathbf{u}_i^\top \mathbf{u}_j)$ of p unit vectors in $\mathbb{R}^p$. (b) For $p = 3$, write $\rho_{ij} = \cos \vartheta_{ij}$ with $\vartheta_{ij} \in [0, \pi]$. Show that [Proposition 2.4.2](#) is equivalent to the spherical triangle inequalities $|\vartheta_{12} - \vartheta_{13}| \leq \vartheta_{23} \leq \min(\vartheta_{12} + \vartheta_{13}, 2\pi - \vartheta_{12} - \vartheta_{13})$.

2.5 Moments of partitioned random vectors

Regression with random regressors studies a response vector $\mathbf{Y}$ ($p \times 1$) together with a regressor vector $\mathbf{X}$ ($q \times 1$) observed on the same unit. In this section $\mathbf{X}$ is a random vector, not a model matrix. We stack the two into one vector and read off what the blocks of its moments mean.

2.5.1 Blocks of the mean and covariance

Theorem 2.5.1 — Partitioned moments

Let $\mathbf{W} = (\mathbf{Y}^\top, \mathbf{X}^\top)^\top$ have finite second moments, and write

$$\mathbf{E}(\mathbf{W}) = \begin{pmatrix} \boldsymbol{\mu}_Y \\ \boldsymbol{\mu}_X \end{pmatrix}, \quad \text{Cov}(\mathbf{W}) = \begin{pmatrix} \boldsymbol{\Sigma}_{YY} & \boldsymbol{\Sigma}_{YX} \\ \boldsymbol{\Sigma}_{XY} & \boldsymbol{\Sigma}_{XX} \end{pmatrix}.$$

- (a) $\boldsymbol{\mu}_Y = \mathbf{E}(\mathbf{Y})$, $\boldsymbol{\mu}_X = \mathbf{E}(\mathbf{X})$, $\boldsymbol{\Sigma}_{YY} = \text{Cov}(\mathbf{Y})$, $\boldsymbol{\Sigma}_{XX} = \text{Cov}(\mathbf{X})$, $\boldsymbol{\Sigma}_{YX} = \text{Cov}(\mathbf{Y}, \mathbf{X})$ is $p \times q$, and $\boldsymbol{\Sigma}_{XY} = \boldsymbol{\Sigma}_{YX}^\top$.
 (b) For constants $\mathbf{A}$, $\mathbf{B}$, $\mathbf{c}$ of conformable sizes, $\text{Cov}(\mathbf{AY}, \mathbf{BX}) = \mathbf{A}\boldsymbol{\Sigma}_{YX}\mathbf{B}^\top$, and

$$\text{Cov}(\mathbf{AY} + \mathbf{BX} + \mathbf{c}) = \mathbf{A}\boldsymbol{\Sigma}_{YY}\mathbf{A}^\top + \mathbf{A}\boldsymbol{\Sigma}_{YX}\mathbf{B}^\top + \mathbf{B}\boldsymbol{\Sigma}_{XY}\mathbf{A}^\top + \mathbf{B}\boldsymbol{\Sigma}_{XX}\mathbf{B}^\top.$$

- (c) $\mathcal{C}(\boldsymbol{\Sigma}_{XY}) \subseteq \mathcal{C}(\boldsymbol{\Sigma}_{XX})$ and $\mathcal{C}(\boldsymbol{\Sigma}_{YX}) \subseteq \mathcal{C}(\boldsymbol{\Sigma}_{YY})$. Equivalently, for every generalized inverse $\boldsymbol{\Sigma}_{XX}^-$,

$$\boldsymbol{\Sigma}_{XX}\boldsymbol{\Sigma}_{XX}^-\boldsymbol{\Sigma}_{XY} = \boldsymbol{\Sigma}_{XY}, \quad \boldsymbol{\Sigma}_{YX}\boldsymbol{\Sigma}_{XX}^-\boldsymbol{\Sigma}_{XX} = \boldsymbol{\Sigma}_{YX},$$

and similarly with the roles of $\mathbf{Y}$ and $\mathbf{X}$ exchanged.

Proof.

- (a) The outer product $(\mathbf{W} - \mathbf{E}\mathbf{W})(\mathbf{W} - \mathbf{E}\mathbf{W})^\top$ has blocks $(\mathbf{Y} - \boldsymbol{\mu}_Y)(\mathbf{Y} - \boldsymbol{\mu}_Y)^\top$, $(\mathbf{Y} - \boldsymbol{\mu}_Y)(\mathbf{X} - \boldsymbol{\mu}_X)^\top$, its transpose, and $(\mathbf{X} - \boldsymbol{\mu}_X)(\mathbf{X} - \boldsymbol{\mu}_X)^\top$. Take expectations blockwise. (b) $\mathbf{A}\mathbf{Y} + \mathbf{B}\mathbf{X} + \mathbf{c} = [\mathbf{A}, \mathbf{B}]\mathbf{W} + \mathbf{c}$, and $\mathbf{A}\mathbf{Y} = [\mathbf{A}, \mathbf{0}]\mathbf{W}$, $\mathbf{B}\mathbf{X} = [\mathbf{0}, \mathbf{B}]\mathbf{W}$. Apply [Theorem 2.2.1](#) and multiply out the blocks.
- (b) Let $\mathbf{a} \in \mathcal{N}(\boldsymbol{\Sigma}_{XX})$. By [Theorem 2.2.3\(c\)](#), $\mathbf{a}^\top \mathbf{X}$ is constant with probability one, so its covariance with $\mathbf{Y}$ vanishes: $\boldsymbol{\Sigma}_{YX}\mathbf{a} = \text{Cov}(\mathbf{Y}, \mathbf{a}^\top \mathbf{X}) = \mathbf{0}$. Thus $\mathcal{N}(\boldsymbol{\Sigma}_{XX}) \subseteq \mathcal{N}(\boldsymbol{\Sigma}_{YX})$. Taking orthogonal complements, and using $\mathcal{N}(\mathbf{B})^\perp = \mathcal{C}(\mathbf{B}^\top)$ and the symmetry of $\boldsymbol{\Sigma}_{XX}$, gives $\mathcal{C}(\boldsymbol{\Sigma}_{XY}) \subseteq \mathcal{C}(\boldsymbol{\Sigma}_{XX})$. The same argument with $\mathbf{Y}$ and $\mathbf{X}$ exchanged gives the second inclusion. A column inclusion $\mathcal{C}(\boldsymbol{\Sigma}_{XY}) \subseteq \mathcal{C}(\boldsymbol{\Sigma}_{XX})$ means $\boldsymbol{\Sigma}_{XY} = \boldsymbol{\Sigma}_{XX}\mathbf{K}^\top$ for some $\mathbf{K}$. Then $\boldsymbol{\Sigma}_{XX}\boldsymbol{\Sigma}_{XX}^-\boldsymbol{\Sigma}_{XY} = \boldsymbol{\Sigma}_{XX}\boldsymbol{\Sigma}_{XX}^-\boldsymbol{\Sigma}_{XX}\mathbf{K}^\top = \boldsymbol{\Sigma}_{XY}$. Conversely, that identity exhibits every column of $\boldsymbol{\Sigma}_{XY}$ as a combination of columns of $\boldsymbol{\Sigma}_{XX}$ ([Theorem 1.4.1](#)). Transposing gives $\boldsymbol{\Sigma}_{YX} = \mathbf{K}\boldsymbol{\Sigma}_{XX}$ and the second identity. ■

Part (c) looks technical, but it removes every difficulty with singular $\boldsymbol{\Sigma}_{XX}$ in what follows. A direction in which $\mathbf{X}$ does not vary cannot covary with anything.

2.5.2 Linear prediction and residual covariance

The most important use of the blocks is to split $\mathbf{Y}$ into a part explained linearly by $\mathbf{X}$ and a part uncorrelated with $\mathbf{X}$.

Theorem 2.5.2 — Best linear predictor

In the setting of [Theorem 2.5.1](#), fix a generalized inverse $\boldsymbol{\Sigma}_{XX}^-$ and put

$$\mathbf{B}_* = \boldsymbol{\Sigma}_{YX}\boldsymbol{\Sigma}_{XX}^-, \quad \mathbf{E} = \mathbf{Y} - \boldsymbol{\mu}_Y - \mathbf{B}_*(\mathbf{X} - \boldsymbol{\mu}_X), \quad \boldsymbol{\Sigma}_{YY \cdot X} = \boldsymbol{\Sigma}_{YY} - \boldsymbol{\Sigma}_{YX}\boldsymbol{\Sigma}_{XX}^-\boldsymbol{\Sigma}_{XY}.$$

- (a) $\mathbf{E}(\mathbf{E}) = \mathbf{0}$ and $\text{Cov}(\mathbf{E}, \mathbf{X}) = \mathbf{0}$.
- (b) $\text{Cov}(\mathbf{E}) = \boldsymbol{\Sigma}_{YY \cdot X}$. This matrix is nonnegative definite and does not depend on the choice of generalized inverse. Neither does $\mathbf{B}_*(\mathbf{X} - \boldsymbol{\mu}_X)$, with probability one.
- (c) For every constant $p \times q$ matrix $\mathbf{B}$ and $p \times 1$ vector $\mathbf{c}$,

$$\mathbf{E}[(\mathbf{Y} - \mathbf{c} - \mathbf{B}\mathbf{X})(\mathbf{Y} - \mathbf{c} - \mathbf{B}\mathbf{X})^\top] = \boldsymbol{\Sigma}_{YY \cdot X} + (\mathbf{B} - \mathbf{B}_*)\boldsymbol{\Sigma}_{XX}(\mathbf{B} - \mathbf{B}_*)^\top + \mathbf{d}\mathbf{d}^\top,$$

where $\mathbf{d} = \boldsymbol{\mu}_Y - \mathbf{c} - \mathbf{B}\boldsymbol{\mu}_X$. So the mean squared error matrix is never smaller, in the nonnegative definite ordering, than $\boldsymbol{\Sigma}_{YY \cdot X}$. Equality holds iff $\mathbf{c} = \boldsymbol{\mu}_Y - \mathbf{B}\boldsymbol{\mu}_X$ and $\mathbf{B}\boldsymbol{\Sigma}_{XX} = \boldsymbol{\Sigma}_{YX}$.

Proof. By [Theorem 2.5.1\(c\)](#), write $\boldsymbol{\Sigma}_{YX} = \mathbf{K}\boldsymbol{\Sigma}_{XX}$. Then $\mathbf{B}_*\boldsymbol{\Sigma}_{XX} = \mathbf{K}\boldsymbol{\Sigma}_{XX}\boldsymbol{\Sigma}_{XX}^-\boldsymbol{\Sigma}_{XX} = \mathbf{K}\boldsymbol{\Sigma}_{XX} = \boldsymbol{\Sigma}_{YX}$.

- (a) The mean is clear. By [Theorem 2.5.1\(b\)](#), $\text{Cov}(\mathbf{E}, \mathbf{X}) = \boldsymbol{\Sigma}_{YX} - \mathbf{B}_*\boldsymbol{\Sigma}_{XX} = \mathbf{0}$.
- (b) Again by [Theorem 2.5.1\(b\)](#), $\text{Cov}(\mathbf{E}) = \boldsymbol{\Sigma}_{YY} - \mathbf{B}_*\boldsymbol{\Sigma}_{XY} - \boldsymbol{\Sigma}_{YX}\mathbf{B}_*^\top + \mathbf{B}_*\boldsymbol{\Sigma}_{XX}\mathbf{B}_*^\top$. Us-

ing $\mathbf{B}_* \Sigma_{XX} = \Sigma_{YX}$, the last term is $\Sigma_{YX} \mathbf{B}_*^\top$, which cancels the third, so $\text{Cov}(\mathbf{E}) = \Sigma_{YY} - \mathbf{B}_* \Sigma_{XY} = \Sigma_{YY.X}$. Next, $\Sigma_{YX} \Sigma_{XX}^{-1} \Sigma_{XY} = \mathbf{K} \Sigma_{XX} \Sigma_{XX}^{-1} \Sigma_{XX} \mathbf{K}^\top = \mathbf{K} \Sigma_{XX} \mathbf{K}^\top$, which involves no generalized inverse. A covariance matrix is nonnegative definite. Finally, by Theorem 2.2.3(d), $\mathbf{X} - \mu_X = \Sigma_{XX} \mathbf{v}$ for some random $\mathbf{v}$, with probability one, and then $\mathbf{B}_*(\mathbf{X} - \mu_X) = \Sigma_{YX} \mathbf{v}$ for every choice of generalized inverse.

- (c) Decompose $\mathbf{Y} - \mathbf{c} - \mathbf{B}\mathbf{X} = \mathbf{E} + (\mathbf{B}_* - \mathbf{B})(\mathbf{X} - \mu_X) + \mathbf{d}$. The three pieces are, respectively, mean zero, mean zero and constant, and the first two are uncorrelated by (a). So the cross products have zero expectation, and the expected outer product is $\text{Cov}(\mathbf{E}) + (\mathbf{B}_* - \mathbf{B}) \Sigma_{XX} (\mathbf{B}_* - \mathbf{B})^\top + \mathbf{d} \mathbf{d}^\top$. Both added matrices are nonnegative definite. They vanish iff $\mathbf{d} = \mathbf{0}$ and $(\mathbf{B} - \mathbf{B}_*) \Sigma_{XX}^{1/2} = \mathbf{0}$, and the latter is equivalent to $(\mathbf{B} - \mathbf{B}_*) \Sigma_{XX} = \mathbf{0}$, that is, to $\mathbf{B} \Sigma_{XX} = \Sigma_{YX}$.

■

The affine function $\mu_Y + \mathbf{B}_*(\mathbf{X} - \mu_X)$ is the **best linear predictor** of $\mathbf{Y}$ from $\mathbf{X}$. Because the ordering in (c) is the nonnegative definite ordering, it is simultaneously best for every linear combination $\mathbf{a}^\top \mathbf{Y}$ and for the total squared error. The matrix $\Sigma_{YY.X}$ is the **partial covariance matrix** of $\mathbf{Y}$ given $\mathbf{X}$. When Σ_{XX} is nonsingular, $\mathbf{B}_* = \Sigma_{YX} \Sigma_{XX}^{-1}$ holds the population regression coefficients, and $\Sigma_{YY.X}$ is a Schur complement. Two consequences follow from Chapter 1. If $\text{Cov}(\mathbf{W})$ is positive definite, the YY block of its inverse is $\Sigma_{YY.X}^{-1}$ (Theorem 1.6.4). And $\det \text{Cov}(\mathbf{W}) = \det(\Sigma_{XX}) \det(\Sigma_{YY.X})$ (Theorem 1.6.3), so the generalized variance factorizes into the part of $\mathbf{X}$ and the part of $\mathbf{Y}$ that $\mathbf{X}$ leaves unexplained.

Nothing here says that $E(\mathbf{Y} | \mathbf{X})$ is linear. The best linear predictor is best among affine functions, whatever the joint distribution. For jointly normal vectors, the conditional mean happens to be this affine function, and $\Sigma_{YY.X}$ is the conditional covariance matrix (Theorem 3.5.2). The partial correlation of Definition 3.6.1 is the correlation computed from $\Sigma_{YY.X}$. Chapter 6 shows that least squares estimates $\mathbf{B}_*$ in large samples, whether or not the regression function is linear. It also recovers the predictor as an orthogonal projection in the space of square-integrable random variables. For scalar Y , the coefficient vector used there is $\mathbf{B}_*^\top$.

Proposition 2.5.3 — Multiple correlation

Let Y be scalar with $\sigma_{YY} = \text{Var}(Y) > 0$, let Σ_{XX} be positive definite, and write $\sigma_{XY} = \text{Cov}(\mathbf{X}, Y)$. Over all $\mathbf{a} \neq \mathbf{0}$,

$$\max_{\mathbf{a}} \text{corr}(Y, \mathbf{a}^\top \mathbf{X})^2 = \rho_{Y.X}^2 = \frac{\sigma_{XY}^\top \Sigma_{XX}^{-1} \sigma_{XY}}{\sigma_{YY}},$$

attained at $\mathbf{a} = \Sigma_{XX}^{-1} \sigma_{XY}$, the coefficients of the best linear predictor (if $\sigma_{XY} = \mathbf{0}$, every $\mathbf{a}$ attains the maximum 0). Moreover $1 - \rho_{Y.X}^2 = \sigma_{YY.X} / \sigma_{YY}$.

Proof. $\text{corr}(Y, \mathbf{a}^\top \mathbf{X})^2 = (\mathbf{a}^\top \sigma_{XY})^2 / (\sigma_{YY} \mathbf{a}^\top \Sigma_{XX} \mathbf{a})$. Write $\mathbf{a}^\top \sigma_{XY} = (\Sigma_{XX}^{1/2} \mathbf{a})^\top (\Sigma_{XX}^{-1/2} \sigma_{XY})$. The Cauchy–Schwarz inequality gives $(\mathbf{a}^\top \sigma_{XY})^2 \leq (\mathbf{a}^\top \Sigma_{XX} \mathbf{a}) (\sigma_{XY}^\top \Sigma_{XX}^{-1} \sigma_{XY})$, with equality iff $\Sigma_{XX}^{1/2} \mathbf{a}$ is proportional to $\Sigma_{XX}^{-1/2} \sigma_{XY}$. Dividing gives the bound and the maximizer. The last identity is the definition of $\sigma_{YY.X} = \sigma_{YY} - \sigma_{XY}^\top \Sigma_{XX}^{-1} \sigma_{XY}$, divided by σ_{YY} .

■

$\rho_{Y \cdot X}^2$ is the population version of R^2 : the fraction of the variance of Y that the best linear predictor accounts for. [Chapter 14](#) studies its sample version.

2.5.3 Conditioning: total expectation and total covariance

When the joint distribution is available, conditioning gives another decomposition of the covariance matrix. Conditional expectations and covariances are taken entry by entry, as in [Definition 2.1.1](#), and $\text{Cov}(\mathbf{Y} \mid \mathbf{X}) = \mathbb{E}[(\mathbf{Y} - \mathbb{E}(\mathbf{Y} \mid \mathbf{X}))(\mathbf{Y} - \mathbb{E}(\mathbf{Y} \mid \mathbf{X}))^\top \mid \mathbf{X}]$.

Proposition 2.5.4 — Laws of total expectation and covariance

If $\mathbf{Y}$ has finite second moments, then $\mathbb{E}(\mathbf{Y}) = \mathbb{E}[\mathbb{E}(\mathbf{Y} \mid \mathbf{X})]$ and

$$\text{Cov}(\mathbf{Y}) = \mathbb{E}[\text{Cov}(\mathbf{Y} \mid \mathbf{X})] + \text{Cov}(\mathbb{E}(\mathbf{Y} \mid \mathbf{X})).$$

Proof. The first identity is the tower property of conditional expectation, applied to each component. For the second, let $\mathbf{m}(\mathbf{X}) = \mathbb{E}(\mathbf{Y} \mid \mathbf{X})$ and write $\mathbf{Y} - \boldsymbol{\mu} = (\mathbf{Y} - \mathbf{m}(\mathbf{X})) + (\mathbf{m}(\mathbf{X}) - \boldsymbol{\mu})$. The outer product has two cross terms. The conditional expectation of $(\mathbf{Y} - \mathbf{m}(\mathbf{X}))(\mathbf{m}(\mathbf{X}) - \boldsymbol{\mu})^\top$ given $\mathbf{X}$ is $[\mathbb{E}(\mathbf{Y} \mid \mathbf{X}) - \mathbf{m}(\mathbf{X})](\mathbf{m}(\mathbf{X}) - \boldsymbol{\mu})^\top = \mathbf{0}$, so its expectation is zero, and likewise for the other. The expectation of $(\mathbf{Y} - \mathbf{m})(\mathbf{Y} - \mathbf{m})^\top$ is $\mathbb{E}[\text{Cov}(\mathbf{Y} \mid \mathbf{X})]$ by the tower property, and that of $(\mathbf{m} - \boldsymbol{\mu})(\mathbf{m} - \boldsymbol{\mu})^\top$ is $\text{Cov}(\mathbf{m}(\mathbf{X}))$ because $\mathbb{E} \mathbf{m}(\mathbf{X}) = \boldsymbol{\mu}$. ■

Example. (A shared random effect)

Measurements $Y_1, \dots, Y_k$ on the same subject share a subject effect: $\mathbf{Y} = \alpha \mathbf{1} + \boldsymbol{\varepsilon}$, where α has variance τ^2 , $\text{Cov}(\boldsymbol{\varepsilon}) = \sigma^2 \mathbf{I}_k$, and $\boldsymbol{\varepsilon}$ is independent of α with mean zero. Given α , the mean is $\alpha \mathbf{1}$ and the covariance is $\sigma^2 \mathbf{I}$. By [Proposition 2.5.4](#),

$$\text{Cov}(\mathbf{Y}) = \sigma^2 \mathbf{I}_k + \tau^2 \mathbf{1} \mathbf{1}^\top,$$

so every pair of measurements has correlation $\tau^2/(\tau^2 + \sigma^2)$. Independent errors combined with a shared random component produce exactly this *equicorrelation* structure. It is the starting point of Chapter 32. Ordinary least squares ignores it, with consequences that Chapter 33 examines.

2.5.4 Exercises

2.5.4.1 A. Check your understanding

Exercise A1.

For scalar Y and X with $\text{Var}(X) > 0$, show that the best linear predictor has slope

σ_{YX}/σ_{XX} , and that its mean squared error is $\sigma_{YY}(1 - \rho^2)$.

Exercise A2.

Let X be uniform on $(-1, 1)$ and $Y = X^2 + \varepsilon$, with ε independent of X , mean 0 and variance σ^2 . Find the best linear predictor of Y from X and its mean squared error, and compare with the mean squared error of $E(Y | X)$.

2.5.4.2 B. Practice

Exercise B1.

Let (Y_1, Y_2, Y_3) have correlation matrix $\mathbf{R}$. Compute the partial covariance matrix of (Y_2, Y_3) given Y_1 from [Theorem 2.5.2](#), and show that the corresponding correlation is

$$\frac{\rho_{23} - \rho_{12}\rho_{13}}{\sqrt{(1 - \rho_{12}^2)(1 - \rho_{13}^2)}}.$$

Exercise B2.

Let $\mathbf{X} = (X_1, X_1)^\top$ with $\text{Var}(X_1) = 1$, and let $\text{Cov}(Y, X_1) = c$. Find two different generalized inverses of Σ_{XX} that give different $\mathbf{B}_*$, and show that both give the same predictor $\mathbf{B}_*(\mathbf{X} - \boldsymbol{\mu}_X)$.

Exercise B3.

A random label K takes values $1, \dots, g$ with probabilities π_k . Given $K = k$, $\mathbf{Y}$ has mean $\mathbf{m}_k$ and covariance Σ_k . Show that $\text{Cov}(\mathbf{Y}) = \sum_k \pi_k \Sigma_k + \sum_k \pi_k (\mathbf{m}_k - \bar{\mathbf{m}})(\mathbf{m}_k - \bar{\mathbf{m}})^\top$, where $\bar{\mathbf{m}} = \sum_k \pi_k \mathbf{m}_k$. Interpret the two terms as within-group and between-group covariance.

2.5.4.3 C. Going deeper

Exercise C1.

Let $\mathbf{N}$ be the symmetric partitioned matrix

$$\mathbf{N} = \begin{pmatrix} \Sigma_{YY} & \Sigma_{YX} \\ \Sigma_{XY} & \Sigma_{XX} \end{pmatrix}.$$

Show that $\mathbf{N}$ is nonnegative definite iff Σ_{XX} is nonnegative definite, $\mathcal{C}(\Sigma_{XY}) \subseteq \mathcal{C}(\Sigma_{XX})$, and $\Sigma_{YY} - \Sigma_{YX}\Sigma_{XX}^- \Sigma_{XY}$ is nonnegative definite for some generalized inverse Σ_{XX}^- (equivalently, given the column-space condition, for every one). Show by a 2×2 example that the column-space condition cannot be dropped.

Solution. If the matrix is nonnegative definite, it is the covariance matrix of some $\mathbf{W} = (\mathbf{Y}^\top, \mathbf{X}^\top)^\top$ ([Theorem 2.2.3\(b\)](#)). Then Σ_{XX} is a covariance matrix, the column condition is [Theorem 2.5.1\(c\)](#), and the Schur complement is $\text{Cov}(\mathbf{E})$ by [Theorem 2.5.2\(b\)](#).

Conversely, the column condition gives $\Sigma_{YX} = \mathbf{K}\Sigma_{XX}$, and then $\Sigma_{YX}\Sigma_{XX}^{-}\Sigma_{XY} = \mathbf{K}\Sigma_{XX}\mathbf{K}^{\top}$. With the nonsingular $\mathbf{T} = \begin{pmatrix} \mathbf{I} & -\mathbf{K} \\ \mathbf{0} & \mathbf{I} \end{pmatrix}$,

$$\mathbf{T} \begin{pmatrix} \Sigma_{YY} & \Sigma_{YX} \\ \Sigma_{XY} & \Sigma_{XX} \end{pmatrix} \mathbf{T}^{\top} = \begin{pmatrix} \Sigma_{YY} - \mathbf{K}\Sigma_{XX}\mathbf{K}^{\top} & \mathbf{0} \\ \mathbf{0} & \Sigma_{XX} \end{pmatrix},$$

which is nonnegative definite. So is the original matrix, since $\mathbf{a}^{\top}\mathbf{N}\mathbf{a} = \mathbf{b}^{\top}(\mathbf{T}\mathbf{N}\mathbf{T}^{\top})\mathbf{b}$ with $\mathbf{b} = (\mathbf{T}^{\top})^{-1}\mathbf{a}$. For the example, take $\begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix}$. Here $\Sigma_{XX} = 0$ is nonnegative definite, and with the generalized inverse 0 the Schur complement is $1 \geq 0$. But the determinant is -1 , so the matrix is not nonnegative definite. The column condition fails, because $\mathcal{C}(1) \not\subseteq \mathcal{C}(0)$. Without that condition the Schur complement also depends on the choice: every scalar t is a generalized inverse of 0, and the Schur complement is then $1 - 1 \cdot t \cdot 1 = 1 - t$, which is negative for $t > 1$. Under the column condition, $\Sigma_{YX}\Sigma_{XX}^{-}\Sigma_{XY} = \mathbf{K}\Sigma_{XX}\mathbf{K}^{\top}$ for every choice, as shown above.

2.6 Moment generating functions and independence

Means and covariance matrices summarize a distribution. They do not determine it, and they cannot certify independence. For that we need a transform of the whole distribution. This section collects the facts that [Chapter 3](#) and [Chapter 4](#) use.

Definition 2.6.1 — Moment generating function

The **moment generating function** of a $p \times 1$ random vector $\mathbf{Y}$ is

$$M_{\mathbf{Y}}(\mathbf{t}) = \mathbb{E} \exp(\mathbf{t}^{\top} \mathbf{Y}), \quad \mathbf{t} \in \mathbb{R}^p,$$

where the expectation may be $+\infty$. We say that $M_{\mathbf{Y}}$ *exists* if it is finite for all $\mathbf{t}$ in some neighbourhood $\{\|\mathbf{t}\| < h\}$ of the origin, $h > 0$.

Theorem 2.6.1 — Properties of moment generating functions

Suppose $M_{\mathbf{Y}}$ exists.

- (a) *Uniqueness.* If $M_{\mathbf{V}}$ also exists and $M_{\mathbf{Y}} = M_{\mathbf{V}}$ on a neighbourhood of $\mathbf{0}$, then $\mathbf{Y}$ and $\mathbf{V}$ have the same distribution.
- (b) *Moments.* All moments of $\mathbf{Y}$ are finite. The gradient of $M_{\mathbf{Y}}$ at $\mathbf{0}$ is $\mathbb{E}(\mathbf{Y})$, and its Hessian at $\mathbf{0}$ is $\mathbb{E}(\mathbf{Y}\mathbf{Y}^{\top})$. The Hessian of $\log M_{\mathbf{Y}}$ at $\mathbf{0}$ is $\text{Cov}(\mathbf{Y})$.
- (c) *Affine maps.* For a constant $k \times p$ matrix $\mathbf{A}$ and $k \times 1$ vector $\mathbf{b}$, $M_{\mathbf{A}\mathbf{Y}+\mathbf{b}}(\mathbf{s}) = \exp(\mathbf{s}^{\top} \mathbf{b}) M_{\mathbf{Y}}(\mathbf{A}^{\top} \mathbf{s})$.
- (d) *Independence.* Partition $\mathbf{Y} = (\mathbf{Y}_1^{\top}, \mathbf{Y}_2^{\top})^{\top}$ and $\mathbf{t} = (\mathbf{t}_1^{\top}, \mathbf{t}_2^{\top})^{\top}$ conformably. Then $\mathbf{Y}_1$ and $\mathbf{Y}_2$ are independent iff $M_{\mathbf{Y}}(\mathbf{t}) = M_{\mathbf{Y}_1}(\mathbf{t}_1) M_{\mathbf{Y}_2}(\mathbf{t}_2)$ for all $\mathbf{t}$ in a neighbourhood of $\mathbf{0}$.

Proof. For (a), the scalar case is a theorem of analysis: a moment generating function that is finite near the origin extends analytically, which pins down the characteristic function and hence the distribution (Billingsley 1995). For vectors, fix $\mathbf{a}$. For small $|s|$, $M_{\mathbf{a}^\top \mathbf{Y}}(s) = M_{\mathbf{Y}}(\mathbf{s}\mathbf{a}) = M_{\mathbf{V}}(\mathbf{s}\mathbf{a}) = M_{\mathbf{a}^\top \mathbf{V}}(s)$, so $\mathbf{a}^\top \mathbf{Y}$ and $\mathbf{a}^\top \mathbf{V}$ have the same distribution by the scalar case, and Proposition 2.6.2 below finishes the argument. For (b), finiteness near $\mathbf{0}$ implies $E \exp(h'|Y_i|) < \infty$ for some $h' > 0$ and each i , so every polynomial in the components is integrable and differentiation under the expectation is justified. Then $\partial M / \partial t_i = E(Y_i e^{\mathbf{t}^\top \mathbf{Y}})$ and $\partial^2 M / \partial t_i \partial t_j = E(Y_i Y_j e^{\mathbf{t}^\top \mathbf{Y}})$. At $\mathbf{t} = \mathbf{0}$ these are $E(Y_i)$ and $E(Y_i Y_j)$. For $K = \log M$, the chain rule gives $\nabla^2 K = M^{-1} \nabla^2 M - M^{-2} (\nabla M)(\nabla M)^\top$, which at $\mathbf{0}$, where $M = 1$, is $E(\mathbf{Y}\mathbf{Y}^\top) - \boldsymbol{\mu}\boldsymbol{\mu}^\top = \text{Cov}(\mathbf{Y})$. For (c), $E \exp(\mathbf{s}^\top (\mathbf{A}\mathbf{Y} + \mathbf{b})) = e^{\mathbf{s}^\top \mathbf{b}} E \exp((\mathbf{A}^\top \mathbf{s})^\top \mathbf{Y})$. For (d), independence gives $E(e^{\mathbf{t}_1^\top \mathbf{Y}_1} e^{\mathbf{t}_2^\top \mathbf{Y}_2}) = E(e^{\mathbf{t}_1^\top \mathbf{Y}_1}) E(e^{\mathbf{t}_2^\top \mathbf{Y}_2})$. Conversely, let $\mathbf{V}_1$ and $\mathbf{V}_2$ be independent with the distributions of $\mathbf{Y}_1$ and $\mathbf{Y}_2$. The vector $(\mathbf{V}_1^\top, \mathbf{V}_2^\top)^\top$ has moment generating function $M_{\mathbf{Y}_1}(\mathbf{t}_1) M_{\mathbf{Y}_2}(\mathbf{t}_2)$, which by hypothesis equals $M_{\mathbf{Y}}$ near $\mathbf{0}$. By (a) it has the same joint distribution as $\mathbf{Y}$, so $\mathbf{Y}_1$ and $\mathbf{Y}_2$ are independent. ■

The next fact explains why Chapter 3 can define the multivariate normal through linear combinations.

Proposition 2.6.2 — Distributions are determined by linear combinations

If $\mathbf{a}^\top \mathbf{Y}$ and $\mathbf{a}^\top \mathbf{V}$ have the same distribution for every $\mathbf{a} \in \mathbb{R}^p$, then $\mathbf{Y}$ and $\mathbf{V}$ have the same distribution.

Proof. The characteristic function $\phi_{\mathbf{Y}}(\mathbf{t}) = E \exp(i\mathbf{t}^\top \mathbf{Y})$ exists for every random vector and determines its distribution (Billingsley 1995). Since $\phi_{\mathbf{Y}}(\mathbf{t}) = \phi_{\mathbf{t}^\top \mathbf{Y}}(1) = \phi_{\mathbf{t}^\top \mathbf{V}}(1) = \phi_{\mathbf{V}}(\mathbf{t})$ for every $\mathbf{t}$, the two distributions coincide. ■

If $\mathbf{Y}_1$ and $\mathbf{Y}_2$ are independent, so are $g(\mathbf{Y}_1)$ and $h(\mathbf{Y}_2)$ for measurable g and h . Independence of statistics is usually proved this way. For normal vectors, Theorem 3.4.1 reduces the independence of two blocks to a zero cross-covariance, and Theorem 4.4.1 and Theorem 4.4.2 build on that.

Independence of *several* vectors means that the joint distribution factors into all the marginals, not merely that each pair is independent.

Example. (Pairwise but not mutually independent)

Let S_1 and S_2 be independent, each equal to ± 1 with probability $\frac{1}{2}$, and put $S_3 = S_1 S_2$. Each S_j is ± 1 with probability $\frac{1}{2}$. Each pair is independent: for instance, $\Pr(S_1 = s, S_3 = u) = \Pr(S_1 = s, S_2 = su) = \frac{1}{4}$ for all signs s, u . So $\text{Cov}(\mathbf{S}) = \mathbf{I}_3$. But the three are not mutually independent, since S_3 is a function of the other two. A third-order moment shows the failure directly: $E(S_1 S_2 S_3) = 1$, whereas mutual independence would force it to be 0. No condition on the covariance matrix can detect this, because covariances involve only pairs.

2.6.1 Exercises

2.6.1.1 A. Check your understanding

Exercise A1.

Find the moment generating function of the multinomial count vector of [Example \(Multinomial counts\)](#), and recover its mean and covariance matrix from [Theorem 2.6.1\(b\)](#).

2.6.1.2 B. Practice

Exercise B1.

Let $S_1, \dots, S_k$ be independent random signs. For each nonempty $A \subseteq \{1, \dots, k\}$ put $T_A = \prod_{i \in A} S_i$. Show that the $2^k - 1$ variables T_A are pairwise independent with covariance matrix $\mathbf{I}$, and find a triple of them that is not mutually independent.

2.7 Sample moments and their computation

The population quantities of this chapter have sample analogues, and the operator rules give their expectations at once. Let $\mathbf{y}_1, \dots, \mathbf{y}_n$ be $p \times 1$ observations, and stack them as the rows of the $n \times p$ data matrix $\mathbf{Y} = [\mathbf{y}_1, \dots, \mathbf{y}_n]^\top$. The script letter keeps the data matrix apart from a single random vector $\mathbf{Y}$. The rows keep the lowercase $\mathbf{y}_i$ of observed data, but in [Proposition 2.7.1](#) they are treated as random. With $\mathbf{C} = \mathbf{I}_n - n^{-1}\mathbf{1}\mathbf{1}^\top$, the sample mean vector and the **sample covariance matrix** are

$$\bar{\mathbf{y}} = \frac{1}{n} \sum_{i=1}^n \mathbf{y}_i = \frac{1}{n} \mathbf{Y}^\top \mathbf{1}, \quad \mathbf{S} = \frac{1}{n-1} \sum_{i=1}^n (\mathbf{y}_i - \bar{\mathbf{y}})(\mathbf{y}_i - \bar{\mathbf{y}})^\top = \frac{1}{n-1} \mathbf{Y}^\top \mathbf{C} \mathbf{Y}.$$

The second form of $\mathbf{S}$ holds because $\mathbf{C}\mathbf{Y}$ is the data matrix with its column means subtracted, and $\mathbf{C}$ is symmetric and idempotent. The sample correlation matrix is $\hat{\mathbf{R}} = \hat{\mathbf{D}}^{-1} \mathbf{S} \hat{\mathbf{D}}^{-1}$, where $\hat{\mathbf{D}}$ holds the sample standard deviations.

Proposition 2.7.1 — Moments of sample moments

Suppose the rows $\mathbf{y}_i$ are pairwise uncorrelated random vectors with common mean $\boldsymbol{\mu}$ and common covariance matrix $\boldsymbol{\Sigma}$. Then

- (a) $E(\bar{\mathbf{y}}) = \boldsymbol{\mu}$ and $\text{Cov}(\bar{\mathbf{y}}) = \boldsymbol{\Sigma}/n$;
- (b) $E(\mathbf{S}) = \boldsymbol{\Sigma}$;
- (c) $\mathbf{S}$ is nonnegative definite, and $\text{rank}(\mathbf{S}) \leq \min(p, n-1)$ for every realization.

Proof.

- (a) is [Proposition 2.2.2\(c\)](#) with $c_i = 1/n$. For (b), let $\mathbf{u}_j$ denote the j th column of $\mathbf{Y}$, the n observations of variable j . The (j, k) entry of $\mathbf{Y}^\top \mathbf{C} \mathbf{Y}$ is the bilinear form $\mathbf{u}_j^\top \mathbf{C} \mathbf{u}_k$. Because different rows are uncorrelated, $\text{Cov}(\mathbf{u}_k, \mathbf{u}_j) = \sigma_{kj} \mathbf{I}_n$, and

$E(\mathbf{u}_j) = \mu_j \mathbf{1}$. By [Corollary 2.3.2\(d\)](#),

$$E(\mathbf{u}_j^\top \mathbf{C} \mathbf{u}_k) = \sigma_{kj} \operatorname{tr}(\mathbf{C}) + \mu_j \mu_k \mathbf{1}^\top \mathbf{C} \mathbf{1} = (n-1) \sigma_{jk},$$

since $\operatorname{tr}(\mathbf{C}) = n-1$ and $\mathbf{C} \mathbf{1} = \mathbf{0}$. (c) For any $\mathbf{a}$, $\mathbf{a}^\top \mathbf{S} \mathbf{a} = \|\mathbf{C} \mathbf{y} \mathbf{a}\|^2 / (n-1) \geq 0$. The rank of $\mathbf{y}^\top \mathbf{C} \mathbf{y} = (\mathbf{C} \mathbf{y})^\top (\mathbf{C} \mathbf{y})$ equals $\operatorname{rank}(\mathbf{C} \mathbf{y})$, which is at most $\operatorname{rank}(\mathbf{C}) = n-1$ and at most p . ■

The divisor $n-1$ is the trace of $\mathbf{C}$, the dimension of the space the centred data live in, exactly as $n-r$ is for regression residuals in [Example \(Residual sum of squares under a wrong mean\)](#). Part (c) has a practical side. With $n \leq p$, as in genomics or finance, the sample covariance matrix is always singular, even when $\mathbf{\Sigma}$ is positive definite. It has a linear combination with zero sample variance whether or not one exists in the population. Chapter 28 returns to this.

Computation. In practice $\mathbf{C}$ is never formed: one subtracts column means. The eigenvalues of $\mathbf{S}$ are the squared singular values of $\mathbf{C} \mathbf{y} / \sqrt{n-1}$ ([Theorem 1.8.1](#)), which can be computed without forming a cross-product matrix ([Chapter 10](#)).

Example. (Macroeconomic series)

Longley's data give $n = 16$ annual observations (1947–1962) on $p = 7$ US series: total employment, the GNP deflator, GNP, unemployment, armed forces, population and the year. The eigenvalues of $\mathbf{S}$ run from 9.94×10^9 down to 1.07×10^{-2} , a ratio of 9.3×10^{11} . That spread says little about the structure of the data. It mostly reflects units: the standard deviation of GNP is 99395, that of the deflator 10.79.

The correlation matrix removes the units ([Proposition 2.4.1\(d\)](#)). Its eigenvalues sum to $p = 7$. The largest is 5.533, so one direction carries about 79% of the standardized variation, and the smallest is 2.59×10^{-4} . By [Theorem 1.7.7](#), the unit vector $\mathbf{v}$ belonging to the smallest eigenvalue minimizes the sample variance of $\mathbf{v}^\top \mathbf{z}$ over standardized combinations. Its two largest weights are 0.681 on year and -0.674 on GNP, and the combination has standard deviation 0.016, against 1 for each standardized variable. In these sixteen years GNP is almost an exact linear function of time (their correlation is 0.995). This is the sample version of [Theorem 2.2.3\(c\)](#): a nearly zero eigenvalue marks a nearly constant linear combination. For regression it signals collinearity, which Chapter 26 studies. Finally, the first 5 rows alone give a sample covariance matrix of rank 4, as [Proposition 2.7.1\(c\)](#) requires.

```
import numpy as np
import statsmodels.api as sm
data = sm.datasets.longley.load_pandas().data
names = list(data.columns)

Y = data.to_numpy(dtype=float)           # n x p data matrix, one row per
    year
n, p = Y.shape
C = np.eye(n) - np.ones((n, n)) / n     # centring matrix I - J/n
S = Y.T @ C @ Y / (n - 1)               # sample covariance matrix
d = np.sqrt(np.diag(S))
R = S / np.outer(d, d)                  # D^{-1} S D^{-1}, D = diag(d)
```

```

lam_S = np.linalg.eigvalsh(S)
lam_R, V = np.linalg.eigh(R)                                # ascending eigenvalues
print("eigenvalues of S:", lam_S)
print("eigenvalues of R:", lam_R)
v = V[:, 0]                                                  # direction of least variance
Z = (Y - Y.mean(0)) / d                                      # standardized variables
print("sample variance of Z v:", np.var(Z @ v, ddof=1))

```

2.7.1 Exercises

2.7.1.1 B. Practice

Exercise B1.

Suppose the rows of $\mathbf{Y}$ are observed in time order, with common mean $\boldsymbol{\mu}$ and covariance $\boldsymbol{\Sigma}$, and that neighbouring rows are correlated: $\text{Cov}(\mathbf{y}_i, \mathbf{y}_{i+1}) = \rho\boldsymbol{\Sigma}$, while $\text{Cov}(\mathbf{y}_i, \mathbf{y}_j) = \mathbf{0}$ for $|i - j| \geq 2$. Show that $E(\mathbf{S}) = (1 - 2\rho/n)\boldsymbol{\Sigma}$ and $\text{Cov}(\bar{\mathbf{y}}) = n^{-2}[n + 2(n - 1)\rho]\boldsymbol{\Sigma}$. Which of the two effects disappears as $n \rightarrow \infty$?

Solution. As in the proof of [Proposition 2.7.1](#), $\text{Cov}(\mathbf{u}_k, \mathbf{u}_j) = \sigma_{kj}\mathbf{T}$, where $\mathbf{T}$ is the $n \times n$ tridiagonal matrix with ones on the diagonal and ρ next to it. Since $\mathbf{C}\mathbf{1} = \mathbf{0}$, [Corollary 2.3.2\(d\)](#) gives $E(\mathbf{u}_j^\top \mathbf{C}\mathbf{u}_k) = \sigma_{kj} \text{tr}(\mathbf{C}\mathbf{T}) = \sigma_{kj} [\text{tr}(\mathbf{T}) - n^{-1}\mathbf{1}^\top \mathbf{T}\mathbf{1}]$. Here $\text{tr}(\mathbf{T}) = n$ and $\mathbf{1}^\top \mathbf{T}\mathbf{1} = n + 2(n - 1)\rho$, so the bracket is $(n - 1) - 2(n - 1)\rho/n$, and dividing by $n - 1$ gives $E(\mathbf{S}) = (1 - 2\rho/n)\boldsymbol{\Sigma}$. For the mean, $\bar{\mathbf{y}} = n^{-1} \sum_i \mathbf{y}_i$ and [Proposition 2.2.2](#) gives n^{-2} times the sum of all $\text{Cov}(\mathbf{y}_i, \mathbf{y}_j)$: n diagonal terms $\boldsymbol{\Sigma}$ and $2(n - 1)$ neighbouring terms $\rho\boldsymbol{\Sigma}$. The bias of $\mathbf{S}$ is $O(1/n)$ and disappears, but $n \text{Cov}(\bar{\mathbf{y}}) \rightarrow (1 + 2\rho)\boldsymbol{\Sigma}$, not $\boldsymbol{\Sigma}$. For $\rho > 0$ the naive standard error $\sqrt{s_{jj}/n}$ stays too small by a fixed factor however large the sample.

2.8 Summary and Notes

2.8.1 Summary

Idea.

1. Expectation of a random matrix is taken entry by entry, and it commutes with constant linear maps, transposes and traces ([Proposition 2.1.1](#)).
2. $\text{Cov}(\mathbf{Y}) = E[(\mathbf{Y} - \boldsymbol{\mu})(\mathbf{Y} - \boldsymbol{\mu})^\top] = E(\mathbf{Y}\mathbf{Y}^\top) - \boldsymbol{\mu}\boldsymbol{\mu}^\top$, and $\text{Cov}(\mathbf{U}, \mathbf{V})$ is the rectangular cross-covariance, with $\text{Cov}(\mathbf{V}, \mathbf{U}) = \text{Cov}(\mathbf{U}, \mathbf{V})^\top$ ([Definition 2.1.2](#) and [Proposition 2.1.2](#)). Uncorrelated does not imply independent ([Example \(A point on a circle\)](#)).
3. Affine maps: $E(\mathbf{A}\mathbf{Y} + \mathbf{b}) = \mathbf{A}\boldsymbol{\mu} + \mathbf{b}$, $\text{Cov}(\mathbf{A}\mathbf{Y} + \mathbf{b}) = \mathbf{A}\boldsymbol{\Sigma}\mathbf{A}^\top$, $\text{Cov}(\mathbf{A}\mathbf{U}, \mathbf{B}\mathbf{V}) = \mathbf{A} \text{Cov}(\mathbf{U}, \mathbf{V}) \mathbf{B}^\top$ ([Theorem 2.2.1](#)). In particular $\text{Var}(\mathbf{a}^\top \mathbf{Y}) = \mathbf{a}^\top \boldsymbol{\Sigma} \mathbf{a}$.
4. Covariance matrices are exactly the nonnegative definite matrices. $\boldsymbol{\Sigma} \mathbf{a} = \mathbf{0}$ iff $\mathbf{a}^\top \mathbf{Y}$ is constant with probability one, and $\mathbf{Y}$ lives in the flat $\boldsymbol{\mu} + \mathcal{C}(\boldsymbol{\Sigma})$ ([Theorem 2.2.3](#)).
5. Whitening by $\mathbf{W}$ with $\mathbf{W}\boldsymbol{\Sigma}\mathbf{W}^\top = \mathbf{I}$ is unique up to an orthogonal factor. Ev-

- ery choice gives the same Mahalanobis distance $(\mathbf{Y} - \boldsymbol{\mu})^\top \boldsymbol{\Sigma}^{-1}(\mathbf{Y} - \boldsymbol{\mu})$ (Proposition 2.2.4).
6. $E(\mathbf{Y}^\top \mathbf{A} \mathbf{Y}) = \text{tr}(\mathbf{A} \boldsymbol{\Sigma}) + \boldsymbol{\mu}^\top \mathbf{A} \boldsymbol{\mu}$ under second moments only (Theorem 2.3.1). Hence $E(\text{RSS}) = \sigma^2(n - r) + \|(\mathbf{I} - \mathbf{M})\boldsymbol{\theta}\|^2$ (Example (Residual sum of squares under a wrong mean)). The variance of a quadratic form needs third and fourth moments (Theorem 2.3.3).
 7. Correlation matrices are the nonnegative definite matrices with unit diagonal. They are invariant to rescaling, and their entries must be jointly consistent (Proposition 2.4.1 and Proposition 2.4.2).
 8. For partitioned vectors, $\mathcal{C}(\boldsymbol{\Sigma}_{XY}) \subseteq \mathcal{C}(\boldsymbol{\Sigma}_{XX})$. The best linear predictor $\boldsymbol{\mu}_Y + \boldsymbol{\Sigma}_{YX} \boldsymbol{\Sigma}_{XX}^{-1}(\mathbf{X} - \boldsymbol{\mu}_X)$ leaves an error uncorrelated with $\mathbf{X}$ with covariance matrix given by the Schur complement $\boldsymbol{\Sigma}_{YY \cdot X}$ (Theorem 2.5.1 and Theorem 2.5.2).
 9. Moment generating functions determine distributions and factor exactly under independence (Theorem 2.6.1). Pairwise independence is weaker than mutual independence (Example (Pairwise but not mutually independent)).
 10. The sample covariance matrix is unbiased under uncorrelated rows and has rank at most $n - 1$. A tiny eigenvalue of the sample correlation matrix flags a nearly constant combination (Proposition 2.7.1 and Example (Macroeconomic series)).

2.8.2 Notes and sources

The operator calculus. Treating expectation and covariance as operators on random vectors and matrices is standard in linear-model texts. Rencher and Schaalje (2008, chapter 3) develops mean vectors, covariance and correlation matrices, partitioned vectors and linear functions in detail and at a gentle pace. Seber and Lee (2003, §§1.4–1.6) give a compact operator treatment, first and second moments of quadratic forms, and moment generating functions with their use in proving independence. This chapter covers the same ground. Rencher and Schaalje already introduce the standardized distance and the transformation by $\boldsymbol{\Sigma}^{-1/2}$. The chapter adds the characterization of singular covariance matrices through their column space (Theorem 2.2.3), the non-uniqueness of whitening and the comparison of symmetric and Cholesky whitening, the best linear predictor with generalized inverses, and the sample covariance matrix. Christensen (2020, chapter 1) states the covariance rules briefly in the notation of linear models. For the multivariate setting, Anderson (2003) remains the standard reference.

Mahalanobis distance and whitening. The distance Eq. 2.2.2 is due to Mahalanobis (1936), who introduced it to compare populations described by several correlated measurements. Kessy et al. (2018) compare whitening transformations. They show that the symmetric choice $\boldsymbol{\Sigma}^{-1/2}$ is the one closest to the original data in mean squared error (Exercise (B3)), and set it against Cholesky and principal-component whitening.

Quadratic forms. The trace argument for Theorem 2.3.1 is folklore. Theorem 2.3.3 is stated without proof by Atiqullah (1962), who used it to study how non-normal errors affect the residual variance estimate and the F -test. The kurtosis term depends on $\mathbf{A}$ only through its diagonal. For the residual sum of squares of a correct model it equals $(\mu_4 - 3\sigma^4) \sum_i (1 - h_{ii})^2$, which takes its simplest form, $(\mu_4 - 3\sigma^4)(n - r)^2/n$, when all leverages h_{ii} are equal. These are the quadratically balanced designs of Atiqullah's title. Seber and Lee (2003, theorem 1.6) give a proof by direct expansion. The proof here organizes the same expansion by splitting the form into quadratic, linear and constant parts. Mathai and Provost (1992) is a book-length treatment of moments and distributions of quadratic forms, including general covariance structures.

Correlation, partial covariance and prediction. The feasible range in [Proposition 2.4.2](#) is a special case of the fact that a symmetric matrix is nonnegative definite iff a Schur complement is, when the leading block is positive definite. Horn and Johnson (2013) give the general Schur-complement and Loewner-order results used in [Theorem 2.5.2](#). Stating the best linear predictor with a generalized inverse, so that nothing needs Σ_{XX} to be nonsingular, is in the spirit of Rao (1973), who uses generalized inverses throughout.

Probability background. Uniqueness of moment generating functions and characteristic functions, and the device of [Proposition 2.6.2](#), are proved in Billingsley (1995). The Longley data come from Longley (1967), who used them to test the numerical accuracy of regression programs. Their near-collinearity, visible in [Example \(Macroeconomic series\)](#), is the reason.

2.8.3 References

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Chapter 3

The Multivariate Normal Distribution

A random vector is normal when every linear combination of its entries is normal. From that definition come the moment generating function, the density, and closure under all linear maps, singular ones included. Two further properties make the normal distribution the working assumption of linear-model theory: zero covariance between jointly normal blocks means independence, and conditioning one block on another gives a linear regression with constant error variance. Partial correlation, the correlation that remains after linear adjustment, connects these facts to least squares.

What you need. [Chapter 1](#) (spectral theorem, square roots, generalized and partitioned inverses), [Chapter 2](#) (mean vectors and covariance matrices). Some examples refer forward to least squares ([Chapter 5](#) and [Chapter 6](#)).

3.0.1 Roadmap

- [Definition through linear combinations](#)
- [Moment generating function and density](#)
- [Linear transformations and marginals](#)
- [Independence and zero covariance](#)
- [Conditional distributions](#)
- [Partial correlation](#)
- [Summary and notes](#)

3.1 Definition through linear combinations

Nearly every exact distributional statement in the first half of this book has the same form. The response is $\mathbf{Y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}$ with normal errors, and the object of interest is a linear function of $\mathbf{Y}$ (a coefficient estimate, a fitted value, a residual) or a quadratic function of $\mathbf{Y}$ (a sum of squares). A usable notion of normality for random vectors must therefore do two things. It must survive every linear map, including maps that collapse dimensions. And it must say when two linear functions of the same vector are independent.

The residual vector shows why the first requirement has to include singular cases. In a model with p linearly independent columns its covariance matrix has rank $n - p$ (Chapter 6), so the residuals have no density on $\mathbb{R}^n$. Defining multivariate normality through a density would exclude it. This chapter uses a definition that does not mention densities at all: a random vector is normal when every linear combination of its entries is normal. The density, the moment generating function and the representation $\boldsymbol{\mu} + \mathbf{A}\mathbf{Z}$ all follow as theorems.

3.1.1 The univariate normal, including the degenerate case

For $\mu \in \mathbb{R}$ and $\sigma^2 > 0$, a random variable X has the $N(\mu, \sigma^2)$ distribution when it has density $(2\pi\sigma^2)^{-1/2} \exp\{-(x - \mu)^2/(2\sigma^2)\}$. We also allow $\sigma^2 = 0$, and take $N(\mu, 0)$ to be the point mass at μ . Treating constants as normal variables with zero variance costs nothing and removes a great many special cases later.

Lemma 3.1.1

Let $X \sim N(\mu, \sigma^2)$ with $\sigma^2 \geq 0$.

- (a) $E e^{sX} = \exp(\mu s + \frac{1}{2}\sigma^2 s^2)$ for every real s .
- (b) A random variable whose moment generating function is $\exp(\mu s + \frac{1}{2}\sigma^2 s^2)$ on an interval around 0 has the $N(\mu, \sigma^2)$ distribution.
- (c) $aX + b \sim N(a\mu + b, a^2\sigma^2)$, and if $X_1 \sim N(\mu_1, \sigma_1^2)$ and $X_2 \sim N(\mu_2, \sigma_2^2)$ are independent, then $X_1 + X_2 \sim N(\mu_1 + \mu_2, \sigma_1^2 + \sigma_2^2)$.

Proof.

- (a) For $\sigma^2 = 0$ this is $e^{\mu s}$. Otherwise, completing the square, $sx - (x - \mu)^2/(2\sigma^2) = \mu s + \sigma^2 s^2/2 - (x - \mu - \sigma^2 s)^2/(2\sigma^2)$, and the remaining Gaussian integral equals $(2\pi\sigma^2)^{1/2}$.
- (b) is the uniqueness theorem for moment generating functions (Theorem 2.6.1(a)), in its scalar form.
- (c) The moment generating function of $aX + b$ is $e^{bs} E e^{(as)X} = \exp\{(a\mu + b)s + \frac{1}{2}a^2\sigma^2 s^2\}$. For the sum, independence gives $E e^{s(X_1 + X_2)} = E e^{sX_1} E e^{sX_2}$, which is the normal moment generating function with the added parameters. Apply (b). ■

3.1.2 The definition

Definition 3.1.1 — Multivariate normal distribution

A random vector $\mathbf{Y} = (Y_1, \dots, Y_n)^\top$ has a **multivariate normal distribution** if $\mathbf{a}^\top \mathbf{Y}$ has a univariate normal distribution (possibly degenerate) for every $\mathbf{a} \in \mathbb{R}^n$. Such a $\mathbf{Y}$ is also said to be **jointly normal**, and its entries are said to be jointly normal random

variables.

Taking $\mathbf{a}$ to be a coordinate vector shows that each Y_i is normal, so $EY_i^2 < \infty$. By the Cauchy–Schwarz inequality every product $Y_i Y_j$ is integrable, so the mean vector $\boldsymbol{\mu} = E\mathbf{Y}$ and the covariance matrix $\boldsymbol{\Sigma} = \text{Cov}(\mathbf{Y})$ of Definition 2.1.2 exist. By Theorem 2.2.1, $E(\mathbf{a}^\top \mathbf{Y}) = \mathbf{a}^\top \boldsymbol{\mu}$ and $\text{Var}(\mathbf{a}^\top \mathbf{Y}) = \mathbf{a}^\top \boldsymbol{\Sigma} \mathbf{a}$, and a normal variable is determined by its mean and variance. Hence

$$\mathbf{a}^\top \mathbf{Y} \sim N(\mathbf{a}^\top \boldsymbol{\mu}, \mathbf{a}^\top \boldsymbol{\Sigma} \mathbf{a}) \quad \text{for every } \mathbf{a} \in \mathbb{R}^n. \quad (3.1.1)$$

We write $\mathbf{Y} \sim N_n(\boldsymbol{\mu}, \boldsymbol{\Sigma})$. The notation claims that the joint distribution of $\mathbf{Y}$ depends only on $\boldsymbol{\mu}$ and $\boldsymbol{\Sigma}$. That is true, but it is not obvious: the definition fixes the law of each one-dimensional projection, and it is not yet clear that these laws pin down the joint law. The claim is proved in Theorem 3.2.1. Until then, $N_n(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ should be read as “normal, with mean $\boldsymbol{\mu}$ and covariance $\boldsymbol{\Sigma}$ ”.

3.1.3 Existence

Theorem 3.1.2 — Existence and the representation $\boldsymbol{\mu} + \mathbf{AZ}$

Let $Z_1, \dots, Z_m$ be independent $N(0, 1)$ variables and $\mathbf{Z} = (Z_1, \dots, Z_m)^\top$.

- (a) $\mathbf{Z} \sim N_m(\mathbf{0}, \mathbf{I}_m)$.
- (b) For any $n \times m$ matrix $\mathbf{A}$ and $\boldsymbol{\mu} \in \mathbb{R}^n$, $\mathbf{Y} = \boldsymbol{\mu} + \mathbf{AZ}$ is multivariate normal with mean $\boldsymbol{\mu}$ and covariance $\mathbf{AA}^\top$.
- (c) For every $\boldsymbol{\mu} \in \mathbb{R}^n$ and every nonnegative definite $n \times n$ matrix $\boldsymbol{\Sigma}$ there is a random vector with the $N_n(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ distribution, for example $\boldsymbol{\mu} + \boldsymbol{\Sigma}^{1/2} \mathbf{Z}$ with $m = n$.

No other matrices occur: the covariance matrix of a normal vector is nonnegative definite.

Proof.

- (a) $\mathbf{a}^\top \mathbf{Z} = \sum_i a_i Z_i$ is a sum of independent normal variables $a_i Z_i \sim N(0, a_i^2)$, which is normal by Lemma 3.1.1(c).
- (b) $\mathbf{a}^\top \mathbf{Y} = \mathbf{a}^\top \boldsymbol{\mu} + (\mathbf{A}^\top \mathbf{a})^\top \mathbf{Z}$ is a constant plus a linear combination of $\mathbf{Z}$, hence normal by (a). Its moments come from Theorem 2.2.1: $E\mathbf{Y} = \boldsymbol{\mu}$ and $\text{Cov}(\mathbf{Y}) = \mathbf{AIA}^\top = \mathbf{AA}^\top$.
- (c) By Theorem 1.7.6 a nonnegative definite $\boldsymbol{\Sigma}$ has a symmetric nonnegative definite square root with $\boldsymbol{\Sigma}^{1/2} \boldsymbol{\Sigma}^{1/2} = \boldsymbol{\Sigma}$. Take $\mathbf{A} = \boldsymbol{\Sigma}^{1/2}$ in (b). The last sentence is Theorem 2.2.3.

■

Any factorization $\boldsymbol{\Sigma} = \mathbf{AA}^\top$ serves in (b). The symmetric square root is convenient in proofs. The Cholesky factor, a lower triangular $\mathbf{L}$ with $\mathbf{LL}^\top = \boldsymbol{\Sigma}$, is cheaper to compute and is what software uses. A factor with fewer columns than rows is natural when $\boldsymbol{\Sigma}$ is singular. Once Theorem 3.2.1 is proved, all of these give the same distribution. The first listing checks this by simulation. It draws 400,000 vectors from a three-dimensional normal distribution with each of

two factors. Both sample covariance matrices agree with Σ to within 0.016 in every entry, and the quantiles of a linear combination match the exact normal quantiles.

```
import numpy as np
from scipy import stats
rng = np.random.default_rng(301)
mu = np.array([1.0, -2.0, 0.5])
Sigma = np.array([[4.0, 1.2, -0.8],
                  [1.2, 2.0, 0.6],
                  [-0.8, 0.6, 1.5]])
N = 400_000

def draw_normal(mu, A, size, rng):
    """Rows are independent copies of mu + A Z with Z ~ N(0, I)."""
    Z = rng.standard_normal((size, A.shape[1]))
    return mu + Z @ A.T

L = np.linalg.cholesky(Sigma)                # lower triangular, L L^T = Sigma
lam, U = np.linalg.eigh(Sigma)
root = U @ np.diag(np.sqrt(lam)) @ U.T       # symmetric square root

Y_chol = draw_normal(mu, L, N, rng)
Y_root = draw_normal(mu, root, N, rng)
for Y in (Y_chol, Y_root):
    print(np.round(np.cov(Y, rowvar=False) - Sigma, 3))
```

3.1.4 Singular normal distributions

When Σ is singular, [Definition 3.1.1](#) still applies, and the distribution is called **singular** or degenerate. If $\Sigma \mathbf{v} = \mathbf{0}$, then $\text{Var}(\mathbf{v}^\top \mathbf{Y}) = \mathbf{v}^\top \Sigma \mathbf{v} = 0$, so $\mathbf{v}^\top \mathbf{Y} = \mathbf{v}^\top \boldsymbol{\mu}$ with probability one. Every vector in $\mathcal{N}(\Sigma)$ gives such a linear constraint, and together they say that $\mathbf{Y}$ lies in the affine subspace $\boldsymbol{\mu} + \mathcal{C}(\Sigma)$ with probability one ([Theorem 2.2.3](#)). A subspace of dimension $\text{rank}(\Sigma) < n$ has Lebesgue measure zero, so no density on $\mathbb{R}^n$ is possible.

Example. (Three overlapping sums)

Let $\mathbf{Z} \sim N_3(\mathbf{0}, \mathbf{I})$ and $\mathbf{U} = (Z_1 + Z_2, Z_2 - Z_3, Z_1 + Z_3)^\top$. Then $\mathbf{U} = \mathbf{A}\mathbf{Z}$ with rows $(1, 1, 0)$, $(0, 1, -1)$ and $(1, 0, 1)$, so $\mathbf{U}$ is normal with mean zero and covariance

$$\mathbf{A}\mathbf{A}^\top = \begin{pmatrix} 2 & 1 & 1 \\ 1 & 2 & -1 \\ 1 & -1 & 2 \end{pmatrix}.$$

The vector $(1, -1, -1)^\top$ lies in $\mathcal{N}(\mathbf{A}\mathbf{A}^\top)$, and the rank is 2. Correspondingly $U_1 - U_2 - U_3 = 0$ for every outcome, not merely with probability one. Each U_i is $N(0, 2)$, each pair (U_i, U_j) has a nonsingular covariance matrix (determinant 3) and a bivariate density, and yet the triple lives on a plane.

In regression the singular case is the rule. With $\mathbf{Y} \sim N_n(\mathbf{X}\boldsymbol{\beta}, \sigma^2\mathbf{I})$, the fitted values and the residuals are normal vectors of ranks $\text{rank}(\mathbf{X})$ and $n - \text{rank}(\mathbf{X})$ ([Example \(Least squares under normal errors\)](#)). The last part of the listing draws a normal vector in $\mathbb{R}^3$ from a 3×2 factor. Every draw lies on the plane $\boldsymbol{\mu} + \mathcal{C}(\mathbf{A})$ to within 2.7×10^{-15} , which is rounding error.

3.1.5 Why linear combinations suffice

Definition 3.1.1 describes a joint distribution entirely through its one-dimensional projections. That this loses no information is a general fact about random vectors, the Cramér–Wold device proved with characteristic functions in [Proposition 2.6.2](#): if $\mathbf{a}^\top \mathbf{Y}$ and $\mathbf{a}^\top \mathbf{W}$ have the same distribution for every $\mathbf{a} \in \mathbb{R}^n$, then $\mathbf{Y}$ and $\mathbf{W}$ have the same distribution. We do not need it in this generality. For normal vectors the moment generating function is finite everywhere, and the next section runs the same argument with it. The device explains why the definition is natural. It also warns that all directions $\mathbf{a}$ matter. Normality of the n coordinate variables, which are only n directions, is not enough ([Section 3.3](#) and [Exercise \(B3\)](#)).

3.1.6 Exercises

3.1.6.1 A. Check your understanding

Exercise A1.

Let $Y_1, \dots, Y_n$ be independent with $Y_i \sim N(\mu_i, \sigma_i^2)$. Show directly from [Definition 3.1.1](#) that $\mathbf{Y} = (Y_1, \dots, Y_n)^\top$ is multivariate normal, and find its mean and covariance. Show also that a constant vector $\mathbf{c}$ is $N_n(\mathbf{c}, \mathbf{0})$.

Exercise A2.

For the bivariate normal of [Example \(The bivariate normal\)](#), find the Cholesky factor $\mathbf{L}$ of Σ and write Y_1 and Y_2 in terms of two independent standard normals. Do the same with the symmetric square root when $\sigma_1 = \sigma_2 = 1$. Why do the two representations define the same distribution?

3.1.6.2 B. Practice

Exercise B1.

Let $Z \sim N(0, 1)$ and $\mathbf{Y} = (Z, 2Z - 1, 3 - Z)^\top$. Show that $\mathbf{Y}$ is normal, find $\boldsymbol{\mu}$ and Σ , and find the rank of Σ . Describe the set in which $\mathbf{Y}$ lies, and write it as the solution set of two linear equations.

Solution. $\mathbf{Y} = \boldsymbol{\mu} + \mathbf{A}Z$ with $\boldsymbol{\mu} = (0, -1, 3)^\top$ and $\mathbf{A} = (1, 2, -1)^\top$, so $\mathbf{Y}$ is normal by [Theorem 3.1.2](#) with

$$\Sigma = \mathbf{A}\mathbf{A}^\top = \begin{pmatrix} 1 & 2 & -1 \\ 2 & 4 & -2 \\ -1 & -2 & 1 \end{pmatrix}, \quad \text{rank}(\Sigma) = 1.$$

$\mathbf{Y}$ lies on the line $\{\boldsymbol{\mu} + s\mathbf{A} : s \in \mathbb{R}\}$. The null space of Σ is $\mathbf{A}^\perp$, spanned by $(2, -1, 0)^\top$ and $(1, 0, 1)^\top$. The corresponding constraints $\mathbf{v}^\top \mathbf{Y} = \mathbf{v}^\top \boldsymbol{\mu}$ are $2Y_1 - Y_2 = 1$ and $Y_1 + Y_3 = 3$, and both hold identically.

3.2 Moment generating function and density

3.2.1 Moment generating functions of random vectors

The moment generating function of a random vector $\mathbf{Y} \in \mathbb{R}^n$ is $M_{\mathbf{Y}}(\mathbf{t}) = E \exp(\mathbf{t}^\top \mathbf{Y})$, and [Theorem 2.6.1](#) proves the two properties we need when it is finite on a neighbourhood of $\mathbf{0}$. *Uniqueness*: two random vectors whose moment generating functions are finite and equal near $\mathbf{0}$ have the same distribution. *Factorization*: two blocks $\mathbf{Y}_1, \mathbf{Y}_2$ are independent iff $M_{\mathbf{Y}}(\mathbf{t}) = M_{\mathbf{Y}_1}(\mathbf{t}_1)M_{\mathbf{Y}_2}(\mathbf{t}_2)$ near $\mathbf{0}$. The second extends to k blocks by induction. If $M_{\mathbf{Y}}$ is the product of the k block functions, setting $\mathbf{t}_1 = \mathbf{0}$ shows that $(\mathbf{Y}_2, \dots, \mathbf{Y}_k)$ has the product of the last $k-1$, so $\mathbf{Y}_1$ is independent of $(\mathbf{Y}_2, \dots, \mathbf{Y}_k)$, whose blocks are mutually independent by the induction hypothesis. The converse is immediate.

Theorem 3.2.1 — Moment generating function

A random vector $\mathbf{Y}$ is $N_n(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ iff

$$M_{\mathbf{Y}}(\mathbf{t}) = \exp(\mathbf{t}^\top \boldsymbol{\mu} + \frac{1}{2} \mathbf{t}^\top \boldsymbol{\Sigma} \mathbf{t}) \quad \text{for every } \mathbf{t} \in \mathbb{R}^n. \quad (3.2.1)$$

Consequently the distribution of a normal vector is determined by its mean vector and covariance matrix.

Proof. Suppose $\mathbf{Y}$ is normal with mean $\boldsymbol{\mu}$ and covariance $\boldsymbol{\Sigma}$. By [Eq. 3.1.1](#), $\mathbf{t}^\top \mathbf{Y} \sim N(\mathbf{t}^\top \boldsymbol{\mu}, \mathbf{t}^\top \boldsymbol{\Sigma} \mathbf{t})$, and $M_{\mathbf{Y}}(\mathbf{t})$ is the moment generating function of this scalar evaluated at $s = 1$. [Lemma 3.1.1\(a\)](#) gives [Eq. 3.2.1](#).

Conversely, suppose [Eq. 3.2.1](#) holds for a symmetric $\boldsymbol{\Sigma}$. Fix $\mathbf{a}$. The moment generating function of $\mathbf{a}^\top \mathbf{Y}$ at s is $M_{\mathbf{Y}}(s\mathbf{a}) = \exp\{(\mathbf{a}^\top \boldsymbol{\mu})s + \frac{1}{2}(\mathbf{a}^\top \boldsymbol{\Sigma} \mathbf{a})s^2\}$. The coefficient $\mathbf{a}^\top \boldsymbol{\Sigma} \mathbf{a}$ is nonnegative, since otherwise this function would tend to zero as $s \rightarrow \pm\infty$, whereas every random variable X has $E e^{sX} + E e^{-sX} \geq E e^{|sX|} \geq 1$. By [Lemma 3.1.1\(b\)](#), $\mathbf{a}^\top \mathbf{Y} \sim N(\mathbf{a}^\top \boldsymbol{\mu}, \mathbf{a}^\top \boldsymbol{\Sigma} \mathbf{a})$. So $\mathbf{Y}$ is normal, and matching means and variances of all linear combinations shows $E \mathbf{Y} = \boldsymbol{\mu}$ and $\text{Cov}(\mathbf{Y}) = \boldsymbol{\Sigma}$.

Finally, two normal vectors with the same $\boldsymbol{\mu}$ and $\boldsymbol{\Sigma}$ have the same moment generating function [Eq. 3.2.1](#), which is finite everywhere, so they have the same distribution by the uniqueness property. ■

The last sentence is what makes $N_n(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ a well-defined distribution. Combined with [Theorem 3.1.2](#), it says that $\mathbf{Y} \sim N_n(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ holds iff $\mathbf{Y}$ has the same distribution as $\boldsymbol{\mu} + \mathbf{A}\mathbf{Z}$ for some, and then every, matrix $\mathbf{A}$ with $\mathbf{A}\mathbf{A}^\top = \boldsymbol{\Sigma}$. The number of columns of $\mathbf{A}$, and so the length of $\mathbf{Z}$, does not matter. Some books take this representation as the definition ([Christensen 2020](#); [Seber and Lee 2003](#)). The proof above shows the two definitions agree.

The moment generating function also encodes moments. Differentiating [Eq. 3.2.1](#) at $\mathbf{t} = \mathbf{0}$ gives $\partial M_{\mathbf{Y}}/\partial \mathbf{t} = \boldsymbol{\mu}$ and $\partial^2 M_{\mathbf{Y}}/\partial \mathbf{t} \partial \mathbf{t}^\top = \boldsymbol{\Sigma} + \boldsymbol{\mu}\boldsymbol{\mu}^\top = \mathbf{E}(\mathbf{Y}\mathbf{Y}^\top)$. Higher derivatives give the fourth moments needed for variances of quadratic forms in [Chapter 4](#) ([Exercise \(B1\)](#)).

3.2.2 The density

Theorem 3.2.2 — Density

Let $\mathbf{Y} \sim N_n(\boldsymbol{\mu}, \boldsymbol{\Sigma})$.

- (a) If $\boldsymbol{\Sigma}$ is positive definite, $\mathbf{Y}$ has the density

$$f(\mathbf{y}) = (2\pi)^{-n/2} \det(\boldsymbol{\Sigma})^{-1/2} \exp\left\{-\frac{1}{2}(\mathbf{y} - \boldsymbol{\mu})^\top \boldsymbol{\Sigma}^{-1}(\mathbf{y} - \boldsymbol{\mu})\right\}, \quad \mathbf{y} \in \mathbb{R}^n. \quad (3.2.2)$$

- (b) Conversely, a random vector with density [Eq. 3.2.2](#), for some positive definite $\boldsymbol{\Sigma}$, is $N_n(\boldsymbol{\mu}, \boldsymbol{\Sigma})$.
- (c) If $\boldsymbol{\Sigma}$ is singular, $\mathbf{Y}$ has no density on $\mathbb{R}^n$.

Proof.

- (a) By [Theorem 3.2.1](#), $\mathbf{Y}$ has the same distribution as $\boldsymbol{\mu} + \boldsymbol{\Sigma}^{1/2}\mathbf{Z}$ with $\mathbf{Z} \sim N_n(\mathbf{0}, \mathbf{I})$, so it suffices to find the density of the latter. The entries of $\mathbf{Z}$ are independent standard normals, so $\mathbf{Z}$ has density $\prod_i (2\pi)^{-1/2} e^{-z_i^2/2} = (2\pi)^{-n/2} \exp(-\frac{1}{2}\mathbf{z}^\top \mathbf{z})$. Since $\boldsymbol{\Sigma}$ is positive definite, so is $\boldsymbol{\Sigma}^{1/2}$ (its eigenvalues are the positive square roots of those of $\boldsymbol{\Sigma}$), and the map $\mathbf{z} \mapsto \mathbf{y} = \boldsymbol{\mu} + \boldsymbol{\Sigma}^{1/2}\mathbf{z}$ is an affine bijection of $\mathbb{R}^n$ with inverse $\mathbf{z} = \boldsymbol{\Sigma}^{-1/2}(\mathbf{y} - \boldsymbol{\mu})$. The change-of-variables formula multiplies the density of $\mathbf{Z}$ at $\mathbf{z}(\mathbf{y})$ by $|\det \boldsymbol{\Sigma}^{-1/2}| = \det(\boldsymbol{\Sigma})^{-1/2}$. Finally $\mathbf{z}^\top \mathbf{z} = (\mathbf{y} - \boldsymbol{\mu})^\top \boldsymbol{\Sigma}^{-1/2} \boldsymbol{\Sigma}^{-1/2}(\mathbf{y} - \boldsymbol{\mu}) = (\mathbf{y} - \boldsymbol{\mu})^\top \boldsymbol{\Sigma}^{-1}(\mathbf{y} - \boldsymbol{\mu})$.
- (b) A density determines a distribution, and by (a) the $N_n(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ distribution has this density.
- (c) $\mathbf{Y}$ lies with probability one in $\boldsymbol{\mu} + \mathcal{C}(\boldsymbol{\Sigma})$, an affine subspace of dimension less than n . It has Lebesgue measure zero, so it would receive probability zero under any distribution with a density. ■

A singular normal vector does have a density with respect to the natural measure on the affine subspace where it lives ([Proposition 3.3.4](#) and [Exercise \(B2\)](#)), but that is rarely needed. In this book, arguments are made with the moment generating function or with linear transformations, which work in every case, and densities are used only for pictures and likelihoods.

Example. (The bivariate normal)

For $n = 2$, write $\sigma_1^2, \sigma_2^2 > 0$ for the variances and ρ for the correlation. Then

$$\boldsymbol{\Sigma} = \begin{pmatrix} \sigma_1^2 & \rho\sigma_1\sigma_2 \\ \rho\sigma_1\sigma_2 & \sigma_2^2 \end{pmatrix}, \quad \det \boldsymbol{\Sigma} = \sigma_1^2\sigma_2^2(1 - \rho^2),$$

so Σ is positive definite iff $|\rho| < 1$. With standardized coordinates $u_i = (y_i - \mu_i)/\sigma_i$, the density Eq. 3.2.2 is

$$f(y_1, y_2) = \frac{1}{2\pi\sigma_1\sigma_2\sqrt{1-\rho^2}} \exp\left\{-\frac{u_1^2 - 2\rho u_1 u_2 + u_2^2}{2(1-\rho^2)}\right\}.$$

When $\rho = 0$ it factors into the two univariate normal densities. When $\rho = \pm 1$ there is no density: $\text{Var}(u_2 \mp u_1) = 0$ in standardized units, so all the probability lies on a line.

3.2.3 Contours and the Mahalanobis distance

For positive definite Σ , the density Eq. 3.2.2 depends on $\mathbf{y}$ only through

$$\Delta^2(\mathbf{y}) = (\mathbf{y} - \boldsymbol{\mu})^\top \Sigma^{-1}(\mathbf{y} - \boldsymbol{\mu}),$$

the squared Mahalanobis distance from $\mathbf{y}$ to $\boldsymbol{\mu}$ introduced in Section 2.2. The density is a decreasing function of Δ^2 , so its contours are the ellipsoids $\{\mathbf{y} : \Delta^2(\mathbf{y}) = c^2\}$. Their shape comes from the spectral decomposition $\Sigma = \sum_i \lambda_i \mathbf{u}_i \mathbf{u}_i^\top$ (Theorem 1.7.2). In the coordinates $w_i = \mathbf{u}_i^\top (\mathbf{y} - \boldsymbol{\mu})$ we have $\Delta^2 = \sum_i w_i^2 / \lambda_i$, so the ellipsoid has principal axes along the eigenvectors $\mathbf{u}_i$, with half-lengths $c\sqrt{\lambda_i}$. Large-variance directions are long axes. The Mahalanobis distance measures distance from the centre in units of standard deviation along each principal direction. The volume of $\{\mathbf{y} : \Delta^2(\mathbf{y}) \leq c^2\}$ is proportional to $\prod_i \sqrt{\lambda_i} = \det(\Sigma)^{1/2}$, so the generalized variance $\det \Sigma$ (Section 2.1) measures the size of the region the distribution occupies. The factor $\det(\Sigma)^{-1/2}$ in Eq. 3.2.2 spreads the same total probability over that region: a larger generalized variance means a flatter density.

Proposition 3.2.3

If $\mathbf{Y} \sim N_n(\boldsymbol{\mu}, \Sigma)$ with Σ positive definite, then $\Delta^2(\mathbf{Y})$ has the chi-squared distribution with n degrees of freedom, $\chi^2(n)$, which is $\chi^2(n, 0)$ in the notation of Definition 4.1.1. So the ellipsoid $\{\mathbf{y} : \Delta^2(\mathbf{y}) \leq q\}$, where q is the $1 - \alpha$ quantile of $\chi^2(n)$, has probability exactly $1 - \alpha$.

Proof. Put $\mathbf{Z} = \Sigma^{-1/2}(\mathbf{Y} - \boldsymbol{\mu})$. For every $\mathbf{a}$, $\mathbf{a}^\top \mathbf{Z} = (\Sigma^{-1/2}\mathbf{a})^\top \mathbf{Y} - \mathbf{a}^\top \Sigma^{-1/2}\boldsymbol{\mu}$ is a linear combination of $\mathbf{Y}$ minus a constant, hence normal by Definition 3.1.1, so $\mathbf{Z}$ is normal. Its mean is zero and its covariance is $\mathbf{I}$ (Proposition 2.2.4(a)). By Theorem 3.2.1 its moment generating function is $\exp(\frac{1}{2}\mathbf{t}^\top \mathbf{t}) = \prod_i \exp(\frac{1}{2}t_i^2)$, which factorizes, so its entries are independent standard normals. $\Delta^2(\mathbf{Y}) = \mathbf{Z}^\top \mathbf{Z} = \sum_i Z_i^2$ is a sum of n independent squared standard normals, which is the definition of $\chi^2(n)$. ■

Chapter 4 generalizes this considerably, to $\mathbf{Y}^\top \mathbf{A} \mathbf{Y}$ with a nonzero mean and a general $\mathbf{A}$ (Theorem 4.3.2). Because the density is largest inside these ellipsoids, they are the smallest regions with their probability content, which is why they reappear as confidence regions for $\boldsymbol{\beta}$ in Chapter 12 (Theorem 12.2.1).

Computing with the density. Neither Σ^{-1} nor $\det \Sigma$ should be formed explicitly. With the Cholesky factorization $\Sigma = \mathbf{L}\mathbf{L}^\top$, we have $\Delta^2(\mathbf{y}) = \|\mathbf{L}^{-1}(\mathbf{y} - \boldsymbol{\mu})\|^2$, computed by a triangular solve, and $\log \det \Sigma = 2 \sum_i \log l_{ii}$. The listing evaluates the log density this way, checks it

against `scipy`, and counts how many of the simulated vectors fall inside the 95% ellipsoid. The fraction is 0.9500. The average of Δ^2 over the draws is 3.004 and its variance is 5.994, against the $\chi^2(3)$ values 3 and 6.

```
import numpy as np
from scipy import stats
rng = np.random.default_rng(301)
mu = np.array([1.0, -2.0, 0.5])
Sigma = np.array([[4.0, 1.2, -0.8],
                  [1.2, 2.0, 0.6],
                  [-0.8, 0.6, 1.5]])

N = 400_000
a = np.array([0.3, -1.0, 2.0])
q = [0.05, 0.25, 0.5, 0.75, 0.95]
exact = stats.norm.ppf(q, loc=a @ mu, scale=np.sqrt(a @ Sigma @ a))
cov_err = max(np.abs(np.cov(Y, rowvar=False) - Sigma).max() for Y in (Y_chol, Y_root))

def draw_normal(mu, A, size, rng):
    """Rows are independent copies of mu + A Z with Z ~ N(0, I)."""
    Z = rng.standard_normal((size, A.shape[1]))
    return mu + Z @ A.T

L = np.linalg.cholesky(Sigma)                # lower triangular, L L^T = Sigma
lam, U = np.linalg.eigh(Sigma)
root = U @ np.diag(np.sqrt(lam)) @ U.T       # symmetric square root

Y_chol = draw_normal(mu, L, N, rng)
Y_root = draw_normal(mu, root, N, rng)
for Y in (Y_chol, Y_root):
    print(np.round(np.cov(Y, rowvar=False) - Sigma, 3))

def normal_logpdf(y, mu, Sigma):
    """log density of N(mu, Sigma) at the rows of y, Sigma positive definite."""
    L = np.linalg.cholesky(Sigma)
    w = np.linalg.solve(L, (y - mu).T)       # w = L^{-1}(y - mu), so w^T w =
    Delta^2
    D2 = np.sum(w**2, axis=0)
    logdet = 2.0 * np.sum(np.log(np.diag(L)))
    return -0.5 * (len(mu) * np.log(2 * np.pi) + logdet + D2), D2

logf, D2 = normal_logpdf(Y_chol, mu, Sigma)
inside = np.mean(D2 <= stats.chi2.ppf(0.95, df=3))
print(f"fraction inside the 95% ellipsoid: {inside:.4f}")
```

3.2.4 Exercises

3.2.4.1 A. Check your understanding

Exercise A1.

Show that $f(y_1, y_2) = k \exp\{-\frac{1}{2}(2y_1^2 + 2y_1y_2 + y_2^2 - 4y_1 - 2y_2)\}$ is a bivariate normal density for a suitable constant k . Find k , μ , Σ and the correlation.

Solution. The quadratic part is $\mathbf{y}^\top \mathbf{P} \mathbf{y}$ with $\mathbf{P} = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix}$, which is positive definite with $\det \mathbf{P} = 1$. Matching $(\mathbf{y} - \boldsymbol{\mu})^\top \mathbf{P} (\mathbf{y} - \boldsymbol{\mu}) = \mathbf{y}^\top \mathbf{P} \mathbf{y} - 2\boldsymbol{\mu}^\top \mathbf{P} \mathbf{y} + \boldsymbol{\mu}^\top \mathbf{P} \boldsymbol{\mu}$ with the linear terms requires $\mathbf{P} \boldsymbol{\mu} = (2, 1)^\top$, so $\boldsymbol{\mu} = (1, 0)^\top$ and $\boldsymbol{\mu}^\top \mathbf{P} \boldsymbol{\mu} = 2$. The exponent is $-\frac{1}{2}(\mathbf{y} - \boldsymbol{\mu})^\top \mathbf{P} (\mathbf{y} - \boldsymbol{\mu}) + 1$. By [Theorem 3.2.2\(b\)](#) this is the $N_2(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ density with $\boldsymbol{\Sigma} = \mathbf{P}^{-1} = \begin{pmatrix} 1 & -1 \\ -1 & 2 \end{pmatrix}$ provided $ke = (2\pi)^{-1}(\det \boldsymbol{\Sigma})^{-1/2} = (2\pi)^{-1}$. So $k = e^{-1}/(2\pi)$, and the correlation is $-1/\sqrt{2}$.

Exercise A2.

Let $\boldsymbol{\Sigma}$ have unit variances and correlation ρ , with $|\rho| < 1$. Find the principal axes and half-lengths of the ellipse $\Delta^2 = c^2$, show that its area is $\pi c^2 \sqrt{1 - \rho^2}$, and show that it contains probability $1 - e^{-c^2/2}$.

3.2.4.2 B. Practice

Exercise B1.

Let $\mathbf{Y} \sim N_n(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ and $\mathbf{X} = \mathbf{Y} - \boldsymbol{\mu}$. Expand the moment generating function of $\mathbf{X}$ in a power series to show that all odd-order moments of $\mathbf{X}$ vanish and

$$E(X_i X_j X_k X_l) = \sigma_{ij} \sigma_{kl} + \sigma_{ik} \sigma_{jl} + \sigma_{il} \sigma_{jk}.$$

Deduce that $\text{Var}(\mathbf{X}^\top \mathbf{A} \mathbf{X}) = 2 \text{tr}(\mathbf{A} \boldsymbol{\Sigma} \mathbf{A} \boldsymbol{\Sigma})$ for symmetric $\mathbf{A}$.

Solution. By [Theorem 3.2.1](#), $M_{\mathbf{X}}(\mathbf{t}) = \exp(\frac{1}{2} \mathbf{t}^\top \boldsymbol{\Sigma} \mathbf{t}) = \sum_{m \geq 0} (\mathbf{t}^\top \boldsymbol{\Sigma} \mathbf{t})^m / (2^m m!)$, a series containing only terms of even degree in $\mathbf{t}$. Mixed moments of order d are the d th partial derivatives at $\mathbf{0}$, which come from the degree- d terms, so odd-order moments vanish. The degree-four term is $p(\mathbf{t}) = \frac{1}{8} (\mathbf{t}^\top \boldsymbol{\Sigma} \mathbf{t})^2 = T(\mathbf{t}, \mathbf{t}, \mathbf{t}, \mathbf{t})$, where $T(\mathbf{u}, \mathbf{v}, \mathbf{w}, \mathbf{x}) = \frac{1}{8} (\mathbf{u}^\top \boldsymbol{\Sigma} \mathbf{v})(\mathbf{w}^\top \boldsymbol{\Sigma} \mathbf{x})$ is 4-linear. Differentiating $T(\mathbf{t}, \mathbf{t}, \mathbf{t}, \mathbf{t})$ once in each of the directions $\mathbf{e}_i, \mathbf{e}_j, \mathbf{e}_k, \mathbf{e}_l$ gives the sum of T over the 24 ways of placing these four vectors in the four slots. This holds whether or not the indices are distinct, since it is the product rule applied to a multilinear form. A placement contributes $\frac{1}{8} \sigma_{ab} \sigma_{cd}$, where the first two slots receive the indices a, b and the last two receive c, d . Each of the three ways of splitting the four *directions* into two pairs, $\{ij, kl\}$, $\{ik, jl\}$, $\{il, jk\}$, arises from 8 placements (choose which pair goes first, and the order within each pair), which gives the formula. Then, for symmetric $\mathbf{A}$,

$$E(\mathbf{X}^\top \mathbf{A} \mathbf{X})^2 = \sum_{i,j,k,l} a_{ij} a_{kl} E(X_i X_j X_k X_l) = (\text{tr} \mathbf{A} \boldsymbol{\Sigma})^2 + 2 \text{tr}(\mathbf{A} \boldsymbol{\Sigma} \mathbf{A} \boldsymbol{\Sigma}),$$

and subtracting $(E \mathbf{X}^\top \mathbf{A} \mathbf{X})^2 = (\text{tr} \mathbf{A} \boldsymbol{\Sigma})^2$ gives the variance.

Exercise B2.

Let $\mathbf{Y} \sim N_n(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ with $\text{rank}(\boldsymbol{\Sigma}) = r < n$, and use the notation of [Proposition 3.3.4](#). Show that $\mathbf{V} = \mathbf{Q}^\top (\mathbf{Y} - \boldsymbol{\mu})$ has a density on $\mathbb{R}^r$, and that this density can be written as $(2\pi)^{-r/2} (\lambda_1 \cdots \lambda_r)^{-1/2} \exp\{-\frac{1}{2}(\mathbf{y} - \boldsymbol{\mu})^\top \boldsymbol{\Sigma}^+ (\mathbf{y} - \boldsymbol{\mu})\}$ with $\mathbf{y} = \boldsymbol{\mu} + \mathbf{Q} \mathbf{v}$, where $\boldsymbol{\Sigma}^+ = \mathbf{Q} \boldsymbol{\Lambda}^{-1} \mathbf{Q}^\top$ is the Moore–Penrose inverse.

Exercise B3.

Let $\mathbf{Y} \sim N_n(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ with $\boldsymbol{\Sigma}$ positive definite and let $\mathbf{v}$ be a fixed vector. Show that $(\mathbf{Y} - \mathbf{v})^\top \boldsymbol{\Sigma}^{-1}(\mathbf{Y} - \mathbf{v})$ has the noncentral chi-squared distribution $\chi^2(n, \gamma)$ with $\gamma = (\boldsymbol{\mu} - \mathbf{v})^\top \boldsymbol{\Sigma}^{-1}(\boldsymbol{\mu} - \mathbf{v})$, in the convention of [Definition 4.1.1](#), and find its mean.

3.2.4.3 C. Going deeper**Exercise C1.**

Let $\mathbf{Y}$ have density [Eq. 3.2.2](#), and let $E = \{\mathbf{y} : \Delta^2(\mathbf{y}) \leq q\}$. Show that every Borel set A with $\Pr(\mathbf{Y} \in A) \geq \Pr(\mathbf{Y} \in E)$ has volume at least that of E . *Hint:* compare $\int_{A \setminus E} f$ with $\int_{E \setminus A} f$.

Solution. The density is a decreasing function of Δ^2 , so with $c = (2\pi)^{-n/2} \det(\boldsymbol{\Sigma})^{-1/2} e^{-q/2} > 0$ we have $f \geq c$ on E and $f < c$ off E . We may assume A has finite volume. Subtracting $\Pr(\mathbf{Y} \in A \cap E)$ from both sides of $\Pr(\mathbf{Y} \in A) \geq \Pr(\mathbf{Y} \in E)$ gives $\int_{A \setminus E} f \geq \int_{E \setminus A} f$. Hence

$$c \operatorname{vol}(A \setminus E) \geq \int_{A \setminus E} f \geq \int_{E \setminus A} f \geq c \operatorname{vol}(E \setminus A),$$

so $\operatorname{vol}(A \setminus E) \geq \operatorname{vol}(E \setminus A)$. Adding $\operatorname{vol}(A \cap E)$ to both sides gives $\operatorname{vol}(A) \geq \operatorname{vol}(E)$. The argument uses only that E is a set on which f is at least as large as anywhere outside it, so it applies to the level sets of any density.

3.3 Linear transformations and marginals

3.3.1 Affine maps

Theorem 3.3.1 — Affine transformations

Let $\mathbf{Y} \sim N_n(\boldsymbol{\mu}, \boldsymbol{\Sigma})$, let $\mathbf{A}$ be any $m \times n$ matrix and $\mathbf{b} \in \mathbb{R}^m$. Then

$$\mathbf{A}\mathbf{Y} + \mathbf{b} \sim N_m(\mathbf{A}\boldsymbol{\mu} + \mathbf{b}, \mathbf{A}\boldsymbol{\Sigma}\mathbf{A}^\top).$$

Proof. For $\mathbf{c} \in \mathbb{R}^m$, $\mathbf{c}^\top(\mathbf{A}\mathbf{Y} + \mathbf{b}) = (\mathbf{A}^\top \mathbf{c})^\top \mathbf{Y} + \mathbf{c}^\top \mathbf{b}$ is a linear combination of $\mathbf{Y}$ plus a constant, which is normal by [Definition 3.1.1](#) and [Lemma 3.1.1\(c\)](#). So $\mathbf{A}\mathbf{Y} + \mathbf{b}$ is normal, and its mean and covariance are given by [Theorem 2.2.1](#). ■

The proof is one line because the definition was chosen to make it one line. There is no rank condition on $\mathbf{A}$. If $\mathbf{A}$ has more rows than $\operatorname{rank}(\boldsymbol{\Sigma})$, the result is a singular normal vector, and that is allowed. With a density-based definition, the same theorem needs $\mathbf{A}$ to have full row rank and $\boldsymbol{\Sigma}$ to be positive definite, and the most useful cases fall outside it. The moment generating function gives a second proof: $E e^{\mathbf{t}^\top(\mathbf{A}\mathbf{Y} + \mathbf{b})} = e^{\mathbf{t}^\top \mathbf{b}} M_{\mathbf{Y}}(\mathbf{A}^\top \mathbf{t}) = \exp\{\mathbf{t}^\top(\mathbf{A}\boldsymbol{\mu} + \mathbf{b}) + \frac{1}{2} \mathbf{t}^\top \mathbf{A} \boldsymbol{\Sigma} \mathbf{A}^\top \mathbf{t}\}.$

Corollary 3.3.2 — Marginal distributions

Every subvector of a normal vector is normal. If $\mathbf{Y} \sim N_n(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ and $\mathbf{Y}_1$ consists of the entries with indices in a set I , then $\mathbf{Y}_1 \sim N(\boldsymbol{\mu}_I, \boldsymbol{\Sigma}_{II})$, where $\boldsymbol{\mu}_I$ and $\boldsymbol{\Sigma}_{II}$ keep the entries, rows and columns indexed by I . In particular $Y_i \sim N(\mu_i, \sigma_{ii})$.

Proof. $\mathbf{Y}_1 = \mathbf{A}\mathbf{Y}$, where the rows of $\mathbf{A}$ are the coordinate vectors $\mathbf{e}_i^\top$, $i \in I$. Then $\mathbf{A}\boldsymbol{\mu} = \boldsymbol{\mu}_I$ and $\mathbf{A}\boldsymbol{\Sigma}\mathbf{A}^\top = \boldsymbol{\Sigma}_{II}$. ■

Corollary 3.3.3 — Standardization and rotation

Let $\mathbf{Y} \sim N_n(\boldsymbol{\mu}, \boldsymbol{\Sigma})$.

- (a) If $\boldsymbol{\Sigma}$ is positive definite and $\boldsymbol{\Sigma} = \mathbf{A}\mathbf{A}^\top$ with $\mathbf{A}$ square, then $\mathbf{A}^{-1}(\mathbf{Y} - \boldsymbol{\mu}) \sim N_n(\mathbf{0}, \mathbf{I})$. Its entries are independent standard normal variables.
- (b) If $\boldsymbol{\Sigma} = \sigma^2\mathbf{I}$ and $\mathbf{Q}$ is an $n \times n$ orthogonal matrix, then $\mathbf{Q}^\top\mathbf{Y} \sim N_n(\mathbf{Q}^\top\boldsymbol{\mu}, \sigma^2\mathbf{I})$.

Proof. Both follow from [Theorem 3.3.1](#): $\mathbf{A}^{-1}\mathbf{A}\mathbf{A}^\top(\mathbf{A}^{-1})^\top = \mathbf{I}$ and $\mathbf{Q}^\top(\sigma^2\mathbf{I})\mathbf{Q} = \sigma^2\mathbf{I}$. Independence in (a) was shown in the proof of [Proposition 3.2.3](#). ■

Part (b) is the probabilistic basis of the canonical form of the linear model. Spherical normal errors look the same in every orthonormal coordinate system, so one may rotate to coordinates adapted to $\mathcal{C}(\mathbf{X})$, in which fitted values and residuals occupy separate independent coordinates. [Chapter 4](#) uses this rotation to find the distributions of sums of squares.

3.3.2 The rank of a normal vector

[Theorem 3.1.2](#) writes a normal vector as an affine function of independent standard normals *in distribution*. When $\boldsymbol{\Sigma}$ is singular the representation can be made exact, with as few standard normals as the rank.

Proposition 3.3.4

Let $\mathbf{Y} \sim N_n(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ with $\text{rank}(\boldsymbol{\Sigma}) = r \geq 1$, and let $\boldsymbol{\Sigma} = \mathbf{Q}\boldsymbol{\Lambda}\mathbf{Q}^\top$, where $\mathbf{Q}$ is $n \times r$ with orthonormal columns and $\boldsymbol{\Lambda} = \text{diag}(\lambda_1, \dots, \lambda_r)$ holds the positive eigenvalues. Put $\mathbf{W} = \boldsymbol{\Lambda}^{-1/2}\mathbf{Q}^\top(\mathbf{Y} - \boldsymbol{\mu})$. Then $\mathbf{W} \sim N_r(\mathbf{0}, \mathbf{I}_r)$, and

$$\mathbf{Y} = \boldsymbol{\mu} + \mathbf{Q}\boldsymbol{\Lambda}^{1/2}\mathbf{W} \quad \text{with probability one.}$$

Proof. By [Theorem 3.3.1](#), $\mathbf{W}$ is normal with covariance $\boldsymbol{\Lambda}^{-1/2}\mathbf{Q}^\top\mathbf{Q}\boldsymbol{\Lambda}\mathbf{Q}^\top\mathbf{Q}\boldsymbol{\Lambda}^{-1/2} = \mathbf{I}_r$. Next, $\mathbf{Q}\boldsymbol{\Lambda}^{1/2}\mathbf{W} = \mathbf{Q}\mathbf{Q}^\top(\mathbf{Y} - \boldsymbol{\mu})$, and $\mathbf{Q}\mathbf{Q}^\top$ is the symmetric idempotent matrix with range $\mathcal{C}(\boldsymbol{\Sigma})$ (the orthogonal projection onto it, in the language of [Chapter 6](#)). Since

$\mathbf{Y} - \boldsymbol{\mu} \in \mathcal{C}(\boldsymbol{\Sigma})$ with probability one (Theorem 2.2.3), $\mathbf{Q}\mathbf{Q}^\top(\mathbf{Y} - \boldsymbol{\mu}) = \mathbf{Y} - \boldsymbol{\mu}$ with probability one. ■

So a normal vector of rank r is exactly r independent standard normals placed in $\mathbb{R}^n$ by an injective affine map. Exercise (B4) gives the same construction for any random vector, with uncorrelated entries in place of independent normal ones. Singular normal vectors are lower-dimensional normal vectors in disguise.

Example. (Least squares under normal errors)

Let $\mathbf{Y} \sim N_n(\mathbf{X}\boldsymbol{\beta}, \sigma^2\mathbf{I})$ with $\mathbf{X}$ of full column rank p , and let $\mathbf{M} = \mathbf{X}(\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{X}^\top$, the symmetric idempotent matrix of rank p with $\mathbf{M}\mathbf{X} = \mathbf{X}$ (Chapter 5 and Chapter 6). By Theorem 3.3.1:

- (a) $\hat{\boldsymbol{\beta}} = (\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{X}^\top\mathbf{Y} \sim N_p(\boldsymbol{\beta}, \sigma^2(\mathbf{X}^\top\mathbf{X})^{-1})$, a nonsingular normal vector;
- (b) $\boldsymbol{\lambda}^\top\hat{\boldsymbol{\beta}} \sim N(\boldsymbol{\lambda}^\top\boldsymbol{\beta}, \sigma^2\boldsymbol{\lambda}^\top(\mathbf{X}^\top\mathbf{X})^{-1}\boldsymbol{\lambda})$ for every $\boldsymbol{\lambda} \in \mathbb{R}^p$;
- (c) the fitted values $\hat{\mathbf{Y}} = \mathbf{M}\mathbf{Y} \sim N_n(\mathbf{X}\boldsymbol{\beta}, \sigma^2\mathbf{M})$, singular of rank p ;
- (d) the residuals $\hat{\boldsymbol{\varepsilon}} = (\mathbf{I} - \mathbf{M})\mathbf{Y} \sim N_n(\mathbf{0}, \sigma^2(\mathbf{I} - \mathbf{M}))$, singular of rank $n - p$.

No density for $\hat{\mathbf{Y}}$ or $\hat{\boldsymbol{\varepsilon}}$ was needed, and none exists.

3.3.3 Independent normal vectors

Lemma 3.3.5

If $\mathbf{Y}_1 \sim N_{n_1}(\boldsymbol{\mu}_1, \boldsymbol{\Sigma}_1)$ and $\mathbf{Y}_2 \sim N_{n_2}(\boldsymbol{\mu}_2, \boldsymbol{\Sigma}_2)$ are independent, then $(\mathbf{Y}_1^\top, \mathbf{Y}_2^\top)^\top$ is jointly normal with mean $(\boldsymbol{\mu}_1^\top, \boldsymbol{\mu}_2^\top)^\top$ and block-diagonal covariance $\text{diag}(\boldsymbol{\Sigma}_1, \boldsymbol{\Sigma}_2)$. Consequently, if also $n_1 = n_2$, then $\mathbf{Y}_1 + \mathbf{Y}_2 \sim N(\boldsymbol{\mu}_1 + \boldsymbol{\mu}_2, \boldsymbol{\Sigma}_1 + \boldsymbol{\Sigma}_2)$.

Proof. $\mathbf{a}_1^\top\mathbf{Y}_1 + \mathbf{a}_2^\top\mathbf{Y}_2$ is a sum of independent normal variables, which is normal by Lemma 3.1.1(c). Independence makes the cross-covariance zero. The sum is $[\mathbf{I}, \mathbf{I}]$ applied to the stacked vector. ■

Independence is essential in this lemma. Two normal vectors defined on the same probability space need not be jointly normal, as we now see.

3.3.4 Normal marginals are not enough

Corollary 3.3.2 cannot be reversed. A vector whose entries are each normal need not be jointly normal. Two examples make the point, one extreme and one realistic.

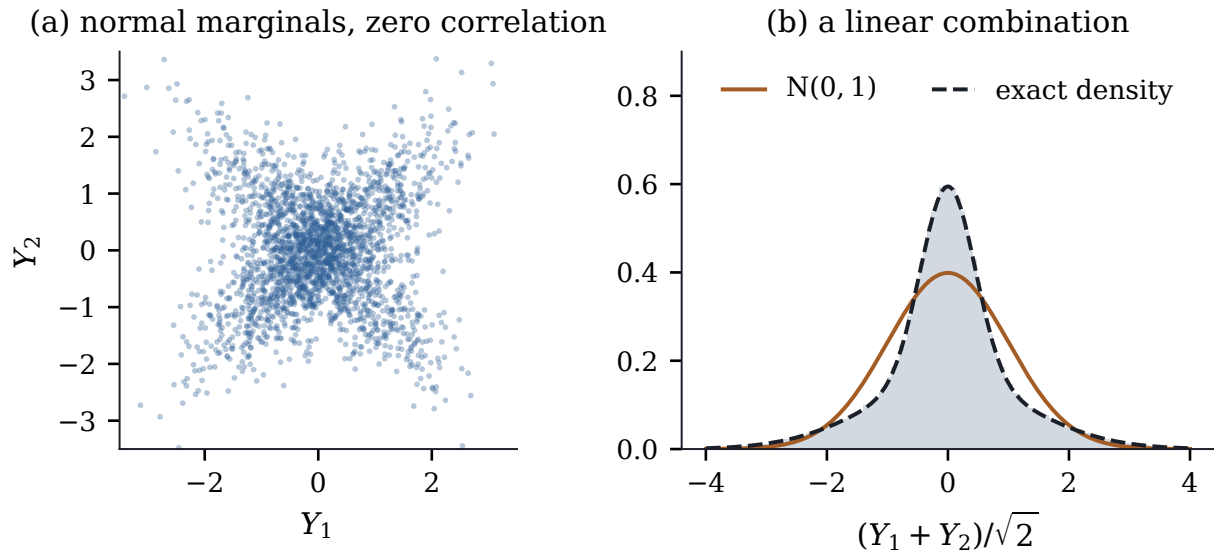


Figure 3.3.1: An equal mixture of bivariate normals with correlations ± 0.8 . (a) Both margins are exactly $N(0, 1)$ and the correlation is zero, but the scatter is an X, not an ellipse. (b) The linear combination $(Y_1 + Y_2)/\sqrt{2}$ has variance one but is sharply peaked and heavy-tailed compared with $N(0, 1)$. By [Definition 3.1.1](#), one non-normal direction is enough to rule out joint normality.

Example. (A random sign)

Let $Z \sim N(0, 1)$ and let S be independent of Z with $\Pr(S = 1) = \Pr(S = -1) = \frac{1}{2}$. Put $W = SZ$. By symmetry of Z , $W \sim N(0, 1)$. Also $\text{Cov}(Z, W) = E(S)E(Z^2) = 0$. But $Z + W = (1 + S)Z$ equals zero with probability $\frac{1}{2}$ and is otherwise $N(0, 4)$. A variable with an atom at zero and a continuous part is not normal, so (Z, W) is not jointly normal. And Z and W are far from independent: $|W| = |Z|$ always.

Example. (A mixture of two correlations)

Fix $0 < \rho < 1$. With probability $\frac{1}{2}$ draw (Y_1, Y_2) from the bivariate normal with unit variances and correlation ρ , and otherwise from the one with correlation $-\rho$. Both components have $N(0, 1)$ margins, so the mixture does too, exactly. The covariance is $\frac{1}{2}\rho + \frac{1}{2}(-\rho) = 0$. The pair is nevertheless dependent: since $E(Y_1^2 Y_2^2) = 1 + 2\rho^2$ for a standard bivariate normal with correlation r , the squares have correlation ρ^2 , not zero. The direction $\mathbf{a} = (1, 1)^\top / \sqrt{2}$ exposes the failure of joint normality. $S = (Y_1 + Y_2)/\sqrt{2}$ is an equal mixture of $N(0, 1 + \rho)$ and $N(0, 1 - \rho)$. It has variance 1 but kurtosis $E S^4 = \frac{3}{2}\{(1 + \rho)^2 + (1 - \rho)^2\} = 3(1 + \rho^2)$, where a normal variable would have 3.

[Figure 3.3.1](#) shows a simulation with $\rho = 0.8$ and 200,000 draws. Kolmogorov–Smirnov tests of normality for the two margins give p -values 0.84 and 0.29. The sample correlation is -0.0033 , the correlation of the squares is 0.640 (theory 0.64), and the kurtosis of S is 4.88 (theory 4.92). The Kolmogorov–Smirnov distance of S from $N(0, 1)$ is 0.067, which with this many draws rejects normality at any conventional level.

```
import numpy as np
from scipy import integrate, stats
rng = np.random.default_rng(302)
```



```

rho, N = 0.8, 200_000

sign = rng.choice([-1.0, 1.0], size=N)          # which component each draw uses
Z1, Z2 = rng.standard_normal((2, N))
Y1 = Z1
Y2 = sign * rho * Z1 + np.sqrt(1 - rho**2) * Z2 # corr(Y1, Y2 | sign) = sign * rho

S = (Y1 + Y2) / np.sqrt(2)                     # a direction that is not normal
print("marginal KS p-values:", stats.kstest(Y1, "norm").pvalue, stats.kstest(Y2, "
    norm").pvalue)
print("correlation:", np.corrcoef(Y1, Y2)[0, 1])
print("corr of squares:", np.corrcoef(Y1**2, Y2**2)[0, 1])
print("kurtosis of S:", stats.kurtosis(S, fisher=False), " theory:", 3 * (1 + rho
    **2))
print("KS p-value of S:", stats.kstest(S, "norm").pvalue)

```

Idea. Normality of a vector is a property of *every direction*, not of the coordinate axes. Normal margins with zero correlation can hide strong dependence. The theorems of the next two sections, which turn zero covariance into independence and give linear conditional means, need *joint* normality. They are not about variables that merely happen to be normal one at a time.

3.3.5 Exercises

3.3.5.1 A. Check your understanding

Exercise A1.

Let $\mathbf{Y} \sim N_3(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ with $\boldsymbol{\mu} = (0, 1, 2)^\top$ and $\boldsymbol{\Sigma} = \begin{pmatrix} 2 & 1 & 0 \\ 1 & 2 & 1 \\ 0 & 1 & 2 \end{pmatrix}$. Find the joint distribution of $U = Y_1 + Y_2 + Y_3$, $V = Y_1 - Y_3$ and $W = Y_1 - 2Y_2 + Y_3$. Which of the pairs are independent? Is V independent of (U, W) ?

Exercise A2.

In [Example \(Least squares under normal errors\)](#), show that the i th residual is $N(0, \sigma^2(1 - m_{ii}))$, where m_{ii} is the i th diagonal entry of $\mathbf{M}$, and that it is zero with probability one iff $m_{ii} = 1$.

3.3.5.2 B. Practice

Exercise B1.

Let $\mathbf{Y} \sim N_n(\boldsymbol{\mu}\mathbf{1}, \sigma^2\mathbf{I})$ and let $\mathbf{Q}$ be an orthogonal matrix whose first row is $n^{-1/2}\mathbf{1}^\top$. Show that the entries of $\mathbf{V} = \mathbf{QY}$ are independent, with $V_1 \sim N(\sqrt{n}\boldsymbol{\mu}, \sigma^2)$ and $V_i \sim N(0, \sigma^2)$ for $i \geq 2$. Show that $\sum_i (Y_i - \bar{Y})^2 = \sum_{i \geq 2} V_i^2$, and conclude again that $\bar{Y}$ and S^2 are independent.

Exercise B2.

Let $Z \sim N(0, 1)$ and, for $c > 0$, let $W = Z$ if $|Z| \leq c$ and $W = -Z$ otherwise. Show that $W \sim N(0, 1)$, that there is a c with $\text{Cov}(Z, W) = 0$, and that (Z, W) is not jointly normal for any c .

Solution. Because Z and $-Z$ have the same distribution and the event $\{|Z| > c\}$ is unchanged by the sign change, $\Pr(W \leq w) = \Pr(Z \leq w, |Z| \leq c) + \Pr(-Z \leq w, |Z| > c) = \Pr(Z \leq w)$. Next, $\text{Cov}(Z, W) = E(Z^2; |Z| \leq c) - E(Z^2; |Z| > c) = 2E(Z^2; |Z| \leq c) - 1$, which is continuous and increasing in c , tends to -1 as $c \rightarrow 0$ and to 1 as $c \rightarrow \infty$, and so vanishes at some c . Finally $Z + W = 2Z \mathbb{I}\{|Z| \leq c\}$ equals 0 with probability $\Pr(|Z| > c) \in (0, 1)$ but is not constant. A normal variable is either constant or has a density, so $Z + W$ is not normal.

Exercise B3.

For the pair (Z, W) of [Example \(A random sign\)](#), find every $\mathbf{a} \neq \mathbf{0}$ for which $a_1 Z + a_2 W$ is normal. *Hint:* an equal mixture of $N(0, v_1)$ and $N(0, v_2)$ has fourth moment $\frac{3}{2}(v_1^2 + v_2^2)$. Conclude that (Z, W) is normal along exactly two lines through the origin, the coordinate axes.

3.4 Independence and zero covariance

For any random vectors with finite second moments, independence implies zero cross-covariance, because the expectation of a product of functions of independent variables factors. The converse fails in general, as [Example \(A random sign\)](#) and [Example \(A mixture of two correlations\)](#) show. For jointly normal vectors it holds. This is the single most used property of the normal distribution in linear-model theory. Every independence statement behind a t or F statistic reduces to it.

Theorem 3.4.1 — Independence

Let $\mathbf{Y} \sim N_n(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ be partitioned into blocks $\mathbf{Y} = (\mathbf{Y}_1^\top, \dots, \mathbf{Y}_k^\top)^\top$, with $\boldsymbol{\mu}$ and $\boldsymbol{\Sigma} = (\boldsymbol{\Sigma}_{ij})$ partitioned to match.

- (a) The blocks $\mathbf{Y}_1, \dots, \mathbf{Y}_k$ are mutually independent iff $\boldsymbol{\Sigma}_{ij} = \text{Cov}(\mathbf{Y}_i, \mathbf{Y}_j) = \mathbf{0}$ for all $i \neq j$.
- (b) In particular, $\mathbf{Y}_1$ and $\mathbf{Y}_2$ are independent iff $\text{Cov}(\mathbf{Y}_1, \mathbf{Y}_2) = \mathbf{0}$, and the scalar entries $Y_1, \dots, Y_n$ are mutually independent iff $\boldsymbol{\Sigma}$ is diagonal.

Proof. If the blocks are independent, every cross-covariance is zero. Conversely, suppose $\boldsymbol{\Sigma}_{ij} = \mathbf{0}$ for $i \neq j$, and partition $\mathbf{t}$ to match. Then

$$\mathbf{t}^\top \boldsymbol{\mu} + \frac{1}{2} \mathbf{t}^\top \boldsymbol{\Sigma} \mathbf{t} = \sum_{i=1}^k \left(\mathbf{t}_i^\top \boldsymbol{\mu}_i + \frac{1}{2} \mathbf{t}_i^\top \boldsymbol{\Sigma}_{ii} \mathbf{t}_i \right),$$

because all the cross terms $\mathbf{t}_i^\top \boldsymbol{\Sigma}_{ij} \mathbf{t}_j$ vanish. Exponentiating, [Theorem 3.2.1](#) gives $M_{\mathbf{Y}}(\mathbf{t}) = \prod_i M_i(\mathbf{t}_i)$, where M_i is the moment generating function of $N(\boldsymbol{\mu}_i, \boldsymbol{\Sigma}_{ii})$. By [Corollary 3.3.2](#) that is the distribution of $\mathbf{Y}_i$, so $M_i = M_{\mathbf{Y}_i}$. The factorization property of [Section 3.2](#) gives mutual independence. Part (b) is the cases $k = 2$ and $k = n$. ■

A consequence worth stating explicitly: for jointly normal blocks, *pairwise* independence implies *mutual* independence, because condition (a) involves only pairs. For general random vectors this is false, as the example of three random signs in [Example \(Pairwise but not mutually independent\)](#) shows. Joint normality of the whole vector matters here: joint normality of each pair is not enough ([Exercise \(C1\)](#)).

The theorem is usually applied to two linear functions of the same normal vector.

Corollary 3.4.2

Let $\mathbf{Y} \sim N_n(\boldsymbol{\mu}, \boldsymbol{\Sigma})$, and let $\mathbf{A}$ and $\mathbf{B}$ be matrices with n columns. Then $\mathbf{AY}$ and $\mathbf{BY}$ are independent iff $\mathbf{A}\boldsymbol{\Sigma}\mathbf{B}^\top = \mathbf{0}$. In that case $g(\mathbf{AY})$ and $h(\mathbf{BY})$ are independent for all measurable functions g and h .

Proof. The stacked vector $((\mathbf{AY})^\top, (\mathbf{BY})^\top)^\top$ is a linear function of $\mathbf{Y}$, so it is normal by [Theorem 3.3.1](#). Its off-diagonal covariance block is $\text{Cov}(\mathbf{AY}, \mathbf{BY}) = \mathbf{A}\boldsymbol{\Sigma}\mathbf{B}^\top$ ([Theorem 2.2.1](#)). Apply [Theorem 3.4.1](#). Functions of independent random vectors are independent. ■

The condition $\mathbf{A}\boldsymbol{\Sigma}\mathbf{B}^\top = \mathbf{0}$ involves only second moments, but the conclusion is full independence. In particular, any quadratic form in $\mathbf{BY}$, such as $\|\mathbf{BY}\|^2$, is independent of $\mathbf{AY}$. [Chapter 4](#) extends this to quadratic forms that are not written as squared lengths ([Theorem 4.4.1](#)).

Example. (Sample mean and deviations)

Let $\mathbf{Y} \sim N_n(\mu\mathbf{1}, \sigma^2\mathbf{I})$, so that $Y_1, \dots, Y_n$ is a random sample from $N(\mu, \sigma^2)$. The mean is $\bar{Y} = n^{-1}\mathbf{1}^\top\mathbf{Y}$, and the vector of deviations is $\mathbf{d} = (\mathbf{I} - n^{-1}\mathbf{1}\mathbf{1}^\top)\mathbf{Y}$. As in [Example \(Centring a sample\)](#),

$$\text{Cov}(\bar{Y}, \mathbf{d}) = \sigma^2 n^{-1} \mathbf{1}^\top (\mathbf{I} - n^{-1} \mathbf{1}\mathbf{1}^\top) = \mathbf{0}^\top,$$

By [Corollary 3.4.2](#), $\bar{Y}$ is independent of $\mathbf{d}$, and so of every function of the deviations: the sample variance $S^2 = \|\mathbf{d}\|^2/(n-1)$, the range, the sample skewness. The distribution of $(n-1)S^2/\sigma^2$ is found in [Corollary 4.4.5](#). The computation used $\text{Cov}(\mathbf{Y}) = \sigma^2\mathbf{I}$ only through $\boldsymbol{\Sigma}\mathbf{1} \in \text{span}(\mathbf{1})$, which is the precise condition for the conclusion ([Exercise \(B1\)](#)).

Example. (Coefficients and residuals)

In the setting of [Example \(Least squares under normal errors\)](#),

$$\text{Cov}(\hat{\boldsymbol{\beta}}, \hat{\boldsymbol{\varepsilon}}) = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top (\sigma^2 \mathbf{I}) (\mathbf{I} - \mathbf{M}) = \sigma^2 (\mathbf{X}^\top \mathbf{X})^{-1} \{(\mathbf{I} - \mathbf{M})\mathbf{X}\}^\top = \mathbf{0},$$

since $(\mathbf{I} - \mathbf{M})\mathbf{X} = \mathbf{0}$. Hence $\hat{\boldsymbol{\beta}}$ is independent of the residual vector, and so of the residual sum of squares $\|\hat{\boldsymbol{\epsilon}}\|^2$ and of the estimate $s^2 = \|\hat{\boldsymbol{\epsilon}}\|^2/(n-p)$. The same argument shows that the fitted values $\hat{\mathbf{Y}}$ and the residuals are independent. This independence is what allows a t statistic $(\boldsymbol{\lambda}^\top \hat{\boldsymbol{\beta}} - \boldsymbol{\lambda}^\top \boldsymbol{\beta})/\{s^2 \boldsymbol{\lambda}^\top (\mathbf{X}^\top \mathbf{X})^{-1} \boldsymbol{\lambda}\}^{1/2}$ to have a t distribution (Theorem 11.6.1).

Remark (What joint normality buys). The two examples use only a covariance calculation. Without joint normality, zero covariance between $\hat{\boldsymbol{\beta}}$ and $\hat{\boldsymbol{\epsilon}}$ still holds under the second-moment assumptions of Definition 5.3.1, but independence does not follow, and neither do exact t and F distributions. Chapter 21 examines what survives approximately.

3.4.1 Exercises

3.4.1.1 A. Check your understanding

Exercise A1.

Let Y_1, Y_2 be independent $N(\mu, \sigma^2)$. Show that $Y_1 + Y_2$ and $Y_1 - Y_2$ are independent. What if the variances differ?

Exercise A2.

Let $\mathbf{Y} \sim N_n(\mathbf{0}, \mathbf{I})$ and let $\mathbf{u}$ be a unit vector. Show that $\mathbf{u}^\top \mathbf{Y}$ and $\|\mathbf{Y} - (\mathbf{u}^\top \mathbf{Y})\mathbf{u}\|^2$ are independent.

3.4.1.2 B. Practice

Exercise B1.

Let $\mathbf{Y} \sim N_n(\mu\mathbf{1}, \boldsymbol{\Sigma})$ with $\boldsymbol{\Sigma}$ arbitrary. Show that $\bar{Y}$ is independent of the deviations $Y_i - \bar{Y}$ iff $\boldsymbol{\Sigma}\mathbf{1} \in \text{span}(\mathbf{1})$, that is, iff all row sums of $\boldsymbol{\Sigma}$ are equal. Check the condition for the equicorrelation matrix $\sigma^2\{(1-\rho)\mathbf{I} + \rho\mathbf{1}\mathbf{1}^\top\}$, and for a two-group matrix: the first m_1 and the last m_2 observations form groups, all variances are 1, the correlation is ρ within a group and τ between groups.

Solution. $\bar{Y} = n^{-1}\mathbf{1}^\top \mathbf{Y}$ and the deviations are $(\mathbf{I} - n^{-1}\mathbf{1}\mathbf{1}^\top)\mathbf{Y}$, so by Corollary 3.4.2 they are independent iff $n^{-1}\mathbf{1}^\top \boldsymbol{\Sigma}(\mathbf{I} - n^{-1}\mathbf{1}\mathbf{1}^\top) = \mathbf{0}^\top$, that is, $(\mathbf{I} - n^{-1}\mathbf{1}\mathbf{1}^\top)\boldsymbol{\Sigma}\mathbf{1} = \mathbf{0}$. The null space of $\mathbf{I} - n^{-1}\mathbf{1}\mathbf{1}^\top$ is $\text{span}(\mathbf{1})$, and $\boldsymbol{\Sigma}\mathbf{1}$ is the vector of row sums. For the equicorrelation matrix, $\boldsymbol{\Sigma}\mathbf{1} = \sigma^2(1-\rho + n\rho)\mathbf{1}$, so independence holds for every admissible ρ . In the two-group matrix, a row in the first group sums to $1 + (m_1 - 1)\rho + m_2\tau$ and a row in the second to $1 + (m_2 - 1)\rho + m_1\tau$. The difference is $(m_1 - m_2)(\rho - \tau)$, so independence holds iff the groups have equal size or $\rho = \tau$, and the latter is the equicorrelated case.

Exercise B2.

Let $\mathbf{Y} \sim N_n(\mathbf{X}\boldsymbol{\beta}, \sigma^2\mathbf{I})$ with $\text{rank}(\mathbf{X}) = r < p$, let $\mathbf{G}$ be a generalized inverse of $\mathbf{X}^\top \mathbf{X}$

and $\mathbf{M} = \mathbf{XGX}^\top$. Show that the least squares estimate $\mathbf{GX}^\top \mathbf{Y}$ and the residual vector $(\mathbf{I} - \mathbf{M})\mathbf{Y}$ are independent, and that every estimable function $\boldsymbol{\lambda}^\top \boldsymbol{\beta}$ with $\boldsymbol{\lambda} \in \mathcal{C}(\mathbf{X}^\top)$ has an estimate independent of the residual sum of squares.

3.4.1.3 C. Going deeper

Exercise C1.

Let φ be the $N(0, 1)$ density, $0 < \varepsilon \leq e^{3/2}$, and

$$f(x, y, z) = \varphi(x)\varphi(y)\varphi(z)\{1 + \varepsilon xyz e^{-(x^2+y^2+z^2)/2}\}.$$

Show that f is a probability density, and that under f every pair of coordinates is $N_2(\mathbf{0}, \mathbf{I})$. So the three variables are pairwise jointly normal and pairwise independent. Show that they are not mutually independent and that (X, Y, Z) is not jointly normal. Which step of [Theorem 3.4.1](#) fails?

Solution. Since $|t|e^{-t^2/2} \leq e^{-1/2}$ for all t , the perturbation satisfies $|\varepsilon xyz e^{-(x^2+y^2+z^2)/2}| \leq \varepsilon e^{-3/2} \leq 1$, so $f \geq 0$. The function $z \mapsto z\varphi(z)e^{-z^2/2}$ is odd and integrable, so integrating f over z removes the perturbation and leaves $\varphi(x)\varphi(y)$. Integrating further shows that f integrates to 1, and by symmetry each pair has density $\varphi \otimes \varphi$, the $N_2(\mathbf{0}, \mathbf{I})$ density. Mutual independence would force $f = \varphi(x)\varphi(y)\varphi(z)$ almost everywhere, but the perturbation is nonzero off the coordinate planes. Directly, $E(XYZ) = \varepsilon(\int t^2 \varphi(t) e^{-t^2/2} dt)^3 > 0$, whereas independence with mean-zero margins gives 0. If (X, Y, Z) were jointly normal, its covariance matrix would be $\mathbf{I}$ (unit variances from the margins, zero covariances from the pairs), and by [Theorem 3.4.1](#) the coordinates would be independent, which they are not. The theorem's hypothesis is joint normality of the whole vector. Here only the pairs are normal, and the moment generating function of the triple does not factor.

3.5 Conditional distributions

Throughout this section $\mathbf{Y} \sim N_n(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ is split into $\mathbf{Y}_1 \in \mathbb{R}^{n_1}$ and $\mathbf{Y}_2 \in \mathbb{R}^{n_2}$, with

$$\boldsymbol{\mu} = \begin{pmatrix} \boldsymbol{\mu}_1 \\ \boldsymbol{\mu}_2 \end{pmatrix}, \quad \boldsymbol{\Sigma} = \begin{pmatrix} \boldsymbol{\Sigma}_{11} & \boldsymbol{\Sigma}_{12} \\ \boldsymbol{\Sigma}_{21} & \boldsymbol{\Sigma}_{22} \end{pmatrix}, \quad \boldsymbol{\Sigma}_{21} = \boldsymbol{\Sigma}_{12}^\top,$$

as in [Theorem 2.5.1](#). The question is what $\mathbf{Y}_1$ looks like once $\mathbf{Y}_2$ has been observed. The answer is the probabilistic core of regression.

3.5.1 Partial covariance with a generalized inverse

The natural answer involves $\boldsymbol{\Sigma}_{22}^{-1}$, but $\boldsymbol{\Sigma}_{22}$ may be singular, for instance when $\mathbf{Y}_2$ contains a sum and its summands. A generalized inverse ([Definition 1.3.1](#)) removes the need for a separate treatment. The facts that make this work were proved for arbitrary random vectors in [Section 2.5](#). We collect them in the notation of this chapter.

Lemma 3.5.1

Let Σ be nonnegative definite and partitioned as above, and let $\mathbf{G}$ be any generalized inverse of Σ_{22} . Then:

- (a) $\mathcal{C}(\Sigma_{21}) \subseteq \mathcal{C}(\Sigma_{22})$, so $\Sigma_{21} = \Sigma_{22}\mathbf{H}$ for some matrix $\mathbf{H}$;
- (b) $\Sigma_{12}\mathbf{G}\Sigma_{22} = \Sigma_{12}$ and $\Sigma_{22}\mathbf{G}\Sigma_{21} = \Sigma_{21}$;
- (c) $\Sigma_{12}\mathbf{G}\Sigma_{21} = \mathbf{H}^\top \Sigma_{22}\mathbf{H}$, which does not depend on the choice of $\mathbf{G}$ and is nonnegative definite.

The matrix $\Sigma_{11.2} = \Sigma_{11} - \Sigma_{12}\Sigma_{22}^-\Sigma_{21}$, the **Schur complement** of Σ_{22} in Σ , is therefore well defined. It is the partial covariance matrix of $\mathbf{Y}_1$ given $\mathbf{Y}_2$ in the sense of [Theorem 2.5.2](#).

Proof. By [Theorem 3.1.2](#), Σ is the covariance matrix of some random vector, so (a) and (b) are [Theorem 2.5.1\(c\)](#). For (c), $\Sigma_{12}\mathbf{G}\Sigma_{21} = \mathbf{H}^\top \Sigma_{22}\mathbf{G}\Sigma_{22}\mathbf{H} = \mathbf{H}^\top \Sigma_{22}\mathbf{H}$. ■

3.5.2 The conditional distribution**Theorem 3.5.2 — Conditional distribution**

Let $\mathbf{Y} \sim N_n(\boldsymbol{\mu}, \Sigma)$ be partitioned as above, let $\mathbf{G}$ be a generalized inverse of Σ_{22} , and put $\mathbf{B} = \Sigma_{12}\mathbf{G}$. Then:

- (a) $\mathbf{W} = \mathbf{Y}_1 - \boldsymbol{\mu}_1 - \mathbf{B}(\mathbf{Y}_2 - \boldsymbol{\mu}_2)$ is independent of $\mathbf{Y}_2$ and $\mathbf{W} \sim N_{n_1}(\mathbf{0}, \Sigma_{11.2})$;
- (b) the conditional distribution of $\mathbf{Y}_1$ given $\mathbf{Y}_2 = \mathbf{y}_2$ is

$$\mathbf{Y}_1 \mid \mathbf{Y}_2 = \mathbf{y}_2 \sim N_{n_1}\left(\boldsymbol{\mu}_1 + \Sigma_{12}\Sigma_{22}^-(\mathbf{y}_2 - \boldsymbol{\mu}_2), \Sigma_{11} - \Sigma_{12}\Sigma_{22}^-\Sigma_{21}\right), \quad (3.5.1)$$

and for $(\mathbf{y}_2)$ in $(\boldsymbol{\mu}_2 + \mathcal{C}(\Sigma_{22}))$, where $(\mathbf{Y}_2)$ lies with one, the conditional mean does not depend on the choice of $(\mathbf{G})$;

- (c) if Σ is positive definite, then $\Sigma_{22}^- = \Sigma_{22}^{-1}$, $\Sigma_{11.2}$ is positive definite, $\Sigma_{11.2}^{-1}$ is the leading $n_1 \times n_1$ block of Σ^{-1} , and $\det \Sigma = \det \Sigma_{22} \det \Sigma_{11.2}$.

Proof.

- (a) $\mathbf{W} = [\mathbf{I}, -\mathbf{B}](\mathbf{Y} - \boldsymbol{\mu})$, so $(\mathbf{W}^\top, \mathbf{Y}_2^\top)^\top$ is an affine function of $\mathbf{Y}$ and is jointly normal ([Theorem 3.3.1](#)). The vector $\mathbf{W}$ is the error of the best linear predictor of $\mathbf{Y}_1$ from $\mathbf{Y}_2$, so by [Theorem 2.5.2\(a,b\)](#) it has mean zero, $\text{Cov}(\mathbf{W}, \mathbf{Y}_2) = \mathbf{0}$ and $\text{Cov}(\mathbf{W}) = \Sigma_{11.2}$. These are moment calculations, valid for any random vector. What normality adds is that zero covariance becomes independence ([Theorem 3.4.1](#)), and that $\mathbf{W}$, a linear function of $\mathbf{Y}$, is normal.
- (b) Write $\mathbf{Y}_1 = \boldsymbol{\mu}_1 + \mathbf{B}(\mathbf{Y}_2 - \boldsymbol{\mu}_2) + \mathbf{W}$. The first two terms are a function of $\mathbf{Y}_2$, and $\mathbf{W}$ is independent of $\mathbf{Y}_2$. For independent $\mathbf{W}$ and $\mathbf{Y}_2$, the conditional distribution

of $h(\mathbf{W}, \mathbf{Y}_2)$ given $\mathbf{Y}_2 = \mathbf{y}_2$ is the distribution of $h(\mathbf{W}, \mathbf{y}_2)$ (Billingsley 1995). With $h(\mathbf{w}, \mathbf{y}_2) = \boldsymbol{\mu}_1 + \mathbf{B}(\mathbf{y}_2 - \boldsymbol{\mu}_2) + \mathbf{w}$, this is the normal distribution Eq. 3.5.1. If $\mathbf{y}_2 - \boldsymbol{\mu}_2 = \boldsymbol{\Sigma}_{22}\mathbf{k}$, then $\mathbf{B}(\mathbf{y}_2 - \boldsymbol{\mu}_2) = \boldsymbol{\Sigma}_{12}\mathbf{G}\boldsymbol{\Sigma}_{22}\mathbf{k} = \boldsymbol{\Sigma}_{12}\mathbf{k}$, which does not involve $\mathbf{G}$.

- (c) If $\boldsymbol{\Sigma}$ is positive definite, so is its principal submatrix $\boldsymbol{\Sigma}_{22}$, and its only generalized inverse is its inverse. By the block determinant formula (Theorem 1.6.3), which needs only $\boldsymbol{\Sigma}_{22}$ nonsingular, $\det \boldsymbol{\Sigma} = \det \boldsymbol{\Sigma}_{22} \det \boldsymbol{\Sigma}_{11.2}$. The left side and $\det \boldsymbol{\Sigma}_{22}$ are positive, so $\boldsymbol{\Sigma}_{11.2}$ is nonsingular. It is nonnegative definite, being $\text{Cov}(\mathbf{W})$ by (a), hence positive definite. With both $\boldsymbol{\Sigma}_{22}$ and its Schur complement nonsingular, the partitioned-inverse formula (Theorem 1.6.4) gives $\boldsymbol{\Sigma}_{11.2}^{-1}$ as the leading block of $\boldsymbol{\Sigma}^{-1}$.

When $\boldsymbol{\Sigma}$ is positive definite, (b) can also be read off the densities. By (c), $\boldsymbol{\Sigma}_{11.2}$ is positive definite, so $\mathbf{W}$ has a density. The map $(\mathbf{y}_1, \mathbf{y}_2) \mapsto (\mathbf{w}, \mathbf{y}_2)$ has unit Jacobian, so $f_{\mathbf{Y}}(\mathbf{y}_1, \mathbf{y}_2) = f_{\mathbf{W}}(\mathbf{y}_1 - \boldsymbol{\mu}_1 - \mathbf{B}(\mathbf{y}_2 - \boldsymbol{\mu}_2)) f_{\mathbf{Y}_2}(\mathbf{y}_2)$ by independence, and dividing by $f_{\mathbf{Y}_2}(\mathbf{y}_2)$ gives the conditional density.

■

Idea. Every jointly normal vector splits as

$$\mathbf{Y}_1 = \underbrace{\boldsymbol{\mu}_1 + \boldsymbol{\Sigma}_{12}\boldsymbol{\Sigma}_{22}^{-1}(\mathbf{Y}_2 - \boldsymbol{\mu}_2)}_{\text{function of } \mathbf{Y}_2} + \underbrace{\mathbf{W}}_{\text{independent of } \mathbf{Y}_2}.$$

The first term is a linear regression of $\mathbf{Y}_1$ on $\mathbf{Y}_2$. The second is a normal error whose distribution is the same whatever value $\mathbf{Y}_2$ takes.

3.5.3 What the theorem says

Three features of Eq. 3.5.1 are special to the normal distribution, and each is an assumption of the classical linear model.

- (i) *The regression is linear.* $E(\mathbf{Y}_1 \mid \mathbf{Y}_2 = \mathbf{y}_2)$ is an affine function of $\mathbf{y}_2$, with coefficient matrix $\boldsymbol{\Sigma}_{12}\boldsymbol{\Sigma}_{22}^{-1}$.
- (ii) *The errors are homoscedastic.* The conditional covariance $\boldsymbol{\Sigma}_{11.2}$ is the same for every $\mathbf{y}_2$. Observing $\mathbf{Y}_2$ tells us where $\mathbf{Y}_1$ is centred, but not how spread out it is.
- (iii) *Conditioning cannot increase uncertainty.* $\boldsymbol{\Sigma}_{11} - \boldsymbol{\Sigma}_{11.2} = \boldsymbol{\Sigma}_{12}\boldsymbol{\Sigma}_{22}^{-1}\boldsymbol{\Sigma}_{21}$ is nonnegative definite (Lemma 3.5.1(c)), so every linear combination $\mathbf{a}^\top \mathbf{Y}_1$ has conditional variance at most its unconditional variance. The reduction is zero iff $\boldsymbol{\Sigma}_{12} = \mathbf{0}$.

None of these holds in general. In the mixture of Example (A mixture of two correlations), given $Y_1 = y_1$ the variable Y_2 is an equal mixture of $N(\rho y_1, 1 - \rho^2)$ and $N(-\rho y_1, 1 - \rho^2)$. Its conditional mean is 0 for every y_1 , but its conditional variance is $1 - \rho^2 + \rho^2 y_1^2$, which grows with $|y_1|$. Linearity and constant variance are not consequences of normal margins or of zero correlation. They are consequences of joint normality.

One response, several predictors. The case used most in this book has a scalar response Y and a vector $\mathbf{X}$ of k predictors, jointly normal, with $\text{Var}(Y) = \sigma_Y^2$, $\boldsymbol{\sigma}_{XY} = \text{Cov}(\mathbf{X}, Y)$ and $\boldsymbol{\Sigma}_{XX} = \text{Cov}(\mathbf{X})$ positive definite. Theorem 3.5.2 says

$$Y = \alpha + \boldsymbol{\beta}^\top \mathbf{X} + e, \quad \boldsymbol{\beta} = \boldsymbol{\Sigma}_{XX}^{-1} \boldsymbol{\sigma}_{XY}, \quad \alpha = \mu_Y - \boldsymbol{\beta}^\top \boldsymbol{\mu}_X, \quad e \sim N(0, \sigma^2) \text{ independent of } \mathbf{X}, \quad (3.5.2)$$

with $\sigma^2 = \sigma_Y^2 - \boldsymbol{\sigma}_{XY}^\top \boldsymbol{\Sigma}_{XX}^{-1} \boldsymbol{\sigma}_{XY} = \sigma_Y^2(1 - \rho_{Y \cdot X}^2)$, where $\rho_{Y \cdot X}^2 = \boldsymbol{\sigma}_{XY}^\top \boldsymbol{\Sigma}_{XX}^{-1} \boldsymbol{\sigma}_{XY} / \sigma_Y^2$ is the population squared multiple correlation ([Proposition 2.5.3](#)). So if observations $(Y_i, \mathbf{X}_i)$ are drawn independently from a joint normal distribution, then conditionally on the predictors, the normal linear model with independent homoscedastic errors holds exactly. Inference that treats the predictors as fixed is valid for random predictors, as [Chapter 14](#) develops ([Theorem 14.1.2](#)).

The bivariate case. With $k = 1$, correlation ρ and standard deviations σ_X, σ_Y ,

$$E(Y | X = x) = \mu_Y + \rho \frac{\sigma_Y}{\sigma_X} (x - \mu_X), \quad \text{Var}(Y | X = x) = \sigma_Y^2(1 - \rho^2).$$

In standard units, $(E(Y | X = x) - \mu_Y) / \sigma_Y = \rho (x - \mu_X) / \sigma_X$. The predicted response is fewer standard deviations from its mean than the predictor is from its mean, whenever $|\rho| < 1$. This is **regression toward the mean**, Galton's observation about the heights of parents and children (Galton 1886), and the origin of the word “regression”. It is also a warning. Regressing X on Y is a different line, $E(X | Y = y) = \mu_X + \rho(\sigma_X / \sigma_Y)(y - \mu_Y)$, and when $0 < |\rho| < 1$ neither line is the major axis of the density contours.

Example. (Two regression lines)

Take $\mu_X = 1$, $\mu_Y = 2$, $\sigma_X = 1$, $\sigma_Y = 1.5$ and $\rho = 0.7$. The regression of Y on X has slope 1.05 and residual standard deviation 1.071. Drawn in the same (x, y) plane, the regression of X on Y has slope 2.14, and the major axis of the ellipses lies between them, with slope 1.76. [Figure 3.5.1](#) shows the three lines. The line $E(Y | X = x)$ passes through the points where each contour ellipse has a vertical tangent ([Exercise \(B2\)](#)): on a vertical line $X = x$, the density is highest at the conditional mean. A least squares line fitted to 60 draws has slope 1.12 (standard error 0.16). It estimates the regression of Y on X , not the major axis.

Example. (Holding a third variable fixed)

Let $\mathbf{Y} \sim N_3(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ with

$$\boldsymbol{\mu} = \begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix}, \quad \boldsymbol{\Sigma} = \begin{pmatrix} 4 & 1 & 2 \\ 1 & 3 & 2 \\ 2 & 2 & 2 \end{pmatrix}.$$

Conditioning Y_1 on Y_2 alone gives slope $\sigma_{12} / \sigma_{22} = \frac{1}{3}$ and conditional variance $\frac{11}{3}$. Conditioning on (Y_2, Y_3) , the listing computes $\mathbf{B} = \boldsymbol{\Sigma}_{12} \boldsymbol{\Sigma}_{22}^{-1}$ in exact arithmetic and finds

$$E(Y_1 | Y_2 = y_2, Y_3 = y_3) = 3 - y_2 + 2y_3, \quad \text{Var}(Y_1 | Y_2, Y_3) = 1.$$

The coefficient of y_2 changes sign. Among outcomes with the same Y_3 , larger Y_2 goes with smaller Y_1 . Across all outcomes, larger Y_2 goes with larger Y_1 , because both tend to be large when Y_3 is large. The conditional variance 1 also equals $\det \boldsymbol{\Sigma} / \det \boldsymbol{\Sigma}_{22} = 2/2$, as [Theorem 3.5.2\(c\)](#) predicts.

```
from fractions import Fraction as F
```

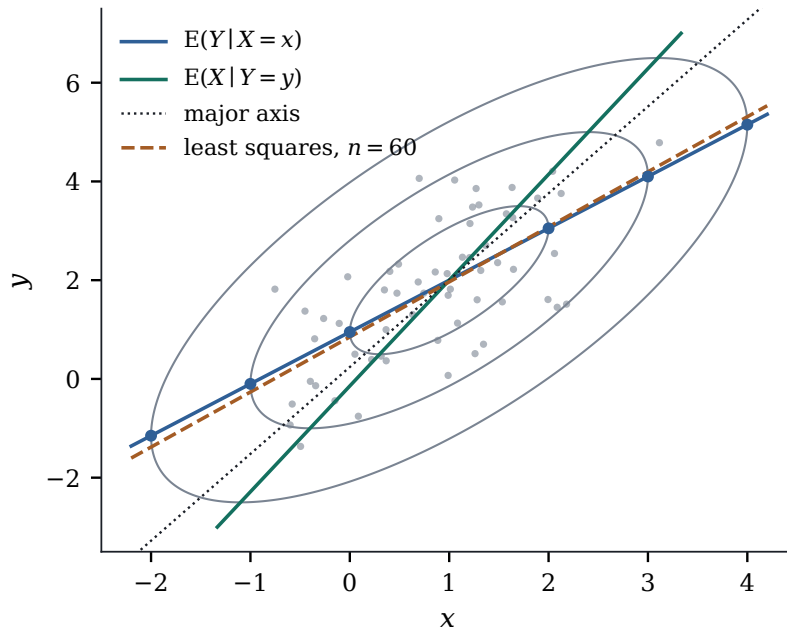



Figure 3.5.1: Contours $\Delta^2 = 1, 4, 9$ of a bivariate normal density with $\rho = 0.7$. The conditional mean line $E(Y | X = x)$ passes through the points of vertical tangency (dots). The line $E(X | Y = y)$ passes through the points of horizontal tangency and is steeper, and the major axis lies between the two. Least squares on a sample of 60 points estimates $E(Y | X = x)$.

```
import numpy as np
rng = np.random.default_rng(305)
mx, my, sx, sy, rho = 1.0, 2.0, 1.0, 1.5, 0.7
Sigma = np.array([[sx**2, rho * sx * sy], [rho * sx * sy, sy**2]])
slope_y_on_x = rho * sy / sx # E(Y | X = x)
slope_x_on_y = rho * sx / sy # E(X | Y = y), as dx/dy
cond_sd = sy * np.sqrt(1 - rho**2)
n = 60
XY = rng.multivariate_normal([mx, my], Sigma, size=n)
b_ls, a_ls = np.polyfit(XY[:, 0], XY[:, 1], 1)
Lin = np.linalg.inv(np.linalg.cholesky(Sigma))
theta = np.linspace(0, 2 * np.pi, 200_001)
circle = np.vstack([np.cos(theta), np.sin(theta)])
L = np.linalg.cholesky(Sigma)
big = rng.multivariate_normal([mx, my], Sigma, size=4_000_000)
x0 = 2.0
s1 = big[np.abs(big[:, 0] - x0) < 0.01, 1]
resid = XY[:, 1] - a_ls - b_ls * XY[:, 0]
se = np.sqrt(resid @ resid / (n - 2) / np.sum((XY[:, 0] - XY[:, 0].mean())**2))
lam, U = np.linalg.eigh(Sigma)
slope_major = U[1, 1] / U[0, 1]
levels = [1, 4, 9]
gx, gy = np.meshgrid(np.linspace(-3, 5, 300), np.linspace(-4, 8, 300))
D = np.stack([gx - mx, gy - my], axis=-1)
D2 = np.einsum("...i,ij,...j->...", D, np.linalg.inv(Sigma), D)
xs = np.linspace(-2.2, 4.2, 2)
ys = np.linspace(-3.0, 7.0, 2)

mu = [F(1), F(0), F(-1)]
S = [[F(4), F(1), F(2)],
      [F(1), F(3), F(2)],
```

```

[F(2), F(2), F(2)]

def inv2(M):
    det = M[0][0] * M[1][1] - M[0][1] * M[1][0]
    return [[M[1][1] / det, -M[0][1] / det], [-M[1][0] / det, M[0][0] / det]]

# Y1 given (Y2, Y3): B = sigma_12 Sigma_22^{-1}, variance sigma_11 - B sigma_21
S22inv = inv2([[S[1][1], S[1][2]], [S[2][1], S[2][2]]])
s12 = [S[0][1], S[0][2]]
B = [s12[0] * S22inv[0][j] + s12[1] * S22inv[1][j] for j in range(2)]
var_1_23 = S[0][0] - (B[0] * s12[0] + B[1] * s12[1])
const = mu[0] - B[0] * mu[1] - B[1] * mu[2]
print("E(Y1 | y2, y3) =", const, "+", B[0], "y2 +", B[1], "y3; variance", var_1_23)

# (Y1, Y2) given Y3, and the partial correlation of Y1 and Y2 given Y3
C = [[S[i][j] - S[i][2] * S[2][j] / S[2][2] for j in range(2)] for i in range(2)]
print("Cov((Y1, Y2) | Y3) =", C)

```

3.5.4 The conditional mean is the best predictor

Section 6.11 defines the best linear predictor $L(\mathbf{Y}_1 | \mathbf{Y}_2)$ as a projection, and shows that the conditional mean is the best predictor among *all* square-integrable functions of $\mathbf{Y}_2$. For a scalar response, the gap between the two is the approximation error of Proposition 6.11.2(c). Under joint normality the gap is zero.

Proposition 3.5.3

Let $\mathbf{Y}$ be jointly normal and partitioned as above, and let $\mathbf{m}(\mathbf{Y}_2) = \boldsymbol{\mu}_1 + \boldsymbol{\Sigma}_{12}\boldsymbol{\Sigma}_{22}^{-1}(\mathbf{Y}_2 - \boldsymbol{\mu}_2)$. For every function $\mathbf{g}$ with $E\|\mathbf{g}(\mathbf{Y}_2)\|^2 < \infty$,

$$E\|\mathbf{Y}_1 - \mathbf{g}(\mathbf{Y}_2)\|^2 = \text{tr}(\boldsymbol{\Sigma}_{11.2}) + E\|\mathbf{m}(\mathbf{Y}_2) - \mathbf{g}(\mathbf{Y}_2)\|^2.$$

So $\mathbf{m}(\mathbf{Y}_2)$ minimizes the mean squared prediction error over all predictors, it equals the best linear predictor of Theorem 6.11.1 (entry by entry, when $\boldsymbol{\Sigma}_{22}$ is positive definite), and the minimum is $\text{tr}(\boldsymbol{\Sigma}_{11.2})$.

Proof. $\mathbf{Y}_1 - \mathbf{g}(\mathbf{Y}_2) = \mathbf{W} + (\mathbf{m}(\mathbf{Y}_2) - \mathbf{g}(\mathbf{Y}_2))$ with $\mathbf{W}$ as in Theorem 3.5.2. The cross term $E[\mathbf{W}^\top(\mathbf{m}(\mathbf{Y}_2) - \mathbf{g}(\mathbf{Y}_2))]$ vanishes, because $\mathbf{W}$ has mean zero and is independent of $\mathbf{Y}_2$. And $E\|\mathbf{W}\|^2 = \text{tr Cov}(\mathbf{W}) = \text{tr}(\boldsymbol{\Sigma}_{11.2})$. The minimizer $\mathbf{m}$ is affine, so it is also the best affine predictor, which is the best linear predictor. ■

The proof is short because independence of $\mathbf{W}$ and $\mathbf{Y}_2$ is much stronger than the zero correlation that defines a projection. For non-normal vectors only zero correlation is available, and the best linear predictor is merely the best *linear* predictor.

3.5.5 Exercises

3.5.5.1 A. Check your understanding

Exercise A1.

For $\boldsymbol{\mu}$ and $\boldsymbol{\Sigma}$ of [Example \(Holding a third variable fixed\)](#), find the conditional distribution of (Y_1, Y_2) given $Y_3 = y_3$ and of Y_3 given $(Y_1, Y_2) = (y_1, y_2)$. Verify that the conditional variances are smaller than the unconditional ones.

Exercise A2.

In the setting of [Eq. 3.5.2](#), let $\boldsymbol{\Omega}$ be the inverse of $\text{Cov}((Y, \mathbf{X}^\top)^\top)$. Show that the row of $\boldsymbol{\Omega}$ belonging to Y is $(1, -\boldsymbol{\beta}^\top)/\sigma^2$, where $\sigma^2 = \sigma_Y^2(1 - \rho_{Y,X}^2)$. So the regression coefficients and the error variance can be read off one row of the precision matrix.

Solution. Put Y first and apply [Exercise \(B5\)](#) with the scalar block Y and $\mathbf{Y}_2 = \mathbf{X}$. The conditional variance is $\omega_{YY}^{-1} = \text{Var}(Y | \mathbf{X}) = \sigma^2$, and the coefficient vector of the conditional mean is $-\omega_{YY}^{-1}\boldsymbol{\Omega}_{YX} = \boldsymbol{\beta}^\top$. Hence $\omega_{YY} = 1/\sigma^2$ and $\boldsymbol{\Omega}_{YX} = -\boldsymbol{\beta}^\top/\sigma^2$. By [Eq. 3.5.2](#), $\sigma^2 = \sigma_Y^2(1 - \rho_{Y,X}^2)$.

3.5.5.2 B. Practice

Exercise B1.

Suppose $\mathbf{Y}_2 \sim N(\boldsymbol{\mu}_2, \boldsymbol{\Sigma}_{22})$ and that, given $\mathbf{Y}_2 = \mathbf{y}_2$, $\mathbf{Y}_1$ is $N(\mathbf{a} + \mathbf{B}\mathbf{y}_2, \mathbf{S})$ with $\mathbf{a}$, $\mathbf{B}$ and $\mathbf{S}$ not depending on $\mathbf{y}_2$. Show that $(\mathbf{Y}_1^\top, \mathbf{Y}_2^\top)^\top$ is jointly normal and find its mean and covariance. Explain why [Example \(A mixture of two correlations\)](#) does not contradict this.

Solution. Condition on $\mathbf{Y}_2$:

$$\mathbb{E} e^{\mathbf{t}_1^\top \mathbf{Y}_1 + \mathbf{t}_2^\top \mathbf{Y}_2} = \mathbb{E} \left[e^{\mathbf{t}_2^\top \mathbf{Y}_2} e^{\mathbf{t}_1^\top (\mathbf{a} + \mathbf{B}\mathbf{Y}_2) + \frac{1}{2} \mathbf{t}_1^\top \mathbf{S} \mathbf{t}_1} \right] = e^{\mathbf{t}_1^\top \mathbf{a} + \frac{1}{2} \mathbf{t}_1^\top \mathbf{S} \mathbf{t}_1} M_{\mathbf{Y}_2}(\mathbf{t}_2 + \mathbf{B}^\top \mathbf{t}_1).$$

Inserting $M_{\mathbf{Y}_2}$ from [Theorem 3.2.1](#), the exponent is linear plus $\frac{1}{2} \mathbf{t}^\top \mathbf{C} \mathbf{t}$ with $\mathbf{C} = \begin{pmatrix} \mathbf{S} + \mathbf{B}\boldsymbol{\Sigma}_{22}\mathbf{B}^\top & \mathbf{B}\boldsymbol{\Sigma}_{22} \\ \boldsymbol{\Sigma}_{22}\mathbf{B}^\top & \boldsymbol{\Sigma}_{22} \end{pmatrix}$, so the pair is normal with mean $((\mathbf{a} + \mathbf{B}\boldsymbol{\mu}_2)^\top, \boldsymbol{\mu}_2^\top)^\top$ and covariance $\mathbf{C}$. In [Example \(A mixture of two correlations\)](#), the conditional distribution of Y_2 given Y_1 is a mixture, not a normal distribution, and its variance depends on y_1 .

Exercise B2.

Show that for a bivariate normal density with $|\rho| < 1$, the points of each contour ellipse at which the tangent is vertical lie on the line $y = \mathbb{E}(Y | X = x)$. Explain the result in terms of the conditional density of Y given $X = x$.

Solution. Write the ellipse as $\{\boldsymbol{\mu} + \mathbf{u} : \mathbf{u}^\top \boldsymbol{\Sigma}^{-1} \mathbf{u} = c^2\}$. At a point with vertical tangent, the normal vector $\boldsymbol{\Sigma}^{-1} \mathbf{u}$ is horizontal, so $\boldsymbol{\Sigma}^{-1} \mathbf{u} = \lambda(1, 0)^\top$ and $\mathbf{u} = \lambda(\sigma_X^2, \rho\sigma_X\sigma_Y)^\top$. The slope $u_2/u_1 = \rho\sigma_Y/\sigma_X$ is that of the conditional mean line, and both such points ($\lambda > 0$

and $\lambda < 0$) lie on it. Along the vertical line $X = x$, the joint density is proportional to the conditional density of Y , which is highest at $E(Y | X = x)$. The contour that just touches the vertical line touches it at the point of highest density on that line.

Exercise B3.

Let $Z_1, \dots, Z_n$ be independent $N(0, 1)$ and $S_k = Z_1 + \dots + Z_k$. Show that $(S_1, \dots, S_n)$ is jointly normal with $\text{Cov}(S_j, S_k) = \min(j, k)$. Find the conditional distribution of S_j given $S_n = s$, and comment on its variance as a function of j .

Solution. $(S_1, \dots, S_n)^\top = \mathbf{L}\mathbf{Z}$ with $\mathbf{L}$ lower triangular and all entries on and below the diagonal equal to one, so the vector is normal with covariance $\mathbf{L}\mathbf{L}^\top$, whose (j, k) entry counts the common indices, $\min(j, k)$. By [Theorem 3.5.2](#) with $\mathbf{Y}_1 = S_j$, $\mathbf{Y}_2 = S_n$,

$$S_j | S_n = s \sim N\left(\frac{j}{n}s, j - \frac{j^2}{n}\right) = N\left(\frac{j}{n}s, \frac{j(n-j)}{n}\right).$$

The conditional mean interpolates linearly between 0 and s . The variance is zero at both ends, where the value is known, and largest in the middle.

Exercise B4.

Let (Y, X_1, X_2) be normal with positive definite covariance, and let $\mathbf{Y}_2 = (X_1, X_2, X_1 + X_2)^\top$. Show that Σ_{22} is singular. Using a generalized inverse, show that the conditional distribution of Y given $\mathbf{Y}_2$ from [Theorem 3.5.2](#) coincides with its conditional distribution given (X_1, X_2) .

Exercise B5.

With $\Omega = \Sigma^{-1}$ partitioned like Σ , show that $E(\mathbf{Y}_1 | \mathbf{Y}_2 = \mathbf{y}_2) = \boldsymbol{\mu}_1 - \Omega_{11}^{-1}\Omega_{12}(\mathbf{y}_2 - \boldsymbol{\mu}_2)$ and $\text{Cov}(\mathbf{Y}_1 | \mathbf{Y}_2) = \Omega_{11}^{-1}$.

Solution. By [Theorem 1.6.4](#), $\Omega_{11} = \Sigma_{11.2}^{-1}$ and $\Omega_{12} = -\Sigma_{11.2}^{-1}\Sigma_{12}\Sigma_{22}^{-1}$. Hence $-\Omega_{11}^{-1}\Omega_{12} = \Sigma_{12}\Sigma_{22}^{-1}$ and $\Omega_{11}^{-1} = \Sigma_{11.2}$. Substitute into [Eq. 3.5.1](#).

3.5.5.3 C. Going deeper

Exercise C1.

Assume Σ is positive definite. Prove [Theorem 3.5.2\(b\)](#) by dividing the joint density by the marginal density of $\mathbf{Y}_2$ directly, using the partitioned inverse and the block determinant formula.

3.6 Partial correlation

Two variables can be strongly correlated only because both depend on a third. [Example \(Holding a third variable fixed\)](#) showed a positive association between Y_1 and Y_2 turning negative once Y_3 is held fixed. The conditional covariance matrix $\Sigma_{11 \cdot 2}$ measures association that remains after conditioning, and its correlation form is the partial correlation.

3.6.1 Definition and interpretations

Definition 3.6.1 — Partial correlation

Let $\mathbf{Y} = (\mathbf{Y}_1^\top, \mathbf{Y}_2^\top)^\top$ be a random vector with covariance matrix Σ , partitioned as in [Section 3.5](#), and write $\Sigma_{11 \cdot 2} = \Sigma_{11} - \Sigma_{12}\Sigma_{22}^{-1}\Sigma_{21} = (\sigma_{ij \cdot 2})$. The **partial covariance** of Y_i and Y_j (two entries of $\mathbf{Y}_1$) given $\mathbf{Y}_2$ is $\sigma_{ij \cdot 2}$, and their **partial correlation** given $\mathbf{Y}_2$ is

$$\rho_{ij \cdot 2} = \frac{\sigma_{ij \cdot 2}}{\sqrt{\sigma_{ii \cdot 2} \sigma_{jj \cdot 2}}},$$

provided $\sigma_{ii \cdot 2}$ and $\sigma_{jj \cdot 2}$ are positive. When the conditioning variables are $Y_k, Y_l, \dots$ we also write $\rho_{ij \cdot kl \dots}$. The matrix of partial correlations is $\mathbf{D}^{-1/2} \Sigma_{11 \cdot 2} \mathbf{D}^{-1/2}$ with $\mathbf{D} = \text{diag}(\Sigma_{11 \cdot 2})$, the correlation matrix ([Definition 2.4.1](#)) built from $\Sigma_{11 \cdot 2}$.

The definition uses only Σ , so partial correlations exist for any random vector with finite second moments. The next result gives them two meanings, one that always applies and one that needs normality.

Proposition 3.6.1

In the setting of [Definition 3.6.1](#):

- (a) $\Sigma_{11 \cdot 2}$ is the covariance matrix of the prediction errors $\mathbf{W} = \mathbf{Y}_1 - \boldsymbol{\mu}_1 - \Sigma_{12}\Sigma_{22}^{-1}(\mathbf{Y}_2 - \boldsymbol{\mu}_2)$, and $\mathbf{W}$ is uncorrelated with $\mathbf{Y}_2$. If Σ_{22} is positive definite, the entries of $\mathbf{W}$ are the errors of the best linear predictors $L(Y_i | \mathbf{Y}_2)$, so $\rho_{ij \cdot 2}$ is the correlation between $Y_i - L(Y_i | \mathbf{Y}_2)$ and $Y_j - L(Y_j | \mathbf{Y}_2)$.
- (b) If $\mathbf{Y}$ is jointly normal, $\rho_{ij \cdot 2}$ is the correlation of Y_i and Y_j in their conditional distribution given $\mathbf{Y}_2 = \mathbf{y}_2$, the same for every $\mathbf{y}_2$. Moreover $\rho_{ij \cdot 2} = 0$ iff Y_i and Y_j are conditionally independent given $\mathbf{Y}_2$.

Proof.

- (a) This is [Theorem 2.5.2](#), restated: $\text{Cov}(\mathbf{W}, \mathbf{Y}_2) = \mathbf{0}$ and $\text{Cov}(\mathbf{W}) = \Sigma_{11 \cdot 2}$ are the moment calculations in the proof of [Theorem 3.5.2\(a\)](#), which use no normality. When Σ_{22} is positive definite, the i th entry of $\boldsymbol{\mu}_1 + \Sigma_{12}\Sigma_{22}^{-1}(\mathbf{Y}_2 - \boldsymbol{\mu}_2)$ is $L(Y_i | \mathbf{Y}_2)$ by [Theorem 6.11.1](#).
- (b) By [Theorem 3.5.2](#), the conditional distribution of (Y_i, Y_j) given $\mathbf{Y}_2 = \mathbf{y}_2$ is bivariate normal with covariance matrix the corresponding 2×2 block of $\Sigma_{11 \cdot 2}$. Its correlation is $\rho_{ij \cdot 2}$, and by [Theorem 3.4.1](#) applied to this conditional normal distribution, the two are independent iff the off-diagonal entry is zero.

Part (a) is the definition to remember: *a partial correlation is the correlation of two residuals*. Remove from each variable the part that is linearly predictable from the conditioning variables, and correlate what is left. This is the population version of the Frisch–Waugh–Lovell construction ([Theorem 6.6.2](#)).

3.6.2 Computing partial correlations

Two routes avoid computing $\Sigma_{11 \cdot 2}$ in full.

With a single conditioning variable, $\Sigma_{11 \cdot 2}$ is a 2×2 matrix with entries $\sigma_{ij} - \sigma_{i3}\sigma_{j3}/\sigma_{33}$, and in correlation units this is the familiar three-variable formula of [Exercise \(B1\)](#),

$$\rho_{12 \cdot 3} = \frac{\rho_{12} - \rho_{13}\rho_{23}}{\sqrt{(1 - \rho_{13}^2)(1 - \rho_{23}^2)}}, \quad |\rho_{13}| < 1, \quad |\rho_{23}| < 1. \quad (3.6.1)$$

The residual interpretation shows that the same formula can be applied one conditioning variable at a time.

Proposition 3.6.2 — Recursion

Let S be a set of indices and $k \notin S \cup \{1, 2\}$ one more. Suppose $\sigma_{11 \cdot S}$, $\sigma_{22 \cdot S}$ and $\sigma_{kk \cdot S}$ are positive and $|\rho_{1k \cdot S}| < 1$, $|\rho_{2k \cdot S}| < 1$. Then

$$\sigma_{12 \cdot Sk} = \sigma_{12 \cdot S} - \frac{\sigma_{1k \cdot S} \sigma_{2k \cdot S}}{\sigma_{kk \cdot S}}, \quad \rho_{12 \cdot Sk} = \frac{\rho_{12 \cdot S} - \rho_{1k \cdot S} \rho_{2k \cdot S}}{\sqrt{(1 - \rho_{1k \cdot S}^2)(1 - \rho_{2k \cdot S}^2)}}.$$

With S empty this is [Eq. 3.6.1](#).

Proof. No normality is needed. For $i \in \{1, 2, k\}$ let e_i be the error of the best linear predictor of Y_i from $\mathbf{Y}_S$, as in [Proposition 3.6.1\(a\)](#). Then each e_i has mean zero, is uncorrelated with $\mathbf{Y}_S$, and $\text{Cov}(e_i, e_j) = \sigma_{ij \cdot S}$. Put $r_i = e_i - c_i e_k$ with $c_i = \sigma_{ik \cdot S}/\sigma_{kk \cdot S}$, for $i = 1, 2$.

We claim that r_i is the error of the best linear predictor of Y_i from $(\mathbf{Y}_S, Y_k)$. First, $Y_i - r_i$ is an affine function of $(\mathbf{Y}_S, Y_k)$, because e_i and e_k differ from Y_i and Y_k by affine functions of $\mathbf{Y}_S$. Second, r_i has mean zero and is uncorrelated with $\mathbf{Y}_S$ and with e_k , by the choice of c_i , hence with $Y_k = e_k + (\text{affine in } \mathbf{Y}_S)$. Now let r'_i be the error from [Proposition 3.6.1\(a\)](#) for predicting Y_i from $(\mathbf{Y}_S, Y_k)$, which has the same two properties. The difference $d = r_i - r'_i$ is an affine function of $(\mathbf{Y}_S, Y_k)$ with mean zero and is uncorrelated with $(\mathbf{Y}_S, Y_k)$, so $\text{Var}(d) = \text{Cov}(d, d) = 0$. Thus $r_i = r'_i$ with probability one.

Therefore $\sigma_{12 \cdot Sk} = \text{Cov}(r_1, r_2) = \sigma_{12 \cdot S} - c_2 \sigma_{1k \cdot S} - c_1 \sigma_{2k \cdot S} + c_1 c_2 \sigma_{kk \cdot S}$, which simplifies to the first formula, and similarly $\sigma_{ii \cdot Sk} = \sigma_{ii \cdot S}(1 - \rho_{ik \cdot S}^2)$, which is positive by hypothesis. Dividing $\sigma_{12 \cdot Sk}$ by $\sqrt{\sigma_{11 \cdot Sk} \sigma_{22 \cdot Sk}}$ and then numerator and denominator by $\sqrt{\sigma_{11 \cdot S} \sigma_{22 \cdot S}}$ gives the second.

Proposition 3.6.3 — Partial correlations from the precision matrix

Let Σ be positive definite and $\Omega = \Sigma^{-1} = (\omega_{ij})$. The partial correlation of Y_i and Y_j given all the other entries of $\mathbf{Y}$ is

$$\rho_{ij \cdot \text{rest}} = -\frac{\omega_{ij}}{\sqrt{\omega_{ii}\omega_{jj}}},$$

and the conditional variance of Y_i given all the others is $1/\omega_{ii}$.

Proof. Permute $\mathbf{Y}$ so that (Y_i, Y_j) comes first. By [Theorem 3.5.2\(c\)](#) (a statement about matrices, valid whether or not $\mathbf{Y}$ is normal), the leading 2×2 block of Ω is $\Sigma_{11 \cdot 2}^{-1}$. Inverting,

$$\Sigma_{11 \cdot 2} = \begin{pmatrix} \omega_{ii} & \omega_{ij} \\ \omega_{ij} & \omega_{jj} \end{pmatrix}^{-1} = \frac{1}{\omega_{ii}\omega_{jj} - \omega_{ij}^2} \begin{pmatrix} \omega_{jj} & -\omega_{ij} \\ -\omega_{ij} & \omega_{ii} \end{pmatrix},$$

whose correlation is $-\omega_{ij}/\sqrt{\omega_{ii}\omega_{jj}}$. The same argument with a 1×1 leading block gives the conditional variance. ■

[Proposition 3.6.3](#) has a striking consequence for normal vectors: *a zero entry ω_{ij} of the inverse covariance matrix means that Y_i and Y_j are conditionally independent given all the other variables.* Zeros in Σ describe marginal independence, zeros in Σ^{-1} describe conditional independence. The pattern of zeros in Σ^{-1} defines a graph on the variables, which is the starting point of Gaussian graphical models (Dempster 1972; Lauritzen 1996).

Example. (The trivariate example, continued)

For Σ of [Example \(Holding a third variable fixed\)](#), the listing gives the covariance of (Y_1, Y_2) given Y_3 as $\begin{pmatrix} 2 & -1 \\ -1 & 1 \end{pmatrix}$. So $\rho_{12} = 0.289$ while $\rho_{12 \cdot 3} = -0.707$. The script confirms that [Eq. 3.6.1](#), [Proposition 3.6.3](#), and the correlation of simulated residuals all give this value.

3.6.3 Sample partial correlations

For data, replace Σ by the sample covariance matrix. Because the empirical distribution of the rows is a probability distribution in its own right ([Section 6.11](#)), all the identities above hold exactly for the sample versions. In particular, for two variables $\mathbf{y}, \mathbf{x}$ and conditioning variables collected with an intercept in $\mathbf{Z}$, the following numbers are *identical*:

- (1) the correlation of the residuals of $\mathbf{y}$ and of $\mathbf{x}$ after least squares regression on $\mathbf{Z}$;
- (2) the recursion [Eq. 3.6.1](#) applied to sample correlations (one conditioning variable) or repeatedly (several);
- (3) $-w_{12}/\sqrt{w_{11}w_{22}}$ computed from the inverse of the sample covariance or correlation matrix of $(\mathbf{y}, \mathbf{x}, \text{conditioning variables})$;
- (4) $\text{sign}(t)\sqrt{t^2/(t^2 + n - p)}$, where t is the t statistic of $\mathbf{x}$ in the regression of $\mathbf{y}$ on $[\mathbf{Z}, \mathbf{x}]$ with p columns ([Exercise \(B3\)](#)).

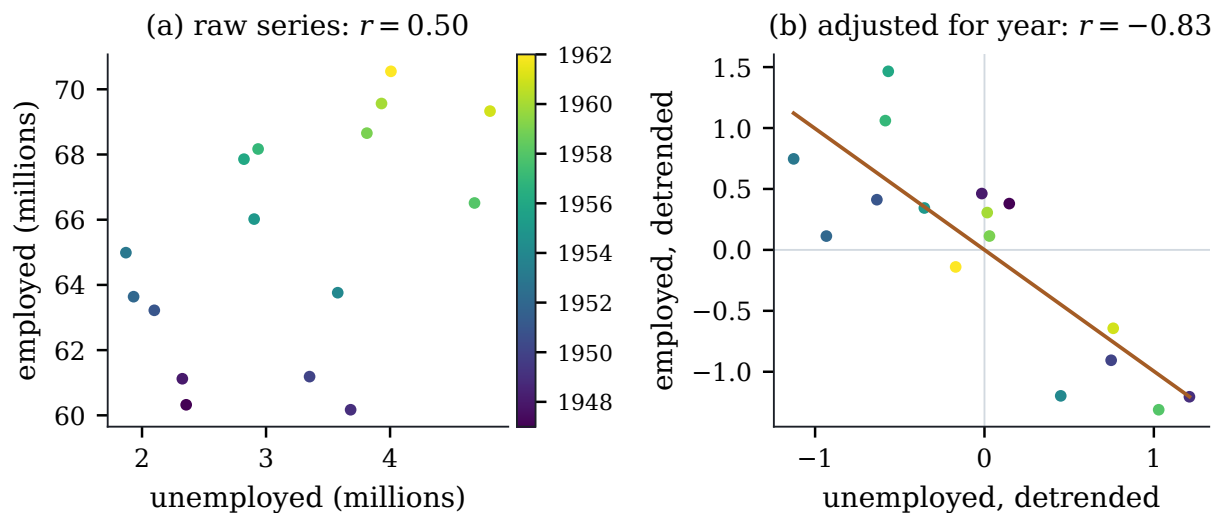


Figure 3.6.1: Employment and unemployment in the United States, 1947–1962, coloured by year. (a) The raw series are positively correlated because both trend upwards. (b) After removing the linear trend in year from each, the residuals are strongly negatively correlated. Their correlation is the partial correlation given year, and the fitted slope is the multiple-regression coefficient.

Example. (Employment and unemployment)

Longley’s macroeconomic series record, among other things, total employment and the number unemployed in the United States for the 16 years 1947–1962. Their correlation is 0.502: years with more people employed had more people unemployed. Both series grew with the population. Their correlations with the calendar year are 0.971 and 0.668. Adjusting for year, the partial correlation is -0.829 . Once the common trend is removed, a year with unemployment above its trend is a year with employment below its trend, as one would expect. Figure 3.6.1 shows both scatterplots. The listing computes the partial correlation in all four ways above. They agree to ten decimal places. The slope in panel (b), -0.996 , is the coefficient of unemployment in the regression of employment on unemployment and year, as Theorem 6.6.2 requires. Its t statistic is -5.34 .

Removing a linear time trend before correlating two series is the problem that Frisch and Waugh (1933) solved, and it is a special case of partial correlation.

```
import numpy as np
import statsmodels.api as sm
data = sm.datasets.longley.load_pandas().data

emp = data["TOTEMP"].to_numpy() / 1000      # thousands -> millions of persons
unemp = data["UNEMP"].to_numpy() / 1000
year = data["YEAR"].to_numpy()
n = len(emp)
Z = np.column_stack([np.ones(n), year])     # the variables we adjust for

def resid(v, Z):
    coef, *_ = np.linalg.lstsq(Z, v, rcond=None)
    return v - Z @ coef

# 1. correlation of residuals (detrended series)
r_resid = np.corrcoef(resid(emp, Z), resid(unemp, Z))[0, 1]
```



```

# 2. the one-variable recursion from ordinary correlations
R = np.corrcoef(np.vstack([emp, unemp, year]))
r12, r13, r23 = R[0, 1], R[0, 2], R[1, 2]
r_formula = (r12 - r13 * r23) / np.sqrt((1 - r13**2) * (1 - r23**2))

# 3. from the inverse of the correlation (or covariance) matrix
W = np.linalg.inv(R)
r_precision = -W[0, 1] / np.sqrt(W[0, 0] * W[1, 1])

# 4. from the t statistic of unemployment in the regression of emp on (1, unemp,
    year)
fit = sm.OLS(emp, np.column_stack([Z, unemp])).fit()
t = fit.tvalues[2]
r_t = np.sign(t) * np.sqrt(t**2 / (t**2 + n - 3))

print(f"marginal r = {r12:.4f}")
print(f"partial r:  {r_resid:.10f} {r_formula:.10f} {r_precision:.10f} {r_t:.10f}")

```

Remark (Adjusting is not always right). A partial correlation answers a precise question: how are the parts of two variables not linearly explained by $\mathbf{Y}_2$ related? Whether that is the *right* question depends on what $\mathbf{Y}_2$ is. Adjusting for a common cause, as the year stands in for population growth here, removes a spurious association. Adjusting for a common effect can *create* an association between variables that are independent. Chapter 25 treats the difference. Under normality, the sampling distribution of a sample partial correlation given q variables is that of an ordinary sample correlation from $n - q$ observations, a result used for tests in [Chapter 14](#).

3.6.4 Exercises

3.6.4.1 A. Check your understanding

Exercise A1.

Show that $\rho_{12 \cdot 3} = 0$ iff $\rho_{12} = \rho_{13}\rho_{23}$. Find $\rho_{12 \cdot 3}$ when $\rho_{12} = 0$ and $\rho_{13} = \rho_{23} = 0.6$, and interpret.

3.6.4.2 B. Practice

Exercise B1.

Let X , Z and e be independent, with $X, Z \sim N(0, 1)$ and $e \sim N(0, \tau^2)$, and let $C = X + Z + e$. Show that $\rho_{XZ} = 0$ but $\rho_{XZ \cdot C} = -1/(1 + \tau^2)$. What happens as $\tau \rightarrow 0$ and as $\tau \rightarrow \infty$?

Solution. The covariance matrix of (X, Z, C) is $\begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 1 & 2 + \tau^2 \end{pmatrix}$. Conditioning on C ,

$$\sigma_{XZ \cdot C} = 0 - \frac{1}{2 + \tau^2}, \quad \sigma_{XX \cdot C} = \sigma_{ZZ \cdot C} = 1 - \frac{1}{2 + \tau^2} = \frac{1 + \tau^2}{2 + \tau^2},$$

so $\rho_{XZ \cdot C} = -1/(1 + \tau^2)$. As $\tau \rightarrow 0$ it tends to -1 : if $X + Z$ is known exactly, a larger

X forces a smaller Z . As $\tau \rightarrow \infty$ it tends to 0, since C carries little information about $X + Z$. Two independent causes become dependent once their common effect is held fixed.

Exercise B2.

Let Σ be positive definite. Show that the coefficient of Y_j in the best linear predictor of Y_1 from all the other entries is $-\omega_{1j}/\omega_{11}$. Deduce that $\rho_{1j \cdot \text{rest}}^2$ equals the product of the coefficient of Y_j for predicting Y_1 and the coefficient of Y_1 for predicting Y_j , each from all the other variables.

Solution. Put Y_1 first. By [Exercise \(B5\)](#) with a scalar first block, the coefficient vector of the best linear predictor of Y_1 from the rest is $-\Omega_{12}/\omega_{11}$, whose j th entry is $-\omega_{1j}/\omega_{11}$. Likewise the coefficient of Y_1 in the predictor of Y_j is $-\omega_{j1}/\omega_{jj}$. Their product is $\omega_{1j}^2/(\omega_{11}\omega_{jj})$, which is $\rho_{1j \cdot \text{rest}}^2$ by [Proposition 3.6.3](#).

Exercise B3.

Let $\mathbf{D}$ be an $n \times k$ data matrix whose columns are centred, and let $\mathbf{S} = \mathbf{D}^\top \mathbf{D}$. Show, using [Theorem 6.6.2](#) and the partitioned inverse, that $-w_{12}/\sqrt{w_{11}w_{22}}$, computed from $\mathbf{S}^{-1} = (w_{ij})$, equals the correlation of the residuals of columns 1 and 2 after regression on the remaining columns.

Exercise B4.

Using the Longley data of [Example \(Employment and unemployment\)](#), compute the partial correlation of employment and unemployment given both year and GNP, in at least two of the four ways. Draw the corresponding residual scatterplot. Does adjusting for GNP as well as year change the conclusion?

3.7 Summary and Notes

3.7.1 Summary

Idea.

1. $\mathbf{Y}$ is multivariate normal when every linear combination $\mathbf{a}^\top \mathbf{Y}$ is normal, constants included ([Definition 3.1.1](#)). The definition covers singular covariance matrices, which regression needs for fitted values and residuals.
2. $\boldsymbol{\mu} + \mathbf{AZ}$, with $\mathbf{Z}$ standard normal, is $N(\boldsymbol{\mu}, \mathbf{AA}^\top)$. Every mean and every nonnegative definite covariance occurs ([Theorem 3.1.2](#)).
3. The moment generating function is $\exp(\mathbf{t}^\top \boldsymbol{\mu} + \frac{1}{2} \mathbf{t}^\top \Sigma \mathbf{t})$, so a normal distribution is determined by $\boldsymbol{\mu}$ and Σ ([Theorem 3.2.1](#)). When Σ is positive definite there is a density whose contours are the ellipsoids of constant Mahalanobis distance, and $\Delta^2(\mathbf{Y}) \sim \chi^2(n)$ ([Theorem 3.2.2](#) and [Proposition 3.2.3](#)).

4. Affine maps of normal vectors are normal, with no rank condition, and so are all subvectors ([Theorem 3.3.1](#) and [Corollary 3.3.2](#)). A normal vector of rank r is an exact affine image of r independent standard normals ([Proposition 3.3.4](#)).
5. Normal margins do not imply joint normality, and zero correlation between normal variables does not imply independence without it ([Example \(A random sign\)](#) and [Example \(A mixture of two correlations\)](#)).
6. For jointly normal blocks, zero cross-covariance is equivalent to independence. $\mathbf{AY}$ and $\mathbf{BY}$ are independent iff $\mathbf{A}\Sigma\mathbf{B}^\top = \mathbf{0}$ ([Theorem 3.4.1](#) and [Corollary 3.4.2](#)).
7. $\mathbf{Y}_1 \mid \mathbf{Y}_2 = \mathbf{y}_2 \sim N(\boldsymbol{\mu}_1 + \Sigma_{12}\Sigma_{22}^-(\mathbf{y}_2 - \boldsymbol{\mu}_2), \Sigma_{11.2})$: linear mean, constant covariance, reduced variance ([Theorem 3.5.2](#)). The conditional mean is the best predictor of all, and coincides with the best linear predictor ([Proposition 3.5.3](#)).
8. The partial correlation is the correlation of the residuals from linear prediction on the conditioning variables. Under normality it is the conditional correlation, and it is zero iff the variables are conditionally independent ([Definition 3.6.1](#) and [Proposition 3.6.1](#)). It can be computed by a recursion, from the inverse covariance matrix, or by FWL residuals ([Proposition 3.6.2](#) and [Proposition 3.6.3](#)).

3.7.2 Notes and sources

Three definitions. Linear-model texts reach the multivariate normal by different routes, and comparing them explains the choices made here. Rencher and Schaalje (2008, chapter 4) start from the positive definite case, obtain the density by transforming independent standard normals with $\Sigma^{1/2}$, and prove the properties of linear functions and independence with moment generating functions, for full-rank transformations. Seber and Lee (2003, chapter 2) also begin with the density, then extend the definition to singular covariance matrices through the representation $\mathbf{AZ} + \boldsymbol{\mu}$ and prove that normality of all linear combinations characterizes the family. Their moment-generating-function proof of the independence theorem is the model for [Section 3.4](#). Christensen (2020, §1.2) takes $\mathbf{AZ} + \boldsymbol{\mu}$ as the definition from the outset, shows with characteristic functions that the law depends only on $\boldsymbol{\mu}$ and Σ , and proves the independence theorem by building an explicit independent version. Defining the distribution through linear combinations, as this chapter does, goes back at least to Rao (1973, chapter 8). It has the advantage that the theorem on affine transformations needs no rank condition and a one-line proof.

Cramér–Wold. The device of reducing a vector to its linear combinations is from Cramér and Wold (1936), where it was used to reduce multivariate limit theorems to univariate ones. [Chapter 2](#) proves it, together with the uniqueness and factorization properties of moment generating functions used in [Section 3.2](#). Billingsley (1995) is the reference for these facts and for the conditioning property of independent variables used in the proof of [Theorem 3.5.2](#).

Conditional distributions and partial correlation. The proof of [Theorem 3.5.2](#) through the independent error $\mathbf{W}$ borrows one idea from Rencher and Schaalje (2008) and Seber and Lee (2003): both pass through the transformed vector $\mathbf{Y}_1 - \mathbf{BY}_2$, which is independent of $\mathbf{Y}_2$, although both still obtain the conditional density as a ratio of densities in the positive definite case. The proof here uses only independence and a generalized inverse, so it covers singular Σ_{22} , an extension that is standard in multivariate analysis (Rao 1973). Anderson (2003) is the classical reference for the multivariate normal in statistics, including the distribution theory of sample partial and multiple correlation coefficients that [Chapter 14](#) uses. Rencher and Schaalje (2008, §4.5) define partial correlation as a conditional correlation, as in [Proposition 3.6.1\(b\)](#).

The residual interpretation (a) does not require normality and is the one that connects to least squares. Regression toward the mean was described by Galton (1886). The link between zeros of the inverse covariance matrix and conditional independence was made the basis of a modelling strategy by Dempster (1972), and Lauritzen (1996) develops the theory of graphical models that grew from it.

Data. [Example \(Employment and unemployment\)](#) uses the macroeconomic series published by Longley (1967) as a test of numerical accuracy, available in `statsmodels`. The same series appear in [Chapter 2](#), as an example of sample moments, and in [Chapter 6](#), to illustrate ill-conditioning.

3.7.3 References

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Chapter 4

Quadratic Forms and Their Distributions

Every sum of squares in a linear model is a quadratic form $\mathbf{Y}^\top \mathbf{A} \mathbf{Y}$ in a normal vector, and every F and t statistic is a ratio built from such forms. This chapter works out their laws: the noncentral chi-squared, F and t distributions and the monotonicity of their tails; the mean, variance and a canonical representation of an arbitrary quadratic form; exactly when a quadratic form is chi-squared; when two forms, or a form and a linear function, are independent; and Cochran's theorem, which says when a sum of squares splits into independent chi-squared pieces.

What you need. [Chapter 1](#) (spectral theorem, symmetric idempotent matrices, trace), [Chapter 2](#) (means and covariances of random vectors) and [Chapter 3](#) (the multivariate normal distribution and its independence properties). Orthogonal projections appear here only as symmetric idempotent matrices. [Chapter 6](#) develops their geometry.

4.0.1 Roadmap

- [Noncentral chi-squared, t and F distributions](#)
- [Mean and variance of quadratic forms](#)
- [When a quadratic form is chi-squared](#)
- [Independence of quadratic forms and linear forms](#)
- [The Cochran theorem](#)
- [Summary and notes](#)

4.1 Noncentral chi-squared, t and F distributions

A regression analysis reports a handful of numbers whose sampling behaviour decides everything: the residual sum of squares, the reduction in it when a term is added, the ratio of the two. Each

is a function of the response vector of the form

$$\mathbf{Y}^\top \mathbf{A} \mathbf{Y} = \sum_{i=1}^n \sum_{j=1}^n a_{ij} Y_i Y_j,$$

a **quadratic form** in $\mathbf{Y}$. The sample variance is one, with $\mathbf{A} = (n-1)^{-1}(\mathbf{I} - n^{-1}\mathbf{1}\mathbf{1}^\top)$. So is the squared length $\|\mathbf{P}\mathbf{Y}\|^2 = \mathbf{Y}^\top \mathbf{P} \mathbf{Y}$ of any symmetric idempotent image of $\mathbf{Y}$, and this is how the sums of squares of [Chapter 6](#) will arise.

Since $\mathbf{Y}^\top \mathbf{A} \mathbf{Y}$ is a scalar, it equals its own transpose $\mathbf{Y}^\top \mathbf{A}^\top \mathbf{Y}$, and so also $\mathbf{Y}^\top \frac{1}{2}(\mathbf{A} + \mathbf{A}^\top) \mathbf{Y}$. Replacing $\mathbf{A}$ by its symmetric part changes nothing, and *throughout this chapter the matrix of a quadratic form is symmetric*. Several formulas below, the variance formula among them, are false for a non-symmetric $\mathbf{A}$ ([Exercise \(A1\)](#)).

When $\mathbf{Y} \sim N_n(\mathbf{0}, \mathbf{I})$ and $\mathbf{A}$ is symmetric idempotent, $\mathbf{Y}^\top \mathbf{A} \mathbf{Y}$ has a chi-squared distribution, which is the null distribution of a test. Under an alternative the mean is not zero, and a slightly larger family appears. We start with that family.

4.1.1 Noncentral chi-squared

Recall that $\chi^2(r)$, the chi-squared distribution with r degrees of freedom, is the law of $Z_1^2 + \dots + Z_r^2$ for independent standard normal Z_i . It is the gamma distribution with shape $r/2$ and scale 2, with density

$$g_r(x) = \frac{x^{r/2-1} e^{-x/2}}{2^{r/2} \Gamma(r/2)}, \quad x > 0, \quad (4.1.1)$$

mean r , variance $2r$ and moment generating function $(1-2t)^{-r/2}$ for $t < 1/2$. By convention $\chi^2(0)$ is the point mass at zero. Everything about the noncentral version follows from one computation.

Lemma 4.1.1

Let $W \sim N(0, 1)$ and $\nu \in \mathbb{R}$. For $t < 1/2$,

$$\mathbb{E}[e^{t(W+\nu)^2}] = (1-2t)^{-1/2} \exp\left(\frac{\nu^2 t}{1-2t}\right).$$

Proof. With $u = w + \nu$ the expectation is $(2\pi)^{-1/2} \int \exp\{tu^2 - (u-\nu)^2/2\} du$. Completing the square in the exponent,

$$tu^2 - \frac{1}{2}(u-\nu)^2 = -\frac{1-2t}{2} \left(u - \frac{\nu}{1-2t}\right)^2 + \frac{\nu^2}{2} \left(\frac{1}{1-2t} - 1\right).$$

The last term equals $\nu^2 t / (1-2t)$. The remaining Gaussian integral is $(2\pi)^{1/2} (1-2t)^{-1/2}$ when $1-2t > 0$. ■

Definition 4.1.1 — Noncentral chi-squared

Let $\mathbf{Z} \sim N_r(\boldsymbol{\mu}, \mathbf{I}_r)$. The distribution of $\mathbf{Z}^\top \mathbf{Z} = \sum_i Z_i^2$ is the **noncentral chi-squared distribution** with r degrees of freedom and **noncentrality** $\gamma = \boldsymbol{\mu}^\top \boldsymbol{\mu} = \|\boldsymbol{\mu}\|^2$, written $\chi^2(r, \gamma)$.

The definition hides a claim: the law of $\|\mathbf{Z}\|^2$ depends on the mean vector $\boldsymbol{\mu}$ only through its length. [Theorem 4.1.2](#) proves it. Intuitively, a rotation $\mathbf{Q}\mathbf{Z}$ of a spherical normal is again spherical normal, with mean $\mathbf{Q}\boldsymbol{\mu}$, and it has the same length as $\mathbf{Z}$, so every mean vector of a given length gives the same answer.

Remark (Convention). In this book the noncentrality is $\gamma = \boldsymbol{\mu}^\top \boldsymbol{\mu}$, with no factor $\frac{1}{2}$. Then $E \chi^2(r, \gamma) = r + \gamma$, which is easy to remember, and γ agrees with the parameter `nc` of `scipy.stats.ncx2`. Several texts, including Rencher and Schaalje (2008) and Christensen (2020), use $\lambda = \boldsymbol{\mu}^\top \boldsymbol{\mu}/2$ instead. When reading them, replace their λ by $\gamma/2$. The same convention applies to the noncentral F distribution below.

Theorem 4.1.2 — Properties of the noncentral chi-squared

Let $U \sim \chi^2(r, \gamma)$.

- (a) The moment generating function is

$$M_U(t) = (1 - 2t)^{-r/2} \exp\left(\frac{\gamma t}{1 - 2t}\right), \quad t < \frac{1}{2}, \quad (4.1.2)$$

so the law depends on $\boldsymbol{\mu}$ only through γ , and $\chi^2(r, 0) = \chi^2(r)$.

- (b) $E(U) = r + \gamma$ and $\text{Var}(U) = 2r + 4\gamma$.
 (c) If $U_1, \dots, U_k$ are independent with $U_i \sim \chi^2(r_i, \gamma_i)$, then $\sum_i U_i \sim \chi^2(\sum_i r_i, \sum_i \gamma_i)$.
 (d) (*Poisson mixture.*) Let $K \sim \text{Poisson}(\gamma/2)$ and, given $K = k$, let $V \sim \chi^2(r + 2k)$. Then $V \sim \chi^2(r, \gamma)$. Consequently U has density

$$f_{r,\gamma}(x) = \sum_{k=0}^{\infty} e^{-\gamma/2} \frac{(\gamma/2)^k}{k!} g_{r+2k}(x), \quad x > 0. \quad (4.1.3)$$

Proof.

- (a) With $\mathbf{Z} \sim N_r(\boldsymbol{\mu}, \mathbf{I})$, the Z_i are independent, and $Z_i = W_i + \mu_i$ with $W_i \sim N(0, 1)$. By [Lemma 4.1.1](#),

$$E e^{t\mathbf{Z}^\top \mathbf{Z}} = \prod_{i=1}^r (1 - 2t)^{-1/2} \exp\left(\frac{\mu_i^2 t}{1 - 2t}\right) = (1 - 2t)^{-r/2} \exp\left(\frac{\gamma t}{1 - 2t}\right).$$

A moment generating function that is finite on an open interval around zero determines the distribution, so the law depends only on (r, γ) . Setting $\gamma = 0$ gives the central mgf.

- (b) The cumulant generating function is $K(t) = \log M_U(t) = -\frac{r}{2} \log(1 - 2t) + \gamma t/(1 - 2t)$. Differentiating, $K'(t) = r(1 - 2t)^{-1} + \gamma(1 - 2t)^{-2}$ and $K''(t) = 2r(1 - 2t)^{-2} + 4\gamma(1 - 2t)^{-3}$. The mean and variance are $K'(0)$ and $K''(0)$.

- (c) By independence the mgf of the sum is the product of the mgfs [Eq. 4.1.2](#), which has the same form with $\sum r_i$ and $\sum \gamma_i$.
- (d) Conditioning on K ,

$$\mathbb{E} e^{tV} = \sum_{k=0}^{\infty} e^{-\gamma/2} \frac{(\gamma/2)^k}{k!} (1-2t)^{-(r+2k)/2} = (1-2t)^{-r/2} e^{-\gamma/2} \exp\left(\frac{\gamma/2}{1-2t}\right),$$

and $-\frac{\gamma}{2} + \frac{\gamma}{2(1-2t)} = \frac{\gamma t}{1-2t}$. This is [Eq. 4.1.2](#). The density of a mixture is the mixture of the densities. ■

The mixture in (d) is more than a formula. It says that noncentrality acts like a random number of extra degrees of freedom, two at a time, with $\gamma/2$ of them on average. Since each extra pair of degrees of freedom pushes the distribution to the right, a larger γ should give a stochastically larger variable. That is the next result, and it is what makes tests based on these distributions sensible.

Example. (A noncentral chi-squared with four degrees of freedom)

Take $r = 4$ and $\gamma = 6$. The Poisson(3) weights of the central densities g_4, g_6, g_8, g_{10} in [Eq. 4.1.3](#) are 0.050, 0.149, 0.224 and 0.224, and the mean and variance are 10 and 32. The listing checks the mixture against a library implementation of the density. The script also squares normal vectors with a mean of length $\sqrt{6}$ in an arbitrary direction, and their average squared length is 10.002. A test that rejects when a $\chi^2(4)$ statistic exceeds its upper 5% point 9.488 rejects with probability 0.470 when the statistic is in fact $\chi^2(4, 6)$. [Figure 4.1.1](#) shows how the density moves right and spreads out as γ grows.

```
import numpy as np
from scipy import stats
rng = np.random.default_rng(20240401)

def ncx2_pdf_mixture(x, r, gamma, terms=200):
    """Density of chi^2(r, gamma) as a Poisson(gamma/2) mixture of central densities
    . """
    k = np.arange(terms)
    weights = stats.poisson.pmf(k, gamma / 2)          # P(K = k)
    dens = stats.chi2.pdf(np.asarray(x)[: , None], r + 2 * k)
    return dens @ weights

r, gamma = 4, 6.0
x = np.linspace(0.05, 40, 400)
print(np.max(np.abs(ncx2_pdf_mixture(x, r, gamma) - stats.ncx2.pdf(x, r, gamma))))
```

Proposition 4.1.3 — Monotonicity in the noncentrality

For fixed $r \geq 1$ and $x > 0$, the tail probability $\gamma \mapsto \Pr\{\chi^2(r, \gamma) > x\}$ is continuous and strictly increasing on $[0, \infty)$, and tends to 1 as $\gamma \rightarrow \infty$.

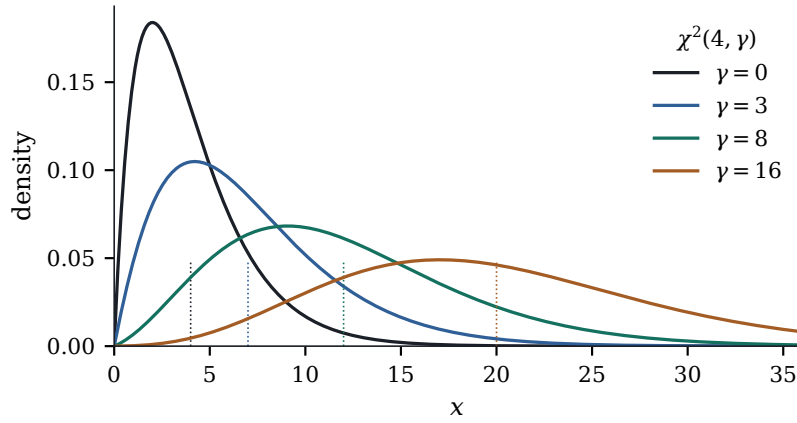


Figure 4.1.1: Densities of $\chi^2(4, \gamma)$. Dotted ticks mark the means $4 + \gamma$. The variance $2r + 4\gamma$ grows with the noncentrality, so the distributions flatten as they move right.

Proof. Let $h(k) = \Pr\{\chi^2(r + 2k) > x\}$ and $p_k(\lambda) = e^{-\lambda}\lambda^k/k!$. By Theorem 4.1.2(d), the tail probability is $\phi(\lambda) = \sum_k p_k(\lambda)h(k)$ with $\lambda = \gamma/2$. First, h is strictly increasing: if $V \sim \chi^2(r + 2k)$ and $E \sim \chi^2(2)$ are independent, then $V + E \sim \chi^2(r + 2k + 2)$ and $h(k + 1) - h(k) = \Pr(V \leq x < V + E) > 0$, since (V, E) has a positive joint density on $(0, \infty)^2$. Next, $p'_k(\lambda) = p_{k-1}(\lambda) - p_k(\lambda)$ (with $p_{-1} = 0$). Because $0 \leq h \leq 1$, the series of derivatives converges uniformly on bounded intervals, so term-by-term differentiation is valid and

$$\phi'(\lambda) = \sum_{k \geq 0} (p_{k-1}(\lambda) - p_k(\lambda))h(k) = \sum_{k \geq 0} p_k(\lambda)(h(k+1) - h(k)) > 0.$$

For the limit, fix $\varepsilon > 0$. Since $\chi^2(r + 2m)$ has mean $r + 2m$ and standard deviation $\sqrt{2r + 4m}$, Chebyshev's inequality gives $h(m) > 1 - \varepsilon$ for some m . Then $\phi(\lambda) \geq h(m) \Pr(K \geq m)$ for $K \sim \text{Poisson}(\lambda)$, and $\Pr(K \geq m) \rightarrow 1$ as $\lambda \rightarrow \infty$. ■

4.1.2 Noncentral F and t

Test statistics divide one quadratic form by another, to remove the unknown scale. The two ratio distributions that result are defined next.

Definition 4.1.2 — Noncentral F

Let $U \sim \chi^2(r, \gamma)$ and $V \sim \chi^2(s)$ be independent, $r, s \geq 1$. The law of

$$F = \frac{U/r}{V/s}$$

is the **noncentral F distribution** $F(r, s, \gamma)$, with numerator and denominator degrees of freedom r and s and noncentrality γ . The central $F(r, s) = F(r, s, 0)$ has upper α point $F_\alpha(r, s)$.

Definition 4.1.3 — Noncentral t

Let $X \sim N(\delta, 1)$ and $V \sim \chi^2(s)$ be independent. The law of $T = X/\sqrt{V/s}$ is the **noncentral t distribution** $t(s, \delta)$, and $t(s) = t(s, 0)$.

Only the numerator is allowed to be noncentral. A noncentral denominator gives a *doubly noncentral* F , which appears when the model used for the error sum of squares is itself wrong (Exercise (B3)). Two relations are immediate. First, $X^2 \sim \chi^2(1, \delta^2)$, so $T^2 \sim F(1, s, \delta^2)$: a two-sided t test is an F test with one numerator degree of freedom. Second, since $E(V^{-1}) = 1/(s-2)$ for $s > 2$, independence gives

$$E(F) = \frac{r + \gamma}{r} \cdot \frac{s}{s - 2}, \quad s > 2.$$

The central F is the null distribution of the tests of Chapter 11, and $F(r, s, \gamma)$ is their distribution under an alternative. The power of such a test is therefore $\Pr\{F(r, s, \gamma) > F_\alpha(r, s)\}$, and the following fact is what guarantees that it behaves as a test should.

Theorem 4.1.4 — Power of the F test increases with γ

For fixed $r, s \geq 1$ and $c > 0$, $\beta(\gamma) = \Pr\{F(r, s, \gamma) > c\}$ is continuous and strictly increasing in $\gamma \geq 0$, and $\beta(\gamma) \rightarrow 1$ as $\gamma \rightarrow \infty$. In particular, the level- α test that rejects when $F > F_\alpha(r, s)$ has power greater than α at every $\gamma > 0$.

Proof. With U, V as in Definition 4.1.2, $F > c$ iff $U > crV/s$. Condition on V , which is independent of U :

$$\beta(\gamma) = E[\psi_\gamma(crV/s)], \quad \psi_\gamma(x) = \Pr\{\chi^2(r, \gamma) > x\}.$$

Since $V > 0$ with probability one, Proposition 4.1.3 gives, for $\gamma < \gamma'$, $\psi_\gamma(crV/s) < \psi_{\gamma'}(crV/s)$ almost surely, so $\beta(\gamma) < \beta(\gamma')$. Continuity and the limit follow from the same proposition by dominated convergence, since $0 \leq \psi_\gamma \leq 1$. ■

Two further monotonicity facts hold for fixed $\gamma > 0$ and fixed level. The power decreases as the numerator degrees of freedom r grow, and increases as the denominator degrees of freedom s grow (Ghosh 1973). The first says that spreading a fixed amount of signal over more dimensions makes it harder to detect. The second says that estimating σ^2 better helps. We use both only qualitatively, and Figure 4.4.1 in Section 4.4 shows the first. Because $T^2 \sim F(1, s, \delta^2)$, Theorem 4.1.4 also shows that the power of the two-sided t test increases with $|\delta|$ (Exercise (A2)).

4.1.3 Exercises**4.1.3.1 A. Check your understanding**

Exercise A1.

Show that $\chi^2(2)$ is the exponential distribution with mean 2. Use [Theorem 4.1.2\(d\)](#) to show that

$$\Pr\{\chi^2(2, \gamma) > x\} = \sum_{k \geq 0} e^{-\gamma/2} \frac{(\gamma/2)^k}{k!} e^{-x/2} \sum_{j=0}^k \frac{(x/2)^j}{j!}.$$

Exercise A2.

Show that $T \sim t(s, \delta)$ implies $T^2 \sim F(1, s, \delta^2)$. Deduce that the power of the two-sided level- α t test depends on δ only through $|\delta|$ and increases with it. Show directly, by conditioning on the denominator, that the power of the one-sided test that rejects for $T > c$ increases with δ .

Solution. $T^2 = X^2/(V/s)$ with $X^2 \sim \chi^2(1, \delta^2)$ independent of V , which is [Definition 4.1.2](#). The two-sided test rejects when $|T| > t_{\alpha/2}(s)$, that is, when $T^2 > F_{\alpha}(1, s)$. Its power is therefore $\Pr\{F(1, s, \delta^2) > F_{\alpha}(1, s)\}$, a strictly increasing function of δ^2 by [Theorem 4.1.4](#). For the one-sided test, $\Pr(T > c) = \mathbb{E} \Pr\{X > c\sqrt{V/s} \mid V\} = \mathbb{E} \Phi(\delta - c\sqrt{V/s})$, and Φ is strictly increasing.

4.1.3.2 B. Practice**Exercise B1.**

Show that the k th cumulant of $\chi^2(r, \gamma)$ is $2^{k-1}(k-1)!(r+k\gamma)$. Deduce that its skewness tends to zero like $3/\sqrt{\gamma}$ as $\gamma \rightarrow \infty$ with r fixed.

Solution. Using $-\frac{1}{2} \log(1-2t) = \sum_{k \geq 1} (2t)^k/(2k)$ and $t/(1-2t) = \sum_{k \geq 1} 2^{k-1}t^k$, the cumulant generating function is

$$K(t) = \sum_{k \geq 1} t^k 2^{k-1} \left(\frac{r}{k} + \gamma \right) = \sum_{k \geq 1} \frac{t^k}{k!} 2^{k-1} (k-1)! (r+k\gamma).$$

The k th cumulant is $k!$ times the coefficient of t^k . In particular $\kappa_2 = 2(r+2\gamma)$ and $\kappa_3 = 8(r+3\gamma)$, so the skewness is $\kappa_3/\kappa_2^{3/2} = 8(r+3\gamma)/\{2(r+2\gamma)\}^{3/2}$. As $\gamma \rightarrow \infty$ this behaves like $24\gamma/(4\gamma)^{3/2} = 3/\sqrt{\gamma}$.

Exercise B2.

Show that $\chi^2(1, \gamma)$ has density $(2\pi x)^{-1/2} e^{-(x+\gamma)/2} \cosh \sqrt{\gamma x}$ for $x > 0$. Verify that expanding $\cosh$ in a power series reproduces [Eq. 4.1.3](#). *Hint:* $\Gamma(k+\frac{1}{2}) = \sqrt{\pi} (2k)!/(4^k k!)$.

Solution. Let $X = (W + \sqrt{\gamma})^2$ with $W \sim N(0, 1)$. For $x > 0$, $X \leq x$ iff $-\sqrt{x} - \sqrt{\gamma} \leq W \leq \sqrt{x} - \sqrt{\gamma}$. Differentiating with ϕ the standard normal density,

$$f(x) = \frac{\phi(\sqrt{x} - \sqrt{\gamma}) + \phi(\sqrt{x} + \sqrt{\gamma})}{2\sqrt{x}} = \frac{e^{-(x+\gamma)/2}}{\sqrt{2\pi x}} \cdot \frac{e^{\sqrt{\gamma x}} + e^{-\sqrt{\gamma x}}}{2}.$$

Expanding, $f(x) = (2\pi x)^{-1/2} e^{-(x+\gamma)/2} \sum_k (\gamma x)^k / (2k)!$. The k th term of Eq. 4.1.3 is

$$\frac{e^{-\gamma/2} (\gamma/2)^k x^{k-1/2} e^{-x/2}}{k! 2^{k+1/2} \Gamma(k + \frac{1}{2})}.$$

By the hint the denominator is $k! 2^{k+1/2} \sqrt{\pi} (2k)! / (4^k k!) = \sqrt{2\pi} 2^{-k} (2k)!$, and the k th term becomes $(2\pi x)^{-1/2} e^{-(x+\gamma)/2} (\gamma x)^k / (2k)!$, as required.

Exercise B3.

Let $U_\gamma \sim \chi^2(r, \gamma)$ with r fixed. Show that $(U_\gamma - r - \gamma) / \sqrt{2r + 4\gamma}$ converges in distribution to $N(0, 1)$ as $\gamma \rightarrow \infty$. *Hint:* U_γ has the law of $(W_1 + \sqrt{\gamma})^2 + W_2^2 + \cdots + W_r^2$ with $W_1, \dots, W_r$ independent standard normal.

Exercise B4.

Let $U \sim \chi^2(a)$ and $V \sim \chi^2(b)$ be independent. Show that $B = U/(U + V)$ has the $\text{Beta}(a/2, b/2)$ distribution and is independent of $U + V$, and that $F = (U/a)/(V/b) = (b/a) B/(1 - B)$. Deduce the null distribution of R^2 asked for in Chapter 6.

Solution. The joint density of (U, V) is proportional to $u^{a/2-1} v^{b/2-1} e^{-(u+v)/2}$. Change variables to $w = u/(u + v) \in (0, 1)$ and $s = u + v > 0$, so that $u = ws$, $v = (1 - w)s$ and the Jacobian is s . The density of (W, S) is proportional to

$$w^{a/2-1} (1 - w)^{b/2-1} \cdot s^{(a+b)/2-1} e^{-s/2},$$

a product of a function of w and a function of s . So B and $U + V$ are independent, $B \sim \text{Beta}(a/2, b/2)$ and $U + V \sim \chi^2(a + b)$. Also $U/V = B/(1 - B)$. For R^2 in a model with intercept, $R^2 = \text{SSR}/(\text{SSR} + \text{SSE})$, where $\text{SSR} = \|(\mathbf{M} - \mathbf{P}_1)\mathbf{Y}\|^2$ and $\text{SSE} = \|(\mathbf{I} - \mathbf{M})\mathbf{Y}\|^2$ come from orthogonal projections of ranks $r - 1$ and $n - r$. If $\boldsymbol{\mu} = \mu\mathbf{1}$, both noncentralities vanish, and Theorem 4.4.4 with $a = r - 1$, $b = n - r$ gives $R^2 \sim \text{Beta}((r - 1)/2, (n - r)/2)$.

Exercise B5.

Show that $f_{r,\gamma}(x)/g_r(x)$ is strictly increasing in $x > 0$ for every $\gamma > 0$. Conclude that among tests based on one observation $U \sim \chi^2(r, \gamma)$ of $\gamma = 0$ against a fixed $\gamma > 0$, rejecting for large U is most powerful.

4.2 Mean and variance of quadratic forms

4.2.1 The mean needs no normality

Theorem 2.3.1 computed $E(\mathbf{Y}^\top \mathbf{A} \mathbf{Y})$ for any random vector with mean $\boldsymbol{\mu}$ and covariance $\boldsymbol{\Sigma}$:

$$E(\mathbf{Y}^\top \mathbf{A} \mathbf{Y}) = \text{tr}(\mathbf{A} \boldsymbol{\Sigma}) + \boldsymbol{\mu}^\top \mathbf{A} \boldsymbol{\mu}. \quad (4.2.1)$$

The formula has two parts with distinct roles. The trace term is what the noise contributes. The term $\boldsymbol{\mu}^\top \mathbf{A} \boldsymbol{\mu}$ is what the signal contributes. For nonnegative definite $\mathbf{A}$ it vanishes exactly when $\mathbf{A} \boldsymbol{\mu} = \mathbf{0}$, which describes the mean vectors a well-designed estimator is meant to ignore.

Example. (Two estimators of σ^2 under a drifting mean)

Let $Y_1, \dots, Y_n$ be uncorrelated with common variance σ^2 , and consider

$$S^2 = \frac{1}{n-1} \sum_{i=1}^n (Y_i - \bar{Y})^2, \quad Q = \frac{1}{2(n-1)} \sum_{i=1}^{n-1} (Y_{i+1} - Y_i)^2,$$

the sample variance and the **mean square successive difference**. In matrix form $S^2 = \mathbf{Y}^\top \mathbf{C} \mathbf{Y} / (n-1)$ with $\mathbf{C} = \mathbf{I} - n^{-1} \mathbf{1} \mathbf{1}^\top$, and $Q = \mathbf{Y}^\top \mathbf{D}^\top \mathbf{D} \mathbf{Y} / \{2(n-1)\}$, where $\mathbf{D}$ is the $(n-1) \times n$ first-difference matrix with rows $(\dots, -1, 1, \dots)$. Both matrices annihilate $\mathbf{1}$, so both estimators ignore a constant mean. Since $\text{tr}(\mathbf{C}) = n-1$ and $\text{tr}(\mathbf{D}^\top \mathbf{D}) = \text{tr}(\mathbf{D} \mathbf{D}^\top) = 2(n-1)$, Eq. 4.2.1 with $\boldsymbol{\Sigma} = \sigma^2 \mathbf{I}$ shows that both are unbiased when the mean is constant.

Now let the mean drift linearly, $\mu_i = \mu_0 + \theta i$. Then $\mathbf{C} \boldsymbol{\mu}$ is the centred trend and $\mathbf{D} \boldsymbol{\mu} = \theta \mathbf{1}_{n-1}$, so

$$E(S^2) = \sigma^2 + \theta^2 \frac{n(n+1)}{12}, \quad E(Q) = \sigma^2 + \frac{\theta^2}{2}.$$

The bias of S^2 grows quadratically with the length of the series, while that of Q does not grow at all. With $n = 20$, $\sigma = 1$ and $\theta = 0.1$, the means are 1.350 and 1.005. Differencing removes smooth trends, and this is why Q was proposed for series whose level may wander (von Neumann 1941). The price of robustness to drift is paid in variance, which we compute below.

4.2.2 A canonical representation

Higher moments and distributions need normality, and one representation answers all such questions at once.

Theorem 4.2.1 — Canonical form of a normal quadratic form

Let $\mathbf{Y} \sim N_n(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ and let $\mathbf{A}$ be symmetric. Write $\boldsymbol{\Sigma} = \mathbf{L} \mathbf{L}^\top$ with $\mathbf{L}$ of size $n \times k$, and take a spectral decomposition $\mathbf{L}^\top \mathbf{A} \mathbf{L} = \mathbf{Q} \boldsymbol{\Lambda} \mathbf{Q}^\top$ with $\boldsymbol{\Lambda} = \text{diag}(\lambda_1, \dots, \lambda_k)$. Put $\mathbf{b} = \mathbf{Q}^\top \mathbf{L}^\top \mathbf{A} \boldsymbol{\mu}$. Then

$$\mathbf{Y}^\top \mathbf{A} \mathbf{Y} \stackrel{d}{=} \boldsymbol{\mu}^\top \mathbf{A} \boldsymbol{\mu} + \sum_{j=1}^k (\lambda_j W_j^2 + 2b_j W_j), \quad \mathbf{W} \sim N_k(\mathbf{0}, \mathbf{I}_k). \quad (4.2.2)$$

For $\lambda_j \neq 0$ the j th term equals $\lambda_j (W_j + b_j/\lambda_j)^2 - b_j^2/\lambda_j$. The nonzero λ_j are the nonzero

eigenvalues of $\mathbf{A}\Sigma$.

Proof. Such an $\mathbf{L}$ exists. By [Theorem 1.7.2](#), $\Sigma = \mathbf{U}\mathbf{D}\mathbf{U}^\top$ with $\mathbf{U}$ orthogonal and $\mathbf{D}$ diagonal with entries $d_i \geq 0$, and $\mathbf{L} = \mathbf{U}\mathbf{D}^{1/2}$ works with $k = n$. Let $\mathbf{Z} \sim N_k(\mathbf{0}, \mathbf{I})$. The vector $\boldsymbol{\mu} + \mathbf{L}\mathbf{Z}$ is normal with mean $\boldsymbol{\mu}$ and covariance $\mathbf{L}\mathbf{L}^\top = \Sigma$ ([Theorem 3.3.1](#)). A normal distribution is determined by its mean and covariance ([Theorem 3.2.1](#)), so $\mathbf{Y}$ and $\boldsymbol{\mu} + \mathbf{L}\mathbf{Z}$ have the same law, and so do $\mathbf{Y}^\top \mathbf{A}\mathbf{Y}$ and

$$(\boldsymbol{\mu} + \mathbf{L}\mathbf{Z})^\top \mathbf{A}(\boldsymbol{\mu} + \mathbf{L}\mathbf{Z}) = \boldsymbol{\mu}^\top \mathbf{A}\boldsymbol{\mu} + 2\boldsymbol{\mu}^\top \mathbf{A}\mathbf{L}\mathbf{Z} + \mathbf{Z}^\top \mathbf{L}^\top \mathbf{A}\mathbf{L}\mathbf{Z}.$$

Put $\mathbf{W} = \mathbf{Q}^\top \mathbf{Z}$, again $N_k(\mathbf{0}, \mathbf{I})$. Then $\mathbf{Z}^\top \mathbf{L}^\top \mathbf{A}\mathbf{L}\mathbf{Z} = \mathbf{W}^\top \boldsymbol{\Lambda} \mathbf{W} = \sum \lambda_j W_j^2$ and $\boldsymbol{\mu}^\top \mathbf{A}\mathbf{L}\mathbf{Z} = \mathbf{b}^\top \mathbf{W}$, which is [Eq. 4.2.2](#). The completed square is algebra. Finally, $\mathbf{L}^\top \mathbf{A}\mathbf{L} = \mathbf{H}\mathbf{F}$ with $\mathbf{F} = \mathbf{A}\mathbf{L}$ and $\mathbf{H} = \mathbf{L}^\top$, and $\mathbf{F}\mathbf{H} = \mathbf{A}\Sigma$. By [Proposition 1.7.1\(d\)](#) these two products have the same nonzero eigenvalues, with the same multiplicities. ■

So a normal quadratic form is a constant plus a sum of independent pieces. Each piece is either a scaled noncentral $\chi^2(1)$ variable, with scale λ_j , or a normal variable with mean zero, when $\lambda_j = 0$ but $b_j \neq 0$. Because the λ_j may be negative, $\mathbf{Y}^\top \mathbf{A}\mathbf{Y}$ need not be positive. The linear pieces are special to singular covariances. If Σ is positive definite, $\mathbf{L}$ can be taken square and nonsingular, and then $\lambda_j = 0$ gives $\mathbf{A}\mathbf{L}\mathbf{q}_j = \mathbf{0}$ for the j th column $\mathbf{q}_j$ of $\mathbf{Q}$, so $b_j = (\mathbf{A}\mathbf{L}\mathbf{q}_j)^\top \boldsymbol{\mu} = 0$. A linear piece needs a direction in which $\mathbf{A}$ responds to the mean while the noise is absent.

Corollary 4.2.2 — Moment generating function

In the setting of [Theorem 4.2.1](#), for $|t| < 1/(2 \max_j |\lambda_j|)$ (any t if all $\lambda_j = 0$),

$$\mathbb{E} e^{t\mathbf{Y}^\top \mathbf{A}\mathbf{Y}} = e^{t\boldsymbol{\mu}^\top \mathbf{A}\boldsymbol{\mu}} \prod_{j=1}^k (1 - 2\lambda_j t)^{-1/2} \exp\left(\frac{2b_j^2 t^2}{1 - 2\lambda_j t}\right).$$

Proof. The terms of [Eq. 4.2.2](#) are independent. If $\lambda_j = 0$, the term $2b_j W_j$ has mgf $e^{2b_j^2 t^2}$. If $\lambda_j \neq 0$, apply [Lemma 4.1.1](#) to $W_j + \nu_j$ with $\nu_j = b_j/\lambda_j$ at the argument $\lambda_j t$, and multiply by $e^{-tb_j^2/\lambda_j}$. The exponent is $b_j^2 t / \{\lambda_j(1 - 2\lambda_j t)\} - b_j^2 t / \lambda_j = 2b_j^2 t^2 / (1 - 2\lambda_j t)$. ■

4.2.3 Variance and covariances

Theorem 4.2.3 — Moments of normal quadratic forms

Let $\mathbf{Y} \sim N_n(\boldsymbol{\mu}, \Sigma)$, let $\mathbf{A}$ and $\mathbf{B}$ be symmetric $n \times n$ matrices, and let $\mathbf{K}$ be an $m \times n$ matrix. Then

- (a) $\mathbb{E}(\mathbf{Y}^\top \mathbf{A}\mathbf{Y}) = \text{tr}(\mathbf{A}\Sigma) + \boldsymbol{\mu}^\top \mathbf{A}\boldsymbol{\mu}$;
- (b) $\text{Var}(\mathbf{Y}^\top \mathbf{A}\mathbf{Y}) = 2 \text{tr}\{(\mathbf{A}\Sigma)^2\} + 4\boldsymbol{\mu}^\top \mathbf{A}\Sigma \mathbf{A}\boldsymbol{\mu}$;


```

def qf_moments(A, mu, Sigma):
    """Mean and variance of  $Y^T A Y$  for  $Y \sim N(\mu, \Sigma)$ ,  $A$  symmetric."""
    AS = A @ Sigma
    mean = np.trace(AS) + mu @ A @ mu
    var = 2 * np.trace(AS @ AS) + 4 * mu @ AS @ A @ mu
    return mean, var

C = np.eye(n) - np.ones((n, n)) / n          # centring matrix
D = np.diff(np.eye(n), axis=0)              # (n-1) x n first differences
A_s2 = C / (n - 1)
A_q = D.T @ D / (2 * (n - 1))
for name, A in [("S2", A_s2), ("Q", A_q)]:
    print(name, qf_moments(A, mu, Sigma), qf_moments(A, 0 * mu, Sigma))

```

Normality enters (b)–(d) through the third and fourth moments of $\mathbf{Y}$. For independent coordinates with excess kurtosis the variance acquires an extra term that involves only the diagonal of $\mathbf{A}$ (Theorem 2.3.3). Variance formulas for sums of squares derived under normality can be badly wrong for heavy-tailed errors, a point taken up in Chapter 21.

4.2.4 Exercises

4.2.4.1 A. Check your understanding

Exercise A1.

Let $\mathbf{Y} \sim N_2(\mathbf{0}, \mathbf{I})$ and $\mathbf{A} = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$. Compute $\text{Var}(\mathbf{Y}^\top \mathbf{A} \mathbf{Y})$ directly, and compare with $2 \text{tr}(\mathbf{A}^2)$ and with $2 \text{tr}(\mathbf{A}_s^2)$, where $\mathbf{A}_s = \frac{1}{2}(\mathbf{A} + \mathbf{A}^\top)$.

4.2.4.2 B. Practice

Exercise B1.

Let $(U_i, W_i), i = 1, \dots, n$, be independent copies of a bivariate normal pair with variances σ_U^2, σ_W^2 and covariance σ_{UW} , and let $s_{UW} = (n-1)^{-1} \sum_i (U_i - \bar{U})(W_i - \bar{W})$. Write $s_{UW} = \mathbf{Y}^\top \mathbf{A} \mathbf{Y}$ for the stacked vector $\mathbf{Y} = (\mathbf{U}^\top, \mathbf{W}^\top)^\top$ and a symmetric $2n \times 2n$ matrix $\mathbf{A}$, and use Theorem 4.2.3 to show that $\text{Var}(s_{UW}) = (\sigma_U^2 \sigma_W^2 + \sigma_{UW}^2) / (n-1)$. Check the answer against $\text{Var}(S^2)$ when $\mathbf{U} = \mathbf{W}$.

Exercise B2.

Let $Y_1, \dots, Y_n$ be independent with mean 0, variance σ^2 and $E Y_i^4 = \mu_4$. By Exercise (B4), for symmetric $\mathbf{A}$ and $\mathbf{B}$,

$$\text{Cov}(\mathbf{Y}^\top \mathbf{A} \mathbf{Y}, \mathbf{Y}^\top \mathbf{B} \mathbf{Y}) = (\mu_4 - 3\sigma^4) \sum_i a_{ii} b_{ii} + 2\sigma^4 \text{tr}(\mathbf{A} \mathbf{B}).$$

Show that in a one-way layout with $g \geq 2$ groups, not all of size one, the between-group and within-group sums of squares are correlated unless $\mu_4 = 3\sigma^4$. Which sign does the correlation have for heavy-tailed errors?

Solution. In the one-way layout the between-group matrix is $\mathbf{A} = \mathbf{M} - \mathbf{P}_1$ and the within-group matrix is $\mathbf{B} = \mathbf{I} - \mathbf{M}$. They satisfy $\mathbf{AB} = \mathbf{0}$, so $\text{tr}(\mathbf{AB}) = 0$ and the covariance reduces to $(\mu_4 - 3\sigma^4) \sum_i a_{ii} b_{ii}$. An observation in group ℓ has $a_{ii} = 1/n_\ell - 1/n$ and $b_{ii} = 1 - 1/n_\ell$, so

$$\sum_i a_{ii} b_{ii} = \sum_\ell n_\ell \left(\frac{1}{n_\ell} - \frac{1}{n} \right) \left(1 - \frac{1}{n_\ell} \right) = \sum_\ell \left(1 - \frac{n_\ell}{n} \right) \left(1 - \frac{1}{n_\ell} \right).$$

With $g \geq 2$ every first factor is positive, and the second is positive for any group with two or more observations. So the sum is positive, and the covariance vanishes iff $\mu_4 = 3\sigma^4$. For heavy-tailed errors, $\mu_4 > 3\sigma^4$, the two sums of squares are positively correlated: a large error inflates both.

Exercise B3.

Let $\mathbf{Y} \sim N_n(\boldsymbol{\mu}, \sigma^2 \mathbf{I})$, and let $\mathbf{M}$ be symmetric idempotent of rank r with $\mathbf{M}\boldsymbol{\mu} \neq \boldsymbol{\mu}$, as when a regression model omits a needed regressor. Show that $\|(\mathbf{I} - \mathbf{M})\mathbf{Y}\|^2/\sigma^2 \sim \chi^2(n - r, \|(\mathbf{I} - \mathbf{M})\boldsymbol{\mu}\|^2/\sigma^2)$, and find the bias of $\|(\mathbf{I} - \mathbf{M})\mathbf{Y}\|^2/(n - r)$ as an estimator of σ^2 . In the setting of [Theorem 4.4.6](#) without the assumption $\mathbf{P}\boldsymbol{\mu} = \boldsymbol{\mu}$, which parts of the conclusion survive?

Exercise B4.

Let $\mathbf{Y} \sim N_n(\mathbf{0}, \boldsymbol{\Sigma})$. Use [Corollary 4.2.2](#) and [Proposition 1.7.1\(d\)](#) to show that $E e^{t\mathbf{Y}^\top \mathbf{A} \mathbf{Y}} = \det(\mathbf{I}_n - 2t\mathbf{A}\boldsymbol{\Sigma})^{-1/2}$ for $|t|$ small.

4.2.4.3 C. Going deeper

Exercise C1.

Let $\mathbf{Y} \sim N_n(\mu \mathbf{1}, \sigma^2 \mathbf{I})$ with μ and σ^2 unknown. Among symmetric $\mathbf{A}$ with $E(\mathbf{Y}^\top \mathbf{A} \mathbf{Y}) = \sigma^2$ for all μ and σ^2 , show that $\mathbf{A} = \mathbf{C}/(n - 1)$, which gives S^2 , uniquely minimizes $\text{Var}(\mathbf{Y}^\top \mathbf{A} \mathbf{Y})$ at every parameter value. (Compare [Exercise \(B5\)](#), where the mean is known to be zero.)

Solution. Since $E(\mathbf{Y}^\top \mathbf{A} \mathbf{Y}) = \sigma^2 \text{tr} \mathbf{A} + \mu^2 \mathbf{1}^\top \mathbf{A} \mathbf{1}$, unbiasedness means $\text{tr} \mathbf{A} = 1$ and $\mathbf{1}^\top \mathbf{A} \mathbf{1} = 0$. By [Theorem 4.2.3](#), $\text{Var}(\mathbf{Y}^\top \mathbf{A} \mathbf{Y}) = 2\sigma^4 \text{tr}(\mathbf{A}^2) + 4\mu^2 \sigma^2 \|\mathbf{A} \mathbf{1}\|^2$. The map $\mathbf{A} \mapsto \mathbf{C} \mathbf{A} \mathbf{C}$ is an orthogonal projection for the inner product $\text{tr}(\mathbf{A} \mathbf{B})$ on symmetric matrices, so $\text{tr}(\mathbf{A}^2) \geq \text{tr}\{(\mathbf{C} \mathbf{A} \mathbf{C})^2\}$, with equality iff $\mathbf{A} = \mathbf{C} \mathbf{A} \mathbf{C}$. Also $\text{tr}(\mathbf{C} \mathbf{A} \mathbf{C}) = \text{tr} \mathbf{A} - \mathbf{1}^\top \mathbf{A} \mathbf{1}/n = 1$. The symmetric matrix $\mathbf{C} \mathbf{A} \mathbf{C}$ has rank at most $n - 1$. If $\lambda_1, \dots, \lambda_{n-1}$ are its eigenvalues on $\mathbf{1}^\perp$, the Cauchy-Schwarz inequality gives $1 = (\sum \lambda_i)^2 \leq (n - 1) \sum \lambda_i^2$. Hence $\text{tr}(\mathbf{A}^2) \geq 1/(n - 1)$, with equality iff $\mathbf{A} = \mathbf{C} \mathbf{A} \mathbf{C}$ and all $\lambda_i = 1/(n - 1)$, that is, iff $\mathbf{A} = \mathbf{C}/(n - 1)$. That matrix also has $\mathbf{A} \mathbf{1} = \mathbf{0}$, so it minimizes both variance terms at once.

4.3 When a quadratic form is chi-squared

A quadratic form is a weighted sum of independent noncentral $\chi^2(1)$ pieces, and the weights are eigenvalues. It is itself chi-squared when every nonzero weight equals one. This section turns that observation into checkable conditions on $\mathbf{A}$ and Σ , and shows that for the spherical case the condition is necessary as well as sufficient.

4.3.1 Spherical normal vectors

Theorem 4.3.1 — Chi-squared forms in a spherical normal vector

Let $\mathbf{Y} \sim N_n(\boldsymbol{\mu}, \mathbf{I}_n)$, let $\mathbf{A}$ be symmetric, and let $r \geq 1$ be an integer and $\gamma \geq 0$. Then $\mathbf{Y}^\top \mathbf{A} \mathbf{Y} \sim \chi^2(r, \gamma)$ if and only if $\mathbf{A}$ is idempotent with $\text{rank}(\mathbf{A}) = r$ and $\gamma = \boldsymbol{\mu}^\top \mathbf{A} \boldsymbol{\mu}$.

Proof. Sufficiency. By Theorem 1.9.2, $\mathbf{A} = \mathbf{Q}_1 \mathbf{Q}_1^\top$, where the $n \times r$ matrix $\mathbf{Q}_1$ has orthonormal columns and $r = \text{rank}(\mathbf{A}) = \text{tr}(\mathbf{A})$. Then $\mathbf{Y}^\top \mathbf{A} \mathbf{Y} = \|\mathbf{Q}_1^\top \mathbf{Y}\|^2$ and $\mathbf{Q}_1^\top \mathbf{Y} \sim N_r(\mathbf{Q}_1^\top \boldsymbol{\mu}, \mathbf{Q}_1^\top \mathbf{Q}_1) = N_r(\mathbf{Q}_1^\top \boldsymbol{\mu}, \mathbf{I}_r)$, so by definition $\mathbf{Y}^\top \mathbf{A} \mathbf{Y} \sim \chi^2(r, \|\mathbf{Q}_1^\top \boldsymbol{\mu}\|^2)$, and $\|\mathbf{Q}_1^\top \boldsymbol{\mu}\|^2 = \boldsymbol{\mu}^\top \mathbf{A} \boldsymbol{\mu}$.

Necessity. Apply Theorem 4.2.1 with $\mathbf{L} = \mathbf{I}$, so that $\mathbf{A} = \mathbf{Q} \boldsymbol{\Lambda} \mathbf{Q}^\top$. Let $\mathbf{v} = \mathbf{Q}^\top \boldsymbol{\mu}$. Then $\mathbf{b} = \mathbf{Q}^\top \mathbf{A} \boldsymbol{\mu} = \boldsymbol{\Lambda} \mathbf{v}$, so $b_j = \lambda_j \nu_j$. The constant left after completing squares is $\boldsymbol{\mu}^\top \mathbf{A} \boldsymbol{\mu} - \sum_{\lambda_j \neq 0} \lambda_j \nu_j^2 = \mathbf{v}^\top \boldsymbol{\Lambda} \mathbf{v} - \sum_j \lambda_j \nu_j^2 = 0$. Hence $\mathbf{Y}^\top \mathbf{A} \mathbf{Y}$ has the law of $\sum_{j \in J} \lambda_j (W_j + \nu_j)^2$, where J indexes the nonzero eigenvalues. J is not empty, because a $\chi^2(r, \gamma)$ variable with $r \geq 1$ is not identically zero.

By Lemma 4.1.1, near $t = 0$ the logarithm of the mgf of this sum is $\sum_{j \in J} \{-\frac{1}{2} \log(1 - 2\lambda_j t) + \lambda_j \nu_j^2 t / (1 - 2\lambda_j t)\}$. Expanding $-\frac{1}{2} \log(1 - x) = \sum_{k \geq 1} x^k / (2k)$ and $x / (1 - x) = \sum_{k \geq 1} x^k$, the coefficient of t^k is $(2^{k-1}/k) \sum_{j \in J} \lambda_j^k (1 + k \nu_j^2)$. The same expansion of Eq. 4.1.2 gives $(2^{k-1}/k)(r + k\gamma)$. Equal distributions have equal mgfs, and two power series that agree on an interval have equal coefficients, so

$$\sum_{j \in J} \lambda_j^k (1 + k \nu_j^2) = r + k\gamma \quad \text{for every } k \geq 1. \quad (4.3.1)$$

Let $\rho = \max_{j \in J} |\lambda_j|$. For even k every term on the left is nonnegative and one of them is at least ρ^k . The right side grows only linearly in k , so $\rho \leq 1$. Split J into $J_+ = \{\lambda_j = 1\}$, $J_- = \{\lambda_j = -1\}$ and $J_0 = \{0 < |\lambda_j| < 1\}$, and write $N_\pm = |J_\pm|$ and $S_\pm = \sum_{j \in J_\pm} \nu_j^2$. The terms over J_0 tend to zero as $k \rightarrow \infty$, so

$$N_+ + (-1)^k N_- + k(S_+ + (-1)^k S_-) - r - k\gamma \rightarrow 0.$$

Dividing by k and letting $k \rightarrow \infty$ through even and through odd values gives $S_+ + S_- = \gamma = S_+ - S_-$. Then the same limit without the factor k gives $N_+ + N_- = r = N_+ - N_-$. So $N_- = S_- = 0$, $N_+ = r$ and $S_+ = \gamma$. The terms over J_+ in Eq. 4.3.1 now account for all of $r + k\gamma$, and with $k = 2$ the remaining sum $\sum_{j \in J_0} \lambda_j^2 (1 + 2\nu_j^2)$ is zero. Every term of that sum is positive, so J_0 is empty. Thus every eigenvalue of $\mathbf{A}$ is 0 or 1, $\mathbf{A}$ is idempotent, $\text{rank}(\mathbf{A}) = N_+ = r$, and $\boldsymbol{\mu}^\top \mathbf{A} \boldsymbol{\mu} = \sum_{j \in J_+} \nu_j^2 = \gamma$.

The necessity half is the one that makes the theorem a tool for *recognizing* distributions. If a quadratic form in a spherical normal vector is chi-squared at all, then it is the squared length of a projection. There is no other way.

4.3.2 General covariance matrices

Theorem 4.3.2 — When a quadratic form is chi-squared

Let $\mathbf{Y} \sim N_n(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ and let $\mathbf{A}$ be symmetric.

- (a) (*Projections.*) If $\boldsymbol{\Sigma} = \sigma^2 \mathbf{I}$ with $\sigma > 0$ and $\mathbf{P}$ is symmetric and idempotent, then

$$\frac{\|\mathbf{PY}\|^2}{\sigma^2} = \frac{\mathbf{Y}^\top \mathbf{PY}}{\sigma^2} \sim \chi^2(\text{rank}(\mathbf{P}), \frac{\|\mathbf{P}\boldsymbol{\mu}\|^2}{\sigma^2}).$$

- (b) (*Positive definite $\boldsymbol{\Sigma}$.*) If $\boldsymbol{\Sigma}$ is positive definite, then $\mathbf{Y}^\top \mathbf{AY}$ has a $\chi^2(r, \gamma)$ distribution for some $r \geq 1$ and $\gamma \geq 0$ iff $\mathbf{A}\boldsymbol{\Sigma}$ is idempotent and $\mathbf{A} \neq \mathbf{0}$. In that case $r = \text{rank}(\mathbf{A}) = \text{tr}(\mathbf{A}\boldsymbol{\Sigma})$ and $\gamma = \boldsymbol{\mu}^\top \mathbf{A}\boldsymbol{\mu}$.

- (c) (*Any $\boldsymbol{\Sigma}$.*) If

$$(i) \ \boldsymbol{\Sigma} \mathbf{A} \boldsymbol{\Sigma} \mathbf{A} \boldsymbol{\Sigma} = \boldsymbol{\Sigma} \mathbf{A} \boldsymbol{\Sigma}, \quad (ii) \ \boldsymbol{\Sigma} \mathbf{A} \boldsymbol{\Sigma} \mathbf{A} \boldsymbol{\mu} = \boldsymbol{\Sigma} \mathbf{A} \boldsymbol{\mu}, \quad (iii) \ \boldsymbol{\mu}^\top \mathbf{A} \boldsymbol{\Sigma} \mathbf{A} \boldsymbol{\mu} = \boldsymbol{\mu}^\top \mathbf{A} \boldsymbol{\mu},$$

then $\mathbf{Y}^\top \mathbf{AY} \sim \chi^2(\text{tr}(\mathbf{A}\boldsymbol{\Sigma}), \boldsymbol{\mu}^\top \mathbf{A} \boldsymbol{\mu})$. The three conditions hold in particular when $\mathbf{A}\boldsymbol{\Sigma} \mathbf{A} = \mathbf{A}$, and when $\mathbf{A}\boldsymbol{\Sigma}$ is idempotent and $\boldsymbol{\mu} \in \mathcal{C}(\boldsymbol{\Sigma})$.

Proof.

- (a) $\sigma^{-1} \mathbf{Y} \sim N_n(\sigma^{-1} \boldsymbol{\mu}, \mathbf{I})$. Apply Theorem 4.3.1 with $\mathbf{A} = \mathbf{P}$, using $\mathbf{P}^\top \mathbf{P} = \mathbf{P}$ to write $\mathbf{Y}^\top \mathbf{PY} = \|\mathbf{PY}\|^2$ and $\boldsymbol{\mu}^\top \mathbf{P} \boldsymbol{\mu} = \|\mathbf{P}\boldsymbol{\mu}\|^2$.
- (b) Let $\boldsymbol{\Sigma}^{1/2}$ be the positive definite square root (Theorem 1.7.6). Then $\mathbf{Z} = \boldsymbol{\Sigma}^{-1/2} \mathbf{Y} \sim N_n(\boldsymbol{\Sigma}^{-1/2} \boldsymbol{\mu}, \mathbf{I})$ and $\mathbf{Y}^\top \mathbf{AY} = \mathbf{Z}^\top \mathbf{BZ}$ with $\mathbf{B} = \boldsymbol{\Sigma}^{1/2} \mathbf{A} \boldsymbol{\Sigma}^{1/2}$ symmetric. By Theorem 4.3.1, $\mathbf{Y}^\top \mathbf{AY}$ is noncentral chi-squared iff $\mathbf{B}$ is idempotent and nonzero. Now $\mathbf{B}^2 = \boldsymbol{\Sigma}^{1/2} \mathbf{A} \boldsymbol{\Sigma} \mathbf{A} \boldsymbol{\Sigma}^{1/2}$, so $\mathbf{B}^2 = \mathbf{B}$ iff $\mathbf{A} \boldsymbol{\Sigma} \mathbf{A} = \mathbf{A}$ (multiply by $\boldsymbol{\Sigma}^{-1/2}$ on both sides) iff $\mathbf{A} \boldsymbol{\Sigma} \mathbf{A} \boldsymbol{\Sigma} = \mathbf{A} \boldsymbol{\Sigma}$ (multiply by $\boldsymbol{\Sigma}$ or $\boldsymbol{\Sigma}^{-1}$ on the right). Also $\mathbf{B} \neq \mathbf{0}$ iff $\mathbf{A} \neq \mathbf{0}$. The degrees of freedom are $\text{rank}(\mathbf{B}) = \text{rank}(\mathbf{A})$, which equals $\text{tr}(\mathbf{B}) = \text{tr}(\mathbf{A}\boldsymbol{\Sigma})$, and the noncentrality is $\boldsymbol{\mu}^\top \boldsymbol{\Sigma}^{-1/2} \mathbf{B} \boldsymbol{\Sigma}^{-1/2} \boldsymbol{\mu} = \boldsymbol{\mu}^\top \mathbf{A} \boldsymbol{\mu}$.
- (c) Take $\mathbf{L}$ of size $n \times k$ with full column rank $k = \text{rank}(\boldsymbol{\Sigma})$ and $\mathbf{L} \mathbf{L}^\top = \boldsymbol{\Sigma}$, for instance $\mathbf{L} = \mathbf{U}_1 \mathbf{D}_1^{1/2}$ from the eigenvectors $\mathbf{U}_1$ for the positive eigenvalues $\mathbf{D}_1$ of $\boldsymbol{\Sigma}$. (If $\boldsymbol{\Sigma} = \mathbf{0}$ the claim is trivial.) As in Theorem 4.2.1, $\mathbf{Y} \stackrel{d}{=} \boldsymbol{\mu} + \mathbf{LZ}$ with $\mathbf{Z} \sim N_k(\mathbf{0}, \mathbf{I})$. Since $\mathbf{L}$ has full column rank, $\mathbf{L}^\top$ has full row rank, so $\mathbf{L} \mathbf{G} \mathbf{L}^\top = \mathbf{L} \mathbf{H} \mathbf{L}^\top$ implies $\mathbf{G} = \mathbf{H}$, and $\mathbf{L} \mathbf{g} = \mathbf{L} \mathbf{h}$ implies $\mathbf{g} = \mathbf{h}$. Put $\mathbf{B} = \mathbf{L}^\top \mathbf{A} \mathbf{L}$ and $\mathbf{c} = \mathbf{L}^\top \mathbf{A} \boldsymbol{\mu}$. Condition (i) reads $\mathbf{L} \mathbf{B}^2 \mathbf{L}^\top = \mathbf{L} \mathbf{B} \mathbf{L}^\top$, so $\mathbf{B}^2 = \mathbf{B}$. Condition (ii) reads $\mathbf{L} \mathbf{B} \mathbf{c} = \mathbf{L} \mathbf{c}$, so $\mathbf{B} \mathbf{c} = \mathbf{c}$. Condition (iii) reads $\mathbf{c}^\top \mathbf{c} = \boldsymbol{\mu}^\top \mathbf{A} \boldsymbol{\mu}$. Hence

$$(\boldsymbol{\mu} + \mathbf{LZ})^\top \mathbf{A} (\boldsymbol{\mu} + \mathbf{LZ}) = \boldsymbol{\mu}^\top \mathbf{A} \boldsymbol{\mu} + 2\mathbf{c}^\top \mathbf{Z} + \mathbf{Z}^\top \mathbf{BZ} = (\mathbf{Z} + \mathbf{c})^\top \mathbf{B} (\mathbf{Z} + \mathbf{c}),$$

because $(\mathbf{Z} + \mathbf{c})^\top \mathbf{B} (\mathbf{Z} + \mathbf{c}) = \mathbf{Z}^\top \mathbf{BZ} + 2\mathbf{c}^\top \mathbf{BZ} + \mathbf{c}^\top \mathbf{B} \mathbf{c}$ and $\mathbf{B} \mathbf{c} = \mathbf{c}$. Now $\mathbf{Z} +$

$\mathbf{c} \sim N_k(\mathbf{c}, \mathbf{I})$, and [Theorem 4.3.1](#) gives the law $\chi^2(\text{rank } \mathbf{B}, \mathbf{c}^\top \mathbf{B} \mathbf{c})$, where $\text{rank } \mathbf{B} = \text{tr } \mathbf{B} = \text{tr}(\mathbf{A}\Sigma)$ and $\mathbf{c}^\top \mathbf{B} \mathbf{c} = \mathbf{c}^\top \mathbf{c} = \boldsymbol{\mu}^\top \mathbf{A} \boldsymbol{\mu}$. If $\mathbf{B} = \mathbf{0}$, then $\mathbf{c} = \mathbf{B} \mathbf{c} = \mathbf{0}$ and the form is identically 0, which is $\chi^2(0)$. If $\mathbf{A}\Sigma\mathbf{A} = \mathbf{A}$, all three conditions follow by substitution. If $\mathbf{A}\Sigma$ is idempotent and $\boldsymbol{\mu} = \Sigma \mathbf{v}$, then (i) follows from multiplying $\mathbf{A}\Sigma\mathbf{A}\Sigma = \mathbf{A}\Sigma$ by Σ on the left, and (ii) and (iii) from the same identity applied to $\mathbf{v}$.

■

Part (a) is the case used most often in this book. For the linear model $\mathbf{Y} \sim N_n(\mathbf{X}\boldsymbol{\beta}, \sigma^2 \mathbf{I})$ and the projection $\mathbf{M}$ onto $\mathcal{C}(\mathbf{X})$, [Chapter 6](#) shows that the residual sum of squares is $\text{SSE} = \|(\mathbf{I} - \mathbf{M})\mathbf{Y}\|^2$. Since $(\mathbf{I} - \mathbf{M})\mathbf{X}\boldsymbol{\beta} = \mathbf{0}$, part (a) gives

$$\text{SSE}/\sigma^2 \sim \chi^2(n - \text{rank}(\mathbf{X})),$$

a central distribution whatever the value of $\boldsymbol{\beta}$. The projections that define test statistics, by contrast, do not annihilate the mean under the alternative, and their noncentrality $\|\mathbf{P}\mathbf{X}\boldsymbol{\beta}\|^2/\sigma^2$ measures how far the truth is from the hypothesis in the metric that the test can see.

In part (c) idempotence of $\mathbf{A}\Sigma$ alone is *not* enough when Σ is singular. A mean with a component outside $\mathcal{C}(\Sigma)$ is a deterministic shift that the form may pick up as a constant or as a linear term ([Exercise \(B2\)](#)). Part (c) also covers the natural generalization of the Mahalanobis distance.

Corollary 4.3.3 — Mahalanobis forms

Let $\mathbf{Y} \sim N_n(\boldsymbol{\mu}, \Sigma)$.

- (a) If Σ is positive definite, $(\mathbf{Y} - \boldsymbol{\mu})^\top \Sigma^{-1} (\mathbf{Y} - \boldsymbol{\mu}) \sim \chi^2(n)$ and $\mathbf{Y}^\top \Sigma^{-1} \mathbf{Y} \sim \chi^2(n, \boldsymbol{\mu}^\top \Sigma^{-1} \boldsymbol{\mu})$.
- (b) If $\mathbf{G}$ is a symmetric generalized inverse of Σ and $\boldsymbol{\mu} \in \mathcal{C}(\Sigma)$, then $\mathbf{Y}^\top \mathbf{G} \mathbf{Y} \sim \chi^2(\text{rank}(\Sigma), \boldsymbol{\mu}^\top \mathbf{G} \boldsymbol{\mu})$.

Proof.

- (a) $\mathbf{A} = \Sigma^{-1}$ satisfies $\mathbf{A}\Sigma\mathbf{A} = \mathbf{A}$, and $\mathbf{Y} - \boldsymbol{\mu} \sim N_n(\mathbf{0}, \Sigma)$. The central statement was already obtained by whitening in [Proposition 3.2.3](#).
- (b) $\mathbf{G}\Sigma$ is idempotent because $\Sigma\mathbf{G}\Sigma = \Sigma$, so part (c) of the theorem applies and the degrees of freedom are $\text{tr}(\mathbf{G}\Sigma) = \text{tr}(\mathbf{L}^\top \mathbf{G} \mathbf{L})$. With $\mathbf{L}$ as in that proof, $\mathbf{L}\mathbf{L}^\top \mathbf{G} \mathbf{L}\mathbf{L}^\top = \mathbf{L}\mathbf{L}^\top$ gives $\mathbf{L}^\top \mathbf{G} \mathbf{L} = \mathbf{I}_k$, whose trace is $k = \text{rank}(\Sigma)$.

■

A symmetric generalized inverse always exists ([Proposition 1.3.3\(d\)](#)). The value of $\mathbf{Y}^\top \mathbf{G} \mathbf{Y}$ does not depend on which one is chosen, with probability one ([Exercise \(B3\)](#)).

Example. (Correlated observations about a common level)

Let $\mathbf{Y} \sim N_n(\boldsymbol{\mu}, \Sigma)$ with Σ positive definite, and suppose we want to measure how far $\mathbf{Y}$ is from a common level $m\mathbf{1}$. The generalized least squares level is $\hat{m} = \mathbf{1}^\top \Sigma^{-1} \mathbf{Y} / \mathbf{1}^\top \Sigma^{-1} \mathbf{1}$,

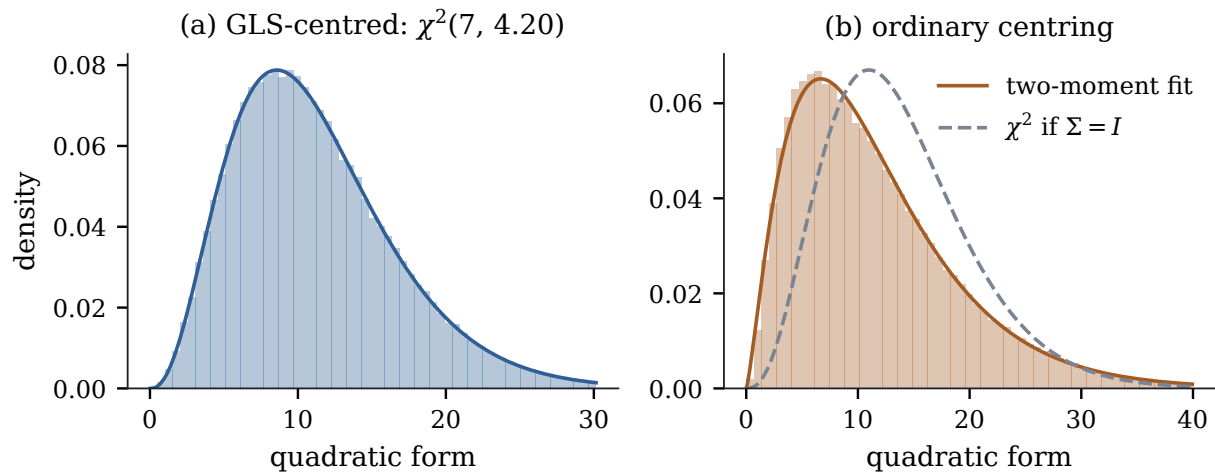


Figure 4.3.1: Two quadratic forms measuring spread about a common level, for $n = 8$ autoregressive observations with a trending mean. (a) With generalized least squares centring, $\mathbf{A}\Sigma$ is idempotent and the form is exactly noncentral chi-squared. (b) With ordinary centring it is a weighted sum of noncentral $\chi^2(1)$ terms. A two-moment scaled chi-squared fits closely. The chi-squared law that would hold for independent observations does not.

and the natural discrepancy is $(\mathbf{Y} - \hat{m}\mathbf{1})^\top \Sigma^{-1}(\mathbf{Y} - \hat{m}\mathbf{1}) = \mathbf{Y}^\top \mathbf{A} \mathbf{Y}$ with

$$\mathbf{A} = \Sigma^{-1} - \frac{\Sigma^{-1}\mathbf{1}\mathbf{1}^\top \Sigma^{-1}}{\mathbf{1}^\top \Sigma^{-1}\mathbf{1}}.$$

A direct multiplication gives $\mathbf{A}\Sigma\mathbf{A} = \mathbf{A}$, and $\text{tr}(\mathbf{A}\Sigma) = n - 1$. By [Theorem 4.3.2\(b\)](#) the form is noncentral chi-squared with $n - 1$ degrees of freedom and noncentrality $\boldsymbol{\mu}^\top \mathbf{A} \boldsymbol{\mu}$, which is zero exactly when $\boldsymbol{\mu}$ is constant. The listing takes $n = 8$, an autoregressive correlation $\Sigma = (\rho^{|i-j|})$ with $\rho = 0.6$, and a linear trend in the mean, for which $\gamma = 4.200$. [Figure 4.3.1\(a\)](#) compares 200,000 simulated values with the $\chi^2(7, 4.200)$ density.

```
import numpy as np
from scipy import integrate, stats
rng = np.random.default_rng(31415)
n, rho = 8, 0.6
idx = np.arange(n)
Sigma = rho ** np.abs(idx[:, None] - idx[None, :]) # AR(1) correlation matrix
mu = 0.4 * (idx - idx.mean()) # linear trend, mean zero
one = np.ones(n)

W = np.linalg.inv(Sigma)
A = W - np.outer(W @ one, W @ one) / (one @ W @ one) # GLS-centred form
AS = A @ Sigma
print("A Sigma idempotent:", np.allclose(AS @ AS, AS))
df, gamma = np.trace(AS), mu @ A @ mu # chi^2(df, gamma)

L = np.linalg.cholesky(Sigma) # Y = mu + L Z
Z = rng.standard_normal((200_000, n))
Y = mu + Z @ L.T
q_gls = np.einsum("ij,jk,ik->i", Y, A, Y)
print(df, gamma, stats.kstest(q_gls, stats.ncx2(df, gamma).cdf).pvalue)
```

4.3.3 Forms that are not chi-squared

Most quadratic forms fail the conditions of [Theorem 4.3.2](#). The ordinary sum of squares about the mean, $\mathbf{Y}^\top \mathbf{C} \mathbf{Y}$ with $\mathbf{C} = \mathbf{I} - n^{-1} \mathbf{1} \mathbf{1}^\top$, is an example as soon as the observations are correlated. Then $\mathbf{C} \boldsymbol{\Sigma}$ is not idempotent, and by [Theorem 4.2.1](#) the law is that of $\sum_j \lambda_j (W_j + \nu_j)^2$, with weights λ_j equal to the nonzero eigenvalues of $\mathbf{C} \boldsymbol{\Sigma}$. In [Example \(Correlated observations about a common level\)](#) these range from 0.259 to 1.832. Two practical ways to handle such a law are the following.

Exact computation. The characteristic function of the weighted sum is known in closed form (put $t = iu$ in [Corollary 4.2.2](#)), and a distribution function can be recovered from a characteristic function φ by the inversion formula of Gil-Pelaez (1951),

$$\Pr(X > x) = \frac{1}{2} + \frac{1}{\pi} \int_0^\infty \frac{\operatorname{Im}\{e^{-iux} \varphi(u)\}}{u} du.$$

For a weighted sum of k chi-squared terms the integrand decays like $u^{-1-k/2}$, so ordinary quadrature works well. Imhof (1961) gave the real-variable form of this integral that most software uses. The listing implements it directly and checks it against simulation.

Two-moment approximation. Match $a\chi^2(\nu)$ to the mean m and variance v from [Theorem 4.2.3](#). Since $a\chi^2(\nu)$ has mean $a\nu$ and variance $2a^2\nu$, this gives $a = v/(2m)$ and $\nu = 2m^2/v$, with ν usually not an integer (Satterthwaite 1946; Patnaik 1949).

In [Example \(Correlated observations about a common level\)](#), the simulated 95th percentile of $\mathbf{Y}^\top \mathbf{C} \mathbf{Y}$ is 26.51. The exact tail probability beyond it is 0.0490, and the approximation with $a = 2.488$ and $\nu = 4.68$ gives 0.0484. Treating the observations as independent, and so using $\chi^2(7, 6.720)$, gives 0.040. [Figure 4.3.1\(b\)](#) shows the three laws. Here the error from ignoring correlation is modest at the 5% point but visible in the shape. It is not a monotone function of ρ , but strong correlation makes it severe: with $\rho = 0.9$ and the same mean, the naive tail probability at the true 95th percentile is 0.197.

```
def canonical(A, mu, L):
    """Y^T A Y = sum_j lam_j (W_j + nu_j)^2 + c (all lam_j nonzero here), W ~ N(0, I)"""
    lam, Qm = np.linalg.eigh(L.T @ A @ L)
    b = Qm.T @ L.T @ A @ mu
    keep = np.abs(lam) > 1e-10
    assert np.allclose(b[~keep], 0) # no linear terms in this example
    c = mu @ A @ mu - np.sum(b[keep] ** 2 / lam[keep])
    return lam[keep], b[keep] / lam[keep], c

def tail_prob(x, lam, nu, c=0.0):
    """P(sum lam_j (W_j + nu_j)^2 + c > x) by Gil-Pelaez inversion."""
    def integrand(u):
        z = 1 - 2j * lam * u
        phi = np.prod(z ** -0.5 * np.exp(1j * u * lam * nu**2 / z))
        return (np.exp(-1j * u * (x - c)) * phi).imag / u
    edges = np.linspace(0, 400, 401) # integrand decays like u
    # ^(-1-k/2)
    val = sum(integrate.quad(integrand, a, b)[0] for a, b in zip(edges[:-1], edges[1:]))
    return 0.5 + val / np.pi
```

4.3.4 Exercises

4.3.4.1 A. Check your understanding

Exercise A1.

Let $\mathbf{Y} \sim N_n(\mathbf{0}, \mathbf{I})$ with $n \geq 2$, and $\mathbf{A} = \alpha \mathbf{I} + \beta \mathbf{1}\mathbf{1}^\top$. Find all (α, β) for which $\mathbf{Y}^\top \mathbf{A} \mathbf{Y}$ has a chi-squared distribution, and identify each form.

Solution. $\mathbf{A}$ has eigenvalue $\alpha + n\beta$ on $\mathbf{1}$ and α on $\mathbf{1}^\perp$. By [Theorem 4.3.1](#), the form is chi-squared iff $\mathbf{A} \neq \mathbf{0}$ and both eigenvalues lie in $\{0, 1\}$. The three solutions are $(1, 0)$, giving $\|\mathbf{Y}\|^2 \sim \chi^2(n)$; $(0, 1/n)$, giving $n\bar{Y}^2 \sim \chi^2(1)$; and $(1, -1/n)$, giving $\sum (Y_i - \bar{Y})^2 \sim \chi^2(n-1)$.

Exercise A2.

Two independent samples of sizes m and k come from $N(\mu_1, \sigma^2)$ and $N(\mu_2, \sigma^2)$. Write the pooled within-sample sum of squares as $\|(\mathbf{I} - \mathbf{M})\mathbf{Y}\|^2$ for a suitable projection, and show that the two-sample t statistic has the $t(m+k-2, (\mu_1 - \mu_2)/(\sigma\sqrt{1/m + 1/k}))$ distribution.

4.3.4.2 B. Practice

Exercise B1.

Let $\mathbf{Y} \sim N_n(\mu\mathbf{1}, \sigma^2\{(1-\rho)\mathbf{I} + \rho\mathbf{1}\mathbf{1}^\top\})$ with $-1/(n-1) < \rho < 1$, and let $\bar{Y}$ and S^2 be the sample mean and variance. Show that $(n-1)S^2/\{\sigma^2(1-\rho)\} \sim \chi^2(n-1)$ and that S^2 is independent of $\bar{Y}$. Deduce that $\sqrt{n}(\bar{Y} - \mu)/S$ has the law of $\{(1 + (n-1)\rho)/(1-\rho)\}^{1/2}$ times a $t(n-1)$ variable, and describe what a positive ρ does to the coverage of the usual 95% t interval for μ .

Exercise B2.

Let $\Sigma = \text{diag}(1, 0)$, $\mu = (0, 1)^\top$, $\mathbf{A} = \mathbf{I}_2$ and $\mathbf{Y} \sim N_2(\mu, \Sigma)$. Show that $\mathbf{A}\Sigma$ is idempotent but $\mathbf{Y}^\top \mathbf{A} \mathbf{Y}$ is not chi-squared. Which of the conditions of [Theorem 4.3.2\(c\)](#) fail? What changes if $\mu = (m, 0)^\top$? Repeat the first part with $\mathbf{A} = \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix}$, where the shift enters through a linear term.

Solution. $\mathbf{A}\Sigma = \Sigma$ is idempotent. But $Y_2 = 1$ with probability one, so $\mathbf{Y}^\top \mathbf{A} \mathbf{Y} = Y_1^2 + 1 \geq 1$, while every $\chi^2(r, \gamma)$ with $r \geq 1$ gives positive probability to $(0, 1)$. Conditions (i) and (ii) hold, since $\Sigma^3 = \Sigma$ and $\Sigma^2\mu = \mathbf{0} = \Sigma\mu$. Condition (iii) fails, since $\mu^\top \Sigma\mu = 0 \neq 1 = \mu^\top \mu$. The part of the mean outside $\mathcal{C}(\Sigma)$ adds a constant. If $\mu = (m, 0)^\top \in \mathcal{C}(\Sigma)$, then $\mathbf{Y}^\top \mathbf{A} \mathbf{Y} = Y_1^2 \sim \chi^2(1, m^2)$, as [Theorem 4.3.2\(c\)](#) predicts. For $\mathbf{A} = \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix}$ and $\mu = (0, 1)^\top$, $\mathbf{A}\Sigma = \begin{pmatrix} 1 & 0 \\ 1 & 0 \end{pmatrix}$ is again idempotent, and $\mathbf{Y}^\top \mathbf{A} \mathbf{Y} = Y_1^2 + 2Y_1Y_2 = Y_1^2 + 2Y_1 = (Y_1 + 1)^2 - 1$, which is negative with positive probability. Conditions (i) and (ii) hold, since $\Sigma\mathbf{A}\Sigma = \Sigma$ and $\Sigma\mathbf{A}\Sigma\mu = \Sigma\mathbf{A}\mu = (1, 0)^\top$, but (iii) fails because $\mu^\top \mathbf{A}\Sigma\mathbf{A}\mu = 1 \neq 0 = \mu^\top \mathbf{A}\mu$. The shift now enters through the linear term $2Y_1$.

Exercise B3.

In [Corollary 4.3.3\(b\)](#), show that $\mathbf{Y}^\top \mathbf{G} \mathbf{Y}$ takes the same value for every generalized inverse $\mathbf{G}$ of Σ , with probability one. *Hint:* [Theorem 2.2.3](#).

Solution. By [Theorem 2.2.3](#), $\mathbf{Y} - \boldsymbol{\mu} \in \mathcal{C}(\Sigma)$ with probability one. Since $\boldsymbol{\mu} \in \mathcal{C}(\Sigma)$, also $\mathbf{Y} \in \mathcal{C}(\Sigma)$, so $\mathbf{Y} = \Sigma \mathbf{v}$ for some (random) $\mathbf{v}$. For any generalized inverse $\mathbf{G}$, $\mathbf{Y}^\top \mathbf{G} \mathbf{Y} = \mathbf{v}^\top \Sigma \mathbf{G} \Sigma \mathbf{v} = \mathbf{v}^\top \Sigma \mathbf{v}$, which does not involve $\mathbf{G}$.

Exercise B4.

Let $\mathbf{Y} \sim N_n(\mathbf{0}, \Sigma)$ with Σ positive definite, and let $\mathbf{A} \neq \mathbf{0}$ be nonnegative definite. Show that the two-moment approximation $a\chi^2(\nu)$ to $\mathbf{Y}^\top \mathbf{A} \mathbf{Y}$ has $\nu = \{\text{tr}(\mathbf{A}\Sigma)\}^2 / \text{tr}\{(\mathbf{A}\Sigma)^2\}$, and that $\nu \leq \text{rank}(\mathbf{A})$, with equality iff all nonzero eigenvalues of $\mathbf{A}\Sigma$ are equal.

4.3.4.3 C. Going deeper

Exercise C1.

Let $\mathbf{Y} \sim N_n(\boldsymbol{\mu}, \Sigma)$ with Σ possibly singular, and suppose $\mathbf{Y}^\top \mathbf{A} \mathbf{Y} \sim \chi^2(r, \gamma)$ with $r \geq 1$. Show that $\Sigma \mathbf{A} \Sigma \mathbf{A} \Sigma = \Sigma \mathbf{A} \Sigma$. *Hint:* in [Corollary 4.2.2](#), compare the coefficients of t^k for $k \geq 3$ with those of [Eq. 4.1.2](#), and adapt the argument of [Theorem 4.3.1](#).

4.4 Independence of quadratic forms and linear forms

An F or t statistic is a ratio, and its distribution is known only if numerator and denominator are independent. For normal vectors, uncorrelated linear functions are independent ([Theorem 3.4.1](#)). Quadratic forms are not linear, but the following observation brings them within reach of that result.

Idea. A quadratic form is a function of a linear form: for any generalized inverse $\mathbf{A}^-$ of $\mathbf{A}$,

$$\mathbf{Y}^\top \mathbf{A} \mathbf{Y} = \mathbf{Y}^\top \mathbf{A} \mathbf{A}^- \mathbf{A} \mathbf{Y} = (\mathbf{A} \mathbf{Y})^\top \mathbf{A}^- (\mathbf{A} \mathbf{Y}).$$

So $\mathbf{Y}^\top \mathbf{A} \mathbf{Y}$ is independent of anything that is independent of the vector $\mathbf{A} \mathbf{Y}$.

4.4.1 The independence theorems

Theorem 4.4.1 — Linear and quadratic forms

Let $\mathbf{Y} \sim N_n(\boldsymbol{\mu}, \Sigma)$, let $\mathbf{B}$ be $m \times n$ and $\mathbf{A}$ symmetric $n \times n$. If $\mathbf{B}\Sigma\mathbf{A} = \mathbf{0}$, then $\mathbf{B}\mathbf{Y}$ and $\mathbf{Y}^\top \mathbf{A} \mathbf{Y}$ are independent. Conversely, if $\mathbf{B}\mathbf{Y}$ and $\mathbf{Y}^\top \mathbf{A} \mathbf{Y}$ are independent for every $\boldsymbol{\mu} \in \mathbb{R}^n$, with Σ fixed, then $\mathbf{B}\Sigma\mathbf{A} = \mathbf{0}$.

Proof. The vector $(\mathbf{BY}, \mathbf{AY})$ is a linear function of $\mathbf{Y}$, hence jointly normal (Theorem 3.3.1), and its cross-covariance is $\text{Cov}(\mathbf{BY}, \mathbf{AY}) = \mathbf{B}\Sigma\mathbf{A}^\top = \mathbf{B}\Sigma\mathbf{A}$ (Theorem 2.2.1). If this is zero, $\mathbf{BY}$ and $\mathbf{AY}$ are independent by Theorem 3.4.1, and so are $\mathbf{BY}$ and the function $(\mathbf{AY})^\top \mathbf{A}^-(\mathbf{AY})$ of $\mathbf{AY}$, which is $\mathbf{Y}^\top \mathbf{AY}$ by the key idea above. Conversely, independent variables with finite variances are uncorrelated, so Theorem 4.2.3(c) gives $2\mathbf{B}\Sigma\mathbf{A}\boldsymbol{\mu} = \mathbf{0}$ for every $\boldsymbol{\mu}$, that is, $\mathbf{B}\Sigma\mathbf{A} = \mathbf{0}$. ■

Theorem 4.4.2 — Two quadratic forms

Let $\mathbf{Y} \sim N_n(\boldsymbol{\mu}, \Sigma)$ and let $\mathbf{A}, \mathbf{B}$ be symmetric. If $\mathbf{A}\Sigma\mathbf{B} = \mathbf{0}$, equivalently $\mathbf{B}\Sigma\mathbf{A} = \mathbf{0}$, then $\mathbf{Y}^\top \mathbf{AY}$ and $\mathbf{Y}^\top \mathbf{BY}$ are independent.

Proof. The two conditions are transposes of each other. As before, $\mathbf{AY}$ and $\mathbf{BY}$ are jointly normal with $\text{Cov}(\mathbf{AY}, \mathbf{BY}) = \mathbf{A}\Sigma\mathbf{B} = \mathbf{0}$, hence independent, and each form is a function of one of them. ■

The condition is sufficient for every $\boldsymbol{\mu}$ and every Σ . When Σ is singular one might hope that the weaker requirement $\Sigma\mathbf{A}\Sigma\mathbf{B}\Sigma = \mathbf{0}$ suffices, since only directions in $\mathcal{C}(\Sigma)$ carry noise. For nonnegative definite $\mathbf{A}$ and $\mathbf{B}$ the two requirements turn out to be equivalent (Exercise (B2)). In the other direction, the condition is necessary under mild assumptions, a result usually credited to Craig (1943). Its general proof is surprisingly delicate, and the history of incorrect published proofs is recounted by Driscoll and Gundberg (1986). The case that matters most for sums of squares has a short proof.

Proposition 4.4.3 — Necessity for nonnegative definite forms

Let $\mathbf{Y} \sim N_n(\mathbf{0}, \mathbf{I})$ and let $\mathbf{A}, \mathbf{B}$ be nonnegative definite. If $\mathbf{Y}^\top \mathbf{AY}$ and $\mathbf{Y}^\top \mathbf{BY}$ are independent, then $\mathbf{AB} = \mathbf{0}$.

Proof. Independence makes the covariance zero, and by Theorem 4.2.3(d) the covariance is $2\text{tr}(\mathbf{AB})$. With the square roots of Theorem 1.7.6,

$$\text{tr}(\mathbf{AB}) = \text{tr}(\mathbf{A}^{1/2}\mathbf{A}^{1/2}\mathbf{B}^{1/2}\mathbf{B}^{1/2}) = \text{tr}\{(\mathbf{A}^{1/2}\mathbf{B}^{1/2})^\top(\mathbf{A}^{1/2}\mathbf{B}^{1/2})\},$$

the sum of squares of the entries of $\mathbf{A}^{1/2}\mathbf{B}^{1/2}$. So $\mathbf{A}^{1/2}\mathbf{B}^{1/2} = \mathbf{0}$, and $\mathbf{AB} = \mathbf{A}^{1/2}(\mathbf{A}^{1/2}\mathbf{B}^{1/2})\mathbf{B}^{1/2} = \mathbf{0}$. ■

For nonnegative definite forms in a standard normal vector, then, zero correlation and independence coincide, just as for jointly normal linear forms. This fails for general symmetric matrices (Exercise (B1)).

4.4.2 Orthogonal projections of a spherical normal vector

The situation in regression is the tidiest one: a spherical normal vector and several projections onto mutually orthogonal subspaces.

Theorem 4.4.4 — Independent projections

Let $\mathbf{Y} \sim N_n(\boldsymbol{\mu}, \sigma^2 \mathbf{I})$, and let $\mathbf{P}_1, \dots, \mathbf{P}_k$ be symmetric idempotent matrices with $\mathbf{P}_i \mathbf{P}_j = \mathbf{0}$ for $i \neq j$, and ranks $r_1, \dots, r_k$. Then $\mathbf{P}_1 \mathbf{Y}, \dots, \mathbf{P}_k \mathbf{Y}$ are mutually independent normal vectors, $\mathbf{P}_i \mathbf{Y} \sim N_n(\mathbf{P}_i \boldsymbol{\mu}, \sigma^2 \mathbf{P}_i)$, and the sums of squares

$$\frac{\|\mathbf{P}_i \mathbf{Y}\|^2}{\sigma^2} \sim \chi^2\left(r_i, \frac{\|\mathbf{P}_i \boldsymbol{\mu}\|^2}{\sigma^2}\right), \quad i = 1, \dots, k,$$

are mutually independent.

Proof. Stack the vectors $\mathbf{P}_i \mathbf{Y}$ into one nk -vector. It is a linear function of $\mathbf{Y}$, so it is normal, and the covariance between blocks i and j is $\mathbf{P}_i(\sigma^2 \mathbf{I})\mathbf{P}_j^\top = \sigma^2 \mathbf{P}_i \mathbf{P}_j$. This is $\mathbf{0}$ for $i \neq j$ and $\sigma^2 \mathbf{P}_i$ for $i = j$. By the block form of [Theorem 3.4.1](#) the blocks are mutually independent, and functions of them, the squared lengths, are then mutually independent. Their laws are [Theorem 4.3.2\(a\)](#). ■

In geometric language, which [Chapter 6](#) develops: the coordinates of a spherical normal vector in mutually orthogonal subspaces are independent. It is the reason the rows of an analysis of variance table can be treated separately.

Corollary 4.4.5 — Sample mean and sample variance

Let $Y_1, \dots, Y_n$ ($n \geq 2$) be independent $N(\mu, \sigma^2)$, with sample mean $\bar{Y}$ and sample variance $S^2 = (n-1)^{-1} \sum_i (Y_i - \bar{Y})^2$. Then

- (a) $\bar{Y}$ and S^2 are independent, and $\bar{Y} \sim N(\mu, \sigma^2/n)$;
- (b) $(n-1)S^2/\sigma^2 \sim \chi^2(n-1)$;
- (c) for any μ_0 , $T = \sqrt{n}(\bar{Y} - \mu_0)/S \sim t(n-1, \delta)$ with $\delta = \sqrt{n}(\mu - \mu_0)/\sigma$. In particular $T \sim t(n-1)$ when $\mu = \mu_0$.

Proof. Here $\mathbf{Y} \sim N_n(\mu \mathbf{1}, \sigma^2 \mathbf{I})$. Let $\mathbf{P}_1 = n^{-1} \mathbf{1} \mathbf{1}^\top$ and $\mathbf{C} = \mathbf{I} - \mathbf{P}_1$. Both are symmetric and idempotent, $\mathbf{P}_1 \mathbf{C} = \mathbf{0}$, and $\text{rank}(\mathbf{C}) = \text{tr}(\mathbf{C}) = n-1$. Since $\mathbf{P}_1 \mathbf{Y} = \bar{Y} \mathbf{1}$ and $\|\mathbf{C} \mathbf{Y}\|^2 = (n-1)S^2$, [Theorem 4.4.4](#) gives (a), which [Example \(Sample mean and deviations\)](#) obtained directly, and, because $\mathbf{C} \mu \mathbf{1} = \mathbf{0}$, also (b). For (c), $X = \sqrt{n}(\bar{Y} - \mu_0)/\sigma \sim N(\delta, 1)$ and $V = (n-1)S^2/\sigma^2 \sim \chi^2(n-1)$ are independent, and $T = X/\sqrt{V/(n-1)}$. Apply [Definition 4.1.3](#). ■

The independence in (a) characterizes the normal distribution: for $n \geq 2$ independent identically distributed observations with finite variance, $\bar{Y}$ and S^2 are independent only under normality (Lukacs 1942). We will not need this, but it shows that normality is used essentially here. It is not a technicality that more careful arguments could remove.

4.4.3 Preview: the F statistic for nested models

The main use of these results comes in [Chapter 11](#). Its core fits in a few lines.

Theorem 4.4.6 — F statistic for nested subspaces

Let $\mathbf{Y} \sim N_n(\boldsymbol{\mu}, \sigma^2 \mathbf{I})$, and let $\mathbf{P}_0$ and $\mathbf{P}$ be symmetric idempotent matrices of ranks $r_0 < r < n$ with $\mathbf{P}\mathbf{P}_0 = \mathbf{P}_0$. Suppose that $\mathbf{P}\boldsymbol{\mu} = \boldsymbol{\mu}$. Then

$$F = \frac{\mathbf{Y}^\top (\mathbf{P} - \mathbf{P}_0) \mathbf{Y} / (r - r_0)}{\mathbf{Y}^\top (\mathbf{I} - \mathbf{P}) \mathbf{Y} / (n - r)} \sim F\left(r - r_0, n - r, \frac{\|(\mathbf{P} - \mathbf{P}_0)\boldsymbol{\mu}\|^2}{\sigma^2}\right),$$

and the noncentrality is zero iff $\mathbf{P}_0\boldsymbol{\mu} = \boldsymbol{\mu}$.

Proof. Transposing $\mathbf{P}\mathbf{P}_0 = \mathbf{P}_0$ gives $\mathbf{P}_0\mathbf{P} = \mathbf{P}_0$. Hence $(\mathbf{P} - \mathbf{P}_0)^2 = \mathbf{P} - 2\mathbf{P}_0 + \mathbf{P}_0 = \mathbf{P} - \mathbf{P}_0$, so $\mathbf{P} - \mathbf{P}_0$ is symmetric idempotent with rank $\text{tr}(\mathbf{P}) - \text{tr}(\mathbf{P}_0) = r - r_0$. Also $(\mathbf{P} - \mathbf{P}_0)(\mathbf{I} - \mathbf{P}) = \mathbf{P} - \mathbf{P} - \mathbf{P}_0 + \mathbf{P}_0 = \mathbf{0}$, and $\mathbf{I} - \mathbf{P}$ is symmetric idempotent of rank $n - r$. By [Theorem 4.4.4](#) the numerator and denominator sums of squares are independent, with σ^{-2} times them distributed as $\chi^2(r - r_0, \|(\mathbf{P} - \mathbf{P}_0)\boldsymbol{\mu}\|^2/\sigma^2)$ and $\chi^2(n - r, \|(\mathbf{I} - \mathbf{P})\boldsymbol{\mu}\|^2/\sigma^2)$. The second noncentrality is 0 since $\mathbf{P}\boldsymbol{\mu} = \boldsymbol{\mu}$. The unknown σ^2 cancels in the ratio, and [Definition 4.1.2](#) applies. Finally, $(\mathbf{P} - \mathbf{P}_0)\boldsymbol{\mu} = \boldsymbol{\mu} - \mathbf{P}_0\boldsymbol{\mu}$. ■

In a linear model, $\mathbf{P} = \mathbf{M}$ projects onto the column space of the full model matrix $\mathbf{X}$ and $\mathbf{P}_0 = \mathbf{M}_0$ onto that of a submodel $\mathbf{X}_0$. The numerator is the drop in residual sum of squares from the submodel to the full model ([Chapter 6](#)). The theorem says three things. The test that rejects for large F has an exactly known null distribution. Its power depends on the unknown parameters only through $\gamma = \|(\mathbf{M} - \mathbf{M}_0)\mathbf{X}\boldsymbol{\beta}\|^2/\sigma^2$. And, by [Theorem 4.1.4](#), that power increases with γ .

Example. (Detecting curvature)

Let $x_1, \dots, x_n$, $n = 15$, be equally spaced on $[0, 1]$, and suppose $Y_i = 1 + 0.5x_i + \beta_2 x_i^2 + \varepsilon_i$ with independent $N(0, 1)$ errors. To test for curvature, compare the straight line ($\mathbf{X}_0 = [\mathbf{1}, \mathbf{x}]$) with the quadratic ($\mathbf{X} = [\mathbf{1}, \mathbf{x}, \mathbf{x}^2]$). The F statistic has 1 and 12 degrees of freedom, and the 5% critical value is 4.747. For $\beta_2 = 8.0$, the listing finds $\gamma = 6.873$ and power 0.673 from the noncentral F , and the rejection rate in 100,000 simulated data sets is 0.674.

The noncentrality has a clear meaning. It is β_2^2 times the squared length of the part of $\mathbf{x}^2$ orthogonal to $\mathbf{1}$ and $\mathbf{x}$, divided by σ^2 . It grows roughly in proportion to n for a fixed design pattern. Keeping the same design on $[0, 1]$, the smallest n with power at least 0.8 is 21, where $\gamma = 8.97$. [Figure 4.4.1](#) places the example on the power curve and shows the loss of power when the same noncentrality is spread over more numerator degrees

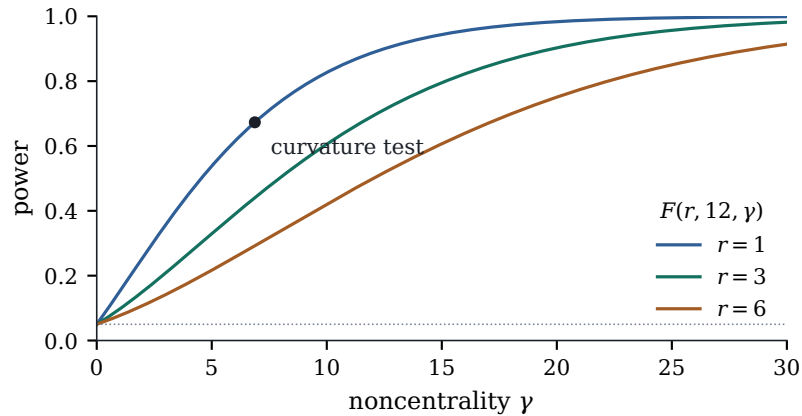


Figure 4.4.1: Power $\Pr\{F(r, 12, \gamma) > F_{0.05}(r, 12)\}$ of the level-0.05 F test as a function of the noncentrality. Every curve starts at 0.05 and increases to 1 (Theorem 4.1.4). For a fixed γ , power falls as the number r of numerator degrees of freedom grows. The dot is the curvature test of Example (Detecting curvature).

of freedom r . At $\gamma = 10$ the power is 0.827, 0.606 and 0.421 for $r = 1, 3, 6$.

```
import numpy as np
from scipy import stats
rng = np.random.default_rng(1104)
alpha, s = 0.05, 12
def power(gamma, q, s, alpha=0.05):
    crit = stats.f.ppf(1 - alpha, q, s)
    return stats.ncf.sf(crit, q, s, np.maximum(gamma, 1e-12))
gammas = np.linspace(0, 30, 301)
gamma_ref, rs = 10.0, (1, 3, 6) # r = numerator degrees of
                                # freedom
curves = {q: power(gammas, q, s) for q in rs}

n, sigma, beta0, beta1, beta2, reps = 15, 1.0, 1.0, 0.5, 8.0, 100_000
x = np.linspace(0, 1, n)
X0 = np.column_stack([np.ones(n), x]) # straight line
X = np.column_stack([X0, x**2]) # adds curvature

def proj(Z):
    Q, _ = np.linalg.qr(Z)
    return Q @ Q.T

M, M0 = proj(X), proj(X0)
mean = X @ np.array([beta0, beta1, beta2])
gamma = mean @ (M - M0) @ mean / sigma**2 # noncentrality
q, df_resid = 1, n - 3
crit = stats.f.ppf(0.95, q, df_resid)
exact_power = stats.ncf.sf(crit, q, df_resid, gamma)

Y = mean + sigma * rng.standard_normal((reps, n))
num = np.einsum("ij,jk,ik->i", Y, M - M0, Y) / q
den = np.einsum("ij,jk,ik->i", Y, np.eye(n) - M, Y) / df_resid
print(gamma, exact_power, np.mean(num / den > crit))
```

4.4.4 Exercises

4.4.4.1 A. Check your understanding

Exercise A1.

In [Example \(Two estimators of \$\sigma^2\$ under a drifting mean\)](#) let $\mathbf{Y} \sim N_n(\mu\mathbf{1}, \sigma^2\mathbf{I})$ with $n \geq 3$. Use [Theorem 4.2.3\(d\)](#) to show that $\text{Cov}(S^2, Q) = \text{Var}(S^2)$, and deduce that S^2 and Q are dependent with correlation $\{2(n-1)/(3n-4)\}^{1/2}$. Which linear forms $\mathbf{a}^\top \mathbf{Y}$ are independent of Q for every mean vector $\mu \in \mathbb{R}^n$?

Solution. With $\Sigma = \sigma^2\mathbf{I}$, part (d) of [Theorem 4.2.3](#) gives $\text{Cov}(S^2, Q) = 2\sigma^4 \text{tr}(\mathbf{C}\mathbf{D}^\top \mathbf{D})/\{2(n-1)^2\}$. Since $\mathbf{D}\mathbf{1} = \mathbf{0}$, $\mathbf{C}\mathbf{D}^\top \mathbf{D} = \mathbf{D}^\top \mathbf{D}$, whose trace is $2(n-1)$, so $\text{Cov}(S^2, Q) = 2\sigma^4/(n-1) = \text{Var}(S^2)$. The mean terms vanish because $\mathbf{C}\mu = \mathbf{D}\mu = \mathbf{0}$. Using [Var\(Q\)](#) from [Example](#), the correlation is $\{\text{Var}(S^2)/\text{Var}(Q)\}^{1/2} = \{2(n-1)/(3n-4)\}^{1/2}$, which is nonzero, so the forms are dependent. By [Theorem 4.4.1](#), $\mathbf{a}^\top \mathbf{Y}$ is independent of Q for every μ iff $\mathbf{a}^\top \mathbf{D}^\top \mathbf{D} = \mathbf{0}^\top$. Multiplying by $\mathbf{a}$ gives $\|\mathbf{D}\mathbf{a}\|^2 = 0$, so $\mathbf{D}\mathbf{a} = \mathbf{0}$ and $\mathbf{a}$ is a multiple of $\mathbf{1}$; conversely such $\mathbf{a}$ work. The only such linear forms are the multiples of $\bar{Y}$.

4.4.4.2 B. Practice

Exercise B1.

Let $\mathbf{Y} \sim N_2(\mathbf{0}, \mathbf{I})$. Show that $Y_1^2 - Y_2^2$ and $2Y_1Y_2$ are uncorrelated but not independent. Why does [Proposition 4.4.3](#) not apply?

Solution. With $\mathbf{A} = \text{diag}(1, -1)$ and $\mathbf{B} = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$, $\text{tr}(\mathbf{AB}) = 0$, so the covariance $2\text{tr}(\mathbf{AB})$ is zero. In polar coordinates $Y_1 = R \cos \Theta$, $Y_2 = R \sin \Theta$, with $R^2 \sim \chi^2(2)$ independent of Θ uniform on $[0, 2\pi)$, the two forms are $R^2 \cos 2\Theta$ and $R^2 \sin 2\Theta$. Now $E(R^4) = 8$ and $E(R^8) = 384$, so $E[(R^2 \cos 2\Theta)^2 (R^2 \sin 2\Theta)^2] = 384/8 = 48$, but the product of the expected squares is $(8/2)^2 = 16$. The forms are dependent. [Proposition 4.4.3](#) needs nonnegative definite matrices, and $\mathbf{A}$ and $\mathbf{B}$ are indefinite.

Exercise B2.

Let $\mathbf{Y} \sim N_n(\mu, \Sigma)$ and let $\mathbf{A}, \mathbf{B}$ be nonnegative definite with $\Sigma\mathbf{A}\Sigma\mathbf{B}\Sigma = \mathbf{0}$. Show that $\mathbf{A}\Sigma\mathbf{B} = \mathbf{0}$, so that $\mathbf{Y}^\top \mathbf{A}\mathbf{Y}$ and $\mathbf{Y}^\top \mathbf{B}\mathbf{Y}$ are independent. Show by an example that the implication can fail when $\mathbf{A}$ is not nonnegative definite.

Solution. Write $\mathbf{A} = \mathbf{R}\mathbf{R}^\top$ and $\mathbf{B} = \mathbf{S}\mathbf{S}^\top$, for instance with symmetric square roots, and $\Sigma = \mathbf{L}\mathbf{L}^\top$ with $\mathbf{L}$ of full column rank. Cancelling $\mathbf{L}$ on both sides of $\mathbf{L}(\mathbf{L}^\top \mathbf{A}\mathbf{L})(\mathbf{L}^\top \mathbf{B}\mathbf{L})\mathbf{L}^\top = \mathbf{0}$ gives $\mathbf{L}^\top \mathbf{A}\mathbf{L}\mathbf{L}^\top \mathbf{B}\mathbf{L} = \mathbf{0}$. The trace of this matrix is

$$\text{tr}(\mathbf{L}^\top \mathbf{R}\mathbf{R}^\top \mathbf{L}\mathbf{L}^\top \mathbf{S}\mathbf{S}^\top \mathbf{L}) = \text{tr}\{(\mathbf{R}^\top \Sigma \mathbf{S})(\mathbf{R}^\top \Sigma \mathbf{S})^\top\} = \|\mathbf{R}^\top \Sigma \mathbf{S}\|_F^2,$$

so $\mathbf{R}^\top \Sigma \mathbf{S} = \mathbf{0}$. Hence $\mathbf{A}\Sigma\mathbf{B} = \mathbf{R}(\mathbf{R}^\top \Sigma \mathbf{S})\mathbf{S}^\top = \mathbf{0}$, and [Theorem 4.4.2](#) applies. For nonnegative definite matrices the two conditions are therefore equivalent, since $\mathbf{A}\Sigma\mathbf{B} = \mathbf{0}$ trivially implies $\Sigma\mathbf{A}\Sigma\mathbf{B}\Sigma = \mathbf{0}$. Without nonnegative definiteness the argument breaks

at the trace step. Take $\Sigma = \text{diag}(1, 0)$, $\mathbf{A} = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$ and $\mathbf{B} = \mathbf{I}_2$. Then $\Sigma\mathbf{A}\Sigma = \mathbf{0}$, so $\Sigma\mathbf{A}\Sigma\mathbf{B}\Sigma = \mathbf{0}$, but $\mathbf{A}\Sigma\mathbf{B} = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix} \neq \mathbf{0}$. With $\mu = (0, m)^\top$ and $m \neq 0$ we have $Y_2 = m$ with probability one, so the forms are $2mY_1$ and $Y_1^2 + m^2$. The second is a nonconstant function of the first, so they are dependent.

Exercise B3.

Let $\mathbf{Y} \sim N_n(\mathbf{X}\beta, \sigma^2\mathbf{I})$ with $\mathbf{X}$ of full column rank $p < n$, let $\hat{\beta} = (\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{X}^\top\mathbf{Y}$, $\mathbf{M} = \mathbf{X}(\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{X}^\top$ and $s^2 = \|(\mathbf{I} - \mathbf{M})\mathbf{Y}\|^2/(n - p)$. With v_{jj} the j th diagonal entry of $(\mathbf{X}^\top\mathbf{X})^{-1}$, show that $(\hat{\beta}_j - b)/(s\sqrt{v_{jj}}) \sim t(n - p, (\beta_j - b)/(\sigma\sqrt{v_{jj}}))$.

Solution. Here $\mathbf{M}$ is symmetric and idempotent with $\text{tr } \mathbf{M} = \text{tr}\{(\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{X}^\top\mathbf{X}\} = p$, and $\mathbf{M}\mathbf{X} = \mathbf{X}$. So $(n - p)s^2/\sigma^2 = \|(\mathbf{I} - \mathbf{M})\mathbf{Y}\|^2/\sigma^2 \sim \chi^2(n - p)$ by Theorem 4.3.2(a), because $(\mathbf{I} - \mathbf{M})\mathbf{X}\beta = \mathbf{0}$. With $\mathbf{B} = (\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{X}^\top$, $\mathbf{B}(\sigma^2\mathbf{I})(\mathbf{I} - \mathbf{M}) = \sigma^2(\mathbf{X}^\top\mathbf{X})^{-1}(\mathbf{X}^\top - \mathbf{X}^\top\mathbf{M}) = \mathbf{0}$, so $\hat{\beta} = \mathbf{B}\mathbf{Y}$ is independent of s^2 (Theorem 4.4.1). Since $\hat{\beta} \sim N_p(\beta, \sigma^2(\mathbf{X}^\top\mathbf{X})^{-1})$, the variable $X = (\hat{\beta}_j - b)/(\sigma\sqrt{v_{jj}})$ is $N((\beta_j - b)/(\sigma\sqrt{v_{jj}}), 1)$. The statistic is $X/\sqrt{V/(n - p)}$ with $V = (n - p)s^2/\sigma^2$, and Definition 4.1.3 applies.

Exercise B4.

Show that $F_\alpha(r, s) \rightarrow \chi_\alpha^2(r)/r$ as $s \rightarrow \infty$, where $\chi_\alpha^2(r)$ is the upper α point of $\chi^2(r)$, and that the power $\Pr\{F(r, s, \gamma) > F_\alpha(r, s)\}$ tends to $\Pr\{\chi^2(r, \gamma) > \chi_\alpha^2(r)\}$, the power of the test that knows σ^2 .

4.5 The Cochran theorem

An analysis of variance starts from an identity

$$\mathbf{Y}^\top\mathbf{Y} = \mathbf{Y}^\top\mathbf{A}_1\mathbf{Y} + \cdots + \mathbf{Y}^\top\mathbf{A}_k\mathbf{Y}, \quad \mathbf{A}_1 + \cdots + \mathbf{A}_k = \mathbf{I},$$

that splits the total sum of squares into pieces attributed to sources of variation. Theorem 4.4.4 says the pieces are independent chi-squared variables when the $\mathbf{A}_i$ are mutually orthogonal projections. Checking that directly means checking k idempotence conditions and $\binom{k}{2}$ orthogonality conditions. Cochran's theorem shows that one count of ranks does all of it.

4.5.1 The algebra

The matrix algebra behind this was done in Chapter 1. For symmetric matrices, any two of the three properties “each $\mathbf{A}_i$ is idempotent”, “ $\mathbf{A}_i\mathbf{A}_j = \mathbf{0}$ for $i \neq j$ ” and “the sum is idempotent” imply the third, and symmetric matrices that add to $\mathbf{I}$ with ranks adding to n are idempotent and mutually orthogonal (Theorem 1.9.3). The next theorem adds the rank count for a sum that is idempotent but not necessarily $\mathbf{I}$, which is what nested models need.

Theorem 4.5.1 — Cochran, algebraic form

Let $\mathbf{A}_1, \dots, \mathbf{A}_k$ be symmetric $n \times n$ matrices and $\mathbf{A} = \sum_i \mathbf{A}_i$. Consider

- (a) each $\mathbf{A}_i$ is idempotent;
- (b) $\mathbf{A}_i \mathbf{A}_j = \mathbf{0}$ for all $i \neq j$;
- (c) $\mathbf{A}$ is idempotent;
- (d) $\text{rank}(\mathbf{A}) = \sum_i \text{rank}(\mathbf{A}_i)$.

Then any two of (a), (b), (c) imply all four, and (c) together with (d) implies all four.

Proof. Any two of (a), (b), (c). By Theorem 1.9.3, any two of them imply the third. When all three hold, rank equals trace for symmetric idempotent matrices (Theorem 1.9.2), so $\text{rank}(\mathbf{A}) = \text{tr}(\mathbf{A}) = \sum_i \text{tr}(\mathbf{A}_i) = \sum_i \text{rank}(\mathbf{A}_i)$, which is (d).

(c) and (d) imply (a) and (b). Let $r = \text{rank}(\mathbf{A})$ and $r_i = \text{rank}(\mathbf{A}_i)$. Every $\mathbf{A}\mathbf{v} = \sum_i \mathbf{A}_i \mathbf{v}$ lies in $\mathcal{C}(\mathbf{A}_1) + \dots + \mathcal{C}(\mathbf{A}_k)$, whose dimension is at most $\sum_i r_i = r = \dim \mathcal{C}(\mathbf{A})$. So the two spaces coincide, and in particular $\mathcal{C}(\mathbf{A}_i) \subseteq \mathcal{C}(\mathbf{A})$. By Theorem 1.9.2(a), $\mathbf{A} = \mathbf{Q}\mathbf{Q}^\top$ with $\mathbf{Q}$ of size $n \times r$ and $\mathbf{Q}^\top \mathbf{Q} = \mathbf{I}_r$. Since $\mathbf{A}$ is symmetric and idempotent it fixes every vector of its column space, so $\mathbf{A}\mathbf{A}_i = \mathbf{A}_i$, and transposing, $\mathbf{A}_i \mathbf{A} = \mathbf{A}_i$. Hence $\mathbf{A}_i = \mathbf{A}\mathbf{A}_i \mathbf{A} = \mathbf{Q}\mathbf{B}_i \mathbf{Q}^\top$ with $\mathbf{B}_i = \mathbf{Q}^\top \mathbf{A}_i \mathbf{Q}$. The $r \times r$ matrices $\mathbf{B}_i$ are symmetric, satisfy $\sum_i \mathbf{B}_i = \mathbf{Q}^\top \mathbf{A} \mathbf{Q} = \mathbf{I}_r$, and have $\text{rank}(\mathbf{B}_i) = r_i$ because $\mathbf{Q}$ has orthonormal columns. By the last statement of Theorem 1.9.3, $\mathbf{B}_i^2 = \mathbf{B}_i$ and $\mathbf{B}_i \mathbf{B}_j = \mathbf{0}$ for $i \neq j$, and multiplying by $\mathbf{Q}$ on the left and $\mathbf{Q}^\top$ on the right transfers both identities to the $\mathbf{A}_i$. ■

4.5.2 The distributional statement**Theorem 4.5.2 — Cochran**

Let $\mathbf{Y} \sim N_n(\boldsymbol{\mu}, \sigma^2 \mathbf{I})$, and let $\mathbf{A}_1, \dots, \mathbf{A}_k$ be symmetric with ranks $r_1, \dots, r_k$, whose sum $\mathbf{A} = \sum_i \mathbf{A}_i$ is idempotent of rank r (for example $\mathbf{A} = \mathbf{I}$, $r = n$). The following are equivalent:

- (i) the forms $\mathbf{Y}^\top \mathbf{A}_i \mathbf{Y} / \sigma^2$ are mutually independent, and $\mathbf{Y}^\top \mathbf{A}_i \mathbf{Y} / \sigma^2 \sim \chi^2(r_i, \boldsymbol{\mu}^\top \mathbf{A}_i \boldsymbol{\mu} / \sigma^2)$ for each i ;
- (ii) $r_1 + \dots + r_k = r$;
- (iii) each $\mathbf{A}_i$ is idempotent;
- (iv) $\mathbf{A}_i \mathbf{A}_j = \mathbf{0}$ for all $i \neq j$.

Proof. Condition (c) of Theorem 4.5.1 holds by assumption, so that theorem shows that (ii), (iii) and (iv) are equivalent, and each implies the other two. If (iii) and (iv) hold, Theorem 4.4.4 gives (i). If (i) holds, apply Theorem 4.3.1 to $\sigma^{-1} \mathbf{Y} \sim N_n(\sigma^{-1} \boldsymbol{\mu}, \mathbf{I})$. Each $\mathbf{A}_i$ with $r_i \geq 1$ is idempotent, and an $\mathbf{A}_i$ of rank zero is $\mathbf{0}$. This is (iii). ■

In words: *if the degrees of freedom add up, the sums of squares are independent chi-squared variables.* Independence in (i) is not needed to reach (iii). The marginal chi-squared laws alone force the whole structure.

Example. (The one-way layout)

Let observations fall into g groups of sizes $n_1, \dots, n_g$, with $Y_{\ell j}$ independent $N(\mu_\ell, \sigma^2)$. Let $\mathbf{M}$ replace each observation by its group mean, and $\mathbf{P}_1 = n^{-1}\mathbf{1}\mathbf{1}^\top$ by the grand mean. Then

$$\mathbf{I} = \mathbf{P}_1 + (\mathbf{M} - \mathbf{P}_1) + (\mathbf{I} - \mathbf{M})$$

splits $\|\mathbf{Y}\|^2$ into $n\bar{Y}^2$, the between-group sum of squares $\sum_\ell n_\ell(\bar{Y}_\ell - \bar{Y})^2$ and the within-group sum of squares $\sum_{\ell,j} (Y_{\ell j} - \bar{Y}_\ell)^2$. The three matrices are symmetric with ranks 1, $g - 1$ and $n - g$, which add to n . Cochran's theorem needs nothing more. The three sums of squares, divided by σ^2 , are independent noncentral chi-squared variables. Their noncentralities are $n\bar{\mu}^2/\sigma^2$, $\sum_\ell n_\ell(\mu_\ell - \bar{\mu})^2/\sigma^2$ with $\bar{\mu} = \sum_\ell n_\ell\mu_\ell/n$, and 0. The last is zero because $\mathbf{M}\boldsymbol{\mu} = \boldsymbol{\mu}$. The ratio of the between and within mean squares is $F(g - 1, n - g, \gamma)$ by [Theorem 4.4.6](#), and it is central iff all group means are equal.

The script uses $g = 3$ groups of sizes 3, 5, 4, means 2, 3, 1.5, and $\sigma = 1$, and draws 200,000 data sets. [Table 4.5.1](#) compares the theoretical means $r_i + \gamma_i$ and variances $2r_i + 4\gamma_i$ with the simulated ones. The largest absolute correlation between the three sums of squares is 0.0013. As a check that goes beyond correlation, the proportion of data sets in which both the between and the within sums of squares exceed their medians is 0.2497, against 0.25 for independent variables.

```
import numpy as np
from scipy import stats
rng = np.random.default_rng(5150)

sizes, means, sigma = [3, 5, 4], [2.0, 3.0, 1.5], 1.0
groups = np.repeat(np.arange(3), sizes)
n = len(groups)
Z = (groups[:, None] == np.arange(3)).astype(float) # group indicators
M = Z @ np.diag(1 / np.array(sizes)) @ Z.T # group-mean projection
J = np.ones((n, n)) / n
pieces = [J, M - J, np.eye(n) - M]
ranks = [np.linalg.matrix_rank(P) for P in pieces]
print("ranks", ranks, "sum", sum(ranks), "n", n)

mu = np.array(means)[groups]
gammas = [mu @ P @ mu / sigma**2 for P in pieces]
Y = mu + sigma * rng.standard_normal((200_000, n))
Q = np.stack([np.einsum("ij,jk,ik->i", Y, P, Y) for P in pieces], axis=1) / sigma**2
print(np.corrcoef(Q, rowvar=False).round(3))
```

Table 4.5.1. Cochran decomposition for a one-way layout with $n = 12$: theory against 200,000 simulated data sets.

Sum of squares	r_i	γ_i	mean	simulated	variance	simulated
grand mean	1	60.750	61.750	61.763	245.00	244.95

Sum of squares	r_i	γ_i	mean	simulated	variance	simulated
between groups	2	5.250	7.250	7.230	25.00	24.85
within groups	9	0.000	9.000	8.994	18.00	17.96

Remark (What the theorem does not cover). Two assumptions carry the weight. The first is normality. Without it, [Theorem 4.2.3\(d\)](#) fails, and with independent non-normal errors the between and within sums of squares are in general correlated, with a correlation driven by the fourth cumulant ([Exercise \(B2\)](#)). The second is the spherical covariance $\sigma^2 \mathbf{I}$. With $\text{Cov}(\mathbf{Y}) = \sigma^2 \mathbf{V}$, $\mathbf{V}$ positive definite, apply the theorem to $\mathbf{V}^{-1/2} \mathbf{Y}$, whose forms have matrices $\mathbf{V}^{1/2} \mathbf{A}_i \mathbf{V}^{1/2}$. Their ranks are those of the $\mathbf{A}_i$, so the rank count is unchanged. What changes is the condition on the sum: $\mathbf{V}^{1/2} \mathbf{A} \mathbf{V}^{1/2}$ is idempotent iff $\mathbf{A} \mathbf{V} \mathbf{A} = \mathbf{A}$ (for example $\sum_i \mathbf{A}_i = \mathbf{V}^{-1}$), and conditions (iii) and (iv) become $\mathbf{A}_i \mathbf{V} \mathbf{A}_i = \mathbf{A}_i$ and $\mathbf{A}_i \mathbf{V} \mathbf{A}_j = \mathbf{0}$. These are the generalized least squares decompositions of Chapter 31.

Most analysis of variance tables in this book come from a chain of nested model spaces. The successive differences of their projections are mutually orthogonal projections whose ranks add up, so Cochran's theorem applies without further work ([Exercise \(B2\)](#), [Theorem 9.1.1](#)).

4.5.3 Exercises

4.5.3.1 A. Check your understanding

Exercise A1.

Let $\mathbf{A}_1 = \text{diag}(2, 0)$ and $\mathbf{A}_2 = \text{diag}(-1, 1)$, so that $\mathbf{A}_1 + \mathbf{A}_2 = \mathbf{I}_2$. Which hypotheses of [Theorem 4.5.2](#) fail, and what are the laws of the two forms when $\mathbf{Y} \sim N_2(\mathbf{0}, \mathbf{I})$?

Solution. The sum $\mathbf{I}_2$ is idempotent, but every one of the conditions (ii)–(iv) fails. The ranks are 1 and 2, which add to $3 \neq 2$. Neither matrix is idempotent, since $\mathbf{A}_1^2 = \text{diag}(4, 0)$ and $\mathbf{A}_2^2 = \mathbf{I}_2$. And $\mathbf{A}_1 \mathbf{A}_2 = \text{diag}(-2, 0) \neq \mathbf{0}$. The forms are $2Y_1^2$, which is twice a $\chi^2(1)$ variable, and $Y_2^2 - Y_1^2$, the difference of two independent $\chi^2(1)$ variables, which takes negative values. Neither is chi-squared. They are dependent, because

$$\text{Cov}(2Y_1^2, Y_2^2 - Y_1^2) = 2 \text{Cov}(Y_1^2, Y_2^2) - 2 \text{Var}(Y_1^2) = 0 - 2 \cdot 2 = -4 \neq 0.$$

4.5.3.2 B. Practice

Exercise B1.

A two-way layout has one observation Y_{ij} for each a rows and b columns. Write $\|\mathbf{Y}\|^2$ as the sum of the grand-mean, row, column and residual sums of squares, identify the four matrices, and use [Theorem 4.5.2](#) to show that under $Y_{ij} \sim N(\mu + \alpha_i + \beta_j, \sigma^2)$

independently the four sums of squares are independent, with degrees of freedom 1, $a - 1$, $b - 1$ and $(a - 1)(b - 1)$.

Exercise B2.

Let $\mathbf{P}_1, \dots, \mathbf{P}_m$ be symmetric idempotent with $\mathcal{C}(\mathbf{P}_1) \subset \mathcal{C}(\mathbf{P}_2) \subset \dots \subset \mathcal{C}(\mathbf{P}_m)$, and put $\mathbf{P}_0 = \mathbf{0}$, $\mathbf{P}_{m+1} = \mathbf{I}$. Show that the matrices $\mathbf{P}_i - \mathbf{P}_{i-1}$, $i = 1, \dots, m+1$, satisfy the conditions of [Theorem 4.5.2](#), so that the sequential sums of squares $\|(\mathbf{P}_i - \mathbf{P}_{i-1})\mathbf{Y}\|^2$ are independent noncentral chi-squared variables when $\mathbf{Y} \sim N_n(\boldsymbol{\mu}, \sigma^2 \mathbf{I})$.

4.5.3.3 C. Going deeper

Exercise C1.

Let $\mathbf{Y} \sim N_n(\mathbf{0}, \mathbf{I})$, and $Q = Q_1 + Q_2$ with $Q = \mathbf{Y}^\top \mathbf{A} \mathbf{Y} \sim \chi^2(r)$, $Q_1 = \mathbf{Y}^\top \mathbf{A}_1 \mathbf{Y} \sim \chi^2(r_1)$ and $Q_2 = \mathbf{Y}^\top \mathbf{A}_2 \mathbf{Y}$ for a nonnegative definite $\mathbf{A}_2$. Show that Q_1 and Q_2 are independent and $Q_2 \sim \chi^2(r - r_1)$.

Solution. By [Theorem 4.3.1](#), $\mathbf{A}$ and $\mathbf{A}_1$ are idempotent, with ranks r and r_1 . If $\mathbf{A}\mathbf{v} = \mathbf{0}$, then $0 = \mathbf{v}^\top \mathbf{A}_1 \mathbf{v} + \mathbf{v}^\top \mathbf{A}_2 \mathbf{v}$ with both terms nonnegative, so $\|\mathbf{A}_1 \mathbf{v}\|^2 = \mathbf{v}^\top \mathbf{A}_1 \mathbf{v} = 0$. Thus $\mathcal{N}(\mathbf{A}) \subseteq \mathcal{N}(\mathbf{A}_1)$, hence $\mathcal{C}(\mathbf{A}_1) \subseteq \mathcal{C}(\mathbf{A})$ and $\mathbf{A}\mathbf{A}_1 = \mathbf{A}_1 = \mathbf{A}_1 \mathbf{A}$. Then $\mathbf{A}_2^2 = (\mathbf{A} - \mathbf{A}_1)^2 = \mathbf{A} - 2\mathbf{A}_1 + \mathbf{A}_1 = \mathbf{A}_2$, so $\mathbf{A}_2$ is idempotent, and $\mathbf{A}_1 \mathbf{A}_2 = \mathbf{A}_1 \mathbf{A} - \mathbf{A}_1 = \mathbf{0}$. By [Theorem 4.4.4](#), Q_1 and Q_2 are independent and $Q_2 \sim \chi^2(\text{tr } \mathbf{A}_2) = \chi^2(r - r_1)$.

4.6 Summary and Notes

4.6.1 Summary

Idea.

1. If $\mathbf{Z} \sim N_r(\boldsymbol{\mu}, \mathbf{I})$ then $\|\mathbf{Z}\|^2 \sim \chi^2(r, \gamma)$ with $\gamma = \|\boldsymbol{\mu}\|^2$, mean $r + \gamma$ and variance $2r + 4\gamma$. The law is a $\text{Poisson}(\gamma/2)$ mixture of $\chi^2(r + 2k)$ laws, and its tails increase strictly with γ ([Theorem 4.1.2](#) and [Proposition 4.1.3](#)).
2. $F(r, s, \gamma)$ and $t(s, \delta)$ put a noncentral numerator over an independent central denominator. The power of an F test increases strictly with γ , and $T^2 \sim F(1, s, \delta^2)$ ([Theorem 4.1.4](#)).
3. $E(\mathbf{Y}^\top \mathbf{A} \mathbf{Y}) = \text{tr}(\mathbf{A}\boldsymbol{\Sigma}) + \boldsymbol{\mu}^\top \mathbf{A} \boldsymbol{\mu}$ for any distribution. Under normality

$$\text{Var}(\mathbf{Y}^\top \mathbf{A} \mathbf{Y}) = 2 \text{tr}\{(\mathbf{A}\boldsymbol{\Sigma})^2\} + 4\boldsymbol{\mu}^\top \mathbf{A} \boldsymbol{\Sigma} \mathbf{A} \boldsymbol{\mu}, \quad \text{Cov}(\mathbf{K} \mathbf{Y}, \mathbf{Y}^\top \mathbf{A} \mathbf{Y}) = 2\mathbf{K} \boldsymbol{\Sigma} \mathbf{A} \boldsymbol{\mu}$$

([Theorem 4.2.3](#)).

4. A normal quadratic form is a constant plus independent terms $\lambda_j(W_j + \nu_j)^2$, and for singular $\boldsymbol{\Sigma}$ possibly normal terms. The weights λ_j are the nonzero eigenvalues of $\mathbf{A}\boldsymbol{\Sigma}$ ([Theorem 4.2.1](#)).
5. For $\mathbf{Y} \sim N_n(\boldsymbol{\mu}, \mathbf{I})$, $\mathbf{Y}^\top \mathbf{A} \mathbf{Y}$ is chi-squared iff $\mathbf{A}$ is idempotent. Then the degrees of freedom are $\text{rank } \mathbf{A} = \text{tr } \mathbf{A}$ and the noncentrality is $\boldsymbol{\mu}^\top \mathbf{A} \boldsymbol{\mu}$ ([Theorem 4.3.1](#)).

For positive definite Σ the condition is that $\mathbf{A}\Sigma$ be idempotent. For projections, $\|\mathbf{P}\mathbf{Y}\|^2/\sigma^2 \sim \chi^2(\text{rank } \mathbf{P}, \|\mathbf{P}\boldsymbol{\mu}\|^2/\sigma^2)$ (Theorem 4.3.2).

6. $\mathbf{B}\mathbf{Y}$ and $\mathbf{Y}^\top \mathbf{A}\mathbf{Y}$ are independent if $\mathbf{B}\Sigma\mathbf{A} = \mathbf{0}$, and $\mathbf{Y}^\top \mathbf{A}\mathbf{Y}$ and $\mathbf{Y}^\top \mathbf{B}\mathbf{Y}$ are independent if $\mathbf{A}\Sigma\mathbf{B} = \mathbf{0}$, because a quadratic form is a function of $\mathbf{A}\mathbf{Y}$ (Theorem 4.4.1 and Theorem 4.4.2).
7. Projections of a spherical normal vector onto mutually orthogonal subspaces are independent. This gives the independence of $\bar{Y}$ and S^2 and the exact $F(r - r_0, n - r, \gamma)$ law of the nested-model F statistic (Theorem 4.4.4, Corollary 4.4.5 and Theorem 4.4.6).
8. Cochran: if symmetric matrices add to an idempotent matrix and their ranks add up, the corresponding sums of squares are independent chi-squared variables (Theorem 4.5.1 and Theorem 4.5.2).

4.6.2 Notes and sources

Source texts. Rencher and Schaalje (2008, chapter 5) cover the material of Section 4.2, Section 4.3, Section 4.4 and Section 4.5 for positive definite covariance matrices. They derive the mgf of a quadratic form by integrating against the normal density and obtain its variance by differentiating the logarithm of the mgf; the converse of the chi-squared theorem is referred to the literature. They use the noncentrality $\boldsymbol{\mu}^\top \boldsymbol{\mu}/2$. Seber and Lee (2003, section 2.4) reduce the general case to a spherical normal vector by an orthogonal diagonalization. They prove necessity of idempotence for central forms by factorizing the moment generating function, and give the eigenvalue criterion for singular covariance matrices. Christensen (2020, section 1.3) states the three conditions of Theorem 4.3.2(c) for arbitrary nonnegative definite Σ , and obtains independence by viewing quadratic forms as functions of linear ones, the approach followed in Section 4.4. The canonical representation (Theorem 4.2.1), the proof of necessity through the coefficients Eq. 4.3.1, the monotonicity proof through the Poisson mixture, and the treatment of the Cochran algebra are this book's own arrangements of standard material.

Noncentral distributions. The noncentral chi-squared distribution first appeared in Fisher (1928). Patnaik (1949) studied its approximation by a scaled central chi-squared variable, and Satterthwaite (1946) used the same two-moment idea for linear combinations of mean squares, which reappears in Chapter 32. Johnson et al. (1995, chapters 29–31) collect the properties of the noncentral chi-squared, F and t families. The monotonicity of power in the noncentrality and in the degrees of freedom is treated systematically by Ghosh (1973).

General quadratic forms. The exact distribution of a quadratic form that is not chi-squared can be computed by numerical inversion of its characteristic function (Gil-Pelaez 1951; Imhof 1961). Mathai and Provost (1992) give a book-length treatment of quadratic forms in normal and non-normal variables. The mean square successive difference of Example (Two estimators of σ^2 under a drifting mean) goes back to von Neumann (1941), who derived the exact distribution of its ratio to S^2 .

Independence and Cochran's theorem. Cochran (1934) proved the decomposition theorem that bears his name, in the context of the analysis of covariance. The necessity of $\mathbf{A}\Sigma\mathbf{B} = \mathbf{0}$ for independence of two forms is attributed to Craig (1943). Its full proof for arbitrary symmetric matrices and arbitrary means turned out to be subtle, and Driscoll and Gundberg (1986) trace the flawed and correct arguments. Searle (1971) gives general proofs of the independence and Cochran theorems in the style most linear-models texts follow. For nonnegative definite $\mathbf{A}$ and $\mathbf{B}$, the condition $\Sigma\mathbf{A}\Sigma\mathbf{B}\Sigma = \mathbf{0}$ of Christensen (2020, Theorem 1.3.8) is in fact equivalent to

$\mathbf{A}\Sigma\mathbf{B} = \mathbf{0}$ (Exercise (B2)), so that theorem adds nothing to Theorem 4.4.2 even for singular Σ , whereas his Theorem 1.3.9, for general symmetric $\mathbf{A}$ and $\mathbf{B}$, does. That the independence of $\bar{Y}$ and S^2 characterizes the normal distribution was shown by Lukacs (1942).

4.6.3 References

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Chapter 5

The Model, Least Squares and the Normal Equations

A regression model is a claim about how the distribution of a response changes with a set of regressors, and the linear model is the simplest useful claim of that kind. It says that the mean of the response is a linear combination of known columns, that the errors around this mean have a common variance and are uncorrelated, and nothing more. This chapter sets the model up and draws out what follows from those assumptions alone. It starts with what the assumptions do and do not say, fits a straight line by calculus, and passes to the matrix form of the model, where polynomials, harmonics, indicator variables and interactions all turn out to be linear models. It then derives the least squares estimator and the normal equations, computes the mean and covariance of the estimator, estimates the error variance without bias, and asks what a fitted coefficient means. The last section surveys the models the rest of the book builds on this one. Throughout, the model matrix has full column rank. [Chapter 6](#) retells the same story geometrically and removes that condition.

What you need. [Chapter 1](#): rank ([Proposition 1.2.3](#)), inverses, nonnegative definite matrices ([Proposition 1.7.5](#)), traces ([Theorem 1.6.1](#)) and the derivatives of linear and quadratic forms ([Theorem 1.11.1](#)). [Chapter 2](#): means and covariances of linear transformations ([Theorem 2.2.1](#)), the mean and variance of a quadratic form ([Theorem 2.3.1](#), [Theorem 2.3.3](#)) and the laws of total expectation and covariance ([Proposition 2.5.4](#)). Normality ([Chapter 3](#)) appears only in remarks.

5.0.1 Roadmap

- [What a regression model claims](#)
- [Simple linear regression as a warm-up](#)
- [The multiple linear regression model](#)
- [Least squares and the normal equations](#)
- [Properties of the estimator under second-moment assumptions](#)
- [Estimating the error variance](#)
- [Interpreting coefficients](#)
- [A panorama of regression models to come](#)
- [Summary and notes](#)

5.1 What a regression model claims

A data set for regression consists of n cases. Each case records a response y_i and a few other measurements, the regressors, on the same unit, such as a patient or a month. The question is how the response depends on the regressors. Two cases with identical regressors almost always have different responses, so a regression model cannot be an equation linking y to the regressors. It is a statement about the *distribution* of the response given the regressors, and usually only about one feature of that distribution, its mean.

This section separates what is automatically true of any response and regressors from what a model actually assumes. The linear model's assumptions come in layers, and each layer buys a different part of the theory.

5.1.1 The regression function

Let Y be a response with finite variance and $\mathbf{X}$ a random vector of regressors, defined on the same probability space. We need only elementary properties of conditional expectation: the tower property $E[E(Y | \mathbf{X})] = E(Y)$, and $E[g(\mathbf{X})Y | \mathbf{X}] = g(\mathbf{X})E(Y | \mathbf{X})$ for functions g of the regressors.

Definition 5.1.1 — Regression function

The **regression function** of Y on $\mathbf{X}$ is the conditional mean

$$m(\mathbf{x}) = E(Y | \mathbf{X} = \mathbf{x}).$$

The **error** is $\varepsilon = Y - m(\mathbf{X})$, so that $Y = m(\mathbf{X}) + \varepsilon$.

The decomposition $Y = m(\mathbf{X}) + \varepsilon$ looks like a model, but it is not one. It holds for every pair $(\mathbf{X}, Y)$ with $EY^2 < \infty$, and the error it defines has two properties that are often listed as “assumptions” even though they are automatic.

Proposition 5.1.1 — The error of the regression function

Let $EY^2 < \infty$, and let m and ε be as in Definition 5.1.1.

- (a) $E(\varepsilon | \mathbf{X}) = 0$, and hence $E(\varepsilon) = 0$.
- (b) $E[\varepsilon g(\mathbf{X})] = 0$ for every function g with $Eg(\mathbf{X})^2 < \infty$. In particular ε is uncorrelated with every such function of the regressors.
- (c) For every such g ,

$$E[(Y - g(\mathbf{X}))^2] = E(\varepsilon^2) + E[(m(\mathbf{X}) - g(\mathbf{X}))^2].$$

So $m(\mathbf{X})$ is the best predictor of Y from $\mathbf{X}$ in mean squared error, and it is the only one up to events of probability zero.

Proof. (a) $E(\varepsilon | \mathbf{X}) = E(Y | \mathbf{X}) - m(\mathbf{X}) = 0$, and the tower property gives $E(\varepsilon) = E[E(\varepsilon | \mathbf{X})] = 0$. (b) The product $\varepsilon g(\mathbf{X})$ is integrable by the Cauchy–Schwarz inequality,

because $E\varepsilon^2 \leq 2EY^2 + 2Em(\mathbf{X})^2$ and $Em(\mathbf{X})^2 \leq EY^2$ by Jensen's inequality for conditional expectations. Then $E[\varepsilon g(\mathbf{X})] = E[g(\mathbf{X}) E(\varepsilon | \mathbf{X})] = 0$. (c) Write $Y - g(\mathbf{X}) = \varepsilon + (m(\mathbf{X}) - g(\mathbf{X}))$ and expand the square. The cross term is $2E[\varepsilon(m(\mathbf{X}) - g(\mathbf{X}))]$, which is zero by (b) applied to the function $m - g$. The second term on the right is zero only if $g(\mathbf{X}) = m(\mathbf{X})$ with probability one. ■

Part (c) is the reason the conditional mean is the natural target. It is what an ideal forecaster would report under squared-error loss. It is also the reason regression is sometimes misread. The function m summarizes the conditional distribution by its centre. Two populations can have the same regression function and very different spreads, shapes and tails around it.

Warning. Part (b) says the error is uncorrelated with the regressors *by construction*. The lack of correlation between errors and regressors therefore cannot be tested from the data or defended as a scientific assumption about the regression function. The real assumption hides elsewhere: in the claim that m has a particular form, or that the quantity of interest is m at all. When someone says “the error is correlated with the regressor”, the error they have in mind is a different object, usually the deviation from a *causal* or *structural* relation rather than from the conditional mean. Chapter 24 and Chapter 25 take this up.

5.1.2 What the linear model adds

A linear regression model makes claims about m and about the errors that $EY^2 < \infty$ does not guarantee. Write $\mathbf{x}_{(i)}$ for the regressors of case i , already transformed into the columns the model uses (a constant 1, the regressors themselves, their squares, indicators of categories, and so on; [Section 5.3](#) gives many examples). The assumptions come in four layers.

- (L1) **Linearity.** $E(Y_i) = \mathbf{x}_{(i)}^\top \boldsymbol{\beta}$ for an unknown $\boldsymbol{\beta} \in \mathbb{R}^p$. The regression function belongs to a fixed p -dimensional family of functions.
- (L2) **Constant variance.** $\text{Var}(Y_i) = \sigma^2$ for every i , with σ^2 unknown.
- (L3) **Uncorrelated errors.** $\text{Cov}(Y_i, Y_j) = 0$ for $i \neq j$.
- (L4) **Normality.** The errors are jointly normal.

The first three layers are the **second-moment assumptions**. They mention only means, variances and covariances, and they are all that [Section 5.5](#) and [Section 5.6](#) use. Adding (L4) gives the normal linear model, and with it exact sampling distributions for estimates and test statistics, which [Chapter 7](#) and [Chapter 11](#) develop. Each layer is used by a specific part of the theory:

Assumptions	What they deliver	Where
(L1)	the least squares estimator is unbiased	Section 5.5
(L1)–(L3)	its covariance matrix; an unbiased estimate of σ^2 ; Gauss–Markov optimality	Section 5.5 , Section 5.6 , Chapter 7

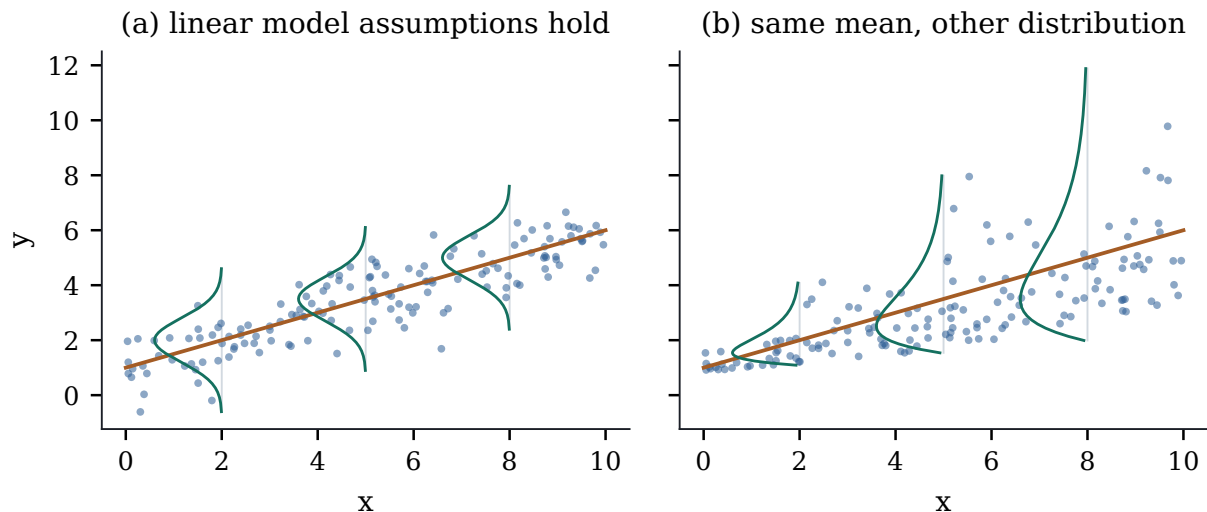


Figure 5.1.1: Two data-generating processes with the same regression function $1 + 0.5x$ (brown line). The curves are conditional densities of Y given x , drawn sideways at $x = 2, 5, 8$. (a) Normal errors with constant variance: all four layers of assumptions hold. (b) Right-skewed errors whose spread grows with x : only linearity holds.

Assumptions	What they deliver	Where
(L1)–(L4)	exact t and F distributions; maximum likelihood; exact intervals	Chapter 7 , Chapter 11 , Chapter 12

Least squares itself needs none of them. The estimator of [Section 5.4](#) is defined for any data at all. The assumptions only tell us how good it is.

Example. (Same regression function, different models)

[Figure 5.1.1](#) shows two simulated data sets with the same regression function $m(x) = 1 + 0.5x$. In panel (a) the errors are normal with constant standard deviation 0.8. In panel (b) they are skewed to the right, and their standard deviation grows linearly in x , from 0.15 at $x = 0$ to 2.65 at $x = 10$. The curves show the conditional density of Y at three values of x . Both processes satisfy (L1), so least squares estimates the line without bias in both. Only process (a) satisfies (L2), so only there do the variance formulas of [Section 5.5](#) apply as stated. In process (b) most points lie below the line, because the median of a right-skewed distribution is below its mean. The line is still the conditional mean.

The assumptions concern the errors ε_i , which are never observed. What we see are residuals, the differences between the responses and a fitted function. Whether the assumptions are credible is judged from residuals, from the design of the study, and from knowledge of the subject. Part V of the book is devoted to the first of these and to what goes wrong when the assumptions fail.

5.1.3 Fixed and random regressors

The layers (L1)–(L4) were written with $E(Y_i)$ and $\text{Var}(Y_i)$, as if the regressors were constants. In a designed experiment they are. The experimenter chooses the temperatures, doses or fertilizer levels, and only the responses are random. In an observational study the regressors are measured on units drawn from a population, so the pairs $(\mathbf{x}_{(i)}, Y_i)$ are random together.

This book's convention, like that of most linear-model theory, is to treat the regressors as fixed, and to read (L1)–(L4) in the random case as statements *conditional on the regressors*: $E(Y_i | \mathbf{X}) = \mathbf{x}_{(i)}^\top \boldsymbol{\beta}$, $\text{Var}(Y_i | \mathbf{X}) = \sigma^2$, and so on, where $\mathbf{X}$ collects all the regressors. There are two reasons for this choice.

First, conditional statements imply unconditional ones where it matters. If the estimator is unbiased given every possible $\mathbf{X}$, it is unbiased when $\mathbf{X}$ is random, by the tower property. [Section 5.5](#) makes this precise ([Corollary 5.5.5](#)). Second, conditioning describes the precision of the data actually in hand. A sample whose regressors happen to be spread out estimates a slope better than one whose regressors happen to be bunched together. The conditional variance says so, and the unconditional variance averages the two cases.

Example. (One slope, two sampling schemes)

Take $Y = 1 + 0.5x + \varepsilon$ with standard normal errors and $n = 15$ cases. In the *fixed design*, the regressor values are drawn once from the uniform distribution on $[0, 10]$ and then kept. They have $S_{xx} = \sum_i (x_i - \bar{x})^2 = 103.73$. In the *random design*, a new set of 15 regressor values is drawn for every sample. Over 50,000 simulated samples, the least squares slope (formula in [Section 5.2](#)) averages 0.5007 under the fixed design and 0.5006 under the random one. Both are unbiased. Their variances differ in meaning. Under the fixed design the simulated variance is 0.00960, matching $\sigma^2/S_{xx} = 0.00964$ for this particular design. Under the random design it is 0.00933, matching the *average* $E(\sigma^2/S_{xx}) = 0.00923$ over designs, estimated from the same simulation. Replacing S_{xx} by its expectation 116.67 would give 0.00857, which is too small. By Jensen's inequality $E(1/S_{xx}) \geq 1/E(S_{xx})$, so random bunching of the regressors costs precision on average.

```
import numpy as np

rng = np.random.default_rng(2005)
n, beta0, beta1, sigma = 15, 1.0, 0.5, 1.0
reps = 50_000

def slope(x, y):
    """Least squares slope of y on x (with an intercept)."""
    xc = x - x.mean()
    return xc @ (y - y.mean()) / (xc @ xc)

x_fixed = rng.uniform(0, 10, size=n)          # drawn once, then held fixed
Sxx_fixed = np.sum((x_fixed - x_fixed.mean()) ** 2)
b_fixed = np.empty(reps)
b_random = np.empty(reps)
Sxx_random = np.empty(reps)
for r in range(reps):
    e = sigma * rng.standard_normal(n)
    b_fixed[r] = slope(x_fixed, beta0 + beta1 * x_fixed + e)
    x = rng.uniform(0, 10, size=n)            # a new design for every sample
```

```

Sxx_random[r] = np.sum((x - x.mean()) ** 2)
b_random[r] = slope(x, beta0 + beta1 * x + e)

print(f"fixed design: mean {b_fixed.mean():.4f} var {b_fixed.var():.5f}"
      f" sigma^2/Sxx = {sigma**2 / Sxx_fixed:.5f}")
print(f"random design: mean {b_random.mean():.4f} var {b_random.var():.5f}"
      f" E(sigma^2/Sxx) = {np.mean(sigma**2 / Sxx_random):.5f}")

```

Conditioning on the regressors is not always harmless. It fails when the regressors are measured with error, because then the regression on the *measured* values is not the one of scientific interest (Chapter 24). It fails when a regressor is a lagged value of the response, as in autoregressive time series, because then $\mathbf{X}$ is not determined before the errors are drawn. It also changes the question when the target is a population quantity, such as the best linear predictor of one variable from others in a population where the regression function is not linear (Section 6.11 and Chapter 14).

5.1.4 What regression is used for

The same fitted equation can serve different purposes, and the purpose decides which assumptions matter. For **prediction** of new cases only the accuracy of the fitted mean matters, and a regressor can be useful without any causal role. For **description** the fitted mean summarizes how the response varies across the population sampled. For **estimation of a parameter**, such as an elasticity or a dose-response slope, one coefficient and its uncertainty are the point. For **control**, regressor values are chosen to achieve a desired response, which requires the fitted relation to hold when the regressors are *set* rather than observed: a causal claim.

Warning. The regression function describes how the mean response differs between cases that *are observed* to have different regressor values. It does not by itself describe what would happen if a regressor were *changed*. Units with high values of one regressor usually differ in other ways too. Randomized assignment of the regressor, as in a designed experiment, is the standard way to make the two questions coincide. Without it, the step from association to effect needs assumptions that the data cannot check. Section 5.7 returns to this point, and Chapter 25 treats it in full.

5.1.5 Exercises

5.1.5.1 A. Check your understanding

Exercise A1.

Let X take the values $-1, 0, 1$ with probability $1/3$ each, and let $Y = X^2 + XZ$, where $Z \sim N(0, 1)$ is independent of X . Find the regression function m and the error ε . Show that $E(\varepsilon | X) = 0$ and $\text{Cov}(\varepsilon, X) = 0$, but that ε and X are not independent. Which of the layers (L1)–(L4) fail for a sample from this distribution, with the columns $1, x, x^2$?

Solution. Since $E(XZ | X) = X E(Z) = 0$, $m(x) = x^2$ and $\varepsilon = XZ$. Then $E(\varepsilon | X) = 0$, and $\text{Cov}(\varepsilon, X) = E(X^2Z) = E(X^2)E(Z) = 0$. But $\text{Var}(\varepsilon | X) = X^2$, which is 0 when $X = 0$ and 1 otherwise. The conditional distribution of ε depends on X , so they are not independent. With columns $1, x, x^2$, (L1) holds with $\beta = (0, 0, 1)^\top$ and (L3) holds

for independent cases. (L2) fails, and so does (L4) as stated, because the error of a case with $x = 0$ is identically zero and the joint distribution is not normal with covariance $\sigma^2 \mathbf{I}$.

Exercise A2.

Let (X, Y) be bivariate normal with means μ_X, μ_Y , standard deviations $\sigma_X, \sigma_Y > 0$ and correlation ρ . Use [Theorem 3.5.2](#) to find $m(x)$ and $\text{Var}(Y \mid X = x)$. Conclude that a random sample from this distribution satisfies (L1)–(L4) conditionally on the x_i , with columns 1, x . What are β and σ^2 ?

5.1.5.2 B. Practice

Exercise B1.

Under the conditions of [Proposition 5.1.1](#), show that $\text{Var}(Y) = \text{Var}(m(\mathbf{X})) + \text{E}[\text{Var}(Y \mid \mathbf{X})]$ and that $\text{E}(\varepsilon^2) = \text{E}[\text{Var}(Y \mid \mathbf{X})]$. The ratio $\eta^2 = \text{Var}(m(\mathbf{X})) / \text{Var}(Y)$ is the proportion of the variance of Y that the regression function explains. Compute it for the distribution of [Exercise \(A1\)](#).

Solution. The first identity is the scalar case of [Proposition 2.5.4](#). For the second, $\text{E}(\varepsilon^2 \mid \mathbf{X}) = \text{E}[(Y - m(\mathbf{X}))^2 \mid \mathbf{X}] = \text{Var}(Y \mid \mathbf{X})$; take expectations. In [Exercise \(A1\)](#), $m(X) = X^2$ takes the value 1 with probability 2/3 and 0 with probability 1/3, so $\text{Var}(m(X)) = 2/9$. Also $\text{E}[\text{Var}(Y \mid X)] = \text{E}(X^2) = 2/3$. Hence $\text{Var}(Y) = 8/9$ and $\eta^2 = 1/4$.

Exercise B2.

In [Example \(One slope, two sampling schemes\)](#) the regressors are independent and uniform on $[0, 10]$. Show that $\text{E}(S_{xx}) = (n-1) \cdot 100/12$. Use Jensen's inequality to show that $\text{E}(\sigma^2 / S_{xx}) \geq \sigma^2 / \text{E}(S_{xx})$, with strict inequality here. Why is the right side of no use for describing the precision of the slope from a given sample?

5.1.5.3 C. Going deeper

Exercise C1.

Let $X \sim N(0, 1)$ and $Y = X^2$. Find the best linear predictor $\beta_0 + \beta_1 X$ of Y ([Theorem 2.5.2](#)) and its error u . Show that $\text{E}(u) = 0$ and $\text{Cov}(u, X) = 0$, but $\text{E}(u \mid X) \neq 0$. Explain why fitting a straight line to a large sample from this distribution estimates a well-defined population quantity even though (L1) fails for the columns 1, x , and why the second-moment theory of this chapter does not describe the sampling variability of the fitted slope.

5.2 Simple linear regression as a warm-up

With a single regressor x and an intercept, the linear model is

$$Y_i = \beta_0 + \beta_1 x_i + \varepsilon_i, \quad i = 1, \dots, n, \quad (5.2.1)$$

where the x_i are known numbers and the errors satisfy the second-moment assumptions (L1)–(L3) of [Section 5.1](#): $E(\varepsilon_i) = 0$, $\text{Var}(\varepsilon_i) = \sigma^2$, and $\text{Cov}(\varepsilon_i, \varepsilon_j) = 0$ for $i \neq j$. The parameter β_1 is the **slope**, the difference in mean response between two values of x one unit apart. The **intercept** β_0 is the mean response at $x = 0$, which may or may not be a value of x that makes sense. Everything in this section is a special case of [Section 5.4](#). We work it out directly because the formulas are short, because they show the structure of the general answer, and because they are the ones used every day.

5.2.1 The least squares line

For a candidate line $b_0 + b_1 x$, the **residual** of case i is $y_i - b_0 - b_1 x_i$, the vertical distance from the data point to the line. The method of least squares chooses the line that minimizes the sum of squared residuals,

$$S(b_0, b_1) = \sum_{i=1}^n (y_i - b_0 - b_1 x_i)^2.$$

Other criteria are possible, for example the sum of absolute residuals (Chapter 45). Squared error is chosen because it leads to linear equations, because it matches the conditional mean of [Proposition 5.1.1\(c\)](#), and because its statistical properties under (L1)–(L3) can be worked out exactly, which is the business of this chapter.

We use the standard abbreviations

$$\begin{aligned} \bar{x} &= \frac{1}{n} \sum_i x_i, & \bar{y} &= \frac{1}{n} \sum_i y_i, & S_{xx} &= \sum_i (x_i - \bar{x})^2, \\ S_{xy} &= \sum_i (x_i - \bar{x})(y_i - \bar{y}), & S_{yy} &= \sum_i (y_i - \bar{y})^2. \end{aligned} \quad (5.2.2)$$

Two identities are used repeatedly: $\sum_i (x_i - \bar{x}) = 0$, and consequently $S_{xy} = \sum_i (x_i - \bar{x})y_i$ and $S_{xx} = \sum_i (x_i - \bar{x})x_i$. Note that $S_{xx} > 0$ unless all the x_i are equal.

Theorem 5.2.1 — Least squares for a straight line

Suppose the x_i are not all equal. Then $S(b_0, b_1)$ has a unique minimizer $(\hat{\beta}_0, \hat{\beta}_1)$, given by

$$\hat{\beta}_1 = \frac{S_{xy}}{S_{xx}}, \quad \hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}. \quad (5.2.3)$$

It is the unique solution of the two **normal equations**

$$\sum_{i=1}^n (y_i - b_0 - b_1 x_i) = 0, \quad \sum_{i=1}^n x_i (y_i - b_0 - b_1 x_i) = 0. \quad (5.2.4)$$

Moreover, for all b_0, b_1 ,

$$\begin{aligned} S(b_0, b_1) &= S(\hat{\beta}_0, \hat{\beta}_1) + n[(b_0 - \hat{\beta}_0) + (b_1 - \hat{\beta}_1)\bar{x}]^2 \\ &\quad + S_{xx}(b_1 - \hat{\beta}_1)^2. \end{aligned} \quad (5.2.5)$$

Proof. S is a differentiable function of (b_0, b_1) , with

$$\frac{\partial S}{\partial b_0} = -2 \sum_i (y_i - b_0 - b_1 x_i), \quad \frac{\partial S}{\partial b_1} = -2 \sum_i x_i (y_i - b_0 - b_1 x_i).$$

A minimizer must make both derivatives zero, which gives [Eq. 5.2.4](#). The first normal equation divided by n says $\bar{y} - b_0 - b_1 \bar{x} = 0$, so $b_0 = \bar{y} - b_1 \bar{x}$. Substituting into the second, and using $y_i - \bar{y} - b_1(x_i - \bar{x})$ for the residual,

$$\sum_i x_i [(y_i - \bar{y}) - b_1(x_i - \bar{x})] = S_{xy} - b_1 S_{xx} = 0,$$

so $b_1 = S_{xy}/S_{xx}$. This is the only stationary point.

A stationary point need not be a minimum, so we check directly. Write $\hat{\varepsilon}_i = y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i$, $d_0 = b_0 - \hat{\beta}_0$ and $d_1 = b_1 - \hat{\beta}_1$. Then $y_i - b_0 - b_1 x_i = \hat{\varepsilon}_i - (d_0 + d_1 x_i)$, and

$$S(b_0, b_1) = \sum_i \hat{\varepsilon}_i^2 - 2d_0 \sum_i \hat{\varepsilon}_i - 2d_1 \sum_i x_i \hat{\varepsilon}_i + \sum_i (d_0 + d_1 x_i)^2.$$

The two middle sums vanish by the normal equations. For the last, $d_0 + d_1 x_i = (d_0 + d_1 \bar{x}) + d_1(x_i - \bar{x})$, and the cross term again sums to zero, so $\sum_i (d_0 + d_1 x_i)^2 = n(d_0 + d_1 \bar{x})^2 + d_1^2 S_{xx}$. This proves [Eq. 5.2.5](#). Both added terms are nonnegative, and since $S_{xx} > 0$ they are both zero only when $d_1 = 0$ and then $d_0 = 0$. So $(\hat{\beta}_0, \hat{\beta}_1)$ is the unique global minimizer. ■

The decomposition [Eq. 5.2.5](#) says more than minimality. It shows the level sets of S are ellipses centred at $(\hat{\beta}_0, \hat{\beta}_1)$, whose shape is set by n , $\bar{x}$ and S_{xx} . When $\bar{x}$ is far from zero the ellipses are long and thin, tilted along the direction where $b_0 + b_1 \bar{x}$ stays constant. Many lines then fit almost equally well, provided they pass near the centre of the data. This is a first look at the correlation between the estimated intercept and slope ([Corollary 5.5.4](#)).

The **fitted values** are $\hat{y}_i = \hat{\beta}_0 + \hat{\beta}_1 x_i$ and the **residuals** are $\hat{\varepsilon}_i = y_i - \hat{y}_i$. The minimum SSE = $\sum_i \hat{\varepsilon}_i^2$ is the **residual sum of squares** (also *error sum of squares*).

Proposition 5.2.2 — Properties of the fitted line

Under the conditions of [Theorem 5.2.1](#):

- (a) $\sum_i \hat{\varepsilon}_i = 0$, $\sum_i x_i \hat{\varepsilon}_i = 0$ and $\sum_i \hat{y}_i \hat{\varepsilon}_i = 0$;

- (b) the line passes through $(\bar{x}, \bar{y})$, and the mean of the fitted values is $\bar{y}$;
 (c) with $SST = S_{yy}$ and $SSR = \sum_i (\hat{y}_i - \bar{y})^2$,

$$SST = SSR + SSE, \quad SSR = \hat{\beta}_1^2 S_{xx} = \frac{S_{xy}^2}{S_{xx}},$$

and so $SSE = S_{yy} - S_{xy}^2/S_{xx}$;

- (d) if also $S_{yy} > 0$, the **coefficient of determination** $R^2 = SSR/SST$ equals r^2 , the square of the sample correlation $r = S_{xy}/\sqrt{S_{xx}S_{yy}}$.

Proof. (a) The first two are the normal equations Eq. 5.2.4. The third follows because $\hat{y}_i$ is a combination of 1 and x_i . (b) The first normal equation is $\bar{y} = \hat{\beta}_0 + \hat{\beta}_1 \bar{x}$, and averaging $\hat{y}_i = y_i - \hat{\varepsilon}_i$ gives $\bar{y}$. (c) By (b), $\hat{y}_i - \bar{y} = \hat{\beta}_1(x_i - \bar{x})$, so $SSR = \hat{\beta}_1^2 S_{xx} = S_{xy}^2/S_{xx}$. Also $y_i - \bar{y} = (\hat{y}_i - \bar{y}) + \hat{\varepsilon}_i$, and the cross term $\sum_i (\hat{y}_i - \bar{y})\hat{\varepsilon}_i$ vanishes by (a). (d) $R^2 = S_{xy}^2/(S_{xx}S_{yy}) = r^2$. ■

Part (c) is the simplest instance of the analysis of variance: the total variation of the response about its mean splits into a part along the fitted line and a part around it. Chapter 9 develops the general decomposition, and Section 6.7 interprets R^2 as a squared cosine.

5.2.2 Regression toward the mean

Divide S_{xy}/S_{xx} above and below by $n-1$. With the sample standard deviations $s_x = \sqrt{S_{xx}/(n-1)}$ and $s_y = \sqrt{S_{yy}/(n-1)}$, the slope is

$$\hat{\beta}_1 = r \frac{s_y}{s_x}, \quad \text{equivalently} \quad \frac{\hat{y} - \bar{y}}{s_y} = r \frac{x - \bar{x}}{s_x}. \quad (5.2.6)$$

In standard units, the fitted value is the regressor's standardized value multiplied by r . Since $|r| \leq 1$ (the Cauchy–Schwarz inequality, Proposition 1.1.1), a case that is two standard deviations above average in x is predicted to be *less* than two standard deviations above average in y , unless the points lie exactly on a line. This is **regression toward the mean**, and it gave the subject its name: Galton observed it in 1886 for the heights of parents and their adult children.

The effect is purely statistical and involves no mechanism. Regressing x on y instead gives the slope $r s_x/s_y$, which predicts x from y with the same shrinkage toward the mean in the other direction. The two fitted lines differ unless $r^2 = 1$. They answer different questions: the mean of y at given x , and the mean of x at given y . It is a common error to read regression toward the mean as evidence that extreme units become more ordinary over time, or that an intervention given to the most extreme units has worked.

5.2.3 A worked example

Example. (Forecasting December sea temperature)

The Niño 1+2 index is the mean sea surface temperature in a region of the Pacific off

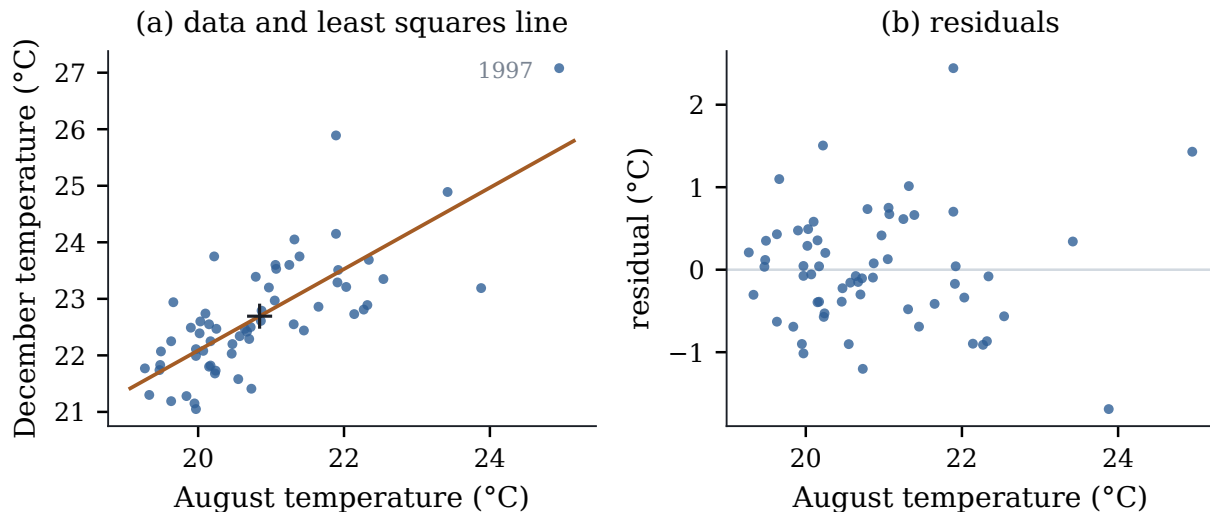


Figure 5.2.1: December against August sea surface temperature in the Niño 1+2 region, 1950–2010. (a) The data and the least squares line, which passes through the point of means (cross). (b) Residuals against the regressor. The El Niño year 1997 is the point at the far right of both panels.

Peru and Ecuador. It is high during El Niño events. We use the monthly values for the 61 years 1950–2010 and ask how well the August value predicts the December value four months later. Here x is the August temperature and y the December temperature, both in degrees Celsius. The summary statistics are

$$\begin{aligned}\bar{x} &= 20.843, & \bar{y} &= 22.693, \\ S_{xx} &= 77.801, & S_{xy} &= 56.008, & S_{yy} &= 70.380,\end{aligned}$$

so by [Theorem 5.2.1](#)

$$\hat{\beta}_1 = \frac{56.008}{77.801} = 0.7199, \quad \hat{\beta}_0 = 22.693 - 0.7199 \times 20.843 = 7.6887.$$

Each extra degree in August goes with 0.7199 of a degree more in the following December, on average. For an August temperature of 22°C the fitted December mean is 23.53°C. The intercept, the fitted December temperature after an August at 0°C, is meaningless: no August in the data is below 19°C.

The residual sum of squares is $SSE = 30.061$, and $R^2 = 0.5729$, the square of the correlation $r = 0.7569$. The slope is less than 1 partly because $r < 1$ and partly because December temperatures vary a little less than August ones: $s_y = 1.083$ against $s_x = 1.139$.

```
import numpy as np
import statsmodels.api as sm
sst = sm.datasets.elnino.load_pandas().data
x = sst["AUG"].to_numpy()           # August temperature
y = sst["DEC"].to_numpy()           # December temperature, same year
n = len(y)

xbar, ybar = x.mean(), y.mean()
Sxx = np.sum((x - xbar) ** 2)
Sxy = np.sum((x - xbar) * (y - ybar))
Syy = np.sum((y - ybar) ** 2)
```

```

b1 = Sxy / Sxx                # slope
b0 = ybar - b1 * xbar         # intercept
fitted = b0 + b1 * x
resid = y - fitted
print(f"n = {n}, slope = {b1:.4f}, intercept = {b0:.4f}")

```

```

print("sum of residuals      ", resid.sum())
print("sum of x * residuals  ", (x * resid).sum())
SSE = resid @ resid
R2 = 1 - SSE / Syy
r = Sxy / np.sqrt(Sxx * Syy)
print(f"R^2 = {R2:.4f}, r^2 = {r**2:.4f}")
print("numpy.polyfit agrees: ", np.polyfit(x, y, 1))

```

The residual plot in [Figure 5.2.1\(b\)](#) is the first diagnostic to draw after any fit. The residuals should show no pattern against x . Here there is no obvious curvature, but the spread is not uniform. The two warmest Augusts, in 1997 and 1983, both fell during strong El Niño events. The 1997 event was still building in December, while the 1982–83 event was over by then, and the two points sit well away from the line on opposite sides. The warmest of all, 1997 (August 24.95°C, December 27.08°C), has a residual of 1.430°C. Because it lies far from $\bar{x}$, it pulls on the slope: without it the slope would be 0.6214. Points with regressor values far from the mean have high *leverage*, a notion made precise in [Section 6.8](#) and Chapter 20.

5.2.4 Regression through the origin

Sometimes theory says the mean response is zero when $x = 0$, and the model $Y_i = \beta_1 x_i + \varepsilon_i$ is fitted without an intercept. Minimizing $\sum_i (y_i - b_1 x_i)^2$ gives one normal equation, $\sum_i x_i (y_i - b_1 x_i) = 0$, with solution

$$\tilde{\beta}_1 = \frac{\sum_i x_i y_i}{\sum_i x_i^2},$$

valid whenever some $x_i \neq 0$. The residuals now satisfy $\sum_i x_i \hat{\varepsilon}_i = 0$ but generally *not* $\sum_i \hat{\varepsilon}_i = 0$, and the decomposition of [Proposition 5.2.2\(c\)](#) about $\bar{y}$ fails. It holds instead about zero, as $\sum_i y_i^2 = \sum_i \hat{y}_i^2 + \sum_i \hat{\varepsilon}_i^2$, which is why software reports an “uncentred” R^2 for such fits. That number is not comparable with the R^2 of a model with an intercept ([Exercise \(A2\)](#)).

Dropping the intercept is a strong assumption. In [Example \(Forecasting December sea temperature\)](#), forcing the line through the origin gives slope 1.0877, a fit much worse than the line with an intercept: the data are nowhere near $x = 0$, and the constraint has nothing to do with how ocean temperatures behave.

5.2.5 Exercises

5.2.5.1 A. Check your understanding

Exercise A1.

For the five points $(x, y) = (0, 1), (1, 1), (2, 3), (3, 4), (4, 6)$, compute $\bar{x}, \bar{y}, S_{xx}, S_{xy}$, the least squares line, the residuals, and R^2 . Check the three identities of [Proposition 5.2.2\(a\)](#).

Solution. $\bar{x} = 2$, $\bar{y} = 3$, $S_{xx} = 10$, $S_{xy} = (-2)(-2) + (-1)(-2) + 0 + 1 \cdot 1 + 2 \cdot 3 = 13$. So $\hat{\beta}_1 = 1.3$ and $\hat{\beta}_0 = 3 - 2.6 = 0.4$. The fitted values are 0.4, 1.7, 3.0, 4.3, 5.6 and the residuals 0.6, -0.7, 0, -0.3, 0.4, which sum to zero. Also $\sum_i x_i \hat{\epsilon}_i = 0 - 0.7 + 0 - 0.9 + 1.6 = 0$. $S_{yy} = 4 + 4 + 0 + 1 + 9 = 18$, $\text{SSE} = 0.36 + 0.49 + 0 + 0.09 + 0.16 = 1.1$, and $R^2 = 1 - 1.1/18 = 0.939$, which equals $13^2/(10 \cdot 18)$.

Exercise A2.

For regression through the origin, show that $\sum_i y_i^2 = \sum_i \hat{y}_i^2 + \sum_i \hat{\epsilon}_i^2$, and give a small data set for which $1 - \sum_i \hat{\epsilon}_i^2 / \sum_i y_i^2$ is close to 1 while $1 - \sum_i \hat{\epsilon}_i^2 / S_{yy}$ is negative.

5.2.5.2 B. Practice

Exercise B1.

Show that $\hat{\beta}_1 = \sum_i c_i y_i$ with $c_i = (x_i - \bar{x})/S_{xx}$, and that $\sum_i c_i = 0$, $\sum_i c_i x_i = 1$ and $\sum_i c_i^2 = 1/S_{xx}$. Write $\hat{\beta}_0$ as $\sum_i d_i y_i$ and find $\sum_i d_i$, $\sum_i d_i x_i$ and $\sum_i d_i^2$.

Solution. $S_{xy} = \sum_i (x_i - \bar{x})y_i$, so $\hat{\beta}_1 = \sum_i c_i y_i$. Then $\sum_i c_i = 0$, $\sum_i c_i x_i = \sum_i (x_i - \bar{x})x_i/S_{xx} = 1$, and $\sum_i c_i^2 = S_{xx}/S_{xx}^2 = 1/S_{xx}$. For the intercept, $\hat{\beta}_0 = \bar{y} - \bar{x}\hat{\beta}_1 = \sum_i d_i y_i$ with $d_i = 1/n - \bar{x}c_i$. Hence $\sum_i d_i = 1$, $\sum_i d_i x_i = \bar{x} - \bar{x} = 0$, and $\sum_i d_i^2 = \sum_i (1/n^2 - 2\bar{x}c_i/n + \bar{x}^2 c_i^2) = 1/n + \bar{x}^2/S_{xx}$. These are the coefficients that give the moments in [Corollary 5.5.4](#).

Exercise B2.

Suppose all x_i are distinct. Show that $\hat{\beta}_1$ is a weighted average of the slopes $(y_j - y_i)/(x_j - x_i)$ of the lines through all pairs of points, with weights proportional to $(x_j - x_i)^2$. *Hint:* show that $\sum_{i < j} (x_j - x_i)(y_j - y_i) = nS_{xy}$ and $\sum_{i < j} (x_j - x_i)^2 = nS_{xx}$.

5.2.5.3 C. Going deeper

Exercise C1.

The **least absolute deviations** line minimizes $\sum_i |y_i - b_0 - b_1 x_i|$. Show that some minimizing line passes through at least two of the data points (assume the x_i are not all equal). *Hint:* for fixed b_1 the best b_0 is a median of the $y_i - b_1 x_i$, and the objective is piecewise linear and convex in b_1 . Compare the effect of the 1997 point in [Example \(Forecasting December sea temperature\)](#) on this line and on the least squares line.

5.3 The multiple linear regression model

A response rarely depends on one regressor alone. The money a country's residents hold depends on their income and on interest rates. The number of physician visits depends on health, age and insurance. Even with one measured regressor, a straight line is often too rigid: the curvature

of a trend or the shape of a seasonal cycle needs several columns to describe. Matrix notation handles all of these cases at once.

5.3.1 Matrix form

Suppose case i has regressor values that have been turned into p numbers $x_{i1}, \dots, x_{ip}$, and consider

$$Y_i = \beta_1 x_{i1} + \beta_2 x_{i2} + \dots + \beta_p x_{ip} + \varepsilon_i, \quad i = 1, \dots, n.$$

Stack the responses, the coefficients and the errors into vectors, and the regressor values into an $n \times p$ matrix:

$$\mathbf{Y} = \begin{pmatrix} Y_1 \\ \vdots \\ Y_n \end{pmatrix}, \quad \boldsymbol{\beta} = \begin{pmatrix} \beta_1 \\ \vdots \\ \beta_p \end{pmatrix}, \quad \boldsymbol{\varepsilon} = \begin{pmatrix} \varepsilon_1 \\ \vdots \\ \varepsilon_n \end{pmatrix},$$

$$\mathbf{X} = \begin{pmatrix} x_{11} & \cdots & x_{1p} \\ \vdots & & \vdots \\ x_{n1} & \cdots & x_{np} \end{pmatrix} = \begin{pmatrix} \mathbf{x}_{(1)}^\top \\ \vdots \\ \mathbf{x}_{(n)}^\top \end{pmatrix}.$$

The n equations become the single equation $\mathbf{Y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}$. Row i of $\mathbf{X}$, written $\mathbf{x}_{(i)}^\top$, holds the regressor values of case i , its **design point**. Column j , written $\mathbf{x}_j$, holds the values of the j th regressor across the cases.

Definition 5.3.1 — The linear model

The **linear model** for a random vector $\mathbf{Y} \in \mathbb{R}^n$ is

$$\mathbf{Y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}, \quad \mathbf{E}(\boldsymbol{\varepsilon}) = \mathbf{0}, \quad \text{Cov}(\boldsymbol{\varepsilon}) = \sigma^2 \mathbf{I}_n, \quad (5.3.1)$$

where the **model matrix** $\mathbf{X}$ is a known $n \times p$ matrix of constants, and the **coefficient vector** $\boldsymbol{\beta} \in \mathbb{R}^p$ and the **error variance** $\sigma^2 > 0$ are unknown parameters. The conditions on $\boldsymbol{\varepsilon}$ are the **second-moment assumptions**. If in addition $\boldsymbol{\varepsilon} \sim N_n(\mathbf{0}, \sigma^2 \mathbf{I}_n)$, equivalently $\mathbf{Y} \sim N_n(\mathbf{X}\boldsymbol{\beta}, \sigma^2 \mathbf{I}_n)$, the model is the **normal linear model**. The model has **full rank** if $\text{rank}(\mathbf{X}) = p$.

The second-moment assumptions are exactly the layers (L1)–(L3) of [Section 5.1](#). By [Theorem 2.2.1](#) they say

$$\mathbf{E}(\mathbf{Y}) = \mathbf{X}\boldsymbol{\beta}, \quad \text{Cov}(\mathbf{Y}) = \sigma^2 \mathbf{I}_n. \quad (5.3.2)$$

The first equation is the whole content of (L1). It says the mean vector $\mathbf{E}(\mathbf{Y})$ is a linear combination of the columns of $\mathbf{X}$, with unknown coefficients. The second combines constant variance and uncorrelated errors. When the regressors are random, both equations are read conditionally on $\mathbf{X}$, as explained in [Section 5.1](#). The normal version is a model for the full distribution of $\mathbf{Y}$, in the sense of [Chapter 3](#). Since a normal vector with covariance $\sigma^2 \mathbf{I}$ has independent components ([Theorem 3.4.1](#)), the errors are then independent and not merely uncorrelated.

The definition does not require full rank; most results in this chapter assume it, which forces $p \leq n$, and [Chapter 6](#) and [Chapter 8](#) treat the general case. What must be *known* is $\mathbf{X}$, and

what must be *linear* is the dependence of $E(\mathbf{Y})$ on $\boldsymbol{\beta}$. Nor does the model say that the errors are identically distributed, only that they have a common variance.

5.3.2 The intercept

Almost every model in practice includes a column of ones, $\mathbf{x}_1 = \mathbf{1}$, so that

$$E(Y_i) = \beta_0 + \beta_1 x_{i1} + \cdots + \beta_k x_{ik}, \quad p = k + 1.$$

In this form it is conventional to number the coefficients from 0, and to call k the number of regressors. The coefficient of the column of ones is the **intercept**. Including it lets the fitted surface find its own level, rather than being forced through the origin, and it has two further consequences that [Section 5.4](#) proves: the residuals sum to zero, and the slopes can be computed from centred data. Leaving out the intercept is justified only when theory requires the mean response to vanish at the origin of every regressor, as in the regression through the origin at the end of [Section 5.2](#).

5.3.3 Linear in the parameters, not in the regressors

The word *linear* in “linear model” refers to $\boldsymbol{\beta}$. The columns of $\mathbf{X}$ can be any known functions of the measured variables. This turns a remarkably wide class of curved relationships into linear models.

Transformations. If the mean response is thought to depend on $\log z$, the column $\log z_i$ enters $\mathbf{X}$, and $E(Y_i) = \beta_0 + \beta_1 \log z_i$ is linear in (β_0, β_1) . The response can be transformed too, as in the money demand model of [Section 5.4](#), where Y is the logarithm of money holdings. A transformation of the response changes the meaning of the error, and hence the assumptions; Chapter 22 treats this.

Polynomials. A quadratic trend $E(Y_i) = \beta_0 + \beta_1 t_i + \beta_2 t_i^2$ uses the columns $\mathbf{1}$, $\mathbf{t}$ and $\mathbf{t}^2$ (squared entrywise). A polynomial of degree d in one variable needs $d + 1$ columns.

Periodic terms. A seasonal cycle of period one year can be written as $a \cos(2\pi t + \phi)$ with unknown amplitude a and phase ϕ . This is not linear in ϕ . But $a \cos(2\pi t + \phi) = \gamma_1 \cos(2\pi t) + \gamma_2 \sin(2\pi t)$ with $\gamma_1 = a \cos \phi$ and $\gamma_2 = -a \sin \phi$, and this *is* linear in (γ_1, γ_2) . The amplitude is recovered as $a = (\gamma_1^2 + \gamma_2^2)^{1/2}$.

Example. (A curved trend and a seasonal cycle)

The monthly mean concentration of carbon dioxide in the air at Mauna Loa, Hawaii, from March 1958 to December 2001, gives $n = 521$ observations in parts per million. Let t be the time in years, at the middle of each month, and $t_c = t - 1980$. The model

$$E(Y) = \beta_0 + \beta_1 t_c + \beta_2 t_c^2 + \gamma_1 \cos 2\pi t + \gamma_2 \sin 2\pi t + \gamma_3 \cos 4\pi t + \gamma_4 \sin 4\pi t$$

has $p = 7$ columns, and the model matrix has full rank. Its least squares fit ([Section 5.4](#)) has

$$\hat{\beta}_0 = 337.6163, \quad \hat{\beta}_1 = 1.3344, \quad \hat{\beta}_2 = 0.0118,$$

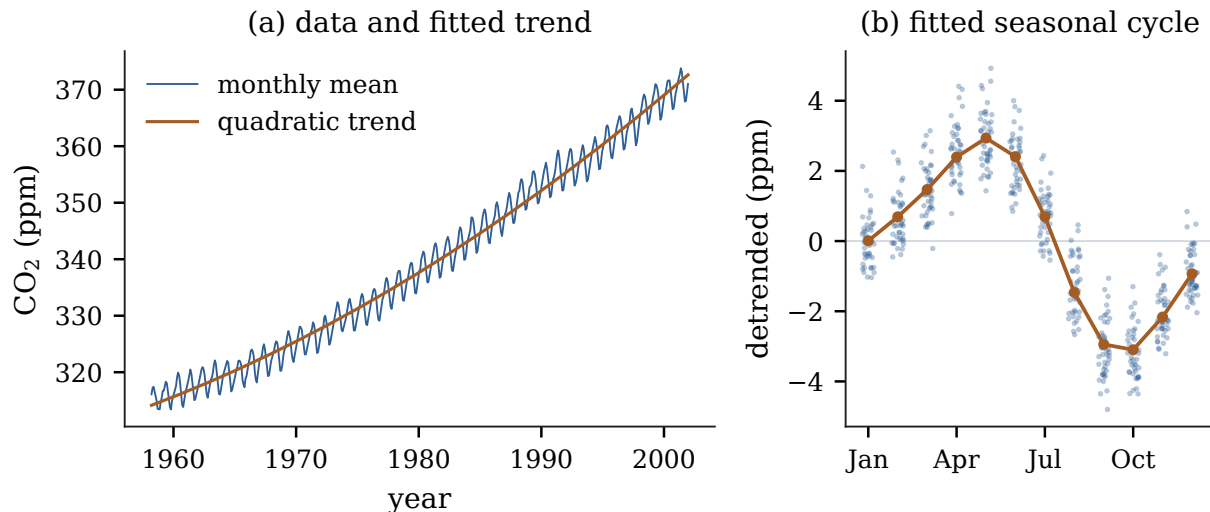


Figure 5.3.1: Monthly carbon dioxide at Mauna Loa, 1958–2001. (a) The data and the fitted quadratic trend. (b) The data with the fitted trend removed, by calendar month, and the fitted seasonal cycle made of two harmonics. The model is curved in time and in the calendar month, but linear in its seven coefficients.

$$\hat{\gamma}_1 = -1.0413, \quad \hat{\gamma}_2 = 2.5641, \quad \hat{\gamma}_3 = 0.6284, \quad \hat{\gamma}_4 = -0.3859.$$

The quadratic term says the rate of increase is itself increasing. The slope of the trend, $\beta_1 + 2\beta_2 t_c$, is estimated as 0.86 ppm per year in 1960 and 1.81 ppm per year in 2000. The annual harmonic has amplitude 2.77 ppm and the half-year harmonic 0.74 ppm. Together they make a seasonal swing of 6.0 ppm from the late-spring peak to the autumn trough, as the vegetation of the northern hemisphere takes up carbon over the summer. The residual standard deviation (estimated as in [Section 5.6](#)) is 0.760 ppm, against 2.743 ppm for a straight line in time alone. [Figure 5.3.1](#) shows the data, the trend and the seasonal cycle.

```
import numpy as np
import statsmodels.api as sm
co2 = sm.datasets.co2.load_pandas().data["co2"]
monthly = co2.resample("MS").mean().dropna()           # weekly values -> monthly means
y = monthly.to_numpy()
t = (monthly.index.year + (monthly.index.month - 0.5) / 12).to_numpy()
tc = t - 1980.0                                         # years since 1980

X = np.column_stack([
    np.ones_like(tc),                                  # intercept
    tc, tc**2,                                         # quadratic trend
    np.cos(2 * np.pi * t), np.sin(2 * np.pi * t),   # annual cycle
    np.cos(4 * np.pi * t), np.sin(4 * np.pi * t),   # half-year harmonic
])
n, p = X.shape
print("n =", n, " p =", p, " rank =", np.linalg.matrix_rank(X))
```

```
beta_hat = np.linalg.solve(X.T @ X, X.T @ y)          # normal equations (fine here)
fitted = X @ beta_hat
resid = y - fitted
print("coefficients:", np.round(beta_hat, 4))
print(f"residual standard deviation {np.sqrt(resid @ resid / (n - p)):.3f} ppm")
```

```
beta_line = np.linalg.solve(X[:, :2].T @ X[:, :2], X[:, :2].T @ y)
print(f"straight line only: slope {beta_line[1]:.3f} ppm per year")
```

The residuals of this model are strongly correlated from month to month: the correlation between consecutive residuals is 0.90, because departures from a smooth trend persist for many months. So (L3) fails and the standard errors of [Section 5.5](#) would be too optimistic. The *fit* is still a sensible description. Chapter 21 treats serially correlated errors.

Models that are not linear. Some relations cannot be rearranged into linear form. Exponential decay toward an unknown asymptote, $E(Y) = \beta_0 + \beta_1 e^{-\beta_2 x}$, is nonlinear in β_2 , and so is a cycle of unknown period. Such models are fitted by iterative nonlinear least squares, which this book does not treat. Others become linear after a transformation: if $Y = \alpha x^\beta \eta$ with a positive multiplicative error η , then $\log Y = \log \alpha + \beta \log x + \log \eta$ is a linear model for $\log Y$, provided the assumptions are made about $\log \eta$ rather than η .

5.3.4 Categorical regressors and indicator variables

A regressor that records a category, such as treatment group, region, or a self-rating on a four-point scale, cannot enter $\mathbf{X}$ as a single column of codes 1, 2, 3, 4. That would force the mean differences between consecutive categories to be equal, which is an arbitrary consequence of the labels. Instead, each category except one gets an **indicator variable** (or *dummy variable*): a column that is 1 for cases in that category and 0 otherwise. The omitted category is the **reference category**.

Example. (Self-rated health)

The RAND Health Insurance Experiment recorded, for 20,190 person-years, the number of outpatient visits to a physician and the person's own rating of their health: excellent, good, fair or poor. With excellent as the reference, the model

$$E(Y_i) = \beta_0 + \beta_1 g_i + \beta_2 f_i + \beta_3 q_i$$

uses three indicator columns **g, f, q** for good, fair and poor health. For a person in excellent health the mean is β_0 ; in good health, $\beta_0 + \beta_1$; in fair health, $\beta_0 + \beta_2$; in poor health, $\beta_0 + \beta_3$. So $\beta_1, \beta_2, \beta_3$ are the differences in mean visits between each category and the reference. [Proposition 5.4.4](#) shows that the least squares estimates are the corresponding differences of sample means. In these data they are 0.268, 1.058 and 3.160 visits a year, on top of $\hat{\beta}_0 = 2.634$ for excellent health.

Why omit a category? With an indicator for *every* category the columns satisfy $\mathbf{e} + \mathbf{g} + \mathbf{f} + \mathbf{q} = \mathbf{1}$, where $\mathbf{e}$ indicates excellent health. The model matrix $[\mathbf{1}, \mathbf{e}, \mathbf{g}, \mathbf{f}, \mathbf{q}]$ then has five columns and rank four, so it does not have full rank. Such a model is not wrong. It describes the same set of mean vectors, but with one more parameter than it can identify, and the least squares coefficients are no longer unique. This situation is sometimes called the *dummy variable trap*. It is not a trap at all once the right theory is in place: [Section 6.4](#) shows that fitted values are unaffected, and [Chapter 8](#) shows which functions of the coefficients, such as the difference between two categories, are still estimated uniquely.

```
import numpy as np
import statsmodels.api as sm
rand = sm.datasets.randhie.load_pandas().data
```

```

y = rand["mdvis"].to_numpy(dtype=float)
D = rand[["hlthg", "hlthf", "hlthp"]].to_numpy(dtype=float)  # good, fair, poor
X = np.column_stack([np.ones(len(y)), D])                  # excellent =
  reference

excellent = 1.0 - D.sum(axis=1)
X_all = np.column_stack([X, excellent])                    # an indicator for every level, and
  1
print("columns:", X_all.shape[1], " rank:", np.linalg.matrix_rank(X_all))
print("1 - (sum of the four indicators) =", np.abs(X_all[:, 0] - X_all[:, 1:].sum(
  axis=1)).max())

```

Reference coding is one choice among several. **Effect coding** replaces the indicator of category k by a column that is 1 in category k , -1 in the reference category, and 0 elsewhere. Its coefficients measure deviations of category means from their unweighted average rather than from a reference. The fitted values are the same under either coding, because the two model matrices have the same column space ([Proposition 5.7.1](#)). Only the meaning of the coefficients changes.

5.3.5 Interactions

In the model $E(Y) = \beta_0 + \beta_1 x_1 + \beta_2 x_2$ the slope in x_1 is β_1 whatever the value of x_2 : the effects are **additive**. An **interaction** lets the slope in one variable depend on another. Adding the product column $x_1 x_2$,

$$E(Y) = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_1 x_2 = (\beta_0 + \beta_2 x_2) + (\beta_1 + \beta_3 x_2) x_1, \quad (5.3.3)$$

gives a slope in x_1 of $\beta_1 + \beta_3 x_2$. The model is still linear in the coefficients. When x_2 is an indicator for a group, [Eq. 5.3.3](#) fits a separate straight line in each group: intercept β_0 and slope β_1 in the reference group, intercept $\beta_0 + \beta_2$ and slope $\beta_1 + \beta_3$ in the other. When both variables are categorical, products of indicators give each combination of categories its own mean, the two-way layout of [Chapter 16](#) ([Definition 16.2.1](#)).

Interactions change what the other coefficients mean. In [Eq. 5.3.3](#), β_1 is the slope in x_1 when $x_2 = 0$, which may be far outside the data. Centring x_2 before forming the product makes β_1 the slope at the average x_2 instead. [Section 5.7](#) returns to this.

5.3.6 When is the model matrix of full rank?

Full rank means the columns of $\mathbf{X}$ are linearly independent: no column is an exact linear combination of the others. By rank-nullity ([Theorem 1.2.2](#)), this holds iff $\mathbf{X}\mathbf{b} = \mathbf{0}$ only for $\mathbf{b} = \mathbf{0}$, that is, iff two different coefficient vectors never give the same mean vector. It fails in three common ways:

1. **Too few cases:** $n < p$. Chapter 28 deals with this setting.
2. **Redundant coding:** an indicator for every category together with an intercept, as above, or the same quantity measured in two units.
3. **Unlucky data:** a design that happens to make columns dependent, such as a quadratic in a variable observed at only two distinct values, or an interaction between two indicators

when some combination of categories never occurs.

Near-failures, where columns are almost dependent, are more common and more troublesome. They make coefficients imprecise (Chapter 26) and computations inaccurate (Chapter 10).

5.3.7 Exercises

5.3.7.1 A. Check your understanding

Exercise A1.

Write out the model matrix for each model, for the data $x = (1, 2, 3, 4)$ and a group label (A, A, B, B) , and give its rank: (a) a quadratic in x ; (b) separate intercepts for the two groups and a common slope; (c) separate lines in the two groups, with reference coding; (d) an intercept, an indicator for A and an indicator for B .

Solution. With g the indicator of group B : (a) rows $(1, x, x^2)$: $(1, 1, 1), (1, 2, 4), (1, 3, 9), (1, 4, 16)$, rank 3. (b) rows $(1, g, x)$: $(1, 0, 1), (1, 0, 2), (1, 1, 3), (1, 1, 4)$, rank 3. (c) rows $(1, g, x, gx)$: $(1, 0, 1, 0), (1, 0, 2, 0), (1, 1, 3, 3), (1, 1, 4, 4)$, rank 4; it is square and nonsingular, so the two lines pass through the data exactly. (d) rows $(1, 1, 0), (1, 1, 0), (1, 0, 1), (1, 0, 1)$, rank 2: the first column is the sum of the other two.

Exercise A2.

Which of these mean functions define linear models in the stated parameters (after a change of parameters if necessary)? (a) $\beta_0 + \beta_1\sqrt{x}$; (b) $\beta_0 e^{\beta_1 x}$; (c) $\beta_0 + \beta_1 x + \beta_2 \max(x - 5, 0)$; (d) $\beta_0 + \beta_1 \sin(x - \beta_2)$; (e) $1/(\beta_0 + \beta_1 x)$.

5.3.7.2 B. Practice

Exercise B1.

For a factor with three levels and an intercept, write the model matrix rows under reference coding (level 3 as reference) and under effect coding. Find the nonsingular 3×3 matrix $\mathbf{K}$ with $\mathbf{X}_{\text{effect}} = \mathbf{X}_{\text{ref}}\mathbf{K}$, and express the effect-coding coefficients in terms of the three level means μ_1, μ_2, μ_3 .

Solution. Reference coding has rows $(1, 1, 0), (1, 0, 1), (1, 0, 0)$ for levels 1, 2, 3. Effect coding has rows $(1, 1, 0), (1, 0, 1), (1, -1, -1)$. Each effect-coded row is the reference-coded row times $\mathbf{K} = \begin{pmatrix} 1 & -1 & -1 \\ 0 & 2 & 1 \\ 0 & 1 & 2 \end{pmatrix}$: for instance $(1, 0, 0)\mathbf{K} = (1, -1, -1)$ and $(1, 1, 0)\mathbf{K} = (1, 1, 0)$. $\det \mathbf{K} = 3 \neq 0$. Under effect coding the level means are $\mu_1 = \alpha_0 + \alpha_1$, $\mu_2 = \alpha_0 + \alpha_2$, $\mu_3 = \alpha_0 - \alpha_1 - \alpha_2$. Adding, $\alpha_0 = (\mu_1 + \mu_2 + \mu_3)/3$, and then $\alpha_k = \mu_k - \alpha_0$ for $k = 1, 2$: deviations from the unweighted average of the means.

Exercise B2.

In Eq. 5.3.3, replace x_2 by $x_2 - c$ in both the main-effect column and the product. Find the new coefficients in terms of the old, and show that the fitted mean function is

unchanged. For which c is the new coefficient of x_1 the slope at the average value of x_2 ?

5.3.7.3 C. Going deeper

Exercise C1.

Let $t_i = i/m$ for $i = 0, \dots, m-1$ (one full period sampled at m equally spaced points, with $m \geq 5$). Show that the columns $\mathbf{1}, \cos 2\pi\mathbf{t}, \sin 2\pi\mathbf{t}, \cos 4\pi\mathbf{t}, \sin 4\pi\mathbf{t}$ are mutually orthogonal, and find their squared lengths. Conclude that for a whole number of periods the harmonic coefficients are estimated separately from each other and from the mean. Why is this not exactly the case in [Example \(A curved trend and a seasonal cycle\)](#)?

5.4 Least squares and the normal equations

The straight-line calculation of [Section 5.2](#) extends to any model matrix. The algebra is shorter in matrix form than it was for two coefficients, because the derivatives of linear and quadratic forms were worked out once and for all in [Chapter 1](#). In this section $\mathbf{y}$ is a fixed data vector. Nothing here uses probability. The statistical properties come in [Section 5.5](#).

5.4.1 The least squares estimate

Definition 5.4.1 — Least squares estimate

For an $n \times p$ matrix $\mathbf{X}$ and $\mathbf{y} \in \mathbb{R}^n$, the **least squares criterion** is

$$S(\mathbf{b}) = \|\mathbf{y} - \mathbf{X}\mathbf{b}\|^2 = \sum_{i=1}^n (y_i - \mathbf{x}_{(i)}^\top \mathbf{b})^2, \quad \mathbf{b} \in \mathbb{R}^p.$$

A **least squares estimate** of $\boldsymbol{\beta}$ is any $\hat{\boldsymbol{\beta}}$ that minimizes S . The corresponding **fitted values** are $\hat{\mathbf{Y}} = \mathbf{X}\hat{\boldsymbol{\beta}}$, the **residuals** are $\hat{\boldsymbol{\varepsilon}} = \mathbf{y} - \mathbf{X}\hat{\boldsymbol{\beta}}$, and the **residual sum of squares** is $\text{SSE} = S(\hat{\boldsymbol{\beta}}) = \|\hat{\boldsymbol{\varepsilon}}\|^2$.

Following the book's notation, we write $\hat{\mathbf{Y}}$ and $\hat{\boldsymbol{\varepsilon}}$ in bold for the vectors, and $\hat{y}_i$ and $\hat{\varepsilon}_i$ for their entries, whether the data are regarded as fixed numbers or as random.

Theorem 5.4.1 — Least squares for a full-rank model

Let $\mathbf{X}$ be an $n \times p$ matrix of rank p . Then:

- (a) $\mathbf{X}^\top \mathbf{X}$ is positive definite, and in particular nonsingular;
- (b) the **normal equations**

$$\mathbf{X}^\top \mathbf{X} \mathbf{b} = \mathbf{X}^\top \mathbf{y} \tag{5.4.1}$$

have exactly one solution,

$$\hat{\boldsymbol{\beta}} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{y}; \quad (5.4.2)$$

(c) for every $\mathbf{b} \in \mathbb{R}^p$,

$$\|\mathbf{y} - \mathbf{X}\mathbf{b}\|^2 = \|\mathbf{y} - \mathbf{X}\hat{\boldsymbol{\beta}}\|^2 + (\mathbf{b} - \hat{\boldsymbol{\beta}})^\top \mathbf{X}^\top \mathbf{X} (\mathbf{b} - \hat{\boldsymbol{\beta}}), \quad (5.4.3)$$

so $\hat{\boldsymbol{\beta}}$ is the unique least squares estimate.

If instead $\text{rank}(\mathbf{X}) < p$, least squares estimates exist but are never unique.

Proof. (a) $\mathbf{X}^\top \mathbf{X}$ is symmetric, and $\mathbf{b}^\top \mathbf{X}^\top \mathbf{X} \mathbf{b} = \|\mathbf{X}\mathbf{b}\|^2 \geq 0$, with equality iff $\mathbf{X}\mathbf{b} = \mathbf{0}$. Since $\mathbf{X}$ has rank p , rank-nullity (Theorem 1.2.2) gives $\mathcal{N}(\mathbf{X}) = \{\mathbf{0}\}$, so the quadratic form is positive for $\mathbf{b} \neq \mathbf{0}$. A positive definite matrix is nonsingular (Proposition 1.7.5(e)).

(b) This is the computation of Example (The least squares criterion): by Theorem 1.11.1 the gradient of S is $-2\mathbf{X}^\top (\mathbf{y} - \mathbf{X}\mathbf{b})$ and its Hessian is $2\mathbf{X}^\top \mathbf{X}$. Every minimizer of the differentiable function S is a stationary point, and the stationary points are the solutions of Eq. 5.4.1. By (a) there is exactly one, Eq. 5.4.2.

(c) Put $\mathbf{d} = \mathbf{b} - \hat{\boldsymbol{\beta}}$. Then $\mathbf{y} - \mathbf{X}\mathbf{b} = \hat{\boldsymbol{\varepsilon}} - \mathbf{X}\mathbf{d}$, and

$$\|\mathbf{y} - \mathbf{X}\mathbf{b}\|^2 = \|\hat{\boldsymbol{\varepsilon}}\|^2 - 2\mathbf{d}^\top \mathbf{X}^\top \hat{\boldsymbol{\varepsilon}} + \mathbf{d}^\top \mathbf{X}^\top \mathbf{X} \mathbf{d}.$$

The middle term vanishes because $\mathbf{X}^\top \hat{\boldsymbol{\varepsilon}} = \mathbf{X}^\top \mathbf{y} - \mathbf{X}^\top \mathbf{X} \hat{\boldsymbol{\beta}} = \mathbf{0}$ by the normal equations. The last term is positive for $\mathbf{d} \neq \mathbf{0}$ by (a). So $S(\mathbf{b}) > S(\hat{\boldsymbol{\beta}})$ whenever $\mathbf{b} \neq \hat{\boldsymbol{\beta}}$.

For the last statement, suppose $\text{rank}(\mathbf{X}) < p$. The vector $\mathbf{X}^\top \mathbf{y}$ lies in $\mathcal{C}(\mathbf{X}^\top) = \mathcal{C}(\mathbf{X}^\top \mathbf{X})$ (Proposition 1.2.3(c)), so by Proposition 1.11.2, applied with $\mathbf{A} = \mathbf{X}^\top \mathbf{X}$ and $S(\mathbf{b}) = \mathbf{y}^\top \mathbf{y} + (\mathbf{b}^\top \mathbf{X}^\top \mathbf{X} \mathbf{b} - 2\mathbf{b}^\top \mathbf{X}^\top \mathbf{y})$, the minimizers of S are exactly the solutions of the normal equations, and they exist. But there is a $\mathbf{v} \neq \mathbf{0}$ with $\mathbf{X}\mathbf{v} = \mathbf{0}$, and then $S(\mathbf{b} + t\mathbf{v}) = S(\mathbf{b})$ for every t , so no minimizer is unique. ■

The calculus proof finds the stationary point and then needs a separate argument, Eq. 5.4.3, to show it is the global minimum. The Hessian $2\mathbf{X}^\top \mathbf{X}$ tells the same story: it is positive definite, so S is a strictly convex quadratic bowl with a single lowest point. The name “normal equations” has nothing to do with the normal distribution. It refers to the orthogonality they express: $\mathbf{X}^\top (\mathbf{y} - \mathbf{X}\hat{\boldsymbol{\beta}}) = \mathbf{0}$ says the residual vector is *normal*, that is, perpendicular, to every column of $\mathbf{X}$. Chapter 6 makes that geometry the starting point, derives the normal equations again without calculus (Theorem 6.4.2), and treats the rank-deficient case in full.

Remark (The rank-deficient case). When $\text{rank}(\mathbf{X}) < p$, the proof shows that the least squares estimates form an affine set $\hat{\boldsymbol{\beta}} + \mathcal{N}(\mathbf{X})$ of dimension $p - \text{rank}(\mathbf{X})$. Different programs report different members of this set, according to which columns they drop or which constraints they impose. The fitted values $\mathbf{X}\hat{\boldsymbol{\beta}}$ are the same for all of them, since $\mathbf{X}(\hat{\boldsymbol{\beta}} + \mathbf{v}) = \mathbf{X}\hat{\boldsymbol{\beta}}$ for $\mathbf{v} \in \mathcal{N}(\mathbf{X})$. Which linear combinations of the coefficients share this invariance is the question of estimability, answered in Chapter 8 (Definition 8.2.1 and Theorem 8.3.1).

Least squares under known linear restrictions $\mathbf{H}\mathbf{b} = \mathbf{h}$, such as shares that add to one, has a

closed form found by Lagrange multipliers in [Exercise \(B1\)](#); the increase in the residual sum of squares it computes drives the test of the restrictions in [Chapter 11](#) ([Theorem 11.3.2](#)).

Example. (The straight line again)

For $\mathbf{X} = [\mathbf{1}, \mathbf{x}]$,

$$\mathbf{X}^\top \mathbf{X} = \begin{pmatrix} n & n\bar{x} \\ n\bar{x} & \sum_i x_i^2 \end{pmatrix}, \quad \mathbf{X}^\top \mathbf{y} = \begin{pmatrix} n\bar{y} \\ \sum_i x_i y_i \end{pmatrix}.$$

The determinant is $n \sum_i x_i^2 - n^2 \bar{x}^2 = nS_{xx}$, positive iff the x_i are not all equal, which is exactly when $\text{rank}(\mathbf{X}) = 2$. Inverting,

$$(\mathbf{X}^\top \mathbf{X})^{-1} = \frac{1}{nS_{xx}} \begin{pmatrix} \sum_i x_i^2 & -n\bar{x} \\ -n\bar{x} & n \end{pmatrix} = \begin{pmatrix} \frac{1}{n} + \frac{\bar{x}^2}{S_{xx}} & -\frac{\bar{x}}{S_{xx}} \\ -\frac{\bar{x}}{S_{xx}} & \frac{1}{S_{xx}} \end{pmatrix}, \quad (5.4.4)$$

using $\sum_i x_i^2 = S_{xx} + n\bar{x}^2$. The second entry of $(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{y}$ is $(\sum_i x_i y_i - n\bar{x}\bar{y})/S_{xx} = S_{xy}/S_{xx}$, and the first is $\bar{y} - \bar{x}S_{xy}/S_{xx}$. This recovers [Theorem 5.2.1](#), and [Eq. 5.4.3](#) reduces to [Eq. 5.2.5](#).

5.4.2 Fitted values, residuals and the hat matrix

Substituting [Eq. 5.4.2](#), the fitted values are

$$\hat{\mathbf{Y}} = \mathbf{X}\hat{\boldsymbol{\beta}} = \mathbf{H}\mathbf{y}, \quad \mathbf{H} = \mathbf{X}(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top, \quad (5.4.5)$$

and the residuals are $\hat{\boldsymbol{\varepsilon}} = (\mathbf{I} - \mathbf{H})\mathbf{y}$. The $n \times n$ matrix $\mathbf{H}$ is the **hat matrix**, so called because it puts the hat on $\mathbf{y}$. It depends only on the regressors, never on the response.

Proposition 5.4.2 — Algebra of the fit

Let $\mathbf{X}$ have rank p , and let $\mathbf{H}$, $\hat{\mathbf{Y}}$ and $\hat{\boldsymbol{\varepsilon}}$ be as above.

- (a) $\mathbf{H}$ is symmetric and idempotent, $\mathbf{H}\mathbf{X} = \mathbf{X}$ and $(\mathbf{I} - \mathbf{H})\mathbf{X} = \mathbf{0}$, and $\text{tr } \mathbf{H} = p$, $\text{tr}(\mathbf{I} - \mathbf{H}) = n - p$.
- (b) $\mathbf{X}^\top \hat{\boldsymbol{\varepsilon}} = \mathbf{0}$ and $\hat{\mathbf{Y}}^\top \hat{\boldsymbol{\varepsilon}} = 0$.
- (c) $\|\mathbf{y}\|^2 = \|\hat{\mathbf{Y}}\|^2 + \|\hat{\boldsymbol{\varepsilon}}\|^2$, and $\text{SSE} = \mathbf{y}^\top (\mathbf{I} - \mathbf{H})\mathbf{y} = \mathbf{y}^\top \mathbf{y} - \hat{\boldsymbol{\beta}}^\top \mathbf{X}^\top \mathbf{y}$.
- (d) If $\mathbf{1} = \mathbf{X}\mathbf{c}$ for some $\mathbf{c}$ (for example, if $\mathbf{X}$ has an intercept column), then $\sum_i \hat{\varepsilon}_i = 0$, the mean of the fitted values is $\bar{y}$, and

$$\sum_i (y_i - \bar{y})^2 = \sum_i (\hat{y}_i - \bar{y})^2 + \sum_i \hat{\varepsilon}_i^2,$$

that is, $\text{SST} = \text{SSR} + \text{SSE}$.

- (e) If $\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}$ for some $\boldsymbol{\beta}$ and $\boldsymbol{\varepsilon}$, then $\hat{\boldsymbol{\beta}} = \boldsymbol{\beta} + (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \boldsymbol{\varepsilon}$ and $\hat{\boldsymbol{\varepsilon}} = (\mathbf{I} - \mathbf{H})\boldsymbol{\varepsilon}$.

Proof. (a) $\mathbf{H}^\top = \mathbf{X}((\mathbf{X}^\top \mathbf{X})^{-1})^\top \mathbf{X}^\top = \mathbf{H}$, since the inverse of a symmetric matrix is symmetric. $\mathbf{H}\mathbf{X} = \mathbf{X}(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{X} = \mathbf{X}$, and then $\mathbf{H}^2 = \mathbf{H}\mathbf{X}(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top = \mathbf{X}(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top = \mathbf{H}$. By the cyclic property (Theorem 1.6.1), $\text{tr } \mathbf{H} = \text{tr}((\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{X}) = \text{tr } \mathbf{I}_p = p$. (b) $\mathbf{X}^\top \hat{\boldsymbol{\varepsilon}} = \mathbf{X}^\top (\mathbf{I} - \mathbf{H})\mathbf{y} = ((\mathbf{I} - \mathbf{H})\mathbf{X})^\top \mathbf{y} = \mathbf{0}$, and $\hat{\mathbf{Y}}^\top \hat{\boldsymbol{\varepsilon}} = \hat{\boldsymbol{\beta}}^\top \mathbf{X}^\top \hat{\boldsymbol{\varepsilon}} = 0$. (c) $\mathbf{y} = \hat{\mathbf{Y}} + \hat{\boldsymbol{\varepsilon}}$ with $\hat{\mathbf{Y}}^\top \hat{\boldsymbol{\varepsilon}} = 0$ gives the first identity. For the second, $(\mathbf{I} - \mathbf{H})$ is symmetric and idempotent, so $\|(\mathbf{I} - \mathbf{H})\mathbf{y}\|^2 = \mathbf{y}^\top (\mathbf{I} - \mathbf{H})\mathbf{y}$; and $\mathbf{y}^\top \mathbf{H}\mathbf{y} = \mathbf{y}^\top \mathbf{X}\hat{\boldsymbol{\beta}} = \hat{\boldsymbol{\beta}}^\top \mathbf{X}^\top \mathbf{y}$. (d) $\mathbf{1}^\top \hat{\boldsymbol{\varepsilon}} = \mathbf{c}^\top \mathbf{X}^\top \hat{\boldsymbol{\varepsilon}} = 0$ by (b). Then $\sum_i \hat{y}_i = \sum_i y_i$. Write $y_i - \bar{y} = (\hat{y}_i - \bar{y}) + \hat{\varepsilon}_i$. The cross term $\sum_i (\hat{y}_i - \bar{y})\hat{\varepsilon}_i = \hat{\mathbf{Y}}^\top \hat{\boldsymbol{\varepsilon}} - \bar{y} \mathbf{1}^\top \hat{\boldsymbol{\varepsilon}} = 0$. (e) Substitute $\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}$ into Eq. 5.4.2 and into $(\mathbf{I} - \mathbf{H})\mathbf{y}$, using (a). ■

Part (e) will carry most of the statistical theory. The estimation error $\hat{\boldsymbol{\beta}} - \boldsymbol{\beta}$ is a fixed linear function of the unobserved errors, and the residual vector does not depend on $\boldsymbol{\beta}$ at all. Part (b) says that the residuals satisfy p independent linear constraints. That is the origin of the divisor $n - p$ in Section 5.6.

For a model with an intercept, part (d) motivates the **coefficient of determination**

$$R^2 = \frac{\text{SSR}}{\text{SST}} = 1 - \frac{\text{SSE}}{\text{SST}},$$

the proportion of the variation of the response about its mean that the fitted values reproduce. It generalizes Proposition 5.2.2(d), and lies in $[0, 1]$. It can only increase when a column is added to $\mathbf{X}$ (Exercise (B1)), which is why Chapter 9 introduces an adjusted version (Proposition 9.5.3). Section 6.7 interprets R as the cosine of an angle.

5.4.3 Centring the regressors

For a model with an intercept, write $\mathbf{X} = [\mathbf{1}, \mathbf{X}_1]$, where $\mathbf{X}_1$ is $n \times k$ with column means $\bar{\mathbf{x}} = \mathbf{X}_1^\top \mathbf{1}/n$. The **centred regressors** are the columns of

$$\mathbf{X}_c = \mathbf{X}_1 - \mathbf{1}\bar{\mathbf{x}}^\top = \left(\mathbf{I} - \frac{1}{n}\mathbf{1}\mathbf{1}^\top\right)\mathbf{X}_1,$$

each of which sums to zero. The next result is the matrix version of the formulas $\hat{\beta}_1 = S_{xy}/S_{xx}$ and $\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}$.

Proposition 5.4.3 — Centred form

Let $\mathbf{X} = [\mathbf{1}, \mathbf{X}_1]$ have rank $p = k + 1$, and partition $\hat{\boldsymbol{\beta}} = (\hat{\beta}_0, \hat{\boldsymbol{\beta}}_1^\top)^\top$. Then $\mathbf{X}_c$ has rank k , and

$$\hat{\boldsymbol{\beta}}_1 = (\mathbf{X}_c^\top \mathbf{X}_c)^{-1} \mathbf{X}_c^\top \mathbf{y} = (\mathbf{X}_c^\top \mathbf{X}_c)^{-1} \mathbf{X}_c^\top (\mathbf{y} - \bar{y}\mathbf{1}), \quad \hat{\beta}_0 = \bar{y} - \bar{\mathbf{x}}^\top \hat{\boldsymbol{\beta}}_1.$$

Proof. For $\mathbf{b} = (b_0, \mathbf{b}_1^\top)^\top$,

$$\mathbf{X}\mathbf{b} = b_0\mathbf{1} + \mathbf{X}_1\mathbf{b}_1 = (b_0 + \bar{\mathbf{x}}^\top \mathbf{b}_1)\mathbf{1} + \mathbf{X}_c\mathbf{b}_1 = [\mathbf{1}, \mathbf{X}_c] \mathbf{T}\mathbf{b}, \quad \mathbf{T} = \begin{pmatrix} 1 & \bar{\mathbf{x}}^\top \\ \mathbf{0} & \mathbf{I}_k \end{pmatrix}.$$

$\mathbf{T}$ is nonsingular (it is upper triangular with unit diagonal), so $[\mathbf{1}, \mathbf{X}_c] = \mathbf{X}\mathbf{T}^{-1}$ has rank

p (Proposition 1.2.3(b)), and hence $\mathbf{X}_c$ has rank k . The criterion satisfies $\|\mathbf{y} - \mathbf{X}\mathbf{b}\|^2 = \|\mathbf{y} - [\mathbf{1}, \mathbf{X}_c]\mathbf{a}\|^2$ with $\mathbf{a} = \mathbf{T}\mathbf{b}$, and $\mathbf{b} \mapsto \mathbf{T}\mathbf{b}$ is a bijection of $\mathbb{R}^p$. So $\hat{\boldsymbol{\beta}}$ minimizes the left side iff $\hat{\mathbf{a}} = \mathbf{T}\hat{\boldsymbol{\beta}}$ minimizes the right side. Because $\mathbf{1}^\top \mathbf{X}_c = \mathbf{0}^\top$, the normal equations for $[\mathbf{1}, \mathbf{X}_c]$ are block diagonal,

$$\begin{pmatrix} n & \mathbf{0}^\top \\ \mathbf{0} & \mathbf{X}_c^\top \mathbf{X}_c \end{pmatrix} \hat{\mathbf{a}} = \begin{pmatrix} n\bar{y} \\ \mathbf{X}_c^\top \mathbf{y} \end{pmatrix},$$

giving $\hat{a}_0 = \bar{y}$ and $\hat{\mathbf{a}}_1 = (\mathbf{X}_c^\top \mathbf{X}_c)^{-1} \mathbf{X}_c^\top \mathbf{y}$. Undoing $\mathbf{T}$: $\hat{\boldsymbol{\beta}}_1 = \hat{\mathbf{a}}_1$ and $\hat{\beta}_0 = \hat{a}_0 - \bar{\mathbf{x}}^\top \hat{\mathbf{a}}_1$. Finally $\mathbf{X}_c^\top \mathbf{1} = \mathbf{0}$, so $\mathbf{X}_c^\top \mathbf{y} = \mathbf{X}_c^\top (\mathbf{y} - \bar{y}\mathbf{1})$.

■

In words: to find the slopes, centre everything and fit without an intercept; the intercept then makes the fitted surface pass through the point of means $(\bar{\mathbf{x}}, \bar{y})$. The matrix $\mathbf{X}_c^\top \mathbf{X}_c / (n-1)$ is the sample covariance matrix of the regressors, and $\mathbf{X}_c^\top \mathbf{y} / (n-1)$ their sample covariances with the response. So the slopes are “covariance divided by variance”, as in the straight-line case, with a matrix inverse in place of the division. This is the simplest case of the Frisch–Waugh–Lovell theorem of Section 6.6, with the intercept playing the part of the regressors that are “partialled out”.

5.4.4 Indicator variables

The least squares estimates in a model with only indicator columns are simple averages.

Proposition 5.4.4 — Coefficients of indicator variables

Suppose each case belongs to one of $c \geq 2$ groups, group k has $n_k \geq 1$ cases with mean response $\bar{y}_k$, and $\mathbf{z}_k$ is the indicator column of group k . Let $\mathbf{X} = [\mathbf{1}, \mathbf{z}_1, \dots, \mathbf{z}_{c-1}]$, so that group c is the reference. Then $\mathbf{X}$ has full rank, and

$$\hat{\beta}_0 = \bar{y}_c, \quad \hat{\beta}_k = \bar{y}_k - \bar{y}_c \quad (k = 1, \dots, c-1).$$

The fitted value of every case is the mean of its group.

Proof. If $\mathbf{X}\mathbf{b} = \mathbf{0}$, the entries for a case in group c give $b_0 = 0$, and those for a case in group $k < c$ give $b_0 + b_k = 0$, so $\mathbf{b} = \mathbf{0}$: the rank is $p = c$. Let $\mathbf{b}$ be the proposed vector. Its fitted value for a case in group $k < c$ is $\bar{y}_c + (\bar{y}_k - \bar{y}_c) = \bar{y}_k$, and for a case in group c it is $\bar{y}_c$. The residuals $y_i - \bar{y}_{k(i)}$ therefore sum to zero within each group, so $\mathbf{z}_k^\top (\mathbf{y} - \mathbf{X}\mathbf{b}) = 0$ for every $k = 1, \dots, c$. Every column of $\mathbf{X}$ is a sum of group indicators, since $\mathbf{1} = \mathbf{z}_1 + \dots + \mathbf{z}_c$, so $\mathbf{X}^\top (\mathbf{y} - \mathbf{X}\mathbf{b}) = \mathbf{0}$: the normal equations hold. By Theorem 5.4.1, $\mathbf{b} = \hat{\boldsymbol{\beta}}$.

■

```
import numpy as np
import statsmodels.api as sm
rand = sm.datasets.randhie.load_pandas().data
y = rand["mdvis"].to_numpy(dtype=float)
D = rand[["hlthg", "hlthf", "hlthp"]].to_numpy(dtype=float) # good, fair, poor
```

```

X = np.column_stack([np.ones(len(y)), D])           # excellent =
    reference

beta_hat = np.linalg.solve(X.T @ X, X.T @ y)
level = D @ np.array([1, 2, 3])                   # 0 = excellent, ...,
    3 = poor
means = np.array([y[level == k].mean() for k in range(4)])
print("coefficients:      ", np.round(beta_hat, 4))
print("group means:      ", np.round(means, 4))
print("means minus reference: ", np.round(means - means[0], 4))

```

For the health data of [Example \(Self-rated health\)](#) the group sizes are 11,019 (excellent), 7,309 (good), 1,560 (fair) and 302 (poor), and the group means are 2.634, 2.902, 3.692 and 5.795 visits. The listing confirms that the coefficients are these means' differences from the reference.

5.4.5 A worked example

Example. (Money demand in Denmark)

Economic theory relates the amount of money people choose to hold to their income, which drives the volume of transactions, and to interest rates, which measure the return given up by holding money rather than bonds. A classic data set records, for $n = 55$ quarters from 1974 to 1987 in Denmark, the logarithm of real money holdings (y), the logarithm of real income, the interest rate on bonds and the interest rate on deposits (rates as fractions per year). The model has $p = 4$ columns: an intercept and the three regressors. Solving the normal equations gives

$$\hat{\beta} = (4.3945, 1.2958, -2.6163, 0.6186)^T,$$

with residual sum of squares $SSE = 0.09253$ and $R^2 = 0.9262$. The listing checks that $\mathbf{X}^T \hat{\mathbf{e}}$ is zero up to rounding error, that the two formulas for SSE in [Proposition 5.4.2\(c\)](#) agree, and that the slopes computed from centred data ([Proposition 5.4.3](#)) are the same. [Section 5.6](#) attaches standard errors to these numbers, and [Section 5.7](#) interprets them.

```

import numpy as np
import statsmodels.api as sm
dk = sm.datasets.danish_data.load_pandas().data
y = dk["lrm"].to_numpy()
X = np.column_stack([np.ones(len(y)), dk["lry"], dk["ibo"], dk["ide"]])
n, p = X.shape
print("n =", n, " p =", p)

```

```

XtX = X.T @ X
Xty = X.T @ y
beta_hat = np.linalg.solve(XtX, Xty)           # solve X^T X b = X^T y
fitted = X @ beta_hat
resid = y - fitted
SSE = resid @ resid

print("beta_hat =", np.round(beta_hat, 4))
print("X^T e     =", X.T @ resid)             # zero up to rounding
print("SSE two ways:", SSE, y @ y - beta_hat @ Xty)

```

```

Z = X[:, 1:] - X[:, 1:].mean(axis=0)          # centred regressors, no intercept column
slopes = np.linalg.solve(Z.T @ Z, Z.T @ (y - y.mean()))
intercept = y.mean() - X[:, 1:].mean(axis=0) @ slopes
print("slopes from centred data:", np.round(slopes, 4))
print("intercept recovered:      ", round(intercept, 4))

```

Warning (Formulas are not algorithms). The formula $\hat{\beta} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{y}$ is for reasoning, not for computing. Forming $\mathbf{X}^\top \mathbf{X}$ squares the condition number of the problem, and inverting it wastes work and accuracy. The listings solve the normal equations with a linear solver, which is acceptable for small, well-conditioned problems like these. Good software computes $\hat{\beta}$ from an orthogonal factorization of $\mathbf{X}$ itself. [Section 6.10](#) shows the loss of accuracy in an example, and [Chapter 10](#) develops the algorithms ([Theorem 10.2.1](#), [Theorem 10.3.2](#)).

5.4.6 Exercises

5.4.6.1 A. Check your understanding

Exercise A1.

Suppose the columns of $\mathbf{X}$ are nonzero and mutually orthogonal. Show that $\hat{\beta}_j = \mathbf{x}_j^\top \mathbf{y} / \mathbf{x}_j^\top \mathbf{x}_j$ for each j , so that each coefficient is the same as in the regression of $\mathbf{y}$ on $\mathbf{x}_j$ alone. Show also that deleting a column does not change the other estimates.

5.4.6.2 B. Practice

Exercise B1.

Let $\mathbf{X} = [\mathbf{X}_1, \mathbf{x}]$ have full rank, and let SSE_1 and SSE be the residual sums of squares from regressing $\mathbf{y}$ on $\mathbf{X}_1$ and on $\mathbf{X}$. Show that $\text{SSE} \leq \text{SSE}_1$, directly from [Definition 5.4.1](#). Deduce that R^2 cannot decrease when a column is added to a model with an intercept.

Solution. Let $\hat{\beta}_1$ be the least squares estimate for $\mathbf{X}_1$. The vector $(\hat{\beta}_1^\top, 0)^\top$ is one candidate in the minimization over $\mathbf{b} \in \mathbb{R}^p$, and it gives the value $\|\mathbf{y} - \mathbf{X}_1 \hat{\beta}_1\|^2 = \text{SSE}_1$. The minimum over all candidates is at most this. With an intercept in $\mathbf{X}_1$, $R^2 = 1 - \text{SSE}/\text{SST}$ and SST does not depend on the model, so R^2 cannot decrease. Equality holds iff the extra coefficient in the larger fit is zero, because by [Eq. 5.4.3](#) the candidate $(\hat{\beta}_1^\top, 0)^\top$ is optimal only if it equals $\hat{\beta}$.

Exercise B2.

Let $\mathbf{W} = \text{diag}(w_1, \dots, w_n)$ with all $w_i > 0$, and $\mathbf{X}$ of full rank. Show that $\sum_i w_i (y_i - \mathbf{x}_{(i)}^\top \mathbf{b})^2$ has the unique minimizer $(\mathbf{X}^\top \mathbf{W} \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{W} \mathbf{y}$, by applying [Theorem 5.4.1](#) to the rescaled data $\mathbf{W}^{1/2} \mathbf{y}$ and $\mathbf{W}^{1/2} \mathbf{X}$. Which cases get most influence when the weights are the reciprocals of the error variances?

Solution. $\sum_i w_i (y_i - \mathbf{x}_i^\top \mathbf{b})^2 = \|\mathbf{W}^{1/2} \mathbf{y} - \mathbf{W}^{1/2} \mathbf{X} \mathbf{b}\|^2$. The matrix $\mathbf{W}^{1/2} \mathbf{X}$ has rank p , because $\mathbf{W}^{1/2}$ is nonsingular (Proposition 1.2.3(b)). By Theorem 5.4.1 the unique minimizer is $((\mathbf{W}^{1/2} \mathbf{X})^\top \mathbf{W}^{1/2} \mathbf{X})^{-1} (\mathbf{W}^{1/2} \mathbf{X})^\top \mathbf{W}^{1/2} \mathbf{y} = (\mathbf{X}^\top \mathbf{W} \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{W} \mathbf{y}$. With $w_i = 1/\text{Var}(Y_i)$, the least variable cases count most. This is weighted least squares, the subject of Chapter 21 and, with a general covariance matrix, of Section 6.9.

Exercise B3.

Show that $\hat{\boldsymbol{\beta}} = \mathbf{A} \mathbf{y}$ with $\mathbf{A} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top$, that $\mathbf{A} \mathbf{X} = \mathbf{I}_p$, and that $\mathbf{A} \mathbf{A}^\top = (\mathbf{X}^\top \mathbf{X})^{-1}$. Give an example of a different $p \times n$ matrix $\mathbf{B}$ with $\mathbf{B} \mathbf{X} = \mathbf{I}_p$ when $n > p$. (Chapter 7 compares all such estimators $\mathbf{B} \mathbf{y}$.)

5.4.6.3 C. Going deeper

Exercise C1.

Let $\mathbf{X}$ have full rank and let a new case $(\mathbf{x}_0, y_0)$ be appended. Using Theorem 1.6.5, show that the new estimate is

$$\hat{\boldsymbol{\beta}}_{\text{new}} = \hat{\boldsymbol{\beta}} + \frac{(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{x}_0 (y_0 - \mathbf{x}_0^\top \hat{\boldsymbol{\beta}})}{1 + \mathbf{x}_0^\top (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{x}_0}.$$

Interpret: the estimate moves in proportion to the new case's prediction error. This is the basis of *recursive least squares*; Chapter 10 treats the numerically stable version.

5.5 Properties of the estimator under second-moment assumptions

Least squares was defined for a fixed data vector. Now let the data be random, generated by the linear model of Definition 5.3.1:

$$\mathbf{Y} = \mathbf{X} \boldsymbol{\beta} + \boldsymbol{\varepsilon}, \quad \mathbf{E}(\boldsymbol{\varepsilon}) = \mathbf{0}, \quad \text{Cov}(\boldsymbol{\varepsilon}) = \sigma^2 \mathbf{I}_n,$$

with $\mathbf{X}$ of full rank p . The estimator $\hat{\boldsymbol{\beta}} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{Y}$ is then a random vector, and so are $\hat{\mathbf{Y}}$ and $\hat{\boldsymbol{\varepsilon}}$. This section computes their means and covariances. Only the second-moment assumptions are used, so the results hold for skewed, heavy-tailed or discrete errors alike, as long as they have finite variance.

5.5.1 Mean and covariance of the estimator

The key observation is that $\hat{\boldsymbol{\beta}}$ is a *linear* function of $\mathbf{Y}$: $\hat{\boldsymbol{\beta}} = \mathbf{A} \mathbf{Y}$ with the constant matrix $\mathbf{A} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top$. The moments of a linear transformation were worked out in Chapter 2.

Theorem 5.5.1 — Moments of the least squares estimator

Let $\mathbf{X}$ have full rank p and $\hat{\boldsymbol{\beta}} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{Y}$.

- (a) If $E(\mathbf{Y}) = \mathbf{X}\boldsymbol{\beta}$, then $E(\hat{\boldsymbol{\beta}}) = \boldsymbol{\beta}$: the least squares estimator is unbiased.
- (b) If moreover $\text{Cov}(\mathbf{Y}) = \sigma^2 \mathbf{I}_n$, then

$$\text{Cov}(\hat{\boldsymbol{\beta}}) = \sigma^2 (\mathbf{X}^\top \mathbf{X})^{-1}. \quad (5.5.1)$$

Proof. Let $\mathbf{A} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top$. By Theorem 2.2.1(a), $E(\hat{\boldsymbol{\beta}}) = \mathbf{A} E(\mathbf{Y}) = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{X} \boldsymbol{\beta} = \boldsymbol{\beta}$. By Theorem 2.2.1(b),

$$\text{Cov}(\hat{\boldsymbol{\beta}}) = \mathbf{A} (\sigma^2 \mathbf{I}_n) \mathbf{A}^\top = \sigma^2 (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{X} (\mathbf{X}^\top \mathbf{X})^{-1} = \sigma^2 (\mathbf{X}^\top \mathbf{X})^{-1},$$

using the symmetry of $(\mathbf{X}^\top \mathbf{X})^{-1}$. ■

The proof is two lines, but the division of labour between the assumptions is worth stating. Unbiasedness needs only (L1): the errors may have unequal variances or be correlated, and $\hat{\boldsymbol{\beta}}$ still centres on $\boldsymbol{\beta}$. The covariance formula needs (L2) and (L3) as well. Neither needs normality. Under normality $\hat{\boldsymbol{\beta}}$ is itself normal, since it is a linear function of a normal vector (Theorem 3.3.1), so $\hat{\boldsymbol{\beta}} \sim N_p(\boldsymbol{\beta}, \sigma^2 (\mathbf{X}^\top \mathbf{X})^{-1})$. Chapter 7 builds exact inference on this (Theorem 7.5.1).

The matrix $(\mathbf{X}^\top \mathbf{X})^{-1}$ carries all the information about how the design affects precision. Its diagonal entries give the variances of the individual coefficients, and its off-diagonal entries their covariances. When two columns are strongly positively correlated, their coefficients are typically estimated with a strong *negative* correlation: the data can tell how large the two effects are together, but not how to split the total between them.

Corollary 5.5.2 — Linear functions of the coefficients

Under the assumptions of Theorem 5.5.1(b), let $\mathbf{a} \in \mathbb{R}^p$ and let $\mathbf{L}$ be a constant $q \times p$ matrix. Then

$$E(\mathbf{a}^\top \hat{\boldsymbol{\beta}}) = \mathbf{a}^\top \boldsymbol{\beta}, \quad \text{Var}(\mathbf{a}^\top \hat{\boldsymbol{\beta}}) = \sigma^2 \mathbf{a}^\top (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{a},$$

and $\text{Cov}(\mathbf{L}\hat{\boldsymbol{\beta}}) = \sigma^2 \mathbf{L} (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{L}^\top$. In particular, the estimated mean response at a design point $\mathbf{x}_0$, $\hat{\mu}_0 = \mathbf{x}_0^\top \hat{\boldsymbol{\beta}}$, is unbiased for $\mu_0 = \mathbf{x}_0^\top \boldsymbol{\beta}$ with $\text{Var}(\hat{\mu}_0) = \sigma^2 \mathbf{x}_0^\top (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{x}_0$, and $\text{Var}(\hat{\beta}_j) = \sigma^2 [(\mathbf{X}^\top \mathbf{X})^{-1}]_{jj}$.

Proof. Apply Theorem 2.2.1 to $\mathbf{L}\hat{\boldsymbol{\beta}}$, with covariance $\mathbf{L} \text{Cov}(\hat{\boldsymbol{\beta}}) \mathbf{L}^\top$, and take $\mathbf{L} = \mathbf{a}^\top$, $\mathbf{L} = \mathbf{x}_0^\top$ and $\mathbf{L} = \mathbf{e}_j^\top$. ■

Linear functions of $\boldsymbol{\beta}$ are what applications usually need: a single coefficient, the difference between two coefficients, the mean response at a new design point, or the change in mean response between two profiles. Chapter 7 shows that for every such function, $\mathbf{a}^\top \hat{\boldsymbol{\beta}}$ has the smallest variance among all linear unbiased estimators (the Gauss–Markov theorem, Theorem 7.2.1).

5.5.2 Fitted values and residuals

Proposition 5.5.3 — Moments of fitted values and residuals

Under the linear model with $\mathbf{X}$ of full rank and $\mathbf{H} = \mathbf{X}(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top$:

- (a) $E(\hat{\mathbf{Y}}) = \mathbf{X}\boldsymbol{\beta}$ and $\text{Cov}(\hat{\mathbf{Y}}) = \sigma^2 \mathbf{H}$;
- (b) $E(\hat{\boldsymbol{\varepsilon}}) = \mathbf{0}$ and $\text{Cov}(\hat{\boldsymbol{\varepsilon}}) = \sigma^2(\mathbf{I} - \mathbf{H})$; in particular $\text{Var}(\hat{\varepsilon}_i) = \sigma^2(1 - h_{ii})$, where h_{ii} is the i th diagonal entry of $\mathbf{H}$;
- (c) $\text{Cov}(\hat{\mathbf{Y}}, \hat{\boldsymbol{\varepsilon}}) = \mathbf{0}$ and $\text{Cov}(\hat{\boldsymbol{\beta}}, \hat{\boldsymbol{\varepsilon}}) = \mathbf{0}$;
- (d) $0 \leq h_{ii} \leq 1$ for every i , and $\sum_i h_{ii} = p$.

Proof. $\hat{\mathbf{Y}} = \mathbf{H}\mathbf{Y}$ and $\hat{\boldsymbol{\varepsilon}} = (\mathbf{I} - \mathbf{H})\mathbf{Y}$, with $\mathbf{H}$ symmetric and idempotent and $\mathbf{H}\mathbf{X} = \mathbf{X}$ (Proposition 5.4.2(a)). By Theorem 2.2.1, $E(\hat{\mathbf{Y}}) = \mathbf{H}\mathbf{X}\boldsymbol{\beta} = \mathbf{X}\boldsymbol{\beta}$ and $\text{Cov}(\hat{\mathbf{Y}}) = \sigma^2 \mathbf{H}\mathbf{H}^\top = \sigma^2 \mathbf{H}$; similarly $E(\hat{\boldsymbol{\varepsilon}}) = (\mathbf{I} - \mathbf{H})\mathbf{X}\boldsymbol{\beta} = \mathbf{0}$ and $\text{Cov}(\hat{\boldsymbol{\varepsilon}}) = \sigma^2(\mathbf{I} - \mathbf{H})^2 = \sigma^2(\mathbf{I} - \mathbf{H})$. The cross-covariances are $\text{Cov}(\mathbf{H}\mathbf{Y}, (\mathbf{I} - \mathbf{H})\mathbf{Y}) = \sigma^2 \mathbf{H}(\mathbf{I} - \mathbf{H}) = \mathbf{0}$ and $\text{Cov}(\mathbf{A}\mathbf{Y}, (\mathbf{I} - \mathbf{H})\mathbf{Y}) = \sigma^2 \mathbf{A}(\mathbf{I} - \mathbf{H}) = \sigma^2(\mathbf{A} - \mathbf{A}\mathbf{H}) = \mathbf{0}$, since $\mathbf{A}\mathbf{H} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{X}(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top = \mathbf{A}$. For (d), $\sigma^2 h_{ii}$ and $\sigma^2(1 - h_{ii})$ are variances, so both are nonnegative, and $\sum_i h_{ii} = \text{tr } \mathbf{H} = p$. ■

Two consequences deserve emphasis. First, residuals are *not* uncorrelated and *not* of equal variance, even when the errors are both. The residual of a case with h_{ii} near 1 has variance near zero: the fit is pulled so close to such a case that its residual says little about its error. The number h_{ii} is the **leverage** of case i (Section 6.8, Chapter 20). Second, the fitted values and the residuals are uncorrelated. Under normality they are then independent (Theorem 3.4.1), which is the key to the exact tests of Chapter 11.

5.5.3 The straight line

Specializing to $\mathbf{X} = [\mathbf{1}, \mathbf{x}]$, the inverse Eq. 5.4.4 gives the classical formulas at once.

Corollary 5.5.4 — Moments for a straight line

In the model Eq. 5.2.1 with $S_{xx} > 0$,

$$\text{Var}(\hat{\beta}_1) = \frac{\sigma^2}{S_{xx}}, \quad \text{Var}(\hat{\beta}_0) = \sigma^2 \left(\frac{1}{n} + \frac{\bar{x}^2}{S_{xx}} \right), \quad \text{Cov}(\hat{\beta}_0, \hat{\beta}_1) = -\frac{\sigma^2 \bar{x}}{S_{xx}},$$

and the estimated mean response at x_0 has

$$\text{Var}(\hat{\beta}_0 + \hat{\beta}_1 x_0) = \sigma^2 \left(\frac{1}{n} + \frac{(x_0 - \bar{x})^2}{S_{xx}} \right). \quad (5.5.2)$$

Both estimates are unbiased, and $\bar{Y}$ is uncorrelated with $\hat{\beta}_1$.

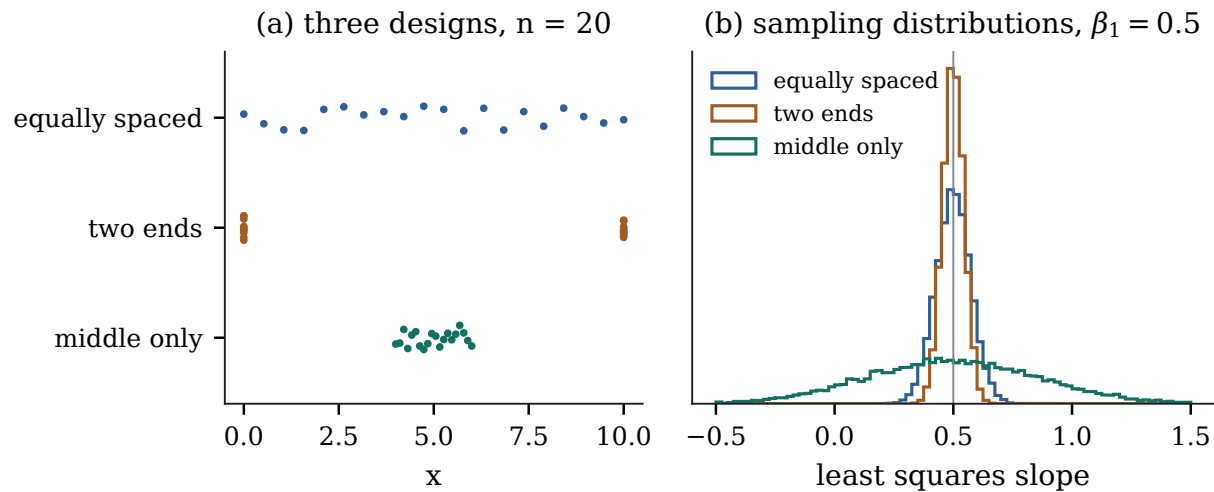


Figure 5.5.1: (a) Three designs with 20 points in $[0, 10]$. (b) Sampling distributions of the least squares slope under each design, with $\beta_1 = 0.5$ and $\sigma = 1$. All three are centred on β_1 ; their spreads are proportional to $S_{xx}^{-1/2}$.

Proof. The first three are the entries of $\sigma^2(\mathbf{X}^\top \mathbf{X})^{-1}$ in Eq. 5.4.4. For the mean response, Corollary 5.5.2 with $\mathbf{x}_0 = (1, x_0)^\top$ gives $\sigma^2\left(\frac{1}{n} + \frac{\bar{x}^2}{S_{xx}} - \frac{2x_0\bar{x}}{S_{xx}} + \frac{x_0^2}{S_{xx}}\right)$, which rearranges to Eq. 5.5.2. Finally $\bar{Y} = \hat{\beta}_0 + \hat{\beta}_1\bar{x}$ (the line passes through the point of means), and $\text{Cov}(\hat{\beta}_0 + \hat{\beta}_1\bar{x}, \hat{\beta}_1) = -\sigma^2\bar{x}/S_{xx} + \bar{x}\sigma^2/S_{xx} = 0$. ■

Three design lessons can be read off. The slope is estimated precisely when the x_i are spread out, since its variance is inversely proportional to S_{xx} . The intercept is estimated precisely only near the data: its variance grows with $\bar{x}^2$, and the estimated intercept and slope are strongly negatively correlated when $\bar{x}$ is large and positive, the tilting of the ellipses noted after Theorem 5.2.1. And the mean response is estimated best at $x_0 = \bar{x}$, with a variance that grows quadratically as x_0 moves away. Extrapolation is doubly risky: the model may fail outside the range of the data, and even if it holds, the estimate is imprecise there.

Example. (Where to put the design points)

Suppose $n = 20$ observations can be taken anywhere in $[0, 10]$ to estimate a slope. Three designs are compared in Figure 5.5.1. Equally spaced points give $S_{xx} = 184.21$ and a slope standard deviation of 0.0737σ . Putting half the points at each end gives the largest possible $S_{xx} = 500.00$ (Exercise B1) and 0.0447σ . Bunching the points in the middle, over $[4, 6]$, gives $S_{xx} = 7.37$ and 0.3684σ , more than eight times worse than the two-ends design. The simulated sampling distributions in panel (b), each from 20,000 samples, match these standard deviations. The two-ends design has a price the variance formula does not show: with only two distinct values of x , the data cannot reveal curvature. The equally spaced design gives up some precision for the ability to check the model.

```
import numpy as np
n = 20
designs = {
    "equally spaced": np.linspace(0, 10, n),
```

```

    "two ends": np.repeat([0.0, 10.0], n // 2),
    "middle only": np.linspace(4, 6, n),
}
for name, xd in designs.items():
    Sxx = np.sum((xd - xd.mean()) ** 2)
    print(f"{name:15s} S_xx = {Sxx:7.2f}    sd(slope) = sigma * {1 / np.sqrt(Sxx):.4f}
    ")

```

5.5.4 Only two moments are needed

Example. (Skewed errors)

To see that normality plays no role in [Theorem 5.5.1](#), fit the quadratic $E(Y) = \beta_0 + \beta_1 x + \beta_2 x^2$ at $x = 1, \dots, 12$, with $\beta = (2, 0.5, -0.03)^\top$ and errors that are centred exponential variables scaled to $\sigma = 1.5$. These errors are strongly skewed. Over 100,000 simulated data sets, the averages of the three estimates are 1.9974, 0.5005 and -0.0300 , and the simulated standard deviations are 1.5421, 0.5457 and 0.0409, against 1.5503, 0.5483 and 0.0411 from [Eq. 5.5.1](#). Every entry of the simulated covariance matrix is within 0.011 of the formula, measured as a correlation. The formula also predicts that $\hat{\beta}_1$ and $\hat{\beta}_2$ have correlation -0.973 , because the columns x and x^2 are nearly collinear on $1, \dots, 12$. A slightly larger linear coefficient can be almost exactly compensated by a slightly more negative quadratic one.

```

import numpy as np
rng = np.random.default_rng(5)
x = np.arange(1.0, 13.0)                # twelve design points
X = np.column_stack([np.ones_like(x), x, x**2])
beta = np.array([2.0, 0.5, -0.03])
sigma = 1.5
reps = 100_000

A = np.linalg.solve(X.T @ X, X.T)        # beta_hat = A y
E = sigma * (rng.exponential(size=(reps, len(x))) - 1.0) # skewed, mean 0, sd
sigma
B = (X @ beta + E) @ A.T                # one estimate per row

print("mean of estimates:", np.round(B.mean(axis=0), 4), " true:", beta)
print("Monte Carlo covariance:\n", np.round(np.cov(B.T), 5))
print("sigma^2 (X^T X)^{-1}:\n", np.round(sigma**2 * np.linalg.inv(X.T @ X), 5))

```

5.5.5 Random regressors

When the regressors are random, the assumptions are read conditionally on $\mathbf{X}$ ([Section 5.1](#)). The conditional results then transfer.

Corollary 5.5.5 — Random regressors

Let $(\mathbf{X}, \mathbf{Y})$ be random, with $\text{rank}(\mathbf{X}) = p$ with probability one, and suppose $E(\mathbf{Y} \mid$

$\mathbf{X}) = \mathbf{X}\boldsymbol{\beta}$ and $\text{Cov}(\mathbf{Y} \mid \mathbf{X}) = \sigma^2 \mathbf{I}_n$. If $E[(\mathbf{X}^\top \mathbf{X})^{-1}]$ is finite, then

$$E(\hat{\boldsymbol{\beta}}) = \boldsymbol{\beta}, \quad \text{Cov}(\hat{\boldsymbol{\beta}}) = \sigma^2 E[(\mathbf{X}^\top \mathbf{X})^{-1}].$$

Proof. Conditionally on $\mathbf{X}$, [Theorem 5.5.1](#) gives $E(\hat{\boldsymbol{\beta}} \mid \mathbf{X}) = \boldsymbol{\beta}$ and $\text{Cov}(\hat{\boldsymbol{\beta}} \mid \mathbf{X}) = \sigma^2 (\mathbf{X}^\top \mathbf{X})^{-1}$. By the laws of total expectation and covariance ([Proposition 2.5.4](#)), $E(\hat{\boldsymbol{\beta}}) = E(\boldsymbol{\beta}) = \boldsymbol{\beta}$ and $\text{Cov}(\hat{\boldsymbol{\beta}}) = E[\sigma^2 (\mathbf{X}^\top \mathbf{X})^{-1}] + \text{Cov}(\boldsymbol{\beta}) = \sigma^2 E[(\mathbf{X}^\top \mathbf{X})^{-1}]$, since $\boldsymbol{\beta}$ is constant. The finiteness assumption guarantees $\hat{\boldsymbol{\beta}}$ has finite second moments, which [Proposition 2.5.4](#) needs. ■

This is what [Example \(One slope, two sampling schemes\)](#) showed by simulation: under the random design, the variance of the slope was $\sigma^2 E(1/S_{xx})$, not σ^2/S_{xx} for any one design. For inference, the conditional covariance $\sigma^2 (\mathbf{X}^\top \mathbf{X})^{-1}$ is the relevant one, because it describes the design actually observed.

5.5.6 Consistency

As the sample grows, a good estimator should approach the truth. For a fixed p and a sequence of model matrices $\mathbf{X}_n$, the mean squared error of $\hat{\boldsymbol{\beta}}$ is

$$E\|\hat{\boldsymbol{\beta}} - \boldsymbol{\beta}\|^2 = \text{tr Cov}(\hat{\boldsymbol{\beta}}) = \sigma^2 \text{tr}(\mathbf{X}_n^\top \mathbf{X}_n)^{-1},$$

by [Theorem 5.5.1](#) and the fact that the trace of a covariance matrix is the sum of the variances.

Proposition 5.5.6 — Consistency in mean square

Under the linear model with $\mathbf{X}_n$ of full rank p for every n , if $\text{tr}(\mathbf{X}_n^\top \mathbf{X}_n)^{-1} \rightarrow 0$, equivalently if the smallest eigenvalue of $\mathbf{X}_n^\top \mathbf{X}_n$ tends to infinity, then $E\|\hat{\boldsymbol{\beta}} - \boldsymbol{\beta}\|^2 \rightarrow 0$, and hence $\Pr(\|\hat{\boldsymbol{\beta}} - \boldsymbol{\beta}\| > \delta) \rightarrow 0$ for every $\delta > 0$.

Proof. The first claim is the display above. The eigenvalues of $(\mathbf{X}_n^\top \mathbf{X}_n)^{-1}$ are the reciprocals $1/\lambda_1, \dots, 1/\lambda_p$ of those of $\mathbf{X}_n^\top \mathbf{X}_n$, all positive, so the trace is $\sum_j 1/\lambda_j$. It tends to zero iff every $\lambda_j \rightarrow \infty$, iff the smallest does. The last claim is Markov's inequality applied to $\|\hat{\boldsymbol{\beta}} - \boldsymbol{\beta}\|^2$: $\Pr(\|\hat{\boldsymbol{\beta}} - \boldsymbol{\beta}\| > \delta) \leq E\|\hat{\boldsymbol{\beta}} - \boldsymbol{\beta}\|^2 / \delta^2$. ■

For a straight line the condition is $S_{xx} \rightarrow \infty$ together with $\bar{x}^2/S_{xx} \rightarrow 0$, which holds for any design that keeps spreading, but fails, for instance, for $x_i = 1/i$, where S_{xx} converges to a finite limit. Then even infinitely many observations do not pin down the slope ([Exercise \(C1\)](#)).

5.5.7 What if the assumptions fail?

The proof of [Theorem 5.5.1](#) shows exactly what each assumption contributes, so it also shows what goes wrong without it. Two cases recur throughout the book.

Proposition 5.5.7 — Moments under departures

Let $\mathbf{X}$ have full rank, $\mathbf{A} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top$ and $\hat{\boldsymbol{\beta}} = \mathbf{A}\mathbf{Y}$.

- (a) If $E(\mathbf{Y}) = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\delta}$ for some $\boldsymbol{\delta} \in \mathbb{R}^n$, then $E(\hat{\boldsymbol{\beta}}) = \boldsymbol{\beta} + (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \boldsymbol{\delta}$.
- (b) If $E(\mathbf{Y}) = \mathbf{X}\boldsymbol{\beta}$ and $\text{Cov}(\mathbf{Y}) = \boldsymbol{\Sigma}$ for some covariance matrix $\boldsymbol{\Sigma}$, then $E(\hat{\boldsymbol{\beta}}) = \boldsymbol{\beta}$ and

$$\text{Cov}(\hat{\boldsymbol{\beta}}) = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \boldsymbol{\Sigma} \mathbf{X} (\mathbf{X}^\top \mathbf{X})^{-1}. \quad (5.5.3)$$

Proof. Both are [Theorem 2.2.1](#) applied to $\mathbf{A}\mathbf{Y}$: $E(\mathbf{A}\mathbf{Y}) = \mathbf{A}\mathbf{X}\boldsymbol{\beta} + \mathbf{A}\boldsymbol{\delta}$ and $\text{Cov}(\mathbf{A}\mathbf{Y}) = \mathbf{A}\boldsymbol{\Sigma}\mathbf{A}^\top$. ■

Part (a) says that when the mean is misspecified, least squares estimates not $\boldsymbol{\beta}$ but $\boldsymbol{\beta}$ plus the coefficients from regressing the missing part $\boldsymbol{\delta}$ on $\mathbf{X}$. The most important case is an omitted regressor, which [Section 5.7](#) examines. Part (b) says that with the wrong covariance, the estimator stays unbiased but the formula $\sigma^2(\mathbf{X}^\top \mathbf{X})^{-1}$ is wrong. The “sandwich” [Eq. 5.5.3](#) is the correct covariance, and Chapter 21 shows how to estimate it when $\boldsymbol{\Sigma}$ is diagonal but unknown. Chapter 19 studies both departures systematically.

5.5.8 Exercises

5.5.8.1 A. Check your understanding

Exercise A1.

In the model of [Proposition 5.4.4](#) with $c = 2$ groups, show that $\text{Var}(\hat{\beta}_1) = \sigma^2(1/n_1 + 1/n_2)$, both from [Eq. 5.5.1](#) and directly from $\hat{\beta}_1 = \bar{Y}_1 - \bar{Y}_2$. For fixed $n = n_1 + n_2$, which split minimizes it?

Exercise A2.

Show that for a straight line $h_{ii} = 1/n + (x_i - \bar{x})^2/S_{xx}$. Which case in [Example \(Forecasting December sea temperature\)](#) has the largest leverage? Check that the leverages sum to 2.

5.5.8.2 B. Practice

Exercise B1.

Suppose n is even and the x_i may be chosen anywhere in $[a, b]$. Show that $S_{xx} \leq n(b-a)^2/4$, with equality iff half the points are at a and half at b . *Hint:* $|x_i - c| \leq (b-a)/2$ with $c = (a+b)/2$, and $S_{xx} = \sum_i (x_i - c)^2 - n(\bar{x} - c)^2$.

Solution. With c the midpoint, $S_{xx} = \sum_i (x_i - c)^2 - n(\bar{x} - c)^2 \leq \sum_i (x_i - c)^2 \leq n(b-a)^2/4$. Equality in the second step needs every $x_i \in \{a, b\}$, and in the first step needs $\bar{x} = c$, that is, equally many points at each end. For $[0, 10]$ and $n = 20$ this gives $S_{xx} = 20 \cdot 25 = 500$, the value in [Example \(Where to put the design points\)](#).

Exercise B2.

In [Example \(Money demand in Denmark\)](#), write the variances of $\hat{\beta}_2 - \hat{\beta}_3$ and $\hat{\beta}_2 + \hat{\beta}_3$, the difference and the sum of the two interest-rate coefficients, in terms of the entries of $\mathbf{G} = (\mathbf{X}^\top \mathbf{X})^{-1}$. Show that if $\hat{\beta}_2$ and $\hat{\beta}_3$ are negatively correlated, the difference is estimated *less* precisely, and the sum *more* precisely, than they would be if the two estimates were uncorrelated. The two rate columns have sample correlation 0.80. Which combination would you expect to be better determined?

Solution. With $\mathbf{a} = (0, 0, 1, \mp 1)^\top$ and indices counting the intercept as the first coefficient, [Corollary 5.5.2](#) gives $\text{Var}(\hat{\beta}_2 \mp \hat{\beta}_3) = \sigma^2(g_{33} + g_{44} \mp 2g_{34})$, and $\text{Cov}(\hat{\beta}_2, \hat{\beta}_3) = \sigma^2 g_{34}$. If $g_{34} < 0$, the variance of the difference exceeds $\sigma^2(g_{33} + g_{44}) = \text{Var}(\hat{\beta}_2) + \text{Var}(\hat{\beta}_3)$, and that of the sum falls below it. Positively correlated columns tend to produce negatively correlated estimates (as in [Example \(Skewed errors\)](#)), because the data can tell how large the combined effect of the two rates is more easily than how to divide it between them. So the sum should be the better determined combination.

5.5.8.3 C. Going deeper

Exercise C1.

Let $x_i = 1/i$ for $i = 1, 2, \dots$ in the straight-line model. Show that S_{xx} converges to a finite limit as $n \rightarrow \infty$, so that $\text{Var}(\hat{\beta}_1)$ does not tend to zero. Then construct a sequence of designs, in which the points may change with n (for example, designs centred farther and farther from 0), with $S_{xx} \rightarrow \infty$ but $\bar{x}^2/S_{xx} \not\rightarrow 0$, and decide whether the slope, the intercept, or both fail to be consistent.

Solution. With $x_i = 1/i$, $S_{xx} = \sum_{i \leq n} 1/i^2 - n\bar{x}^2$. The sum increases to $\pi^2/6$, and $n\bar{x}^2 = H_n^2/n \rightarrow 0$ because the harmonic number $H_n = \sum_{i \leq n} 1/i$ grows like $\log n$. So $S_{xx} \rightarrow \pi^2/6$ and $\text{Var}(\hat{\beta}_1) = \sigma^2/S_{xx}$ stays above $6\sigma^2/\pi^2$.

For even n take $x_i = n + (-1)^i$, $i = 1, \dots, n$: half the points at $n-1$, half at $n+1$. Then $\bar{x} = n$ and $S_{xx} = n$, so $\bar{x}^2/S_{xx} = n \rightarrow \infty$. By [Corollary 5.5.4](#), $\text{Var}(\hat{\beta}_1) = \sigma^2/n \rightarrow 0$, so the slope is consistent (Chebyshev's inequality), while $\text{Var}(\hat{\beta}_0) = \sigma^2(1/n + n) \rightarrow \infty$. The intercept, the mean response at $x = 0$, ever farther from the data, is not consistent: with normal errors, $\hat{\beta}_0 - \beta_0$ is normal with unbounded variance.

5.6 Estimating the error variance

The covariance formula $\text{Cov}(\hat{\boldsymbol{\beta}}) = \sigma^2(\mathbf{X}^\top \mathbf{X})^{-1}$ is of little use until σ^2 is estimated. Least squares does not estimate it: the criterion $\|\mathbf{y} - \mathbf{X}\mathbf{b}\|^2$ involves only $\boldsymbol{\beta}$. But the size of the residuals clearly carries information about the size of the errors. This section finds the right way to turn one into

an estimate of the other, and shows why the obvious way is wrong.

5.6.1 The residual sum of squares

Since $\sigma^2 = \text{Var}(\varepsilon_i) = E(\varepsilon_i^2)$, a natural first idea is to replace each unobserved error by its residual and average: SSE/n . This underestimates σ^2 . The fitted values were chosen to make the residuals as small as possible, so the residuals are systematically smaller than the errors. The next theorem measures exactly how much smaller.

Theorem 5.6.1 — Unbiased estimation of the error variance

In the linear model [Definition 5.3.1](#) with $\text{rank}(\mathbf{X}) = p < n$,

$$E(\text{SSE}) = \sigma^2(n - p).$$

Hence

$$s^2 = \frac{\text{SSE}}{n - p} = \frac{\|\mathbf{Y} - \mathbf{X}\hat{\boldsymbol{\beta}}\|^2}{n - p} \quad (5.6.1)$$

is an unbiased estimator of σ^2 .

Proof. By [Proposition 5.4.2\(e\)](#) the residual vector is $\hat{\boldsymbol{\varepsilon}} = (\mathbf{I} - \mathbf{H})\boldsymbol{\varepsilon}$, whatever $\boldsymbol{\beta}$ is. Since $\mathbf{I} - \mathbf{H}$ is symmetric and idempotent, $\text{SSE} = \hat{\boldsymbol{\varepsilon}}^\top \hat{\boldsymbol{\varepsilon}} = \boldsymbol{\varepsilon}^\top (\mathbf{I} - \mathbf{H})\boldsymbol{\varepsilon}$, a quadratic form in the error vector, which has mean $\mathbf{0}$ and covariance $\sigma^2 \mathbf{I}_n$. By [Corollary 2.3.2\(b\)](#),

$$E(\text{SSE}) = \sigma^2 \text{tr}(\mathbf{I} - \mathbf{H}) + \mathbf{0}^\top (\mathbf{I} - \mathbf{H})\mathbf{0} = \sigma^2(n - p),$$

using $\text{tr}(\mathbf{I} - \mathbf{H}) = n - p$ from [Proposition 5.4.2\(a\)](#). ■

The computation is the case $\boldsymbol{\theta} = \mathbf{X}\boldsymbol{\beta}$ of [Example \(Residual sum of squares under a wrong mean\)](#) in [Chapter 2](#), specialized to full rank.

The same result follows from [Proposition 5.5.3](#) without quadratic forms. Each residual has mean zero and variance $\sigma^2(1 - h_{ii})$, so $E(\text{SSE}) = \sum_i E(\hat{\varepsilon}_i^2) = \sigma^2 \sum_i (1 - h_{ii}) = \sigma^2(n - p)$. This version shows *where* the shortfall comes from: each residual is less variable than its error by the factor $1 - h_{ii}$, and the leverages add up to p .

The divisor $n - p$ is the number of **residual degrees of freedom**. The name reflects the constraints $\mathbf{X}^\top \hat{\boldsymbol{\varepsilon}} = \mathbf{0}$ of [Proposition 5.4.2\(b\)](#): the n residuals satisfy p independent linear equations, so only $n - p$ of them can vary freely. [Chapter 6](#) makes this precise: the residual vector lies in a subspace of dimension $n - p$, and the degrees of freedom of a sum of squares is always the dimension of the space it lives in. For the straight line $p = 2$, so $s^2 = \text{SSE}/(n - 2)$. For the sample variance of a single sample, the model matrix is $\mathbf{1}$, $p = 1$, and [Eq. 5.6.1](#) is the familiar divisor $n - 1$.

Corollary 5.6.2 — Estimated covariance and standard errors

Under the conditions of [Theorem 5.6.1](#), $s^2(\mathbf{X}^\top \mathbf{X})^{-1}$ is an unbiased estimator of $\text{Cov}(\hat{\boldsymbol{\beta}})$, and $s^2 \mathbf{a}^\top (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{a}$ is an unbiased estimator of $\text{Var}(\mathbf{a}^\top \hat{\boldsymbol{\beta}})$. The **standard error** of $\hat{\beta}_j$ is

$$\text{se}(\hat{\beta}_j) = s \sqrt{[(\mathbf{X}^\top \mathbf{X})^{-1}]_{jj}}.$$

Proof. $(\mathbf{X}^\top \mathbf{X})^{-1}$ and $\mathbf{a}$ are constants, so the expectations are $E(s^2)(\mathbf{X}^\top \mathbf{X})^{-1} = \sigma^2(\mathbf{X}^\top \mathbf{X})^{-1}$ and similarly for the scalar. ■

The standard error is not unbiased for the standard deviation of $\hat{\beta}_j$. By Jensen's inequality $E(s) \leq \sqrt{E(s^2)} = \sigma$, with strict inequality unless s is constant. The bias is small when $n - p$ is large, and it matters little, since the standard error is used mainly through the ratio $(\hat{\beta}_j - \beta_j) / \text{se}(\hat{\beta}_j)$, whose exact distribution under normality is the t distribution with $n - p$ degrees of freedom ([Chapter 7](#), [Theorem 7.5.1](#)).

Example. (Standard errors for money demand)

For the money demand model of [Example \(Money demand in Denmark\)](#), $n = 55$ and $p = 4$, so

$$s^2 = \frac{0.09253}{51} = 0.001814, \quad s = 0.04259.$$

Since the response is a logarithm, s says that money holdings typically deviate from the fitted relation by about 4 percent. The standard errors of the four coefficients are 0.5811, 0.0940, 0.3282 and 0.6911. The income coefficient 1.2958 is almost fourteen standard errors from zero and the bond rate coefficient -2.6163 about eight, but the deposit rate coefficient 0.6186 is smaller than its standard error. Whether the deposit rate belongs in the model at all is a testing question for [Chapter 11](#).

These standard errors rest on (L2) and (L3). For a quarterly economic series (L3) is doubtful. The correlation between consecutive residuals is 0.62, which suggests that the errors are positively correlated in time, and then the formula $\sigma^2(\mathbf{X}^\top \mathbf{X})^{-1}$ usually understates the true variability. Chapter 21 shows how to detect and allow for this.

```
import numpy as np
import statsmodels.api as sm
dk = sm.datasets.danish_data.load_pandas().data
y = dk["lrm"].to_numpy()
X = np.column_stack([np.ones(len(y)), dk["lry"], dk["ibo"], dk["ide"]])
n, p = X.shape
XtX = X.T @ X
Xty = X.T @ y
beta_hat = np.linalg.solve(XtX, Xty)          # solve X^T X b = X^T y
fitted = X @ beta_hat
resid = y - fitted
SSE = resid @ resid

s2 = SSE / (n - p)                            # unbiased estimate of sigma^2
cov_hat = s2 * np.linalg.inv(XtX)             # estimated Cov(beta_hat)
```

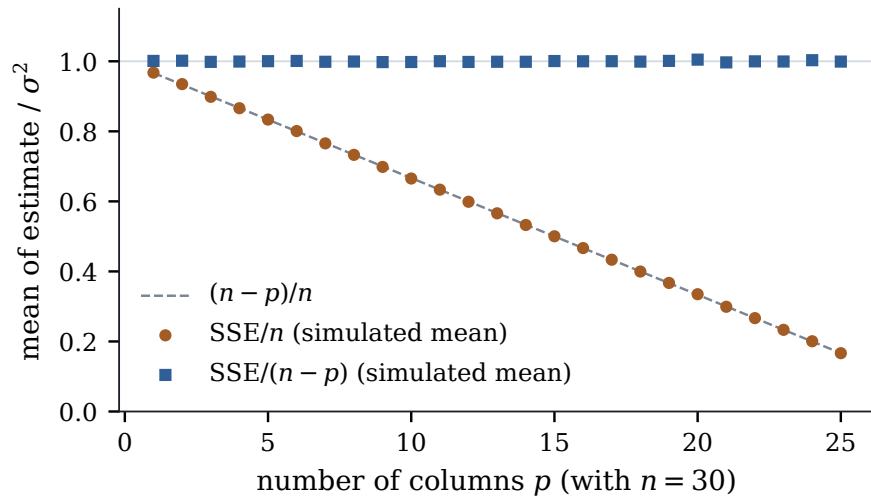



Figure 5.6.1: Average of two estimators of $\sigma^2 = 1$ over 20,000 simulated data sets, with $n = 30$ and p columns. Dividing the residual sum of squares by n underestimates σ^2 by the factor $(n - p)/n$; dividing by $n - p$ is unbiased for every p .

```
se = np.sqrt(np.diag(cov_hat))
print(f"s^2 = {s2:.6f}, s = {np.sqrt(s2):.5f}")
print("standard errors:", np.round(se, 4))
print("SSE / n would give", SSE / n)
```

5.6.2 Dividing by n

The estimator $\hat{\sigma}^2 = \text{SSE}/n$ is not unbiased, but it is not arbitrary either: under normality it is the maximum likelihood estimator ([Theorem 7.3.1](#)). Its expectation is

$$E(\hat{\sigma}^2) = \frac{n-p}{n} \sigma^2 = \sigma^2 - \frac{p}{n} \sigma^2.$$

The bias is negligible when p is a small fraction of n , and severe when it is not. For the money demand data the two estimates of σ are 0.04259 and 0.04102, a difference of about 4 percent. But a model with many columns relative to the number of cases, such as a two-way layout with a few replicates per cell or a regression with many indicator variables, can make SSE/n badly misleading.

Example. (How bad is dividing by n?)

Take $n = 30$ cases and model matrices with an intercept and $p - 1$ further columns of standard normal numbers, for $p = 1, \dots, 25$. For each p , 20,000 error vectors are simulated with $\sigma^2 = 1$. [Figure 5.6.1](#) plots the average of SSE/n and of $\text{SSE}/(n - p)$. The first follows the line $(n - p)/n$: at $p = 5$ it averages 0.834, at $p = 15$ it averages 0.500, half the true value, and at $p = 25$ only 0.167. The unbiased estimator averages 1.000, 1.001 and 0.999 at the same values of p .

```
import numpy as np
rng = np.random.default_rng(506)
n, sigma2, reps = 30, 1.0, 20_000
Xfull = np.column_stack([np.ones(n), rng.normal(size=(n, 24))])
```

```

ps = np.arange(1, 26)
mean_n, mean_np = [], []
for p in ps:
    X = Xfull[:, :p]
    R = np.eye(n) - X @ np.linalg.solve(X.T @ X, X.T)      # I - H
    E = rng.standard_normal((reps, n))                      # errors, sigma^2 = 1
    SSE = np.einsum("ri,ij,rj->r", E, R, E)                 # e^T (I - H) e
    mean_n.append(SSE.mean() / n)
    mean_np.append(SSE.mean() / (n - p))
print("p      E(SSE/n)  E(SSE/(n-p))")
for p in (1, 5, 15, 25):
    print(f"{p:2d}      {mean_n[p - 1]:.3f}      {mean_np[p - 1]:.3f}")

```

5.6.3 The variance of s^2

Unbiasedness says where s^2 is centred, not how much it varies. Its variance depends on the fourth moment of the errors, so the second-moment assumptions are not enough to find it. With independent errors, [Chapter 2](#) supplies the formula.

Proposition 5.6.3 — Variance of s^2

In the linear model with $\text{rank}(\mathbf{X}) = p < n$, suppose the errors $\varepsilon_1, \dots, \varepsilon_n$ are independent with mean 0, variance σ^2 , common third moment $\mu_3 = E(\varepsilon_i^3)$ and common fourth moment $\mu_4 = E(\varepsilon_i^4) < \infty$. Then

$$\text{Var}(\text{SSE}) = 2\sigma^4(n - p) + (\mu_4 - 3\sigma^4) \sum_{i=1}^n (1 - h_{ii})^2.$$

For normal errors, $\mu_4 = 3\sigma^4$ and $\text{Var}(s^2) = 2\sigma^4/(n - p)$.

Proof. Apply [Theorem 2.3.3](#) to the vector $\boldsymbol{\varepsilon}$, whose means θ_i are all zero, with $\mathbf{A} = \mathbf{I} - \mathbf{H}$. The terms involving $\boldsymbol{\theta}$ vanish (in particular μ_3 enters only multiplied by $\boldsymbol{\theta} = \mathbf{0}$, so its value is irrelevant). The diagonal of $\mathbf{A}$ has entries $1 - h_{ii}$, and $\text{tr}(\mathbf{A}^2) = \text{tr}(\mathbf{A}) = n - p$ because $\mathbf{A}$ is idempotent. For the normal case, the fourth moment of a $N(0, \sigma^2)$ variable is $3\sigma^4$, and $\text{Var}(s^2) = \text{Var}(\text{SSE})/(n - p)^2$. ■

The kurtosis term can be large. Heavy-tailed errors, with μ_4 much larger than $3\sigma^4$, make s^2 much less reliable than the normal-theory formula suggests, even though it remains unbiased. Since $0 \leq 1 - h_{ii} \leq 1$, the sum in the kurtosis term is at most $\sum_i (1 - h_{ii}) = n - p$, which gives the following.

Corollary 5.6.4 — Consistency of s^2

Under the conditions of [Proposition 5.6.3](#),

$$\text{Var}(s^2) \leq \frac{2\sigma^4 + |\mu_4 - 3\sigma^4|}{n - p}.$$

So if $n - p \rightarrow \infty$ along a sequence of models, $s^2 \rightarrow \sigma^2$ in mean square and in probability.

Proof. Divide the formula of [Proposition 5.6.3](#) by $(n - p)^2$ and bound the sum by $n - p$. Since $E(s^2) = \sigma^2$, $E(s^2 - \sigma^2)^2 = \text{Var}(s^2) \rightarrow 0$, and Chebyshev's inequality gives convergence in probability. ■

Example. (Skewed errors inflate the variance of s^2)

With $n = 30$ and the first $p = 6$ columns of the model matrix of [Example \(How bad is dividing by \$n\$?\)](#), the leverages give $\sum_i (1 - h_{ii})^2 = 19.431$. For normal errors with $\sigma^2 = 1$, $\text{Var}(\text{SSE}) = 2(n - p) = 48.00$. For centred exponential errors, which have $\sigma^2 = 1$ and $\mu_4 = 9$, [Proposition 5.6.3](#) gives $48 + 6 \times 19.431 = 164.59$, and a simulation with 200,000 data sets gives 165.19. The standard deviation of s^2 is 0.289 under normal errors and 0.535 under exponential errors, nearly twice as large, although s^2 is unbiased in both cases.

```
p = 6
X = Xfull[:, :p]
H = X @ np.linalg.solve(X.T @ X, X.T)
R = np.eye(n) - H
mu4 = 9.0                                     # fourth moment of a centred Exp(1)
variable
var_normal = 2 * (n - p)
var_exp = (mu4 - 3) * np.sum(np.diag(R) ** 2) + 2 * (n - p)
reps2 = 200_000
E = rng.exponential(size=(reps2, n)) - 1.0
SSE = np.einsum("ri,ij,rj->r", E, R, E)
print(f"Var(SSE): normal errors {var_normal:.2f}, exponential errors {var_exp:.2f},"
      f" simulated {SSE.var():.2f}")
```

Under normality much more is true: SSE/σ^2 has the chi-squared distribution with $n - p$ degrees of freedom and is independent of $\hat{\beta}$ ([Theorem 7.5.1](#), using [Theorem 4.3.2](#) and [Theorem 4.4.1](#)). That is what turns $\hat{\beta}$, s^2 and the standard errors into exact confidence intervals and tests. Among unbiased estimators of σ^2 , s^2 then has the smallest variance ([Theorem 7.4.6](#)).

5.6.4 When the mean is wrong

[Theorem 5.6.1](#) assumed that $E(\mathbf{Y}) = \mathbf{X}\beta$. If instead $E(\mathbf{Y}) = \boldsymbol{\theta}$ is not of this form, [Example \(Residual sum of squares under a wrong mean\)](#) in [Chapter 2](#) shows that

$$E(\text{SSE}) = \sigma^2(n - p) + \|(\mathbf{I} - \mathbf{H})\boldsymbol{\theta}\|^2.$$

The extra term is the squared distance from the true mean vector to the nearest mean vector the model can produce. So an incomplete model *overestimates* σ^2 : variation that the model fails

to capture is counted as error. This is one reason to prefer a model that includes all relevant regressors, and it is the basis of the lack-of-fit tests of [Chapter 14 \(Theorem 14.6.2\)](#), which compare s^2 with an estimate of σ^2 that does not depend on the form of the mean.

5.6.5 Exercises

5.6.5.1 A. Check your understanding

Exercise A1.

For regression through the origin, $E(Y_i) = \beta x_i$, what is the unbiased estimator of σ^2 ? Verify $E(\text{SSE}) = \sigma^2(n-1)$ directly, by writing $\text{SSE} = \sum_i \varepsilon_i^2 - (\sum_i x_i \varepsilon_i)^2 / \sum_i x_i^2$.

Exercise A2.

For [Example \(Forecasting December sea temperature\)](#), compute s and the standard error of the slope. Is the slope more than two standard errors away from 1, the value that would mean December simply tracks August one for one?

5.6.5.2 B. Practice

Exercise B1.

Show that $\text{SSE} = \mathbf{y}^\top \mathbf{y} - \hat{\boldsymbol{\beta}}^\top \mathbf{X}^\top \mathbf{X} \hat{\boldsymbol{\beta}}$, and that for a model with an intercept $\text{SSE} = S_{yy} - \hat{\boldsymbol{\beta}}_1^\top \mathbf{X}_c^\top \mathbf{X}_c \hat{\boldsymbol{\beta}}_1$ in the notation of [Proposition 5.4.3](#). Explain why the second form is numerically preferable when the responses are large and nearly equal, as in [Example \(A curved trend and a seasonal cycle\)](#).

Solution. By [Proposition 5.4.2\(c\)](#), $\text{SSE} = \mathbf{y}^\top \mathbf{y} - \hat{\boldsymbol{\beta}}^\top \mathbf{X}^\top \mathbf{y}$, and $\mathbf{X}^\top \mathbf{y} = \mathbf{X}^\top \mathbf{X} \hat{\boldsymbol{\beta}}$ by the normal equations. For the centred form, the model with columns $[\mathbf{1}, \mathbf{X}_c]$ has the same fitted values (the proof of [Proposition 5.4.3](#)), with coefficients $(\bar{y}, \hat{\boldsymbol{\beta}}_1)$. So $\text{SSE} = \mathbf{y}^\top \mathbf{y} - n\bar{y}^2 - \hat{\boldsymbol{\beta}}_1^\top \mathbf{X}_c^\top \mathbf{X}_c \hat{\boldsymbol{\beta}}_1$, because the cross-product matrix is block diagonal, and $\mathbf{y}^\top \mathbf{y} - n\bar{y}^2 = S_{yy}$. When the responses are about 340 with variation of a few units, $\mathbf{y}^\top \mathbf{y}$ and $\hat{\boldsymbol{\beta}}^\top \mathbf{X}^\top \mathbf{y}$ are both near $n \cdot 340^2$ and agree in their leading digits; their difference loses those digits to cancellation. The centred form subtracts numbers of the size of the variation itself.

Exercise B2.

Under normality SSE/σ^2 has mean $n-p$ and variance $2(n-p)$. Among estimators $c \cdot \text{SSE}$, find the c that minimizes the mean squared error $E(c \text{SSE} - \sigma^2)^2$. Compare with $1/(n-p)$ and $1/n$.

Solution. Write $\text{SSE} = \sigma^2 Q$ with $E Q = m = n-p$ and $E Q^2 = 2m + m^2$. Then $E(cQ - 1)^2 \sigma^4 = \sigma^4 (c^2(m^2 + 2m) - 2cm + 1)$, minimized at $c = m/(m^2 + 2m) = 1/(n-p+2)$. The minimum-MSE estimator divides by $n-p+2$: it is biased downward but less variable. The ordering of divisors is $n-p < n-p+2$ and, when $p \leq 2$, $n \leq n-p+2$, so for small p the maximum likelihood divisor n lies between the unbiased and the minimum-MSE

choice. Unbiasedness is a convention, not an optimality property; it is kept because it makes s^2 combine cleanly in the t and F statistics.

5.6.5.3 C. Going deeper

Exercise C1.

Show that a quadratic form $\mathbf{Y}^\top \mathbf{A} \mathbf{Y}$ with $\mathbf{A}$ symmetric is unbiased for σ^2 for every $\boldsymbol{\beta}$ and σ^2 iff $\mathbf{X}^\top \mathbf{A} \mathbf{X} = \mathbf{0}$ and $\text{tr } \mathbf{A} = 1$. Verify that $\mathbf{A} = (\mathbf{I} - \mathbf{H})/(n - p)$ qualifies, and find another $\mathbf{A}$ that does. (Under normality, Theorem 7.4.6 shows that s^2 has the smallest variance among all unbiased estimators.)

5.7 Interpreting coefficients

A fitted regression is usually reported as a table of coefficients. The short answer, “the change in the mean response per unit change in the regressor, holding the other regressors fixed”, is a useful slogan and a frequent source of error. This section makes the slogan precise within the model, shows how the meaning of a coefficient depends on units, centring, transformations and the other columns of the model, and marks where the slogan stops being true.

5.7.1 What a coefficient measures

Definition 5.7.1 — Interpretation of a coefficient

In a linear model $E(Y | \mathbf{x}) = \beta_0 + \beta_1 x_1 + \cdots + \beta_k x_k$, suppose the regressor x_j enters the model only through its own column, so that no other column is a function of x_j . For any two regressor profiles $\mathbf{x}$ and $\mathbf{x}'$ that agree in every regressor except the j th, and have $x'_j = x_j + 1$,

$$E(Y | \mathbf{x}') - E(Y | \mathbf{x}) = \beta_j. \quad (5.7.1)$$

The coefficient β_j is the **partial effect** of x_j in this model: the difference in mean response per unit difference in x_j , between profiles that agree on all the other regressors in the model. Its estimate $\hat{\beta}_j$ is the least squares estimate of this difference.

Three qualifications are built into the definition.

1. **It compares profiles, not interventions.** Eq. 5.7.1 compares the mean response of two kinds of case that exist in the population the model describes. Whether changing x_j in a given case would change its response by β_j is a causal question, which the model alone cannot answer (Chapter 25).
2. **“The other regressors” means the ones in the model.** Adding or removing a regressor changes the comparison being made, and so changes β_j , unless the regressors are orthogonal. Proposition 5.7.2 gives the exact relation.
3. **It needs x_j to vary alone.** If x_j also appears in other columns, as in a polynomial or

an interaction, there are no two profiles that differ only in the x_j column, and β_j alone does not measure the effect of x_j .

The geometric content of “holding the others fixed” in a data set, where nobody held anything fixed, is the Frisch–Waugh–Lovell theorem of [Section 6.6](#): $\hat{\beta}_j$ is the slope of the response on the part of x_j that is not linearly explained by the other regressors.

5.7.2 Units, rescaling and reparameterization

The coefficient β_j is measured in units of the response per unit of x_j . Changing units rescales it. More generally, any invertible linear recoding of the columns changes the coefficients in a predictable way and leaves the fit untouched.

Proposition 5.7.1 — Reparameterization

Let $\mathbf{X}$ be $n \times p$ of rank p , let $\mathbf{K}$ be a nonsingular $p \times p$ matrix, and put $\mathbf{Z} = \mathbf{X}\mathbf{K}$. Then $\mathbf{Z}$ has rank p , and the least squares estimates from the two model matrices are related by

$$\hat{\beta}_Z = \mathbf{K}^{-1} \hat{\beta}_X.$$

The fitted values, residuals, SSE and s^2 are the same for both, and the estimated covariance matrices satisfy $s^2(\mathbf{Z}^\top \mathbf{Z})^{-1} = \mathbf{K}^{-1} [s^2(\mathbf{X}^\top \mathbf{X})^{-1}] (\mathbf{K}^{-1})^\top$.

Proof. $\text{rank}(\mathbf{Z}) = \text{rank}(\mathbf{X}) = p$ by [Proposition 1.2.3\(b\)](#). Then

$$\begin{aligned} \hat{\beta}_Z &= (\mathbf{K}^\top \mathbf{X}^\top \mathbf{X} \mathbf{K})^{-1} \mathbf{K}^\top \mathbf{X}^\top \mathbf{y} \\ &= \mathbf{K}^{-1} (\mathbf{X}^\top \mathbf{X})^{-1} (\mathbf{K}^\top)^{-1} \mathbf{K}^\top \mathbf{X}^\top \mathbf{y} = \mathbf{K}^{-1} \hat{\beta}_X, \end{aligned}$$

and $\mathbf{Z} \hat{\beta}_Z = \mathbf{X} \mathbf{K} \mathbf{K}^{-1} \hat{\beta}_X = \mathbf{X} \hat{\beta}_X$. Equal fitted values give equal residuals, SSE and s^2 . The covariance identity follows from $(\mathbf{Z}^\top \mathbf{Z})^{-1} = \mathbf{K}^{-1} (\mathbf{X}^\top \mathbf{X})^{-1} (\mathbf{K}^{-1})^\top$. ■

With $\mathbf{K} = \text{diag}(1, c_1, \dots, c_k)$, measuring x_j in units $1/c_j$ times as large divides its coefficient and its standard error by c_j , and leaves everything else alone. Centring, effect coding of a factor ([Exercise \(B1\)](#)) and many other changes of coding are also of the form $\mathbf{X}\mathbf{K}$. [Section 6.7](#) shows that what really stays fixed is the column space of $\mathbf{X}$.

The intercept and centring. The intercept β_0 is the mean response at the profile where every regressor is zero. That profile may be impossible (zero income) or far outside the data (an August sea temperature of 0°C in [Example \(Forecasting December sea temperature\)](#)). If the regressors are centred, the intercept becomes the mean response at the average profile $\bar{\mathbf{x}}$, and its estimate is $\bar{y}$ ([Proposition 5.4.3](#)). Centring does not change the slopes. It makes the intercept interpretable, and it makes $\hat{\beta}_0$ uncorrelated with the slopes ([Exercise \(B2\)](#)). With interactions or polynomials, centring also changes what the lower-order coefficients mean, as the next subsections show.

Standardized coefficients. Measuring every variable in units of its own sample standard deviation gives the **standardized coefficients** $\hat{\beta}_j^* = \hat{\beta}_j s_{x_j}/s_y$: the difference in the response, in standard deviations, per standard deviation of x_j . They make coefficients unit-free and are


```

beta_pct = np.linalg.solve(X_pct.T @ X_pct, X_pct.T @ y)
print("rates as fractions:", np.round(beta_hat, 4))
print("rates in percent:  ", np.round(beta_pct, 5))
print("same fitted values:", np.allclose(X @ beta_hat, X_pct @ beta_pct))

sd = X[:, 1:].std(axis=0, ddof=1)
standardized = beta_hat[1:] * sd / y.std(ddof=1)
print("standardized slopes:", np.round(standardized, 3))

```

5.7.4 Terms that move together

If x enters through both x and x^2 , the mean response $\beta_0 + \beta_1 x + \beta_2 x^2$ changes at the rate

$$\frac{\partial E(Y | x)}{\partial x} = \beta_1 + 2\beta_2 x,$$

which depends on x . The coefficient β_1 alone is this rate at $x = 0$, and β_2 is half the rate at which the slope itself changes. Neither is “the effect of x ”. In [Example \(A curved trend and a seasonal cycle\)](#) the growth rate of carbon dioxide was 0.86 ppm per year in 1960 and 1.81 ppm per year in 2000. The coefficient $\hat{\beta}_1 = 1.3344$ is the growth rate in 1980, the year chosen as the origin of t_c , and would have been different had the origin been chosen differently.

Interactions behave the same way. In [Eq. 5.3.3](#) the slope in x_1 is $\beta_1 + \beta_3 x_2$. Reporting “the effect of x_1 ” requires choosing a value of x_2 , and β_1 is the choice $x_2 = 0$. Centring x_2 at a representative value before forming the product makes β_1 the slope at that value. The fitted surface does not change ([Exercise \(B2\)](#)).

5.7.5 Indicator variables

With reference coding, the coefficient of an indicator is the difference in mean response between its category and the reference category, among profiles that agree on the other regressors. Changing the reference changes every indicator coefficient but not the fit, since the reference coding for one reference category is a reparameterization of the coding for any other.

Example. (Self-rated health, with and without adjustment)

In [Example \(Self-rated health\)](#), people reporting poor health made 3.160 more physician visits a year, on average, than those reporting excellent health. Add two regressors: a chronic-disease score (the variable `disea`) and an indicator of a physical limitation. The coefficient of poor health falls to 1.554, and that of fair health from 1.058 to 0.202. The good-health coefficient changes sign, to -0.081 . The adjusted comparison is between people with the same disease score and limitation status. Much of the difference in visits between health categories goes with differences in measured disease, and the self-rating adds less once disease is taken into account. Neither set of numbers is “the” effect of self-rated health. They answer different questions.

The data record the same people in several years, so the cases are not independent and the second-moment assumptions do not hold exactly; Chapter 33 treats such clustered data.


```

import numpy as np
import statsmodels.api as sm
rand = sm.datasets.randhie.load_pandas().data
y = rand["mdvis"].to_numpy(dtype=float)
D = rand[["hlthg", "hlthf", "hlthp"]].to_numpy(dtype=float) # good, fair, poor
X = np.column_stack([np.ones(len(y)), D]) # excellent =
    reference

Xa = np.column_stack([X, rand["disea"], rand["physlm"]])
beta_a = np.linalg.solve(Xa.T @ Xa, Xa.T @ y)
print("adjusted for chronic-disease score and physical limitation:")
print(np.round(beta_a, 4))

```

5.7.6 Omitted regressors

How does a coefficient change when other regressors are added or removed? Partition the model matrix as $\mathbf{X} = [\mathbf{X}_1, \mathbf{X}_2]$, with $\hat{\boldsymbol{\beta}} = (\hat{\boldsymbol{\beta}}_1^\top, \hat{\boldsymbol{\beta}}_2^\top)^\top$ the estimate from the full (“long”) regression, and let $\tilde{\boldsymbol{\beta}}_1 = (\mathbf{X}_1^\top \mathbf{X}_1)^{-1} \mathbf{X}_1^\top \mathbf{y}$ be the estimate from the “short” regression on $\mathbf{X}_1$ alone.

Proposition 5.7.2 — Omitted regressors

Let $\mathbf{X} = [\mathbf{X}_1, \mathbf{X}_2]$ have full rank, and let $\boldsymbol{\Pi} = (\mathbf{X}_1^\top \mathbf{X}_1)^{-1} \mathbf{X}_1^\top \mathbf{X}_2$ be the coefficient matrix from regressing each column of $\mathbf{X}_2$ on $\mathbf{X}_1$. Then:

- (a) for every data vector $\mathbf{y}$, $\tilde{\boldsymbol{\beta}}_1 = \hat{\boldsymbol{\beta}}_1 + \boldsymbol{\Pi} \hat{\boldsymbol{\beta}}_2$;
- (b) if $E(\mathbf{Y}) = \mathbf{X}_1 \boldsymbol{\beta}_1 + \mathbf{X}_2 \boldsymbol{\beta}_2$, then $E(\tilde{\boldsymbol{\beta}}_1) = \boldsymbol{\beta}_1 + \boldsymbol{\Pi} \boldsymbol{\beta}_2$.

In particular the short and long coefficients agree, for all $\mathbf{y}$, iff $\mathbf{X}_1^\top \mathbf{X}_2 = \mathbf{0}$.

Proof. (a) Write $\mathbf{y} = \mathbf{X}_1 \hat{\boldsymbol{\beta}}_1 + \mathbf{X}_2 \hat{\boldsymbol{\beta}}_2 + \hat{\boldsymbol{\varepsilon}}$, where $\mathbf{X}^\top \hat{\boldsymbol{\varepsilon}} = \mathbf{0}$ and so $\mathbf{X}_1^\top \hat{\boldsymbol{\varepsilon}} = \mathbf{0}$. Then

$$\begin{aligned} \tilde{\boldsymbol{\beta}}_1 &= (\mathbf{X}_1^\top \mathbf{X}_1)^{-1} \mathbf{X}_1^\top \mathbf{y} \\ &= \hat{\boldsymbol{\beta}}_1 + (\mathbf{X}_1^\top \mathbf{X}_1)^{-1} \mathbf{X}_1^\top \mathbf{X}_2 \hat{\boldsymbol{\beta}}_2 + (\mathbf{X}_1^\top \mathbf{X}_1)^{-1} \mathbf{X}_1^\top \hat{\boldsymbol{\varepsilon}} = \hat{\boldsymbol{\beta}}_1 + \boldsymbol{\Pi} \hat{\boldsymbol{\beta}}_2. \end{aligned}$$

(b) Take expectations in (a), using $E(\hat{\boldsymbol{\beta}}) = \boldsymbol{\beta}$ (Theorem 5.5.1), or apply Proposition 5.5.7(a) with $\boldsymbol{\delta} = \mathbf{X}_2 \boldsymbol{\beta}_2$. For the last claim, if $\mathbf{X}_1^\top \mathbf{X}_2 = \mathbf{0}$ then $\boldsymbol{\Pi} = \mathbf{0}$. Conversely, if $\boldsymbol{\Pi} \neq \mathbf{0}$, take $\mathbf{y} = \mathbf{X}_2 \mathbf{c}$ with $\boldsymbol{\Pi} \mathbf{c} \neq \mathbf{0}$: the long regression gives $\hat{\boldsymbol{\beta}}_2 = \mathbf{c}$, and the two coefficients differ by $\boldsymbol{\Pi} \mathbf{c}$. ■

Part (b) is the **omitted-variable bias** formula. The short regression does not estimate $\boldsymbol{\beta}_1$ but $\boldsymbol{\beta}_1 + \boldsymbol{\Pi} \boldsymbol{\beta}_2$. Each omitted regressor contributes its own coefficient times its association with the included ones. The bias is zero if the omitted regressors have no effect or are uncorrelated with the included ones, in the sense $\mathbf{X}_1^\top \mathbf{X}_2 = \mathbf{0}$; otherwise it is there, and it can have either sign. Part (a) says the same thing about the numbers actually computed from a data set, with no model assumptions at all.

Example. (Income without interest rates)

Regressing log money on log income alone, with an intercept, gives an income elasticity of 1.819, against 1.2958 when both interest rates are included. [Proposition 5.7.2\(a\)](#) accounts for the whole difference. Regressing each rate on log income gives slopes -0.2215 (bond rate) and -0.0917 (deposit rate): in these data, quarters with higher income had lower interest rates, and the correlation of log income with the bond rate is -0.52 . The short-regression elasticity therefore absorbs

$$(-0.2215)(-2.6163) + (-0.0917)(0.6186) = 0.5229,$$

mostly the association of higher income with lower bond rates, which in turn go with more money. Which elasticity is the “right” one depends on the question. The long regression compares quarters with equal interest rates. The short one compares quarters as they came, with whatever rates accompanied a given income.

```
X1 = X[:, :2]                                # intercept and log income
X2 = X[:, 2:]                                # the two interest rates
b_short = np.linalg.solve(X1.T @ X1, X1.T @ y)
A = np.linalg.solve(X1.T @ X1, X1.T @ X2)    # regress each rate on (1, lry)
print("short regression slope on income:", round(b_short[1], 4))
print("long slope + A @ (rate coefs): ", round(beta_hat[1] + A[1] @ beta_hat[2:],
4))
print("auxiliary slopes of the rates on income:", np.round(A[1], 4))
```

Warning (Adjustment is not a cure). [Proposition 5.7.2](#) suggests a tempting strategy: include every variable that might be relevant, so that nothing is omitted. Two problems remain. First, some variables should *not* be held fixed. If a treatment works partly by changing an intermediate variable, adjusting for that variable removes part of the effect one wants to measure. Adjusting for a common consequence of the regressor and the response can create an association where there is none. Second, adding regressors that are nearly collinear with x_j inflates the variance of $\hat{\beta}_j$ (Chapter 26). Which variables to adjust for is a question about how the data were generated, not about the fit. Chapter 25 develops the tools for answering it.

5.7.7 Exercises

5.7.7.1 A. Check your understanding

Exercise A1.

In [Example \(Forecasting December sea temperature\)](#), suppose both temperatures are converted to degrees Fahrenheit, $F = 32 + 1.8C$. What happens to the slope, the intercept, R^2 and s ? Answer using [Proposition 5.7.1](#), noting that the response is also transformed.

5.7.7.2 B. Practice

Exercise B1.

Show that the standardized coefficients are the slopes obtained by regressing the standardized response $(y_i - \bar{y})/s_y$ on the standardized regressors $(x_{ij} - \bar{x}_j)/s_{x_j}$, without an intercept. For a single regressor, show that the standardized slope is the correlation r .

Solution. Let $\mathbf{X}_c$ be the centred regressors and $\mathbf{D} = \text{diag}(s_{x_1}, \dots, s_{x_k})$. The standardized regressors are $\mathbf{X}_c \mathbf{D}^{-1}$ and the standardized response is $(\mathbf{y} - \bar{y}\mathbf{1})/s_y$. By [Proposition 5.4.3](#) the slopes of $\mathbf{y}$ on $\mathbf{X}_c$ are $\hat{\boldsymbol{\beta}}_1$. Dividing the response by s_y divides them by s_y , since least squares is linear in the response; replacing $\mathbf{X}_c$ by $\mathbf{X}_c \mathbf{D}^{-1}$ multiplies them by $\mathbf{D}$ ([Proposition 5.7.1](#) with $\mathbf{K} = \mathbf{D}^{-1}$). The result is $\mathbf{D}\hat{\boldsymbol{\beta}}_1/s_y$, whose entries are $\hat{\beta}_j s_{x_j}/s_y$. For one regressor this is $(S_{xy}/S_{xx})\sqrt{S_{xx}/S_{yy}} = r$, as in [Eq. 5.2.6](#).

Exercise B2.

In the centred parameterization $E(\mathbf{Y}) = \alpha\mathbf{1} + \mathbf{X}_c\boldsymbol{\beta}_1$ of [Proposition 5.4.3](#), show that $\text{Cov}(\hat{\alpha}, \hat{\boldsymbol{\beta}}_1) = \mathbf{0}$, $\text{Var}(\hat{\alpha}) = \sigma^2/n$ and $\text{Cov}(\hat{\boldsymbol{\beta}}_1) = \sigma^2(\mathbf{X}_c^\top \mathbf{X}_c)^{-1}$. Deduce that $\text{Var}(\hat{\beta}_0) = \sigma^2(1/n + \bar{\mathbf{x}}^\top (\mathbf{X}_c^\top \mathbf{X}_c)^{-1} \bar{\mathbf{x}})$, which generalizes [Corollary 5.5.4](#).

Solution. The matrix $[\mathbf{1}, \mathbf{X}_c]$ has full rank and $[\mathbf{1}, \mathbf{X}_c]^\top [\mathbf{1}, \mathbf{X}_c] = \text{diag}(n, \mathbf{X}_c^\top \mathbf{X}_c)$, because $\mathbf{1}^\top \mathbf{X}_c = \mathbf{0}^\top$. By [Theorem 5.5.1](#) the covariance of $(\hat{\alpha}, \hat{\boldsymbol{\beta}}_1)^\top$ is $\sigma^2 \text{diag}(1/n, (\mathbf{X}_c^\top \mathbf{X}_c)^{-1})$, which gives all three claims. Since $\hat{\beta}_0 = \hat{\alpha} - \bar{\mathbf{x}}^\top \hat{\boldsymbol{\beta}}_1$, [Corollary 5.5.2](#) gives $\text{Var}(\hat{\beta}_0) = \sigma^2/n + \bar{\mathbf{x}}^\top \sigma^2(\mathbf{X}_c^\top \mathbf{X}_c)^{-1} \bar{\mathbf{x}}$, the cross term being zero. With one regressor, $\mathbf{X}_c^\top \mathbf{X}_c = S_{xx}$.

Exercise B3.

In a model with an intercept and two regressors, let the short regression omit x_2 . Show that the bias of the slope on x_1 is $\beta_2 S_{12}/S_{11}$, where $S_{12} = \sum_i (x_{i1} - \bar{x}_1)(x_{i2} - \bar{x}_2)$ and $S_{11} = \sum_i (x_{i1} - \bar{x}_1)^2$. Tabulate the sign of the bias according to the signs of β_2 and the correlation of the two regressors.

5.7.7.3 C. Going deeper**Exercise C1.**

Construct a data set with $n = 8$ cases, two regressors and an intercept, in which the slope of y on x_1 is positive in the simple regression and negative in the regression on both x_1 and x_2 . Explain the reversal with [Proposition 5.7.2](#), and draw the scatterplot of y against x_1 with the cases labelled by x_2 .

5.8 A panorama of regression models to come

The linear model of this chapter makes three kinds of claim: about the *mean* of the response (a linear combination of known columns), about its *variability* (a common variance, no correlation) and, in the normal version, about its *distribution*. Every model in the rest of the book relaxes

one or more of these claims, or keeps them and asks harder questions about them. This section surveys the route.

5.8.1 Three components of a regression model

It helps to see any regression model as three separate choices.

1. **A distribution for the response given the regressors:** normal, binomial, Poisson, gamma, or something less specified, such as “any distribution with this mean and variance”.
2. **A predictor** that combines the regressors: the linear predictor $\eta_i = \mathbf{x}_{(i)}^\top \boldsymbol{\beta}$, or a sum of smooth functions, or a sum that includes random terms.
3. **A link** between the predictor and the mean, $g(E(Y_i)) = \eta_i$, where g is a known monotone function.

The normal linear model chooses the normal distribution with constant variance, the linear predictor, and the identity link $g(\mu) = \mu$. Changing one choice at a time generates the models below.

5.8.2 The ladder

Model	What changes	Where
Linear model, second moments	the starting point: geometry, optimality, estimability, computation	Chapters 5–10
Normal linear model	exact tests, intervals, multiple comparisons, designed experiments	Chapters 11–18
Departures from the assumptions	diagnostics, transformations, resampling, measurement error, causation	Chapters 19–25
Many regressors	collinearity, shrinkage, $p > n$, selection, boosting	Chapters 26–30
General covariance and random effects	$\text{Cov}(\boldsymbol{\epsilon}) = \sigma^2 \mathbf{V}$; $\mathbf{Y} = \mathbf{X}\boldsymbol{\beta} + \mathbf{Z}\mathbf{b} + \boldsymbol{\epsilon}$	Chapters 31–33
Generalized linear models	non-normal distribution and a link	Chapters 34–41
Nonparametric and additive models	smooth functions in the predictor	Chapters 42–44
Distributional regression	regressors act on more than the mean	Chapter 45

The normal linear model and its designs. With normal errors, the least squares estimator has an exact normal distribution and the residual sum of squares an exact chi-squared distribution ([Theorem 7.5.1](#)). This gives t and F tests and exact confidence regions (Chapters 11 and 12), and procedures for many comparisons at once ([Chapter 13](#)). Analysis of variance and designed experiments (Chapters 15–18) are linear models whose columns are indicator variables, often not of full rank; their theory rests on [Chapter 6](#) and [Chapter 8](#).

When the assumptions fail, or there are many regressors. Part V asks what happens when the mean is misspecified, the variance is not constant, the errors are correlated or not normal, the regressors are measured with error, or the question is causal; some remedies, such as robust standard errors based on [Eq. 5.5.3](#) (Chapter 21), stay inside the linear model. Part VI treats large p and nearly dependent columns, where ridge regression, principal components and the lasso trade a little bias for less variance. Regression trees, random forests and neural networks also estimate m , judged by prediction error rather than by the meaning of coefficients; the book does not develop them, but Chapters 29 and 30 compare predictors and treat boosting.

General covariance and mixed models. The general Gauss–Markov model replaces $\sigma^2\mathbf{I}$ by $\sigma^2\mathbf{V}$ (Chapter 31), and generalized least squares is least squares in a different inner product ([Section 6.9](#)). Linear mixed models write the response as

$$\mathbf{Y} = \mathbf{X}\boldsymbol{\beta} + \mathbf{Z}\mathbf{b} + \boldsymbol{\varepsilon},$$

where $\mathbf{b}$ is a vector of random effects shared by related cases, such as repeated measurements on one person or pupils in one school (Chapters 32–33). The RAND data of [Example \(Self-rated health\)](#), with the same people observed in several years, are of this kind.

Generalized linear models. For a binary response the mean is a probability, and a linear predictor can leave $[0, 1]$. For a count the mean must be positive and the variance grows with the mean. Generalized linear models keep the linear predictor, choose a distribution from an exponential family, and connect the two by a link: the logit $\log\{\mu/(1 - \mu)\}$ for binary responses (Chapter 35), the logarithm for counts (Chapter 37). Coefficients act additively on the scale of the link, so in a Poisson model with log link e^{β_j} is the factor by which the mean changes per unit of x_j . Chapters 34–41 develop the theory, overdispersion, correlated responses and missing data.

Nonparametric and additive models. Polynomials, as in [Example \(A curved trend and a seasonal cycle\)](#), force a global shape on a curve. Splines and smoothers let the data choose the shape locally (Chapters 42–43), and additive models replace each term $\beta_j x_j$ by an unknown smooth function $f_j(x_j)$ (Chapter 44). Much of this is still penalized least squares on a model matrix of basis functions.

Beyond the mean. Quantile regression models a chosen quantile of Y , such as the median or the tenth percentile, as a linear function of the regressors, and so describes how the whole conditional distribution shifts. Distributional regression lets the regressors act on the location, the scale and the shape of the response distribution through separate predictors, turning process (b) of [Figure 5.1.1](#) into a model rather than a violation (Chapter 45).

The Bayesian thread. A prior distribution on $(\boldsymbol{\beta}, \sigma^2)$ turns the likelihood into a posterior. For the normal linear model with a conjugate prior the posterior is available in closed form ([Theorem 7.6.2](#)), and it reappears for intervals ([Theorem 12.6.2](#)) and for generalized linear models (Chapter 39).

5.8.3 A first look beyond the linear model

Example. (Physician visits are counts)

The number of physician visits in the RAND data is a count. It is zero for 31.2 percent of person-years, and its variance is 7.09 times its mean. Group the person-years into ten groups by the chronic-disease score, and compute the mean and variance of visits in each ([Figure 5.8.1](#)). The variance rises with the mean, and in every group it is between

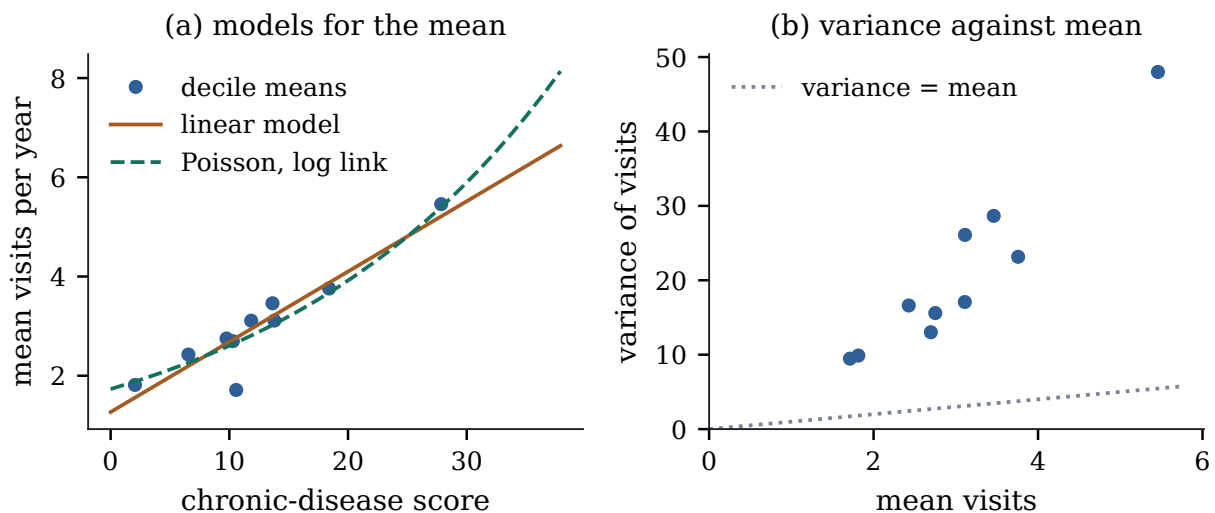


Figure 5.8.1: Annual physician visits in the RAND Health Insurance Experiment, grouped into ten groups by a chronic-disease score. (a) Group means with a linear model and a Poisson regression with log link. (b) Group variances against group means; the dotted line is the Poisson relation variance = mean.

4.8 and 8.8 times the mean. (L2) fails badly.

Two models for the mean are fitted to the disease score x . The linear model gives $1.268 + 0.1416x$ visits. The Poisson regression with log link (Chapter 37) gives $\exp(0.549 + 0.0408x)$: each unit of the index multiplies the mean by $e^{0.0408}$, 4.17 percent more visits. Over the bulk of the data the two curves are close, and both follow the group means. They differ in shape at high values of the index, where the log-linear mean grows ever faster and the linear one at a constant rate, and where the data are too sparse to decide between them. They differ more in what they assume about variability. The linear model assumes a constant variance; the Poisson model assumes variance equal to the mean. The data contradict both. Chapter 38 shows how to keep the log-linear mean and let the variance be a multiple of it.

The Poisson fit is not computed by ordinary least squares, but its estimating equations have the same form as the normal equations: $\mathbf{X}^\top(\mathbf{y} - \hat{\boldsymbol{\mu}}) = \mathbf{0}$, where $\hat{\boldsymbol{\mu}}$ is the vector of fitted means. The residuals are again orthogonal to every column of $\mathbf{X}$. The listing checks this.

```
import numpy as np
import pandas as pd
import statsmodels.api as sm
rand = sm.datasets.randhie.load_pandas().data
y = rand["mdvis"].to_numpy(dtype=float)
x = rand["disea"].to_numpy()
X = sm.add_constant(x)

bins = pd.qcut(x, 10, duplicates="drop")          # ten groups by disease score
groups = pd.DataFrame({"y": y, "x": x}).groupby(bins, observed=True)
table = groups.agg(x=("x", "mean"), mean=("y", "mean"), var=("y", "var"))
print(table.round(2).to_string(index=False))
print(f"share of zeros {np.mean(y == 0):.3f}; variance/mean overall {y.var() / y.mean():.2f}")

linear = sm.OLS(y, X).fit()
```

```
poisson = sm.GLM(y, X, family=sm.families.Poisson()).fit()
print("linear model:      ", np.round(linear.params, 4))
print("Poisson, log link:", np.round(poisson.params, 4))
mu = poisson.fittedvalues
print("X^T (y - mu_hat) =", X.T @ (y - mu))
```

5.8.4 Exercises

5.8.4.1 A. Check your understanding

Exercise A1.

For each response, name the rung of the ladder you would start from, and say which of the three components (distribution, predictor, link) departs from the normal linear model: (a) whether a loan is repaid; (b) blood pressure measured monthly on the same patients; (c) the number of insurance claims per policy; (d) the 90th percentile of birth weight as a function of the mother's age; (e) crop yield in a randomized block experiment.

5.8.4.2 B. Practice

Exercise B1.

Let Y_i be binary with $\Pr(Y_i = 1) = \mathbf{x}_{(i)}^\top \boldsymbol{\beta}$ (the *linear probability model*). Show that (L1) holds but (L2) fails, with $\text{Var}(Y_i) = \mu_i(1 - \mu_i)$. Show that least squares is still unbiased, find its covariance matrix from [Proposition 5.5.7\(b\)](#), and explain why fitted values outside $[0, 1]$ are possible.

Solution. $E(Y_i) = \Pr(Y_i = 1) = \mu_i = \mathbf{x}_{(i)}^\top \boldsymbol{\beta}$, which is (L1). $\text{Var}(Y_i) = \mu_i - \mu_i^2$, which varies with i unless all the μ_i are equal or symmetric about $1/2$, so (L2) fails. Unbiasedness needs only (L1) ([Theorem 5.5.1\(a\)](#)). With independent cases, $\boldsymbol{\Sigma} = \text{diag}(\mu_i(1 - \mu_i))$ and [Eq. 5.5.3](#) gives $\text{Cov}(\hat{\boldsymbol{\beta}}) = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \boldsymbol{\Sigma} \mathbf{X} (\mathbf{X}^\top \mathbf{X})^{-1}$. Nothing in least squares constrains $\mathbf{x}^\top \hat{\boldsymbol{\beta}}$ to $[0, 1]$: the estimate is a linear function of $\mathbf{y}$, and at design points far from the centre of the data the fitted line can cross 0 or 1 even when every true probability is inside. The logit link of Chapter 35 removes the problem.

Exercise B2.

For independent $Y_i \sim \text{Poisson}(\mu_i)$ with $\log \mu_i = \mathbf{x}_{(i)}^\top \boldsymbol{\beta}$, write the log-likelihood and show that its gradient is $\mathbf{X}^\top (\mathbf{y} - \boldsymbol{\mu})$. Deduce that when $\mathbf{X}$ contains an intercept the fitted means sum to $\sum_i y_i$, and that for the intercept-only model the fitted mean is $\bar{y}$.

5.8.4.3 C. Going deeper

Exercise C1.

In process (b) of [Example \(Same regression function, different models\)](#) the error standard deviation is $0.15 + 0.25x$. Suppose instead $\log \text{Var}(Y_i) = \gamma_0 + \gamma_1 x_i$ with unknown γ .

Propose a two-step procedure: fit the mean by least squares, then regress $\log \hat{\varepsilon}_i^2$ on x_i . Show that, when the errors are normal and $n \rightarrow \infty$ with p fixed, the slope of the second regression estimates γ_1 consistently, while its intercept estimates $\gamma_0 + E(\log \chi_1^2)$. (Argue first with the errors ε_i in place of the residuals, then explain why the difference does not matter in the limit.) Compute this bias, $-\gamma_E - \log 2 \approx -1.27$, where γ_E is Euler's constant. (Chapter 45 treats models of this kind by likelihood.)

5.9 Summary and Notes

5.9.1 Summary

Idea.

1. The regression function $m(\mathbf{x}) = E(Y \mid \mathbf{X} = \mathbf{x})$ is the best predictor in mean squared error, and its error is automatically uncorrelated with every function of the regressors (Proposition 5.1.1). A regression model is a claim about m and about the distribution around it, made in layers: linearity, constant variance, uncorrelated errors, normality (Section 5.1).
2. The least squares line has slope S_{xy}/S_{xx} and passes through the point of means (Theorem 5.2.1, Proposition 5.2.2). In standard units its slope is the correlation, which is regression toward the mean (Eq. 5.2.6).
3. The linear model $\mathbf{Y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}$, $E(\boldsymbol{\varepsilon}) = \mathbf{0}$, $\text{Cov}(\boldsymbol{\varepsilon}) = \sigma^2\mathbf{I}$, is linear in $\boldsymbol{\beta}$, not in the regressors. Polynomials, harmonics, transformations, indicator variables and interactions are all linear models (Definition 5.3.1).
4. For a full-rank model matrix, least squares has the unique solution $\hat{\boldsymbol{\beta}} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{y}$ of the normal equations $\mathbf{X}^\top \mathbf{X} \mathbf{b} = \mathbf{X}^\top \mathbf{y}$, and the criterion exceeds its minimum by $(\mathbf{b} - \hat{\boldsymbol{\beta}})^\top \mathbf{X}^\top \mathbf{X} (\mathbf{b} - \hat{\boldsymbol{\beta}})$ (Theorem 5.4.1). The residuals are orthogonal to every column (Proposition 5.4.2). Slopes can be computed from centred data (Proposition 5.4.3), and indicator coefficients are differences of group means (Proposition 5.4.4).
5. Under the second-moment assumptions, $\hat{\boldsymbol{\beta}}$ is unbiased with covariance $\sigma^2(\mathbf{X}^\top \mathbf{X})^{-1}$, and so is every linear function $\mathbf{a}^\top \hat{\boldsymbol{\beta}}$ (Theorem 5.5.1, Corollary 5.5.2). Unbiasedness needs only the mean assumption; a wrong mean produces bias and a wrong covariance produces the sandwich formula (Proposition 5.5.7).
6. $s^2 = \text{SSE}/(n - p)$ is unbiased for σ^2 , because the residuals are the errors projected into a space with $n - p$ degrees of freedom (Theorem 5.6.1). Its variance depends on the kurtosis of the errors (Proposition 5.6.3).
7. A coefficient is the difference in mean response per unit of its regressor between profiles that agree on the *other regressors in the model* (Definition 5.7.1). It changes with the units (Proposition 5.7.1) and with the set of other regressors, by exactly the amount given by the omitted-variable formula (Proposition 5.7.2). It is not by itself a causal effect.
8. Everything later in the book changes one of three components of this model: the distribution of the response, the form of the predictor, or the link between predictor and mean (Section 5.8).

5.9.2 Notes and sources

History. Least squares was published by Legendre (1805) as a method for fitting orbits, and claimed by Gauss (1809), who gave it a probabilistic justification through the normal distribution. Half a century earlier Boscovich had fitted a line to measurements of meridian arcs by making the absolute deviations as small as possible, an idea Laplace took up and that returns in this book as quantile regression. The priority dispute and the early history are told by Stigler (1981; 1986). The word *regression* comes from Galton (1886), whose study of the heights of parents and children found the phenomenon now called regression toward the mean. For Galton the slope of a fitted line measured an inherited tendency. Only later did the least squares machinery of the astronomers and the regression of the biometricians become one subject.

The source books. The chapter covers the material of Rencher and Schaalje (2008, chapter 1, chapter 6 and sections 7.1–7.7) and of Sen and Srivastava (1990, chapters 1 and 2), whose presentations of simple and multiple regression by calculus and matrix algebra are the closest to the one here. Sen and Srivastava’s emphasis on the Gauss–Markov conditions as a benchmark, and their discussion of centring and scaling, are reflected in [Section 5.1](#) and [Section 5.7](#). Rencher and Schaalje also treat the geometry, the normal model and R^2 in their chapter 7; in this book those topics are in [Chapter 6](#), [Chapter 7](#) and [Chapter 9](#). The view of a model as a distribution, a linear predictor and a link, and the discussion of indicator variables and model matrices, follow Agresti (2015, chapter 1). The panorama of [Section 5.8](#) is organized like the survey in Fahrmeir, Kneib, Lang and Marx (2021, chapter 2), whose chapter 3 also discusses what the standard assumptions mean in applications and how covariates are coded. Seber and Lee (2003, chapter 3) and Christensen (2020, chapters 1–2) give terser, more general treatments that start from projections.

Fixed and random regressors. Conditioning on the regressors is standard in linear-model theory. Freedman (2009) is a careful discussion of what regression models assume and of the gap between association and causation that [Section 5.1](#) and [Section 5.7](#) point to.

Standard errors under departures. The sandwich formula [Eq. 5.5.3](#), with the unknown covariance matrix estimated from squared residuals, is due to White (1980) in econometrics; it is developed in [Chapter 21](#).

Beyond the linear model. Generalized linear models were introduced by Nelder and Wedderburn (1972) and quantile regression by Koenker and Bassett (1978). Additive models were developed and popularized by Hastie and Tibshirani (1986; 1990), and generalized additive models for location, scale and shape were proposed by Rigby and Stasinopoulos (2005). Each is the subject of a later part of the book.

Data. The Niño 1+2 sea surface temperatures are from the US National Oceanic and Atmospheric Administration. The carbon dioxide record is that of Keeling and Whorf (2004). The Danish money demand data are those analysed by Johansen and Juselius (1990). The RAND Health Insurance Experiment is described by Manning et al. (1987); the extract used here is the one analysed by Cameron and Trivedi (2005). All four data sets are distributed with the Python package statsmodels.

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Chapter 6

Projections: The Geometry of Least Squares

Least squares chooses the point of a subspace that is closest to the data. Everything else in this chapter follows from that one statement: the normal equations, why fitted values are unique when coefficients are not, the orthogonal decompositions behind every sum of squares, the Frisch–Waugh–Lovell theorem, R^2 as a squared cosine, leverage, generalized least squares, and how the projection should actually be computed.

What you need. [Chapter 1](#) (rank, inverses, eigenvalues), [Chapter 5](#) (the linear model and least squares). Random vectors ([Chapter 2](#)) appear only in a few remarks.

6.0.1 Roadmap

- [Least squares as a nearest-point problem](#)
- [The subspaces we need](#)
- [Orthogonal projection](#)
- [Least squares via projection](#)
- [Nested models and sequential projection](#)
- [The Frisch–Waugh–Lovell theorem](#)
- [Reparameterization, centring and \$R^2\$](#)
- [Leverage as geometry](#)
- [Other inner products](#)
- [Computation](#)
- [Projection in the population: best linear prediction](#)
- [Summary and notes](#)

6.1 Least squares as a nearest-point problem

[Chapter 5](#) introduced least squares as a calculus problem: write down the sum of squared errors as a function of the coefficients, differentiate, set the derivative to zero. That route gets the right answer, but it hides the structure of the answer. It treats the coefficient vector $\boldsymbol{\beta}$ as the basic unknown. It needs $\mathbf{X}^\top \mathbf{X}$ to be invertible before it can say anything. And it gives no hint

of why some quantities, such as fitted values, residuals and sums of squares, behave well in every model, while others, such as individual coefficients, can be fragile or not defined at all.

This chapter starts from a different place. Instead of the p -dimensional space where β lives, it works in the n -dimensional space where the data live.

6.1.1 Two pictures of the same data

A regression data set with n observations and a single regressor is usually drawn as a scatterplot: n points in the plane, one for each observation, with the regressor on one axis and the response on the other. Call this the **variable space** picture. It is the right picture for seeing the shape of a relationship, but it scales badly. With k regressors it needs $k + 1$ dimensions, and it has no natural place for the model's coefficients.

The other picture turns this around. Stack the n responses into a single vector $\mathbf{y} = (y_1, \dots, y_n)^\top$, and regard it as *one* point in $\mathbb{R}^n$. Each column of the model matrix $\mathbf{X}$ is also a vector in $\mathbb{R}^n$: the intercept column is $\mathbf{1}$, and the column for a regressor holds its n observed values. This is the **observation space** picture. Its dimension is the sample size, which is usually far more than three. But it contains only $p + 1$ vectors of interest, and those always span a subspace of dimension at most $p + 1$. Most of the geometry happens in a subspace small enough to draw.

In observation space, the linear model

$$\mathbf{E}(\mathbf{Y}) = \mathbf{X}\beta, \quad \beta \in \mathbb{R}^p,$$

says one thing: the mean vector $\mathbf{E}(\mathbf{Y})$ lies in the set

$$\mathcal{C}(\mathbf{X}) = \{\mathbf{X}\mathbf{b} : \mathbf{b} \in \mathbb{R}^p\},$$

the **column space** of $\mathbf{X}$. That set is a subspace of $\mathbb{R}^n$. The coefficients β are the coordinates of $\mathbf{E}(\mathbf{Y})$ with respect to the columns of $\mathbf{X}$, but the claim itself is only about *where* $\mathbf{E}(\mathbf{Y})$ is. It does not depend on which spanning vectors are used to describe the subspace.

6.1.2 The least squares principle, restated

The observed $\mathbf{y}$ is almost never in $\mathcal{C}(\mathbf{X})$, because noise moves it off the subspace. A natural estimate of $\mathbf{E}(\mathbf{Y})$ is the point of $\mathcal{C}(\mathbf{X})$ that is closest to $\mathbf{y}$. With Euclidean distance, this is exactly least squares:

$$\|\mathbf{y} - \mathbf{X}\mathbf{b}\|^2 = \sum_{i=1}^n (y_i - \mathbf{x}_{(i)}^\top \mathbf{b})^2,$$

where $\mathbf{x}_{(i)}^\top$ is the i th row of $\mathbf{X}$. Minimizing over $\mathbf{b}$ is the same as searching $\mathcal{C}(\mathbf{X})$ for the point nearest to $\mathbf{y}$.

Definition 6.1.1 — Least squares

A vector $\hat{\beta} \in \mathbb{R}^p$ is a **least squares estimate** of β if

$$\|\mathbf{y} - \mathbf{X}\hat{\beta}\| \leq \|\mathbf{y} - \mathbf{X}\mathbf{b}\| \quad \text{for every } \mathbf{b} \in \mathbb{R}^p.$$

The vector $\hat{\mathbf{Y}} = \mathbf{X}\hat{\boldsymbol{\beta}}$ is the **fitted value** vector, and $\hat{\boldsymbol{\varepsilon}} = \mathbf{y} - \hat{\mathbf{Y}}$ is the **residual** vector.

The definition does not assume that the minimizer is unique, and deliberately so. In [Section 6.4](#) we will see that the *nearest point* $\hat{\mathbf{Y}}$ is always unique. The coefficient vector $\hat{\boldsymbol{\beta}}$ that produces it is unique only when the columns of $\mathbf{X}$ are linearly independent.

6.1.3 A picture with three observations

The smallest case that still shows all the geometry has $n = 3$ and $p = 2$. Then $\mathcal{C}(\mathbf{X})$ is a plane through the origin in $\mathbb{R}^3$, and $\mathbf{y}$ is a point somewhere off the plane.

Example. (Three observations)

Take responses $\mathbf{y} = (1, 4, 2)^\top$ at regressor values $\mathbf{x} = (0, 1, 3)^\top$, with an intercept, so that $\mathbf{X} = [\mathbf{1}, \mathbf{x}]$. The least squares fit (computed by the listing below) is

$$\hat{\mathbf{Y}} = (2.143, 2.286, 2.571)^\top, \quad \hat{\boldsymbol{\varepsilon}} = (-1.143, 1.714, -0.571)^\top,$$

with intercept 2.143 and slope 0.143. Two facts stand out. First, the residual is orthogonal to both columns of $\mathbf{X}$: the entries of $\hat{\boldsymbol{\varepsilon}}$ sum to zero, and $\mathbf{x}^\top \hat{\boldsymbol{\varepsilon}} = 0$. Second, the squared lengths add:

$$\|\mathbf{y}\|^2 = 21.000 = 16.429 + 4.571 = \|\hat{\mathbf{Y}}\|^2 + \|\hat{\boldsymbol{\varepsilon}}\|^2.$$

[Figure 6.1.1](#) shows the configuration. The fitted vector is the foot of the perpendicular dropped from $\mathbf{y}$ to the plane.

```
import numpy as np

y = np.array([1.0, 4.0, 2.0])
X = np.column_stack([np.ones(3), [0.0, 1.0, 3.0]]) # intercept and one regressor

beta_hat, *_ = np.linalg.lstsq(X, y, rcond=None)
y_hat = X @ beta_hat # the point of C(X) nearest to y
e_hat = y - y_hat # the residual vector

print("fitted ", y_hat)
print("residual", e_hat)
print("X^T e ", X.T @ e_hat) # zero: the residual is orthogonal to C(X)
```

The two facts in [Example \(Three observations\)](#) are not coincidences of this data set. They are the whole theory in miniature. The residual is orthogonal to the subspace *because* $\hat{\mathbf{Y}}$ is the nearest point, and the squared lengths add *because* the two pieces are orthogonal. The rest of the chapter makes these two statements precise and follows their consequences.

Idea. Fitted values are determined by the *subspace* $\mathcal{C}(\mathbf{X})$. Coefficients are determined by the *basis* used to describe it. Anything that depends only on $\mathcal{C}(\mathbf{X})$ is unchanged by reparameterization, by rescaling or recoding regressors, and by rank deficiency. This covers fitted values, residuals, sums of squares, R^2 and leverages.

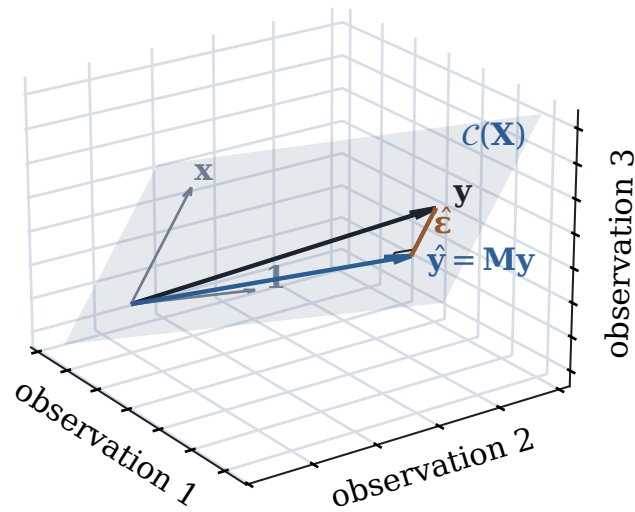


Figure 6.1.1: Least squares with $n = 3$ observations. The model's column space $\mathcal{C}(\mathbf{X})$ is the plane spanned by $\mathbf{1}$ and $\mathbf{x}$. The fitted vector $\hat{\mathbf{Y}} = \mathbf{M}\mathbf{y}$ is the point of the plane nearest to $\mathbf{y}$, and the residual $\hat{\boldsymbol{\epsilon}} = \mathbf{y} - \hat{\mathbf{Y}}$ is perpendicular to the plane.

6.1.4 Plan of the chapter

[Section 6.2](#) collects the facts about subspaces that the geometry needs. [Section 6.3](#) proves that nearest points exist and are unique, and identifies the matrices that compute them: the symmetric idempotent matrices, including $\mathbf{X}(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top$ for every generalized inverse. [Section 6.4](#) applies this to least squares. [Section 6.5](#) studies nested subspaces, which is where sums of squares come from. [Section 6.6](#) proves the Frisch–Waugh–Lovell theorem, which says what a single coefficient in a multiple regression measures. [Section 6.7](#) covers reparameterization and R^2 , and [Section 6.8](#) covers leverage. [Section 6.9](#) replaces the Euclidean inner product with a general one, the step that leads to generalized least squares. [Section 6.10](#) shows how the projection should be computed in floating point.

6.2 The subspaces we need

This section gathers the vector-space facts used in the rest of the chapter. Readers who have worked through [Chapter 1](#) can skim the definitions. The results on orthogonal complements, especially [Lemma 6.2.1](#) and [Theorem 6.2.3](#), are used constantly afterwards and are proved in full.

6.2.1 Subspaces, spans and dimension

A nonempty set $\mathcal{S} \subseteq \mathbb{R}^n$ is a **subspace** if it is closed under addition and scalar multiplication: $\mathbf{u}, \mathbf{v} \in \mathcal{S}$ and $a, b \in \mathbb{R}$ imply $a\mathbf{u} + b\mathbf{v} \in \mathcal{S}$. Every subspace contains $\mathbf{0}$. The **span** of vectors

$\mathbf{v}_1, \dots, \mathbf{v}_k$ is the set of all their linear combinations. It is the smallest subspace containing them. A set of vectors is *linearly independent* if the only linear combination equal to $\mathbf{0}$ is the one with all coefficients zero. A **basis** for $\mathcal{S}$ is a linearly independent set that spans $\mathcal{S}$. All bases of $\mathcal{S}$ have the same number of elements, the **dimension** $\dim \mathcal{S}$.

For an $n \times p$ matrix $\mathbf{A}$ there are two subspaces we use constantly:

$$\begin{aligned}\mathcal{C}(\mathbf{A}) &= \{\mathbf{A}\mathbf{b} : \mathbf{b} \in \mathbb{R}^p\} \subseteq \mathbb{R}^n && \text{(column space),} \\ \mathcal{N}(\mathbf{A}) &= \{\mathbf{b} \in \mathbb{R}^p : \mathbf{A}\mathbf{b} = \mathbf{0}\} \subseteq \mathbb{R}^p && \text{(null space).}\end{aligned}$$

The **rank** of $\mathbf{A}$ is $\text{rank}(\mathbf{A}) = \dim \mathcal{C}(\mathbf{A})$. Two standard facts from [Chapter 1](#): row rank equals column rank, so $\text{rank}(\mathbf{A}) = \text{rank}(\mathbf{A}^\top)$; and the rank-nullity theorem

$$\text{rank}(\mathbf{A}) + \dim \mathcal{N}(\mathbf{A}) = p. \quad (6.2.1)$$

In regression, $\mathcal{C}(\mathbf{X})$ is the **model space**. $\mathcal{N}(\mathbf{X})$ is the set of directions in which $\boldsymbol{\beta}$ can move without changing $\mathbf{X}\boldsymbol{\beta}$. It is $\{\mathbf{0}\}$ exactly when $\mathbf{X}$ has full column rank.

Example. (A rank-deficient layout)

Suppose n observations fall into g groups and $\mathbf{X} = [\mathbf{1}, \mathbf{z}_1, \dots, \mathbf{z}_g]$, where $\mathbf{z}_k$ is the indicator of group k . Since $\mathbf{z}_1 + \dots + \mathbf{z}_g = \mathbf{1}$, the vector $(1, -1, \dots, -1)^\top$ lies in $\mathcal{N}(\mathbf{X})$. The g indicators are linearly independent (their supports are disjoint), so $\text{rank}(\mathbf{X}) = g$ while $p = g + 1$. By [Eq. 6.2.1](#), $\mathcal{N}(\mathbf{X})$ is exactly the line spanned by $(1, -1, \dots, -1)^\top$. The model space $\mathcal{C}(\mathbf{X}) = \mathcal{C}([\mathbf{z}_1, \dots, \mathbf{z}_g])$ is the set of vectors that are constant within groups.

6.2.2 Inner products, orthogonality and complements

Unless stated otherwise $\mathbb{R}^n$ carries the Euclidean inner product $\langle \mathbf{u}, \mathbf{v} \rangle = \mathbf{u}^\top \mathbf{v}$ and norm $\|\mathbf{u}\| = (\mathbf{u}^\top \mathbf{u})^{1/2}$. Vectors are **orthogonal**, written $\mathbf{u} \perp \mathbf{v}$, if $\mathbf{u}^\top \mathbf{v} = 0$. The Cauchy–Schwarz inequality $|\mathbf{u}^\top \mathbf{v}| \leq \|\mathbf{u}\| \|\mathbf{v}\|$, with equality iff one vector is a multiple of the other, lets us define the angle $\theta \in [0, \pi]$ between nonzero vectors by

$$\cos \theta = \frac{\mathbf{u}^\top \mathbf{v}}{\|\mathbf{u}\| \|\mathbf{v}\|}. \quad (6.2.2)$$

The angle comes back in [Section 6.7](#), where R^2 turns out to be a squared cosine. The Pythagorean identity

$$\mathbf{u} \perp \mathbf{v} \implies \|\mathbf{u} + \mathbf{v}\|^2 = \|\mathbf{u}\|^2 + \|\mathbf{v}\|^2 \quad (6.2.3)$$

follows by expanding $(\mathbf{u} + \mathbf{v})^\top (\mathbf{u} + \mathbf{v})$. The cross terms are the ones that vanish.

Definition 6.2.1 — Orthogonal complement

The **orthogonal complement** of a subspace $\mathcal{S} \subseteq \mathbb{R}^n$ is

$$\mathcal{S}^\perp = \{\mathbf{v} \in \mathbb{R}^n : \mathbf{v}^\top \mathbf{u} = 0 \text{ for all } \mathbf{u} \in \mathcal{S}\}.$$

It is immediate that $\mathcal{S}^\perp$ is a subspace and that $\mathcal{S} \cap \mathcal{S}^\perp = \{\mathbf{0}\}$: a vector orthogonal to itself has zero length. To test whether $\mathbf{v} \in \mathcal{S}^\perp$ it is enough to check orthogonality to a spanning set of $\mathcal{S}$. In particular, $\mathbf{v} \in \mathcal{C}(\mathbf{A})^\perp$ iff $\mathbf{A}^\top \mathbf{v} = \mathbf{0}$.

Lemma 6.2.1

For any $n \times p$ matrix $\mathbf{A}$,

$$\mathcal{N}(\mathbf{A}^\top) = \mathcal{C}(\mathbf{A})^\perp \quad \text{and} \quad \mathcal{N}(\mathbf{A}) = \mathcal{C}(\mathbf{A}^\top)^\perp.$$

Proof. $\mathbf{v} \in \mathcal{N}(\mathbf{A}^\top)$ means $\mathbf{A}^\top \mathbf{v} = \mathbf{0}$, that is, $\mathbf{v}$ is orthogonal to every column of $\mathbf{A}$, which is the same as $\mathbf{v} \perp \mathcal{C}(\mathbf{A})$. The second identity is the first applied to $\mathbf{A}^\top$. ■

6.2.3 Orthonormal bases and Gram–Schmidt

Vectors $\mathbf{u}_1, \dots, \mathbf{u}_k$ are **orthonormal** if $\mathbf{u}_i^\top \mathbf{u}_j$ is 1 for $i = j$ and 0 otherwise. Orthonormal vectors are linearly independent: taking the inner product of $\sum_j a_j \mathbf{u}_j = \mathbf{0}$ with $\mathbf{u}_i$ gives $a_i = 0$. The coordinates of a vector $\mathbf{v} \in \text{span}(\mathbf{u}_1, \dots, \mathbf{u}_k)$ are simply the inner products $a_i = \mathbf{u}_i^\top \mathbf{v}$, with no linear system to solve. This is why orthonormal bases are so convenient.

Proposition 6.2.2 — Gram–Schmidt

Let $\mathbf{v}_1, \dots, \mathbf{v}_k$ be linearly independent. Define recursively

$$\mathbf{w}_j = \mathbf{v}_j - \sum_{i < j} (\mathbf{u}_i^\top \mathbf{v}_j) \mathbf{u}_i, \quad \mathbf{u}_j = \mathbf{w}_j / \|\mathbf{w}_j\|, \quad j = 1, \dots, k.$$

Then each $\mathbf{w}_j \neq \mathbf{0}$. The vectors $\mathbf{u}_1, \dots, \mathbf{u}_k$ are orthonormal, and $\text{span}(\mathbf{u}_1, \dots, \mathbf{u}_j) = \text{span}(\mathbf{v}_1, \dots, \mathbf{v}_j)$ for every j .

Proof. Induction on j . Suppose the claims hold for $j - 1$. The vector $\mathbf{w}_j$ is $\mathbf{v}_j$ minus an element of $\text{span}(\mathbf{u}_1, \dots, \mathbf{u}_{j-1}) = \text{span}(\mathbf{v}_1, \dots, \mathbf{v}_{j-1})$. If $\mathbf{w}_j$ were zero, $\mathbf{v}_j$ would lie in that span, contradicting independence. For $l < j$, $\mathbf{u}_l^\top \mathbf{w}_j = \mathbf{u}_l^\top \mathbf{v}_j - \sum_{i < j} (\mathbf{u}_i^\top \mathbf{v}_j) \mathbf{u}_l^\top \mathbf{u}_i = \mathbf{u}_l^\top \mathbf{v}_j - \mathbf{u}_l^\top \mathbf{v}_j = 0$, so $\mathbf{u}_j$ is orthogonal to its predecessors. Finally, $\mathbf{u}_j \in \text{span}(\mathbf{u}_1, \dots, \mathbf{u}_{j-1}, \mathbf{v}_j) = \text{span}(\mathbf{v}_1, \dots, \mathbf{v}_j)$ and $\mathbf{v}_j \in \text{span}(\mathbf{u}_1, \dots, \mathbf{u}_j)$, so the two spans agree. ■

Two consequences are used repeatedly. Every subspace has an orthonormal basis (apply Gram–Schmidt to any basis). And any orthonormal basis of a subspace $\mathcal{S}$ can be extended to an orthonormal basis of $\mathbb{R}^n$: extend it to a basis of $\mathbb{R}^n$ in any way, then apply Gram–Schmidt, which leaves the first vectors unchanged. [Section 6.10](#) shows that Gram–Schmidt, written in matrix form, is the QR factorization.

6.2.4 Direct sums and the orthogonal decomposition

For subspaces $\mathcal{S}, \mathcal{T}$, the sum $\mathcal{S} + \mathcal{T} = \{\mathbf{u} + \mathbf{v} : \mathbf{u} \in \mathcal{S}, \mathbf{v} \in \mathcal{T}\}$ is a subspace. When $\mathcal{S} \cap \mathcal{T} = \{\mathbf{0}\}$ the sum is *direct*, written $\mathcal{S} \oplus \mathcal{T}$. Every element then decomposes as $\mathbf{u} + \mathbf{v}$ in exactly one way: if $\mathbf{u} + \mathbf{v} = \mathbf{u}' + \mathbf{v}'$ then $\mathbf{u} - \mathbf{u}' = \mathbf{v}' - \mathbf{v}$ lies in both subspaces and so is zero. When in addition $\mathcal{S} \perp \mathcal{T}$, the sum is an *orthogonal* direct sum.

Theorem 6.2.3 — Orthogonal decomposition

For every subspace $\mathcal{S} \subseteq \mathbb{R}^n$,

$$\mathbb{R}^n = \mathcal{S} \oplus \mathcal{S}^\perp, \quad \dim \mathcal{S} + \dim \mathcal{S}^\perp = n, \quad (\mathcal{S}^\perp)^\perp = \mathcal{S}.$$

Hence every $\mathbf{y} \in \mathbb{R}^n$ can be written uniquely as $\mathbf{y} = \mathbf{u} + \mathbf{v}$ with $\mathbf{u} \in \mathcal{S}$ and $\mathbf{v} \perp \mathcal{S}$.

Proof. Let $\mathbf{u}_1, \dots, \mathbf{u}_r$ be an orthonormal basis of $\mathcal{S}$, extended to an orthonormal basis $\mathbf{u}_1, \dots, \mathbf{u}_n$ of $\mathbb{R}^n$. A vector $\mathbf{y} = \sum_{i=1}^n a_i \mathbf{u}_i$ is orthogonal to $\mathcal{S}$ iff $a_1 = \dots = a_r = 0$, so $\mathcal{S}^\perp = \text{span}(\mathbf{u}_{r+1}, \dots, \mathbf{u}_n)$ has dimension $n - r$. Writing $\mathbf{y} = \sum_{i \leq r} a_i \mathbf{u}_i + \sum_{i > r} a_i \mathbf{u}_i$ exhibits the decomposition, and it is unique because $\mathcal{S} \cap \mathcal{S}^\perp = \{\mathbf{0}\}$. By the same argument with the roles of the two index sets swapped, $(\mathcal{S}^\perp)^\perp = \text{span}(\mathbf{u}_1, \dots, \mathbf{u}_r) = \mathcal{S}$. ■

Combining [Theorem 6.2.3](#) with [Lemma 6.2.1](#) gives a matrix fact used in almost every later chapter.

Corollary 6.2.4

For any $n \times p$ matrix $\mathbf{A}$:

- (a) $\mathcal{N}(\mathbf{A}^\top \mathbf{A}) = \mathcal{N}(\mathbf{A})$;
- (b) $\mathcal{C}(\mathbf{A}^\top \mathbf{A}) = \mathcal{C}(\mathbf{A}^\top)$ and $\mathcal{C}(\mathbf{A} \mathbf{A}^\top) = \mathcal{C}(\mathbf{A})$;
- (c) $\text{rank}(\mathbf{A}^\top \mathbf{A}) = \text{rank}(\mathbf{A} \mathbf{A}^\top) = \text{rank}(\mathbf{A})$;
- (d) $\mathcal{C}(\mathbf{A})^\perp = \mathcal{N}(\mathbf{A}^\top)$ and $\mathbb{R}^p = \mathcal{C}(\mathbf{A}^\top) \oplus \mathcal{N}(\mathbf{A})$.

Proof.

- (a) If $\mathbf{A}\mathbf{b} = \mathbf{0}$ then $\mathbf{A}^\top \mathbf{A}\mathbf{b} = \mathbf{0}$. Conversely, $\mathbf{A}^\top \mathbf{A}\mathbf{b} = \mathbf{0}$ gives $\mathbf{b}^\top \mathbf{A}^\top \mathbf{A}\mathbf{b} = \|\mathbf{A}\mathbf{b}\|^2 = 0$, so $\mathbf{A}\mathbf{b} = \mathbf{0}$.
- (b) By [Lemma 6.2.1](#) applied to the symmetric matrix $\mathbf{A}^\top \mathbf{A}$, and then (a) and [Lemma 6.2.1](#)

again, $\mathcal{C}(\mathbf{A}^\top \mathbf{A}) = \mathcal{N}(\mathbf{A}^\top \mathbf{A})^\perp = \mathcal{N}(\mathbf{A})^\perp = (\mathcal{C}(\mathbf{A}^\top)^\perp)^\perp = \mathcal{C}(\mathbf{A}^\top)$, using $(\mathcal{S}^\perp)^\perp = \mathcal{S}$ in the last step. The second identity is the first applied to $\mathbf{A}^\top$.

- (c) Take dimensions in (b) and use $\text{rank}(\mathbf{A}^\top) = \text{rank}(\mathbf{A})$.
- (d) This is [Lemma 6.2.1](#) and [Theorem 6.2.3](#).

■

Part (b) is the reason the normal equations $\mathbf{X}^\top \mathbf{X} \mathbf{b} = \mathbf{X}^\top \mathbf{y}$ can always be solved: the right-hand side lies in $\mathcal{C}(\mathbf{X}^\top) = \mathcal{C}(\mathbf{X}^\top \mathbf{X})$. We come back to this in [Section 6.4](#).

Lemma 6.2.5

For subspaces $\mathcal{S}_1, \mathcal{S}_2 \subseteq \mathbb{R}^n$,

$$(\mathcal{S}_1 + \mathcal{S}_2)^\perp = \mathcal{S}_1^\perp \cap \mathcal{S}_2^\perp \quad \text{and} \quad (\mathcal{S}_1 \cap \mathcal{S}_2)^\perp = \mathcal{S}_1^\perp + \mathcal{S}_2^\perp.$$

Proof. A vector is orthogonal to $\mathcal{S}_1 + \mathcal{S}_2$ iff it is orthogonal to every $\mathbf{u}_1 + \mathbf{u}_2$, which (taking $\mathbf{u}_2 = \mathbf{0}$ or $\mathbf{u}_1 = \mathbf{0}$) is the case iff it is orthogonal to both subspaces. For the second identity, apply the first to $\mathcal{S}_1^\perp$ and $\mathcal{S}_2^\perp$: $(\mathcal{S}_1^\perp + \mathcal{S}_2^\perp)^\perp = \mathcal{S}_1 \cap \mathcal{S}_2$, and take complements of both sides.

■

[Lemma 6.2.5](#) is what we need to describe reduced models defined by linear constraints ([Theorem 6.5.2](#)).

6.2.5 Exercises

6.2.5.1 A. Check your understanding

Exercise A1.

Show that $\mathcal{S}^\perp$ is a subspace for any set $\mathcal{S}$ (not necessarily a subspace), and that $\mathcal{S} \subseteq \mathcal{T}$ implies $\mathcal{T}^\perp \subseteq \mathcal{S}^\perp$. Show that $\mathcal{S}^\perp = \text{span}(\mathcal{S})^\perp$.

6.2.5.2 B. Practice

Exercise B1.

A one-way layout has three groups of sizes 2, 2, 1 and model matrix $\mathbf{X} = [\mathbf{1}, \mathbf{z}_1, \mathbf{z}_2, \mathbf{z}_3]$. Find bases for $\mathcal{C}(\mathbf{X})$, $\mathcal{N}(\mathbf{X})$, $\mathcal{C}(\mathbf{X}^\top)$ and $\mathcal{C}(\mathbf{X})^\perp$, and check that the dimensions agree with [Eq. 6.2.1](#) and [Theorem 6.2.3](#).

Exercise B2.

Show that $\mathcal{C}(\mathbf{AB}) \subseteq \mathcal{C}(\mathbf{A})$, and deduce $\text{rank}(\mathbf{AB}) \leq \min\{\text{rank}(\mathbf{A}), \text{rank}(\mathbf{B})\}$. Show that $\mathcal{C}(\mathbf{AB}) = \mathcal{C}(\mathbf{A})$ whenever $\mathbf{B}$ has full row rank.

6.2.5.3 C. Going deeper

Exercise C1.

For subspaces $\mathcal{S}, \mathcal{T} \subseteq \mathbb{R}^n$, prove $\dim(\mathcal{S} + \mathcal{T}) = \dim \mathcal{S} + \dim \mathcal{T} - \dim(\mathcal{S} \cap \mathcal{T})$. Use it to show that two r -dimensional subspaces of $\mathbb{R}^n$ with $2r > n$ must share a nonzero vector.

6.3 Orthogonal projection

6.3.1 Nearest points

Theorem 6.3.1 — Projection theorem

Let $\mathcal{S}$ be a subspace of $\mathbb{R}^n$ and $\mathbf{y} \in \mathbb{R}^n$. There is exactly one $\hat{\mathbf{y}} \in \mathcal{S}$ such that

$$\|\mathbf{y} - \hat{\mathbf{y}}\| \leq \|\mathbf{y} - \mathbf{u}\| \quad \text{for all } \mathbf{u} \in \mathcal{S}.$$

It is the unique $\hat{\mathbf{y}} \in \mathcal{S}$ with $\mathbf{y} - \hat{\mathbf{y}} \perp \mathcal{S}$. Moreover, for every $\mathbf{u} \in \mathcal{S}$,

$$\|\mathbf{y} - \mathbf{u}\|^2 = \|\mathbf{y} - \hat{\mathbf{y}}\|^2 + \|\hat{\mathbf{y}} - \mathbf{u}\|^2. \quad (6.3.1)$$

Proof. By Theorem 6.2.3 there is a unique decomposition $\mathbf{y} = \hat{\mathbf{y}} + \mathbf{v}$ with $\hat{\mathbf{y}} \in \mathcal{S}$ and $\mathbf{v} \in \mathcal{S}^\perp$. For $\mathbf{u} \in \mathcal{S}$, $\mathbf{y} - \mathbf{u} = \mathbf{v} + (\hat{\mathbf{y}} - \mathbf{u})$, where $\mathbf{v} \perp \mathcal{S}$ and $\hat{\mathbf{y}} - \mathbf{u} \in \mathcal{S}$. The Pythagorean identity Eq. 6.2.3 then gives Eq. 6.3.1. The right-hand side is at least $\|\mathbf{y} - \hat{\mathbf{y}}\|^2$, with equality iff $\mathbf{u} = \hat{\mathbf{y}}$. So $\hat{\mathbf{y}}$ is the unique nearest point. If some other $\mathbf{w} \in \mathcal{S}$ also had $\mathbf{y} - \mathbf{w} \perp \mathcal{S}$, then $\mathbf{y} = \mathbf{w} + (\mathbf{y} - \mathbf{w})$ would be a second decomposition, contradicting uniqueness. ■

The theorem has two halves, and it helps to keep them apart. The *variational* half says $\hat{\mathbf{y}}$ minimizes a distance. The *orthogonality* half says the error $\mathbf{y} - \hat{\mathbf{y}}$ is perpendicular to the subspace. The first half explains why least squares is a sensible estimation principle. The second is what we compute with, because orthogonality is a set of linear equations.

6.3.2 The projection as a matrix

Write $\mathbf{P}_{\mathcal{S}}\mathbf{y}$ for the nearest point $\hat{\mathbf{y}}$. The map $\mathbf{y} \mapsto \mathbf{P}_{\mathcal{S}}\mathbf{y}$ is linear: if $\mathbf{y}_k = \hat{\mathbf{y}}_k + \mathbf{v}_k$ are the decompositions of $\mathbf{y}_1$ and $\mathbf{y}_2$, then $a\mathbf{y}_1 + b\mathbf{y}_2 = (a\hat{\mathbf{y}}_1 + b\hat{\mathbf{y}}_2) + (a\mathbf{v}_1 + b\mathbf{v}_2)$ is a decomposition of the same kind, and by uniqueness it is *the* decomposition. A linear map on $\mathbb{R}^n$ is given by a unique $n \times n$ matrix, since two matrices that agree on every vector agree on the standard basis vectors and are therefore equal.

Definition 6.3.1 — Orthogonal projection matrix

The **orthogonal projection** onto a subspace $\mathcal{S}$ is the unique $n \times n$ matrix $\mathbf{P}_{\mathcal{S}}$ such that, for every $\mathbf{y}$, $\mathbf{P}_{\mathcal{S}}\mathbf{y} \in \mathcal{S}$ and $\mathbf{y} - \mathbf{P}_{\mathcal{S}}\mathbf{y} \perp \mathcal{S}$. When $\mathcal{S} = \mathcal{C}(\mathbf{X})$ for the model matrix of a linear model, we write $\mathbf{M} = \mathbf{P}_{\mathcal{C}(\mathbf{X})}$.

The most useful formula for $\mathbf{P}_{\mathcal{S}}$ uses an orthonormal basis.

Proposition 6.3.2

Let the columns of the $n \times r$ matrix $\mathbf{Q}$ be an orthonormal basis for $\mathcal{S}$, so that $\mathbf{Q}^{\top}\mathbf{Q} = \mathbf{I}_r$. Then

$$\mathbf{P}_{\mathcal{S}} = \mathbf{Q}\mathbf{Q}^{\top} = \sum_{i=1}^r \mathbf{u}_i\mathbf{u}_i^{\top},$$

where $\mathbf{u}_1, \dots, \mathbf{u}_r$ are the columns of $\mathbf{Q}$.

Proof. For any $\mathbf{y}$, $\mathbf{Q}\mathbf{Q}^{\top}\mathbf{y} = \mathbf{Q}(\mathbf{Q}^{\top}\mathbf{y}) \in \mathcal{C}(\mathbf{Q}) = \mathcal{S}$, and $\mathbf{Q}^{\top}(\mathbf{y} - \mathbf{Q}\mathbf{Q}^{\top}\mathbf{y}) = \mathbf{Q}^{\top}\mathbf{y} - \mathbf{Q}^{\top}\mathbf{y} = \mathbf{0}$, so the difference is orthogonal to every column of $\mathbf{Q}$ and hence to $\mathcal{S}$. By uniqueness, $\mathbf{Q}\mathbf{Q}^{\top} = \mathbf{P}_{\mathcal{S}}$. ■

Example. (Projections onto a line)

For $\mathcal{S} = \text{span}(\mathbf{v})$ with $\mathbf{v} \neq \mathbf{0}$, the single vector $\mathbf{v}/\|\mathbf{v}\|$ is an orthonormal basis, so

$$\mathbf{P}_{\mathcal{S}} = \frac{\mathbf{v}\mathbf{v}^{\top}}{\mathbf{v}^{\top}\mathbf{v}}, \quad \mathbf{P}_{\mathcal{S}}\mathbf{y} = \frac{\mathbf{v}^{\top}\mathbf{y}}{\mathbf{v}^{\top}\mathbf{v}} \mathbf{v}.$$

With $\mathbf{v} = \mathbf{1}$ this gives $\mathbf{P} = n^{-1}\mathbf{1}\mathbf{1}^{\top}$, the matrix with every entry $1/n$, and $\mathbf{P}\mathbf{y} = \bar{y}\mathbf{1}$. *The sample mean is a projection:* fitting the intercept-only model replaces every observation by $\bar{y}$, and the residuals $y_i - \bar{y}$ form a vector in $\mathbf{1}^{\perp}$, the set of vectors whose entries sum to zero.

6.3.3 Characterizing projection matrices

Which matrices are orthogonal projections? [Proposition 6.3.2](#) shows that every projection has the form $\mathbf{Q}\mathbf{Q}^{\top}$, so it is *symmetric*, and $\mathbf{Q}\mathbf{Q}^{\top}\mathbf{Q}\mathbf{Q}^{\top} = \mathbf{Q}\mathbf{I}\mathbf{Q}^{\top} = \mathbf{Q}\mathbf{Q}^{\top}$, so it is *idempotent*. The converse is the basic recognition theorem.

Theorem 6.3.3

An $n \times n$ matrix $\mathbf{P}$ is the orthogonal projection onto some subspace iff $\mathbf{P}^{\top} = \mathbf{P}$ and $\mathbf{P}^2 = \mathbf{P}$. In that case the subspace is $\mathcal{C}(\mathbf{P})$, and $\mathbf{I} - \mathbf{P}$ is the orthogonal projection onto $\mathcal{C}(\mathbf{P})^{\perp}$.

Proof. Necessity was shown above. For sufficiency, let $\mathbf{P}$ be symmetric and idempotent and let $\mathbf{y} \in \mathbb{R}^n$. Certainly $\mathbf{P}\mathbf{y} \in \mathcal{C}(\mathbf{P})$. For any $\mathbf{u} = \mathbf{P}\mathbf{b} \in \mathcal{C}(\mathbf{P})$,

$$\mathbf{u}^\top (\mathbf{y} - \mathbf{P}\mathbf{y}) = \mathbf{b}^\top \mathbf{P}^\top (\mathbf{I} - \mathbf{P})\mathbf{y} = \mathbf{b}^\top (\mathbf{P} - \mathbf{P}^2)\mathbf{y} = 0,$$

so $\mathbf{y} - \mathbf{P}\mathbf{y} \perp \mathcal{C}(\mathbf{P})$ and $\mathbf{P} = \mathbf{P}_{\mathcal{C}(\mathbf{P})}$. For the last claim, $\mathbf{I} - \mathbf{P}$ is symmetric and idempotent, since $(\mathbf{I} - \mathbf{P})^2 = \mathbf{I} - 2\mathbf{P} + \mathbf{P}^2 = \mathbf{I} - \mathbf{P}$. It sends $\mathbf{y}$ to $\mathbf{y} - \mathbf{P}\mathbf{y} \in \mathcal{C}(\mathbf{P})^\perp$, and $\mathbf{y} - (\mathbf{I} - \mathbf{P})\mathbf{y} = \mathbf{P}\mathbf{y} \perp \mathcal{C}(\mathbf{P})^\perp$. ■

Before this theorem, checking that a matrix computes a projection meant thinking about subspaces. Afterwards it is two matrix identities. Also, a projection is determined by its range: if $\mathbf{P}_1, \mathbf{P}_2$ are symmetric idempotent with $\mathcal{C}(\mathbf{P}_1) = \mathcal{C}(\mathbf{P}_2)$, then $\mathbf{P}_1 = \mathbf{P}_2$, because both are the projection onto the same subspace.

Proposition 6.3.4 — Spectrum, rank and trace

Let $\mathbf{P}$ be the orthogonal projection onto an r -dimensional subspace $\mathcal{S}$. Then:

- (a) every eigenvalue of $\mathbf{P}$ is 0 or 1, and 1 has multiplicity r ;
- (b) $\text{rank}(\mathbf{P}) = \text{tr}(\mathbf{P}) = r$;
- (c) $\mathbf{P}\mathbf{v} = \mathbf{v}$ iff $\mathbf{v} \in \mathcal{S}$, and $\mathbf{P}\mathbf{v} = \mathbf{0}$ iff $\mathbf{v} \in \mathcal{S}^\perp$;
- (d) $\mathbf{v}^\top \mathbf{P}\mathbf{v} = \|\mathbf{P}\mathbf{v}\|^2 \leq \|\mathbf{v}\|^2$, so $\mathbf{P}$ and $\mathbf{I} - \mathbf{P}$ are nonnegative definite.

Proof. Write $\mathbf{P} = \mathbf{Q}\mathbf{Q}^\top$ as in Proposition 6.3.2, and extend $\mathbf{Q}$ to an orthogonal matrix $\mathbf{U} = [\mathbf{Q}, \mathbf{Q}_\perp]$. Then $\mathbf{P} = \mathbf{U} \text{diag}(\mathbf{I}_r, \mathbf{0}) \mathbf{U}^\top$ is a spectral decomposition, which gives (a). Rank and trace of a symmetric matrix are the number of nonzero eigenvalues and their sum, which gives (b). Directly, $\text{tr}(\mathbf{Q}\mathbf{Q}^\top) = \text{tr}(\mathbf{Q}^\top \mathbf{Q}) = \text{tr}(\mathbf{I}_r) = r$. Part (c) restates the projection theorem: the nearest point of $\mathcal{S}$ to a vector already in $\mathcal{S}$ is the vector itself, and $\mathbf{v} \perp \mathcal{S}$ iff $\mathbf{0}$ is its nearest point. For (d), $\mathbf{v}^\top \mathbf{P}\mathbf{v} = \mathbf{v}^\top \mathbf{P}^\top \mathbf{P}\mathbf{v}$, and $\|\mathbf{v}\|^2 = \|\mathbf{P}\mathbf{v}\|^2 + \|(\mathbf{I} - \mathbf{P})\mathbf{v}\|^2$ by Pythagoras. ■

The identity $\text{rank} = \text{tr}$ gets a lot of use. The dimension of a subspace, which is the “degrees of freedom” of every sum of squares in this book, can be read off as a sum of diagonal entries without finding a basis.

6.3.4 Generalized inverses

Proposition 6.3.2 needs an orthonormal basis. In statistics the subspace usually comes as the column space of a model matrix $\mathbf{X}$ whose columns are not orthonormal and may not even be independent. To write $\mathbf{M}$ directly in terms of $\mathbf{X}$ we need a substitute for $(\mathbf{X}^\top \mathbf{X})^{-1}$ that exists even when $\mathbf{X}^\top \mathbf{X}$ is singular.

Definition 6.3.2 — Generalized inverse

A **generalized inverse** of an $m \times k$ matrix $\mathbf{A}$ is any $k \times m$ matrix $\mathbf{A}^-$ such that $\mathbf{A}\mathbf{A}^-\mathbf{A} = \mathbf{A}$.

If $\mathbf{A}$ is square and nonsingular, multiplying $\mathbf{A}\mathbf{A}^-\mathbf{A} = \mathbf{A}$ on both sides by $\mathbf{A}^{-1}$ gives $\mathbf{A}^- = \mathbf{A}^{-1}$, so the generalized inverse is unique and equal to the inverse. When $\mathbf{A}$ is singular there are infinitely many.

Proposition 6.3.5 — Existence

Every matrix has a generalized inverse. If $\mathbf{A}$ has rank r and is permuted so that

$$\mathbf{A} = \begin{pmatrix} \mathbf{A}_{11} & \mathbf{A}_{12} \\ \mathbf{A}_{21} & \mathbf{A}_{22} \end{pmatrix}$$

with $\mathbf{A}_{11}$ an $r \times r$ nonsingular block, then $\mathbf{A}^- = \begin{pmatrix} \mathbf{A}_{11}^{-1} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} \end{pmatrix}$ (permuted back) is a generalized inverse.

Proof. A rank- r matrix has r independent rows and r independent columns, and the submatrix where they cross is nonsingular (Chapter 1). This guarantees the block form. Since the first r columns are a basis for the column space, $\begin{pmatrix} \mathbf{A}_{12} \\ \mathbf{A}_{22} \end{pmatrix} = \begin{pmatrix} \mathbf{A}_{11} \\ \mathbf{A}_{21} \end{pmatrix} \mathbf{B}$ for some $\mathbf{B}$, so $\mathbf{B} = \mathbf{A}_{11}^{-1} \mathbf{A}_{12}$ and $\mathbf{A}_{22} = \mathbf{A}_{21} \mathbf{A}_{11}^{-1} \mathbf{A}_{12}$. Multiplying out,

$$\mathbf{A}\mathbf{A}^-\mathbf{A} = \begin{pmatrix} \mathbf{I} & \mathbf{0} \\ \mathbf{A}_{21}\mathbf{A}_{11}^{-1} & \mathbf{0} \end{pmatrix} \mathbf{A} = \begin{pmatrix} \mathbf{A}_{11} & \mathbf{A}_{12} \\ \mathbf{A}_{21} & \mathbf{A}_{21}\mathbf{A}_{11}^{-1}\mathbf{A}_{12} \end{pmatrix} = \mathbf{A}.$$

■

Among the many generalized inverses, one is singled out by extra symmetry conditions. The **Moore–Penrose inverse** $\mathbf{A}^+$ is the unique matrix satisfying $\mathbf{A}\mathbf{A}^+\mathbf{A} = \mathbf{A}$, $\mathbf{A}^+\mathbf{A}\mathbf{A}^+ = \mathbf{A}^+$, and both $\mathbf{A}\mathbf{A}^+$ and $\mathbf{A}^+\mathbf{A}$ symmetric (Penrose 1955). Section 6.10 computes it from the singular value decomposition. For projections we will need none of these extra conditions, and that is the point of the next result.

Lemma 6.3.6

Let $\mathbf{X}$ be $n \times p$ and let $\mathbf{G}$ be any generalized inverse of $\mathbf{X}^\top \mathbf{X}$. Then:

- (a) $\mathbf{X}\mathbf{G}\mathbf{X}^\top \mathbf{X} = \mathbf{X}$ and $\mathbf{X}^\top \mathbf{X}\mathbf{G}\mathbf{X}^\top = \mathbf{X}^\top$;
- (b) $\mathbf{X}\mathbf{G}\mathbf{X}^\top$ does not depend on the choice of generalized inverse;
- (c) $\mathbf{X}\mathbf{G}\mathbf{X}^\top$ is symmetric, even when $\mathbf{G}$ is not.

Proof. By Corollary 6.2.4(b), $\mathcal{C}(\mathbf{X}^\top) = \mathcal{C}(\mathbf{X}^\top \mathbf{X})$, so there is a $p \times n$ matrix $\mathbf{B}$ with

$\mathbf{X}^\top = \mathbf{X}^\top \mathbf{X} \mathbf{B}$. Transposing, $\mathbf{X} = \mathbf{B}^\top \mathbf{X}^\top \mathbf{X}$. Then

$$\mathbf{X} \mathbf{G} \mathbf{X}^\top \mathbf{X} = \mathbf{B}^\top (\mathbf{X}^\top \mathbf{X}) \mathbf{G} (\mathbf{X}^\top \mathbf{X}) = \mathbf{B}^\top \mathbf{X}^\top \mathbf{X} = \mathbf{X},$$

which is the first identity in (a). The second follows the same way, using $\mathbf{X}^\top = \mathbf{X}^\top \mathbf{X} \mathbf{B}$ on the right. For (b) and (c),

$$\mathbf{X} \mathbf{G} \mathbf{X}^\top = \mathbf{B}^\top (\mathbf{X}^\top \mathbf{X}) \mathbf{G} (\mathbf{X}^\top \mathbf{X}) \mathbf{B} = \mathbf{B}^\top \mathbf{X}^\top \mathbf{X} \mathbf{B},$$

which involves no $\mathbf{G}$ at all, and $(\mathbf{B}^\top \mathbf{X}^\top \mathbf{X} \mathbf{B})^\top = \mathbf{B}^\top \mathbf{X}^\top \mathbf{X} \mathbf{B}$.

■

Theorem 6.3.7

For any $n \times p$ matrix $\mathbf{X}$ and any generalized inverse $(\mathbf{X}^\top \mathbf{X})^-$,

$$\mathbf{M} = \mathbf{X}(\mathbf{X}^\top \mathbf{X})^- \mathbf{X}^\top$$

is the orthogonal projection onto $\mathcal{C}(\mathbf{X})$. In particular $\text{rank}(\mathbf{M}) = \text{tr}(\mathbf{M}) = \text{rank}(\mathbf{X})$. If $\mathbf{X}$ has full column rank, $\mathbf{M} = \mathbf{X}(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top$.

Proof. By Lemma 6.3.6, $\mathbf{M}$ is symmetric and $\mathbf{M}^2 = \mathbf{X}(\mathbf{X}^\top \mathbf{X})^- [\mathbf{X}^\top \mathbf{X}(\mathbf{X}^\top \mathbf{X})^- \mathbf{X}^\top] = \mathbf{X}(\mathbf{X}^\top \mathbf{X})^- \mathbf{X}^\top = \mathbf{M}$. By Theorem 6.3.3, $\mathbf{M}$ is the projection onto $\mathcal{C}(\mathbf{M})$. Clearly $\mathcal{C}(\mathbf{M}) \subseteq \mathcal{C}(\mathbf{X})$, and $\mathbf{M}\mathbf{X} = \mathbf{X}$ (again by the lemma) shows $\mathcal{C}(\mathbf{X}) \subseteq \mathcal{C}(\mathbf{M})$. The rank statement is Proposition 6.3.4(b).

■

It is worth pausing on how strong Theorem 6.3.7 is. The matrix $(\mathbf{X}^\top \mathbf{X})^-$ can be almost anything: symmetric or not, reflexive or not, with entries of any size. The sandwich $\mathbf{X}(\cdot)\mathbf{X}^\top$ removes all of that freedom. We will see in Section 6.4 that this is exactly the difference between coefficients and fitted values, made concrete.

Example. (The one-way layout)

In the setting of Example (A rank-deficient layout), sort the observations by group and let group k have n_k members. The model space is spanned by the orthogonal vectors $\mathbf{z}_1, \dots, \mathbf{z}_g$, so by Proposition 6.3.2 with $\mathbf{u}_k = \mathbf{z}_k / \sqrt{n_k}$,

$$\mathbf{M} = \sum_{k=1}^g \frac{\mathbf{z}_k \mathbf{z}_k^\top}{n_k} = \begin{pmatrix} n_1^{-1} \mathbf{1}_{n_1} \mathbf{1}_{n_1}^\top & & \\ & \ddots & \\ & & n_g^{-1} \mathbf{1}_{n_g} \mathbf{1}_{n_g}^\top \end{pmatrix},$$

a block-diagonal matrix of within-group averaging operators. $\mathbf{M}\mathbf{y}$ replaces each observation by its group mean. The intercept column adds nothing to $\mathcal{C}(\mathbf{X})$, so it does not appear, and $\text{tr}(\mathbf{M}) = \sum_k n_k/n_k = g = \text{rank}(\mathbf{X})$. We got here without a generalized inverse. Theorem 6.3.7 guarantees that any generalized inverse of the singular $(g+1) \times (g+1)$ matrix $\mathbf{X}^\top \mathbf{X}$ produces this same matrix. Section 6.4 checks this numerically.

6.3.5 Sums and products of projections

Complicated subspaces are often built from simpler ones, and their projections are built the same way.

Theorem 6.3.8

Let $\mathbf{P}_1, \mathbf{P}_2$ be orthogonal projections. Then $\mathbf{P}_1 + \mathbf{P}_2$ is an orthogonal projection iff $\mathbf{P}_1\mathbf{P}_2 = \mathbf{0}$, that is, iff $\mathcal{C}(\mathbf{P}_1) \perp \mathcal{C}(\mathbf{P}_2)$. In that case it projects onto $\mathcal{C}(\mathbf{P}_1) \oplus \mathcal{C}(\mathbf{P}_2)$.

Proof. $\mathbf{P}_1 + \mathbf{P}_2$ is always symmetric, and $(\mathbf{P}_1 + \mathbf{P}_2)^2 = \mathbf{P}_1 + \mathbf{P}_2 + \mathbf{P}_1\mathbf{P}_2 + \mathbf{P}_2\mathbf{P}_1$. So it is idempotent iff $\mathbf{P}_1\mathbf{P}_2 + \mathbf{P}_2\mathbf{P}_1 = \mathbf{0}$. If $\mathbf{P}_1\mathbf{P}_2 = \mathbf{0}$, then transposing gives $\mathbf{P}_2\mathbf{P}_1 = \mathbf{0}$ and the condition holds. Conversely, suppose $\mathbf{P}_1\mathbf{P}_2 + \mathbf{P}_2\mathbf{P}_1 = \mathbf{0}$. Multiplying on the left by $\mathbf{P}_1$ gives $\mathbf{P}_1\mathbf{P}_2 + \mathbf{P}_1\mathbf{P}_2\mathbf{P}_1 = \mathbf{0}$, and on the right by $\mathbf{P}_1$ gives $\mathbf{P}_1\mathbf{P}_2\mathbf{P}_1 + \mathbf{P}_2\mathbf{P}_1 = \mathbf{0}$. Subtracting, $\mathbf{P}_1\mathbf{P}_2 = \mathbf{P}_2\mathbf{P}_1$, and then $2\mathbf{P}_1\mathbf{P}_2 = \mathbf{0}$. The equivalence with orthogonality of ranges holds because $\mathbf{P}_1\mathbf{P}_2 = \mathbf{0}$ says exactly that every vector of $\mathcal{C}(\mathbf{P}_2)$ is sent to $\mathbf{0}$ by $\mathbf{P}_1$, that is (Proposition 6.3.4(c)), lies in $\mathcal{C}(\mathbf{P}_1)^\perp$. Finally, $\mathcal{C}(\mathbf{P}_1 + \mathbf{P}_2) \subseteq \mathcal{C}(\mathbf{P}_1) + \mathcal{C}(\mathbf{P}_2)$, and for $\mathbf{u} = \mathbf{P}_1\mathbf{u} + \mathbf{P}_2\mathbf{u}$ in that sum we have $(\mathbf{P}_1 + \mathbf{P}_2)\mathbf{u} = \mathbf{u}$, so the ranges agree. ■

By induction, if $\mathbf{P}_1, \dots, \mathbf{P}_k$ are projections with $\mathbf{P}_i\mathbf{P}_j = \mathbf{0}$ for $i \neq j$, their sum projects onto the orthogonal direct sum of their ranges. When the sum is $\mathbf{I}$, we get an orthogonal decomposition

$$\mathbf{y} = \mathbf{P}_1\mathbf{y} + \dots + \mathbf{P}_k\mathbf{y}, \quad \|\mathbf{y}\|^2 = \|\mathbf{P}_1\mathbf{y}\|^2 + \dots + \|\mathbf{P}_k\mathbf{y}\|^2.$$

Every analysis of variance table in this book is a table of such a decomposition.

Products behave differently. $\mathbf{P}_1\mathbf{P}_2$ need not be symmetric or idempotent, and it is generally not a projection. Exercise (B1) shows that it is one iff $\mathbf{P}_1$ and $\mathbf{P}_2$ commute, in which case it projects onto $\mathcal{C}(\mathbf{P}_1) \cap \mathcal{C}(\mathbf{P}_2)$. The most important instance is the nested case $\mathcal{C}(\mathbf{P}_2) \subseteq \mathcal{C}(\mathbf{P}_1)$, treated in Section 6.5.

6.3.6 Exercises

6.3.6.1 A. Check your understanding

Exercise A1.

Let $\mathcal{S} = \text{span}\{(1, 1, 0)^\top, (0, 1, 1)^\top\}$. Compute $\mathbf{P}_{\mathcal{S}}$ in two ways: by Gram-Schmidt and Proposition 6.3.2, and by Theorem 6.3.7. Verify symmetry, idempotence and $\text{tr } \mathbf{P}_{\mathcal{S}} = 2$. Find $\mathcal{S}^\perp$ and check that $\mathbf{I} - \mathbf{P}_{\mathcal{S}}$ projects onto it.

Exercise A2.

Show that a symmetric matrix whose eigenvalues all lie in $\{0, 1\}$ is an orthogonal projection. Give a non-symmetric 2×2 matrix with eigenvalues 0 and 1 that is not idempotent, and one that is idempotent.

Exercise A3.

Let $\mathbf{P}$ be an orthogonal projection and a, b real numbers. Show that the eigenvalues of $a\mathbf{P} + b\mathbf{I}$ are b and $a + b$. Show that when $b \neq 0$ and $a + b \neq 0$,

$$(a\mathbf{P} + b\mathbf{I})^{-1} = \frac{1}{b} \left(\mathbf{I} - \frac{a}{a+b} \mathbf{P} \right).$$

Use this to invert the equicorrelation matrix $(1 - \rho)\mathbf{I} + \rho\mathbf{1}\mathbf{1}^\top$.

Exercise A4.

Show that if $\mathbf{G}$ is a generalized inverse of the symmetric matrix $\mathbf{X}^\top \mathbf{X}$, then so is $\mathbf{G}^\top$, and so is $\mathbf{G}\mathbf{X}^\top \mathbf{X}\mathbf{G}^\top$. Which of the four Penrose conditions does the last one satisfy?

6.3.6.2 B. Practice**Exercise B1.**

Let $\mathbf{P}_1, \mathbf{P}_2$ be orthogonal projections. Show that $\mathbf{P}_1\mathbf{P}_2$ is an orthogonal projection iff $\mathbf{P}_1\mathbf{P}_2 = \mathbf{P}_2\mathbf{P}_1$, and that in this case it projects onto $\mathcal{C}(\mathbf{P}_1) \cap \mathcal{C}(\mathbf{P}_2)$. *Hint:* for the range, show $\mathbf{P}_1\mathbf{P}_2\mathbf{u} = \mathbf{u}$ iff $\mathbf{u}$ lies in both subspaces.

Solution. If $\mathbf{P}_1\mathbf{P}_2 = \mathbf{P}_2\mathbf{P}_1$, then $(\mathbf{P}_1\mathbf{P}_2)^\top = \mathbf{P}_2\mathbf{P}_1 = \mathbf{P}_1\mathbf{P}_2$, and $(\mathbf{P}_1\mathbf{P}_2)^2 = \mathbf{P}_1\mathbf{P}_2\mathbf{P}_2\mathbf{P}_1 = \mathbf{P}_1\mathbf{P}_2\mathbf{P}_1 = \mathbf{P}_1\mathbf{P}_1\mathbf{P}_2 = \mathbf{P}_1\mathbf{P}_2$. So the product is symmetric and idempotent. Conversely, if $\mathbf{P}_1\mathbf{P}_2$ is an orthogonal projection, it is symmetric, so $\mathbf{P}_1\mathbf{P}_2 = (\mathbf{P}_1\mathbf{P}_2)^\top = \mathbf{P}_2\mathbf{P}_1$. For the range: if $\mathbf{u}$ lies in both subspaces, then $\mathbf{P}_1\mathbf{P}_2\mathbf{u} = \mathbf{P}_1\mathbf{u} = \mathbf{u}$. If $\mathbf{P}_1\mathbf{P}_2\mathbf{u} = \mathbf{u}$, then $\mathbf{u} \in \mathcal{C}(\mathbf{P}_1)$, and also $\mathbf{u} = \mathbf{P}_2\mathbf{P}_1\mathbf{u} \in \mathcal{C}(\mathbf{P}_2)$. A projection's range is the set of vectors it fixes, so $\mathcal{C}(\mathbf{P}_1\mathbf{P}_2) = \mathcal{C}(\mathbf{P}_1) \cap \mathcal{C}(\mathbf{P}_2)$.

Exercise B2.

Let $\mathbf{G}_0$ be a generalized inverse of $\mathbf{A}$.

- Show that $\mathbf{G}_0 + (\mathbf{I} - \mathbf{G}_0\mathbf{A})\mathbf{U} + \mathbf{V}(\mathbf{I} - \mathbf{A}\mathbf{G}_0)$ is a generalized inverse of $\mathbf{A}$ for all conformable $\mathbf{U}, \mathbf{V}$.
- Show that every generalized inverse $\mathbf{G}$ of $\mathbf{A}$ has this form. *Hint:* take $\mathbf{U} = \mathbf{G} - \mathbf{G}_0$ and $\mathbf{V} = \mathbf{G}_0\mathbf{A}(\mathbf{G} - \mathbf{G}_0)$, and use $\mathbf{A}(\mathbf{G} - \mathbf{G}_0)\mathbf{A} = \mathbf{0}$.
- Conclude that a matrix has a unique generalized inverse iff it is square and non-singular.

Solution. Parts a,b. For (a), multiply out: $\mathbf{A}[\mathbf{G}_0 + (\mathbf{I} - \mathbf{G}_0\mathbf{A})\mathbf{U} + \mathbf{V}(\mathbf{I} - \mathbf{A}\mathbf{G}_0)]\mathbf{A} = \mathbf{A}\mathbf{G}_0\mathbf{A} + (\mathbf{A} - \mathbf{A}\mathbf{G}_0\mathbf{A})\mathbf{U}\mathbf{A} + \mathbf{A}\mathbf{V}(\mathbf{A} - \mathbf{A}\mathbf{G}_0\mathbf{A}) = \mathbf{A}$. For (b), put $\mathbf{\Delta} = \mathbf{G} - \mathbf{G}_0$, so that $\mathbf{A}\mathbf{\Delta}\mathbf{A} = \mathbf{A} - \mathbf{A} = \mathbf{0}$. With $\mathbf{U} = \mathbf{\Delta}$ and $\mathbf{V} = \mathbf{G}_0\mathbf{A}\mathbf{\Delta}$,

$$(\mathbf{I} - \mathbf{G}_0\mathbf{A})\mathbf{\Delta} + \mathbf{G}_0\mathbf{A}\mathbf{\Delta}(\mathbf{I} - \mathbf{A}\mathbf{G}_0) = \mathbf{\Delta} - \mathbf{G}_0\mathbf{A}\mathbf{\Delta} + \mathbf{G}_0\mathbf{A}\mathbf{\Delta} - \mathbf{G}_0(\mathbf{A}\mathbf{\Delta}\mathbf{A})\mathbf{G}_0 = \mathbf{\Delta},$$

so the formula reproduces $\mathbf{G}_0 + \mathbf{\Delta} = \mathbf{G}$.

Exercise B3.

Using the singular value decomposition, show that $(\mathbf{X}^\top \mathbf{X})^+ \mathbf{X}^\top = \mathbf{X}^+$ and $\mathbf{X} \mathbf{X}^+ = \mathbf{M}$. Show also that $\mathbf{X}^+ = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top$ when $\mathbf{X}$ has full column rank.

6.3.6.3 C. Going deeper**Exercise C1.**

Show that $\mathbf{M}$ is the only $n \times n$ matrix $\mathbf{P}$ satisfying $\mathbf{P}\mathbf{X} = \mathbf{X}$, $\mathbf{P}^\top = \mathbf{P}$ and $\text{rank}(\mathbf{P}) = \text{rank}(\mathbf{X})$. Is the rank condition necessary? Is symmetry?

6.4 Least squares via projection

We now return to the linear model $\mathbf{Y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}$, with $\mathbf{X}$ an $n \times p$ matrix of rank $r \leq p$, and $\mathbf{M}$ the orthogonal projection onto $\mathcal{C}(\mathbf{X})$. Nothing in this section uses probability. The results hold for every data vector $\mathbf{y}$ and so apply whatever the distribution of the errors.

6.4.1 The characterization of least squares estimates

Theorem 6.4.1 — Least squares is projection

A vector $\hat{\boldsymbol{\beta}}$ is a least squares estimate of $\boldsymbol{\beta}$ iff

$$\mathbf{X}\hat{\boldsymbol{\beta}} = \mathbf{M}\mathbf{y}.$$

Consequently the fitted values $\hat{\mathbf{Y}} = \mathbf{M}\mathbf{y}$ and residuals $\hat{\boldsymbol{\varepsilon}} = (\mathbf{I} - \mathbf{M})\mathbf{y}$ are the same for every least squares estimate.

Proof. Apply the projection theorem (Theorem 6.3.1) with $\mathcal{S} = \mathcal{C}(\mathbf{X})$ and $\mathbf{u} = \mathbf{X}\mathbf{b}$. By Eq. 6.3.1,

$$\|\mathbf{y} - \mathbf{X}\mathbf{b}\|^2 = \|\mathbf{y} - \mathbf{M}\mathbf{y}\|^2 + \|\mathbf{M}\mathbf{y} - \mathbf{X}\mathbf{b}\|^2 \quad \text{for every } \mathbf{b} \in \mathbb{R}^p.$$

The first term does not involve $\mathbf{b}$, and the second is zero iff $\mathbf{X}\mathbf{b} = \mathbf{M}\mathbf{y}$. Such a $\mathbf{b}$ exists because $\mathbf{M}\mathbf{y} \in \mathcal{C}(\mathbf{X})$. ■

The orthogonality half of the projection theorem turns this into linear equations. The residual $\mathbf{y} - \mathbf{X}\mathbf{b}$ is orthogonal to $\mathcal{C}(\mathbf{X})$ iff it is orthogonal to every column of $\mathbf{X}$, that is, iff $\mathbf{X}^\top(\mathbf{y} - \mathbf{X}\mathbf{b}) = \mathbf{0}$.

Theorem 6.4.2 — Normal equations

The **normal equations**

$$\mathbf{X}^\top \mathbf{X} \mathbf{b} = \mathbf{X}^\top \mathbf{y} \quad (6.4.1)$$

are consistent for every $\mathbf{y}$, and their solutions are exactly the least squares estimates. One solution is $\hat{\boldsymbol{\beta}} = (\mathbf{X}^\top \mathbf{X})^- \mathbf{X}^\top \mathbf{y}$, for any generalized inverse, and the set of all solutions is

$$\hat{\boldsymbol{\beta}} + \mathcal{N}(\mathbf{X}) = \{\hat{\boldsymbol{\beta}} + \mathbf{v} : \mathbf{X}\mathbf{v} = \mathbf{0}\}.$$

In particular, the least squares estimate is unique iff $\text{rank}(\mathbf{X}) = p$, and then $\hat{\boldsymbol{\beta}} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{y}$.

Proof. A vector $\mathbf{b}$ solves Eq. 6.4.1 iff $\mathbf{y} - \mathbf{X}\mathbf{b} \perp \mathcal{C}(\mathbf{X})$. Since $\mathbf{X}\mathbf{b} \in \mathcal{C}(\mathbf{X})$ automatically, this holds iff $\mathbf{y} = \mathbf{X}\mathbf{b} + (\mathbf{y} - \mathbf{X}\mathbf{b})$ is the orthogonal decomposition of $\mathbf{y}$, that is, iff $\mathbf{X}\mathbf{b} = \mathbf{M}\mathbf{y}$. By Theorem 6.4.1 this is the same as $\mathbf{b}$ being a least squares estimate. For the particular solution, $\mathbf{X}(\mathbf{X}^\top \mathbf{X})^- \mathbf{X}^\top \mathbf{y} = \mathbf{M}\mathbf{y}$ by Theorem 6.3.7. Finally, $\mathbf{X}\mathbf{b} = \mathbf{M}\mathbf{y} = \mathbf{X}\hat{\boldsymbol{\beta}}$ iff $\mathbf{X}(\mathbf{b} - \hat{\boldsymbol{\beta}}) = \mathbf{0}$, and $\mathcal{N}(\mathbf{X}) = \{\mathbf{0}\}$ iff $\text{rank}(\mathbf{X}) = p$ by Eq. 6.2.1. ■

The logic of this proof runs opposite to the usual calculus derivation, and the difference matters. Setting a gradient to zero finds stationary points. One then has to argue separately that they are minima and that one exists. Here existence, minimality and the full solution set all come from the decomposition $\mathbb{R}^n = \mathcal{C}(\mathbf{X}) \oplus \mathcal{C}(\mathbf{X})^\perp$. The consistency of the normal equations, which can look surprising when $\mathbf{X}^\top \mathbf{X}$ is singular, is just the statement $\mathbf{X}^\top \mathbf{y} \in \mathcal{C}(\mathbf{X}^\top) = \mathcal{C}(\mathbf{X}^\top \mathbf{X})$ from Corollary 6.2.4.

6.4.2 Fitted values and residuals

Collecting the consequences for $\hat{\mathbf{Y}} = \mathbf{M}\mathbf{y}$ and $\hat{\boldsymbol{\varepsilon}} = (\mathbf{I} - \mathbf{M})\mathbf{y}$:

Proposition 6.4.3

For any data vector $\mathbf{y}$:

- (a) $\mathbf{X}^\top \hat{\boldsymbol{\varepsilon}} = \mathbf{0}$ and $\hat{\mathbf{Y}}^\top \hat{\boldsymbol{\varepsilon}} = 0$;
- (b) if $\mathbf{1} \in \mathcal{C}(\mathbf{X})$ (for example, the model has an intercept), then $\sum_i \hat{\varepsilon}_i = 0$ and the mean of the fitted values is $\bar{y}$;
- (c) $\|\mathbf{y}\|^2 = \|\hat{\mathbf{Y}}\|^2 + \|\hat{\boldsymbol{\varepsilon}}\|^2$;
- (d) $\|\hat{\boldsymbol{\varepsilon}}\|^2 = \mathbf{y}^\top (\mathbf{I} - \mathbf{M})\mathbf{y}$ and $\|\hat{\mathbf{Y}}\|^2 = \mathbf{y}^\top \mathbf{M}\mathbf{y}$;
- (e) if $\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}$ for some $\boldsymbol{\beta}$, then $\hat{\mathbf{Y}} = \mathbf{X}\boldsymbol{\beta} + \mathbf{M}\boldsymbol{\varepsilon}$ and $\hat{\boldsymbol{\varepsilon}} = (\mathbf{I} - \mathbf{M})\boldsymbol{\varepsilon}$.

Proof.

- (a) is the orthogonality of $\hat{\boldsymbol{\varepsilon}}$ to $\mathcal{C}(\mathbf{X})$, and $\hat{\mathbf{Y}} \in \mathcal{C}(\mathbf{X})$. For (b), $\mathbf{1} \in \mathcal{C}(\mathbf{X})$ gives $\mathbf{1}^\top \hat{\boldsymbol{\varepsilon}} = 0$, and then $\mathbf{1}^\top \hat{\mathbf{Y}} = \mathbf{1}^\top \mathbf{y}$. (c) is Pythagoras. (d) uses symmetry and idempotence: $\|(\mathbf{I} - \mathbf{M})\mathbf{y}\|^2 = \mathbf{y}^\top (\mathbf{I} - \mathbf{M})^\top (\mathbf{I} - \mathbf{M})\mathbf{y} = \mathbf{y}^\top (\mathbf{I} - \mathbf{M})\mathbf{y}$. For (e), $\mathbf{M}\mathbf{X} = \mathbf{X}$ and $(\mathbf{I} - \mathbf{M})\mathbf{X} = \mathbf{0}$.

Part (e) is simple but important. The residual vector does not depend on $\boldsymbol{\beta}$ at all. It is the projection of the unobservable error vector onto the $(n-r)$ -dimensional space $\mathcal{C}(\mathbf{X})^\perp$. Everything about σ^2 that the data can reveal must therefore come from $(\mathbf{I} - \mathbf{M})\boldsymbol{\varepsilon}$. If $E(\boldsymbol{\varepsilon}) = \mathbf{0}$ and $\text{Cov}(\boldsymbol{\varepsilon}) = \sigma^2\mathbf{I}$, the expectation of a quadratic form (Chapter 2) gives

$$E\|\hat{\boldsymbol{\varepsilon}}\|^2 = E[\boldsymbol{\varepsilon}^\top (\mathbf{I} - \mathbf{M})\boldsymbol{\varepsilon}] = \sigma^2 \text{tr}(\mathbf{I} - \mathbf{M}) = \sigma^2(n - r),$$

using Proposition 6.3.4. So $\|\hat{\boldsymbol{\varepsilon}}\|^2/(n - r)$ is unbiased for σ^2 . The divisor is the dimension of the space in which the residuals live, not the number of observations and not $n - p$. Chapter 7 develops the sampling theory, and Chapter 9 develops the decomposition.

6.4.3 Coefficients are not unique; fitted values are

When $\text{rank}(\mathbf{X}) < p$, Theorem 6.4.2 gives a whole affine subspace of least squares estimates, one for each generalized inverse and more besides. Statistical software has to pick one. Some programs drop columns that are linearly dependent on earlier ones. Others impose “side conditions” such as a zero coefficient for a reference level or effects that sum to zero. Others report the minimum-norm solution. These choices look very different in the printed coefficient table. The geometry says they are all describing the same fitted vector.

Example. (Four answers, one fit)

Nine observations fall in three groups of sizes 3, 2, 4 with group means 5.700, 7.800 and 4.300. The model matrix $\mathbf{X} = [\mathbf{1}, \mathbf{z}_1, \mathbf{z}_2, \mathbf{z}_3]$ has $p = 4$ columns and rank 3, so $\mathbf{X}^\top \mathbf{X}$ is singular. We compute $\hat{\boldsymbol{\beta}} = \mathbf{G}\mathbf{X}^\top \mathbf{y}$ for four generalized inverses $\mathbf{G}$ of $\mathbf{X}^\top \mathbf{X}$: the Moore–Penrose inverse; the inverse obtained by deleting the intercept row and column (Proposition 6.3.5); the one obtained by deleting the first group’s row and column; and a deliberately strange, non-symmetric one built from the general formula of Exercise (B2).

Generalized inverse	$\hat{\beta}_0$	$\hat{\beta}_1$	$\hat{\beta}_2$	$\hat{\beta}_3$	$\hat{\beta}_1 - \hat{\beta}_2$	$\max \mathbf{M} - \mathbf{M}_{\text{MP}} $
Moore– Penrose	4.450	1.250	3.350	-0.150	-2.100	0
drop intercept	0.000	5.700	7.800	4.300	-2.100	1.1×10^{-16}
drop group 1	5.700	0.000	2.100	-1.400	-2.100	1.1×10^{-16}
random	-9.102	14.802	16.902	13.402	-2.100	1.8×10^{-15}

The coefficient vectors disagree wildly, including in sign. Yet each produces the same projection matrix up to rounding error, and hence the same fitted values (the group means) and residuals. The difference $\hat{\beta}_1 - \hat{\beta}_2$ is the same in every row, and so is $\hat{\beta}_0 + \hat{\beta}_k$ for each k . Individual coefficients are not. The trace of $\mathbf{M}$ is 3.000, the rank of $\mathbf{X}$.

```
import numpy as np
```

```

groups = np.array([0, 0, 0, 1, 1, 2, 2, 2, 2])    # three groups, sizes 3, 2, 4
y = np.array([5.1, 6.3, 5.7, 8.2, 7.4, 4.0, 4.9, 3.8, 4.5])
n, g = len(y), 3
X = np.column_stack([np.ones(n), np.eye(g)[groups]]) # [1, indicator columns]
A = X.T @ X
print("n =", n, " p =", X.shape[1], " rank(X) =", np.linalg.matrix_rank(X))

```

```

def g_drop(j):
    """Generalized inverse obtained by deleting row and column j of A."""
    keep = [k for k in range(A.shape[0]) if k != j]
    G = np.zeros_like(A)
    G[np.ix_(keep, keep)] = np.linalg.inv(A[np.ix_(keep, keep)])
    return G

rng = np.random.default_rng(6)
G0 = np.linalg.pinv(A)                                # Moore-Penrose inverse
U, V = rng.normal(size=A.shape), rng.normal(size=A.shape)
G_random = G0 + (np.eye(4) - G0 @ A) @ U + V @ (np.eye(4) - A @ G0)

ginverses = {
    "Moore-Penrose": G0,
    "drop intercept": g_drop(0),
    "drop group 1": g_drop(1),
    "random": G_random,
}

```

```

M_ref = X @ G0 @ X.T
for name, G in ginverses.items():
    assert np.allclose(A @ G @ A, A)                # G really is a g-inverse
    beta = G @ X.T @ y
    M = X @ G @ X.T
    print(f"{name:15s} beta = {np.round(beta, 3)}",
          f" max|M - M_ref| = {np.abs(M - M_ref).max():.1e}",
          f" group1-group2 = {beta[1] - beta[2]:.3f}")

```

Which linear functions of $\hat{\beta}$ come out the same for every least squares estimate? The answer is a first look at estimability, the subject of [Chapter 8](#).

Theorem 6.4.4

For $\lambda \in \mathbb{R}^p$, the value $\lambda^\top \hat{\beta}$ is the same for every least squares estimate $\hat{\beta}$ (and every $\mathbf{y}$) iff $\lambda \in \mathcal{C}(\mathbf{X}^\top)$, that is, iff $\lambda^\top = \rho^\top \mathbf{X}$ for some $\rho \in \mathbb{R}^n$. In that case $\lambda^\top \hat{\beta} = \rho^\top \mathbf{M}\mathbf{y}$.

Proof. By [Theorem 6.4.2](#) the least squares estimates form $\hat{\beta} + \mathcal{N}(\mathbf{X})$. So $\lambda^\top \hat{\beta}$ is invariant iff $\lambda^\top \mathbf{v} = 0$ for all $\mathbf{v} \in \mathcal{N}(\mathbf{X})$, that is, iff $\lambda \in \mathcal{N}(\mathbf{X})^\perp = \mathcal{C}(\mathbf{X}^\top)$ ([Corollary 6.2.4](#)). If $\lambda^\top = \rho^\top \mathbf{X}$ then $\lambda^\top \hat{\beta} = \rho^\top \mathbf{X}\hat{\beta} = \rho^\top \mathbf{M}\mathbf{y}$. ■

In [Example \(Four answers, one fit\)](#), $\lambda = (0, 1, -1, 0)^\top$ is the difference between any row of $\mathbf{X}$ from group 1, $(1, 1, 0, 0)$, and any row from group 2, $(1, 0, 1, 0)$. So $\lambda \in \mathcal{C}(\mathbf{X}^\top)$ and $\hat{\beta}_1 - \hat{\beta}_2$ is invariant. For $\lambda = (0, 1, 0, 0)^\top$, note that $\rho^\top \mathbf{X}$ has first entry $\sum_i \rho_i$ and remaining entries equal

to the sums of ρ_i within each group. Matching $(0, 1, 0, 0)$ would need the group sums to be $1, 0, 0$ and their total to be 0 , which is impossible. So $\hat{\beta}_1$ is not invariant.

Remark (Side conditions choose coordinates). Imposing $\beta_0 = 0$ or $\beta_1 = 0$ does not add information about the data. It picks one point from the affine set $\hat{\beta} + \mathcal{N}(\mathbf{X})$. A side condition $\mathbf{c}^\top \beta = 0$ picks exactly one point iff $\mathbf{c} \notin \mathcal{C}(\mathbf{X}^\top)$. A condition inside $\mathcal{C}(\mathbf{X}^\top)$ would constrain the fitted values and change the model rather than just its coordinates.

[Theorem 8.4.2](#) makes this precise.

6.4.4 Exercises

6.4.4.1 A. Check your understanding

Exercise A1.

For [Example \(Three observations\)](#), write out the 3×3 matrix $\mathbf{M}$ with exact fractional entries, verify $\text{tr } \mathbf{M} = 2$, and recover the fitted values and residuals quoted in the text.

Exercise A2.

Show that the residuals sum to zero whenever $\mathbf{1} \in \mathcal{C}(\mathbf{X})$, even if no column of $\mathbf{X}$ is constant. Give an example of a model without such a column in which this happens, and one in which the residuals do not sum to zero.

6.4.4.2 B. Practice

Exercise B1.

Suppose $E(\mathbf{Y}) = \mathbf{X}\beta$ and $\text{Cov}(\mathbf{Y}) = \sigma^2 \mathbf{I}$. Show that $E(\hat{\mathbf{Y}}) = \mathbf{X}\beta$, $\text{Cov}(\hat{\mathbf{Y}}) = \sigma^2 \mathbf{M}$, $\text{Cov}(\hat{\epsilon}) = \sigma^2(\mathbf{I} - \mathbf{M})$ and $\text{Cov}(\hat{\mathbf{Y}}, \hat{\epsilon}) = \mathbf{0}$. Which of these statements survive when $\text{Cov}(\mathbf{Y}) = \sigma^2 \mathbf{V}$ for a general positive definite $\mathbf{V}$?

Exercise B2.

Let $\text{rank}(\mathbf{X}) = r < p$ and let $\mathbf{C}$ be a $(p - r) \times p$ matrix. Show that the constrained problem “minimize $\|\mathbf{y} - \mathbf{X}\mathbf{b}\|$ subject to $\mathbf{C}\mathbf{b} = \mathbf{0}$ ” has a unique solution for every $\mathbf{y}$, and that its fitted values are $\mathbf{M}\mathbf{y}$, iff $\text{rank}(\frac{\mathbf{X}}{\mathbf{C}}) = p$. Show that the condition implies $\mathcal{C}(\mathbf{C}^\top) \cap \mathcal{C}(\mathbf{X}^\top) = \{\mathbf{0}\}$, and interpret: the constraints must not restrict any estimable function.

6.4.4.3 C. Going deeper

Exercise C1.

Suppose $\mathbf{X}^\top \mathbf{y} \neq \mathbf{0}$. Show that every least squares estimate $\mathbf{b}$ can be written as $\mathbf{G}\mathbf{X}^\top \mathbf{y}$ for some generalized inverse $\mathbf{G}$ of $\mathbf{A} = \mathbf{X}^\top \mathbf{X}$. *Hint:* with $\mathbf{c} = \mathbf{X}^\top \mathbf{y}$, try $\mathbf{G} = \mathbf{G}_0 + (\mathbf{b} - \mathbf{G}_0 \mathbf{c})\mathbf{c}^\top / (\mathbf{c}^\top \mathbf{c})$. What goes wrong when $\mathbf{X}^\top \mathbf{y} = \mathbf{0}$?

6.5 Nested models and sequential projection

Statistical questions about linear models are almost always comparisons. Does a regressor matter? Are several group means equal? Is a curve needed, or will a line do? Each comparison sets a full model against a *reduced* model whose mean space is a subspace of the full one. This section studies the geometry of such nested pairs. [Chapter 9](#) and [Chapter 11](#) build the analysis of variance and the F test on it.

6.5.1 Two nested subspaces

Let $\mathcal{C}(\mathbf{X}_0) \subseteq \mathcal{C}(\mathbf{X})$, with ranks $r_0 \leq r$. Write $\mathbf{M}_0$ and $\mathbf{M}$ for the two projections.

Theorem 6.5.1 — Nested projections

If $\mathcal{C}(\mathbf{X}_0) \subseteq \mathcal{C}(\mathbf{X})$, then:

- (a) $\mathbf{M}\mathbf{M}_0 = \mathbf{M}_0\mathbf{M} = \mathbf{M}_0$;
- (b) $\mathbf{M} - \mathbf{M}_0$ is the orthogonal projection onto $\mathcal{C}(\mathbf{X}_0)^\perp \cap \mathcal{C}(\mathbf{X})$, the orthogonal complement of $\mathcal{C}(\mathbf{X}_0)$ within $\mathcal{C}(\mathbf{X})$;
- (c) $\text{rank}(\mathbf{M} - \mathbf{M}_0) = r - r_0$;
- (d) $(\mathbf{I} - \mathbf{M}_0) - (\mathbf{I} - \mathbf{M}) = \mathbf{M} - \mathbf{M}_0$, and $\mathbf{M}_0(\mathbf{M} - \mathbf{M}_0) = (\mathbf{M} - \mathbf{M}_0)(\mathbf{I} - \mathbf{M}) = \mathbf{0}$.

Proof.

- (a) Every column of $\mathbf{M}_0$ lies in $\mathcal{C}(\mathbf{X}_0) \subseteq \mathcal{C}(\mathbf{X})$ and is left unchanged by $\mathbf{M}$, so $\mathbf{M}\mathbf{M}_0 = \mathbf{M}_0$. Transposing gives $\mathbf{M}_0\mathbf{M} = \mathbf{M}_0$.
- (b) $\mathbf{M} - \mathbf{M}_0$ is symmetric, and by (a) $(\mathbf{M} - \mathbf{M}_0)^2 = \mathbf{M} - \mathbf{M}\mathbf{M}_0 - \mathbf{M}_0\mathbf{M} + \mathbf{M}_0 = \mathbf{M} - \mathbf{M}_0$. By [Theorem 6.3.3](#) it projects onto its column space. If $\mathbf{v} \in \mathcal{C}(\mathbf{X})$ and $\mathbf{v} \perp \mathcal{C}(\mathbf{X}_0)$, then $(\mathbf{M} - \mathbf{M}_0)\mathbf{v} = \mathbf{v} - \mathbf{0} = \mathbf{v}$, so $\mathbf{v} \in \mathcal{C}(\mathbf{M} - \mathbf{M}_0)$. Conversely, a vector $\mathbf{v} = (\mathbf{M} - \mathbf{M}_0)\mathbf{u}$ lies in $\mathcal{C}(\mathbf{X})$ and satisfies $\mathbf{M}_0\mathbf{v} = (\mathbf{M}_0 - \mathbf{M}_0)\mathbf{u} = \mathbf{0}$.
- (c) $\text{tr}(\mathbf{M} - \mathbf{M}_0) = r - r_0$ and [Proposition 6.3.4](#).
- (d) Direct multiplication using (a).

■

The theorem gives a three-way orthogonal decomposition of $\mathbb{R}^n$,

$$\mathbb{R}^n = \underbrace{\mathcal{C}(\mathbf{X}_0)}_{\dim r_0} \oplus \underbrace{\mathcal{C}(\mathbf{X}_0)^\perp \cap \mathcal{C}(\mathbf{X})}_{\dim r - r_0} \oplus \underbrace{\mathcal{C}(\mathbf{X})^\perp}_{\dim n - r},$$

and correspondingly $\mathbf{I} = \mathbf{M}_0 + (\mathbf{M} - \mathbf{M}_0) + (\mathbf{I} - \mathbf{M})$, with the three terms mutually orthogonal projections. Applying the decomposition to $\mathbf{y}$ and using Pythagoras,

$$\|\mathbf{y}\|^2 = \|\mathbf{M}_0\mathbf{y}\|^2 + \|(\mathbf{M} - \mathbf{M}_0)\mathbf{y}\|^2 + \|(\mathbf{I} - \mathbf{M})\mathbf{y}\|^2. \quad (6.5.1)$$

The middle term has a second reading that is the key to model comparison. Since $(\mathbf{I} - \mathbf{M}_0) = (\mathbf{M} - \mathbf{M}_0) + (\mathbf{I} - \mathbf{M})$ is an orthogonal sum,

$$\|(\mathbf{M} - \mathbf{M}_0)\mathbf{y}\|^2 = \|(\mathbf{I} - \mathbf{M}_0)\mathbf{y}\|^2 - \|(\mathbf{I} - \mathbf{M})\mathbf{y}\|^2 = \text{SSE}_0 - \text{SSE}. \quad (6.5.2)$$

The *drop in residual sum of squares* from the reduced to the full model is the squared length of a projection onto an $(r - r_0)$ -dimensional space. That is why it has $r - r_0$ degrees of freedom, and why it is independent of SSE under normality: the two lie in orthogonal subspaces (Chapter 4). It also shows that SSE can never increase when the model space grows, with equality iff $\mathbf{M}\mathbf{y} = \mathbf{M}_0\mathbf{y}$.

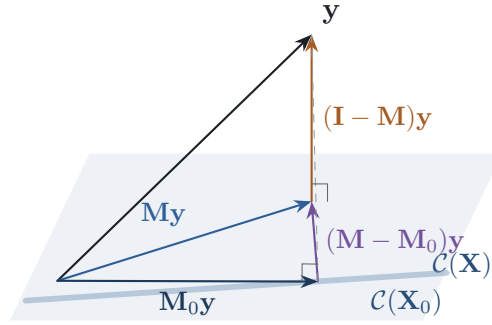


Figure 6.5.1. Nested projections. The fitted vector of the reduced model, $\mathbf{M}_0\mathbf{y}$, is also the projection of $\mathbf{M}\mathbf{y}$ onto $\mathcal{C}(\mathbf{X}_0)$. The three coloured pieces $\mathbf{M}_0\mathbf{y}$, $(\mathbf{M} - \mathbf{M}_0)\mathbf{y}$ and $(\mathbf{I} - \mathbf{M})\mathbf{y}$ are mutually orthogonal. The dashed segment is the reduced model's residual, whose squared length is the sum of the squared lengths of the two segments meeting at $\mathbf{M}\mathbf{y}$.

Part (a) also says $\mathbf{M}_0\mathbf{y} = \mathbf{M}_0(\mathbf{M}\mathbf{y})$. *Fitting the reduced model to the data gives the same result as fitting it to the full model's fitted values.* Projection can be done in stages (Figure 6.5.1).

Example. (Intercept inside simple regression)

Let $\mathbf{X}_0 = \mathbf{1}$ and $\mathbf{X} = [\mathbf{1}, \mathbf{x}]$ with $\mathbf{x}$ not constant. Then $\mathcal{C}(\mathbf{X}_0)^\perp \cap \mathcal{C}(\mathbf{X})$ is spanned by the centred regressor $\mathbf{x} - \bar{x}\mathbf{1} = (\mathbf{I} - n^{-1}\mathbf{1}\mathbf{1}^\top)\mathbf{x}$, which lies in $\mathcal{C}(\mathbf{X})$ and is orthogonal to $\mathbf{1}$. By Example (Projections onto a line),

$$\mathbf{M} - \mathbf{M}_0 = \frac{(\mathbf{x} - \bar{x}\mathbf{1})(\mathbf{x} - \bar{x}\mathbf{1})^\top}{S_{xx}}, \quad S_{xx} = \sum_i (x_i - \bar{x})^2,$$

and so

$$\mathbf{M}\mathbf{y} = \bar{y}\mathbf{1} + \frac{S_{xy}}{S_{xx}}(\mathbf{x} - \bar{x}\mathbf{1}), \quad S_{xy} = \sum_i (x_i - \bar{x})(y_i - \bar{y}).$$

Reading off coordinates, the slope is S_{xy}/S_{xx} and the intercept is $\bar{y} - \bar{x} S_{xy}/S_{xx}$. These are the familiar formulas, obtained without solving any equations, and the decomposition Eq. 6.5.1 becomes

$$\sum_i y_i^2 = n\bar{y}^2 + \frac{S_{xy}^2}{S_{xx}} + \text{SSE}.$$

6.5.2 Chains of subspaces

The argument extends to any chain $\mathcal{C}(\mathbf{X}_1) \subseteq \mathcal{C}(\mathbf{X}_2) \subseteq \cdots \subseteq \mathcal{C}(\mathbf{X}_k)$ with projections $\mathbf{M}_1, \dots, \mathbf{M}_k$. Setting $\mathbf{M}_0 = \mathbf{0}$ and $\mathbf{M}_{k+1} = \mathbf{I}$, the differences $\mathbf{M}_j - \mathbf{M}_{j-1}$ for $j = 1, \dots, k+1$ are mutually orthogonal projections summing to $\mathbf{I}$. The corresponding lengths

$$\|(\mathbf{M}_j - \mathbf{M}_{j-1})\mathbf{y}\|^2 = \text{SSE}_{j-1} - \text{SSE}_j$$

are the **sequential sums of squares** of [Definition 9.3.1](#). Their values depend on the *order* in which the subspaces are nested. A regressor added early can take credit that it would not get if added last. The Frisch–Waugh–Lovell theorem of [Section 6.6](#) describes what the regressor added last actually contributes.

Example. (Order matters)

For the state data of [Example \(Poverty and murder rates\)](#), enter the three regressors after the intercept in two different orders. The listing computes each squared length $\|(\mathbf{M}_j - \mathbf{M}_{j-1})\mathbf{y}\|^2$ directly from the projections.

Order A		Order B	
poverty	102.49	single-parent	162.84
single-parent	80.04	urban	3.59
urban	0.27	poverty	16.37
residual	101.96	residual	101.96

Entered first, poverty accounts for 102.49 of the corrected total sum of squares 284.76. Entered last, it accounts for only 16.37. The residual sum of squares, and the total explained by all three regressors together (182.80), do not depend on the order, because the final subspace does not. The individual pieces do depend on the order, because the intermediate subspaces differ. Poverty and single parenthood are correlated, so whichever enters first takes the variation they share. The piece for the *last* regressor is the only one with an order-free meaning. By [Theorem 6.6.2](#), it is the squared length of the projection of $\mathbf{y}$ onto the part of that regressor orthogonal to all the others.

```
import numpy as np
import statsmodels.api as sm
data = sm.datasets.statcrime.load_pandas().data.drop(index="District of Columbia")
y = data["murder"].to_numpy()
cols = {"poverty": data["poverty"], "single": data["single"], "urban": data["urban"]}

def projection(Z):
    Q, _ = np.linalg.qr(Z)
    return Q @ Q.T
    # fine for n = 50; never do this for large n

def sequential_ss(y, columns):
    """Squared lengths ||(M_j - M_{j-1}) y||^2 as columns enter one at a time."""
    n = len(y)
    Z = np.ones((n, 1))
    M_prev = projection(Z)
```

```

pieces = {"mean": y @ M_prev @ y}
for name, x in columns:
    Z = np.column_stack([Z, x])
    M = projection(Z)
    pieces[name] = y @ (M - M_prev) @ y
    M_prev = M
pieces["residual"] = y @ (np.eye(n) - M_prev) @ y
return pieces

order_a = sequential_ss(y, [(k, cols[k]) for k in ["poverty", "single", "urban"]])
order_b = sequential_ss(y, [(k, cols[k]) for k in ["single", "urban", "poverty"]])
for name in ["poverty", "single", "urban", "residual"]:
    print(f"{name:9s}  order A: {order_a[name]:8.2f}  order B: {order_b[name]:8.2f}")
    )

```

6.5.3 Two factors and when the pieces are orthogonal

A sum of projections is a projection exactly when the pieces are orthogonal ([Theorem 6.3.8](#)). In designed experiments, orthogonality comes from balance, and when it fails, it fails in ways that later chapters have to face.

Example. (Additive two-factor model, one observation per cell)

Let factor A have a levels and factor B have b levels, with one observation y_{ij} for each of the $n = ab$ combinations. Let $\mathbf{P}_A$ project onto vectors that depend only on the A level, so $\mathbf{P}_A \mathbf{y}$ has entries $\bar{y}_{i\cdot}$, and define $\mathbf{P}_B$ similarly. Let $\mathbf{P}_0 = n^{-1} \mathbf{1} \mathbf{1}^\top$. Both $\mathcal{C}(\mathbf{P}_A)$ and $\mathcal{C}(\mathbf{P}_B)$ contain $\mathbf{1}$, so $\mathbf{P}_A \mathbf{P}_0 = \mathbf{P}_B \mathbf{P}_0 = \mathbf{P}_0$ by [Theorem 6.5.1](#).

The crucial computation is $\mathbf{P}_A \mathbf{P}_B$. Averaging over j with i fixed a vector that depends only on j gives the overall average of that vector. So for every $\mathbf{y}$ the entries of $\mathbf{P}_A(\mathbf{P}_B \mathbf{y})$ are all equal to $b^{-1} \sum_j \bar{y}_{\cdot j} = \bar{y}_{\cdot\cdot}$. Therefore

$$\mathbf{P}_A \mathbf{P}_B = \mathbf{P}_0,$$

and so $(\mathbf{P}_A - \mathbf{P}_0)(\mathbf{P}_B - \mathbf{P}_0) = \mathbf{P}_0 - \mathbf{P}_0 - \mathbf{P}_0 + \mathbf{P}_0 = \mathbf{0}$. The two *centred* factor spaces are orthogonal. By [Theorem 6.3.8](#), the additive model space has projection $\mathbf{P}_0 + (\mathbf{P}_A - \mathbf{P}_0) + (\mathbf{P}_B - \mathbf{P}_0) = \mathbf{P}_A + \mathbf{P}_B - \mathbf{P}_0$, and

$$\mathbf{I} = \mathbf{P}_0 + (\mathbf{P}_A - \mathbf{P}_0) + (\mathbf{P}_B - \mathbf{P}_0) + (\mathbf{I} - \mathbf{P}_A - \mathbf{P}_B + \mathbf{P}_0)$$

is an orthogonal decomposition with ranks 1, $a-1$, $b-1$ and $(a-1)(b-1)$. The residual projection sends y_{ij} to $y_{ij} - \bar{y}_{i\cdot} - \bar{y}_{\cdot j} + \bar{y}_{\cdot\cdot}$.

The key step used the fact that every A level appears with every B level equally often. If cells are missing or unequally replicated, $\mathbf{P}_A \mathbf{P}_B \neq \mathbf{P}_0$, the centred factor spaces are no longer orthogonal, and the sum of squares for A depends on whether B was fitted first. [Chapter 17](#) is about that situation.

6.5.4 Reduced models defined by constraints

Reduced models often come as *constraints* on the parameters of the full model, for example $\beta_2 = \beta_3$ or $\beta_4 = 0$, rather than as a smaller model matrix. In mean space, a constraint $\mathbf{A}^\top \boldsymbol{\beta} = \mathbf{0}$ on an estimable function $\mathbf{A}^\top = \mathbf{P}^\top \mathbf{X}$ restricts $\boldsymbol{\mu} = \mathbf{X}\boldsymbol{\beta}$ to the subspace

$$\mathcal{S}_0 = \mathcal{C}(\mathbf{X}) \cap \mathcal{N}(\mathbf{P}^\top) = \{\boldsymbol{\mu} \in \mathcal{C}(\mathbf{X}) : \mathbf{P}^\top \boldsymbol{\mu} = \mathbf{0}\}.$$

To test the constraint we need the piece of $\mathcal{C}(\mathbf{X})$ that $\mathcal{S}_0$ removes.

Theorem 6.5.2

Let $\mathbf{M}$ project onto $\mathcal{C}(\mathbf{X})$ and let $\mathbf{P}$ be an $n \times q$ matrix. If $\mathcal{S}_0 = \mathcal{C}(\mathbf{X}) \cap \mathcal{N}(\mathbf{P}^\top)$, then

$$\mathcal{S}_0^\perp \cap \mathcal{C}(\mathbf{X}) = \mathcal{C}(\mathbf{MP}).$$

Hence the projection onto $\mathcal{S}_0$ is $\mathbf{M} - \mathbf{P}_{\mathcal{C}(\mathbf{MP})}$, and $\dim \mathcal{S}_0 = r - \text{rank}(\mathbf{MP})$.

Proof. By Lemma 6.2.5 and Lemma 6.2.1, $\mathcal{S}_0^\perp = \mathcal{C}(\mathbf{X})^\perp + \mathcal{N}(\mathbf{P}^\top)^\perp = \mathcal{C}(\mathbf{X})^\perp + \mathcal{C}(\mathbf{P})$. If $\mathbf{v} \in \mathcal{S}_0^\perp \cap \mathcal{C}(\mathbf{X})$, write $\mathbf{v} = \mathbf{w} + \mathbf{P}\mathbf{a}$ with $\mathbf{w} \perp \mathcal{C}(\mathbf{X})$. Applying $\mathbf{M}$ gives $\mathbf{v} = \mathbf{M}\mathbf{v} = \mathbf{MP}\mathbf{a} \in \mathcal{C}(\mathbf{MP})$. Conversely, $\mathbf{MP}\mathbf{a}$ lies in $\mathcal{C}(\mathbf{X})$ and equals $\mathbf{P}\mathbf{a} - (\mathbf{I} - \mathbf{M})\mathbf{P}\mathbf{a} \in \mathcal{C}(\mathbf{P}) + \mathcal{C}(\mathbf{X})^\perp = \mathcal{S}_0^\perp$. The remaining claims follow from Theorem 6.5.1 with the roles $\mathcal{C}(\mathbf{X}_0) = \mathcal{S}_0$. ■

The subspace $\mathcal{C}(\mathbf{MP})$ is the *test space* of the hypothesis. Its dimension $\text{rank}(\mathbf{MP})$ counts how many independent restrictions the constraint really places on the mean. It can be smaller than q if some rows of the constraint are redundant. Chapter 11 uses exactly this space for the numerator of the F statistic.

6.5.5 Exercises

6.5.5.1 A. Check your understanding

Exercise A1.

Construct the projections $\mathbf{P}_A$, $\mathbf{P}_B$ and $\mathbf{P}_0$ of Example (Additive two-factor model, one observation per cell) for $a = 3$, $b = 2$ and verify $\mathbf{P}_A\mathbf{P}_B = \mathbf{P}_0$ numerically. Delete one observation and recompute. Show that the centred factor spaces are no longer orthogonal, and find the largest entry of $(\mathbf{P}_A - \mathbf{P}_0)(\mathbf{P}_B - \mathbf{P}_0)$.

6.5.5.2 B. Practice

Exercise B1.

With $\mathcal{C}(\mathbf{X}_0) \subseteq \mathcal{C}(\mathbf{X})$, show that $\text{SSE}_0 - \text{SSE} = \|\hat{\mathbf{Y}} - \hat{\mathbf{Y}}_0\|^2$, the squared distance between the two fitted vectors.

Solution. $\hat{\mathbf{Y}} - \hat{\mathbf{Y}}_0 = (\mathbf{M} - \mathbf{M}_0)\mathbf{y}$, and $\|(\mathbf{M} - \mathbf{M}_0)\mathbf{y}\|^2 = \text{SSE}_0 - \text{SSE}$ by Eq. 6.5.2.

Exercise B2.

In the model $E(\mathbf{Y}) = \beta_0\mathbf{1} + \beta_1\mathbf{x}_1 + \beta_2\mathbf{x}_2 + \beta_3\mathbf{x}_3$ with full-rank $\mathbf{X}$, the hypothesis $\beta_2 = \beta_3$ defines a reduced model with model matrix $\mathbf{X}_0 = [\mathbf{1}, \mathbf{x}_1, \mathbf{x}_2 + \mathbf{x}_3]$. Show that the test space $\mathcal{C}(\mathbf{X}_0)^\perp \cap \mathcal{C}(\mathbf{X})$ is spanned by $(\mathbf{I} - \mathbf{M}_0)\mathbf{x}_2$, and that $(\mathbf{I} - \mathbf{M}_0)\mathbf{x}_2 = -(\mathbf{I} - \mathbf{M}_0)\mathbf{x}_3$. Verify that this agrees with Theorem 6.5.2 for a suitable $\mathbf{P}$.

Solution. The two model spaces differ in dimension by one, so the test space is a line. The vector $(\mathbf{I} - \mathbf{M}_0)\mathbf{x}_2$ lies in $\mathcal{C}(\mathbf{X})$ and is orthogonal to $\mathcal{C}(\mathbf{X}_0)$. It is nonzero, because $\mathbf{x}_2 \in \mathcal{C}(\mathbf{X}_0)$ would make the columns of $\mathbf{X}$ dependent. Since $\mathbf{x}_2 + \mathbf{x}_3 \in \mathcal{C}(\mathbf{X}_0)$, $(\mathbf{I} - \mathbf{M}_0)(\mathbf{x}_2 + \mathbf{x}_3) = \mathbf{0}$. For Theorem 6.5.2, write the constraint as $\boldsymbol{\lambda}^\top \boldsymbol{\beta} = 0$ with $\boldsymbol{\lambda} = (0, 0, 1, -1)^\top$, and take $\mathbf{P} = \mathbf{X}(\mathbf{X}^\top \mathbf{X})^{-1} \boldsymbol{\lambda}$, so that $\mathbf{P}^\top \mathbf{X} = \boldsymbol{\lambda}^\top$. Then $\mathbf{M}\mathbf{P} = \mathbf{P}$. Each column of $\mathbf{X}_0$ is $\mathbf{X}\mathbf{k}$ with $\boldsymbol{\lambda}^\top \mathbf{k} = 0$, so $\mathbf{P} \perp \mathcal{C}(\mathbf{X}_0)$ and $\mathbf{P}$ spans the same line.

6.5.5.3 C. Going deeper

Exercise C1.

Extend Example (Additive two-factor model, one observation per cell) to $m \geq 2$ observations in every cell and a model with interaction. Find five mutually orthogonal projections summing to $\mathbf{I}$, corresponding to the grand mean, factor A , factor B , the interaction, and within-cell error, and find their ranks. Describe the action of each on $\mathbf{y}$ in terms of cell, row, column and grand means.

6.6 The Frisch–Waugh–Lovell theorem

A coefficient in a multiple regression is often described as the effect of its regressor “holding the other regressors fixed”. In observational data nobody held anything fixed, so the phrase needs a precise meaning. The Frisch–Waugh–Lovell (FWL) theorem supplies one. A coefficient in a multiple regression equals the slope of a simple regression in which the other regressors have been *projected out* of both the response and the regressor of interest.

6.6.1 Statement and proof

Partition the model matrix and coefficients as

$$\mathbf{X} = [\mathbf{X}_1, \mathbf{X}_2], \quad \boldsymbol{\beta} = \begin{pmatrix} \beta_1 \\ \beta_2 \end{pmatrix},$$

with $\mathbf{X}_1$ of size $n \times p_1$ and $\mathbf{X}_2$ of size $n \times p_2$, and assume $\mathbf{X}$ has full column rank. Let $\mathbf{M}_2$ project onto $\mathcal{C}(\mathbf{X}_2)$, and write

$$\tilde{\mathbf{X}}_1 = (\mathbf{I} - \mathbf{M}_2)\mathbf{X}_1, \quad \tilde{\mathbf{y}} = (\mathbf{I} - \mathbf{M}_2)\mathbf{y}$$

for the parts of $\mathbf{X}_1$ and $\mathbf{y}$ that $\mathbf{X}_2$ cannot explain linearly. Each column of $\tilde{\mathbf{X}}_1$ is the residual vector from regressing the corresponding column of $\mathbf{X}_1$ on $\mathbf{X}_2$.

Lemma 6.6.1 — Orthogonalizing a partitioned model space

Under the assumptions above:

- (a) $\mathcal{C}(\mathbf{X}) = \mathcal{C}(\mathbf{X}_2) \oplus \mathcal{C}(\tilde{\mathbf{X}}_1)$, and the two summands are orthogonal;
- (b) $\tilde{\mathbf{X}}_1$ has full column rank;
- (c) $\mathbf{M} = \mathbf{M}_2 + \tilde{\mathbf{M}}_1$, where $\tilde{\mathbf{M}}_1 = \tilde{\mathbf{X}}_1(\tilde{\mathbf{X}}_1^\top \tilde{\mathbf{X}}_1)^{-1} \tilde{\mathbf{X}}_1^\top$.

Proof.

- (a) The columns of $\tilde{\mathbf{X}}_1 = \mathbf{X}_1 - \mathbf{M}_2 \mathbf{X}_1$ lie in $\mathcal{C}(\mathbf{X})$ and are orthogonal to $\mathcal{C}(\mathbf{X}_2)$. Conversely, for any $\mathbf{b} = (\mathbf{b}_1^\top, \mathbf{b}_2^\top)^\top$,

$$\mathbf{X}\mathbf{b} = \mathbf{X}_1\mathbf{b}_1 + \mathbf{X}_2\mathbf{b}_2 = \tilde{\mathbf{X}}_1\mathbf{b}_1 + (\mathbf{M}_2\mathbf{X}_1\mathbf{b}_1 + \mathbf{X}_2\mathbf{b}_2),$$

with the second term in $\mathcal{C}(\mathbf{X}_2)$. So $\mathcal{C}(\mathbf{X}) \subseteq \mathcal{C}(\tilde{\mathbf{X}}_1) + \mathcal{C}(\mathbf{X}_2)$.

- (b) If $\tilde{\mathbf{X}}_1\mathbf{a} = \mathbf{0}$, then $\mathbf{X}_1\mathbf{a} = \mathbf{M}_2\mathbf{X}_1\mathbf{a} = \mathbf{X}_2\mathbf{c}$ for some $\mathbf{c}$, so $\mathbf{X}(\mathbf{a}^\top, -\mathbf{c}^\top)^\top = \mathbf{0}$. Full column rank of $\mathbf{X}$ forces $\mathbf{a} = \mathbf{0}$.
- (c) Combine (a) with [Theorem 6.3.8](#) and [Theorem 6.3.7](#).

■

Theorem 6.6.2 — Frisch–Waugh–Lovell

Let $\hat{\boldsymbol{\beta}} = (\hat{\boldsymbol{\beta}}_1^\top, \hat{\boldsymbol{\beta}}_2^\top)^\top$ be the least squares estimate in the full model. Then:

- (a) $\hat{\boldsymbol{\beta}}_1 = (\tilde{\mathbf{X}}_1^\top \tilde{\mathbf{X}}_1)^{-1} \tilde{\mathbf{X}}_1^\top \tilde{\mathbf{y}} = (\tilde{\mathbf{X}}_1^\top \tilde{\mathbf{X}}_1)^{-1} \tilde{\mathbf{X}}_1^\top \mathbf{y}$. So $\hat{\boldsymbol{\beta}}_1$ is the coefficient from regressing $\tilde{\mathbf{y}}$ (or $\mathbf{y}$ itself) on $\tilde{\mathbf{X}}_1$;
- (b) the residual vector from regressing $\tilde{\mathbf{y}}$ on $\tilde{\mathbf{X}}_1$ equals the residual vector $(\mathbf{I} - \mathbf{M})\mathbf{y}$ of the full model;
- (c) $\hat{\boldsymbol{\beta}}_2 = (\mathbf{X}_2^\top \mathbf{X}_2)^{-1} \mathbf{X}_2^\top (\mathbf{y} - \mathbf{X}_1 \hat{\boldsymbol{\beta}}_1)$.

Proof. Let $\tilde{\mathbf{b}} = (\tilde{\mathbf{X}}_1^\top \tilde{\mathbf{X}}_1)^{-1} \tilde{\mathbf{X}}_1^\top \mathbf{y}$. Since $\mathbf{I} - \mathbf{M}_2$ is symmetric and idempotent, $\tilde{\mathbf{X}}_1^\top \mathbf{y} = \mathbf{X}_1^\top (\mathbf{I} - \mathbf{M}_2) \mathbf{y} = \tilde{\mathbf{X}}_1^\top \tilde{\mathbf{y}}$, so the two expressions in (a) agree. By [Lemma 6.6.1\(c\)](#),

$$\mathbf{M}\mathbf{y} = \mathbf{M}_2\mathbf{y} + \tilde{\mathbf{X}}_1\tilde{\mathbf{b}} = \mathbf{M}_2\mathbf{y} + \mathbf{X}_1\tilde{\mathbf{b}} - \mathbf{M}_2\mathbf{X}_1\tilde{\mathbf{b}} = \mathbf{X}_1\tilde{\mathbf{b}} + \mathbf{M}_2(\mathbf{y} - \mathbf{X}_1\tilde{\mathbf{b}}).$$

The last term lies in $\mathcal{C}(\mathbf{X}_2)$, so it equals $\mathbf{X}_2\mathbf{c}$ with $\mathbf{c} = (\mathbf{X}_2^\top \mathbf{X}_2)^{-1} \mathbf{X}_2^\top (\mathbf{y} - \mathbf{X}_1\tilde{\mathbf{b}})$. Thus $\mathbf{M}\mathbf{y} = \mathbf{X}_1\tilde{\mathbf{b}} + \mathbf{X}_2\mathbf{c}$. Because $\mathbf{X}$ has full column rank, the coefficient vector representing $\mathbf{M}\mathbf{y}$ is unique, so $\hat{\boldsymbol{\beta}}_1 = \tilde{\mathbf{b}}$ and $\hat{\boldsymbol{\beta}}_2 = \mathbf{c}$. This proves (a) and (c). For (b), $\mathbf{y} - \mathbf{M}\mathbf{y} = (\mathbf{I} - \mathbf{M}_2)\mathbf{y} - \tilde{\mathbf{M}}_1\mathbf{y} = \tilde{\mathbf{y}} - \tilde{\mathbf{X}}_1\tilde{\mathbf{b}}$, which is the residual of the regression of $\tilde{\mathbf{y}}$ on $\tilde{\mathbf{X}}_1$, because $\tilde{\mathbf{X}}_1\tilde{\mathbf{b}} = \tilde{\mathbf{M}}_1\mathbf{y} = \tilde{\mathbf{M}}_1\tilde{\mathbf{y}}$.

■

The proof uses nothing beyond the orthogonal decomposition of the model space. Part (a) contains a small surprise. Residualizing $\mathbf{y}$ is optional, because the part of $\mathbf{y}$ lying in $\mathcal{C}(\mathbf{X}_2)$ is orthogonal to $\tilde{\mathbf{X}}_1$ and drops out of $\tilde{\mathbf{X}}_1^\top \mathbf{y}$ anyway. Residualizing the regressor of interest is

essential.

Idea. In a multiple regression, $\hat{\beta}_j$ measures how $\mathbf{y}$ moves with *the part of $\mathbf{x}_j$ that is orthogonal to the other regressors*. A regressor that is almost a linear combination of the others has a tiny orthogonal part, so its coefficient rests on very little information. That is collinearity, seen geometrically.

6.6.2 Consequences

Variances. The FWL formula writes $\hat{\beta}_1$ as a linear function of $\mathbf{y}$. If $\text{Cov}(\mathbf{Y}) = \sigma^2 \mathbf{I}$, then

$$\text{Cov}(\hat{\beta}_1) = \sigma^2 (\tilde{\mathbf{X}}_1^\top \tilde{\mathbf{X}}_1)^{-1} \tilde{\mathbf{X}}_1^\top \tilde{\mathbf{X}}_1 (\tilde{\mathbf{X}}_1^\top \tilde{\mathbf{X}}_1)^{-1} = \sigma^2 (\tilde{\mathbf{X}}_1^\top \tilde{\mathbf{X}}_1)^{-1}.$$

Comparing with $\text{Cov}(\hat{\beta}) = \sigma^2 (\mathbf{X}^\top \mathbf{X})^{-1}$ (Theorem 5.5.1) identifies the leading block of the inverse of a partitioned matrix,

$$[(\mathbf{X}^\top \mathbf{X})^{-1}]_{11} = (\mathbf{X}_1^\top (\mathbf{I} - \mathbf{M}_2) \mathbf{X}_1)^{-1} = (\mathbf{X}_1^\top \mathbf{X}_1 - \mathbf{X}_1^\top \mathbf{X}_2 (\mathbf{X}_2^\top \mathbf{X}_2)^{-1} \mathbf{X}_2^\top \mathbf{X}_1)^{-1}, \quad (6.6.1)$$

a Schur-complement identity that Chapter 1 proves by algebra. For a single regressor $\mathbf{x}_j$, with all other columns (including the intercept) in $\mathbf{X}_2$,

$$\text{Var}(\hat{\beta}_j) = \frac{\sigma^2}{\|(\mathbf{I} - \mathbf{M}_{(j)})\mathbf{x}_j\|^2} = \frac{\sigma^2}{S_{jj}(1 - R_j^2)}, \quad (6.6.2)$$

where $S_{jj} = \sum_i (x_{ij} - \bar{x}_j)^2$ and R_j^2 is the coefficient of determination from regressing $\mathbf{x}_j$ on the other regressors. The second equality is Exercise (B1). The factor $1/(1 - R_j^2)$ is the *variance inflation factor* of Chapter 26.

Centring. Take $\mathbf{X}_2 = \mathbf{1}$. Then $\mathbf{M}_2 = n^{-1} \mathbf{1} \mathbf{1}^\top$ and $\tilde{\mathbf{X}}_1, \tilde{\mathbf{y}}$ are the centred regressors and response. FWL says the slopes in a model with an intercept are the coefficients of a no-intercept regression of centred $\mathbf{y}$ on centred regressors, and part (c) gives the intercept $\bar{y} - \bar{\mathbf{x}}^\top \hat{\beta}_1$. Example (Intercept inside simple regression) was the case $p_1 = 1$.

Detrending and deseasonalizing. The original application was economic time series. Frisch and Waugh (1933) showed that regressing on a time trend together with other variables gives the same slope as first detrending every series. Lovell (1963) proved the analogous statement for seasonal dummy variables, and noted a trap: if only the *response* is seasonally adjusted and the regressors are not, the result is generally wrong. Part (a) of the theorem shows why. What must be residualized is the regressor.

A warning about standard errors. If the residualized regression of $\tilde{\mathbf{y}}$ on $\tilde{\mathbf{X}}_1$ is fitted with ordinary software, the coefficient and residuals are right, but the software divides the residual sum of squares by $n - p_1$ instead of $n - p_1 - p_2$. It does not know that p_2 dimensions were used up in forming $\tilde{\mathbf{y}}$. The reported standard error is too small by the factor $\sqrt{(n - p_1)/(n - p)}$.

Example. (Poverty and murder rates)

For the 50 US states in 2009, regress the murder rate (per 100,000) on the poverty rate, the percentage of single-parent households, and the percentage of the population living in urban areas, with an intercept. The coefficient of poverty is 0.2538. On its own,

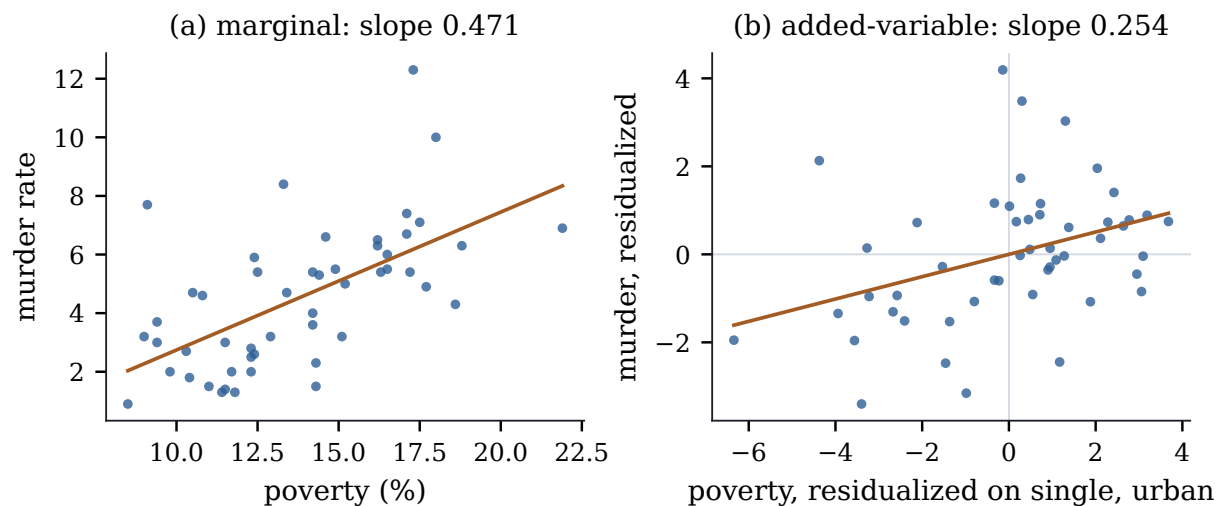


Figure 6.6.1: Murder rate and poverty for 50 US states. (a) The marginal relationship. (b) The added-variable plot: both variables residualized on the single-parent percentage, urban percentage and intercept. The slope in (b) is the multiple-regression coefficient of poverty (Theorem 6.6.2).

a simple regression of murder rate on poverty has slope 0.4706, nearly twice as large. Poverty and single parenthood are correlated, and the simple regression gives poverty credit for part of their shared association.

Figure 6.6.1 shows both regressions. Panel (b) is an **added-variable plot**: the residual of murder rate after regression on the other two regressors and the intercept, against the corresponding residual of poverty. By FWL its least squares slope is exactly the multiple-regression coefficient. The listing confirms the equality to machine precision.

The standard error of the coefficient in the full model is 0.0934. Fitting the residualized regression naively gives 0.0905, smaller by the factor 1.0321, which is $\sqrt{(n-1)/(n-4)}$ as predicted.

```
import numpy as np
import statsmodels.api as sm
data = sm.datasets.statcrime.load_pandas().data
data = data.drop(index="District of Columbia")      # an extreme point; see the
                                                    # exercises of Section 6.6

y = data["murder"].to_numpy()
x1 = data["poverty"].to_numpy()                    # the coefficient we care
                                                    # about
X2 = np.column_stack([np.ones(len(y)), data["single"], data["urban"]])
X = np.column_stack([x1, X2])

def resid(v, Z):
    """Residual of v after projecting onto C(Z)."""
    coef, *_ = np.linalg.lstsq(Z, v, rcond=None)
    return v - Z @ coef

beta_full, *_ = np.linalg.lstsq(X, y, rcond=None)  # regress y on everything

y_tilde = resid(y, X2)                            # (I - M2) y
x_tilde = resid(x1, X2)                           # (I - M2) x1
```

```

beta_fwl = (x_tilde @ y_tilde) / (x_tilde @ x_tilde) # simple regression, no
            intercept

print(f"full regression:  {beta_full[0]:.6f}")
print(f"FWL regression:   {beta_fwl:.6f}")

```

Remark (What FWL does not say). FWL is an algebraic identity about fitted coefficients. It says what the number $\hat{\beta}_j$ is: an association with the unexplained part of $\mathbf{x}_j$. It does not say that changing x_j in the world, with the other regressors held fixed, would change the response by $\hat{\beta}_j$. That is a causal claim, and it needs assumptions about how the data arose. Chapter 25 takes up this question, including cases where adding a regressor to $\mathbf{X}_2$ makes the coefficient *less* interpretable rather than more.

6.6.3 Fixed effects are a projection

One of the most widely used estimators in applied statistics is the FWL theorem in disguise. Suppose units $k = 1, \dots, K$ are observed repeatedly: firms over years, patients over visits, schools over cohorts. The analyst wants the effect of time-varying regressors $\mathbf{X}_1$ while allowing each unit its own intercept:

$$E(\mathbf{Y}) = \mathbf{X}_1\boldsymbol{\beta}_1 + \mathbf{D}\boldsymbol{\alpha},$$

where $\mathbf{D}$ is the $n \times K$ matrix of unit indicators. Fitting this directly means a regression with K extra columns, and K may be in the thousands.

FWL with $\mathbf{X}_2 = \mathbf{D}$ removes that burden. The projection onto $\mathcal{C}(\mathbf{D})$ replaces each entry by its unit's average (Example (The one-way layout)), so $\mathbf{I} - \mathbf{M}_D$ subtracts unit means. Therefore $\hat{\boldsymbol{\beta}}_1$ is the least squares coefficient from regressing the *within-unit deviations* $y_{kt} - \bar{y}_k$ on $\mathbf{x}_{kt} - \bar{\mathbf{x}}_k$, with no indicators at all. This is the **within estimator** of panel-data econometrics. Its standard error needs the degrees-of-freedom correction described above, with $p_2 = K$, which is large.

Example. (Grunfeld's investment data)

Grunfeld's classic panel follows 11 US firms for 20 years, giving $n = 220$ observations of gross investment, market value and capital stock. With a separate intercept for each firm, the coefficients of value and capital are 0.1101 and 0.3100. The same numbers come from demeaning every variable within firm and fitting two columns. Ignoring the firms altogether (one common intercept) gives 0.1145 and 0.2275 instead. That comparison uses variation *between* firms, of the sort "large firms invest more", which the within estimator deliberately discards.

```

import numpy as np
import pandas as pd
import statsmodels.api as sm
df = sm.datasets.grunfeld.load_pandas().data
y = df["invest"].to_numpy()
X1 = df[["value", "capital"]].to_numpy()

# (i) least squares with one indicator column per firm ("dummy variables")
D = pd.get_dummies(df["firm"]).to_numpy(dtype=float)
beta_dummies, *_ = np.linalg.lstsq(np.column_stack([X1, D]), y, rcond=None)

```



```

# (ii) FWL: projecting onto  $\mathcal{C}(D)$  replaces each value by its firm mean,
#      so  $(I - M_D)$  subtracts firm means -- the "within" transformation
def demean_by(v, groups):
    v = pd.DataFrame(v)
    return (v - v.groupby(groups.to_numpy()).transform("mean")).to_numpy()

X1_within = demean_by(X1, df["firm"])
y_within = demean_by(y, df["firm"]).ravel()
beta_within, *_ = np.linalg.lstsq(X1_within, y_within, rcond=None)

print("with firm indicators:", beta_dummies[:2])
print("within transformation:", beta_within)

```

Whether discarding between-unit variation is wise depends on why units differ. That question is taken up for random-effects alternatives in Chapter 32 and Chapter 33.

Added-variable plots are among the most useful diagnostics for multiple regression. They show the evidence for a single coefficient in two dimensions, together with any points that dominate it. They reappear in Chapter 20.

6.6.4 Exercises

6.6.4.1 A. Check your understanding

Exercise A1.

Refit the model of [Example \(Poverty and murder rates\)](#) with the District of Columbia included. How does the poverty coefficient change? Draw the added-variable plot and locate the District. Relate what you see to its leverage ([Example \(High-leverage states\)](#)).

6.6.4.2 B. Practice

Exercise B1.

Let $\mathbf{x}_j$ be a column of $\mathbf{X}$, let $\mathbf{X}_{(j)}$ be $\mathbf{X}$ with that column removed, and assume $\mathbf{1} \in \mathcal{C}(\mathbf{X}_{(j)})$. Show that $\|(\mathbf{I} - \mathbf{M}_{(j)})\mathbf{x}_j\|^2 = S_{jj}(1 - R_j^2)$, where R_j^2 is the R^2 from regressing $\mathbf{x}_j$ on $\mathbf{X}_{(j)}$. Deduce [Eq. 6.6.2](#).

Solution. Since $\mathbf{1} \in \mathcal{C}(\mathbf{X}_{(j)})$, the vector $(\mathbf{I} - \mathbf{M}_{(j)})\mathbf{x}_j$ is the residual from regressing $\mathbf{x}_j$ on $\mathbf{X}_{(j)}$. Its squared length is that regression's residual sum of squares, which equals $\text{SST}(1 - R_j^2)$ with $\text{SST} = S_{jj}$ by the definition of R^2 . Substituting into the variance formula from FWL gives [Eq. 6.6.2](#).

Exercise B2.

In [Theorem 6.6.2](#), regressing $\mathbf{y}$ (not $\tilde{\mathbf{y}}$) on $\tilde{\mathbf{X}}_1$ gives the coefficient $\hat{\beta}_1$. Show that its residual vector is $(\mathbf{I} - \mathbf{M})\mathbf{y} + \mathbf{M}_2\mathbf{y}$, so the residuals are generally wrong even though the coefficient is right.

Solution. The fitted vector of the regression of $\mathbf{y}$ on $\tilde{\mathbf{X}}_1$ is $\tilde{\mathbf{M}}_1\mathbf{y}$. By Lemma 6.6.1, $\mathbf{y} - \tilde{\mathbf{M}}_1\mathbf{y} = \mathbf{y} - (\mathbf{M} - \mathbf{M}_2)\mathbf{y} = (\mathbf{I} - \mathbf{M})\mathbf{y} + \mathbf{M}_2\mathbf{y}$. The extra term $\mathbf{M}_2\mathbf{y}$ is zero only if $\mathbf{y} \perp \mathcal{C}(\mathbf{X}_2)$.

Exercise B3.

For a single regressor $\mathbf{x}_j$, the *partial correlation* of $\mathbf{y}$ and $\mathbf{x}_j$ given the other regressors is the sample correlation of $\tilde{\mathbf{y}}$ and $\tilde{\mathbf{x}}_j$ (both residualized on $\mathbf{X}_{(j)}$, which contains $\mathbf{1}$). Let $t = \hat{\beta}_j / \text{se}(\hat{\beta}_j)$ with $\text{se}(\hat{\beta}_j)^2 = \text{SSE} / [(n - p) \|\tilde{\mathbf{x}}_j\|^2]$. Show that the squared partial correlation equals $t^2 / (t^2 + n - p)$.

Solution. Write $\tilde{\mathbf{x}} = \tilde{\mathbf{x}}_j$ and $\tilde{\mathbf{y}} = \tilde{\mathbf{y}}$. Both have mean zero because $\mathbf{1} \in \mathcal{C}(\mathbf{X}_{(j)})$, so their correlation is $r = \tilde{\mathbf{x}}^\top \tilde{\mathbf{y}} / (\|\tilde{\mathbf{x}}\| \|\tilde{\mathbf{y}}\|)$. By FWL, $\hat{\beta}_j = \tilde{\mathbf{x}}^\top \tilde{\mathbf{y}} / \|\tilde{\mathbf{x}}\|^2$, and the full-model residual sum of squares equals that of the residualized regression: $\text{SSE} = \|\tilde{\mathbf{y}}\|^2 - (\tilde{\mathbf{x}}^\top \tilde{\mathbf{y}})^2 / \|\tilde{\mathbf{x}}\|^2 = \|\tilde{\mathbf{y}}\|^2 (1 - r^2)$. Hence

$$t^2 = \frac{\hat{\beta}_j^2 (n - p) \|\tilde{\mathbf{x}}\|^2}{\text{SSE}} = \frac{(n - p) r^2}{1 - r^2},$$

and solving gives $r^2 = t^2 / (t^2 + n - p)$.

6.6.4.3 C. Going deeper

Exercise C1.

Drop the full-rank assumption in Theorem 6.6.2. Show that $\mathbf{b}_1$ solves the normal equations of the regression of $\tilde{\mathbf{y}}$ on $\tilde{\mathbf{X}}_1$ iff there exists $\mathbf{b}_2$ such that $(\mathbf{b}_1^\top, \mathbf{b}_2^\top)^\top$ is a least squares estimate in the full model.

6.7 Reparameterization, centring and R^2

6.7.1 Same subspace, same fit

The key idea of Section 6.1 now becomes a theorem.

Theorem 6.7.1 — Invariance under reparameterization

Let $\mathbf{X}$ ($n \times p$) and $\mathbf{W}$ ($n \times q$) satisfy $\mathcal{C}(\mathbf{X}) = \mathcal{C}(\mathbf{W})$. Then the two linear models $E(\mathbf{Y}) = \mathbf{X}\boldsymbol{\beta}$ and $E(\mathbf{Y}) = \mathbf{W}\boldsymbol{\gamma}$ have the same projection $\mathbf{M}$, the same fitted values and residuals for every $\mathbf{y}$, and the same residual sum of squares. There are a $p \times q$ matrix $\mathbf{T}$ with $\mathbf{W} = \mathbf{X}\mathbf{T}$ and a $q \times p$ matrix $\mathbf{S}$ with $\mathbf{X} = \mathbf{W}\mathbf{S}$. If $\hat{\boldsymbol{\gamma}}$ is a least squares estimate for $\mathbf{W}$, then $\mathbf{T}\hat{\boldsymbol{\gamma}}$ is one for $\mathbf{X}$, and $\mathbf{S}\hat{\boldsymbol{\beta}}$ is one for $\mathbf{W}$ whenever $\hat{\boldsymbol{\beta}}$ is one for $\mathbf{X}$. If both matrices have full column rank, then $p = q$, $\mathbf{T} = \mathbf{S}^{-1}$, and $\hat{\boldsymbol{\gamma}} = \mathbf{S}\hat{\boldsymbol{\beta}}$.

Proof. A projection is determined by its range (Theorem 6.3.3), so both models have projection $\mathbf{M}$, and fitted values and residuals follow from Theorem 6.4.1. Each column

of $\mathbf{W}$ lies in $\mathcal{C}(\mathbf{X})$, so $\mathbf{W} = \mathbf{X}\mathbf{T}$ for some $\mathbf{T}$, and likewise $\mathbf{X} = \mathbf{W}\mathbf{S}$. If $\mathbf{W}\hat{\boldsymbol{\gamma}} = \mathbf{M}\mathbf{y}$, then $\mathbf{X}(\mathbf{T}\hat{\boldsymbol{\gamma}}) = \mathbf{M}\mathbf{y}$, so $\mathbf{T}\hat{\boldsymbol{\gamma}}$ is a least squares estimate for $\mathbf{X}$. The other direction is symmetric. With full column rank, $\mathbf{X} = \mathbf{X}\mathbf{T}\mathbf{S}$ gives $\mathbf{X}(\mathbf{I} - \mathbf{T}\mathbf{S}) = \mathbf{0}$ and so $\mathbf{T}\mathbf{S} = \mathbf{I}$, and similarly $\mathbf{S}\mathbf{T} = \mathbf{I}$. Uniqueness of least squares estimates then gives $\hat{\boldsymbol{\gamma}} = \mathbf{S}\hat{\boldsymbol{\beta}}$. ■

Many operations that change the printed coefficient table leave the fit exactly unchanged:

- **Changing units** of a regressor (metres to centimetres, dollars to thousands of dollars) rescales one column: $\mathbf{W} = \mathbf{X}\mathbf{D}$ with $\mathbf{D}$ diagonal. The coefficient is divided by the scale factor, and nothing else changes.
- **Centring** regressors, $\mathbf{x}_j \mapsto \mathbf{x}_j - \bar{x}_j\mathbf{1}$, when the model has an intercept. The slopes are unchanged, and the intercept becomes the fitted value at the mean of the regressors instead of at zero, which is often far outside the data.
- **Recoding a factor**. Reference-level dummies, sum-to-zero effect coding, and cell-means coding without an intercept all span the space of vectors that are constant within levels.
- **Replacing polynomial terms by orthogonal polynomials** (Chapter 42), which span the same space with much better numerical behaviour (Section 6.10).

Operations that *do* change the fit are exactly those that change $\mathcal{C}(\mathbf{X})$. Examples are dropping the intercept when no other combination of columns produces $\mathbf{1}$, replacing a regressor by its logarithm, and adding an interaction.

6.7.2 Centring as a projection

Suppose $\mathbf{1} \in \mathcal{C}(\mathbf{X})$. By Theorem 6.5.1 with $\mathbf{X}_0 = \mathbf{1}$, the fitted vector splits as

$$\mathbf{M}\mathbf{y} = \bar{y}\mathbf{1} + (\mathbf{M} - n^{-1}\mathbf{1}\mathbf{1}^\top)\mathbf{y},$$

and the second projection is onto $\mathcal{C}(\mathbf{X}) \cap \mathbf{1}^\perp$, the vectors in the model space with entries summing to zero. By Lemma 6.6.1 with $\mathbf{X}_2 = \mathbf{1}$, that subspace is spanned by the centred columns $(\mathbf{I} - n^{-1}\mathbf{1}\mathbf{1}^\top)\mathbf{X}$. Applying Pythagoras to the orthogonal pieces $\hat{\mathbf{Y}} - \bar{y}\mathbf{1}$ and $\hat{\boldsymbol{\epsilon}}$ of $\mathbf{y} - \bar{y}\mathbf{1}$ gives the *corrected* decomposition

$$\underbrace{\sum_i (y_i - \bar{y})^2}_{\text{SST}} = \underbrace{\sum_i (\hat{y}_i - \bar{y})^2}_{\text{SSR}} + \underbrace{\sum_i (y_i - \hat{y}_i)^2}_{\text{SSE}}, \quad (6.7.1)$$

with dimensions $n - 1 = (r - 1) + (n - r)$. Chapter 9 builds the analysis of variance on this identity.

6.7.3 R^2 is a squared cosine

The **coefficient of determination** of a model with $\mathbf{1} \in \mathcal{C}(\mathbf{X})$ is

$$R^2 = \frac{\text{SSR}}{\text{SST}} = 1 - \frac{\text{SSE}}{\text{SST}}.$$

It depends on the model only through $\mathcal{C}(\mathbf{X})$, so by Theorem 6.7.1 it is invariant under every reparameterization. Its geometric meaning is simple.

inside $\mathbf{1}^\perp$: $R^2 = \cos^2 \theta = 0.617$

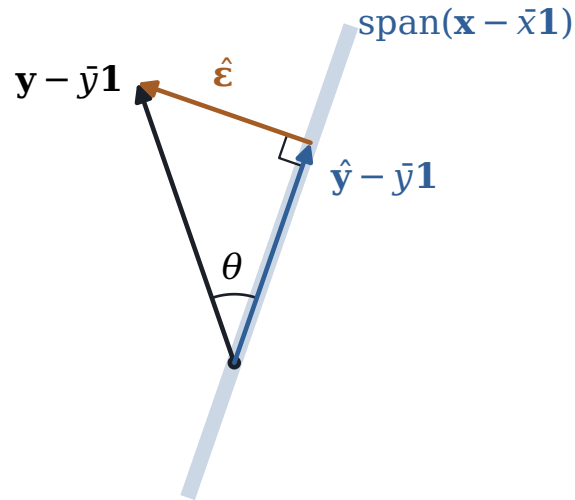


Figure 6.7.1: R^2 as a squared cosine, for $n = 3$ and one regressor. Centred vectors lie in the plane $\mathbf{1}^\perp$, so the whole configuration is two-dimensional. The centred fitted vector is the projection of the centred response onto the line spanned by the centred regressor.

Theorem 6.7.2

Assume $\mathbf{1} \in \mathcal{C}(\mathbf{X})$ and $\mathbf{y} \notin \text{span}(\mathbf{1})$. Let θ be the angle between the centred response $\mathbf{y} - \bar{y}\mathbf{1}$ and the centred fitted vector $\hat{\mathbf{Y}} - \bar{y}\mathbf{1}$. Then

$$R^2 = \cos^2 \theta = \text{corr}(\mathbf{y}, \hat{\mathbf{Y}})^2,$$

where corr is the sample correlation. For simple regression on one regressor $\mathbf{x}$, $R^2 = \text{corr}(\mathbf{y}, \mathbf{x})^2$.

Proof. Write $\mathbf{u} = \mathbf{y} - \bar{y}\mathbf{1}$ and $\mathbf{v} = \hat{\mathbf{Y}} - \bar{y}\mathbf{1}$. Then $\mathbf{u} = \mathbf{v} + \hat{\mathbf{e}}$ with $\mathbf{v} \perp \hat{\mathbf{e}}$, so $\mathbf{u}^\top \mathbf{v} = \|\mathbf{v}\|^2$ and

$$\cos^2 \theta = \frac{(\mathbf{u}^\top \mathbf{v})^2}{\|\mathbf{u}\|^2 \|\mathbf{v}\|^2} = \frac{\|\mathbf{v}\|^2}{\|\mathbf{u}\|^2} = \frac{\text{SSR}}{\text{SST}}.$$

(If $\mathbf{v} = \mathbf{0}$ the angle is undefined, but $R^2 = 0$ and the correlation is taken to be zero.) Since $\hat{\mathbf{Y}}$ has mean $\bar{y}$ (Proposition 6.4.3(b)), $\cos \theta$ is the sample correlation of $\mathbf{y}$ and $\hat{\mathbf{Y}}$. In simple regression $\mathbf{v} = (S_{xy}/S_{xx})(\mathbf{x} - \bar{x}\mathbf{1})$ by Example (Intercept inside simple regression), a scalar multiple of the centred regressor. Rescaling a vector by a nonzero constant does not change a squared cosine, so $\cos^2 \theta$ is the squared correlation of $\mathbf{y}$ and $\mathbf{x}$. When $S_{xy} = 0$ both sides are zero. ■

So R^2 measures how closely the model space lines up with the centred data. Figure 6.7.1 draws the case $n = 3$. For the three-regressor model of Example (Poverty and murder rates), $R^2 = 0.6419$, meaning the centred response makes an angle of 36.8° with the model space. The simple regression on poverty alone has $R^2 = 0.3599$, the square of the correlation 0.5999.

```

import numpy as np
import statsmodels.api as sm

def r_squared_as_cosine(y, X):
    """X must contain the intercept column. Returns (R^2 from sums of squares, cos^2
    of angle)."""
    beta, *_ = np.linalg.lstsq(X, y, rcond=None)
    y_hat = X @ beta
    yc, yhc = y - y.mean(), y_hat - y.mean()          # centre both vectors
    r2 = 1 - np.sum((y - y_hat) ** 2) / np.sum(yc ** 2)
    cos = (yc @ yhc) / (np.linalg.norm(yc) * np.linalg.norm(yhc))
    return r2, cos ** 2

```

Three consequences follow directly from the geometry.

1. R^2 never decreases when the model space grows. If $\mathcal{C}(\mathbf{X}_0) \subseteq \mathcal{C}(\mathbf{X})$ and both contain $\mathbf{1}$, then $\text{SSE}_0 - \text{SSE} = \|(\mathbf{M} - \mathbf{M}_0)\mathbf{y}\|^2 \geq 0$ by [Eq. 6.5.2](#). Adding a regressor of pure noise increases R^2 almost surely. This is why R^2 cannot be used for model selection (Chapter 29).
2. $R^2 = 1$ once the model space contains $\mathbf{y}$, which always happens when $\text{rank}(\mathbf{X}) = n$. A model with as many free parameters as observations interpolates, and its R^2 says nothing.
3. Without an intercept, the definition changes. If $\mathbf{1} \notin \mathcal{C}(\mathbf{X})$, the corrected decomposition [Eq. 6.7.1](#) fails, because $\hat{\mathbf{Y}} - \bar{y}\mathbf{1}$ and $\hat{\mathbf{e}}$ need not be orthogonal. Software then usually reports the *unc centred* $R_0^2 = \|\hat{\mathbf{Y}}\|^2 / \|\mathbf{y}\|^2$, the squared cosine of the angle between $\mathbf{y}$ and $\mathcal{C}(\mathbf{X})$ itself. It is typically much larger and cannot be compared with the centred R^2 of a model with an intercept ([Exercise \(B1\)](#)).

6.7.4 Exercises

6.7.4.1 A. Check your understanding

Exercise A1.

A factor with three levels is coded with an intercept and indicators for levels 2 and 3. Write down the matrices $\mathbf{T}$ and $\mathbf{S}$ of [Theorem 6.7.1](#) that convert to the coding in which level 3, rather than level 1, is the omitted baseline, and to sum-to-zero effect coding. Express each new coefficient in terms of the old ones.

Exercise A2.

Show that, for a model with $\mathbf{1} \in \mathcal{C}(\mathbf{X})$, R^2 is unchanged when $\mathbf{y}$ is replaced by $a\mathbf{1} + b\mathbf{y}$ with $b \neq 0$, and when any column of $\mathbf{X}$ is replaced by an affine function of itself. Is it unchanged when a regressor is replaced by its logarithm?

6.7.4.2 B. Practice

Exercise B1.

Show that if $\mathbf{1} \in \mathcal{C}(\mathbf{X})$ then $R_0^2 = \|\hat{\mathbf{Y}}\|^2 / \|\mathbf{y}\|^2 = (n\bar{y}^2 + \text{SSR}) / (n\bar{y}^2 + \text{SST}) \geq R^2$.

Construct a data set and a model without an intercept for which $R_0^2 > 0.95$ but the model's residual sum of squares exceeds that of the intercept-only model.

Solution. With $\mathbf{1} \in \mathcal{C}(\mathbf{X})$, the fitted vector splits orthogonally as $\bar{y}\mathbf{1} + (\hat{\mathbf{Y}} - \bar{y}\mathbf{1})$, so $\|\hat{\mathbf{Y}}\|^2 = n\bar{y}^2 + \text{SSR}$, and likewise $\|\mathbf{y}\|^2 = n\bar{y}^2 + \text{SST}$. For $a \geq 0$ and $0 \leq \text{SSR} \leq \text{SST}$, $(a + \text{SSR})/(a + \text{SST}) \geq \text{SSR}/\text{SST}$. For the example, take responses near 100 with small noise and a regressor unrelated to them, and fit through the origin. The fitted line captures the large mean, so R_0^2 is near 1, but its residuals exceed those around $\bar{y}$.

6.7.4.3 C. Going deeper

Exercise C1.

Suppose $\mathbf{Y} \sim N_n(\mu\mathbf{1}, \sigma^2\mathbf{I})$ and $\mathbf{1} \in \mathcal{C}(\mathbf{X})$ with $\text{rank}(\mathbf{X}) = r$. Using the results of Chapter 4, show that $R^2 \sim \text{Beta}((r-1)/2, (n-r)/2)$, so that $E(R^2) = (r-1)/(n-1)$ even though no regressor is related to the response. Use this to motivate the adjusted $R^2 = 1 - (1 - R^2)(n-1)/(n-r)$.

6.8 Leverage as geometry

The projection $\mathbf{M}$ is an $n \times n$ matrix whose entries have a direct interpretation. Since $\hat{y}_i = \sum_j m_{ij}y_j$, the entry m_{ij} is the weight observation j receives in the fitted value for observation i . So

$$m_{ij} = \frac{\partial \hat{y}_i}{\partial y_j}.$$

The diagonal entries measure how strongly each observation pulls the fitted surface towards itself.

Definition 6.8.1 — Leverage

The **leverage** of observation i is $h_{ii} = m_{ii}$, the i th diagonal entry of $\mathbf{M}$. In this context $\mathbf{M}$ is often called the **hat matrix** and written $\mathbf{H}$, because it “puts the hat on” $\mathbf{y}$.

Leverage is a property of the design, not of the response. It depends on $\mathbf{y}$ not at all, and on $\mathbf{X}$ only through $\mathcal{C}(\mathbf{X})$, so it is unchanged by reparameterization.

Proposition 6.8.1 — Properties of leverage

Let $\text{rank}(\mathbf{X}) = r$, and let $\mathbf{e}_i$ denote the i th standard basis vector. Then:

- (a) $h_{ii} = \|\mathbf{M}\mathbf{e}_i\|^2 = \sum_j m_{ij}^2$, the squared length of the projection of $\mathbf{e}_i$ onto $\mathcal{C}(\mathbf{X})$;
- (b) $0 \leq h_{ii} \leq 1$, and $\sum_i h_{ii} = r$, so the average leverage is r/n ;
- (c) if $\mathbf{1} \in \mathcal{C}(\mathbf{X})$, then $h_{ii} \geq 1/n$;
- (d) $h_{ii} = 1$ iff $\mathbf{e}_i \in \mathcal{C}(\mathbf{X})$. In that case $\hat{y}_i = y_i$ for every $\mathbf{y}$, the i th residual is always zero, and $m_{ij} = 0$ for $j \neq i$;

- (e) if $\mathbf{X} = \mathbf{QR}$ with $\mathbf{Q}$ having orthonormal columns spanning $\mathcal{C}(\mathbf{X})$, then h_{ii} is the squared length of the i th row of $\mathbf{Q}$.

Proof.

- (a) $h_{ii} = \mathbf{e}_i^\top \mathbf{M} \mathbf{e}_i = \mathbf{e}_i^\top \mathbf{M}^\top \mathbf{M} \mathbf{e}_i$, and $\mathbf{M} \mathbf{e}_i$ is the i th column of $\mathbf{M}$.
 (b) By [Proposition 6.3.4\(d\)](#), $0 \leq \|\mathbf{M} \mathbf{e}_i\|^2 \leq \|\mathbf{e}_i\|^2 = 1$. The sum is $\text{tr}(\mathbf{M}) = r$.
 (c) With $\mathbf{M}_0 = n^{-1} \mathbf{1} \mathbf{1}^\top$, the difference $\mathbf{M} - \mathbf{M}_0$ is a projection ([Theorem 6.5.1](#)), hence nonnegative definite, so $h_{ii} \geq (\mathbf{M}_0)_{ii} = 1/n$.
 (d) Equality holds in $\|\mathbf{M} \mathbf{e}_i\| \leq \|\mathbf{e}_i\|$ iff $(\mathbf{I} - \mathbf{M}) \mathbf{e}_i = \mathbf{0}$, that is, iff $\mathbf{e}_i \in \mathcal{C}(\mathbf{X})$. Then $\mathbf{M} \mathbf{e}_i = \mathbf{e}_i$, so column i of $\mathbf{M}$ is $\mathbf{e}_i$, and by symmetry so is row i . Hence $\hat{y}_i = y_i$.
 (e) $\mathbf{M} = \mathbf{Q} \mathbf{Q}^\top$, so $m_{ii} = \sum_k q_{ik}^2$.

■

Part (d) describes the extreme case. An observation with leverage one has a coordinate direction of its own in the model space. This happens, for example, when it is the only member of a group in a one-way layout, or when a column of $\mathbf{X}$ is an indicator for that single observation. The model then fits the observation exactly, whatever its value, and the data give no check on it. High leverage is a weaker form of the same thing. If h_{ii} is close to one, then $\hat{y}_i \approx y_i$, and a gross error in y_i barely shows up in its residual.

This shows up in the residual variances. If $\text{Cov}(\boldsymbol{\varepsilon}) = \sigma^2 \mathbf{I}$, then by [Proposition 6.4.3\(e\)](#)

$$\text{Var}(\hat{\varepsilon}_i) = \sigma^2(1 - h_{ii}),$$

so residuals at high-leverage points are *less* variable than others, even though those are exactly the points where an error would do the most damage. Chapter 20 standardizes residuals to correct for this.

6.8.1 Leverage as distance from the centre

When the model has an intercept, leverage has a second description in variable space: distance from the centroid of the regressors. Write $\mathbf{X} = [\mathbf{1}, \mathbf{X}_1]$, let $\bar{\mathbf{x}}$ be the vector of column means of $\mathbf{X}_1$, and let $\tilde{\mathbf{X}}_1 = (\mathbf{I} - n^{-1} \mathbf{1} \mathbf{1}^\top) \mathbf{X}_1$ be the centred regressors, assumed to have full column rank.

Proposition 6.8.2

Under these assumptions,

$$h_{ii} = \frac{1}{n} + (\mathbf{x}_{(i)} - \bar{\mathbf{x}})^\top (\tilde{\mathbf{X}}_1^\top \tilde{\mathbf{X}}_1)^{-1} (\mathbf{x}_{(i)} - \bar{\mathbf{x}}),$$

where $\mathbf{x}_{(i)}$ is the i th row of $\mathbf{X}_1$, written as a column vector.

Proof. By [Lemma 6.6.1](#) with $\mathbf{X}_2 = \mathbf{1}$, $\mathbf{M} = n^{-1} \mathbf{1} \mathbf{1}^\top + \tilde{\mathbf{X}}_1 (\tilde{\mathbf{X}}_1^\top \tilde{\mathbf{X}}_1)^{-1} \tilde{\mathbf{X}}_1^\top$. The i th row of $\tilde{\mathbf{X}}_1$ is $(\mathbf{x}_{(i)} - \bar{\mathbf{x}})^\top$.

■

The quadratic form is $(n-1)^{-1}$ times the squared Mahalanobis distance of $\mathbf{x}_{(i)}$ from the centroid, measured in the metric of the sample covariance matrix of the regressors. Leverage is therefore high for points that are unusual *jointly*, even when no single coordinate is extreme. A state with moderate poverty and moderate single parenthood, but an unusual *combination* of the two, can have high leverage. For simple regression the formula reduces to $h_{ii} = n^{-1} + (x_i - \bar{x})^2 / S_{xx}$.

Example. (High-leverage states)

Return to the state data, now with all 51 jurisdictions including the District of Columbia, and the model of [Example \(Poverty and murder rates\)](#). The average leverage is $p/n = 0.078$, and the smallest is 0.021, just above the bound $1/n$. The three largest are

Jurisdiction	h_{ii}	multiple of average
District of Columbia	0.506	6.5
Mississippi	0.212	2.7
Alaska	0.177	2.3

The District of Columbia is entirely urban, with by far the highest proportion of single-parent households. It sits at a corner of the regressor cloud that no state comes near, and its leverage is about half the maximum possible. It was omitted from [Example \(Poverty and murder rates\)](#). [Exercise \(A1\)](#) asks what happens to the poverty coefficient when it is put back.

```
import numpy as np
import statsmodels.api as sm
data = sm.datasets.statelcrime.load_pandas().data           # District of Columbia
# kept here
X = np.column_stack([np.ones(len(data)), data["poverty"], data["single"], data["urban"]])

Q, _ = np.linalg.qr(X)                                     # orthonormal basis for C(X)
h = np.sum(Q ** 2, axis=1)                                 # h_ii = m_ii = ||row i of Q||^2
n, p = X.shape
print("sum of leverages:", h.sum(), " (= p =", p, ")")
print("range:", h.min(), h.max(), " bounds:", 1 / n, 1)
top = np.argsort(h)[::-1][:3]
for i in top:
    print(f"{data.index[i]:22s} h = {h[i]:.3f}    ({h[i] / (p / n):.1f} x average)")
```

The listing computes leverages from a QR factorization rather than by forming $\mathbf{M}$. For large n the $n \times n$ matrix $\mathbf{M}$ should never be formed. [Section 6.10](#) explains why.

6.8.2 Exercises

6.8.2.1 A. Check your understanding

Exercise A1.

For simple regression, derive $h_{ii} = n^{-1} + (x_i - \bar{x})^2 / S_{xx}$ directly from [Example \(Intercept](#)

inside simple regression), and show that the largest leverage belongs to the observation with x_i farthest from $\bar{x}$.

6.8.2.2 B. Practice

Exercise B1.

Show that $m_{ij}^2 \leq m_{ii}(1 - m_{ii})$ for $i \neq j$, and hence $|m_{ij}| \leq 1/2$. When is the bound attained?

Solution. From [Proposition 6.8.1\(a\)](#), $m_{ii} = \sum_k m_{ik}^2 \geq m_{ii}^2 + m_{ij}^2$. So $m_{ij}^2 \leq m_{ii}(1 - m_{ii}) \leq 1/4$. Equality needs $m_{ii} = 1/2$ and m_{ij} the only other nonzero entry of row i . The intercept-only model with $n = 2$, where every entry of $\mathbf{M}$ is $1/2$, attains it.

Exercise B2.

Show that adding a column to $\mathbf{X}$ cannot decrease any leverage. Show that the leverages in a one-way layout are $1/n_k$ for members of group k , and explain why an observation forming a group on its own has leverage one.

6.9 Other inner products

Everything so far has used the Euclidean inner product $\mathbf{u}^\top \mathbf{v}$. That choice treats the n coordinates of observation space symmetrically: an error of a given size counts the same in every observation, and errors in different observations are unrelated. When observations have different variances or are correlated, Euclidean distance is the wrong way to measure how far $\mathbf{y}$ is from the model space. This section shows that the whole theory carries over to any inner product, and that the result is generalized least squares.

6.9.1 Projection in a general inner product

Let $\mathbf{A}$ be an $n \times n$ symmetric positive definite matrix, and define

$$\langle \mathbf{u}, \mathbf{v} \rangle_{\mathbf{A}} = \mathbf{u}^\top \mathbf{A} \mathbf{v}, \quad \|\mathbf{u}\|_{\mathbf{A}} = (\mathbf{u}^\top \mathbf{A} \mathbf{u})^{1/2}.$$

This is an inner product: bilinear, symmetric, and positive on nonzero vectors. Say $\mathbf{u}$ and $\mathbf{v}$ are $\mathbf{A}$ -orthogonal if $\mathbf{u}^\top \mathbf{A} \mathbf{v} = 0$. Nothing in the proofs of [Proposition 6.2.2](#), [Theorem 6.2.3](#) or [Theorem 6.3.1](#) used anything about $\mathbf{u}^\top \mathbf{v}$ except these three properties. So those results hold verbatim with $\langle \cdot, \cdot \rangle_{\mathbf{A}}$ in place of the Euclidean inner product. In particular, every $\mathbf{y}$ has a unique $\mathbf{A}$ -nearest point in a subspace $\mathcal{S}$, characterized by $\mathbf{A}$ -orthogonality of the error to $\mathcal{S}$. The map sending $\mathbf{y}$ to it is linear, and we write it $\mathbf{P}_{\mathcal{S}, \mathbf{A}}$.

Theorem 6.9.1

Let $\mathbf{A}$ be positive definite and $\mathbf{X}$ any $n \times p$ matrix. The $\mathbf{A}$ -orthogonal projection onto

$\mathcal{C}(\mathbf{X})$ is

$$\mathbf{P}_{\mathbf{X},\mathbf{A}} = \mathbf{X}(\mathbf{X}^\top \mathbf{A} \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{A},$$

for any generalized inverse. It satisfies $\mathbf{P}_{\mathbf{X},\mathbf{A}}^2 = \mathbf{P}_{\mathbf{X},\mathbf{A}}$ and $\mathcal{C}(\mathbf{P}_{\mathbf{X},\mathbf{A}}) = \mathcal{C}(\mathbf{X})$, and the matrix $\mathbf{A}\mathbf{P}_{\mathbf{X},\mathbf{A}}$ is symmetric. The matrix $\mathbf{P}_{\mathbf{X},\mathbf{A}}$ itself is symmetric iff $\mathcal{A}\mathcal{C}(\mathbf{X}) \subseteq \mathcal{C}(\mathbf{X})$, and in general it is not.

Proof. Let $\mathbf{A} = \mathbf{L}\mathbf{L}^\top$ with $\mathbf{L}$ nonsingular (for instance the Cholesky factor), and put $\mathbf{X}_* = \mathbf{L}^\top \mathbf{X}$ and $\mathbf{y}_* = \mathbf{L}^\top \mathbf{y}$. Then $\|\mathbf{y} - \mathbf{X}\mathbf{b}\|_{\mathbf{A}}^2 = \|\mathbf{y}_* - \mathbf{X}_*\mathbf{b}\|^2$, so $\mathbf{b}$ minimizes the $\mathbf{A}$ -distance iff it is an ordinary least squares estimate for $(\mathbf{y}_*, \mathbf{X}_*)$. By [Theorem 6.3.7](#) and [Theorem 6.4.2](#), one such estimate is $(\mathbf{X}_*^\top \mathbf{X}_*)^{-1} \mathbf{X}_*^\top \mathbf{y}_* = (\mathbf{X}^\top \mathbf{A} \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{A} \mathbf{y}$, and $\mathbf{X}_*(\mathbf{X}_*^\top \mathbf{X}_*)^{-1} \mathbf{X}_*^\top$ is invariant to the choice of generalized inverse. Multiplying on the left by $\mathbf{L}^{-\top}$ and on the right by $\mathbf{L}^\top$ shows that $\mathbf{P}_{\mathbf{X},\mathbf{A}} = \mathbf{L}^{-\top} \mathbf{M}_* \mathbf{L}^\top$, where $\mathbf{M}_*$ projects onto $\mathcal{C}(\mathbf{X}_*)$. Idempotence and $\mathcal{C}(\mathbf{P}_{\mathbf{X},\mathbf{A}}) = \mathbf{L}^{-\top} \mathcal{C}(\mathbf{X}_*) = \mathcal{C}(\mathbf{X})$ follow, and $\mathbf{A}\mathbf{P}_{\mathbf{X},\mathbf{A}} = \mathbf{L}\mathbf{M}_*\mathbf{L}^\top$ is symmetric. For the last claim, see [Exercise \(B1\)](#). ■

The proof also gives the general principle: *$\mathbf{A}$ -geometry is Euclidean geometry after the change of variables $\mathbf{y} \mapsto \mathbf{L}^\top \mathbf{y}$.* Any fact about orthogonal projections has an $\mathbf{A}$ -version, obtained by transporting it through $\mathbf{L}$.

Symmetry is the Euclidean signature of an orthogonal projection. The $\mathbf{A}$ -version of [Theorem 6.3.3](#) replaces it with self-adjointness.

Proposition 6.9.2

An idempotent matrix $\mathbf{P}$ is the $\mathbf{A}$ -orthogonal projection onto $\mathcal{C}(\mathbf{P})$ iff $\mathbf{A}\mathbf{P}$ is symmetric, that is, iff $\langle \mathbf{P}\mathbf{u}, \mathbf{v} \rangle_{\mathbf{A}} = \langle \mathbf{u}, \mathbf{P}\mathbf{v} \rangle_{\mathbf{A}}$ for all $\mathbf{u}, \mathbf{v}$.

Proof. For idempotent $\mathbf{P}$, the residual $(\mathbf{I} - \mathbf{P})\mathbf{y}$ is $\mathbf{A}$ -orthogonal to $\mathcal{C}(\mathbf{P})$ for every $\mathbf{y}$ iff $\mathbf{P}^\top \mathbf{A}(\mathbf{I} - \mathbf{P}) = \mathbf{0}$, that is, iff $\mathbf{P}^\top \mathbf{A} = \mathbf{P}^\top \mathbf{A}\mathbf{P}$. If $\mathbf{A}\mathbf{P}$ is symmetric, then $\mathbf{P}^\top \mathbf{A} = \mathbf{A}\mathbf{P}$, and $\mathbf{P}^\top \mathbf{A}\mathbf{P} = \mathbf{A}\mathbf{P}\mathbf{P} = \mathbf{A}\mathbf{P} = \mathbf{P}^\top \mathbf{A}$. Conversely, $\mathbf{P}^\top \mathbf{A} = \mathbf{P}^\top \mathbf{A}\mathbf{P}$ has a symmetric right-hand side, so $\mathbf{P}^\top \mathbf{A}$ and hence its transpose $\mathbf{A}\mathbf{P}$ are symmetric. ■

6.9.2 Generalized least squares

Suppose $\text{Cov}(\boldsymbol{\varepsilon}) = \sigma^2 \mathbf{V}$ with $\mathbf{V}$ known and positive definite. Observations with large variance should count for less, and correlated observations partly repeat each other. The natural distance is the one that makes the errors look uncorrelated: $\|\cdot\|_{\mathbf{V}^{-1}}$. If $\mathbf{V} = \mathbf{L}\mathbf{L}^\top$, then $\mathbf{L}^{-1}\boldsymbol{\varepsilon}$ has covariance $\sigma^2 \mathbf{I}$, and $\|\mathbf{u}\|_{\mathbf{V}^{-1}} = \|\mathbf{L}^{-1}\mathbf{u}\|$.

Definition 6.9.1 — Generalized least squares

A **generalized least squares** (GLS) estimate minimizes $(\mathbf{y} - \mathbf{X}\mathbf{b})^\top \mathbf{V}^{-1}(\mathbf{y} - \mathbf{X}\mathbf{b})$. Equivalently, $\mathbf{X}\hat{\boldsymbol{\beta}}_{\text{GLS}} = \mathbf{P}_{\mathbf{X},\mathbf{V}^{-1}}\mathbf{y}$, and $\hat{\boldsymbol{\beta}}_{\text{GLS}}$ solves the generalized normal equations

$$\mathbf{X}^\top \mathbf{V}^{-1} \mathbf{X} \mathbf{b} = \mathbf{X}^\top \mathbf{V}^{-1} \mathbf{y}.$$

Theorem 6.9.1 with $\mathbf{A} = \mathbf{V}^{-1}$ is the whole of the algebra. Computationally, GLS is ordinary least squares applied to the *whitened* data $(\mathbf{L}^{-1}\mathbf{y}, \mathbf{L}^{-1}\mathbf{X})$. Aitken (1935) showed that GLS gives the best linear unbiased estimator under this covariance, which is the Gauss–Markov theorem transported through $\mathbf{L}$. Chapter 31 proves this, and handles singular $\mathbf{V}$ and the more realistic case where $\mathbf{V}$ must be estimated.

When is the ordinary least squares fit already the GLS fit? Geometrically, the two projections onto $\mathcal{C}(\mathbf{X})$ coincide iff the Euclidean and the $\mathbf{V}^{-1}$ notions of orthogonality agree on this particular subspace.

Theorem 6.9.3 — Kruskal

For positive definite $\mathbf{V}$, $\mathbf{M}\mathbf{y} = \mathbf{P}_{\mathbf{X}, \mathbf{V}^{-1}}\mathbf{y}$ for every $\mathbf{y}$ iff $\mathcal{C}(\mathbf{V}\mathbf{X}) \subseteq \mathcal{C}(\mathbf{X})$.

Proof. The two projections have the same range. They agree iff they have the same null space, since an idempotent matrix is determined by its range and null space (**Proposition 6.9.4**). The null space of $\mathbf{M}$ is $\mathcal{C}(\mathbf{X})^\perp = \mathcal{N}(\mathbf{X}^\top)$. The null space of $\mathbf{P}_{\mathbf{X}, \mathbf{V}^{-1}}$ is the set of $\mathbf{u}$ with $\mathbf{X}^\top \mathbf{V}^{-1} \mathbf{u} = \mathbf{0}$, that is, $\mathbf{V} \mathcal{N}(\mathbf{X}^\top)$. So the projections agree iff $\mathbf{V} \mathcal{N}(\mathbf{X}^\top) = \mathcal{N}(\mathbf{X}^\top)$, and since $\mathbf{V}$ is nonsingular and the spaces have equal dimension, iff $\mathbf{V} \mathcal{N}(\mathbf{X}^\top) \subseteq \mathcal{N}(\mathbf{X}^\top)$. By symmetry of $\mathbf{V}$, this says $\mathbf{X}^\top \mathbf{V} \mathbf{u} = \mathbf{0}$ whenever $\mathbf{X}^\top \mathbf{u} = \mathbf{0}$, that is, $\mathcal{C}(\mathbf{V}\mathbf{X}) \perp \mathcal{C}(\mathbf{X})^\perp$, which is $\mathcal{C}(\mathbf{V}\mathbf{X}) \subseteq \mathcal{C}(\mathbf{X})$. ■

The condition holds, for example, for equicorrelated errors $\mathbf{V} = (1 - \rho)\mathbf{I} + \rho\mathbf{1}\mathbf{1}^\top$ whenever $\mathbf{1} \in \mathcal{C}(\mathbf{X})$ (**Exercise (A1)**). It also underlies the analysis of balanced split-plot designs in Chapter 33. The result is due to Kruskal (1968), and the coordinate-free argument is his.

Example. (Autocorrelated errors)

Take eight equally spaced observations with a linear trend and error covariance $v_{st} = 0.7^{|s-t|}$. The $\mathbf{V}^{-1}$ -projection onto $\mathcal{C}(\mathbf{X})$ is idempotent and self-adjoint for $\langle \cdot, \cdot \rangle_{\mathbf{V}^{-1}}$ but not symmetric: its largest asymmetry $\max |p_{st} - p_{ts}|$ is 0.444. The ordinary and generalized slope estimates are 0.648 and 0.650. The listing checks each claim, including that the GLS fit is ordinary least squares on whitened data.

```
import numpy as np
rng = np.random.default_rng(9)
n = 8
X = np.column_stack([np.ones(n), np.arange(n, dtype=float)])
y = X @ np.array([1.0, 0.5]) + rng.normal(size=n)
t = np.arange(n)
V = 0.7 ** np.abs(t[:, None] - t[None, :])           # AR(1)-type covariance

Vinv = np.linalg.inv(V)
P_V = X @ np.linalg.solve(X.T @ Vinv @ X, X.T @ Vinv) # V^{-1}-orthogonal
               projection onto C(X)

print("idempotent:", np.allclose(P_V @ P_V, P_V))
```

```

print("symmetric: ", np.allclose(P_V, P_V.T))                # False: oblique in
                    the usual geometry
print("self-adjoint for <u,v> = u'V^{-1}v:", np.allclose(Vinv @ P_V, (Vinv @ P_V).T)
    )

# GLS = ordinary least squares after whitening with a square root of V^{-1}
L = np.linalg.cholesky(V)                                     # V = L L'
Xw, yw = np.linalg.solve(L, X), np.linalg.solve(L, y)
beta_whitened, *_ = np.linalg.lstsq(Xw, yw, rcond=None)
beta_gls = np.linalg.solve(X.T @ Vinv @ X, X.T @ Vinv @ y)
print("GLS via P_V equals OLS on whitened data:", np.allclose(beta_gls,
    beta_whitened))

```

6.9.3 Weighted least squares

The simplest non-Euclidean geometry is a diagonal one. If the errors are uncorrelated with variances σ^2/w_i , then $\mathbf{V}^{-1} = \mathbf{W} = \text{diag}(w_1, \dots, w_n)$, and GLS minimizes $\sum_i w_i (y_i - \mathbf{x}_{(i)}^\top \mathbf{b})^2$. Observations believed to be noisier get less weight in the distance. Whitening multiplies row i of $\mathbf{X}$ and $\mathbf{y}$ by $\sqrt{w_i}$. The fitted values are $\mathbf{P}_{\mathbf{X}, \mathbf{W}} \mathbf{y}$, and the natural leverages are the diagonal entries of $\mathbf{P}_{\mathbf{X}, \mathbf{W}}$. These equal the diagonal of the orthogonal projection onto $\mathcal{C}(\mathbf{W}^{1/2} \mathbf{X})$ and still sum to $\text{rank}(\mathbf{X})$ (Exercise (B3)).

Example. (Engel's food expenditure data)

Engel's 1857 survey of 235 Belgian working-class households records annual income and food expenditure. The spread of expenditure grows with income, as Figure 6.9.1(a) shows. Weighting by $w_i = 1/\text{income}_i^2$ (standard deviation proportional to income) changes the fitted slope from 0.4852 to 0.5740. The change in geometry shows up more clearly in the leverages, panel (b). Under ordinary least squares the richest households have the most leverage, with a maximum of 0.255. In the weighted geometry their large variance shrinks their influence, and no household has leverage above 0.037. Whether this is the *right* geometry depends on whether the variance model holds. Chapter 21 discusses how to check it, and how to get valid standard errors when it fails.

```

import numpy as np
import statsmodels.api as sm
df = sm.datasets.engel.load_pandas().data
x, y = df["income"].to_numpy(), df["foodexp"].to_numpy()
X = np.column_stack([np.ones(len(x)), x])

w = 1.0 / x**2                # working model: sd proportional to
    income
sw = np.sqrt(w)
beta_ols, *_ = np.linalg.lstsq(X, y, rcond=None)
beta_wls, *_ = np.linalg.lstsq(X * sw[:, None], y * sw, rcond=None) # OLS on (W
    ^{1/2} X, W^{1/2} y)

# leverages in the weighted geometry: squared row norms of an orthonormal basis of C
    (W^{1/2} X)
Qw, _ = np.linalg.qr(X * sw[:, None])
h_wls = np.sum(Qw**2, axis=1)
Q, _ = np.linalg.qr(X)

```

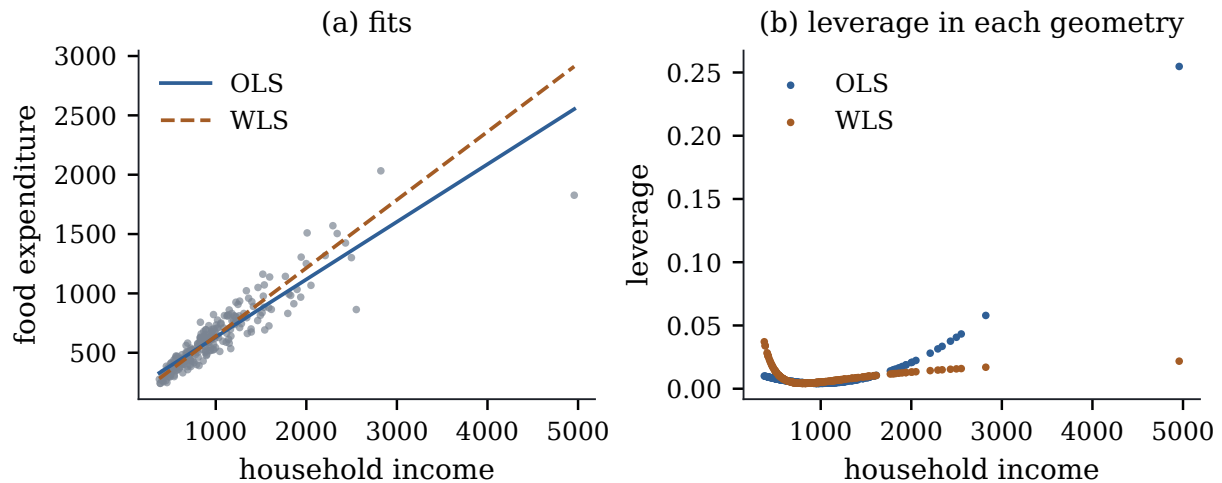


Figure 6.9.1: Engel's data. (a) Ordinary and weighted least squares fits, with weights proportional to $1/\text{income}^2$. (b) Leverages under each geometry. The weighted projection spreads influence away from the high-variance, high-income households.

```
h_ols = np.sum(Q**2, axis=1)
print("OLS:", beta_ols, " WLS:", beta_wls)
print("max leverage OLS %.3f, WLS %.3f" % (h_ols.max(), h_wls.max()))
```

6.9.4 Oblique projections

The matrices $\mathbf{P}_{\mathbf{X}, \mathbf{A}}$ are idempotent without being symmetric. In the Euclidean geometry of $\mathbb{R}^n$ they are **oblique projections**. Idempotent matrices in general have a clean description.

Proposition 6.9.4

Let $\mathbf{P}$ be idempotent. Then $\mathbb{R}^n = \mathcal{C}(\mathbf{P}) \oplus \mathcal{N}(\mathbf{P})$, and $\mathbf{P}$ maps each $\mathbf{y} = \mathbf{u} + \mathbf{v}$ ($\mathbf{u} \in \mathcal{C}(\mathbf{P})$, $\mathbf{v} \in \mathcal{N}(\mathbf{P})$) to $\mathbf{u}$. Conversely, for any decomposition $\mathbb{R}^n = \mathcal{S} \oplus \mathcal{T}$ there is exactly one idempotent $\mathbf{P}$ with $\mathcal{C}(\mathbf{P}) = \mathcal{S}$ and $\mathcal{N}(\mathbf{P}) = \mathcal{T}$, the *projection onto $\mathcal{S}$ along $\mathcal{T}$* . It is an orthogonal projection iff $\mathcal{T} = \mathcal{S}^\perp$, that is, iff $\mathbf{P}$ is symmetric.

Proof. Write $\mathbf{y} = \mathbf{P}\mathbf{y} + (\mathbf{I} - \mathbf{P})\mathbf{y}$. The first term lies in $\mathcal{C}(\mathbf{P})$, and $\mathbf{P}(\mathbf{I} - \mathbf{P})\mathbf{y} = \mathbf{0}$ puts the second in $\mathcal{N}(\mathbf{P})$. If $\mathbf{u} = \mathbf{P}\mathbf{w} \in \mathcal{N}(\mathbf{P})$ then $\mathbf{u} = \mathbf{P}^2\mathbf{w} = \mathbf{P}\mathbf{u} = \mathbf{0}$, so the sum is direct. For $\mathbf{u} \in \mathcal{C}(\mathbf{P})$, $\mathbf{P}\mathbf{u} = \mathbf{u}$ (write $\mathbf{u} = \mathbf{P}\mathbf{w}$), so $\mathbf{P}(\mathbf{u} + \mathbf{v}) = \mathbf{u}$. Conversely, given $\mathcal{S} \oplus \mathcal{T}$, the map $\mathbf{u} + \mathbf{v} \mapsto \mathbf{u}$ is well defined and linear by uniqueness of the decomposition, and it is idempotent with the required range and null space. It is unique because its values are forced on all of $\mathbb{R}^n$. If $\mathcal{T} = \mathcal{S}^\perp$, the map is the orthogonal projection, which is symmetric. If $\mathbf{P}$ is symmetric, then $\mathcal{N}(\mathbf{P}) = \mathcal{N}(\mathbf{P}^\top) = \mathcal{C}(\mathbf{P})^\perp$ by Lemma 6.2.1. ■

Figure 6.9.2 shows the difference. The nearest-point property belongs to the orthogonal projection alone. An oblique projection can move a vector *farther* away: its operator norm exceeds one unless it is orthogonal (Exercise (B2)). The GLS projection is still a nearest-point map, but in the $\mathbf{V}^{-1}$ norm, not the Euclidean one. Which geometry is “right” is not a mathematical

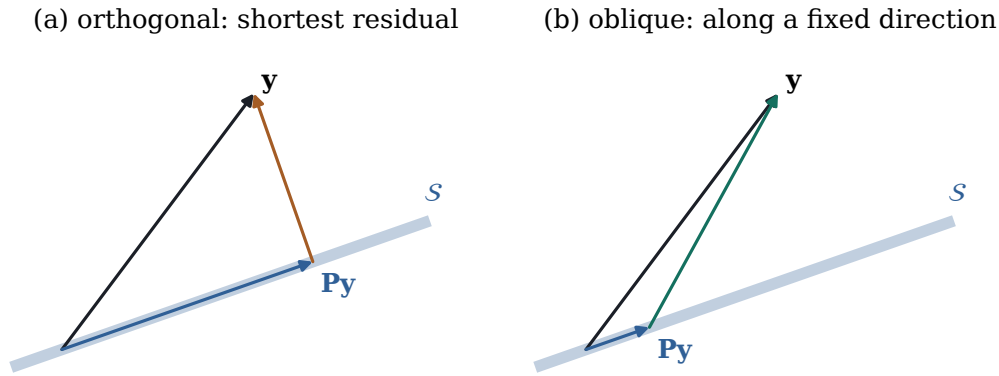


Figure 6.9.2: Projections onto a line $\mathcal{S}$ in $\mathbb{R}^2$. (a) The orthogonal projection: the residual is perpendicular to $\mathcal{S}$, and $\mathbf{P}\mathbf{y}$ is the nearest point. (b) An oblique projection moves $\mathbf{y}$ along a fixed direction that is not perpendicular to $\mathcal{S}$. The result is idempotent but is not the nearest point, and the residual is longer.

question. It is decided by the covariance of the errors.

6.9.5 Exercises

6.9.5.1 A. Check your understanding

Exercise A1.

Let $\mathbf{V} = (1 - \rho)\mathbf{I} + \rho\mathbf{1}\mathbf{1}^\top$ with $-1/(n-1) < \rho < 1$. Show that $\mathbf{V}$ is positive definite and that $\mathcal{C}(\mathbf{V}\mathbf{X}) \subseteq \mathcal{C}(\mathbf{X})$ whenever $\mathbf{1} \in \mathcal{C}(\mathbf{X})$. Conclude that ordinary least squares is generalized least squares for equicorrelated errors in any model with an intercept.

Solution. $\mathbf{V}\mathbf{1} = (1 + (n-1)\rho)\mathbf{1}$ and $\mathbf{V}\mathbf{u} = (1 - \rho)\mathbf{u}$ for $\mathbf{u} \perp \mathbf{1}$. So the eigenvalues are $1 + (n-1)\rho$ and $1 - \rho$, both positive exactly when $-1/(n-1) < \rho < 1$. Then $\mathbf{V}\mathbf{X} = (1 - \rho)\mathbf{X} + \rho\mathbf{1}(\mathbf{1}^\top\mathbf{X})$, whose columns lie in $\mathcal{C}(\mathbf{X})$ when $\mathbf{1}$ does. [Theorem 6.9.3](#) applies.

6.9.5.2 B. Practice

Exercise B1.

Complete the proof of [Theorem 6.9.1](#). Show that $\mathbf{P}_{\mathbf{X},\mathbf{A}}$ is symmetric iff it equals $\mathbf{M}$, and deduce from [Theorem 6.9.3](#) (with $\mathbf{V} = \mathbf{A}^{-1}$) that this happens iff $\mathbf{A}\mathcal{C}(\mathbf{X}) \subseteq \mathcal{C}(\mathbf{X})$.

Exercise B2.

Let $\mathbf{P} \neq \mathbf{0}$ be idempotent. Show that the operator norm $\|\mathbf{P}\|_2 = \max_{\|\mathbf{u}\|=1} \|\mathbf{P}\mathbf{u}\|$ is at least 1, with equality iff $\mathbf{P}$ is symmetric. *Hint:* if $\mathbf{P}$ is not symmetric, find $\mathbf{u} \in \mathcal{C}(\mathbf{P})$ and $\mathbf{v} \in \mathcal{N}(\mathbf{P})$ with $\mathbf{u}^\top \mathbf{v} \neq 0$, and consider $\mathbf{u} + t\mathbf{v}$ for small t .

Solution. For $\mathbf{u} \in \mathcal{C}(\mathbf{P})$, $\mathbf{P}\mathbf{u} = \mathbf{u}$, so $\|\mathbf{P}\|_2 \geq 1$. If $\mathbf{P}$ is symmetric, $\|\mathbf{P}\mathbf{u}\| \leq \|\mathbf{u}\|$ by Proposition 6.3.4(d). Suppose $\mathbf{P}$ is not symmetric. Then $\mathcal{N}(\mathbf{P}) \neq \mathcal{C}(\mathbf{P})^\perp$ by Proposition 6.9.4. The two spaces have the same dimension, so some $\mathbf{v} \in \mathcal{N}(\mathbf{P})$ is not orthogonal to $\mathcal{C}(\mathbf{P})$. Choose a unit $\mathbf{u} \in \mathcal{C}(\mathbf{P})$ with $\mathbf{u}^\top \mathbf{v} > 0$. For $t > 0$, $\mathbf{P}(\mathbf{u} - t\mathbf{v}) = \mathbf{u}$, while $\|\mathbf{u} - t\mathbf{v}\|^2 = 1 - 2t\mathbf{u}^\top \mathbf{v} + t^2\|\mathbf{v}\|^2 < 1$ for small t . So $\mathbf{P}$ stretches $\mathbf{u} - t\mathbf{v}$, and $\|\mathbf{P}\|_2 > 1$.

Exercise B3.

For weighted least squares, $\mathbf{V} = \text{diag}(1/w_1, \dots, 1/w_n)$ with $w_i > 0$. Show that the diagonal entries of $\mathbf{P}_{\mathbf{X}, \mathbf{V}^{-1}}$ equal those of the orthogonal projection onto $\mathcal{C}(\mathbf{W}^{1/2}\mathbf{X})$, where $\mathbf{W} = \text{diag}(w_i)$, and that they sum to $\text{rank}(\mathbf{X})$. Interpret them as leverages.

6.10 Computation

The formulas $\mathbf{M} = \mathbf{X}(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top$ and $\hat{\boldsymbol{\beta}} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{y}$ are the right way to *reason* about least squares and the wrong way to *compute* it. There are two problems. $\mathbf{M}$ is $n \times n$, so forming it costs $O(n^2)$ memory for a quantity that is never needed in full. And forming $\mathbf{X}^\top \mathbf{X}$ squares the condition number of the problem, which can throw away half of the available floating-point digits before any equation is solved. This section shows how the geometry itself suggests the fix, which is to build an orthonormal basis for $\mathcal{C}(\mathbf{X})$. Chapter 10 develops the numerical linear algebra in full.

6.10.1 QR: Gram–Schmidt as a factorization

Apply Gram–Schmidt (Proposition 6.2.2) to the columns $\mathbf{x}_1, \dots, \mathbf{x}_p$ of a full-column-rank $\mathbf{X}$. Each $\mathbf{x}_j$ is a combination of $\mathbf{u}_1, \dots, \mathbf{u}_j$ with coefficients $r_{ij} = \mathbf{u}_i^\top \mathbf{x}_j$ for $i < j$ and $r_{jj} = \|\mathbf{w}_j\| > 0$. In matrix form,

$$\mathbf{X} = \mathbf{Q}\mathbf{R},$$

where $\mathbf{Q} = [\mathbf{u}_1, \dots, \mathbf{u}_p]$ is $n \times p$ with $\mathbf{Q}^\top \mathbf{Q} = \mathbf{I}_p$, and $\mathbf{R}$ is $p \times p$ upper triangular with positive diagonal. This is the **thin QR factorization**. Since $\mathcal{C}(\mathbf{Q}) = \mathcal{C}(\mathbf{X})$, Proposition 6.3.2 gives $\mathbf{M} = \mathbf{Q}\mathbf{Q}^\top$ at once, and the rest of least squares follows:

$$\hat{\mathbf{Y}} = \mathbf{Q}(\mathbf{Q}^\top \mathbf{y}), \quad \mathbf{R}\hat{\boldsymbol{\beta}} = \mathbf{Q}^\top \mathbf{y}, \quad \text{SSE} = \|\mathbf{y}\|^2 - \|\mathbf{Q}^\top \mathbf{y}\|^2, \quad h_{ii} = \sum_k q_{ik}^2. \quad (6.10.1)$$

The coefficient equation holds because the normal equations become $\mathbf{R}^\top \mathbf{R}\mathbf{b} = \mathbf{R}^\top \mathbf{Q}^\top \mathbf{y}$, and $\mathbf{R}^\top$ is nonsingular. It is solved by back substitution, with no matrix inverse. Nothing of size $n \times n$ is ever formed.

In floating point, Gram–Schmidt as written in Proposition 6.2.2 gradually loses orthogonality among the computed $\mathbf{u}_j$. Production software computes the same factorization with Householder reflections, which are backward stable (Golub and Van Loan 2013; Higham 2002). The geometry does not change: $\mathbf{Q}$ is an orthonormal basis for the model space.

6.10.2 Conditioning

The sensitivity of least squares to rounding is governed by the **condition number**

$$\kappa(\mathbf{X}) = \frac{\sigma_{\max}(\mathbf{X})}{\sigma_{\min}(\mathbf{X})},$$

the ratio of the largest and smallest singular values of $\mathbf{X}$ (Chapter 1). Geometrically, $\kappa(\mathbf{X})$ is large when some combination of columns of unit coefficient norm is nearly zero. The columns are then close to linearly dependent, and the coordinates of a point in $\mathcal{C}(\mathbf{X})$ are poorly determined, even though the point itself is not. The singular values of $\mathbf{X}^\top \mathbf{X}$ are the squares of those of $\mathbf{X}$, so

$$\kappa(\mathbf{X}^\top \mathbf{X}) = \kappa(\mathbf{X})^2.$$

A useful rule of thumb from numerical analysis says that solving a linear system with condition number κ in double precision loses about $\log_{10} \kappa$ of the roughly 16 available significant digits. The normal equations solve a system whose matrix is $\mathbf{X}^\top \mathbf{X}$, so they lose about $2 \log_{10} \kappa(\mathbf{X})$ digits. QR solves the triangular system $\mathbf{R}\mathbf{b} = \mathbf{Q}^\top \mathbf{y}$, and $\kappa(\mathbf{R}) = \kappa(\mathbf{X})$ because $\mathbf{Q}$ has orthonormal columns, so it loses about $\log_{10} \kappa(\mathbf{X})$. The precise perturbation theory has one more subtlety. When the residual is large, the least squares *problem* itself has a sensitivity term proportional to $\kappa(\mathbf{X})^2 \|\hat{\mathbf{e}}\|$, and no algorithm can avoid it (Higham 2002, ch. 20). QR's advantage is largest when the model fits well.

Example. (Polynomial regression)

The columns $1, t, t^2, \dots, t^d$ evaluated at 200 equally spaced points in $[0, 1]$ become nearly dependent as d grows. To isolate the numerical error, the listing sets $\mathbf{y} = \mathbf{X}\boldsymbol{\beta}$ exactly for a random $\boldsymbol{\beta}$, so that the true least squares solution is $\boldsymbol{\beta}$ itself. For $d = 8$ the condition number is 6.8×10^5 . The normal equations return $\boldsymbol{\beta}$ with relative error 1.0×10^{-5} , while QR achieves 1.7×10^{-11} . At $d = 13$, with $\kappa(\mathbf{X}) = 4.1 \times 10^9$, the normal-equation answer has relative error 9.3×10^{-1} and not one correct digit, while QR still gets 4.4×10^{-9} . Figure 6.10.1 plots the whole sequence against the reference lines $\kappa\epsilon$ and $\kappa^2\epsilon$, where $\epsilon \approx 2.2 \times 10^{-16}$ is machine precision.

```
import numpy as np
import statsmodels.api as sm

def fit_normal_equations(X, y):
    return np.linalg.solve(X.T @ X, X.T @ y)

def fit_qr(X, y):
    Q, R = np.linalg.qr(X)                # X = QR, Q has orthonormal columns
    return np.linalg.solve(R, Q.T @ y)    # triangular system R b = Q'y
```

```
rng = np.random.default_rng(10)
n = 200
t = np.linspace(0, 1, n)
rows = []

for degree in range(1, 14):
    X = np.vander(t, degree + 1, increasing=True)    # 1, t, t^2, ..., t^degree
    beta = rng.normal(size=degree + 1)
```

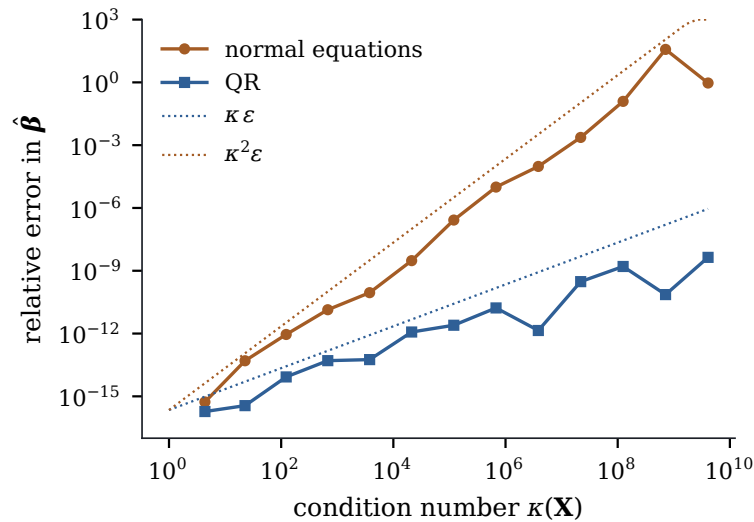



Figure 6.10.1: Accuracy of least squares coefficients for polynomial regressions of degree 1 to 13, against the condition number of the model matrix. The error of the normal equations tracks $\kappa^2\epsilon$ until it reaches order one. The error of QR tracks $\kappa\epsilon$.

```

y = X @ beta                                     # exact fit: the true
answer is beta
kappa = np.linalg.cond(X)
err_ne = np.linalg.norm(fit_normal_equations(X, y) - beta) / np.linalg.norm(beta)
)
err_qr = np.linalg.norm(fit_qr(X, y) - beta) / np.linalg.norm(beta)
rows.append((degree, kappa, err_ne, err_qr))
print(f"degree {degree:2d}   cond(X) = {kappa:9.2e}   normal eq. {err_ne:8.1e}
      QR {err_qr:8.1e}")

```

The polynomial example is extreme by design, but ill-conditioning is common in real data. The macroeconomic series of Longley's classic test problem (in `statsmodels.datasets.longley`) give a model matrix with $\kappa(\mathbf{X}) \approx 4.86 \times 10^9$. Then $\kappa(\mathbf{X}^\top \mathbf{X}) \approx 2.36 \times 10^{19}$, which is larger than $1/\epsilon$. The rule of thumb predicts about 6 correct digits from QR and about 0 from the normal equations. Part of that condition number is only a matter of units: one column is a year, another is population in thousands. Rescaling columns ([Theorem 6.7.1](#)) changes κ without changing the fit ([Exercise \(B1\)](#)). The part that remains after rescaling reflects genuine near-dependence among the regressors, the subject of Chapter 26.

Two practical lessons follow. First, never compute $(\mathbf{X}^\top \mathbf{X})^{-1}$ just to obtain $\hat{\beta}$. Solve with QR, or call a least squares routine that does. Second, reparameterizations that leave $\mathcal{C}(\mathbf{X})$ unchanged, such as centring, scaling and orthogonal polynomials, are statistically neutral but can be numerically decisive.

6.10.3 Rank deficiency and the singular value decomposition

QR with positive diagonal needs full column rank. For rank-deficient or nearly rank-deficient $\mathbf{X}$, the most informative tool is the **singular value decomposition**

$$\mathbf{X} = \mathbf{U}\mathbf{D}\mathbf{V}^\top = \mathbf{U}_r\mathbf{D}_r\mathbf{V}_r^\top,$$

where $\mathbf{U}$ ($n \times n$) and $\mathbf{V}$ ($p \times p$) are orthogonal, $\mathbf{D}$ is $n \times p$ with nonnegative diagonal entries $\sigma_1 \geq \dots \geq \sigma_r > 0$, and the subscript r keeps only the first r columns. Then:

- $\mathbf{U}_r$ is an orthonormal basis for $\mathcal{C}(\mathbf{X})$, so $\mathbf{M} = \mathbf{U}_r \mathbf{U}_r^\top$;
- the last $p - r$ columns of $\mathbf{V}$ form an orthonormal basis for $\mathcal{N}(\mathbf{X})$;
- the Moore–Penrose inverse is $\mathbf{X}^+ = \mathbf{V}_r \mathbf{D}_r^{-1} \mathbf{U}_r^\top$, and $\mathbf{M} = \mathbf{X} \mathbf{X}^+$.

Proposition 6.10.1 — Minimum-norm least squares

$\hat{\boldsymbol{\beta}}^+ = \mathbf{X}^+ \mathbf{y}$ is a least squares estimate, and it has the smallest Euclidean norm among all least squares estimates.

Proof. $\mathbf{X} \mathbf{X}^+ \mathbf{y} = \mathbf{U}_r \mathbf{D}_r \mathbf{V}_r^\top \mathbf{V}_r \mathbf{D}_r^{-1} \mathbf{U}_r^\top \mathbf{y} = \mathbf{U}_r \mathbf{U}_r^\top \mathbf{y} = \mathbf{M} \mathbf{y}$, so $\hat{\boldsymbol{\beta}}^+$ is a least squares estimate. It lies in $\mathcal{C}(\mathbf{V}_r) = \mathcal{C}(\mathbf{X}^\top) = \mathcal{N}(\mathbf{X})^\perp$. Every other least squares estimate is $\hat{\boldsymbol{\beta}}^+ + \mathbf{v}$ with $\mathbf{v} \in \mathcal{N}(\mathbf{X})$ (Theorem 6.4.2), and $\|\hat{\boldsymbol{\beta}}^+ + \mathbf{v}\|^2 = \|\hat{\boldsymbol{\beta}}^+\|^2 + \|\mathbf{v}\|^2$. ■

In Example (Four answers, one fit), the Moore–Penrose row is this minimum-norm solution. In floating point, “ $\sigma_j > 0$ ” becomes “ σ_j larger than a tolerance”, and choosing the tolerance decides the numerical rank. That choice is a statistical decision in disguise. Chapter 10 discusses it, together with pivoted QR (Theorem 10.7.2), which gives most of the SVD’s rank-revealing ability at lower cost.

6.10.4 Exercises

6.10.4.1 A. Check your understanding

Exercise A1.

Prove the four identities in Eq. 6.10.1.

6.10.4.2 B. Practice

Exercise B1.

Compute $\kappa(\mathbf{X})$ for the Longley data (with intercept) as given, after scaling every non-intercept column to unit Euclidean norm, and after centring and scaling. Confirm that the fitted values agree to the precision you expect in all three cases. Which part of the original condition number is “just units”?

Exercise B2.

Implement classical Gram–Schmidt as in Proposition 6.2.2, and the *modified* version that subtracts each projection from the working vector before computing the next inner

product. For the degree-10 polynomial design of [Example \(Polynomial regression\)](#), compare $\|\mathbf{Q}^\top \mathbf{Q} - \mathbf{I}\|$ for the two and for `numpy.linalg.qr`.

6.10.4.3 C. Going deeper

Exercise C1.

Let $\mathbf{X} = [\mathbf{1}, \mathbf{x}]$ with $\bar{x} = c$ and $S_{xx} = s^2 n$. Find $\kappa(\mathbf{X})$ as a function of c , s and n , and show that it grows without bound as $c \rightarrow \infty$ with s fixed. Show that centring the regressor, a reparameterization, reduces the condition number to $\max(1, s)/\min(1, s)$.

Solution. Write $\mathbf{x} = c\mathbf{1} + \mathbf{d}$ with $\mathbf{d} \perp \mathbf{1}$ and $\|\mathbf{d}\|^2 = ns^2$. Then $\mathbf{X}^\top \mathbf{X} = n \begin{pmatrix} 1 & c \\ c & c^2 + s^2 \end{pmatrix}$. The matrix in parentheses has trace $T = 1 + c^2 + s^2$ and determinant s^2 , so its eigenvalues are $\lambda_{\pm} = (T \pm \sqrt{T^2 - 4s^2})/2$, with $\lambda_+ \lambda_- = s^2$ and $\lambda_+ \geq T/2$. Hence

$$\kappa(\mathbf{X})^2 = \frac{\lambda_+}{\lambda_-} = \frac{\lambda_+^2}{s^2} \geq \frac{T^2}{4s^2} \rightarrow \infty \quad \text{as } c \rightarrow \infty.$$

After centring, $\mathbf{X} = [\mathbf{1}, \mathbf{d}]$ has orthogonal columns of lengths $\sqrt{n}$ and $s\sqrt{n}$, so its singular values are these lengths and $\kappa = \max(1, s)/\min(1, s)$.

6.11 Projection in the population: best linear prediction

So far the geometry has lived in observation space $\mathbb{R}^n$, and every statement was about one fixed data vector. There is a parallel geometry one level up, in which the vectors are *random variables* and the inner product is an expectation. In that geometry, least squares estimates something even when the linear model is false, namely the population projection of Y onto a span of regressors. The conditional expectation turns out to be a projection too. This section sets up the correspondence. [Chapter 14](#) develops prediction theory from it, and [Chapter 19](#) uses it to study what least squares estimates under misspecification.

6.11.1 Random variables as vectors

Consider random variables with finite second moments on a common probability space. They form a vector space, and

$$\langle U, V \rangle = \mathbb{E}(UV), \quad \|U\| = (\mathbb{E}U^2)^{1/2}$$

is an inner product, provided we identify variables that are equal with probability one, so that $\|U\| = 0$ forces $U = 0$. This space is usually written L^2 . Two random variables are orthogonal when $\mathbb{E}(UV) = 0$. For variables with mean zero, this is the same as being uncorrelated. The Cauchy–Schwarz inequality [Eq. 6.2.2](#) becomes the statement that correlations lie in $[-1, 1]$. The *cosine* of the angle between two centred random variables is their correlation.

For a random k -vector $\mathbf{X}$ with finite second moments, the set $\mathcal{L} = \{a + \mathbf{b}^\top \mathbf{X} : a \in \mathbb{R}, \mathbf{b} \in \mathbb{R}^k\}$ is a subspace of dimension at most $k + 1$. The proofs of [Proposition 6.2.2](#), [Theorem 6.2.3](#) and

Theorem 6.3.1 used only the axioms of an inner product and a finite spanning set, so they apply unchanged. Every Y with $EY^2 < \infty$ has a unique nearest point in $\mathcal{L}$, characterized by orthogonality of the error to $\mathcal{L}$.

Definition 6.11.1 — Best linear predictor

The **best linear predictor** of Y given $\mathbf{X}$, written $L(Y | \mathbf{X})$, is the projection of Y onto $\mathcal{L}$. It is the $a + \mathbf{b}^\top \mathbf{X}$ that minimizes the mean squared error $E(Y - a - \mathbf{b}^\top \mathbf{X})^2$.

Theorem 6.11.1

Let $\mu_X = E\mathbf{X}$, $\mu_Y = EY$, $\Sigma_{XX} = \text{Cov}(\mathbf{X})$ and $\sigma_{XY} = \text{Cov}(\mathbf{X}, Y)$, and suppose Σ_{XX} is positive definite. Then

$$L(Y | \mathbf{X}) = \alpha^* + \beta^{*\top} \mathbf{X}, \quad \beta^* = \Sigma_{XX}^{-1} \sigma_{XY}, \quad \alpha^* = \mu_Y - \beta^{*\top} \mu_X.$$

The prediction error $U = Y - L(Y | \mathbf{X})$ has mean zero and is uncorrelated with every component of $\mathbf{X}$. Moreover

$$\text{Var}(Y) = \text{Var}(L(Y | \mathbf{X})) + E U^2, \quad \frac{\text{Var}(L(Y | \mathbf{X}))}{\text{Var}(Y)} = \frac{\sigma_{XY}^\top \Sigma_{XX}^{-1} \sigma_{XY}}{\text{Var}(Y)}. \quad (6.11.1)$$

Proof. Orthogonality of $U = Y - a - \mathbf{b}^\top \mathbf{X}$ to the spanning set $\{1, X_1, \dots, X_k\}$ of $\mathcal{L}$ gives the *population normal equations*

$$E(U) = 0, \quad E(\mathbf{X}U) = \mathbf{0}.$$

The first gives $a = \mu_Y - \mathbf{b}^\top \mu_X$. Substituting into the second and subtracting $\mu_X E(U) = \mathbf{0}$ gives $\text{Cov}(\mathbf{X}, Y) - \text{Cov}(\mathbf{X})\mathbf{b} = \mathbf{0}$, so $\mathbf{b} = \Sigma_{XX}^{-1} \sigma_{XY}$. By the projection theorem the solution is the unique minimizer. Since U has mean zero and is uncorrelated with $\mathbf{X}$, it is uncorrelated with $L(Y | \mathbf{X})$, so the variances add. Finally, $\text{Var}(\beta^{*\top} \mathbf{X}) = \beta^{*\top} \Sigma_{XX} \beta^* = \sigma_{XY}^\top \Sigma_{XX}^{-1} \sigma_{XY}$. ■

The ratio in Eq. 6.11.1 is the population *squared multiple correlation*, the population counterpart of R^2 . By the same cosine argument as Theorem 6.7.2, it is the squared correlation between Y and $L(Y | \mathbf{X})$.

6.11.2 Conditional expectation is also a projection

The linear span $\mathcal{L}$ is one subspace of L^2 . A much larger one is $\mathcal{G} = \{g(\mathbf{X}) : E g(\mathbf{X})^2 < \infty\}$, all square-integrable functions of $\mathbf{X}$. It is infinite-dimensional, but it is closed, and the projection theorem still holds in it. We accept that fact from measure-theoretic probability and only use its conclusion.

Proposition 6.11.2

Let $m(\mathbf{X}) = E(Y | \mathbf{X})$ with $EY^2 < \infty$. Then:

- (a) $m(\mathbf{X})$ is the projection of Y onto $\mathcal{G}$. It minimizes $E(Y - g(\mathbf{X}))^2$ over g , and $E[(Y - m(\mathbf{X}))g(\mathbf{X})] = 0$ for all $g(\mathbf{X}) \in \mathcal{G}$;
- (b) $L(Y | \mathbf{X}) = L(m(\mathbf{X}) | \mathbf{X})$: the best linear predictor of Y equals the best linear predictor of the regression function;
- (c) $E(Y - L(Y | \mathbf{X}))^2 = E(Y - m(\mathbf{X}))^2 + E(m(\mathbf{X}) - L(Y | \mathbf{X}))^2$;
- (d) if m is affine, $m(\mathbf{X}) = L(Y | \mathbf{X})$.

Proof.

- (a) The orthogonality property follows from the tower rule: $E[(Y - m(\mathbf{X}))g(\mathbf{X})] = E[g(\mathbf{X}) E(Y - m(\mathbf{X}) | \mathbf{X})] = 0$. Orthogonality to the subspace characterizes the nearest point, by the argument of Theorem 6.3.1, which needs only Pythagoras.
- (b) This is Theorem 6.5.1(a) in L^2 . Since $\mathcal{L} \subseteq \mathcal{G}$, projecting onto $\mathcal{L}$ can be done in two stages.
- (c) $Y - L = (Y - m) + (m - L)$, and $m - L \in \mathcal{G}$ is orthogonal to $Y - m$ by (a).
- (d) If $m \in \mathcal{L}$, the second term in (c) is zero.

■

Part (c) is the population version of Figure 6.5.1. The error of the best linear predictor splits into two orthogonal parts. One is irreducible noise, which no function of $\mathbf{X}$ can remove. The other is *approximation error*, the price of insisting on linearity. Chapters in Part IX reduce the second part by enlarging $\mathcal{L}$ towards $\mathcal{G}$.

6.11.3 Least squares estimates the population projection

Here is the connection with the rest of the chapter. Given data $(\mathbf{x}_{(i)}, y_i)$, put probability $1/n$ on each row. For this *empirical distribution*, the L^2 inner product of two variables is $n^{-1} \sum_i u_i v_i = n^{-1} \mathbf{u}^\top \mathbf{v}$, which is the Euclidean inner product of observation space up to the factor $1/n$. So the best linear predictor under the empirical distribution is the least squares fit, and the population normal equations become the normal equations Eq. 6.4.1. Observation-space geometry is L^2 geometry for the empirical distribution.

As the sample grows, the empirical distribution approaches the population distribution, and the projections converge with it.

Proposition 6.11.3 — Consistency for the projection

Let $(\mathbf{X}_i, Y_i)$, $i = 1, 2, \dots$, be independent copies of $(\mathbf{X}, Y)$ with finite second moments and Σ_{XX} positive definite. Let $(\hat{\alpha}_n, \hat{\beta}_n)$ be the least squares coefficients from the first n observations, with an intercept. Then $(\hat{\alpha}_n, \hat{\beta}_n) \rightarrow (\alpha^*, \beta^*)$ with probability one. No

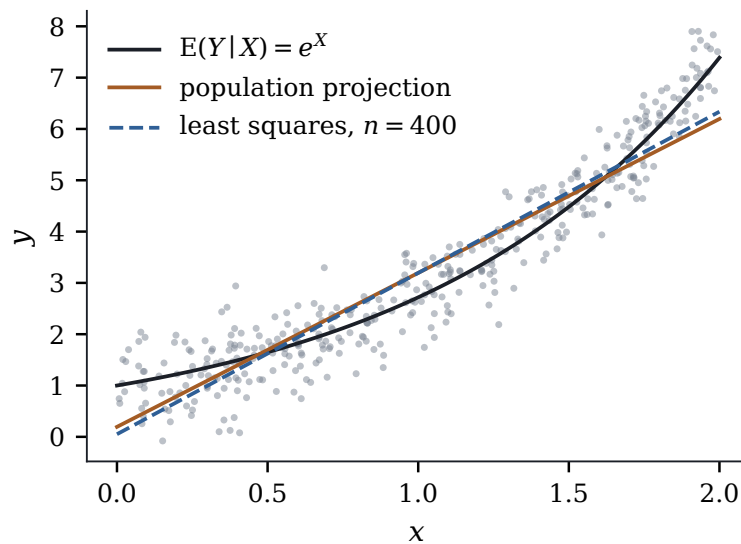


Figure 6.11.1: When the regression function is curved, least squares estimates its best linear approximation in the L^2 sense, the population projection. The approximation error is largest where e^x bends away from the line.

assumption about the form of $E(Y | \mathbf{X})$, the distribution of the errors, or their variance is needed.

Proof. For large n , $\hat{\beta}_n = \mathbf{S}_{XX}^{-1} \mathbf{s}_{XY}$, where $\mathbf{S}_{XX}$ and $\mathbf{s}_{XY}$ are sample covariances with divisor n (Section 6.6, centring). By the strong law of large numbers the sample means, second moments and cross-moments converge almost surely to their population values. So $\mathbf{S}_{XX} \rightarrow \Sigma_{XX}$ and $\mathbf{s}_{XY} \rightarrow \sigma_{XY}$. Matrix inversion is continuous at the nonsingular Σ_{XX} , so eventually $\mathbf{S}_{XX}$ is invertible and $\hat{\beta}_n \rightarrow \beta^*$. The intercept follows the same way. ■

Example. (A curved mean)

Let $X \sim \text{Uniform}(0, 2)$ and $Y = e^X + \varepsilon$ with independent $\varepsilon \sim N(0, 0.25)$. The regression function is not linear, but Theorem 6.11.1 still identifies a best line. A short calculation gives $\text{Cov}(X, Y) = 1$ and $\text{Var}(X) = 1/3$, so the population projection has slope 3.000 and intercept 0.195. The listing fits least squares lines to samples of increasing size. The estimates settle on these values, not on any feature of e^x . Figure 6.11.1 shows the regression function, the population projection, and a least squares fit.

```
import numpy as np
rng = np.random.default_rng(611)

# X ~ Uniform(0, 2), E(Y | X) = exp(X), Var(Y | X) = 0.25.
# Population projection of Y onto span{1, X}: slope Cov(X, Y)/Var(X), intercept E Y
# - slope * E X.
EX, VarX = 1.0, 4.0 / 12.0
EY = (np.exp(2) - 1) / 2
EXY = (np.exp(2) + 1) / 2
slope_pop = (EXY - EX * EY) / VarX
```

```

intercept_pop = EY - slope_pop * EX

for n in [20, 200, 2000, 20000]:
    x = rng.uniform(0, 2, n)
    y = np.exp(x) + rng.normal(0, 0.5, n)
    b, *_ = np.linalg.lstsq(np.column_stack([np.ones(n), x]), y, rcond=None)
    print(f"n = {n:6d}    intercept {b[0]:.3f}    slope {b[1]:.3f}")
print(f"population    intercept {intercept_pop:.3f}    slope {slope_pop:.3f}")

```

Remark (What changes under misspecification). [Proposition 6.11.3](#) is reassuring about the *target*. A least squares line always estimates a well-defined population quantity. It says nothing about the *uncertainty* of the estimate. When $E(Y | \mathbf{X})$ is not linear, or the variance is not constant, the approximation error $m(\mathbf{X}) - L(Y | \mathbf{X})$ behaves like extra noise that depends on $\mathbf{X}$. The textbook formula $\sigma^2(\mathbf{X}^\top \mathbf{X})^{-1}$ for the covariance of $\hat{\boldsymbol{\beta}}$ is then wrong, even in large samples. The “sandwich” covariance estimators of Chapter 21 are designed for exactly this situation.

6.12 Summary and Notes

6.12.1 Summary

Idea.

1. Least squares is the nearest-point problem in observation space. The fitted vector $\hat{\mathbf{Y}} = \mathbf{M}\mathbf{y}$ is the orthogonal projection of $\mathbf{y}$ onto $\mathcal{C}(\mathbf{X})$, and the residual is orthogonal to $\mathcal{C}(\mathbf{X})$ ([Theorem 6.4.1](#)).
2. Orthogonal projections are exactly the symmetric idempotent matrices. Their rank equals their trace ([Theorem 6.3.3](#) and [Proposition 6.3.4](#)).
3. $\mathbf{M} = \mathbf{X}(\mathbf{X}^\top \mathbf{X})^{-} \mathbf{X}^\top$ for *every* generalized inverse ([Theorem 6.3.7](#)). Fitted values, residuals and every function of them are unique, whatever the rank. Coefficients are unique only when $\text{rank}(\mathbf{X}) = p$, and $\boldsymbol{\lambda}^\top \hat{\boldsymbol{\beta}}$ is invariant iff $\boldsymbol{\lambda} \in \mathcal{C}(\mathbf{X}^\top)$ ([Theorem 6.4.2](#) and [Theorem 6.4.4](#)).
4. Nested model spaces give orthogonal decompositions $\mathbf{I} = \mathbf{M}_0 + (\mathbf{M} - \mathbf{M}_0) + (\mathbf{I} - \mathbf{M})$. The drop in residual sum of squares is $\|(\mathbf{M} - \mathbf{M}_0)\mathbf{y}\|^2$, with $r - r_0$ degrees of freedom ([Theorem 6.5.1](#)).
5. A multiple-regression coefficient is the slope on the part of its regressor orthogonal to the others. Fixed-effects estimators and added-variable plots are applications ([Theorem 6.6.2](#)).
6. Anything that depends only on $\mathcal{C}(\mathbf{X})$ is invariant to reparameterization. R^2 is the squared cosine between centred data and centred fit ([Theorem 6.7.1](#) and [Theorem 6.7.2](#)).
7. Leverage $h_{ii} = m_{ii}$ is the squared length of the projection of a coordinate vector. It lies in $[1/n, 1]$ with an intercept and averages r/n ([Proposition 6.8.1](#)).
8. Replacing the inner product by $\mathbf{u}^\top \mathbf{V}^{-1} \mathbf{v}$ turns least squares into generalized least squares. The projection is then idempotent and $\mathbf{V}^{-1}$ -self-adjoint, but not symmetric ([Theorem 6.9.1](#) and [Theorem 6.9.3](#)).

9. Compute projections with orthonormal bases (QR, SVD), never by inverting $\mathbf{X}^\top \mathbf{X}$, which squares the condition number ([Section 6.10](#)).
10. In the population, least squares estimates the L^2 projection of Y onto linear functions of $\mathbf{X}$, whether or not the regression function is linear ([Theorem 6.11.1](#) and [Proposition 6.11.3](#)).

6.12.2 Notes and sources

The geometric tradition. Seeing least squares as projection is as old as least squares itself, but it became the organizing principle of linear-model theory only in the twentieth century. Herr (1980) traces the history, from Fisher’s geometric derivations of sampling distributions through the coordinate-free treatments of the 1960s. Book-length geometric developments at different levels include Saville and Wood (1991), written for a first course, and Wickens (1995), for multivariate methods. Christensen (2020) is the most sustained vector-space treatment of linear model theory. Its chapter 2 and appendices A–B cover the material of [Section 6.2](#), [Section 6.3](#) and [Section 6.4](#) for the rank-deficient case throughout, and influenced this chapter’s choice to state results for general rank from the start. Seber and Lee (2003, appendix B) gives a compact account of projections, including the constraint-space result that appears here as [Theorem 6.5.2](#). Agresti (2015, chapter 2) develops projection matrices through worked null-model, one-way and two-way examples, and is a gentle companion to [Section 6.5](#). For the underlying linear algebra, Halmos (1958) remains a model of coordinate-free exposition.

Generalized inverses. The four-condition inverse is due to Penrose (1955). Its statistical uses, and the theory of general $\mathbf{A}^-$ including the characterization in [Exercise \(B2\)](#), are developed by Rao (1973). The invariance of $\mathbf{X}(\mathbf{X}^\top \mathbf{X})^- \mathbf{X}^\top$ ([Lemma 6.3.6](#)) is the fact that makes generalized inverses harmless in linear models. Every quantity of statistical interest is sandwiched in this way.

Frisch–Waugh–Lovell. The theorem is due to Frisch and Waugh (1933) for time trends and Lovell (1963) for seasonal dummies and general partitions. It is also known under Lovell’s name alone, and in econometrics as the “partialling-out” result. It underlies added-variable plots, double/debiased machine-learning estimators, and the fixed-effects (“within”) estimator for panel data.

Leverage. Hoaglin and Welsch (1978) introduced the name “hat matrix” into common use and established the basic properties in [Proposition 6.8.1](#). Diagnostic use of leverage is taken up in Chapter 20.

Other inner products. Generalized least squares is due to Aitken (1935). The coordinate-free condition for equality of ordinary and generalized least squares ([Theorem 6.9.3](#)) is from Kruskal (1968). Christensen (2020, chapter 10) develops the consequences for the general Gauss–Markov model, which Chapter 31 follows.

Computation. Golub and Van Loan (2013) is the standard reference for QR, the SVD and their perturbation theory. Higham (2002) gives the precise statement of the conditioning of least squares problems, including the $\kappa^2 \|\hat{\mathbf{e}}\|$ term mentioned in [Section 6.10](#). Seber and Lee (2003, chapter 11) surveys algorithms from a statistician’s perspective, and [Chapter 10](#) builds on it.

6.12.3 References

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Chapter 7

Optimality: Gauss–Markov, Maximum Likelihood and Minimum Variance

Least squares was introduced as a way of fitting, and [Chapter 6](#) showed that it is a projection. Neither fact says it is a *good* way of estimating. This chapter asks in what precise senses least squares is the best one can do, and where those senses end. With only means and covariances assumed, least squares has the smallest variance among linear unbiased estimators (Gauss–Markov), and the proof is one more application of the projection theorem. With normal errors three more things hold. Least squares is maximum likelihood. It has the smallest variance among *all* unbiased estimators (Lehmann–Scheffé). And its sampling distribution is known exactly, which is what Part III builds tests and intervals on. The same assumptions also show where optimality stops. Nonlinear estimators can win when the errors are not normal, and biased estimators can win on mean squared error. A Bayesian analysis with a conjugate prior is one principled biased estimator, and it recovers least squares as the prior becomes flat.

What you need. Covariances of linear functions and the mean of a quadratic form ([Chapter 2: Theorem 2.2.1](#) and [Theorem 2.3.1](#)). The multivariate normal and its density ([Chapter 3: Theorem 3.3.1](#) and [Theorem 3.2.2](#)). Chi-squared quadratic forms and their independence ([Chapter 4: Theorem 4.3.2](#) and [Theorem 4.4.1](#)). The linear model and its first two moments ([Chapter 5: Definition 5.3.1](#) and [Theorem 5.5.1](#)). Least squares as projection, including the rank-deficient case ([Chapter 6: Theorem 6.4.1](#), [Theorem 6.4.2](#) and [Theorem 6.4.4](#)).

7.0.1 Roadmap

- [Linear unbiased estimation](#)
- [The Gauss–Markov theorem](#)
- [Maximum likelihood under normality](#)
- [Sufficiency, completeness and minimum variance](#)
- [Sampling distributions of the estimates](#)
- [Bayesian thread I: the conjugate linear model](#)
- [Summary and notes](#)

7.1 Linear unbiased estimation

To call an estimator the best, we have to say what it is compared with and by what criterion. The comparison can't be made against every estimator. The constant estimator $\tilde{\beta}_1 \equiv 0.5$ of a slope has variance zero, and it is perfect when the true slope happens to be 0.5 and useless otherwise. No estimator has smaller variance, yet nobody would call it good. Some restriction on the class of competitors is unavoidable. The classical restriction has two parts. The estimator must be *unbiased*, which rules out estimators that buy precision by ignoring the data. And it must be *linear* in the response, which makes its variance computable from the first two moments of the errors alone. This section describes the class these two conditions carve out. The description is geometric, and the Gauss–Markov theorem of [Section 7.2](#) will follow from it in a few lines.

7.1.1 The setting

Throughout this chapter the response satisfies the linear model of [Chapter 5](#) ([Definition 5.3.1](#)),

$$\mathbf{Y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}, \quad \mathbf{E}(\boldsymbol{\varepsilon}) = \mathbf{0}, \quad \text{Cov}(\boldsymbol{\varepsilon}) = \sigma^2 \mathbf{I}, \quad (7.1.1)$$

with $\mathbf{X}$ a fixed $n \times p$ matrix of rank $r \leq p$, $\boldsymbol{\beta} \in \mathbb{R}^p$ unknown and $\sigma^2 > 0$ unknown. These are the **second-moment assumptions**. The errors have mean zero, a common variance, and are uncorrelated. Nothing is assumed about their distribution beyond that. From [Section 7.3](#) on we add normality, and we will say so each time. As in [Chapter 6](#), $\mathbf{M}$ denotes the orthogonal projection onto $\mathcal{C}(\mathbf{X})$, $\hat{\boldsymbol{\beta}}$ is any least squares estimate, and $\text{SSE} = \|(\mathbf{I} - \mathbf{M})\mathbf{Y}\|^2$ is the residual sum of squares.

The quantities we want to estimate are linear functions $\boldsymbol{\lambda}^\top \boldsymbol{\beta}$, for fixed $\boldsymbol{\lambda} \in \mathbb{R}^p$. This covers a single coefficient ($\boldsymbol{\lambda}$ a coordinate vector), a difference of two coefficients, and the mean response $\mathbf{x}_0^\top \boldsymbol{\beta}$ at a new point $\mathbf{x}_0$. Several such functions at once are collected as $\boldsymbol{\Lambda}^\top \boldsymbol{\beta}$, where $\boldsymbol{\Lambda}$ is a $p \times q$ matrix.

7.1.2 Linear and unbiased

Definition 7.1.1 — Linear unbiased and best linear unbiased estimators

Let $\boldsymbol{\lambda} \in \mathbb{R}^p$.

- (a) A **linear estimator** is a statistic of the form $c + \mathbf{a}^\top \mathbf{Y}$, with a constant $c \in \mathbb{R}$ and a constant vector $\mathbf{a} \in \mathbb{R}^n$ (constant means not depending on $\mathbf{Y}$ or on the parameters).
- (b) A linear estimator is a **linear unbiased estimator** (LUE) of $\boldsymbol{\lambda}^\top \boldsymbol{\beta}$ if $\mathbf{E}(c + \mathbf{a}^\top \mathbf{Y}) = \boldsymbol{\lambda}^\top \boldsymbol{\beta}$ for every $\boldsymbol{\beta} \in \mathbb{R}^p$ and every $\sigma^2 > 0$.
- (c) An LUE $c + \mathbf{a}^\top \mathbf{Y}$ of $\boldsymbol{\lambda}^\top \boldsymbol{\beta}$ is a **best linear unbiased estimator** (BLUE) if $\text{Var}(c + \mathbf{a}^\top \mathbf{Y}) \leq \text{Var}(d + \mathbf{b}^\top \mathbf{Y})$ for every LUE $d + \mathbf{b}^\top \mathbf{Y}$ of $\boldsymbol{\lambda}^\top \boldsymbol{\beta}$, for every $\boldsymbol{\beta}$ and every $\sigma^2 > 0$.
- (d) For q functions $\boldsymbol{\Lambda}^\top \boldsymbol{\beta}$, a **linear estimator** is $\mathbf{c} + \mathbf{A}\mathbf{Y}$ with a constant $q \times n$ matrix $\mathbf{A}$

and $\mathbf{c} \in \mathbb{R}^q$. It is an LUE of $\mathbf{\Lambda}^\top \boldsymbol{\beta}$ if each coordinate is an LUE of the corresponding coordinate of $\mathbf{\Lambda}^\top \boldsymbol{\beta}$, and a BLUE if $\text{Cov}(\mathbf{BY} + \mathbf{d}) - \text{Cov}(\mathbf{AY} + \mathbf{c})$ is nonnegative definite for every LUE $\mathbf{BY} + \mathbf{d}$ (Definition 1.7.1).

The words “for every $\boldsymbol{\beta}$ ” in (b) carry all the weight. An estimator that is unbiased at one parameter value only, like the constant 0.5 when $\beta_1 = 0.5$, does not qualify. The first result (Proposition 7.1.1) turns unbiasedness into linear equations.

Proposition 7.1.1 — Unbiasedness as a linear equation

The linear estimator $c + \mathbf{a}^\top \mathbf{Y}$ is an LUE of $\lambda^\top \boldsymbol{\beta}$ iff $c = 0$ and $\mathbf{X}^\top \mathbf{a} = \lambda$. Consequently $\lambda^\top \boldsymbol{\beta}$ has an LUE iff $\lambda \in \mathcal{C}(\mathbf{X}^\top)$.

Proof. By Theorem 2.2.1, $E(c + \mathbf{a}^\top \mathbf{Y}) = c + \mathbf{a}^\top \mathbf{X}\boldsymbol{\beta}$. This equals $\lambda^\top \boldsymbol{\beta}$ for all $\boldsymbol{\beta}$ iff the affine function $\boldsymbol{\beta} \mapsto c + (\mathbf{X}^\top \mathbf{a} - \lambda)^\top \boldsymbol{\beta}$ vanishes identically. Putting $\boldsymbol{\beta} = \mathbf{0}$ gives $c = 0$. Putting $\boldsymbol{\beta}$ equal to each coordinate vector in turn then gives $\mathbf{X}^\top \mathbf{a} = \lambda$. The converse is immediate. An LUE therefore exists iff the equation $\mathbf{X}^\top \mathbf{a} = \lambda$ has a solution, that is, iff $\lambda \in \mathcal{C}(\mathbf{X}^\top)$. ■

So the constant c is always zero, and from now on we drop it. A linear function $\lambda^\top \boldsymbol{\beta}$ with $\lambda \in \mathcal{C}(\mathbf{X}^\top)$ is called **estimable**. For these functions the data carry linear information. For the others no linear combination of the observations is centred on the target. Chapter 8 takes this as the definition of estimability (Definition 8.2.1) and develops its many equivalent forms. For now we need only Proposition 7.1.1 and one comparison. Theorem 6.4.4 of Chapter 6 found the same condition $\lambda \in \mathcal{C}(\mathbf{X}^\top)$ as the answer to a different question: when is $\lambda^\top \hat{\boldsymbol{\beta}}$ the same for every least squares estimate $\hat{\boldsymbol{\beta}}$? The functions that least squares estimates unambiguously are exactly the functions that have linear unbiased estimators at all. When $\mathbf{X}$ has full column rank, $\mathcal{C}(\mathbf{X}^\top) = \mathbb{R}^p$, so every linear function of $\boldsymbol{\beta}$ is estimable.

7.1.3 The geometry of the unbiased class

Fix an estimable $\lambda^\top \boldsymbol{\beta}$ and write $\lambda = \mathbf{X}^\top \boldsymbol{\rho}$ for some $\boldsymbol{\rho} \in \mathbb{R}^n$. The coefficient vectors of its LUEs form the set

$$\mathcal{A}_\lambda = \{\mathbf{a} \in \mathbb{R}^n : \mathbf{X}^\top \mathbf{a} = \lambda\}.$$

This set is the key object of the chapter’s first half.

Proposition 7.1.2 — The set of unbiased coefficient vectors

Let $\lambda = \mathbf{X}^\top \boldsymbol{\rho}$. Then:

- (a) $\mathcal{A}_\lambda = \mathbf{M}\boldsymbol{\rho} + \mathcal{C}(\mathbf{X})^\perp$, an affine subspace of $\mathbb{R}^n$ of dimension $n - r$, parallel to $\mathcal{C}(\mathbf{X})^\perp$;
- (b) every $\mathbf{a} \in \mathcal{A}_\lambda$ has the same projection $\mathbf{M}\mathbf{a} = \mathbf{M}\boldsymbol{\rho}$, and $\mathbf{M}\boldsymbol{\rho}$ is the only element of $\mathcal{A}_\lambda$ that lies in $\mathcal{C}(\mathbf{X})$;

- (c) $\mathbf{M}\boldsymbol{\rho} = \mathbf{X}(\mathbf{X}^\top \mathbf{X})^{-1} \boldsymbol{\lambda}$ for every generalized inverse, and $(\mathbf{M}\boldsymbol{\rho})^\top \mathbf{Y} = \boldsymbol{\lambda}^\top \hat{\boldsymbol{\beta}}$ for every least squares estimate $\hat{\boldsymbol{\beta}}$.

Proof.

- (a) By [Proposition 7.1.1](#), $\mathbf{a} \in \mathcal{A}_\lambda$ means $\mathbf{X}^\top \mathbf{a} = \boldsymbol{\lambda}$. Now $\mathbf{X}^\top \mathbf{a} = \mathbf{X}^\top \boldsymbol{\rho}$ iff $\mathbf{X}^\top (\mathbf{a} - \boldsymbol{\rho}) = \mathbf{0}$, iff $\mathbf{a} - \boldsymbol{\rho} \in \mathcal{N}(\mathbf{X}^\top) = \mathcal{C}(\mathbf{X})^\perp$ ([Lemma 6.2.1](#)). So $\mathcal{A}_\lambda = \boldsymbol{\rho} + \mathcal{C}(\mathbf{X})^\perp$. Since $\boldsymbol{\rho} - \mathbf{M}\boldsymbol{\rho} = (\mathbf{I} - \mathbf{M})\boldsymbol{\rho} \in \mathcal{C}(\mathbf{X})^\perp$, the same set is $\mathbf{M}\boldsymbol{\rho} + \mathcal{C}(\mathbf{X})^\perp$. Its dimension is $\dim \mathcal{C}(\mathbf{X})^\perp = n - r$.
- (b) If $\mathbf{a} = \mathbf{M}\boldsymbol{\rho} + \mathbf{v}$ with $\mathbf{v} \in \mathcal{C}(\mathbf{X})^\perp$, then $\mathbf{M}\mathbf{a} = \mathbf{M}\boldsymbol{\rho} + \mathbf{M}\mathbf{v} = \mathbf{M}\boldsymbol{\rho}$. If moreover $\mathbf{a} \in \mathcal{C}(\mathbf{X})$, then $\mathbf{a} = \mathbf{M}\mathbf{a} = \mathbf{M}\boldsymbol{\rho}$.
- (c) By [Theorem 6.3.7](#), $\mathbf{M}\boldsymbol{\rho} = \mathbf{X}(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \boldsymbol{\rho} = \mathbf{X}(\mathbf{X}^\top \mathbf{X})^{-1} \boldsymbol{\lambda}$. By [Theorem 6.4.4](#), $\boldsymbol{\lambda}^\top \hat{\boldsymbol{\beta}} = \boldsymbol{\rho}^\top \mathbf{M}\mathbf{Y} = (\mathbf{M}\boldsymbol{\rho})^\top \mathbf{Y}$. ■

Part (c) says that least squares is itself one of the linear unbiased estimators. Its coefficient vector is $\mathbf{a}_* = \mathbf{M}\boldsymbol{\rho}$, the one member of the class that lies in the model space. It does not depend on which $\boldsymbol{\rho}$ was used to represent $\boldsymbol{\lambda}$, because every such $\boldsymbol{\rho}$ has the same projection. That $\mathbf{a}_*^\top \mathbf{Y}$ is unbiased can also be seen directly: $E(\mathbf{a}_*^\top \mathbf{Y}) = \boldsymbol{\rho}^\top \mathbf{M}\mathbf{X}\boldsymbol{\beta} = \boldsymbol{\rho}^\top \mathbf{X}\boldsymbol{\beta} = \boldsymbol{\lambda}^\top \boldsymbol{\beta}$.

The variance of any linear estimator follows from [Theorem 2.2.1](#):

$$\text{Var}(\mathbf{a}^\top \mathbf{Y}) = \mathbf{a}^\top (\sigma^2 \mathbf{I}) \mathbf{a} = \sigma^2 \|\mathbf{a}\|^2. \quad (7.1.2)$$

Under the second-moment assumptions, then, the variance of a linear estimator is σ^2 times the squared length of its coefficient vector. Finding the BLUE means finding the shortest vector in the affine set $\mathcal{A}_\lambda$.

Idea. The linear unbiased estimators of $\boldsymbol{\lambda}^\top \boldsymbol{\beta}$ correspond to the points of a flat $\mathcal{A}_\lambda = \mathbf{a}_* + \mathcal{C}(\mathbf{X})^\perp$ in $\mathbb{R}^n$. The variance of each is σ^2 times its squared distance from the origin. The question “which LUE is best?” is a nearest-point problem, the same kind of problem that least squares solves in [Chapter 6](#), now posed in the space of coefficient vectors.

A second description of the class is useful too. Every $\mathbf{a} \in \mathcal{A}_\lambda$ is $\mathbf{a}_* + (\mathbf{I} - \mathbf{M})\mathbf{w}$ for some $\mathbf{w}$, so every LUE has the form

$$\mathbf{a}^\top \mathbf{Y} = \boldsymbol{\lambda}^\top \hat{\boldsymbol{\beta}} + \mathbf{w}^\top (\mathbf{I} - \mathbf{M})\mathbf{Y} = \boldsymbol{\lambda}^\top \hat{\boldsymbol{\beta}} + \mathbf{w}^\top \hat{\boldsymbol{\epsilon}}. \quad (7.1.3)$$

An LUE is the least squares estimate plus a linear function of the residuals. The linear functions of the residuals are exactly the linear unbiased estimators of zero ([Exercise B1](#)). Adding one to an estimator keeps it unbiased and can only add noise. [Section 7.2](#) makes “can only add noise” precise.

7.1.4 Three estimators of a slope

Example. (Three linear unbiased estimators of a slope)

A straight line $E(Y_i) = \beta_0 + \beta_1 x_i$ is fitted to $n = 12$ observations at the regressor values $x = 1, 2, 2.5, 3, 4, 5, 5.5, 7, 8, 9, 10.5, 12$, for which $S_{xx} = \sum_i (x_i - \bar{x})^2 = 137.229$. Here $\mathbf{X} = [\mathbf{1}, \mathbf{x}]$ has full rank and the target is $\boldsymbol{\lambda} = (0, 1)^\top$. Three natural estimators of the slope are linear in $\mathbf{Y}$.

- (i) *Least squares*, $\hat{\beta}_1 = \sum_i (x_i - \bar{x}) Y_i / S_{xx}$ ([Theorem 5.2.1](#)), with $a_i = (x_i - \bar{x}) / S_{xx}$.
- (ii) *End points*, the slope of the line through the first and last observations, $(Y_{12} - Y_1) / (x_{12} - x_1)$.
- (iii) *Group means*, which splits the data into the six smallest and six largest x values and takes $(\bar{Y}_{\text{high}} - \bar{Y}_{\text{low}}) / (\bar{x}_{\text{high}} - \bar{x}_{\text{low}})$.

Each coefficient vector satisfies $\mathbf{1}^\top \mathbf{a} = 0$ and $\mathbf{x}^\top \mathbf{a} = 1$, which is the condition $\mathbf{X}^\top \mathbf{a} = \boldsymbol{\lambda}$ of [Proposition 7.1.1](#). So all three are unbiased, whatever the distribution of the errors. By [Eq. 7.1.2](#) their variances are $\sigma^2 \|\mathbf{a}\|^2$:

$$0.00729 \sigma^2, \quad 0.01653 \sigma^2, \quad 0.01008 \sigma^2.$$

Least squares has the smallest, and the other two have efficiencies 0.441 and 0.723 relative to it. The listing also confirms what [Proposition 7.1.2\(b\)](#) predicts: the three coefficient vectors differ, but their projections onto $\mathcal{C}(\mathbf{X})$ all equal the least squares vector.

```
import numpy as np

x = np.array([1.0, 2.0, 2.5, 3.0, 4.0, 5.0, 5.5, 7.0, 8.0, 9.0, 10.5, 12.0])
n = len(x)
X = np.column_stack([np.ones(n), x])
lam = np.array([0.0, 1.0])           # target: the slope beta_1
M = X @ np.linalg.solve(X.T @ X, X.T) # projection onto C(X)

# coefficient vectors a of three linear estimators a^T y of the slope
a_ls = X @ np.linalg.solve(X.T @ X, lam) # least squares: a = X (X^T X)^{-1} lam
a_end = np.zeros(n)
a_end[[0, -1]] = [-1.0, 1.0]
a_end /= x[-1] - x[0]                # slope through the two end points
low, high = np.arange(n) < n // 2, np.arange(n) >= n // 2
a_grp = (high / high.sum() - low / low.sum()) / (x[high].mean() - x[low].mean()) # group means

for name, a in [("least squares", a_ls), ("end points", a_end), ("group means", a_grp)]:
    print(f"{name:14s} X^T a = {X.T @ a}    Var/sigma^2 = {a @ a:.5f}    Ma = a_ls: {np.allclose(M @ a, a_ls)}")
```

The end-point estimator throws away ten of the twelve observations, so its poor showing is no surprise. The group-means estimator uses every observation, and it is still clearly worse than least squares. It weights all observations in a half equally, while least squares weights each observation in proportion to its distance from $\bar{x}$. Estimators of this grouping type were proposed by Wald (1940) for regressions in which x itself is measured with error. Chapter 24 returns to that problem.

7.1.5 Why these restrictions?

Linearity and unbiasedness are restrictions of convenience as well as of principle, and it is worth being clear about what each one does.

Unbiasedness removes estimators that are good only near one parameter value. It is not sacred. [Section 7.2](#) shows biased estimators with smaller mean squared error, and [Section 7.6](#) derives a whole family of them from prior information. Unbiasedness can also be too demanding: a nonestimable $\lambda^\top \beta$ has no linear unbiased estimator, and under normality no unbiased estimator of any kind ([Exercise \(C1\)](#)).

Linearity is what makes the second-moment assumptions enough. The variance of a nonlinear estimator, such as a median or a trimmed mean, depends on the whole error distribution. Its variance can't be compared with that of least squares until that distribution is specified. For linear estimators [Eq. 7.1.2](#) needs only σ^2 . The restriction therefore has a cost: optimality among linear estimators says nothing about nonlinear ones. [Section 7.2](#) gives an error distribution under which a simple nonlinear estimator is far better than least squares. [Section 7.4](#) shows that under normality this cannot happen.

7.1.6 Exercises

7.1.6.1 A. Check your understanding

Exercise A1.

In the location model $Y_i = \mu + \varepsilon_i$ ($\mathbf{X} = \mathbf{1}$), show that the LUEs of μ are the weighted averages $\sum_i a_i Y_i$ with $\sum_i a_i = 1$. Identify $\mathcal{A}_\lambda$ and $\mathbf{a}_*$ and find the LUE of smallest variance directly.

Exercise A2.

In the one-way layout $E(Y_{ij}) = \mu + \alpha_i$ with three groups and an intercept ([Example \(Four answers, one fit\)](#)), show that α_1 has no linear unbiased estimator. Write down two different LUEs of $\alpha_1 - \alpha_2$, and verify that their coefficient vectors have the same projection onto $\mathcal{C}(\mathbf{X})$.

7.1.6.2 B. Practice

Exercise B1.

A linear estimator $\mathbf{a}^\top \mathbf{Y}$ is an **unbiased estimator of zero** if $E(\mathbf{a}^\top \mathbf{Y}) = 0$ for every β . Show that this holds iff $\mathbf{a} \in \mathcal{C}(\mathbf{X})^\perp$, iff $\mathbf{a}^\top \mathbf{Y}$ is a linear function of the residual vector $\hat{\varepsilon}$. Show that every unbiased estimator of zero is uncorrelated with $\lambda^\top \hat{\beta}$ for every estimable $\lambda^\top \beta$.

Solution. By [Proposition 7.1.1](#) with $\lambda = \mathbf{0}$, $\mathbf{a}^\top \mathbf{Y}$ is unbiased for zero iff $\mathbf{X}^\top \mathbf{a} = \mathbf{0}$, iff $\mathbf{a} \in \mathcal{C}(\mathbf{X})^\perp$. If so, $\mathbf{a} = (\mathbf{I} - \mathbf{M})\mathbf{a}$ and $\mathbf{a}^\top \mathbf{Y} = \mathbf{a}^\top (\mathbf{I} - \mathbf{M})\mathbf{Y} = \mathbf{a}^\top \hat{\varepsilon}$. Conversely, $\mathbf{w}^\top \hat{\varepsilon} =$

$((\mathbf{I} - \mathbf{M})\mathbf{w})^\top \mathbf{Y}$ has coefficient vector in $\mathcal{C}(\mathbf{X})^\perp$. For the last claim, write $\boldsymbol{\lambda}^\top \hat{\boldsymbol{\beta}} = \boldsymbol{\rho}^\top \mathbf{M}\mathbf{Y}$. By [Theorem 2.2.1](#), $\text{Cov}(\boldsymbol{\rho}^\top \mathbf{M}\mathbf{Y}, \mathbf{a}^\top \mathbf{Y}) = \sigma^2 \boldsymbol{\rho}^\top \mathbf{M}\mathbf{a} = 0$, because $\mathbf{M}\mathbf{a} = \mathbf{0}$.

Exercise B2.

Let $\mathbf{A}$ be $p \times q$ with $\mathcal{C}(\mathbf{A}) \subseteq \mathcal{C}(\mathbf{X}^\top)$. Show that $\mathbf{A}\mathbf{Y}$ is an LUE of $\mathbf{A}^\top \boldsymbol{\beta}$ iff $\mathbf{A}\mathbf{X} = \mathbf{A}^\top$, and that the set of such $\mathbf{A}$ is $\{\mathbf{A}^\top (\mathbf{X}^\top \mathbf{X})^- \mathbf{X}^\top + \mathbf{W}(\mathbf{I} - \mathbf{M}) : \mathbf{W} \in \mathbb{R}^{q \times n}\}$.

7.1.6.3 C. Going deeper

Exercise C1.

Assume $\mathbf{Y} \sim N_n(\mathbf{X}\boldsymbol{\beta}, \sigma^2 \mathbf{I})$ and $\boldsymbol{\lambda} \notin \mathcal{C}(\mathbf{X}^\top)$. Show that no statistic $\delta(\mathbf{Y})$, linear or not, satisfies $E \delta(\mathbf{Y}) = \boldsymbol{\lambda}^\top \boldsymbol{\beta}$ for all $\boldsymbol{\beta}$. *Hint:* the distribution of $\mathbf{Y}$ depends on $\boldsymbol{\beta}$ only through $\mathbf{X}\boldsymbol{\beta}$.

Solution. Since $\boldsymbol{\lambda} \notin \mathcal{C}(\mathbf{X}^\top) = \mathcal{N}(\mathbf{X})^\perp$, there is $\mathbf{v} \in \mathcal{N}(\mathbf{X})$ with $\boldsymbol{\lambda}^\top \mathbf{v} \neq 0$. The parameters $\boldsymbol{\beta}$ and $\boldsymbol{\beta} + \mathbf{v}$ give the same mean $\mathbf{X}\boldsymbol{\beta}$, hence the same distribution of $\mathbf{Y}$, hence the same value of $E \delta(\mathbf{Y})$. An unbiased δ would need $\boldsymbol{\lambda}^\top \boldsymbol{\beta} = \boldsymbol{\lambda}^\top (\boldsymbol{\beta} + \mathbf{v})$, which is false. The argument uses only that the law of $\mathbf{Y}$ is a function of $\mathbf{X}\boldsymbol{\beta}$ and σ^2 . [Chapter 8](#) calls this *nonidentifiability* ([Definition 8.1.1](#)).

Exercise C2.

For the group-means estimator of [Example \(Three linear unbiased estimators of a slope\)](#) with general $x_1 \leq \dot{s} \leq x_n$ ($n = 2m$), find its variance in terms of $\bar{x}_{\text{high}} - \bar{x}_{\text{low}}$. Show that its efficiency relative to least squares is $(\bar{x}_{\text{high}} - \bar{x}_{\text{low}})^2 n / (4S_{xx})$, and that this is 1 iff the x values take only two distinct values, with half of the observations at each.

7.2 The Gauss–Markov theorem

[Section 7.1](#) turned the search for a best linear unbiased estimator into a geometric problem. The unbiased coefficient vectors form the flat $\mathcal{A}_{\boldsymbol{\lambda}} = \mathbf{a}_* + \mathcal{C}(\mathbf{X})^\perp$, and the variance of each is σ^2 times its squared length. The shortest vector in a flat parallel to $\mathcal{C}(\mathbf{X})^\perp$ is the one in $\mathcal{C}(\mathbf{X})$, and that vector is the least squares coefficient vector $\mathbf{a}_*$. The Gauss–Markov theorem says exactly this, together with its matrix form and its consequences.

7.2.1 The theorem

We keep the second-moment assumptions [Eq. 7.1.1](#), with $\mathbf{X}$ of any rank r , and write $(\mathbf{X}^\top \mathbf{X})^-$ for any generalized inverse of $\mathbf{X}^\top \mathbf{X}$.

Theorem 7.2.1 — Gauss–Markov

Assume [Eq. 7.1.1](#). Let $\lambda \in \mathcal{C}(\mathbf{X}^\top)$, say $\lambda = \mathbf{X}^\top \rho$, and let $\hat{\beta}$ be any least squares estimate.

- (a) $\lambda^\top \hat{\beta}$ is a linear unbiased estimator of $\lambda^\top \beta$, with

$$\text{Var}(\lambda^\top \hat{\beta}) = \sigma^2 \lambda^\top (\mathbf{X}^\top \mathbf{X})^{-1} \lambda.$$

- (b) For every linear unbiased estimator $\mathbf{a}^\top \mathbf{Y}$ of $\lambda^\top \beta$,

$$\text{Var}(\mathbf{a}^\top \mathbf{Y}) = \text{Var}(\lambda^\top \hat{\beta}) + \sigma^2 \|(\mathbf{I} - \mathbf{M})\mathbf{a}\|^2. \quad (7.2.1)$$

- (c) Consequently $\lambda^\top \hat{\beta}$ is a BLUE of $\lambda^\top \beta$, and it is the only one: an LUE $\mathbf{a}^\top \mathbf{Y}$ is a BLUE iff $\mathbf{a} = \mathbf{X}(\mathbf{X}^\top \mathbf{X})^{-1} \lambda$, that is, iff $\mathbf{a}^\top \mathbf{y} = \lambda^\top \hat{\beta}$ for every $\mathbf{y}$.

- (d) (*Matrix form.*) Let $\mathbf{A}$ be $p \times q$ with $\mathcal{C}(\mathbf{A}) \subseteq \mathcal{C}(\mathbf{X}^\top)$. For every linear unbiased estimator $\mathbf{A}\mathbf{Y}$ of $\mathbf{A}^\top \beta$,

$$\text{Cov}(\mathbf{A}\mathbf{Y}) = \text{Cov}(\mathbf{A}^\top \hat{\beta}) + \sigma^2 \mathbf{A}(\mathbf{I} - \mathbf{M})\mathbf{A}^\top, \quad (7.2.2)$$

where $\text{Cov}(\mathbf{A}^\top \hat{\beta}) = \sigma^2 \mathbf{A}^\top (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{A}$. So $\text{Cov}(\mathbf{A}\mathbf{Y}) - \text{Cov}(\mathbf{A}^\top \hat{\beta})$ is nonnegative definite, and it is zero iff $\mathbf{A}\mathbf{Y} = \mathbf{A}^\top \hat{\beta}$ for every $\mathbf{y}$.

Proof. Put $\mathbf{a}_* = \mathbf{M}\rho = \mathbf{X}(\mathbf{X}^\top \mathbf{X})^{-1} \lambda$. By [Proposition 7.1.2\(c\)](#), $\mathbf{a}_*^\top \mathbf{Y} = \lambda^\top \hat{\beta}$. Since $\mathbf{a}_* \in \mathcal{A}_\lambda$, it is an LUE by [Proposition 7.1.1](#). By [Eq. 7.1.2](#) its variance is

$$\sigma^2 \|\mathbf{M}\rho\|^2 = \sigma^2 \rho^\top \mathbf{M}\rho = \sigma^2 \rho^\top \mathbf{X}(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \rho = \sigma^2 \lambda^\top (\mathbf{X}^\top \mathbf{X})^{-1} \lambda,$$

using [Theorem 6.3.7](#). This proves (a).

For (b), let $\mathbf{a}^\top \mathbf{Y}$ be any LUE. Then $\mathbf{a} \in \mathcal{A}_\lambda$, so $\mathbf{M}\mathbf{a} = \mathbf{a}_*$ by [Proposition 7.1.2\(b\)](#). The decomposition $\mathbf{a} = \mathbf{M}\mathbf{a} + (\mathbf{I} - \mathbf{M})\mathbf{a} = \mathbf{a}_* + (\mathbf{I} - \mathbf{M})\mathbf{a}$ is orthogonal, so $\|\mathbf{a}\|^2 = \|\mathbf{a}_*\|^2 + \|(\mathbf{I} - \mathbf{M})\mathbf{a}\|^2$. Multiplying by σ^2 and using [Eq. 7.1.2](#) twice gives [Eq. 7.2.1](#).

For (c), the second term of [Eq. 7.2.1](#) is nonnegative, so $\lambda^\top \hat{\beta}$ is a BLUE. Because $\sigma^2 > 0$, equality holds iff $(\mathbf{I} - \mathbf{M})\mathbf{a} = \mathbf{0}$, that is, iff $\mathbf{a} \in \mathcal{C}(\mathbf{X})$. By [Proposition 7.1.2\(b\)](#) the only point of $\mathcal{A}_\lambda$ in $\mathcal{C}(\mathbf{X})$ is $\mathbf{a}_*$. Two linear statistics $\mathbf{a}^\top \mathbf{y}$ and $\mathbf{a}_*^\top \mathbf{y}$ agree for every $\mathbf{y}$ iff $\mathbf{a} = \mathbf{a}_*$.

For (d), write $\mathbf{A} = \mathbf{X}^\top \mathbf{P}$ for an $n \times q$ matrix $\mathbf{P}$, and put $\mathbf{A}_* = \mathbf{P}^\top \mathbf{M}$, so that $\mathbf{A}_* \mathbf{Y} = \mathbf{A}^\top \hat{\beta}$ by [Proposition 7.1.2\(c\)](#) applied column by column. Each row $\mathbf{a}_k^\top$ of an LUE matrix $\mathbf{A}$ has $\mathbf{a}_k \in \mathcal{A}_{\lambda_k}$, where λ_k is the k th column of $\mathbf{A}$. So $\mathbf{M}\mathbf{a}_k = \mathbf{M}\mathbf{p}_k$, and hence $\mathbf{A}\mathbf{M} = \mathbf{A}_*$ (the k th row of $\mathbf{A}\mathbf{M}$ is $(\mathbf{M}\mathbf{a}_k)^\top$, since $\mathbf{M}$ is symmetric). Now split the identity:

$$\begin{aligned} \mathbf{A}\mathbf{A}^\top &= \mathbf{A}\mathbf{M}\mathbf{A}^\top + \mathbf{A}(\mathbf{I} - \mathbf{M})\mathbf{A}^\top \\ &= (\mathbf{A}\mathbf{M})(\mathbf{A}\mathbf{M})^\top + \mathbf{A}(\mathbf{I} - \mathbf{M})\mathbf{A}^\top = \mathbf{A}_* \mathbf{A}_*^\top + \mathbf{A}(\mathbf{I} - \mathbf{M})\mathbf{A}^\top, \end{aligned}$$

using $\mathbf{M} = \mathbf{M}\mathbf{M}^\top$. Multiplying by σ^2 and using [Theorem 2.2.1](#) gives $\text{Cov}(\mathbf{A}\mathbf{Y}) = \text{Cov}(\mathbf{A}_* \mathbf{Y}) + \sigma^2 \mathbf{A}(\mathbf{I} - \mathbf{M})\mathbf{A}^\top$. Also $\mathbf{A}_* \mathbf{A}_*^\top = \mathbf{P}^\top \mathbf{M}\mathbf{P} = \mathbf{A}^\top (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{A}$, as in (a). The difference $\sigma^2 \mathbf{A}(\mathbf{I} - \mathbf{M})\mathbf{A}^\top = \sigma^2 [\mathbf{A}(\mathbf{I} - \mathbf{M})][\mathbf{A}(\mathbf{I} - \mathbf{M})]^\top$ is nonnegative definite. It is zero iff $\mathbf{A}(\mathbf{I} - \mathbf{M}) = \mathbf{0}$, iff $\mathbf{A} = \mathbf{A}\mathbf{M} = \mathbf{A}_*$. ■

Three features of the theorem deserve comment.

It is distribution-free. The proof used $E(\boldsymbol{\varepsilon}) = \mathbf{0}$ and $\text{Cov}(\boldsymbol{\varepsilon}) = \sigma^2 \mathbf{I}$, and nothing else. The errors may be skewed, heavy-tailed, discrete, or dependent in any way that leaves them uncorrelated. They need not even be identically distributed, as long as their variances are equal.

It holds in every rank. When $\mathbf{X}$ is rank deficient, $\hat{\boldsymbol{\beta}}$ is not unique and neither is the generalized inverse. But for estimable $\boldsymbol{\lambda}$, both $\boldsymbol{\lambda}^\top \hat{\boldsymbol{\beta}}$ and $\boldsymbol{\lambda}^\top (\mathbf{X}^\top \mathbf{X})^- \boldsymbol{\lambda}$ are the same for every choice (Theorem 6.4.4 and Lemma 6.3.6). A nonestimable $\boldsymbol{\lambda}^\top \boldsymbol{\beta}$ has no LUE at all, so the theorem covers every function for which the question makes sense.

The proof is a projection. Equation Eq. 7.2.1 is Pythagoras in $\mathbb{R}^n$, applied to the coefficient vector and not to the data.

Idea. The least squares coefficient vector $\mathbf{a}_*$ is the orthogonal projection onto $\mathcal{C}(\mathbf{X})$ of the coefficient vector of *any* linear unbiased estimator. Projecting a coefficient vector onto $\mathcal{C}(\mathbf{X})$ keeps the estimator unbiased and removes the part that only adds noise. That part, $(\mathbf{I} - \mathbf{M})\mathbf{a}$, multiplies the residuals, and the residuals carry no information about $\boldsymbol{\beta}$.

The decomposition Eq. 7.1.3 gives the same result in the language of estimators. Every LUE is $\boldsymbol{\lambda}^\top \hat{\boldsymbol{\beta}} + \mathbf{w}^\top \hat{\boldsymbol{\varepsilon}}$, and the two terms are uncorrelated (Exercise (B1)), so their variances add. An estimator that uses the residuals only adds uncorrelated noise to least squares.

7.2.2 What the matrix form buys

The matrix form Eq. 7.2.2 is stronger than the scalar statement applied coordinate by coordinate, and it is the form usually quoted for vector estimators.

Corollary 7.2.2 — Consequences of the Gauss–Markov theorem

Assume Eq. 7.1.1, let $\mathcal{C}(\boldsymbol{\Lambda}) \subseteq \mathcal{C}(\mathbf{X}^\top)$, and let $\tilde{\boldsymbol{\theta}} = \mathbf{A}\mathbf{Y}$ be any LUE of $\boldsymbol{\theta} = \boldsymbol{\Lambda}^\top \boldsymbol{\beta}$, with $\hat{\boldsymbol{\theta}} = \boldsymbol{\Lambda}^\top \hat{\boldsymbol{\beta}}$.

- (a) $\text{Var}(\mathbf{b}^\top \tilde{\boldsymbol{\theta}}) \geq \text{Var}(\mathbf{b}^\top \hat{\boldsymbol{\theta}})$ for every $\mathbf{b} \in \mathbb{R}^q$; in particular each coordinate of $\hat{\boldsymbol{\theta}}$ has the smallest variance.
- (b) $E\|\tilde{\boldsymbol{\theta}} - \boldsymbol{\theta}\|^2 \geq E\|\hat{\boldsymbol{\theta}} - \boldsymbol{\theta}\|^2$.
- (c) $\det \text{Cov}(\tilde{\boldsymbol{\theta}}) \geq \det \text{Cov}(\hat{\boldsymbol{\theta}})$.
- (d) If $\text{rank}(\mathbf{X}) = p$, then $\hat{\boldsymbol{\beta}} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{Y}$ is the BLUE of $\boldsymbol{\beta}$, and $\text{Cov}(\hat{\boldsymbol{\beta}}) - \sigma^2(\mathbf{X}^\top \mathbf{X})^{-1}$ is nonnegative definite for every LUE $\tilde{\boldsymbol{\beta}}$ of $\boldsymbol{\beta}$.
- (e) The fitted vector $\hat{\mathbf{Y}} = \mathbf{M}\mathbf{Y}$ is the BLUE of $\mathbf{X}\boldsymbol{\beta}$, and the fitted value $\hat{y}_i$ is the BLUE of $\mathbf{x}_{(i)}^\top \boldsymbol{\beta}$, with variance $\sigma^2 h_{ii}$, where h_{ii} is the leverage.

Proof. Let $\mathbf{D} = \text{Cov}(\tilde{\boldsymbol{\theta}}) - \text{Cov}(\hat{\boldsymbol{\theta}}) \succeq \mathbf{0}$ (Theorem 7.2.1(d)). (a) $\text{Var}(\mathbf{b}^\top \tilde{\boldsymbol{\theta}}) - \text{Var}(\mathbf{b}^\top \hat{\boldsymbol{\theta}}) = \mathbf{b}^\top \mathbf{D} \mathbf{b} \geq 0$ by Definition 1.7.1. The coordinates are the case $\mathbf{b} = \mathbf{e}_k$. (b) Both estimators are unbiased, so the mean squared errors are the traces of the covariance matrices, and $\text{tr } \mathbf{D} = \sum_k \mathbf{e}_k^\top \mathbf{D} \mathbf{e}_k \geq 0$. (c) Write $\mathbf{C} = \text{Cov}(\hat{\boldsymbol{\theta}})$. If $\mathbf{C}$ is singular, its determinant is zero and there is nothing to prove, because covariance matrices are nonnegative definite (Theorem 2.2.3). Otherwise $\mathbf{C}$ is positive definite, with a symmetric positive definite

square root (Theorem 1.7.6). The matrix $\mathbf{C}^{-1/2} \text{Cov}(\tilde{\boldsymbol{\theta}}) \mathbf{C}^{-1/2} = \mathbf{I} + \mathbf{C}^{-1/2} \mathbf{D} \mathbf{C}^{-1/2}$ has all eigenvalues at least 1: the matrix $\mathbf{C}^{-1/2} \mathbf{D} \mathbf{C}^{-1/2}$ is nonnegative definite by Proposition 1.7.5(c), so its eigenvalues are nonnegative by Theorem 1.7.4, and adding $\mathbf{I}$ adds 1 to each of them. Its determinant $\det \text{Cov}(\tilde{\boldsymbol{\theta}}) / \det \mathbf{C}$ is therefore at least 1. (d) Take $\boldsymbol{\Lambda} = \mathbf{I}_p$, which is allowed because $\mathcal{C}(\mathbf{X}^\top) = \mathbb{R}^p$, and use $\text{Cov}(\hat{\boldsymbol{\beta}}) = \sigma^2 (\mathbf{X}^\top \mathbf{X})^{-1}$ (Theorem 5.5.1). (e) Take $\boldsymbol{\Lambda} = \mathbf{X}^\top$. Then $\boldsymbol{\Lambda}^\top \hat{\boldsymbol{\beta}} = \mathbf{X} \hat{\boldsymbol{\beta}} = \mathbf{M} \mathbf{Y}$ (Theorem 6.4.1), with covariance $\sigma^2 \mathbf{M}$. Its i th diagonal entry is $\sigma^2 h_{ii}$ (Definition 6.8.1).

■

Part (b) says least squares minimizes the total mean squared error among unbiased linear estimators. Part (c) has a geometric meaning that Chapter 12 will use. For normal errors, the confidence ellipsoid for $\boldsymbol{\theta}$ has volume proportional to $\sqrt{\det \text{Cov}}$, so least squares gives the smallest ellipsoids.

Example. (The slope estimators under skewed errors)

Return to the three slope estimators of Example (Three linear unbiased estimators of a slope) and simulate 200,000 data sets with $\beta_0 = 2$, $\beta_1 = 0.5$ and errors $\varepsilon_i = E_i - 1$, where the E_i are independent exponential with mean one. These errors have mean zero and variance one, but they are strongly skewed: most are slightly negative and a few are large and positive. The average estimates are 0.5000, 0.4998 and 0.5001, and the simulated variances are 0.00724, 0.01643 and 0.01004, against the exact values 0.00729, 0.01653 and 0.01008. Figure 7.2.1(a) shows the three sampling distributions. All three are centred on the truth, and least squares is the most concentrated: its ± 2 standard deviation bar is the shortest. The end-point estimator, a difference of just two skewed errors, has a sharp peak but long tails, so the height of a density is a poor guide to variance. The errors are far from normal and the theorem does not care.

```
import numpy as np

x = np.array([1.0, 2.0, 2.5, 3.0, 4.0, 5.0, 5.5, 7.0, 8.0, 9.0, 10.5, 12.0])
n = len(x)
X = np.column_stack([np.ones(n), x])
lam = np.array([0.0, 1.0]) # target: the slope beta_1
a_ls = X @ np.linalg.solve(X.T @ X, lam) # least squares: a = X (X^T X)^{-1}
    lambda
a_end = np.zeros(n)
a_end[[0, -1]] = [-1.0, 1.0]
a_end /= x[-1] - x[0] # slope through the two end points
low, high = np.arange(n) < n // 2, np.arange(n) >= n // 2
a_grp = (high / high.sum() - low / low.sum()) / (x[high].mean() - x[low].mean()) #
    group means

estimators = {"ls": a_ls, "end": a_end, "grp": a_grp}
rng = np.random.default_rng(7)
beta, sigma, reps = np.array([2.0, 0.5]), 1.0, 200_000
E = sigma * (rng.exponential(size=(reps, n)) - 1.0) # skewed errors, mean 0,
    variance sigma^2
Y = X @ beta + E # one simulated data set per
    row
sims = {name: Y @ a for name, a in estimators.items()}
for name, est in sims.items():
    print(f"{name:4s} mean {est.mean():.4f}    variance {est.var():.4f}")
```

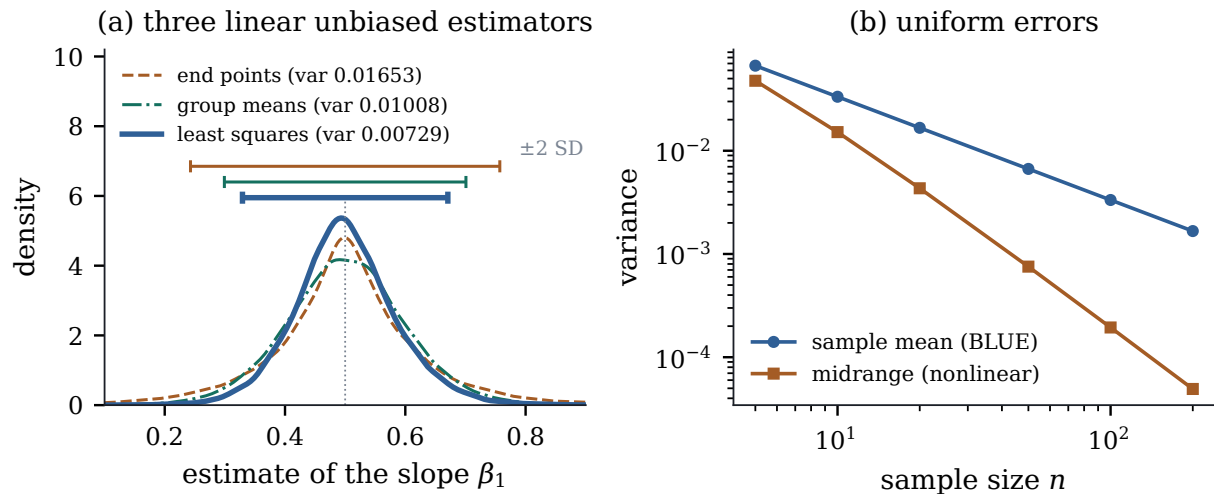


Figure 7.2.1: (a) Sampling distributions of three linear unbiased estimators of a slope, with skewed errors (200,000 simulated data sets), as smoothed densities, with bars marking the true value ± 2 standard deviations. Least squares has the smallest variance, as the Gauss–Markov theorem requires. (b) In the location model with uniform errors, the midrange, which is unbiased but not linear, has far smaller variance than the sample mean, which is the BLUE.

7.2.3 Correlated errors: Aitken’s theorem

If $\text{Cov}(\boldsymbol{\varepsilon}) = \sigma^2 \mathbf{V}$ with $\mathbf{V}$ known and positive definite, the theorem applies after a change of coordinates, and the BLUE becomes the generalized least squares estimate of Definition 6.9.1.

Corollary 7.2.3 — Aitken

Assume $E(\mathbf{Y}) = \mathbf{X}\boldsymbol{\beta}$ and $\text{Cov}(\mathbf{Y}) = \sigma^2 \mathbf{V}$, with $\mathbf{V}$ known and positive definite. For $\boldsymbol{\lambda} \in \mathcal{C}(\mathbf{X}^\top)$, the BLUE of $\boldsymbol{\lambda}^\top \boldsymbol{\beta}$ is $\boldsymbol{\lambda}^\top \hat{\boldsymbol{\beta}}_{\text{GLS}}$, where $\hat{\boldsymbol{\beta}}_{\text{GLS}}$ is any solution of $\mathbf{X}^\top \mathbf{V}^{-1} \mathbf{X} \mathbf{b} = \mathbf{X}^\top \mathbf{V}^{-1} \mathbf{Y}$. Its variance is $\sigma^2 \boldsymbol{\lambda}^\top (\mathbf{X}^\top \mathbf{V}^{-1} \mathbf{X})^{-1} \boldsymbol{\lambda}$.

Proof. Let $\mathbf{V}^{1/2}$ be the symmetric positive definite square root of $\mathbf{V}$ (Theorem 1.7.6). Put $\mathbf{Z} = \mathbf{V}^{-1/2} \mathbf{Y}$ and $\mathbf{X}_* = \mathbf{V}^{-1/2} \mathbf{X}$. By Theorem 2.2.1, $E(\mathbf{Z}) = \mathbf{X}_* \boldsymbol{\beta}$ and $\text{Cov}(\mathbf{Z}) = \sigma^2 \mathbf{I}$, so $\mathbf{Z}$ satisfies Eq. 7.1.1 with model matrix $\mathbf{X}_*$. Since $\mathbf{V}^{-1/2}$ is nonsingular, $\mathcal{C}(\mathbf{X}_*^\top) = \mathcal{C}(\mathbf{X}^\top \mathbf{V}^{-1/2}) = \mathcal{C}(\mathbf{X}^\top)$, so the same functions are estimable. The map $\mathbf{a}^\top \mathbf{Y} = (\mathbf{V}^{1/2} \mathbf{a})^\top \mathbf{Z}$ is a bijection between linear estimators based on $\mathbf{Y}$ and those based on $\mathbf{Z}$. It preserves means and variances, since both sides are the same random variable. So a BLUE in one model is a BLUE in the other. By Theorem 7.2.1 applied to $\mathbf{Z}$, the BLUE is $\boldsymbol{\lambda}^\top \mathbf{b}$ for any least squares solution $\mathbf{b}$ of the $\mathbf{Z}$ problem. That is, $\mathbf{b}$ minimizes $\|\mathbf{Z} - \mathbf{X}_* \mathbf{b}\|^2 = (\mathbf{Y} - \mathbf{X} \mathbf{b})^\top \mathbf{V}^{-1} (\mathbf{Y} - \mathbf{X} \mathbf{b})$, which is the generalized least squares criterion. Its normal equations (Theorem 6.4.2) are $\mathbf{X}_*^\top \mathbf{X}_* \mathbf{b} = \mathbf{X}_*^\top \mathbf{Z}$, that is, $\mathbf{X}^\top \mathbf{V}^{-1} \mathbf{X} \mathbf{b} = \mathbf{X}^\top \mathbf{V}^{-1} \mathbf{Y}$. The variance is part (a) of the theorem with $\mathbf{X}_*$ in place of $\mathbf{X}$. ■

Ordinary least squares remains unbiased when $\text{Cov}(\mathbf{Y}) = \sigma^2 \mathbf{V}$, but in general it is no longer

best. [Theorem 6.9.3](#) characterized the models in which the two estimators coincide, and [Exercise \(C3\)](#) shows that exactly these models keep ordinary least squares optimal. Chapter 31 develops the general theory, including singular $\mathbf{V}$.

The whitening in the proof carries the rest of the least squares toolkit with it. The fitted vector $\mathbf{X}\hat{\boldsymbol{\beta}}_{\text{GLS}} = \mathbf{V}^{1/2}\mathbf{M}_*\mathbf{Z}$, where $\mathbf{M}_*$ is the orthogonal projection onto $\mathcal{C}(\mathbf{X}_*)$, is the generalized least squares fit of [Definition 6.9.1](#): it is idempotent but not symmetric, an oblique projection onto $\mathcal{C}(\mathbf{X})$ in the sense of [Proposition 6.9.4](#). The residual sum of squares of the whitened problem is $\|(\mathbf{I} - \mathbf{M}_*)\mathbf{Z}\|^2 = (\mathbf{Y} - \mathbf{X}\hat{\boldsymbol{\beta}}_{\text{GLS}})^\top \mathbf{V}^{-1}(\mathbf{Y} - \mathbf{X}\hat{\boldsymbol{\beta}}_{\text{GLS}})$, and since $\text{rank}(\mathbf{X}_*) = r$, [Theorem 2.3.1](#) and [Proposition 6.3.4](#) give it expectation $\sigma^2(n - r)$. So

$$s_{\text{GLS}}^2 = \frac{(\mathbf{Y} - \mathbf{X}\hat{\boldsymbol{\beta}}_{\text{GLS}})^\top \mathbf{V}^{-1}(\mathbf{Y} - \mathbf{X}\hat{\boldsymbol{\beta}}_{\text{GLS}})}{n - r}$$

is unbiased for σ^2 .

7.2.4 Design: when is a coefficient estimated best?

The theorem fixes $\mathbf{X}$ and optimizes over estimators. The variance it leaves, $\sigma^2 \boldsymbol{\lambda}^\top (\mathbf{X}^\top \mathbf{X})^{-1} \boldsymbol{\lambda}$, still depends on $\mathbf{X}$. When the regressor values can be chosen, one can ask which designs make it small. The Frisch–Waugh–Lovell theorem gives a clean answer for single coefficients.

Proposition 7.2.4 — Orthogonal columns are best

Let $\mathbf{X}$ have full column rank, let $\mathbf{x}_j$ be its j th column and $\mathbf{X}_{(j)}$ the other columns, with projection $\mathbf{M}_{(j)}$ onto $\mathcal{C}(\mathbf{X}_{(j)})$. Under [Eq. 7.1.1](#),

$$\text{Var}(\hat{\beta}_j) = \frac{\sigma^2}{\|(\mathbf{I} - \mathbf{M}_{(j)})\mathbf{x}_j\|^2} \geq \frac{\sigma^2}{\|\mathbf{x}_j\|^2},$$

with equality iff $\mathbf{x}_j$ is orthogonal to every other column. If all columns are mutually orthogonal, then $\hat{\beta}_j = \mathbf{x}_j^\top \mathbf{Y} / \|\mathbf{x}_j\|^2$, and it is unchanged when any other columns are deleted from the model.

Proof. Put $\tilde{\mathbf{x}}_j = (\mathbf{I} - \mathbf{M}_{(j)})\mathbf{x}_j$. By [Theorem 6.6.2\(a\)](#), with $\mathbf{X}_1 = \mathbf{x}_j$ and $\mathbf{X}_2 = \mathbf{X}_{(j)}$, $\hat{\beta}_j = \tilde{\mathbf{x}}_j^\top \mathbf{Y} / \|\tilde{\mathbf{x}}_j\|^2$. So $\text{Var}(\hat{\beta}_j) = \sigma^2 / \|\tilde{\mathbf{x}}_j\|^2$ by [Theorem 2.2.1](#). By [Proposition 6.3.4\(d\)](#) applied to $\mathbf{I} - \mathbf{M}_{(j)}$, $\|\tilde{\mathbf{x}}_j\| \leq \|\mathbf{x}_j\|$, with equality iff $\mathbf{M}_{(j)}\mathbf{x}_j = \mathbf{0}$, that is, iff $\mathbf{x}_j \perp \mathcal{C}(\mathbf{X}_{(j)})$. If all columns are orthogonal, $\tilde{\mathbf{x}}_j = \mathbf{x}_j$, and the formula for $\hat{\beta}_j$ involves no other column, so deleting columns does not change it. ■

So among designs whose columns have given lengths, orthogonal designs minimize the variance of every coefficient at once. This is one reason designed experiments are built from orthogonal contrasts ([Part IV](#)), and why collinearity, meaning near-dependence among columns, inflates variances (Chapter 26). [Exercise \(C1\)](#) applies the proposition to a problem of weighing objects on a balance.

7.2.5 What the theorem does not say

The Gauss–Markov theorem is often quoted as “least squares is optimal”. The theorem is narrower than that. It compares least squares only with estimators that are linear and unbiased, and it assumes the model’s mean and covariance are right. Relax any of these and least squares can lose.

Nonlinear estimators. When the errors are not normal, a nonlinear unbiased estimator can beat least squares by a wide margin.

Example. (Uniform errors and the midrange)

In the location model $Y_i = \mu + \varepsilon_i$, let the errors be independent and uniform on $(-1, 1)$, so $\sigma^2 = 1/3$. The BLUE of μ is $\bar{Y}$, with variance $1/(3n)$. The **midrange** $(Y_{(1)} + Y_{(n)})/2$, the average of the smallest and largest observation, is not linear. It is unbiased, because the error distribution is symmetric about zero, and its variance is $2/\{(n+1)(n+2)\}$ ([Exercise \(C2\)](#)). For $n = 20$ the two variances are 0.0167 and 0.00433, a ratio of 3.9. For $n = 100$ they are 0.0033 and 0.00019, a ratio of 17.2. The variance of the mean falls like $1/n$ and that of the midrange like $1/n^2$ ([Figure 7.2.1\(b\)](#)).

```
ns = np.array([5, 10, 20, 50, 100, 200])
var_mean = 1 / (3 * ns)                                # errors uniform on (-1, 1):
    variance 1/3
var_mid = 2 / ((ns + 1) * (ns + 2))                    # exact variance of the midrange
for m, v1, v2 in zip(ns, var_mean, var_mid):
    print(f"n = {m:3d}    Var(mean) = {v1:.5f}    Var(midrange) = {v2:.6f}    ratio = {
        v1 / v2:.1f}")
```

Uniform errors have sharp edges, and the extreme observations locate those edges very precisely. A linear estimator must give every observation a fixed weight whatever the sample looks like, so it cannot exploit this. Heavy-tailed errors produce the opposite phenomenon: there the extremes are unreliable, and estimators that downweight them, such as medians and M-estimators, beat least squares ([Exercise \(C1\)](#); Chapter 20 and Chapter 45 return to such estimators). [Section 7.4](#) shows that under *normal* errors neither phenomenon occurs, and least squares is best among all unbiased estimators.

Biased estimators. Mean squared error is variance plus squared bias. Accepting a little bias can reduce variance by more. For an estimator $\tilde{\beta}$ of β , the natural matrix criterion is

$$E[(\tilde{\beta} - \beta)(\tilde{\beta} - \beta)^\top] = \text{Cov}(\tilde{\beta}) + (E\tilde{\beta} - \beta)(E\tilde{\beta} - \beta)^\top,$$

and the Gauss–Markov theorem ranks estimators by it only within the unbiased class. The ridge estimator $(\mathbf{X}^\top \mathbf{X} + k\mathbf{I})^{-1} \mathbf{X}^\top \mathbf{Y}$ is linear but biased, and some $k > 0$ always gives a smaller total mean squared error than least squares ([Exercise \(B1\)](#)). Stein (1956) and James and Stein (1961) showed more: in the model $\mathbf{Y} \sim N_n(\beta, \sigma^2 \mathbf{I})$ with known σ^2 and $n \geq 3$, a nonlinear shrinkage estimator has smaller total mean squared error than $\mathbf{Y}$ itself *for every* β . Chapter 27 develops shrinkage. [Section 7.6](#) derives a family of biased linear estimators from prior information.

Wrong assumptions. If $\text{Cov}(\varepsilon) \neq \sigma^2 \mathbf{I}$, [Corollary 7.2.3](#) replaces least squares by generalized least squares. If the mean is misspecified, so that $E(\mathbf{Y}) \notin \mathcal{C}(\mathbf{X})$, then $E(\hat{\mathbf{Y}}) = \mathbf{M}E(\mathbf{Y})$ is the projection of the true mean. Least squares then estimates the best approximation within the model, not the truth (compare [Theorem 6.11.1](#)). Part V is about diagnosing and repairing these departures.

7.2.6 Exercises

7.2.6.1 A. Check your understanding

Exercise A1.

Show that $\sum_i \text{Var}(\hat{y}_i) = \sigma^2 r$ under Eq. 7.1.1. Using Corollary 7.2.2(e), deduce that no linear unbiased estimator of $\mathbf{X}\boldsymbol{\beta}$ has total variance less than $\sigma^2 r$. Interpret this as “each estimated dimension of the mean costs one σ^2 ”.

7.2.6.2 B. Practice

Exercise B1.

Let $\mathbf{X}$ have full column rank and, for $k \geq 0$, let $\tilde{\boldsymbol{\beta}}_k = (\mathbf{X}^\top \mathbf{X} + k\mathbf{I})^{-1} \mathbf{X}^\top \mathbf{Y}$. Under Eq. 7.1.1, find the bias and covariance matrix of $\tilde{\boldsymbol{\beta}}_k$. Using a spectral decomposition $\mathbf{X}^\top \mathbf{X} = \mathbf{Q}\mathbf{D}\mathbf{Q}^\top$, show that the total mean squared error is

$$f(k) = \sum_{j=1}^p \frac{\sigma^2 d_j + k^2 \gamma_j^2}{(d_j + k)^2}, \quad \boldsymbol{\gamma} = \mathbf{Q}^\top \boldsymbol{\beta}, \quad \mathbf{D} = \text{diag}(d_1, \dots, d_p),$$

and that $f'(0) < 0$. Conclude that some $k > 0$ beats least squares in total mean squared error, for every $\boldsymbol{\beta}$ and σ^2 , although the best k depends on them.

Solution. Write $\mathbf{S} = \mathbf{X}^\top \mathbf{X}$ and $\mathbf{W}_k = (\mathbf{S} + k\mathbf{I})^{-1}$. Then $\mathbb{E} \tilde{\boldsymbol{\beta}}_k = \mathbf{W}_k \mathbf{S} \boldsymbol{\beta} = \boldsymbol{\beta} - k \mathbf{W}_k \boldsymbol{\beta}$, so the bias is $-k \mathbf{W}_k \boldsymbol{\beta}$, and $\text{Cov}(\tilde{\boldsymbol{\beta}}_k) = \sigma^2 \mathbf{W}_k \mathbf{S} \mathbf{W}_k$. With $\mathbf{S} = \mathbf{Q}\mathbf{D}\mathbf{Q}^\top$, $\mathbf{W}_k = \mathbf{Q}(\mathbf{D} + k\mathbf{I})^{-1} \mathbf{Q}^\top$, so $\text{tr Cov} = \sigma^2 \sum_j d_j / (d_j + k)^2$ and $\|k \mathbf{W}_k \boldsymbol{\beta}\|^2 = k^2 \sum_j \gamma_j^2 / (d_j + k)^2$. Adding gives $f(k)$. Differentiating,

$$f'(k) = \sum_j \frac{2d_j(k\gamma_j^2 - \sigma^2)}{(d_j + k)^3}, \quad f'(0) = -2\sigma^2 \sum_j d_j^{-2} < 0.$$

Since f is differentiable at 0 with negative derivative, $f(k) < f(0)$ for all small $k > 0$, and $f(0)$ is the total mean squared error of $\hat{\boldsymbol{\beta}}$. This does not contradict Theorem 7.2.1, because $\tilde{\boldsymbol{\beta}}_k$ is biased for $k > 0$.

Exercise B2.

Suppose the design grows with n and $\mathbf{X}^\top \mathbf{X}$ has full rank for all large n . Show that $\text{Var}(\boldsymbol{\lambda}^\top \hat{\boldsymbol{\beta}}) \leq \sigma^2 \|\boldsymbol{\lambda}\|^2 / \lambda_{\min}(\mathbf{X}^\top \mathbf{X})$. Deduce that if the smallest eigenvalue of $\mathbf{X}^\top \mathbf{X}$ tends to infinity, then $\boldsymbol{\lambda}^\top \hat{\boldsymbol{\beta}} \rightarrow \boldsymbol{\lambda}^\top \boldsymbol{\beta}$ in mean square, for every $\boldsymbol{\lambda}$.

Exercise B3.

In the location model with $\text{Cov}(\mathbf{Y}) = \sigma^2 \text{diag}(v_1, \dots, v_n)$, compare the variance of the ordinary mean $\bar{Y}$ with that of the BLUE given by Corollary 7.2.3. Show that the ratio is at least one, with equality iff all v_i are equal.

7.2.6.3 C. Going deeper

Exercise C1.

Each of k objects with unknown weights $\beta_1, \dots, \beta_k$ is weighed in n weighings on a two-pan balance. In weighing i every object is put on the left pan or the right pan, and the reading Y_i is the difference of the pan loads plus an error. So $\mathbf{X}$ is $n \times k$ with entries ± 1 , and Eq. 7.1.1 holds.

- (a) Show that $\text{Var}(\hat{\beta}_j) \geq \sigma^2/n$ for every j , with equality for all j iff $\mathbf{X}^\top \mathbf{X} = n\mathbf{I}$.
 (b) Take $k = 3$, $n = 4$, and the two designs whose rows are

$$\mathbf{X}_1 : (1, 1, 1), (1, -1, -1), (-1, 1, -1), (-1, -1, 1); \quad \mathbf{X}_2 : (1, 1, 1), (1, 1, -1), (1, -1, 1), (1, 1, 1).$$

For each, compute $\text{Cov}(\hat{\boldsymbol{\beta}})/\sigma^2$, its trace and its determinant. Which design is better by each criterion, and what do the criteria measure (compare Corollary 7.2.2(b) and (c))?

- (c) Show that if a ± 1 matrix with $\mathbf{X}^\top \mathbf{X} = n\mathbf{I}$ exists with $k = n \geq 3$, then n is a multiple of 4. (Whether every multiple of 4 occurs is the Hadamard conjecture, which is open.)

Solution.

- (a) Every column has $\|\mathbf{x}_j\|^2 = n$, so Proposition 7.2.4 gives $\text{Var}(\hat{\beta}_j) \geq \sigma^2/n$, with equality iff $\mathbf{x}_j$ is orthogonal to the other columns. Equality for all j means all columns are mutually orthogonal, that is, $\mathbf{X}^\top \mathbf{X} = n\mathbf{I}$.
 (b) $\mathbf{X}_1^\top \mathbf{X}_1 = 4\mathbf{I}$, so $\text{Cov}(\hat{\boldsymbol{\beta}})/\sigma^2 = \mathbf{I}/4$, with trace $3/4$ and determinant $1/64$. For $\mathbf{X}_2$, the columns are $(1, 1, 1, 1)$, $(1, 1, -1, 1)$, $(1, -1, 1, 1)$, so

$$\mathbf{X}_2^\top \mathbf{X}_2 = \begin{pmatrix} 4 & 2 & 2 \\ 2 & 4 & 0 \\ 2 & 0 & 4 \end{pmatrix}, \quad (\mathbf{X}_2^\top \mathbf{X}_2)^{-1} = \frac{1}{32} \begin{pmatrix} 16 & -8 & -8 \\ -8 & 12 & 4 \\ -8 & 4 & 12 \end{pmatrix},$$

with trace $40/32 = 5/4$ and determinant $1/32$. $\mathbf{X}_1$ wins on both counts, and on every single variance ($1/4$ against $1/2$, $3/8$, $3/8$). The trace is the total mean squared error of $\hat{\boldsymbol{\beta}}$ divided by σ^2 , and the determinant governs the volume of confidence ellipsoids.

- (c) Here $\mathbf{X}$ is square and $\mathbf{X}^\top \mathbf{X} = n\mathbf{I}$ gives $\mathbf{X}\mathbf{X}^\top = n\mathbf{I}$, so the rows are mutually orthogonal too. Changing the sign of a column keeps this property, so we may take the first row to be all ones. For the first three rows, let a, b, c, d count the columns in which rows 2 and 3 have signs $(+, +)$, $(+, -)$, $(-, +)$, $(-, -)$. Then $a + b + c + d = n$, orthogonality of row 1 to rows 2 and 3 gives $a + b - c - d = 0$ and $a - b + c - d = 0$, and orthogonality of rows 2 and 3 gives $a - b - c + d = 0$. Adding the four equations gives $4a = n$.

Exercise C2.

Let $U_1, \dots, U_n$ be independent and uniform on $(0, 1)$. The k th smallest has a $\text{Beta}(k, n - k + 1)$ distribution, and $\text{Cov}(U_{(1)}, U_{(n)}) = 1/\{(n + 1)^2(n + 2)\}$. Derive the variance $2/\{(n + 1)(n + 2)\}$ of the midrange in Example (Uniform errors and the midrange). Show also that the midrange is unbiased for μ whenever the errors have a distribution

symmetric about zero with finite mean.

Solution. $\text{Var}(U_{(1)}) = \text{Var}(U_{(n)}) = n/\{(n+1)^2(n+2)\}$ from the beta variance. Hence

$$\text{Var}\left(\frac{U_{(1)} + U_{(n)}}{2}\right) = \frac{1}{4} \cdot \frac{2n+2}{(n+1)^2(n+2)} = \frac{1}{2(n+1)(n+2)}.$$

Errors uniform on $(-1, 1)$ are $2U_i - 1$, which multiplies variances by 4 and gives $2/\{(n+1)(n+2)\}$. For unbiasedness, if ε and $-\varepsilon$ have the same distribution, then $\varepsilon_{(1)} + \varepsilon_{(n)}$ and $-(\varepsilon_{(n)} + \varepsilon_{(1)})$ have the same distribution. So the midrange minus μ is symmetric about zero, and it has mean zero because $|\varepsilon_{(1)} + \varepsilon_{(n)}| \leq 2 \max_i |\varepsilon_i|$ is integrable.

Exercise C3.

Let $E(\mathbf{Y}) = \mathbf{X}\boldsymbol{\beta}$ and $\text{Cov}(\mathbf{Y}) = \sigma^2\mathbf{V}$ with $\mathbf{V}$ positive definite. Show that an LUE $\mathbf{a}^\top \mathbf{Y}$ of $\boldsymbol{\lambda}^\top \boldsymbol{\beta}$ is a BLUE iff it is uncorrelated with every linear unbiased estimator of zero. Deduce that $\boldsymbol{\lambda}^\top \hat{\boldsymbol{\beta}}$ (ordinary least squares) is the BLUE for every estimable $\boldsymbol{\lambda}^\top \boldsymbol{\beta}$ iff $\mathcal{C}(\mathbf{V}\mathbf{X}) \subseteq \mathcal{C}(\mathbf{X})$, in agreement with [Theorem 6.9.3](#).

Solution. Every LUE is $\mathbf{a}^\top \mathbf{Y} + \mathbf{u}^\top \mathbf{Y}$ with $\mathbf{u}^\top \mathbf{Y}$ an unbiased estimator of zero, and $\text{Var}(\mathbf{a}^\top \mathbf{Y} + t\mathbf{u}^\top \mathbf{Y}) = \text{Var}(\mathbf{a}^\top \mathbf{Y}) + 2t \text{Cov}(\mathbf{a}^\top \mathbf{Y}, \mathbf{u}^\top \mathbf{Y}) + t^2 \text{Var}(\mathbf{u}^\top \mathbf{Y})$ for every real t . If $\mathbf{a}^\top \mathbf{Y}$ is a BLUE, this quadratic in t is minimized at $t = 0$, which forces the covariance to vanish. Conversely, if every such covariance vanishes, the variance of every competitor is $\text{Var}(\mathbf{a}^\top \mathbf{Y}) + \text{Var}(\mathbf{u}^\top \mathbf{Y}) \geq \text{Var}(\mathbf{a}^\top \mathbf{Y})$. By [Exercise \(B1\)](#), the unbiased estimators of zero are $\mathbf{w}^\top (\mathbf{I} - \mathbf{M})\mathbf{Y}$. So ordinary least squares $\boldsymbol{\rho}^\top \mathbf{M}\mathbf{Y}$ is a BLUE for every $\boldsymbol{\rho}$ iff $\sigma^2 \boldsymbol{\rho}^\top \mathbf{M}\mathbf{V}(\mathbf{I} - \mathbf{M})\mathbf{w} = 0$ for all $\boldsymbol{\rho}, \mathbf{w}$, that is, iff $\mathbf{M}\mathbf{V}(\mathbf{I} - \mathbf{M}) = \mathbf{0}$. Transposing, this says $(\mathbf{I} - \mathbf{M})\mathbf{V}\mathbf{M} = \mathbf{0}$, so every column of $\mathbf{V}\mathbf{M}$ lies in $\mathcal{C}(\mathbf{X})$. Since $\mathcal{C}(\mathbf{V}\mathbf{M}) = \mathcal{C}(\mathbf{V}\mathbf{X})$, this is $\mathcal{C}(\mathbf{V}\mathbf{X}) \subseteq \mathcal{C}(\mathbf{X})$.

7.3 Maximum likelihood under normality

The Gauss–Markov theorem needs only two moments. That is its strength, and also the reason it can say nothing about σ^2 , nothing about nonlinear estimators, and nothing about the distribution of the estimates. From now on we assume more: the errors are normal. The linear model becomes a fully specified family of distributions,

$$\mathbf{Y} \sim N_n(\mathbf{X}\boldsymbol{\beta}, \sigma^2\mathbf{I}), \quad \boldsymbol{\beta} \in \mathbb{R}^p, \sigma^2 > 0, \quad (7.3.1)$$

the normal linear model of [Definition 5.3.1](#). The first general-purpose method of estimation in such a family is maximum likelihood. This section shows that it gives back least squares for $\boldsymbol{\beta}$, and gives a slightly different estimate of σ^2 .

7.3.1 The likelihood

By [Theorem 3.2.2](#) with $\Sigma = \sigma^2 \mathbf{I}$, which is positive definite with $\det(\sigma^2 \mathbf{I}) = \sigma^{2n}$, the density of $\mathbf{Y}$ at the observed $\mathbf{y}$ is

$$L(\boldsymbol{\beta}, \sigma^2) = (2\pi\sigma^2)^{-n/2} \exp\left\{-\frac{\|\mathbf{y} - \mathbf{X}\boldsymbol{\beta}\|^2}{2\sigma^2}\right\}.$$

As a function of the parameters with $\mathbf{y}$ held fixed, this is the **likelihood**. Its logarithm is

$$\ell(\boldsymbol{\beta}, \sigma^2) = -\frac{n}{2} \log(2\pi) - \frac{n}{2} \log \sigma^2 - \frac{\|\mathbf{y} - \mathbf{X}\boldsymbol{\beta}\|^2}{2\sigma^2}. \quad (7.3.2)$$

The data enter the log-likelihood only through the sum of squares $\|\mathbf{y} - \mathbf{X}\boldsymbol{\beta}\|^2$. That is the first sign that normality and least squares belong together. The Gaussian density penalizes a residual by its square, so the most likely parameter values are those with the smallest squared residuals.

One technical point matters. If $\mathbf{y} \in \mathcal{C}(\mathbf{X})$, the residual sum of squares at the least squares fit is zero, and $\ell \rightarrow \infty$ as $\sigma^2 \rightarrow 0$, so no maximum exists. This happens with probability zero when $n > r$. The subspace $\mathcal{C}(\mathbf{X})$ has dimension $r < n$, so it has Lebesgue measure zero, and $\mathbf{Y}$ has a density. We therefore assume $n > r$ throughout, and $\text{SSE} > 0$ for the observed data.

7.3.2 The maximum likelihood estimates

Theorem 7.3.1 — Maximum likelihood in the normal linear model

Assume [Eq. 7.3.1](#) with $n > r$, and let $\mathbf{y}$ be such that $\text{SSE} = \|(\mathbf{I} - \mathbf{M})\mathbf{y}\|^2 > 0$. Then $(\tilde{\boldsymbol{\beta}}, \tilde{\sigma}^2)$ maximizes the likelihood iff $\tilde{\boldsymbol{\beta}}$ is a least squares estimate and

$$\tilde{\sigma}^2 = \hat{\sigma}_{\text{ML}}^2 = \frac{\text{SSE}}{n}.$$

The maximized log-likelihood is $\ell_{\max} = -\frac{n}{2} \{\log(2\pi \text{SSE}/n) + 1\}$. The maximum likelihood estimate of $\mathbf{X}\boldsymbol{\beta}$ is $\mathbf{M}\mathbf{y}$, and for estimable $\boldsymbol{\lambda}^\top \boldsymbol{\beta}$ it is $\boldsymbol{\lambda}^\top \hat{\boldsymbol{\beta}}$. Both are unique.

Proof. Fix $\sigma^2 > 0$. By [Eq. 7.3.2](#), $\ell(\mathbf{b}, \sigma^2)$ is a decreasing function of $\|\mathbf{y} - \mathbf{X}\mathbf{b}\|^2$. So it is maximized over $\mathbf{b}$ exactly at the least squares estimates, and there $\|\mathbf{y} - \mathbf{X}\mathbf{b}\|^2 = \text{SSE}$ ([Theorem 6.4.1](#)). This gives

$$\begin{aligned} \ell(\mathbf{b}, \sigma^2) &\leq \ell(\hat{\boldsymbol{\beta}}, \sigma^2) = g(\sigma^2) \\ &:= -\frac{n}{2} \log(2\pi) - \frac{n}{2} \log \sigma^2 - \frac{\text{SSE}}{2\sigma^2}, \end{aligned}$$

with equality iff $\mathbf{b}$ is a least squares estimate. Now put $v_0 = \text{SSE}/n$ and $u = v_0/\sigma^2 > 0$. Then $\text{SSE}/(2\sigma^2) = nu/2$ and $\log \sigma^2 = \log v_0 - \log u$, so

$$g(v_0) - g(\sigma^2) = \frac{n}{2} (u - 1 - \log u) \geq 0,$$

because $\log u \leq u - 1$ for all $u > 0$, with equality only at $u = 1$. Hence $\ell(\mathbf{b}, \sigma^2) \leq g(\sigma^2) \leq g(v_0)$, with equality throughout iff $\mathbf{b}$ is a least squares estimate and $\sigma^2 = v_0$. The value

$g(v_0)$ is the stated maximum. The least squares estimates all give the fitted vector $\mathbf{My}$ (Theorem 6.4.1), and all give the same value of $\lambda^\top \hat{\beta}$ for $\lambda \in \mathcal{C}(\mathbf{X}^\top)$ (Theorem 6.4.4). ■

The proof finds a global maximum without solving any equations. Calculus gives the same answer and is the usual route in a first course: set the gradient $\partial\ell/\partial\beta = \mathbf{X}^\top(\mathbf{y} - \mathbf{X}\beta)/\sigma^2$ to zero, which gives the normal equations, and $\partial\ell/\partial\sigma^2 = -n/(2\sigma^2) + \|\mathbf{y} - \mathbf{X}\beta\|^2/(2\sigma^4)$ to zero. But the stationarity argument still has to be completed by showing that the stationary point is a maximum (Exercise (B1)). The inequality $\log u \leq u - 1$ handles existence, maximality and uniqueness together.

When $\mathbf{X}$ is rank deficient, the likelihood is constant along the affine set of least squares estimates $\hat{\beta} + \mathcal{N}(\mathbf{X})$, so the MLE of β is not unique. This is a general fact about nonidentified parameters. If $\mathbf{X}\beta_1 = \mathbf{X}\beta_2$, the two parameter values give the same distribution of $\mathbf{Y}$ and hence the same likelihood for every data set. Maximum likelihood cannot separate what the model itself does not separate. What it does determine uniquely, $\mathbf{X}\beta$ and the estimable functions, is exactly what least squares determines uniquely.

Example. (Stack loss)

Brownlee's stack loss data record $n = 21$ days of operation of a plant that oxidizes ammonia to nitric acid. The response is the percentage of ammonia lost (times ten). The regressors are air flow, the temperature of the cooling water, and the acid concentration. With an intercept, $p = 4$, and least squares gives

$$\hat{\beta} = (-39.9197, 0.7156, 1.2953, -0.1521)^\top, \quad \text{SSE} = 178.830.$$

By Theorem 7.3.1 these are also the maximum likelihood estimates of β . The MLE of σ^2 is $178.830/21 = 8.516$, while the unbiased estimate is $s^2 = 178.830/17 = 10.519$. The maximized log-likelihood is -52.288 . A general-purpose numerical optimizer, started far from the answer and knowing nothing about least squares, arrives at the same point.

```
import numpy as np
import statsmodels.api as sm
from scipy import optimize, stats

data = sm.datasets.stackloss.load_pandas().data
y = data["STACKLOSS"].to_numpy()
X = np.column_stack([np.ones(len(y)), data[["AIRFLOW", "WATERTEMP", "ACIDCONC"]]])
n, p = X.shape

beta_hat, *_ = np.linalg.lstsq(X, y, rcond=None)
sse = np.sum((y - X @ beta_hat) ** 2)
sigma2_mle, s2 = sse / n, sse / (n - p)
loglik_max = -n / 2 * (np.log(2 * np.pi * sigma2_mle) + 1)
print("beta_hat", np.round(beta_hat, 4))
print(f"SSE = {sse:.3f}    MLE of sigma^2 = {sigma2_mle:.3f}    s^2 = {s2:.3f}")
print(f"maximized log-likelihood = {loglik_max:.3f}")
```

```
def negloglik(theta):
    """Minus the normal log-likelihood; theta = (beta, log sigma^2)."""
    b, log_s2 = theta[:p], theta[p]
    r = y - X @ b
```

```

    return n / 2 * (np.log(2 * np.pi) + log_s2) + r @ r / (2 * np.exp(log_s2))

start = np.r_[np.zeros(p), np.log(np.var(y))]
opt = optimize.minimize(negloglik, start, method="BFGS", options={"gtol": 1e-8})
print("numerical MLE of beta  ", np.round(opt.x[:p], 4))
print("numerical MLE sigma^2  ", round(float(np.exp(opt.x[p])), 3))

```

7.3.3 The bias of the variance estimate

Unlike least squares for β , the MLE of σ^2 is biased. Since $E(\text{SSE}) = \sigma^2(n - r)$ (Theorem 5.6.1 in full rank, and Section 6.4 in general),

$$E \hat{\sigma}_{\text{ML}}^2 = \frac{n - r}{n} \sigma^2.$$

The bias is downward, by the fraction r/n . The residual vector lives in the $(n - r)$ -dimensional space $\mathcal{C}(\mathbf{X})^\perp$, and the MLE divides its squared length by n as if it had n dimensions. The r dimensions spent on fitting the mean are the dimensions in which the fit has pulled the data toward itself. With a few regressors and many observations the bias is negligible. It is not negligible when the number of parameters grows with the sample size.

Example. (Many means, one variance)

Suppose m laboratories each make two measurements, $Y_{i1}, Y_{i2} \stackrel{\text{iid}}{\sim} N(\mu_i, \sigma^2)$, with a different μ_i for each laboratory and a common measurement variance. This is a linear model with $n = 2m$ and $r = m$, one indicator column per laboratory. The fitted value for each pair is its mean, and

$$\begin{aligned} \hat{\sigma}_{\text{ML}}^2 &= \frac{1}{2m} \sum_{i=1}^m \sum_{j=1}^2 (Y_{ij} - \bar{Y}_{i\cdot})^2 = \frac{1}{4m} \sum_{i=1}^m (Y_{i1} - Y_{i2})^2, \\ E \hat{\sigma}_{\text{ML}}^2 &= \frac{\sigma^2}{2}. \end{aligned}$$

The bias does not disappear as $m \rightarrow \infty$. By the law of large numbers, $\hat{\sigma}_{\text{ML}}^2 \rightarrow \sigma^2/2$ with probability one, so the MLE is *inconsistent*. The unbiased $s^2 = \text{SSE}/(n - r)$ is exactly twice as large and is consistent. This is the example of Neyman and Scott (1948). It shows that the good large-sample behaviour of maximum likelihood can fail when the number of nuisance parameters grows with the data. Restricted maximum likelihood, which removes the mean before estimating the variance (Exercise (C2)), repairs it, and Chapter 32 builds variance-component estimation on that idea.

Unbiasedness is not the only possible criterion for choosing a divisor. Under normality we can compare estimators of the form SSE/d by mean squared error.

Proposition 7.3.2 — The best multiple of SSE

Assume Eq. 7.3.1 with $k = n - r \geq 1$. Among estimators SSE/d with $d > 0$,

$$\frac{E(\text{SSE}/d - \sigma^2)^2}{\sigma^4} = \frac{2k}{d^2} + \left(\frac{k}{d} - 1\right)^2,$$

which is minimized at $d = k + 2$, with minimum value $2/(k + 2)$. The unbiased choice $d = k$ gives $2/k$, and the maximum likelihood choice $d = n$ gives $(2k + r^2)/n^2$.

Proof. Write $\text{SSE} = \mathbf{Y}^\top (\mathbf{I} - \mathbf{M}) \mathbf{Y}$. By [Theorem 4.2.3](#) with $\mathbf{A} = \mathbf{I} - \mathbf{M}$, $\Sigma = \sigma^2 \mathbf{I}$ and $\boldsymbol{\mu} = \mathbf{X}\boldsymbol{\beta}$, for which $\mathbf{A}\boldsymbol{\mu} = \mathbf{0}$,

$$E(\text{SSE}) = \sigma^2 \text{tr}(\mathbf{I} - \mathbf{M}) = \sigma^2 k, \quad \text{Var}(\text{SSE}) = 2\sigma^4 \text{tr}((\mathbf{I} - \mathbf{M})^2) = 2\sigma^4 k.$$

The mean squared error is variance plus squared bias: $2\sigma^4 k/d^2 + (k/d - 1)^2 \sigma^4$. As a function of $t = 1/d$, this is $\sigma^4 \{(2k + k^2)t^2 - 2kt + 1\}$, a convex quadratic minimized at $t = k/(2k + k^2) = 1/(k + 2)$. Substituting $d = k + 2$, k and $n = k + r$ gives the three values. For the last, $2k/n^2 + (k - n)^2/n^2 = (2k + r^2)/n^2$. ■

For the stack loss data, $k = 17$, and the three relative mean squared errors are 0.1176 for s^2 , 0.1134 for the MLE and 0.1053 for $\text{SSE}/(k + 2)$. The unbiased estimator is the worst of the three by this criterion. Comparing $(2k + r^2)/n^2$ with $2/k$ shows that the MLE beats s^2 iff $(r - 4)k < 2r$. So it always does when $r \leq 4$, and it loses once many parameters are fitted relative to the residual degrees of freedom, as in [Example \(Many means, one variance\)](#). In practice s^2 is used anyway, because it is what makes the t and F statistics of [Section 7.5](#) exact. The choice of divisor matters for point estimation of σ^2 and for nothing else. Any fixed multiple of SSE leads to the same tests and intervals once the matching distribution is used.

7.3.4 Profile likelihood

In most problems only some parameters are of interest. The others, called **nuisance parameters**, still appear in the likelihood. A simple and general way to eliminate them is to maximize them out.

Definition 7.3.1 — Profile likelihood

Let the parameter be $\boldsymbol{\theta} = (\psi, \boldsymbol{\eta})$, where ψ is of interest and $\boldsymbol{\eta}$ is a nuisance parameter. The **profile log-likelihood** of ψ is $\ell_p(\psi) = \sup_{\boldsymbol{\eta}} \ell(\psi, \boldsymbol{\eta})$.

The profile log-likelihood is maximized at the MLE of ψ , with maximum $\ell_{\max}$. The drop $\ell_{\max} - \ell_p(\psi)$ measures how much less plausible the value ψ is than the best value. In the linear model the profiles have closed forms. The proof of [Theorem 7.3.1](#) computed two of them. With σ^2 as nuisance, the profile of $\boldsymbol{\beta}$ is

$$\ell_p(\boldsymbol{\beta}) = -\frac{n}{2} \left\{ \log \left(\frac{2\pi \|\mathbf{y} - \mathbf{X}\boldsymbol{\beta}\|^2}{n} \right) + 1 \right\},$$

a decreasing function of $\|\mathbf{y} - \mathbf{X}\boldsymbol{\beta}\|^2$. Its level sets are the level sets of the residual sum of squares. When $\mathbf{X}$ has full rank they are ellipsoids centred at $\hat{\boldsymbol{\beta}}$, and they are the basis of the confidence ellipsoids of [Chapter 12](#) ([Theorem 12.2.1](#)). With $\boldsymbol{\beta}$ as nuisance, the profile of σ^2 is the function g in the proof.

The most useful case is a single estimable function. Here the profile likelihood turns out to be a function of the familiar t statistic.

Proposition 7.3.3 — Profile likelihood of an estimable function

Assume Eq. 7.3.1 with $n > r$, and let $\lambda \in \mathcal{C}(\mathbf{X}^\top)$, $\lambda \neq \mathbf{0}$. For $\psi \in \mathbb{R}$, put

$$t(\psi) = \frac{\lambda^\top \hat{\beta} - \psi}{s \sqrt{\lambda^\top (\mathbf{X}^\top \mathbf{X})^{-1} \lambda}}, \quad s^2 = \frac{\text{SSE}}{n - r}.$$

Then the smallest residual sum of squares subject to $\lambda^\top \beta = \psi$ is

$$\text{SSE}(\psi) = \text{SSE} + \frac{(\lambda^\top \hat{\beta} - \psi)^2}{\lambda^\top (\mathbf{X}^\top \mathbf{X})^{-1} \lambda}, \quad (7.3.3)$$

and the profile log-likelihood of $\psi = \lambda^\top \beta$, with (β, σ^2) otherwise free, satisfies

$$2\{\ell_{\max} - \ell_p(\psi)\} = n \log \left(1 + \frac{t(\psi)^2}{n - r} \right). \quad (7.3.4)$$

Proof. Write $\lambda = \mathbf{X}^\top \rho$ and $\mathbf{c} = \mathbf{M}\rho$. Then $\mathbf{c} \neq \mathbf{0}$, since $\mathbf{X}^\top \mathbf{c} = \mathbf{X}^\top \rho = \lambda \neq \mathbf{0}$, and $\|\mathbf{c}\|^2 = \lambda^\top (\mathbf{X}^\top \mathbf{X})^{-1} \lambda$ as in the proof of Theorem 7.2.1(a). For any $\mathbf{b}$, Eq. 6.3.1 gives $\|\mathbf{y} - \mathbf{X}\mathbf{b}\|^2 = \text{SSE} + \|\mathbf{X}\hat{\beta} - \mathbf{X}\mathbf{b}\|^2$. Put $\mathbf{u} = \mathbf{X}(\hat{\beta} - \mathbf{b}) \in \mathcal{C}(\mathbf{X})$. The constraint $\lambda^\top \mathbf{b} = \psi$ reads $\rho^\top \mathbf{X}(\hat{\beta} - \mathbf{b}) = \lambda^\top \hat{\beta} - \psi =: \delta$, that is, $\rho^\top \mathbf{u} = \delta$. Since $\mathbf{u} \in \mathcal{C}(\mathbf{X})$, $\rho^\top \mathbf{u} = \rho^\top \mathbf{M}\mathbf{u} = \mathbf{c}^\top \mathbf{u}$. By the Cauchy–Schwarz inequality, $|\delta| = |\mathbf{c}^\top \mathbf{u}| \leq \|\mathbf{c}\| \|\mathbf{u}\|$, so $\|\mathbf{u}\|^2 \geq \delta^2 / \|\mathbf{c}\|^2$. Equality holds for $\mathbf{u} = \delta \mathbf{c} / \|\mathbf{c}\|^2$, which lies in $\mathcal{C}(\mathbf{X})$ and so equals $\mathbf{X}(\hat{\beta} - \mathbf{b})$ for some $\mathbf{b}$. That $\mathbf{b}$ satisfies the constraint. This proves Eq. 7.3.3. Maximizing over σ^2 as in the proof of Theorem 7.3.1, with SSE replaced by SSE(ψ), gives $\ell_p(\psi) = -\frac{n}{2} \{\log(2\pi \text{SSE}(\psi)/n) + 1\}$. Hence

$$\begin{aligned} 2\{\ell_{\max} - \ell_p(\psi)\} &= n \log \frac{\text{SSE}(\psi)}{\text{SSE}} = n \log \left(1 + \frac{\delta^2}{\|\mathbf{c}\|^2 \text{SSE}} \right) \\ &= n \log \left(1 + \frac{t(\psi)^2}{n - r} \right), \end{aligned}$$

since $\delta^2 / (\|\mathbf{c}\|^2 \text{SSE}) = t(\psi)^2 / (n - r)$. ■

Two consequences show up immediately. First, any procedure based on the profile likelihood of $\lambda^\top \beta$ is a procedure based on $|t|$, because Eq. 7.3.4 is increasing in t^2 . The likelihood ratio test of $\lambda^\top \beta = \psi_0$ is therefore the t test, a fact Chapter 11 extends to general hypotheses (Theorem 11.4.1). Second, the large-sample calibration of the likelihood ratio, “reject when $2\{\ell_{\max} - \ell_p\}$ exceeds a $\chi^2(1)$ quantile”, is only approximate. The exact calibration uses the $t(n - r)$ distribution of Section 7.5.

Example. (Profile likelihood for the air-flow coefficient)

For the stack loss model, Figure 7.3.1 plots $\ell_p(\beta_1) - \ell_{\max}$ for the air-flow coefficient β_1 , computed by refitting the model with β_1 fixed on a grid. The listing confirms Eq. 7.3.4 to rounding error. The standard error of $\hat{\beta}_1$ is 0.1349. The set of β_1 with

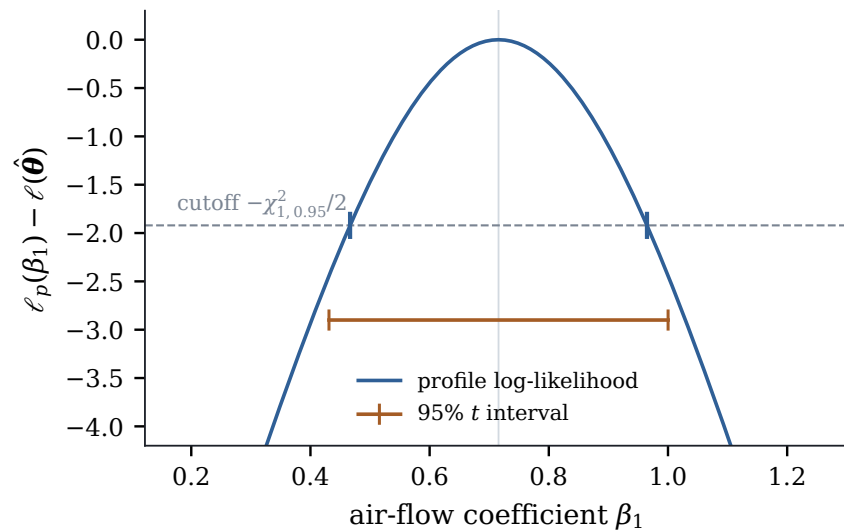


Figure 7.3.1: Profile log-likelihood of the air-flow coefficient in the stack loss regression, relative to its maximum. The ticks on the dashed cutoff line mark the χ^2 -calibrated likelihood interval, and the bar below it marks the exact 95% t interval, which is wider.

$2\{\ell_{\max} - \ell_p\} \leq 3.841$, the 95% point of $\chi^2(1)$, is the likelihood interval (0.467, 0.965). By Eq. 7.3.4 it consists of the values with $|t| \leq 1.847$. The exact 95% interval uses the $t(17)$ quantile 2.110 instead, and is (0.431, 1.000). With only 17 residual degrees of freedom, the χ^2 calibration makes the interval about 12% too short.

```
def profile_loglik(b1):
    """Profile log-likelihood of the air-flow coefficient: maximize over everything
    else."""
    others = np.delete(X, 1, axis=1)
    z = y - b1 * X[:, 1]
    coef, *_ = np.linalg.lstsq(others, z, rcond=None)
    sse_b = np.sum((z - others @ coef) ** 2)
    with beta_1 = b1
    return -n / 2 * (np.log(2 * np.pi * sse_b / n) + 1)

se1 = np.sqrt(s2 * np.linalg.inv(X.T @ X)[1, 1])
grid = np.linspace(beta_hat[1] - 4 * se1, beta_hat[1] + 4 * se1, 201)
lp = np.array([profile_loglik(b) for b in grid])
t = (beta_hat[1] - grid) / se1
print("max |2(l_max - l_p) - n log(1 + t^2/(n-p))| =",
      np.abs(2 * (loglik_max - lp) - n * np.log1p(t ** 2 / (n - p))).max())
```

Remark (Other error distributions). The agreement between maximum likelihood and least squares is special to normal errors. If the errors have the Laplace density $(2\tau)^{-1}e^{-|\varepsilon|/\tau}$, the log-likelihood is $-n \log(2\tau) - \sum_i |y_i - \mathbf{x}_{(i)}^\top \boldsymbol{\beta}|/\tau$, and maximum likelihood minimizes the sum of *absolute* residuals; Exercise (C1) works out the location case. For errors with a t distribution it downweights large residuals. For each error distribution the likelihood picks its own loss function. Least squares is the one that goes with the Gaussian. Chapter 20 uses robust fitting as a diagnostic, and Chapter 45 treats median and quantile regression.

7.3.5 Exercises

7.3.5.1 A. Check your understanding

Exercise A1.

For $Y_1, \dots, Y_n \stackrel{\text{iid}}{\sim} N(\mu, \sigma^2)$, write the model in the form [Eq. 7.3.1](#) and read off the MLEs from [Theorem 7.3.1](#). Show that $\hat{\sigma}_{\text{ML}}^2 = (n-1)s^2/n$, where s^2 is the sample variance.

Exercise A2.

Let $\mathcal{C}(\mathbf{X}_0) \subseteq \mathcal{C}(\mathbf{X})$ with residual sums of squares $\text{SSE}_0 \geq \text{SSE}$. Show that the ratio of maximized likelihoods is $L_{0,\text{max}}/L_{\text{max}} = (\text{SSE}/\text{SSE}_0)^{n/2}$, and write it as a decreasing function of $(\text{SSE}_0 - \text{SSE})/\text{SSE}$.

Solution. By [Theorem 7.3.1](#), $L_{\text{max}} = (2\pi \text{SSE}/n)^{-n/2} e^{-n/2}$, and similarly for the smaller model. The ratio is $(\text{SSE}_0/\text{SSE})^{-n/2} = \{1 + (\text{SSE}_0 - \text{SSE})/\text{SSE}\}^{-n/2}$, which decreases in $(\text{SSE}_0 - \text{SSE})/\text{SSE}$. That quantity is, up to degrees of freedom, the F statistic of [Theorem 4.4.6](#). So the likelihood ratio test for nested normal linear models is the F test. [Chapter 11](#) develops this ([Theorem 11.4.1](#)).

7.3.5.2 B. Practice

Exercise B1.

Assume $\text{rank}(\mathbf{X}) = p$. Compute the gradient and the Hessian of $\ell(\boldsymbol{\beta}, \sigma^2)$, show that the only stationary point is $(\hat{\boldsymbol{\beta}}, \text{SSE}/n)$, and show that the Hessian there is negative definite. Why does this not by itself prove that the stationary point is a global maximum? What additional argument completes the proof?

Exercise B2.

In [Example \(Many means, one variance\)](#), verify the identity $\sum_j (Y_{ij} - \bar{Y}_{i\cdot})^2 = (Y_{i1} - Y_{i2})^2/2$, compute $E \hat{\sigma}_{\text{ML}}^2$ and $\text{Var} \hat{\sigma}_{\text{ML}}^2$, and prove that $\hat{\sigma}_{\text{ML}}^2 \rightarrow \sigma^2/2$ in probability as $m \rightarrow \infty$.

Solution. $Y_{i1} - \bar{Y}_{i\cdot} = (Y_{i1} - Y_{i2})/2 = -(Y_{i2} - \bar{Y}_{i\cdot})$, so the sum of squares is $2 \cdot (Y_{i1} - Y_{i2})^2/4$. Now $D_i = Y_{i1} - Y_{i2} \sim N(0, 2\sigma^2)$ independently, so $D_i^2/(2\sigma^2) \sim \chi^2(1)$, with mean 1 and variance 2. Hence $\hat{\sigma}_{\text{ML}}^2 = \frac{1}{4m} \sum_i D_i^2$ has mean $2\sigma^2/4 = \sigma^2/2$ and variance $m \cdot 4\sigma^4 \cdot 2/(16m^2) = \sigma^4/(2m)$. The variance tends to zero, so by Chebyshev's inequality $\hat{\sigma}_{\text{ML}}^2 \rightarrow \sigma^2/2$ in probability.

Exercise B3.

Suppose $\mathbf{Y} \sim N_n(\mathbf{X}\boldsymbol{\beta}, \sigma^2\mathbf{V})$ with $\mathbf{V}$ known and positive definite and $\mathbf{X}$ of full rank. Find the MLEs of $\boldsymbol{\beta}$ and σ^2 .

7.3.5.3 C. Going deeper

Exercise C1.

In the location model $Y_i = \mu + \varepsilon_i$, $i = 1, \dots, n$, let the errors be independent with the Laplace density $(2\tau)^{-1}e^{-|\varepsilon|/\tau}$, so $\text{Var}(\varepsilon_i) = 2\tau^2$.

- Show that every sample median maximizes the likelihood in μ , whatever τ is, and that the MLE of τ is the mean absolute deviation $n^{-1} \sum_i |Y_i - \tilde{\mu}|$ about a median $\tilde{\mu}$.
- The sample median of n observations from a density f that is positive and continuous at its median μ has asymptotic variance $1/\{4nf(\mu)^2\}$. Compute this for the Laplace density and compare it with $\text{Var}(\bar{Y})$. Which is larger, and by what factor?
- $\bar{Y}$ is the BLUE of μ (Corollary 7.2.2). Explain why (b) does not contradict Theorem 7.2.1, and contrast the answer with Example (Uniform errors and the midrange), where the winning nonlinear estimator used the extremes rather than the middle of the sample.

Solution.

- The log-likelihood is $-n \log(2\tau) - \tau^{-1} \sum_i |Y_i - \mu|$. For fixed τ it is maximized by minimizing $g(\mu) = \sum_i |Y_i - \mu|$. The function g is convex and piecewise linear, with slope $\#\{i : Y_i < \mu\} - \#\{i : Y_i > \mu\}$ between data points. The slope is ≤ 0 to the left of every median and ≥ 0 to the right, so the minimizers are exactly the sample medians. Substituting $\tilde{\mu}$ and maximizing $-n \log \tau - g(\tilde{\mu})/\tau$ over $\tau > 0$ gives $\hat{\tau} = g(\tilde{\mu})/n$.
- $f(\mu) = 1/(2\tau)$, so the median has asymptotic variance τ^2/n , while $\text{Var}(\bar{Y}) = 2\tau^2/n$. The mean has twice the variance of the median.
- The median is not a linear function of $\mathbf{Y}$: its weights depend on the ordering of the sample. The Gauss–Markov theorem compares $\bar{Y}$ only with linear unbiased estimators. With Laplace errors, which have heavier tails than the normal, the extremes are unreliable and the median gains by ignoring them; with uniform errors the extremes are the most informative observations and the midrange gains by using only them.

Exercise C2.

Let $\mathbf{Q}_2$ be an $n \times (n - r)$ matrix whose columns are an orthonormal basis of $\mathcal{C}(\mathbf{X})^\perp$, and let $\mathbf{Z} = \mathbf{Q}_2^\top \mathbf{Y}$ (“error contrasts”). Show that $\mathbf{Z} \sim N_{n-r}(\mathbf{0}, \sigma^2 \mathbf{I})$, whatever $\boldsymbol{\beta}$ is, and that $\|\mathbf{Z}\|^2 = \text{SSE}$. Show that the MLE of σ^2 based on $\mathbf{Z}$ alone is $\text{SSE}/(n - r)$. This is **restricted** (or residual) maximum likelihood.

Solution. $\mathbf{Q}_2^\top \mathbf{X} = \mathbf{0}$, so $\mathbf{Z} = \mathbf{Q}_2^\top \boldsymbol{\varepsilon} \sim N_{n-r}(\mathbf{0}, \sigma^2 \mathbf{Q}_2^\top \mathbf{Q}_2) = N_{n-r}(\mathbf{0}, \sigma^2 \mathbf{I})$ by Theorem 3.3.1. Also $\mathbf{Q}_2 \mathbf{Q}_2^\top = \mathbf{I} - \mathbf{M}$ (Proposition 6.3.2), so $\|\mathbf{Z}\|^2 = \mathbf{Y}^\top \mathbf{Q}_2 \mathbf{Q}_2^\top \mathbf{Y} = \mathbf{Y}^\top (\mathbf{I} - \mathbf{M}) \mathbf{Y} = \text{SSE}$. The log-likelihood of $\mathbf{Z}$ is $-\frac{n-r}{2} \log(2\pi\sigma^2) - \|\mathbf{Z}\|^2/(2\sigma^2)$, which is the function g of the proof of Theorem 7.3.1 with n replaced by $n - r$. It is maximized at $\|\mathbf{Z}\|^2/(n - r) = \text{SSE}/(n - r)$. In Example (Many means, one variance) this gives s^2 , which is consistent.

7.4 Sufficiency, completeness and minimum variance

The Gauss–Markov theorem compares least squares only with *linear* unbiased estimators, and [Example \(Uniform errors and the midrange\)](#) showed that the restriction matters: with uniform errors a nonlinear estimator wins easily. This section shows that under normality the restriction can be dropped. In the normal linear model, $\lambda^\top \hat{\beta}$ has the smallest variance among *all* unbiased estimators of an estimable $\lambda^\top \beta$, and s^2 has the smallest variance among all unbiased estimators of σ^2 . The argument is the classical one through sufficiency and completeness. It also produces best unbiased estimators of nonlinear functions, such as σ itself. At the end we compare the variances with the Cramér–Rao lower bound, which least squares attains for β and s^2 does not attain for σ^2 .

Two facts from mathematical statistics are quoted without proof: the factorization criterion for sufficiency and the completeness of full-rank exponential families. Both are proved in standard texts such as Lehmann and Casella (1998) and Lehmann and Romano (2005). Everything built on them is proved here.

7.4.1 Sufficiency

Definition 7.4.1 — Sufficient and complete statistics

Let $\mathbf{Y}$ have a distribution P_θ from a family indexed by $\theta \in \Theta$, and let $\mathbf{T} = \mathbf{T}(\mathbf{Y})$ be a statistic.

- (a) $\mathbf{T}$ is **sufficient** for θ if the conditional law of $\mathbf{Y}$ given $\mathbf{T}$ is the same for every θ .
- (b) $\mathbf{T}$ is **complete** if, for every function h with $E_\theta |h(\mathbf{T})| < \infty$ and $E_\theta h(\mathbf{T}) = 0$ for all $\theta \in \Theta$, we have $\Pr_\theta\{h(\mathbf{T}) = 0\} = 1$ for all θ .

A sufficient statistic carries all the information about θ in the data. Once $\mathbf{T}$ is known, the rest of $\mathbf{Y}$ is noise whose distribution does not involve the parameter. Completeness says that $\mathbf{T}$ carries no redundant information either: no nontrivial function of $\mathbf{T}$ has expectation zero for every parameter value. Equivalently, an unbiased estimator that is a function of $\mathbf{T}$ is unique.

Lemma 7.4.1 — Factorization criterion

Suppose $\mathbf{Y}$ has a density $f_\theta(\mathbf{y})$ on $\mathbb{R}^n$. Then $\mathbf{T}(\mathbf{Y})$ is sufficient iff there are functions g_θ and h , with h not depending on θ , such that $f_\theta(\mathbf{y}) = g_\theta(\mathbf{T}(\mathbf{y})) h(\mathbf{y})$ for all $\mathbf{y}$ and θ .

Lemma 7.4.2 — Completeness in exponential families

Suppose $\mathbf{Y}$ has a density of the form

$$f_\eta(\mathbf{y}) = c(\eta) h(\mathbf{y}) \exp\{\eta^\top \mathbf{T}(\mathbf{y})\}, \quad \eta \in \mathcal{E} \subseteq \mathbb{R}^k,$$

with $\mathbf{T}(\mathbf{y}) \in \mathbb{R}^k$. If $\mathcal{E}$ contains a nonempty open subset of $\mathbb{R}^k$, then $\mathbf{T}(\mathbf{Y})$ is complete

and sufficient.

Sufficiency follows from [Lemma 7.4.1](#) with $g_{\boldsymbol{\eta}}(\mathbf{T}) = c(\boldsymbol{\eta})e^{\boldsymbol{\eta}^\top \mathbf{T}}$. Completeness rests on the uniqueness of Laplace transforms: $E_{\boldsymbol{\eta}} h(\mathbf{T}) = 0$ on an open set of $\boldsymbol{\eta}$ forces a certain two-sided Laplace transform to vanish on an open set, and hence $h(\mathbf{T}) = 0$ almost surely.

For the normal linear model, expand the exponent of the density:

$$\|\mathbf{y} - \mathbf{X}\boldsymbol{\beta}\|^2 = \mathbf{y}^\top \mathbf{y} - 2\boldsymbol{\beta}^\top \mathbf{X}^\top \mathbf{y} + \boldsymbol{\beta}^\top \mathbf{X}^\top \mathbf{X} \boldsymbol{\beta}.$$

So the density of [Eq. 7.3.1](#) is

$$f(\mathbf{y}) = (2\pi\sigma^2)^{-n/2} \exp\left\{-\frac{\boldsymbol{\beta}^\top \mathbf{X}^\top \mathbf{X} \boldsymbol{\beta}}{2\sigma^2}\right\} \exp\left\{\frac{\boldsymbol{\beta}^\top \mathbf{X}^\top \mathbf{y}}{\sigma^2} - \frac{\mathbf{y}^\top \mathbf{y}}{2\sigma^2}\right\}, \quad (7.4.1)$$

and the data enter only through $\mathbf{X}^\top \mathbf{y}$ and $\mathbf{y}^\top \mathbf{y}$.

Proposition 7.4.3 — A complete sufficient statistic

In the normal linear model [Eq. 7.3.1](#), let $\mathbf{Q}$ be an $n \times r$ matrix whose columns are an orthonormal basis of $\mathcal{C}(\mathbf{X})$. Then $\mathbf{T} = (\mathbf{Q}^\top \mathbf{Y}, \mathbf{Y}^\top \mathbf{Y})$ is complete and sufficient for $(\boldsymbol{\beta}, \sigma^2)$. The statistics $(\mathbf{X}^\top \mathbf{Y}, \mathbf{Y}^\top \mathbf{Y})$ and $(\mathbf{M}\mathbf{Y}, \text{SSE})$ are one-to-one functions of $\mathbf{T}$, so they are complete and sufficient too.

Proof. Since $\mathbf{X}\boldsymbol{\beta} \in \mathcal{C}(\mathbf{X})$, $\mathbf{X}\boldsymbol{\beta} = \mathbf{Q}\boldsymbol{\gamma}$ with $\boldsymbol{\gamma} = \mathbf{Q}^\top \mathbf{X}\boldsymbol{\beta} \in \mathbb{R}^r$, and $\boldsymbol{\beta}^\top \mathbf{X}^\top \mathbf{y} = \boldsymbol{\gamma}^\top \mathbf{Q}^\top \mathbf{y}$. By [Eq. 7.4.1](#) the density has the form of [Lemma 7.4.2](#) with $k = r + 1$,

$$\mathbf{T}(\mathbf{y}) = (\mathbf{Q}^\top \mathbf{y}, \mathbf{y}^\top \mathbf{y}), \quad \boldsymbol{\eta} = (\boldsymbol{\gamma}/\sigma^2, -1/(2\sigma^2)).$$

As $\boldsymbol{\beta}$ ranges over $\mathbb{R}^p$, $\boldsymbol{\gamma} = \mathbf{Q}^\top \mathbf{X}\boldsymbol{\beta}$ ranges over all of $\mathbb{R}^r$: the $r \times p$ matrix $\mathbf{Q}^\top \mathbf{X}$ has rank r , because $\mathbf{Q}\mathbf{Q}^\top \mathbf{X} = \mathbf{M}\mathbf{X} = \mathbf{X}$ has rank r . Hence $\mathcal{E} = \mathbb{R}^r \times (-\infty, 0)$, which is open, and [Lemma 7.4.2](#) applies. For the equivalent statistics: $\mathbf{M}\mathbf{Y} = \mathbf{Q}\mathbf{Q}^\top \mathbf{Y}$ ([Proposition 6.3.2](#)) and $\text{SSE} = \mathbf{Y}^\top \mathbf{Y} - \|\mathbf{Q}^\top \mathbf{Y}\|^2$ are functions of $\mathbf{T}$, and conversely $\mathbf{Q}^\top \mathbf{Y} = \mathbf{Q}^\top \mathbf{M}\mathbf{Y}$ and $\mathbf{Y}^\top \mathbf{Y} = \text{SSE} + \|\mathbf{M}\mathbf{Y}\|^2$. Also $\mathbf{X}^\top \mathbf{Y} = \mathbf{X}^\top \mathbf{Q}(\mathbf{Q}^\top \mathbf{Y})$, and $\mathbf{Q} = \mathbf{X}\mathbf{K}$ for some $\mathbf{K}$, since the columns of $\mathbf{Q}$ lie in $\mathcal{C}(\mathbf{X})$, so $\mathbf{Q}^\top \mathbf{Y} = \mathbf{K}^\top \mathbf{X}^\top \mathbf{Y}$. A one-to-one function of a complete sufficient statistic is again complete and sufficient, since both definitions refer only to the information (the σ -field) the statistic generates. ■

The reduction is dramatic. An n -dimensional data vector is replaced by $r+1$ numbers: the fitted values (equivalently, the least squares estimates of the estimable functions) and the residual sum of squares. Under normality the residual *vector*, apart from its length, carries no information about the parameters. Its direction is uniformly distributed over the unit sphere of $\mathcal{C}(\mathbf{X})^\perp$ whatever $(\boldsymbol{\beta}, \sigma^2)$ is ([Exercise \(C1\)](#)). This is the precise sense in which the residuals are “just noise”, and it is also why they are useful for checking the model. Their pattern can reveal departures from normality precisely because, under normality, it has a known and parameter-free distribution.

7.4.2 Rao–Blackwell and Lehmann–Scheffé

Theorem 7.4.4 — Rao–Blackwell

Let $\mathbf{T}$ be sufficient for $\boldsymbol{\theta}$, and let $\delta(\mathbf{Y})$ be an unbiased estimator of $g(\boldsymbol{\theta})$ with $\text{Var}_{\boldsymbol{\theta}} \delta < \infty$ for all $\boldsymbol{\theta}$. Then $\delta^* = E(\delta \mid \mathbf{T})$ is a statistic, it is unbiased for $g(\boldsymbol{\theta})$, and

$$\begin{aligned} \text{Var}_{\boldsymbol{\theta}}(\delta) &= \text{Var}_{\boldsymbol{\theta}}(\delta^*) + E_{\boldsymbol{\theta}}[(\delta - \delta^*)^2] \\ &\geq \text{Var}_{\boldsymbol{\theta}}(\delta^*) \end{aligned}$$

for every $\boldsymbol{\theta}$, with equality iff $\delta = \delta^*$ with probability one.

Proof. The law of $\mathbf{Y}$ given $\mathbf{T}$ does not depend on $\boldsymbol{\theta}$, so neither does $E(\delta(\mathbf{Y}) \mid \mathbf{T})$. It is therefore computable from the data, which makes it a statistic. By the law of total expectation (Proposition 2.5.4), $E \delta^* = E \delta = g(\boldsymbol{\theta})$. By the law of total variance in the same result, $\text{Var} \delta = E\{\text{Var}(\delta \mid \mathbf{T})\} + \text{Var}(\delta^*)$, and $E\{\text{Var}(\delta \mid \mathbf{T})\} = E\{E[(\delta - \delta^*)^2 \mid \mathbf{T}]\} = E(\delta - \delta^*)^2$. This last term is zero iff $\delta = \delta^*$ almost surely. ■

Conditioning on a sufficient statistic averages out the part of an estimator that responds to parameter-free noise. It never hurts, and it helps unless the estimator was already a function of $\mathbf{T}$.

Theorem 7.4.5 — Lehmann–Scheffé

Let $\mathbf{T}$ be complete and sufficient for $\boldsymbol{\theta}$. If $\phi(\mathbf{T})$ is unbiased for $g(\boldsymbol{\theta})$ and has finite variance, then $\phi(\mathbf{T})$ is a **uniformly minimum variance unbiased estimator** (UMVUE) of $g(\boldsymbol{\theta})$: $\text{Var}_{\boldsymbol{\theta}} \phi(\mathbf{T}) \leq \text{Var}_{\boldsymbol{\theta}} \delta$ for every unbiased δ and every $\boldsymbol{\theta}$. It is the only unbiased function of $\mathbf{T}$ and, up to sets of probability zero, the only UMVUE.

Proof. If $\phi_1(\mathbf{T})$ and $\phi_2(\mathbf{T})$ are both unbiased, then $E_{\boldsymbol{\theta}}\{\phi_1(\mathbf{T}) - \phi_2(\mathbf{T})\} = 0$ for all $\boldsymbol{\theta}$, so $\phi_1(\mathbf{T}) = \phi_2(\mathbf{T})$ almost surely by completeness. Now let δ be any unbiased estimator with finite variance. By Theorem 7.4.4, $E(\delta \mid \mathbf{T})$ is an unbiased function of $\mathbf{T}$, so it equals $\phi(\mathbf{T})$ almost surely, and $\text{Var} \phi(\mathbf{T}) \leq \text{Var} \delta$, with equality iff $\delta = \phi(\mathbf{T})$ almost surely. (If δ has infinite variance there is nothing to prove.) ■

7.4.3 Best unbiased estimators in the normal linear model

Theorem 7.4.6 — Minimum variance unbiased estimation

Assume the normal linear model Eq. 7.3.1 with $n > r$, and let $s^2 = \text{SSE}/(n - r)$.

- (a) For every $\boldsymbol{\lambda} \in \mathcal{C}(\mathbf{X}^\top)$, $\boldsymbol{\lambda}^\top \hat{\boldsymbol{\beta}}$ is the UMVUE of $\boldsymbol{\lambda}^\top \boldsymbol{\beta}$.
- (b) s^2 is the UMVUE of σ^2 .

- (c) More generally, every estimator of the form $\phi(\mathbf{MY}, \text{SSE})$ with finite variance is the UMVUE of its expectation. In particular, for $k = n - r$,

$$b_k = \sqrt{\frac{2}{k}} \frac{\Gamma(\frac{k+1}{2})}{\Gamma(\frac{k}{2})},$$

s/b_k is the UMVUE of σ , and $\mathbf{x}_0^\top \hat{\boldsymbol{\beta}} + z s/b_k$ is the UMVUE of $\mathbf{x}_0^\top \boldsymbol{\beta} + z\sigma$ for every constant z and every $\mathbf{x}_0 \in \mathcal{C}(\mathbf{X}^\top)$.

Proof. By Proposition 7.4.3, $(\mathbf{MY}, \text{SSE})$ is complete and sufficient, so by Theorem 7.4.5 it suffices to show that each estimator is an unbiased function of it with finite variance.

(a) Write $\boldsymbol{\lambda} = \mathbf{X}^\top \boldsymbol{\rho}$. Then $\boldsymbol{\lambda}^\top \hat{\boldsymbol{\beta}} = \boldsymbol{\rho}^\top \mathbf{MY}$ is a function of $\mathbf{MY}$, it is unbiased by Theorem 7.2.1(a), and its variance is finite. (b) s^2 is a function of SSE, it is unbiased because $E(\text{SSE}) = \sigma^2(n - r)$ (Theorem 2.3.1), and it has finite variance. (c) The first statement is Theorem 7.4.5 again. For σ , note that $\text{SSE}/\sigma^2 = \mathbf{Y}^\top (\mathbf{I} - \mathbf{M}) \mathbf{Y} / \sigma^2 \sim \chi^2(k)$ by Theorem 4.3.2(a), since $\mathbf{I} - \mathbf{M}$ is a projection of rank k and $(\mathbf{I} - \mathbf{M})\mathbf{X}\boldsymbol{\beta} = \mathbf{0}$. If $W \sim \chi^2(k)$, with density $w^{k/2-1}e^{-w/2}/\{2^{k/2}\Gamma(k/2)\}$, then

$$\begin{aligned} E\sqrt{W} &= \frac{1}{2^{k/2}\Gamma(k/2)} \int_0^\infty w^{(k+1)/2-1} e^{-w/2} dw \\ &= \frac{2^{(k+1)/2}\Gamma(\frac{k+1}{2})}{2^{k/2}\Gamma(\frac{k}{2})} = \sqrt{2} \frac{\Gamma(\frac{k+1}{2})}{\Gamma(\frac{k}{2})}. \end{aligned}$$

Hence $Es = \sigma E\sqrt{W/k} = b_k\sigma$, and s/b_k is unbiased for σ , with variance at most $E s^2 / b_k^2 < \infty$. The last estimator is a sum of two unbiased functions of $(\mathbf{MY}, \text{SSE})$. ■

Part (a) is the promised strengthening of Gauss–Markov. Under normality no unbiased estimator of $\boldsymbol{\lambda}^\top \boldsymbol{\beta}$, linear or not, beats least squares. The midrange of Example (Uniform errors and the midrange) cannot help when the errors are normal. Part (c) shows the method at work beyond linear estimators. The obvious estimate s of σ is biased, because $Es < \sqrt{E s^2}$ by Jensen’s inequality, and dividing by b_k removes the bias.

Example. (Estimating σ without bias)

The constant $b_k = E(s)/\sigma$ depends only on the residual degrees of freedom. For the stack loss regression, $k = 17$ and $b_{17} = 0.9854$, so the UMVUE of σ is $s/b_{17} = 1.0148 s$. The correction is 1.5% here. It is large for very small k , with $b_2 = 0.8862$, and negligible for large k , with $b_{200} = 0.9988$.

```
import numpy as np
from scipy import special

def mean_s_factor(k):
    """b_k = E(s)/sigma when k s^2 / sigma^2 ~ chi^2(k)."""
    return np.sqrt(2 / k) * np.exp(special.gammaln((k + 1) / 2) - special.gammaln(k / 2))

for k in [2, 5, 10, 17, 50, 200]:
    b = mean_s_factor(k)
```

```
print(f"k = {k:3d}    E(s)/sigma = {b:.4f}    UMVUE of sigma = s / {b:.4f}")
```

7.4.4 Rao–Blackwellization by reflection

The Rao–Blackwell theorem is not only a device for proofs. In the normal linear model we can carry out the conditioning explicitly, and it takes every linear unbiased estimator to least squares.

Example. (Conditioning on the sufficient statistic)

Let $\mathbf{a}^\top \mathbf{Y}$ be any LUE of an estimable $\lambda^\top \beta$, such as the end-point slope of [Example \(Three linear unbiased estimators of a slope\)](#). Let $\mathbf{R} = 2\mathbf{M} - \mathbf{I}$, the reflection that fixes $\mathcal{C}(\mathbf{X})$ and reverses $\mathcal{C}(\mathbf{X})^\perp$. It is symmetric and orthogonal ($\mathbf{R}^2 = \mathbf{I}$) and $\mathbf{R}\mathbf{X} = \mathbf{X}$. Consequently:

- $\mathbf{R}\mathbf{Y} \sim N_n(\mathbf{R}\mathbf{X}\beta, \sigma^2 \mathbf{R}\mathbf{R}^\top) = N_n(\mathbf{X}\beta, \sigma^2 \mathbf{I})$ by [Theorem 3.3.1](#), so $\mathbf{R}\mathbf{Y}$ has the same distribution as $\mathbf{Y}$;
- the sufficient statistic is unchanged: $\mathbf{X}^\top \mathbf{R}\mathbf{Y} = \mathbf{X}^\top \mathbf{Y}$ and $\|\mathbf{R}\mathbf{Y}\| = \|\mathbf{Y}\|$.

So the pairs $(\mathbf{a}^\top \mathbf{Y}, \mathbf{T}(\mathbf{Y}))$ and $(\mathbf{a}^\top \mathbf{R}\mathbf{Y}, \mathbf{T}(\mathbf{Y}))$ have the same joint distribution, where $\mathbf{T} = (\mathbf{X}^\top \mathbf{Y}, \mathbf{Y}^\top \mathbf{Y})$. Hence $E(\mathbf{a}^\top \mathbf{Y} \mid \mathbf{T}) = E(\mathbf{a}^\top \mathbf{R}\mathbf{Y} \mid \mathbf{T})$, and averaging,

$$E(\mathbf{a}^\top \mathbf{Y} \mid \mathbf{T}) = E\left(\mathbf{a}^\top \frac{\mathbf{Y} + \mathbf{R}\mathbf{Y}}{2} \mid \mathbf{T}\right) = E(\mathbf{a}^\top \mathbf{M}\mathbf{Y} \mid \mathbf{T}) = \mathbf{a}^\top \mathbf{M}\mathbf{Y} = \lambda^\top \hat{\beta},$$

because $\mathbf{a}^\top \mathbf{M}\mathbf{Y}$ is a function of $\mathbf{T}$ and $\mathbf{M}\mathbf{a} = \mathbf{a}_*$ ([Proposition 7.1.2](#)). The Rao–Blackwell improvement of *every* LUE is the least squares estimate. The listing checks the identity $\frac{1}{2}(\mathbf{a}^\top \mathbf{y} + \mathbf{a}^\top \mathbf{R}\mathbf{y}) = \lambda^\top \hat{\beta}$ for simulated data and the end-point slope.

```
import numpy as np

x = np.array([1.0, 2.0, 2.5, 3.0, 4.0, 5.0, 5.5, 7.0, 8.0, 9.0, 10.5, 12.0])
n = len(x)
X = np.column_stack([np.ones(n), x])
M = X @ np.linalg.solve(X.T @ X, X.T)
R = 2 * M - np.eye(n)                                # reflection: keeps C(X), flips C(X)
                                                        )-perp

rng = np.random.default_rng(71)
y = X @ np.array([2.0, 0.5]) + rng.normal(size=n)
y_ref = R @ y
print("X^T y unchanged:", np.allclose(X.T @ y_ref, X.T @ y))
print("y^T y unchanged:", np.isclose(y_ref @ y_ref, y @ y))

a_end = np.zeros(n)
a_end[[0, -1]] = [-1.0, 1.0]
a_end /= x[-1] - x[0]                                # the end-point slope estimator
a_ls = X @ np.linalg.solve(X.T @ X, [0.0, 1.0])
print("average of a^T y over the two points:", (a_end @ y + a_end @ y_ref) / 2)
print("least squares slope:", a_ls @ y)
```

The reflection argument uses the normal distribution only through its invariance under orthogo-

nal maps that fix the mean. It makes concrete what “the residuals are noise” means. Reversing the residual vector changes neither the likelihood nor the sufficient statistic, so an estimator should not depend on the residuals’ direction. An LUE that uses the residuals is replaced by the average of its value at $\mathbf{y}$ and at the reflected data, and that average no longer involves the residuals.

7.4.5 The Cramér–Rao bound

A different route to optimality bounds the variance of every unbiased estimator from below and checks whether the bound is attained. We state the bound for the full-rank normal linear model, where every ingredient can be computed.

Write $\boldsymbol{\theta} = (\boldsymbol{\beta}^\top, v)^\top$ with $v = \sigma^2$. The **score** is the gradient of the log-likelihood Eq. 7.3.2, evaluated at the random $\mathbf{Y}$:

$$\mathbf{S} = \begin{pmatrix} \partial \ell / \partial \boldsymbol{\beta} \\ \partial \ell / \partial v \end{pmatrix} = \begin{pmatrix} \mathbf{X}^\top \boldsymbol{\varepsilon} / v \\ -\frac{n}{2v} + \frac{\boldsymbol{\varepsilon}^\top \boldsymbol{\varepsilon}}{2v^2} \end{pmatrix}, \quad \boldsymbol{\varepsilon} = \mathbf{Y} - \mathbf{X}\boldsymbol{\beta}.$$

Proposition 7.4.7 — Score and information

In Eq. 7.3.1 with $\text{rank}(\mathbf{X}) = p$, $\mathbb{E} \mathbf{S} = \mathbf{0}$ and

$$\mathcal{I}(\boldsymbol{\theta}) = \text{Cov}(\mathbf{S}) = \begin{pmatrix} \mathbf{X}^\top \mathbf{X} / \sigma^2 & \mathbf{0} \\ \mathbf{0}^\top & n / (2\sigma^4) \end{pmatrix}.$$

Proof. $\boldsymbol{\varepsilon} \sim N_n(\mathbf{0}, v\mathbf{I})$. Hence $\mathbb{E}(\mathbf{X}^\top \boldsymbol{\varepsilon} / v) = \mathbf{0}$ and $\text{Cov}(\mathbf{X}^\top \boldsymbol{\varepsilon} / v) = \mathbf{X}^\top \mathbf{X} / v$ by Theorem 2.2.1. Next, $\boldsymbol{\varepsilon}^\top \boldsymbol{\varepsilon} / v \sim \chi^2(n)$ by Theorem 4.3.2(a) with $\mathbf{P} = \mathbf{I}$, so it has mean n and variance $2n$. So the last coordinate has mean $-n/(2v) + nv/(2v^2) = 0$ and variance $2n/(4v^2) = n/(2v^2)$. Finally, by Theorem 4.2.3(c) with $\mathbf{K} = \mathbf{X}^\top$, $\mathbf{A} = \mathbf{I}$ and $\boldsymbol{\mu} = \mathbf{0}$, $\text{Cov}(\mathbf{X}^\top \boldsymbol{\varepsilon}, \boldsymbol{\varepsilon}^\top \boldsymbol{\varepsilon}) = \mathbf{0}$. ■

Theorem 7.4.8 — Cramér–Rao bound in the normal linear model

Assume Eq. 7.3.1 with $\text{rank}(\mathbf{X}) = p$. Let $\delta(\mathbf{Y})$ have finite variance and satisfy $\mathbb{E}_{\boldsymbol{\theta}} \delta = g(\boldsymbol{\theta})$ for all $\boldsymbol{\theta}$, with g differentiable. Then

$$\begin{aligned} \text{Var}_{\boldsymbol{\theta}}(\delta) &\geq \nabla g(\boldsymbol{\theta})^\top \mathcal{I}(\boldsymbol{\theta})^{-1} \nabla g(\boldsymbol{\theta}) \\ &= \sigma^2 \mathbf{g}_{\boldsymbol{\beta}}^\top (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{g}_{\boldsymbol{\beta}} + \frac{2\sigma^4}{n} g_v^2, \end{aligned}$$

where $\mathbf{g}_{\boldsymbol{\beta}} = \partial g / \partial \boldsymbol{\beta}$ and $g_v = \partial g / \partial v$.

Proof. Write $f_{\boldsymbol{\theta}}$ for the density. Differentiating $g(\boldsymbol{\theta}) = \int \delta(\mathbf{y}) f_{\boldsymbol{\theta}}(\mathbf{y}) d\mathbf{y}$ under the integral sign, which is justified because the normal family is an exponential family (Lehmann and

Casella 1998, chapter 1), gives

$$\begin{aligned}\nabla g(\boldsymbol{\theta}) &= \int \delta(\mathbf{y}) \nabla \log f_{\boldsymbol{\theta}}(\mathbf{y}) f_{\boldsymbol{\theta}}(\mathbf{y}) d\mathbf{y} \\ &= E(\delta \mathbf{S}) = \text{Cov}(\delta, \mathbf{S}),\end{aligned}$$

the last step because $E\mathbf{S} = \mathbf{0}$. For any constant vector $\mathbf{c}$, $0 \leq \text{Var}(\delta - \mathbf{c}^\top \mathbf{S}) = \text{Var} \delta - 2\mathbf{c}^\top \nabla g + \mathbf{c}^\top \mathcal{I} \mathbf{c}$. Choosing $\mathbf{c} = \mathcal{I}^{-1} \nabla g$ gives $\text{Var} \delta \geq \nabla g^\top \mathcal{I}^{-1} \nabla g$. The block-diagonal form of [Proposition 7.4.7](#) gives the right-hand side.

■

For $g(\boldsymbol{\theta}) = \boldsymbol{\lambda}^\top \boldsymbol{\beta}$ the bound is $\sigma^2 \boldsymbol{\lambda}^\top (\mathbf{X}^\top \mathbf{X})^{-1} \boldsymbol{\lambda}$, exactly the variance of $\boldsymbol{\lambda}^\top \hat{\boldsymbol{\beta}}$. Least squares **attains** the bound, which gives a second proof of [Theorem 7.4.6\(a\)](#) in full rank. For $g = \sigma^2$ the bound is $2\sigma^4/n$, while

$$\text{Var}(s^2) = \frac{\sigma^4}{(n-p)^2} \text{Var}\{\chi^2(n-p)\} = \frac{2\sigma^4}{n-p} > \frac{2\sigma^4}{n}.$$

So s^2 does not attain the bound, although by [Theorem 7.4.6\(b\)](#) no unbiased estimator does better. The bound is simply not sharp here. Its ratio to the attained minimum, the **efficiency** of s^2 , is $(n-p)/n$. That is 0.810 for the stack loss regression, and it tends to one as $n \rightarrow \infty$ with p fixed. The same happens for σ : the UMVUE s/b_k has variance $0.0298 \sigma^2$ when $k = 17$, against the bound $\sigma^2/(2n) = 0.0238 \sigma^2$, an efficiency of 0.798. The equality condition in the proof explains why. The bound is attained iff $\text{Var}(\delta - \mathbf{c}^\top \mathbf{S}) = 0$, that is, iff $\delta = g(\boldsymbol{\theta}) + \nabla g^\top \mathcal{I}^{-1} \mathbf{S}$ with probability one. For $g = \sigma^2$ the right-hand side is $\sigma^2 + (2\sigma^4/n)\{-n/(2\sigma^2) + \boldsymbol{\varepsilon}^\top \boldsymbol{\varepsilon}/(2\sigma^4)\} = \|\mathbf{Y} - \mathbf{X}\boldsymbol{\beta}\|^2/n$. This is not a statistic, because it involves the unknown $\boldsymbol{\beta}$, so no unbiased estimator of σ^2 attains the bound.

Warning. The Cramér–Rao bound is a lower bound, not a target. An estimator that fails to attain it may still be the best unbiased estimator, as s^2 is. When an estimator does attain it, the bound certifies optimality without any appeal to completeness. The Lehmann–Scheffé route is stronger in one respect: it needs no differentiability and covers every estimable function in every rank.

7.4.6 What changes without normality

Every result of this section uses the normal density. Without it the conclusions can fail. Under uniform errors the midrange beats $\bar{Y}$ ([Example \(Uniform errors and the midrange\)](#)). Under Laplace errors the sample median has asymptotic variance $\sigma^2/(2n)$, half that of $\bar{Y}$. Under the second-moment assumptions alone, the best that can be said for s^2 is a result in the spirit of Gauss–Markov. Among quadratic unbiased estimators $\mathbf{Y}^\top \mathbf{A} \mathbf{Y}$ of σ^2 with $\mathbf{A}$ nonnegative definite, s^2 has the smallest variance when the errors have zero excess kurtosis, or when all leverages are equal. [Exercise \(C2\)](#) outlines this result of Atiqullah (1962).

7.4.7 Exercises

7.4.7.1 A. Check your understanding

Exercise A1.

Assume $\text{rank}(\mathbf{X}) = p$. Use [Lemma 7.4.1](#) and the identity $\|\mathbf{y} - \mathbf{X}\boldsymbol{\beta}\|^2 = \text{SSE} + (\hat{\boldsymbol{\beta}} - \boldsymbol{\beta})^\top \mathbf{X}^\top \mathbf{X} (\hat{\boldsymbol{\beta}} - \boldsymbol{\beta})$ to show directly that $(\hat{\boldsymbol{\beta}}, \text{SSE})$ is sufficient for $(\boldsymbol{\beta}, \sigma^2)$. Is $\hat{\boldsymbol{\beta}}$ alone sufficient?

Exercise A2.

Suppose $\sigma^2 = \sigma_0^2$ is known. Show that $\mathbf{X}^\top \mathbf{Y}$ alone is complete and sufficient for $\boldsymbol{\beta}$, and that $\boldsymbol{\lambda}^\top \hat{\boldsymbol{\beta}}$ is still the UMVUE of every estimable $\boldsymbol{\lambda}^\top \boldsymbol{\beta}$. Why is $\mathbf{X}^\top \mathbf{Y}$ not sufficient when σ^2 is unknown?

7.4.7.2 B. Practice**Exercise B1.**

For estimable $\boldsymbol{\lambda}^\top \boldsymbol{\beta}$, find the UMVUE of $(\boldsymbol{\lambda}^\top \boldsymbol{\beta})^2$ in the normal linear model. Can it be negative? Comment.

Solution. $E(\boldsymbol{\lambda}^\top \hat{\boldsymbol{\beta}})^2 = (\boldsymbol{\lambda}^\top \boldsymbol{\beta})^2 + \sigma^2 \boldsymbol{\lambda}^\top (\mathbf{X}^\top \mathbf{X})^{-1} \boldsymbol{\lambda}$ and $E s^2 = \sigma^2$, so $(\boldsymbol{\lambda}^\top \hat{\boldsymbol{\beta}})^2 - s^2 \boldsymbol{\lambda}^\top (\mathbf{X}^\top \mathbf{X})^{-1} \boldsymbol{\lambda}$ is unbiased. It is a function of $(\mathbf{M}\mathbf{Y}, \text{SSE})$, so it is the UMVUE by [Theorem 7.4.6\(c\)](#). It is negative whenever $|t| < 1$ for the t statistic of [Proposition 7.3.3](#) at $\psi = 0$, which has positive probability. An estimator of a nonnegative quantity that can be negative is clearly improvable, for example by its positive part. The positive part is biased but has smaller mean squared error. Unbiasedness can be a costly requirement.

Exercise B2.

Compute the observed information $-\nabla^2 \ell$ at $(\boldsymbol{\beta}, v)$, take its expectation, and confirm [Proposition 7.4.7](#) by this second route.

Exercise B3.

For the stack loss model, the 90% quantile of the response at a new operating condition $\mathbf{x}_0$ is $\mathbf{x}_0^\top \boldsymbol{\beta} + 1.2816 \sigma$. Write down its UMVUE, and explain why $\mathbf{x}_0^\top \hat{\boldsymbol{\beta}} + 1.2816 s$ is biased. By how much, in units of σ ?

7.4.7.3 C. Going deeper**Exercise C1.**

In the normal linear model with $n - r \geq 2$, let $\mathbf{U} = (\mathbf{I} - \mathbf{M})\mathbf{Y} / \|(\mathbf{I} - \mathbf{M})\mathbf{Y}\|$. Show that $\mathbf{U}$ is independent of $(\mathbf{M}\mathbf{Y}, \text{SSE})$, and that its distribution (uniform on the unit sphere of $\mathcal{C}(\mathbf{X})^\perp$) does not depend on $(\boldsymbol{\beta}, \sigma^2)$. *Hint:* write $(\mathbf{I} - \mathbf{M})\mathbf{Y} = \mathbf{Q}_2 \mathbf{Z}$ as in [Exercise \(C2\)](#), and use the rotation invariance of $N(\mathbf{0}, \sigma^2 \mathbf{I})$.

Solution. With $\mathbf{Q}_2$ as in [Exercise \(C2\)](#), $(\mathbf{I} - \mathbf{M})\mathbf{Y} = \mathbf{Q}_2\mathbf{Z}$ with $\mathbf{Z} \sim N_{n-r}(\mathbf{0}, \sigma^2\mathbf{I})$, and $\mathbf{Z}$ is independent of $\mathbf{MY}$ by [Theorem 3.4.1](#), since $\text{Cov}(\mathbf{MY}, \mathbf{Z}) = \sigma^2\mathbf{MQ}_2 = \mathbf{0}$. Then $\text{SSE} = \|\mathbf{Z}\|^2$ and $\mathbf{U} = \mathbf{Q}_2\mathbf{Z}/\|\mathbf{Z}\|$. For any orthogonal $\mathbf{H}$, $\mathbf{HZ}$ has the same law as $\mathbf{Z}$, so the law of $\mathbf{Z}/\|\mathbf{Z}\|$ is invariant under all rotations. The only such law on the sphere is the uniform one. The independence of the direction $\mathbf{Z}/\|\mathbf{Z}\|$ and the length $\|\mathbf{Z}\|$ follows from the polar form of the density of $\mathbf{Z}$, which is a function of $\|\mathbf{z}\|$ only. Hence $\mathbf{U}$ is independent of $(\mathbf{MY}, \text{SSE})$, and its law involves neither $\boldsymbol{\beta}$ nor σ^2 .

Exercise C2.

Assume only that the errors are independent with common variance σ^2 , zero third moment and fourth moment μ_4 . Let $\mathbf{Y}^\top \mathbf{A} \mathbf{Y}$ be an unbiased estimator of σ^2 with $\mathbf{A}$ symmetric nonnegative definite. (a) Show that $\mathbf{A}\mathbf{X} = \mathbf{0}$ and $\text{tr } \mathbf{A} = 1$. (b) Using [Theorem 2.3.3](#), show that $\text{Var}(\mathbf{Y}^\top \mathbf{A} \mathbf{Y}) = (\mu_4 - 3\sigma^4) \sum_i a_{ii}^2 + 2\sigma^4 \text{tr}(\mathbf{A}^2)$. (c) Show that $\text{tr}(\mathbf{A}^2) \geq 1/(n-r)$, with equality iff $\mathbf{A} = (\mathbf{I} - \mathbf{M})/(n-r)$. (d) Conclude that s^2 has the smallest variance in this class when $\mu_4 = 3\sigma^4$. *Hint for (c):* $\mathbf{A} = (\mathbf{I} - \mathbf{M})\mathbf{A}(\mathbf{I} - \mathbf{M})$, so the eigenvalues of $\mathbf{A}$ live on $\mathcal{C}(\mathbf{X})^\perp$.

7.5 Sampling distributions of the estimates

A point estimate is of little use without a measure of its uncertainty. Under the second-moment assumptions we know the mean and variance of $\boldsymbol{\lambda}^\top \hat{\boldsymbol{\beta}}$ and the mean of s^2 , but not their distributions. Under normality we know everything. The estimates of the estimable functions are jointly normal, the residual sum of squares is a scaled chi-squared variable, and the two are independent. These three facts are what the whole of Part III rests on. Tests, confidence intervals, confidence ellipsoids and prediction intervals are all built from them. The machinery was assembled in [Chapter 3](#) and [Chapter 4](#). Here we apply it.

7.5.1 The main theorem

Theorem 7.5.1 — Sampling distributions in the normal linear model

Assume [Eq. 7.3.1](#) with $r = \text{rank}(\mathbf{X}) < n$. Let $\boldsymbol{\Lambda}$ be $p \times q$ with $\mathcal{C}(\boldsymbol{\Lambda}) \subseteq \mathcal{C}(\mathbf{X}^\top)$, let $(\mathbf{X}^\top \mathbf{X})^-$ be any generalized inverse, and let $s^2 = \text{SSE}/(n-r)$.

- (a) $\boldsymbol{\Lambda}^\top \hat{\boldsymbol{\beta}} \sim N_q(\boldsymbol{\Lambda}^\top \boldsymbol{\beta}, \sigma^2 \boldsymbol{\Lambda}^\top (\mathbf{X}^\top \mathbf{X})^- \boldsymbol{\Lambda})$. In particular $\hat{\mathbf{Y}} = \mathbf{MY} \sim N_n(\mathbf{X}\boldsymbol{\beta}, \sigma^2 \mathbf{M})$, and if $\text{rank}(\mathbf{X}) = p$, $\hat{\boldsymbol{\beta}} \sim N_p(\boldsymbol{\beta}, \sigma^2 (\mathbf{X}^\top \mathbf{X})^{-1})$.
- (b) $\text{SSE}/\sigma^2 = (n-r)s^2/\sigma^2 \sim \chi^2(n-r)$.
- (c) $\mathbf{MY}$ and SSE are independent. Hence $\boldsymbol{\Lambda}^\top \hat{\boldsymbol{\beta}}$, and every function of the fitted values, is independent of s^2 .
- (d) $E(s^2) = \sigma^2$ and $\text{Var}(s^2) = 2\sigma^4/(n-r)$.

Proof. Write $\mathbf{\Lambda} = \mathbf{X}^\top \mathbf{P}$ for an $n \times q$ matrix $\mathbf{P}$. As in the proof of [Theorem 7.2.1\(d\)](#), $\mathbf{\Lambda}^\top \hat{\boldsymbol{\beta}} = \mathbf{P}^\top \mathbf{M} \mathbf{Y}$ for every least squares estimate $\hat{\boldsymbol{\beta}}$.

- (a) By [Theorem 3.3.1](#), $\mathbf{P}^\top \mathbf{M} \mathbf{Y}$ is normal with mean $\mathbf{P}^\top \mathbf{M} \mathbf{X} \boldsymbol{\beta} = \mathbf{P}^\top \mathbf{X} \boldsymbol{\beta} = \mathbf{\Lambda}^\top \boldsymbol{\beta}$ and covariance $\sigma^2 \mathbf{P}^\top \mathbf{M} \mathbf{M}^\top \mathbf{P} = \sigma^2 \mathbf{P}^\top \mathbf{M} \mathbf{P} = \sigma^2 \mathbf{\Lambda}^\top (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{\Lambda}$, by [Theorem 6.3.7](#). The special cases are $\mathbf{\Lambda} = \mathbf{X}^\top$, with $\mathbf{P} = \mathbf{I}$, and $\mathbf{\Lambda} = \mathbf{I}_p$ in full rank.
- (b) $\text{SSE} = \mathbf{Y}^\top (\mathbf{I} - \mathbf{M}) \mathbf{Y}$, and $\mathbf{I} - \mathbf{M}$ is symmetric and idempotent of rank $n - r$ ([Theorem 6.3.3](#) and [Proposition 6.3.4](#)). By [Theorem 4.3.2\(a\)](#), $\text{SSE}/\sigma^2 \sim \chi^2(n - r, \|(\mathbf{I} - \mathbf{M})\mathbf{X}\boldsymbol{\beta}\|^2/\sigma^2)$, and the noncentrality is zero because $(\mathbf{I} - \mathbf{M})\mathbf{X} = \mathbf{0}$.
- (c) Apply [Theorem 4.4.1](#) with $\mathbf{B} = \mathbf{M}$, $\mathbf{A} = \mathbf{I} - \mathbf{M}$ and $\boldsymbol{\Sigma} = \sigma^2 \mathbf{I}$. Then $\mathbf{B} \boldsymbol{\Sigma} \mathbf{A} = \sigma^2 \mathbf{M}(\mathbf{I} - \mathbf{M}) = \mathbf{0}$, so $\mathbf{M} \mathbf{Y}$ and $\mathbf{Y}^\top (\mathbf{I} - \mathbf{M}) \mathbf{Y}$ are independent. Functions of independent random vectors are independent.
- (d) A $\chi^2(k)$ variable has mean k and variance $2k$ ([Theorem 4.1.2\(b\)](#) with $\gamma = 0$). So $\text{E SSE} = \sigma^2(n - r)$ and $\text{Var}(\text{SSE}) = 2\sigma^4(n - r)$. Divide by $n - r$ and $(n - r)^2$.

■

The proof is short because the work was done earlier. It is worth seeing where each ingredient comes from geometrically. The fitted vector $\mathbf{M} \mathbf{Y}$ and the residual vector $(\mathbf{I} - \mathbf{M}) \mathbf{Y}$ are projections of $\mathbf{Y}$ onto orthogonal subspaces. For a spherical normal vector, orthogonal projections are independent ([Theorem 4.4.4](#)). Everything about $\boldsymbol{\beta}$ lives in the first projection. The second is pure noise, and its squared length, measured in units of σ^2 , counts $n - r$ independent squared standard normals.

The independence in (c) has no analogue under the second-moment assumptions. If the errors are independent with zero third moment, $\mathbf{\Lambda}^\top \hat{\boldsymbol{\beta}}$ and s^2 are still uncorrelated, but in general they are not independent. The simulation at the end of the section shows the difference.

7.5.2 Standardizing: the t statistic

The distribution in (a) involves the unknown σ . Since $\text{E}(s^2) = \sigma^2$, the matrix $s^2 \mathbf{\Lambda}^\top (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{\Lambda}$ is an unbiased estimate of $\text{Cov}(\mathbf{\Lambda}^\top \hat{\boldsymbol{\beta}})$, and in full rank $s^2 (\mathbf{X}^\top \mathbf{X})^{-1}$ is unbiased for $\text{Cov}(\hat{\boldsymbol{\beta}})$; this needs only the second-moment assumptions. The square roots of its diagonal entries are the standard errors reported by regression software. Replacing σ by s changes the normal distribution into a t distribution, and independence is what makes the change exact.

Corollary 7.5.2 — The t statistic for an estimable function

Assume [Eq. 7.3.1](#) with $r < n$, let $\boldsymbol{\lambda} \in \mathcal{C}(\mathbf{X}^\top)$ with $\boldsymbol{\lambda} \neq \mathbf{0}$, and put $\text{se}(\boldsymbol{\lambda}^\top \hat{\boldsymbol{\beta}}) = s \sqrt{\boldsymbol{\lambda}^\top (\mathbf{X}^\top \mathbf{X})^{-1} \boldsymbol{\lambda}}$. For every constant d ,

$$T_d = \frac{\boldsymbol{\lambda}^\top \hat{\boldsymbol{\beta}} - d}{\text{se}(\boldsymbol{\lambda}^\top \hat{\boldsymbol{\beta}})} \sim t(n - r, \delta), \quad \delta = \frac{\boldsymbol{\lambda}^\top \boldsymbol{\beta} - d}{\sigma \sqrt{\boldsymbol{\lambda}^\top (\mathbf{X}^\top \mathbf{X})^{-1} \boldsymbol{\lambda}}}.$$

In particular $T = (\boldsymbol{\lambda}^\top \hat{\boldsymbol{\beta}} - \boldsymbol{\lambda}^\top \boldsymbol{\beta})/\text{se}(\boldsymbol{\lambda}^\top \hat{\boldsymbol{\beta}}) \sim t(n - r)$, and

$$\Pr\{|\lambda^\top \hat{\beta} - \lambda^\top \beta| \leq t_{n-r, \alpha/2} \text{se}(\lambda^\top \hat{\beta})\} = 1 - \alpha, \quad (7.5.1)$$

where $t_{n-r, \alpha/2}$ is the upper $\alpha/2$ point of $t(n-r)$.

Proof. Put $c^2 = \lambda^\top (\mathbf{X}^\top \mathbf{X})^{-1} \lambda$. Then $c > 0$, since $c^2 = \|\mathbf{M}\rho\|^2$ with $\mathbf{X}^\top \mathbf{M}\rho = \lambda \neq \mathbf{0}$. By Theorem 7.5.1(a), $Z = (\lambda^\top \hat{\beta} - d)/(\sigma c) \sim N(\delta, 1)$. By (b), $V = (n-r)s^2/\sigma^2 \sim \chi^2(n-r)$, and by (c) Z and V are independent. Now $T_d = Z/\sqrt{V/(n-r)}$, which is $t(n-r, \delta)$ by Definition 4.1.3. With $d = \lambda^\top \beta$, $\delta = 0$, and Eq. 7.5.1 restates $\Pr\{|T| \leq t_{n-r, \alpha/2}\} = 1 - \alpha$. ■

Eq. 7.5.1 is the familiar confidence interval $\lambda^\top \hat{\beta} \pm t_{n-r, \alpha/2} \text{se}$, and T_d is the statistic for testing $\lambda^\top \beta = d$. Its noncentrality δ governs the power. Chapter 11 and Chapter 12 develop both. For the air-flow coefficient in the stack loss regression this interval was (0.431, 1.000) (Example (Profile likelihood for the air-flow coefficient)). Two features of the corollary are easy to overlook. The degrees of freedom are $n-r$, the dimension of the residual space. They are not n , and when $\mathbf{X}$ is rank deficient they are not $n-p$ either. And the interval is exact for every sample size, which no large-sample argument delivers.

7.5.3 Several functions at once

For a vector of estimable functions, the natural summary of Theorem 7.5.1(a) is a quadratic form. The covariance matrix $\Lambda^\top (\mathbf{X}^\top \mathbf{X})^{-1} \Lambda$ may be singular when the columns of Λ are linearly dependent, so a generalized inverse is needed.

Corollary 7.5.3 — A chi-squared pivot for several functions

In the setting of Theorem 7.5.1, let $\mathbf{W} = \Lambda^\top (\mathbf{X}^\top \mathbf{X})^{-1} \Lambda$, let $\mathbf{W}^-$ be any generalized inverse of $\mathbf{W}$, and put $\mathbf{D} = \Lambda^\top \hat{\beta} - \Lambda^\top \beta$. Then

$$Q = \frac{\mathbf{D}^\top \mathbf{W}^- \mathbf{D}}{\sigma^2} \sim \chi^2(\text{rank}(\Lambda)),$$

Q does not depend on the choice of $\mathbf{W}^-$ (with probability one), and Q is independent of SSE. Consequently

$$\frac{\mathbf{D}^\top \mathbf{W}^- \mathbf{D} / \text{rank}(\Lambda)}{s^2} \sim F(\text{rank}(\Lambda), n-r).$$

Proof. By Theorem 7.5.1(a), $\mathbf{D} \sim N_q(\mathbf{0}, \sigma^2 \mathbf{W})$. The quadratic form is unchanged if $\mathbf{W}^-$ is replaced by its symmetric part $\frac{1}{2}(\mathbf{W}^- + \mathbf{W}^{-\top})$, which is again a generalized inverse of the symmetric matrix $\mathbf{W}$ (Proposition 1.3.3(d)). So we may assume $\mathbf{W}^-$ symmetric. Apply Theorem 4.3.2(c) with $\Sigma = \sigma^2 \mathbf{W}$, $\mu = \mathbf{0}$ and $\mathbf{A} = \mathbf{W}^-/\sigma^2$. Conditions (ii) and (iii) hold trivially because $\mu = \mathbf{0}$. Condition (i) is $\Sigma \mathbf{A} \Sigma \mathbf{A} \Sigma = \sigma^2 \mathbf{W} \mathbf{W}^- \mathbf{W} \mathbf{W}^- \mathbf{W} = \sigma^2 \mathbf{W} = \Sigma \mathbf{A} \Sigma$, by two uses of $\mathbf{W} \mathbf{W}^- \mathbf{W} = \mathbf{W}$. So $Q \sim \chi^2(\text{tr}(\mathbf{A} \Sigma))$, where $\text{tr}(\mathbf{A} \Sigma) = \text{tr}(\mathbf{W}^- \mathbf{W}) = \text{rank}(\mathbf{W})$, because $\mathbf{W}^- \mathbf{W}$ is idempotent with the same rank as $\mathbf{W}$ (Proposition 1.3.3(a)), and the trace of an idempotent matrix is its rank (Proposition 1.9.1).

It remains to show $\text{rank}(\mathbf{W}) = \text{rank}(\mathbf{A})$. With $\mathbf{A} = \mathbf{X}^\top \mathbf{P}$, $\mathbf{W} = \mathbf{P}^\top \mathbf{M} \mathbf{P} = (\mathbf{M} \mathbf{P})^\top (\mathbf{M} \mathbf{P})$, so $\text{rank} \mathbf{W} = \text{rank}(\mathbf{M} \mathbf{P})$. Also $\mathbf{A} = \mathbf{X}^\top \mathbf{P} = \mathbf{X}^\top \mathbf{M} \mathbf{P}$. If $\mathbf{X}^\top \mathbf{M} \mathbf{P} \mathbf{u} = \mathbf{0}$, then $\mathbf{M} \mathbf{P} \mathbf{u} \in \mathcal{C}(\mathbf{X}) \cap \mathcal{C}(\mathbf{X})^\perp = \{\mathbf{0}\}$, so $\mathbf{X}^\top$ is one-to-one on $\mathcal{C}(\mathbf{M} \mathbf{P})$ and $\text{rank} \mathbf{A} = \text{rank}(\mathbf{M} \mathbf{P})$.

For invariance: $\mathbf{D} \in \mathcal{C}(\mathbf{W})$ with probability one by [Theorem 2.2.3\(d\)](#). If $\mathbf{D} = \mathbf{W} \mathbf{u}$, then $\mathbf{D}^\top \mathbf{W}^- \mathbf{D} = \mathbf{u}^\top \mathbf{W} \mathbf{W}^- \mathbf{W} \mathbf{u} = \mathbf{u}^\top \mathbf{W} \mathbf{u}$, whatever $\mathbf{W}^-$ is. Independence follows from [Theorem 7.5.1\(c\)](#), since $\mathbf{D}$ is a function of $\mathbf{M} \mathbf{Y}$. The F ratio is then a ratio of independent chi-squared variables divided by their degrees of freedom ([Definition 4.1.2](#)). ■

The set of $\mathbf{A}^\top \boldsymbol{\beta}$ values for which this F ratio stays below an F quantile is a confidence ellipsoid. [Chapter 12](#) studies these regions, and [Chapter 11](#) turns the same statistic into the test of the general linear hypothesis. Here we note only that the degrees of freedom in the numerator are the number of *linearly independent* functions being estimated, not the number of rows of $\mathbf{A}^\top$.

Example. (A rank-deficient model, simulated)

Consider a one-way layout with three groups of sizes 4, 5, 6, an intercept, one indicator column per group, and a covariate. So $n = 15$ and $p = 5$, but the rank is $r = 4$, because the indicators add up to the intercept. The contrast $\boldsymbol{\lambda}^\top \boldsymbol{\beta} = \alpha_1 - \alpha_2$, with $\boldsymbol{\lambda} = (0, 1, -1, 0, 0)^\top$, is estimable. With $\sigma = 2$, [Theorem 7.5.1](#) gives $\text{Var}(\boldsymbol{\lambda}^\top \hat{\boldsymbol{\beta}}) = \sigma^2 \boldsymbol{\lambda}^\top (\mathbf{X}^\top \mathbf{X})^- \boldsymbol{\lambda} = 1.9007$, computed with the Moore–Penrose inverse. In 100,000 simulated data sets the variance of the estimated contrast is 1.9017, and the average of SSE/σ^2 is 11.001, against $n - r = 11$. The sample correlation between the contrast estimate and s^2 is 0.0036. The standardized contrast T falls within ± 2.201 , the 97.5% point of $t(11)$, in a proportion 0.9498 of the samples. [Figure 7.5.1](#) compares the simulated distributions with the theory. Near the centre the $t(11)$ and $N(0, 1)$ densities are hard to tell apart, so panel (b) uses a logarithmic density scale, on which the tails separate: the simulated points follow $t(11)$, and the normal density falls away too fast. The tail mass says the same in numbers. A proportion 0.0758 of the samples have $|T| > 1.96$, against 0.05 under $N(0, 1)$ and 0.0758 under $t(11)$. A nominal 95% interval built with the normal point 1.96 would therefore undercover.

```
import numpy as np
from scipy import stats

groups = np.repeat([0, 1, 2], [4, 5, 6])
z = np.array([2.1, 3.4, 1.8, 4.0, 2.9, 3.3, 5.1, 2.2, 3.8, 1.5, 4.4, 2.7, 3.9, 5.0,
              3.1])
n = len(z)
X = np.column_stack([np.ones(n), np.eye(3)[groups], z]) # p = 5 columns, rank 4
r = np.linalg.matrix_rank(X)
G = np.linalg.pinv(X.T @ X) # one generalized inverse
M = X @ G @ X.T
lam = np.array([0.0, 1.0, -1.0, 0.0, 0.0]) # group 1 minus group 2:
    estimable
beta, sigma = np.array([1.0, 0.5, -0.5, 0.0, 0.8]), 2.0
var_lam = sigma ** 2 * lam @ G @ lam # Var(lambda^T beta_hat)
print(f"n = {n}, p = {X.shape[1]}, rank = {r}, Var(lambda^T beta_hat) = {var_lam:.4f}
      ")
```

```
rng = np.random.default_rng(75)
```

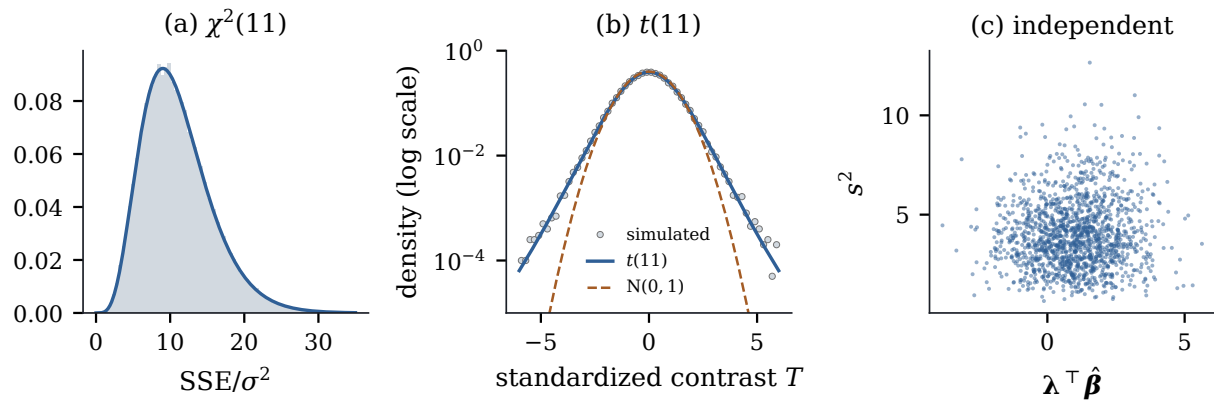


Figure 7.5.1: Sampling distributions in a rank-deficient normal linear model ($n = 15$, $r = 4$; 100,000 simulated data sets). (a) SSE/σ^2 with the $\chi^2(11)$ density. (b) The standardized contrast (binned density, points) with the $t(11)$ and standard normal densities, on a logarithmic scale. (c) The contrast estimate against s^2 : no relationship, as independence requires.

```

reps = 100_000
Y = X @ beta + sigma * rng.normal(size=(reps, n))           # one data set per row
est = Y @ (X @ G @ lam)                                     # lambda^T beta_hat =
    lambda^T G X^T y
sse = np.sum((Y - Y @ M) ** 2, axis=1)                       # y^T (I - M) y
s2 = sse / (n - r)
t = (est - lam @ beta) / np.sqrt(s2 * (lam @ G @ lam))
print(f"mean {est.mean():.4f}, variance {est.var():.4f}")
print(f"E(SSE)/sigma^2 = {np.mean(sse) / sigma**2:.3f} (n - r = {n - r})")
print(f"corr(estimate, s^2) = {np.corrcoef(est, s2)[0, 1]:.4f}")
print("KS p-value, SSE/sigma^2 vs chi^2(n-r):", round(stats.kstest(sse / sigma**2, "
    chi2", args=(n - r,)).pvalue, 3))
print("KS p-value, t vs t(n-r):", round(stats.kstest(t, "t", args=(n -
    r,)).pvalue, 3))

```

7.5.4 Without normality

Which parts of [Theorem 7.5.1](#) survive when the errors are merely uncorrelated with common variance? The first two moments of $\lambda^\top \hat{\beta}$ do, by [Theorem 7.2.1](#), and so does $E(s^2) = \sigma^2$. The normality of $\lambda^\top \hat{\beta}$ is lost, but it often returns approximately. $\lambda^\top \hat{\beta}$ is a weighted sum of independent errors, and a central limit theorem applies when no single observation dominates the weights. Chapter 19 makes this precise. The distribution of SSE is less forgiving, because its variance depends on the fourth moment of the errors.

Proposition 7.5.4 — Variance of SSE without normality

Let the errors be independent with mean zero, variance σ^2 , third moment μ_3 and fourth moment μ_4 , and let h_{ii} be the leverages. Then

$$\text{Var}(\text{SSE}) = 2\sigma^4(n - r) + (\mu_4 - 3\sigma^4) \sum_{i=1}^n (1 - h_{ii})^2.$$

Proof. Apply [Theorem 2.3.3](#) to $\text{SSE} = \mathbf{Y}^\top (\mathbf{I} - \mathbf{M}) \mathbf{Y}$, with $\boldsymbol{\theta} = \mathbf{X}\boldsymbol{\beta}$ and $\mathbf{A} = \mathbf{I} - \mathbf{M}$, whose diagonal entries are $1 - h_{ii}$. The terms $4\sigma^2 \boldsymbol{\theta}^\top \mathbf{A}^2 \boldsymbol{\theta}$ and $4\mu_3 \boldsymbol{\theta}^\top \mathbf{A} \mathbf{a}$ vanish because $\mathbf{A}\boldsymbol{\theta} = (\mathbf{I} - \mathbf{M})\mathbf{X}\boldsymbol{\beta} = \mathbf{0}$, and $\text{tr}(\mathbf{A}^2) = \text{tr}(\mathbf{A}) = n - r$. ■

The factor $\mu_4 - 3\sigma^4$ is σ^4 times the excess kurtosis. For normal errors it vanishes and we recover $2\sigma^4(n - r)$. For heavy-tailed errors it is positive, and SSE is more variable than the chi-squared theory says. Intervals for σ^2 built from [Theorem 7.5.1\(b\)](#) are then too short, and this does not improve with sample size: the ratio of the true variance to the nominal one tends to $1 + \frac{1}{2}(\mu_4/\sigma^4 - 3)$ when the leverages are small. The t interval for $\boldsymbol{\lambda}^\top \boldsymbol{\beta}$ is much less affected, because s^2 enters only through s , and s converges to σ whatever the kurtosis.

Example. (Laplace errors)

Rerun the simulation of [Example \(A rank-deficient model, simulated\)](#) with Laplace (double-exponential) errors of the same variance $\sigma^2 = 4$. For these $\mu_4 = 6\sigma^4$. The contrast still has variance 1.9007. For SSE, $\sum_i (1 - h_{ii})^2 = 8.138$, and [Proposition 7.5.4](#) gives $\text{Var}(\text{SSE}) = 742.6$, more than twice the normal-theory value 352.0. The simulated variance is 741.5. Independence fails too. The contrast and SSE are still uncorrelated, because the errors are symmetric, but the *squared* error of the contrast has correlation 0.211 with SSE. Samples with a few large errors inflate both.

```
E_lap = rng.laplace(scale=sigma / np.sqrt(2), size=(reps, n)) # variance sigma^2,
mu_4 = 6 * sigma^4
Y_lap = X @ beta + E_lap
est_lap = Y_lap @ (X @ G @ lam)
sse_lap = np.sum((Y_lap - Y_lap @ M) ** 2, axis=1)
h = np.diag(np.eye(n) - M)
var_sse_lap = (6 - 3) * sigma**4 * h @ h + 2 * sigma**4 * (n - r)
print(f"Var(SSE): normal theory {2 * sigma**4 * (n - r):.1f}, Laplace formula {
    var_sse_lap:.1f}, "
      f"simulated {sse_lap.var():.1f}")
print(f"corr(squared error of estimate, SSE) under Laplace errors: "
      f"{np.corrcoef((est_lap - lam @ beta) ** 2, sse_lap)[0, 1]:.3f}")
```

In the listing, $\mathbf{h}$ holds the diagonal of $\mathbf{I} - \mathbf{M}$, that is, $1 - h_{ii}$.

7.5.5 Exercises

7.5.5.1 A. Check your understanding

Exercise A1.

Under [Eq. 7.3.1](#), find the joint distribution of $(\hat{\mathbf{Y}}, \hat{\boldsymbol{\varepsilon}})$. Show that $\hat{\mathbf{Y}}$ and $\hat{\boldsymbol{\varepsilon}}$ are independent, and that $\hat{\boldsymbol{\varepsilon}} \sim N_n(\mathbf{0}, \sigma^2(\mathbf{I} - \mathbf{M}))$ whatever $\boldsymbol{\beta}$ is. Is $\hat{\boldsymbol{\varepsilon}}$ independent of SSE?

Exercise A2.

In simple linear regression, write out T of [Corollary 7.5.2](#) for the slope, in terms of S_{xx} and s , and state its distribution. What is r ?

7.5.5.2 B. Practice**Exercise B1.**

Let $\lambda_1, \lambda_2 \in \mathcal{C}(\mathbf{X}^\top)$. Show that $\text{Cov}(\lambda_1^\top \hat{\boldsymbol{\beta}}, \lambda_2^\top \hat{\boldsymbol{\beta}}) = \sigma^2 \lambda_1^\top (\mathbf{X}^\top \mathbf{X})^{-1} \lambda_2$, and that under normality the two estimates are independent iff $\lambda_1^\top (\mathbf{X}^\top \mathbf{X})^{-1} \lambda_2 = 0$. In a one-way layout with a groups of m observations each and cell-means parameterization $E(Y_{ij}) = \mu_i$, show that the estimates of $\sum_i c_i \mu_i$ and $\sum_i d_i \mu_i$ are independent iff $\sum_i c_i d_i = 0$.

Solution. The covariance is the off-diagonal entry of [Theorem 7.5.1\(a\)](#) with $\boldsymbol{\Lambda} = [\lambda_1, \lambda_2]$. Under normality the pair is bivariate normal, so zero covariance is equivalent to independence ([Theorem 3.4.1](#)). In the cell-means model $\mathbf{X}^\top \mathbf{X} = m \mathbf{I}_a$, so $\lambda_1^\top (\mathbf{X}^\top \mathbf{X})^{-1} \lambda_2 = \sum_i c_i d_i / m$, which vanishes iff $\sum_i c_i d_i = 0$.

Exercise B2.

A new observation $Y_0 = \mathbf{x}_0^\top \boldsymbol{\beta} + \varepsilon_0$ will be taken, with $\varepsilon_0 \sim N(0, \sigma^2)$ independent of $\mathbf{Y}$ and $\mathbf{x}_0 \in \mathcal{C}(\mathbf{X}^\top)$. Show that

$$\frac{Y_0 - \mathbf{x}_0^\top \hat{\boldsymbol{\beta}}}{s \sqrt{1 + \mathbf{x}_0^\top (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{x}_0}} \sim t(n - r),$$

and deduce a prediction interval for Y_0 . Why is the extra 1 in the denominator never negligible?

Solution. Y_0 is independent of $\mathbf{Y}$, so $Y_0 - \mathbf{x}_0^\top \hat{\boldsymbol{\beta}}$ is normal with mean $\mathbf{x}_0^\top \boldsymbol{\beta} - \mathbf{x}_0^\top \boldsymbol{\beta} = 0$ and variance $\sigma^2 + \sigma^2 \mathbf{x}_0^\top (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{x}_0$. It is a function of $(Y_0, \mathbf{M}\mathbf{Y})$, which is independent of SSE. Dividing by $\sigma \sqrt{1 + \mathbf{x}_0^\top (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{x}_0}$ gives a standard normal, independent of $(n - r)s^2/\sigma^2 \sim \chi^2(n - r)$, and the ratio is $t(n - r)$ by [Definition 4.1.3](#). The interval is $\mathbf{x}_0^\top \hat{\boldsymbol{\beta}} \pm t_{n-r, \alpha/2} s \sqrt{1 + \mathbf{x}_0^\top (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{x}_0}$. The 1 is the variance of the new error itself, which no amount of data about $\boldsymbol{\beta}$ can reduce. The second term tends to zero as the design grows. The first does not.

Exercise B3.

Use [Theorem 7.5.1\(b\)](#) to construct an exact $1 - \alpha$ confidence interval for σ^2 . Then, using [Proposition 7.5.4](#) with small leverages, explain why its coverage does not tend to $1 - \alpha$ as $n \rightarrow \infty$ when the errors have nonzero excess kurtosis.

7.5.5.3 C. Going deeper

Exercise C1.

Give a second proof of [Theorem 7.5.1](#)(b) and (c) using [Theorem 4.4.4](#) with the two projections $\mathbf{M}$ and $\mathbf{I} - \mathbf{M}$. Extend it: if $\mathcal{C}(\mathbf{X}_0) \subseteq \mathcal{C}(\mathbf{X})$, show that $\|(\mathbf{M} - \mathbf{M}_0)\mathbf{Y}\|^2$ and SSE are independent, and find the distribution of $\|(\mathbf{M} - \mathbf{M}_0)\mathbf{Y}\|^2/\sigma^2$ when $E(\mathbf{Y}) = \mathbf{X}\boldsymbol{\beta}$ is arbitrary.

Exercise C2.

For the test that rejects $\boldsymbol{\lambda}^\top \boldsymbol{\beta} = d$ when $|T_d| > t_{n-r, \alpha/2}$, show that the power depends on the parameters only through $|\delta|$ of [Corollary 7.5.2](#). For a fixed design, how does the power change when the sample is replicated m times ($\mathbf{X}$ replaced by m stacked copies)?

7.6 Bayesian thread I: the conjugate linear model

The previous sections judged estimators by their behaviour over repeated samples, for each fixed value of the parameters. The Bayesian approach treats $(\boldsymbol{\beta}, \sigma^2)$ as uncertain quantities with a probability distribution. It combines a **prior** distribution, expressing what is known before the data, with the likelihood of [Section 7.3](#), and the result is a **posterior** distribution. For the normal linear model there is a family of priors for which the posterior is available in closed form, and it connects directly to the rest of the chapter. The posterior mean is a matrix-weighted combination of the prior mean and the least squares estimate. It is biased in the frequentist sense, which is how it escapes the Gauss–Markov bound. And as the prior becomes flat, the posterior reproduces least squares and the t intervals of [Section 7.5](#) exactly. This is the first of several Bayesian threads in the book. [Section 12.6](#) continues it with credible regions and predictive distributions, Chapter 27 with shrinkage priors, and Chapter 39 with generalized linear models.

7.6.1 The normal-inverse-gamma family

Definition 7.6.1 — Inverse gamma and normal-inverse-gamma distributions (a)

A positive random variable V has the **inverse gamma** distribution $\text{IG}(a, b)$, with $a, b > 0$, if $1/V$ has the gamma distribution with shape a and rate b . Its density is

$$\pi(v) = \frac{b^a}{\Gamma(a)} v^{-a-1} e^{-b/v}, \quad v > 0.$$

- (b) The pair $(\boldsymbol{\beta}, \sigma^2)$ has the **normal-inverse-gamma** distribution $\text{NIG}(\mathbf{m}, \mathbf{V}, a, b)$, with $\mathbf{m} \in \mathbb{R}^p$, $\mathbf{V}$ positive definite and $a, b > 0$, if $\sigma^2 \sim \text{IG}(a, b)$ and, given σ^2 , $\boldsymbol{\beta} \sim N_p(\mathbf{m}, \sigma^2 \mathbf{V})$. Its joint density is

$$\pi(\boldsymbol{\beta}, \sigma^2) = \frac{b^a}{\Gamma(a)(2\pi)^{p/2} \det(\mathbf{V})^{1/2}} (\sigma^2)^{-(a+p/2)-1} \times \exp\left\{-\frac{2b + (\boldsymbol{\beta} - \mathbf{m})^\top \mathbf{V}^{-1}(\boldsymbol{\beta} - \mathbf{m})}{2\sigma^2}\right\}. \quad (7.6.1)$$

The prior on $\boldsymbol{\beta}$ is scaled by σ^2 : prior uncertainty about the coefficients is measured in units of the noise level. This is a modelling choice, and it is what makes the family conjugate. It also makes $\mathbf{V}$ dimensionless relative to the likelihood's $(\mathbf{X}^\top \mathbf{X})^{-1}$, which is the comparison that matters below. The marginal distribution of $\boldsymbol{\beta}$ is a multivariate t .

Definition 7.6.2 — Multivariate t

Let $\mathbf{Z} \sim N_p(\mathbf{0}, \mathbf{S})$ and $W \sim \chi^2(\nu)$ be independent, with $\mathbf{S}$ positive definite. The law of $\mathbf{m} + \mathbf{Z}/\sqrt{W/\nu}$ is the **multivariate t distribution** $t_\nu(\mathbf{m}, \mathbf{S})$.

For $p = 1$ and $\mathbf{S} = 1$, this is the $t(\nu)$ distribution of Definition 4.1.3 shifted by $\mathbf{m}$. For every $\mathbf{c} \neq \mathbf{0}$, $\mathbf{c}^\top \mathbf{u}$ with $\mathbf{u} \sim t_\nu(\mathbf{m}, \mathbf{S})$ is $\mathbf{c}^\top \mathbf{m} + \sqrt{\mathbf{c}^\top \mathbf{S} \mathbf{c}} T$ with $T \sim t(\nu)$, because $\mathbf{c}^\top \mathbf{Z}/\sqrt{\mathbf{c}^\top \mathbf{S} \mathbf{c}} \sim N(0, 1)$ is independent of W .

Lemma 7.6.1 — Marginals of the normal-inverse-gamma

If $(\boldsymbol{\beta}, \sigma^2) \sim \text{NIG}(\mathbf{m}, \mathbf{V}, a, b)$, then $2b/\sigma^2 \sim \chi^2(2a)$, $E(\sigma^2) = b/(a - 1)$ for $a > 1$, and

$$\boldsymbol{\beta} \sim t_{2a}(\mathbf{m}, (b/a)\mathbf{V}).$$

In particular, for every $\boldsymbol{\lambda} \neq \mathbf{0}$, $(\boldsymbol{\lambda}^\top \boldsymbol{\beta} - \boldsymbol{\lambda}^\top \mathbf{m})/\sqrt{(b/a)\boldsymbol{\lambda}^\top \mathbf{V} \boldsymbol{\lambda}} \sim t(2a)$.

Proof. Put $W = 2b/\sigma^2$. Changing variables $v = 2b/w$ in the inverse gamma density, with $|dv/dw| = 2b/w^2$, gives the density of W :

$$\frac{b^a}{\Gamma(a)} \left(\frac{2b}{w}\right)^{-a-1} e^{-w/2} \frac{2b}{w^2} = \frac{w^{a-1} e^{-w/2}}{2^a \Gamma(a)}, \quad w > 0,$$

which is the $\chi^2(2a)$ density (Eq. 4.1.1). Then $E\sigma^2 = 2b E(1/W) = 2b/(2a - 2)$, since $E(1/W) = 1/(\nu - 2)$ for $W \sim \chi^2(\nu)$, $\nu > 2$. Next let $\mathbf{Z}_0 = (\boldsymbol{\beta} - \mathbf{m})/\sigma$. Given σ^2 , $\mathbf{Z}_0 \sim N_p(\mathbf{0}, \mathbf{V})$, a law that does not depend on σ^2 . So $\mathbf{Z}_0$ is independent of σ^2 , and hence of W , and

$$\boldsymbol{\beta} = \mathbf{m} + \sigma \mathbf{Z}_0 = \mathbf{m} + \sqrt{\frac{2b}{W}} \mathbf{Z}_0 = \mathbf{m} + \frac{\sqrt{b/a} \mathbf{Z}_0}{\sqrt{W/(2a)}}.$$

Since $\sqrt{b/a} \mathbf{Z}_0 \sim N_p(\mathbf{0}, (b/a)\mathbf{V})$, this is $t_{2a}(\mathbf{m}, (b/a)\mathbf{V})$ by Definition 7.6.2. The last statement is the remark after Definition 7.6.2. ■

7.6.2 The conjugate update

Theorem 7.6.2 — Conjugate posterior

Let $\mathbf{y}$ follow the normal linear model Eq. 7.3.1, with $\mathbf{X}$ of any rank, and let the prior be $(\boldsymbol{\beta}, \sigma^2) \sim \text{NIG}(\mathbf{m}_0, \mathbf{V}_0, a_0, b_0)$. Then the posterior is $\text{NIG}(\mathbf{m}_n, \mathbf{V}_n, a_n, b_n)$, where

$$\begin{aligned} \mathbf{V}_n &= (\mathbf{V}_0^{-1} + \mathbf{X}^\top \mathbf{X})^{-1}, & \mathbf{m}_n &= \mathbf{V}_n(\mathbf{V}_0^{-1} \mathbf{m}_0 + \mathbf{X}^\top \mathbf{y}), \\ a_n &= a_0 + \frac{n}{2}, \end{aligned} \quad (7.6.2)$$

$$\begin{aligned} b_n &= b_0 + \frac{1}{2}(\mathbf{y}^\top \mathbf{y} + \mathbf{m}_0^\top \mathbf{V}_0^{-1} \mathbf{m}_0 - \mathbf{m}_n^\top \mathbf{V}_n^{-1} \mathbf{m}_n) \\ &= b_0 + \frac{1}{2}\{\|\mathbf{y} - \mathbf{X}\mathbf{m}_n\|^2 + (\mathbf{m}_n - \mathbf{m}_0)^\top \mathbf{V}_0^{-1}(\mathbf{m}_n - \mathbf{m}_0)\}. \end{aligned} \quad (7.6.3)$$

Consequently $\boldsymbol{\beta} \mid \mathbf{y} \sim t_{2a_n}(\mathbf{m}_n, (b_n/a_n)\mathbf{V}_n)$, $\sigma^2 \mid \mathbf{y} \sim \text{IG}(a_n, b_n)$, and $E(\boldsymbol{\beta} \mid \mathbf{y}) = \mathbf{m}_n$.

Proof. Write $\mathbf{P}_0 = \mathbf{V}_0^{-1}$ and $\mathbf{P}_n = \mathbf{P}_0 + \mathbf{X}^\top \mathbf{X}$. The matrix $\mathbf{P}_n$ is positive definite, because $\mathbf{P}_0$ is and $\mathbf{X}^\top \mathbf{X}$ is nonnegative definite. This holds whatever the rank of $\mathbf{X}$. Expanding both sides and using $\mathbf{P}_n \mathbf{m}_n = \mathbf{P}_0 \mathbf{m}_0 + \mathbf{X}^\top \mathbf{y}$ gives the completion of the square

$$\|\mathbf{y} - \mathbf{X}\boldsymbol{\beta}\|^2 + (\boldsymbol{\beta} - \mathbf{m}_0)^\top \mathbf{P}_0(\boldsymbol{\beta} - \mathbf{m}_0) = (\boldsymbol{\beta} - \mathbf{m}_n)^\top \mathbf{P}_n(\boldsymbol{\beta} - \mathbf{m}_n) + C,$$

with $C = \mathbf{y}^\top \mathbf{y} + \mathbf{m}_0^\top \mathbf{P}_0 \mathbf{m}_0 - \mathbf{m}_n^\top \mathbf{P}_n \mathbf{m}_n$. Indeed, the left side is $\boldsymbol{\beta}^\top \mathbf{P}_n \boldsymbol{\beta} - 2\boldsymbol{\beta}^\top (\mathbf{X}^\top \mathbf{y} + \mathbf{P}_0 \mathbf{m}_0) + \mathbf{y}^\top \mathbf{y} + \mathbf{m}_0^\top \mathbf{P}_0 \mathbf{m}_0$, and $(\boldsymbol{\beta} - \mathbf{m}_n)^\top \mathbf{P}_n(\boldsymbol{\beta} - \mathbf{m}_n) = \boldsymbol{\beta}^\top \mathbf{P}_n \boldsymbol{\beta} - 2\boldsymbol{\beta}^\top \mathbf{P}_n \mathbf{m}_n + \mathbf{m}_n^\top \mathbf{P}_n \mathbf{m}_n$. Putting $\boldsymbol{\beta} = \mathbf{m}_n$ in the identity shows that $C = \|\mathbf{y} - \mathbf{X}\mathbf{m}_n\|^2 + (\mathbf{m}_n - \mathbf{m}_0)^\top \mathbf{P}_0(\mathbf{m}_n - \mathbf{m}_0) \geq 0$, which is the second form of Eq. 7.6.3.

By Bayes' theorem the posterior density is proportional, as a function of $(\boldsymbol{\beta}, \sigma^2)$, to prior times likelihood. Using Eq. 7.6.1 and dropping factors free of the parameters,

$$\begin{aligned} \pi(\boldsymbol{\beta}, \sigma^2 \mid \mathbf{y}) &\propto (\sigma^2)^{-(a_0+p/2)-1} e^{-\{2b_0 + (\boldsymbol{\beta} - \mathbf{m}_0)^\top \mathbf{P}_0(\boldsymbol{\beta} - \mathbf{m}_0)\}/(2\sigma^2)} \\ &\quad \times (\sigma^2)^{-n/2} e^{-\|\mathbf{y} - \mathbf{X}\boldsymbol{\beta}\|^2/(2\sigma^2)}. \end{aligned}$$

By the completion of the square, the exponent is $-\{2b_0 + C + (\boldsymbol{\beta} - \mathbf{m}_n)^\top \mathbf{P}_n(\boldsymbol{\beta} - \mathbf{m}_n)\}/(2\sigma^2)$, and $2b_0 + C = 2b_n$. The power of σ^2 is $-(a_0 + n/2 + p/2) - 1 = -(a_n + p/2) - 1$. So the posterior is proportional to the $\text{NIG}(\mathbf{m}_n, \mathbf{V}_n, a_n, b_n)$ density Eq. 7.6.1. Two densities that are proportional are equal, since both integrate to one. The marginals follow from Lemma 7.6.1. The posterior mean of $\boldsymbol{\beta}$ is $\mathbf{m}_n$: by the tower property, $E(\boldsymbol{\beta} \mid \mathbf{y}) = E\{E(\boldsymbol{\beta} \mid \sigma^2, \mathbf{y}) \mid \mathbf{y}\} = \mathbf{m}_n$. ■

The update has a simple structure. **Precisions add:** the posterior precision (per unit σ^2) is the prior precision $\mathbf{V}_0^{-1}$ plus the data precision $\mathbf{X}^\top \mathbf{X}$. **Shape parameters count observations:** each observation adds 1/2 to a . **Scale parameters accumulate squared discrepancies:** b_n adds half of the residual sum of squares at the posterior mean and half of the prior-weighted distance between the posterior and prior means. Because $\mathbf{P}_n$ is positive definite even when $\mathbf{X}$ is rank deficient, the posterior is proper for every design. The prior supplies information about

the directions of $\mathcal{N}(\mathbf{X})$ that the data cannot, and there the posterior equals the prior ([Exercise \(B2\)](#)).

7.6.3 The posterior mean as shrinkage

Proposition 7.6.3 — Posterior mean as a combination

In [Theorem 7.6.2](#), let $\mathbf{X}$ have full column rank and put $\mathbf{W} = \mathbf{V}_n \mathbf{V}_0^{-1}$. Then:

- (a) $\mathbf{m}_n = \mathbf{W}\mathbf{m}_0 + (\mathbf{I} - \mathbf{W})\hat{\boldsymbol{\beta}}$, with $\mathbf{I} - \mathbf{W} = \mathbf{V}_n \mathbf{X}^\top \mathbf{X}$. Equivalently $\mathbf{V}_n^{-1} \mathbf{m}_n = \mathbf{V}_0^{-1} \mathbf{m}_0 + (\mathbf{X}^\top \mathbf{X})\hat{\boldsymbol{\beta}}$: the posterior mean is the precision-weighted average of the prior mean and the least squares estimate.
- (b) $b_n = b_0 + \frac{1}{2} \{ \text{SSE} + (\hat{\boldsymbol{\beta}} - \mathbf{m}_0)^\top (\mathbf{V}_0 + (\mathbf{X}^\top \mathbf{X})^{-1})^{-1} (\hat{\boldsymbol{\beta}} - \mathbf{m}_0) \}$.
- (c) (*g-prior.*) If $\mathbf{V}_0 = g(\mathbf{X}^\top \mathbf{X})^{-1}$ with $g > 0$, then $\mathbf{W} = \mathbf{I}/(1 + g)$ and $\mathbf{m}_n = (\mathbf{m}_0 + g\hat{\boldsymbol{\beta}})/(1 + g)$.
- (d) (*Ridge.*) If $\mathbf{m}_0 = \mathbf{0}$ and $\mathbf{V}_0 = k^{-1}\mathbf{I}$ with $k > 0$, then $\mathbf{m}_n = (\mathbf{X}^\top \mathbf{X} + k\mathbf{I})^{-1} \mathbf{X}^\top \mathbf{y}$.

Proof.

- (a) $\mathbf{X}^\top \mathbf{y} = \mathbf{X}^\top \mathbf{X} \hat{\boldsymbol{\beta}}$ by the normal equations ([Theorem 6.4.2](#)), so $\mathbf{m}_n = \mathbf{V}_n \mathbf{V}_0^{-1} \mathbf{m}_0 + \mathbf{V}_n \mathbf{X}^\top \mathbf{X} \hat{\boldsymbol{\beta}}$, and $\mathbf{V}_n \mathbf{V}_0^{-1} + \mathbf{V}_n \mathbf{X}^\top \mathbf{X} = \mathbf{V}_n \mathbf{V}_n^{-1} = \mathbf{I}$.
- (b) Write $\mathbf{P}_0 = \mathbf{V}_0^{-1}$ and $\mathbf{K} = \mathbf{X}^\top \mathbf{X}$. The completion of the square in the proof of [Theorem 7.6.2](#) shows that $2(b_n - b_0)$ is the minimum over $\boldsymbol{\beta}$ of $\|\mathbf{y} - \mathbf{X}\boldsymbol{\beta}\|^2 + (\boldsymbol{\beta} - \mathbf{m}_0)^\top \mathbf{P}_0 (\boldsymbol{\beta} - \mathbf{m}_0)$, because the remaining term $(\boldsymbol{\beta} - \mathbf{m}_n)^\top \mathbf{P}_n (\boldsymbol{\beta} - \mathbf{m}_n)$ is nonnegative and vanishes at $\mathbf{m}_n$. By [Eq. 6.3.1](#), $\|\mathbf{y} - \mathbf{X}\boldsymbol{\beta}\|^2 = \text{SSE} + (\boldsymbol{\beta} - \hat{\boldsymbol{\beta}})^\top \mathbf{K} (\boldsymbol{\beta} - \hat{\boldsymbol{\beta}})$. Put $\mathbf{d} = \hat{\boldsymbol{\beta}} - \mathbf{m}_0$ and $\boldsymbol{\beta} = \mathbf{m}_0 + \mathbf{u}$. The quantity to minimize becomes $\text{SSE} + (\mathbf{u} - \mathbf{d})^\top \mathbf{K} (\mathbf{u} - \mathbf{d}) + \mathbf{u}^\top \mathbf{P}_0 \mathbf{u}$, whose minimum over $\mathbf{u}$, attained at $\mathbf{u} = (\mathbf{K} + \mathbf{P}_0)^{-1} \mathbf{K} \mathbf{d}$, is $\text{SSE} + \mathbf{d}^\top \{ \mathbf{K} - \mathbf{K}(\mathbf{K} + \mathbf{P}_0)^{-1} \mathbf{K} \} \mathbf{d}$. By [Theorem 1.6.5](#), $\mathbf{K} - \mathbf{K}(\mathbf{K} + \mathbf{P}_0)^{-1} \mathbf{K} = (\mathbf{K}^{-1} + \mathbf{P}_0^{-1})^{-1} = (\mathbf{V}_0 + (\mathbf{X}^\top \mathbf{X})^{-1})^{-1}$.
- (c) With $\mathbf{V}_0^{-1} = \mathbf{X}^\top \mathbf{X}/g$, $\mathbf{V}_n^{-1} = (1 + 1/g)\mathbf{X}^\top \mathbf{X}$, so $\mathbf{W} = \mathbf{V}_n \mathbf{V}_0^{-1} = \{g/(1 + g)\}(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{X}/g = \mathbf{I}/(1 + g)$. (d) Substitute into [Eq. 7.6.2](#).

■

Part (a) is the Bayesian counterpart of Gauss–Markov. The posterior mean is linear in $\mathbf{y}$, and it is biased unless $\mathbf{m}_0 = \boldsymbol{\beta}$: $E(\mathbf{m}_n) = \mathbf{W}\mathbf{m}_0 + (\mathbf{I} - \mathbf{W})\boldsymbol{\beta}$. So the Gauss–Markov theorem does not apply to it, and it can have smaller mean squared error than least squares when the prior mean is close to the truth. Part (d) identifies ridge regression, whose mean squared error was studied in [Exercise \(B1\)](#), as a posterior mean. Chapter 27 develops the connection. When the prior is weak relative to the data, $\mathbf{W}$ is close to $\mathbf{0}$ and the posterior mean is close to $\hat{\boldsymbol{\beta}}$. When it is strong, $\mathbf{W}$ is close to $\mathbf{I}$ and the data barely move it.

In general $\mathbf{W}$ is a full matrix, not a multiple of the identity. Shrinkage then happens along the eigenvectors of the problem, not coordinate by coordinate. A single coefficient’s posterior mean need not lie between its prior mean and its least squares estimate, as the next example shows. Only the *g*-prior of part (c) shrinks every coordinate by the same factor. Its prior covariance copies the shape of the sampling covariance of $\hat{\boldsymbol{\beta}}$, and that makes it convenient for model comparison. Zellner (1986) introduced it.

7.6.4 The flat-prior limit

What happens when the prior carries no information? Letting $\mathbf{V}_0^{-1} \rightarrow \mathbf{0}$ in Eq. 7.6.2 gives $\mathbf{m}_n \rightarrow \hat{\boldsymbol{\beta}}$ when $\mathbf{X}$ has full rank. The shape parameter needs more care. The NIG density Eq. 7.6.1 carries the factor $(\sigma^2)^{-(a_0+p/2)-1}$, so letting only $a_0, b_0 \rightarrow 0$ leads to the prior $(\sigma^2)^{-p/2-1}$, with $a_n = n/2$ and t marginals on n degrees of freedom rather than $n - p$. The improper prior $\pi(\boldsymbol{\beta}, \sigma^2) \propto 1/\sigma^2$, uniform in $\boldsymbol{\beta}$ and in $\log \sigma$, is the limit $\mathbf{V}_0^{-1} \rightarrow \mathbf{0}$, $a_0 \rightarrow -p/2$, $b_0 \rightarrow 0$. A negative shape $a_0 = -p/2$ lies outside the proper NIG family, so this prior cannot be reached inside Theorem 7.6.2 and has to be handled directly. The result is clean.

Corollary 7.6.4 — The flat prior reproduces least squares

Let $\text{rank}(\mathbf{X}) = p < n$, let $\text{SSE} > 0$, and take the improper prior $\pi(\boldsymbol{\beta}, \sigma^2) \propto 1/\sigma^2$. Then the posterior is proper and equals $\text{NIG}(\hat{\boldsymbol{\beta}}, (\mathbf{X}^\top \mathbf{X})^{-1}, (n-p)/2, \text{SSE}/2)$. In particular, for every $\boldsymbol{\lambda} \neq \mathbf{0}$,

$$\frac{\boldsymbol{\lambda}^\top \boldsymbol{\beta} - \boldsymbol{\lambda}^\top \hat{\boldsymbol{\beta}}}{s \sqrt{\boldsymbol{\lambda}^\top (\mathbf{X}^\top \mathbf{X})^{-1} \boldsymbol{\lambda}}} \mid \mathbf{y} \sim t(n-p),$$

so the $1 - \alpha$ equal-tailed credible interval for $\boldsymbol{\lambda}^\top \boldsymbol{\beta}$ is the t interval Eq. 7.5.1.

Proof. By Eq. 6.3.1, $\|\mathbf{y} - \mathbf{X}\boldsymbol{\beta}\|^2 = \text{SSE} + (\boldsymbol{\beta} - \hat{\boldsymbol{\beta}})^\top \mathbf{X}^\top \mathbf{X} (\boldsymbol{\beta} - \hat{\boldsymbol{\beta}})$. So prior times likelihood is proportional to

$$\begin{aligned} & (\sigma^2)^{-1-n/2} \exp\left\{-\frac{\text{SSE} + (\boldsymbol{\beta} - \hat{\boldsymbol{\beta}})^\top \mathbf{X}^\top \mathbf{X} (\boldsymbol{\beta} - \hat{\boldsymbol{\beta}})}{2\sigma^2}\right\} \\ &= (\sigma^2)^{-(a+p/2)-1} \exp\left\{-\frac{2b + (\boldsymbol{\beta} - \hat{\boldsymbol{\beta}})^\top \mathbf{X}^\top \mathbf{X} (\boldsymbol{\beta} - \hat{\boldsymbol{\beta}})}{2\sigma^2}\right\}, \end{aligned}$$

with $a = (n-p)/2 > 0$ and $b = \text{SSE}/2 > 0$. This is proportional to the density Eq. 7.6.1 of $\text{NIG}(\hat{\boldsymbol{\beta}}, (\mathbf{X}^\top \mathbf{X})^{-1}, a, b)$, which is integrable. So the posterior is proper and equal to it. By Lemma 7.6.1 with $2a = n - p$ and $b/a = \text{SSE}/(n-p) = s^2$, the stated ratio has the $t(n-p)$ distribution. ■

The frequentist and Bayesian statements now read alike, but they say different things. The t interval of Section 7.5 has probability $1 - \alpha$ of covering the fixed $\boldsymbol{\lambda}^\top \boldsymbol{\beta}$ *before the data are seen*. The credible interval has posterior probability $1 - \alpha$ of containing the random $\boldsymbol{\lambda}^\top \boldsymbol{\beta}$ *given the observed data*. That the two coincide is a special property of this model and this prior. It is one reason why $1/\sigma^2$ is regarded as the natural “noninformative” prior for the normal linear model.

7.6.5 A worked example

Example. (A prior for the stack loss plant)

Return to the stack loss regression of Example (Stack loss). Suppose an engineer, before seeing these 21 days, believed from the chemistry of the process that stack loss should rise roughly one for one with air flow and with cooling water temperature, and should

not depend on acid concentration. This belief is hypothetical and is used only for illustration. We encode it as

$$\mathbf{m}_0 = (0, 1, 1, 0)^\top, \quad \mathbf{V}_0 = \text{diag}(100^2, 0.1^2, 0.2^2, 0.1^2), \quad a_0 = 3, \quad b_0 = 18.$$

So the prior standard deviations of the slopes are 0.1σ , 0.2σ and 0.1σ , the intercept is left essentially free, and the prior mean of σ^2 is $b_0/(a_0 - 1) = 9$.

The posterior has $a_n = 13.5$, so the marginal posteriors are t with 27 degrees of freedom, and $b_n = 112.523$. The posterior means of the coefficients are

$$\mathbf{m}_n = (-41.1490, 0.7616, 1.1606, -0.1371)^\top,$$

against the least squares estimates $(-39.9197, 0.7156, 1.2953, -0.1521)$. Each slope has moved toward its prior mean. The intercept has moved too, though its prior is vague, because it is strongly correlated with the slopes when the regressors are not centred. The 95% credible interval for the air-flow coefficient, using the $t(27)$ quantile 2.052, is $(0.550, 0.973)$, narrower than the least squares t interval $(0.431, 1.000)$, because the prior adds information. For water temperature and acid concentration the intervals are $(0.594, 1.727)$ and $(-0.392, 0.118)$. The posterior mean of σ^2 is 9.002, almost exactly the prior mean 9, although $s^2 = 10.519$. The two parts of [Eq. 7.6.3](#) explain why. The lack of fit at the posterior mean is $\|\mathbf{y} - \mathbf{X}\mathbf{m}_n\|^2 = 180.667$, and on its own, with b_0 , it would give $(b_0 + 180.667/2)/(a_n - 1) = 8.667$: the shape $a_n = a_0 + n/2$ counts all n observations, while the residuals have spent some of them on $\boldsymbol{\beta}$. The prior–data conflict term $(\mathbf{m}_n - \mathbf{m}_0)^\top \mathbf{V}_0^{-1}(\mathbf{m}_n - \mathbf{m}_0) = 8.378$ adds to the scale and makes up the difference, so the data barely move σ^2 from where the prior put it.

[Figure 7.6.1\(a\)](#) shows the three marginal densities for the air-flow coefficient. Panel (b) traces the posterior means of the slopes as the prior precision of the slopes is multiplied by a factor τ , from 10^{-3} (essentially least squares) to 10^3 (essentially the prior mean). The water-temperature coefficient does not move monotonically. It overshoots its prior mean of 1, down to 0.946, before returning. Matrix shrinkage acts on combinations of coefficients, and a single coordinate can pass beyond its prior mean on the way.

```
import numpy as np
import statsmodels.api as sm
from scipy import stats

data = sm.datasets.stackloss.load_pandas().data
y = data["STACKLOSS"].to_numpy()
X = np.column_stack([np.ones(len(y)), data[["AIRFLOW", "WATERTEMP", "ACIDCONC"]]])
n, p = X.shape
XtX, Xty = X.T @ X, X.T @ y
beta_hat = np.linalg.solve(XtX, Xty)
sse = np.sum((y - X @ beta_hat) ** 2)
```

```
def nig_posterior(m0, V0, a0, b0):
    """Conjugate update of the normal-inverse-gamma prior NIG(m0, V0, a0, b0)."""
    P0 = np.linalg.inv(V0) # prior precision (per unit sigma
    ~2)
    Vn = np.linalg.inv(P0 + XtX)
    mn = Vn @ (P0 @ m0 + Xty)
    an = a0 + n / 2
    bn = b0 + (y @ y + m0 @ P0 @ m0 - mn @ (P0 + XtX) @ mn) / 2
    return mn, Vn, an, bn
```

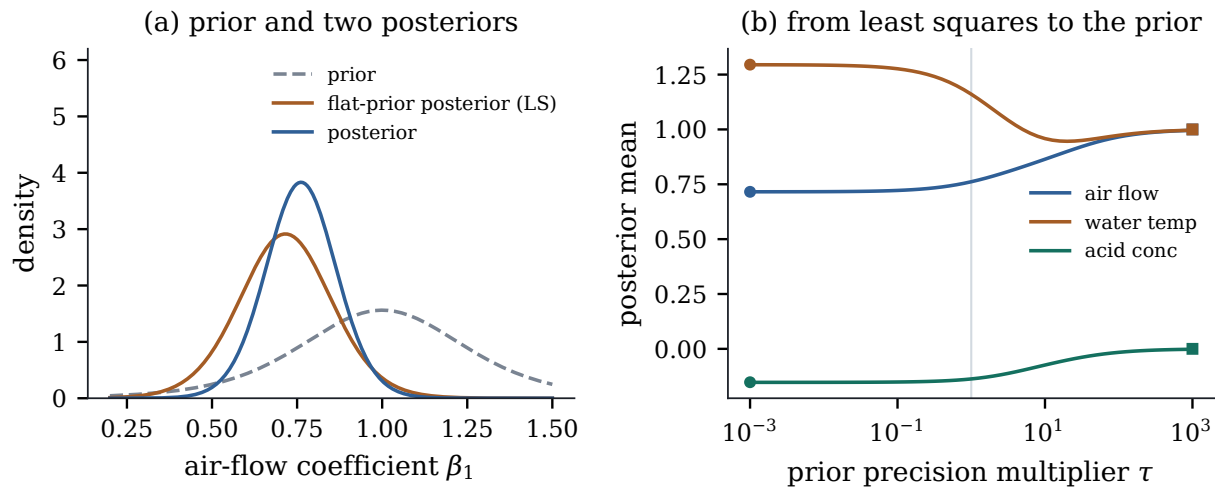


Figure 7.6.1: Conjugate analysis of the stack loss regression. (a) Marginal prior, flat-prior posterior (which is the least squares t distribution) and posterior for the air-flow coefficient. (b) Posterior means of the three slopes as the prior precision of the slopes is scaled by τ . Circles mark least squares, squares mark the prior means, and the grey line marks the prior of the example ($\tau = 1$).

```
# prior: vague intercept; slopes centred on 1, 1, 0 with sd 0.1, 0.2, 0.1 in units
# of sigma
m0 = np.array([0.0, 1.0, 1.0, 0.0])
V0 = np.diag([100.0**2, 0.1**2, 0.2**2, 0.1**2])
a0, b0 = 3.0, 18.0                                     # prior mean of sigma^2 is b0/(a0
                                                         # -1) = 9
mn, Vn, an, bn = nig_posterior(m0, V0, a0, b0)
scale = np.sqrt(np.diag(Vn) * bn / an)                 # marginal posterior: t with 2
                                                         # a_n d.f.
q = stats.t.ppf(0.975, 2 * an)
for j, name in enumerate(["intercept", "air flow", "water temp", "acid conc"]):
    print(f"{name:10s} LS {beta_hat[j]:8.4f} posterior mean {mn[j]:8.4f} "
          f"95% interval ({mn[j] - q * scale[j]:.3f}, {mn[j] + q * scale[j]:.3f})")
print(f"posterior mean of sigma^2 = {bn / (an - 1):.3f}")
```

The script behind this example also checks the theorem numerically. The log posterior computed from Eq. 7.6.2 differs from log prior plus log-likelihood by the same constant at 50 random parameter values. The two forms of b_n agree. A Monte Carlo sample drawn from σ^2 and then $\beta \mid \sigma^2$ reproduces the marginal t quantiles. And a nearly flat prior reproduces least squares and the t interval.

Remark (How the Bayesian answer relates to the optimality results). Seen from the frequentist side, the posterior mean is a linear estimator with bias $\mathbf{W}(\mathbf{m}_0 - \beta)$ and covariance $\sigma^2(\mathbf{I} - \mathbf{W})(\mathbf{X}^\top \mathbf{X})^{-1}(\mathbf{I} - \mathbf{W})^\top$, which is smaller than that of $\hat{\beta}$ in the non-negative definite order. Whether its mean squared error is smaller depends on how far $\mathbf{m}_0$ is from β , in the metric that $\mathbf{W}$ defines. Averaged over the prior, the posterior mean is the estimator with the smallest expected squared error, because a conditional mean minimizes mean squared error. Different criteria single out different estimators. The Gauss–Markov theorem, the Lehmann–Scheffé theorem and the conjugate posterior (Theorem 7.6.2) answer three different questions, and in the flat-prior limit their answers agree.

7.6.6 Exercises

7.6.6.1 A. Check your understanding

Exercise A1.

Show that the $\text{IG}(a, b)$ distribution has mean $b/(a - 1)$ for $a > 1$ and mode $b/(a + 1)$. What are the mean and mode of $\sigma^2 \mid \mathbf{y}$ under the flat prior of [Corollary 7.6.4](#), and how do they compare with s^2 and with the MLE of [Theorem 7.3.1](#)?

Exercise A2.

Suppose σ^2 is known and the prior is $\boldsymbol{\beta} \sim N_p(\mathbf{m}_0, \sigma^2 \mathbf{V}_0)$. Show that the posterior is $N_p(\mathbf{m}_n, \sigma^2 \mathbf{V}_n)$ with $\mathbf{m}_n$ and $\mathbf{V}_n$ as in [Eq. 7.6.2](#).

7.6.6.2 B. Practice

Exercise B1.

Under the g -prior of [Proposition 7.6.3\(c\)](#) with $\mathbf{m}_0 = \mathbf{0}$, show that $b_n = b_0 + \frac{1}{2}\{\text{SSE} + \hat{\boldsymbol{\beta}}^\top \mathbf{X}^\top \mathbf{X} \hat{\boldsymbol{\beta}} / (1 + g)\}$. Show that as $g \rightarrow \infty$ the posterior mean tends to $\hat{\boldsymbol{\beta}}$, and that for finite g the frequentist mean squared error of $\mathbf{m}_n$ is $\{g^2 \sigma^2 \text{tr}(\mathbf{X}^\top \mathbf{X})^{-1} + \|\boldsymbol{\beta}\|^2\} / (1 + g)^2$.

Solution. By [Proposition 7.6.3\(b\)](#), with $\mathbf{V}_0 + (\mathbf{X}^\top \mathbf{X})^{-1} = (1 + g)(\mathbf{X}^\top \mathbf{X})^{-1}$, the quadratic term is $\hat{\boldsymbol{\beta}}^\top \mathbf{X}^\top \mathbf{X} \hat{\boldsymbol{\beta}} / (1 + g)$. As $g \rightarrow \infty$, $\mathbf{m}_n = g\hat{\boldsymbol{\beta}} / (1 + g) \rightarrow \hat{\boldsymbol{\beta}}$. For the mean squared error, $\mathbf{m}_n - \boldsymbol{\beta} = \{g(\hat{\boldsymbol{\beta}} - \boldsymbol{\beta}) - \boldsymbol{\beta}\} / (1 + g)$, and $\hat{\boldsymbol{\beta}} - \boldsymbol{\beta}$ has mean zero and covariance $\sigma^2(\mathbf{X}^\top \mathbf{X})^{-1}$. So the expected squared norm is $\{g^2 \sigma^2 \text{tr}(\mathbf{X}^\top \mathbf{X})^{-1} + \|\boldsymbol{\beta}\|^2\} / (1 + g)^2$. It is smaller than $\sigma^2 \text{tr}(\mathbf{X}^\top \mathbf{X})^{-1}$, the value for $\hat{\boldsymbol{\beta}}$, iff $\|\boldsymbol{\beta}\|^2 < (1 + 2g)\sigma^2 \text{tr}(\mathbf{X}^\top \mathbf{X})^{-1}$.

Exercise B2.

Let $\mathbf{X}$ be rank deficient, $\mathbf{m}_0 = \mathbf{0}$ and $\mathbf{V}_0 = \tau^{-1} \mathbf{I}$. For $\mathbf{v} \in \mathcal{N}(\mathbf{X})$, show that $\mathbf{v}^\top \mathbf{m}_n = 0$ and $\mathbf{v}^\top \mathbf{V}_n \mathbf{v} = \|\mathbf{v}\|^2 / \tau$. So the conditional posterior of $\mathbf{v}^\top \boldsymbol{\beta}$ given σ^2 equals its prior: the data do not update nonidentified directions. What happens to $\boldsymbol{\lambda}^\top \mathbf{m}_n$ for $\boldsymbol{\lambda} \in \mathcal{C}(\mathbf{X}^\top)$ as $\tau \rightarrow 0$?

Solution. $\mathbf{V}_n^{-1} \mathbf{v} = (\tau \mathbf{I} + \mathbf{X}^\top \mathbf{X}) \mathbf{v} = \tau \mathbf{v}$, so $\mathbf{V}_n \mathbf{v} = \mathbf{v} / \tau$. Then $\mathbf{v}^\top \mathbf{m}_n = \mathbf{v}^\top \mathbf{V}_n \mathbf{X}^\top \mathbf{y} = (\mathbf{V}_n \mathbf{v})^\top \mathbf{X}^\top \mathbf{y} = (\mathbf{X} \mathbf{v})^\top \mathbf{y} / \tau = 0$, and $\mathbf{v}^\top \mathbf{V}_n \mathbf{v} = \|\mathbf{v}\|^2 / \tau$. These are the prior mean and variance (per unit σ^2) of $\mathbf{v}^\top \boldsymbol{\beta}$. For the second question, $\mathbf{m}_n = (\mathbf{X}^\top \mathbf{X} + \tau \mathbf{I})^{-1} \mathbf{X}^\top \mathbf{y}$ converges as $\tau \rightarrow 0$ to the minimum-norm least squares estimate $\mathbf{X}^+ \mathbf{y}$ ([Proposition 6.10.1](#)). This follows from the spectral decomposition of $\mathbf{X}^\top \mathbf{X}$: eigenvectors with eigenvalue $d > 0$ get the factor $1/(d + \tau) \rightarrow 1/d$, and those with eigenvalue 0 get zero. Hence $\boldsymbol{\lambda}^\top \mathbf{m}_n \rightarrow \boldsymbol{\lambda}^\top \mathbf{X}^+ \mathbf{y} = \boldsymbol{\lambda}^\top \hat{\boldsymbol{\beta}}$, the least squares estimate of the estimable function.

7.6.6.3 C. Going deeper

Exercise C1.

Under the posterior of [Theorem 7.6.2](#), let $Y_0 = \mathbf{x}_0^\top \boldsymbol{\beta} + \varepsilon_0$ be a future observation, with $\varepsilon_0 \sim N(0, \sigma^2)$ independent of everything given σ^2 . Show that the posterior predictive distribution of Y_0 is $t_{2a_n}(\mathbf{x}_0^\top \mathbf{m}_n, (b_n/a_n)(1 + \mathbf{x}_0^\top \mathbf{V}_n \mathbf{x}_0))$. Compare it with the frequentist prediction interval of [Exercise \(B2\)](#) under the flat prior.

Solution. Given σ^2 and $\mathbf{y}$, $\mathbf{x}_0^\top \boldsymbol{\beta} \sim N(\mathbf{x}_0^\top \mathbf{m}_n, \sigma^2 \mathbf{x}_0^\top \mathbf{V}_n \mathbf{x}_0)$, and ε_0 is independent of it. So $Y_0 | \sigma^2, \mathbf{y} \sim N(\mathbf{x}_0^\top \mathbf{m}_n, \sigma^2(1 + \mathbf{x}_0^\top \mathbf{V}_n \mathbf{x}_0))$. Thus $(Y_0, \sigma^2) | \mathbf{y}$ is normal-inverse-gamma with $p = 1$, mean $\mathbf{x}_0^\top \mathbf{m}_n$ and $\mathbf{V} = 1 + \mathbf{x}_0^\top \mathbf{V}_n \mathbf{x}_0$, and [Lemma 7.6.1](#) gives the t_{2a_n} law. Under the flat prior, $\mathbf{m}_n = \hat{\boldsymbol{\beta}}$, $\mathbf{V}_n = (\mathbf{X}^\top \mathbf{X})^{-1}$, $2a_n = n - p$ and $b_n/a_n = s^2$. The equal-tailed predictive interval is then exactly the prediction interval of [Exercise \(B2\)](#).

Exercise C2.

Split the data into two batches $(\mathbf{y}_1, \mathbf{X}_1)$ and $(\mathbf{y}_2, \mathbf{X}_2)$. Show that updating the prior with the first batch, and then using that posterior as the prior for the second batch, gives the same posterior as a single update with all the data. Which quantities in [Eq. 7.6.2](#) make this obvious?

7.7 Summary and Notes

7.7.1 Summary

Idea.

1. A linear estimator $\mathbf{a}^\top \mathbf{Y}$ is unbiased for $\boldsymbol{\lambda}^\top \boldsymbol{\beta}$ iff $\mathbf{X}^\top \mathbf{a} = \boldsymbol{\lambda}$. Such estimators exist iff $\boldsymbol{\lambda} \in \mathcal{C}(\mathbf{X}^\top)$, the same condition under which least squares estimates $\boldsymbol{\lambda}^\top \hat{\boldsymbol{\beta}}$ uniquely. Their coefficient vectors form the flat $\mathbf{M}\boldsymbol{\rho} + \mathcal{C}(\mathbf{X})^\perp$ ([Proposition 7.1.1](#) and [Proposition 7.1.2](#)).
2. **Gauss–Markov.** Under $E(\boldsymbol{\varepsilon}) = \mathbf{0}$ and $\text{Cov}(\boldsymbol{\varepsilon}) = \sigma^2 \mathbf{I}$, in any rank, $\boldsymbol{\lambda}^\top \hat{\boldsymbol{\beta}}$ is the unique BLUE of every estimable $\boldsymbol{\lambda}^\top \boldsymbol{\beta}$. For any LUE, $\text{Var}(\mathbf{a}^\top \mathbf{Y}) = \text{Var}(\boldsymbol{\lambda}^\top \hat{\boldsymbol{\beta}}) + \sigma^2 \|(\mathbf{I} - \mathbf{M})\mathbf{a}\|^2$. In matrix form, $\text{Cov}(\mathbf{A}\mathbf{Y}) - \text{Cov}(\mathbf{A}^\top \hat{\boldsymbol{\beta}}) = \sigma^2 \mathbf{A}(\mathbf{I} - \mathbf{M})\mathbf{A}^\top \succeq \mathbf{0}$ ([Theorem 7.2.1](#) and [Corollary 7.2.2](#)). With $\text{Cov}(\boldsymbol{\varepsilon}) = \sigma^2 \mathbf{V}$, generalized least squares is the BLUE ([Corollary 7.2.3](#)).
3. The theorem does not cover nonlinear estimators (the midrange beats the mean under uniform errors, [Example \(Uniform errors and the midrange\)](#)) or biased ones (ridge, [Exercise \(B1\)](#)). Orthogonal columns minimize coefficient variances ([Proposition 7.2.4](#)).
4. Under normality, the MLE of $\boldsymbol{\beta}$ is least squares and the MLE of σ^2 is SSE/n , biased by the factor $(n - r)/n$ and inconsistent when r grows with n ([Theorem 7.3.1](#) and [Example \(Many means, one variance\)](#)). The profile likelihood of an estimable function is a decreasing function of t^2 ([Proposition 7.3.3](#)).
5. $(\mathbf{M}\mathbf{Y}, \text{SSE})$ is complete and sufficient. By Lehmann–Scheffé, $\boldsymbol{\lambda}^\top \hat{\boldsymbol{\beta}}$ and s^2 are the UMVUEs, with no linearity restriction ([Proposition 7.4.3](#) and [Theorem 7.4.6](#)). Least squares attains the Cramér–Rao bound for $\boldsymbol{\beta}$, and s^2 has efficiency $(n - p)/n$.

(Theorem 7.4.8).

6. Under normality, $\mathbf{\Lambda}^\top \hat{\boldsymbol{\beta}} \sim N(\mathbf{\Lambda}^\top \boldsymbol{\beta}, \sigma^2 \mathbf{\Lambda}^\top (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{\Lambda})$, $\text{SSE}/\sigma^2 \sim \chi^2(n-r)$, the two are independent, and standardized estimable functions are $t(n-r)$ (Theorem 7.5.1, Corollary 7.5.2 and Corollary 7.5.3). Without normality the variance of SSE depends on the kurtosis (Proposition 7.5.4).
7. The normal-inverse-gamma prior is conjugate. The posterior mean is a precision-weighted combination of the prior mean and $\hat{\boldsymbol{\beta}}$, and the flat prior $1/\sigma^2$ reproduces least squares and the t intervals (Theorem 7.6.2, Proposition 7.6.3 and Corollary 7.6.4).

7.7.2 Notes and sources

Gauss and Markov. Gauss (1823) proved that least squares gives the linear unbiased combination of observations with the smallest variance, without assuming normal errors. It was his second justification of the method, after the normal-theory argument of 1809. Markov's lectures on probability (1900 and later editions) restated the result, and Neyman (1934) attached Markov's name to it. The history is traced by Plackett (1949, 1972) and Stigler (1986). The projection proof of Section 7.2 splits the coefficient vector of a competing estimator by Pythagoras, so that its excess variance is $\sigma^2 \|(\mathbf{I} - \mathbf{M})\mathbf{a}\|^2$. This is the proof of Seber and Lee (2003, section 3.2, Theorem 3.2), stated there for linear functions of the mean vector; we apply it to estimable functions in the rank-deficient case. Christensen (2020, section 2.3) argues instead through the covariance with $\mathbf{p}^\top \mathbf{M}\mathbf{Y}$. Rencher and Schaalje (2008, section 7.3) give the matrix form for full-rank models, and Agresti (2015, section 2.7) discusses its scope and its extension to generalized least squares. The generalized least squares version (Corollary 7.2.3) is due to Aitken (1935). The characterization of models in which ordinary least squares remains best (Exercise C3) is due to Zyskind (1967) and, in coordinate-free form, to Kruskal (1968).

What Gauss–Markov does not cover. That shrinkage can improve on unbiased estimation in three or more dimensions was discovered by Stein (1956), and James and Stein (1961) gave the explicit estimator. Hoerl and Kennard (1970) introduced ridge regression. The existence of a better ridge estimator (Exercise B1) is their theorem. Wald (1940) proposed the group-means slope estimator of Example (Three linear unbiased estimators of a slope) for regressions with errors in the regressor. The optimality of orthogonal weighing designs goes back to Hotelling (1944).

Likelihood. The inconsistency of the maximum likelihood estimate of a variance in the presence of many nuisance means is the example of Neyman and Scott (1948). Restricted maximum likelihood (Exercise C2) was developed by Patterson and Thompson (1971) for variance components. Christensen (2020, section 2.4), Seber and Lee (2003, section 3.5) and Rencher and Schaalje (2008, section 7.6) derive the normal-theory MLEs. The maximization through $\log u \leq u - 1$ used in Theorem 7.3.1 follows Seber and Lee (2003, section 3.5).

Minimum variance. The Rao–Blackwell theorem is due to Rao (1945) and Blackwell (1947). Completeness and the uniqueness theorem are due to Lehmann and Scheffé (1950). The two results we quoted, the factorization criterion and the completeness of full-rank exponential families, are proved in Lehmann and Casella (1998) and Lehmann and Romano (2005). The same variance bound was found independently by Rao (1945) and Cramér (1946). Christensen (2020, section 2.5) outlines the minimum variance unbiased argument for the linear model by reparameterizing to full rank. Seber and Lee (2003, section 3.5) compare the Cramér–Rao bound with the variance of s^2 . The optimality of s^2 among nonnegative quadratic unbiased estimators

(Exercise (C2)) is due to Atiqullah (1962). That the divisor $k + 2$ minimizes the mean squared error among multiples of SSE (Proposition 7.3.2) is due to Theil and Schweitzer (1961). The reflection argument of Example (Conditioning on the sufficient statistic) is our own illustration.

Sampling distributions. The distribution theory of Section 7.5 follows Christensen (2020, section 2.6), Seber and Lee (2003, section 3.4) and Rencher and Schaalje (2008, section 7.6.3), with the rank-deficient forms throughout. The effect of kurtosis on inference about variances, and the relative robustness of inference about means, were emphasized by Box (1953).

Bayesian thread. The conjugate normal-inverse-gamma analysis goes back to Raiffa and Schlaifer (1961) and is developed for regression by Zellner (1971) and Box and Tiao (1973). O’Hagan and Forster (2004) give a modern account. Zellner (1986) introduced the g -prior. Seber and Lee (2003, section 3.12), Christensen (2020, section 2.10) and Rencher and Schaalje (2008, chapter 11) treat Bayesian estimation in the linear model, and Fahrmeir, Kneib, Lang and Marx (2021, section 4.4) present the conjugate analysis as the starting point for richer priors. For an applied account of the Gauss–Markov conditions and what follows from them, see Sen and Srivastava (1990, section 2.5).

Data. The stack loss data are from Brownlee (1965) and are distributed with statsmodels.

7.7.3 References

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Chapter 8

Estimability, Identifiability and Indicator Variables

When the columns of the model matrix are linearly dependent, the data cannot tell some coefficient vectors apart, because different vectors produce the same mean. This chapter asks which questions about β still have answers. The answer is the same from every direction. A linear function $\lambda^\top \beta$ can be estimated without bias exactly when it is determined by the mean vector $\mathbf{X}\beta$, exactly when λ lies in the row space of $\mathbf{X}$, and exactly when every least squares solution gives it the same value. The rest of the chapter uses this criterion. Reparameterizations and side conditions turn out to be two ways of choosing coordinates, and neither can change what is estimable. Factors enter a regression through indicator variables, and each coding of a factor amounts to a choice of which linear functions of the cell means the coefficients report. Contrasts are the estimable functions of a factor, and orthogonal sets of contrasts split the factor's sum of squares into separate pieces.

What you need. [Chapter 1](#) (rank, generalized inverses and the consistency of linear systems, especially [Theorem 1.4.1](#) and [Corollary 1.4.2](#)), [Chapter 2](#) (means and covariances of linear functions, [Theorem 2.2.1](#)), [Chapter 4](#) (for the distribution of contrast sums of squares under normality, [Theorem 4.4.4](#)), [Chapter 5](#) (the linear model, [Definition 5.3.1](#)), and [Chapter 6](#), above all [Section 6.2](#) and [Section 6.4](#), where rank-deficient least squares and the invariance of $\lambda^\top \hat{\beta}$ for $\lambda \in \mathcal{C}(\mathbf{X}^\top)$ ([Theorem 6.4.4](#)) first appeared. A few remarks use the Gauss–Markov theorem of [Chapter 7](#).

8.0.1 Roadmap

- [Identifiability](#)
- [Estimable functions](#)
- [Characterizations of estimability](#)
- [Reparameterization and side conditions](#)
- [Indicator variables and the coding of a factor](#)
- [Several factors, interactions and covariates](#)
- [Contrasts](#)
- [Summary and notes](#)

8.1 Identifiability

Section 6.4 ended with a table of four coefficient vectors that disagreed in every entry, some even in sign, and still produced the same fitted values. Nothing had gone wrong with the computation. The model matrix had linearly dependent columns, and so the least squares problem had a whole affine subspace of solutions. This chapter asks what can be learned about β in that situation. We begin with a question that comes before estimation and does not involve data at all. Suppose we knew the distribution of $\mathbf{Y}$ exactly, as if we had infinitely many replications. Would we then know β ? If not, which features of β would we know?

8.1.1 What the distribution of the data can reveal

Throughout, $\mathbf{Y} = \mathbf{X}\beta + \epsilon$ is the linear model of Definition 5.3.1: $\mathbf{X}$ is a fixed $n \times p$ matrix of rank r , and $E(\epsilon) = \mathbf{0}$. We do not assume $r = p$. The coefficient vector enters the model only through the product $\mathbf{X}\beta$. Suppose the distribution of the error vector ϵ is allowed to depend on other parameters, such as σ^2 , but not on β . Then the distribution of $\mathbf{Y}$ is the distribution of ϵ shifted by the vector $\mathbf{X}\beta$. Two coefficient vectors with the same product $\mathbf{X}\beta$ therefore give the same distribution of $\mathbf{Y}$, with the other parameters held fixed. Conversely, the distribution of $\mathbf{Y}$ determines its mean $E(\mathbf{Y}) = \mathbf{X}\beta$. So the most we can learn about β from the distribution is the value of $\mathbf{X}\beta$. The mean vector

$$\mu = \mathbf{X}\beta \in \mathcal{C}(\mathbf{X})$$

is the object the data speak about. A value of β is one way of writing μ in coordinates.

This observation reduces the question to linear algebra. The map $\beta \mapsto \mathbf{X}\beta$ from $\mathbb{R}^p$ to $\mathbb{R}^n$ is linear. Two coefficient vectors have the same image iff their difference lies in the null space $\mathcal{N}(\mathbf{X})$. So

$$\{\mathbf{b} \in \mathbb{R}^p : \mathbf{X}\mathbf{b} = \mathbf{X}\beta\} = \beta + \mathcal{N}(\mathbf{X}), \quad (8.1.1)$$

an affine subspace of dimension $p - r$ by rank-nullity (Eq. 6.2.1). All vectors in Eq. 8.1.1 are observationally equivalent: no experiment with this model matrix can distinguish between them.

Definition 8.1.1 — Identifiability

In the linear model $E(\mathbf{Y}) = \mathbf{X}\beta$, $\beta \in \mathbb{R}^p$, the parameter β is **identifiable** if $\mathbf{X}\beta_1 = \mathbf{X}\beta_2$ implies $\beta_1 = \beta_2$. A function g defined on $\mathbb{R}^p$, with values in any set, is **identifiable** if $\mathbf{X}\beta_1 = \mathbf{X}\beta_2$ implies $g(\beta_1) = g(\beta_2)$.

The definition is phrased through the mean, not the whole distribution. By the argument above the two versions agree whenever the error distribution does not involve β , which is the situation throughout this book until the generalized linear models of Part VIII. There the mean is $h(\mathbf{X}\beta)$ for a known one-to-one function h , and the same definition applies unchanged. It says that $g(\beta)$ is a well-defined function of the point μ , whichever coordinates were used to describe that point.

Theorem 8.1.1 — Identifiable means “a function of the mean”

A function g on $\mathbb{R}^p$ is identifiable iff there is a function h on $\mathcal{C}(\mathbf{X})$ with

$$g(\boldsymbol{\beta}) = h(\mathbf{X}\boldsymbol{\beta}) \quad \text{for every } \boldsymbol{\beta} \in \mathbb{R}^p.$$

Equivalently, g is constant on each affine subspace $\boldsymbol{\beta} + \mathcal{N}(\mathbf{X})$.

Proof. If $g = h \circ \mathbf{X}$ and $\mathbf{X}\boldsymbol{\beta}_1 = \mathbf{X}\boldsymbol{\beta}_2$, then $g(\boldsymbol{\beta}_1) = h(\mathbf{X}\boldsymbol{\beta}_1) = h(\mathbf{X}\boldsymbol{\beta}_2) = g(\boldsymbol{\beta}_2)$. Conversely, let g be identifiable. For $\boldsymbol{\mu} \in \mathcal{C}(\mathbf{X})$ choose any $\boldsymbol{\beta}$ with $\mathbf{X}\boldsymbol{\beta} = \boldsymbol{\mu}$ and put $h(\boldsymbol{\mu}) = g(\boldsymbol{\beta})$. The value does not depend on the choice, because any two choices have the same image under $\mathbf{X}$. By construction $g(\boldsymbol{\beta}) = h(\mathbf{X}\boldsymbol{\beta})$. The last sentence restates the definition using [Eq. 8.1.1](#). ■

The theorem makes identifiability a statement about information. A function that can be computed from $\boldsymbol{\mu}$ is one that could, in principle, be estimated consistently, provided that $\boldsymbol{\mu}$ itself can be estimated. A function that cannot be computed from $\boldsymbol{\mu}$ is one about which the model is silent. Its value can be anything along the affine subspace, and more data from the same design cannot help.

8.1.2 When the parameter is identifiable**Proposition 8.1.2**

The following are equivalent: (a) $\boldsymbol{\beta}$ is identifiable; (b) $\mathcal{N}(\mathbf{X}) = \{\mathbf{0}\}$; (c) $\text{rank}(\mathbf{X}) = p$; (d) $\mathbf{X}^\top \mathbf{X}$ is nonsingular. In that case the function h of [Theorem 8.1.1](#) for $g(\boldsymbol{\beta}) = \boldsymbol{\beta}$ is $h(\boldsymbol{\mu}) = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \boldsymbol{\mu}$.

Proof. By [Eq. 8.1.1](#), $\boldsymbol{\beta}$ is identifiable iff every affine subspace $\boldsymbol{\beta} + \mathcal{N}(\mathbf{X})$ is a single point, which is (b). (b) $\Leftrightarrow$ (c) is rank-nullity, and (c) $\Leftrightarrow$ (d) is [Corollary 6.2.4\(c\)](#). If $\mathbf{X}^\top \mathbf{X}$ is nonsingular and $\boldsymbol{\mu} = \mathbf{X}\boldsymbol{\beta}$, then $(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \boldsymbol{\mu} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{X}\boldsymbol{\beta} = \boldsymbol{\beta}$. ■

When $\text{rank}(\mathbf{X}) = r < p$ the parameter is not identifiable, but parts of it may be. The simplest parts are the individual coordinates.

Proposition 8.1.3 — Identifiability of a single coefficient

Write $\mathbf{x}_1, \dots, \mathbf{x}_p$ for the columns of $\mathbf{X}$. For each j the following are equivalent:

- (a) β_j is identifiable;
- (b) every $\mathbf{v} \in \mathcal{N}(\mathbf{X})$ has $v_j = 0$;

- (c) $\mathbf{x}_j$ is not a linear combination of the other columns of $\mathbf{X}$;
- (d) deleting column j from $\mathbf{X}$ lowers the rank.

Proof. (a) $\Leftrightarrow$ (b): by Eq. 8.1.1, the j th coordinates of the vectors observationally equivalent to $\boldsymbol{\beta}$ are $\beta_j + v_j$, $\mathbf{v} \in \mathcal{N}(\mathbf{X})$. They all equal β_j iff $v_j = 0$ for every null vector. (b) $\Leftrightarrow$ (c): a relation $\mathbf{x}_j = \sum_{k \neq j} a_k \mathbf{x}_k$ is the same as a null vector with j th entry -1 , and any null vector with $v_j \neq 0$ can be rescaled to have $v_j = -1$. (c) $\Leftrightarrow$ (d): the column space loses a dimension when $\mathbf{x}_j$ is removed iff $\mathbf{x}_j$ is not in the span of the remaining columns. ■

Part (c) gives a practical test. A coefficient is lost exactly when its regressor is *aliased*, that is, when it can be rebuilt from the others. In a model that is only partly identifiable, some coefficients can still be perfectly well identified. If $\mathbf{X} = [\mathbf{X}_1, \mathbf{x}_j]$ and $\mathbf{x}_j \notin \mathcal{C}(\mathbf{X}_1)$, then β_j is identifiable however badly $\mathbf{X}_1$ is parameterized. A typical case is a numeric covariate added to an overparameterized factor model: the covariate's coefficient is identifiable as long as the covariate is not constant within every level of the factor (Section 8.6).

8.1.3 Where rank deficiency comes from

Linear dependence among the columns of $\mathbf{X}$ arises in three quite different ways, and it helps to distinguish them.

By construction. A factor with g levels entered as an intercept together with all g indicator columns gives $g+1$ columns spanning a g -dimensional space (Example (A rank-deficient layout)). Such *overparameterized* models are written down deliberately, because they treat the levels symmetrically. The dependence is harmless once it is understood, and most of this chapter is about understanding it.

By the logic of the variables. Some sets of variables satisfy an identity that holds for every observation. Examples are shares of a whole that add to one, a total entered together with all its parts, and, the best-known case, age, period and birth cohort.

By the design. Two regressors that are logically unrelated may be exactly collinear in the sample at hand. A factor level may never be observed. The design may have fewer distinct rows than parameters. In each case another data set with the same model could identify the parameter, but this one cannot.

Identifiability is therefore a property of the model matrix, and through it of the design, not of the population being studied. The next example shows the second kind of dependence.

Example. (Age, period and cohort)

A survey is fielded every five years from 2000 to 2020, and its respondents are grouped in eleven five-year age bands from 20 to 70. For each of the 55 combinations we have one average response. A demographer wants to separate three linear trends: in age (people change as they grow older), in period (everyone changes with the times), and in birth cohort (generations differ). With age, period and cohort centred at 45, 2010 and 1965, the model is

$$E(y) = \beta_0 + \beta_A \text{age} + \beta_P \text{period} + \beta_C \text{cohort}.$$

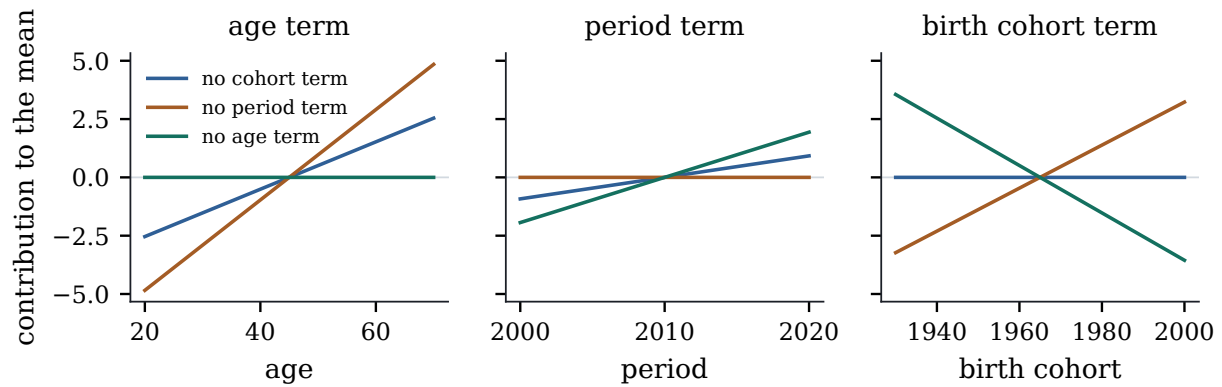


Figure 8.1.1: Three equally good fits of the age–period–cohort model. Each panel shows the contribution of one term to the fitted mean under the three fits of [Example \(Age, period and cohort\)](#). The sum of the three contributions is the same for every respondent, so the data cannot choose between the fits.

But cohort is period minus age, for every respondent. The four columns satisfy age – period + cohort = $\mathbf{0}$, so $\mathbf{v} = (0, 1, -1, 1)^\top$ spans $\mathcal{N}(\mathbf{X})$ and $\text{rank}(\mathbf{X}) = 3$. By [Proposition 8.1.3](#), none of the three slopes is identifiable.

The listing fits the model three times, each time dropping one of the three columns. All three fits have residual sum of squares 157.00, because they have the same column space. They tell three different stories:

Fit	age slope	period slope	cohort slope
no cohort term	0.102	0.092	0.000
no period term	0.194	0.000	0.092
no age term	0.000	0.194	-0.102

One analyst reports a steady ageing effect and no generational change. Another reports a strong ageing effect and a positive cohort trend. A third reports no ageing at all, a strong secular trend, and younger generations scoring lower. The data were in fact simulated with slopes 0.30, -0.10 and 0.20. That vector lies on the parallel line $\boldsymbol{\beta} + t\mathbf{v}$ of equally valid truths, and the data cannot prefer it. Only the identifiable combinations are recovered: the fitted 0.194 and 0.092 estimate the true 0.20 and 0.10.

What the three fits share are the linear functions orthogonal to $\mathbf{v} = (0, 1, -1, 1)^\top$:

$$\beta_A + \beta_P = 0.194, \quad \beta_P + \beta_C = 0.092, \quad \beta_A - \beta_C = 0.102,$$

in every row of the table, and β_0 . Any statement about the separate trends requires information from outside these data, for instance a constraint justified by subject knowledge, and the conclusions then rest on that constraint ([Exercise \(C1\)](#)). [Figure 8.1.1](#) shows how each fit divides the same fitted surface among the three terms.

```
import numpy as np

ages = np.arange(20, 75, 5)           # 11 age bands: 20, 25, ..., 70
periods = np.arange(2000, 2025, 5)   # 5 survey waves: 2000, ..., 2020
A, P = np.meshgrid(ages, periods, indexing="ij")
age, period = A.ravel() - 45.0, P.ravel() - 2010.0 # centred at 45 and 2010
```

```

cohort = period - age                                     # birth year minus 1965, exactly
X = np.column_stack([np.ones(age.size), age, period, cohort])
print("n =", X.shape[0], " p =", X.shape[1], " rank =", np.linalg.matrix_rank(X))
print("X @ (0, 1, -1, 1) =", np.abs(X @ [0, 1, -1, 1]).max())

```

```

rng = np.random.default_rng(8)
truth = np.array([50.0, 0.30, -0.10, 0.20])
y = X @ truth + rng.normal(scale=1.5, size=age.size)

def fit_without(j):
    """Least squares after deleting column j (one of age, period, cohort)."""
    keep = [k for k in range(4) if k != j]
    b = np.zeros(4)
    b[keep] = np.linalg.lstsq(X[:, keep], y, rcond=None)[0]
    return b

stories = {"no cohort term": fit_without(3),
           "no period term": fit_without(2),
           "no age term": fit_without(1)}
for name, b in stories.items():
    sse = np.sum((y - X @ b) ** 2)
    print(f"{name:15s} b = {np.round(b, 3)}   SSE = {sse:.3f}",
          f" age+period = {b[1] + b[2]:.3f}   period+cohort = {b[2] + b[3]:.3f}")

```

8.1.4 Linear and nonlinear identifiable functions

In the example the identifiable linear functions were exactly those whose coefficient vectors are orthogonal to the null space. That is true in general.

Proposition 8.1.4

A linear function $\lambda^\top \beta$ is identifiable iff $\lambda \perp \mathcal{N}(\mathbf{X})$, that is, iff $\lambda \in \mathcal{C}(\mathbf{X}^\top)$. The identifiable linear functions form a subspace of dimension $r = \text{rank}(\mathbf{X})$.

Proof. By [Theorem 8.1.1](#), $\lambda^\top \beta$ is identifiable iff $\lambda^\top (\beta + \mathbf{v}) = \lambda^\top \beta$ for all β and all $\mathbf{v} \in \mathcal{N}(\mathbf{X})$, that is, iff $\lambda^\top \mathbf{v} = 0$ for all $\mathbf{v} \in \mathcal{N}(\mathbf{X})$. By [Corollary 6.2.4\(d\)](#), $\mathcal{N}(\mathbf{X})^\perp = \mathcal{C}(\mathbf{X}^\top)$, which has dimension $\text{rank}(\mathbf{X}^\top) = r$. ■

Identifiable functions need not be linear. If $\lambda_1^\top \beta$ and $\lambda_2^\top \beta$ are identifiable, so are their ratio (where defined), their product, and $\max\{\lambda_1^\top \beta, 0\}$. Every function of μ is identifiable, for example $\|\mu\|^2$ or the largest mean. For instance, in a quadratic regression $E(y) = \beta_0 + \beta_1 x + \beta_2 x^2$ with a full-rank design, the location $-\beta_1/(2\beta_2)$ of the turning point is identifiable. It is not linear, and [Section 12.5](#) has to estimate it with care. The next section shows that among the *linear* functions, identifiability coincides with a much more concrete property, the existence of a linear unbiased estimator.

Idea. The data determine the mean vector $\mu = \mathbf{X}\beta$ and nothing else about β . A function of β is identifiable exactly when it is a function of μ . A *linear* function $\lambda^\top \beta$ is identifiable exactly when λ lies in the row space $\mathcal{C}(\mathbf{X}^\top)$, the orthogonal complement of

the invisible directions $\mathcal{N}(\mathbf{X})$.

8.1.5 Exercises

8.1.5.1 A. Check your understanding

Exercise A1.

In [Example \(Age, period and cohort\)](#), show directly that β_0 , $\beta_A + \beta_P$ and $\beta_P + \beta_C$ are identifiable, that every identifiable linear function is a combination of these three, and that $\beta_A - \beta_C$ is one of them. Is $\beta_A + \beta_C$ identifiable?

Exercise A2.

A household's spending is split into food, housing and other, with shares $s_1 + s_2 + s_3 = 1$. A regression of savings on an intercept and all three shares is proposed. Find $\mathcal{N}(\mathbf{X})$ (assuming the shares vary enough that there is no other relation among the columns), decide which coefficients are identifiable, and describe two standard ways to make the model identifiable.

8.1.5.2 B. Practice

Exercise B1.

Let $\mathbf{Z} = [\mathbf{z}_1, \dots, \mathbf{z}_g]$ be the indicator matrix of a factor and $\mathbf{x}$ a numeric covariate, and consider $\mathbf{X} = [\mathbf{1}, \mathbf{Z}, \mathbf{x}]$. Show that the slope of $\mathbf{x}$ is identifiable iff $\mathbf{x} \notin \mathcal{C}(\mathbf{Z})$, that is, iff $\mathbf{x}$ is not constant within every level. Show that no intercept-type parameter β_0 or α_k is identifiable.

Solution. The columns $\mathbf{1}, \mathbf{z}_1, \dots, \mathbf{z}_g$ satisfy $\mathbf{1} - \sum_k \mathbf{z}_k = \mathbf{0}$, so they are dependent, and $\mathcal{C}([\mathbf{1}, \mathbf{Z}]) = \mathcal{C}(\mathbf{Z})$. By [Proposition 8.1.3\(c\)](#), the slope is identifiable iff $\mathbf{x}$ is not a combination of $\mathbf{1}, \mathbf{z}_1, \dots, \mathbf{z}_g$, that is, iff $\mathbf{x} \notin \mathcal{C}(\mathbf{Z})$. A vector lies in $\mathcal{C}(\mathbf{Z})$ iff it is constant within each level. For the other coefficients, $\mathbf{v} = (1, -1, \dots, -1, 0)^\top$ is a null vector with nonzero entries in positions $0, 1, \dots, g$, so by [Proposition 8.1.3\(b\)](#) none of $\beta_0, \alpha_1, \dots, \alpha_g$ is identifiable.

Exercise B2.

Let $\mathbf{P}$ be the orthogonal projection onto $\mathcal{C}(\mathbf{X}^\top)$. Show that g is identifiable iff $g(\boldsymbol{\beta}) = g(\mathbf{P}\boldsymbol{\beta})$ for every $\boldsymbol{\beta}$. Deduce that the identifiable functions are exactly the functions of $\mathbf{P}\boldsymbol{\beta}$, and that $\mathbf{P}\boldsymbol{\beta}$ is the unique coefficient vector in $\mathcal{C}(\mathbf{X}^\top)$ representing the mean $\mathbf{X}\boldsymbol{\beta}$.

Exercise B3.

A design with model matrix $\mathbf{X}$ of rank $r < p$ is augmented by one more observation at a point $\mathbf{x}_0$. Show that the rank increases iff $\mathbf{x}_0 \notin \mathcal{C}(\mathbf{X}^\top)$. Show that, in the age–period–cohort design, no additional observation can make the slopes identifiable, and explain

why in words.

8.1.5.3 C. Going deeper

Exercise C1.

Replace the three linear terms of [Example \(Age, period and cohort\)](#) by three factors: an indicator for each age band, each period and each cohort (with an intercept). Show that the null space of the resulting model matrix has dimension 3 plus exactly one extra dimension. Identify the extra null vector: it assigns the linear sequences $a - \bar{a}$, $-(t - \bar{t})$ and $c - \bar{c}$ to the age, period and cohort effects. Deduce that every second difference of age effects, such as $\alpha_{a+1} - 2\alpha_a + \alpha_{a-1}$, is identifiable, while no linear trend in any single one of the three factors is.

Solution. Each factor's indicators sum to $\mathbf{1}$, which gives the usual three null vectors (intercept minus all indicators of one factor). The identity age – period + cohort = const holds for every observation. Writing the age of an observation as $\sum_a a z_a^A$, and similarly for period and cohort, it becomes a relation $\sum_a a \mathbf{z}_a^A - \sum_t t \mathbf{z}_t^P + \sum_c c \mathbf{z}_c^C - \kappa \mathbf{1} = \mathbf{0}$ among the columns. Subtracting suitable multiples of the three standard relations centres the three sequences, and this gives the stated vector. It is not a combination of the other three, because those are constant on each factor's effects and this one is linear and nonconstant. To see that there are no more, check that the matrix has rank $1 + (A - 1) + (P - 1) + (C - 1) - 1$. This rank check can be done directly or numerically for the design of [Example \(Age, period and cohort\)](#). A function $\sum_a d_a \alpha_a$ is identifiable iff $\mathbf{d}$ is orthogonal to all four null vectors, that is, iff $\sum_a d_a = 0$ and $\sum_a d_a a = 0$. A second difference satisfies both. A linear trend $\sum_a (a - \bar{a}) \alpha_a$ fails the second condition.

8.2 Estimable functions

Identifiability asks whether a quantity is determined by the distribution of the data. Estimation asks for more. We want a statistic whose distribution is centred on the quantity, however the parameter is set. For linear functions of $\boldsymbol{\beta}$, the natural candidates are linear functions of $\mathbf{Y}$, and the natural centring requirement is unbiasedness. This leads to the classical notion of an estimable function.

8.2.1 Linear unbiased estimators

Here and below the model is $E(\mathbf{Y}) = \mathbf{X}\boldsymbol{\beta}$ with $\mathbf{X}$ an $n \times p$ matrix of rank r ; no assumption on $\text{Cov}(\mathbf{Y})$ is needed until variances are computed.

Definition 8.2.1 — Estimable function

A linear function $\boldsymbol{\lambda}^\top \boldsymbol{\beta}$, $\boldsymbol{\lambda} \in \mathbb{R}^p$, is **estimable** if there is a vector $\mathbf{a} \in \mathbb{R}^n$ such that

$$E(\mathbf{a}^\top \mathbf{Y}) = \boldsymbol{\lambda}^\top \boldsymbol{\beta} \quad \text{for every } \boldsymbol{\beta} \in \mathbb{R}^p.$$

A vector $\Lambda^\top \beta$ of linear functions, with Λ a $p \times q$ matrix, is estimable if each of its q components is. Equivalently, there is an $n \times q$ matrix $\mathbf{A}$ with $E(\mathbf{A}^\top \mathbf{Y}) = \Lambda^\top \beta$ for all β .

The statistic $\mathbf{a}^\top \mathbf{Y}$ is a **linear unbiased estimator** of $\lambda^\top \beta$. The requirement “for every β ” is essential. For a single fixed β , almost any statistic can be made unbiased by adding a constant. What the definition demands is one statistic that is centred correctly whatever the true coefficients are. Chapter 7 asked which linear unbiased estimator is best (Definition 7.1.1). Here we ask only whether there is one.

One might allow an *affine* estimator $a_0 + \mathbf{a}^\top \mathbf{Y}$. This gains nothing, as Chapter 7 already showed.

Lemma 8.2.1

The statistic $a_0 + \mathbf{a}^\top \mathbf{Y}$ is unbiased for $\lambda^\top \beta$, for every β , iff $a_0 = 0$ and $\mathbf{X}^\top \mathbf{a} = \lambda$.

Proof. This is Proposition 7.1.1, proved in Chapter 7 by evaluating the expectation $a_0 + \mathbf{a}^\top \mathbf{X}\beta$ at $\beta = \mathbf{0}$ and at the coordinate vectors. ■

So the linear unbiased estimators of $\lambda^\top \beta$ correspond exactly to the solutions $\mathbf{a}$ of the linear system $\mathbf{X}^\top \mathbf{a} = \lambda$. There is at least one iff the system is consistent, and when it is, the solutions form the affine subspace $\mathbf{a}_0 + \mathcal{N}(\mathbf{X}^\top)$ (Theorem 1.4.1).

8.2.2 Estimable is the same as identifiable

Theorem 8.2.2

For $\lambda \in \mathbb{R}^p$ the following are equivalent:

- (a) $\lambda^\top \beta$ is estimable;
- (b) $\lambda^\top \beta$ is identifiable;
- (c) $\lambda \in \mathcal{C}(\mathbf{X}^\top)$, that is, $\lambda^\top = \rho^\top \mathbf{X}$ for some $\rho \in \mathbb{R}^n$.

For a $p \times q$ matrix Λ , $\Lambda^\top \beta$ is estimable iff $\Lambda = \mathbf{X}^\top \mathbf{P}$ for some $n \times q$ matrix $\mathbf{P}$, that is, iff $\mathcal{C}(\Lambda) \subseteq \mathcal{C}(\mathbf{X}^\top)$.

Proof. (a) $\Leftrightarrow$ (c) is Lemma 8.2.1: an unbiased $\mathbf{a}^\top \mathbf{Y}$ exists iff $\mathbf{X}^\top \mathbf{a} = \lambda$ has a solution, which is the statement $\lambda \in \mathcal{C}(\mathbf{X}^\top)$. (b) $\Leftrightarrow$ (c) is Proposition 8.1.4. The matrix version applies the vector version to each column of Λ and collects the vectors ρ as the columns of $\mathbf{P}$. ■

The theorem connects two different ideas. Identifiability is about what the model could reveal with unlimited data, and it involves no estimator. Estimability is about a concrete property of a concrete statistic. For linear functions they are the same thing, and both reduce to membership

of λ in the row space of $\mathbf{X}$. The row space is spanned by the rows $\mathbf{x}_{(1)}^\top, \dots, \mathbf{x}_{(n)}^\top$, so the theorem says:

A linear function of β is estimable iff it is a linear combination of the expected responses $E(y_i) = \mathbf{x}_{(i)}^\top \beta$.

This reading is the most useful one in practice. To show that $\lambda^\top \beta$ is estimable, write it as a combination of the means of observations that were actually made. To show that it is not, find a direction $\mathbf{v}$ with $\mathbf{X}\mathbf{v} = \mathbf{0}$ and $\lambda^\top \mathbf{v} \neq 0$.

Corollary 8.2.3 — The estimable space

The set of λ for which $\lambda^\top \beta$ is estimable is the subspace $\mathcal{C}(\mathbf{X}^\top)$ of dimension $r = \text{rank}(\mathbf{X})$. In particular:

- (a) every expected response $\mathbf{x}_{(i)}^\top \beta$, and hence the whole mean vector $\mathbf{X}\beta$, is estimable;
- (b) every entry of $\mathbf{X}^\top \mathbf{X}\beta$ is estimable;
- (c) at most r estimable functions can be linearly independent, and a set of r linearly independent ones determines all the others as linear combinations;
- (d) if $r = p$, every linear function of β is estimable.

Proof. The first sentence is [Theorem 8.2.2](#). For (a) take $\rho = \mathbf{e}_i$, the i th coordinate vector. For (b), $\mathbf{X}^\top \mathbf{X}\beta = \mathbf{X}^\top (\mathbf{X}\beta)$ and the rows of $\mathbf{X}^\top \mathbf{X}$ lie in $\mathcal{C}(\mathbf{X}^\top)$. Part (c) is the dimension count, and (d) is the case $\mathcal{C}(\mathbf{X}^\top) = \mathbb{R}^p$. ■

8.2.3 The mean at a new point

A frequent use of a fitted model is to estimate the mean response at a setting $\mathbf{x}_0$ of the regressors, which may or may not have been observed. The quantity $\mathbf{x}_0^\top \beta$ is a linear function of β , so the theorem applies. It is estimable iff $\mathbf{x}_0 \in \mathcal{C}(\mathbf{X}^\top)$. In a full-rank model this is automatic, and the only danger in predicting at a new point is extrapolation, meaning that the model may not hold there. In a rank-deficient model there is a sharper limit. Some settings are *algebraically* out of reach. The data contain no information about the mean there, even if the model is exactly true.

Example. (A fertilizer trial with a fixed ratio)

Ten plots receive nitrogen at 0, 20, 40, 60, 80 kg/ha (two plots each), and phosphorus is always applied at half the nitrogen rate, as in a commercial blend. The model

$$E(y) = \beta_0 + \beta_1 N + \beta_2 P$$

is natural agronomically, but in this design $\mathbf{x}_P = \frac{1}{2}\mathbf{x}_N$, so $\mathbf{v} = (0, 1, -2)^\top$ spans $\mathcal{N}(\mathbf{X})$ and $\text{rank}(\mathbf{X}) = 2$. A linear function is estimable iff its coefficient vector satisfies $\lambda_1 - 2\lambda_2 = 0$. So:

- β_0 , the mean yield without fertilizer, is estimable. Plots with $N = P = 0$ were observed.

- The mean yield at $(N, P) = (40, 20)$ is estimable, since $40 - 2 \cdot 20 = 0$, and so is the mean at any dose of the blend, observed or not, such as $(50, 25)$.
- The mean yield at $(40, 0)$, nitrogen alone, is not estimable, since $40 - 0 \neq 0$.
- Neither β_1 nor β_2 is estimable, but $\beta_1 + \frac{1}{2}\beta_2$, the gain per kilogram of nitrogen *when it comes with its phosphorus*, is.

The listing computes three least squares solutions: the minimum-norm one, the one that omits P , and the one that omits N . They give intercepts 2.9579, 2.9579 and 2.9579, all equal as they must be. The pairs of slopes $(0.0296, 0.0148)$, $(0.0370, 0.0000)$ and $(0.0000, 0.0740)$ all disagree. At the blend dose $(40, 20)$ all three predict 4.439 t/ha. At the nitrogen-only dose $(40, 0)$ they predict 4.143, 4.439 and 2.958. The third value is just $\hat{\beta}_0$, because that solution puts the whole fertilizer effect on phosphorus. Each of these is a legitimate least squares answer, and none of them estimates anything.

The two kinds of prediction differ in kind, not in degree. Nothing in this experiment says what nitrogen does without phosphorus. A larger experiment of the same design would not help. What is needed is at least one plot off the line $P = N/2$ ([Exercise \(B3\)](#)).

```
import numpy as np

N = np.array([0, 0, 20, 20, 40, 40, 60, 60, 80, 80], dtype=float) # kg/ha nitrogen
P = N / 2 # phosphorus,
    always half
X = np.column_stack([np.ones(N.size), N, P])
rng = np.random.default_rng(80)
y = 3.0 + 0.020 * N + 0.030 * P + rng.normal(scale=0.25, size=N.size) # yield, t/
    ha
print("rank(X) =", np.linalg.matrix_rank(X), " null vector (0, 1, -2):", X @ [0, 1,
    -2])
```

```
b_min = np.linalg.pinv(X) @ y # minimum-norm
    solution
b_noP = np.append(np.linalg.lstsq(X[:, :2], y, rcond=None)[0], 0.0) # drop P
b_noN = np.insert(np.linalg.lstsq(X[:, [0, 2]], y, rcond=None)[0], 1, 0.0) # drop N
new_points = {"N=40, P=20 (on the line)": [1, 40, 20],
    "N=40, P=0 (off the line)": [1, 40, 0]}
for name, x0 in new_points.items():
    print(name, [round(float(np.dot(x0, b)), 3) for b in (b_min, b_noP, b_noN)])
```

8.2.4 Many unbiased estimators, one least squares estimator

An estimable function usually has many linear unbiased estimators. In [Example \(A fertilizer trial with a fixed ratio\)](#), the mean at the blend dose $(40, 20)$ is estimated without bias by the average of the two plots that received exactly that dose, $\mathbf{a}_1^\top \mathbf{Y}$ with $\mathbf{a}_1$ putting weight $\frac{1}{2}$ on each. It is also estimated without bias by the value of any least squares fit at that dose. If $\text{Cov}(\mathbf{Y}) = \sigma^2 \mathbf{I}$, the first estimator has variance $\sigma^2 \|\mathbf{a}_1\|^2 = 0.500 \sigma^2$, while the least squares estimator has variance $0.100 \sigma^2$, five times smaller, because it uses all ten plots.

The least squares estimator can be written in the same form. If $\mathbf{a}$ is any solution of $\mathbf{X}^\top \mathbf{a} = \boldsymbol{\lambda}$,

then [Theorem 6.4.4](#) gives

$$\boldsymbol{\lambda}^\top \hat{\boldsymbol{\beta}} = \mathbf{a}^\top \mathbf{X} \hat{\boldsymbol{\beta}} = \mathbf{a}^\top \mathbf{M} \mathbf{y} = (\mathbf{M} \mathbf{a})^\top \mathbf{y}, \quad (8.2.1)$$

for every least squares solution $\hat{\boldsymbol{\beta}}$. So the least squares estimator is the linear unbiased estimator whose coefficient vector is the projection $\mathbf{M} \mathbf{a}$ of any one coefficient vector onto $\mathcal{C}(\mathbf{X})$. It is unbiased, because $\mathbf{X}^\top \mathbf{M} \mathbf{a} = \mathbf{X}^\top \mathbf{a} = \boldsymbol{\lambda}$. Every solution of $\mathbf{X}^\top \mathbf{a} = \boldsymbol{\lambda}$ has the form $\mathbf{M} \mathbf{a} + \mathbf{w}$ with $\mathbf{w} \in \mathcal{N}(\mathbf{X}^\top) = \mathcal{C}(\mathbf{X})^\perp$, and Pythagoras gives $\|\mathbf{M} \mathbf{a} + \mathbf{w}\|^2 = \|\mathbf{M} \mathbf{a}\|^2 + \|\mathbf{w}\|^2$. This is the Gauss–Markov theorem in its simplest form: among linear unbiased estimators of an estimable function, least squares has the smallest variance when $\text{Cov}(\mathbf{Y}) = \sigma^2 \mathbf{I}$. [Chapter 7](#) states and proves it in general ([Theorem 7.2.1](#)).

8.2.5 What estimability does not depend on

Three remarks keep the concept in its place.

It does not depend on the covariance. Estimability involves only $E(\mathbf{Y}) = \mathbf{X} \boldsymbol{\beta}$, so it is the same whether the errors are uncorrelated, correlated, heteroscedastic or normal, provided $\text{Cov}(\mathbf{Y})$ is free of $\boldsymbol{\beta}$. Generalized least squares in Part VII has the same estimable functions as ordinary least squares.

It is a property of the design, not the response. Whether $\boldsymbol{\lambda}^\top \boldsymbol{\beta}$ is estimable can be decided before any response is measured, from $\mathbf{X}$ alone. This makes estimability a planning tool. If a comparison matters, the design must contain the rows that make it estimable.

It is a statement about linear functions. A turning point $-\beta_1/(2\beta_2)$ or a quotient of two estimable quantities is not “estimable” in the technical sense, and usually has no unbiased estimator of any kind. Yet when it is identifiable it is a perfectly sensible target, to be estimated by other means. The line that must not be crossed is identifiability, not linearity. In this chapter “nonestimable” is used only for linear functions, and for them it means the same as “not identifiable”.

Warning. Software fitted to a rank-deficient model reports a number for every coefficient and every requested combination, estimable or not. For a nonestimable function the number is an artefact of the particular solution the software chose, as the three predictions at $(40, 0)$ show. It comes with a standard error that is equally arbitrary ([Section 8.3](#)).

8.2.6 Exercises

8.2.6.1 A. Check your understanding

Exercise A1.

In [Example \(A fertilizer trial with a fixed ratio\)](#), decide which of the following are estimable: $\beta_1 - 2\beta_2$; $2\beta_1 + \beta_2$; the mean yield at $(N, P) = (100, 50)$; the mean yield at $(0, 10)$; the difference between the mean yields at $(60, 30)$ and $(20, 10)$. For the estimable ones give a vector $\mathbf{a}$ with $\mathbf{X}^\top \mathbf{a} = \boldsymbol{\lambda}$.

Exercise A2.

In simple linear regression $E(y_i) = \beta_0 + \beta_1 x_i$, show that β_1 is estimable iff the x_i are not all equal. If all $x_i = c$, which linear functions are estimable, and what is their least squares estimator?

8.2.6.2 B. Practice**Exercise B1.**

Let $\lambda^\top \beta$ be estimable and let $\mathbf{a}^* = \mathbf{M}\mathbf{a}$ as in Eq. 8.2.1. Show that $\mathbf{a}^*$ does not depend on which solution $\mathbf{a}$ of $\mathbf{X}^\top \mathbf{a} = \lambda$ is used, that $\mathbf{a}^* = \mathbf{X}\mathbf{G}\lambda$ for any generalized inverse $\mathbf{G}$ of $\mathbf{X}^\top \mathbf{X}$, and that $\|\mathbf{a}^*\|^2 = \lambda^\top \mathbf{G}\lambda$.

Solution. Two solutions differ by a vector of $\mathcal{N}(\mathbf{X}^\top) = \mathcal{C}(\mathbf{X})^\perp$, which $\mathbf{M}$ sends to $\mathbf{0}$, so $\mathbf{M}\mathbf{a}$ is the same for both. With $\mathbf{M} = \mathbf{X}\mathbf{G}\mathbf{X}^\top$ (Theorem 6.3.7), $\mathbf{M}\mathbf{a} = \mathbf{X}\mathbf{G}\mathbf{X}^\top \mathbf{a} = \mathbf{X}\mathbf{G}\lambda$. Finally, $\|\mathbf{M}\mathbf{a}\|^2 = \mathbf{a}^\top \mathbf{M}\mathbf{a} = \mathbf{a}^\top \mathbf{X}\mathbf{G}\mathbf{X}^\top \mathbf{a} = \lambda^\top \mathbf{G}\lambda$.

Exercise B2.

Show that $\Lambda^\top \beta$ is estimable iff $\Lambda^\top \mathbf{b}_1 = \Lambda^\top \mathbf{b}_2$ whenever $\mathbf{X}\mathbf{b}_1 = \mathbf{X}\mathbf{b}_2$, and that in that case $\Lambda^\top \hat{\beta} = \mathbf{P}^\top \mathbf{M}\mathbf{y}$ for any $\mathbf{P}$ with $\mathbf{X}^\top \mathbf{P} = \Lambda$. Show also that $\mathbf{P}^\top \mathbf{M}$ does not depend on the choice of $\mathbf{P}$.

Exercise B3.

A study will compare three treatments, A , B and C , in the additive model $E(y) = \mu + \tau_{\text{trt}} + \delta_{\text{site}}$. Two sites are available, and each site can host two of the three treatments. Which allocations make every difference $\tau_s - \tau_t$ estimable? Show that putting $\{A, B\}$ at both sites leaves $\tau_A - \tau_C$ nonestimable, while $\{A, B\}$ at one site and $\{B, C\}$ at the other makes all three differences estimable.

8.2.6.3 C. Going deeper**Exercise C1.**

Suppose the distribution of ε does not depend on β . Show that if some statistic $f(\mathbf{Y})$, linear or not, satisfies $E f(\mathbf{Y}) = g(\beta)$ for every β , then g is identifiable. Conclude that no statistic whatever is unbiased for a nonestimable linear function.

Solution. The distribution of $\mathbf{Y} = \mathbf{X}\beta + \varepsilon$ depends on β only through $\mathbf{X}\beta$. If $\mathbf{X}\beta_1 = \mathbf{X}\beta_2$, then $\mathbf{Y}$ has the same distribution under both, so $g(\beta_1) = E_{\beta_1} f(\mathbf{Y}) = E_{\beta_2} f(\mathbf{Y}) = g(\beta_2)$. A linear function with an unbiased estimator is therefore identifiable, hence estimable by Theorem 8.2.2.

8.3 Characterizations of estimability

[Theorem 8.2.2](#) reduced estimability to a single condition, $\lambda \in \mathcal{C}(\mathbf{X}^\top)$. In practice that condition appears in several forms, depending on what is at hand. We may have the rows of $\mathbf{X}$, the matrix $\mathbf{X}^\top \mathbf{X}$ and one of its generalized inverses, a collection of least squares solutions, or an SVD. This section collects the equivalent forms in one theorem, derives the estimator of an estimable function with its variance, and then turns to the practical question of deciding estimability in floating-point arithmetic, where exact rank is not available.

8.3.1 Eight equivalent conditions

Theorem 8.3.1 — Characterizations of estimability

Let $\mathbf{X}$ be $n \times p$, let $\mathbf{G}$ denote a generalized inverse of $\mathbf{X}^\top \mathbf{X}$, and let $\lambda \in \mathbb{R}^p$. The following are equivalent:

- (a) $\lambda^\top \beta$ is estimable;
- (b) $\lambda \in \mathcal{C}(\mathbf{X}^\top)$: there is $\rho \in \mathbb{R}^n$ with $\lambda^\top = \rho^\top \mathbf{X}$;
- (c) $\lambda \perp \mathcal{N}(\mathbf{X})$: $\lambda^\top \mathbf{v} = 0$ whenever $\mathbf{X}\mathbf{v} = \mathbf{0}$;
- (d) $\lambda \in \mathcal{C}(\mathbf{X}^\top \mathbf{X})$: the system $\mathbf{X}^\top \mathbf{X}\mathbf{r} = \lambda$ has a solution;
- (e) $\lambda^\top \mathbf{G}\mathbf{X}^\top \mathbf{X} = \lambda^\top$ for some generalized inverse $\mathbf{G}$;
- (f) $\lambda^\top \mathbf{G}\mathbf{X}^\top \mathbf{X} = \lambda^\top$ for every generalized inverse $\mathbf{G}$;
- (g) for every $\mathbf{y}$, $\lambda^\top \mathbf{b}$ has the same value for all solutions $\mathbf{b}$ of the normal equations $\mathbf{X}^\top \mathbf{X}\mathbf{b} = \mathbf{X}^\top \mathbf{y}$ ([Theorem 6.4.2](#));
- (h) $\text{rank} \begin{pmatrix} \mathbf{X} \\ \lambda^\top \end{pmatrix} = \text{rank}(\mathbf{X})$.

Proof. (a) $\Leftrightarrow$ (b) is [Theorem 8.2.2](#). (b) $\Leftrightarrow$ (c) holds because $\mathcal{C}(\mathbf{X}^\top) = \mathcal{N}(\mathbf{X})^\perp$ ([Corollary 6.2.4\(d\)](#)). (b) $\Leftrightarrow$ (d) holds because $\mathcal{C}(\mathbf{X}^\top \mathbf{X}) = \mathcal{C}(\mathbf{X}^\top)$ ([Corollary 6.2.4\(b\)](#)). (d) $\Rightarrow$ (f): if $\lambda = \mathbf{X}^\top \mathbf{X}\mathbf{r}$, then for every generalized inverse $\lambda^\top \mathbf{G}\mathbf{X}^\top \mathbf{X} = \mathbf{r}^\top (\mathbf{X}^\top \mathbf{X}\mathbf{G}\mathbf{X}^\top \mathbf{X}) = \mathbf{r}^\top \mathbf{X}^\top \mathbf{X} = \lambda^\top$. (f) $\Rightarrow$ (e) holds because generalized inverses exist ([Proposition 6.3.5](#)). (e) $\Rightarrow$ (d): transposing $\lambda^\top \mathbf{G}\mathbf{X}^\top \mathbf{X} = \lambda^\top$ gives $\mathbf{X}^\top \mathbf{X}(\mathbf{G}^\top \lambda) = \lambda$, so $\mathbf{r} = \mathbf{G}^\top \lambda$ solves the system. (b) $\Leftrightarrow$ (g) is [Theorem 6.4.4](#). (b) $\Leftrightarrow$ (h): the rank of the stacked matrix is the dimension of its row space, $\mathcal{C}([\mathbf{X}^\top, \lambda])$. This contains $\mathcal{C}(\mathbf{X}^\top)$, and it has the same dimension iff it equals $\mathcal{C}(\mathbf{X}^\top)$, that is, iff $\lambda \in \mathcal{C}(\mathbf{X}^\top)$. ■

Conditions (e) and (f) are [Corollary 1.4.2](#) applied to the normal equations. Condition (g) is the one that matters for interpretation: an estimable function is one whose estimate does not depend on the arbitrary choice of solution. Condition (c) is the one to use when proving that something is *not* estimable, since a single null vector $\mathbf{v}$ with $\lambda^\top \mathbf{v} \neq 0$ settles the question. Condition (h) is the most direct computational test, but in floating point it inherits all the difficulties of deciding a rank.

8.3.2 The estimator and its variance

For an estimable function, every least squares solution produces the same estimate, and that estimate has a clean description.

Proposition 8.3.2 — Least squares estimation of estimable functions

Let $\lambda^\top \beta$ be estimable, with $\lambda = \mathbf{X}^\top \rho = \mathbf{X}^\top \mathbf{X} \mathbf{r}$, let $\hat{\beta}$ be any least squares solution and $\mathbf{G}$ any generalized inverse of $\mathbf{X}^\top \mathbf{X}$. Then:

- (a) $\lambda^\top \hat{\beta} = \rho^\top \mathbf{M} \mathbf{y} = \mathbf{r}^\top \mathbf{X}^\top \mathbf{y} = \lambda^\top \mathbf{G} \mathbf{X}^\top \mathbf{y}$;
- (b) $E(\lambda^\top \hat{\beta}) = \lambda^\top \beta$;
- (c) if $\text{Cov}(\mathbf{Y}) = \sigma^2 \mathbf{I}$, then

$$\text{Var}(\lambda^\top \hat{\beta}) = \sigma^2 \lambda^\top \mathbf{G} \lambda = \sigma^2 \mathbf{r}^\top \lambda = \sigma^2 \|\mathbf{M} \rho\|^2,$$

the same for every $\mathbf{G}$;

- (d) more generally, if $\Lambda^\top \beta$ is estimable, then $\text{Cov}(\Lambda^\top \hat{\beta}) = \sigma^2 \Lambda^\top \mathbf{G} \Lambda$ for every $\mathbf{G}$.

Proof.

- (a) The first equality is [Theorem 6.4.4](#). For the second, the normal equations give $\mathbf{r}^\top \mathbf{X}^\top \mathbf{y} = \mathbf{r}^\top \mathbf{X}^\top \mathbf{X} \hat{\beta} = \lambda^\top \hat{\beta}$. The last is the second with the solution $\hat{\beta} = \mathbf{G} \mathbf{X}^\top \mathbf{y}$ and $\mathbf{r} = \mathbf{G}^\top \lambda$ (see the proof of [Theorem 8.3.1](#)). (b)–(d) were proved in [Chapter 7](#): unbiasedness and $\text{Var}(\lambda^\top \hat{\beta}) = \sigma^2 \lambda^\top \mathbf{G} \lambda$ are [Theorem 7.2.1\(a\)](#), and the matrix form $\text{Cov}(\Lambda^\top \hat{\beta}) = \sigma^2 \Lambda^\top \mathbf{G} \Lambda$ is part (d) of the same theorem. Neither depends on $\mathbf{G}$, because $\mathbf{M} = \mathbf{X} \mathbf{G} \mathbf{X}^\top$ for every $\mathbf{G}$ ([Theorem 6.3.7](#)). The second form in (c) follows from $\lambda^\top \mathbf{G} \lambda = \mathbf{r}^\top \mathbf{X}^\top \mathbf{X} \mathbf{G} \mathbf{X}^\top \mathbf{X} \mathbf{r} = \mathbf{r}^\top \mathbf{X}^\top \mathbf{X} \mathbf{r} = \mathbf{r}^\top \lambda$. The third is $\rho^\top \mathbf{M} \rho = \|\mathbf{M} \rho\|^2$. ■

The formula $\sigma^2 \lambda^\top \mathbf{G} \lambda$ is the rank-deficient version of the familiar $\sigma^2 \lambda^\top (\mathbf{X}^\top \mathbf{X})^{-1} \lambda$ ([Theorem 5.5.1](#)). The generalized inverse is arbitrary, but the quadratic form in an estimable λ is not. The condition $\lambda \in \mathcal{C}(\mathbf{X}^\top)$ under which [Chapter 7](#) proved optimality is exactly estimability, so the Gauss–Markov theorem and the normal theory ([Theorem 7.5.1](#)) apply to every estimable $\lambda^\top \hat{\beta}$ unchanged.

8.3.3 What a generalized inverse reports for a non-estimable function

Software that solves the normal equations with a particular generalized inverse will print $\lambda^\top \mathbf{G} \mathbf{X}^\top \mathbf{y}$ and $\hat{\sigma}^2 \lambda^\top \mathbf{G} \lambda$ for any λ whatever. To see what these numbers mean when λ is not estimable, look at the matrix

$$\mathbf{H}_\mathbf{G} = \mathbf{G} \mathbf{X}^\top \mathbf{X}. \quad (8.3.1)$$

Proposition 8.3.3 — The matrix $\mathbf{GX}^\top \mathbf{X}$

For any generalized inverse $\mathbf{G}$ of $\mathbf{X}^\top \mathbf{X}$, with $\mathbf{H}_\mathbf{G}$ as in Eq. 8.3.1:

- (a) $\mathbf{H}_\mathbf{G}$ is idempotent, $\mathcal{N}(\mathbf{H}_\mathbf{G}) = \mathcal{N}(\mathbf{X})$, $\text{rank}(\mathbf{H}_\mathbf{G}) = r$, and the rows of $\mathbf{H}_\mathbf{G}$ span $\mathcal{C}(\mathbf{X}^\top)$;
- (b) $\lambda^\top \beta$ is estimable iff $\lambda^\top \mathbf{H}_\mathbf{G} = \lambda^\top$;
- (c) the solution $\hat{\beta}_\mathbf{G} = \mathbf{GX}^\top \mathbf{Y}$ has $E(\hat{\beta}_\mathbf{G}) = \mathbf{H}_\mathbf{G} \beta$. So for every λ , $\lambda^\top \hat{\beta}_\mathbf{G}$ is unbiased for the estimable function $\lambda^\top \mathbf{H}_\mathbf{G} \beta$, which equals $\lambda^\top \beta$ iff $\lambda^\top \beta$ is estimable;
- (d) for the Moore–Penrose inverse $\mathbf{G} = (\mathbf{X}^\top \mathbf{X})^+$, $\mathbf{H}_\mathbf{G}$ is the orthogonal projection onto $\mathcal{C}(\mathbf{X}^\top)$.

Proof.

- (a) $\mathbf{H}_\mathbf{G}^2 = \mathbf{G}(\mathbf{X}^\top \mathbf{X} \mathbf{G} \mathbf{X}^\top \mathbf{X}) = \mathbf{GX}^\top \mathbf{X} = \mathbf{H}_\mathbf{G}$. By Proposition 1.3.3(a), $\mathcal{N}(\mathbf{GX}^\top \mathbf{X}) = \mathcal{N}(\mathbf{X}^\top \mathbf{X})$, which is $\mathcal{N}(\mathbf{X})$ by Corollary 6.2.4(a). Rank-nullity gives rank r . The row space of $\mathbf{H}_\mathbf{G}$ is $\mathcal{C}(\mathbf{H}_\mathbf{G}) = \mathcal{N}(\mathbf{H}_\mathbf{G})^\perp = \mathcal{N}(\mathbf{X})^\perp = \mathcal{C}(\mathbf{X}^\top)$.
- (b) is Theorem 8.3.1(e),(f).
- (c) $E(\mathbf{GX}^\top \mathbf{Y}) = \mathbf{GX}^\top \mathbf{X} \beta$. The coefficient vector $\mathbf{H}_\mathbf{G}^\top \lambda$ lies in the row space of $\mathbf{H}_\mathbf{G}$, which is $\mathcal{C}(\mathbf{X}^\top)$, so the function is estimable. It equals $\lambda^\top \beta$ for all β iff $\mathbf{H}_\mathbf{G}^\top \lambda = \lambda$, which is (b).
- (d) By the Penrose conditions (Definition 1.3.2), $(\mathbf{X}^\top \mathbf{X})^+ \mathbf{X}^\top \mathbf{X}$ is symmetric and idempotent. It is therefore an orthogonal projection (Theorem 6.3.3), and its null space is $\mathcal{N}(\mathbf{X})$ by (a). So it projects onto $\mathcal{N}(\mathbf{X})^\perp = \mathcal{C}(\mathbf{X}^\top)$.

■

Part (c) answers the question. For a nonestimable λ , the printed number is a perfectly good unbiased estimate, but of a *different* function, $\lambda^\top \mathbf{H}_\mathbf{G} \beta$, and which function depends on the software's choice of $\mathbf{G}$. The printed “variance” $\sigma^2 \lambda^\top \mathbf{G} \lambda$ is the variance of that statistic when $\mathbf{G}$ is symmetric and reflexive, as the Moore–Penrose inverse is, but in general it need not be the variance of anything (Exercise (C1)). Part (d) gives the cleanest picture. With the Moore–Penrose inverse, $\beta = \mathbf{H} \beta + (\mathbf{I} - \mathbf{H}) \beta$ splits the coefficient vector into its component in the estimable space, which the data determine, and its component in $\mathcal{N}(\mathbf{X})$, which they cannot see. The minimum-norm solution estimates the first component and sets the second to zero.

Example. (Checking estimability in a one-way layout)

Eighteen observations fall into four groups of sizes 4, 6, 3, 5, with $\mathbf{X} = [\mathbf{1}, \mathbf{z}_1, \mathbf{z}_2, \mathbf{z}_3, \mathbf{z}_4]$ and $\beta = (\mu, \alpha_1, \dots, \alpha_4)^\top$, so $\text{rank}(\mathbf{X}) = 4$ and $\mathcal{N}(\mathbf{X})$ is spanned by $(1, -1, -1, -1, -1)^\top$. By condition (c), $\lambda^\top \beta$ is estimable iff $\lambda_0 = \lambda_1 + \lambda_2 + \lambda_3 + \lambda_4$. The listing tests five vectors in two ways: condition (c) through the SVD of $\mathbf{X}$, and condition (e) with three generalized inverses (Moore–Penrose, the one that deletes the intercept row and column, and a non-symmetric one built as in Proposition 1.3.3(e)). The two tests always agree, and condition (e) holds for all three inverses or for none, as (f) requires.

$\lambda^\top \beta$	estimable	$\lambda^\top \mathbf{G} \lambda$:		
		Moore–Penrose	drop intercept	non-symmetric
$\mu + \alpha_1$	yes	0.250	0.250	0.250
$\alpha_1 - \alpha_2$	yes	0.417	0.417	0.417
$\alpha_1 + \alpha_2 - 2\alpha_3$	yes	1.750	1.750	1.750
α_1	no	0.188	0.250	0.630
$\alpha_1 + \alpha_2 + \alpha_3 + \alpha_4$	no	0.038	0.950	2.767

For the estimable rows the quadratic form is the same for all three and matches the direct computation: $\text{Var}(\bar{y}_1 - \bar{y}_2) = \sigma^2(\frac{1}{4} + \frac{1}{6})$, for instance. For α_1 the “variances” disagree. With the drop-intercept inverse, the solution is $\hat{\beta}_{\mathbf{G}} = (0, \bar{y}_1, \dots, \bar{y}_4)^\top$, so the reported “ $\hat{\alpha}_1$ ” is $\bar{y}_1$, which estimates $\mu + \alpha_1$. Its reported variance $0.250 \sigma^2$ is the variance of $\bar{y}_1$, which is correct, but $\bar{y}_1$ estimates $\mu + \alpha_1$, not α_1 .

```
import numpy as np

groups = np.repeat([0, 1, 2, 3], [4, 6, 3, 5])          # four groups, sizes 4, 6, 3,
5
X = np.column_stack([np.ones(groups.size), np.eye(4)[groups]]) # [1, z1, z2, z3, z4
]
A = X.T @ X

def is_estimable(lam, X, tol=None):
    """lambda is estimable iff it is orthogonal to N(X); use the SVD of X."""
    U, s, Vt = np.linalg.svd(X)
    if tol is None:
        tol = max(X.shape) * np.finfo(float).eps * s[0]
    r = int(np.sum(s > tol))
    V0 = Vt[r:].T                                     # orthonormal basis of N(X)
    return np.linalg.norm(V0.T @ lam) <= 1e-8 * max(1.0, np.linalg.norm(lam))

G_mp = np.linalg.pinv(A)                             # Moore-Penrose inverse of X^T X
G_ref = np.zeros((5, 5))
G_ref[1:, 1:] = np.linalg.inv(A[1:, 1:])              # delete the intercept row and
column
rng = np.random.default_rng(3)
G_odd = G_mp + (np.eye(5) - G_mp @ A) @ rng.normal(size=(5, 5)) # non-symmetric g-
inverse

candidates = {"mu + alpha_1": [1, 1, 0, 0, 0],
              "alpha_1 - alpha_2": [0, 1, -1, 0, 0],
              "alpha_1 + alpha_2 - 2 alpha_3": [0, 1, 1, -2, 0],
              "alpha_1": [0, 1, 0, 0, 0],
              "alpha_1 + ... + alpha_4": [0, 1, 1, 1, 1]}
for name, lam in candidates.items():
    lam = np.array(lam, dtype=float)
    H_checks = [np.allclose(lam @ G @ A, lam) for G in (G_mp, G_ref, G_odd)]
    variances = [lam @ G @ lam for G in (G_mp, G_ref, G_odd)]
    print(f"{name:30s} SVD test: {is_estimable(lam, X)!s:5s}",
          f" lambda'GX'X = lambda': {H_checks}",
          f" lambda'G lambda: {np.round(variances, 3)}")
```

8.3.4 Deciding estimability in floating point

Every condition in [Theorem 8.3.1](#) is a statement of exact linear algebra. On a computer, the rank of $\mathbf{X}$ must be decided with a tolerance, and the answer to “is λ estimable?” inherits that tolerance. The most reliable test works with $\mathbf{X}$ itself, not with $\mathbf{X}^\top \mathbf{X}$. Compute the singular value decomposition $\mathbf{X} = \mathbf{U}\mathbf{D}\mathbf{V}^\top$. Declare the singular values below a threshold τ to be zero. With the usual choice $\tau = \max(n, p) \epsilon d_1$, where d_1 is the largest singular value and $\epsilon \approx 2.2 \times 10^{-16}$ is the machine precision, the columns $\mathbf{V}_0$ of $\mathbf{V}$ that belong to the discarded singular values form an orthonormal basis of the numerical null space. Then λ is declared estimable iff $\|\mathbf{V}_0^\top \lambda\|$ is negligible relative to $\|\lambda\|$. This is condition (c), and it is what the function `is_estimable` in the listing does. [Chapter 10](#) treats the SVD and its rank decisions in detail ([Theorem 10.4.1](#)).

Two things can go wrong, and both matter.

Exact dependence can be blurred by rounding. If a column is computed, say as a sum of other columns, rounding may leave a singular value of order 10^{-15} rather than zero. The tolerance exists to absorb this.

Near dependence is not dependence. If the columns are nearly but not exactly dependent, every λ is estimable in the exact sense, but some are estimable only with enormous variance. Estimability is then a matter of degree, not a yes-or-no property.

Example. (Estimable in principle, not in practice)

Take twelve equally spaced values t_i on $[-1, 1]$ and the model matrix with columns $\mathbf{1}$, $\mathbf{t}$ and $\mathbf{t} + \delta \mathbf{w}$, where $w_i = \cos(\pi t_i)$ and δ is a small perturbation. At $\delta = 0$ the last two columns coincide. Then $\beta_1 + \beta_2$ is estimable and β_1 is not. For $\delta > 0$ the matrix has full rank, so everything is estimable. But [Figure 8.3.1](#) shows how the variance of $\hat{\beta}_1$ grows as $\delta \rightarrow 0$: in proportion to $1/\delta^2$ (the fitted slope on the log–log scale is -2.00). At $\delta = 10^{-4}$, the smallest singular value of $\mathbf{X}$ is 1.8×10^{-4} , and $\text{Var}(\hat{\beta}_1) = 1.56 \times 10^7 \sigma^2$, while $\text{Var}(\hat{\beta}_1 + \hat{\beta}_2) = 0.212 \sigma^2$ is unaffected. The function that is estimable at $\delta = 0$ stays well estimated, and the one that is not becomes useless long before it becomes nonestimable.

The listing adds a warning about computation. At $\delta = 10^{-8}$, the SVD of $\mathbf{X}$ still sees three nonzero singular values, so β_1 is (barely) estimable. But the matrix $\mathbf{X}^\top \mathbf{X}$ has eigenvalues equal to the *squared* singular values. Its smallest eigenvalue is about 3×10^{-16} , below any sensible tolerance, and `pinv(X.T @ X)` silently treats $\mathbf{X}^\top \mathbf{X}$ as having rank 2. It then reports 0.0529 as the “variance” of $\hat{\beta}_1$, which is the Moore–Penrose quadratic form of a nonestimable function. The true variance, computed from the SVD of $\mathbf{X}$ itself, is more than 10^{16} times larger. Forming $\mathbf{X}^\top \mathbf{X}$ squares the condition number, as [Section 6.10](#) showed, and here that turns a nearly nonestimable function into a numerically nonestimable one.

```
t = np.linspace(-1, 1, 12)
w = np.cos(np.pi * t)                                # a fixed direction to perturb
               along
def near_design(delta):
    """Columns 1, t and t + delta*w: exactly rank deficient at delta = 0."""
    return np.column_stack([np.ones_like(t), t, t + delta * w])

for delta in [1.0, 1e-2, 1e-4, 1e-8, 0.0]:
    Xd = near_design(delta)
```

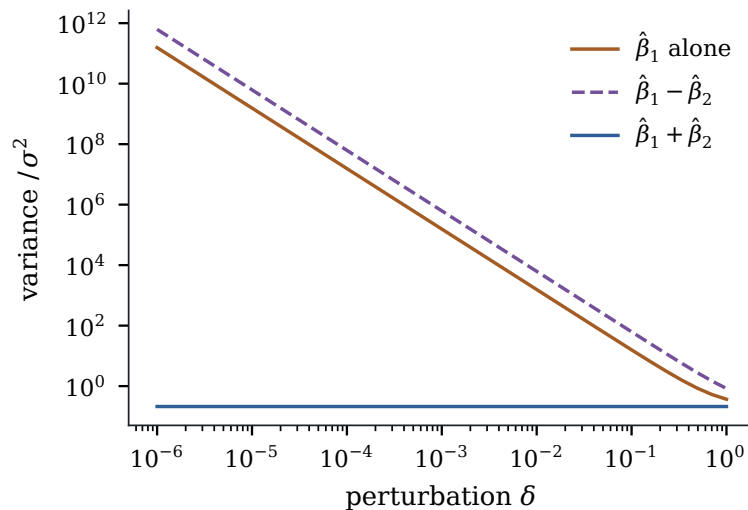


Figure 8.3.1: Variances of three linear functions of $\hat{\beta}$ in the design with columns $\mathbf{1}, \mathbf{t}, \mathbf{t} + \delta \mathbf{w}$. As $\delta \rightarrow 0$, the functions that are not estimable at $\delta = 0$ (β_1 alone and $\beta_1 - \beta_2$) have variances growing like δ^{-2} , while $\beta_1 + \beta_2$, which is estimable at $\delta = 0$, is unaffected.

```
s = np.linalg.svd(Xd, compute_uv=False)
G = np.linalg.pinv(Xd.T @ Xd)
v_sum = np.array([0, 1, 1]) @ G @ [0, 1, 1]      # beta_1 + beta_2
v_one = np.array([0, 1, 0]) @ G @ [0, 1, 0]      # beta_1 alone
print(f"delta={delta:7.0e}  smallest singular value {s[-1]:8.1e}",
      f" Var(b1+b2)/s2 = {v_sum:8.3f}  Var(b1)/s2 = {v_one:9.2e}",
      f" beta_1 estimable? {is_estimable(np.array([0., 1, 0]), Xd)}")
```

The practical rule has two parts. Decide estimability with the SVD of $\mathbf{X}$, never with $\mathbf{X}^\top \mathbf{X}$. Then look at the variance, not only the verdict. A function that passes the test with a variance of $10^7 \sigma^2$ has, for practical purposes, not been estimated. Collinearity, in Chapter 26, is the study of this middle ground.

8.3.5 Exercises

8.3.5.1 A. Check your understanding

Exercise A1.

In [Example \(Checking estimability in a one-way layout\)](#), decide which of the following are estimable, using condition (c): $2\mu + \alpha_1 + \alpha_2$; $\alpha_1 + \alpha_2 - \alpha_3 - \alpha_4$; μ ; $4\mu + \alpha_1 + \alpha_2 + \alpha_3 + \alpha_4$. For each estimable one give its least squares estimator in terms of the group means and its variance.

8.3.5.2 B. Practice

Exercise B1.

In [Example \(Checking estimability in a one-way layout\)](#), compute $\mathbf{H}_G$ for the drop-intercept generalized inverse, and identify the estimable function $\lambda^\top \mathbf{H}_G \beta$ that the soft-

ware is really estimating when it reports “ $\hat{\alpha}_k$ ”. Do the same for the generalized inverse that deletes the row and column of α_4 .

Solution. With $\mathbf{G}$ holding $\text{diag}(1/n_1, \dots, 1/n_4)$ in its lower right block, $\mathbf{GX}^\top\mathbf{X}$ has first row zero and row $k+1$ equal to $(1, \mathbf{e}_k^\top)$, because the row of $\mathbf{X}^\top\mathbf{X}$ for α_k is $n_k(1, \mathbf{e}_k^\top)$. So $E(\hat{\alpha}_k) = \mu + \alpha_k$: the reported “effect” estimates the group mean. If instead α_4 is deleted, the solution sets $\hat{\alpha}_4 = 0$, $\hat{\mu} = \bar{y}_4$ and $\hat{\alpha}_k = \bar{y}_k - \bar{y}_4$, so $E(\hat{\mu}) = \mu + \alpha_4$ and $E(\hat{\alpha}_k) = \alpha_k - \alpha_4$ for $k \leq 3$.

Exercise B2.

Let $\mathbf{\Lambda}^\top\boldsymbol{\beta}$ be estimable with $\mathbf{\Lambda}$ of full column rank q , and suppose $\text{Cov}(\mathbf{Y}) = \sigma^2\mathbf{I}$ with $\sigma^2 > 0$. Show that $\text{Cov}(\mathbf{\Lambda}^\top\hat{\boldsymbol{\beta}}) = \sigma^2\mathbf{\Lambda}^\top\mathbf{G}\mathbf{\Lambda}$ is positive definite, whichever generalized inverse is used. *Hint:* write $\mathbf{\Lambda} = \mathbf{X}^\top\mathbf{P}$ and show $\mathbf{MP}$ has full column rank.

8.3.5.3 C. Going deeper

Exercise C1.

Let $\mathbf{\Lambda}^\top\boldsymbol{\beta}$ be nonestimable. Show that as $\mathbf{G}$ ranges over all generalized inverses of $\mathbf{X}^\top\mathbf{X}$, the number $\boldsymbol{\lambda}^\top\mathbf{G}\boldsymbol{\lambda}$ takes *every* real value, including negative ones. So the “variance” that software attaches to a nonestimable function can be anything at all.

Solution. Let $\mathbf{A} = \mathbf{X}^\top\mathbf{X}$ and $\mathbf{G}_0$ be one generalized inverse. By Proposition 1.3.3(e), $\mathbf{G} = \mathbf{G}_0 + (\mathbf{I} - \mathbf{G}_0\mathbf{A})\mathbf{U}$ is a generalized inverse for every $\mathbf{U}$, and

$$\boldsymbol{\lambda}^\top\mathbf{G}\boldsymbol{\lambda} = \boldsymbol{\lambda}^\top\mathbf{G}_0\boldsymbol{\lambda} + \mathbf{u}^\top\mathbf{U}\boldsymbol{\lambda}, \quad \mathbf{u} = (\mathbf{I} - \mathbf{G}_0\mathbf{A})^\top\boldsymbol{\lambda}.$$

Because $\boldsymbol{\lambda}$ is not estimable, $\boldsymbol{\lambda}^\top\mathbf{G}_0\mathbf{A} \neq \boldsymbol{\lambda}^\top$ (by (a) $\Leftrightarrow$ (e) of Theorem 8.3.1), so $\mathbf{u} \neq \mathbf{0}$. Also $\boldsymbol{\lambda} \neq \mathbf{0}$, since $\mathbf{0}$ is in $\mathcal{C}(\mathbf{X}^\top)$. Take $\mathbf{U} = t\mathbf{u}\boldsymbol{\lambda}^\top/(\|\mathbf{u}\|^2\|\boldsymbol{\lambda}\|^2)$. Then $\boldsymbol{\lambda}^\top\mathbf{G}\boldsymbol{\lambda} = \boldsymbol{\lambda}^\top\mathbf{G}_0\boldsymbol{\lambda} + t$, which takes every real value as t varies.

Exercise C2.

Let $\mathbf{P}$ be the orthogonal projection onto $\mathcal{C}(\mathbf{X}^\top)$. Show that the minimum-norm least squares solution $\hat{\boldsymbol{\beta}}^+$ satisfies $E(\hat{\boldsymbol{\beta}}^+) = \mathbf{P}\boldsymbol{\beta}$, and that for any $\boldsymbol{\lambda}$, $\boldsymbol{\lambda}^\top\hat{\boldsymbol{\beta}}^+$ is the least squares estimator of the estimable function $(\mathbf{P}\boldsymbol{\lambda})^\top\boldsymbol{\beta}$. Interpret $\mathbf{P}\boldsymbol{\lambda}$ as the estimable function “closest” to $\boldsymbol{\lambda}^\top\boldsymbol{\beta}$.

8.4 Reparameterization and side conditions

A rank-deficient model can be handled in three ways. One can work with it as it stands, estimating only estimable functions, which is what the last two sections did. One can *reparameterize*, replacing $\mathbf{X}$ by a full-rank matrix with the same column space. Or one can impose *side conditions*, linear restrictions on $\boldsymbol{\beta}$ that select a single coefficient vector from each set $\boldsymbol{\beta} + \mathcal{N}(\mathbf{X})$. This section shows that the second and third are two descriptions of the same act: choosing

coordinates for the mean vector. Neither changes any estimable quantity. It also records what the most common software does by default.

8.4.1 Full-rank reparameterizations

By [Corollary 8.2.3](#), at most r estimable functions are linearly independent. A set of exactly r of them can serve as a new parameter vector.

Theorem 8.4.1 — Reparameterizing to full rank

Let $\text{rank}(\mathbf{X}) = r$. Let $\mathbf{U}$ be an $r \times p$ matrix of rank r whose rows lie in $\mathcal{C}(\mathbf{X}^\top)$, so that $\boldsymbol{\gamma} = \mathbf{U}\boldsymbol{\beta}$ is a vector of r linearly independent estimable functions, and put

$$\mathbf{Z} = \mathbf{X}\mathbf{U}^\top(\mathbf{U}\mathbf{U}^\top)^{-1}. \quad (8.4.1)$$

Then:

- (a) $\mathbf{X} = \mathbf{Z}\mathbf{U}$, $\mathbf{Z}$ has full column rank r , and $\mathcal{C}(\mathbf{Z}) = \mathcal{C}(\mathbf{X})$;
- (b) the models $E(\mathbf{Y}) = \mathbf{X}\boldsymbol{\beta}$, $\boldsymbol{\beta} \in \mathbb{R}^p$, and $E(\mathbf{Y}) = \mathbf{Z}\boldsymbol{\gamma}$, $\boldsymbol{\gamma} \in \mathbb{R}^r$, describe the same set of mean vectors, with $\boldsymbol{\gamma} = \mathbf{U}\boldsymbol{\beta}$;
- (c) the least squares estimate in the new model is $\hat{\boldsymbol{\gamma}} = (\mathbf{Z}^\top\mathbf{Z})^{-1}\mathbf{Z}^\top\mathbf{y} = \mathbf{U}\hat{\boldsymbol{\beta}}$ for every least squares solution $\hat{\boldsymbol{\beta}}$ of the old one;
- (d) $\boldsymbol{\lambda}^\top\boldsymbol{\beta}$ is estimable iff $\boldsymbol{\lambda} = \mathbf{U}^\top\mathbf{c}$ for some $\mathbf{c} \in \mathbb{R}^r$, which is then unique, and in that case $\boldsymbol{\lambda}^\top\hat{\boldsymbol{\beta}} = \mathbf{c}^\top\hat{\boldsymbol{\gamma}}$;
- (e) conversely, if $\mathbf{Z}$ is any $n \times r$ matrix of full column rank with $\mathcal{C}(\mathbf{Z}) = \mathcal{C}(\mathbf{X})$, then $\mathbf{X} = \mathbf{Z}\mathbf{U}$ for a unique $\mathbf{U}$, and $\mathbf{U}$ has rank r and estimable rows.

Proof. Since $\mathcal{C}(\mathbf{U}^\top) \subseteq \mathcal{C}(\mathbf{X}^\top)$ and both have dimension r , $\mathcal{C}(\mathbf{U}^\top) = \mathcal{C}(\mathbf{X}^\top)$. The matrix $\mathbf{Q} = \mathbf{U}^\top(\mathbf{U}\mathbf{U}^\top)^{-1}\mathbf{U}$ is symmetric and idempotent with column space $\mathcal{C}(\mathbf{U}^\top)$, so it is the orthogonal projection onto $\mathcal{C}(\mathbf{X}^\top)$ ([Theorem 6.3.3](#)). Hence $\mathbf{I} - \mathbf{Q}$ projects onto $\mathcal{C}(\mathbf{X}^\top)^\perp = \mathcal{N}(\mathbf{X})$, and $\mathbf{X}(\mathbf{I} - \mathbf{Q}) = \mathbf{0}$, that is, $\mathbf{X} = \mathbf{X}\mathbf{Q} = \mathbf{Z}\mathbf{U}$. (a) Now $r = \text{rank}(\mathbf{X}) = \text{rank}(\mathbf{Z}\mathbf{U}) \leq \text{rank}(\mathbf{Z}) \leq r$, so $\mathbf{Z}$ has full column rank. $\mathcal{C}(\mathbf{Z}) \subseteq \mathcal{C}(\mathbf{X})$ because $\mathbf{Z} = \mathbf{X}(\cdot)$, and $\mathcal{C}(\mathbf{X}) \subseteq \mathcal{C}(\mathbf{Z})$ because $\mathbf{X} = \mathbf{Z}\mathbf{U}$. (b) $\mathbf{X}\boldsymbol{\beta} = \mathbf{Z}(\mathbf{U}\boldsymbol{\beta})$, and $\boldsymbol{\beta} \mapsto \mathbf{U}\boldsymbol{\beta}$ maps $\mathbb{R}^p$ onto $\mathbb{R}^r$ because $\mathbf{U}$ has full row rank. So both models describe $\mathcal{C}(\mathbf{X})$. (c) $\mathbf{Z}\hat{\boldsymbol{\gamma}} = \mathbf{M}\mathbf{y} = \mathbf{X}\hat{\boldsymbol{\beta}} = \mathbf{Z}(\mathbf{U}\hat{\boldsymbol{\beta}})$ by [Theorem 6.4.1](#), and $\mathbf{Z}$ has full column rank. (d) $\boldsymbol{\lambda}$ is estimable iff $\boldsymbol{\lambda} \in \mathcal{C}(\mathbf{X}^\top) = \mathcal{C}(\mathbf{U}^\top)$. The vector $\mathbf{c}$ is unique because $\mathbf{U}^\top$ has full column rank, and $\boldsymbol{\lambda}^\top\hat{\boldsymbol{\beta}} = \mathbf{c}^\top\mathbf{U}\hat{\boldsymbol{\beta}} = \mathbf{c}^\top\hat{\boldsymbol{\gamma}}$. (e) Each column of $\mathbf{X}$ lies in $\mathcal{C}(\mathbf{Z})$, so $\mathbf{X} = \mathbf{Z}\mathbf{U}$ with $\mathbf{U} = (\mathbf{Z}^\top\mathbf{Z})^{-1}\mathbf{Z}^\top\mathbf{X}$, unique because $\mathbf{Z}$ has full column rank. Then $\mathbf{U}\boldsymbol{\beta} = (\mathbf{Z}^\top\mathbf{Z})^{-1}\mathbf{Z}^\top(\mathbf{X}\boldsymbol{\beta})$ is a linear function of $\mathbf{X}\boldsymbol{\beta}$, so each row of $\mathbf{U}$ is estimable. Finally $r = \text{rank}(\mathbf{X}) \leq \text{rank}(\mathbf{U}) \leq r$. ■

So reparameterization is the same thing as choosing a basis for the estimable space. [Theorem 6.7.1](#) of [Chapter 6](#) already showed that fitted values and residuals do not care which basis is chosen. Part (d) adds that estimable functions do not care either: each is a fixed linear combination of the new parameters, and its estimate is the same combination of their estimates. Everything in the full-rank theory of [Chapter 5](#) now applies to $\hat{\boldsymbol{\gamma}}$, including [Theorem 5.5.1](#) and

the variance estimator of [Theorem 5.6.1](#), with p replaced by r .

For the one-way layout with $\boldsymbol{\beta} = (\mu, \alpha_1, \dots, \alpha_g)^\top$, two choices of $\mathbf{U}$ stand out. The rows $(1, \mathbf{e}_k^\top)$, $k = 1, \dots, g$, give $\boldsymbol{\gamma} = (\mu + \alpha_1, \dots, \mu + \alpha_g)^\top$, the cell means, and $\mathbf{Z}$ is the indicator matrix. The rows $(1, \mathbf{e}_1^\top)$ and $(0, \mathbf{e}_k^\top - \mathbf{e}_1^\top)$, $k = 2, \dots, g$, give the first group's mean and the differences from it. This is the reference coding of [Section 8.5](#).

8.4.2 Side conditions

The other classical device leaves $\mathbf{X}$ alone and adds equations. A **side condition** is a set of linear restrictions $\mathbf{T}\mathbf{b} = \mathbf{0}$, with $\mathbf{T}$ a $q \times p$ matrix, imposed on the solutions of the normal equations. The aim is to single out one solution. Examples in the one-way layout are $\alpha_1 = 0$, $\sum_k \alpha_k = 0$ and $\mu = 0$. The theorem says exactly which restrictions achieve this.

Theorem 8.4.2 — Side conditions

Let $\mathbf{X}$ be $n \times p$ of rank r and $\mathbf{T}$ be $q \times p$. The following are equivalent:

- (a) for every $\mathbf{y} \in \mathbb{R}^n$ there is exactly one $\mathbf{b}$ with $\mathbf{X}^\top \mathbf{X}\mathbf{b} = \mathbf{X}^\top \mathbf{y}$ and $\mathbf{T}\mathbf{b} = \mathbf{0}$;
- (b) for every $\boldsymbol{\mu} \in \mathcal{C}(\mathbf{X})$ there is exactly one $\mathbf{b}$ with $\mathbf{X}\mathbf{b} = \boldsymbol{\mu}$ and $\mathbf{T}\mathbf{b} = \mathbf{0}$;
- (c) $\mathbb{R}^p = \mathcal{N}(\mathbf{X}) \oplus \mathcal{N}(\mathbf{T})$;
- (d) $\text{rank} \begin{pmatrix} \mathbf{X} \\ \mathbf{T} \end{pmatrix} = p$ and $\mathcal{C}(\mathbf{T}^\top) \cap \mathcal{C}(\mathbf{X}^\top) = \{\mathbf{0}\}$;
- (e) $\text{rank}(\mathbf{T}) = p - r$, and no nonzero linear combination of the rows of $\mathbf{T}$ is estimable.

When these hold, $\mathbf{X}^\top \mathbf{X} + \mathbf{T}^\top \mathbf{T}$ is nonsingular, the solution in (a) is

$$\hat{\boldsymbol{\beta}}_{\mathbf{T}} = (\mathbf{X}^\top \mathbf{X} + \mathbf{T}^\top \mathbf{T})^{-1} \mathbf{X}^\top \mathbf{y}, \quad (8.4.2)$$

and $(\mathbf{X}^\top \mathbf{X} + \mathbf{T}^\top \mathbf{T})^{-1}$ is a generalized inverse of $\mathbf{X}^\top \mathbf{X}$.

Proof. (a) $\Leftrightarrow$ (b). The solutions of the normal equations for $\mathbf{y}$ are the $\mathbf{b}$ with $\mathbf{X}\mathbf{b} = \mathbf{M}\mathbf{y}$ ([Theorem 6.4.2](#)), and $\mathbf{M}\mathbf{y}$ runs over all of $\mathcal{C}(\mathbf{X})$ as $\mathbf{y}$ does.

(b) $\Leftrightarrow$ (c). The set $\{\mathbf{b} : \mathbf{X}\mathbf{b} = \boldsymbol{\mu}\}$ is a coset $\mathbf{b}_0 + \mathcal{N}(\mathbf{X})$, and every $\mathbf{b}_0 \in \mathbb{R}^p$ arises (take $\boldsymbol{\mu} = \mathbf{X}\mathbf{b}_0$). So (b) says that every coset of $\mathcal{N}(\mathbf{X})$ meets $\mathcal{N}(\mathbf{T})$ in exactly one point. Some point $\mathbf{b}_0 + \mathbf{v}$ of the coset lies in $\mathcal{N}(\mathbf{T})$ iff $\mathbf{b}_0 \in \mathcal{N}(\mathbf{T}) + \mathcal{N}(\mathbf{X})$, so existence for every coset means $\mathbb{R}^p = \mathcal{N}(\mathbf{X}) + \mathcal{N}(\mathbf{T})$. Two points of the same coset that both lie in $\mathcal{N}(\mathbf{T})$ differ by an element of $\mathcal{N}(\mathbf{X}) \cap \mathcal{N}(\mathbf{T})$, and a nonzero element of that intersection added to one such point gives a second. So uniqueness means $\mathcal{N}(\mathbf{X}) \cap \mathcal{N}(\mathbf{T}) = \{\mathbf{0}\}$. Together, these say the sum is direct.

(c) $\Leftrightarrow$ (d). By [Lemma 6.2.1](#) and [Lemma 6.2.5](#),

$$(\mathcal{N}(\mathbf{X}) + \mathcal{N}(\mathbf{T}))^\perp = \mathcal{C}(\mathbf{X}^\top) \cap \mathcal{C}(\mathbf{T}^\top), \quad (\mathcal{N}(\mathbf{X}) \cap \mathcal{N}(\mathbf{T}))^\perp = \mathcal{C}(\mathbf{X}^\top) + \mathcal{C}(\mathbf{T}^\top).$$

So $\mathcal{N}(\mathbf{X}) + \mathcal{N}(\mathbf{T}) = \mathbb{R}^p$ iff $\mathcal{C}(\mathbf{X}^\top) \cap \mathcal{C}(\mathbf{T}^\top) = \{\mathbf{0}\}$, and $\mathcal{N}(\mathbf{X}) \cap \mathcal{N}(\mathbf{T}) = \{\mathbf{0}\}$ iff $\mathcal{C}(\mathbf{X}^\top) + \mathcal{C}(\mathbf{T}^\top) = \mathbb{R}^p$. The last space is the row space of the stacked matrix, so this holds iff that matrix has rank p .

(d) $\Leftrightarrow$ (e). A vector in $\mathcal{C}(\mathbf{T}^\top)$ is a combination of the rows of $\mathbf{T}$, and it is estimable iff it lies in $\mathcal{C}(\mathbf{X}^\top)$. So the second halves of (d) and (e) say the same thing. When they hold, the sum $\mathcal{C}(\mathbf{X}^\top) + \mathcal{C}(\mathbf{T}^\top)$ is direct, with dimension $r + \text{rank}(\mathbf{T})$. This equals p iff $\text{rank}(\mathbf{T}) = p - r$.

For the formula, suppose (d). If $(\mathbf{X}^\top \mathbf{X} + \mathbf{T}^\top \mathbf{T})\mathbf{v} = \mathbf{0}$, then $0 = \mathbf{v}^\top \mathbf{X}^\top \mathbf{X} \mathbf{v} + \mathbf{v}^\top \mathbf{T}^\top \mathbf{T} \mathbf{v} = \|\mathbf{X} \mathbf{v}\|^2 + \|\mathbf{T} \mathbf{v}\|^2$, so $\mathbf{v}$ lies in $\mathcal{N}(\mathbf{X}) \cap \mathcal{N}(\mathbf{T}) = \{\mathbf{0}\}$. Hence the matrix is nonsingular. The solution $\mathbf{b}$ of (a) satisfies $(\mathbf{X}^\top \mathbf{X} + \mathbf{T}^\top \mathbf{T})\mathbf{b} = \mathbf{X}^\top \mathbf{y} + \mathbf{T}^\top \mathbf{0}$, which gives Eq. 8.4.2. Finally, write $\mathbf{G}_\mathbf{T} = (\mathbf{X}^\top \mathbf{X} + \mathbf{T}^\top \mathbf{T})^{-1}$. For every $\mathbf{y}$, $\mathbf{G}_\mathbf{T} \mathbf{X}^\top \mathbf{y}$ solves the normal equations, so $\mathbf{X}^\top \mathbf{X} \mathbf{G}_\mathbf{T} \mathbf{X}^\top = \mathbf{X}^\top$. Multiplying on the right by $\mathbf{X}$ gives $\mathbf{X}^\top \mathbf{X} \mathbf{G}_\mathbf{T} \mathbf{X}^\top \mathbf{X} = \mathbf{X}^\top \mathbf{X}$. ■

Condition (e) is the familiar rule of thumb, made exact: a side condition must supply exactly as many restrictions as the rank deficiency, and none of them, singly or in combination, may be estimable. The reason for the second requirement is clear from (d). An estimable restriction constrains the mean vector itself, and so it changes the model instead of merely choosing coordinates.

Remark (Existence and uniqueness separately). The proof shows more than the theorem states. The condition $\mathcal{C}(\mathbf{T}^\top) \cap \mathcal{C}(\mathbf{X}^\top) = \{\mathbf{0}\}$ alone guarantees that the restricted normal equations always have *at least* one solution. The condition $\text{rank}\left(\begin{smallmatrix} \mathbf{X} \\ \mathbf{T} \end{smallmatrix}\right) = p$ alone guarantees *at most* one. A constraint that is too weak, such as a single restriction when $p - r = 2$, leaves many solutions. One with an estimable part leaves none for most $\mathbf{y}$ (Exercise (C2)).

8.4.3 What the constrained coefficients mean

Under a side condition every coefficient becomes identifiable, and it acquires a meaning.

Proposition 8.4.3 — Constrained coefficients are estimable functions

Suppose $\mathbf{T}$ satisfies the conditions of Theorem 8.4.2, and let $\mathbf{P}_\mathbf{T} = (\mathbf{X}^\top \mathbf{X} + \mathbf{T}^\top \mathbf{T})^{-1} \mathbf{X}^\top \mathbf{X}$. For each $\boldsymbol{\beta}$, $\mathbf{P}_\mathbf{T} \boldsymbol{\beta}$ is the unique vector $\mathbf{b}$ with $\mathbf{X} \mathbf{b} = \mathbf{X} \boldsymbol{\beta}$ and $\mathbf{T} \mathbf{b} = \mathbf{0}$. The matrix $\mathbf{P}_\mathbf{T}$ is idempotent, with range $\mathcal{N}(\mathbf{T})$ and null space $\mathcal{N}(\mathbf{X})$. Every row of $\mathbf{P}_\mathbf{T}$ lies in $\mathcal{C}(\mathbf{X}^\top)$. So each constrained coefficient $\mathbf{e}_j^\top \mathbf{P}_\mathbf{T} \boldsymbol{\beta}$ is an estimable function of the original parameters, and $\hat{\boldsymbol{\beta}}_\mathbf{T}$ estimates it without bias.

Proof. With $\mathbf{y} = \mathbf{X} \boldsymbol{\beta}$, the vector $\mathbf{P}_\mathbf{T} \boldsymbol{\beta} = \mathbf{G}_\mathbf{T} \mathbf{X}^\top (\mathbf{X} \boldsymbol{\beta})$ is the unique solution of Theorem 8.4.2(a) for that $\mathbf{y}$. It satisfies $\mathbf{X} \mathbf{b} = \mathbf{M} \mathbf{X} \boldsymbol{\beta} = \mathbf{X} \boldsymbol{\beta}$ and $\mathbf{T} \mathbf{b} = \mathbf{0}$. The matrix $\mathbf{P}_\mathbf{T}$ is $\mathbf{H}_\mathbf{G}$ of Eq. 8.3.1 for the generalized inverse $\mathbf{G}_\mathbf{T}$, so Proposition 8.3.3 gives idempotence, the null space $\mathcal{N}(\mathbf{X})$ and rows in $\mathcal{C}(\mathbf{X}^\top)$. Its range lies in $\mathcal{N}(\mathbf{T})$. The range has dimension r , and so does $\mathcal{N}(\mathbf{T})$, because $\text{rank}(\mathbf{T}) = p - r$. So the two coincide. Unbiasedness is Proposition 8.3.3(c). ■

In words, $\mathbf{P}_\mathbf{T}$ is the projection of $\mathbb{R}^p$ onto $\mathcal{N}(\mathbf{T})$ along $\mathcal{N}(\mathbf{X})$, generally an oblique one. It sends

the true β to the unique representative of $\beta + \mathcal{N}(\mathbf{X})$ that satisfies the side condition. That representative's coordinates are functions of the mean, so they can be estimated. Their *meaning* is fixed by $\mathbf{T}$. In the one-way layout, the condition $\sum_k \alpha_k = 0$ makes μ the unweighted average of the g cell means and α_k the deviation of cell k from it. The condition $\sum_k n_k \alpha_k = 0$ makes μ the size-weighted average instead. The condition $\alpha_1 = 0$ makes μ the first cell mean. The data do not choose among these. The analyst chooses, by deciding which question the coefficients should answer.

One side condition deserves special mention, because it is the one many programs impose without saying so.

Proposition 8.4.4 — Minimum norm as a side condition

Let the columns of the $p \times (p-r)$ matrix $\mathbf{V}_0$ be a basis of $\mathcal{N}(\mathbf{X})$. Then $\mathbf{T} = \mathbf{V}_0^\top$ satisfies the conditions of [Theorem 8.4.2](#), and the resulting solution $\hat{\beta}_{\mathbf{T}}$ is the minimum-norm least squares solution $\mathbf{X}^+\mathbf{y}$.

Proof. $\mathcal{C}(\mathbf{T}^\top) = \mathcal{C}(\mathbf{V}_0) = \mathcal{N}(\mathbf{X})$, which has dimension $p-r$ and meets $\mathcal{C}(\mathbf{X}^\top) = \mathcal{N}(\mathbf{X})^\perp$ only in $\mathbf{0}$. So (d) holds. The least squares solutions are $\hat{\beta}_{\mathbf{T}} + \mathcal{N}(\mathbf{X})$, and $\hat{\beta}_{\mathbf{T}} \in \mathcal{N}(\mathbf{T}) = \mathcal{C}(\mathbf{V}_0)^\perp = \mathcal{N}(\mathbf{X})^\perp$. For $\mathbf{v} \in \mathcal{N}(\mathbf{X})$, Pythagoras gives $\|\hat{\beta}_{\mathbf{T}} + \mathbf{v}\|^2 = \|\hat{\beta}_{\mathbf{T}}\|^2 + \|\mathbf{v}\|^2$, so $\hat{\beta}_{\mathbf{T}}$ has the smallest norm. By [Proposition 6.10.1](#) this is $\mathbf{X}^+\mathbf{y}$. ■

In the one-way layout the null space is spanned by $(1, -1, \dots, -1)^\top$, so the minimum-norm solution is the one satisfying $\mu = \alpha_1 + \dots + \alpha_g$. No one would choose this restriction on purpose, and its coefficients have no natural interpretation.

Example. (Five side conditions, one fit)

Eighteen simulated observations fall into four groups of sizes 5, 3, 6, 4, with group means 11.111, 15.204, 10.558 and 13.624. In the model $E(y_{kj}) = \mu + \alpha_k$, $p-r=1$, so a side condition is a single nonestimable restriction. The listing solves [Eq. 8.4.2](#) for five of them and computes the minimum-norm solution.

Side condition	$\hat{\mu}$	$\hat{\alpha}_1$	$\hat{\alpha}_2$	$\hat{\alpha}_3$	$\hat{\alpha}_4$
$\alpha_1 = 0$	11.111	0.000	4.093	-0.553	2.513
$\alpha_4 = 0$	13.624	-2.513	1.580	-3.066	0.000
$\sum_k \alpha_k = 0$	12.624	-1.513	2.580	-2.066	1.000
$\sum_k n_k \alpha_k = 0$	12.167	-1.056	3.037	-1.609	1.457
$\mu = 0$	0.000	11.111	15.204	10.558	13.624
minimum norm ($\mu = \sum_k \alpha_k$)	10.100	1.012	5.105	0.459	3.525

Every row solves the normal equations. In every row $\hat{\mu} + \hat{\alpha}_k$ is the k th group mean, and $\hat{\alpha}_1 - \hat{\alpha}_2$ is the same. The value of $\hat{\mu}$ is, in turn, the first group mean, the last group mean, the average of the four group means (12.624), the overall mean $\bar{y} = 12.167$, zero, and an uninterpretable 10.100.

Compare a restriction that *is* estimable, $\alpha_1 = \alpha_2$. It is not a side condition. Imposing it (by solving the Lagrange system for least squares under a linear restriction) forces the first two group means to be equal. The residual sum of squares rises from 14.665 to 46.078. The fit has changed, which is what testing the hypothesis $\alpha_1 = \alpha_2$ in [Chapter 11](#) measures.

```
import numpy as np

sizes = [5, 3, 6, 4]
groups = np.repeat(np.arange(4), sizes)
rng = np.random.default_rng(84)
y = np.array([12.0, 15.0, 11.0, 14.0])[groups] + rng.normal(scale=1.0, size=groups.size)
X = np.column_stack([np.ones(groups.size), np.eye(4)[groups]]) # (mu, alpha_1..
                      alpha_4)
A, c = X.T @ X, X.T @ y
```

```
n_k = np.array(sizes, dtype=float)
side = {"alpha_1 = 0": [0, 1, 0, 0, 0],
        "alpha_4 = 0": [0, 0, 0, 0, 1],
        "sum alpha_k = 0": [0, 1, 1, 1, 1],
        "sum n_k alpha_k = 0": [0, *n_k],
        "mu = 0": [1, 0, 0, 0, 0]}
solutions = {}
for name, t in side.items():
    T = np.array([t], dtype=float)
    G = np.linalg.inv(A + T.T @ T) # (X'X + T'T)^{-1}: a g-inverse of X'X
    b = G @ c
    solutions[name] = b
    print(f"{name:20s} b = {np.round(b, 3)}   T b = {(T @ b)[0]:.1e}")
b_mn = np.linalg.pinv(X) @ y
print(f"{'minimum norm':20s} b = {np.round(b_mn, 3)}",
      f"   mu - sum(alpha) = {b_mn[0] - b_mn[1:].sum():.1e}")
```

```
# "alpha_1 = alpha_2" is a restriction on an estimable function: it changes the
  model
T_bad = np.array([[0, 1, -1, 0, 0]], dtype=float)
K = np.block([[A, T_bad.T], [T_bad, np.zeros((1, 1))]]) # Lagrange (bordered)
                      system
b_bad = np.linalg.lstsq(K, np.append(c, 0.0), rcond=None)[0][:5]
sse_free = np.sum((y - X @ solutions["alpha_1 = 0"]) ** 2)
sse_bad = np.sum((y - X @ b_bad) ** 2)
print(f"SSE without restriction {sse_free:.3f};   with alpha_1 = alpha_2 {sse_bad:.3f}
      ")
```

Reparameterization and side conditions are two languages for the same choice. Deleting the column of α_1 from $\mathbf{X}$ and fitting the full-rank remainder gives exactly the solution with $\alpha_1 = 0$. More generally, deleting a set of columns whose removal leaves a full-rank matrix with the same column space is the side condition that sets those coefficients to zero, and it satisfies [Theorem 8.4.2\(c\)](#) ([Exercise \(B1\)](#)).

8.4.4 What software does

Programs treat a rank-deficient $\mathbf{X}$ differently.

- *Code the factor at full rank before fitting.* Formula interfaces such as R's `lm()` and statsmodels' formula API expand a factor with g levels into an intercept and $g - 1$ columns, by default the reference coding of [Section 8.5](#). The overparameterized matrix never appears. This is a reparameterization in the sense of [Theorem 8.4.1](#).
- *Drop columns found to be dependent.* If the columns supplied are dependent anyway, R's `lm()` computes a QR decomposition in which a column found to be (numerically) a combination of earlier ones is moved to the end and left out. Its coefficient is reported as NA. This is the side condition setting that coefficient to zero, and which column is dropped depends on the order of the columns. SAS's GLM procedure similarly solves the normal equations with a generalized inverse that zeroes the coefficients of dependent columns, and it flags the resulting estimates as not unique.
- *Return the minimum-norm solution.* `numpy.linalg.lstsq` returns $\mathbf{X}^+\mathbf{y}$ silently (it reports the numerical rank only as a second output), and statsmodels' OLS, whose `fit` uses the pseudoinverse by default, returns the same vector with a `SingularMatrixWarning`. This is the side condition of [Proposition 8.4.4](#), and it does not depend on the order of the columns.
- *Pivot greedily.* A QR decomposition with column pivoting ([Chapter 10](#), [Theorem 10.7.2](#)) chooses at each step the remaining column with the largest component orthogonal to those already chosen, and the columns left at the end are dropped. In [Example \(Five side conditions, one fit\)](#) it takes the intercept, then α_3 (the largest group), α_1 and α_4 , and drops α_2 , the smallest group, a choice that depends on the group sizes.

```
import scipy.linalg
import statsmodels.api as sm
b_sm = sm.OLS(y, X).fit().params          # statsmodels default: method="pinv"
"
b_np = np.linalg.lstsq(X, y, rcond=None)[0]  # LAPACK gelsd: minimum-norm
solution
_, R, piv = scipy.linalg.qr(X, mode="economic", pivoting=True)
print("statsmodels:", np.round(b_sm, 3))
print("numpy lstsq:", np.round(b_np, 3))
print("pivoted QR column order:", piv, " |R_kk| =", np.round(np.abs(np.diag(R)), 3))
```

The lesson of this section is that all these outputs are correct, and none is more correct than the others. They are different coordinate systems for one fitted mean vector. Any conclusion that changes when the software changes is a conclusion about a nonestimable function.

Warning. Before interpreting a coefficient from a rank-deficient fit, find out which side condition produced it. A coefficient reported as “the effect of level 3” may be a difference from level 1, a deviation from an average of cell means, a deviation from the overall mean, or, under the minimum-norm convention, a quantity with no interpretation at all.

8.4.5 Exercises

8.4.5.1 A. Check your understanding

Exercise A1.

In the one-way model with four groups and $\boldsymbol{\beta} = (\mu, \alpha_1, \dots, \alpha_4)^\top$, decide which of the following are side conditions in the sense of [Theorem 8.4.2](#): (i) $\alpha_2 = 0$; (ii) $\mu = \alpha_3$; (iii) $\alpha_1 - \alpha_2 = 0$; (iv) $\alpha_1 = 0$ together with $\alpha_2 = 0$; (v) $2\mu + \alpha_1 + \alpha_2 = 0$.

8.4.5.2 B. Practice**Exercise B1.**

Let $\mathbf{X} = [\mathbf{X}_1, \mathbf{X}_2]$, where $\mathbf{X}_1$ has full column rank and $\mathcal{C}(\mathbf{X}_1) = \mathcal{C}(\mathbf{X})$. Show that the least squares fit using $\mathbf{X}_1$ alone, padded with zeros for the coefficients of $\mathbf{X}_2$, is the solution of the side condition $\mathbf{T} = [\mathbf{0}, \mathbf{I}]$. Check the conditions of [Theorem 8.4.2](#) directly.

Solution. The padded vector $(\hat{\boldsymbol{\beta}}_1^\top, \mathbf{0}^\top)^\top$ satisfies $\mathbf{X}(\hat{\boldsymbol{\beta}}_1^\top, \mathbf{0}^\top)^\top = \mathbf{X}_1 \hat{\boldsymbol{\beta}}_1 = \mathbf{M}\mathbf{y}$, because $\mathcal{C}(\mathbf{X}_1) = \mathcal{C}(\mathbf{X})$. So it solves the normal equations and satisfies $\mathbf{T}\mathbf{b} = \mathbf{0}$. For the conditions: $\text{rank}(\mathbf{T}) = p_2$, and $\text{rank}(\mathbf{X}) = \text{rank}(\mathbf{X}_1) = p_1$, so $\text{rank}(\mathbf{T}) = p - r$. If $\mathbf{T}^\top \mathbf{c} = (\mathbf{0}^\top, \mathbf{c}^\top)^\top$ were in $\mathcal{C}(\mathbf{X}^\top)$, say $\mathbf{X}_1^\top \boldsymbol{\rho} = \mathbf{0}$ and $\mathbf{X}_2^\top \boldsymbol{\rho} = \mathbf{c}$, then $\boldsymbol{\rho} \perp \mathcal{C}(\mathbf{X}_1) = \mathcal{C}(\mathbf{X}) \supseteq \mathcal{C}(\mathbf{X}_2)$, so $\mathbf{c} = \mathbf{0}$. Condition (e) holds.

Exercise B2.

For the additive model $E(y_{ij}) = \mu + \alpha_i + \beta_j$ with $i = 1, 2, j = 1, 2, 3$ and one observation per cell, take $\mathbf{U}$ with rows giving $\mu + \alpha_1 + \beta_1, \alpha_2 - \alpha_1, \beta_2 - \beta_1$ and $\beta_3 - \beta_1$. Compute $\mathbf{Z}$ from [Eq. 8.4.1](#), and check that it is the reference-coded model matrix.

Exercise B3.

Let $\mathbf{T}$ satisfy the conditions of [Theorem 8.4.2](#) and have full row rank $p - r$. Show that the bordered matrix $\begin{pmatrix} \mathbf{X}^\top \mathbf{X} & \mathbf{T}^\top \\ \mathbf{T} & \mathbf{0} \end{pmatrix}$ is nonsingular, that the solution of the bordered system with right-hand side $(\mathbf{X}^\top \mathbf{y}, \mathbf{0})$ has first block $\hat{\boldsymbol{\beta}}_{\mathbf{T}}$ and second block $\mathbf{0}$, and interpret the zero Lagrange multiplier. Relate this to [Exercise \(B2\)](#).

8.4.5.3 C. Going deeper**Exercise C1.**

Replace $\mathbf{X}$ by $\mathbf{X}\mathbf{D}$ for a nonsingular diagonal $\mathbf{D}$ (a change of units). Show that the minimum-norm solution for $\mathbf{X}\mathbf{D}$, converted back to the original units, is the solution for the side condition $\mathbf{V}_0^\top \mathbf{D}^{-2} \mathbf{b} = \mathbf{0}$, which differs in general from $\mathbf{V}_0^\top \mathbf{b} = \mathbf{0}$. Conclude that the minimum-norm convention depends on the units of measurement, while no estimable function does.

Exercise C2.

Suppose $\text{rank}(\mathbf{T}) = p - r$ but $\mathcal{C}(\mathbf{T}^\top) \cap \mathcal{C}(\mathbf{X}^\top) \neq \{\mathbf{0}\}$. Show that for some $\mathbf{y}$ the restricted normal equations of [Theorem 8.4.2\(a\)](#) have no solution, and that whenever they have

one they have infinitely many.

Solution. By the dimension count, $\mathcal{C}(\mathbf{X}^\top) + \mathcal{C}(\mathbf{T}^\top)$ has dimension less than $r + (p-r) = p$. So $\mathcal{N}(\mathbf{X}) \cap \mathcal{N}(\mathbf{T}) \neq \{\mathbf{0}\}$, and any solution can be moved along a nonzero vector of that intersection. For existence, the proof of [Theorem 8.4.2](#) shows that a solution exists for every $\mathbf{y}$ iff $\mathcal{N}(\mathbf{X}) + \mathcal{N}(\mathbf{T}) = \mathbb{R}^p$, which fails because its orthogonal complement $\mathcal{C}(\mathbf{X}^\top) \cap \mathcal{C}(\mathbf{T}^\top)$ is nonzero. So some coset $\mathbf{b}_0 + \mathcal{N}(\mathbf{X})$ misses $\mathcal{N}(\mathbf{T})$, and the response $\mathbf{y} = \mathbf{X}\mathbf{b}_0$ has no solution.

8.5 Indicator variables and the coding of a factor

Most regressors in practice are not measurements on a continuous scale. They record a category: treatment arm, region, occupation, party, a yes-or-no answer. Such a variable is a **factor**, and its possible values are its **levels**. A factor enters a linear model through *indicator variables* (also called dummy variables), columns of zeros and ones that mark membership of a level. There are many ways to build these columns, and they give quite different coefficient tables. This section shows that they all describe the same fitted model. Each choice of coding is a choice of which linear functions of the level means the coefficients report. It gives a single formula that says exactly which ones.

8.5.1 One binary regressor

Suppose each observation belongs to one of two groups, and let $d_i = 1$ if observation i is in group 1 and $d_i = 0$ if it is in group 0. The model

$$E(y_i) = \beta_0 + \beta_1 d_i$$

says that the mean is $\mu_0 = \beta_0$ in group 0 and $\mu_1 = \beta_0 + \beta_1$ in group 1. So $\beta_1 = \mu_1 - \mu_0$ is the difference between the two group means. The least squares fit sets the fitted value in each group to that group's mean (the model space is the set of vectors constant within groups, [Example \(The one-way layout\)](#)), so $\hat{\beta}_0 = \bar{y}_0$ and $\hat{\beta}_1 = \bar{y}_1 - \bar{y}_0$. The t statistic for $\beta_1 = 0$ is the pooled two-sample t statistic ([Exercise \(A1\)](#)). The familiar two-sample comparison is a regression on one indicator.

It is tempting to add a second indicator, $1 - d_i$, for group 0. Then $\mathbf{X} = [\mathbf{1}, \mathbf{d}, \mathbf{1} - \mathbf{d}]$ has three columns, but they satisfy $\mathbf{1} - \mathbf{d} - (\mathbf{1} - \mathbf{d}) = \mathbf{0}$, so $\text{rank}(\mathbf{X}) = 2$. This is the **dummy variable trap**: an intercept together with an indicator for every level produces an exactly collinear model matrix. By [Section 8.1](#), none of the three coefficients is then identifiable. The trap is not a mistake in the model, which still describes two means, but in the coordinates.

8.5.2 A factor with g levels

Let a factor have levels $1, \dots, g$, with $n_k \geq 1$ observations at level k . Its **indicator matrix** is the $n \times g$ matrix $\mathbf{Z} = [\mathbf{z}_1, \dots, \mathbf{z}_g]$ with $z_{ik} = 1$ if observation i is at level k and 0 otherwise. Each row of $\mathbf{Z}$ contains a single 1, so

$$\mathbf{Z}\mathbf{1}_g = \mathbf{1}_n, \quad \mathbf{Z}^\top \mathbf{Z} = \mathbf{D} = \text{diag}(n_1, \dots, n_g), \quad \mathbf{Z}^\top \mathbf{y} = (y_{1\cdot}, \dots, y_{g\cdot})^\top, \quad (8.5.1)$$

where $y_{k\cdot}$ is the total of the responses at level k . The columns have disjoint supports, so $\mathbf{Z}$ has full column rank g , and $\mathcal{C}(\mathbf{Z})$ is the set of vectors that are constant within levels.

The *cell-means model* $E(\mathbf{Y}) = \mathbf{Z}\boldsymbol{\mu}$, $\boldsymbol{\mu} = (\mu_1, \dots, \mu_g)^\top$, gives each level its own mean. It has full rank, and by [Eq. 8.5.1](#)

$$\hat{\boldsymbol{\mu}} = \mathbf{D}^{-1} \mathbf{Z}^\top \mathbf{y} = (\bar{y}_1, \dots, \bar{y}_g)^\top,$$

the vector of level means. Any other model containing $\mathcal{C}(\mathbf{Z})$ and nothing more has the same fit. The overparameterized model $[\mathbf{1}, \mathbf{Z}]$ of [Example \(A rank-deficient layout\)](#) is one example. The full-rank codings below are others.

Definition 8.5.1 — Coding of a factor

A **coding** of a factor with g levels is a nonsingular $g \times g$ matrix $\mathbf{K}$. The coded model matrix is $\mathbf{ZK}$, and the model is $E(\mathbf{Y}) = \mathbf{ZK}\boldsymbol{\gamma}$. When the first column of $\mathbf{K}$ is $\mathbf{1}_g$, so that $\mathbf{K} = [\mathbf{1}_g, \mathbf{C}]$, the coded matrix is $[\mathbf{1}_n, \mathbf{ZC}]$: an intercept and the $g - 1$ columns $\mathbf{ZC}$. The $g \times (g - 1)$ matrix $\mathbf{C}$ is then called the **coding matrix** of the factor.

Row k of $\mathbf{K}$ lists the values that the coded columns take for an observation at level k . In the software of [Section 8.4](#) the matrix $\mathbf{C}$ is what one specifies (in R, the “contrasts” attribute of a factor), and the program builds the columns $\mathbf{ZC}$ from it.

Proposition 8.5.1 — What a coding estimates

Let $\mathbf{K}$ be a coding and $\hat{\boldsymbol{\mu}} = \mathbf{D}^{-1} \mathbf{Z}^\top \mathbf{y}$ the vector of level means. Then:

- (a) $\mathcal{C}(\mathbf{ZK}) = \mathcal{C}(\mathbf{Z})$, so every coding has the same fitted values, residuals and residual sum of squares;
- (b) the coefficients are $\boldsymbol{\gamma} = \mathbf{K}^{-1}\boldsymbol{\mu}$, and their least squares estimates are $\hat{\boldsymbol{\gamma}} = \mathbf{K}^{-1}\hat{\boldsymbol{\mu}}$;
- (c) $[\mathbf{1}_g, \mathbf{C}]$ is nonsingular iff the columns of $\mathbf{C}$ are linearly independent and $\mathbf{1}_g \notin \mathcal{C}(\mathbf{C})$;
- (d) to make the coefficients estimate prescribed linear functions $\mathbf{L}\boldsymbol{\mu}$ of the level means, with $\mathbf{L}$ a nonsingular $g \times g$ matrix, use the coding $\mathbf{K} = \mathbf{L}^{-1}$.

Proof.

- (a) The columns of $\mathbf{ZK}$ lie in $\mathcal{C}(\mathbf{Z})$, and $\text{rank}(\mathbf{ZK}) = \text{rank}(\mathbf{Z}) = g$ because $\mathbf{K}$ is nonsingular ([Proposition 1.2.3\(b\)](#)). A subspace of $\mathcal{C}(\mathbf{Z})$ of the same dimension is $\mathcal{C}(\mathbf{Z})$ itself. The fit depends only on the column space ([Theorem 6.7.1](#)).
- (b) The two models describe the same means iff $\mathbf{ZK}\boldsymbol{\gamma} = \mathbf{Z}\boldsymbol{\mu}$, and since $\mathbf{Z}$ has full column rank this means $\mathbf{K}\boldsymbol{\gamma} = \boldsymbol{\mu}$. For the estimates, $\mathbf{ZK}\hat{\boldsymbol{\gamma}} = \mathbf{M}\mathbf{y} = \mathbf{Z}\hat{\boldsymbol{\mu}}$, and the same cancellation gives $\mathbf{K}\hat{\boldsymbol{\gamma}} = \hat{\boldsymbol{\mu}}$.
- (c) A square matrix is nonsingular iff its columns are linearly independent. The columns $\mathbf{1}_g, \mathbf{c}_1, \dots, \mathbf{c}_{g-1}$ are independent iff the $\mathbf{c}_j$ are independent and $\mathbf{1}_g$ is not in their span.
- (d) With $\mathbf{K} = \mathbf{L}^{-1}$, (b) gives $\boldsymbol{\gamma} = \mathbf{L}\boldsymbol{\mu}$.

Part (b) is the whole story. Whatever the coding, the coefficients are fixed linear combinations of the level means, and these combinations are the *rows of* $\mathbf{K}^{-1}$, not the columns of $\mathbf{K}$. Each such combination is estimable in the overparameterized model, since it is a function of the means. Part (d) reverses the logic, and in practice it is the useful direction. Decide which questions the coefficients should answer, write them as the rows of $\mathbf{L}$, and invert.

8.5.3 The standard codings

For a factor with g levels, the codings in common use are the following. For each we give the meaning of the intercept γ_0 and of the other coefficients $\gamma_1, \dots, \gamma_{g-1}$, that is, the rows of $\mathbf{K}^{-1}$.

- **Reference (treatment) coding.** $\mathbf{C} = \begin{pmatrix} \mathbf{0}^\top \\ \mathbf{I}_{g-1}^\top \end{pmatrix}$: the columns are the indicators of levels $2, \dots, g$. Then $\gamma_0 = \mu_1$, and $\gamma_j = \mu_{j+1} - \mu_1$ is the difference between level $j+1$ and the *reference level* 1. The level chosen as reference is arbitrary.
- **Sum-to-zero (deviation or effect) coding.** $\mathbf{C} = \begin{pmatrix} \mathbf{I}_{g-1}^\top \\ -\mathbf{1}^\top \end{pmatrix}$. Level $k < g$ has mean $\gamma_0 + \gamma_k$ and level g has mean $\gamma_0 - \sum_{j < g} \gamma_j$. Summing over the g levels gives $\gamma_0 = \bar{\mu} = g^{-1} \sum_k \mu_k$, the unweighted average of the level means. Then $\gamma_k = \mu_k - \bar{\mu}$ is the deviation of level k from that average. The deviation of the last level is $-\sum_{j < g} \gamma_j$.
- **Weighted effect coding.** $\mathbf{C} = \begin{pmatrix} \mathbf{I}_{g-1}^\top \\ -(n_1, \dots, n_{g-1})/n_g \end{pmatrix}$. Now $\sum_k n_k \mu_k = n\gamma_0$, so $\gamma_0 = \sum_k n_k \mu_k / n$, the size-weighted average, estimated by the overall mean $\bar{y}$. The coefficient $\gamma_k = \mu_k - \gamma_0$ is the deviation from it.
- **Successive differences.** With $\mathbf{L}$ having first row $g^{-1}\mathbf{1}^\top$ and rows $\mathbf{e}_{j+1}^\top - \mathbf{e}_j^\top$, part (d) gives a coding in which $\gamma_0 = \bar{\mu}$ and $\gamma_j = \mu_{j+1} - \mu_j$. This is natural for ordered levels.
- **Helmert and orthogonal polynomial codings**, whose columns are contrasts, are treated in [Section 8.7](#).

The first three are the side conditions of [Example \(Five side conditions, one fit\)](#) in another language. In the overparameterized model $\mu + \alpha_k$, reference coding is the side condition $\alpha_1 = 0$, sum-to-zero coding is $\sum_k \alpha_k = 0$, and weighted effect coding is $\sum_k n_k \alpha_k = 0$. In each case the constrained $(\mu, \alpha_2, \dots, \alpha_g)$ or $(\mu, \alpha_1, \dots, \alpha_{g-1})$ equals $\boldsymbol{\gamma}$ ([Exercise \(C1\)](#)).

Example. (Party identification and ideology)

An extract of the 1996 American National Election Study, distributed with statsmodels as a public-domain data set, contains 944 respondents who placed themselves on a seven-point scale from extremely liberal (1) to extremely conservative (7). It also records their party identification in seven levels from strong Democrat to strong Republican. The level means and sizes are

party identifi- cation	strong Dem	weak Dem	ind. Dem	ind. independenRep	weak Rep	strong Rep
n_k	200	180	108	37	94	150
$\bar{y}_k$	3.335	3.617	3.685	4.054	4.957	5.691

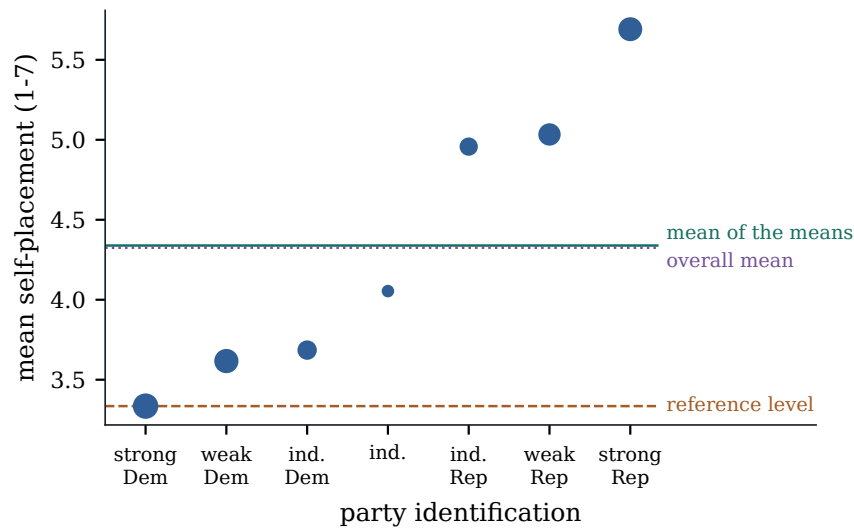


Figure 8.5.1: Mean self-placement by party identification in the 1996 ANES (point area proportional to group size), with the three baselines used by the codings of [Example \(Party identification and ideology\)](#): the reference level, the unweighted mean of the seven means, and the overall mean.

Here are the coefficients of four codings of the same factor ($\gamma_1, \dots, \gamma_6$ refer to the columns of $\mathbf{C}$ in the order defined above; for the two deviation codings they belong to the first six levels):

coding	$\hat{\gamma}_0$	$\hat{\gamma}_1$	$\hat{\gamma}_2$	$\hat{\gamma}_3$	$\hat{\gamma}_4$	$\hat{\gamma}_5$	$\hat{\gamma}_6$
reference (strong Dem)	3.335	0.282	0.350	0.719	1.622	1.698	2.356
sum to zero	4.339	-1.004	-0.722	-0.654	-0.285	0.618	0.694
weighted effect	4.325	-0.990	-0.709	-0.640	-0.271	0.632	0.708
successive differ- ences	4.339	0.282	0.069	0.369	0.903	0.076	0.658

Each row is $\mathbf{K}^{-1}\hat{\boldsymbol{\mu}}$ for its own $\mathbf{K}$, as [Proposition 8.5.1\(b\)](#) says, and all four have residual sum of squares 1178.3. The intercepts are the reference mean 3.335, the unweighted mean of the seven means 4.339, the overall mean $\bar{y} = 4.325$ and again 4.339. [Figure 8.5.1](#) shows the three reference points. The successive differences show that the largest step in ideology, 0.903, falls between pure independents and independents leaning Republican, and the step between weak and strong Republicans is 0.658. The reference coding cannot show this directly. The two deviation codings nearly agree here because the groups are of similar size, apart from the small group of pure independents.

```
import numpy as np
import statsmodels.api as sm

anes = sm.datasets.anes96.load_pandas().data
y = anes["selfLR"].to_numpy()      # 1 = very liberal ... 7 = very conservative
```

```

level = anes["PID"].to_numpy().astype(int)    # 0 = strong Dem. ... 6 = strong Rep.
g = 7
Z = np.eye(g)[level]                         # n x 7 indicator (cell-means) matrix
n_k = Z.sum(axis=0)
cell_means = (Z.T @ y) / n_k
print("group sizes:", n_k.astype(int))
print("cell means: ", np.round(cell_means, 3))

```

```

def coding_matrices(n_k):
    """Each coding is  $X = ZK$  for a nonsingular  $g \times g$   $K$ ; the coefficients are  $K^{-1}\mu$ ."""
    g = len(n_k)
    one = np.ones((g, 1))
    reference = np.vstack([np.zeros(g - 1), np.eye(g - 1)])    # level 0 is baseline
    deviation = np.vstack([np.eye(g - 1), -np.ones(g - 1)])    # sum-to-zero
    weighted = np.vstack([np.eye(g - 1), -n_k[:-1] / n_k[-1]])  # sum  $n_k \times$ 
    effect = 0
    successive = np.array([(j + 1 - g if i <= j else j + 1) / g for j in range(g - 1)]
                           for i in range(g))                    # backward differences
    return {"cell means": np.eye(g),
            "reference": np.hstack([one, reference]),
            "sum to zero": np.hstack([one, deviation]),
            "weighted effect": np.hstack([one, weighted]),
            "successive differences": np.hstack([one, successive])}

for name, K in coding_matrices(n_k).items():
    Xc = Z @ K
    gamma = np.linalg.lstsq(Xc, y, rcond=None)[0]
    print(f"{name:23s}", np.round(gamma, 3))

```

8.5.4 Choosing a coding

Since every coding gives the same fit, the choice is about communication and, to a lesser extent, precision. Four points are worth keeping in mind.

Anything that depends on the fit is invariant. This includes fitted values, residuals, R^2 , $\hat{\sigma}^2$, and the F test that all levels have the same mean. That test compares $\mathcal{C}(\mathbf{Z})$ with $\mathcal{C}(\mathbf{1})$, and the coding does not enter it ([Theorem 11.1.1](#)).

Coefficient tests answer different questions. The t test for a reference-coded coefficient tests whether level k differs from the reference level. The t test for a sum-coded coefficient tests whether level k differs from the average of all levels, a question that changes if a level is added. Neither question is wrong, but they are different questions. A table of coefficients from an unfamiliar coding should never be read as if it came from the familiar one.

Precision depends on the reference. In reference coding $\text{Var}(\hat{\gamma}_j) = \sigma^2(1/n_{j+1} + 1/n_1)$. A small reference group inflates the variance of every coefficient. In [Example \(Party identification and ideology\)](#), comparing each of the other five levels with the 37 pure independents instead of the 200 strong Democrats would multiply the standard errors by factors between 1.55 and 1.76. A common default is to use the largest group, or a natural control condition, as the reference.

With other regressors, coefficients are adjusted comparisons. When the model also contains other variables, a reference-coded coefficient compares level k with the reference level at equal values of those variables. That is the “holding the others fixed” reading of [Definition 5.7.1](#), with all its caveats. [Section 8.6](#) gives an example in which adjustment changes the answer a great deal.

Idea. A coding is a choice of coordinates for the g -dimensional space of level means. The coefficients are $\mathbf{K}^{-1}\hat{\boldsymbol{\mu}}$. To get the comparisons you want, write them down first as the rows of a nonsingular matrix $\mathbf{L}$, then code with $\mathbf{K} = \mathbf{L}^{-1}$.

For an ordered factor, a useful instance of this principle is the *cumulative* coding, with columns $\mathbf{1}\{\text{level} \geq k\}$ for $k = 2, \dots, g$. Then $\mathbf{K}$ is lower triangular with ones on and below the diagonal. Its inverse takes successive differences, so the intercept is μ_1 and the coefficient of the k th column is $\mu_k - \mu_{k-1}$ ([Exercise \(B2\)](#)). An observation’s mean is built up as a sum of steps, which is often how an ordered factor is best described.

8.5.5 Exercises

8.5.5.1 A. Check your understanding

Exercise A1.

In the model $E(y_i) = \beta_0 + \beta_1 d_i$, with n_0 observations in group 0 (level 1) and n_1 in group 1 (level 2), write down the coding matrix $\mathbf{K}$ with $\boldsymbol{\mu} = \mathbf{K}\boldsymbol{\beta}$. Use [Proposition 8.5.1\(b\)](#), not an inverse of $\mathbf{X}^\top \mathbf{X}$, to find $\hat{\boldsymbol{\beta}}$. Then show that $\text{Var}(\hat{\beta}_1) = \sigma^2(1/n_0 + 1/n_1)$ and that $\hat{\beta}_1 / \text{se}(\hat{\beta}_1)$ is the pooled two-sample t statistic, where $s^2 = \text{SSE}/(n-2)$ replaces σ^2 .

Solution. The two means are $\mu_0 = \beta_0$ and $\mu_1 = \beta_0 + \beta_1$, so $\mathbf{K} = \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}$ and $\mathbf{K}^{-1} = \begin{pmatrix} 1 & 0 \\ -1 & 1 \end{pmatrix}$. By [Proposition 8.5.1\(b\)](#), $\hat{\boldsymbol{\beta}} = \mathbf{K}^{-1}\hat{\boldsymbol{\mu}} = (\bar{y}_0, \bar{y}_1 - \bar{y}_0)^\top$. The level means are the estimates of the full-rank cell-means model, with $\text{Cov}(\hat{\boldsymbol{\mu}}) = \sigma^2 \mathbf{D}^{-1} = \sigma^2 \text{diag}(1/n_0, 1/n_1)$ ([Theorem 5.5.1](#)), so $\text{Var}(\hat{\beta}_1) = (-1, 1) \sigma^2 \mathbf{D}^{-1} (-1, 1)^\top = \sigma^2(1/n_0 + 1/n_1)$. The residuals are deviations from the group means ([Proposition 8.5.1\(a\)](#)), so s^2 is the pooled variance, and the ratio is $(\bar{y}_1 - \bar{y}_0) / \{s\sqrt{1/n_0 + 1/n_1}\}$.

Exercise A2.

For a factor with three levels, write down the null space of $[\mathbf{1}, \mathbf{z}_1, \mathbf{z}_2, \mathbf{z}_3]$. Show that adding a second factor’s full set of indicators, $[\mathbf{1}, \mathbf{Z}_A, \mathbf{Z}_B]$, creates a null space of dimension at least two, and give two independent null vectors.

8.5.5.2 B. Practice

Exercise B1.

Let $\mathbf{X}$ have full column rank, let $\mathbf{e}_i$ be the i th coordinate vector with $\mathbf{e}_i \notin \mathcal{C}(\mathbf{X})$, and fit $[\mathbf{X}, \mathbf{e}_i]$ with coefficients $(\boldsymbol{\beta}, \delta)$. Write $\hat{e}_i$ and h_{ii} for the residual and the leverage ([Definition 6.8.1](#)) of observation i in the fit of $\mathbf{X}$ alone.

- (a) Use [Theorem 6.6.2](#), with $\mathbf{e}_i$ as the regressor of interest and $\mathbf{X}$ as the one partialled

out, to show that $\hat{\delta} = \hat{e}_i/(1 - h_{ii})$, and explain why $h_{ii} < 1$.

- (b) Show that the enlarged fit reproduces y_i exactly, and that its $\hat{\boldsymbol{\beta}}$ is the estimate $\hat{\boldsymbol{\beta}}_{(i)}$ without observation i .
- (c) Deduce the deletion identity $y_i - \mathbf{x}_{(i)}^\top \hat{\boldsymbol{\beta}}_{(i)} = \hat{e}_i/(1 - h_{ii})$: the error in predicting y_i from the other observations is the ordinary residual inflated by $1/(1 - h_{ii})$.

Solution.

- (a) Apply Theorem 6.6.2(a) with $\mathbf{X}_1 = \mathbf{e}_i$ and $\mathbf{X}_2 = \mathbf{X}$, and let $\mathbf{M}$ project onto $\mathcal{C}(\mathbf{X})$. Then $\hat{\delta}$ is the coefficient in the regression of $\mathbf{y}$ on $(\mathbf{I} - \mathbf{M})\mathbf{e}_i$: $\hat{\delta} = \mathbf{e}_i^\top (\mathbf{I} - \mathbf{M})\mathbf{y} / \mathbf{e}_i^\top (\mathbf{I} - \mathbf{M})\mathbf{e}_i = \hat{e}_i/(1 - h_{ii})$. The denominator is $\|(\mathbf{I} - \mathbf{M})\mathbf{e}_i\|^2$, which is positive because $\mathbf{e}_i \notin \mathcal{C}(\mathbf{X})$.
- (b) The residual vector of the enlarged fit is orthogonal to every column, in particular to $\mathbf{e}_i$, so its i th entry is zero. For fixed $\boldsymbol{\beta}$, the residual sum of squares $\sum_{j \neq i} (y_j - \mathbf{x}_{(j)}^\top \boldsymbol{\beta})^2 + (y_i - \mathbf{x}_{(i)}^\top \boldsymbol{\beta} - \delta)^2$ is minimized over δ by making the last term zero. What remains is the deletion criterion, and its minimizer $\hat{\boldsymbol{\beta}}_{(i)}$ is unique: if $\mathbf{X}\mathbf{b} = c\mathbf{e}_i$ with $\mathbf{b} \neq \mathbf{0}$, then $c \neq 0$ by full rank, and $\mathbf{e}_i \in \mathcal{C}(\mathbf{X})$. So $\hat{\boldsymbol{\beta}} = \hat{\boldsymbol{\beta}}_{(i)}$.
- (c) By (b), $\hat{\delta} = y_i - \mathbf{x}_{(i)}^\top \hat{\boldsymbol{\beta}}_{(i)}$; compare with (a). The t statistic for δ is the externally studentized residual of Chapter 20.

Exercise B2.

For an ordered factor with g levels, let the coded columns be the indicators of $\{\text{level} \geq k\}$, $k = 2, \dots, g$, together with an intercept. Show that $\mathbf{K}$ is lower triangular with ones on and below the diagonal, that $\mathbf{K}^{-1}$ is the first-difference matrix, and hence that the coefficients are μ_1 and $\mu_k - \mu_{k-1}$.

Exercise B3.

On each of m days a warehouse runs N_{1i} machines of type 1 and N_{2i} of type 2, and meters its total energy use V_i . Suppose each running machine of type k uses an amount of energy with mean θ_k and variance τ_k^2 , independently of the other machines and the same on every day. Show that $E(V_i) = \theta_1 N_{1i} + \theta_2 N_{2i}$, a linear model *without* intercept, and that (θ_1, θ_2) is identifiable iff the vectors (N_{1i}) and (N_{2i}) are not proportional. Find $\text{Cov}(\mathbf{V})$; why is it not a multiple of $\mathbf{I}$? Why is it essential that θ_k does not vary by day?

8.5.5.3 C. Going deeper

Exercise C1.

Show that fitting the reference-coded model is the same as solving the normal equations of the overparameterized one-way model under the side condition $\alpha_1 = 0$, with $\gamma_0 = \mu$ and $\gamma_j = \alpha_{j+1}$. State and prove the corresponding facts for the sum-to-zero and weighted effect codings.

8.6 Several factors, interactions and covariates

With one factor, rank deficiency is a nuisance, and a coding removes it once and for all. With two or more factors, or a factor together with a numeric regressor, it becomes informative. The rank of the model matrix now depends on which combinations of levels were observed. Whether a comparison is estimable depends on how the design links the levels involved. This section works out the three cases that occur most often: two factors without interaction, two factors with interaction, and a factor with a numeric covariate.

8.6.1 Two factors without interaction

Let factor A have levels $i = 1, \dots, a$ and factor B levels $j = 1, \dots, b$. Call a pair (i, j) a **cell**. A cell is *occupied* if at least one observation has that combination, and we assume every level of each factor occurs in some occupied cell. The *additive* model is

$$E(y) = \mu + \alpha_i + \beta_j \quad \text{for an observation in cell } (i, j), \quad (8.6.1)$$

with model matrix $\mathbf{X} = [\mathbf{1}, \mathbf{Z}_A, \mathbf{Z}_B]$ built from the two indicator matrices. The number of parameters is $p = 1 + a + b$. Since $\mathbf{Z}_A \mathbf{1}_a = \mathbf{Z}_B \mathbf{1}_b = \mathbf{1}$, the vectors $(1, -\mathbf{1}_a^\top, \mathbf{0}^\top)^\top$ and $(1, \mathbf{0}^\top, -\mathbf{1}_b^\top)^\top$ always lie in $\mathcal{N}(\mathbf{X})$, so $\text{rank}(\mathbf{X}) \leq a + b - 1$. Whether equality holds, and which differences $\alpha_i - \alpha_k$ are estimable, depends on the pattern of occupied cells.

Picture the design as a graph. The vertices are the a levels of A and the b levels of B , and each occupied cell (i, j) is an edge joining level i of A to level j of B . The design is **connected** if this graph is connected, that is, if one can walk from any level to any other through occupied cells.

Theorem 8.6.1 — Connectedness and estimability in the additive model

In the additive model Eq. 8.6.1, let the design graph have c connected components. Then:

- (a) $\text{rank}(\mathbf{X}) = a + b - c$;
- (b) $\alpha_i - \alpha_k$ is estimable iff levels i and k of A lie in the same component, and likewise for $\beta_j - \beta_l$;
- (c) if the design is connected, the estimable functions are exactly the linear functions of the cell means $\mu + \alpha_i + \beta_j$ over *all* cells, occupied or not. In particular every difference between levels of the same factor is estimable.

Proof. A vector $(m, \mathbf{u}^\top, \mathbf{v}^\top)^\top$ lies in $\mathcal{N}(\mathbf{X})$ iff $m + u_i + v_j = 0$ for every occupied cell (i, j) . Fix such a vector. If (i, j) and (k, j) are both occupied, then $u_i = u_k$. If (i, j) and (i, l) are both occupied, then $v_j = v_l$. Following edges, u is constant on the A -levels of each component, say $u_i = t_C$ for all A -levels i in component C , and then $v_j = -m - t_C$ for the B -levels of C . Conversely, any choice of m and of numbers $t_1, \dots, t_c$, one for each component, defines in this way a vector satisfying all the equations. So $\mathcal{N}(\mathbf{X})$

is parameterized linearly and injectively by $(m, t_1, \dots, t_c)$, and it has dimension $c + 1$. Rank-nullity gives $\text{rank}(\mathbf{X}) = 1 + a + b - (c + 1)$, which is (a).

For (b), by [Theorem 8.3.1\(c\)](#), $\alpha_i - \alpha_k$ is estimable iff $u_i - u_k = 0$ for every null vector, that is, iff $t_{C(i)} = t_{C(k)}$ for all choices of the t 's. This holds iff $C(i) = C(k)$. The argument for B is the same.

For (c), if $c = 1$, then (b) makes every $\alpha_i - \alpha_k$ and $\beta_j - \beta_l$ estimable. Pick an occupied cell (i_0, j_0) . Then $\mu + \alpha_i + \beta_j = (\mu + \alpha_{i_0} + \beta_{j_0}) + (\alpha_i - \alpha_{i_0}) + (\beta_j - \beta_{j_0})$ is estimable for every (i, j) . Conversely, every estimable function is a combination of the means of occupied cells ([Theorem 8.2.2](#)), and these are among the cell means.

■

Part (c) contains a point worth stating separately. In a connected additive design, the mean of an *empty* cell is estimable. The data never observed that combination, but additivity lets it be pieced together from cells that were observed: $\mu + \alpha_i + \beta_j$ is the mean of an occupied cell (i, j') , plus the difference $\beta_j - \beta_{j'}$ learned elsewhere. The estimate is only as good as the additivity assumption, which cannot be checked at the empty cell. This is extrapolation of a subtle kind.

Example. (Two islands)

Take $a = 4, b = 5$ and seven occupied cells: $(1, 1), (1, 2), (2, 2), (2, 3)$ and $(3, 4), (4, 4), (4, 5)$ (levels numbered from 1). The design graph has two components: rows 1–2 with columns 1–3, and rows 3–4 with columns 4–5. By [Theorem 8.6.1](#), $\text{rank}(\mathbf{X}) = 4 + 5 - 2 = 7$, one less than $a + b - 1 = 8$. The difference $\alpha_1 - \alpha_2$ is estimable, since rows 1 and 2 are linked through column 2, but $\alpha_1 - \alpha_3$ is not. The two islands could sit at any vertical offset from each other without changing any fitted value. Occupying a single further cell, $(3, 3)$, joins the islands. The rank rises to 8, and $\alpha_1 - \alpha_3$ becomes estimable. The listing checks each statement with the rank test [Theorem 8.3.1\(h\)](#). The script also checks the rank formula on 200 random designs.

```
import numpy as np

def additive_matrix(cells, a, b):
    """Model matrix [1, row indicators, column indicators] for occupied (i, j) cells
    . """
    rows = np.array([i for i, j in cells])
    cols = np.array([j for i, j in cells])
    return np.column_stack([np.ones(len(cells)), np.eye(a)[rows], np.eye(b)[cols]])

def estimable(lam, X):
    """lambda is estimable iff appending it to the rows of X does not raise the rank
    . """
    return np.linalg.matrix_rank(np.vstack([X, lam])) == np.linalg.matrix_rank(X)

a, b = 4, 5
cells = [(0, 0), (0, 1), (1, 1), (1, 2), (2, 3), (3, 3), (3, 4)] # two separate
    blocks
X = additive_matrix(cells, a, b)
row_diff = lambda i, k: np.r_[0, np.eye(a)[i] - np.eye(a)[k], np.zeros(b)]
print("rank:", np.linalg.matrix_rank(X), " (a + b - 1 =", a + b - 1, ")")
print("alpha_1 - alpha_2 estimable:", estimable(row_diff(0, 1), X))
print("alpha_1 - alpha_3 estimable:", estimable(row_diff(0, 2), X))
```



```

X_linked = additive_matrix(cells + [(2, 2)], a, b)                                # one more
      occupied cell
print("after adding cell (3, 3): rank", np.linalg.matrix_rank(X_linked),
      " alpha_1 - alpha_3 estimable:", estimable(row_diff(0, 2), X_linked))

```

A complete design, with every cell occupied, is connected. So is any design in which one level of B appears with every level of A and one level of A with every level of B . Disconnected designs arise in practice more often than one might expect: in observational data with many sparse levels (workers and firms, students and schools, players and teams), and in experiments where blocks were filled carelessly. The estimable functions of a disconnected design are characterized in [Exercise \(C1\)](#).

8.6.2 Two factors with interaction

The additive model assumes that the difference between two levels of A is the same at every level of B . The *interaction model* drops that assumption:

$$E(y) = \mu + \alpha_i + \beta_j + \gamma_{ij} \quad \text{for an observation in cell } (i, j), \quad (8.6.2)$$

with an interaction parameter γ_{ij} for each of the ab cells. Write $\mathbf{d}$ for the $n \times ab$ indicator matrix of the cells. The column of an empty cell is zero. The model matrix is $\mathbf{X} = [\mathbf{1}, \mathbf{Z}_A, \mathbf{Z}_B, \mathbf{d}]$, with $p = 1 + a + b + ab$ columns, and m denotes the number of occupied cells.

Proposition 8.6.2 — Estimability in the interaction model

In the interaction model [Eq. 8.6.2](#) with m occupied cells, write $\mu_{ij} = \mu + \alpha_i + \beta_j + \gamma_{ij}$ for the mean of an occupied cell.

- (a) $\mathcal{C}(\mathbf{X}) = \mathcal{C}()$ and $\text{rank}(\mathbf{X}) = m$: the model is the cell-means model in disguise.
- (b) The estimable functions are exactly the linear combinations $\sum c_{ij}\mu_{ij}$ over occupied cells.
- (c) An interaction contrast $\sum_{i,j} d_{ij}\gamma_{ij}$ whose coefficient table (d_{ij}) has zero row and column sums is estimable iff $d_{ij} = 0$ for every empty cell. In particular, $\gamma_{ij} - \gamma_{il} - \gamma_{kj} + \gamma_{kl}$ is estimable iff all four cells are occupied.

Proof.

- (a) Every column of $\mathbf{1}$, $\mathbf{Z}_A$ and $\mathbf{Z}_B$ is a sum of columns of $\mathbf{d}$. For example, the indicator of level i of A is the sum of the indicators of the cells in row i . So $\mathcal{C}(\mathbf{X}) = \mathcal{C}()$. The nonzero columns of $\mathbf{d}$ are the m indicators of occupied cells, which have disjoint supports.
- (b) By [Theorem 8.2.2](#), the estimable functions are the combinations of the expected responses, and these are the μ_{ij} .
- (c) If $d_{ij} = 0$ off the occupied cells, then $\sum_{\text{occ}} d_{ij}\mu_{ij} = \sum d_{ij}\mu + \sum_i \alpha_i \sum_j d_{ij} + \sum_j \beta_j \sum_i d_{ij} + \sum d_{ij}\gamma_{ij} = \sum d_{ij}\gamma_{ij}$, because the row and column sums vanish. So the contrast is estimable. Conversely, if $d_{i_0j_0} \neq 0$ for an empty cell (i_0, j_0) , the column of $\gamma_{i_0j_0}$ in $\mathbf{X}$ is zero, so the coordinate vector $\mathbf{v}$ of that parameter lies in $\mathcal{N}(\mathbf{X})$. The contrast's coefficient vector has inner product $d_{i_0j_0} \neq 0$ with $\mathbf{v}$, so it

is not estimable ([Theorem 8.3.1\(c\)](#)).

■

In the interaction model the fitted value of each occupied cell is its own mean, and an empty cell has no estimable mean at all. The two models treat empty cells in opposite ways. The additive model fills them by assumption, while the interaction model declines to say anything about them. Which is appropriate is a substantive question, and [Chapter 17](#) returns to it ([Theorem 17.4.1](#)).

Example. (Empty cells in a survey)

Cross-classify the 944 ANES respondents of [Example \(Party identification and ideology\)](#) by party identification (seven levels) and education (seven levels, labelled in the statsmodels documentation from “grades 1–8” to “PhD”). Two of the 49 cells are empty: no pure independent and no weak Republican in the sample has only an elementary-school education. Other cells are thin: only 3 pure independents report some high school. The overparameterized interaction model has $p = 64$ columns and rank 47, the number of occupied cells, as [Proposition 8.6.2\(a\)](#) says. The interaction contrast comparing strong Democrats with pure independents across “some high school” and “high-school graduate” involves four occupied cells and is estimable. It rests on only 3 respondents in one cell, so it is imprecise, but estimable. The same comparison across “grades 1–8” and “high-school graduate” involves an empty cell and is not estimable. No amount of computation will produce it, and software that prints a number for it is printing an artefact of its side condition. The additive model, with rank $7 + 7 - 1 = 13$, treats both comparisons as zero by assumption.

```
import statsmodels.api as sm

anes = sm.datasets.anes96.load_pandas().data
pid = anes["PID"].to_numpy().astype(int)          # 7 levels
educ = anes["educ"].to_numpy().astype(int) - 1    # 7 levels, now 0..6
counts = np.zeros((7, 7), dtype=int)
np.add.at(counts, (pid, educ), 1)
print("empty (PID, educ) cells:",
      [(int(i), int(j) + 1) for i, j in zip(*np.nonzero(counts == 0))])

n = len(pid)
cell = pid * 7 + educ
X_int = np.column_stack([np.ones(n), np.eye(7)[pid], np.eye(7)[educ], np.eye(49)[
    cell]])
print("interaction model: p =", X_int.shape[1], " rank =", np.linalg.matrix_rank(
    X_int),
      " occupied cells =", int((counts > 0).sum()))

def interaction_contrast(i, k, j, l):
    """gamma_ij - gamma_il - gamma_kj + gamma_kl, as a coefficient vector on X_int.
    """
    lam = np.zeros(X_int.shape[1])
    for (r, c), s in [((i, j), 1), ((i, l), -1), ((k, j), -1), ((k, l), 1)]:
        lam[15 + r * 7 + c] += s
    return lam

print("PID 0 vs 3, educ 2 vs 3:", estimable(interaction_contrast(0, 3, 1, 2), X_int)
      )
```

```
print("PID 0 vs 3, educ 1 vs 3:", estimable(interaction_contrast(0, 3, 0, 2), X_int)
)
```

8.6.3 A factor and a numeric covariate

Now let a factor with g levels, with indicator matrix $\mathbf{Z}$, share the model with a numeric covariate $\mathbf{x}$. The two basic models are the **parallel lines** model

$$E(y) = \mu_k + \theta x \quad \text{for an observation at level } k, \quad (8.6.3)$$

with model matrix $[\mathbf{Z}, \mathbf{x}]$, and the **separate lines** model

$$E(y) = \mu_k + \theta_k x, \quad (8.6.4)$$

with model matrix $[\mathbf{Z}, \mathbf{Z} \circ \mathbf{x}]$, where $\mathbf{Z} \circ \mathbf{x}$ multiplies each indicator column entrywise by $\mathbf{x}$, so its k th column is $\mathbf{x}$ on level k and zero elsewhere. The same models can be written with an intercept and any coding of the factor. For example, the separate lines model in reference coding has columns $\mathbf{1}$, $\mathbf{ZC}$, $\mathbf{x}$, $(\mathbf{ZC}) \circ \mathbf{x}$. The coefficients of the last block are then differences of slopes from the reference level, and the column space is unchanged.

By [Exercise \(B1\)](#), the common slope θ is identifiable iff $\mathbf{x} \notin \mathcal{C}(\mathbf{Z})$, that is, iff x varies within at least one level. In the separate lines model, the slope θ_k is identifiable iff x varies within level k , since the k th column of $\mathbf{Z} \circ \mathbf{x}$ is a multiple of $\mathbf{z}_k$ exactly when x is constant on level k . When the slopes are identifiable, the separate lines model is simply g simple regressions fitted at once. Its coefficients are those of the separate fits, because the columns for different levels have disjoint supports.

The parallel lines model is where adjustment happens.

Proposition 8.6.3 — Adjusted differences

In the parallel lines model [Eq. 8.6.3](#) with $\mathbf{x} \notin \mathcal{C}(\mathbf{Z})$, let $\bar{y}_k$ and $\bar{x}_k$ be the level means. Then

$$\hat{\theta} = \frac{\sum_k \sum_{i \in k} (x_i - \bar{x}_k)(y_i - \bar{y}_k)}{\sum_k \sum_{i \in k} (x_i - \bar{x}_k)^2}, \quad \hat{\mu}_k - \hat{\mu}_l = (\bar{y}_k - \bar{y}_l) - \hat{\theta}(\bar{x}_k - \bar{x}_l).$$

Proof. The model matrix $[\mathbf{Z}, \mathbf{x}]$ has full column rank because $\mathbf{x} \notin \mathcal{C}(\mathbf{Z})$. Apply [Theorem 6.6.2](#) with $\mathbf{X}_1 = \mathbf{x}$ and $\mathbf{X}_2 = \mathbf{Z}$. Projecting onto $\mathcal{C}(\mathbf{Z})$ replaces each entry by its level mean ([Example \(The one-way layout\)](#)), so $(\mathbf{I} - \mathbf{M}_Z)\mathbf{x}$ and $(\mathbf{I} - \mathbf{M}_Z)\mathbf{y}$ are the within-level deviations, and $\hat{\theta}$ is the no-intercept slope of one on the other. This is the first formula. By [Theorem 6.6.2\(c\)](#), $\hat{\boldsymbol{\mu}} = (\mathbf{Z}^\top \mathbf{Z})^{-1} \mathbf{Z}^\top (\mathbf{y} - \hat{\theta} \mathbf{x})$, whose k th entry is $\bar{y}_k - \hat{\theta} \bar{x}_k$. Subtracting gives the second formula. ■

The difference $\bar{y}_k - \bar{y}_l$ compares the levels as they happen to be. The adjusted difference compares them at equal values of the covariate, correcting for the fact that the groups differ in x . The

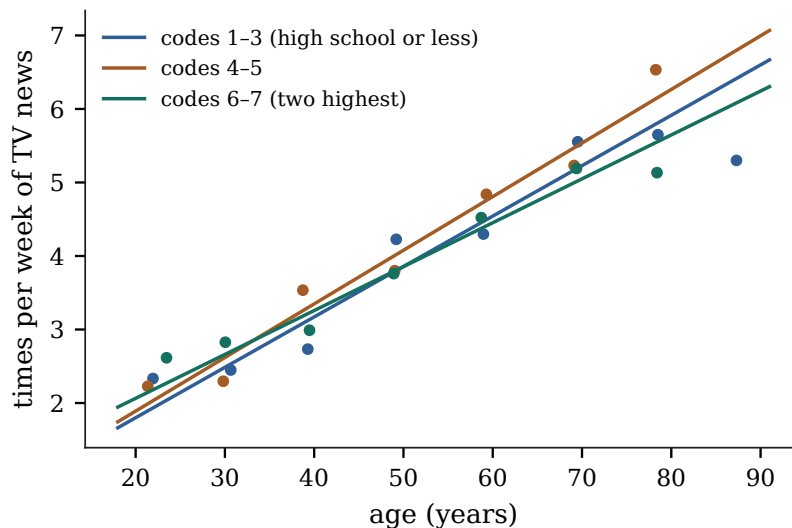


Figure 8.6.1: Times per week of television news against age for three education groups in the 1996 ANES: separate least squares lines (Eq. 8.6.4) and the group means within ten-year age bins (points).

correction is the slope times the difference in covariate means. It is large when the groups are unbalanced on a covariate that matters. The estimated slope uses only variation *within* levels, so that differences between levels in both x and y cannot be mistaken for an effect of x .

Example. (Education and television news, adjusted for age)

In the ANES data, respondents report how many times per week they watch television news. Coding education with seven levels and “high-school graduate” as the reference, respondents with some high school but no diploma watch 1.005 times a week more than high-school graduates. They are also older, with mean ages 59.8 and 48.2, and news watching rises with age. Adding age to the model, the within-level slope is 0.0674 times per week per year of age, and the adjusted difference shrinks to 0.224. More than three quarters of the raw difference is an age difference, and Proposition 8.6.3 accounts for it exactly. Whether the adjusted figure is the “effect of education” is a different question. Here age plausibly influences both schooling (older cohorts had less of it) and viewing habits, which argues for adjusting, but Chapter 25 shows that adjustment is not always appropriate.

Allowing separate age slopes for three education groups (codes 1–3, high school or less, 313 respondents; codes 4–5, 277; and the two highest levels, codes 6–7, 354) gives slopes 0.0686, 0.0729 and 0.0597, close to the common slope 0.0672 of the parallel lines model with the same three groups. The residual sum of squares falls only from 5623.6 to 5616.7 for the two extra parameters. Figure 8.6.1 shows the three fitted lines with binned means. The lines are nearly parallel, and the highest education group starts higher and rises a little more slowly.

```
tv = anes["TVnews"].to_numpy()           # times per week watching TV
news
age = anes["age"].to_numpy()
E = np.eye(7)[educ][:, [0, 1, 3, 4, 5, 6]] # reference: high-school
graduate
unadjusted = np.linalg.lstsq(np.column_stack([np.ones(n), E]), tv, rcond=None)[0]
```

```
adjusted = np.linalg.lstsq(np.column_stack([np.ones(n), E, age]), tv, rcond=None)[0]
print("some high school minus high-school graduate:",
      f"unadjusted {unadjusted[2]:.3f}, adjusted for age {adjusted[2]:.3f}")
print("mean age: some high school", age[educ == 1].mean().round(1),
      " high-school graduate", age[educ == 2].mean().round(1),
      " common age slope", adjusted[-1].round(4))
```

```
grp = np.select([educ <= 2, educ <= 4], [0, 1], 2) # education codes 1-3 / 4-5 /
6-7
D = np.eye(3)[grp]
X_par = np.column_stack([D, age]) # parallel lines: 3 intercepts,
one slope
X_sep = np.column_stack([D, D * age[:, None]]) # separate lines: 3 intercepts,
3 slopes
b_par = np.linalg.lstsq(X_par, tv, rcond=None)[0]
b_sep = np.linalg.lstsq(X_sep, tv, rcond=None)[0]
sse_par = np.sum((tv - X_par @ b_par) ** 2)
sse_sep = np.sum((tv - X_sep @ b_sep) ** 2)
print("separate slopes:", b_sep[3:].round(4),
      f" SSE parallel {sse_par:.1f}, separate {sse_sep:.1f}")
```

Warning. In a separate lines model the difference between two levels is $(\mu_k - \mu_l) + (\theta_k - \theta_l)x$, which depends on x . The coefficient that software labels as the “main effect” of level k is this difference at $x = 0$. For a covariate like age, that is a comparison of newborns, far outside the data. Centre the covariate at a meaningful value before reading such coefficients, or report the difference at several values of x .

8.6.4 Broken lines

Indicators can also switch a slope on. Let c be a known value of x and put $(x-c)_+ = \max(x-c, 0)$, which is $x - c$ times the indicator of $x > c$. The model

$$E(y) = \beta_0 + \beta_1 x + \beta_2 (x - c)_+$$

is a continuous line with a bend at c : slope β_1 to the left of c and $\beta_1 + \beta_2$ to the right. Adding the indicator $\mathbf{1}\{x > c\}$ itself allows a jump at c as well. With the bend point known, these are ordinary linear models, and all of this chapter applies. For example, β_2 is identifiable iff the data contain values of x on both sides of c (strictly below and strictly above) and at least three distinct values of x in all ([Exercise \(B2\)](#)). When c is unknown the model is nonlinear in c . Such piecewise linear terms are the simplest splines, the subject of Part IX.

8.6.5 Exercises

8.6.5.1 A. Check your understanding

Exercise A1.

In the additive model with $a = b = 2$, suppose cells $(1, 1)$, $(2, 1)$ and $(2, 2)$ are occupied and $(1, 2)$ is empty. Find the rank of $\mathbf{X}$ and a basis of $\mathcal{N}(\mathbf{X})$. Show that $\mu + \alpha_1 + \beta_2$, the mean of the empty cell, is estimable by writing it as a combination of the means of

occupied cells.

Exercise A2.

A factor has three levels, and the covariate x is constant on level 2 but varies on levels 1 and 3. Find the rank of the separate lines model matrix $[\mathbf{Z}, \mathbf{Z} \circ \mathbf{x}]$, and say which of $\mu_1, \mu_2, \mu_3, \theta_1, \theta_2, \theta_3$ are identifiable.

8.6.5.2 B. Practice

Exercise B1.

In the separate lines model, show that the mean at level k and covariate value x_0 , $\mu_k + \theta_k x_0$, is estimable iff either x varies within level k , or x_0 equals the common value of x on level k . Interpret this as a statement about extrapolation.

Solution. The mean is $\boldsymbol{\lambda}^\top \boldsymbol{\beta}$ with $\boldsymbol{\lambda}$ having 1 in the position of μ_k and x_0 in the position of θ_k . The columns for μ_k and θ_k are $\mathbf{z}_k$ and $\mathbf{z}_k \circ \mathbf{x}$, and they involve only level k . If x varies on level k , both coefficients are identifiable ([Proposition 8.1.3](#)), so $\boldsymbol{\lambda}^\top \boldsymbol{\beta}$ is estimable. If $x \equiv x^*$ on level k , then $\mathbf{z}_k \circ \mathbf{x} = x^* \mathbf{z}_k$, and the null space contains the vector with $-x^*$ at μ_k and 1 at θ_k . Then $\boldsymbol{\lambda}^\top \mathbf{v} = x_0 - x^*$, which vanishes iff $x_0 = x^*$. A single covariate value within a level supports no statement about other values, however many observations share it.

Exercise B2.

Write the broken-line model $E(y) = \beta_0 + \beta_1 x + \beta_2(x - c)_+$ in matrix form and prove the identifiability condition stated in the text. Show that the continuous broken line is the same as the model with two separate lines, one on each side of c , constrained to meet at c , and count parameters to explain the difference of one.

Exercise B3.

In the interaction model, suppose all cells are occupied except $(1, 1)$. Show that the $(a - 1)(b - 1)$ contrasts $\gamma_{ij} - \gamma_{i1} - \gamma_{1j} + \gamma_{11}$ are no longer estimable, but that $\gamma_{ij} - \gamma_{i2} - \gamma_{1j} + \gamma_{12}$ is estimable for $i \geq 2, j \geq 3$. How many linearly independent estimable interaction contrasts remain?

8.6.5.3 C. Going deeper

Exercise C1.

In the additive model, let the design graph have components $C_1, \dots, C_c$. Show that $\lambda_0 \mu + \sum_i \lambda_i \alpha_i + \sum_j \kappa_j \beta_j$ is estimable iff $\lambda_0 = \sum_j \kappa_j$ and, for every component C , $\sum_{i \in C} \lambda_i = \sum_{j \in C} \kappa_j$. Deduce [Theorem 8.6.1\(b\)](#).

Solution. From the proof of [Theorem 8.6.1](#), the null vectors are $(m, \mathbf{u}, \mathbf{v})$ with $u_i = t_{C(i)}$ and $v_j = -m - t_{C(j)}$, for arbitrary $m, t_1, \dots, t_c$. The inner product with the coefficient vector is

$$\lambda_0 m + \sum_i \lambda_i t_{C(i)} - \sum_j \kappa_j (m + t_{C(j)}) = m \left(\lambda_0 - \sum_j \kappa_j \right) + \sum_C t_C \left(\sum_{i \in C} \lambda_i - \sum_{j \in C} \kappa_j \right).$$

This vanishes for all m, t iff every bracket is zero, which is the stated condition ([Theorem 8.3.1\(c\)](#)). For $\alpha_i - \alpha_k$, $\lambda_0 = 0$, all $\kappa_j = 0$, and the component sums of λ are zero iff i and k lie in the same component.

8.7 Contrasts

In the one-way model the individual effects α_k are not estimable, but their differences are. The general form of such a difference is a contrast. Contrasts are the natural targets of inference about a factor. They are estimable, they do not depend on the side condition or coding, and they are what a scientific question about a factor usually asks: does treatment beat control, is there a trend across doses, do two groups of levels differ? This section characterizes them, shows how an orthogonal set of contrasts splits the factor's sum of squares, and constructs the Helmert and polynomial families.

8.7.1 Definition and estimability

Throughout, the one-way model is $E(y_{kj}) = \mu + \alpha_k$ for observation j at level k , with $n_k \geq 1$ observations at level $k = 1, \dots, g$, level means $\mu_k = \mu + \alpha_k$ and sample means $\bar{y}_k$.

Definition 8.7.1 — Contrast

In a factor model, a **contrast** in the effects of a factor is a linear function $\sum_k c_k \alpha_k$ whose coefficients satisfy $\sum_k c_k = 0$, not all zero. In the one-way model it equals $\sum_k c_k \mu_k$, and we call either form a contrast in the level means. Two contrasts are **orthogonal** (for the given design) if $\sum_k c_k d_k / n_k = 0$.

The two forms agree because $\sum_k c_k \mu_k = \mu \sum_k c_k + \sum_k c_k \alpha_k$. This is where the condition $\sum_k c_k = 0$ comes from: it makes the unidentified μ drop out. Simple differences $\alpha_k - \alpha_l$, comparisons of one level with the average of others such as $\alpha_1 - \frac{1}{2}(\alpha_2 + \alpha_3)$, and trend coefficients are all contrasts.

Theorem 8.7.1 — Estimable functions in the one-way model

In the one-way model with every $n_k \geq 1$, the function $\lambda_0 \mu + \sum_k \lambda_k \alpha_k$ is estimable iff $\lambda_0 = \sum_k \lambda_k$. In particular, a linear function of the effects alone, $\sum_k c_k \alpha_k$, is estimable iff it is a contrast (or zero). For a contrast, the least squares estimator and its variance under $\text{Cov}(\mathbf{Y}) = \sigma^2 \mathbf{I}$ are

$$\hat{\psi} = \sum_k c_k \bar{y}_k, \quad \text{Var}(\hat{\psi}) = \sigma^2 \sum_k \frac{c_k^2}{n_k}. \quad (8.7.1)$$

Proof. The null space of $\mathbf{X} = [\mathbf{1}, \mathbf{Z}]$ is spanned by $\mathbf{v} = (1, -1, \dots, -1)^\top$ (Example (A rank-deficient layout)). By Theorem 8.3.1(c), the function is estimable iff $\lambda_0 - \sum_k \lambda_k = 0$. For an estimable function, $\lambda_0 \mu + \sum_k \lambda_k \alpha_k = \sum_k \lambda_k (\mu + \alpha_k)$. Every least squares solution has $\hat{\mu} + \hat{\alpha}_k = \bar{y}_k$, since the fitted values are the level means (Example (The one-way layout)). So the estimate is $\sum_k \lambda_k \bar{y}_k$. The level means are uncorrelated with variances σ^2/n_k , by Theorem 2.2.1, which gives the variance. ■

The estimator in Eq. 8.7.1 is the same whatever coding or side condition was used to fit the model. Proposition 8.3.2 guarantees this, and it is the reason contrasts, rather than coefficients, should be reported.

8.7.2 Sums of squares and orthogonal contrasts

Each contrast defines a direction in the observation space, and its estimate is the component of $\mathbf{y}$ along that direction. With $\mathbf{D} = \text{diag}(n_1, \dots, n_g)$ and $\mathbf{Z}$ the indicator matrix, put

$$\mathbf{u}_c = \mathbf{ZD}^{-1}\mathbf{c}, \quad (8.7.2)$$

the vector whose entries equal c_k/n_k for the observations at level k .

Proposition 8.7.2 — Contrasts as projections

Let $\mathbf{c}$ and $\mathbf{d}$ be contrasts, and write $\hat{\psi}_c = \mathbf{c}^\top \hat{\boldsymbol{\mu}}$. Then:

- (a) $\hat{\psi}_c = \mathbf{u}_c^\top \mathbf{y}$, and $\mathbf{u}_c$ is a nonzero vector in $\mathcal{C}(\mathbf{Z}) \cap \mathbf{1}^\perp$;
- (b) the **sum of squares for the contrast**,

$$\text{SS}(\mathbf{c}) = \frac{\hat{\psi}_c^2}{\sum_k c_k^2/n_k},$$

equals $\|\mathbf{P}\mathbf{y}\|^2$, where $\mathbf{P}$ is the orthogonal projection onto the line spanned by $\mathbf{u}_c$;

- (c) $\text{Cov}(\hat{\psi}_c, \hat{\psi}_d) = \sigma^2 \sum_k c_k d_k / n_k = \sigma^2 \mathbf{u}_c^\top \mathbf{u}_d$ when $\text{Cov}(\mathbf{Y}) = \sigma^2 \mathbf{I}$, so orthogonal contrasts have uncorrelated estimates and orthogonal vectors $\mathbf{u}$;
- (d) if $\mathbf{c}_1, \dots, \mathbf{c}_{g-1}$ are pairwise orthogonal contrasts, then

$$\sum_{j=1}^{g-1} \text{SS}(\mathbf{c}_j) = \sum_{k=1}^g n_k (\bar{y}_k - \bar{y})^2,$$

the sum of squares between levels.

Proof.

- (a) $\hat{\boldsymbol{\mu}} = \mathbf{D}^{-1}\mathbf{Z}^\top \mathbf{y}$ (Eq. 8.5.1), so $\mathbf{c}^\top \hat{\boldsymbol{\mu}} = (\mathbf{ZD}^{-1}\mathbf{c})^\top \mathbf{y}$. Clearly $\mathbf{u}_{\mathbf{c}} \in \mathcal{C}(\mathbf{Z})$, and $\mathbf{1}^\top \mathbf{u}_{\mathbf{c}} = \mathbf{1}^\top \mathbf{ZD}^{-1}\mathbf{c} = (n_1, \dots, n_g)\mathbf{D}^{-1}\mathbf{c} = \sum_k c_k = 0$. It is nonzero because $\mathbf{ZD}^{-1}$ has full column rank and $\mathbf{c} \neq \mathbf{0}$.
- (b) By Example (Projections onto a line), $\|\mathbf{Py}\|^2 = (\mathbf{u}_{\mathbf{c}}^\top \mathbf{y})^2 / \|\mathbf{u}_{\mathbf{c}}\|^2$, and $\|\mathbf{u}_{\mathbf{c}}\|^2 = \mathbf{c}^\top \mathbf{D}^{-1}\mathbf{Z}^\top \mathbf{ZD}^{-1}\mathbf{c} = \mathbf{c}^\top \mathbf{D}^{-1}\mathbf{c} = \sum_k c_k^2 / n_k$.
- (c) Theorem 2.2.1 gives $\text{Cov}(\mathbf{u}_{\mathbf{c}}^\top \mathbf{Y}, \mathbf{u}_{\mathbf{d}}^\top \mathbf{Y}) = \sigma^2 \mathbf{u}_{\mathbf{c}}^\top \mathbf{u}_{\mathbf{d}}$, and the same computation as in (b) gives $\mathbf{u}_{\mathbf{c}}^\top \mathbf{u}_{\mathbf{d}} = \mathbf{c}^\top \mathbf{D}^{-1}\mathbf{d}$.
- (d) The space $\mathcal{C}(\mathbf{Z}) \cap \mathbf{1}^\perp$ has dimension $g - 1$ (Theorem 6.5.1(c), with $\mathcal{C}(\mathbf{1}) \subseteq \mathcal{C}(\mathbf{Z})$). By (a) and (c) the vectors $\mathbf{u}_{\mathbf{c}_1}, \dots, \mathbf{u}_{\mathbf{c}_{g-1}}$ are $g - 1$ nonzero, mutually orthogonal vectors in it, so they form an orthogonal basis. By Theorem 6.3.8 the sum of the projections onto the lines they span is the projection onto that space, which is $\mathbf{M} - n^{-1}\mathbf{1}\mathbf{1}^\top$ (Theorem 6.5.1(b)). Applying it to $\mathbf{y}$ and using Pythagoras gives $\sum_j \text{SS}(\mathbf{c}_j) = \|(\mathbf{M} - n^{-1}\mathbf{1}\mathbf{1}^\top)\mathbf{y}\|^2$. The vector $(\mathbf{M} - n^{-1}\mathbf{1}\mathbf{1}^\top)\mathbf{y}$ has entry $\bar{y}_k - \bar{y}$ at every observation of level k .

■

So a complete set of orthogonal contrasts decomposes the between-level sum of squares into $g - 1$ one-dimensional pieces. Under normality the pieces are independent, each is σ^2 times a noncentral χ^2 with one degree of freedom, and each is independent of the residual sum of squares (Theorem 4.4.4). Each contrast can therefore be tested on its own. Chapter 9 places this decomposition in the general theory of orthogonal sums of squares, and Chapter 15 uses it for the one-way analysis of variance (Theorem 15.4.1). In a *balanced* design, with all n_k equal, orthogonality reduces to the ordinary $\sum_k c_k d_k = 0$. In an unbalanced design, contrasts that look orthogonal on paper need not be orthogonal for the data.

8.7.3 Helmert contrasts

The Helmert matrix of Example (Helmert matrices) has rows proportional to

$$\mathbf{h}_k = (\underbrace{1, \dots, 1}_{k-1}, -(k-1), 0, \dots, 0), \quad k = 2, \dots, g.$$

The contrast $\mathbf{h}_k^\top \boldsymbol{\mu}$ compares level k with the average of the levels before it, since it is $(k - 1)$ times $\frac{1}{k-1} \sum_{l < k} \mu_l - \mu_k$. The $\mathbf{h}_k$ are mutually orthogonal in the ordinary sense, so in a balanced design they are orthogonal contrasts. They suit factors whose levels are added in a meaningful order, such as successive refinements of a treatment, where each new level is compared with everything before it.

8.7.4 Polynomial contrasts

When the levels of a factor are ordered and equally spaced, such as doses 0, 10, 20, 30 or the seven points of a party scale, the natural questions are about the *shape* of the sequence of means. Is there a linear trend? Curvature? Give the levels scores $s_k = k$. Apply Gram–Schmidt (Proposition 6.2.2) to the vectors $\mathbf{1}, \mathbf{s}, \mathbf{s}^2, \dots, \mathbf{s}^{g-1}$ in $\mathbb{R}^g$, where $\mathbf{s}^j$ has entries s_k^j . This gives orthogonal vectors $\mathbf{p}_0, \dots, \mathbf{p}_{g-1}$, and $\mathbf{p}_j$ is a polynomial of degree j in the score. For $j \geq 1$,

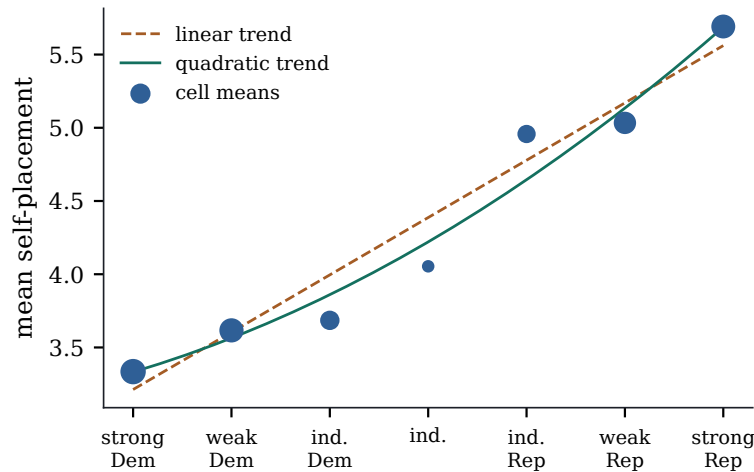


Figure 8.7.1: Mean ideological self-placement by party identification (point area proportional to group size), with the weighted least squares linear and quadratic trends in the score. The linear contrast accounts for almost all of the variation between groups.

$\mathbf{p}_j \perp \mathbf{1}$, so $\mathbf{p}_j$ is a contrast. Rescaled to integers, the first three for $g = 7$ are

linear $(-3, -2, -1, 0, 1, 2, 3)$, quadratic $(5, 0, -3, -4, -3, 0, 5)$, cubic $(-1, 1, 1, 0, -1, -1, 1)$.

By construction $\mathbf{p}_j$ is orthogonal to every polynomial of degree less than j . So if the level means follow a polynomial of degree d in the score, every contrast of degree above d is zero. The linear contrast measures the trend, the quadratic the curvature beyond it, and so on. This is the reason for their use.

With unequal group sizes the classical integer contrasts are no longer orthogonal for the design. The fix follows [Proposition 8.7.2\(c\)](#). Orthogonality is needed in the inner product $\sum_k c_k d_k / n_k$, which suggests setting $\mathbf{c}_j = \mathbf{D}\mathbf{p}_j$, where now the $\mathbf{p}_j$ are orthogonal polynomials in the weighted inner product $\sum_k n_k p_k q_k$. Then $\sum_k c_{jk} = \sum_k n_k p_{jk} = 0$, and $\sum_k c_{jk} c_{lk} / n_k = \sum_k n_k p_{jk} p_{lk} = 0$ for $j \neq l$. In observation space, $\mathbf{u}_{\mathbf{c}_j} = \mathbf{Z}\mathbf{p}_j$ is the polynomial of degree j in each observation's score, orthogonalized over the observations. The weighted linear contrast gives exactly the regression sum of squares of $\mathbf{y}$ on the score ([Exercise \(C1\)](#)).

Example. (Trend in ideology across party identification)

Return to the seven party-identification groups of [Example \(Party identification and ideology\)](#). The between-group sum of squares is 772.87. The classical linear contrast has estimate $\hat{\psi} = \sum_k (k-4)\bar{y}_k = 11.175$, variance $0.1652 \sigma^2$ by [Eq. 8.7.1](#), and sum of squares 755.85, which is 97.8 percent of the between-group sum of squares. The quadratic and cubic contrasts have sums of squares 10.15 and 2.58. Ideology rises almost linearly across the seven-point party scale, with some upward curvature ([Figure 8.7.1](#)).

The group sizes range from 37 to 200, so the six classical contrasts are not orthogonal for this design. Their sums of squares add to 784.02, which exceeds the between-group total. Part of the variation is counted twice. The weighted orthogonal polynomial contrasts decompose the total exactly: 746.57 for the linear trend and 26.30 for the five higher-degree contrasts together.

```
import numpy as np
import statsmodels.api as sm
```

```

anes = sm.datasets.anes96.load_pandas().data
y = anes["selfLR"].to_numpy()
level = anes["PID"].to_numpy().astype(int)          # 0 = strong Dem. ... 6 = strong
Rep.
g = 7
n_k = np.bincount(level).astype(float)
means = np.bincount(level, weights=y) / n_k

```

```

def orthogonal_polynomials(scores, weights):
    """Columns 1..g-1: polynomials in the scores, orthogonal in sum_k w_k u_k v_k.
    """
    V = np.vander(scores, len(scores), increasing=True)      # 1, s, s^2, ...
    W = np.sqrt(weights)[: , None]
    Q, _ = np.linalg.qr(W * V)                               # weighted Gram-Schmidt
    P = Q / W                                                 # back to the original
    scale                                                    #
    P = P / np.abs(P).max(axis=0)                             # cosmetic rescaling
    return P[:, 1:]                                           # drop the constant
    column

def contrast_ss(c, means, n_k):
    """Estimate c'μ-hat and its sum of squares (c'μ-hat)^2 / sum(c_k^2 / n_k)."""
    psi = c @ means
    return psi, psi ** 2 / np.sum(c ** 2 / n_k)

scores = np.arange(g, dtype=float)
C_unw = orthogonal_polynomials(scores, np.ones(g))           # the classical table (
    equal n)
C_wtd = n_k[:, None] * orthogonal_polynomials(scores, n_k)   # c_k = n_k p(k)
ss_between = np.sum(n_k * (means - y.mean()) ** 2)
for name, C in [("unweighted", C_unw), ("weighted", C_wtd)]:
    ss = [contrast_ss(C[:, j], means, n_k)[1] for j in range(g - 1)]
    print(f"{name:10s} SS by degree: {np.round(ss, 2)} total {sum(ss):.2f}",
          f"(between groups {ss_between:.2f})")

```

8.7.5 Coding matrices are not contrast matrices

Section 8.5 coded a factor by an intercept and the columns $\mathbf{ZC}$. When the columns of $\mathbf{C}$ are contrasts, as for Helmert or polynomial coding, it is tempting to read the coefficient of column j as the estimate of the contrast $\mathbf{c}_j^\top \hat{\boldsymbol{\mu}}$. That reading is wrong in general. By Proposition 8.5.1(b) the coefficients are the rows of $\mathbf{K}^{-1}$, with $\mathbf{K} = [\mathbf{1}, \mathbf{C}]$, applied to $\hat{\boldsymbol{\mu}}$. The rows of $\mathbf{K}^{-1}$ and the columns of $\mathbf{K}$ are related simply only when the columns are orthogonal.

Proposition 8.7.3 — Orthogonal codings

If the columns of $\mathbf{K} = [\mathbf{1}_g, \mathbf{c}_1, \dots, \mathbf{c}_{g-1}]$ are mutually orthogonal, then the coefficients of the coded model are

$$\hat{\gamma}_0 = \frac{1}{g} \sum_k \bar{y}_k, \quad \hat{\gamma}_j = \frac{\mathbf{c}_j^\top \hat{\boldsymbol{\mu}}}{\mathbf{c}_j^\top \mathbf{c}_j}, \quad j = 1, \dots, g-1,$$

whatever the group sizes.

Proof. With orthogonal columns, $\mathbf{K}^\top \mathbf{K} = \text{diag}(g, \mathbf{c}_1^\top \mathbf{c}_1, \dots, \mathbf{c}_{g-1}^\top \mathbf{c}_{g-1})$, so $\mathbf{K}^{-1} = (\mathbf{K}^\top \mathbf{K})^{-1} \mathbf{K}^\top$. Apply [Proposition 8.5.1\(b\)](#). ■

The orthogonality required here is the plain one, $\mathbf{c}_j^\top \mathbf{c}_l = 0$, not the design orthogonality of [Definition 8.7.1](#), because $\hat{\boldsymbol{\gamma}} = \mathbf{K}^{-1} \hat{\boldsymbol{\mu}}$ is pure algebra and does not involve the n_k . Helmert coding, in the usual software convention, uses the columns $\mathbf{c}_j = -\mathbf{h}_{j+1}$, which compare level $j+1$ with the levels before it, and $\mathbf{c}_j^\top \mathbf{c}_j = j(j+1)$. In the party data the first three Helmert coefficients are 0.1408, 0.0698 and 0.1271, while the contrasts themselves are 0.2817, 0.4187 and 1.5253, larger by the factors 2, 6 and 12. For a non-orthogonal coding there is no such shortcut. Reference coding has $\mathbf{c}_j = \mathbf{e}_{j+1}$, yet its coefficients are $\mu_{j+1} - \mu_1$, not μ_{j+1} .

```
def helmert(g):
    """Coding matrix whose column k compares level k+1 with the mean of levels 1..k.
    """
    C = np.zeros((g, g - 1))
    for k in range(1, g):
        C[:k, k - 1] = -1.0
        C[k, k - 1] = k
    return C

H = helmert(g)
K = np.column_stack([np.ones(g), H])
gamma = np.linalg.solve(K, means)           # Helmert-coded coefficients
psi = H.T @ means                           # the contrasts c_k' mu-hat
print("coefficients:", np.round(gamma[1:], 4))
print("contrasts:    ", np.round(psi, 4))
print("ratio:        ", np.round(psi / gamma[1:], 1))  # = c_k' c_k = k(k+1)
```

Warning. A coefficient from a contrast-coded regression estimates a contrast only up to a scale factor, and only when the coding columns are orthogonal. To test or report a contrast, compute $\mathbf{c}^\top \hat{\boldsymbol{\mu}}$ and its variance [Eq. 8.7.1](#) directly, or read it from the coded fit using $\mathbf{K}^{-1}$. Do not read it off the coefficient table.

8.7.6 Contrasts in the two-way additive model

In the additive model $\mu + \alpha_i + \beta_j$ of [Section 8.6](#), a **row contrast** is $\sum_i c_i \alpha_i$ with $\sum_i c_i = 0$. It is a combination of the differences $\alpha_i - \alpha_1$, so in a connected design it is estimable ([Theorem 8.6.1](#)). In a balanced design, with $m \geq 1$ observations in every one of the ab cells, the least squares fit is $\bar{y}_{i..} + \bar{y}_{.j.} - \bar{y}_{...}$. This is shown in [Example \(Additive two-factor model, one observation per cell\)](#) for $m = 1$, and the averaging argument there is unchanged for $m > 1$. Hence

$$\widehat{\sum_i c_i \alpha_i} = \sum_i c_i (\bar{y}_{i..} + \bar{y}_{.j.} - \bar{y}_{...}) = \sum_i c_i \bar{y}_{i..}, \quad \text{Var} = \frac{\sigma^2}{bm} \sum_i c_i^2,$$

since the terms not depending on i are multiplied by $\sum_i c_i = 0$. The row contrast is estimated from the row means alone, as if factor B were absent. In an unbalanced design this is false. The row means are then contaminated by the column effects ([Exercise \(B2\)](#)), and the least squares

estimate has to adjust for them. That adjustment is the subject of [Chapter 17](#). Interaction contrasts, and their estimability when cells are empty, were treated in [Proposition 8.6.2](#).

8.7.7 Exercises

8.7.7.1 A. Check your understanding

Exercise A1.

Find the linear, quadratic and cubic orthogonal polynomial contrasts for $g = 4$ equally spaced levels, scaled to integers. Check that they are mutually orthogonal and that the quadratic contrast vanishes when $\mu_k = a + bk$.

Exercise A2.

Show that the Helmert contrasts $\mathbf{h}_2, \dots, \mathbf{h}_g$ are orthogonal for the design iff $n_1 = n_2 = \dots = n_{g-1}$, whatever n_g is. *Hint:* compute $\sum_k h_{jk}h_{lk}/n_k$ for $j < l$.

8.7.7.2 B. Practice

Exercise B1.

Show that for every contrast $\mathbf{c}$, $\text{SS}(\mathbf{c}) \leq \sum_k n_k (\bar{y}_k - \bar{y})^2$, with equality when $c_k = n_k(\bar{y}_k - \bar{y})$. This fact is the basis of Scheffé's method in [Chapter 13](#) ([Theorem 13.2.2](#)).

Solution. By [Proposition 8.7.2\(b\)](#), $\text{SS}(\mathbf{c}) = \|\mathbf{P}\mathbf{y}\|^2$ with $\mathbf{P}$ projecting onto a line inside $\mathcal{S} = \mathcal{C}(\mathbf{Z}) \cap \mathbf{1}^\perp$. Let $\mathbf{P}_{\mathcal{S}} = \mathbf{M} - n^{-1}\mathbf{1}\mathbf{1}^\top$. Because the line lies in $\mathcal{S}$, $\mathbf{P}\mathbf{y} = \mathbf{P}\mathbf{P}_{\mathcal{S}}\mathbf{y}$ ([Theorem 6.5.1\(a\)](#)), and so $\|\mathbf{P}\mathbf{y}\|^2 \leq \|\mathbf{P}_{\mathcal{S}}\mathbf{y}\|^2 = \sum_k n_k (\bar{y}_k - \bar{y})^2$. Equality holds when the line contains $\mathbf{P}_{\mathcal{S}}\mathbf{y}$, whose entries are $\bar{y}_k - \bar{y}$ at level k . That vector is $\mathbf{u}_{\mathbf{c}}$ for $c_k = n_k(\bar{y}_k - \bar{y})$, and this $\mathbf{c}$ is a contrast because $\sum_k n_k (\bar{y}_k - \bar{y}) = 0$.

Exercise B2.

In the additive two-way model with n_{ij} observations in cell (i, j) , show that $E(\bar{y}_{i..}) = \mu + \alpha_i + \sum_j n_{ij}\beta_j/n_{i..}$. Deduce that $\sum_i c_i \bar{y}_{i..}$ is unbiased for every row contrast and every β iff the proportions $n_{ij}/n_{i..}$ do not depend on i , that is, iff the cell counts are *proportional*, $n_{ij} = n_{i..}n_{.j}/n$.

8.7.7.3 C. Going deeper

Exercise C1.

Let $\mathbf{x}$ be the n -vector giving each observation's score s_k . Show that the sum of squares of the weighted linear contrast $\mathbf{c}_1 = \mathbf{D}\mathbf{p}_1$ equals the regression sum of squares $\|(\mathbf{M}_1 - n^{-1}\mathbf{1}\mathbf{1}^\top)\mathbf{y}\|^2$ of the simple regression of $\mathbf{y}$ on $\mathbf{x}$ with intercept, where $\mathbf{M}_1$ projects onto

$\mathcal{C}([\mathbf{1}, \mathbf{x}])$. Conclude that the remaining weighted contrasts together measure the lack of fit of the straight line to the level means.

Solution. In the weighted inner product, $\mathbf{p}_1$ is $\mathbf{s}$ minus its weighted mean, $\mathbf{s} - \bar{s}_w \mathbf{1}$ with $\bar{s}_w = \sum_k n_k s_k / n$, up to scale. Then $\mathbf{u}_{\mathbf{c}_1} = \mathbf{ZD}^{-1}\mathbf{Dp}_1 = \mathbf{Zp}_1$, whose entries are $s_k - \bar{s}_w$. That is $\mathbf{x} - \bar{x}\mathbf{1}$, the centred score vector. By Proposition 8.7.2(b), $SS(\mathbf{c}_1)$ is the squared length of the projection of $\mathbf{y}$ onto the line spanned by $\mathbf{x} - \bar{x}\mathbf{1}$. By Example (Intercept inside simple regression) this line is $\mathcal{C}([\mathbf{1}, \mathbf{x}]) \cap \mathbf{1}^\perp$, and the projection onto it is $\mathbf{M}_1 - n^{-1}\mathbf{1}\mathbf{1}^\top$. The remaining weighted contrasts span the orthogonal complement of that line in $\mathcal{C}(\mathbf{Z}) \cap \mathbf{1}^\perp$, which is $\mathcal{C}(\mathbf{Z}) \cap \mathcal{C}([\mathbf{1}, \mathbf{x}])^\perp$. Their total sum of squares is $\|(\mathbf{M} - \mathbf{M}_1)\mathbf{y}\|^2$, the reduction in residual sum of squares from the straight-line model to the model with a separate mean for each level.

8.8 Summary and Notes

8.8.1 Summary

Idea.

1. The data determine the mean vector $\mathbf{X}\boldsymbol{\beta}$ and nothing more. A function of $\boldsymbol{\beta}$ is identifiable iff it is a function of $\mathbf{X}\boldsymbol{\beta}$, and $\boldsymbol{\beta}$ itself is identifiable iff $\text{rank}(\mathbf{X}) = p$ (Definition 8.1.1, Theorem 8.1.1 and Proposition 8.1.2).
2. A single coefficient is identifiable iff its column is not a combination of the others (Proposition 8.1.3). Rank deficiency comes from construction, from identities among variables (age, period and cohort), or from the design (Example (Age, period and cohort)).
3. A linear function $\boldsymbol{\lambda}^\top \boldsymbol{\beta}$ has a linear unbiased estimator iff it is identifiable iff $\boldsymbol{\lambda} \in \mathcal{C}(\mathbf{X}^\top)$ (Definition 8.2.1 and Theorem 8.2.2). The estimable functions form a space of dimension $\text{rank}(\mathbf{X})$ (Corollary 8.2.3).
4. Eight equivalent tests of estimability, including $\boldsymbol{\lambda} \perp \mathcal{N}(\mathbf{X})$, $\boldsymbol{\lambda}^\top \mathbf{G}\mathbf{X}^\top \mathbf{X} = \boldsymbol{\lambda}^\top$, and invariance over least squares solutions (Theorem 8.3.1). The estimate is $\boldsymbol{\lambda}^\top \hat{\boldsymbol{\beta}}$ for any solution, with variance $\sigma^2 \boldsymbol{\lambda}^\top \mathbf{G}\boldsymbol{\lambda}$ for any generalized inverse (Proposition 8.3.2).
5. For a nonestimable $\boldsymbol{\lambda}$, a generalized-inverse solution estimates a different, estimable function $\boldsymbol{\lambda}^\top \mathbf{G}\mathbf{X}^\top \mathbf{X}\boldsymbol{\beta}$, and the reported variance can be any number (Proposition 8.3.3 and Exercise (C1)). Decide estimability with the SVD of $\mathbf{X}$, and treat near-nonestimable functions as unestimated (Example (Estimable in principle, not in practice)).
6. Full-rank reparameterizations are choices of basis for the estimable space (Theorem 8.4.1). Side conditions $\mathbf{T}\mathbf{b} = \mathbf{0}$ select a unique solution iff $\text{rank}(\mathbf{T}) = p - r$ and no combination of them is estimable (Theorem 8.4.2). The constrained coefficients are then particular estimable functions (Proposition 8.4.3). The minimum-norm solution is itself a side condition (Proposition 8.4.4).
7. A coding of a factor is a nonsingular matrix $\mathbf{K}$, and its coefficients estimate $\mathbf{K}^{-1}\boldsymbol{\mu}$, the rows of $\mathbf{K}^{-1}$ applied to the level means (Definition 8.5.1 and Proposition 8.5.1).
8. In the additive two-way model the rank is $a + b$ minus the number of connected components of the design, and a difference of levels is estimable iff the levels are

connected ([Theorem 8.6.1](#)). In the interaction model, exactly the combinations of occupied cell means are estimable ([Proposition 8.6.2](#)).

9. With a covariate, adjusted differences between levels subtract the slope times the difference in covariate means, the slope being estimated within levels ([Proposition 8.6.3](#)).
10. Contrasts are the estimable functions of a factor's effects ([Definition 8.7.1](#) and [Theorem 8.7.1](#)). Orthogonal contrasts, in the design inner product $\sum_k c_k d_k / n_k$, decompose the between-level sum of squares ([Proposition 8.7.2](#)). Coding coefficients equal scaled contrasts only for orthogonal codings ([Proposition 8.7.3](#)).

8.8.2 Notes and sources

Identifiability and estimability. Christensen (2020, §2.1) develops estimability from identifiability, as this chapter does, and stresses that the estimable functions are exactly the identifiable linear ones. Agresti (2015, §1.4) gives a short treatment aimed at generalized linear models, where the same definitions apply to the linear predictor. Rencher and Schaalje (2008, ch. 12) is a detailed textbook account of the non-full-rank model: estimable functions and their tests, the estimators $\mathbf{r}^\top \mathbf{X}^\top \mathbf{y}$ and $\boldsymbol{\lambda}^\top \hat{\boldsymbol{\beta}}$, reparameterization and side conditions, with small worked layouts. The notion of an estimable function goes back to R. C. Bose's work on the design of experiments in the 1940s. Scheffé (1959) and Searle (1971) made it standard, and Rao (1973) treats it through generalized inverses. The characterization [Theorem 8.3.1](#) collects conditions that are scattered through these sources. That a generalized-inverse solution estimates the estimable function $\boldsymbol{\lambda}^\top \mathbf{G} \mathbf{X}^\top \mathbf{X} \boldsymbol{\beta}$ ([Proposition 8.3.3](#)) follows at once from $E(\mathbf{G} \mathbf{X}^\top \mathbf{Y}) = \mathbf{G} \mathbf{X}^\top \mathbf{X} \boldsymbol{\beta}$, but its consequence for software output deserves more emphasis than it usually gets.

Side conditions and software. The exact conditions of [Theorem 8.4.2](#), and the formula $(\mathbf{X}^\top \mathbf{X} + \mathbf{T}^\top \mathbf{T})^{-1} \mathbf{X}^\top \mathbf{y}$, are standard (see Rencher and Schaalje 2008, §12.6, and Searle 1971). The coding matrices, called “contrasts”, that S and R use for factors are described by Chambers and Hastie (1992), and Venables and Ripley (2002) discuss the treatment, sum, Helmert and polynomial codings of R. The minimum-norm solution rests on the Moore–Penrose inverse (Penrose 1955), and column-pivoted QR is described in Golub and Van Loan (2013).

Indicator variables. Sen and Srivastava (1990, ch. 4) is a practical account of indicator variables: the two-sample comparison as a regression, polychotomous factors, hierarchical codings, factors with continuous covariates, broken-line regression and aggregated indicators as responses. Where this chapter meets those themes it takes its own route: the two-sample comparison through the coding proposition ([Exercise \(A1\)](#)), the indicator of a single observation through the Frisch–Waugh–Lovell theorem ([Exercise \(B1\)](#)), and count regressors without intercept in a new setting ([Exercise \(B3\)](#)).

Connectedness, empty cells and cohorts. Connectedness of two-way designs, and its role in estimability, is treated at length by Searle (1971, 1987). Searle (1987) is also the standard reference for unbalanced data and empty cells, which [Chapter 17](#) takes up. The age–period–cohort problem of [Example \(Age, period and cohort\)](#) is a well-known instance of a dependency among variables. Holford (1983) showed which functions of the age, period and cohort effects are estimable, among them the second differences of [Exercise \(C1\)](#).

Data. The survey examples use the extract of the 1996 American National Election Study distributed with statsmodels (Seabold and Perktold 2010) as a public-domain data set. The other examples use simulated data, generated by the scripts in `code/ch08/`.

8.8.3 References

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Chapter 9

Sums of Squares and Orthogonal Decomposition

An analysis of variance table looks like bookkeeping. Behind every row there is a subspace, and behind the whole table an orthogonal decomposition of the space in which the data live: sums of squares are squared lengths of projections, and degrees of freedom are dimensions. Sequential and partial sums of squares, and the Type I, II and III sums of squares printed by software, differ only in which subspaces they project onto, and so in which hypotheses they address. When the subspaces are orthogonal, as in balanced designs and with orthogonal polynomials, the distinctions disappear. The coefficient of determination is a ratio of two of these squared lengths, and we mark what it does not measure. Finally, the expectation of each sum of squares shows what an F ratio compares, preparing the tests of Part III.

What you need. Chapter 6 throughout, especially projections and their ranks ([Theorem 6.3.8](#), [Proposition 6.3.4](#)), nested models ([Theorem 6.5.1](#), [Theorem 6.5.2](#)), the Frisch–Waugh–Lovell theorem ([Theorem 6.6.2](#)) and R^2 as a squared cosine ([Theorem 6.7.2](#)). The linear model and the moments of least squares estimates from [Chapter 5](#) ([Definition 5.3.1](#), [Theorem 5.5.1](#)). The mean of a quadratic form ([Theorem 2.3.1](#)) and the distribution theory of [Chapter 4](#) ([Theorem 4.4.4](#), [Theorem 4.5.1](#)). Estimable functions and contrasts from [Chapter 8](#) appear in [Sections 9.2 to 9.4](#).

9.0.1 Roadmap

- [Decomposing the total sum of squares](#)
- [Extra sums of squares](#)
- [Sequential and partial sums of squares](#)
- [Orthogonal designs](#)
- [The coefficient of determination and its limits](#)
- [Expected mean squares](#)
- [Summary and notes](#)

9.1 Decomposing the total sum of squares

Every sum of squares in a linear model is the squared length of a projection of the data. Chapter 6 showed this for the residual sum of squares, $\|(\mathbf{I} - \mathbf{M})\mathbf{y}\|^2$, and for the drop in residual sum of squares between nested models, $\|(\mathbf{M} - \mathbf{M}_0)\mathbf{y}\|^2$ (Eq. 6.5.2). This section turns these observations into a general framework. An analysis of variance is a choice of mutually orthogonal subspaces that together fill the model space. Once that choice is made, the whole table follows from Pythagoras: the sums of squares add up because the pieces are orthogonal, and the degrees of freedom add up because dimensions of orthogonal subspaces add.

Throughout the chapter the model is $E(\mathbf{Y}) = \mathbf{X}\boldsymbol{\beta}$ with $\text{Cov}(\mathbf{Y}) = \sigma^2\mathbf{I}$ (Definition 5.3.1). $\mathbf{X}$ is $n \times p$ of rank r , which may be less than p , and $\mathbf{M}$ is the orthogonal projection onto $\mathcal{C}(\mathbf{X})$. The identities of the chapter hold for every data vector $\mathbf{y}$. Probability enters only where a covariance, an expectation or a distribution is stated, and systematically in Section 9.6.

9.1.1 Orthogonal decompositions of the model space

Definition 9.1.1 — Analysis of variance decomposition

An **analysis of variance decomposition** of the model space is a list of subspaces $\mathcal{V}_1, \dots, \mathcal{V}_k$ of $\mathcal{C}(\mathbf{X})$ that are mutually orthogonal and whose sum is $\mathcal{C}(\mathbf{X})$:

$$\mathcal{C}(\mathbf{X}) = \mathcal{V}_1 \oplus \mathcal{V}_2 \oplus \dots \oplus \mathcal{V}_k, \quad \mathcal{V}_i \perp \mathcal{V}_j \ (i \neq j).$$

The subspace $\mathcal{V}_{k+1} = \mathcal{C}(\mathbf{X})^\perp$ is the **error space**. Writing $\mathbf{P}_i$ for the orthogonal projection onto $\mathcal{V}_i$, the **sum of squares** for $\mathcal{V}_i$ is $\|\mathbf{P}_i\mathbf{y}\|^2 = \mathbf{y}^\top \mathbf{P}_i \mathbf{y}$, its **degrees of freedom** are $r_i = \dim \mathcal{V}_i$, and its **mean square** is $\|\mathbf{P}_i\mathbf{y}\|^2/r_i$.

The same space can be decomposed in many ways, and the choice is the statistician's: the constants, a regressor of interest, a block of nuisance regressors, the linear and quadratic parts of a trend. The theorem collects what is true of all decompositions.

Theorem 9.1.1 — The analysis of variance identity

Let $\mathcal{V}_1, \dots, \mathcal{V}_k$ be an analysis of variance decomposition of $\mathcal{C}(\mathbf{X})$, with projections $\mathbf{P}_1, \dots, \mathbf{P}_k$ and dimensions $r_1, \dots, r_k$. Then:

- (a) $\mathbf{P}_i\mathbf{P}_j = \mathbf{0}$ for $i \neq j$, $\mathbf{P}_1 + \dots + \mathbf{P}_k = \mathbf{M}$, and

$$\mathbf{I} = \mathbf{P}_1 + \mathbf{P}_2 + \dots + \mathbf{P}_k + (\mathbf{I} - \mathbf{M}),$$

a sum of $k + 1$ mutually orthogonal projections;

- (b) for every $\mathbf{y} \in \mathbb{R}^n$ the vectors $\mathbf{P}_1\mathbf{y}, \dots, \mathbf{P}_k\mathbf{y}, (\mathbf{I} - \mathbf{M})\mathbf{y}$ are mutually orthogonal and

$$\|\mathbf{y}\|^2 = \|\mathbf{P}_1\mathbf{y}\|^2 + \dots + \|\mathbf{P}_k\mathbf{y}\|^2 + \|(\mathbf{I} - \mathbf{M})\mathbf{y}\|^2; \quad (9.1.1)$$

- (c) $r_i = \text{rank}(\mathbf{P}_i) = \text{tr}(\mathbf{P}_i)$, $r_1 + \dots + r_k = r$, and the error space has dimension $\text{tr}(\mathbf{I} - \mathbf{M}) = n - r$;

- (d) if the columns of $\mathbf{Q}_i$ ($n \times r_i$) form an orthonormal basis of $\mathcal{V}_i$, then $\|\mathbf{P}_i \mathbf{y}\|^2 = \|\mathbf{Q}_i^\top \mathbf{y}\|^2$ is a sum of r_i squares, and together with an orthonormal basis $\mathbf{Q}_{k+1}$ of the error space the matrix $\mathbf{U} = [\mathbf{Q}_1, \dots, \mathbf{Q}_{k+1}]$ is orthogonal;
- (e) the subspaces $\mathcal{W}_j = \mathcal{V}_1 \oplus \dots \oplus \mathcal{V}_j$ form a chain $\mathcal{W}_1 \subseteq \dots \subseteq \mathcal{W}_k = \mathcal{C}(\mathbf{X})$ with projections $\mathbf{N}_j = \mathbf{P}_1 + \dots + \mathbf{P}_j$, and $\mathbf{P}_j = \mathbf{N}_j - \mathbf{N}_{j-1}$ (with $\mathbf{N}_0 = \mathbf{0}$). Conversely, every chain $\{\mathbf{0}\} = \mathcal{W}_0 \subseteq \mathcal{W}_1 \subseteq \dots \subseteq \mathcal{W}_k = \mathcal{C}(\mathbf{X})$ gives an analysis of variance decomposition with $\mathcal{V}_j = \mathcal{W}_j^\perp \cap \mathcal{W}_{j-1}$.

Proof.

- (a) For any $\mathbf{u}$, $\mathbf{P}_j \mathbf{u} \in \mathcal{V}_j \subseteq \mathcal{V}_i^\perp$, so $\mathbf{P}_i \mathbf{P}_j \mathbf{u} = \mathbf{0}$ by [Proposition 6.3.4\(c\)](#). Applying [Theorem 6.3.8](#) repeatedly, the sum $\mathbf{P}_1 + \dots + \mathbf{P}_k$ is the orthogonal projection onto $\mathcal{V}_1 \oplus \dots \oplus \mathcal{V}_k = \mathcal{C}(\mathbf{X})$, and a projection is determined by its range, so the sum is $\mathbf{M}$. Each $\mathbf{P}_i$ satisfies $\mathbf{M} \mathbf{P}_i = \mathbf{P}_i$, because its columns lie in $\mathcal{C}(\mathbf{X})$, and therefore $(\mathbf{I} - \mathbf{M}) \mathbf{P}_i = \mathbf{0}$. The projection $\mathbf{I} - \mathbf{M}$ onto $\mathcal{C}(\mathbf{X})^\perp$ ([Theorem 6.3.3](#)) is thus orthogonal to every $\mathbf{P}_i$.
- (b) Multiply $\mathbf{I} = \sum_i \mathbf{P}_i + (\mathbf{I} - \mathbf{M})$ by $\mathbf{y}$. For two different pieces, $(\mathbf{P}_i \mathbf{y})^\top (\mathbf{P}_j \mathbf{y}) = \mathbf{y}^\top \mathbf{P}_i \mathbf{P}_j \mathbf{y} = 0$, using symmetry and (a). Pythagoras gives the identity.
- (c) Rank equals trace for projections ([Proposition 6.3.4\(b\)](#)), and the trace is additive, so $\sum_i r_i = \text{tr}(\mathbf{M}) = r$ by [Theorem 6.3.7](#), and $\text{tr}(\mathbf{I} - \mathbf{M}) = n - r$.
- (d) By [Proposition 6.3.2](#), $\mathbf{P}_i = \mathbf{Q}_i \mathbf{Q}_i^\top$, so $\mathbf{y}^\top \mathbf{P}_i \mathbf{y} = \|\mathbf{Q}_i^\top \mathbf{y}\|^2$. Columns taken from different $\mathbf{Q}_i$ are orthogonal because the subspaces are, so $\mathbf{U}$ has orthonormal columns. It has $r_1 + \dots + r_k + (n - r) = n$ of them, so it is square and hence orthogonal.
- (e) The $\mathbf{N}_j$ are projections onto $\mathcal{W}_j$ by [Theorem 6.3.8](#), and the chain is nested by construction. Conversely, given a chain, [Theorem 6.5.1\(b\)](#) shows that $\mathbf{N}_j - \mathbf{N}_{j-1}$ is the projection onto $\mathcal{V}_j = \mathcal{W}_j^\perp \cap \mathcal{W}_{j-1}$. For $i < j$ we have $\mathcal{V}_i \subseteq \mathcal{W}_i \subseteq \mathcal{W}_{j-1}$, which is orthogonal to $\mathcal{V}_j$. The projections telescope to $\mathbf{N}_k = \mathbf{M}$, so by [Theorem 6.3.8](#) the orthogonal sum of the $\mathcal{V}_j$ is $\mathcal{C}(\mathbf{M}) = \mathcal{C}(\mathbf{X})$.

■

Part (e) says that decompositions and chains are the same thing. Listing the rows of an analysis of variance table from the top is the same as fitting a sequence of ever larger models and recording what each step adds. [Section 9.3](#) builds on this.

9.1.2 Degrees of freedom are dimensions

“Degrees of freedom” can seem like a rule of thumb (“subtract one for every parameter estimated”). Part (d) of [Theorem 9.1.1](#) gives the phrase an exact meaning. Rotate the data into the orthonormal coordinates $\mathbf{z} = \mathbf{U}^\top \mathbf{y}$. The rotation preserves length, and it splits $\mathbf{z}$ into consecutive blocks $\mathbf{Q}_1^\top \mathbf{y}, \dots, \mathbf{Q}_{k+1}^\top \mathbf{y}$ of sizes $r_1, \dots, r_k, n - r$. The sum of squares for $\mathcal{V}_i$ is the sum of the squares of the r_i coordinates in its block. *A sum of squares has r_i degrees of freedom because it is a sum of r_i squares of orthonormal linear combinations of the data, and it cannot be written with fewer.*

The rotation also explains why degrees of freedom govern the size of a sum of squares. If $\text{Cov}(\mathbf{Y}) = \sigma^2 \mathbf{I}$, the coordinates $\mathbf{U}^\top \mathbf{Y}$ are uncorrelated with variance σ^2 ([Theorem 2.2.1](#)), so a block of r_i of them has expected sum of squares $r_i \sigma^2$ plus the squared length of its mean

(Section 9.6). The divisor $n - 1$ of the sample variance is the case $\mathbb{R}^n = \text{span}(\mathbf{1}) \oplus \mathbf{1}^\perp$: the n deviations $y_i - \bar{y}$ lie in an $(n - 1)$ -dimensional subspace, and the Helmert matrix of [Example \(Helmert matrices\)](#) writes their sum of squares as $n - 1$ squared orthonormal contrasts.

9.1.3 The regression analysis of variance table

The most common decomposition comes from the chain $\text{span}(\mathbf{1}) \subseteq \mathcal{C}(\mathbf{X})$, available whenever $\mathbf{1} \in \mathcal{C}(\mathbf{X})$, which is the case for every model with an intercept. Write $\mathbf{P}_1 = n^{-1}\mathbf{1}\mathbf{1}^\top$ for the projection onto the constant vectors ([Example \(Projections onto a line\)](#)). The chain gives the three-way decomposition of [Eq. 6.5.1](#),

$$\mathbf{I} = \mathbf{P}_1 + (\mathbf{M} - \mathbf{P}_1) + (\mathbf{I} - \mathbf{M}),$$

with ranks 1, $r - 1$ and $n - r$. The squared lengths are

$$\|\mathbf{P}_1\mathbf{y}\|^2 = n\bar{y}^2, \quad \text{SSR} = \|(\mathbf{M} - \mathbf{P}_1)\mathbf{y}\|^2 = \sum_i (\hat{y}_i - \bar{y})^2, \quad \text{SSE} = \|(\mathbf{I} - \mathbf{M})\mathbf{y}\|^2 = \sum_i (y_i - \hat{y}_i)^2.$$

They are usually arranged as follows.

Source	df	Sum of squares	Mean square
Mean	1	$n\bar{y}^2$	$n\bar{y}^2$
Regression (after the mean)	$r - 1$	SSR	$\text{SSR}/(r - 1)$
Residual	$n - r$	SSE	$s^2 = \text{SSE}/(n - r)$
Total	n	$\mathbf{y}^\top \mathbf{y}$	

This is the **uncorrected** table. Its total is the raw sum of squares $\sum_i y_i^2$, with n degrees of freedom. The row for the mean is rarely of interest, because it records only that the average response is not zero. Deleting it gives the **corrected** table, whose total is $\text{SST} = \sum_i (y_i - \bar{y})^2 = \|(\mathbf{I} - \mathbf{P}_1)\mathbf{y}\|^2$ on $n - 1$ degrees of freedom. The corrected identity $\text{SST} = \text{SSR} + \text{SSE}$ is [Eq. 6.7.1](#). “Correcting for the mean” is thus a projection. We first project the data onto $\mathbf{1}^\perp$ and then decompose what is left.

The ratio of the two mean squares,

$$F = \frac{\text{SSR}/(r - 1)}{\text{SSE}/(n - r)},$$

compares the variation along the regression directions with the variation in the error space. Under normality and when no regressor matters, it has the $F(r - 1, n - r)$ distribution ([Theorem 4.4.6](#)). [Section 9.6](#) explains why this ratio is the right comparison, and [Chapter 11](#) develops the test.

A few formulas make the table easy to compute from any least squares solution.

Proposition 9.1.2 — Computing the sums of squares

Let $\hat{\boldsymbol{\beta}}$ be any least squares estimate. Then:

- (a) $\|\mathbf{M}\mathbf{y}\|^2 = \hat{\boldsymbol{\beta}}^\top \mathbf{X}^\top \mathbf{y}$ and $\text{SSE} = \mathbf{y}^\top \mathbf{y} - \hat{\boldsymbol{\beta}}^\top \mathbf{X}^\top \mathbf{y}$;
- (b) if $\mathbf{1} \in \mathcal{C}(\mathbf{X})$, then $\text{SSR} = \hat{\boldsymbol{\beta}}^\top \mathbf{X}^\top \mathbf{y} - n\bar{y}^2$;

- (c) if $\mathbf{X} = [\mathbf{1}, \mathbf{X}_1]$ has full column rank and $\tilde{\mathbf{X}}_1 = (\mathbf{I} - \mathbf{P}_1)\mathbf{X}_1$ is the matrix of centred regressors, then $\text{SSR} = \hat{\boldsymbol{\beta}}_1^\top \tilde{\mathbf{X}}_1^\top \tilde{\mathbf{X}}_1 \hat{\boldsymbol{\beta}}_1 = \hat{\boldsymbol{\beta}}_1^\top \tilde{\mathbf{X}}_1^\top \mathbf{y}$, where $\hat{\boldsymbol{\beta}}_1$ holds the slopes.

Proof.

- (a) $\mathbf{X}\hat{\boldsymbol{\beta}} = \mathbf{M}\mathbf{y}$ (Theorem 6.4.1), so $\|\mathbf{M}\mathbf{y}\|^2 = \mathbf{y}^\top \mathbf{M}\mathbf{y} = \mathbf{y}^\top \mathbf{X}\hat{\boldsymbol{\beta}} = \hat{\boldsymbol{\beta}}^\top \mathbf{X}^\top \mathbf{y}$. The second formula follows from $\text{SSE} = \mathbf{y}^\top \mathbf{y} - \mathbf{y}^\top \mathbf{M}\mathbf{y}$ (Proposition 6.4.3). (b) $\mathbf{M} - \mathbf{P}_1$ is a projection (Theorem 6.5.1), so $\text{SSR} = \mathbf{y}^\top \mathbf{M}\mathbf{y} - \mathbf{y}^\top \mathbf{P}_1 \mathbf{y} = \|\mathbf{M}\mathbf{y}\|^2 - n\bar{y}^2$; now use (a). (c) By Lemma 6.6.1 with $\mathbf{X}_2 = \mathbf{1}$, $\mathbf{M} - \mathbf{P}_1$ is the projection onto $\mathcal{C}(\tilde{\mathbf{X}}_1)$, and by Theorem 6.6.2 the slopes are the coefficients of $(\mathbf{M} - \mathbf{P}_1)\mathbf{y}$ in terms of the columns of $\tilde{\mathbf{X}}_1$. So $(\mathbf{M} - \mathbf{P}_1)\mathbf{y} = \tilde{\mathbf{X}}_1 \hat{\boldsymbol{\beta}}_1$ and $\text{SSR} = \|\tilde{\mathbf{X}}_1 \hat{\boldsymbol{\beta}}_1\|^2$. The last equality uses $\tilde{\mathbf{X}}_1^\top \mathbf{y} = \tilde{\mathbf{X}}_1^\top (\mathbf{M} - \mathbf{P}_1)\mathbf{y} = \tilde{\mathbf{X}}_1^\top \tilde{\mathbf{X}}_1 \hat{\boldsymbol{\beta}}_1$. ■

In (a), $\hat{\boldsymbol{\beta}}^\top \mathbf{X}^\top \mathbf{y}$ is the same for every solution of the normal equations, even though $\hat{\boldsymbol{\beta}}$ is not unique when $r < p$. That is no accident. It equals $\|\mathbf{M}\mathbf{y}\|^2$, which depends only on the model space.

Example. (An analysis of variance table for murder rates)

Return to the regression of the 2009 murder rate of the 50 US states on poverty, single parenthood and urbanization, with an intercept (Example (Poverty and murder rates)). Here $r = 4$ and $\bar{y} = 4.514$. The listing computes every sum of squares from the coordinates of $\mathbf{y}$ in an orthonormal basis of $\mathcal{C}(\mathbf{X})$, as in Theorem 9.1.1(d): the first coordinate gives the mean, the next three the regression.

Source	df	Sum of squares	Mean square
Mean	1	1018.81	1018.81
Regression	3	182.80	60.93
Residual	46	101.96	2.217
Total (uncorrected)	50	1303.57	
Total (corrected)	49	284.76	

The regression mean square is 27.49 times the residual mean square. If none of the three regressors were related to the murder rate and the errors were normal, a ratio this large would occur with probability about 2.5×10^{-10} . The proportion of the corrected total that the regression accounts for is $182.80/284.76 = 0.6419$, the R^2 of Theorem 6.7.2.

```
import numpy as np
import statsmodels.api as sm
from scipy import stats
data = sm.datasets.statelcrime.load_pandas().data.drop(index="District of Columbia")

y = data["murder"].to_numpy()
n = len(y)
X = np.column_stack([np.ones(n), data["poverty"], data["single"], data["urban"]])
r = np.linalg.matrix_rank(X)

Q, _ = np.linalg.qr(X)                                # orthonormal basis of C(X); first column
```

```

    spans 1
z = Q.T @ y                                # coordinates of My in that basis
ss_mean = z[0] ** 2                        # n * ybar^2
ss_reg = np.sum(z[1:] ** 2)                # ||(M - P1) y||^2, r - 1 squared coordinates
ss_res = y @ y - np.sum(z ** 2)           # ||(I - M) y||^2

rows = [("mean", 1, ss_mean), ("regression", r - 1, ss_reg),
        ("residual", n - r, ss_res), ("total", n, y @ y)]
for name, df, ss in rows:
    print(f"{name:11s} df = {df:3d}    SS = {ss:9.2f}    MS = {ss / df:8.3f}")
F = (ss_reg / (r - 1)) / (ss_res / (n - r))
print(f"corrected total = {ss_reg + ss_res:.2f},    F = {F:.2f}")

```

The QR factorization supplies the orthonormal basis of [Theorem 9.1.1\(d\)](#); its first column is $\pm \mathbf{1}/\sqrt{n}$ because Gram–Schmidt starts with the intercept ([Proposition 6.2.2](#)). No $n \times n$ projection matrix is formed.

9.1.4 Models without an intercept

The corrected table depends on $\mathbf{1} \in \mathcal{C}(\mathbf{X})$. Without it, the chain $\text{span}(\mathbf{1}) \subseteq \mathcal{C}(\mathbf{X})$ is not available, and the corrected “identity” fails.

Proposition 9.1.3

For any model matrix $\mathbf{X}$, with $\hat{\mathbf{Y}} = \mathbf{M}\mathbf{y}$ and $\hat{\mathbf{e}} = (\mathbf{I} - \mathbf{M})\mathbf{y}$,

$$\sum_i (y_i - \bar{y})^2 = \sum_i (\hat{y}_i - \bar{y})^2 + \sum_i \hat{e}_i^2 - 2\bar{y} \sum_i \hat{e}_i.$$

The last term vanishes for every $\mathbf{y}$ iff $\mathbf{1} \in \mathcal{C}(\mathbf{X})$.

Proof. Write $\mathbf{y} - \bar{y}\mathbf{1} = (\hat{\mathbf{Y}} - \bar{y}\mathbf{1}) + \hat{\mathbf{e}}$ and expand the squared length. The cross term is $2(\hat{\mathbf{Y}} - \bar{y}\mathbf{1})^\top \hat{\mathbf{e}} = 2\hat{\mathbf{Y}}^\top \hat{\mathbf{e}} - 2\bar{y}\mathbf{1}^\top \hat{\mathbf{e}} = -2\bar{y}\mathbf{1}^\top \hat{\mathbf{e}}$, since $\hat{\mathbf{Y}} \perp \hat{\mathbf{e}}$. If $\mathbf{1} \in \mathcal{C}(\mathbf{X})$, then $\mathbf{1}^\top \hat{\mathbf{e}} = 0$ ([Proposition 6.4.3\(b\)](#)). If $\mathbf{1} \notin \mathcal{C}(\mathbf{X})$, take $\mathbf{y} = \mathbf{1}$. Then $\bar{y} = 1$ and $\mathbf{1}^\top \hat{\mathbf{e}} = \mathbf{1}^\top (\mathbf{I} - \mathbf{M})\mathbf{1} = \|(\mathbf{I} - \mathbf{M})\mathbf{1}\|^2 > 0$, so the last term is not zero. ■

Without an intercept the valid decomposition is the uncorrected one, $\|\mathbf{y}\|^2 = \|\mathbf{M}\mathbf{y}\|^2 + \text{SSE}$, whose regression row has r degrees of freedom. That is appropriate only when the mean response is known to vanish when every regressor is zero.

Example. (Forcing the murder regression through the origin)

Refit the model of [Example \(An analysis of variance table for murder rates\)](#) without the intercept. The residual sum of squares rises from 101.96 to 177.73, so the fit is much worse. Yet the residuals now sum to -8.332 , and the would-be corrected identity is off by $-2\bar{y} \sum_i \hat{e}_i = 75.22$: the corrected total 284.76 is not $31.81 + 177.73 = 209.54$. The uncentred ratio $\|\hat{\mathbf{Y}}\|^2/\|\mathbf{y}\|^2$, which some software reports as R^2 for such models, is 0.8637, much larger than the 0.6419 of the better model with an intercept ([Exercise](#)

(B1)).

```

X0 = X[:, 1:]                                # drop the intercept column
b0, *_ = np.linalg.lstsq(X0, y, rcond=None)
e0 = y - X0 @ b0
ybar = y.mean()
sst = np.sum((y - ybar) ** 2)
ssr_star = np.sum((X0 @ b0 - ybar) ** 2)
sse0 = e0 @ e0
print(f"SST = {sst:.2f} but SSR* + SSE = {ssr_star + sse0:.2f}")
print(f"sum of residuals = {e0.sum():.3f}, -2 ybar * sum = {-2 * ybar * e0.sum():.2f}")

```

9.1.5 Coarsening and refining

Coarsening merges rows: replacing $\mathcal{V}_1$ and $\mathcal{V}_2$ by $\mathcal{V}_1 \oplus \mathcal{V}_2$ gives another decomposition, whose sums of squares and degrees of freedom are sums of the old ones. The regression row is a coarsening of the table that gives each regressor its own row (Section 9.3). **Refining** splits a row: any subspace of dimension r_i can be cut into r_i orthogonal lines, each carrying one degree of freedom. A refinement is useful only when its lines answer questions of interest. Section 9.4 chooses them with orthogonal contrasts and orthogonal polynomials.

Idea. An analysis of variance table is a list of mutually orthogonal subspaces that fill $\mathbb{R}^n$. Its sums of squares are the squared lengths of the projections of $\mathbf{y}$, its degrees of freedom are the dimensions, and both columns add up because of Pythagoras.

9.1.6 Exercises

9.1.6.1 A. Check your understanding

Exercise A1.

For simple regression, $\mathbf{X} = [\mathbf{1}, \mathbf{x}]$ with $\mathbf{x}$ not constant, show that $\text{SSR} = \hat{\beta}_1^2 S_{xx} = S_{xy}^2 / S_{xx}$ and that the F ratio of the corrected table equals t^2 , where $t = \hat{\beta}_1 \sqrt{S_{xx}} / s$.

Exercise A2.

A regression with an intercept has a corrected table in which the regression row has 3 degrees of freedom and sum of squares 120, the residual mean square is 2.5, and the corrected total sum of squares is 172.5. Find n , the residual degrees of freedom, the F ratio and R^2 .

9.1.6.2 B. Practice

Exercise B1.

Let $\mathbf{U} = [\mathbf{Q}_1, \dots, \mathbf{Q}_{k+1}]$ be the orthogonal matrix of Theorem 9.1.1(d) and $\mathbf{z} = \mathbf{U}^\top \mathbf{Y}$. Suppose $E(\mathbf{Y}) = \boldsymbol{\mu} \in \mathcal{C}(\mathbf{X})$ and $\text{Cov}(\mathbf{Y}) = \sigma^2 \mathbf{I}$. Show that the coordinates of $\mathbf{z}$ are

uncorrelated with variance σ^2 , that the last $n - r$ of them have mean zero, and deduce $E(\text{SSE}) = (n - r)\sigma^2$. Show that if in addition $\mathbf{Y}$ is normal, the coordinates are independent.

Solution. By [Theorem 2.2.1](#), $\text{Cov}(\mathbf{z}) = \mathbf{U}^\top(\sigma^2\mathbf{I})\mathbf{U} = \sigma^2\mathbf{I}$ and $E(\mathbf{z}) = \mathbf{U}^\top\boldsymbol{\mu}$. The last block is $\mathbf{Q}_{k+1}^\top\boldsymbol{\mu}$, and $\boldsymbol{\mu} \in \mathcal{C}(\mathbf{X})$ is orthogonal to the error space, so it is zero. Then $\text{SSE} = \|\mathbf{Q}_{k+1}^\top\mathbf{Y}\|^2$ is a sum of $n - r$ squares of variables with mean zero and variance σ^2 , so its expectation is $(n - r)\sigma^2$. Under normality $\mathbf{z}$ is normal ([Theorem 3.3.1](#)) with diagonal covariance, and uncorrelated jointly normal variables are independent ([Theorem 3.4.1](#)).

Exercise B2.

For a model without an intercept, the quantity $1 - \text{SSE}/\text{SST}$ is sometimes reported as R^2 . Show that it is negative iff the model fits worse than the intercept-only model. Construct a data set with $n = 4$ and one regressor for which it is below -1 .

9.1.6.3 C. Going deeper

Exercise C1.

Let $\mathcal{V}$ be a subspace with projection $\mathbf{P}$. Show that

$$\|\mathbf{Py}\|^2 = \max_{\mathbf{v} \in \mathcal{V}, \mathbf{v} \neq \mathbf{0}} \frac{(\mathbf{v}^\top \mathbf{y})^2}{\mathbf{v}^\top \mathbf{v}},$$

with the maximum attained at $\mathbf{v} = \mathbf{Py}$ when $\mathbf{Py} \neq \mathbf{0}$. Interpret: the sum of squares of a row with r_i degrees of freedom is the largest one-degree-of-freedom sum of squares obtainable by refining that row. [Chapter 13](#) uses this fact to build Scheffé's simultaneous intervals.

Solution. For $\mathbf{v} \in \mathcal{V}$, $\mathbf{v}^\top \mathbf{y} = \mathbf{v}^\top \mathbf{Py}$, so by Cauchy–Schwarz ([Proposition 1.1.1](#)) $(\mathbf{v}^\top \mathbf{y})^2 \leq \mathbf{v}^\top \mathbf{v} \|\mathbf{Py}\|^2$, with equality iff $\mathbf{v}$ is a multiple of $\mathbf{Py}$. Taking $\mathbf{v} = \mathbf{Py} \in \mathcal{V}$ attains the bound. A line $\text{span}(\mathbf{v})$ inside $\mathcal{V}$ carries the one-degree-of-freedom sum of squares $(\mathbf{v}^\top \mathbf{y})^2 / \mathbf{v}^\top \mathbf{v}$ ([Example \(Projections onto a line\)](#)), so no refinement of the row gives a line with a larger sum of squares than the whole row.

9.2 Extra sums of squares

Most questions asked of a regression compare two models. Each question names a reduced model inside the full one, and the natural measure of what the full model adds is how much it lowers the residual sum of squares. This section develops that measure, its computation when regressors are added to a fitted model, and a closed form when the reduced model is defined by linear constraints.

9.2.1 Notation

Definition 9.2.1 — Extra sum of squares

Let $\mathbf{X}_1$ and $\mathbf{X}_2$ be matrices with n rows, let $\mathbf{X} = [\mathbf{X}_1, \mathbf{X}_2]$, and let $\mathbf{M}_1$ and $\mathbf{M}$ be the projections onto $\mathcal{C}(\mathbf{X}_1)$ and $\mathcal{C}(\mathbf{X})$. The **extra sum of squares** of $\mathbf{X}_2$ given $\mathbf{X}_1$ is

$$\text{SS}(\mathbf{X}_2 \mid \mathbf{X}_1) = \mathbf{y}^\top (\mathbf{M} - \mathbf{M}_1) \mathbf{y} = \text{SSE}(\mathbf{X}_1) - \text{SSE}(\mathbf{X}),$$

with $\text{rank}(\mathbf{X}) - \text{rank}(\mathbf{X}_1)$ degrees of freedom. Here $\text{SSE}(\cdot)$ is the residual sum of squares of the model with the given model matrix.

The two expressions agree by [Eq. 6.5.2](#). One also writes $\text{SS}(\boldsymbol{\beta}_2 \mid \boldsymbol{\beta}_1)$, or $\text{SS}(B \mid \mu, A)$ in factor models; some books write R , for “reduction”. Four properties follow from [Theorem 6.5.1](#) and [Theorem 6.7.1](#).

1. *It depends only on the two column spaces.* Rescaling, recoding, centring, replacing $\mathbf{X}_2$ by $\mathbf{X}_2 + \mathbf{X}_1 \mathbf{B}$, and rank deficiency change nothing.
2. *It is nonnegative*, and zero iff $\mathbf{M} \mathbf{y} = \mathbf{M}_1 \mathbf{y}$, that is, iff adding $\mathbf{X}_2$ leaves the fitted values unchanged.
3. *It obeys a chain rule:* $\text{SS}(\mathbf{X}_2, \mathbf{X}_3 \mid \mathbf{X}_1) = \text{SS}(\mathbf{X}_2 \mid \mathbf{X}_1) + \text{SS}(\mathbf{X}_3 \mid \mathbf{X}_1, \mathbf{X}_2)$, because the projections telescope.
4. *It is the sum of squares of a residualized regression.* The next lemma makes this precise.

Lemma 9.2.1 — Extra sums of squares by residualizing

For any $\mathbf{X}_1$ and $\mathbf{X}_2$, let $\tilde{\mathbf{X}}_2 = (\mathbf{I} - \mathbf{M}_1) \mathbf{X}_2$. Then $\mathcal{C}([\mathbf{X}_1, \mathbf{X}_2]) = \mathcal{C}(\mathbf{X}_1) \oplus \mathcal{C}(\tilde{\mathbf{X}}_2)$ with the two summands orthogonal, so $\mathbf{M} = \mathbf{M}_1 + \mathbf{P}_{\mathcal{C}(\tilde{\mathbf{X}}_2)}$ and

$$\text{SS}(\mathbf{X}_2 \mid \mathbf{X}_1) = \|\mathbf{P}_{\mathcal{C}(\tilde{\mathbf{X}}_2)} \mathbf{y}\|^2 = \mathbf{y}^\top \tilde{\mathbf{X}}_2 (\tilde{\mathbf{X}}_2^\top \tilde{\mathbf{X}}_2)^{-1} \tilde{\mathbf{X}}_2^\top \mathbf{y}.$$

Proof. The argument for part (a) of [Lemma 6.6.1](#) uses no rank condition, so it applies here: the columns of $\tilde{\mathbf{X}}_2$ lie in $\mathcal{C}(\mathbf{X})$ and are orthogonal to $\mathcal{C}(\mathbf{X}_1)$, and every $\mathbf{X}_1 \mathbf{b}_1 + \mathbf{X}_2 \mathbf{b}_2$ equals $\tilde{\mathbf{X}}_2 \mathbf{b}_2 + (\mathbf{X}_1 \mathbf{b}_1 + \mathbf{M}_1 \mathbf{X}_2 \mathbf{b}_2)$. By [Theorem 6.3.8](#), $\mathbf{M} = \mathbf{M}_1 + \mathbf{P}_{\mathcal{C}(\tilde{\mathbf{X}}_2)}$. The formula is [Theorem 6.3.7](#) applied to $\tilde{\mathbf{X}}_2$. ■

9.2.2 Adding regressors to a fitted model

Suppose $E(\mathbf{Y}) = \mathbf{X} \boldsymbol{\beta}$ has been fitted and we add the columns of an $n \times t$ matrix $\mathbf{Z}$, giving $E(\mathbf{Y}) = \mathbf{X} \boldsymbol{\beta} + \mathbf{Z} \boldsymbol{\gamma}$ with model matrix $\mathbf{W} = [\mathbf{X}, \mathbf{Z}]$. Only a $t \times t$ matrix has to be inverted.

Theorem 9.2.2 — Adding regressors

Let $\mathbf{W} = [\mathbf{X}, \mathbf{Z}]$ have full column rank $p + t$, let $\mathbf{C} = (\mathbf{X}^\top \mathbf{X})^{-1}$, $\hat{\boldsymbol{\beta}} = \mathbf{C} \mathbf{X}^\top \mathbf{y}$, and $\mathbf{R} = \mathbf{I} - \mathbf{X} \mathbf{C} \mathbf{X}^\top$. Then:

- (a) $\mathbf{Z}^\top \mathbf{R} \mathbf{Z}$ is positive definite;
 (b) the least squares coefficients of the enlarged model are

$$\hat{\gamma} = (\mathbf{Z}^\top \mathbf{R} \mathbf{Z})^{-1} \mathbf{Z}^\top \mathbf{R} \mathbf{y}, \quad \hat{\beta}_W = \hat{\beta} - \mathbf{L} \hat{\gamma}, \quad \mathbf{L} = \mathbf{C} \mathbf{X}^\top \mathbf{Z};$$

- (c) the residual sum of squares falls by

$$\text{SS}(\mathbf{Z} \mid \mathbf{X}) = \mathbf{y}^\top \mathbf{R} \mathbf{Z} (\mathbf{Z}^\top \mathbf{R} \mathbf{Z})^{-1} \mathbf{Z}^\top \mathbf{R} \mathbf{y} = \hat{\gamma}^\top \mathbf{Z}^\top \mathbf{R} \mathbf{y} = \hat{\gamma}^\top (\mathbf{Z}^\top \mathbf{R} \mathbf{Z}) \hat{\gamma};$$

- (d) if $\text{Cov}(\mathbf{Y}) = \sigma^2 \mathbf{I}$, then $\text{Cov}(\hat{\gamma}) = \sigma^2 (\mathbf{Z}^\top \mathbf{R} \mathbf{Z})^{-1}$, $\text{Cov}(\hat{\beta}_W, \hat{\gamma}) = -\mathbf{L} \text{Cov}(\hat{\gamma})$ and

$$\text{Cov}(\hat{\beta}_W) = \sigma^2 [\mathbf{C} + \mathbf{L} (\mathbf{Z}^\top \mathbf{R} \mathbf{Z})^{-1} \mathbf{L}^\top].$$

Proof.

- (a) $\mathbf{R}$ is the symmetric idempotent residual projection of the model $\mathbf{X}$, so $\mathbf{a}^\top \mathbf{Z}^\top \mathbf{R} \mathbf{Z} \mathbf{a} = \|\mathbf{R} \mathbf{Z} \mathbf{a}\|^2 \geq 0$. If $\mathbf{R} \mathbf{Z} \mathbf{a} = \mathbf{0}$, then $\mathbf{Z} \mathbf{a} \in \mathcal{C}(\mathbf{X})$, say $\mathbf{Z} \mathbf{a} = \mathbf{X} \mathbf{b}$, so $\mathbf{W}(-\mathbf{b}^\top, \mathbf{a}^\top)^\top = \mathbf{0}$ and full column rank forces $\mathbf{a} = \mathbf{0}$.
 (b) Apply the Frisch–Waugh–Lovell theorem ([Theorem 6.6.2](#)) with $\mathbf{X}_1 = \mathbf{Z}$ and $\mathbf{X}_2 = \mathbf{X}$. Then $\mathbf{M}_2 = \mathbf{I} - \mathbf{R}$ and $\tilde{\mathbf{X}}_1 = \mathbf{R} \mathbf{Z}$, so part (a) of that theorem gives $\hat{\gamma} = (\mathbf{Z}^\top \mathbf{R} \mathbf{R} \mathbf{Z})^{-1} \mathbf{Z}^\top \mathbf{R} \mathbf{y}$, which is the stated formula because $\mathbf{R}$ is symmetric and idempotent. Part (c) of the theorem gives $\hat{\beta}_W = \mathbf{C} \mathbf{X}^\top (\mathbf{y} - \mathbf{Z} \hat{\gamma}) = \hat{\beta} - \mathbf{L} \hat{\gamma}$.
 (c) By [Lemma 9.2.1](#), the extra sum of squares is the squared length of the projection of $\mathbf{y}$ onto $\mathcal{C}(\mathbf{R} \mathbf{Z})$, which is $\mathbf{y}^\top \mathbf{R} \mathbf{Z} (\mathbf{Z}^\top \mathbf{R} \mathbf{Z})^{-1} \mathbf{Z}^\top \mathbf{R} \mathbf{y}$ by [Theorem 6.3.7](#). Substituting (b) gives the other two forms.
 (d) Both $\hat{\gamma}$ and $\hat{\beta}$ are linear in $\mathbf{y}$, and by [Theorem 2.2.1](#)

$$\text{Cov}(\hat{\gamma}) = \sigma^2 (\mathbf{Z}^\top \mathbf{R} \mathbf{Z})^{-1} \mathbf{Z}^\top \mathbf{R} \mathbf{R} \mathbf{Z} (\mathbf{Z}^\top \mathbf{R} \mathbf{Z})^{-1} = \sigma^2 (\mathbf{Z}^\top \mathbf{R} \mathbf{Z})^{-1}, \quad \text{Cov}(\hat{\beta}, \hat{\gamma}) = \sigma^2 \mathbf{C} \mathbf{X}^\top \mathbf{R} \mathbf{Z} (\mathbf{Z}^\top \mathbf{R} \mathbf{Z})^{-1},$$

since $\mathbf{X}^\top \mathbf{R} = \mathbf{0}$. With $\text{Cov}(\hat{\beta}) = \sigma^2 \mathbf{C}$ ([Theorem 5.5.1](#)), the formula $\hat{\beta}_W = \hat{\beta} - \mathbf{L} \hat{\gamma}$ gives $\text{Cov}(\hat{\beta}_W, \hat{\gamma}) = -\mathbf{L} \text{Cov}(\hat{\gamma})$ and $\text{Cov}(\hat{\beta}_W) = \text{Cov}(\hat{\beta}) + \mathbf{L} \text{Cov}(\hat{\gamma}) \mathbf{L}^\top$.

■

Since $\mathbf{L} (\mathbf{Z}^\top \mathbf{R} \mathbf{Z})^{-1} \mathbf{L}^\top$ is nonnegative definite, part (d) shows that *adding regressors never decreases the variances of the coefficients already in the model*, when both models are correct (compare [Exercise \(B2\)](#), and see [Exercise \(C2\)](#)). Chapter 26 measures this price of adjustment as variance inflation. In computation one updates a QR factorization instead of using these formulas ([Chapter 10](#)).

Example. (Adding poverty and urbanization)

In the state data, fit the murder rate on single parenthood first: $\hat{y} = -8.2146 + 0.5147 x_{\text{single}}$, with residual sum of squares 121.92. Now add $\mathbf{Z} = [\mathbf{x}_{\text{poverty}}, \mathbf{x}_{\text{urban}}]$. The listing computes $\hat{\gamma} = (0.2538, 0.0046)$ from the old residual projection alone, and corrects the old coefficients to -9.0946 and 0.3980 . These agree with a fresh fit of the enlarged model. The residual sum of squares falls by 19.96, on 2 degrees of freedom, to 101.96. Single parenthood's coefficient drops by almost a quarter once poverty is allowed to take its share, the pattern seen in [Example \(Order matters\)](#).

```

import numpy as np
import statsmodels.api as sm
data = sm.datasets.statcrime.load_pandas().data.drop(index="District of Columbia")
y = data["murder"].to_numpy()
n = len(y)
one = np.ones(n)
cols = {"poverty": data["poverty"].to_numpy(), "single": data["single"].to_numpy(),
        "urban": data["urban"].to_numpy()}

X = np.column_stack([one, cols["single"]])           # the model already
        fitted
Z = np.column_stack([cols["poverty"], cols["urban"]]) # the regressors to add
XtX_inv = np.linalg.inv(X.T @ X)
beta_X = XtX_inv @ X.T @ y
R = np.eye(n) - X @ XtX_inv @ X.T                 # residual projection of
        the old model

gamma = np.linalg.solve(Z.T @ R @ Z, Z.T @ R @ y)   # new coefficients
beta_new = beta_X - XtX_inv @ X.T @ Z @ gamma       # old coefficients,
        corrected
extra_ss = gamma @ Z.T @ R @ y                     # SS(Z | X) = SSE_X -
        SSE_W
print("gamma =", gamma.round(4), " corrected beta =", beta_new.round(4))
print(f"SSE_X = {y @ R @ y:.2f}, extra SS = {extra_ss:.2f}")

```

9.2.3 One column at a time

With $t = 1$, [Theorem 9.2.2](#) links the extra sum of squares of a single regressor to its coefficient and standard error.

Corollary 9.2.3 — The extra sum of squares of one column

Let $\mathbf{X}$ have full column rank p , write c_{jj} for the j th diagonal entry of $(\mathbf{X}^\top \mathbf{X})^{-1}$, and let $\mathbf{X}_{(j)}$ be $\mathbf{X}$ without its j th column. Then

$$SS(\mathbf{x}_j | \mathbf{X}_{(j)}) = \frac{\hat{\beta}_j^2}{c_{jj}} = t_j^2 s^2, \quad t_j = \frac{\hat{\beta}_j}{s\sqrt{c_{jj}}}, \quad s^2 = \frac{SSE}{n-p}.$$

Consequently, the ratio of this one-degree-of-freedom sum of squares to the residual mean square is t_j^2 .

Proof. Permute the columns so that $\mathbf{x}_j$ comes last; this carries c_{jj} to the last diagonal entry of the inverse. Write $\mathbf{X} = [\mathbf{X}_{(j)}, \mathbf{x}_j]$ and $\mathbf{R} = \mathbf{I} - \mathbf{X}_{(j)}(\mathbf{X}_{(j)}^\top \mathbf{X}_{(j)})^{-1} \mathbf{X}_{(j)}^\top$ for the residual projection of $\mathbf{X}_{(j)}$. By [Theorem 1.6.4](#) the last diagonal entry of $(\mathbf{X}^\top \mathbf{X})^{-1}$ is the inverse of the Schur complement $\mathbf{x}_j^\top \mathbf{x}_j - \mathbf{x}_j^\top \mathbf{X}_{(j)}(\mathbf{X}_{(j)}^\top \mathbf{X}_{(j)})^{-1} \mathbf{X}_{(j)}^\top \mathbf{x}_j = \mathbf{x}_j^\top \mathbf{R} \mathbf{x}_j$, so $\mathbf{x}_j^\top \mathbf{R} \mathbf{x}_j = 1/c_{jj}$. Now apply [Theorem 9.2.2](#) with $\mathbf{X}_{(j)}$ as the old model and $\mathbf{Z} = \mathbf{x}_j$: then $\hat{\gamma} = \hat{\beta}_j$, and part (c) gives $SS(\mathbf{x}_j | \mathbf{X}_{(j)}) = \hat{\beta}_j^2 \mathbf{x}_j^\top \mathbf{R} \mathbf{x}_j = \hat{\beta}_j^2 / c_{jj}$. The definition of t_j gives the last form. ■

For poverty in the three-regressor model, $\hat{\beta} = 0.2538$ and $c_{jj} = 0.003935$, so the extra sum of squares is $0.2538^2 / 0.003935 = 16.37$. This is exactly the last entry for poverty in Order B of [Example \(Order matters\)](#). The t statistic is 2.718, and $t^2 s^2 = 7.386 \times 2.217 = 16.37$.

```
Xf = np.column_stack([one, cols["poverty"], cols["single"], cols["urban"]])
fit = sm.OLS(y, Xf).fit()
C = np.linalg.inv(Xf.T @ Xf)
j = 1 # poverty
partial_ss = fit.params[j] ** 2 / C[j, j]
print(f"beta = {fit.params[j]:.4f}, c_jj = {C[j, j]:.6f}, beta^2/c_jj = {partial_ss:.2f}")
print(f"t = {fit.tvalues[j]:.3f}, t^2 * s^2 = {fit.tvalues[j] ** 2 * fit.scale:.2f}")
```

9.2.4 Sums of squares for linear hypotheses

Reduced models often come as constraints: two coefficients are equal, a set of coefficients vanishes, a contrast among group means is zero. [Section 6.5](#) identified the test space of such a constraint ([Theorem 6.5.2](#)). The next theorem gives its sum of squares in terms of the least squares estimate, with no reduced model fitted.

Theorem 9.2.4 — Sum of squares of a linear hypothesis

Let $\mathbf{\Lambda}$ be a $p \times q$ matrix of rank q such that each entry of $\mathbf{\Lambda}^\top \boldsymbol{\beta}$ is estimable, so that $\mathbf{\Lambda} = \mathbf{X}^\top \mathbf{T}$ for some $n \times q$ matrix $\mathbf{T} = [\boldsymbol{\rho}_1, \dots, \boldsymbol{\rho}_q]$ ([Definition 8.2.1](#), [Theorem 6.4.4](#)). Let $\mathbf{G}$ be any generalized inverse of $\mathbf{X}^\top \mathbf{X}$, let $\hat{\boldsymbol{\beta}}$ be any least squares estimate, and let $\mathbf{d} \in \mathbb{R}^q$. Then:

- (a) $\mathbf{\Lambda}^\top \hat{\boldsymbol{\beta}}$ and $\mathbf{\Lambda}^\top \mathbf{G} \mathbf{\Lambda}$ do not depend on the choices of $\hat{\boldsymbol{\beta}}$ and $\mathbf{G}$, and $\mathbf{\Lambda}^\top \mathbf{G} \mathbf{\Lambda}$ is positive definite;
- (b) the subspace $\mathcal{S}_0 = \{\mathbf{Xb} : \mathbf{\Lambda}^\top \mathbf{b} = \mathbf{0}\}$ of $\mathcal{C}(\mathbf{X})$ has orthogonal complement in $\mathcal{C}(\mathbf{X})$ equal to $\mathcal{C}(\mathbf{MT})$, of dimension q , and the projection onto $\mathcal{C}(\mathbf{MT})$ is $\mathbf{MT}(\mathbf{\Lambda}^\top \mathbf{G} \mathbf{\Lambda})^{-1} \mathbf{T}^\top \mathbf{M}$;
- (c) the minimum of $\|\mathbf{y} - \mathbf{Xb}\|^2$ over $\{\mathbf{b} : \mathbf{\Lambda}^\top \mathbf{b} = \mathbf{d}\}$ exceeds SSE by

$$SS_H = (\mathbf{\Lambda}^\top \hat{\boldsymbol{\beta}} - \mathbf{d})^\top (\mathbf{\Lambda}^\top \mathbf{G} \mathbf{\Lambda})^{-1} (\mathbf{\Lambda}^\top \hat{\boldsymbol{\beta}} - \mathbf{d}). \quad (9.2.1)$$

Proof.

- (a) $\mathbf{\Lambda}^\top \hat{\boldsymbol{\beta}} = \mathbf{T}^\top \mathbf{X} \hat{\boldsymbol{\beta}} = \mathbf{T}^\top \mathbf{M} \mathbf{y}$ and $\mathbf{\Lambda}^\top \mathbf{G} \mathbf{\Lambda} = \mathbf{T}^\top \mathbf{X} \mathbf{G} \mathbf{X}^\top \mathbf{T} = \mathbf{T}^\top \mathbf{M} \mathbf{T}$ by [Theorem 6.3.7](#), and neither expression involves the choices. Since $\mathbf{X}^\top \mathbf{M} = \mathbf{X}^\top$, we have $\mathbf{X}^\top (\mathbf{MT}) = \mathbf{\Lambda}$, which has rank q , so the q columns of $\mathbf{MT}$ are linearly independent. Hence $\mathbf{T}^\top \mathbf{M} \mathbf{T} = (\mathbf{MT})^\top (\mathbf{MT})$ is positive definite.
- (b) $\mathbf{\Lambda}^\top \mathbf{b} = \mathbf{T}^\top \mathbf{Xb}$, so $\mathcal{S}_0 = \mathcal{C}(\mathbf{X}) \cap \mathcal{N}(\mathbf{T}^\top)$. [Theorem 6.5.2](#) identifies its orthogonal complement in $\mathcal{C}(\mathbf{X})$ as $\mathcal{C}(\mathbf{MT})$. The columns of $\mathbf{MT}$ are independent, so [Theorem 6.3.7](#) gives the projection $\mathbf{MT}(\mathbf{T}^\top \mathbf{M} \mathbf{T})^{-1} \mathbf{T}^\top \mathbf{M}$, and $\mathbf{T}^\top \mathbf{M} \mathbf{T} = \mathbf{\Lambda}^\top \mathbf{G} \mathbf{\Lambda}$ by (a).
- (c) First let $\mathbf{d} = \mathbf{0}$. Minimizing over $\mathbf{\Lambda}^\top \mathbf{b} = \mathbf{0}$ is minimizing $\|\mathbf{y} - \boldsymbol{\mu}\|^2$ over $\boldsymbol{\mu} \in \mathcal{S}_0$, so by [Eq. 6.5.2](#) the excess over SSE is the squared length of the projection of $\mathbf{y}$ onto $\mathcal{C}(\mathbf{X}) \ominus \mathcal{S}_0 = \mathcal{C}(\mathbf{MT})$, where $\ominus$ denotes the orthogonal complement within

$\mathcal{C}(\mathbf{X})$. By (b) this is $\mathbf{y}^\top \mathbf{M} \mathbf{T} (\mathbf{\Lambda}^\top \mathbf{G} \mathbf{\Lambda})^{-1} \mathbf{T}^\top \mathbf{M} \mathbf{y}$, and $\mathbf{T}^\top \mathbf{M} \mathbf{y} = \mathbf{\Lambda}^\top \hat{\boldsymbol{\beta}}$. For general $\mathbf{d}$, the rows of $\mathbf{\Lambda}^\top$ are independent, so there is a $\mathbf{b}_0$ with $\mathbf{\Lambda}^\top \mathbf{b}_0 = \mathbf{d}$. Substituting $\mathbf{b} = \mathbf{b}_0 + \mathbf{b}_*$ turns the problem into minimizing $\|\mathbf{y}_* - \mathbf{X} \mathbf{b}_*\|^2$ subject to $\mathbf{\Lambda}^\top \mathbf{b}_* = \mathbf{0}$, with $\mathbf{y}_* = \mathbf{y} - \mathbf{X} \mathbf{b}_0$. The data $\mathbf{y}_*$ have the same residual sum of squares as $\mathbf{y}$, and $\hat{\boldsymbol{\beta}} - \mathbf{b}_0$ is a least squares estimate for them, since $\mathbf{X}(\hat{\boldsymbol{\beta}} - \mathbf{b}_0) = \mathbf{M} \mathbf{y}_*$. Applying the case $\mathbf{d} = \mathbf{0}$ to $\mathbf{y}_*$ gives Eq. 9.2.1, because $\mathbf{\Lambda}^\top (\hat{\boldsymbol{\beta}} - \mathbf{b}_0) = \mathbf{\Lambda}^\top \hat{\boldsymbol{\beta}} - \mathbf{d}$.

■

In Eq. 9.2.1, $\mathbf{\Lambda}^\top \hat{\boldsymbol{\beta}} - \mathbf{d}$ measures how far the estimate is from the hypothesis, and $\sigma^2 \mathbf{\Lambda}^\top \mathbf{G} \mathbf{\Lambda}$ is the covariance of $\mathbf{\Lambda}^\top \hat{\boldsymbol{\beta}}$ (Chapter 7), so SS_H / σ^2 is the squared Mahalanobis length of the discrepancy (Proposition 3.2.3). No reduced model has to be fitted.

In the full-rank case the constrained minimizer itself has a closed form, which recovers the classical restricted least squares estimator.

Theorem 9.2.5 — Restricted least squares

Let $\mathbf{X}$ have full column rank, $\mathbf{C} = (\mathbf{X}^\top \mathbf{X})^{-1}$, and let $\mathbf{\Lambda}$ ($p \times q$) have rank q . For any $\mathbf{d} \in \mathbb{R}^q$, the unique minimizer of $\|\mathbf{y} - \mathbf{X} \mathbf{b}\|^2$ subject to $\mathbf{\Lambda}^\top \mathbf{b} = \mathbf{d}$ is

$$\hat{\boldsymbol{\beta}}_H = \hat{\boldsymbol{\beta}} - \mathbf{C} \mathbf{\Lambda} (\mathbf{\Lambda}^\top \mathbf{C} \mathbf{\Lambda})^{-1} (\mathbf{\Lambda}^\top \hat{\boldsymbol{\beta}} - \mathbf{d}),$$

and $\|\mathbf{y} - \mathbf{X} \hat{\boldsymbol{\beta}}_H\|^2 = \text{SSE} + \|\mathbf{X}(\hat{\boldsymbol{\beta}} - \hat{\boldsymbol{\beta}}_H)\|^2$, where the last term equals SS_H of Eq. 9.2.1 with $\mathbf{G} = \mathbf{C}$.

Proof. $\mathbf{\Lambda}^\top \mathbf{C} \mathbf{\Lambda}$ is positive definite because $\mathbf{C}$ is and $\mathbf{\Lambda}$ has full column rank, and $\mathbf{\Lambda}^\top \hat{\boldsymbol{\beta}}_H = \mathbf{\Lambda}^\top \hat{\boldsymbol{\beta}} - (\mathbf{\Lambda}^\top \hat{\boldsymbol{\beta}} - \mathbf{d}) = \mathbf{d}$. For any $\mathbf{b}$, Eq. 6.3.1 with $\mathbf{u} = \mathbf{X} \mathbf{b}$ gives $\|\mathbf{y} - \mathbf{X} \mathbf{b}\|^2 = \text{SSE} + \|\mathbf{X}(\hat{\boldsymbol{\beta}} - \mathbf{b})\|^2$. Every feasible $\mathbf{b}$ has the form $\hat{\boldsymbol{\beta}}_H + \mathbf{v}$ with $\mathbf{\Lambda}^\top \mathbf{v} = \mathbf{0}$, and then

$$\|\mathbf{X}(\hat{\boldsymbol{\beta}} - \mathbf{b})\|^2 = \|\mathbf{X}(\hat{\boldsymbol{\beta}} - \hat{\boldsymbol{\beta}}_H)\|^2 - 2(\hat{\boldsymbol{\beta}} - \hat{\boldsymbol{\beta}}_H)^\top \mathbf{X}^\top \mathbf{X} \mathbf{v} + \|\mathbf{X} \mathbf{v}\|^2.$$

The cross term vanishes: $(\hat{\boldsymbol{\beta}} - \hat{\boldsymbol{\beta}}_H)^\top \mathbf{X}^\top \mathbf{X} \mathbf{v} = (\mathbf{\Lambda}^\top \hat{\boldsymbol{\beta}} - \mathbf{d})^\top (\mathbf{\Lambda}^\top \mathbf{C} \mathbf{\Lambda})^{-1} \mathbf{\Lambda}^\top \mathbf{C} \mathbf{X}^\top \mathbf{X} \mathbf{v} = (\mathbf{\Lambda}^\top \hat{\boldsymbol{\beta}} - \mathbf{d})^\top (\mathbf{\Lambda}^\top \mathbf{C} \mathbf{\Lambda})^{-1} \mathbf{\Lambda}^\top \mathbf{v} = 0$. So the objective is $\text{SSE} + \|\mathbf{X}(\hat{\boldsymbol{\beta}} - \hat{\boldsymbol{\beta}}_H)\|^2 + \|\mathbf{X} \mathbf{v}\|^2$, minimized exactly when $\mathbf{X} \mathbf{v} = \mathbf{0}$, that is, $\mathbf{v} = \mathbf{0}$. Finally $\|\mathbf{X}(\hat{\boldsymbol{\beta}} - \hat{\boldsymbol{\beta}}_H)\|^2 = (\mathbf{\Lambda}^\top \hat{\boldsymbol{\beta}} - \mathbf{d})^\top (\mathbf{\Lambda}^\top \mathbf{C} \mathbf{\Lambda})^{-1} \mathbf{\Lambda}^\top \mathbf{C} \mathbf{X}^\top \mathbf{X} \mathbf{C} \mathbf{\Lambda} (\mathbf{\Lambda}^\top \mathbf{C} \mathbf{\Lambda})^{-1} (\mathbf{\Lambda}^\top \hat{\boldsymbol{\beta}} - \mathbf{d})$, which simplifies to SS_H .

■

The same estimator comes out of Lagrange multipliers (Proposition 1.11.3 with $\mathbf{A} = \mathbf{X}^\top \mathbf{X}$). The proof above shows instead that the constrained fit is the feasible point nearest to $\hat{\boldsymbol{\beta}}$ in the geometry of $\mathbf{X}^\top \mathbf{X}$.

Example. (Equal effects of poverty and single parenthood)

Both poverty and single parenthood are percentages of the population. Is one percentage point of either associated with the same change in the murder rate? The hypothesis is $\mathbf{\Lambda}^\top \boldsymbol{\beta} = \mathbf{0}$ with $\boldsymbol{\lambda} = (0, 1, -1, 0)^\top$. Formula Eq. 9.2.1 gives $\text{SS}_H = 1.807$ on one degree of freedom, identical to the drop in residual sum of squares when the two regressors are replaced by their sum. Its ratio to the residual mean square is 0.815, so the data are entirely compatible with equal effects. For a hypothesis with a nonzero right-hand side,

$\beta_{\text{single}} = 0.5$, [Theorem 9.2.5](#) gives the constrained coefficients 0.1803 for poverty and -0.0044 for urbanization, and a sum of squares of 3.357, both confirmed by a direct fit of the reduced model.

```
def hypothesis_ss(X, y, Lam, d=None):
    """(L'b - d)' [L' (X'X)^- L]^-1 (L'b - d) for estimable L'beta = d."""
    G = np.linalg.pinv(X.T @ X)
    b = G @ X.T @ y
    u = Lam.T @ b - (0 if d is None else d)
    return u @ np.linalg.solve(Lam.T @ G @ Lam, u)

def sse(Z):
    b, *_ = np.linalg.lstsq(Z, y, rcond=None)
    e = y - Z @ b
    return e @ e

Lam_equal = np.array([[0.0, 1.0, -1.0, 0.0]]).T          # beta_poverty =
beta_single
ss_equal = hypothesis_ss(Xf, y, Lam_equal)
X_equal = np.column_stack([one, cols["poverty"] + cols["single"], cols["urban"]])
print(f"H: equal slopes    formula {ss_equal:.3f}    SSE_0 - SSE {sse(X_equal) - sse(
Xf):.3f}")
```

9.2.5 Constraints that are not estimable

Without estimability, the reduced model $\mathcal{S}_0 = \{\mathbf{X}\mathbf{b} : \mathbf{\Lambda}^\top \mathbf{b} = \mathbf{0}\}$ is still defined, but it may be larger than the number of constraints suggests.

Proposition 9.2.6 — How much a constraint restricts

For any $p \times q$ matrix $\mathbf{\Lambda}$, the subspace $\mathcal{S}_0 = \{\mathbf{X}\mathbf{b} : \mathbf{\Lambda}^\top \mathbf{b} = \mathbf{0}\}$ has dimension $r - \dim(\mathcal{C}(\mathbf{\Lambda}) \cap \mathcal{C}(\mathbf{X}^\top))$. So the extra sum of squares for the constraint has $\dim(\mathcal{C}(\mathbf{\Lambda}) \cap \mathcal{C}(\mathbf{X}^\top))$ degrees of freedom. It is identically zero iff no nonzero linear combination of the constrained functions is estimable.

Proof. $\mathcal{S}_0$ is the image of $\mathcal{N}(\mathbf{\Lambda}^\top)$ under $\mathbf{b} \mapsto \mathbf{X}\mathbf{b}$, and the kernel of this map restricted to $\mathcal{N}(\mathbf{\Lambda}^\top)$ is $\mathcal{N}(\mathbf{\Lambda}^\top) \cap \mathcal{N}(\mathbf{X})$, the null space of the stacked matrix $[\mathbf{\Lambda}, \mathbf{X}^\top]^\top$. By rank-nullity ([Theorem 1.2.2](#)),

$$\dim \mathcal{S}_0 = \dim \mathcal{N}(\mathbf{\Lambda}^\top) - \dim(\mathcal{N}(\mathbf{\Lambda}^\top) \cap \mathcal{N}(\mathbf{X})) = (p - \text{rank } \mathbf{\Lambda}) - (p - \text{rank}[\mathbf{\Lambda}, \mathbf{X}^\top]).$$

The dimension formula for a sum of subspaces ([Section 6.2](#)) gives $\text{rank}[\mathbf{\Lambda}, \mathbf{X}^\top] = \dim(\mathcal{C}(\mathbf{\Lambda}) + \mathcal{C}(\mathbf{X}^\top)) = \text{rank } \mathbf{\Lambda} + r - \dim(\mathcal{C}(\mathbf{\Lambda}) \cap \mathcal{C}(\mathbf{X}^\top))$. Substituting, $\dim \mathcal{S}_0 = r - \dim(\mathcal{C}(\mathbf{\Lambda}) \cap \mathcal{C}(\mathbf{X}^\top))$. The degrees of freedom of the extra sum of squares are $r - \dim \mathcal{S}_0$. A combination $\mathbf{\Lambda}\mathbf{a}$ is estimable iff it lies in $\mathcal{C}(\mathbf{X}^\top)$ ([Theorem 6.4.4](#)), which gives the last statement. ■

In the one-way layout $E(Y_{ij}) = \mu + \alpha_i$, the constraint $\alpha_1 = \alpha_2$ has one degree of freedom. The constraint $\alpha_1 = 0$ has none: it does not restrict the mean vector, and only picks one of the least squares solutions ([Theorem 8.4.2](#)). Software asked to “test” a nonestimable hypoth-

esis either refuses or silently tests its estimable part. [Chapter 17](#) meets this with empty cells ([Theorem 17.4.1](#)).

9.2.6 Exercises

9.2.6.1 A. Check your understanding

Exercise A1.

In simple regression, compute SS_H for $\beta_1 = 0$ from [Eq. 9.2.1](#) and show that it equals $\hat{\beta}_1^2 S_{xx}$, in agreement with [Exercise \(A1\)](#).

9.2.6.2 B. Practice

Exercise B1.

Prove the chain rule $SS(\mathbf{X}_2, \mathbf{X}_3 \mid \mathbf{X}_1) = SS(\mathbf{X}_2 \mid \mathbf{X}_1) + SS(\mathbf{X}_3 \mid \mathbf{X}_1, \mathbf{X}_2)$, and deduce $SS(\mathbf{X}_3 \mid \mathbf{X}_1, \mathbf{X}_2) \leq SS(\mathbf{X}_2, \mathbf{X}_3 \mid \mathbf{X}_1)$. Show by an example from [Example \(Adding poverty and urbanization\)](#) or its listing that $SS(\mathbf{X}_3 \mid \mathbf{X}_1, \mathbf{X}_2) \leq SS(\mathbf{X}_3 \mid \mathbf{X}_1)$ can fail.

Solution. Let $\mathbf{M}_1$, $\mathbf{M}_{12}$ and $\mathbf{M}_{123}$ be the projections onto the nested column spaces. Then $\mathbf{M}_{123} - \mathbf{M}_1 = (\mathbf{M}_{12} - \mathbf{M}_1) + (\mathbf{M}_{123} - \mathbf{M}_{12})$, and the chain rule follows by taking $\mathbf{y}^\top(\cdot)\mathbf{y}$. Both terms are nonnegative, which gives the inequality. For the failure, the listing of [Section 9.3](#) reports $SS(\mathbf{x}_{\text{urban}} \mid \mathbf{1}) = 6.30$ but $SS(\mathbf{x}_{\text{urban}} \mid \mathbf{1}, \mathbf{x}_{\text{poverty}}) = 29.26$. Adjusting for poverty more than quadruples the sum of squares for urbanization.

Exercise B2.

In [Theorem 9.2.5](#) with $\text{Cov}(\mathbf{Y}) = \sigma^2 \mathbf{I}$, show that

$$\text{Cov}(\hat{\boldsymbol{\beta}}_H) = \sigma^2 [\mathbf{C} - \mathbf{C}\mathbf{A}(\mathbf{A}^\top \mathbf{C}\mathbf{A})^{-1} \mathbf{A}^\top \mathbf{C}],$$

that $\text{Cov}(\hat{\boldsymbol{\beta}}) - \text{Cov}(\hat{\boldsymbol{\beta}}_H)$ is nonnegative definite, and that $E(\hat{\boldsymbol{\beta}}_H) = \boldsymbol{\beta}$ when the constraint is true. What is $E(\hat{\boldsymbol{\beta}}_H)$ when it is false?

Solution. Write $\mathbf{K} = \mathbf{C}\mathbf{A}(\mathbf{A}^\top \mathbf{C}\mathbf{A})^{-1}$, so $\hat{\boldsymbol{\beta}}_H = (\mathbf{I} - \mathbf{K}\mathbf{A}^\top)\hat{\boldsymbol{\beta}} + \mathbf{K}\mathbf{d}$. By [Theorem 2.2.1](#) and $\text{Cov}(\hat{\boldsymbol{\beta}}) = \sigma^2 \mathbf{C}$, $\text{Cov}(\hat{\boldsymbol{\beta}}_H) = \sigma^2 (\mathbf{I} - \mathbf{K}\mathbf{A}^\top)\mathbf{C}(\mathbf{I} - \mathbf{A}\mathbf{K}^\top)$. Expanding and using $\mathbf{A}^\top \mathbf{C}\mathbf{A}(\mathbf{A}^\top \mathbf{C}\mathbf{A})^{-1} = \mathbf{I}$, the two cross terms and the last term all equal $\mathbf{K}\mathbf{A}^\top \mathbf{C} = \mathbf{C}\mathbf{A}(\mathbf{A}^\top \mathbf{C}\mathbf{A})^{-1} \mathbf{A}^\top \mathbf{C}$, with signs $-, -, +$, which gives the formula. The difference $\sigma^2 \mathbf{C}\mathbf{A}(\mathbf{A}^\top \mathbf{C}\mathbf{A})^{-1} \mathbf{A}^\top \mathbf{C}$ is of the form $\mathbf{B}\mathbf{A}\mathbf{B}^\top$ with $\mathbf{A}$ positive definite, hence nonnegative definite. Finally $E(\hat{\boldsymbol{\beta}}_H) = \boldsymbol{\beta} - \mathbf{K}(\mathbf{A}^\top \boldsymbol{\beta} - \mathbf{d})$, which is $\boldsymbol{\beta}$ iff $\mathbf{A}^\top \boldsymbol{\beta} = \mathbf{d}$. A false constraint buys the smaller variance at the price of a bias.

Exercise B3.

In the one-way layout with model matrix $\mathbf{X} = [\mathbf{1}, \mathbf{z}_1, \dots, \mathbf{z}_g]$ and group sizes $n_1, \dots, n_g$, use the generalized inverse $\mathbf{G} = \text{diag}(0, 1/n_1, \dots, 1/n_g)$ in [Eq. 9.2.1](#) to show that the sum of squares for $\alpha_1 = \alpha_2$ is $(\bar{y}_1 - \bar{y}_2)^2 / (1/n_1 + 1/n_2)$. Verify with [Proposition 9.2.6](#)

that the constraint $\alpha_1 = 0$ has zero degrees of freedom.

Solution. $\mathbf{G}$ is a generalized inverse because $\mathbf{X}^\top \mathbf{X}$ has the block form with first row $(n, n_1, \dots, n_g)$ and lower right block $\text{diag}(n_1, \dots, n_g)$, and a direct multiplication gives $\mathbf{X}^\top \mathbf{X} \mathbf{G} \mathbf{X}^\top \mathbf{X} = \mathbf{X}^\top \mathbf{X}$. With $\boldsymbol{\lambda} = (0, 1, -1, 0, \dots, 0)^\top$, $\mathbf{G} \mathbf{X}^\top \mathbf{y} = (0, \bar{y}_1, \dots, \bar{y}_g)^\top$, so $\boldsymbol{\lambda}^\top \hat{\boldsymbol{\beta}} = \bar{y}_1 - \bar{y}_2$ and $\boldsymbol{\lambda}^\top \mathbf{G} \boldsymbol{\lambda} = 1/n_1 + 1/n_2$. For $\alpha_1 = 0$, $\boldsymbol{\lambda} = (0, 1, 0, \dots, 0)^\top$ is not in $\mathcal{C}(\mathbf{X}^\top)$ (Example (Four answers, one fit)), and $\mathcal{C}(\boldsymbol{\lambda})$ is a line, so $\mathcal{C}(\boldsymbol{\lambda}) \cap \mathcal{C}(\mathbf{X}^\top) = \{\mathbf{0}\}$.

9.2.6.3 C. Going deeper

Exercise C1.

Using Exercise (C1) and Lemma 9.2.1, show that

$$\text{SS}(\mathbf{X}_2 \mid \mathbf{X}_1) = \max_{\mathbf{a}} \frac{(\mathbf{a}^\top \tilde{\mathbf{X}}_2^\top \mathbf{y})^2}{\mathbf{a}^\top \tilde{\mathbf{X}}_2^\top \tilde{\mathbf{X}}_2 \mathbf{a}},$$

the maximum being over $\mathbf{a}$ with $\tilde{\mathbf{X}}_2 \mathbf{a} \neq \mathbf{0}$. Deduce that the extra sum of squares of a block is at least that of any single column of the block, given $\mathbf{X}_1$, and that the sum of squares of an estimable hypothesis $\boldsymbol{\Lambda}^\top \boldsymbol{\beta} = \mathbf{0}$ equals $\max_{\mathbf{a}} (\mathbf{a}^\top \boldsymbol{\Lambda}^\top \boldsymbol{\beta})^2 / (\mathbf{a}^\top \boldsymbol{\Lambda}^\top \mathbf{G} \boldsymbol{\Lambda} \mathbf{a})$.

Exercise C2.

Exercise (B2) gives the matrix inequality behind the remark after Theorem 9.2.2. Using part (d) of that theorem instead, show that $\text{Var}(\hat{\beta}_{W,j}) = \text{Var}(\hat{\beta}_j)$ iff the j th row of $\mathbf{L}$ is zero, and that this holds for every j iff $\mathbf{X}^\top \mathbf{Z} = \mathbf{0}$. Show that when $\mathbf{X}^\top \mathbf{Z} = \mathbf{0}$ the estimates $\hat{\boldsymbol{\beta}}$ do not change at all.

Exercise C3.

In the additive two-way model $E(Y_{ijk}) = \mu + \alpha_i + \beta_j$ with all cells occupied, use Proposition 9.2.6 to find the degrees of freedom of the extra sums of squares for the constraints (i) $\sum_i \alpha_i = 0$ and $\sum_j \beta_j = 0$; (ii) $\alpha_1 = \alpha_2$; (iii) $\alpha_1 = 0$ and $\alpha_1 = \alpha_2$.

9.3 Sequential and partial sums of squares

How should the variation explained by a model be shared among its terms? A term is a block of columns representing one idea: a regressor, the coding columns of a factor, or an interaction. One answer enters the terms one at a time and credits each with what it adds to those already present. Another credits each term with what it adds when entered last. This section defines both, explains the three “types” of sums of squares that software prints, and identifies the hypothesis that each of them addresses.

9.3.1 Sequential sums of squares

Write the model matrix as a sequence of terms,

$$\mathbf{X} = [\mathbf{X}_0, \mathbf{X}_1, \dots, \mathbf{X}_k],$$

where $\mathbf{X}_0$ is usually the intercept column $\mathbf{1}$. Let $\mathbf{M}_j$ be the projection onto $\mathcal{C}(\mathbf{X}_0, \mathbf{X}_1, \dots, \mathbf{X}_j)$, so that $\mathbf{M}_k = \mathbf{M}$.

Definition 9.3.1 — Sequential sums of squares

The **sequential** (or **Type I**) sum of squares of the term $\mathbf{X}_j$, for the order $\mathbf{X}_0, \mathbf{X}_1, \dots, \mathbf{X}_k$, is

$$\text{SS}(\mathbf{X}_j \mid \mathbf{X}_0, \dots, \mathbf{X}_{j-1}) = \mathbf{y}^\top (\mathbf{M}_j - \mathbf{M}_{j-1}) \mathbf{y}, \quad j = 1, \dots, k,$$

with $d_j = \text{rank}(\mathbf{M}_j) - \text{rank}(\mathbf{M}_{j-1})$ degrees of freedom.

The sequential sums of squares are the analysis of variance decomposition of [Theorem 9.1.1\(e\)](#) for the chain $\mathcal{C}(\mathbf{X}_0) \subseteq \mathcal{C}(\mathbf{X}_0, \mathbf{X}_1) \subseteq \dots \subseteq \mathcal{C}(\mathbf{X})$, after the first piece $\mathbf{M}_0$ has been set aside. They add up to the regression sum of squares $\text{SS}(\mathbf{X}_1, \dots, \mathbf{X}_k \mid \mathbf{X}_0)$, their degrees of freedom add up to $r - \text{rank}(\mathbf{X}_0)$, and under normality they are independent of each other and of the residual sum of squares ([Theorem 4.4.4](#)). A term linearly dependent on earlier ones receives zero degrees of freedom. The price is that each piece depends on the order: $\mathbf{X}_j$ is credited only with the part of its span orthogonal to *the terms that happen to precede it*. For the murder rates of [Section 6.5](#), poverty received 102.49 first and 16.37 last ([Example \(Order matters\)](#)).

The order dependence can be more surprising than “entering first gives more credit”. The listing computes $\text{SS}(\mathbf{x} \mid \mathbf{1}, S)$ for each of the three regressors and every set S of the other two.

Regressor	given $\mathbf{1}$ only	given $\mathbf{1}$ and one other (named)	given $\mathbf{1}$ and the remaining other (named)	given $\mathbf{1}$ and both others
poverty	102.49	19.69 (single)	125.45 (urban)	16.37
single	162.84	80.04 (poverty)	160.13 (urban)	51.06
urban	6.30	29.26 (poverty)	3.59 (single)	0.27

Adjusting for urbanization *increases* the sum of squares of poverty from 102.49 to 125.45, and adjusting for poverty more than quadruples that of urbanization. Urban states have somewhat less poverty but more murder, so holding one fixed sharpens the association of the other; such a regressor is called a **suppressor** ([Exercise \(B1\)](#)). A sequential sum of squares is not a property of a regressor, and not even a monotone function of the set adjusted for.

```
from itertools import combinations
import numpy as np
import statsmodels.api as sm
data = sm.datasets.statelcrime.load_pandas().data.drop(index="District of Columbia")
y = data["murder"].to_numpy()
n = len(y)
one = np.ones(n)
cols = {"poverty": data["poverty"].to_numpy(), "single": data["single"].to_numpy(),
        "urban": data["urban"].to_numpy()}
```

```

def sse(Z):
    b, *_ = np.linalg.lstsq(Z, y, rcond=None)
    e = y - Z @ b
    return e @ e

table = {}
for name, x in cols.items():
    others = [k for k in cols if k != name]
    for size in range(3):
        for S in combinations(others, size):
            base = np.column_stack([one] + [cols[k] for k in S])
            table[name, S] = sse(base) - sse(np.column_stack([base, x]))
            print(f"SS({name} | 1{''.join(', ' + k for k in S)}) = {table[name, S]
                  }:7.2f}")

```

9.3.2 Partial sums of squares: Types II and III

The second answer credits a term with what it adds last. For models whose terms are factors and their interactions, “last” needs care, because a main effect is part of every interaction that contains it. Say that a term $\mathbf{X}_i$ **contains** a term $\mathbf{X}_j$ if the factors of $\mathbf{X}_j$ are a proper subset of the factors of $\mathbf{X}_i$. For example the interaction AB contains the main effects A and B .

Definition 9.3.2 — Partial sums of squares; Types II and III

For a model with terms $\mathbf{X}_0, \mathbf{X}_1, \dots, \mathbf{X}_k$, write $\mathbf{X}_{(-j)}$ for $\mathbf{X}$ with the columns of $\mathbf{X}_j$ removed.

- (a) The **partial** (or **last-in**) sum of squares of $\mathbf{X}_j$ is $SS(\mathbf{X}_j \mid \mathbf{X}_{(-j)}) = \mathbf{y}^\top (\mathbf{M} - \mathbf{M}_{(-j)}) \mathbf{y}$, where $\mathbf{M}_{(-j)}$ projects onto $\mathcal{C}(\mathbf{X}_{(-j)})$.
- (b) The **Type II** sum of squares of $\mathbf{X}_j$ is its extra sum of squares given all terms that do not contain it: $SS(\mathbf{X}_j \mid \mathbf{X}_i : \mathbf{X}_i \text{ does not contain } \mathbf{X}_j, i \neq j)$.
- (c) The **Type III** sum of squares of $\mathbf{X}_j$ is its partial sum of squares (a) computed when every factor is coded by sum-to-zero contrast columns and every interaction by the products of the columns of its factors.

When no term contains another, as in a regression on numerical regressors without interactions, the three coincide, and they equal the sequential sum of squares of $\mathbf{X}_j$ when it is entered last. They also coincide for a term that is not contained in any other, such as the highest-order interaction. They differ for main effects in models with interactions.

Partial sums of squares are not a decomposition. The subspaces $\mathcal{C}(\mathbf{X}) \ominus \mathcal{C}(\mathbf{X}_{(-j)})$ are generally not orthogonal, so the partial sums of squares can add up to more or less than the regression sum of squares ([Exercise \(C1\)](#)), and they need not be independent of each other, though each is independent of the residual sum of squares under normality.

Why does the definition of Type III name a coding? Because in models with interactions the partial sum of squares of a main effect depends on the coding. Deleting the columns of A while keeping those of AB leaves the reduced space $\mathcal{C}(\mathbf{1}, \mathbf{X}_B, \mathbf{X}_{AB})$, which depends on which columns represent AB , although the full space does not. Different codings give different reduced models and so test different hypotheses.

9.3.3 What each sum of squares tests

A sum of squares $\mathbf{y}^\top \mathbf{P} \mathbf{y}$ with $\mathbf{P}$ a projection of rank d has expectation

$$\mathbf{E}(\mathbf{Y}^\top \mathbf{P} \mathbf{Y}) = \sigma^2 d + \|\mathbf{P} \boldsymbol{\mu}\|^2, \quad \boldsymbol{\mu} = \mathbf{E}(\mathbf{Y}),$$

by [Theorem 2.3.1](#) with $\text{Cov}(\mathbf{Y}) = \sigma^2 \mathbf{I}$. So the hypothesis a sum of squares addresses is $\mathbf{P} \boldsymbol{\mu} = \mathbf{0}$: *the mean has no component in the subspace onto which it projects*. Under normality this is exactly the condition for its F ratio to be central ([Section 9.6](#)). We translate it into statements about parameters.

Proposition 9.3.1 — Hypotheses of sequential and partial sums of squares

Let $\mathbf{X} = [\mathbf{X}_0, \dots, \mathbf{X}_k]$ have full column rank and $\boldsymbol{\mu} = \sum_l \mathbf{X}_l \beta_l$.

- (a) The partial sum of squares of $\mathbf{X}_j$ addresses $\beta_j = \mathbf{0}$: $(\mathbf{M} - \mathbf{M}_{(-j)})\boldsymbol{\mu} = \mathbf{0}$ iff $\beta_j = \mathbf{0}$.
- (b) The sequential sum of squares of $\mathbf{X}_j$ addresses

$$\beta_j + (\tilde{\mathbf{X}}_j^\top \tilde{\mathbf{X}}_j)^{-1} \tilde{\mathbf{X}}_j^\top \sum_{l>j} \mathbf{X}_l \beta_l = \mathbf{0}, \quad \tilde{\mathbf{X}}_j = (\mathbf{I} - \mathbf{M}_{j-1})\mathbf{X}_j.$$

This is $\beta_j = \mathbf{0}$ whenever the later terms have zero coefficients, or whenever $\tilde{\mathbf{X}}_j^\top \mathbf{X}_l = \mathbf{0}$ for all $l > j$.

Proof.

- (a) For $l \neq j$, $\mathbf{X}_l \beta_l$ lies in $\mathcal{C}(\mathbf{X}_{(-j)}) \subseteq \mathcal{C}(\mathbf{X})$, so $(\mathbf{M} - \mathbf{M}_{(-j)})\mathbf{X}_l \beta_l = \mathbf{0}$. Let $\tilde{\mathbf{X}}_j = (\mathbf{I} - \mathbf{M}_{(-j)})\mathbf{X}_j$. By [Lemma 9.2.1](#), $\mathbf{M} - \mathbf{M}_{(-j)}$ projects onto $\mathcal{C}(\tilde{\mathbf{X}}_j)$, so $(\mathbf{M} - \mathbf{M}_{(-j)})\mathbf{X}_j \beta_j$ is the projection of $\mathbf{X}_j \beta_j$ onto that space. Since $\mathbf{X}_j \beta_j - \tilde{\mathbf{X}}_j \beta_j = \mathbf{M}_{(-j)}\mathbf{X}_j \beta_j$ is orthogonal to $\mathcal{C}(\tilde{\mathbf{X}}_j)$, the projection is $\tilde{\mathbf{X}}_j \beta_j$. The columns of $\tilde{\mathbf{X}}_j$ are independent ([Lemma 6.6.1\(b\)](#)), so it vanishes iff $\beta_j = \mathbf{0}$.
- (b) Terms with $l < j$ are annihilated by $\mathbf{M}_j - \mathbf{M}_{j-1}$, which projects onto $\mathcal{C}(\tilde{\mathbf{X}}_j)$ by [Lemma 9.2.1](#). As in (a), $\mathbf{X}_j \beta_j$ projects to $\tilde{\mathbf{X}}_j \beta_j$. A later term projects to $\tilde{\mathbf{X}}_j (\tilde{\mathbf{X}}_j^\top \tilde{\mathbf{X}}_j)^{-1} \tilde{\mathbf{X}}_j^\top \mathbf{X}_l \beta_l$ by [Theorem 6.3.7](#). Adding, $(\mathbf{M}_j - \mathbf{M}_{j-1})\boldsymbol{\mu} = \tilde{\mathbf{X}}_j \mathbf{v}$, where $\mathbf{v}$ is the vector in the display, and this is zero iff $\mathbf{v} = \mathbf{0}$ because $\tilde{\mathbf{X}}_j$ has independent columns. ■

So a sequential sum of squares answers the question “does $\mathbf{X}_j$ matter?” only if the terms fitted after it do not matter, or are orthogonal to it after adjusting for the terms before it. Otherwise the effects of later terms leak into it, through the coefficients $(\tilde{\mathbf{X}}_j^\top \tilde{\mathbf{X}}_j)^{-1} \tilde{\mathbf{X}}_j^\top \mathbf{X}_l$ of the regressions of later terms on $\tilde{\mathbf{X}}_j$.

For factors, the natural parameters are the cell means. Consider a two-way layout with factor A at a levels and B at b levels, $n_{ij} \geq 1$ observations in cell (i, j) , row totals $n_{i.}$, and cell means μ_{ij} . The full model with interaction has model space $\mathcal{C}(\mathbf{X}) =$ all vectors that are constant within

cells, of dimension ab . Define the **weighted** and **unweighted** row means

$$\bar{\mu}_i^w = \frac{1}{n_{i.}} \sum_j n_{ij} \mu_{ij}, \quad \bar{\mu}_{i.} = \frac{1}{b} \sum_j \mu_{ij}.$$

Theorem 9.3.2 — What the sums of squares for a main effect test

In the two-way layout with all $n_{ij} \geq 1$:

- (a) the sequential sum of squares $SS(A \mid \mu)$ of A entered first addresses $\bar{\mu}_1^w = \cdots = \bar{\mu}_a^w$;
- (b) the Type III sum of squares of A addresses $\bar{\mu}_{1.} = \cdots = \bar{\mu}_{a.}$;
- (c) the partial sum of squares of the columns of A under reference-level coding (indicators of levels $2, \dots, a$ of A and $2, \dots, b$ of B , and their products for AB) addresses $\mu_{11} = \mu_{21} = \cdots = \mu_{a1}$, the equality of the A means at the first level of B ;
- (d) the Type II sum of squares $SS(A \mid \mu, B)$ addresses a hypothesis that, when the cell means are additive ($\mu_{ij} = \alpha_i + \beta_j$), is equivalent to $\alpha_1 = \cdots = \alpha_a$.

Each sum of squares equals the sum of squares Eq. 9.2.1 of the hypothesis it addresses, in the cell-means model.

Proof. Let $\mathbf{P}_0$, $\mathbf{P}_A$, $\mathbf{P}_B$ and $\mathbf{P}_{A+B}$ be the projections onto the constant vectors, the vectors constant within rows, those constant within columns, and their sum. Each hypothesis below defines a subspace $\mathcal{S}_0$ of $\mathcal{C}(\mathbf{X})$, and each sum of squares is $\|\mathbf{P}\mathbf{y}\|^2$ with $\mathbf{P}$ the projection onto $\mathcal{C}(\mathbf{X}) \ominus \mathcal{S}_0$. Then $\mathbf{P}\boldsymbol{\mu} = \mathbf{0}$ iff $\boldsymbol{\mu} \in \mathcal{S}_0$, and the sum of squares equals $SSE_0 - SSE$ for $\mathcal{S}_0$, which is the sum of squares of the hypothesis by Theorem 9.2.4.

- (a) $\mathbf{P}_A$ replaces each entry of a vector by the average of its row, so $\mathbf{P}_A\boldsymbol{\mu}$ has entry $\bar{\mu}_i^w$ in row i , and $\mathbf{P}_0\boldsymbol{\mu}$ has every entry equal to the overall weighted mean $\sum_i n_{i.} \bar{\mu}_i^w / n$. So $(\mathbf{P}_A - \mathbf{P}_0)\boldsymbol{\mu} = \mathbf{0}$ iff all $\bar{\mu}_i^w$ equal their weighted average, that is, iff they are all equal. The relevant $\mathcal{S}_0$ is $\{\boldsymbol{\mu} \in \mathcal{C}(\mathbf{X}) : (\mathbf{P}_A - \mathbf{P}_0)\boldsymbol{\mu} = \mathbf{0}\}$, whose orthogonal complement in $\mathcal{C}(\mathbf{X})$ is $\mathcal{C}(\mathbf{P}_A - \mathbf{P}_0)$, because $\mathcal{C}(\mathbf{P}_A - \mathbf{P}_0) \subseteq \mathcal{C}(\mathbf{X})$.
- (b) Any $a \times b$ array of cell means decomposes uniquely as

$$\mu_{ij} = \bar{\mu}_{..} + (\bar{\mu}_{i.} - \bar{\mu}_{..}) + (\bar{\mu}_{.j} - \bar{\mu}_{..}) + (\mu_{ij} - \bar{\mu}_{i.} - \bar{\mu}_{.j} + \bar{\mu}_{..}),$$

where bars denote unweighted averages. The four arrays lie in the mutually orthogonal subspaces (for the unweighted inner product on arrays) of constants, row effects summing to zero, column effects summing to zero, and arrays with all row and column sums zero, of dimensions $1, a-1, b-1, (a-1)(b-1)$. Sum-to-zero coding spans exactly these: the columns $\mathbf{u}_i - \mathbf{u}_1$ ($i = 2, \dots, a$), where $\mathbf{u}_i$ indicates row i , give all row effects α_i with $\sum_i \alpha_i = 0$; similarly for B ; and the $(a-1)(b-1)$ products are linearly independent arrays with zero row and column sums, so they span that subspace. Deleting the columns of A removes exactly the row-effect component, so the reduced model is $\mathcal{S}_0 = \{\boldsymbol{\mu} : \bar{\mu}_{i.} = \bar{\mu}_{..} \text{ for all } i\}$.

- (c) With reference coding, every cell-mean array is uniquely $\mu_{ij} = m + \alpha_i + \beta_j + \gamma_{ij}$ with $\alpha_1 = \beta_1 = 0$ and $\gamma_{1j} = \gamma_{i1} = 0$; explicitly $m = \mu_{11}$, $\alpha_i = \mu_{i1} - \mu_{11}$, $\beta_j = \mu_{1j} - \mu_{11}$ and $\gamma_{ij} = \mu_{ij} - \mu_{i1} - \mu_{1j} + \mu_{11}$. Deleting the columns of A forces $\alpha_i = 0$, that is, $\mu_{i1} = \mu_{11}$ for all i .
- (d) The Type II sum of squares is $\mathbf{y}^\top (\mathbf{P}_{A+B} - \mathbf{P}_B) \mathbf{y}$. If $\boldsymbol{\mu}$ is additive, then $\boldsymbol{\mu} \in \mathcal{C}(\mathbf{P}_{A+B})$ and $(\mathbf{P}_{A+B} - \mathbf{P}_B)\boldsymbol{\mu} = \boldsymbol{\mu} - \mathbf{P}_B\boldsymbol{\mu}$, which is zero iff $\boldsymbol{\mu}$ is constant within columns,

that is, iff α_i does not depend on i .

■

These are four different questions. Part (a) averages over B with the design's weights, so if B matters and the rows have different mixes of B levels, it can detect “an effect of A ” that is entirely an effect of B ([Exercise B3](#)). Part (b) averages with equal weights. Part (c) compares the rows at one level of B : a simple effect, not a main effect, and one that changes with the reference level. Part (d) is the natural test of A without interaction; with interaction its hypothesis has weights that depend on all the n_{ij} .

Warning (“Type III” with the default coding). Many regression programs code factors by reference levels unless told otherwise. Asking them for last-in sums of squares of main effects in a model with interactions then produces the hypotheses of [Theorem 9.3.2\(c\)](#), which concern a single level of the other factor. They are not Type III sums of squares, although they are sometimes reported under that name. Use sum-to-zero coding, or better, write down the hypothesis and use [Eq. 9.2.1](#).

9.3.4 An unbalanced example

Example. (Party, education and political self-placement)

In the 1996 American National Election Study, 944 respondents placed themselves on a seven-point scale from extremely liberal (1) to extremely conservative (7). Classify them by party identification (Democrat, Independent, Republican, counting those who only lean to a party as Independents) and by education (high school or less, college, graduate study). The cell counts and means are:

	high school	college	graduate
Democrat	146, 3.97	108, 3.43	126, 2.92
Independent	80, 4.39	64, 4.28	95, 4.09
Republican	87, 5.40	105, 5.36	133, 5.40

The design is unbalanced, and education is associated with party: 38% of the Democrats but only 27% of the Republicans have at most a high school education. The pattern of means shows an interaction. Among Democrats self-placement moves left with education by more than a scale point, while among Republicans it does not move at all. The sums of squares for the terms are:

Term	df	Type I, party first	Type I, education first	Type II	Type III	reference coding
party	2	647.46	674.42	674.42	650.41	112.27
education	2	43.26	16.31	43.26	32.03	75.12
interaction	4	35.81	35.81	35.81	35.81	35.81
residual	935	1224.62	1224.62	1224.62	1224.62	1224.62

The two Type I columns each add up to the model sum of squares 726.54, up to rounding. The other columns do not: the Type II entries add up to more. Education receives 16.31 when entered first and 43.26 after party. Since the least educated group contains the most Democrats, who are the most liberal, the marginal comparison mixes the leftward drift of educated Democrats with the partisan composition of each education group, and the two partly cancel. The Type III sum of squares 32.03 tests whether the unweighted averages over the three parties of the education means are equal. The last column, computed with reference coding, answers different questions: 75.12 for education measures the education gradient among *Democrats only* (the first party level), which is the steepest, and 112.27 for party compares the parties among respondents with a high school education only. The interaction row is the same in every column because the interaction is the highest-order term.

```
import numpy as np
import pandas as pd
import statsmodels.api as sm
d = sm.datasets.anes96.load_pandas().data
party = pd.cut(d["PID"], [-1, 1.5, 4.5, 6], labels=["Dem", "Ind", "Rep"]).rename("party")
educ = pd.cut(d["educ"], [0, 3.5, 5.5, 7], labels=["HS", "College", "Graduate"]).rename("educ")
y = d["selfLR"].to_numpy()
n = len(y)
print(pd.crosstab(party, educ))
print(d.groupby([party, educ], observed=True)["selfLR"].mean().unstack().round(3))
```

```
def proj(*blocks):
    """Orthogonal projection onto the span of the given column blocks."""
    Z = np.column_stack(blocks)
    U, s, _ = np.linalg.svd(Z, full_matrices=False)
    U = U[:, s > 1e-10 * s[0]] # orthonormal basis of C(Z), any
    rank # rank
    return U @ U.T

one = np.ones((n, 1))
A = pd.get_dummies(party).to_numpy(float) # indicators of party
B = pd.get_dummies(educ).to_numpy(float) # indicators of education
AB = np.column_stack([A[:, [i]] * B for i in range(3)]) # the nine cells
PO, PA, PB = proj(one), proj(one, A), proj(one, B)
```

```

PAB, M = proj(one, A, B), proj(AB)                                # additive model, cell-means
                        model
I = np.eye(n)

def ss(P):
    return y @ P @ y

type1_AB = {"party": ss(PA - P0), "educ": ss(PAB - PA), "inter": ss(M - PAB)}
type1_BA = {"educ": ss(PB - P0), "party": ss(PAB - PB), "inter": ss(M - PAB)}
type2 = {"party": ss(PAB - PB), "educ": ss(PAB - PA), "inter": ss(M - PAB)}
sse = ss(I - M)

def coded(labels, kind):
    """Columns for a factor: sum-to-zero ("sum") or reference-level ("treatment")
    coding."""
    D = pd.get_dummies(labels).to_numpy(float)
    return D[:, 1:] - D[:, [0]] if kind == "sum" else D[:, 1:]

type3 = {}
for kind in ["sum", "treatment"]:
    Ca, Cb = coded(party, kind), coded(educ, kind)
    Cab = np.column_stack([Ca[:, [i]] * Cb for i in range(2)])
    full = proj(one, Ca, Cb, Cab)                                # always the cell-means space
    type3[kind] = {"party": ss(full - proj(one, Cb, Cab)),
                  "educ": ss(full - proj(one, Ca, Cab))}
    assert np.allclose(full, M)
for name, table in [("Type I, party first", type1_AB), ("Type I, education first",
type1_BA),
                    ("Type II", type2), ("Type III, sum coding", type3["sum"]),
                    ("Type III, treatment coding", type3["treatment"])] :
    print(f"{name:30s}", {k: round(float(v), 2) for k, v in table.items()})

```

The script behind these cells also checks every entry against statsmodels, and it confirms [Theorem 9.3.2](#) numerically: the party entries of the first, fourth and fifth columns equal [Eq. 9.2.1](#) for the hypotheses of equal weighted row means, equal unweighted row means, and equal means in the high-school column.

9.3.5 Which one to use

There is no single right sum of squares for a main effect in an unbalanced design with interaction, because there is no single main-effect hypothesis. Decide the hypothesis first, from the four above, and compute its sum of squares with [Eq. 9.2.1](#). If A and B interact, a single “main effect of A ” summarizes a heterogeneous comparison with arbitrary weights; Nelder (1977) argued that such tests are rarely meaningful, and Type II sums of squares respect this marginality by adjusting A for B but not for AB . Without interaction the Type II and Type III hypotheses coincide, and Type II, which does not adjust for the interaction columns, generally gives the more powerful test. In regression without interactions all partial types agree; the distinction matters only for terms contained in other terms.

[Chapter 17](#) treats empty cells, where even Type III hypotheses may be nonestimable ([Exercise \(B4\)](#), [Theorem 17.4.1](#)).

9.3.6 Exercises

9.3.6.1 A. Check your understanding

Exercise A1.

For a two-way model with terms A , B and AB , write each of the Type I sums of squares (both orders), the Type II sums of squares and the partial sums of squares in the notation $SS(\cdot | \cdot)$. Which of them are equal for every data set, whatever the design?

Solution. Type I, A first: $SS(A | \mu)$, $SS(B | \mu, A)$, $SS(AB | \mu, A, B)$. Type I, B first: $SS(B | \mu)$, $SS(A | \mu, B)$, $SS(AB | \mu, A, B)$. Type II: $SS(A | \mu, B)$, $SS(B | \mu, A)$, $SS(AB | \mu, A, B)$. Partial: $SS(A | \mu, B, AB)$, $SS(B | \mu, A, AB)$, $SS(AB | \mu, A, B)$, where the first two depend on the coding. Always equal: the interaction entries of all four lists; the Type II entry for A and the Type I entry for A with B first; the Type II entry for B and the Type I entry for B with A first.

9.3.6.2 B. Practice

Exercise B1.

Let $\mathbf{X} = [\mathbf{1}, \mathbf{x}_1, \mathbf{x}_2]$, and let r_{y1} , r_{y2} and r_{12} be the sample correlations among $\mathbf{y}$, $\mathbf{x}_1$ and $\mathbf{x}_2$, with $|r_{12}| < 1$. Show that

$$\frac{SS(\mathbf{x}_2 | \mathbf{1})}{SST} = r_{y2}^2, \quad \frac{SS(\mathbf{x}_2 | \mathbf{1}, \mathbf{x}_1)}{SST} = \frac{(r_{y2} - r_{y1}r_{12})^2}{1 - r_{12}^2}.$$

Show that adjusting for $\mathbf{x}_1$ increases the sum of squares of $\mathbf{x}_2$ whenever $r_{y2} = 0$ and $r_{y1}r_{12} \neq 0$, and find the condition for it in general.

Solution. Work in $\mathbf{1}^\perp$ with the centred vectors $\mathbf{u} = \mathbf{y} - \bar{y}\mathbf{1}$, $\mathbf{v}_1$, $\mathbf{v}_2$, scaled to unit length, so that inner products are correlations and $SST = \|\mathbf{y} - \bar{y}\mathbf{1}\|^2$. By Lemma 9.2.1, $SS(\mathbf{x}_2 | \mathbf{1}) = SST(\mathbf{u}^\top \mathbf{v}_2)^2 = SST r_{y2}^2$. Adjusting for $\mathbf{x}_1$ replaces $\mathbf{v}_2$ by $\mathbf{w} = \mathbf{v}_2 - r_{12}\mathbf{v}_1$, with $\|\mathbf{w}\|^2 = 1 - r_{12}^2$ and $\mathbf{u}^\top \mathbf{w} = r_{y2} - r_{y1}r_{12}$, so $SS(\mathbf{x}_2 | \mathbf{1}, \mathbf{x}_1) = SST(\mathbf{u}^\top \mathbf{w})^2 / \|\mathbf{w}\|^2$. If $r_{y2} = 0$ the first ratio is zero and the second is $r_{y1}^2 r_{12}^2 / (1 - r_{12}^2) > 0$. In general the sum of squares increases iff $(r_{y2} - r_{y1}r_{12})^2 > r_{y2}^2(1 - r_{12}^2)$.

Exercise B2.

Show that the hypotheses of Theorem 9.3.2(a) and (b) are the same subspace of cell means iff $n_{ij} = n_{i.}/b$ for all i, j , that is, iff within every row all cells have the same count. Show that under proportional frequencies, $n_{ij} = n_{i.}n_{.j}/n$ (Theorem 9.4.3), the hypothesis (a) compares row means weighted by the column proportions $n_{.j}/n$, the same weights in every row.

Solution. Write $w_{ij} = n_{ij}/n_{i.}$, so $\sum_j w_{ij} = 1$, and let $f_i(\boldsymbol{\mu}) = \sum_j w_{ij}\mu_{ij}$ and $g_i(\boldsymbol{\mu}) = \sum_j \mu_{ij}/b$. Hypothesis (a) is the null space of the linear functionals $\sum_i c_i f_i$ with $\sum_i c_i = 0$, and (b) that of the functionals $\sum_i d_i g_i$ with $\sum_i d_i = 0$. Two subspaces defined as null spaces coincide iff the spaces of functionals coincide. The functional $f_i - f_1$ has coefficient

w_{ij} on cell (i, j) and $-w_{1j}$ on cell $(1, j)$. If it equals some $\sum_i d_i g_i$, whose coefficients are constant across each row, then w_{ij} does not depend on j , so $w_{ij} = 1/b$; likewise $w_{1j} = 1/b$. Hence coincidence forces $n_{ij} = n_{i\cdot}/b$. Conversely, if all $w_{ij} = 1/b$, then $f_i = g_i$. Under proportional frequencies $w_{ij} = n_{\cdot j}/n$, which does not depend on i , so (a) compares $\sum_j (n_{\cdot j}/n) \mu_{ij}$ across rows.

Exercise B3.

Take $a = b = 2$, counts $n_{11} = n_{22} = 9$ and $n_{12} = n_{21} = 1$, and additive cell means $\mu_{ij} = \beta_j$ with $\beta_1 = 0$ and $\beta_2 = 1$, so that A has no effect. Compute the weighted row means $\bar{\mu}_i^w$ and show that the hypothesis addressed by the Type I sum of squares of A entered first is false. What do the Type II and Type III hypotheses say here?

Exercise B4.

Suppose that in a 2×2 layout the cell $(2, 2)$ is empty. Show that $\bar{\mu}_1 - \bar{\mu}_2$ is not estimable. Compute what the last-in sum of squares of A under sum-to-zero coding tests in this case. *Hint:* with a cell missing, the interaction columns restricted to the observed cells are no longer independent of the main-effect columns.

9.3.6.3 C. Going deeper

Exercise C1.

With the notation of [Exercise \(B1\)](#), show that

$$\frac{SS(\mathbf{x}_1 \mid \mathbf{1}, \mathbf{x}_2) + SS(\mathbf{x}_2 \mid \mathbf{1}, \mathbf{x}_1) - SSR}{SST} = \frac{r_{12}[r_{12}(r_{y1}^2 + r_{y2}^2) - 2r_{y1}r_{y2}]}{1 - r_{12}^2}.$$

Deduce that the partial sums of squares add up to the regression sum of squares when $r_{12} = 0$, that they add up to less when $r_{y1} = r_{y2} \neq 0$ and $0 < r_{12} < 1$, and give correlations for which they add up to more.

9.4 Orthogonal designs

[Section 9.3](#) showed how much the sums of squares for a term can depend on the other terms. This section asks when they do not. The answer is geometric: after the mean is removed, the terms must span mutually orthogonal subspaces. Balanced experiments achieve this by design and observational data essentially never do, but orthogonality can be manufactured inside a term by a change of basis, as with orthogonal polynomials and orthogonal contrasts.

9.4.1 When the order does not matter

Let the model matrix be $\mathbf{X} = [\mathbf{1}, \mathbf{X}_1, \dots, \mathbf{X}_k]$ with an intercept and k terms. Write $\mathbf{P}_0 = n^{-1}\mathbf{1}\mathbf{1}^\top$, and let

$$\tilde{\mathbf{X}}_j = (\mathbf{I} - \mathbf{P}_0)\mathbf{X}_j, \quad \mathcal{V}_j = \mathcal{C}(\tilde{\mathbf{X}}_j),$$

be the centred columns of term j and the subspace they span. By [Lemma 9.2.1](#) with $\mathbf{X}_1 = \mathbf{1}$, $\mathcal{C}(\mathbf{1}, \mathbf{X}_j) = \text{span}(\mathbf{1}) \oplus \mathcal{V}_j$, so $\text{SS}(\mathbf{X}_j | \mathbf{1}) = \|\mathbf{P}_{\mathcal{V}_j}\mathbf{y}\|^2$, the sum of squares of the term fitted on its own after the mean. Call the design **orthogonal** (for these terms) if the subspaces $\mathcal{V}_1, \dots, \mathcal{V}_k$ are mutually orthogonal, that is, if $\tilde{\mathbf{X}}_i^\top \tilde{\mathbf{X}}_j = \mathbf{0}$ for $i \neq j$. For numerical regressors this says that every pair of regressors from different terms has sample correlation zero.

Theorem 9.4.1 — Orthogonal designs

For a model $\mathbf{X} = [\mathbf{1}, \mathbf{X}_1, \dots, \mathbf{X}_k]$, the following are equivalent:

- (a) the centred subspaces $\mathcal{V}_1, \dots, \mathcal{V}_k$ are mutually orthogonal;
- (b) $\mathbf{M} = \mathbf{P}_0 + \mathbf{P}_{\mathcal{V}_1} + \dots + \mathbf{P}_{\mathcal{V}_k}$;
- (c) for every j and every $\mathbf{y}$, the marginal and partial sums of squares agree: $\text{SS}(\mathbf{X}_j | \mathbf{1}) = \text{SS}(\mathbf{X}_j | \mathbf{X}_{(-j)})$;
- (d) for every $\mathbf{y}$, the marginal sums of squares add up to the regression sum of squares: $\sum_j \text{SS}(\mathbf{X}_j | \mathbf{1}) = \text{SS}(\mathbf{X}_1, \dots, \mathbf{X}_k | \mathbf{1})$.

When they hold, $\text{SS}(\mathbf{X}_j | \mathbf{1}, \mathbf{X}_S) = \|\mathbf{P}_{\mathcal{V}_j}\mathbf{y}\|^2$ for every set S of other terms. In particular the sequential sum of squares of $\mathbf{X}_j$ is the same in every order, and it equals the partial and the Type II sums of squares of $\mathbf{X}_j$.

Proof. (a) implies (b) and the final statement. Let S be any set of terms. The subspaces $\text{span}(\mathbf{1})$ and $\mathcal{V}_l$, $l \in S$, are mutually orthogonal, because each $\mathcal{V}_l \subseteq \mathbf{1}^\perp$, and they sum to $\mathcal{C}(\mathbf{1}, \mathbf{X}_S)$, because each $\mathcal{C}(\mathbf{1}, \mathbf{X}_l) = \text{span}(\mathbf{1}) \oplus \mathcal{V}_l$. By [Theorem 6.3.8](#) the projection onto $\mathcal{C}(\mathbf{1}, \mathbf{X}_S)$ is $\mathbf{P}_0 + \sum_{l \in S} \mathbf{P}_{\mathcal{V}_l}$. With S all terms this is (b). With S and $S \cup \{j\}$, the difference of the two projections is $\mathbf{P}_{\mathcal{V}_j}$, which gives $\text{SS}(\mathbf{X}_j | \mathbf{1}, \mathbf{X}_S) = \|\mathbf{P}_{\mathcal{V}_j}\mathbf{y}\|^2$. Sequential, partial and Type II sums of squares of $\mathbf{X}_j$ are all of this form.

(b) implies (d): $\text{SS}(\mathbf{X}_1, \dots, \mathbf{X}_k | \mathbf{1}) = \mathbf{y}^\top (\mathbf{M} - \mathbf{P}_0) \mathbf{y} = \sum_j \mathbf{y}^\top \mathbf{P}_{\mathcal{V}_j} \mathbf{y}$.

(d) implies (a). The two sides of (d) are the quadratic forms of the symmetric matrices $\sum_j \mathbf{P}_{\mathcal{V}_j}$ and $\mathbf{M} - \mathbf{P}_0$. Two symmetric matrices with the same quadratic form are equal, because $2\mathbf{u}^\top \mathbf{A} \mathbf{v} = (\mathbf{u} + \mathbf{v})^\top \mathbf{A} (\mathbf{u} + \mathbf{v}) - \mathbf{u}^\top \mathbf{A} \mathbf{u} - \mathbf{v}^\top \mathbf{A} \mathbf{v}$. So $\sum_j \mathbf{P}_{\mathcal{V}_j} = \mathbf{M} - \mathbf{P}_0$, which is idempotent ([Theorem 6.5.1](#)). By the algebraic form of Cochran's theorem ([Theorem 4.5.1](#)), projections whose sum is idempotent satisfy $\mathbf{P}_{\mathcal{V}_i} \mathbf{P}_{\mathcal{V}_j} = \mathbf{0}$ for $i \neq j$, which is (a).

(a) implies (c) is part of the final statement.

(c) implies (a). As before, equality of the quadratic forms gives $\mathbf{P}_{\mathcal{V}_j} = \mathbf{M} - \mathbf{M}_{(-j)}$. The right-hand side projects onto $\mathcal{C}(\mathbf{X}) \ominus \mathcal{C}(\mathbf{X}_{(-j)})$ ([Theorem 6.5.1](#)), which is orthogonal to $\mathcal{C}(\mathbf{X}_{(-j)})$. The space $\mathcal{C}(\mathbf{X}_{(-j)})$ contains $\mathcal{V}_i$ for every $i \neq j$, since $\tilde{\mathbf{X}}_i = \mathbf{X}_i - \mathbf{P}_0 \mathbf{X}_i$ has columns in $\mathcal{C}(\mathbf{1}, \mathbf{X}_i)$. So $\mathcal{V}_j \perp \mathcal{V}_i$. ■

Idea. Orthogonality is a property of the *centred* subspaces. The raw columns of two indicator variables are never orthogonal to the intercept, and a regressor and its square are rarely orthogonal. What matters is whether the parts of the terms that vary around their means point in perpendicular directions.

Orthogonality also stabilizes the coefficients.

Corollary 9.4.2 — Coefficients in orthogonal designs

Suppose $\mathbf{X} = [\mathbf{1}, \mathbf{X}_1, \dots, \mathbf{X}_k]$ has full column rank and satisfies the conditions of [Theorem 9.4.1](#). Then the least squares coefficients of the columns of $\mathbf{X}_j$ are the same in every model that contains $\mathbf{1}$ and $\mathbf{X}_j$, whichever other terms it contains. Only the intercept changes.

Proof. In the model with terms $S \ni j$, the fitted vector is $\mathbf{P}_0 \mathbf{y} + \sum_{l \in S} \mathbf{P}_{\mathbf{X}_l} \mathbf{y}$ by the proof of the theorem, and $\mathbf{P}_{\mathbf{X}_l} \mathbf{y} = \tilde{\mathbf{X}}_l \mathbf{b}_l$ with $\mathbf{b}_l = (\tilde{\mathbf{X}}_l^\top \tilde{\mathbf{X}}_l)^{-1} \tilde{\mathbf{X}}_l^\top \mathbf{y}$, which does not depend on S . Since $\tilde{\mathbf{X}}_l \mathbf{b}_l = \mathbf{X}_l \mathbf{b}_l - \mathbf{P}_0 \mathbf{X}_l \mathbf{b}_l$, the fitted vector equals $\sum_{l \in S} \mathbf{X}_l \mathbf{b}_l$ plus a multiple of $\mathbf{1}$. The coefficients are unique by full rank, so the coefficient vector of $\mathbf{X}_j$ is $\mathbf{b}_j$ in every such model. ■

With uncorrelated regressors, dropping one leaves the others' estimates as they were, and each regressor has an order-free sum of squares. This is one reason factorial experiments are laid out on orthogonal arrays (Chapters 16 and 18).

9.4.2 Balanced and proportional layouts

For two factors the orthogonality condition reduces to a condition on the cell counts.

Theorem 9.4.3 — Proportional frequencies

In a two-way layout with factor A at a levels, factor B at b levels, cell counts n_{ij} , and all row totals $n_{i\cdot}$ and column totals $n_{\cdot j}$ positive, the centred subspaces of A and B are orthogonal iff

$$n_{ij} = \frac{n_{i\cdot} n_{\cdot j}}{n} \quad \text{for all } i, j. \quad (9.4.1)$$

Proof. A vector in the centred subspace of A takes a value α_i on every observation in row i , with $\sum_i n_{i\cdot} \alpha_i = 0$. A vector in that of B takes the value β_j in column j , with $\sum_j n_{\cdot j} \beta_j = 0$. Their inner product is $\sum_{i,j} n_{ij} \alpha_i \beta_j = \boldsymbol{\alpha}^\top \mathbf{N} \boldsymbol{\beta}$, with $\mathbf{N} = (n_{ij})$. Write $\mathbf{r} = (n_{1\cdot}, \dots, n_{a\cdot})^\top$ and $\mathbf{c} = (n_{\cdot 1}, \dots, n_{\cdot b})^\top$. The subspaces are orthogonal iff $\boldsymbol{\alpha}^\top \mathbf{N} \boldsymbol{\beta} = 0$ whenever $\boldsymbol{\alpha}^\top \mathbf{r} = 0$ and $\mathbf{c}^\top \boldsymbol{\beta} = 0$.

If [Eq. 9.4.1](#) holds, $\mathbf{N} = \mathbf{r} \mathbf{c}^\top / n$ and $\boldsymbol{\alpha}^\top \mathbf{N} \boldsymbol{\beta} = (\boldsymbol{\alpha}^\top \mathbf{r})(\mathbf{c}^\top \boldsymbol{\beta}) / n = 0$. Conversely, let $\mathbf{E} = \mathbf{N} - \mathbf{r} \mathbf{c}^\top / n$. Its row sums are $\mathbf{r} - \mathbf{r}(\mathbf{c}^\top \mathbf{1}) / n = \mathbf{0}$ and its column sums are $\mathbf{c}^\top - (\mathbf{1}^\top \mathbf{r}) \mathbf{c}^\top / n = \mathbf{0}^\top$, since $\mathbf{1}^\top \mathbf{r} = \mathbf{c}^\top \mathbf{1} = n$. So $\boldsymbol{\alpha}^\top \mathbf{E} \boldsymbol{\beta}$ does not change when $\boldsymbol{\alpha}$ or $\boldsymbol{\beta}$ is shifted by a multiple

of **1**. Given any α and β , shift them to $\alpha' = \alpha - (\alpha^\top \mathbf{r}/n)\mathbf{1}$ and $\beta' = \beta - (\mathbf{c}^\top \beta/n)\mathbf{1}$, which satisfy $\alpha'^\top \mathbf{r} = 0$ and $\mathbf{c}^\top \beta' = 0$. Then $\alpha^\top \mathbf{E}\beta = \alpha'^\top \mathbf{E}\beta' = \alpha'^\top \mathbf{N}\beta' - 0 = 0$ by assumption. As α and β are arbitrary, $\mathbf{E} = \mathbf{0}$. ■

Equal replication, $n_{ij} = m$, is the common case ([Example \(Additive two-factor model, one observation per cell\)](#), [Exercise \(C1\)](#)). If the interaction subspace is defined as the orthogonal complement of the additive model within the cell-constant vectors, then under [Eq. 9.4.1](#) the subspaces for the mean, A , B and AB form a decomposition whatever the order.

Orthogonality settles Type I against Type II, but not Type III. Under proportional frequencies the Type I and Type II sums of squares of A coincide and address equality of row means weighted by the column proportions $n_{.j}/n$ ([Exercise \(B2\)](#)), while Type III weights the columns equally. They agree with equal replication or without interaction.

Example. (Firms and years)

Grunfeld's investment panel ([Example \(Grunfeld's investment data\)](#)) records 11 firms in each of 20 years, one observation per firm and year. As a two-way layout with factors firm and year, it has $n_{ij} = 1$, so [Eq. 9.4.1](#) holds, and the additive model's sums of squares for the logarithm of investment are order-free:

Source	df	Sum of squares
firm	10	473.482
year	19	26.749
residual	190	14.667
corrected total	219	514.898

Firms differ enormously in the scale of their investment, and years matter much less. Adding the logarithm of market value as a covariate destroys the orthogonality, because value varies with both firm and year. The sum of squares for firm then depends on what else is in the model: it is 90.701 after year and value, and 88.443 after value alone. Most of the variation between firms is variation in size, which value already measures. This is the setting of analysis of covariance ([Chapter 18](#), [Theorem 18.1.1](#)).

```
import numpy as np
import pandas as pd
import statsmodels.api as sm

def proj(*blocks):
    Z = np.column_stack(blocks)
    U, s, _ = np.linalg.svd(Z, full_matrices=False)
    U = U[:, s > 1e-10 * s[0]]
    return U @ U.T

g = sm.datasets.grunfeld.load_pandas().data
y = np.log(g["invest"].to_numpy())
n = len(y)
one = np.ones((n, 1))
F = pd.get_dummies(g["firm"]).to_numpy(float)      # 11 firms
T = pd.get_dummies(g["year"]).to_numpy(float)      # 20 years
```

```

P0, PF, PT, PFT = proj(one), proj(one, F), proj(one, T), proj(one, F, T)

print("centred spaces orthogonal:", np.abs((PF - P0) @ (PT - P0)).max() < 1e-10)
print(f"firm:  first {y @ (PF - P0) @ y:8.3f}    after year {y @ (PFT - PT) @ y:8.3f}"
      ")
print(f"year:  first {y @ (PT - P0) @ y:8.3f}    after firm {y @ (PFT - PF) @ y:8.3f}"
      ")

v = np.log(g["value"].to_numpy())[:, None] # log market value
PFv, PTv, PFTv = proj(one, F, v), proj(one, T, v), proj(one, F, T, v)
print(f"with log value: firm after year, value {y @ (PFTv - PTv) @ y:8.3f}"
      f"    firm after value {y @ (PFv - proj(one, v)) @ y:8.3f}")

```

9.4.3 Orthogonal polynomials

A polynomial regression in a single variable x , with distinct values $x_1, \dots, x_n$, has a built-in chain of models,

$$\mathcal{P}_0 \subseteq \mathcal{P}_1 \subseteq \dots \subseteq \mathcal{P}_d, \quad \mathcal{P}_k = \text{span}(\mathbf{1}, \mathbf{x}, \mathbf{x}^2, \dots, \mathbf{x}^k),$$

where $\mathbf{x}^k$ is the vector with entries x_i^k . The decomposition of interest is that of the chain (Theorem 9.1.1(e)). Its k th subspace, $\mathcal{P}_{k-1}^\perp \cap \mathcal{P}_k$, is a line. Applying Gram–Schmidt (Proposition 6.2.2) to $\mathbf{1}, \mathbf{x}, \mathbf{x}^2, \dots$ produces unit vectors $\mathbf{q}_0, \mathbf{q}_1, \dots, \mathbf{q}_d$ with $\text{span}(\mathbf{q}_0, \dots, \mathbf{q}_k) = \mathcal{P}_k$ for every k . So $\mathbf{q}_k$ spans that line. The vectors $\mathbf{q}_k$, evaluated at the design points, are the **orthogonal polynomials** of the design. Three consequences follow at once.

1. The sequential sum of squares for degree k , the drop in residual sum of squares from degree $k-1$ to degree k , is $(\mathbf{q}_k^\top \mathbf{y})^2$, with one degree of freedom.
2. The columns $\mathbf{q}_1, \dots, \mathbf{q}_d$ are orthogonal and centred, so as separate terms they form an orthogonal design (Theorem 9.4.1), and their sums of squares do not depend on the order of entry.
3. The coefficient of $\mathbf{q}_k$ is $\mathbf{q}_k^\top \mathbf{y}$ in every fit that includes it (Corollary 9.4.2). Raising the degree never changes the lower coefficients.

For equally spaced x the orthogonal polynomials have integer multiples that are widely tabulated. For three points they are $(-1, 0, 1)$ and $(1, -2, 1)$. For four points they are $(-3, -1, 1, 3)$, $(1, -1, -1, 1)$ and $(-1, 3, -3, 1)$. They serve as contrasts for a quantitative factor with equally spaced levels, splitting the treatment sum of squares into linear, quadratic and higher components. Their numerical merits and the recurrence that generates them belong to Chapter 42.

Example. (A trend in the Nile)

The annual flow of the Nile at Aswan was recorded for the 100 years 1871–1970. How much of its variation is a smooth trend? The listing builds the orthonormal polynomials $\mathbf{q}_0, \dots, \mathbf{q}_4$ for the years by a QR factorization of the matrix of powers, and reads off the sequential sums of squares (in units of $(10^8 \text{ m}^3)^2$):

Degree	1	2	3	4
Sequential sum of squares	613893	309415	1894	134142

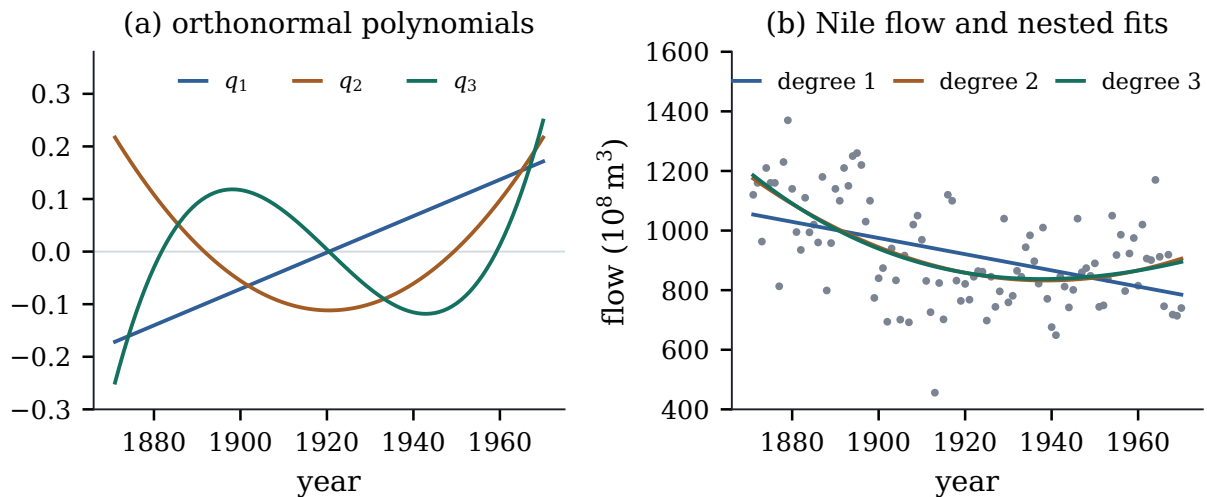


Figure 9.4.1: (a) The orthonormal polynomials q_1, q_2, q_3 for the years 1871–1970. (b) The Nile flow with the nested least squares fits of degrees 1, 2 and 3. The quadratic and cubic fits nearly coincide, because the cubic sum of squares is tiny.

The corrected total is 2835157. The linear term accounts for a fraction 0.217 of it, and the first three terms together for 0.326. Against the residual mean square of the cubic fit, 19895 on 96 degrees of freedom, the quadratic component has $F = 15.55$ and the cubic only 0.10. The quartic sum of squares is large again, a sign that the pattern is not a gentle polynomial at all. The flow dropped abruptly around the end of the nineteenth century (Cobb 1978), and a low-degree polynomial can only approximate a step (Figure 9.4.1).

With the raw powers of the centred and scaled year, the sums of squares depend on the order: the linear column entered after the quadratic and cubic ones receives 74751 instead of 613893. The raw linear coefficient is -134.4 in the linear and quadratic fits, but -117.3 in the cubic and quartic ones. It survives the quadratic term because the years are symmetric about their mean, which makes odd and even centred powers orthogonal (Exercise (C1)), but not the cubic term. The coefficient of q_1 is -783.5 in every fit.

```
import numpy as np
import statsmodels.api as sm
nile = sm.datasets.nile.load_pandas().data
year = nile["year"].to_numpy()
y = nile["volume"].to_numpy()

t = year - year.mean()
V = np.vander(t / t.max(), 5, increasing=True)
Q, R = np.linalg.qr(V)
Q = Q * np.sign(np.diag(R))
    with t^k
coef = Q.T @ y
    q_4
seq_ss = coef[1:] ** 2
    quartic
for k, s in enumerate(seq_ss, start=1):
    print(f"degree {k}: sequential SS = {s:10.0f}")
print(f"residual after degree 4: {y @ y - np.sum(coef ** 2):10.0f}")
```

centring only helps rounding
1, t, t^2, t^3, t^4 (scaled)
Gram-Schmidt in matrix form
make each q_k agree in sign

coordinates of y on $q_0, \dots$

sequential SS: linear, $\dots$

9.4.4 Single-degree-of-freedom decompositions

Orthogonal polynomials split a sum of squares into one-dimensional pieces along directions chosen for their meaning. The same can be done for any hypothesis, by choosing single estimable functions whose estimates are uncorrelated.

Theorem 9.4.4 — Orthogonal single-degree-of-freedom sums of squares

Let $\lambda_1, \dots, \lambda_s$ be nonzero vectors with $\lambda_k^\top \beta$ estimable, so that $\lambda_k = \mathbf{X}^\top \rho_k$, and let $\mathbf{G}$ be a generalized inverse of $\mathbf{X}^\top \mathbf{X}$.

- (a) The sum of squares for $\lambda_k^\top \beta = 0$ is

$$SS_k = \frac{(\lambda_k^\top \hat{\beta})^2}{\lambda_k^\top \mathbf{G} \lambda_k} = \|\mathbf{P}_{\mathbf{v}_k} \mathbf{y}\|^2, \quad \mathbf{v}_k = \mathbf{M} \rho_k,$$

the squared length of the projection of $\mathbf{y}$ onto the line spanned by $\mathbf{v}_k$.

- (b) $\mathbf{v}_h^\top \mathbf{v}_k = \lambda_h^\top \mathbf{G} \lambda_k$. If $\lambda_h^\top \mathbf{G} \lambda_k = 0$ for all $h \neq k$, the lines are mutually orthogonal, and $\sum_k SS_k$ is the sum of squares for the joint hypothesis $\lambda_1^\top \beta = \dots = \lambda_s^\top \beta = 0$, on s degrees of freedom.
- (c) Conversely, every orthonormal basis $\mathbf{u}_1, \dots, \mathbf{u}_s$ of a subspace of $\mathcal{C}(\mathbf{X})$ arises in this way, with $\lambda_k = \mathbf{X}^\top \mathbf{u}_k$.

Proof.

- (a) is Theorem 9.2.4 with $q = 1$ and $\mathbf{d} = 0$: the test space is spanned by $\mathbf{M} \rho_k = \mathbf{v}_k$, and the projection of $\mathbf{y}$ onto it has squared length $(\mathbf{v}_k^\top \mathbf{y})^2 / \mathbf{v}_k^\top \mathbf{v}_k$, where $\mathbf{v}_k^\top \mathbf{y} = \rho_k^\top \mathbf{M} \mathbf{y} = \lambda_k^\top \hat{\beta}$ and $\mathbf{v}_k^\top \mathbf{v}_k = \lambda_k^\top \mathbf{G} \lambda_k$. (b) $\mathbf{v}_h^\top \mathbf{v}_k = \rho_h^\top \mathbf{M} \rho_k = \rho_h^\top \mathbf{X} \mathbf{G} \mathbf{X}^\top \rho_k = \lambda_h^\top \mathbf{G} \lambda_k$ by Theorem 6.3.7. Orthogonal nonzero vectors are independent, so $\mathbf{\Lambda} = [\lambda_1, \dots, \lambda_s]$ has rank s (because $\mathbf{X}^\top \mathbf{V} = \mathbf{\Lambda}$ with $\mathbf{V} = [\mathbf{v}_1, \dots, \mathbf{v}_s]$, and $\mathbf{X}^\top$ is one-to-one on $\mathcal{C}(\mathbf{X})$). The test space of the joint hypothesis is $\mathcal{C}(\mathbf{M} \mathbf{T}) = \mathcal{C}(\mathbf{V})$, with $\mathbf{T} = [\rho_1, \dots, \rho_s]$, by Theorem 9.2.4(b), and the projection onto a span of orthogonal lines is the sum of the projections onto the lines (Theorem 6.3.8). (c) With $\rho_k = \mathbf{u}_k$, $\mathbf{v}_k = \mathbf{M} \mathbf{u}_k = \mathbf{u}_k$, and $\lambda_h^\top \mathbf{G} \lambda_k = \mathbf{u}_h^\top \mathbf{u}_k$.

■

In the one-way layout with group sizes $n_1, \dots, n_g$, take the generalized inverse of Exercise (B3). A contrast $\sum_i c_i \alpha_i$ with $\sum_i c_i = 0$ (Definition 8.7.1) is estimated by $\sum_i c_i \bar{y}_i$, and $\lambda^\top \mathbf{G} \lambda = \sum_i c_i^2 / n_i$. So two contrasts $\mathbf{c}$ and $\mathbf{c}'$ give orthogonal sums of squares iff

$$\sum_{i=1}^g \frac{c_i c'_i}{n_i} = 0,$$

and $g - 1$ such contrasts split the between-groups sum of squares into single degrees of freedom. With equal group sizes this is ordinary orthogonality, $\sum_i c_i c'_i = 0$; with unequal sizes, contrasts that look orthogonal need not be.

Example. (Splitting the education sum of squares)

Consider the 944 self-placements of [Example \(Party, education and political self-placement\)](#) by education alone. The group sizes are 313, 277 and 354, and the group means are 4.476, 4.357 and 4.167. The between-groups sum of squares is 16.306 on two degrees of freedom. The contrast of college against graduate, $\mathbf{c}_2 = (0, 1, -1)$, has sum of squares 5.653. The contrast of high school against the other two, weighted by their sizes, $\mathbf{c}_1 = (1, -0.439, -0.561)$, is orthogonal to $\mathbf{c}_2$ in the sense above, and has sum of squares 10.652. The two add up to the between-groups sum of squares, up to rounding. The more obvious contrast $(1, -\frac{1}{2}, -\frac{1}{2})$, high school against the plain average of the others, has sum of squares 9.534, and together with $\mathbf{c}_2$ it gives 15.187: with unequal group sizes it is not orthogonal to $\mathbf{c}_2$.

```
import numpy as np
import pandas as pd
import statsmodels.api as sm
d = sm.datasets.anes96.load_pandas().data
educ = pd.cut(d["educ"], [0, 3.5, 5.5, 7], labels=["HS", "College", "Graduate"])
y = d["selfLR"].to_numpy()

counts = educ.value_counts(sort=False).to_numpy().astype(float)
means = d.groupby(educ, observed=True)["selfLR"].mean().to_numpy()
grand = y.mean()
ss_treat = np.sum(counts * (means - grand) ** 2)          # SS(educ | 1), 2 df

def contrast_ss(c):
    """(c' ybar)^2 / sum(c_i^2 / n_i): the SS for the hypothesis c' mu = 0."""
    return (c @ means) ** 2 / np.sum(c ** 2 / counts)

def orthogonal(c, h):
    return np.isclose(np.sum(c * h / counts), 0.0)

n2, n3 = counts[1], counts[2]
c1 = np.array([1.0, -n2 / (n2 + n3), -n3 / (n2 + n3)]) # HS against the rest,
    weighted
c2 = np.array([0.0, 1.0, -1.0])                      # College against Graduate
c1_plain = np.array([1.0, -0.5, -0.5])                # HS against the plain
    average
print("orthogonal:", orthogonal(c1, c2), orthogonal(c1_plain, c2))
print(f"treatment SS {ss_treat:.3f} = {contrast_ss(c1):.3f} + {contrast_ss(c2):.3f}"
      )
print(f"with the plain contrast: {contrast_ss(c1_plain):.3f} + {contrast_ss(c2):.3f}"
      )
```

Single-degree-of-freedom sums of squares reappear in [Chapter 13](#), where many of them are tested at once, and in [Chapter 15](#), where orthogonal contrasts organize the one-way analysis of variance ([Theorem 15.4.1](#)).

9.4.5 Exercises

9.4.5.1 A. Check your understanding

Exercise A1.

With $x = 1, 2, 3, 4$ and $\mathbf{y} = (3, 5, 4, 8)^\top$, use the orthogonal polynomial contrasts for four equally spaced points to compute the linear, quadratic and cubic sums of squares, and check that they add up to the corrected total sum of squares.

Exercise A2.

Which of the 2×2 count tables $\begin{pmatrix} 2 & 4 \\ 3 & 6 \end{pmatrix}$, $\begin{pmatrix} 2 & 4 \\ 4 & 2 \end{pmatrix}$ and $\begin{pmatrix} 1 & 3 \\ 3 & 9 \end{pmatrix}$ give orthogonal centred factor spaces? For the one that does not, compute the inner product of the two centred indicator vectors of the first levels.

Exercise A3.

Show that if the regressors $\mathbf{x}_1, \dots, \mathbf{x}_k$ of a model with an intercept are pairwise uncorrelated, then $R^2 = \sum_j r_{yj}^2$, where r_{yj} is the sample correlation of $\mathbf{y}$ and $\mathbf{x}_j$.

9.4.5.2 B. Practice**Exercise B1.**

In a two-way layout with interaction and $n_{ij} = m \geq 1$ for all cells, show that the Type I sums of squares of A in either order, and its Type II and Type III sums of squares, are all equal to $bm \sum_i (\bar{y}_{i..} - \bar{y}_{...})^2$.

Solution. Equal counts satisfy [Eq. 9.4.1](#), so by [Theorem 9.4.1](#) the Type I sums of squares of A in both orders and its Type II sum of squares equal $\text{SS}(A \mid \mu) = \|(\mathbf{P}_A - \mathbf{P}_0)\mathbf{y}\|^2$. Each row has bm observations, and $\mathbf{P}_A - \mathbf{P}_0$ replaces an observation in row i by $\bar{y}_{i..} - \bar{y}_{...}$, which gives the formula. For Type III, the hypothesis of [Theorem 9.3.2\(b\)](#) is that of (a), because with equal counts weighted and unweighted row means coincide. A sum of squares for a hypothesis depends only on the hypothesis (it is $\text{SSE}_0 - \text{SSE}$), so the Type III sum of squares equals $\text{SS}(A \mid \mu)$ too.

Exercise B2.

Take counts $n_{11} = 1, n_{12} = 2, n_{21} = 2, n_{22} = 4$. Check [Eq. 9.4.1](#). Find cell means for which the Type III hypothesis for A holds but the Type I (and Type II) hypothesis fails.

Exercise B3.

In a one-way layout with g groups of equal size m , show that the Helmert contrasts $\mathbf{c}_k = (1, \dots, 1, -(k-1), 0, \dots, 0)$, with $k-1$ leading ones, $k = 2, \dots, g$, are mutually orthogonal, and write the between-groups sum of squares as a sum of $g-1$ single-degree-of-freedom sums of squares. What goes wrong with unequal group sizes?

9.4.5.3 C. Going deeper

Exercise C1.

Suppose the design points are symmetric about their mean: after sorting, $x_i - \bar{x} = -(x_{n+1-i} - \bar{x})$. Show that every odd power of the centred variable is orthogonal to every even power (including the zeroth), that $\mathbf{q}_k$ is an even or odd vector according to the parity of k , and explain the pattern of raw coefficients in [Example \(A trend in the Nile\)](#).

Solution. Let $\mathbf{t}$ have entries $t_i = x_i - \bar{x}$ and let $\mathbf{J}$ be the reversal permutation, so that $\mathbf{J}\mathbf{t} = -\mathbf{t}$. A vector $\mathbf{u}$ is even if $\mathbf{J}\mathbf{u} = \mathbf{u}$ and odd if $\mathbf{J}\mathbf{u} = -\mathbf{u}$. Powers $\mathbf{t}^k$ have the parity of k , and an even vector $\mathbf{u}$ and an odd vector $\mathbf{v}$ are orthogonal because $\mathbf{u}^\top \mathbf{v} = (\mathbf{J}\mathbf{u})^\top (\mathbf{J}\mathbf{v}) = -\mathbf{u}^\top \mathbf{v}$. Gram–Schmidt on $\mathbf{1}, \mathbf{t}, \mathbf{t}^2, \dots$ subtracts from $\mathbf{t}^k$ only its projections on the earlier $\mathbf{q}_j$ of the same parity, by induction, so $\mathbf{q}_k$ has the parity of k . Hence the column $\mathbf{t}$ is orthogonal to the column $\mathbf{t}^2$ and to $\mathbf{1}$, and adding the quadratic term does not change the linear coefficient ([Corollary 9.4.2](#), applied to the terms $\mathbf{t}$ and $\mathbf{t}^2 - \bar{\mathbf{t}}^2\mathbf{1}$). The cubic column is odd and not orthogonal to $\mathbf{t}$, so it changes the linear coefficient. The quartic column is even. By [Theorem 9.2.2\(b\)](#), adding it changes the cubic fit’s coefficients by $-\mathbf{L}\hat{\gamma}$ with $\mathbf{L} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{t}^4$, where $\mathbf{X} = [\mathbf{1}, \mathbf{t}, \mathbf{t}^2, \mathbf{t}^3]$. Inner products between odd and even columns vanish, so $\mathbf{X}^\top \mathbf{X}$ and its inverse do not link the odd and even columns, and the entries of $\mathbf{X}^\top \mathbf{t}^4$ for the odd columns are zero. Hence the rows of $\mathbf{L}$ for $\mathbf{t}$ and $\mathbf{t}^3$ vanish, and the linear coefficient is unchanged.

9.5 The coefficient of determination and its limits

The coefficient of determination is the most widely reported summary of a regression, and probably the most widely misread. It is a ratio of two rows of the corrected table,

$$R^2 = \frac{\text{SSR}}{\text{SST}} = 1 - \frac{\text{SSE}}{\text{SST}},$$

defined when $\mathbf{1} \in \mathcal{C}(\mathbf{X})$ and $\mathbf{y}$ is not constant. [Section 6.7](#) showed that it is a squared cosine ([Theorem 6.7.2](#)) that cannot decrease as the model space grows. This section adds R^2 as a maximal correlation, its null distribution, the adjusted and partial versions, and what it does not measure.

9.5.1 A maximal correlation

The square root $R = \sqrt{R^2}$ is called the **multiple correlation coefficient**. The name comes from the following property, the sample counterpart of the population result [Proposition 2.5.3](#).

Theorem 9.5.1 — R is the largest correlation with a linear combination

Suppose $\mathbf{1} \in \mathcal{C}(\mathbf{X})$ and $\mathbf{y}$ is not constant. For every $\mathbf{b}$ such that $\mathbf{Xb}$ is not constant,

$$\text{corr}(\mathbf{y}, \mathbf{Xb})^2 \leq R^2.$$

If $R^2 > 0$, equality holds iff $\mathbf{Xb} = a\hat{\mathbf{Y}} + c\mathbf{1}$ for some $a \neq 0$ and some c , and the correlation $\text{corr}(\mathbf{y}, \mathbf{Xb})$ is maximized, with value R , by the fitted values themselves. If $R^2 = 0$, every such correlation is zero.

Proof. Let $\mathbf{u} = \mathbf{y} - \bar{y}\mathbf{1}$ and $\mathcal{V} = \mathcal{C}(\mathbf{X}) \cap \mathbf{1}^\perp$, whose projection is $\mathbf{M} - \mathbf{P}_1$ (Theorem 6.5.1), and let $\mathbf{v} = \mathbf{Xb} - \bar{\mathbf{Xb}}\mathbf{1}$. Since $\mathbf{1} \in \mathcal{C}(\mathbf{X})$, $\mathbf{v} \in \mathcal{V}$, so $\mathbf{u}^\top \mathbf{v} = \mathbf{u}^\top (\mathbf{M} - \mathbf{P}_1)\mathbf{v} = ((\mathbf{M} - \mathbf{P}_1)\mathbf{u})^\top \mathbf{v}$. The sample correlation is $\mathbf{u}^\top \mathbf{v} / (\|\mathbf{u}\| \|\mathbf{v}\|)$, and by Cauchy–Schwarz (Proposition 1.1.1)

$$\text{corr}(\mathbf{y}, \mathbf{Xb})^2 = \frac{((\mathbf{M} - \mathbf{P}_1)\mathbf{u})^\top \mathbf{v})^2}{\|\mathbf{u}\|^2 \|\mathbf{v}\|^2} \leq \frac{\|(\mathbf{M} - \mathbf{P}_1)\mathbf{u}\|^2}{\|\mathbf{u}\|^2} = \frac{\text{SSR}}{\text{SST}},$$

with equality iff $\mathbf{v}$ and $(\mathbf{M} - \mathbf{P}_1)\mathbf{u} = \hat{\mathbf{Y}} - \bar{y}\mathbf{1}$ are linearly dependent. If $R^2 > 0$ the second vector is nonzero, and since $\mathbf{v} \neq \mathbf{0}$ dependence means $\mathbf{v} = a(\hat{\mathbf{Y}} - \bar{y}\mathbf{1})$ with $a \neq 0$, which is the stated condition; for $a > 0$ the correlation itself is $+R$. If $R^2 = 0$, then $(\mathbf{M} - \mathbf{P}_1)\mathbf{u} = \mathbf{0}$ and $\mathbf{u}^\top \mathbf{v} = 0$. ■

Among all linear combinations of the regressors, the fitted values are the one most correlated with the response. When the regressors are pairwise uncorrelated, R^2 is the sum of the squared simple correlations (Exercise A3). In general it can be smaller or larger than that sum (Exercise C2).

9.5.2 R^2 when nothing is going on

Since R^2 never decreases as regressors are added, a model with many regressors has a sizeable R^2 even if none of them matters. The theory of Chapter 4 says how sizeable.

Theorem 9.5.2 — The distribution of R^2

Let $\mathbf{Y} \sim N_n(\boldsymbol{\mu}, \sigma^2 \mathbf{I})$ with $\boldsymbol{\mu} \in \mathcal{C}(\mathbf{X})$, $\mathbf{1} \in \mathcal{C}(\mathbf{X})$ and $1 < r = \text{rank}(\mathbf{X}) < n$. Let $F = [\text{SSR}/(r-1)]/[\text{SSE}/(n-r)]$ and $\gamma = \|(\mathbf{M} - \mathbf{P}_1)\boldsymbol{\mu}\|^2/\sigma^2$. Then:

- (a) $R^2 = \frac{(r-1)F}{(r-1)F + n-r}$, and $F \sim F(r-1, n-r, \gamma)$;
- (b) if $\boldsymbol{\mu} \in \text{span}(\mathbf{1})$, so that no regressor matters, then $R^2 \sim \text{Beta}(\frac{r-1}{2}, \frac{n-r}{2})$ and $E(R^2) = \frac{r-1}{n-1}$.

Proof.

- (a) $R^2/(1-R^2) = \text{SSR}/\text{SSE} = (r-1)F/(n-r)$, and solving for R^2 gives the formula. The projections $\mathbf{M} - \mathbf{P}_1$ and $\mathbf{I} - \mathbf{M}$ are orthogonal with ranks $r-1$ and $n-r$, and $(\mathbf{I} - \mathbf{M})\boldsymbol{\mu} = \mathbf{0}$, so by Theorem 4.4.4 $\text{SSR}/\sigma^2 \sim \chi^2(r-1, \gamma)$ and $\text{SSE}/\sigma^2 \sim \chi^2(n-r)$,

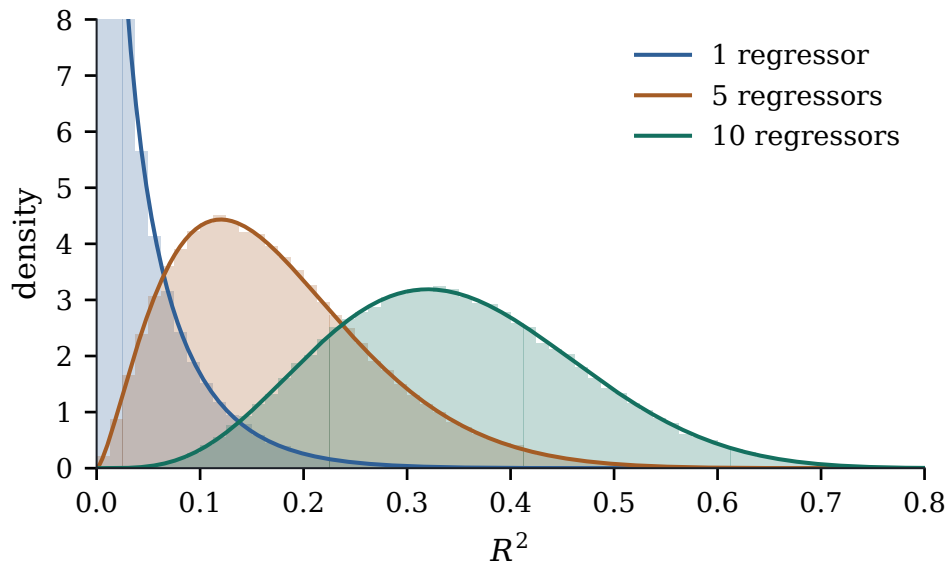


Figure 9.5.1: Null distribution of R^2 for $n = 30$ observations and 1, 5 or 10 regressors unrelated to the response. Histograms: 100,000 simulated data sets each. Curves: the Beta densities of [Theorem 9.5.2](#).

independently. The ratio F is therefore $F(r-1, n-r, \gamma)$ ([Definition 4.1.2](#)).

- (b) Now $\gamma = 0$, and $R^2 = U/(U+V)$ with $U = \text{SSR}/\sigma^2 \sim \chi^2(r-1)$ and $V = \text{SSE}/\sigma^2 \sim \chi^2(n-r)$ independent. By [Exercise \(B4\)](#) this is $\text{Beta}(\frac{r-1}{2}, \frac{n-r}{2})$, whose mean is $\frac{r-1}{2} / \frac{n-1}{2} = (r-1)/(n-1)$.

■

Part (b) answers the exercise [Exercise \(C1\)](#) of [Chapter 6](#). With $k = r - 1$ regressors of pure noise, R^2 is on average $k/(n-1)$. A model with 10 useless regressors fitted to 30 observations has expected R^2 about one third. [Figure 9.5.1](#) compares the Beta densities with 100,000 simulated data sets with $n = 30$ and $k = 1, 5, 10$ regressors drawn at random. The simulated means are 0.0345, 0.1725 and 0.3445, against the exact 0.0345, 0.1724 and 0.3448. The upper 5% points of the three null distributions are 0.130, 0.353 and 0.556, so with ten regressors an R^2 of one half is unremarkable. Part (a) shows that R^2 and F carry the same information: testing R^2 against its null distribution is the overall F test of the regression.

```
import numpy as np
rng = np.random.default_rng(20260919)
n, reps = 30, 100_000
results = {}
for k in (1, 5, 10):
    # number of regressors
    # besides 1
    X = np.column_stack([np.ones(n), rng.normal(size=(n, k))])
    r = k + 1
    Q, _ = np.linalg.qr(X)
    Y = 3.0 + rng.normal(size=(n, reps))
    Z = Q.T @ Y
    sst = np.sum(Y ** 2, axis=0) - Z[0] ** 2
    ssr = np.sum(Z[1:] ** 2, axis=0)
    R2 = ssr / sst
    R2adj = 1 - (1 - R2) * (n - 1) / (n - r)
    # no regressor matters
    print(f"k = {k:2d}: mean R^2 = {R2.mean():.4f} (theory {k / (n - 1):.4f}),"
```

```
f" mean adjusted R^2 = {R2adj.mean():+.4f}"
results[k] = R2
```

9.5.3 Adjusted R^2

The bias of R^2 under the null suggests a correction. Write $s^2 = \text{SSE}/(n-r)$ and $s_y^2 = \text{SST}/(n-1)$ for the unbiased estimators of the error variance in the fitted model and in the intercept-only model. Replacing sums of squares by these mean squares defines the **adjusted coefficient of determination**

$$\bar{R}^2 = 1 - \frac{s^2}{s_y^2} = 1 - (1 - R^2) \frac{n-1}{n-r}. \quad (9.5.1)$$

Proposition 9.5.3 — Properties of the adjusted R^2

Let $\mathbf{1} \in \mathcal{C}(\mathbf{X})$, $1 \leq r = \text{rank}(\mathbf{X}) < n$, and let F be the overall F ratio of [Theorem 9.5.2](#).

- (a) $\bar{R}^2 \leq R^2$, with equality iff $r = 1$ or $R^2 = 1$.
- (b) For $r > 1$, $\bar{R}^2 < 0$ iff $F < 1$.
- (c) If a model with rank r_0 is enlarged to one of rank $r = r_0 + q$ containing it, then $\bar{R}^2$ increases iff the F ratio of the added terms, $[\text{SS}(\text{added} \mid \text{old})/q]/s^2$, exceeds 1. For a single added column this means $|t| > 1$.
- (d) Among models for the same response, $\bar{R}^2$ is largest for the model with the smallest residual mean square s^2 .
- (e) Under the normal model with $\boldsymbol{\mu} \in \text{span}(\mathbf{1})$, $E(\bar{R}^2) = 0$.

Proof.

- (a) $1 - \bar{R}^2 = (1 - R^2)(n-1)/(n-r)$, and $(n-1)/(n-r) \geq 1$ with equality iff $r = 1$, so $1 - \bar{R}^2 \geq 1 - R^2$, with equality iff $r = 1$ or $1 - R^2 = 0$.
- (b) $\bar{R}^2 < 0$ iff $s^2 > s_y^2$, that is, $\text{SSE}(n-1) > (\text{SSR} + \text{SSE})(n-r)$, that is, $\text{SSE}(r-1) > \text{SSR}(n-r)$, which for $r > 1$ is $F < 1$.
- (c) s_y^2 does not depend on the model, so $\bar{R}^2$ increases iff s^2 decreases. Write $\text{SSE}_0 = \text{SSE} + D$ with D the extra sum of squares. Then $\text{SSE}/(n-r) < (\text{SSE} + D)/(n-r_0)$ iff $\text{SSE}(n-r_0) < (\text{SSE} + D)(n-r)$, iff $q\text{SSE} < D(n-r)$, iff $(D/q)/s^2 > 1$. For one column the ratio is t^2 ([Corollary 9.2.3](#)).
- (d) is immediate from [Eq. 9.5.1](#).
- (e) By [Theorem 9.5.2\(b\)](#), $E(1 - R^2) = 1 - (r-1)/(n-1) = (n-r)/(n-1)$, so $E(1 - \bar{R}^2) = 1$.

■

Part (e) is the point of the adjustment. In the simulation the averages of $\bar{R}^2$ are 0.0000, 0.0001 and -0.0005 , but $\bar{R}^2$ is negative in a proportion 0.674 of the data sets with one regressor: it is not a proportion of anything. Part (c) shows what selection by $\bar{R}^2$ amounts to: a regressor is

admitted when $|t| > 1$, far below the threshold near two of a 5% test, so the chosen models tend to be too large (Chapter 29).

Example. (Adjusted R^2 for the state data)

For all seven models built from poverty, single parenthood and urbanization (each with an intercept):

Regressors	R^2	$\bar{R}^2$
poverty	0.3599	0.3466
single	0.5719	0.5629
urban	0.0221	0.0017
poverty, single	0.6410	0.6257
poverty, urban	0.4626	0.4398
single, urban	0.5844	0.5668
all three	0.6419	0.6186

R^2 is largest for the full model, as it must be. $\bar{R}^2$ prefers poverty and single parenthood without urbanization. Adding urbanization to that model raises R^2 slightly but lowers $\bar{R}^2$, because its t statistic in the full model is only 0.349, so its F ratio is $0.121 < 1$.

9.5.4 Partial R^2

Applied to what a reduced model leaves unexplained, the idea of R^2 gives a measure for a term.

Definition 9.5.1 — Partial coefficient of determination

Let $\mathcal{C}(\mathbf{X}_1) \subseteq \mathcal{C}(\mathbf{X})$ with $\mathbf{1} \in \mathcal{C}(\mathbf{X}_1)$. The **partial coefficient of determination** of $\mathbf{X}$ over $\mathbf{X}_1$ is

$$R_{\mathbf{X}|\mathbf{X}_1}^2 = \frac{\text{SSE}(\mathbf{X}_1) - \text{SSE}(\mathbf{X})}{\text{SSE}(\mathbf{X}_1)} = \frac{\text{SS}(\mathbf{X} | \mathbf{X}_1)}{\text{SSE}(\mathbf{X}_1)},$$

the proportion of the residual variation of the reduced model that the larger model explains.

Theorem 9.5.4 — Partial R^2 , F and t

With the notation of Definition 9.5.1, let $r_1 = \text{rank}(\mathbf{X}_1)$, $r = \text{rank}(\mathbf{X})$, $q = r - r_1 > 0$, $s^2 = \text{SSE}(\mathbf{X})/(n - r)$, and let $F = [\text{SS}(\mathbf{X} | \mathbf{X}_1)/q]/s^2$. Then:

- (a) $1 - R^2 = (1 - R_1^2)(1 - R_{\mathbf{X}|\mathbf{X}_1}^2)$, where R^2 and R_1^2 belong to the two models;
- (b) $R_{\mathbf{X}|\mathbf{X}_1}^2 = \frac{qF}{qF + n - r}$;
- (c) if $\mathbf{X} = [\mathbf{X}_1, \mathbf{x}_j]$ adds one column, then $R_{\mathbf{X}|\mathbf{X}_1}^2 = \frac{t_j^2}{t_j^2 + n - r}$, and it equals the squared sample correlation between the residuals $(\mathbf{I} - \mathbf{M}_1)\mathbf{y}$ and $(\mathbf{I} - \mathbf{M}_1)\mathbf{x}_j$;

(d) for a chain $\mathcal{C}(\mathbf{1}) = \mathcal{C}(\mathbf{X}_0) \subseteq \mathcal{C}(\mathbf{X}_1) \subseteq \cdots \subseteq \mathcal{C}(\mathbf{X}_k)$, $1 - R_k^2 = \prod_{j=1}^k (1 - R_{\mathbf{X}_j|\mathbf{X}_{j-1}}^2)$.

Proof.

- (a) $1 - R^2 = \text{SSE}(\mathbf{X})/\text{SST} = (\text{SSE}(\mathbf{X}_1)/\text{SST})(\text{SSE}(\mathbf{X})/\text{SSE}(\mathbf{X}_1))$, and the two factors are $1 - R_1^2$ and $1 - R_{\mathbf{X}|\mathbf{X}_1}^2$.
- (b) Write $D = \text{SS}(\mathbf{X} | \mathbf{X}_1) = qFs^2$ and $\text{SSE}(\mathbf{X}) = (n - r)s^2$. Then $R_{\mathbf{X}|\mathbf{X}_1}^2 = D/(D + \text{SSE}(\mathbf{X})) = qF/(qF + n - r)$.
- (c) With $q = 1$, $F = t_j^2$ by [Corollary 9.2.3](#), and (b) gives the first formula. For the correlation, let $\tilde{\mathbf{y}} = (\mathbf{I} - \mathbf{M}_1)\mathbf{y}$ and $\tilde{\mathbf{x}} = (\mathbf{I} - \mathbf{M}_1)\mathbf{x}_j$. Then $\text{SSE}(\mathbf{X}_1) = \|\tilde{\mathbf{y}}\|^2$, and by [Lemma 9.2.1](#) $\text{SS}(\mathbf{X} | \mathbf{X}_1) = (\tilde{\mathbf{x}}^\top \tilde{\mathbf{y}})^2 / \|\tilde{\mathbf{x}}\|^2$. Their ratio is the squared cosine of the angle between $\tilde{\mathbf{y}}$ and $\tilde{\mathbf{x}}$. Both have mean zero because $\mathbf{1} \in \mathcal{C}(\mathbf{X}_1)$, so the cosine is their sample correlation ([Exercise \(B3\)](#)).
- (d) Apply (a) repeatedly along the chain, with $R_0^2 = 0$ for the intercept-only model.

■

Part (c) identifies the partial R^2 of one regressor with a squared **sample partial correlation**, the sample counterpart of [Definition 3.6.1](#); [Chapter 14](#) studies its sampling distribution. Within one model, ranking regressors by $|t|$ and by partial R^2 is the same thing.

Example. (Partial R^2 in the state data)

In the three-regressor model, with $n - r = 46$:

Regressor	t	partial R^2	R^2 without it
poverty	2.718	0.1383	0.5844
single	4.799	0.3337	0.4626
urban	0.349	0.0026	0.6410

Each partial R^2 equals $t^2/(t^2 + 46)$, and each row satisfies part (a): for poverty, $1 - 0.6419 = (1 - 0.5844)(1 - 0.1383)$. Poverty explains 14% of the variation that single parenthood and urbanization leave unexplained, although on its own it explains 36% of the total ($R^2 = 0.3599$). For the pair poverty and urbanization over single parenthood, the F ratio on 2 and 46 degrees of freedom is 4.502 and the partial R^2 is 0.1637.

```
import numpy as np
import statsmodels.api as sm
data = sm.datasets.statcrime.load_pandas().data.drop(index="District of Columbia")

y = data["murder"].to_numpy()
names = ["poverty", "single", "urban"]
n = len(y)

def fit(cols):
    X = np.column_stack([np.ones(n)] + [data[c].to_numpy() for c in cols])
    return sm.OLS(y, X).fit()

full = fit(names)
```

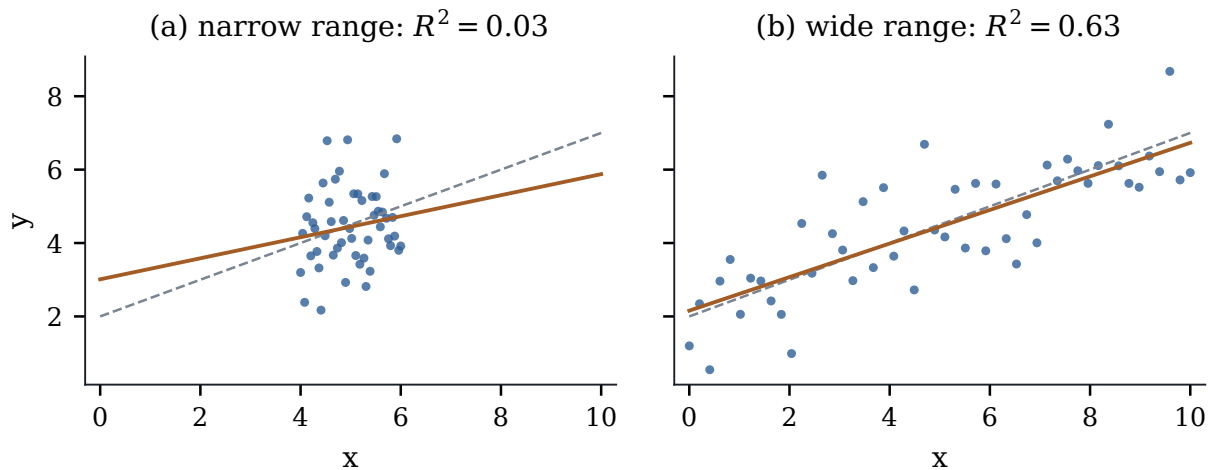


Figure 9.5.2: One line, two designs. The dashed line is the true mean $2 + 0.5x$, the solid line the least squares fit. The errors and the residual mean square are the same in both panels.

```
df_res = full.df_resid
for j, name in enumerate(names, start=1):
    reduced = fit([c for c in names if c != name])
    partial_r2 = (reduced.ssr - full.ssr) / reduced.ssr
    t = full.tvalues[j]
    print(f"{name:8s} partial R^2 = {partial_r2:.4f}, t = {t:6.3f},"
          f" t^2/(t^2 + df) = {t ** 2 / (t ** 2 + df_res):.4f}")
print(f"R^2 = {full.rsquared:.4f}, adjusted R^2 = {full.rsquared_adj:.4f}")
```

9.5.5 What R^2 does not measure

A high R^2 is often read as “a good model” and a low one as “a useless model”; neither reading is safe.

It is not a property of the relationship alone. R^2 depends on the spread of the regressors, which is a property of the design. In simple regression with fixed x values, the expectations of the two sums of squares are ([Exercise \(B1\)](#))

$$E(\text{SSR}) = \sigma^2 + \beta_1^2 S_{xx}, \quad E(\text{SST}) = (n-1)\sigma^2 + \beta_1^2 S_{xx},$$

so R^2 is roughly $\beta_1^2 S_{xx} / (\beta_1^2 S_{xx} + (n-1)\sigma^2)$. In [Figure 9.5.2](#) the line $E(y) = 2 + 0.5x$ and the same 50 errors with $\sigma = 1$ are combined with x values spread evenly over $[4, 6]$ and over $[0, 10]$. The two designs span the same $\mathcal{C}(\mathbf{X})$, so the residuals are identical and the residual mean square is 1.101 in both fits, but R^2 is 0.026 in one case and 0.632 in the other. The approximate values from the formula are 0.081 and 0.689. Restricting the range of a regressor lowers R^2 without changing the relationship.

```
import numpy as np
rng = np.random.default_rng(9)
n, beta0, beta1, sigma = 50, 2.0, 0.5, 1.0
eps = sigma * rng.normal(size=n)
# the same errors in both designs

def fit(x):
    y = beta0 + beta1 * x + eps
```



```

X = np.column_stack([np.ones(n), x])
b, *_ = np.linalg.lstsq(X, y, rcond=None)
sse = np.sum((y - X @ b) ** 2)
r2 = 1 - sse / np.sum((y - y.mean()) ** 2)
return y, b, sse / (n - 2), r2

for label, x in [("narrow", np.linspace(4, 6, n)), ("wide", np.linspace(0, 10, n))]:
    y, b, s2, r2 = fit(x)
    Sxx = np.sum((x - x.mean()) ** 2)
    expected = beta1 ** 2 * Sxx / (beta1 ** 2 * Sxx + (n - 1) * sigma ** 2)
    print(f"{label:6s}: slope {b[1]:.3f}, s^2 {s2:.3f}, R^2 {r2:.3f}, design value {
        expected:.3f}")

```

It is not a test of the model. A line fitted to points on a smooth curve can have a high R^2 while missing the curvature; residual plots and lack-of-fit tests ([Theorem 14.6.2](#)) reveal it. Conversely, a correct model has a low R^2 whenever the error variance is large, and its coefficients can still be precisely estimated and important.

It is not comparable across responses. Models for y and $\log y$, or for individuals and for group averages, have different denominators SST. Averaging replicates inflates R^2 ([Exercise \(B2\)](#)).

It is not comparable without an intercept ([Proposition 9.1.3](#), [Example \(Forcing the murder regression through the origin\)](#)).

It is not a measure of prediction or of cause. Computed on the fitting data, it overstates predictive accuracy for new data (Chapter 29), and variation “explained” is linear association, not the effect of changing a regressor (Chapter 25).

9.5.6 Exercises

9.5.6.1 A. Check your understanding

Exercise A1.

A regression with an intercept, $n = 20$ and $r = 6$ has $R^2 = 0.40$. Compute $\bar{R}^2$ and the overall F ratio. Is $\bar{R}^2$ positive? Is it consistent with part (b) of [Proposition 9.5.3](#)?

9.5.6.2 B. Practice

Exercise B1.

In simple regression with fixed x values, $E(\mathbf{Y}) = \beta_0 \mathbf{1} + \beta_1 \mathbf{x}$ and $\text{Cov}(\mathbf{Y}) = \sigma^2 \mathbf{I}$, use [Theorem 2.3.1](#) to show that $E(\text{SSR}) = \sigma^2 + \beta_1^2 S_{xx}$ and $E(\text{SST}) = (n - 1)\sigma^2 + \beta_1^2 S_{xx}$. Show that the ratio of these expectations tends to 1 as $S_{xx} \rightarrow \infty$ with n fixed, and to $1/(n - 1)$ as $S_{xx} \rightarrow 0$, and interpret the second limit.

Solution. $\text{SSR} = \mathbf{y}^\top (\mathbf{M} - \mathbf{P}_1) \mathbf{y}$, where $\mathbf{M} - \mathbf{P}_1$ projects onto the line spanned by $\mathbf{x} - \bar{x} \mathbf{1}$ ([Example \(Intercept inside simple regression\)](#)) and has trace 1. With $\boldsymbol{\mu} = \beta_0 \mathbf{1} + \beta_1 \mathbf{x}$, $(\mathbf{M} - \mathbf{P}_1) \boldsymbol{\mu} = \beta_1 (\mathbf{x} - \bar{x} \mathbf{1})$, whose squared length is $\beta_1^2 S_{xx}$. [Theorem 2.3.1](#) gives $E(\text{SSR}) = \sigma^2 + \beta_1^2 S_{xx}$. Similarly $\text{SST} = \mathbf{y}^\top (\mathbf{I} - \mathbf{P}_1) \mathbf{y}$, with trace $n - 1$ and $(\mathbf{I} - \mathbf{P}_1) \boldsymbol{\mu} = \beta_1 (\mathbf{x} - \bar{x} \mathbf{1})$.

The ratio $(\sigma^2 + \beta_1^2 S_{xx})/((n-1)\sigma^2 + \beta_1^2 S_{xx})$ tends to 1 as $S_{xx} \rightarrow \infty$ and to $1/(n-1)$ as $S_{xx} \rightarrow 0$. The second limit is the null value of $E(R^2)$ for $r = 2$ ([Theorem 9.5.2](#)): with almost no spread in x , the slope contributes almost nothing and the regression sum of squares is almost pure noise.

Exercise B2.

Suppose each of g distinct values $x_1, \dots, x_g$ carries m observations. Show that the least squares line fitted to all gm observations is the same as the line fitted to the g group means, and that its R^2 for the individual data is at most its R^2 for the means, with equality iff every observation equals its group mean.

Solution. The model space for the individual data is spanned by $\mathbf{1}$ and $\mathbf{x}$, both constant within groups, so it lies inside the space of group-constant vectors. By [Theorem 6.5.1](#), projecting $\mathbf{y}$ onto it equals projecting the vector of group means (repeated m times) onto it, which is the fit to the means with each point weighted equally. Write SSR_g and SST_g for the sums over the g means, and W for the within-group sum of squares. For the individual data $\text{SSR} = m \text{SSR}_g$ and $\text{SST} = m \text{SST}_g + W$. Hence $R^2 = m \text{SSR}_g / (m \text{SST}_g + W) \leq \text{SSR}_g / \text{SST}_g$, with equality iff $W = 0$.

Exercise B3.

Using part (d) of [Theorem 9.5.4](#) and the sequential sums of squares of [Example \(Order matters\)](#) (order A), express $1 - R^2$ for the three-regressor model as a product of three factors, and identify each factor as one minus a partial R^2 .

9.5.6.3 C. Going deeper

Exercise C1.

Under the conditions of [Theorem 9.5.2](#), show that for every $c \in (0, 1)$ the probability $\Pr(R^2 > c)$ is a strictly increasing function of the noncentrality γ . *Hint:* part (a) of the theorem and [Theorem 4.1.4](#).

Exercise C2.

For two regressors with sample correlations $r_{y1} = 0.5$, $r_{y2} = 0$ and $r_{12} = 0.6$, compute R^2 and compare it with $r_{y1}^2 + r_{y2}^2$. Use the formulas of [Exercise \(B1\)](#). Explain geometrically why a regressor that is uncorrelated with the response can raise R^2 .

9.6 Expected mean squares

This section computes the expectations of sums of squares when the data are random, using only two moments, and their distributions under normality. Each mean square estimates σ^2 plus a term that vanishes exactly when its hypothesis is true, while the residual mean square estimates σ^2 alone. That is the structure of the F ratio, on which Part III builds.

9.6.1 Expected sums of squares for a decomposition

In this section $E(\mathbf{Y}) = \boldsymbol{\mu}$ and $\text{Cov}(\mathbf{Y}) = \sigma^2 \mathbf{I}$. We do not assume at first that $\boldsymbol{\mu} \in \mathcal{C}(\mathbf{X})$, because the effect of a wrong model on the mean squares is part of the story.

Theorem 9.6.1 — Expected mean squares

Let $\mathcal{V}_1, \dots, \mathcal{V}_k$ be an analysis of variance decomposition of $\mathcal{C}(\mathbf{X})$, with projections $\mathbf{P}_i$ and dimensions r_i , and let $E(\mathbf{Y}) = \boldsymbol{\mu}$, $\text{Cov}(\mathbf{Y}) = \sigma^2 \mathbf{I}$.

- (a) $E(\mathbf{Y}^\top \mathbf{P}_i \mathbf{Y}) = \sigma^2 r_i + \|\mathbf{P}_i \boldsymbol{\mu}\|^2$, so the mean square of $\mathcal{V}_i$ has expectation

$$E(\text{MS}_i) = \sigma^2 + \frac{\|\mathbf{P}_i \boldsymbol{\mu}\|^2}{r_i}. \quad (9.6.1)$$

- (b) $E(\text{SSE}) = \sigma^2(n - r) + \|(\mathbf{I} - \mathbf{M})\boldsymbol{\mu}\|^2$, so $E(s^2) = \sigma^2$ iff $\boldsymbol{\mu} \in \mathcal{C}(\mathbf{X})$.
 (c) If $\boldsymbol{\mu} = \mathbf{X}\boldsymbol{\beta}$ with $\mathbf{X} = [\mathbf{X}_0, \dots, \mathbf{X}_k]$ partitioned into terms, the partial sum of squares of $\mathbf{X}_j$ has

$$E \text{ SS}(\mathbf{X}_j \mid \mathbf{X}_{(-j)}) = \sigma^2 d_j + \boldsymbol{\beta}_j^\top \mathbf{X}_j^\top (\mathbf{I} - \mathbf{M}_{(-j)}) \mathbf{X}_j \boldsymbol{\beta}_j,$$

with $d_j = \text{rank}(\mathbf{M} - \mathbf{M}_{(-j)})$, and the sequential sum of squares of $\mathbf{X}_j$ has $E \text{ SS}(\mathbf{X}_j \mid \mathbf{X}_0, \dots, \mathbf{X}_{j-1}) = \sigma^2 d'_j + \|(\mathbf{M}_j - \mathbf{M}_{j-1})\mathbf{X}\boldsymbol{\beta}\|^2$, with $d'_j = \text{rank}(\mathbf{M}_j - \mathbf{M}_{j-1})$.

- (d) If moreover $\mathbf{Y}$ is normal, then $\text{Var}(\mathbf{Y}^\top \mathbf{P}_i \mathbf{Y}) = 2\sigma^4 r_i + 4\sigma^2 \|\mathbf{P}_i \boldsymbol{\mu}\|^2$, and the sums of squares of different subspaces, including the error space, are uncorrelated.

Proof.

- (a) By Theorem 2.3.1 with $\mathbf{A} = \mathbf{P}_i$ and $\boldsymbol{\Sigma} = \sigma^2 \mathbf{I}$, $E(\mathbf{Y}^\top \mathbf{P}_i \mathbf{Y}) = \sigma^2 \text{tr}(\mathbf{P}_i) + \boldsymbol{\mu}^\top \mathbf{P}_i \boldsymbol{\mu}$. The trace is r_i (Proposition 6.3.4), and $\boldsymbol{\mu}^\top \mathbf{P}_i \boldsymbol{\mu} = \boldsymbol{\mu}^\top \mathbf{P}_i^\top \mathbf{P}_i \boldsymbol{\mu} = \|\mathbf{P}_i \boldsymbol{\mu}\|^2$. Divide by r_i .
 (b) The same with $\mathbf{A} = \mathbf{I} - \mathbf{M}$, of trace $n - r$. The second term vanishes iff $\boldsymbol{\mu} \in \mathcal{C}(\mathbf{X})$.
 (c) For the partial sum of squares, $(\mathbf{M} - \mathbf{M}_{(-j)})\mathbf{X}_l \boldsymbol{\beta}_l = \mathbf{0}$ for $l \neq j$, because $\mathbf{X}_l \boldsymbol{\beta}_l$ lies in $\mathcal{C}(\mathbf{X}_{(-j)})$ and is fixed by both projections. And $(\mathbf{M} - \mathbf{M}_{(-j)})\mathbf{X}_j \boldsymbol{\beta}_j = \mathbf{X}_j \boldsymbol{\beta}_j - \mathbf{M}_{(-j)}\mathbf{X}_j \boldsymbol{\beta}_j = (\mathbf{I} - \mathbf{M}_{(-j)})\mathbf{X}_j \boldsymbol{\beta}_j$. Its squared length is $\boldsymbol{\beta}_j^\top \mathbf{X}_j^\top (\mathbf{I} - \mathbf{M}_{(-j)}) \mathbf{X}_j \boldsymbol{\beta}_j$, since $\mathbf{I} - \mathbf{M}_{(-j)}$ is symmetric and idempotent. The sequential case is (a) for the chain decomposition (Definition 9.3.1).
 (d) By Theorem 4.2.3(b) with $\boldsymbol{\Sigma} = \sigma^2 \mathbf{I}$, $\text{Var}(\mathbf{Y}^\top \mathbf{P}_i \mathbf{Y}) = 2 \text{tr}\{(\sigma^2 \mathbf{P}_i)^2\} + 4\sigma^2 \boldsymbol{\mu}^\top \mathbf{P}_i \mathbf{P}_i \boldsymbol{\mu} = 2\sigma^4 r_i + 4\sigma^2 \|\mathbf{P}_i \boldsymbol{\mu}\|^2$. By part (d) of the same theorem the covariance of two forms is $2\sigma^4 \text{tr}(\mathbf{P}_i \mathbf{P}_j) + 4\sigma^2 \boldsymbol{\mu}^\top \mathbf{P}_i \mathbf{P}_j \boldsymbol{\mu} = 0$ when $\mathbf{P}_i \mathbf{P}_j = \mathbf{0}$.

■

Parts (a) to (c) use nothing beyond second moments. They hold for skewed or heavy-tailed errors, provided the errors are uncorrelated with a common variance. Part (d) uses normality, and without it both the variances and the zero correlations fail in general (Exercise C1).

Part (c) confirms Proposition 9.3.1. For full-rank $\mathbf{X}$ the term $\|(\mathbf{I} - \mathbf{M}_{(-j)})\mathbf{X}_j \boldsymbol{\beta}_j\|^2$ is positive

unless $\beta_j = \mathbf{0}$, and it is large only if $\mathbf{X}_j$ has a large part orthogonal to the other terms. A nearly collinear term has a small expected partial sum of squares even when its coefficients are substantial.

Under normality the expectations become full distributions.

Corollary 9.6.2 — Distributions of the mean squares

Let $\mathbf{Y} \sim N_n(\boldsymbol{\mu}, \sigma^2 \mathbf{I})$ with $\boldsymbol{\mu} \in \mathcal{C}(\mathbf{X})$, and let $\mathcal{V}_1, \dots, \mathcal{V}_k$ be an analysis of variance decomposition of $\mathcal{C}(\mathbf{X})$. Then the sums of squares $\mathbf{Y}^\top \mathbf{P}_1 \mathbf{Y}, \dots, \mathbf{Y}^\top \mathbf{P}_k \mathbf{Y}$ and SSE are mutually independent,

$$\frac{\mathbf{Y}^\top \mathbf{P}_i \mathbf{Y}}{\sigma^2} \sim \chi^2(r_i, \gamma_i), \quad \gamma_i = \frac{\|\mathbf{P}_i \boldsymbol{\mu}\|^2}{\sigma^2}, \quad \frac{\text{SSE}}{\sigma^2} \sim \chi^2(n - r),$$

and $F_i = \text{MS}_i / s^2 \sim F(r_i, n - r, \gamma_i)$, which is central iff $\mathbf{P}_i \boldsymbol{\mu} = \mathbf{0}$. The same holds for a single projection $\mathbf{P}$ onto any subspace of $\mathcal{C}(\mathbf{X})$, such as a partial sum of squares, paired with SSE.

Proof. The projections $\mathbf{P}_1, \dots, \mathbf{P}_k, \mathbf{I} - \mathbf{M}$ are mutually orthogonal (Theorem 9.1.1(a)), so Theorem 4.4.4 gives independence and the noncentral chi-squared laws, with $\|(\mathbf{I} - \mathbf{M})\boldsymbol{\mu}\|^2 = 0$ for the error space. The ratio of two independent chi-squared variables, each divided by its degrees of freedom and the numerator noncentral, is the noncentral F (Definition 4.1.2). A single projection $\mathbf{P}$ onto a subspace of $\mathcal{C}(\mathbf{X})$ satisfies $\mathbf{P}(\mathbf{I} - \mathbf{M}) = \mathbf{0}$, and the same argument applies to the pair. ■

The residual part is the familiar Theorem 7.5.1; the rest applies the same argument to the other pieces.

9.6.2 The logic of the F ratio

By Eq. 9.6.1 every mean square estimates σ^2 plus $\|\mathbf{P}_i \boldsymbol{\mu}\|^2 / r_i$, the squared size per dimension of the mean's component in $\mathcal{V}_i$, while under a correct model the residual mean square estimates σ^2 alone. So the ratio

$$F_i = \frac{\text{MS}_i}{s^2}$$

compares two estimates of σ^2 that agree in expectation exactly when $\mathbf{P}_i \boldsymbol{\mu} = \mathbf{0}$. This is the logic of the analysis of variance: *choose a denominator whose expectation equals that of the numerator when the hypothesis holds*. Independence gives the distribution exactly, and the noncentrality $\gamma_i = \|\mathbf{P}_i \boldsymbol{\mu}\|^2 / \sigma^2$ governs the power. The probability that F_i exceeds any fixed critical value increases strictly with γ_i (Theorem 4.1.4). The mean of the ratio follows from independence and $E(1/V) = 1/(\nu - 2)$ for $V \sim \chi^2(\nu)$, $\nu > 2$:

$$E(F_i) = \frac{n - r}{n - r - 2} \cdot \frac{r_i + \gamma_i}{r_i}. \quad (9.6.2)$$

The principle is more general than the choice of s^2 . With random effects the expected mean squares contain variance components, and the right denominator may be another mean square

(Exercise (B1); Chapters 32 and 33). Also, when the model is wrong, Theorem 9.6.1(b) shows that the denominator itself is inflated, by $\|(\mathbf{I} - \mathbf{M})\boldsymbol{\mu}\|^2 / (n - r)$ (compare Example (Residual sum of squares under a wrong mean)). Omitted terms inflate the denominator of every F ratio, which pushes the ratios towards zero, and they can also shift the numerators, because the omitted part of $\boldsymbol{\mu}$ may have components in the $\mathcal{V}_i$ (Exercise (B2)).

Chapter 11 derives the F test from the likelihood ratio principle (Theorem 11.4.1) and studies its power (Theorem 11.5.1).

9.6.3 Expected mean squares in an unbalanced layout

In the two-way layout of Section 9.3, Eq. 9.6.1 makes the hypotheses of Theorem 9.3.2 concrete. For the sequential sum of squares of A entered first, $(\mathbf{P}_A - \mathbf{P}_0)\boldsymbol{\mu}$ has entry $\bar{\mu}_i^w - \bar{\mu}^w$ on each of the n_i observations of row i , where $\bar{\mu}^w = \sum_i n_i \bar{\mu}_i^w / n$. So

$$\text{E MS}(A \mid \mu) = \sigma^2 + \frac{1}{a-1} \sum_i n_i (\bar{\mu}_i^w - \bar{\mu}^w)^2.$$

For the Type III sum of squares, the noncentrality is the sum of squares Eq. 9.2.1 evaluated at the true means, divided by σ^2 (Exercise (B3)):

$$\gamma = \frac{(\boldsymbol{\Lambda}^\top \boldsymbol{\mu}_c)^\top (\boldsymbol{\Lambda}^\top \mathbf{N}^{-1} \boldsymbol{\Lambda})^{-1} (\boldsymbol{\Lambda}^\top \boldsymbol{\mu}_c)}{\sigma^2},$$

where $\boldsymbol{\mu}_c$ is the vector of ab cell means, $\mathbf{N} = \text{diag}(n_{ij})$, and the columns of $\boldsymbol{\Lambda}$ express the $a-1$ contrasts of unweighted row means. In the first, the design enters through the weights n_{ij}/n_i , which also define the hypothesis. In the second, the hypothesis does not depend on the design, which enters only through the precision matrix $\mathbf{N}^{-1}$ of the cell means.

Example. (Checking expected mean squares by simulation)

Take the design of Example (Party, education and political self-placement): $n = 944$ respondents in the nine cells of party by education, with the observed counts. Set the true cell means equal to the observed cell means, and $\sigma^2 = 1.21$. The script generates 40,000 data sets with normal errors and 40,000 with centred exponential errors of the same variance, which are strongly skewed. For each it computes the sequential decomposition (party, then education, then interaction), the residual, and the Type III sum of squares for party, and averages the mean squares:

Source	df	γ	E(MS)	normal	skewed	Var(MS)	simulated
mean	1	14594.91	17661.050	17660.423	17661.259	85476.55	86357.17
party	2	535.09	324.942	324.915	324.787	784.89	786.39
education	2	35.75	22.840	22.836	22.891	53.81	53.13
given							
party							
interaction	4	29.60	10.163	10.174	10.179	11.57	11.75
residual	935	0.00	1.210	1.210	1.210	0.0031	0.0031

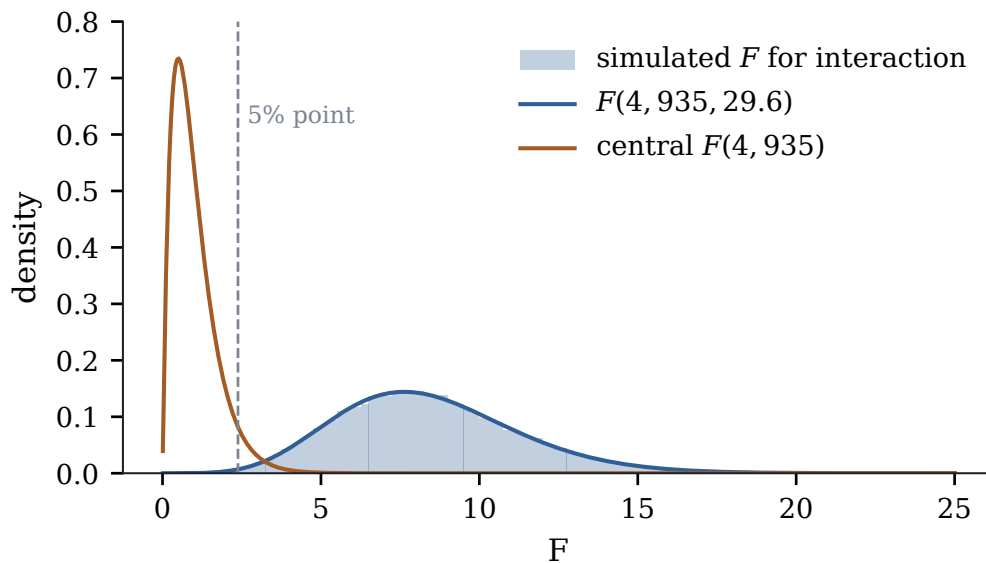


Figure 9.6.1: The F ratio for the interaction in [Example \(Checking expected mean squares by simulation\)](#) over 40,000 simulated data sets with normal errors, with the noncentral $F(4, 935, 29.6)$ density of [Corollary 9.6.2](#) and the central null density $F(4, 935)$ with its upper 5% point.

party,	2	537.53	326.414	326.362	326.259	788.46	789.47
Type III							

The columns $E(\text{MS})$ and $\text{Var}(\text{MS})$ come from [Eq. 9.6.1](#) and [Theorem 9.6.1\(d\)](#), with $\text{Var}(\text{MS}_i) = \sigma^4(2r_i + 4\gamma_i)/r_i^2$; the last column is the simulated variance under normal errors. The averages agree with the expectations under both error distributions, as parts (a) and (b) promise. The variances are another matter. Under skewed errors the variance of the residual mean square is 0.0124, 3.97 times the normal-theory value, because the variance of a quadratic form involves the fourth moment of the errors ([Theorem 2.3.3](#)). And the correlation between the interaction and residual mean squares, 0.008 under normal errors (zero in theory), is 0.020 under skewed errors, against a theoretical 0.025 from the third and fourth moments: the two mean squares are no longer independent.

For the interaction, the noncentrality $\gamma = 29.60$ is large. By [Eq. 9.6.2](#), the F ratio has mean 8.417, and the simulated mean is 8.427. Its 5% critical value is 2.381, and the power is 0.9967 in theory and 0.9967 in the simulation. [Figure 9.6.1](#) shows the simulated distribution with the noncentral F density.

```
import numpy as np
import pandas as pd
import statsmodels.api as sm
d = sm.datasets.anes96.load_pandas().data
party = pd.cut(d["PID"], [-1, 1.5, 4.5, 6], labels=["Dem", "Ind", "Rep"])
educ = pd.cut(d["educ"], [0, 3.5, 5.5, 7], labels=["HS", "College", "Graduate"])
cell = party.cat.codes.to_numpy() * 3 + educ.cat.codes.to_numpy()
n = len(cell)

def basis(*blocks):
    Z = np.column_stack(blocks)
    U, s, _ = np.linalg.svd(Z, full_matrices=False)
```

```

    return U[:, s > 1e-10 * s[0]]

one = np.ones((n, 1))
A = np.eye(3)[party.cat.codes.to_numpy()]
B = np.eye(3)[educ.cat.codes.to_numpy()]
Cell = np.eye(9)[cell]
U0, UA, UAB, UC = basis(one), basis(one, A), basis(one, A, B), basis(Cell)

def complement(U_big, U_small):
    """Orthonormal basis of  $C(U\_small)$ -perp within  $C(U\_big)$ ."""
    R = U_big - U_small @ (U_small.T @ U_big)
    return basis(R)

pieces = {"mean": U0, "party": complement(UA, U0),
          "educ | party": complement(UAB, UA), "interaction": complement(UC, UAB)}
Ub = basis(one, B, np.column_stack([(A[:, [i]] - A[:, [0]]) * (B[:, [j]] - B[:,
[0]])
                                     for i in (1, 2) for j in (1, 2)]))
pieces["party (Type III)"] = complement(UC, Ub) # unweighted-means hypothesis

sigma = 1.1
mu = d.groupby(cell)["selfLR"].mean().to_numpy()[cell] # true means: observed cell
means
for name, U in pieces.items():
    r = U.shape[1]
    gamma = np.sum((U.T @ mu) ** 2) / sigma ** 2 # noncentrality  $\|P \mu\|^2 / \sigma^2$ 
    print(f"{name:17s} rank {r}  gamma = {gamma:9.2f}"
          f"  E(MS) = {sigma ** 2 * (1 + gamma / r):9.3f}")

```

```

rng = np.random.default_rng(99)
reps, chunk = 40_000, 4_000
dfe = n - 9
sums = {"normal": [], "skewed": []}
for kind in sums:
    out = []
    for _ in range(reps // chunk):
        if kind == "normal":
            E = sigma * rng.normal(size=(n, chunk))
        else:
            # centred exponential,
            # variance  $\sigma^2$ 
            E = sigma * (rng.exponential(size=(n, chunk)) - 1.0)
        Y = mu[:, None] + E
        ms = {name: np.sum((U.T @ Y) ** 2, axis=0) / U.shape[1] for name, U in
              pieces.items()}
        ms["residual"] = (np.sum(Y ** 2, axis=0) - np.sum((UC.T @ Y) ** 2, axis=0))
        / dfe
        out.append(pd.DataFrame(ms))
    sums[kind] = pd.concat(out, ignore_index=True)
print(pd.DataFrame({k: v.mean() for k, v in sums.items()}).round(3))

```

The simulation never forms an $n \times n$ matrix: each sum of squares comes from the coordinates of $\mathbf{Y}$ in an orthonormal basis of its subspace (Theorem 9.1.1(d)).

9.6.4 Exercises

9.6.4.1 A. Check your understanding

Exercise A1.

In simple regression with $E(\mathbf{Y}) = \beta_0 \mathbf{1} + \beta_1 \mathbf{x}$, show that the regression mean square has expectation $\sigma^2 + \beta_1^2 S_{xx}$, and that the F ratio of the corrected table has noncentrality $\beta_1^2 S_{xx} / \sigma^2$ under normality.

Exercise A2.

In a one-way layout with g groups, group sizes n_i and means μ_i , show that the between-groups mean square has expectation $\sigma^2 + \sum_i n_i (\mu_i - \bar{\mu})^2 / (g - 1)$, where $\bar{\mu} = \sum_i n_i \mu_i / n$. Compare with [Example \(The one-way layout\)](#).

9.6.4.2 B. Practice

Exercise B1.

In a balanced one-way layout with g groups of m observations, let $Y_{ij} = \mu + a_i + \varepsilon_{ij}$, where the a_i and ε_{ij} are uncorrelated random variables with mean zero and variances σ_a^2 and σ^2 . Show that $\text{Cov}(\mathbf{Y}) = \sigma^2 \mathbf{I} + \sigma_a^2 \mathbf{Z}\mathbf{Z}^\top$, with $\mathbf{Z}$ the matrix of group indicators, and use [Theorem 2.3.1](#) to show that

$$E(\text{MS}_{\text{between}}) = \sigma^2 + m\sigma_a^2, \quad E(\text{MS}_{\text{within}}) = \sigma^2.$$

Which ratio tests $\sigma_a^2 = 0$, and what unbiased estimator of σ_a^2 do the two mean squares suggest?

Solution. $\mathbf{Y} = \mu \mathbf{1} + \mathbf{Z}\mathbf{a} + \boldsymbol{\varepsilon}$, so $E(\mathbf{Y}) = \mu \mathbf{1}$ and $\text{Cov}(\mathbf{Y}) = \mathbf{Z}(\sigma_a^2 \mathbf{I}_g) \mathbf{Z}^\top + \sigma^2 \mathbf{I}$ by [Theorem 2.2.1](#). The between-groups projection is $\mathbf{M} - \mathbf{P}_1$, where $\mathbf{M} = m^{-1} \mathbf{Z}\mathbf{Z}^\top$ averages within groups, and the within-groups projection is $\mathbf{I} - \mathbf{M}$. The mean $\mu \mathbf{1}$ is annihilated by both. By [Theorem 2.3.1](#), $E(\mathbf{Y}^\top (\mathbf{M} - \mathbf{P}_1) \mathbf{Y}) = \text{tr}\{(\mathbf{M} - \mathbf{P}_1)(\sigma^2 \mathbf{I} + \sigma_a^2 \mathbf{Z}\mathbf{Z}^\top)\}$. Now $\mathbf{Z}\mathbf{Z}^\top = m\mathbf{M}$, and $(\mathbf{M} - \mathbf{P}_1)\mathbf{M} = \mathbf{M} - \mathbf{P}_1$, so the trace is $\sigma^2(g - 1) + \sigma_a^2 m(g - 1)$. For the within-groups projection, $(\mathbf{I} - \mathbf{M})\mathbf{Z} = \mathbf{0}$, so the trace is $\sigma^2(n - g)$. Dividing by the degrees of freedom gives the two expectations. The ratio $\text{MS}_{\text{between}}/\text{MS}_{\text{within}}$ has expectations that agree iff $\sigma_a^2 = 0$, and $(\text{MS}_{\text{between}} - \text{MS}_{\text{within}})/m$ is unbiased for σ_a^2 (though it can be negative).

Exercise B2.

Suppose $E(\mathbf{Y}) = \mathbf{X}\boldsymbol{\beta} + \mathbf{z}\delta$ but the model $E(\mathbf{Y}) = \mathbf{X}\boldsymbol{\beta}$ is fitted, with $\mathbf{1} \in \mathcal{C}(\mathbf{X})$. Show that

$$E(\text{SSE}) = \sigma^2(n - r) + \delta^2 \|(\mathbf{I} - \mathbf{M})\mathbf{z}\|^2, \quad E(\text{SSR}) = \sigma^2(r - 1) + \|(\mathbf{M} - \mathbf{P}_1)(\mathbf{X}\boldsymbol{\beta} + \mathbf{z}\delta)\|^2.$$

Give an example in which $\boldsymbol{\beta}$ has all slopes zero but the regression mean square is inflated.

Exercise B3.

In the setting of [Theorem 9.2.4](#) with $\mathbf{d} = \mathbf{0}$, show that the expected sum of squares of the hypothesis $\mathbf{\Lambda}^\top \boldsymbol{\beta} = \mathbf{0}$ is $\sigma^2 q + (\mathbf{\Lambda}^\top \boldsymbol{\beta})^\top (\mathbf{\Lambda}^\top \mathbf{G} \mathbf{\Lambda})^{-1} \mathbf{\Lambda}^\top \boldsymbol{\beta}$, so the noncentrality is obtained by evaluating [Eq. 9.2.1](#) at the true parameter.

Solution. By [Theorem 9.2.4](#)(b), the sum of squares is $\mathbf{Y}^\top \mathbf{K} \mathbf{Y}$ with $\mathbf{K} = \mathbf{M} \mathbf{T} (\mathbf{\Lambda}^\top \mathbf{G} \mathbf{\Lambda})^{-1} \mathbf{T}^\top \mathbf{M}$, a projection of rank q , where $\mathbf{T}$ is the $n \times q$ matrix with $\mathbf{\Lambda} = \mathbf{X}^\top \mathbf{T}$. By [Theorem 9.6.1](#)(a) its expectation is $\sigma^2 q + \boldsymbol{\beta}^\top \mathbf{X}^\top \mathbf{K} \mathbf{X} \boldsymbol{\beta}$. Since $\mathbf{T}^\top \mathbf{M} \mathbf{X} \boldsymbol{\beta} = \mathbf{T}^\top \mathbf{X} \boldsymbol{\beta} = \mathbf{\Lambda}^\top \boldsymbol{\beta}$, the second term is $(\mathbf{\Lambda}^\top \boldsymbol{\beta})^\top (\mathbf{\Lambda}^\top \mathbf{G} \mathbf{\Lambda})^{-1} \mathbf{\Lambda}^\top \boldsymbol{\beta}$.

9.6.4.3 C. Going deeper**Exercise C1.**

Let the errors be independent with mean zero, variance σ^2 and fourth moment $\mu_4 = E(\varepsilon_i^4)$, and let $\boldsymbol{\mu} \in \mathcal{C}(\mathbf{X})$. Use [Theorem 2.3.3](#) to show that

$$\text{Var}(\text{SSE}) = (\mu_4 - 3\sigma^4) \sum_i (1 - h_{ii})^2 + 2\sigma^4(n - r),$$

where h_{ii} are the leverages. For the centred exponential errors of [Example \(Checking expected mean squares by simulation\)](#), $\mu_4 = 9\sigma^4$. Show that $\text{Var}(s^2) \approx 8\sigma^4/(n - r)$ when all leverages are small, and compare with the simulated variance of the residual mean square.

Solution. With $\mathbf{A} = \mathbf{I} - \mathbf{M}$ and $\boldsymbol{\theta} = \boldsymbol{\mu}$, the terms of [Theorem 2.3.3](#) involving $\boldsymbol{\theta}$ vanish because $(\mathbf{I} - \mathbf{M})\boldsymbol{\mu} = \mathbf{0}$. The diagonal of $\mathbf{I} - \mathbf{M}$ is $1 - h_{ii}$, and $\text{tr}\{(\mathbf{I} - \mathbf{M})^2\} = n - r$, which gives the formula. If the leverages are small, $\sum_i (1 - h_{ii})^2 \approx n - r$ (it equals $n - 2r + \sum_i h_{ii}^2$), so $\text{Var}(\text{SSE}) \approx (6 + 2)\sigma^4(n - r)$ and $\text{Var}(s^2) \approx 8\sigma^4/(n - r)$, four times the normal value $2\sigma^4/(n - r)$. This is the factor seen in the simulation.

Exercise C2.

For the Type I sum of squares of A entered first in a two-way layout, show that its noncentrality can be positive when $\mu_{ij} = \beta_j$ depends on B only, and find the condition on the counts n_{ij} under which it is zero for all such $\boldsymbol{\mu}$. Relate your answer to [Theorem 9.4.3](#).

9.7 Summary and notes

9.7.1 Summary

Idea.

1. An analysis of variance is a choice of mutually orthogonal subspaces filling the model space. Sums of squares are squared lengths of the projections of $\mathbf{y}$, degrees

- of freedom are dimensions, and both add up by Pythagoras. Decompositions and chains of nested models are the same thing (Definition 9.1.1, Theorem 9.1.1).
2. The regression table comes from the chain $\text{span}(\mathbf{1}) \subseteq \mathcal{C}(\mathbf{X})$. The corrected identity $\text{SST} = \text{SSR} + \text{SSE}$ needs $\mathbf{1} \in \mathcal{C}(\mathbf{X})$; without it an extra term $-2\bar{y} \sum_i \hat{\varepsilon}_i$ appears (Proposition 9.1.2, Proposition 9.1.3).
 3. The extra sum of squares $\text{SS}(\mathbf{X}_2 \mid \mathbf{X}_1) = \mathbf{y}^\top (\mathbf{M} - \mathbf{M}_1) \mathbf{y}$ depends only on column spaces. Adding regressors to a fitted model needs only a $t \times t$ inverse, and never lowers the variances of the old coefficients. For one column it equals $t_j^2 s^2$ (Definition 9.2.1, Theorem 9.2.2, Corollary 9.2.3).
 4. The sum of squares of an estimable hypothesis $\mathbf{A}^\top \boldsymbol{\beta} = \mathbf{d}$ is $(\mathbf{A}^\top \hat{\boldsymbol{\beta}} - \mathbf{d})^\top (\mathbf{A}^\top \mathbf{G} \mathbf{A})^{-1} (\mathbf{A}^\top \hat{\boldsymbol{\beta}} - \mathbf{d})$, whatever the rank. A nonestimable constraint restricts only through its estimable part (Theorem 9.2.4, Theorem 9.2.5, Proposition 9.2.6).
 5. Sequential (Type I) sums of squares decompose the regression sum of squares but depend on the order. Partial, Type II and Type III sums of squares do not decompose anything, and in models with interactions they address different hypotheses: weighted row means, additive main effects, unweighted row means, or, with reference coding, means at a single level of the other factor (Definition 9.3.1, Definition 9.3.2, Proposition 9.3.1, Theorem 9.3.2).
 6. Order stops mattering exactly when the centred subspaces of the terms are orthogonal. For two factors this is the proportional-frequency condition. Orthogonal polynomials and orthogonal contrasts manufacture orthogonality inside a single term (Theorem 9.4.1, Theorem 9.4.3, Theorem 9.4.4).
 7. R is the largest correlation of $\mathbf{y}$ with a linear combination of the regressors. Under the null $R^2 \sim \text{Beta}(\frac{r-1}{2}, \frac{n-r}{2})$, with mean $(r-1)/(n-1)$. The adjusted R^2 has mean zero under the null and rises exactly when an added term has $F > 1$. Partial R^2 equals $t^2/(t^2 + n - r)$ (Theorem 9.5.1, Theorem 9.5.2, Proposition 9.5.3, Theorem 9.5.4).
 8. R^2 depends on the design, and it is not a test of the model, a comparison across responses, or a measure of predictive accuracy or of cause (Section 9.5).
 9. Every mean square has expectation $\sigma^2 + \|\mathbf{P}\boldsymbol{\mu}\|^2 / \text{rank}(\mathbf{P})$, using only second moments. Under normality the pieces of a decomposition are independent noncentral chi-squared variables, and an F ratio compares two estimates of σ^2 that agree exactly when its hypothesis $\mathbf{P}\boldsymbol{\mu} = \mathbf{0}$ holds (Theorem 9.6.1, Corollary 9.6.2).

9.7.2 Notes and sources

Analysis of variance. The decomposition of a total sum of squares into components with their own degrees of freedom is Fisher's. An analysis of variance already appears in Fisher and Mackenzie (1923), and Fisher set out the analysis of variance table for general use in *Statistical Methods for Research Workers* (Fisher 1925), after making the notion of degrees of freedom precise in Fisher (1922). The distribution theory that makes the components independent chi-squared variables is Cochran's theorem (Cochran 1934), proved in Chapter 4. The coordinate-free view, in which a table is a list of orthogonal subspaces, follows Christensen (2020). His section 3.6, which splits a test space into orthogonal lines, is the source of Theorem 9.4.4, and his section 7.4 treats proportional numbers. Scheffé (1959) remains a clear account of the classical theory, including expected mean squares for random effects.

Extra sums of squares, adding regressors and restrictions. Seber and Lee (2003, sections

3.7–3.9) give the updating formulas of [Theorem 9.2.2](#), restricted least squares, and their rank-deficient versions. Our proofs go through the Frisch–Waugh–Lovell theorem and Pythagoras instead of Lagrange multipliers. [Proposition 9.2.6](#) makes precise the remark, common to the sources, that only the estimable part of a hypothesis can be tested.

Types of sums of squares. Agresti (2015, sections 2.3–2.4) presents sequential and partial sums of squares through projection matrices, with the decomposition of the regression sum of squares when the regressors are uncorrelated, and notes the Type I, II and III terminology. The problem of main effects in unbalanced layouts with interaction goes back to Yates (1934), whose “weighted squares of means” is closely related to what software now calls Type III. The names of the types come from statistical software, and programs have used them inconsistently, which is why this chapter defines each type by its projection. Nelder (1977) argued that tests of main effects in the presence of interactions violate the marginality of the model terms; his position underlies the preference for Type II sums of squares when interactions are absent. Speed, Hocking and Hackney (1978) identify the cell-mean hypotheses tested by the various sums of squares, and Searle (1987) treats unbalanced data exhaustively in the same cell-mean terms.

The coefficient of determination. The distribution of the sample multiple correlation coefficient, under the null and in general for random normal regressors, is due to Fisher (1928); [Theorem 9.5.2](#) is the simpler fixed-regressor version. Kvralseth (1985) surveys the many different formulas that software has used for R^2 , especially for models without an intercept, and Anscombe (1973) gives four data sets with the same R^2 and very different relationships, a standing warning that R^2 is not a test of the model.

Data. The state crime, election study, Grunfeld and Nile data are public-domain data sets distributed with statsmodels. The Nile series is the one analysed for a change point by Cobb (1978).

What is new here. The organization of the chapter around subspaces, the proofs of [Theorem 9.4.1](#) and [Theorem 9.4.3](#), the reference-coding case of [Theorem 9.3.2](#), and the numerical examples and simulations are this book’s.

9.7.3 References

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Chapter 10

Computation: QR, SVD and Conditioning

Every least squares estimate in this book is written as $\hat{\beta} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{y}$ or as $\mathbf{X}(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{y}$. These are the right formulas to reason with. They are not how a computer should produce the numbers. This chapter explains what a computer should do instead, and how accurate the result is. It separates two questions: how sensitive the least squares problem is to small changes in the data, which depends on $\kappa(\mathbf{X})$ and the size of the residual, and whether an algorithm adds more error than that. The answers give a practical ranking. The Cholesky factorization of $\mathbf{X}^\top \mathbf{X}$ is fast and is enough when $\mathbf{X}$ is well conditioned. Householder or Givens QR is the default. The singular value decomposition is the tool for rank deficiency and near dependence. Pivoted QR decides the rank cheaply, which is how statistical software chooses what to do with aliased columns.

What you need. [Chapter 1](#): the Cholesky factorization ([Theorem 1.7.4](#)), QR ([Theorem 1.5.2](#)), the singular value decomposition ([Theorem 1.8.1](#)), the Eckart–Young inequality ([Proposition 1.8.2](#)) and the condition number ([Definition 1.8.1](#)). [Chapter 5](#): the linear model and the moments of $\hat{\beta}$ ([Theorem 5.5.1](#)). [Chapter 6](#): least squares as projection, and the first look at computation in [Section 6.10](#). A few remarks use the sampling theory of [Chapter 7](#) and the sums of squares of [Chapter 9](#).

10.0.1 Roadmap

- [Why not invert the cross-product matrix](#)
- [The Cholesky factorization](#)
- [Householder and Givens QR](#)
- [The singular value decomposition](#)
- [Conditioning and perturbation bounds](#)
- [Updating and downdating](#)
- [Rank-revealing factorizations](#)
- [Summary and notes](#)

10.1 Why not invert the cross-product matrix

Chapter 5 found the least squares estimate by setting a gradient to zero, which gave $\hat{\beta} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{y}$ when $\mathbf{X}$ has full column rank (Theorem 5.4.1). Read as a recipe, the formula says: form the $p \times p$ matrix $\mathbf{X}^\top \mathbf{X}$, invert it, and multiply. Section 6.10 already showed that the recipe can lose twice as many digits as necessary. This section explains why.

10.1.1 Floating-point numbers

A double-precision number is zero or a number $\pm m \times 2^{e-52}$, where m is an integer with $2^{52} \leq m < 2^{53}$ and e is an integer in a fixed range ($-1022 \leq e \leq 1023$). Between 2^e and 2^{e+1} the doubles are therefore equally spaced, 2^{e-52} apart. Just above 1 the spacing is $\epsilon_M = 2^{-52} \approx 2.22 \times 10^{-16}$, the **machine epsilon**. Rounding a real number x in the range of the doubles to the nearest double gives a number $\text{fl}(x)$ with

$$\text{fl}(x) = x(1 + \delta), \quad |\delta| \leq u = 2^{-53} \approx 1.11 \times 10^{-16}.$$

The **unit roundoff** $u = \epsilon_M/2$ is half the gap, because the nearest double is at most half a gap away. Rounding errors are *relative*: a double represents large and small numbers with the same number of significant digits, about sixteen.

The IEEE 754 standard, which essentially all current hardware follows, requires the basic operations to be correctly rounded. The computed result of $x + y$ is the exact sum, rounded. That is the only property of the hardware that the analysis below uses.

Definition 10.1.1 — Standard model of floating-point arithmetic

For floating-point numbers x, y and $\circ \in \{+, -, \times, /\}$, the computed result satisfies

$$\text{fl}(x \circ y) = (x \circ y)(1 + \delta), \quad |\delta| \leq u,$$

and $\text{fl}(\sqrt{x}) = \sqrt{x}(1 + \delta)$ with $|\delta| \leq u$, provided no result overflows or underflows.

Two consequences are worth seeing at once. The number $1 + u$ lies exactly halfway between the doubles 1 and $1 + \epsilon_M$, and the tie is broken towards 1, so adding u to 1 changes nothing. And most decimal fractions are not doubles, so familiar identities fail at the level of u :

```
import numpy as np

eps = np.finfo(float).eps          # spacing of the doubles just above 1: 2^-52
u = eps / 2                        # unit roundoff: 2^-53
print("eps =", eps, " u =", u)
print("1 + u == 1:", 1.0 + u == 1.0, " 1 + eps == 1:", 1.0 + eps == 1.0)
print("0.1 + 0.2 == 0.3:", 0.1 + 0.2 == 0.3)
```

Trouble comes from accumulation and above all from subtraction. The difference of nearly equal numbers is computed exactly but exposes the errors they already carried: if a and b agree in

ten digits, a rounding error in either becomes a large relative error in $a - b$. This **cancellation** is behind every numerical difficulty in this chapter.

10.1.2 Accumulating rounding errors

Products of many factors $1 + \delta_i$ appear whenever errors accumulate. The following lemma packages them.

Lemma 10.1.1

Let $|\delta_i| \leq u$ and $\rho_i \in \{1, -1\}$ for $i = 1, \dots, k$, and suppose $ku < 1$. Then

$$\prod_{i=1}^k (1 + \delta_i)^{\rho_i} = 1 + \theta_k, \quad |\theta_k| \leq \gamma_k = \frac{ku}{1 - ku}. \quad (10.1.1)$$

Proof. Each factor lies between $1 - u$ and $1/(1 - u)$: for $\rho_i = 1$ because $1 + u \leq 1/(1 - u)$, and for $\rho_i = -1$ because $1/(1 + \delta_i) \geq 1/(1 + u) \geq 1 - u$. So the product lies between $(1 - u)^k$ and $(1 - u)^{-k}$. By Bernoulli's inequality $(1 - u)^k \geq 1 - ku$, which gives the lower end $1 - ku \geq 1 - \gamma_k$, and also the upper end $(1 - u)^{-k} \leq 1/(1 - ku) = 1 + \gamma_k$. ■

The constant γ_k is essentially ku : for $k = 10^6$ it is about 1.1×10^{-10} . This is a worst case; in practice errors of different signs partly cancel and grow more like $\sqrt{k}u$.

The basic step of every least squares algorithm is an inner product.

Proposition 10.1.2 — Rounding error in an inner product

Let $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$ have floating-point entries and $nu < 1$. The inner product computed by recursive summation, $s_1 = \text{fl}(x_1 y_1)$ and $s_k = \text{fl}(s_{k-1} + \text{fl}(x_k y_k))$, satisfies

$$s_n = \sum_{i=1}^n x_i y_i (1 + \theta_i), \quad |\theta_i| \leq \gamma_n.$$

Consequently $|s_n - \mathbf{x}^\top \mathbf{y}| \leq \gamma_n \sum_i |x_i y_i|$, and $s_n = (\mathbf{x} + \Delta \mathbf{x})^\top \mathbf{y}$ for a vector $\Delta \mathbf{x}$ with $|\Delta x_i| \leq \gamma_n |x_i|$.

Proof. By the standard model, $\text{fl}(x_k y_k) = x_k y_k (1 + \mu_k)$ and $s_k = (s_{k-1} + x_k y_k (1 + \mu_k))(1 + \sigma_k)$ for $k \geq 2$, with all $|\mu_k|, |\sigma_k| \leq u$. Unrolling the recursion,

$$s_n = \sum_{i=1}^n x_i y_i (1 + \mu_i) \prod_{k=\max(i, 2)}^n (1 + \sigma_k).$$

The i th term carries at most n factors of the form $1 + \delta$, so by Lemma 10.1.1 it equals $x_i y_i (1 + \theta_i)$ with $|\theta_i| \leq \gamma_n$ (since $\gamma_k \leq \gamma_n$ for $k \leq n$). The two consequences follow by taking $\Delta x_i = \theta_i x_i$.

The first conclusion is a *forward* bound, which can be large relative to a result whose terms cancel. The second is a *backward* statement: the computed number is the exact inner product of slightly perturbed data. Over 200 random inner products of length up to 400, checked in rational arithmetic, the largest ratio of error to bound is 0.149.

```
from fractions import Fraction

def gamma(k, u=2.0 ** -53):
    return k * u / (1 - k * u)

def dot_recursive(x, y):
    """Inner product by recursive summation, one rounding per operation."""
    s = 0.0
    for a, b in zip(x, y):
        s = s + a * b
    return s

rng = np.random.default_rng(101)
worst = 0.0
for trial in range(200):
    n = int(rng.integers(2, 400))
    x = rng.normal(size=n) * 10.0 ** rng.integers(-3, 4, size=n)
    y = rng.normal(size=n)
    exact = sum(Fraction(a) * Fraction(b) for a, b in zip(x, y))      # exact
    rational value
    error = abs(Fraction(dot_recursive(x, y)) - exact)
    bound = Fraction(gamma(n)) * sum(abs(Fraction(a) * Fraction(b)) for a, b in zip(
        x, y))
    worst = max(worst, float(error / bound))
print(f"largest error / bound over 200 inner products: {worst:.3f}")
```

10.1.3 Conditioning and stability

A computation turns data d into a result $f(d)$. The computed result can be inaccurate because the *problem* is sensitive, or because the *algorithm* adds more error than the data justify. The first is measured by a **condition number**, the largest factor by which a small relative change in d can be magnified in $f(d)$; the second by the following standard.

Definition 10.1.2 — Backward stability

An algorithm that computes $f(d)$ is **backward stable** if, for every input d , its computed output $\hat{f}(d)$ is the exact result for nearby data: $\hat{f}(d) = f(d + \Delta d)$ with $\|\Delta d\| \leq cu\|d\|$, where c is a modest constant depending only on the dimensions of the problem.

Proposition 10.1.2 says that recursive summation is a backward stable way to compute inner products, with $c = \gamma_n/u \approx n$. Backward stability is the most one can ask, because storing the data as doubles already perturbs them by u . The two notions combine, to first order in u , as

$$\text{relative forward error} \lesssim \text{condition number} \times \text{backward error}. \quad (10.1.2)$$

For least squares the condition number, which involves $\kappa(\mathbf{X})$ and sometimes $\kappa(\mathbf{X})^2$, is the subject of [Section 10.5](#). Sections 10.2 to 10.4 ask whether algorithms are backward stable *for the least squares problem*, with perturbations of $\mathbf{X}$ and $\mathbf{y}$.

10.1.4 What forming the cross-product matrix loses

The normal-equations recipe is not backward stable for least squares, and the damage is done in its very first step. Here is what rounding does to $\mathbf{X}^\top \mathbf{X}$.

Proposition 10.1.3 — Rounding in the cross-product matrix

Let $\mathbf{X}$ be $n \times p$ with full column rank and let $\hat{\mathbf{A}}$ be $\mathbf{X}^\top \mathbf{X}$ computed entry by entry as in [Proposition 10.1.2](#). Then $\hat{\mathbf{A}} = \mathbf{X}^\top \mathbf{X} + \Delta \mathbf{A}$ with $|\Delta \mathbf{A}| \leq \gamma_n |\mathbf{X}|^\top |\mathbf{X}|$ entrywise, and

$$\|\Delta \mathbf{A}\|_2 \leq p \gamma_n \|\mathbf{X}\|_2^2.$$

In particular $\hat{\mathbf{A}}$ is positive definite whenever $p \gamma_n \kappa(\mathbf{X})^2 < 1$.

Proof. The entrywise bound is [Proposition 10.1.2](#) applied to each pair of columns. The spectral norm is at most the Frobenius norm, the Frobenius norm is monotone in the absolute values of the entries and submultiplicative, and $\|\mathbf{X}\|_F^2 = \sum_i \sigma_i^2 \leq p \sigma_1^2$ by [Proposition 1.8.2\(a\)](#). Hence

$$\|\Delta \mathbf{A}\|_2 \leq \gamma_n \|\mathbf{X}\|^\top \|\mathbf{X}\|_F \leq \gamma_n \|\mathbf{X}\|_F^2 \leq p \gamma_n \sigma_1^2.$$

For a unit vector $\mathbf{v}$, $\mathbf{v}^\top \hat{\mathbf{A}} \mathbf{v} = \|\mathbf{X} \mathbf{v}\|^2 + \mathbf{v}^\top \Delta \mathbf{A} \mathbf{v} \geq \sigma_p^2 - p \gamma_n \sigma_1^2$, which is positive when $p \gamma_n \sigma_1^2 / \sigma_p^2 < 1$. ■

The information about $\mathbf{X}$ that sits in its smallest singular value σ_p appears in $\mathbf{X}^\top \mathbf{X}$ as an eigenvalue σ_p^2 . Rounding perturbs $\mathbf{X}^\top \mathbf{X}$ by an amount of order $u \sigma_1^2$. Once σ_p^2 is below that level, the computed cross-product matrix may not determine the direction $\mathbf{v}_p$ at all. Even rounding the *exact* $\mathbf{X}^\top \mathbf{X}$ to doubles does this much damage. The loss becomes total around $\kappa(\mathbf{X}) \approx u^{-1/2} = 9.5 \times 10^7$, and it is already serious well before that: the normal equations are then a problem with condition number $\kappa(\mathbf{X}^\top \mathbf{X}) = \kappa(\mathbf{X})^2$ ([Eq. 1.8.2](#)), and [Eq. 10.1.2](#) predicts relative errors of order $\kappa(\mathbf{X})^2 u$ in $\hat{\boldsymbol{\beta}}$ whatever method is then used to solve them.

Example. (A cross-product matrix that rounds to a singular one)

Take $\delta = 10^{-9}$ and

$$\mathbf{X} = \begin{pmatrix} 1 & 1 \\ \delta & 0 \\ 0 & \delta \end{pmatrix}, \quad \mathbf{X}^\top \mathbf{X} = \begin{pmatrix} 1 + \delta^2 & 1 \\ 1 & 1 + \delta^2 \end{pmatrix}.$$

The columns of $\mathbf{X}$ are independent, and $\kappa(\mathbf{X}) = \sqrt{2 + \delta^2} / \delta = 1.41 \times 10^9$. But $\delta^2 = 10^{-18}$ is smaller than u , so $1 + \delta^2$ rounds to 1 and the computed $\mathbf{X}^\top \mathbf{X}$ is the singular matrix of ones. The Cholesky factorization of [Section 10.2](#) fails, and no method applied to the computed normal equations can recover the solution. For $\mathbf{y} = \mathbf{X}(1, 1)^\top$, the exact least

squares estimate is $(1, 1)^\top$ with zero residual. QR, which never forms $\mathbf{X}^\top \mathbf{X}$, returns it with relative error 3.9×10^{-16} .

```
delta = 1e-9
X = np.array([[1.0, 1.0],
              [delta, 0.0],
              [0.0, delta]])
y = X @ np.array([1.0, 1.0])          # exact data: the solution is b = (1, 1),
                                     # residual 0

A = X.T @ X                          # 1 + delta^2 rounds to 1
print("computed X'X =\n", A)
print("cond(X) =", np.linalg.cond(X))
try:
    np.linalg.cholesky(A)
except np.linalg.LinAlgError as err:
    print("Cholesky of the computed X'X fails:", err)

Q, R = np.linalg.qr(X)
b_qr = np.linalg.solve(R, Q.T @ y)
print("QR solution:", b_qr)
```

10.1.5 Why not the inverse either

Even when $\mathbf{X}^\top \mathbf{X}$ is safely positive definite, the inverse costs p^3 flops against $p^3/3$ for a Cholesky factorization plus $2p^2$ for two triangular solves, and it is less accurate. Statistics needs only the diagonal of $\text{Cov}(\hat{\boldsymbol{\beta}}) = \sigma^2(\mathbf{X}^\top \mathbf{X})^{-1}$ (Theorem 5.5.1): if $\mathbf{X}^\top \mathbf{X} = \mathbf{R}^\top \mathbf{R}$, its j th entry is σ^2 times the squared length of row j of $\mathbf{R}^{-1}$.

10.1.6 Counting the work

Cost is traditionally counted in floating-point operations (**flops**), each addition, subtraction, multiplication and division counting one. Flop counts ignore memory traffic and parallelism, so they are only a first guide.

Task	Leading flop count	Where
Form $\mathbf{X}^\top \mathbf{X}$ and $\mathbf{X}^\top \mathbf{y}$ (using symmetry)	np^2	this section
Cholesky factorization of $\mathbf{X}^\top \mathbf{X}$	$p^3/3$	Section 10.2
Two triangular solves	$2p^2$	Section 10.2
Householder QR of $\mathbf{X}$, with $\mathbf{Q}^\top \mathbf{y}$	$2np^2 - 2p^3/3$	Section 10.3
Givens QR of $\mathbf{X}$	$3np^2 - p^3$	Section 10.3
Modified Gram–Schmidt	$2np^2$	Section 10.3
Singular values and $\mathbf{V}$, QR first (R-SVD)	$2np^2 + 11p^3$	Section 10.4

When $n \gg p$, as usual in regression, the normal equations cost about np^2 and Householder

QR about $2np^2$: the accurate method is at most twice as expensive. And the one-pass, $O(p^2)$ -memory accumulation of cross-products has a QR counterpart in row-wise Givens QR (Section 10.3).

Idea. Forming $\mathbf{X}^\top \mathbf{X}$ squares the condition number before any equation is solved, and inverting it adds cost and error. Factor $\mathbf{X}$ itself, by an orthogonal transformation, and read everything off the factors.

10.1.7 Exercises

10.1.7.1 A. Check your understanding

Exercise A1.

Show that the doubles in $[1, 2)$ are exactly the numbers $1 + k 2^{-52}$, $k = 0, 1, \dots, 2^{52} - 1$. What is the smallest double greater than 1? What is the largest integer N such that every integer from 0 to N is a double?

Exercise A2.

For $x = 10^{-10}$, the computer evaluates $(1 + x) - 1$. Use the standard model to show that the result is $x + \delta(1 + x)$ for some $|\delta| \leq u$, where the subtraction itself is exact. Estimate the relative error, and check it numerically. Which of the two operations is responsible?

Solution. The addition gives $\text{fl}(1 + x) = (1 + x)(1 + \delta)$. This number is within a factor of two of 1, and the difference of two doubles within a factor of two of each other is itself a double, so the subtraction is exact: $\text{fl}(1 + x) - 1 = x + \delta(1 + x)$. The relative error is $|\delta|(1 + x)/x \approx u/x \approx 10^{-6}$. All of it comes from rounding $1 + x$, which kept only about six significant digits of x . The subtraction then exposed that loss.

10.1.7.2 B. Practice

Exercise B1.

Show that recursive summation of $x_1, \dots, x_n$ returns $\sum_i x_i(1 + \theta_i)$ with $|\theta_i| \leq \gamma_{n-1}$, and that pairwise summation returns such a sum with $|\theta_i| \leq \gamma_{\lceil \log_2 n \rceil}$.

Solution. With $s_1 = x_1$ and $s_k = (s_{k-1} + x_k)(1 + \sigma_k)$, $s_n = \sum_i x_i \prod_{k=\max(i,2)}^n (1 + \sigma_k)$. Term i has at most $n - 1$ factors, so Lemma 10.1.1 gives $|\theta_i| \leq \gamma_{n-1}$. In pairwise summation each x_i passes through one addition per level of the tree, and a balanced tree over n leaves has $\lceil \log_2 n \rceil$ levels. The same unrolling gives at most $\lceil \log_2 n \rceil$ factors per term.

Exercise B2.

Let $\mathbf{A}$ be $m \times n$ and $\mathbf{b} \in \mathbb{R}^n$. Show that the computed product $\text{fl}(\mathbf{A}\mathbf{b})$, each entry an inner product, equals $(\mathbf{A} + \Delta\mathbf{A})\mathbf{b}$ with $|\Delta\mathbf{A}| \leq \gamma_n |\mathbf{A}|$. Deduce $\|\text{fl}(\mathbf{A}\mathbf{b}) - \mathbf{A}\mathbf{b}\| \leq$

$$\gamma_n \sqrt{n} \|\mathbf{A}\|_2 \|\mathbf{b}\|.$$

10.1.7.3 C. Going deeper

Exercise C1.

For $0 < \delta \leq 2^{-27}$ let $\mathbf{X}_\delta$ be the matrix of [Example \(A cross-product matrix that rounds to a singular one\)](#), and let $\mathbf{y} = \mathbf{X}_\delta(3, 1)^\top$ and $\mathbf{y}' = \mathbf{X}_\delta(1, 3)^\top$. Show that the computed cross-product matrices of $[\mathbf{X}_\delta, \mathbf{y}]$ and $[\mathbf{X}_\delta, \mathbf{y}']$ are identical (and do not depend on δ), although the exact least squares estimates are $(3, 1)^\top$ and $(1, 3)^\top$. Conclude that no algorithm that starts from the computed cross-product matrix can solve these problems. Is anything special about 2^{-27} ?

10.2 The Cholesky factorization

If the cross-product matrix is to be used at all, the normal equations should be solved by factoring it as a triangular matrix times its transpose. One factorization of the augmented cross-product matrix then delivers every standard regression quantity.

10.2.1 The algorithm

[Chapter 1](#) proved that a positive definite matrix has a unique factorization $\mathbf{A} = \mathbf{L}\mathbf{L}^\top$ with $\mathbf{L}$ lower triangular and positive on the diagonal ([Theorem 1.7.4\(d\)](#)). Numerical work usually writes it with the upper triangular factor $\mathbf{R} = \mathbf{L}^\top$.

Theorem 10.2.1 — Cholesky factorization

Let $\mathbf{A}$ be a $p \times p$ positive definite matrix.

- (a) There is exactly one upper triangular $\mathbf{R}$ with positive diagonal entries such that $\mathbf{A} = \mathbf{R}^\top \mathbf{R}$.
- (b) Its rows are given, for $j = 1, \dots, p$ in turn, by

$$\begin{aligned} r_{jj} &= \left(a_{jj} - \sum_{k < j} r_{kj}^2 \right)^{1/2}, \\ r_{ji} &= \frac{1}{r_{jj}} \left(a_{ji} - \sum_{k < j} r_{kj} r_{ki} \right), \quad i > j, \end{aligned} \tag{10.2.1}$$

and every radicand in it is positive.

- (c) The computation takes $p^3/3 + O(p^2)$ flops.
- (d) If $\mathbf{A} = \mathbf{X}^\top \mathbf{X}$ and $\mathbf{X} = \mathbf{Q}\mathbf{R}_0$ is a QR factorization with $\mathbf{Q}^\top \mathbf{Q} = \mathbf{I}_p$ and $\mathbf{R}_0$ upper triangular with positive diagonal, then $\mathbf{R}_0 = \mathbf{R}$.

Proof.

- (a) is [Theorem 1.7.4\(d\)](#) with $\mathbf{R} = \mathbf{L}^\top$.
- (b) Entry (j, i) , $i \geq j$, of $\mathbf{R}^\top \mathbf{R}$ is $\sum_{k \leq j} r_{kj} r_{ki}$, because $r_{kj} = 0$ for $k > j$. Setting it equal to a_{ji} and isolating the $k = j$ term gives the recursion [Eq. 10.2.1](#). So the entries of the unique factor of (a) satisfy [Eq. 10.2.1](#), and [Eq. 10.2.1](#) determines row j from rows $1, \dots, j-1$. By induction on j , the rows computed by [Eq. 10.2.1](#) are those of the factor in (a). In particular the j th radicand is $r_{jj}^2 > 0$.
- (c) Row j needs about $2(j-1)$ flops for each of its $p-j+1$ entries, so the total is $\sum_j 2(j-1)(p-j+1) = p^3/3 + O(p^2)$.
- (d) $\mathbf{X}^\top \mathbf{X} = \mathbf{R}_0^\top \mathbf{Q}^\top \mathbf{Q} \mathbf{R}_0 = \mathbf{R}_0^\top \mathbf{R}_0$, and uniqueness in (a).

■

By part (d), the triangle of a QR factorization of $\mathbf{X}$ and the Cholesky factor of $\mathbf{X}^\top \mathbf{X}$ are the same matrix (see also [Exercise \(B2\)](#)). The two routes of this chapter compute the same object, one from $\mathbf{X}$, the other from $\mathbf{X}^\top \mathbf{X}$ after the small singular values may already have been damaged ([Proposition 10.1.3](#)).

The algorithm itself is backward stable *for the factorization*. The computed $\hat{\mathbf{R}}$ satisfies $\hat{\mathbf{R}}^\top \hat{\mathbf{R}} = \mathbf{A} + \Delta \mathbf{A}$ with $|\Delta \mathbf{A}| \leq \gamma_{p+1} |\hat{\mathbf{R}}|^\top |\hat{\mathbf{R}}|$ entrywise, and it runs to completion whenever $\mathbf{A}$ is not too close to singular (Higham 2002, chapter 10), with no pivoting. The weakness of the normal equations lies entirely in forming $\mathbf{A} = \mathbf{X}^\top \mathbf{X}$, not in factoring it.

```
import numpy as np
import statsmodels.api as sm

def cholesky_upper(A):
    """Upper triangular R with positive diagonal and R'R = A (A positive definite)."""
    A = np.array(A, dtype=float)
    p = A.shape[0]
    R = np.zeros_like(A)
    for j in range(p):
        d = A[j, j] - R[:j, j] @ R[:j, j]
        if d <= 0:
            raise np.linalg.LinAlgError(f"not positive definite at step {j + 1}")
        R[j, j] = np.sqrt(d)
        R[j, j + 1:] = (A[j, j + 1:] - R[:j, j] @ R[:j, j + 1:]) / R[j, j]
    return R

def solve_upper(R, b):
    """Back substitution for R x = b, R upper triangular."""
    x = np.zeros(np.shape(b))
    for i in range(len(b) - 1, -1, -1):
        x[i] = (b[i] - R[i, i + 1:] @ x[i + 1:]) / R[i, i]
    return x

def solve_lower(L, b):
    """Forward substitution for L x = b, L lower triangular."""
    x = np.zeros(np.shape(b))
    for i in range(len(b)):
        x[i] = (b[i] - L[i, :i] @ x[:i]) / L[i, i]
    return x
```

10.2.2 Everything from one factorization

Solving the normal equations $\mathbf{R}^\top \mathbf{R} \mathbf{b} = \mathbf{X}^\top \mathbf{y}$ takes two triangular solves: $\mathbf{R}^\top \mathbf{z} = \mathbf{X}^\top \mathbf{y}$ for $\mathbf{z}$, then $\mathbf{R} \mathbf{b} = \mathbf{z}$. Each costs p^2 flops. It is neater to factor the cross-product matrix of the augmented matrix $\mathbf{Z} = [\mathbf{X}, \mathbf{y}]$. The factor then contains the solution of the first triangular system and the residual sum of squares.

Proposition 10.2.2 — The augmented factorization

Let $\mathbf{X}$ be $n \times p$ of full column rank, suppose $\mathbf{y} \notin \mathcal{C}(\mathbf{X})$, and let

$$\mathbf{Z}^\top \mathbf{Z} = \begin{pmatrix} \mathbf{X}^\top \mathbf{X} & \mathbf{X}^\top \mathbf{y} \\ \mathbf{y}^\top \mathbf{X} & \mathbf{y}^\top \mathbf{y} \end{pmatrix} = \mathbf{T}^\top \mathbf{T}, \quad \mathbf{T} = \begin{pmatrix} \mathbf{R} & \mathbf{z} \\ \mathbf{0}^\top & d \end{pmatrix},$$

be the Cholesky factorization (Theorem 10.2.1) of the cross-product matrix of $\mathbf{Z} = [\mathbf{X}, \mathbf{y}]$. Then:

- (a) $\mathbf{R}$ is the Cholesky factor of $\mathbf{X}^\top \mathbf{X}$, the least squares estimate solves $\mathbf{R} \hat{\boldsymbol{\beta}} = \mathbf{z}$, and $d^2 = \text{SSE} = \|\mathbf{y} - \mathbf{X} \hat{\boldsymbol{\beta}}\|^2$;
- (b) for each $k \leq p$, the residual sum of squares of the regression of $\mathbf{y}$ on the first k columns of $\mathbf{X}$ is $d^2 + \sum_{j>k} z_j^2$. The drop in residual sum of squares when column k joins columns $1, \dots, k-1$ is z_k^2 ;
- (c) $\det(\mathbf{X}^\top \mathbf{X}) = \prod_j r_{jj}^2$, $(\mathbf{X}^\top \mathbf{X})^{-1} = \mathbf{R}^{-1} \mathbf{R}^{-\top}$, and the leverage of row $\mathbf{x}_{(i)}^\top$ is $h_{ii} = \|\mathbf{R}^{-\top} \mathbf{x}_{(i)}\|^2$.

Proof. $\mathbf{Z}$ has full column rank because $\mathbf{y} \notin \mathcal{C}(\mathbf{X})$, so $\mathbf{Z}^\top \mathbf{Z}$ is positive definite and $\mathbf{T}$ exists. Multiplying out $\mathbf{T}^\top \mathbf{T}$ and comparing blocks with $\mathbf{Z}^\top \mathbf{Z}$ gives

$$\mathbf{R}^\top \mathbf{R} = \mathbf{X}^\top \mathbf{X}, \quad \mathbf{R}^\top \mathbf{z} = \mathbf{X}^\top \mathbf{y}, \quad \mathbf{z}^\top \mathbf{z} + d^2 = \mathbf{y}^\top \mathbf{y}.$$

The first identity and uniqueness make $\mathbf{R}$ the Cholesky factor of $\mathbf{X}^\top \mathbf{X}$. The second gives $\mathbf{R}^\top \mathbf{R} \hat{\boldsymbol{\beta}} = \mathbf{R}^\top \mathbf{z}$ for the solution of the normal equations, so $\mathbf{R} \hat{\boldsymbol{\beta}} = \mathbf{z}$.

For (b), and for the rest of (a), let $\mathbf{Q} = \mathbf{Z} \mathbf{T}^{-1}$. Then $\mathbf{Q}^\top \mathbf{Q} = \mathbf{T}^{-\top} \mathbf{Z}^\top \mathbf{Z} \mathbf{T}^{-1} = \mathbf{I}$, so the columns $\mathbf{q}_1, \dots, \mathbf{q}_{p+1}$ of $\mathbf{Q}$ are orthonormal, and $\mathbf{Z} = \mathbf{Q} \mathbf{T}$. Because $\mathbf{T}$ is upper triangular with nonzero diagonal, the first k columns of $\mathbf{X}$ span the same subspace as $\mathbf{q}_1, \dots, \mathbf{q}_k$. The last column of $\mathbf{Z} = \mathbf{Q} \mathbf{T}$ reads $\mathbf{y} = \sum_{j \leq p} z_j \mathbf{q}_j + d \mathbf{q}_{p+1}$. By Proposition 6.3.2 the projection of $\mathbf{y}$ onto the span of the first k columns is $\sum_{j \leq k} z_j \mathbf{q}_j$, and the residual sum of squares is $\|\mathbf{y}\|^2 - \sum_{j \leq k} z_j^2 = d^2 + \sum_{j>k} z_j^2$. With $k = p$ this is d^2 , which completes (a). Taking differences gives the rest of (b).

- (c) The determinant of a triangular matrix is the product of its diagonal. The inverse formula follows from $\mathbf{X}^\top \mathbf{X} = \mathbf{R}^\top \mathbf{R}$. Finally $h_{ii} = \mathbf{x}_{(i)}^\top (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{x}_{(i)} = \mathbf{x}_{(i)}^\top \mathbf{R}^{-1} \mathbf{R}^{-\top} \mathbf{x}_{(i)} = \|\mathbf{R}^{-\top} \mathbf{x}_{(i)}\|^2$.

■

Part (b) says that the squares $z_1^2, \dots, z_p^2$ are the *sequential sums of squares* of the regressors in the order of the columns (Definition 9.3.1). A single factorization gives the whole nested

sequence of fits. This is the content of [Theorem 6.5.1](#), now in coordinates. Part (c) costs about np^2 flops for all n leverages, each a triangular solve of p^2 flops.

Example. (Murder rates by the augmented factorization)

For the state data of [Example \(Poverty and murder rates\)](#) ($n = 50$, with an intercept and the percentages in poverty, in single-parent households and urban, so $p = 4$), the augmented factorization ([Proposition 10.2.2](#)) gives the coefficients -9.0946 , 0.2538 , 0.3980 and 0.0046 , and the corner entry $d = 10.0976$, so $\text{SSE} = d^2 = 101.96$. The condition number of $\mathbf{X}$ is only 509, and the solution agrees with a QR solution to a relative difference of 2×10^{-14} . The largest leverage is 0.260, and $\log_{10} \det(\mathbf{X}^\top \mathbf{X}) = 11.14$. The script also checks the standard errors $s \| \mathbf{e}_j^\top \mathbf{R}^{-1} \|$ against statsmodels.

```
data = sm.datasets.statelcrime.load_pandas().data.drop(index="District of Columbia")
y = data["murder"].to_numpy()
X = np.column_stack([np.ones(len(y)), data[["poverty", "single", "urban"]]])
n, p = X.shape
```

```
Z = np.column_stack([X, y])          # augmented matrix [X, y]
S = Z.T @ Z                          # (p+1) x (p+1) cross-product matrix
T = cholesky_upper(S)                # T = [[R, z], [0, d]]
R, z, d = T[:p, :p], T[:p, p], T[p, p]

beta = solve_upper(R, z)              # R b = z
sse = d ** 2                         # the last diagonal entry is sqrt(SSE)
K = solve_upper(R, np.eye(p))        # R^{-1}
XtX_inv = K @ K.T                    # (X'X)^{-1} = R^{-1} R^{-T}
h = np.array([np.sum(solve_lower(R.T, x) ** 2) for x in X]) # leverages
print("coefficients:", np.round(beta, 4))
print(f"SSE = {sse:.4f}, d = {d:.4f}")
```

10.2.3 When the normal equations are good enough

[Proposition 10.1.3](#) and [Eq. 10.1.2](#) predict a relative error of order $\kappa^2 u$ in $\hat{\beta}$ from the normal equations, where, as [Section 10.5](#) shows, κ is the condition number of $\mathbf{X}$ after its columns are scaled to equal length. With $\kappa \leq 10^4$ about eight digits survive, more than any coefficient is known statistically. The normal equations are therefore sound for well-conditioned designs, when only cross-products are available, and inside iterations such as the reweighted least squares of Chapter 34. Nearly collinear observational data are where the κ^2 loss bites.

10.2.4 The sweep operator

Before QR became standard, regression programs worked on the augmented cross-product matrix with an operation that adds one variable at a time to the fit. It is still used for fitting many subsets of variables.

Definition 10.2.1 — Sweep

Let $\mathbf{A}$ be symmetric with $a_{kk} \neq 0$. **Sweeping** $\mathbf{A}$ on pivot k produces the symmetric matrix $\mathbf{B}$ with

$$b_{kk} = -\frac{1}{a_{kk}}, \quad b_{ik} = b_{ki} = \frac{a_{ik}}{a_{kk}}, \quad b_{ij} = a_{ij} - \frac{a_{ik}a_{kj}}{a_{kk}} \quad (i, j \neq k).$$

Proposition 10.2.3 — What sweeping computes

Let $\mathbf{S} = \mathbf{Z}^\top \mathbf{Z}$ be the cross-product matrix of $\mathbf{Z} = [\mathbf{z}_1, \dots, \mathbf{z}_m]$. Let K be a set of column indices such that $\mathbf{Z}_K$ has full column rank, and let L be the remaining indices. Sweeping $\mathbf{S}$ on the pivots in K , one at a time and in any order, meets only positive pivots and produces the matrix $\mathbf{W}$ with blocks

$$\mathbf{W}_{KK} = -(\mathbf{Z}_K^\top \mathbf{Z}_K)^{-1}, \quad \mathbf{W}_{KL} = \mathbf{W}_{LK}^\top = (\mathbf{Z}_K^\top \mathbf{Z}_K)^{-1} \mathbf{Z}_K^\top \mathbf{Z}_L, \quad \mathbf{W}_{LL} = \mathbf{Z}_L^\top (\mathbf{I} - \mathbf{M}_K) \mathbf{Z}_L,$$

where $\mathbf{M}_K$ is the projection onto $\mathcal{C}(\mathbf{Z}_K)$. Column $j \in L$ of $\mathbf{W}_{KL}$ holds the least squares coefficients of $\mathbf{z}_j$ regressed on $\mathbf{Z}_K$, and $\mathbf{W}_{LL}$ holds the residual cross-products. With $\mathbf{Z} = [\mathbf{X}, \mathbf{y}]$ and K the columns of $\mathbf{X}$, the last column of $\mathbf{W}$ contains $\hat{\boldsymbol{\beta}}$ and, in its corner, the residual sum of squares.

Proof. We induct on the size of K ; the case $K = \emptyset$ is $\mathbf{W} = \mathbf{S}$. Add a pivot $m \in L$ with $\mathbf{Z}_{K \cup \{m\}}$ of full column rank, and write $\mathbf{G} = (\mathbf{Z}_K^\top \mathbf{Z}_K)^{-1}$, $\mathbf{B}_j = \mathbf{G} \mathbf{Z}_K^\top \mathbf{z}_j$ and $\mathbf{w} = (\mathbf{I} - \mathbf{M}_K) \mathbf{z}_m$. The pivot is $\mathbf{z}_m^\top (\mathbf{I} - \mathbf{M}_K) \mathbf{z}_m = \|\mathbf{w}\|^2 > 0$. For $i, j \in L \setminus \{m\}$ the new entries are $\mathbf{z}_i^\top (\mathbf{I} - \mathbf{M}_K - \mathbf{w} \mathbf{w}^\top / \|\mathbf{w}\|^2) \mathbf{z}_j = \mathbf{z}_i^\top (\mathbf{I} - \mathbf{M}_{K \cup \{m\}}) \mathbf{z}_j$ by Lemma 6.6.1. The new entries $\mathbf{w}^\top \mathbf{z}_j / \|\mathbf{w}\|^2$ in row m , and $\mathbf{B}_j - \mathbf{B}_m \mathbf{w}^\top \mathbf{z}_j / \|\mathbf{w}\|^2$ in the rows of K , are the coefficients of $\mathbf{z}_j$ on $[\mathbf{Z}_K, \mathbf{z}_m]$ by Theorem 6.6.2. The partitioned inverse (Theorem 1.6.4), with Schur complement $\|\mathbf{w}\|^2$, shows that $(\mathbf{Z}_{K \cup \{m\}}^\top \mathbf{Z}_{K \cup \{m\}})^{-1}$ has blocks $\mathbf{G} + \mathbf{B}_m \mathbf{B}_m^\top / \|\mathbf{w}\|^2$, $-\mathbf{B}_m / \|\mathbf{w}\|^2$ and $1 / \|\mathbf{w}\|^2$, whose negatives are what the sweep produces: $-\mathbf{G} - \mathbf{B}_m \mathbf{B}_m^\top / \|\mathbf{w}\|^2$, $\mathbf{B}_m / \|\mathbf{w}\|^2$ and $-1 / \|\mathbf{w}\|^2$. The result depends only on the set K , so the order of the sweeps does not matter. ■

Sweeping a pivot costs about $(p+1)^2$ flops, so a new variable enters a fitted model for $O(p^2)$ work. The price is the same as for the normal equations: the sweep works with cross-products and inherits their κ^2 sensitivity. For the state data, sweeping all four columns reproduces the coefficients and SSE of Example (Murder rates by the augmented factorization). Sweeping only the intercept and the single-parent column puts the SSE of that two-column model, 121.92, in the corner.

```
def sweep(A, k):
    """Sweep the symmetric matrix A on pivot k."""
    A = np.array(A, dtype=float)
    a = A[k, k]
    B = A - np.outer(A[:, k], A[k, :]) / a
    B[k, :] = A[k, :] / a
    B[:, k] = A[:, k] / a
    B[k, k] = -1 / a
    return B
```



```

W = S.copy()
for k in range(p):
    W = sweep(W, k)
print("coefficients from the sweep:", np.round(W[:p, p], 4))
print(f"SSE from the sweep: {W[p, p]:.4f}")

```

10.2.5 Forming cross-products: the centring problem

With an intercept in the model, the cross-product matrix is usually computed in centred form, $\sum_i (x_{ij} - \bar{x}_j)(x_{ik} - \bar{x}_k)$, which by [Theorem 6.7.1](#) changes the parameterization but not the fit. How it is computed matters a great deal. The “textbook” formula $\sum_i x_i^2 - n\bar{x}^2$ subtracts two large, nearly equal numbers whenever the mean is large compared with the spread. The loss is governed by the ratio

$$\kappa_{\mathbf{x}} = \frac{\|\mathbf{x}\|}{\|\mathbf{x} - \bar{x}\mathbf{1}\|} = \left(1 + \frac{n\bar{x}^2}{\sum_i (x_i - \bar{x})^2}\right)^{1/2},$$

which is the condition number of the column, and it is large when the coefficient of variation is small. Three algorithms are in common use:

- *textbook*: accumulate $\sum x_i$ and $\sum x_i^2$ in one pass, then subtract;
- *two-pass*: compute $\bar{x}$ in a first pass, then sum $(x_i - \bar{x})^2$ in a second;
- *updating*: one pass, maintaining the running mean and the running centred sum by the recursions below.

Proposition 10.2.4 — Updating means and centred cross-products

For vectors $\mathbf{x}_1, \mathbf{x}_2, \dots$ in $\mathbb{R}^q$ let $\mathbf{m}_k$ be the mean of the first k and $\mathbf{C}_k = \sum_{i \leq k} (\mathbf{x}_i - \mathbf{m}_k)(\mathbf{x}_i - \mathbf{m}_k)^\top$. With $\mathbf{d}_k = \mathbf{x}_k - \mathbf{m}_{k-1}$,

$$\mathbf{m}_k = \mathbf{m}_{k-1} + \frac{\mathbf{d}_k}{k}, \quad \mathbf{C}_k = \mathbf{C}_{k-1} + \frac{k-1}{k} \mathbf{d}_k \mathbf{d}_k^\top.$$

Proof. The first identity is $k\mathbf{m}_k = (k-1)\mathbf{m}_{k-1} + \mathbf{x}_k$. For the second, write $\mathbf{C}_k = \sum_{i \leq k} \mathbf{x}_i \mathbf{x}_i^\top - k\mathbf{m}_k \mathbf{m}_k^\top$ and put $\mathbf{m} = \mathbf{m}_{k-1}$, $\mathbf{d} = \mathbf{d}_k$. Then

$$\mathbf{C}_k - \mathbf{C}_{k-1} = \mathbf{x}_k \mathbf{x}_k^\top - k\mathbf{m}_k \mathbf{m}_k^\top + (k-1)\mathbf{m} \mathbf{m}^\top.$$

Substituting $\mathbf{x}_k = \mathbf{m} + \mathbf{d}$ and $k\mathbf{m}_k \mathbf{m}_k^\top = k\mathbf{m} \mathbf{m}^\top + \mathbf{m} \mathbf{d}^\top + \mathbf{d} \mathbf{m}^\top + \mathbf{d} \mathbf{d}^\top / k$, all the terms in $\mathbf{m}$ cancel and $\mathbf{d} \mathbf{d}^\top (1 - 1/k)$ remains. ■

The update adds a *centred* rank-one term, so it never subtracts large quantities. It is the cross-product analogue of the row updates of [Section 10.6](#). Chan, Golub and LeVeque (1983) give error bounds of roughly $n\kappa_{\mathbf{x}}^2 u$ for the textbook formula, $n\kappa_{\mathbf{x}} u$ for updating and $nu + n^2 \kappa_{\mathbf{x}}^2 u^2$ for the two-pass algorithm.

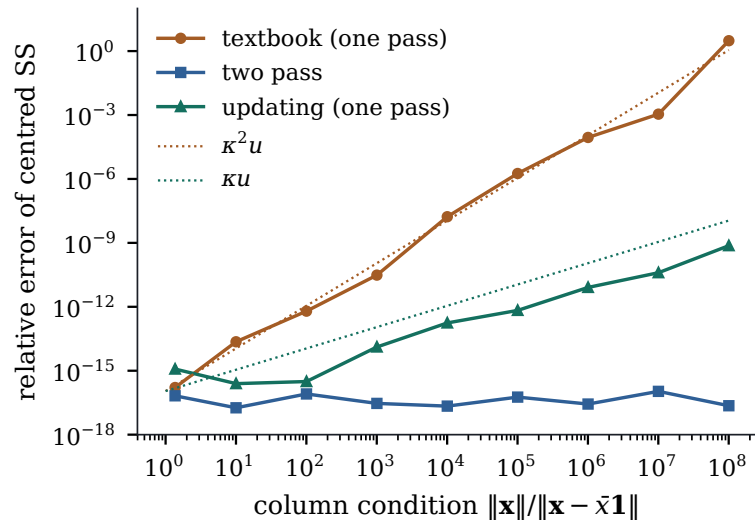


Figure 10.2.1: Relative error of the centred sum of squares of a column of 1000 values with standard deviation one and mean up to 10^8 , against the column condition number $\kappa_{\mathbf{x}}$. The one-pass textbook formula loses accuracy like $\kappa^2 u$, the one-pass updating formula like κu , and the two-pass algorithm not at all.

Example. (Centring a column with a large mean)

A sample of 1000 standard normal values is shifted by $c = 1, 10, \dots, 10^8$, and the centred sum of squares is computed by the three algorithms and compared with its exact value (found in rational arithmetic). At $c = 10^6$, where $\kappa_{\mathbf{x}} = 1.0 \times 10^6$, the relative errors are 8.9×10^{-5} for the textbook formula, 8.1×10^{-12} for updating and 2.7×10^{-17} for two passes. At $c = 10^8$ the textbook formula has relative error 3.1: not a single correct digit, and possibly a negative “sum of squares”. Updating gives 7.4×10^{-10} and two passes 2.3×10^{-17} . Figure 10.2.1 shows the three growth rates $\kappa^2 u$, κu and u .

```
def ss_textbook(x):
    return np.sum(x * x) - len(x) * np.mean(x) ** 2      # one pass, then subtract

def ss_two_pass(x):
    m = np.mean(x)                                       # first pass: the mean
    return np.sum((x - m) ** 2)                         # second pass: deviations

def ss Updating(x):
    m, s = 0.0, 0.0
    for k, xk in enumerate(x, start=1):                # one pass, no
        # cancellation
        d = xk - m
        m = m + d / k
        s = s + (k - 1) / k * d * d
    return s
```

Calendar years, population totals and price indices have large means and small spread. A program that forms cross-products should centre them with two passes (or update), solve for the slopes, and recover the intercept as $\bar{y} - \bar{\mathbf{x}}^\top \hat{\boldsymbol{\beta}}_{\text{slopes}}$.

10.2.6 Exercises

10.2.6.1 A. Check your understanding

Exercise A1.

Compute the Cholesky factor of $\mathbf{A} = \begin{pmatrix} 4 & 2 & -2 \\ 2 & 10 & 2 \\ -2 & 2 & 6 \end{pmatrix}$ by Eq. 10.2.1, and use it to solve $\mathbf{A}\mathbf{b} = (2, 14, 6)^\top$ and to find $\det \mathbf{A}$.

Solution. Row 1: $r_{11} = 2$, $r_{12} = 1$, $r_{13} = -1$. Row 2: $r_{22} = \sqrt{10-1} = 3$, $r_{23} = (2 - (1)(-1))/3 = 1$. Row 3: $r_{33} = \sqrt{6-1-1} = 2$. So $\mathbf{R} = \begin{pmatrix} 2 & 1 & -1 \\ 0 & 3 & 1 \\ 0 & 0 & 2 \end{pmatrix}$ and $\det \mathbf{A} = (2 \cdot 3 \cdot 2)^2 = 144$. Forward substitution in $\mathbf{R}^\top \mathbf{z} = (2, 14, 6)^\top$ gives $z_1 = 1$, $z_2 = (14-1)/3 = 13/3$, $z_3 = (6 + 1 - 13/3)/2 = 4/3$. Back substitution in $\mathbf{R}\mathbf{b} = \mathbf{z}$ gives $b_3 = 2/3$, $b_2 = (13/3 - 2/3)/3 = 11/9$ and $b_1 = (1 - 11/9 + 2/3)/2 = 2/9$. Check: $\mathbf{A}\mathbf{b} = (8/9 + 22/9 - 12/9, 4/9 + 110/9 + 12/9, -4/9 + 22/9 + 36/9) = (2, 14, 6)$.

Exercise A2.

For simple regression, $\mathbf{X} = [\mathbf{1}, \mathbf{x}]$, find the Cholesky factor (Theorem 10.2.1) of $\mathbf{X}^\top \mathbf{X}$ and of the augmented matrix, and show that it reproduces the slope S_{xy}/S_{xx} of Theorem 5.2.1. Where does the centring happen?

Solution. $\mathbf{X}^\top \mathbf{X} = \begin{pmatrix} n & n\bar{x} \\ n\bar{x} & \sum x_i^2 \end{pmatrix}$ gives $r_{11} = \sqrt{n}$, $r_{12} = \sqrt{n}\bar{x}$ and $r_{22} = (\sum x_i^2 - n\bar{x}^2)^{1/2} = S_{xx}^{1/2}$. The augmented column gives $z_1 = n\bar{y}/\sqrt{n} = \sqrt{n}\bar{y}$ and $z_2 = (\sum x_i y_i - n\bar{x}\bar{y})/S_{xx}^{1/2} = S_{xy}/S_{xx}^{1/2}$. Back substitution gives the slope $b_2 = z_2/r_{22} = S_{xy}/S_{xx}$ and the intercept $b_1 = (z_1 - r_{12}b_2)/r_{11} = \bar{y} - \bar{x}b_2$. The centring happens in the subtraction $a_{22} - r_{12}^2$, which is the textbook formula of Example (Centring a column with a large mean). So the Cholesky factorization of an uncentred cross-product matrix inherits its cancellation.

10.2.6.2 B. Practice

Exercise B1.

Show that the standard error of $\hat{\beta}_j$ is $s \|\mathbf{e}_j^\top \mathbf{R}^{-1}\|$, where $s^2 = \text{SSE}/(n-p)$ (Theorem 5.6.1). Show that all p standard errors cost $p^3/3 + O(p^2)$ flops, and all n leverages $np^2 + O(np)$ flops. Why is computing $\mathbf{M}$ itself a bad idea?

Solution. By Theorem 5.5.1 the estimated variance of $\hat{\beta}_j$ is $s^2[(\mathbf{X}^\top \mathbf{X})^{-1}]_{jj}$, and $[(\mathbf{X}^\top \mathbf{X})^{-1}]_{jj} = \mathbf{e}_j^\top \mathbf{R}^{-1} \mathbf{R}^{-\top} \mathbf{e}_j = \|\mathbf{e}_j^\top \mathbf{R}^{-1}\|^2$. The inverse of a triangular matrix is triangular, and column j of $\mathbf{R}^{-1}$ is found by back substitution on its leading $j \times j$ block, at about j^2 flops. The total is $\sum_j j^2 \approx p^3/3$. Each leverage is a triangular solve with $\mathbf{R}^\top$ (p^2 flops) and a squared norm. The projection $\mathbf{M}$ has n^2 entries, which for $n = 10^6$ is too many to store, and only its diagonal is ever needed.

Exercise B2.

Two groups of observations have sizes n_a, n_b , means $\mathbf{m}_a, \mathbf{m}_b$ and centred cross-product matrices $\mathbf{C}_a, \mathbf{C}_b$. Show that the pooled data have centred cross-product matrix

$$\mathbf{C} = \mathbf{C}_a + \mathbf{C}_b + \frac{n_a n_b}{n_a + n_b} (\mathbf{m}_a - \mathbf{m}_b)(\mathbf{m}_a - \mathbf{m}_b)^\top.$$

Show that [Proposition 10.2.4](#) is the special case $n_b = 1$, and explain how the formula allows cross-products to be computed in parallel on separate blocks of data.

10.2.6.3 C. Going deeper**Exercise C1.**

Let $\mathbf{A}$ be nonnegative definite of rank r . Show that the Cholesky recursion with *symmetric pivoting*, which at step j moves the largest remaining diagonal entry of the current Schur complement to position j , produces a permutation $\mathbf{\Pi}$ and $\mathbf{\Pi}^\top \mathbf{A} \mathbf{\Pi} = \mathbf{R}^\top \mathbf{R}$ with $\mathbf{R} = \begin{pmatrix} \mathbf{R}_{11} & \mathbf{R}_{12} \\ \mathbf{0} & \mathbf{0} \end{pmatrix}$ and $\mathbf{R}_{11}$ an $r \times r$ nonsingular upper triangle. (Use [Proposition 1.7.5\(a\)](#).) How is this related to the pivoted QR of [Section 10.7](#)?

10.3 Householder and Givens QR

Orthogonal matrices preserve lengths ([Proposition 1.5.3\(a\)](#)). For any orthogonal $\mathbf{Q}$,

$$\|\mathbf{y} - \mathbf{X}\mathbf{b}\| = \|\mathbf{Q}^\top \mathbf{y} - \mathbf{Q}^\top \mathbf{X}\mathbf{b}\| \quad \text{for every } \mathbf{b}.$$

So a least squares problem can be replaced by an equivalent one in which $\mathbf{Q}^\top \mathbf{X}$ is upper triangular, without ever forming $\mathbf{X}^\top \mathbf{X}$. Such a $\mathbf{Q}$ is built from reflections, which zero a whole column below the diagonal at once, or from rotations, which zero one entry at a time.

10.3.1 Householder reflections

A reflection $\mathbf{H} = \mathbf{I} - 2\mathbf{v}\mathbf{v}^\top/(\mathbf{v}^\top \mathbf{v})$ is symmetric and orthogonal, and it maps $\mathbf{v}$ to $-\mathbf{v}$ and fixes every vector orthogonal to $\mathbf{v}$ ([Exercise \(B1\)](#)). The whole numerical content lies in choosing $\mathbf{v}$ so that a given vector is mapped onto a coordinate axis, and in choosing its sign so that no cancellation occurs.

Lemma 10.3.1 — Householder vector

Let $\mathbf{x} \in \mathbb{R}^m$, $\mathbf{x} \neq \mathbf{0}$, let $\alpha = -\text{sign}(x_1)\|\mathbf{x}\|$ (with $\text{sign}(0) = 1$), and let $\mathbf{v} = \mathbf{x} - \alpha\mathbf{e}_1$. Then $\mathbf{v} \neq \mathbf{0}$, and $\mathbf{H} = \mathbf{I} - 2\mathbf{v}\mathbf{v}^\top/(\mathbf{v}^\top \mathbf{v})$ satisfies $\mathbf{H}\mathbf{x} = \alpha\mathbf{e}_1$. The first entry $v_1 = x_1 - \alpha$ is a sum of two numbers of the same sign, so $|v_1| = |x_1| + \|\mathbf{x}\|$. Applying $\mathbf{H}$ to a vector $\mathbf{w}$ as $\mathbf{w} - 2\mathbf{v}(\mathbf{v}^\top \mathbf{w})/(\mathbf{v}^\top \mathbf{v})$ costs about $4m$ flops, and $\mathbf{H}$ is never formed.

Proof. α and x_1 have opposite signs (or $x_1 = 0$), so $v_1 = x_1 - \alpha$ adds two numbers of the same sign and $|v_1| = |x_1| + \|\mathbf{x}\| > 0$. Next, $\mathbf{v}^\top \mathbf{x} = \|\mathbf{x}\|^2 - \alpha x_1$ and $\mathbf{v}^\top \mathbf{v} = \|\mathbf{x}\|^2 - 2\alpha x_1 + \alpha^2 = 2(\|\mathbf{x}\|^2 - \alpha x_1)$, using $\alpha^2 = \|\mathbf{x}\|^2$. Hence $2\mathbf{v}^\top \mathbf{x} / \mathbf{v}^\top \mathbf{v} = 1$ and $\mathbf{H}\mathbf{x} = \mathbf{x} - \mathbf{v} = \alpha \mathbf{e}_1$. ■

The opposite sign choice, $\alpha = +\text{sign}(x_1)\|\mathbf{x}\|$, also gives a reflection onto the axis. But then $v_1 = x_1 - \alpha$ subtracts two nearly equal numbers whenever $\mathbf{x}$ is nearly parallel to $\mathbf{e}_1$. This is the cancellation of [Section 10.1](#), and it can make $\mathbf{v}$ inaccurate in every digit.

Theorem 10.3.2 — Householder QR

Let $\mathbf{X}$ be $n \times p$ with $n \geq p$.

- (a) There are reflections $\mathbf{H}_1, \dots, \mathbf{H}_p$, with $\mathbf{H}_j$ acting only on coordinates $j, \dots, n$ (or equal to $\mathbf{I}$), such that

$$\mathbf{H}_p \cdots \mathbf{H}_1 \mathbf{X} = \begin{pmatrix} \mathbf{R} \\ \mathbf{0} \end{pmatrix}$$

with $\mathbf{R}$ upper triangular of size $p \times p$. Thus $\mathbf{X} = \mathbf{Q} \begin{pmatrix} \mathbf{R} \\ \mathbf{0} \end{pmatrix}$ with $\mathbf{Q} = \mathbf{H}_1 \cdots \mathbf{H}_p$ orthogonal, and $\mathbf{X} = \mathbf{Q}_1 \mathbf{R}$ where $\mathbf{Q}_1$ holds the first p columns of $\mathbf{Q}$. If $\text{rank } \mathbf{X} = p$, then $\mathbf{R}$ is nonsingular, and changing the signs of the rows of $\mathbf{R}$ with negative diagonal entries gives the Cholesky factor of $\mathbf{X}^\top \mathbf{X}$.

- (b) The reduction takes $2np^2 - 2p^3/3$ flops to leading order. Applying $\mathbf{Q}^\top$ (or $\mathbf{Q}$) to a vector takes $4np - 2p^2$, and forming $\mathbf{Q}_1$ explicitly takes another $2np^2 - 2p^3/3$.
- (c) (Backward stability.) In floating-point arithmetic, the computed $\hat{\mathbf{R}}$ is the exact triangular factor of $\mathbf{X} + \Delta\mathbf{X}$, where each column satisfies $\|\Delta\mathbf{x}_j\| \leq c_{np}u\|\mathbf{x}_j\|$. The least squares solution $\hat{\mathbf{b}}$ computed from $\hat{\mathbf{R}}$ and the computed $\mathbf{Q}^\top \mathbf{y}$ is the exact least squares solution for data $(\mathbf{X} + \Delta\mathbf{X}, \mathbf{y} + \Delta\mathbf{y})$ with perturbations of the same kind, $\|\Delta\mathbf{y}\| \leq c_{np}u\|\mathbf{y}\|$. Here c_{np} is a modest multiple of np .

Proof.

- (a) Induction on the column. Suppose $\mathbf{H}_{j-1} \cdots \mathbf{H}_1 \mathbf{X}$ has zeros below the diagonal in its first $j-1$ columns. Let $\mathbf{x}$ be the part of column j in rows $j, \dots, n$. If $\mathbf{x} = \mathbf{0}$, take $\mathbf{H}_j = \mathbf{I}$. Otherwise take the reflection of [Lemma 10.3.1](#) for $\mathbf{x}$, acting on rows $j, \dots, n$. It zeroes column j below the diagonal. It leaves rows $1, \dots, j-1$ alone. It maps the first $j-1$ columns, which are zero in rows $j, \dots, n$, to themselves. After p steps the matrix is $\begin{pmatrix} \mathbf{R} \\ \mathbf{0} \end{pmatrix}$, and $\mathbf{Q}$ is a product of orthogonal matrices. Orthogonal factors do not change rank, so $\text{rank } \mathbf{R} = \text{rank } \mathbf{X}$. If $\mathbf{D}$ is the diagonal matrix of signs of the diagonal of $\mathbf{R}$, then $\mathbf{X} = (\mathbf{Q}_1 \mathbf{D})(\mathbf{D}\mathbf{R})$ is a QR factorization with positive diagonal, and $\mathbf{D}\mathbf{R}$ is the Cholesky factor by [Theorem 10.2.1\(d\)](#).
- (b) Step j builds $\mathbf{v}$ (about $2(n-j)$ flops) and applies the reflection to the $p-j$ remaining columns of length $n-j+1$, at about $4(n-j)(p-j)$ flops. Summing,

$$\sum_{j=1}^p 4(n-j)(p-j) \approx 4 \int_0^p (n-t)(p-t) dt = 2np^2 - \frac{2}{3}p^3.$$

Applying the p reflections to a vector costs $\sum_j 4(n-j+1) \approx 4np - 2p^2$. Forming

$\mathbf{Q}_1$ means applying the reflections in reverse order to the first p columns of $\mathbf{I}_n$. Reflection j then acts on columns $j, \dots, p$ only, which gives the same integral again.

- (c) This is a theorem of rounding-error analysis, not of linear algebra, and we state it without proof. Its essence is that a reflection applied in floating point is an exact reflection applied to a slightly perturbed matrix, that orthogonal matrices do not amplify perturbations, and that the p steps add their perturbations. See Higham (2002, chapters 19 and 20) and Björck (1996, chapter 2).

■

Part (c) is why QR is the default. Its backward error is small relative to each column separately, which [Section 10.5](#) uses to explain why column scaling does not change the accuracy of QR. Software never forms $\mathbf{Q}$; it stores the vectors $\mathbf{v}_j$ in the zeroed part of $\mathbf{X}$.

```
import numpy as np
import statsmodels.api as sm

def house(x):
    """Unit vector v with (I - 2 v v') x = alpha e_1, alpha = -sign(x_1) ||x||."""
    alpha = -np.copysign(np.linalg.norm(x), x[0])
    v = x.astype(float).copy()
    v[0] -= alpha                                # x_1 - alpha adds two numbers of the same
    sign                                         sign
    return v / np.linalg.norm(v), alpha

def householder_qr(X):
    """Return the reflection vectors and the p x p triangle R of X = Q [R; 0]."""
    A = np.array(X, dtype=float)
    n, p = A.shape
    vs = []
    for j in range(p):
        v, alpha = house(A[j:, j])
        A[j:, j:] -= 2.0 * np.outer(v, v @ A[j:, j:])    # apply I - 2vv' to the
        trailing block
        vs.append(v)
    return vs, np.triu(A[:p, :])

def apply_Qt(vs, y):
    """Q'y, applying the reflections H_1, ..., H_p in turn (Q itself is never formed)."""
    z = np.array(y, dtype=float)
    for j, v in enumerate(vs):
        z[j:] -= 2.0 * v * (v @ z[j:])
    return z

def apply_Q(vs, z):
    """Q z, applying the reflections in reverse order."""
    w = np.array(z, dtype=float)
    for j in range(len(vs) - 1, -1, -1):
        v = vs[j]
        w[j:] -= 2.0 * v * (v @ w[j:])
    return w
```

10.3.2 Regression quantities from the factorization

Section 6.10 derived the regression quantities from the thin factorization $\mathbf{X} = \mathbf{Q}_1\mathbf{R}$ (Eq. 6.10.1). The full factorization, with $\mathbf{Q}$ square, adds one thing: the part of $\mathbf{Q}^\top \mathbf{y}$ that the model cannot reach, which is the residual in rotated coordinates.

Proposition 10.3.3 — Least squares from QR

Let $\mathbf{X} = \mathbf{Q} \begin{pmatrix} \mathbf{R} \\ \mathbf{0} \end{pmatrix}$ with $\mathbf{Q}$ orthogonal and $\text{rank } \mathbf{X} = p$, and partition $\mathbf{c} = \mathbf{Q}^\top \mathbf{y}$ as $(\mathbf{c}_1^\top, \mathbf{c}_2^\top)^\top$ with $\mathbf{c}_1 \in \mathbb{R}^p$. Then:

- (a) $\hat{\boldsymbol{\beta}}$ is the solution of the triangular system $\mathbf{R}\mathbf{b} = \mathbf{c}_1$, and $\text{SSE} = \|\mathbf{c}_2\|^2$;
- (b) $\hat{\mathbf{Y}} = \mathbf{Q} \begin{pmatrix} \mathbf{c}_1 \\ \mathbf{0} \end{pmatrix}$ and $\hat{\boldsymbol{\varepsilon}} = \mathbf{Q} \begin{pmatrix} \mathbf{0} \\ \mathbf{c}_2 \end{pmatrix}$;
- (c) $(\mathbf{X}^\top \mathbf{X})^{-1} = \mathbf{R}^{-1} \mathbf{R}^{-\top}$ and $h_{ii} = \|\mathbf{e}_i^\top \mathbf{Q}_1\|^2$;
- (d) c_k^2 is the drop in residual sum of squares when column k is added to columns $1, \dots, k-1$, the sequential sum of squares of Definition 9.3.1.

Proof.

- (a) With $\mathbf{Q}_1$ the first p columns of $\mathbf{Q}$, $\mathbf{X} = \mathbf{Q}_1\mathbf{R}$ is a thin QR factorization and $\mathbf{c}_1 = \mathbf{Q}_1^\top \mathbf{y}$, so $\mathbf{R}\hat{\boldsymbol{\beta}} = \mathbf{c}_1$ and $\text{SSE} = \|\mathbf{y}\|^2 - \|\mathbf{c}_1\|^2$ are Eq. 6.10.1. What the full $\mathbf{Q}$ adds is $\|\mathbf{c}\| = \|\mathbf{y}\|$, hence $\text{SSE} = \|\mathbf{c}_2\|^2$.
- (b) $\mathbf{X}\hat{\boldsymbol{\beta}} = \mathbf{Q} \begin{pmatrix} \mathbf{R}\hat{\boldsymbol{\beta}} \\ \mathbf{0} \end{pmatrix} = \mathbf{Q} \begin{pmatrix} \mathbf{c}_1 \\ \mathbf{0} \end{pmatrix}$, and $\mathbf{y} = \mathbf{Q}\mathbf{c}$.
- (c) $\mathbf{X}^\top \mathbf{X} = \mathbf{R}^\top \mathbf{Q}_1^\top \mathbf{Q}_1 \mathbf{R} = \mathbf{R}^\top \mathbf{R}$, and the leverage formula is Eq. 6.10.1.
- (d) The first k columns of $\mathbf{X}$ equal the first k columns of $\mathbf{Q}$ times the leading $k \times k$ block of $\mathbf{R}$, which is nonsingular. So they span the same subspace as $\mathbf{q}_1, \dots, \mathbf{q}_k$, and the squared length of the projection of $\mathbf{y}$ onto it is $\sum_{j \leq k} (\mathbf{q}_j^\top \mathbf{y})^2 = \sum_{j \leq k} c_j^2$ (Proposition 6.3.2).

■

Part (d) is the QR form of Proposition 10.2.2(b): by uniqueness of the triangle, $\mathbf{c}_1$ equals the vector $\mathbf{z}$ of that proposition up to signs. The vector $\mathbf{c}_2$ replaces the n correlated residuals by $n - p$ uncorrelated ones with the same sum of squares (Exercise (B1)).

Example. (Sequential sums of squares from one factorization)

For the state data of Example (Murder rates by the augmented factorization), Householder QR (Theorem 10.3.2) gives $\mathbf{R}$ with diagonal $-7.071, 21.511, 21.609$ and 113.413 . The first entry is $-\sqrt{50}$: the sign convention of Lemma 10.3.1 makes it negative. The squares of the first four entries of $\mathbf{Q}^\top \mathbf{y}$ are the sequential sums of squares: 1018.81 for the intercept (which is $n\bar{y}^2$), 102.49 for poverty, 80.04 for single-parent households given poverty, and 0.27 for urbanization given the other two. The remaining 46 entries have squared length $\text{SSE} = 101.96$. The urbanization variable adds almost nothing once the other two are in the model, which is what its small coefficient in Example (Poverty and murder rates) suggested. The script checks every number against nested fits, and checks the leverages against the squared row lengths of $\mathbf{Q}_1$.

```

data = sm.datasets.statelcrime.load_pandas().data.drop(index="District of Columbia")
y = data["murder"].to_numpy()
X = np.column_stack([np.ones(len(y)), data[["poverty", "single", "urban"]]])
n, p = X.shape

vs, R = householder_qr(X)
c = apply_Qt(vs, y)                                # c = Q'y = (c1, c2)
c1, c2 = c[:p], c[p:]
beta = np.linalg.solve(R, c1)                       # triangular system R b = c1
sse = c2 @ c2                                       # SSE = ||c2||^2
resid = apply_Q(vs, np.concatenate([np.zeros(p), c2]))
seq_ss = c1 ** 2                                   # sequential sums of squares, in column
                                                order
print("R diagonal:", np.round(np.diag(R), 4))
print("coefficients:", np.round(beta, 4), f" SSE = {sse:.4f}")
print("sequential SS (intercept, poverty, single, urban):", np.round(seq_ss, 3))

```

10.3.3 Givens rotations

A reflection zeroes a whole column at once. Sometimes it is better to zero one entry at a time, touching only two rows. This is the case when $\mathbf{X}$ is sparse, when rows arrive one by one, or when an existing factorization must be modified.

- Theorem 10.3.4 — Givens rotations** (a) For $(a, b) \neq (0, 0)$ let $r = (a^2 + b^2)^{1/2}$, $c = a/r$ and $s = b/r$. The rotation $\mathbf{G} = \begin{pmatrix} c & s \\ -s & c \end{pmatrix}$ is orthogonal and $\mathbf{G}(a, b)^\top = (r, 0)^\top$. Embedded in rows i and k of the identity, it changes only rows i and k of a matrix it multiplies, at 6 flops per column.
- (b) Zeroing the subdiagonal of $\mathbf{X}$ column by column, each entry against the row above it, produces $\mathbf{GX} = \begin{pmatrix} \mathbf{R} \\ \mathbf{0} \end{pmatrix}$ with $\mathbf{G}$ orthogonal. It takes $np - p(p+1)/2$ rotations and $3np^2 - p^3$ flops to leading order.
- (c) (Row-wise QR.) Let $\mathbf{R}$ be upper triangular and $\mathbf{x} \in \mathbb{R}^p$. There are p rotations, the k th acting on row k of $\mathbf{R}$ and on the new row, that reduce $\begin{pmatrix} \mathbf{R} \\ \mathbf{x}^\top \end{pmatrix}$ to $\begin{pmatrix} \mathbf{R}_+ \\ \mathbf{0}^\top \end{pmatrix}$ with $\mathbf{R}_+$ upper triangular and $\mathbf{R}_+^\top \mathbf{R}_+ = \mathbf{R}^\top \mathbf{R} + \mathbf{x}\mathbf{x}^\top$. The cost is $3p^2$ flops. If $\mathbf{R}$ and $\mathbf{x}$ are augmented by the response, as in Proposition 10.2.2, the new row ends as $(0, \dots, 0, t)$, and the residual sum of squares increases by t^2 .
- (d) A rotation of rows i and k creates a nonzero in column j only if row i or row k already had one there.

Proof.

- (a) $c^2 + s^2 = 1$ gives $\mathbf{G}^\top \mathbf{G} = \mathbf{I}$. Also $ca + sb = (a^2 + b^2)/r = r$ and $-sa + cb = 0$. Each column of the product needs four multiplications and two additions.
- (b) When entry (i, j) is zeroed against row $i - 1$, working upwards from row n , both rows are already zero in columns $1, \dots, j - 1$, so no earlier zero is destroyed. Column j needs $n - j$ rotations, each acting on $p - j + 1$ columns, so the count is $\sum_j 6(n - j)(p - j + 1) \approx 6 \int_0^p (n - t)(p - t) dt = 3np^2 - p^3$.
- (c) Before rotation k , the new row is zero in columns $1, \dots, k - 1$, and so is row k of $\mathbf{R}$. The rotation chosen by (a) to zero the k th entry of the new row keeps both

rows zero in those columns. After p rotations the new row is zero, and row k of $\mathbf{R}$ has been replaced by a row that is zero before column k . With $\mathbf{G}$ the product of the rotations,

$$\mathbf{R}_+^\top \mathbf{R}_+ = \begin{pmatrix} \mathbf{R} \\ \mathbf{x}^\top \end{pmatrix}^\top \mathbf{G}^\top \mathbf{G} \begin{pmatrix} \mathbf{R} \\ \mathbf{x}^\top \end{pmatrix} = \mathbf{R}^\top \mathbf{R} + \mathbf{x} \mathbf{x}^\top.$$

Rotation k acts on $p - k + 1$ columns, which gives $\sum_k 6(p - k + 1) \approx 3p^2$. In the augmented case, apply p rotations to the augmented $(p + 1) \times (p + 1)$ triangle $\mathbf{T}$ and the row $(\mathbf{x}^\top, y)$. The new row becomes $(\mathbf{0}^\top, t)$, and $\mathbf{T}_+^\top \mathbf{T}_+ = \mathbf{T}^\top \mathbf{T} + (\mathbf{x}^\top, y)^\top (\mathbf{x}^\top, y) - (\mathbf{0}^\top, t)^\top (\mathbf{0}^\top, t)$. The corner entry of $\mathbf{T}_+$ is still d , so the augmented triangle with the new row included is obtained by one more rotation, of d against t . This gives a corner $(d^2 + t^2)^{1/2}$, and the claim follows from [Proposition 10.2.2\(a\)](#).

(d) Each new row is a combination of the two old ones. ■

Part (c) turns QR into a streaming algorithm (Gentleman 1973): feed the rows of $[\mathbf{X}, \mathbf{y}]$ one at a time into $\mathbf{R} = \mathbf{0}$. Memory is $O(p^2)$, one pass suffices, and the result is as accurate as Householder QR, so the one practical advantage of accumulating cross-products is kept without the κ^2 loss. Part (d) keeps $\mathbf{R}$ sparse for sparse designs, such as factors with many levels.

```
def givens(a, b):
    """c, s, r with [[c, s], [-s, c]] @ [a, b] = [r, 0]."""
    if b == 0.0:
        return 1.0, 0.0, a
    r = np.hypot(a, b)                                # sqrt(a^2 + b^2) without overflow
    return a / r, b / r, r
```

```
def add_row(R, z, x, y):
    """Rotate one observation (x, y) into the triangle R and the vector z = (Q'y)[:p].

    Returns the new R, z and the entry that leaves: its square is the increase in SSE.
    """
    R, z, x, y = R.copy(), z.copy(), np.array(x, dtype=float), float(y)
    for k in range(len(x)):
        c, s, r = givens(R[k, k], x[k])
        Rk, xk = R[k, k:].copy(), x[k:].copy()
        R[k, k:], x[k:] = c * Rk + s * xk, -s * Rk + c * xk
        z[k], y = c * z[k] + s * y, -s * z[k] + c * y
    return R, z, y
```

```
def rowwise_ls(X, y):
    """Least squares with one pass over the rows and O(p^2) memory."""
    p = X.shape[1]
    R, z, sse = np.zeros((p, p)), np.zeros(p), 0.0
    for xi, yi in zip(X, y):
        R, z, leftover = add_row(R, z, xi, yi)
        sse += leftover ** 2
    return R, z, sse
```

```
R_row, z_row, sse_row = rowwise_ls(X, y)
```

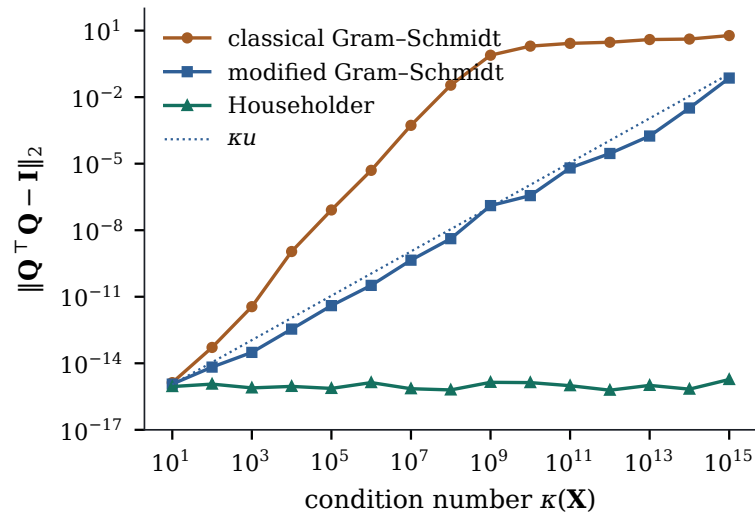


Figure 10.3.1: Departure from orthogonality of the computed $\mathbf{Q}$ against the condition number of $\mathbf{X}$, for classical and modified Gram–Schmidt and for Householder QR. The dotted line is κu .

```
print("coefficients (row-wise Givens):", np.round(np.linalg.solve(R_row, z_row), 4))
print(f"SSE (row-wise Givens): {sse_row:.4f}")
```

The rotation in `givens` uses `hypot`, which computes $(a^2 + b^2)^{1/2}$ without overflow or underflow. Square-root-free “fast” variants (Gentleman 1973) matter less on modern hardware.

10.3.4 Gram–Schmidt revisited

Gram–Schmidt (Theorem 1.5.2) also computes a QR factorization, at $2np^2$ flops, and forms $\mathbf{Q}_1$ explicitly, which gives leverages directly. The classical form computes $r_{kj} = \mathbf{q}_k^\top \mathbf{x}_j$ from the *original* column; the modified form subtracts each new direction from all remaining columns at once. The two agree in exact arithmetic and differ greatly in floating point.

Example. (Loss of orthogonality)

For 100×12 matrices with prescribed condition numbers from 10 to 10^{15} , Figure 10.3.1 plots $\|\hat{\mathbf{Q}}^\top \hat{\mathbf{Q}} - \mathbf{I}\|_2$ for the computed $\hat{\mathbf{Q}}$. At $\kappa = 10^8$ classical Gram–Schmidt gives 3.5×10^{-2} , and its columns are no longer an orthonormal basis in any useful sense. Modified Gram–Schmidt gives 4.2×10^{-9} , in line with the bound $O(\kappa u)$ of Björck (1967). The $\mathbf{Q}$ accumulated from Householder reflections gives 6.3×10^{-16} , orthogonal to working precision whatever the condition number.

```
def cgs(X):
    """Classical Gram-Schmidt: inner products with the original column."""
    n, p = X.shape
    Q = np.zeros((n, p))
    for j in range(p):
        w = X[:, j] - Q[:, :j] @ (Q[:, :j].T @ X[:, j])
        Q[:, j] = w / np.linalg.norm(w)
    return Q

def mgs(X):
```

```

"""Modified Gram-Schmidt: subtract each direction from the working vector at
once."""
Q = np.array(X, dtype=float)
n, p = Q.shape
for j in range(p):
    Q[:, j] /= np.linalg.norm(Q[:, j])
    Q[:, j + 1:] -= np.outer(Q[:, j], Q[:, j] @ Q[:, j + 1:])
return Q

```

Loss of orthogonality does not make modified Gram–Schmidt useless. It is numerically equivalent to Householder QR applied to $\mathbf{X}$ with p rows of zeros on top (Björck and Paige 1992), so applied to $[\mathbf{X}, \mathbf{y}]$, with $\mathbf{Q}_1^\top \mathbf{y}$ taken from the last column, it gives a backward stable least squares solution. Leverages computed from the rows of $\hat{\mathbf{Q}}_1$ are not reliable when κ is large.

Idea. Orthogonal transformations reduce $\mathbf{X}$ to a triangle while preserving every residual length. Householder reflections do it fastest; Givens rotations do it one entry at a time, which suits streaming, updating and sparse data.

10.3.5 Exercises

10.3.5.1 A. Check your understanding

Exercise A1.

Find the Householder vector and α of Lemma 10.3.1 for $\mathbf{x} = (3, 4)^\top$, and verify $\mathbf{H}\mathbf{x} = \alpha \mathbf{e}_1$. For $\mathbf{x} = (1, 10^{-9})^\top$, compute v_1 with both sign choices in double precision, and explain what goes wrong with the unstable one.

Solution. $\|\mathbf{x}\| = 5$, $\alpha = -5$, $\mathbf{v} = (8, 4)^\top$, $\mathbf{v}^\top \mathbf{v} = 80$ and $\mathbf{v}^\top \mathbf{x} = 40$, so $\mathbf{H}\mathbf{x} = \mathbf{x} - 2 \cdot \frac{40}{80} \mathbf{v} = (3, 4) - (8, 4) = (-5, 0)$. For $\mathbf{x} = (1, 10^{-9})^\top$, $\|\mathbf{x}\| = (1 + 10^{-18})^{1/2}$ rounds to 1. The stable choice gives $v_1 = 1 + 1 = 2$. The unstable one gives $v_1 = 1 - 1 = 0$ in floating point, while the true value is $1 - \|\mathbf{x}\| \approx -5 \times 10^{-19}$. With $\mathbf{v} = (0, 10^{-9})$ the “reflection” swaps the sign of the second coordinate and maps $\mathbf{x}$ to $(1, -10^{-9})$, not onto the axis. An algebraically equivalent formula, $v_1 = -(x_2^2 + \cdots + x_m^2)/(x_1 + \|\mathbf{x}\|)$, rescues that choice.

10.3.5.2 B. Practice

Exercise B1.

In the setting of Proposition 10.3.3, let $\mathbf{Q} = [\mathbf{Q}_1, \mathbf{Q}_2]$. Show that under the linear model with $\text{Cov}(\boldsymbol{\varepsilon}) = \sigma^2 \mathbf{I}$, the vector $\mathbf{c}_2 = \mathbf{Q}_2^\top \mathbf{y}$ has mean $\mathbf{0}$ and covariance $\sigma^2 \mathbf{I}_{n-p}$, and that $\hat{\boldsymbol{\varepsilon}} = \mathbf{Q}_2 \mathbf{c}_2$. Under normality, deduce from $\mathbf{c}_2$ alone that $\text{SSE}/\sigma^2 \sim \chi^2(n-p)$ independently of $\hat{\boldsymbol{\beta}}$ (compare Theorem 7.5.1). Is $\mathbf{c}_2$ unique?

Solution. $\mathbf{Q}_2^\top \mathbf{X} = \mathbf{0}$, because $\mathbf{Q}^\top \mathbf{X} = \begin{pmatrix} \mathbf{R} \\ \mathbf{0} \end{pmatrix}$. So $\mathbf{c}_2 = \mathbf{Q}_2^\top \mathbf{X} \boldsymbol{\beta} + \mathbf{Q}_2^\top \boldsymbol{\varepsilon} = \mathbf{Q}_2^\top \boldsymbol{\varepsilon}$, with mean $\mathbf{0}$ and covariance $\sigma^2 \mathbf{Q}_2^\top \mathbf{Q}_2 = \sigma^2 \mathbf{I}$. By Proposition 10.3.3(b), $\hat{\boldsymbol{\varepsilon}} = \mathbf{Q} \begin{pmatrix} \mathbf{0} \\ \mathbf{c}_2 \end{pmatrix} = \mathbf{Q}_2 \mathbf{c}_2$. Under normality, $\mathbf{Q}^\top \boldsymbol{\varepsilon} \sim N_n(\mathbf{0}, \sigma^2 \mathbf{I})$ (Theorem 3.3.1), so $\mathbf{c}_1 - \mathbf{R} \boldsymbol{\beta} = \mathbf{Q}_1^\top \boldsymbol{\varepsilon}$ and $\mathbf{c}_2 = \mathbf{Q}_2^\top \boldsymbol{\varepsilon}$ are independent normal vectors with identity covariance times σ^2 . Then $\text{SSE}/\sigma^2 = \|\mathbf{c}_2/\sigma\|^2 \sim \chi^2(n-p)$, and $\hat{\boldsymbol{\beta}} = \mathbf{R}^{-1} \mathbf{c}_1$ is a function of $\mathbf{c}_1$ alone. $\mathbf{c}_2$ is not unique: $\mathbf{Q}_2$

can be replaced by $\mathbf{Q}_2\mathbf{O}$ for any orthogonal $\mathbf{O}$, which changes $\mathbf{c}_2$ but not its length or its distribution.

Exercise B2.

Weighted least squares with a diagonal weight matrix $\mathbf{W}$ of positive weights is ordinary least squares for $\mathbf{W}^{1/2}\mathbf{X}$ and $\mathbf{W}^{1/2}\mathbf{y}$. Show that $\kappa(\mathbf{W}^{1/2}\mathbf{X}) \leq \kappa(\mathbf{W}^{1/2})\kappa(\mathbf{X})$. Give an example with two very different weights in which $\kappa(\mathbf{W}^{1/2}\mathbf{X})$ is large although $\kappa(\mathbf{X})$ is small, and one in which it is small although $\kappa(\mathbf{W}^{1/2})$ is large.

Solution. $\sigma_{\max}(\mathbf{W}^{1/2}\mathbf{X}) \leq \sigma_{\max}(\mathbf{W}^{1/2})\sigma_{\max}(\mathbf{X})$ and $\|\mathbf{W}^{1/2}\mathbf{X}\mathbf{v}\| \geq \sigma_{\min}(\mathbf{W}^{1/2})\|\mathbf{X}\mathbf{v}\| \geq \sigma_{\min}(\mathbf{W}^{1/2})\sigma_{\min}(\mathbf{X})$ for unit $\mathbf{v}$, by Proposition 1.8.2. Divide. For the first example take $\mathbf{X} = \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$ (with $\kappa = 1$) and $\mathbf{W} = \text{diag}(1, w^2)$ with w large. Then $\mathbf{W}^{1/2}\mathbf{X}$ has rows $(1, 1)$ and $(w, -w)$, whose singular values are $\sqrt{2}$ and $\sqrt{2}w$, so $\kappa = w$. For the second take $\mathbf{X} = \begin{pmatrix} 1 & 0 \\ 0 & 1/w \end{pmatrix}$ with the same $\mathbf{W}$. Then $\mathbf{W}^{1/2}\mathbf{X} = \mathbf{I}$.

10.3.5.3 C. Going deeper

Exercise C1.

Show that, in exact arithmetic, Householder QR (Theorem 10.3.2) applied to $\begin{pmatrix} \mathbf{0}_{p \times p} \\ \mathbf{X} \end{pmatrix}$ produces the same $\mathbf{R}$ as modified Gram–Schmidt applied to $\mathbf{X}$ up to the signs of its rows, and that with the sign convention of Lemma 10.3.1 its reflection vectors are $\mathbf{v}_j = (\mathbf{e}_j / \mathbf{q}_j) / \sqrt{2}$. This is the observation of Björck and Paige (1992), who take $\alpha = +\|\mathbf{x}\|$ and so obtain $(-\mathbf{e}_j / \mathbf{q}_j) / \sqrt{2}$.

10.4 The singular value decomposition

QR is the workhorse when $\mathbf{X}$ has full column rank. When it does not, or nearly does not, the singular value decomposition tells the whole story: every least squares solution, the one of minimum norm, and, in the smallest singular values, the combinations of the columns that the data cannot resolve. Theorem 1.8.1 established the decomposition, and Proposition 6.10.1 previewed its use for least squares.

10.4.1 Least squares through the SVD

Throughout, $\mathbf{X}$ is $n \times p$ of rank $r \geq 1$, with singular value decomposition $\mathbf{X} = \sum_{i \leq r} \sigma_i \mathbf{u}_i \mathbf{v}_i^\top$ (Theorem 1.8.1). We complete $\mathbf{u}_1, \dots, \mathbf{u}_r$ and $\mathbf{v}_1, \dots, \mathbf{v}_r$ to orthonormal bases $\mathbf{u}_1, \dots, \mathbf{u}_n$ of $\mathbb{R}^n$ and $\mathbf{v}_1, \dots, \mathbf{v}_p$ of $\mathbb{R}^p$, and write $c_i = \mathbf{u}_i^\top \mathbf{y}$ for the coordinates of the data in the left singular basis.

Theorem 10.4.1 — Least squares via the SVD (a) $\mathbf{b}$ is a least squares estimate iff $\mathbf{v}_i^\top \mathbf{b} = c_i / \sigma_i$ for $i = 1, \dots, r$. The coordinates $\mathbf{v}_i^\top \mathbf{b}$ for $i > r$ are

arbitrary, and the set of least squares estimates is $\mathbf{X}^+\mathbf{y} + \mathcal{N}(\mathbf{X})$ with $\mathcal{N}(\mathbf{X}) = \text{span}(\mathbf{v}_{r+1}, \dots, \mathbf{v}_p)$.

- (b) $\mathbf{X}^+\mathbf{y} = \sum_{i \leq r} (c_i/\sigma_i) \mathbf{v}_i$ is the unique least squares estimate of smallest Euclidean norm.
- (c) $\hat{\mathbf{Y}} = \sum_{i \leq r} c_i \mathbf{u}_i$ and $\text{SSE} = \sum_{i > r} c_i^2 = \|\mathbf{y}\|^2 - \sum_{i \leq r} c_i^2$.
- (d) If $r = p$ and $\text{Cov}(\mathbf{Y}) = \sigma^2 \mathbf{I}$, the estimate $\hat{\boldsymbol{\beta}} = \mathbf{X}^+\mathbf{Y}$ has $\text{Cov}(\hat{\boldsymbol{\beta}}) = \sigma^2 \sum_{i \leq p} \sigma_i^{-2} \mathbf{v}_i \mathbf{v}_i^\top$. In particular $\text{Var}(\mathbf{v}_i^\top \hat{\boldsymbol{\beta}}) = \sigma^2/\sigma_i^2$, and the total variance is $\text{tr Cov}(\hat{\boldsymbol{\beta}}) = \sigma^2 \sum_i \sigma_i^{-2}$.

Proof. Let $\mathbf{U} = [\mathbf{u}_1, \dots, \mathbf{u}_n]$, an orthogonal matrix. For any $\mathbf{b}$, $\mathbf{X}\mathbf{b} = \sum_{i \leq r} \sigma_i (\mathbf{v}_i^\top \mathbf{b}) \mathbf{u}_i$, so the coordinates of $\mathbf{X}\mathbf{b}$ in the basis $\mathbf{u}_1, \dots, \mathbf{u}_n$ are $\sigma_i \mathbf{v}_i^\top \mathbf{b}$ for $i \leq r$ and zero for $i > r$. By orthogonal invariance,

$$\|\mathbf{y} - \mathbf{X}\mathbf{b}\|^2 = \|\mathbf{U}^\top \mathbf{y} - \mathbf{U}^\top \mathbf{X}\mathbf{b}\|^2 = \sum_{i \leq r} (c_i - \sigma_i \mathbf{v}_i^\top \mathbf{b})^2 + \sum_{i > r} c_i^2. \quad (10.4.1)$$

The second sum does not involve $\mathbf{b}$, and the first vanishes exactly when $\mathbf{v}_i^\top \mathbf{b} = c_i/\sigma_i$ for $i \leq r$. This proves (a) and (c), since $\mathbf{b} = \sum_i (\mathbf{v}_i^\top \mathbf{b}) \mathbf{v}_i$ and $\mathcal{N}(\mathbf{X})$ is spanned by the $\mathbf{v}_i$ with $i > r$ (Theorem 1.8.1). Part (b) is Proposition 6.10.1. In coordinates, $\|\mathbf{b}\|^2 = \sum_{i \leq p} (\mathbf{v}_i^\top \mathbf{b})^2$, so the minimum-norm estimate is the one whose free coordinates are zero, $\sum_{i \leq r} (c_i/\sigma_i) \mathbf{v}_i$. For (d), $\hat{\boldsymbol{\beta}} = \sum_i \sigma_i^{-1} \mathbf{v}_i \mathbf{u}_i^\top \mathbf{Y}$, and $\text{Cov}(\mathbf{u}_i^\top \mathbf{Y}, \mathbf{u}_j^\top \mathbf{Y}) = \sigma^2 \mathbf{u}_i^\top \mathbf{u}_j$ is σ^2 or 0. By Theorem 2.2.1, $\text{Cov}(\hat{\boldsymbol{\beta}}) = \sigma^2 \sum_i \sigma_i^{-2} \mathbf{v}_i \mathbf{v}_i^\top$. Its trace is $\sigma^2 \sum_i \sigma_i^{-2}$ because $\|\mathbf{v}_i\| = 1$. ■

Part (d) is the statistical face of ill-conditioning. The combination $\mathbf{v}_p^\top \boldsymbol{\beta}$ of the coefficients, along the direction in which the columns of $\mathbf{X}$ are nearly dependent, is estimated with standard deviation σ/σ_p . When σ_p is tiny, the data carry almost no information about that combination, and no algorithm can change this.

```
import numpy as np

def svd_ls(X, y, tol=None):
    """Minimum-norm least squares solution, treating singular values <= tol as zero.
    """
    U, s, Vt = np.linalg.svd(X, full_matrices=False)
    if tol is None:
        tol = max(X.shape) * np.finfo(float).eps * s[0] # the usual default
    r = int(np.sum(s > tol)) # numerical rank
    c = U[:, :r].T @ y # coordinates of the fit
    b = Vt[:r].T @ (c / s[:r])
    sse = y @ y - c @ c
    return b, r, sse, s
```

10.4.2 Computing the SVD

The SVD is not computed from the eigenvalues of $\mathbf{X}^\top \mathbf{X}$, which has already lost the small singular values (Section 10.1). The standard method (Golub and Kahan 1965; Golub and Reinsch 1970)

first applies Householder reflections alternately from the left and the right to reduce $\mathbf{X}$ to upper bidiagonal form, at about twice the cost of QR, and then drives the superdiagonal to zero by an implicitly shifted QR-type iteration that never forms a cross-product. When $n \gg p$, it pays to compute $\mathbf{X} = \mathbf{Q}_1 \mathbf{R}$ first and then an SVD $\mathbf{R} = \mathbf{U}_R \mathbf{\Sigma} \mathbf{V}^\top$ of the triangle, since $\mathbf{X} = (\mathbf{Q}_1 \mathbf{U}_R) \mathbf{\Sigma} \mathbf{V}^\top$ (Exercise (B2)). Least squares needs only $\mathbf{U}^\top \mathbf{y}$, never $\mathbf{U}$ itself, which gives the count $2np^2 + 11p^3$ of Section 10.1 (Golub and Van Loan 2013).

The computed singular values are those of a nearby matrix $\mathbf{X} + \mathbf{E}$ with $\|\mathbf{E}\|_2 \leq cu\|\mathbf{X}\|_2$. The next result turns this into accuracy.

Proposition 10.4.2 — Singular values are well conditioned

For any matrices $\mathbf{A}$ and $\mathbf{E}$ of the same size and every k ,

$$|\sigma_k(\mathbf{A} + \mathbf{E}) - \sigma_k(\mathbf{A})| \leq \|\mathbf{E}\|_2.$$

Proof. For any matrix $\mathbf{C}$ and any $\mathbf{B}$ of rank at most $k - 1$, $\sigma_k(\mathbf{C}) \leq \|\mathbf{C} - \mathbf{B}\|_2$. When $k - 1 < \text{rank } \mathbf{C}$ this is the Eckart–Young inequality (Proposition 1.8.2(c)), and otherwise $\sigma_k(\mathbf{C}) = 0$. Let $\mathbf{A}_{k-1}$ be the truncated SVD of $\mathbf{A}$ with $k - 1$ terms, or $\mathbf{A}$ itself if $\text{rank } \mathbf{A} < k$, so that $\|\mathbf{A} - \mathbf{A}_{k-1}\|_2 = \sigma_k(\mathbf{A})$. Then

$$\sigma_k(\mathbf{A} + \mathbf{E}) \leq \|\mathbf{A} + \mathbf{E} - \mathbf{A}_{k-1}\|_2 \leq \sigma_k(\mathbf{A}) + \|\mathbf{E}\|_2.$$

Exchanging the roles of $\mathbf{A}$ and $\mathbf{A} + \mathbf{E}$ gives the other inequality. ■

So a backward stable SVD computes every singular value with *absolute* error of order $u\sigma_1$, and a singular value of that size or below is indistinguishable from zero. An exactly rank-deficient design, such as a one-way layout with an intercept and all its indicators, therefore shows a smallest singular value near $10^{-16}\sigma_1$ rather than zero: 5×10^{-17} times the largest in the script's example.

10.4.3 Numerical rank

In floating point, and with data measured to finite precision, the meaningful question is how many singular values are *clearly* nonzero.

Definition 10.4.1 — Numerical rank

For a tolerance $\tau > 0$, the **numerical rank** of $\mathbf{X}$ is $r_\tau = \#\{i : \sigma_i > \tau\}$. The **truncated SVD** with k terms is $\mathbf{X}_k = \sum_{i \leq k} \sigma_i \mathbf{u}_i \mathbf{v}_i^\top$.

By the Eckart–Young inequality, r_τ is the least rank of any matrix within spectral distance τ of $\mathbf{X}$ (Exercise (B1)), so τ should reflect how well $\mathbf{X}$ is known. The default in numpy and LAPACK-based software, $\tau = \max(n, p) \epsilon_M \sigma_1$, accounts for rounding only; data recorded to six significant figures cannot resolve singular values below about $10^{-6}\sigma_1$. The choice matters, because the minimum-norm solution is not a continuous function of $\mathbf{X}$.

Example. (The pseudoinverse jumps)

Let $\mathbf{x}_1 = (1, 2, 3, 4)^\top$, let $\mathbf{d} = (1, -1, -1, 1)^\top$ be orthogonal to it, and let $\mathbf{X}_t = [\mathbf{x}_1, \mathbf{x}_1 + t\mathbf{d}]$, with $\mathbf{y} = (1, 3, 2, 5)^\top$. For every $t \neq 0$ the column space is the same plane $\text{span}(\mathbf{x}_1, \mathbf{d})$, so the fitted values do not depend on t . The coefficients do. At $t = 10^{-6}$ they are -249999 and 250000 , because the coefficients must reproduce a fixed multiple of $\mathbf{d}$ using the small difference $t\mathbf{d}$ of the two columns. At $t = 0$ the rank drops to one. The minimum-norm solution jumps to 0.550 for both columns, and the fitted values jump too, because the $\mathbf{d}$ -direction has disappeared from the model. Treating singular values below a tolerance as zero makes the two answers agree. Whether it should be a statistical question: is the difference between the columns signal, or recording error?

```
x1 = np.array([1.0, 2.0, 3.0, 4.0])
d = np.array([1.0, -1.0, -1.0, 1.0])          # orthogonal to x1
yd = np.array([1.0, 3.0, 2.0, 5.0])
for t in [1e-2, 1e-6, 1e-10, 0.0]:
    Xt = np.column_stack([x1, x1 + t * d])      # second column tends to the first
    b = np.linalg.pinv(Xt, rcond=1e-15) @ yd
    print(f"t = {t:7.0e}    b = {np.round(b, 3)}    fit = {np.round(Xt @ b, 3)}")
```

10.4.4 Truncated SVD as an estimator

The truncated solution $\mathbf{b}_k = \sum_{i \leq k} (c_i / \sigma_i) \mathbf{v}_i$ is more than a numerical safeguard. It is a statistical estimator in its own right, and it trades bias for variance.

Proposition 10.4.3 — Bias and variance of truncated SVD

Suppose $\text{rank } \mathbf{X} = p$, $E(\mathbf{Y}) = \mathbf{X}\boldsymbol{\beta}$ and $\text{Cov}(\mathbf{Y}) = \sigma^2 \mathbf{I}$, and let $\mathbf{b}_k = \sum_{i \leq k} (\mathbf{u}_i^\top \mathbf{Y} / \sigma_i) \mathbf{v}_i$ for $k \leq p$. Then

$$E(\mathbf{b}_k) = \sum_{i \leq k} (\mathbf{v}_i^\top \boldsymbol{\beta}) \mathbf{v}_i, \quad \text{Cov}(\mathbf{b}_k) = \sigma^2 \sum_{i \leq k} \frac{\mathbf{v}_i \mathbf{v}_i^\top}{\sigma_i^2},$$

$$E\|\mathbf{b}_k - \boldsymbol{\beta}\|^2 = \sum_{i > k} (\mathbf{v}_i^\top \boldsymbol{\beta})^2 + \sigma^2 \sum_{i \leq k} \frac{1}{\sigma_i^2}. \quad (10.4.2)$$

Hence $\mathbf{b}_k$ has smaller mean squared error than the least squares estimate $\mathbf{b}_p$ iff $\sum_{i > k} (\mathbf{v}_i^\top \boldsymbol{\beta})^2 < \sigma^2 \sum_{i > k} \sigma_i^{-2}$.

Proof. $E(\mathbf{u}_i^\top \mathbf{Y}) = \mathbf{u}_i^\top \mathbf{X}\boldsymbol{\beta} = \sigma_i \mathbf{v}_i^\top \boldsymbol{\beta}$, which gives the mean. The covariance is part (d) of [Theorem 10.4.1](#) with the sum stopped at k . For the mean squared error, write $E\|\mathbf{b}_k - \boldsymbol{\beta}\|^2 = \|E\mathbf{b}_k - \boldsymbol{\beta}\|^2 + \text{tr Cov}(\mathbf{b}_k)$. The bias $E\mathbf{b}_k - \boldsymbol{\beta}$ is $-\sum_{i > k} (\mathbf{v}_i^\top \boldsymbol{\beta}) \mathbf{v}_i$, with squared length $\sum_{i > k} (\mathbf{v}_i^\top \boldsymbol{\beta})^2$. Comparing [Eq. 10.4.2](#) for k and p gives the last claim. ■

Each dropped component costs its squared coefficient $(\mathbf{v}_i^\top \boldsymbol{\beta})^2$ in bias and saves σ^2 / σ_i^2 in variance.

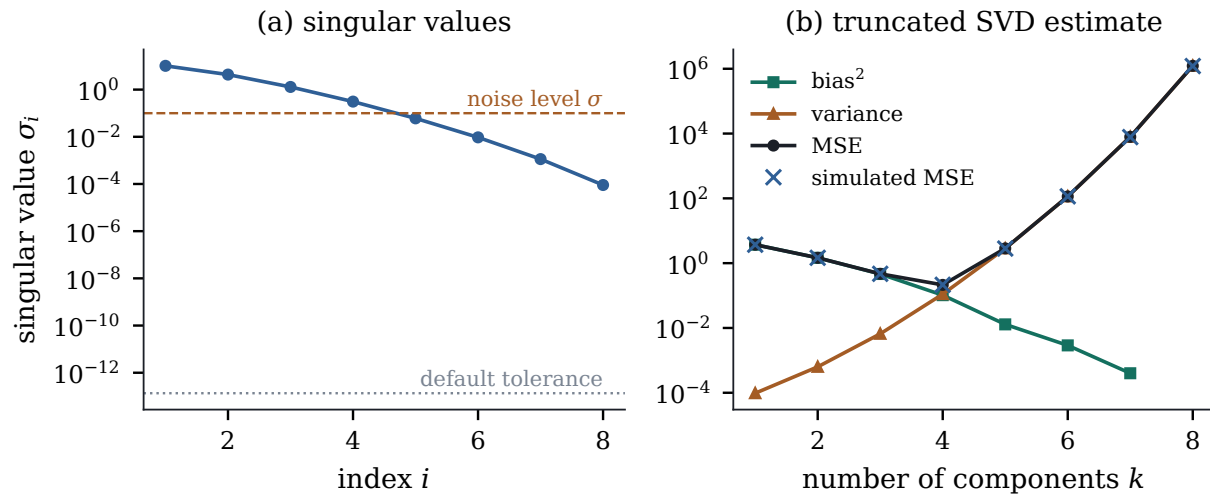


Figure 10.4.1: Truncated SVD for a degree-7 polynomial design. (a) Singular values, with the default numerical-rank tolerance and the noise level σ . (b) Squared bias, variance and mean squared error of the truncated estimate $\mathbf{b}_k$ (Eq. 10.4.2), with simulated values. Least squares is $k = 8$.

It is worth dropping exactly when the signal along $\mathbf{v}_i$ is weaker than the noise in its estimate, that is, when $\sigma_i |\mathbf{v}_i^\top \boldsymbol{\beta}| < \sigma$. The estimator $\mathbf{b}_k$ is principal components regression in its uncentred form. Ridge regression replaces the hard cut-off by weights $\sigma_i^2 / (\sigma_i^2 + \lambda)$. Both are developed in Chapter 27, and choosing k from data is a model-selection problem (Chapter 29).

Example. (Truncating a polynomial design)

The powers $1, t, \dots, t^7$ at 60 equally spaced points in $[0, 1]$ form a design with $\kappa(\mathbf{X}) = 1.1 \times 10^5$ and smallest singular value 9.0×10^{-5} . Take $\sigma = 0.1$ and a coefficient vector whose components along $\mathbf{v}_1, \dots, \mathbf{v}_8$ decay $(2, -1.5, 1, -0.6, 0.3, -0.1, 0.05, -0.02)$, as they do for smooth regression functions. By Eq. 10.4.2, least squares has mean squared error 1231055. The best truncation keeps 4 components and has mean squared error 0.21, made up of squared bias 0.10 and variance 0.11. A simulation with 4000 replicates agrees with the formula at every k (Figure 10.4.1). The default numerical tolerance treats the design as having full rank, which it does to working precision. The statistical “effective rank” is set by the noise level instead.

```
n, p, sigma = 60, 8, 0.1
t = np.linspace(0, 1, n)
X = np.vander(t, p, increasing=True) # 1, t, ..., t^7: badly conditioned
U, s, Vt = np.linalg.svd(X, full_matrices=False)
a = np.array([2.0, -1.5, 1.0, -0.6, 0.3, -0.1, 0.05, -0.02]) # decaying components
    v_i' beta
beta = Vt.T @ a
print("condition number:", s[0] / s[-1])
print("singular values:", np.array2string(s, precision=2))
```

```
bias2 = np.array([np.sum((Vt[k:] @ beta) ** 2) for k in range(1, p + 1)]) # ||(I -
    V_k V_k') beta||^2
var = np.array([sigma ** 2 * np.sum(1 / s[:k] ** 2) for k in range(1, p + 1)])
mse = bias2 + var
print("k    bias^2    variance    MSE")
for k in range(p):
```



```
print(f"{k + 1} {bias2[k]:10.3e} {var[k]:10.3e} {mse[k]:10.3e}")
```

Idea. The SVD gives every least squares solution, the precision σ/σ_i of each combination $\mathbf{v}_i^\top \boldsymbol{\beta}$, and, relative to a tolerance, how many combinations are determined at all.

10.4.5 Exercises

10.4.5.1 A. Check your understanding

Exercise A1.

Let $\mathbf{X} = \begin{pmatrix} 1 & 1 \\ 1 & 1 \\ 0 & 0 \end{pmatrix}$ and $\mathbf{y} = (1, 3, 5)^\top$. Find the SVD of $\mathbf{X}$, the set of least squares estimates, the minimum-norm estimate and the SSE.

Solution. $\mathbf{X} = 2\mathbf{u}_1\mathbf{v}_1^\top$ with $\mathbf{u}_1 = (1, 1, 0)^\top/\sqrt{2}$, $\mathbf{v}_1 = (1, 1)^\top/\sqrt{2}$, $\sigma_1 = 2$ and $r = 1$. Here $c_1 = \mathbf{u}_1^\top \mathbf{y} = 4/\sqrt{2}$, so the least squares estimates are the $\mathbf{b}$ with $\mathbf{v}_1^\top \mathbf{b} = c_1/\sigma_1 = \sqrt{2}$, that is, $b_1 + b_2 = 2$. The minimum-norm estimate is $\sqrt{2}\mathbf{v}_1 = (1, 1)^\top$. The fitted vector is $c_1\mathbf{u}_1 = (2, 2, 0)^\top$, and $\text{SSE} = \|\mathbf{y}\|^2 - c_1^2 = 35 - 8 = 27$, which is $(1-2)^2 + (3-2)^2 + 5^2$.

10.4.5.2 B. Practice

Exercise B1.

Show that if τ is not a singular value of $\mathbf{X}$, then $r_\tau = \min\{\text{rank } \mathbf{B} : \|\mathbf{X} - \mathbf{B}\|_2 \leq \tau\}$, and that the minimum is attained by the truncated SVD $\mathbf{X}_{r_\tau}$.

Solution. $\|\mathbf{X} - \mathbf{X}_k\|_2 = \sigma_{k+1}$. For $k = r_\tau$, $\sigma_{k+1} \leq \tau$ (with $\sigma_{k+1} = 0$ if $k = \text{rank } \mathbf{X}$), so $\mathbf{X}_{r_\tau}$ is within τ and has rank r_τ . If $\text{rank } \mathbf{B} = k < r_\tau$, then by Proposition 1.8.2(c) $\|\mathbf{X} - \mathbf{B}\|_2 \geq \sigma_{k+1} \geq \sigma_{r_\tau} > \tau$. So no matrix of smaller rank is within τ .

Exercise B2.

Let $\mathbf{X} = \mathbf{Q}_1\mathbf{R}$ be a thin QR and $\mathbf{R} = \mathbf{U}_R\boldsymbol{\Sigma}\mathbf{V}^\top$ an SVD of the triangle. Show that $\mathbf{X} = (\mathbf{Q}_1\mathbf{U}_R)\boldsymbol{\Sigma}\mathbf{V}^\top$ is an SVD of $\mathbf{X}$, and that the minimum-norm least squares solution can be computed from $\mathbf{R}$ and $\mathbf{Q}_1^\top \mathbf{y}$ alone. Why is this attractive when n is much larger than p ?

Exercise B3.

Show that $\mathbf{b}_k$ minimizes $\|\mathbf{y} - \mathbf{X}\mathbf{b}\|$ over $\mathbf{b} \in \text{span}(\mathbf{v}_1, \dots, \mathbf{v}_k)$, and that it is the minimum-norm least squares estimate for the matrix $\mathbf{X}_k$. Interpret $\mathbf{b}_k$ as least squares after deleting the directions in which $\mathbf{X}$ is nearly degenerate.

10.4.5.3 C. Going deeper

Exercise C1.

Show that the ridge estimate $(\mathbf{X}^\top \mathbf{X} + \lambda \mathbf{I})^{-1} \mathbf{X}^\top \mathbf{y}$ equals $\sum_i \frac{\sigma_i}{\sigma_i^2 + \lambda} c_i \mathbf{v}_i$. Derive its bias and variance in the form of [Proposition 10.4.3](#), and show that for every $\boldsymbol{\beta}$ there is a $\lambda > 0$ (depending on $\boldsymbol{\beta}$ and σ^2) for which its mean squared error is below that of least squares.

10.5 Conditioning and perturbation bounds

[Sections 10.2](#) to [10.4](#) were about algorithms. This section is about the problem itself: if the data $(\mathbf{X}, \mathbf{y})$ change a little, how much can the least squares solution and the residuals change? The answer does not depend on the algorithm. Combined with backward stability through [Eq. 10.1.2](#), it predicts the accuracy of any computed solution.

10.5.1 The perturbation theorem

Throughout this section $\mathbf{X}$ is $n \times p$ with full column rank, $\hat{\boldsymbol{\beta}} = \mathbf{X}^+ \mathbf{y}$ is its least squares solution (assumed nonzero) and $\hat{\boldsymbol{\epsilon}} = \mathbf{y} - \mathbf{X} \hat{\boldsymbol{\beta}}$ its residual. Write $\kappa = \kappa(\mathbf{X}) = \sigma_1 / \sigma_p$. The perturbed data are $\tilde{\mathbf{X}} = \mathbf{X} + \mathbf{E}$ and $\tilde{\mathbf{y}} = \mathbf{y} + \mathbf{f}$, with

$$\|\mathbf{E}\|_2 \leq \epsilon \|\mathbf{X}\|_2, \quad \|\mathbf{f}\| \leq \epsilon \|\mathbf{y}\|. \quad (10.5.1)$$

The solution and residual for the perturbed data are $\tilde{\mathbf{b}}$ and $\tilde{\boldsymbol{\epsilon}}$. A single number measures how badly the model fits relative to the size of the problem:

$$\eta = \frac{\|\hat{\boldsymbol{\epsilon}}\|}{\|\mathbf{X}\|_2 \|\hat{\boldsymbol{\beta}}\|}.$$

Since $\|\mathbf{X} \hat{\boldsymbol{\beta}}\| \leq \|\mathbf{X}\|_2 \|\hat{\boldsymbol{\beta}}\|$, $\eta \leq \tan \theta$, where θ is the angle between $\mathbf{y}$ and $\mathcal{C}(\mathbf{X})$. A model that fits well has small η .

Theorem 10.5.1 — Perturbation of least squares

Under [Eq. 10.5.1](#) with $\kappa\epsilon < 1$, the matrix $\tilde{\mathbf{X}}$ has full column rank, and

$$\frac{\|\tilde{\mathbf{b}} - \hat{\boldsymbol{\beta}}\|}{\|\hat{\boldsymbol{\beta}}\|} \leq \frac{\kappa\epsilon}{1 - \kappa\epsilon} (2 + \eta) + \frac{\kappa^2\epsilon}{(1 - \kappa\epsilon)^2} \eta, \quad (10.5.2)$$

$$\frac{\|\tilde{\boldsymbol{\epsilon}} - \hat{\boldsymbol{\epsilon}}\|}{\|\mathbf{y}\|} \leq \epsilon \left(1 + \frac{2\kappa}{1 - \kappa\epsilon} \right). \quad (10.5.3)$$

To first order in ϵ , the coefficient bound is $\epsilon(2\kappa + \kappa(\kappa + 1)\eta)$.

Proof. By [Proposition 10.4.2](#), $\sigma_p(\tilde{\mathbf{X}}) \geq \sigma_p - \|\mathbf{E}\|_2 \geq \sigma_p(1 - \kappa\epsilon) > 0$, so $\tilde{\mathbf{X}}$ has full column rank and $\tilde{\mathbf{X}}^+ \tilde{\mathbf{X}} = \mathbf{I}$. Hence

$$\tilde{\mathbf{b}} - \hat{\boldsymbol{\beta}} = \tilde{\mathbf{X}}^+ \tilde{\mathbf{y}} - \tilde{\mathbf{X}}^+ \tilde{\mathbf{X}} \hat{\boldsymbol{\beta}} = \tilde{\mathbf{X}}^+ (\tilde{\mathbf{y}} - \tilde{\mathbf{X}} \hat{\boldsymbol{\beta}}) = \tilde{\mathbf{X}}^+ (\hat{\boldsymbol{\epsilon}} + \mathbf{f} - \mathbf{E} \hat{\boldsymbol{\beta}}).$$

The residual term simplifies because $\mathbf{X}^\top \hat{\boldsymbol{\epsilon}} = \mathbf{0}$: $\tilde{\mathbf{X}}^+ \hat{\boldsymbol{\epsilon}} = (\tilde{\mathbf{X}}^\top \tilde{\mathbf{X}})^{-1} (\mathbf{X} + \mathbf{E})^\top \hat{\boldsymbol{\epsilon}} = (\tilde{\mathbf{X}}^\top \tilde{\mathbf{X}})^{-1} \mathbf{E}^\top \hat{\boldsymbol{\epsilon}}$. So

$$\tilde{\mathbf{b}} - \hat{\boldsymbol{\beta}} = \tilde{\mathbf{X}}^+ (\mathbf{f} - \mathbf{E} \hat{\boldsymbol{\beta}}) + (\tilde{\mathbf{X}}^\top \tilde{\mathbf{X}})^{-1} \mathbf{E}^\top \hat{\boldsymbol{\epsilon}}. \quad (10.5.4)$$

Now $\|\tilde{\mathbf{X}}^+\|_2 = 1/\sigma_p(\tilde{\mathbf{X}})$ and $\|(\tilde{\mathbf{X}}^\top \tilde{\mathbf{X}})^{-1}\|_2 = 1/\sigma_p(\tilde{\mathbf{X}})^2$ ([Theorem 1.8.1](#)). Also $\|\mathbf{y}\| \leq \|\mathbf{X} \hat{\boldsymbol{\beta}}\| + \|\hat{\boldsymbol{\epsilon}}\| \leq \sigma_1 \|\hat{\boldsymbol{\beta}}\| + \|\hat{\boldsymbol{\epsilon}}\|$, so $\|\mathbf{f} - \mathbf{E} \hat{\boldsymbol{\beta}}\| \leq \epsilon(\|\mathbf{y}\| + \sigma_1 \|\hat{\boldsymbol{\beta}}\|) \leq \epsilon \sigma_1 \|\hat{\boldsymbol{\beta}}\| (2 + \eta)$. Therefore

$$\|\tilde{\mathbf{b}} - \hat{\boldsymbol{\beta}}\| \leq \frac{\epsilon \sigma_1 \|\hat{\boldsymbol{\beta}}\| (2 + \eta)}{\sigma_p(1 - \kappa\epsilon)} + \frac{\epsilon \sigma_1 \|\hat{\boldsymbol{\epsilon}}\|}{\sigma_p^2(1 - \kappa\epsilon)^2}.$$

Dividing by $\|\hat{\boldsymbol{\beta}}\|$ and using $\|\hat{\boldsymbol{\epsilon}}\| = \eta \sigma_1 \|\hat{\boldsymbol{\beta}}\|$ gives [Eq. 10.5.2](#).

For the residual, $\tilde{\boldsymbol{\epsilon}} = (\mathbf{I} - \tilde{\mathbf{M}}) \tilde{\mathbf{y}}$, where $\tilde{\mathbf{M}}$ projects onto $\mathcal{C}(\tilde{\mathbf{X}})$. Write $\tilde{\mathbf{y}} = \tilde{\mathbf{X}} \hat{\boldsymbol{\beta}} + (\mathbf{f} - \mathbf{E} \hat{\boldsymbol{\beta}}) + \hat{\boldsymbol{\epsilon}}$. The first term is annihilated by $\mathbf{I} - \tilde{\mathbf{M}}$, and $\tilde{\mathbf{M}} \hat{\boldsymbol{\epsilon}} = (\tilde{\mathbf{X}}^+)^{\top} \tilde{\mathbf{X}}^{\top} \hat{\boldsymbol{\epsilon}} = (\tilde{\mathbf{X}}^+)^{\top} \mathbf{E}^{\top} \hat{\boldsymbol{\epsilon}}$ as before. So

$$\tilde{\boldsymbol{\epsilon}} - \hat{\boldsymbol{\epsilon}} = (\mathbf{I} - \tilde{\mathbf{M}})(\mathbf{f} - \mathbf{E} \hat{\boldsymbol{\beta}}) - (\tilde{\mathbf{X}}^+)^{\top} \mathbf{E}^{\top} \hat{\boldsymbol{\epsilon}}.$$

The first term has norm at most $\epsilon(\|\mathbf{y}\| + \sigma_1 \|\hat{\boldsymbol{\beta}}\|) \leq \epsilon(1 + \kappa)\|\mathbf{y}\|$, because $\|\hat{\boldsymbol{\beta}}\| = \|\mathbf{X}^+ \mathbf{y}\| \leq \|\mathbf{y}\|/\sigma_p$. The second has norm at most $\epsilon \sigma_1 \|\hat{\boldsymbol{\epsilon}}\|/(\sigma_p(1 - \kappa\epsilon)) \leq \epsilon \kappa \|\mathbf{y}\|/(1 - \kappa\epsilon)$. Adding and using $1 + \kappa + \kappa/(1 - \kappa\epsilon) \leq 1 + 2\kappa/(1 - \kappa\epsilon)$ gives the bound [Eq. 10.5.3](#). ■

The exact identity [Eq. 10.5.4](#) shows where each term comes from. Perturbing the data along $\mathcal{C}(\mathbf{X})$ moves the solution through $\tilde{\mathbf{X}}^+$, with amplification $1/\sigma_p$. This is the familiar κ of linear systems ([Eq. 1.8.3](#)). The second term is new. A perturbation $\mathbf{E}$ that tilts a column of $\mathbf{X}$ towards the residual direction changes the subspace, and the residual, however large, then “leaks” into the solution through $(\tilde{\mathbf{X}}^\top \tilde{\mathbf{X}})^{-1}$, with amplification $1/\sigma_p^2$. The residuals themselves, and hence the fitted values, are only κ -sensitive.

The κ^2 term is not an artefact of the proof.

Example. (The squared condition number is attained)

Let $\delta = 10^{-3}$ and

$$\mathbf{X} = \begin{pmatrix} 1 & 0 \\ 0 & \delta \\ 0 & 0 \end{pmatrix}, \quad \mathbf{y} = \begin{pmatrix} 0 \\ \delta \\ 1 \end{pmatrix},$$

so that $\hat{\boldsymbol{\beta}} = (0, 1)^\top$, $\hat{\boldsymbol{\epsilon}} = (0, 0, 1)^\top$, $\kappa = 1000$ and $\eta = 1$. Perturb only the $(3, 2)$ entry of $\mathbf{X}$, by $\epsilon = 10^{-10}$. This tilts the weak second column towards the residual. Its coefficient becomes $(\delta^2 + \epsilon)/(\delta^2 + \epsilon^2)$, and the relative change in $\hat{\boldsymbol{\beta}}$ is 1.00×10^{-4} , equal to $\kappa^2 \epsilon = 1.00 \times 10^{-4}$. A perturbation of relative size 10^{-10} has changed the answer in the fourth digit.

```
import numpy as np
```

```

import statsmodels.api as sm

def ls(X, y):
    return np.linalg.lstsq(X, y, rcond=None)[0]

delta, eps = 1e-3, 1e-10
X = np.array([[1.0, 0.0], [0.0, delta], [0.0, 0.0]])
y = np.array([0.0, delta, 1.0])          # b = (0, 1), residual e = (0, 0, 1)
E = np.array([[0.0, 0.0], [0.0, 0.0], [0.0, eps]]) # ||E|| = eps ||X||
b = ls(X, y)
b_tilde = ls(X + E, y)
print("kappa =", np.linalg.cond(X))
print("relative change in b:", np.linalg.norm(b_tilde - b) / np.linalg.norm(b))
print("kappa^2 * eps      :", np.linalg.cond(X) ** 2 * eps)

```

The script also draws 300 random problems with random perturbations and checks [Eq. 10.5.2](#) on each. The worst ratio of the actual change to the bound is 0.31.

10.5.2 What the theorem says about algorithms

A backward stable algorithm (Householder QR, Givens QR, modified Gram–Schmidt on the augmented matrix, the SVD) returns the exact solution for data perturbed as in [Eq. 10.5.1](#) with $\epsilon \approx cu$, where c grows modestly with the dimensions. Its relative error is therefore about

$$cu(\kappa + \kappa^2\eta).$$

The normal equations incur a relative error of about $cu\kappa^2$ *whatever* η is, because forming $\mathbf{X}^\top \mathbf{X}$ already commits it ([Proposition 10.1.3](#)). Two regimes follow:

- **Small residual** ($\eta \ll 1/\kappa$): QR gains a full factor κ over the normal equations. This is the regime of the experiment in [Section 6.10](#), where $\mathbf{y} = \mathbf{X}\boldsymbol{\beta}$ exactly.
- **Large residual** (η of order one): every method, however stable, loses about $2\log_{10}\kappa$ digits, because the problem is that sensitive. The normal equations are then worse only by a modest factor.

[Figure 10.5.1](#) shows both regimes on test problems with prescribed condition numbers. With zero residual and $\kappa = 10^6$, the normal equations give relative error 1.1×10^{-5} and QR 7.8×10^{-12} . With $\eta = 1$, that is $\|\hat{\mathbf{e}}\| = \|\mathbf{X}\|_2 \|\boldsymbol{\beta}\|$, QR gives 1.9×10^{-7} and the SVD 1.9×10^{-7} . The κ^2 term has caught up with the stable methods.

```

def by_cholesky(X, y):
    L = np.linalg.cholesky(X.T @ X)
    return np.linalg.solve(L.T, np.linalg.solve(L, X.T @ y))

def by_qr(X, y):
    Q, R = np.linalg.qr(X)
    return np.linalg.solve(R, Q.T @ y)

def by_svd(X, y):
    U, s, Vt = np.linalg.svd(X, full_matrices=False)
    return Vt.T @ ((U.T @ y) / s)

```

In regression η is rarely tiny, but QR keeps an advantage: its $\kappa^2\eta$ term is a property of the problem, not of the method.

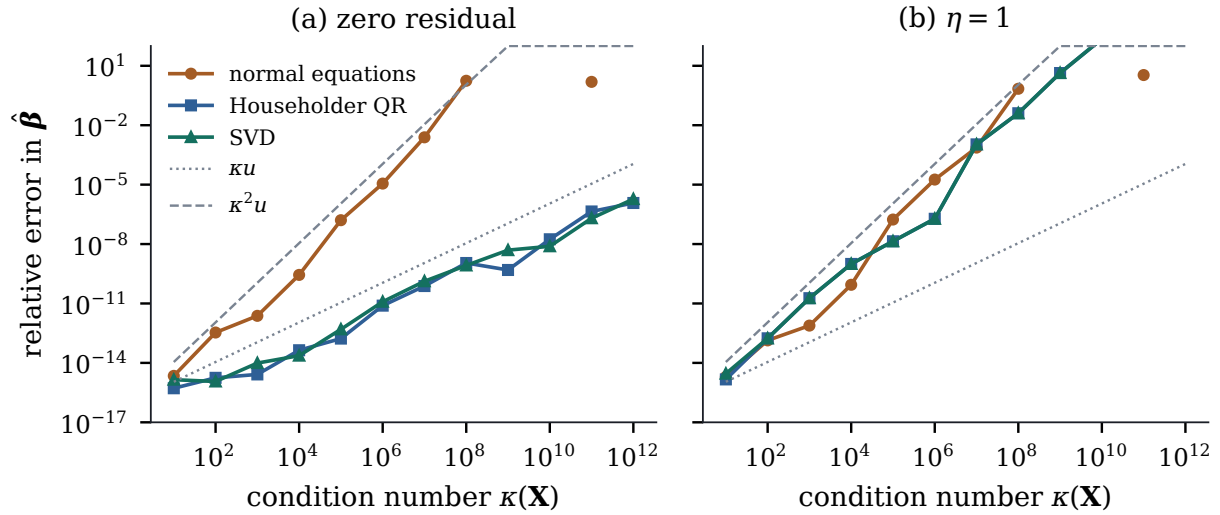


Figure 10.5.1: Relative error of the coefficients computed by the normal equations (Cholesky), Householder QR and the SVD, against $\kappa(\mathbf{X})$. (a) Consistent data, $\hat{\epsilon} = \mathbf{0}$: QR and SVD track κu , the normal equations track $\kappa^2 u$. (b) A large residual, $\eta = 1$: all three track $\kappa^2 u$, as [Theorem 10.5.1](#) predicts.

10.5.3 Column scaling

The normwise condition number depends on the units of the columns: measuring one regressor in millimetres instead of metres can multiply $\kappa(\mathbf{X})$ by up to 1000 ([Exercise \(B2\)](#)), although the fit is unchanged. Equalizing the column lengths is essentially the best diagonal scaling.

Proposition 10.5.2 — Equilibrating the columns

Let $\mathbf{X}$ have full column rank and let $\mathbf{D}_0 = \text{diag}(1/\|\mathbf{x}_1\|, \dots, 1/\|\mathbf{x}_p\|)$. For every nonsingular diagonal $\mathbf{D}$,

$$\kappa(\mathbf{XD}_0) \leq \sqrt{p} \kappa(\mathbf{XD}).$$

Proof. The columns of $\mathbf{XD}_0$ have unit length, so $\sigma_1(\mathbf{XD}_0) \leq \|\mathbf{XD}_0\|_F = \sqrt{p}$. For the smallest singular value, let $\mathbf{z}$ be any nonzero vector and put $\mathbf{w} = \mathbf{D}^{-1}\mathbf{D}_0\mathbf{z}$. Then $\|\mathbf{XD}_0\mathbf{z}\| = \|\mathbf{XD}\mathbf{w}\| \geq \sigma_p(\mathbf{XD})\|\mathbf{w}\|$ and $\|\mathbf{w}\| \geq \min_j (1/(|d_j|\|\mathbf{x}_j\|))\|\mathbf{z}\|$. The largest column length of $\mathbf{XD}$, $\max_j |d_j|\|\mathbf{x}_j\|$, is at most $\sigma_1(\mathbf{XD})$, because the columns are images of unit vectors. So $\sigma_p(\mathbf{XD}_0) \geq \sigma_p(\mathbf{XD})/\sigma_1(\mathbf{XD}) = 1/\kappa(\mathbf{XD})$, and $\kappa(\mathbf{XD}_0) \leq \sqrt{p} \kappa(\mathbf{XD})$. ■

This is due to van der Sluis (1969). What makes it more than a remark about units is the form of the backward errors. Householder QR has a *columnwise* backward error ([Theorem 10.3.2\(c\)](#)). Rescaling the columns rescales the perturbations with them, so the computed solution is as accurate as if the columns had been equilibrated first. The rounding errors in $\mathbf{X}^\top \mathbf{X}$ are entrywise small relative to $|\mathbf{X}|^\top |\mathbf{X}|$ ([Proposition 10.1.3](#)), which has the same property. Both methods therefore behave according to the condition number of the *scaled* matrix, $\kappa(\mathbf{XD}_0)$, whatever the units. The rule of thumb should be applied to $\kappa(\mathbf{XD}_0)$, not to $\kappa(\mathbf{X})$ as recorded.

Example. (Longley’s data, exactly)

Longley (1967) built his macroeconomic data to test regression programs. With an intercept and six regressors, $\kappa(\mathbf{X}) = 4.86 \times 10^9$ (Section 6.10; the data are described in Example (Macroeconomic series)). The data are published as decimals, so the exact least squares solution can be computed in rational arithmetic. The script does so and counts the correct significant digits of the worst coefficient for each method:

Method	Correct digits
Normal equations (Cholesky)	7.2
Householder QR	10.9
SVD	10.8
statsmodels OLS	10.8
Normal equations after scaling the columns to unit length	7.0

The raw κ predicts about 6 digits from QR and none from the normal equations (Section 6.10). Both predictions are far too pessimistic. After scaling the columns to unit length, $\kappa(\mathbf{X}\mathbf{D}_0) = 4.33 \times 10^4$. Then $\kappa u \approx 5 \times 10^{-12}$ predicts about eleven digits from QR, and $\kappa^2 u \approx 2 \times 10^{-7}$ about seven from the normal equations. That is what is observed. Scaling the columns explicitly before forming the normal equations changes nothing, as the argument above predicts. Centring as well reduces the condition number to 1.11×10^2 . That is a reparameterization rather than a scaling (Theorem 6.7.1), and what remains is genuine collinearity among the regressors (Chapter 26).

```
lon = sm.datasets.longley.load_pandas()
XL = sm.add_constant(lon.exog.to_numpy())
yL = lon.endog.to_numpy()
names = ["const"] + list(lon.exog.columns)
print("kappa(X) as recorded:", f"{np.linalg.cond(XL):.3e}")
```

10.5.4 Numerical error and statistical error

The perturbation theorem applies to any perturbation of the data, including the rounding of recorded values, which is far above u . Longley’s GNP deflator is given to one decimal place and the other series to whole units. Perturbing each regressor entry (the year is exact) uniformly within half a unit of its last recorded digit, 1000 times, the coefficients vary with standard deviations between 0.005 and 0.024 of their standard errors. The numerical error of the normal equations, the least accurate method, is at most 4×10^{-8} standard errors. Sampling error dominates, then comes the rounding of the published data, and the arithmetic is last by far. Beaton, Rubin and Barone (1976) made this point about Longley’s data. A problem can be numerically delicate and statistically ordinary, or the reverse.

```
rng = np.random.default_rng(106)
half_unit = np.array([0.0, 0.05, 0.5, 0.5, 0.5, 0.5, 0.0]) # half a unit in the
last digit
fitL = sm.OLS(yL, XL).fit() # year and constant are
exact
reps = 1000
shifts = []
```

```

for _ in range(reps):
    Xp = XL + rng.uniform(-1, 1, size=XL.shape) * half_unit
    shifts.append(ls(Xp, yL) - fitL.params)
spread = np.std(shifts, axis=0)
ratio = spread / fitL.bse
for name, r in zip(names, ratio):
    print(f"{name:8s} sd of coefficient under rounding / standard error = {r:.2f}")

```

Idea. Least squares has a κ sensitivity, and a $\kappa^2\eta$ one when the residual is large. Stable algorithms add nothing beyond this; the normal equations add κ^2 regardless. Measure κ after equilibrating the columns.

10.5.5 Exercises

10.5.5.1 A. Check your understanding

Exercise A1.

In [Example \(The squared condition number is attained\)](#), find $\tilde{\mathbf{b}}$ exactly and show that the relative change is $\epsilon(1 - \epsilon)/(\delta^2 + \epsilon^2)$. For which ϵ is it close to $\kappa^2\epsilon$?

10.5.5.2 B. Practice

Exercise B1.

Deduce from [Eq. 10.5.3](#) that the fitted values satisfy $\|\tilde{\mathbf{X}}\tilde{\mathbf{b}} - \mathbf{X}\hat{\boldsymbol{\beta}}\| \leq \epsilon(2 + 2\kappa/(1 - \kappa\epsilon))\|\mathbf{y}\|$. Why are fitted values never κ^2 -sensitive, even though coefficients can be?

Solution. $\tilde{\mathbf{X}}\tilde{\mathbf{b}} = \tilde{\mathbf{y}} - \tilde{\boldsymbol{\epsilon}}$ and $\mathbf{X}\hat{\boldsymbol{\beta}} = \mathbf{y} - \hat{\boldsymbol{\epsilon}}$, so the difference is $\mathbf{f} - (\tilde{\boldsymbol{\epsilon}} - \hat{\boldsymbol{\epsilon}})$, with norm at most $\epsilon\|\mathbf{y}\| + \epsilon(1 + 2\kappa/(1 - \kappa\epsilon))\|\mathbf{y}\|$. The κ^2 term in the coefficients comes from $(\tilde{\mathbf{X}}^\top \tilde{\mathbf{X}})^{-1} \mathbf{E}^\top \hat{\boldsymbol{\epsilon}}$, a change of coordinates *within* the slightly rotated column space. Multiplying by $\tilde{\mathbf{X}}$ turns the factor $1/\sigma_p^2$ into $1/\sigma_p$. A direction that the columns barely span is a direction in which large coefficient changes produce only small changes in the fit.

Exercise B2.

Show that for consistent data ($\hat{\boldsymbol{\epsilon}} = \mathbf{0}$) the bound reduces to $2\kappa\epsilon/(1 - \kappa\epsilon)$. Construct a perturbation $\mathbf{f}$ of $\mathbf{y}$ alone that changes $\hat{\boldsymbol{\beta}}$ by a relative amount $\kappa\epsilon$, so that the bound is attained up to a factor of two.

10.5.5.3 C. Going deeper

Exercise C1.

Take data with $\hat{\boldsymbol{\beta}} = \mathbf{v}_p$ and $\hat{\boldsymbol{\epsilon}} \neq \mathbf{0}$, and the perturbation $\mathbf{E} = \epsilon\sigma_1 \hat{\boldsymbol{\epsilon}} \mathbf{v}_p^\top / \|\hat{\boldsymbol{\epsilon}}\|$, with $\mathbf{f} = \mathbf{0}$. Use [Eq. 10.5.4](#) to show that, to first order in ϵ , $\tilde{\mathbf{b}} - \hat{\boldsymbol{\beta}} = \epsilon\sigma_1 \|\hat{\boldsymbol{\epsilon}}\| \sigma_p^{-2} \mathbf{v}_p$, so that the relative change is $\epsilon\kappa^2\eta$. Compare with the first-order bound $\epsilon(2\kappa + \kappa(\kappa + 1)\eta)$.

10.6 Updating and downdating

A fitted regression often has to be changed slightly: new observations arrive, a suspicious case is set aside, a window slides along a time series, a variable is added or removed. Refitting from scratch costs $O(np^2)$ each time. Modifying the factorization costs $O(p^2)$ for a change of one row, and $O(np)$ for a change of one column. This section develops those modifications and uses them to compute every leave-one-out quantity from a single fit.

10.6.1 Adding and deleting observations

Adding a row is part (c) of [Theorem 10.3.4](#): p rotations fold the new row into the triangle. Removing a row is harder, because orthogonal transformations can only combine rows that are present. The trick is to run the addition backwards.

Theorem 10.6.1 — Updating and downdating a triangular factor

Let $\mathbf{T}$ be an $m \times m$ nonsingular upper triangular matrix with $\mathbf{T}^\top \mathbf{T} = \mathbf{A}$, and let $\mathbf{z} \in \mathbb{R}^m$.

- (a) (*Adding a row.*) There are m rotations, the k th acting on row k of $\mathbf{T}$ and on an extra row, that turn $\begin{pmatrix} \mathbf{T} \\ \mathbf{z}^\top \end{pmatrix}$ into $\begin{pmatrix} \mathbf{T}_+ \\ \mathbf{0}^\top \end{pmatrix}$, with $\mathbf{T}_+$ upper triangular and $\mathbf{T}_+^\top \mathbf{T}_+ = \mathbf{A} + \mathbf{z}\mathbf{z}^\top$.
- (b) (*Deleting a row.*) Let $\mathbf{a}$ solve $\mathbf{T}^\top \mathbf{a} = \mathbf{z}$. The matrix $\mathbf{A} - \mathbf{z}\mathbf{z}^\top$ is positive definite iff $\|\mathbf{a}\| < 1$. In that case put $\alpha = (1 - \|\mathbf{a}\|^2)^{1/2}$. Choose rotations $\mathbf{G}_m, \dots, \mathbf{G}_1$, with $\mathbf{G}_k$ acting on coordinates k and $m+1$, such that $\mathbf{G}_1 \cdots \mathbf{G}_m (\mathbf{a}^\top, \alpha)^\top = (\mathbf{0}^\top, 1)^\top$. Then

$$\mathbf{G}_1 \cdots \mathbf{G}_m \begin{pmatrix} \mathbf{T} \\ \mathbf{0}^\top \end{pmatrix} = \begin{pmatrix} \mathbf{T}_- \\ \mathbf{z}^\top \end{pmatrix}$$

with $\mathbf{T}_-$ upper triangular and $\mathbf{T}_-^\top \mathbf{T}_- = \mathbf{A} - \mathbf{z}\mathbf{z}^\top$.

- (c) Each operation costs $O(m^2)$ flops: $3m^2$ for (a), and $3m^2$ plus a triangular solve of m^2 for (b).

Proof.

- (a) is [Theorem 10.3.4\(c\)](#).
- (b) $\mathbf{A} - \mathbf{z}\mathbf{z}^\top = \mathbf{T}^\top (\mathbf{I} - \mathbf{a}\mathbf{a}^\top) \mathbf{T}$. The matrix $\mathbf{I} - \mathbf{a}\mathbf{a}^\top$ has eigenvalue $1 - \|\mathbf{a}\|^2$ on $\mathbf{a}$ and 1 on $\mathbf{a}^\perp$. So it is positive definite iff $\|\mathbf{a}\| < 1$, and then so is $\mathbf{A} - \mathbf{z}\mathbf{z}^\top$ ([Proposition 1.7.5\(c\)](#)). The rotations exist: $\mathbf{G}_m$ combines a_m with α and leaves 0 in position m and $(a_m^2 + \alpha^2)^{1/2}$ in position $m+1$. Then $\mathbf{G}_{m-1}$ does the same with a_{m-1} , and so on ([Theorem 10.3.4\(a\)](#)). The final vector has norm $(\|\mathbf{a}\|^2 + \alpha^2)^{1/2} = 1$ and is zero except in the last position. That entry is positive, so it is 1. Apply the same rotations, in the same order, to $\begin{pmatrix} \mathbf{T} \\ \mathbf{0}^\top \end{pmatrix}$. Before $\mathbf{G}_k$ is applied, the extra row is zero in columns $1, \dots, k$, because it has only been mixed with rows $k+1, \dots, m$ of $\mathbf{T}$, which are zero there. Row k of $\mathbf{T}$ is zero in columns $1, \dots, k-1$. So $\mathbf{G}_k$ produces a new row k that is zero before column k , and the result is $\begin{pmatrix} \mathbf{T}_- \\ \mathbf{z}^\top \end{pmatrix}$.

with $\mathbf{T}_-$ upper triangular. Write $\mathbf{G} = \mathbf{G}_1 \cdots \mathbf{G}_m$, which is orthogonal. Then

$$\mathbf{z} = \mathbf{T}^\top \mathbf{a} = \begin{pmatrix} \mathbf{T} \\ \mathbf{0}^\top \end{pmatrix}^\top \begin{pmatrix} \mathbf{a} \\ \alpha \end{pmatrix} = \left(\mathbf{G} \begin{pmatrix} \mathbf{T} \\ \mathbf{0}^\top \end{pmatrix} \right)^\top \mathbf{G} \begin{pmatrix} \mathbf{a} \\ \alpha \end{pmatrix} = \begin{pmatrix} \mathbf{T}_- \\ \mathbf{w}^\top \end{pmatrix}^\top \begin{pmatrix} \mathbf{0} \\ 1 \end{pmatrix} = \mathbf{w},$$

and in the same way $\mathbf{T}^\top \mathbf{T} = \mathbf{T}_-^\top \mathbf{T}_- + \mathbf{w} \mathbf{w}^\top$. Hence $\mathbf{T}_-^\top \mathbf{T}_- = \mathbf{A} - \mathbf{z} \mathbf{z}^\top$.

(c) The rotations act on rows of length at most m , and there are m of them. ■

If $\mathbf{A} = \mathbf{X}^\top \mathbf{X}$ and $\mathbf{z} = \mathbf{x}_{(i)}$, then $\|\mathbf{a}\|^2 = \mathbf{z}^\top \mathbf{A}^{-1} \mathbf{z}$ is the leverage h_{ii} (Definition 6.8.1): a case can be deleted exactly when the others still determine every coefficient. Downdating the augmented triangle of $[\mathbf{X}, \mathbf{y}]$ (Proposition 10.2.2) gives the fit without case i , with the square root of the new SSE in the corner. Downdating is less stable than updating: $1 - \|\mathbf{a}\|^2$ cancels when $h_{ii} \approx 1$, and the relative error can reach about $u/(1 - h_{ii})$ (Stewart 1979).

```
import numpy as np
import statsmodels.api as sm

def givens(a, b):
    """c, s, r with [[c, s], [-s, c]] @ [a, b] = [r, 0]."""
    if b == 0.0:
        return 1.0, 0.0, a
    r = np.hypot(a, b)
    return a / r, b / r, r

def update(T, z):
    """Triangle of [A; z'] from the triangle T of A: rotate the new row into T."""
    T, z = T.copy(), np.array(z, dtype=float)
    for k in range(len(z)):
        c, s, r = givens(T[k, k], z[k])
        Tk, zk = T[k, k:].copy(), z[k:].copy()
        T[k, k:], z[k:] = c * Tk + s * zk, -s * Tk + c * zk
    return T

def downdate(T, z):
    """Triangle of A with the row z' removed, from the triangle T of A (T'T = A'A)."""
    m = len(z)
    a = np.linalg.solve(T.T, z) # T'a = z; ||a||^2 is the leverage of z
    alpha2 = 1.0 - a @ a
    if alpha2 <= 0:
        raise np.linalg.LinAlgError("the row cannot be removed (leverage >= 1)")
    alpha = np.sqrt(alpha2)
    T, w = T.copy(), np.zeros(m) # w: the extra row, initially zero
    for k in range(m - 1, -1, -1): # rotate (a_k, alpha) into (0, new alpha)
        c, s, r = givens(alpha, a[k])
        alpha = r
        Tk = T[k, k:].copy()
        T[k, k:], w[k:] = c * Tk - s * w[k:], s * Tk + c * w[k:]
    return T # on exit w equals z
```

The alternative, **recursive least squares** (Plackett 1950), updates $\mathbf{K} = (\mathbf{X}^\top \mathbf{X})^{-1}$ by the

Sherman–Morrison formula ([Theorem 1.6.5](#), [Example \(Deleting an observation\)](#)): adding $(\mathbf{x}, y)$ gives

$$\mathbf{K}_+ = \mathbf{K} - \frac{\mathbf{K}\mathbf{x}\mathbf{x}^\top\mathbf{K}}{1 + \mathbf{x}^\top\mathbf{K}\mathbf{x}}, \quad \hat{\boldsymbol{\beta}}_+ = \hat{\boldsymbol{\beta}} + \frac{\mathbf{K}\mathbf{x}(y - \mathbf{x}^\top\hat{\boldsymbol{\beta}})}{1 + \mathbf{x}^\top\mathbf{K}\mathbf{x}}.$$

It also costs $O(p^2)$, but it carries an explicit inverse of the cross-product matrix, with its κ^2 sensitivity, and nothing corrects its accumulated errors ([Example \(Ten-year regressions of consumption\)](#)).

10.6.2 Leave-one-out quantities from one fit

Many diagnostics ask what would happen if observation i were left out. Computing them by n separate fits costs $O(n^2p^2)$, but all of them follow from one fit.

Proposition 10.6.2 — Deletion formulas

Let $\mathbf{X}$ have full column rank, let $h_{ii} < 1$, and let $\hat{\boldsymbol{\beta}}_{(i)}$ and $\text{SSE}_{(i)}$ denote the least squares estimate and residual sum of squares computed without observation i . Then

- (a) $\hat{\boldsymbol{\beta}} - \hat{\boldsymbol{\beta}}_{(i)} = (\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{x}_{(i)}\hat{\varepsilon}_i/(1 - h_{ii})$;
- (b) $y_i - \mathbf{x}_{(i)}^\top\hat{\boldsymbol{\beta}}_{(i)} = \hat{\varepsilon}_i/(1 - h_{ii})$;
- (c) $\text{SSE}_{(i)} = \text{SSE} - \hat{\varepsilon}_i^2/(1 - h_{ii})$.

Proof.

- (a) is [Exercise \(B1\)](#), which follows from the deletion formula of [Example \(Deleting an observation\)](#).
- (b) Multiply (a) by $\mathbf{x}_{(i)}^\top$: $\mathbf{x}_{(i)}^\top\hat{\boldsymbol{\beta}}_{(i)} = \mathbf{x}_{(i)}^\top\hat{\boldsymbol{\beta}} - h_{ii}\hat{\varepsilon}_i/(1 - h_{ii})$, so $y_i - \mathbf{x}_{(i)}^\top\hat{\boldsymbol{\beta}}_{(i)} = \hat{\varepsilon}_i + h_{ii}\hat{\varepsilon}_i/(1 - h_{ii}) = \hat{\varepsilon}_i/(1 - h_{ii})$.
- (c) Put $g = \hat{\varepsilon}_i/(1 - h_{ii})$ and let $h_{ji} = \mathbf{x}_{(j)}^\top(\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{x}_{(i)}$ be the entries of $\mathbf{M}$. By (a), $y_j - \mathbf{x}_{(j)}^\top\hat{\boldsymbol{\beta}}_{(i)} = \hat{\varepsilon}_j + h_{ji}g$ for every j , including $j = i$, where it gives g by (b). Summing squares over all j ,

$$\sum_j (\hat{\varepsilon}_j + h_{ji}g)^2 = \text{SSE} + 2g(\mathbf{M}\hat{\boldsymbol{\varepsilon}})_i + g^2(\mathbf{M}^2)_{ii} = \text{SSE} + g^2h_{ii},$$

because $\mathbf{M}\hat{\boldsymbol{\varepsilon}} = \mathbf{0}$ and $\mathbf{M}^2 = \mathbf{M}$. Removing the $j = i$ term, which is g^2 , leaves $\text{SSE}_{(i)} = \text{SSE} - g^2(1 - h_{ii}) = \text{SSE} - \hat{\varepsilon}_i^2/(1 - h_{ii})$. ■

With a QR factorization the leverages cost $O(np)$ ([Proposition 10.3.3\(c\)](#)), and the rest $O(n)$. This gives the PRESS residuals $\hat{\varepsilon}_i/(1 - h_{ii})$, whose sum of squares is leave-one-out cross-validation (Allen 1974; Chapter 29), and the deleted variances $s_{(i)}^2 = \text{SSE}_{(i)}/(n - p - 1)$ behind the externally studentized residuals $t_i = \hat{\varepsilon}_i/(s_{(i)}\sqrt{1 - h_{ii}})$ of Chapter 20.

Example. (Leaving out the District of Columbia)

Refit the state murder-rate regression with all 51 jurisdictions, including the District of Columbia, which [Chapter 6](#) set aside. The residual sum of squares is 142.53 and the PRESS statistic is 230.52. The largest PRESS residual belongs to the District of Columbia. Its ordinary residual is 4.475 and its leverage 0.506, so its prediction error when it is left out is $\hat{\varepsilon}_i/(1 - h_{ii}) = 9.066$. Its externally studentized residual is 4.28. By [Proposition 10.6.2\(c\)](#), deleting it reduces the residual sum of squares to 101.96, the value found for the 50 states in [Example \(Murder rates by the augmented factorization\)](#). The script confirms all 51 deletions three ways: by the formulas, by downdating the augmented triangle, and by refitting.

```
data = sm.datasets.statecrime.load_pandas().data           # all 50 states and DC
y = data["murder"].to_numpy()
X = np.column_stack([np.ones(len(y)), data[["poverty", "single", "urban"]]])
n, p = X.shape

Q, R = np.linalg.qr(X)
beta = np.linalg.solve(R, Q.T @ y)
e = y - X @ beta
sse = e @ e
h = np.sum(Q ** 2, axis=1)                                # leverages
press_resid = e / (1 - h)                                  # y_i minus its prediction
                                without case i
sse_loo = sse - e ** 2 / (1 - h)                           # SSE after deleting case i
s_loo = np.sqrt(sse_loo / (n - 1 - p))
t_ext = e / (s_loo * np.sqrt(1 - h))                       # externally studentized
                                residuals
i = int(np.argmax(np.abs(press_resid)))
print(f"largest PRESS residual: {data.index[i]}, e = {e[i]:.3f}, h = {h[i]:.3f}, "
      f" e/(1-h) = {press_resid[i]:.3f}, t = {t_ext[i]:.2f}")
print(f"PRESS = {np.sum(press_resid ** 2):.2f}, SSE = {sse:.2f}")
```

10.6.3 Adding and deleting variables

Changing the columns of $\mathbf{X}$ is a Gram–Schmidt step, or its reverse.

Proposition 10.6.3 — Adding a column

Let $\mathbf{X} = \mathbf{Q}_1\mathbf{R}$ be a thin QR factorization of a full-column-rank $\mathbf{X}$, and let $\mathbf{z} \notin \mathcal{C}(\mathbf{X})$. Put $\mathbf{r} = \mathbf{Q}_1^\top \mathbf{z}$, $\mathbf{w} = \mathbf{z} - \mathbf{Q}_1\mathbf{r}$, $\rho = \|\mathbf{w}\|$ and $\mathbf{q} = \mathbf{w}/\rho$. Then

$$[\mathbf{X}, \mathbf{z}] = [\mathbf{Q}_1, \mathbf{q}] \begin{pmatrix} \mathbf{R} & \mathbf{r} \\ \mathbf{0}^\top & \rho \end{pmatrix}$$

is a thin QR factorization. The residual sum of squares falls by $(\mathbf{q}^\top \mathbf{y})^2$ when $\mathbf{z}$ is added, and the coefficient of $\mathbf{z}$ in the enlarged model is $\mathbf{q}^\top \mathbf{y}/\rho = \mathbf{w}^\top \mathbf{y}/\mathbf{w}^\top \mathbf{w}$.

Proof. $\mathbf{w} = (\mathbf{I} - \mathbf{M})\mathbf{z} \neq \mathbf{0}$ because $\mathbf{z} \notin \mathcal{C}(\mathbf{X})$, and $\mathbf{q}$ is a unit vector orthogonal to $\mathcal{C}(\mathbf{Q}_1)$. The product of the factors is $[\mathbf{Q}_1\mathbf{R}, \mathbf{Q}_1\mathbf{r} + \rho\mathbf{q}] = [\mathbf{X}, \mathbf{z}]$. By [Proposition 10.3.3\(d\)](#), applied

to the enlarged factorization, the new column's sequential sum of squares is the square of its coordinate $\mathbf{q}^\top \mathbf{y}$. The last row of the triangular system of [Proposition 10.3.3\(a\)](#) reads $\rho b_z = \mathbf{q}^\top \mathbf{y}$. This is also the Frisch–Waugh–Lovell coefficient $\mathbf{w}^\top \mathbf{y} / \mathbf{w}^\top \mathbf{w}$ of [Theorem 6.6.2](#). ■

If $\mathbf{z}$ is nearly in $\mathcal{C}(\mathbf{X})$, the computed $\mathbf{w}$ loses orthogonality to $\mathbf{Q}_1$; repeating the step once restores it (Björck 1996, chapter 2). Deleting column j of $\mathbf{R}$ leaves an upper Hessenberg block that $p-j$ rotations restore to triangular form, at $O(p^2)$ cost ([Exercise \(B2\)](#)). So stepwise procedures move between models at $O(p^2)$ or $O(np)$ cost per step.

```
def add_column(Q1, R, x):
    """Thin QR of [X, x] from the thin QR of X (one Gram-Schmidt step, repeated once)."""
    r = Q1.T @ x
    w = x - Q1 @ r
    r2 = Q1.T @ w                                # reorthogonalize: guards against
    cancellation
    w, r = w - Q1 @ r2, r + r2
    rho = np.linalg.norm(w)
    Rn = np.block([[R, r[:, None]], [np.zeros((1, R.shape[1])), np.array([[rho]])]])
    return np.column_stack([Q1, w / rho]), Rn
```

10.6.4 A moving window

Example. (Ten-year regressions of consumption)

Quarterly US real consumption is regressed on an intercept, real disposable income, real investment, population and unemployment (the public-domain macroeconomic series of [Example \(A macroeconomic design\)](#)), in a window of 40 quarters that slides forward one quarter at a time. Each step adds one row and deletes one. Over the 163 steps, whose windows have condition numbers up to 2.9×10^6 , the fit is maintained in two ways. The first applies two Sherman–Morrison updates to $(\mathbf{X}^\top \mathbf{X})^{-1}$. The second updates and then downdates the augmented triangle. After each step both are compared with a fresh QR fit of the window ([Figure 10.6.1](#)). Inverse updating reaches a relative error of 6.2×10^{-7} and ends at 8.4×10^{-8} . Updating the triangle stays below 1.1×10^{-10} and ends at 3.6×10^{-11} . Both are cheap. Only one keeps QR's accuracy.

```
macro = sm.datasets.macrodta.load_pandas().data
yw = macro["realcons"].to_numpy()
Xw = np.column_stack([np.ones(len(yw)), macro[["realdpi", "realinv", "pop", "unemp"]]])
w = 40                                # ten years of quarterly data
print("kappa of the full design:", f"{np.linalg.cond(Xw):.2e}")
```

```
K = np.linalg.inv(Xw[:w].T @ Xw[:w])          # (a) rank-one updates of the
inverse
g = Xw[:w].T @ yw[:w]
Zw = np.column_stack([Xw, yw])
Tw = np.linalg.qr(Zw[:w], mode="r")           # (b) triangle of [X, y]
```

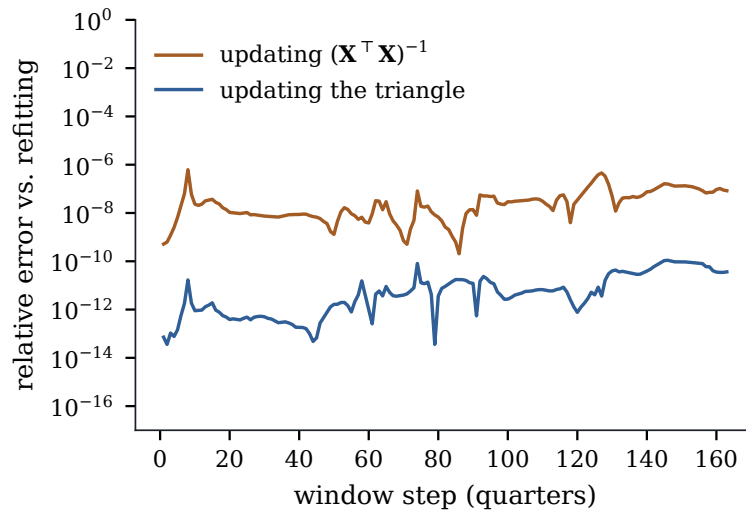


Figure 10.6.1: A moving-window regression maintained by rank-one updates of $(\mathbf{X}^T \mathbf{X})^{-1}$ and by Givens updating and downdating of the augmented triangle. The vertical axis is the relative difference in the coefficients from a fresh QR fit of each window.

```

p1 = Xw.shape[1]
err_sm, err_giv = [], []
for t in range(w, len(yw)):
    new, old = Xw[t], Xw[t - w]
    Kx = K @ new                                     # add the new quarter (Sherman-
    K = K - np.outer(Kx, Kx) / (1 + new @ Kx)        Morrison)
    Kx = K @ old                                     # delete the oldest quarter
    K = K + np.outer(Kx, Kx) / (1 - old @ Kx)
    g = g + new * yw[t] - old * yw[t - w]
    b_sm = K @ g
    Tw = downdate(update(Tw, Zw[t]), Zw[t - w])
    b_giv = np.linalg.solve(Tw[:p1, :p1], Tw[:p1, p1])
    b_ref = np.linalg.lstsq(Xw[t - w + 1:t + 1], yw[t - w + 1:t + 1], rcond=None)[0]
    # (c) refit
    err_sm.append(np.linalg.norm(b_sm - b_ref) / np.linalg.norm(b_ref))
    err_giv.append(np.linalg.norm(b_giv - b_ref) / np.linalg.norm(b_ref))
print(f"final relative error: inverse updating {err_sm[-1]:.1e}, Givens {err_giv
[-1]:.1e}")

```

Idea. A change of one row or column costs $O(p^2)$ or $O(np)$ on the triangular factor, not $O(np^2)$. All n leave-one-out fits cost one, and the leverage says whether, and how accurately, a case can be removed.

10.6.5 Exercises

10.6.5.1 A. Check your understanding

Exercise A1.

Show that $\text{PRESS} = \sum_i \hat{\epsilon}_i^2 / (1 - h_{ii})^2 \geq \text{SSE}$. Show that if all leverages equal p/n , then $\text{PRESS} = \text{SSE } n^2 / (n - p)^2$. Evaluate this for the state data with $n = 51$ and $p = 4$, and

compare with the actual ratio in [Example \(Leaving out the District of Columbia\)](#).

10.6.5.2 B. Practice

Exercise B1.

Show that the leverage of row i of the augmented matrix $\mathbf{Z} = [\mathbf{X}, \mathbf{y}]$ is $h_{ii} + \hat{\varepsilon}_i^2/\text{SSE}$. Deduce the condition under which the augmented triangle can be downdated, and show that it fails only if $h_{ii} = 1$ or deleting case i leaves a model that fits the remaining data exactly.

Solution. $\mathcal{C}(\mathbf{Z}) = \mathcal{C}(\mathbf{X}) \oplus \text{span}(\hat{\mathbf{e}})$, with the two summands orthogonal, so by [Theorem 6.3.8](#) the projection onto $\mathcal{C}(\mathbf{Z})$ is $\mathbf{M} + \hat{\mathbf{e}}\hat{\mathbf{e}}^\top / \|\hat{\mathbf{e}}\|^2$. Its i th diagonal entry is $h_{ii} + \hat{\varepsilon}_i^2/\text{SSE}$. Downdating requires this to be below 1. By [Proposition 10.6.2\(c\)](#), $1 - h_{ii} - \hat{\varepsilon}_i^2/\text{SSE} = (1 - h_{ii})\text{SSE}_{(i)}/\text{SSE}$, which vanishes iff $h_{ii} = 1$ or $\text{SSE}_{(i)} = 0$.

Exercise B2.

Let $\mathbf{X} = \mathbf{Q}_1\mathbf{R}$ and let $\mathbf{X}_{-j}$ be $\mathbf{X}$ without column j . Show that $\mathbf{X}_{-j} = \mathbf{Q}_1\mathbf{H}$, where $\mathbf{H}$ is $\mathbf{R}$ without column j , which is upper Hessenberg in its last $p - j$ columns. Show that $p - j$ rotations of adjacent rows reduce $\mathbf{H}$ to $\begin{pmatrix} \mathbf{R}_{-j} \\ \mathbf{0}^\top \end{pmatrix}$, and that $\mathbf{R}_{-j}$ is the triangle of $\mathbf{X}_{-j}$. How much does the residual sum of squares rise?

Solution. Columns of $\mathbf{X}$ are $\mathbf{Q}_1$ times columns of $\mathbf{R}$, so deleting a column of both gives $\mathbf{X}_{-j} = \mathbf{Q}_1\mathbf{H}$. Column $k \geq j$ of $\mathbf{H}$ is column $k + 1$ of $\mathbf{R}$, which has nonzeros down to row $k + 1$: one below the diagonal. Rotate rows $j, j + 1$ to zero entry $(j + 1, j)$. This mixes two rows that are zero before column j , so nothing earlier fills in. Continue with rows $(j + 1, j + 2)$ and so on. After $p - j$ rotations, $\mathbf{GH} = \begin{pmatrix} \mathbf{R}_{-j} \\ \mathbf{0}^\top \end{pmatrix}$ with $\mathbf{R}_{-j}$ upper triangular, and $\mathbf{X}_{-j} = (\mathbf{Q}_1\mathbf{G}^\top) \begin{pmatrix} \mathbf{R}_{-j} \\ \mathbf{0}^\top \end{pmatrix}$, where $\mathbf{Q}_1\mathbf{G}^\top$ has orthonormal columns. Apply the same rotations to $\mathbf{c}_1 = \mathbf{Q}_1^\top \mathbf{y}$. The last entry of the rotated vector is the coordinate of $\mathbf{y}$ along the direction that has dropped out of the model, and its square is the rise in the residual sum of squares. This is the partial sum of squares of column j ([Definition 9.3.2](#)).

10.6.5.3 C. Going deeper

Exercise C1.

Feed the rows of $[\mathbf{X}, \mathbf{y}]$ one at a time into the augmented triangle, as in [Theorem 10.3.4\(c\)](#). For $k > p$, let t_k be the entry left in the new row when case k is added. Show that

$$t_k = \pm \frac{y_k - \mathbf{x}_{(k)}^\top \hat{\boldsymbol{\beta}}_{k-1}}{(1 + \mathbf{x}_{(k)}^\top (\mathbf{X}_{k-1}^\top \mathbf{X}_{k-1})^{-1} \mathbf{x}_{(k)})^{1/2}},$$

where $\hat{\boldsymbol{\beta}}_{k-1}$ and $\mathbf{X}_{k-1}$ use the first $k - 1$ cases. Show that under the normal linear model the t_k are independent $N(0, \sigma^2)$. These are the recursive residuals of Brown, Durbin and Evans (1975), used to test the constancy of a regression over time.

10.7 Rank-revealing factorizations

A model matrix with an intercept and indicators for every level of a factor has dependent, or **aliased**, columns. Software must notice them and decide which to set aside, but in floating point it can only ask whether a column is *nearly* a combination of others, relative to a tolerance. The SVD answers that (Section 10.4) but does not say which columns to drop, and it costs more. The standard tool is QR with column pivoting.

10.7.1 QR with column pivoting

At step k of Householder QR, the trailing columns (rows k to n , columns k to p of the partly reduced matrix) are what remains of the original columns after the parts explained by the first $k - 1$ chosen columns have been removed. **Column pivoting** (Businger and Golub 1965) moves the trailing column of largest norm, the one least explained by the columns already chosen, into position k before reflecting. The result is

$$\mathbf{X}\mathbf{\Pi} = \mathbf{Q} \begin{pmatrix} \mathbf{R} \\ \mathbf{0} \end{pmatrix}, \quad \mathbf{R} = \begin{pmatrix} \mathbf{R}_{11} & \mathbf{R}_{12} \\ \mathbf{0} & \mathbf{R}_{22} \end{pmatrix},$$

with $\mathbf{\Pi}$ a permutation matrix, $\mathbf{R}_{11}$ the leading $k \times k$ block and $\mathbf{R}_{22}$ the trailing $(p - k) \times (p - k)$ block, for any k . We need one fact about submatrices.

Lemma 10.7.1 — Singular values of a submatrix

If $\mathbf{X}_1$ is formed from some of the columns of $\mathbf{X}$, then $\sigma_i(\mathbf{X}_1) \leq \sigma_i(\mathbf{X})$ for every i up to the number of columns of $\mathbf{X}_1$. The same holds for a submatrix formed from some of the rows.

Proof. Write $\mathbf{X}_1 = \mathbf{X}\mathbf{S}$, where $\mathbf{S}$ holds the corresponding columns of the identity, so $\|\mathbf{S}\|_2 = 1$. Let $\mathbf{X}_{i-1}$ be the truncated SVD with $i - 1$ terms, or $\mathbf{X}$ itself if $\text{rank } \mathbf{X} < i$, so that $\|\mathbf{X} - \mathbf{X}_{i-1}\|_2 = \sigma_i(\mathbf{X})$. The matrix $\mathbf{X}_{i-1}\mathbf{S}$ has rank at most $i - 1$, so by the Eckart–Young inequality, in the form used in the proof of Proposition 10.4.2,

$$\sigma_i(\mathbf{X}_1) \leq \|\mathbf{X}\mathbf{S} - \mathbf{X}_{i-1}\mathbf{S}\|_2 \leq \|\mathbf{X} - \mathbf{X}_{i-1}\|_2 \|\mathbf{S}\|_2 = \sigma_i(\mathbf{X}).$$

For rows, apply this to $\mathbf{X}^\top$, which has the same singular values. ■

Theorem 10.7.2 — Column-pivoted QR

Let $\mathbf{X}$ be $n \times p$ with $n \geq p$, and let $\mathbf{X}\mathbf{\Pi} = \mathbf{Q} \begin{pmatrix} \mathbf{R} \\ \mathbf{0} \end{pmatrix}$ be computed by Householder QR (Theorem 10.3.2) with column pivoting.

- (a) For every $k \leq j$, $r_{kk}^2 \geq \sum_{i=k}^j r_{ij}^2$. In particular $|r_{11}| \geq |r_{22}| \geq \cdots \geq |r_{pp}|$.
- (b) For every $k < p$,

$$\sigma_{k+1}(\mathbf{X}) \leq \|\mathbf{R}_{22}\|_2 \leq \sqrt{p - k} |r_{k+1,k+1}| \quad \text{and} \quad \sigma_{\min}(\mathbf{R}_{11}) \leq \sigma_k(\mathbf{X}).$$

- (c) In exact arithmetic, if $\text{rank } \mathbf{X} = r$, then $r_{kk} \neq 0$ for $k \leq r$ and $\mathbf{R}_{22} = \mathbf{0}$ for $k = r$. The first r columns of $\mathbf{X}\mathbf{\Pi}$ are a basis of $\mathcal{C}(\mathbf{X})$, and $\mathbf{X}\mathbf{\Pi} = \mathbf{Q} \begin{pmatrix} \mathbf{R}_{11} & \mathbf{R}_{12} \\ \mathbf{0} & \mathbf{0} \end{pmatrix}$ with $\mathbf{R}_{11}$ nonsingular.
- (d) The cost is that of the unpivoted factorization, $2np^2 - 2p^3/3$ flops, plus $O(np)$ for keeping track of the column norms.

Proof.

- (a) Reflections preserve the length of each column segment they act on. When column k is chosen, its segment in rows $k, \dots, n$ has the largest norm among the trailing columns, and the reflection maps it to $(r_{kk}, 0, \dots, 0)^\top$, so $|r_{kk}|$ equals that norm. Every column that ends in a position $j > k$ was among the trailing columns at step k , so its segment in rows $k, \dots, n$ had norm at most $|r_{kk}|$. The reflections at steps $k, \dots, p$ act on rows $k, \dots, n$ only, so they preserve this norm, and at the end the segment is $(r_{kj}, \dots, r_{jj}, 0, \dots, 0)^\top$. Hence $\sum_{i=k}^j r_{ij}^2 \leq r_{kk}^2$. With $j = k+1$ this gives $r_{k+1,k+1}^2 \leq r_{kk}^2$.
- (b) $\mathbf{X}\mathbf{\Pi}$ has the same singular values as $\mathbf{X}$, because $\mathbf{\Pi}$ is orthogonal. The matrix $\mathbf{Q} \begin{pmatrix} \mathbf{R}_{11} & \mathbf{R}_{12} \\ \mathbf{0} & \mathbf{0} \end{pmatrix}$ has rank at most k and differs from $\mathbf{X}\mathbf{\Pi}$ by $\mathbf{Q} \begin{pmatrix} \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \mathbf{R}_{22} \end{pmatrix}$, whose norm is $\|\mathbf{R}_{22}\|_2$. The Eckart–Young inequality (Proposition 1.8.2(c)) gives $\sigma_{k+1}(\mathbf{X}) \leq \|\mathbf{R}_{22}\|_2$. Next, $\|\mathbf{R}_{22}\|_2 \leq \|\mathbf{R}_{22}\|_F \leq \sqrt{p-k} \max_j \|\text{column } j \text{ of } \mathbf{R}_{22}\|$, and by (a) with $k+1$ in place of k , each column of $\mathbf{R}_{22}$ has norm at most $|r_{k+1,k+1}|$. Finally, the first k columns of $\mathbf{X}\mathbf{\Pi}$ equal $\mathbf{Q}_1 \begin{pmatrix} \mathbf{R}_{11} \\ \mathbf{0} \end{pmatrix}$, so their singular values are those of $\mathbf{R}_{11}$. Then Lemma 10.7.1 gives $\sigma_k(\mathbf{R}_{11}) \leq \sigma_k(\mathbf{X})$.
- (c) Because $\mathbf{R}_{11}$ is triangular, $\text{rank } \mathbf{R} = k + \text{rank } \mathbf{R}_{22}$ whenever $r_{11}, \dots, r_{kk}$ are nonzero (subtract multiples of the first k columns from the others to clear $\mathbf{R}_{12}$). Also $\text{rank } \mathbf{R} = \text{rank } \mathbf{X} = r$. So while $k < r$, $\mathbf{R}_{22} \neq \mathbf{0}$, some trailing column is nonzero, and the next pivot $r_{k+1,k+1}$ is nonzero. At $k = r$, $\text{rank } \mathbf{R}_{22} = 0$. The first r columns of $\mathbf{X}\mathbf{\Pi}$ are $\mathbf{Q}_1$ times the independent columns $\begin{pmatrix} \mathbf{R}_{11} \\ \mathbf{0} \end{pmatrix}$, so they are independent, and there are $r = \dim \mathcal{C}(\mathbf{X})$ of them.
- (d) The norms of the trailing column segments are updated at each step by subtracting r_{kj}^2 , at $O(p)$ flops per step. When cancellation makes the update inaccurate, the norm is recomputed.

■

Part (b) is the rank-revealing property, and it is one-sided. A small pivot $|r_{k+1,k+1}|$ certifies a small singular value σ_{k+1} : pivoted QR never reports a near dependence that is not there. The converse can fail. A matrix can have a tiny singular value although no pivot is small.

Example. (Kahan's matrix)

Kahan (1966) gave the upper triangular matrix with diagonal $1, s, s^2, \dots, s^{m-1}$ and entries $-cs^{i-1}$ above the diagonal in row i , where $c^2 + s^2 = 1$. Its columns are already in pivoted order (a tiny perturbation of the column scaling breaks ties in favour of that order), so pivoted QR leaves it unchanged. With $m = 100$ and $c = 0.2$, the smallest diagonal entry is 0.133, yet the smallest singular value is 3.7×10^{-9} . No diagonal entry hints at this (Figure 10.7.1(a)). Such matrices are rare in practice. Chan (1987) gave the first rank-revealing QR algorithm; factorizations whose bounds in both directions are polynomial in p exist (Hong and Pan 1992) and can be computed efficiently (Gu

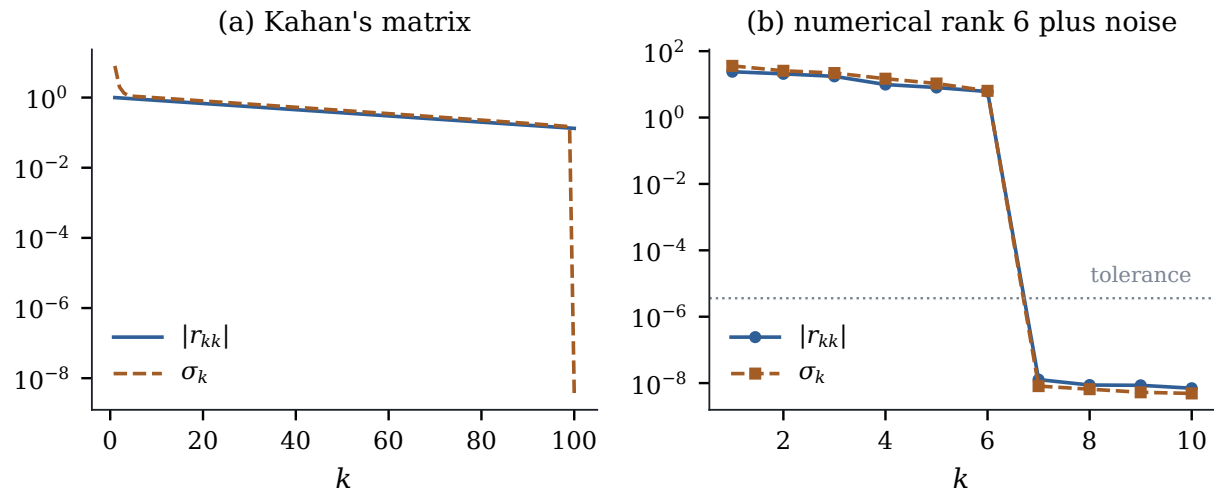


Figure 10.7.1: Diagonal of the pivoted triangle, $|r_{kk}|$, and singular values σ_k . (a) Kahan's matrix: the diagonal decreases slowly while the last singular value collapses. (b) A matrix of numerical rank six: both drop at $k = 7$, well below the tolerance.

and Eisenstat 1996). Panel (b) shows the typical case: a 50×10 matrix of rank six plus noise of relative size 10^{-9} . The pivoted diagonal drops by eight orders of magnitude exactly where the singular values do.

```
import numpy as np
import statsmodels.api as sm

def pivoted_qr(X):
    """Householder QR with column pivoting:  $X[:, \text{perm}] = Q R$ ,  $|r_{11}| \geq |r_{22}| \geq \dots$ """
    A = np.array(X, dtype=float)
    n, p = A.shape
    perm = np.arange(p)
    for k in range(min(n, p)):
        norms = np.sum(A[k:, k:] ** 2, axis=0)  # squared norms of the
        trailing columns
        j = k + int(np.argmax(norms))
        A[:, [k, j]], perm[[k, j]] = A[:, [j, k]], perm[[j, k]]
        x = A[k:, k]
        alpha = -np.copysign(np.linalg.norm(x), x[0])
        if alpha == 0.0:
            break  # the trailing block is
            exactly zero
        v = x.copy()
        v[0] -= alpha
        v /= np.linalg.norm(v)
        A[k:, k:] -= 2.0 * np.outer(v, v @ A[k:, k:])
    return np.triu(A[:p, :]), perm

def numerical_rank(R, tol):
    """Number of diagonal entries of the pivoted triangle above  $\text{tol} * |r_{11}|$ ."""
    d = np.abs(np.diag(R))
    return int(np.sum(d > tol * d[0]))
```

```
m, c = 100, 0.2
sn = np.sqrt(1 - c ** 2)
```

```

K = np.diag(sn ** np.arange(m)) @ (np.eye(m) - c * np.triu(np.ones((m, m)), 1))
K = K @ np.diag(1 - 1e-10 * np.arange(m))          # tiny tilt so that pivoting
              keeps the order
RK, permK = pivoted_qr(K)
sK = np.linalg.svd(K, compute_uv=False)
print("pivot order unchanged:", np.array_equal(permK, np.arange(m)))
print(f"smallest |r_kk| = {np.abs(RK[-1, -1]):.3f}, smallest singular value = {sK
      [-1]:.2e}")

```

10.7.2 Basic solutions and aliased columns

Suppose pivoted QR finds numerical rank $r < p$, so that $\mathbf{X}\Pi \approx \mathbf{Q} \begin{pmatrix} \mathbf{R}_{11} & \mathbf{R}_{12} \\ \mathbf{0} & \mathbf{0} \end{pmatrix}$. Write $\mathbf{Q}^\top \mathbf{y} = (\mathbf{c}_1^\top, \mathbf{c}_2^\top)^\top$ with $\mathbf{c}_1 \in \mathbb{R}^r$, and split the permuted coefficients as $(\mathbf{b}_1, \mathbf{b}_2)$. As in [Proposition 10.3.3](#), the least squares estimates are the solutions of

$$\mathbf{R}_{11}\mathbf{b}_1 + \mathbf{R}_{12}\mathbf{b}_2 = \mathbf{c}_1,$$

with $\mathbf{b}_2$ free. Setting $\mathbf{b}_2 = \mathbf{0}$ gives the **basic solution**, which simply drops the last $p - r$ pivoted columns. It has at most r nonzero coefficients. It differs from the minimum-norm solution of [Theorem 10.4.1](#). That solution can also be computed at QR cost, by a second orthogonal reduction from the right that turns $[\mathbf{R}_{11}, \mathbf{R}_{12}]$ into $[\mathbf{T}, \mathbf{0}]$ (a *complete orthogonal decomposition*; Golub and Van Loan 2013). Every least squares solution gives the same fitted values. By [Theorem 6.4.4](#), and in the language of [Chapter 8](#) ([Theorem 8.3.1](#)), every estimable function $\lambda^\top \beta$ has the same estimate under every convention. The coefficients of aliased columns are not estimable.

Software differs in the choice. R's `lm` takes the columns in order, moves to the end any column whose remaining norm falls below a relative tolerance (by default 10^{-7}), and reports its coefficient as `NA`. statsmodels' OLS uses the pseudoinverse and so reports the minimum-norm solution. Programs built on the sweep operator ([Proposition 10.2.3](#)), such as SAS's GLM procedure, skip numerically zero pivots and set those coefficients to zero, another basic solution.

Example. (Firm effects in investment data)

Grunfeld's investment data ($n = 220$: 11 US firms over 20 years, public domain in statsmodels) are modelled with an intercept, the firm's market value and capital stock, and indicators for all 11 firms: $p = 14$ columns of rank 13. Pivoted QR with a tolerance of 10^{-10} finds rank 13 and leaves out the indicator for General Motors. Three conventions give three different intercepts: -50.67 for the minimum-norm solution (as statsmodels reports), -20.58 for the in-order rule that marks the last indicator as aliased (as R would), and -70.30 for the basic solution from full pivoting. They agree to every printed digit on everything estimable: the coefficient 0.1101 of market value, 0.3100 of capital, the difference -172.20 between the General Motors and US Steel effects, and the residual sum of squares 523719.

```

gr = sm.datasets.grunfeld.load_pandas().data
firms = list(dict.fromkeys(gr["firm"]))
D = np.column_stack([(gr["firm"] == f).to_numpy(float) for f in firms])  # 11
    indicators
y = gr["invest"].to_numpy()
X = np.column_stack([np.ones(len(y)), gr[["value", "capital"]], D])      # 14
    columns, rank 13

```

```

n, p = X.shape
R, perm = pivoted_qr(X)
r = numerical_rank(R, 1e-10)
print("p =", p, " numerical rank =", r, " column left out:", perm[r:])

```

```

def basic_solution(X, y, tol=1e-10):
    """Drop the columns that pivoted QR puts last; fit the rest (zeros for dropped
    columns)."""
    R, perm = pivoted_qr(X)
    r = numerical_rank(R, tol)
    b = np.zeros(X.shape[1])
    b[perm[:r]] = np.linalg.lstsq(X[:, perm[:r]], y, rcond=None)[0]
    return b

def in_order_solution(X, y, tol=1e-7):
    """Keep a column only if it is not (nearly) a combination of the columns kept
    before it."""
    kept = []
    for j in range(X.shape[1]):
        Z = X[:, kept + [j]]
        Q, R = np.linalg.qr(Z)
        if abs(R[-1, -1]) > tol * np.linalg.norm(X[:, j]):
            kept.append(j)
    b = np.full(X.shape[1], np.nan) # aliased coefficients reported
    # as missing
    b[kept] = np.linalg.lstsq(X[:, kept], y, rcond=None)[0]
    return b

solutions = {
    "minimum norm (pinv)": np.linalg.pinv(X) @ y,
    "pivoted QR, basic": basic_solution(X, y),
    "in order, aliased = NA": in_order_solution(X, y),
}
for name, b in solutions.items():
    fitted = X @ np.nan_to_num(b)
    print(f"{name:24s} value {b[1]:.5f} capital {b[2]:.5f} const {b[0]:9.3f}"
          f" GM - US Steel {b[3] - b[4]:9.3f} SSE {np.sum((y - fitted) ** 2):.1f}"
          )

```

10.7.3 Choosing a tolerance

Exact aliasing, as in the example, is easy: the offending pivot is at the level of rounding error, far below any reasonable tolerance. The hard case is a column that is nearly, but not exactly, a combination of others. The numerical rank then depends on the tolerance, and so do the coefficients, dramatically.

Example. (A nearly aliased column)

Add to the Grunfeld regressors (intercept, value, capital) a fourth column equal to value plus capital plus noise of relative size 10^{-9} . Its smallest singular value, relative to the largest, is 4.0×10^{-10} . The default tolerance of `numpy.linalg.lstsq`, $\max(n, p) \epsilon_M$ relative to σ_1 , treats the matrix as having full rank. It returns coefficients as large as 8.1×10^5 , which spend one more degree of freedom to reduce the residual sum of squares

from 1768678.4 to 1768252.5, by 0.02 per cent. With the tolerance set to 10^{-8} , the fourth column is treated as aliased, and the largest coefficient is 38.41.

```
rng = np.random.default_rng(108)
z = X[:, 1] + X[:, 2] + 1e-9 * np.linalg.norm(X[:, 1]) * rng.normal(size=n) / np.
    sqrt(n)
Xz = np.column_stack([X[:, :3], z]) # constant, value, capital, value +
    capital + tiny
s = np.linalg.svd(Xz, compute_uv=False)
print("relative singular values:", np.array2string(s / s[0], precision=2))
for tol in [None, 1e-8]:
    b, _, rank, _ = np.linalg.lstsq(Xz, y, rcond=tol)
    print(f"rcond = {tol}: rank {rank}, largest |b_j| = {np.abs(b).max():.3g},"
          f" SSE = {np.sum((y - Xz @ b) ** 2):.1f}")
```

Neither answer is wrong as arithmetic. But a column that differs from a combination of the others only at the level of 10^{-9} of its size carries no information a real measurement could support. Three rules follow. Scale the columns to comparable lengths before deciding the rank (Proposition 10.5.2); R's relative test does this implicitly. Tie the tolerance to the accuracy of the data, not of the machine: regressors recorded to six significant figures cannot resolve near dependences below about 10^{-6} of the column lengths (Proposition 10.4.2, applied to the recording error). And look for a gap, as in Figure 10.7.1(b). Without one the rank decision is fragile, the data are showing collinearity (Chapter 26), and shrinkage is often a better response than a sharp cut-off (Chapter 27).

Idea. Pivoted QR brings the most independent columns to the front. A small trailing pivot proves a near dependence, and setting those coefficients aside changes nothing estimable. The tolerance that decides “small” is a statement about the accuracy of the data.

10.7.4 Exercises

10.7.4.1 A. Check your understanding

Exercise A1.

Apply column-pivoted QR (Theorem 10.7.2) by hand to the matrix with columns $\mathbf{x}_1 = (1, 1, 0, 0)^\top$, $\mathbf{x}_2 = (0, 0, 1, 1)^\top$ and $\mathbf{x}_3 = (1, 1, 1, 1)^\top$ (use Gram–Schmidt on the pivoted columns, which gives the same $|r_{ij}|$). Find the pivot order, $\mathbf{R}$, and the rank. Give the basic solution for $\mathbf{y} = (1, 2, 3, 6)^\top$.

Solution. The norms are $\sqrt{2}, \sqrt{2}, 2$, so $\mathbf{x}_3$ is chosen first, with $|r_{11}| = 2$ and $\mathbf{q}_1 = \mathbf{x}_3/2$. Then $\mathbf{q}_1^\top \mathbf{x}_1 = \mathbf{q}_1^\top \mathbf{x}_2 = 1$, and the remainders are $\mathbf{x}_1 - \mathbf{q}_1 = (\frac{1}{2}, \frac{1}{2}, -\frac{1}{2}, -\frac{1}{2})^\top$ and $\mathbf{x}_2 - \mathbf{q}_1$, its negative. Both have norm 1, so the tie goes to $\mathbf{x}_1$, with $|r_{22}| = 1$. The remainder of $\mathbf{x}_2$ is then zero, so $r_{33} = 0$ and the rank is 2. The pivot order is $(3, 1, 2)$, and up to signs $\mathbf{R} = \begin{pmatrix} 2 & 1 & 1 \\ 0 & 1 & -1 \\ 0 & 0 & 0 \end{pmatrix}$. The basic solution regresses $\mathbf{y}$ on $\mathbf{x}_3$ and $\mathbf{x}_1$ with $b_2 = 0$. The fit is the pair of group means, 1.5 for the first two observations and 4.5 for the last two. So $b_3 = 4.5$ and $b_1 = 1.5 - 4.5 = -3$.

10.7.4.2 B. Practice

Exercise B1.

In the setting of this section, show that the basic solution and the minimum-norm solution have the same fitted values. Show that the minimum-norm solution is orthogonal to $\mathcal{N}(\mathbf{X})$, while the basic solution in general is not. For the one-way layout of [Example \(Four answers, one fit\)](#), which generalized inverse in that example gives the basic solution that drops the intercept?

Solution. Both satisfy $\mathbf{R}_{11}\mathbf{b}_1 + \mathbf{R}_{12}\mathbf{b}_2 = \mathbf{c}_1$, so both are least squares estimates ([Proposition 10.3.3](#)), and by [Theorem 6.4.1](#) both have fitted values $\mathbf{M}\mathbf{y}$. The minimum-norm solution lies in $\mathcal{C}(\mathbf{X}^\top) = \mathcal{N}(\mathbf{X})^\perp$ ([Theorem 10.4.1\(b\)](#)). The basic solution has zeros in the positions of the dropped columns. A null vector with a nonzero entry in one of those positions is generally not orthogonal to it. For the one-way layout, dropping the intercept column and fitting the three indicators is the “drop intercept” generalized inverse of [Example \(Four answers, one fit\)](#), which gives the group means as coefficients and a zero intercept.

Exercise B2.

Show that $|r_{11}| = \max_j \|\mathbf{x}_j\|$, and deduce $|r_{11}| \leq \sigma_1(\mathbf{X}) \leq \sqrt{p}|r_{11}|$. Combine with [Theorem 10.7.2\(b\)](#) to show that $\kappa(\mathbf{X}) \geq |r_{11}/r_{pp}|$, so that a large ratio of pivots proves ill-conditioning.

10.7.4.3 C. Going deeper

Exercise C1.

For Kahan’s matrix $\mathbf{K}_m(c)$, find a vector $\mathbf{x}$ with $\|\mathbf{x}\| = 1$ and $\|\mathbf{K}_m\mathbf{x}\|$ exponentially small in m when c is fixed. (Solve $\mathbf{K}_m\mathbf{x} = \mathbf{e}_m s^{m-1}$ by back substitution and examine the growth of the entries.) Why does column pivoting not detect this?

10.8 Summary and Notes

10.8.1 Summary

Idea.

1. Floating-point arithmetic commits a relative error of at most $u = 2^{-53}$ per operation ([Definition 10.1.1](#)). Inner products are backward stable ([Proposition 10.1.2](#)), but forming $\mathbf{X}^\top \mathbf{X}$ destroys the information in singular values below about $u^{1/2}\sigma_1$ ([Proposition 10.1.3](#), [Example \(A cross-product matrix that rounds to a singular one\)](#)).
2. The Cholesky factorization of the augmented cross-product matrix gives $\hat{\boldsymbol{\beta}}$, the SSE, the sequential sums of squares, the leverages and the standard errors in

- $np^2 + p^3/3$ flops ([Theorem 10.2.1](#), [Proposition 10.2.2](#)), accurately enough when the column-scaled condition number is modest. Sweeping fits nested models ([Proposition 10.2.3](#)); centring needs two passes or updating ([Proposition 10.2.4](#)).
3. Householder QR reduces $\mathbf{X}$ to a triangle by reflections in $2np^2 - 2p^3/3$ flops and is backward stable with columnwise errors ([Theorem 10.3.2](#), [Proposition 10.3.3](#)). Givens rotations work one row at a time, for streaming and sparse data ([Theorem 10.3.4](#)).
 4. The SVD gives every least squares solution, the minimum-norm one, and the precision σ/σ_i of each combination $\mathbf{v}_i^\top \boldsymbol{\beta}$ ([Theorem 10.4.1](#)). Singular values are perturbed by at most $\|\mathbf{E}\|_2$ ([Proposition 10.4.2](#)), so rank is meaningful only relative to a tolerance ([Definition 10.4.1](#)). Truncated SVD trades bias for variance ([Proposition 10.4.3](#)).
 5. The least squares solution has sensitivity $\kappa + \kappa^2\eta$, where η measures the residual relative to the fit; residuals have sensitivity κ only ([Theorem 10.5.1](#)). The normal equations add κ^2 regardless. Equilibrating the columns is nearly optimal among scalings ([Proposition 10.5.2](#), [Example \(Longley's data, exactly\)](#)).
 6. A triangular factor can be updated or downdated by one row in $O(p^2)$ ([Theorem 10.6.1](#)), and by one column in $O(np)$ ([Proposition 10.6.3](#)). All leave-one-out residuals, coefficients and sums of squares follow from a single fit ([Proposition 10.6.2](#)).
 7. QR with column pivoting orders the columns by how much each adds to the others, and a small trailing pivot certifies a small singular value ([Theorem 10.7.2](#)). Basic and minimum-norm solutions agree on everything estimable ([Example \(Firm effects in investment data\)](#)); the tolerance should reflect the accuracy of the data ([Example \(A nearly aliased column\)](#)).

10.8.2 Notes and sources

Source coverage. The chapter covers Seber and Lee (2003, chapter 11) on solving the normal equations, QR by Gram–Schmidt, Householder and Givens, the SVD, centring and leverages (sections 11.2–11.4, 11.7 and 11.10), with parts of sections 11.6, 11.8 and 11.9 on updating, comparing methods and rank deficiency.

Floating point and stability. The standard model and the γ_n notation follow Higham (2002), whose chapters 3, 10, 19 and 20 are the definitive treatment of inner products, Cholesky, QR and least squares. Goldberg (1991) is a readable introduction to IEEE arithmetic, now specified by IEEE Std 754-2019. The distinction between a problem's condition and an algorithm's backward stability goes back to Wilkinson (1963). Trefethen and Bau (1997) give a short modern account. Läuchli (1961) introduced the matrix of [Example \(A cross-product matrix that rounds to a singular one\)](#) to show the loss in forming normal equations.

Algorithms. Householder (1958) introduced the reflections and Golub (1965) applied them to least squares; Givens (1958) used plane rotations. Gentleman (1973) developed square-root-free rotations and the row-wise algorithm, turned into widely used routines by Miller (1992). Björck (1967) analysed modified Gram–Schmidt, and Björck and Paige (1992) explained its good behaviour for least squares. The SVD algorithm is due to Golub and Kahan (1965) and Golub and Reinsch (1970), column pivoting to Businger and Golub (1965), and the counterexample of [Example \(Kahan's matrix\)](#) to Kahan (1966). Chan (1987) gave the first rank-revealing QR algorithm, Hong and Pan (1992) proved that factorizations with polynomial bounds exist, and

Gu and Eisenstat (1996) showed how to compute them efficiently. Goodnight (1979) describes the sweep operator for statisticians. Welford (1962) gave the updating recursion, and Chan, Golub and LeVeque (1983) compare variance algorithms. Downdating appears in LINPACK (Dongarra et al. 1979) and was analysed by Stewart (1979). The standard monographs are Björck (1996), Lawson and Hanson (1974) and Golub and Van Loan (2013); Thisted (1988) and Kennedy and Gentle (1980) take the statistician's side.

Perturbation theory. Golub and Wilkinson (1966) first observed that the least squares solution can be sensitive to κ^2 when the residual is large. Wedin (1973) gave sharp perturbation bounds for pseudoinverses and least squares. The equilibration result of [Proposition 10.5.2](#) is from van der Sluis (1969). The inequality of [Proposition 10.4.2](#) is the singular-value form of Weyl's (1912) inequality for eigenvalues. Beaton, Rubin and Barone (1976) argued that errors from rounding the data should be the yardstick for computational accuracy, the point made at the end of [Section 10.5](#).

Statistics. Recursive least squares via the Sherman–Morrison formula goes back to Plackett (1950). The PRESS statistic is due to Allen (1974). The recursive residuals of [Exercise \(C1\)](#) are those of Brown, Durbin and Evans (1975). Deletion diagnostics are developed in Chapter 20, collinearity in Chapter 26, and ridge and principal components regression in Chapter 27.

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Chapter 11

Testing Models: The General Linear Hypothesis

Almost every question put to a linear model asks whether a simpler model would do: whether a group of regressors can be dropped, whether two coefficients are equal, whether regions differ once other variables are accounted for. Each such question names a subspace of the model space. The whole theory of testing it rests on one picture. The part of the data that the simpler model cannot explain splits orthogonally into a piece in the *test space* and a piece in the residual space. The F statistic compares the squared lengths of the two pieces, per dimension. Normality turns the comparison into an exact test. This chapter derives that test in the three forms met in practice, as a comparison of fitted models, as a statement about estimable functions $\mathbf{A}^\top \boldsymbol{\beta} = \mathbf{d}$, and as a likelihood ratio. It shows that the Wald and score tests are the same test in disguise, and proves that no invariant test is more powerful. It then studies power, which depends on the unknown parameters only through one noncentrality, and ends with the t test, the overall F test and tests of groups of coefficients, including the ways a group test and the individual tests within it can disagree. Warnings about what a nonsignificant F does and does not show run through the chapter.

What you need. The normal linear model and its sampling distributions from [Chapter 7](#) (Eq. 7.3.1, [Theorem 7.5.1](#), [Corollary 7.5.2](#), [Theorem 7.3.1](#)). The noncentral χ^2 , F and t distributions and the F statistic for nested subspaces from [Chapter 4](#) ([Definition 4.1.2](#), [Theorem 4.1.2](#), [Theorem 4.1.4](#), [Theorem 4.4.6](#)). Nested projections and the constraint space from [Chapter 6](#) ([Theorem 6.5.1](#), [Theorem 6.5.2](#), [Theorem 6.3.7](#)). Estimable functions from [Chapter 8](#) ([Definition 8.2.1](#), [Theorem 8.3.1](#)). Extra sums of squares and the sum of squares of a linear hypothesis from [Chapter 9](#) ([Definition 9.2.1](#), [Theorem 9.2.4](#), [Theorem 9.2.5](#), [Proposition 9.2.6](#)).

11.0.1 Roadmap

- [Testing reduced models](#)
- [The F statistic as a comparison of projections](#)
- [Testable hypotheses](#)
- [The likelihood ratio test and its relatives](#)
- [Power and noncentrality](#)
- [Testing single coefficients and groups of coefficients](#)
- [Summary and notes](#)

11.1 Testing reduced models

Part II fitted linear models without asking whether a fitted term was needed. Part III adds normal errors, and with them exact answers to that question. Does the murder rate of a state depend on its region once poverty and family structure are accounted for? Is a straight line enough, or is there curvature? Do two groups share a regression line? Is a coefficient equal to the value a theory predicts? Each asks whether a *reduced model*, a special case of the model already fitted, is adequate. This section sets up that comparison and derives its test. The rest of the chapter looks at the same test from other directions.

11.1.1 Hypotheses are subspaces

Throughout the chapter the data follow the normal linear model

$$\mathbf{Y} \sim N_n(\mathbf{X}\boldsymbol{\beta}, \sigma^2\mathbf{I}), \quad \boldsymbol{\beta} \in \mathbb{R}^p, \sigma^2 > 0,$$

of Eq. 7.3.1, with $r = \text{rank}(\mathbf{X}) < n$. We call it the **full model**. Its content is that the mean vector $\boldsymbol{\mu} = E(\mathbf{Y})$ lies in the r -dimensional subspace $\mathcal{C}(\mathbf{X})$ (Section 6.1). A reduced model says more: the mean lies in a smaller subspace.

Definition 11.1.1 — Reduced model and linear hypothesis

Let $\mathbf{X}_0$ be an $n \times p_0$ matrix with $\mathcal{C}(\mathbf{X}_0) \subseteq \mathcal{C}(\mathbf{X})$ and $r_0 = \text{rank}(\mathbf{X}_0) < r$. The **reduced model** is $E(\mathbf{Y}) = \mathbf{X}_0\mathbf{c}$, $\mathbf{c} \in \mathbb{R}^{p_0}$. Inside the full model it corresponds to the **linear hypothesis**

$$H_0 : \mathbf{X}\boldsymbol{\beta} \in \mathcal{C}(\mathbf{X}_0) \quad \text{against} \quad H_1 : \mathbf{X}\boldsymbol{\beta} \in \mathcal{C}(\mathbf{X}), \mathbf{X}\boldsymbol{\beta} \notin \mathcal{C}(\mathbf{X}_0).$$

The number $q = r - r_0$ is the number of **degrees of freedom of the hypothesis**, and we write $\mathbf{M}$ and $\mathbf{M}_0$ for the orthogonal projections onto $\mathcal{C}(\mathbf{X})$ and $\mathcal{C}(\mathbf{X}_0)$.

Three features of the definition deserve comment. First, the hypothesis is a statement about the mean vector, not about coefficients. Coefficients enter only through the column spaces, so recoding the regressors of either model changes nothing (Theorem 6.7.1). Statements about coefficients, such as $\beta_2 = \beta_3$, are translated into subspaces in Section 11.3. Second, the models must be *nested*. Two models neither of which contains the other, say one with poverty and one with urbanization, are not compared by the tests of this chapter. Third, both models share the same σ^2 and the same error law. The alternative is not “anything other than the reduced model”. It is “the full model, but not the reduced one”.

The most common reduced models come from three operations on $\mathbf{X}$:

- *Deleting columns.* Dropping a block $\mathbf{X}_2$ from $\mathbf{X} = [\mathbf{X}_1, \mathbf{X}_2]$ tests whether those regressors are needed given the others. With a single column this is the familiar question about one coefficient. With all columns except the intercept it is the “overall” test of Section 11.6.
- *Merging columns.* Replacing $\mathbf{x}_2$ and $\mathbf{x}_3$ by their sum $\mathbf{x}_2 + \mathbf{x}_3$ forces equal coefficients. Merging the indicator columns of several groups forces those groups to share a mean.
- *Fixing a component.* Moving a known multiple of a column into an offset, for example testing that a slope equals 1, gives a reduced model of the form $E(\mathbf{Y}) = \mathbf{X}_0\mathbf{c} + \mathbf{X}\mathbf{b}_0$ with $\mathbf{b}_0$ known. It is handled by subtracting $\mathbf{X}\mathbf{b}_0$ from the data (Section 11.3).

11.1.2 The F statistic

If the reduced model is adequate, forcing the fit into $\mathcal{C}(\mathbf{X}_0)$ should cost little: the residual sum of squares $\text{SSE}_0 = \|(\mathbf{I} - \mathbf{M}_0)\mathbf{y}\|^2$ should exceed $\text{SSE} = \|(\mathbf{I} - \mathbf{M})\mathbf{y}\|^2$ by no more than noise would produce. The difference is the extra sum of squares of [Section 9.2](#), and by [Eq. 6.5.2](#) it is a squared length,

$$\text{SSE}_0 - \text{SSE} = \|(\mathbf{M} - \mathbf{M}_0)\mathbf{y}\|^2 = \mathbf{y}^\top (\mathbf{M} - \mathbf{M}_0)\mathbf{y},$$

the squared length of the projection of $\mathbf{y}$ onto the orthogonal complement of $\mathcal{C}(\mathbf{X}_0)$ within $\mathcal{C}(\mathbf{X})$. How large is “more than noise”? Noise contributes σ^2 per dimension ([Theorem 9.6.1](#)), and $\mathbf{M} - \mathbf{M}_0$ has rank q , so we compare the **hypothesis mean square** $(\text{SSE}_0 - \text{SSE})/q$ with the unbiased estimate $s^2 = \text{SSE}/(n - r)$ of σ^2 from the full model. The ratio

$$F = \frac{(\text{SSE}_0 - \text{SSE})/(r - r_0)}{\text{SSE}/(n - r)} = \frac{\mathbf{y}^\top (\mathbf{M} - \mathbf{M}_0)\mathbf{y}/q}{\mathbf{y}^\top (\mathbf{I} - \mathbf{M})\mathbf{y}/(n - r)} \quad (11.1.1)$$

is the F **statistic** for the reduced model. It is scale-free: multiplying $\mathbf{y}$ by a constant, or changing the units of the response, does not change it. Its exact distribution is the content of the first theorem of the chapter.

Theorem 11.1.1 — The F test of a reduced model

Assume the normal linear model [Eq. 7.3.1](#) and the setting of [Definition 11.1.1](#), so that $0 \leq r_0 < r < n$. Then:

- (a) $\text{SSE}_0 - \text{SSE}$ and SSE are independent, with

$$\frac{\text{SSE}_0 - \text{SSE}}{\sigma^2} \sim \chi^2(r - r_0, \gamma), \quad \frac{\text{SSE}}{\sigma^2} \sim \chi^2(n - r);$$

- (b) the statistic [Eq. 11.1.1](#) has the noncentral F distribution

$$F \sim F(r - r_0, n - r, \gamma), \quad \gamma = \frac{\|(\mathbf{M} - \mathbf{M}_0)\mathbf{X}\boldsymbol{\beta}\|^2}{\sigma^2}; \quad (11.1.2)$$

- (c) $\gamma = \|(\mathbf{I} - \mathbf{M}_0)\mathbf{X}\boldsymbol{\beta}\|^2/\sigma^2 = \min_{\mathbf{c}} \|\mathbf{X}\boldsymbol{\beta} - \mathbf{X}_0\mathbf{c}\|^2/\sigma^2$, the squared distance from the mean vector to the reduced model space in units of σ^2 . In particular $\gamma = 0$ iff H_0 holds;

- (d) under H_0 , $F \sim F(r - r_0, n - r)$, the central F distribution, whatever the values of $\boldsymbol{\beta}$ and σ^2 .

Proof. By [Theorem 6.5.1](#), $\mathbf{M} - \mathbf{M}_0$ and $\mathbf{I} - \mathbf{M}$ are symmetric idempotent with ranks $r - r_0$ and $n - r$, and $(\mathbf{M} - \mathbf{M}_0)(\mathbf{I} - \mathbf{M}) = \mathbf{0}$. [Theorem 4.4.4](#), applied with $\boldsymbol{\mu} = \mathbf{X}\boldsymbol{\beta}$, shows that $\|(\mathbf{M} - \mathbf{M}_0)\mathbf{Y}\|^2/\sigma^2$ and $\|(\mathbf{I} - \mathbf{M})\mathbf{Y}\|^2/\sigma^2$ are independent with distributions $\chi^2(r - r_0, \|(\mathbf{M} - \mathbf{M}_0)\mathbf{X}\boldsymbol{\beta}\|^2/\sigma^2)$ and $\chi^2(n - r, \|(\mathbf{I} - \mathbf{M})\mathbf{X}\boldsymbol{\beta}\|^2/\sigma^2)$. The second noncentrality is zero because $\mathbf{M}\mathbf{X}\boldsymbol{\beta} = \mathbf{X}\boldsymbol{\beta}$. With [Eq. 6.5.2](#) this is (a). Part (b) follows from (a) and [Definition 4.1.2](#), since σ^2 cancels in the ratio. (It is also [Theorem 4.4.6](#) with $\mathbf{P} = \mathbf{M}$ and $\mathbf{P}_0 = \mathbf{M}_0$.) For (c), $\mathbf{M}\mathbf{X}\boldsymbol{\beta} = \mathbf{X}\boldsymbol{\beta}$ gives $(\mathbf{M} - \mathbf{M}_0)\mathbf{X}\boldsymbol{\beta} = (\mathbf{I} - \mathbf{M}_0)\mathbf{X}\boldsymbol{\beta}$, which is the difference between $\mathbf{X}\boldsymbol{\beta}$ and its nearest point in $\mathcal{C}(\mathbf{X}_0)$ ([Theorem 6.3.1](#)). Its length is therefore the distance from $\mathbf{X}\boldsymbol{\beta}$ to $\mathcal{C}(\mathbf{X}_0)$, which is zero iff $\mathbf{X}\boldsymbol{\beta} \in \mathcal{C}(\mathbf{X}_0)$. Part (d) is (b) with $\gamma = 0$.

Part (d) is what makes a test possible. The reduced model is a *composite* hypothesis: it leaves $\mathbf{c}$ and σ^2 unspecified. Yet the null distribution of F is the same at every point of it. The unknown coefficients cancel because $(\mathbf{M} - \mathbf{M}_0)\mathbf{X}_0\mathbf{c} = \mathbf{0}$ and $(\mathbf{I} - \mathbf{M})\mathbf{X}_0\mathbf{c} = \mathbf{0}$, and the unknown scale cancels in the ratio. A statistic with this property is called *pivotal* under the hypothesis.

Part (c) says what the test detects. The alternative is measured by one number, the squared distance of the true mean from the reduced model, relative to the noise variance. Two alternatives with the same γ are equally hard to detect, however different their coefficients. [Section 11.5](#) is built on this.

11.1.3 The test and its p-value

Large values of F are evidence against H_0 : under the alternative the numerator mean square estimates $\sigma^2 + \sigma^2\gamma/q$ while the denominator still estimates σ^2 ([Eq. 9.6.1](#)). The F test at level α rejects H_0 when

$$F > F_\alpha(r - r_0, n - r),$$

where $F_\alpha(a, b)$ is the upper α point of the central $F(a, b)$ distribution. By [Theorem 11.1.1\(d\)](#) its probability of rejecting is exactly α at every point of the reduced model. It rejects with probability greater than α at every point of the alternative, by (c) and [Theorem 4.1.4](#). Instead of a fixed level, one usually reports the **p-value**

$$p(\mathbf{y}) = \Pr\{F(r - r_0, n - r) \geq F_{\text{obs}}\},$$

the probability under the reduced model of a statistic at least as extreme as the one observed.

Proposition 11.1.2 — The p-value is uniform under the hypothesis

Under H_0 , the p-value $p(\mathbf{Y})$ has the uniform distribution on $(0, 1)$. For every $\alpha \in (0, 1)$, rejecting when $p(\mathbf{Y}) \leq \alpha$ is the F test at level α .

Proof. Let Φ_F be the distribution function of $F(q, n - r)$. It is continuous and strictly increasing on $(0, \infty)$, since the distribution has a positive density there, and $p(\mathbf{y}) = 1 - \Phi_F(F_{\text{obs}})$. Under H_0 the statistic F has distribution function Φ_F , so for $u \in (0, 1)$, $\Pr\{p(\mathbf{Y}) \leq u\} = \Pr\{\Phi_F(F) \geq 1 - u\} = \Pr\{F \geq \Phi_F^{-1}(1 - u)\} = u$. Finally, $p(\mathbf{y}) \leq \alpha$ iff $\Phi_F(F_{\text{obs}}) \geq 1 - \alpha$ iff $F_{\text{obs}} \geq F_\alpha(q, n - r)$, and the event $F = F_\alpha$ has probability zero.

The p-value is a statement about the data under H_0 . It is *not* the probability that H_0 is true, and it is not the probability that the result is “due to chance”. A small p-value says that data like these would be unusual if the reduced model held.

Example. (Do regions matter?)

In the 2009 data on 50 US states used in [Chapter 6](#) and [Chapter 9](#), regress the murder rate on the poverty rate, the percentage of single-parent households and the percentage living in urbanized areas, with an intercept. This is the reduced model, with $r_0 = 4$.

The full model adds indicators of three of the four Census regions (Northeast, South, West, with the Midwest as reference), so $r = 7$. The question is whether the regions differ in murder rate once the three covariates are accounted for. The listing gives

Source	Degrees of freedom	Sum of squares	Mean square
Regions, given the covariates	3	14.83	4.944
Residual, full model	43	87.13	2.026
Residual, reduced model	46	101.96	

so $F = 4.944/2.026 = 2.440$ on 3 and 43 degrees of freedom. The 5% critical value is 2.822 and the p-value is 0.077. At the conventional level the reduced model is not rejected. The data do not show convincingly that the regions differ, given the covariates.

Yet in the full model the Northeast coefficient is -1.736 murders per 100,000, with a t statistic of -2.519 and a p-value of 0.016. One of the three contrasts looks clearly nonzero while the joint test does not reject. There is no contradiction: [Section 11.6](#) shows how a joint test can dilute one strong effect among weak ones, and [Chapter 13](#) explains why a contrast singled out after looking at the estimates cannot be judged by its own p-value.

```
import numpy as np
import statsmodels.api as sm
from scipy import stats

data = sm.datasets.statecrime.load_pandas().data
data.index = data.index.str.strip()
data = data.drop(index="District of Columbia")
# Census region of each state, in the data's alphabetical order (N, M, S, W)
codes = "SWWSWWNSSSWMMMMSSNSNMMSMWNNWNSMMSWNNMSWNSWSMW"
region = np.array(list(codes))
y = data["murder"].to_numpy()
n = len(y)
covariates = np.column_stack([data["poverty"], data["single"], data["urban"]])
X0 = np.column_stack([np.ones(n), covariates]) # reduced model
D = np.column_stack([(region == g).astype(float) for g in "NSW"]) # Midwest is the
# reference
X = np.column_stack([X0, D]) # full model

def sse(Z):
    """Residual sum of squares of the least squares fit of y on the columns of Z."""
    b, *_ = np.linalg.lstsq(Z, y, rcond=None)
    return np.sum((y - Z @ b) ** 2)

r, r0 = np.linalg.matrix_rank(X), np.linalg.matrix_rank(X0)
sse_full, sse_reduced = sse(X), sse(X0)
F = ((sse_reduced - sse_full) / (r - r0)) / (sse_full / (n - r))
p_value = stats.f.sf(F, r - r0, n - r)
crit = stats.f.ppf(0.95, r - r0, n - r)
print(f"SSE0 = {sse_reduced:.2f}, SSE = {sse_full:.2f}, df = ({r - r0}, {n - r})")
print(f"F = {F:.3f}, p = {p_value:.4f}, 5% critical value = {crit:.3f}")
```

11.1.4 What the test assumes

[Theorem 11.1.1](#) rests on three assumptions.

The full model must hold. The denominator s^2 estimates σ^2 only if $E(\mathbf{Y}) \in \mathcal{C}(\mathbf{X})$. If the full model omits something, $E(s^2)$ exceeds σ^2 by $\|(\mathbf{I} - \mathbf{M})\boldsymbol{\mu}\|^2/(n - r)$ ([Theorem 9.6.1\(b\)](#)), the denominator is inflated, and F is pushed towards zero. The test then loses power, and a nonsignificant result says even less than usual. The test of a reduced model is a test *within* the full model. It cannot certify the full model itself. [Chapter 14](#) tests the full model against a nonparametric alternative when there are replicate observations.

The errors must be independent with a common variance. Correlated or heteroscedastic errors change both numerator and denominator, and the null distribution is no longer F . [Chapter 21](#) and [Chapter 31](#) treat these cases.

Normality gives the exact distribution, and it is the least critical of the three. The null distribution is exactly F for every spherically symmetric error law ([Section 11.2](#)), and approximately so for independent non-normal errors unless the sample is small or some points have high leverage ([Chapter 19](#)).

Warning (What a nonsignificant F does not show). Failing to reject the reduced model is *not* evidence that the reduced model is true. The test controls the probability of wrongly rejecting H_0 . It says nothing on its own about the probability of wrongly accepting it, which depends on the unknown noncentrality γ and may be large. In [Example \(Do regions matter?\)](#), $p = 0.077$ is compatible with no regional differences. It is equally compatible with sizeable regional differences. [Section 11.5](#) computes that the test had only about an even chance of detecting a Northeast shortfall of 1.5 murders per 100,000, about the size estimated in [Section 11.3](#) (1.379 below the other three regions). To support “the reduced model is adequate”, one must show that every alternative of practical importance would have been rejected. That calls for a power calculation made before the data were seen, or better, a confidence region for the omitted effects that excludes all important values ([Chapter 12](#)), or a test of equivalence in which the roles of hypothesis and alternative are reversed. Nor does a nonsignificant F license dropping the terms and then reporting the reduced fit as if it had been specified in advance.

Warning (What a significant F does not show). A significant F says that the mean vector is not in $\mathcal{C}(\mathbf{X}_0)$, assuming the full model holds. It does not say which of the q dimensions of the test space carries the departure, nor that every omitted coefficient is nonzero. It does not say that the departure is large enough to matter: with large n , a tiny γ/n is detected with certainty ([Section 11.5](#)). And it does not say that the full model is correct. The statistic only compares two models, and a third, better model may exist.

11.1.5 Exercises

11.1.5.1 A. Check your understanding

Exercise A1.

Show that the F statistic [Eq. 11.1.1](#) does not change when $\mathbf{y}$ is replaced by $a\mathbf{y} + \mathbf{X}_0\mathbf{c}$

for any $a \neq 0$ and any vector $\mathbf{c}$. Interpret the result for the region test of [Example \(Do regions matter?\)](#) when murder rates are expressed per million instead of per 100,000.

Exercise A2.

In the model $E(Y_i) = \beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \beta_3 x_{i3}$, write down a matrix $\mathbf{X}_0$ (or an offset and an $\mathbf{X}_0$) for each reduced model and give q : (i) $\beta_2 = \beta_3 = 0$; (ii) $\beta_1 = \beta_2 = \beta_3$; (iii) $\beta_1 = 2$; (iv) $\beta_2 + \beta_3 = 0$.

11.1.5.2 B. Practice

Exercise B1.

Suppose $\sigma^2 = \sigma_0^2$ is known. Show that $X^2 = (\text{SSE}_0 - \text{SSE})/\sigma_0^2$ has the $\chi^2(q, \gamma)$ distribution and that the test rejecting when X^2 exceeds the upper α point of $\chi^2(q)$ has level α . Show that $qF/X^2 \rightarrow 1$ in probability as $n \rightarrow \infty$ with q fixed and the full model true, and explain why the F test and this test then behave alike.

Solution. The distribution is [Theorem 11.1.1\(a\)](#) with $\sigma^2 = \sigma_0^2$; under H_0 it is $\chi^2(q)$, so the level is exact. Now $qF/X^2 = \sigma_0^2/s^2$, and $s^2 \rightarrow \sigma^2 = \sigma_0^2$ in probability, since $E(s^2) = \sigma^2$ and $\text{Var}(s^2) = 2\sigma^4/(n-r) \rightarrow 0$ ([Theorem 7.5.1\(d\)](#)) and Chebyshev's inequality applies. So qF and X^2 differ by a factor tending to one, and $qF_\alpha(q, n-r)$ tends to the upper α point of $\chi^2(q)$ for the same reason. With few residual degrees of freedom the two tests differ, and only the F test has the right level when σ^2 must be estimated.

Exercise B2.

Suppose the full model is wrong: $E(\mathbf{Y}) = \boldsymbol{\mu}$ with $(\mathbf{I} - \mathbf{M})\boldsymbol{\mu} = \boldsymbol{\delta} \neq \mathbf{0}$, but $(\mathbf{M} - \mathbf{M}_0)\boldsymbol{\mu} = \mathbf{0}$. Show that the numerator and denominator of [Eq. 11.1.1](#) are still independent, that $E(\text{numerator}) = \sigma^2$ and $E(\text{denominator}) = \sigma^2 + \|\boldsymbol{\delta}\|^2/(n-r)$, and that $\Pr(F \leq c) \geq \Pr\{F(q, n-r) \leq c\}$ for every $c > 0$. Explain why an F statistic far *below* one, with a p-value near one, is a hint that the full model is inadequate.

Solution. Independence and the distributions follow from [Theorem 4.4.4](#) as in the proof of [Theorem 11.1.1](#): the numerator is $\sigma^2 \chi^2(q)/q$, and $\text{SSE}/\sigma^2 \sim \chi^2(n-r, \|\boldsymbol{\delta}\|^2/\sigma^2)$, with mean $n-r + \|\boldsymbol{\delta}\|^2/\sigma^2$ ([Theorem 4.1.2\(b\)](#)). For the probability, condition on the numerator U : $F \leq c$ iff $\text{SSE}/\sigma^2 \geq (n-r)U/(qc\sigma^2)$, and a noncentral χ^2 exceeds any fixed value with at least the central probability ([Proposition 4.1.3](#)). Averaging over U gives the inequality. So omitted structure in the residual space makes small values of F more likely. Under the full model, $\Pr(F \leq c)$ is small for small c , and an observed F in the far lower tail is surprising.

Exercise B3.

Let $\mathcal{C}(\mathbf{X}_0) \subseteq \mathcal{C}(\mathbf{X}_1) \subseteq \mathcal{C}(\mathbf{X})$ with ranks $r_0 < r_1 < r < n$, and let $s^2 = \text{SSE}/(n-r)$ come

from the largest model. Show that

$$F_1 = \frac{\mathbf{y}^\top (\mathbf{M}_1 - \mathbf{M}_0) \mathbf{y} / (r_1 - r_0)}{s^2}$$

has the $F(r_1 - r_0, n - r, \gamma_1)$ distribution with $\gamma_1 = \|(\mathbf{M}_1 - \mathbf{M}_0) \mathbf{X} \boldsymbol{\beta}\|^2 / \sigma^2$ whenever the largest model holds, and that its numerator is independent of that of the test of $\mathbf{X}_1$ within $\mathbf{X}$. Why is this preferable to using the residual mean square of $\mathbf{X}_1$?

Exercise B4.

Two samples of sizes n_1 and n_2 follow $E(\mathbf{Y}_k) = \mathbf{X}_k \boldsymbol{\beta}_k$, $k = 1, 2$, with a common σ^2 and $\mathbf{X}_k$ of full column rank p . Write the model for both samples together as a full model with block-diagonal model matrix, and the hypothesis $\boldsymbol{\beta}_1 = \boldsymbol{\beta}_2$ as a reduced model. Show that

$$F = \frac{(\text{SSE}_{\text{pooled}} - \text{SSE}_1 - \text{SSE}_2)/p}{(\text{SSE}_1 + \text{SSE}_2)/(n_1 + n_2 - 2p)},$$

where SSE_k comes from fitting sample k alone and $\text{SSE}_{\text{pooled}}$ from one fit to all the data.

Apply the test to the question whether two firms of the Grunfeld panel (`sm.datasets.grunfeld`) share their investment equation, and comment on the assumption of a common σ^2 .

11.1.5.3 C. Going deeper

Exercise C1.

In [Exercise \(B4\)](#) suppose instead that $n_2 \leq p$ and $\text{rank}(\mathbf{X}_2) = n_2$, so that $\boldsymbol{\beta}_2$ cannot be estimated from the second sample. Show that the full model then fits the second sample exactly, identify the test space, and show that the F statistic of [Theorem 11.1.1](#) becomes

$$F = \frac{(\text{SSE}_{\text{pooled}} - \text{SSE}_1)/n_2}{\text{SSE}_1/(n_1 - p)},$$

a test of whether the second sample is predicted by the fit to the first, on n_2 and $n_1 - p$ degrees of freedom.

11.2 The F statistic as a comparison of projections

The formula [Eq. 11.1.1](#) mentions two residual sums of squares, but the test is really about three subspaces and the way the data vector sits among them. Seen this way, the F statistic is a function of a single angle, its rejection region is a cone, and its null distribution survives far beyond normal errors. The same view gives a canonical form in which every linear hypothesis looks the same. [Section 11.4](#) needs that form to prove that the F test is optimal.

11.2.1 Three orthogonal pieces

By [Theorem 6.5.1](#), nested model spaces split $\mathbb{R}^n$ into three mutually orthogonal subspaces:

$$\mathbb{R}^n = \underbrace{\mathcal{C}(\mathbf{X}_0)}_{\dim r_0} \oplus \underbrace{\mathcal{C}(\mathbf{X}) \cap \mathcal{C}(\mathbf{X}_0)^\perp}_{\dim q=r-r_0} \oplus \underbrace{\mathcal{C}(\mathbf{X})^\perp}_{\dim n-r}, \quad \mathbf{I} = \mathbf{M}_0 + (\mathbf{M} - \mathbf{M}_0) + (\mathbf{I} - \mathbf{M}). \quad (11.2.1)$$

The middle subspace, $\mathcal{C}(\mathbf{M} - \mathbf{M}_0)$, is the **test space** of the hypothesis. It consists of the directions in which the full model can move the mean and the reduced model cannot. Every data vector splits accordingly,

$$\mathbf{y} = \mathbf{M}_0\mathbf{y} + (\mathbf{M} - \mathbf{M}_0)\mathbf{y} + (\mathbf{I} - \mathbf{M})\mathbf{y},$$

and so does the mean vector, $\mathbf{X}\boldsymbol{\beta} = \mathbf{M}_0\mathbf{X}\boldsymbol{\beta} + (\mathbf{M} - \mathbf{M}_0)\mathbf{X}\boldsymbol{\beta} + \mathbf{0}$. The hypothesis says that the middle component of the *mean* is zero. The middle component of the *data*, $(\mathbf{M} - \mathbf{M}_0)\mathbf{Y}$, is its unbiased estimate, with covariance $\sigma^2(\mathbf{M} - \mathbf{M}_0)$. The residual component $(\mathbf{I} - \mathbf{M})\mathbf{Y}$ has mean zero whether or not the hypothesis holds, and it serves as a yardstick for noise. The first component $\mathbf{M}_0\mathbf{Y}$ estimates the part of the mean that both models allow. It carries no information about the hypothesis, and the F statistic ignores it.

What the statistic does use is $(\mathbf{I} - \mathbf{M}_0)\mathbf{y}$, the residual vector of the reduced model. It is split into a piece in the test space and a piece in the residual space, and

$$F = \frac{\|(\mathbf{M} - \mathbf{M}_0)\mathbf{y}\|^2/q}{\|(\mathbf{I} - \mathbf{M})\mathbf{y}\|^2/(n-r)}$$

compares their squared lengths per dimension. Under the hypothesis both pieces are pure noise, and each dimension carries on average σ^2 . Under the alternative the first piece also carries the signal $(\mathbf{M} - \mathbf{M}_0)\mathbf{X}\boldsymbol{\beta}$.

The test space can be described by a spanning set. Since $\mathbf{M} - \mathbf{M}_0 = (\mathbf{I} - \mathbf{M}_0)\mathbf{M}$ by [Theorem 6.5.1\(a\)](#), and $\mathcal{C}(\mathbf{M}) = \mathcal{C}(\mathbf{X})$,

$$\mathcal{C}(\mathbf{M} - \mathbf{M}_0) = \mathcal{C}((\mathbf{I} - \mathbf{M}_0)\mathbf{X}).$$

The test space is spanned by the columns of $\mathbf{X}$ after the reduced model has been regressed out of them. When $\mathbf{X} = [\mathbf{X}_0, \mathbf{X}_2]$ adds a block $\mathbf{X}_2$, this is $\mathcal{C}((\mathbf{I} - \mathbf{M}_0)\mathbf{X}_2)$, the space of [Lemma 9.2.1](#). For a hypothesis on coefficients it is the constraint space $\mathcal{C}(\mathbf{M}\mathbf{T})$ of [Theorem 6.5.2](#), as [Section 11.3](#) shows.

11.2.2 One angle

Because the two pieces of $(\mathbf{I} - \mathbf{M}_0)\mathbf{y}$ are orthogonal, their lengths are the legs of a right triangle whose hypotenuse is the reduced model's residual vector.

Proposition 11.2.1 — F as a function of an angle

Let $\mathbf{w} = (\mathbf{I} - \mathbf{M}_0)\mathbf{y} \neq \mathbf{0}$ and let $\theta \in [0, \pi/2]$ be the angle between $\mathbf{w}$ and the test space, defined by $\cos \theta = \|(\mathbf{M} - \mathbf{M}_0)\mathbf{y}\|/\|\mathbf{w}\|$. Then

$$\|(\mathbf{M} - \mathbf{M}_0)\mathbf{y}\|^2 = \|\mathbf{w}\|^2 \cos^2 \theta, \quad \text{SSE} = \|\mathbf{w}\|^2 \sin^2 \theta, \quad F = \frac{n-r}{q} \cot^2 \theta.$$

The partial coefficient of determination of [Definition 9.5.1](#) is $\cos^2 \theta$, and the F test rejects exactly when $\theta < \theta_\alpha$, where $\tan^2 \theta_\alpha = (n - r) / \{q F_\alpha(q, n - r)\}$.

Proof. $\mathbf{w} = (\mathbf{M} - \mathbf{M}_0)\mathbf{y} + (\mathbf{I} - \mathbf{M})\mathbf{y}$ with orthogonal summands, since $(\mathbf{M} - \mathbf{M}_0)(\mathbf{I} - \mathbf{M}) = \mathbf{0}$. Pythagoras gives $\|(\mathbf{I} - \mathbf{M})\mathbf{y}\|^2 = \|\mathbf{w}\|^2 - \|\mathbf{w}\|^2 \cos^2 \theta = \|\mathbf{w}\|^2 \sin^2 \theta$. Since $(\mathbf{M} - \mathbf{M}_0)\mathbf{y}$ is the projection of $\mathbf{w}$ onto the test space, θ is also the angle between $\mathbf{w}$ and its projection, the smallest angle between $\mathbf{w}$ and any vector of the test space. The ratio of the two squared lengths is $\cot^2 \theta$, which gives F . The partial coefficient of determination is $(\text{SSE}_0 - \text{SSE}) / \text{SSE}_0 = \|(\mathbf{M} - \mathbf{M}_0)\mathbf{y}\|^2 / \|\mathbf{w}\|^2 = \cos^2 \theta$. Finally $\cot^2$ is strictly decreasing on $(0, \pi/2]$, so $F > F_\alpha$ iff $\cot^2 \theta > q F_\alpha / (n - r)$ iff $\theta < \theta_\alpha$. ■

So the F test asks a geometric question: *does the reduced model's residual vector point suspiciously close to the test space?* Under the hypothesis, $\mathbf{w}$ is noise spread over $n - r_0$ dimensions, only q of which belong to the test space, so it typically points well away from it. The rejection region $\{\mathbf{y} : \theta < \theta_\alpha\}$ is a circular cone around the test space inside $\mathcal{C}(\mathbf{X}_0)^\perp$, extended by arbitrary components in $\mathcal{C}(\mathbf{X}_0)$. That this is the same as the partial R^2 test of [Theorem 9.5.4\(b\)](#) is no coincidence. The partial R^2 is a squared cosine, just as [Theorem 6.7.2](#) showed for the ordinary R^2 .

Example. (The region test as an angle)

In [Example \(Do regions matter?\)](#), the reduced model's residual vector has a component of length 3.851 in the three-dimensional test space and a component of length 9.334 in the 43-dimensional residual space. The angle between the residual vector and the test space is $\theta = 67.6^\circ$, and the partial R^2 is $\cos^2 \theta = 0.145$. The 5% test rejects when $\theta < 66.1^\circ$, so the data fall just outside the cone.

[Figure 11.2.1](#) plots the two lengths for the data and for 4000 simulated data sets. With the reduced model true, 0.053 of them fall in the rejection region. With the Northeast lowered by two murders per 100,000, and all else as fitted, the fraction is 0.774.

```
import numpy as np
import statsmodels.api as sm
from scipy import stats

data = sm.datasets.statecrime.load_pandas().data
data.index = data.index.str.strip()
data = data.drop(index="District of Columbia")
codes = "SWWSWNSSSWMMMMSSNSNMMSMWNNWNSMMSWNNSSWNSWSMW"
region = np.array(list(codes))
y = data["murder"].to_numpy()
n = len(y)
covariates = np.column_stack([data["poverty"], data["single"], data["urban"]])
X0 = np.column_stack([np.ones(n), covariates])
D = np.column_stack([(region == g).astype(float) for g in "NSW"])
X = np.column_stack([X0, D])

def proj(Z):
    """Orthogonal projection onto C(Z), from an orthonormal basis."""
    Q, _ = np.linalg.qr(Z)
    return Q @ Q.T
```

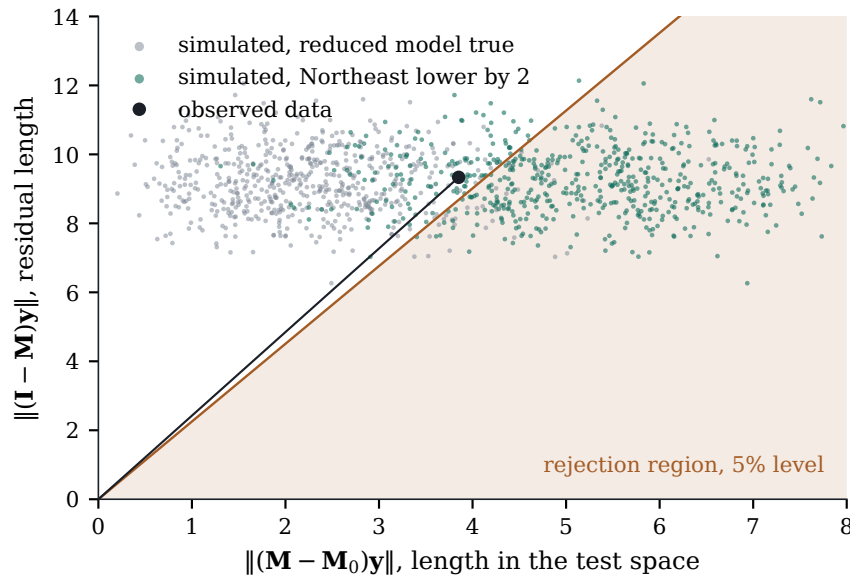


Figure 11.2.1: The region test of [Example \(Do regions matter?\)](#) in two coordinates: the length of $(\mathbf{M} - \mathbf{M}_0)\mathbf{y}$ in the test space and the length of the residual vector $(\mathbf{I} - \mathbf{M})\mathbf{y}$. The F statistic depends only on the angle of the point, and the test rejects inside the shaded wedge. The grey points are simulated with the reduced model true, the green points with the Northeast lowered by two murders per 100,000. The observed data are the black point, just above the wedge.

```
M, M0 = proj(X), proj(X0)
q, nu = 3, n - 7
u = (M - M0) @ y
e = y - M @ y
w = y - M0 @ y
# r - r0 and n - r
# component in the test space
# residual vector
# what the reduced model leaves
unexplained
theta = np.arccos(np.linalg.norm(u) / np.linalg.norm(w))
F = (nu / q) / np.tan(theta) ** 2
theta_crit = np.arctan(np.sqrt(nu / (q * stats.f.ppf(0.95, q, nu))))
print(f"angle = {np.degrees(theta):.1f} degrees, F = {F:.3f}")
print(f"reject when the angle is below {np.degrees(theta_crit):.1f} degrees")
```

11.2.3 What the statistic ignores

Since F depends on $\mathbf{y}$ only through the angle θ , many transformations of the data leave it unchanged. They form the symmetry group of the testing problem, and [Section 11.4](#) uses them to characterize the F test.

Proposition 11.2.2 — Invariance of the F statistic

The statistic [Eq. 11.1.1](#) depends on $\mathbf{X}$ and $\mathbf{X}_0$ only through $\mathcal{C}(\mathbf{X})$ and $\mathcal{C}(\mathbf{X}_0)$. Where it is defined, that is, when $(\mathbf{I} - \mathbf{M})\mathbf{y} \neq \mathbf{0}$, it is unchanged when $\mathbf{y}$ is replaced by

$$a\mathbf{H}\mathbf{y} + \mathbf{v}, \quad a \neq 0, \quad \mathbf{v} \in \mathcal{C}(\mathbf{X}_0),$$

where $\mathbf{H}$ is any orthogonal matrix that maps $\mathcal{C}(\mathbf{X}_0)$ onto itself and $\mathcal{C}(\mathbf{X})$ onto itself.

Proof. The first claim holds because $\mathbf{M}$ and $\mathbf{M}_0$ are determined by the column spaces. For the second, the transformations can be applied one at a time. Adding $\mathbf{v}$ changes nothing, because $\mathbf{M} - \mathbf{M}_0$ and $\mathbf{I} - \mathbf{M}$ annihilate $\mathcal{C}(\mathbf{X}_0)$. The factor a^2 cancels in the ratio. For $\mathbf{H}$: the matrix $\mathbf{HMH}^\top$ is symmetric and idempotent with column space $\mathbf{HC}(\mathbf{X}) = \mathcal{C}(\mathbf{X})$, so it equals $\mathbf{M}$ (Theorem 6.3.3). Hence $\mathbf{HM} = \mathbf{MH}$, and likewise $\mathbf{HM}_0 = \mathbf{M}_0\mathbf{H}$. Then $\|(\mathbf{M} - \mathbf{M}_0)\mathbf{H}\mathbf{y}\| = \|\mathbf{H}(\mathbf{M} - \mathbf{M}_0)\mathbf{y}\| = \|(\mathbf{M} - \mathbf{M}_0)\mathbf{y}\|$, because $\mathbf{H}$ preserves lengths, and the same holds for $\mathbf{I} - \mathbf{M}$. ■

The orthogonal matrices allowed here are exactly those that rotate the test space within itself and the residual space within itself, and fix $\mathcal{C}(\mathbf{X}_0)$ as a set. The invariance says that no direction within the test space is favoured. A departure of given size $\|(\mathbf{M} - \mathbf{M}_0)\mathbf{X}\boldsymbol{\beta}\|$ is equally detectable in every direction. When some directions matter more than others, for instance when one contrast is of primary interest, a test directed at them can have more power there. It pays for this with less power elsewhere (Section 11.6).

11.2.4 Canonical form

Orthonormal bases adapted to Eq. 11.2.1 turn every linear hypothesis into the same problem about independent normal coordinates.

Proposition 11.2.3 — Canonical form of a linear hypothesis

Let $\mathbf{Q} = [\mathbf{Q}_0, \mathbf{Q}_1, \mathbf{Q}_2]$ be an orthogonal $n \times n$ matrix whose blocks of r_0 , q and $n - r$ columns are orthonormal bases of $\mathcal{C}(\mathbf{X}_0)$, of the test space and of $\mathcal{C}(\mathbf{X})^\perp$. Put $\mathbf{z}_k = \mathbf{Q}_k^\top \mathbf{Y}$ and $\boldsymbol{\eta}_k = \mathbf{Q}_k^\top \mathbf{X}\boldsymbol{\beta}$, $k = 0, 1, 2$. Under Eq. 7.3.1:

- (a) $\mathbf{z}_0, \mathbf{z}_1, \mathbf{z}_2$ are independent, $\mathbf{z}_k \sim N(\boldsymbol{\eta}_k, \sigma^2 \mathbf{I})$, and $\boldsymbol{\eta}_2 = \mathbf{0}$;
- (b) $\|\mathbf{z}_1\|^2 = \text{SSE}_0 - \text{SSE}$ and $\|\mathbf{z}_2\|^2 = \text{SSE}$, so $F = \{\|\mathbf{z}_1\|^2/q\} / \{\|\mathbf{z}_2\|^2/(n - r)\}$;
- (c) $\|\boldsymbol{\eta}_1\|^2 = \sigma^2\gamma$, and H_0 holds iff $\boldsymbol{\eta}_1 = \mathbf{0}$.

Proof.

- (a) $\mathbf{Q}^\top \mathbf{Y} \sim N_n(\mathbf{Q}^\top \mathbf{X}\boldsymbol{\beta}, \sigma^2 \mathbf{Q}^\top \mathbf{Q}) = N_n(\mathbf{Q}^\top \mathbf{X}\boldsymbol{\beta}, \sigma^2 \mathbf{I})$ by Theorem 3.3.1. Its blocks are uncorrelated, hence independent (Theorem 3.4.1). The columns of $\mathbf{Q}_2$ are orthogonal to $\mathcal{C}(\mathbf{X})$, so $\boldsymbol{\eta}_2 = \mathbf{Q}_2^\top \mathbf{X}\boldsymbol{\beta} = \mathbf{0}$. (b) By Proposition 6.3.2, $\mathbf{Q}_1 \mathbf{Q}_1^\top = \mathbf{M} - \mathbf{M}_0$ and $\mathbf{Q}_2 \mathbf{Q}_2^\top = \mathbf{I} - \mathbf{M}$, so $\|\mathbf{z}_1\|^2 = \mathbf{Y}^\top \mathbf{Q}_1 \mathbf{Q}_1^\top \mathbf{Y} = \mathbf{Y}^\top (\mathbf{M} - \mathbf{M}_0) \mathbf{Y}$ and similarly for $\mathbf{z}_2$. (c) In the same way $\|\boldsymbol{\eta}_1\|^2 = \|(\mathbf{M} - \mathbf{M}_0)\mathbf{X}\boldsymbol{\beta}\|^2$, and Theorem 11.1.1(c) applies. ■

In canonical form the problem is stripped to its essentials. One observes a q -vector $\mathbf{z}_1$, normal with unknown mean $\boldsymbol{\eta}_1$ and covariance $\sigma^2 \mathbf{I}$, and wishes to test $\boldsymbol{\eta}_1 = \mathbf{0}$. An independent vector $\mathbf{z}_2$ of pure noise, of dimension $n - r$, measures σ^2 . The vector $\mathbf{z}_0$ has an unknown mean that neither hypothesis restricts, so it is of no use. Every hypothesis of this chapter, whether about coefficients, contrasts or whole models, is an instance of this problem with particular q and $n - r$.

11.2.5 Beyond normal errors

The angle picture shows why normality is less important for the null distribution than it first appears. Under H_0 , $(\mathbf{I} - \mathbf{M}_0)\mathbf{Y} = (\mathbf{I} - \mathbf{M}_0)\boldsymbol{\varepsilon}$, so F is a function of the error vector alone, and it is a function of its *direction* only: replacing $\boldsymbol{\varepsilon}$ by $c\boldsymbol{\varepsilon}$, $c > 0$, leaves every angle unchanged. Any error law under which the direction of $\boldsymbol{\varepsilon}$ behaves as it does for normal errors therefore gives the same null distribution. We call $\mathbf{U}$ **uniformly distributed on the unit sphere** if it has the distribution of $\mathbf{Z}/\|\mathbf{Z}\|$ for $\mathbf{Z} \sim N_n(\mathbf{0}, \mathbf{I})$.

Proposition 11.2.4 — Spherical errors

Let $\mathbf{Y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}$ with $\mathbf{X}\boldsymbol{\beta} \in \mathcal{C}(\mathbf{X}_0)$, where $\boldsymbol{\varepsilon} = R\mathbf{U}$ for a random variable $R > 0$ and a vector $\mathbf{U}$ uniformly distributed on the unit sphere. The variables R and $\mathbf{U}$ need not be independent. Then $F \sim F(r - r_0, n - r)$.

Proof. Write $g(\mathbf{v}) = \{\|(\mathbf{M} - \mathbf{M}_0)\mathbf{v}\|^2/q\} / \{\|(\mathbf{I} - \mathbf{M})\mathbf{v}\|^2/(n - r)\}$ for vectors with $(\mathbf{I} - \mathbf{M})\mathbf{v} \neq \mathbf{0}$. Under H_0 , $(\mathbf{M} - \mathbf{M}_0)\mathbf{Y} = (\mathbf{M} - \mathbf{M}_0)\boldsymbol{\varepsilon}$ and $(\mathbf{I} - \mathbf{M})\mathbf{Y} = (\mathbf{I} - \mathbf{M})\boldsymbol{\varepsilon}$, since both projections annihilate $\mathbf{X}\boldsymbol{\beta} \in \mathcal{C}(\mathbf{X}_0)$. So $F = g(\boldsymbol{\varepsilon})$. Because $g(c\mathbf{v}) = g(\mathbf{v})$ for $c > 0$, $F = g(R\mathbf{U}) = g(\mathbf{U})$. The law of $g(\mathbf{U})$ is the law of $g(\mathbf{Z}/\|\mathbf{Z}\|) = g(\mathbf{Z})$ with $\mathbf{Z} \sim N_n(\mathbf{0}, \mathbf{I})$. By Theorem 11.1.1(d) with $\sigma = 1$, that law is $F(r - r_0, n - r)$. (The events $\mathbf{Z} = \mathbf{0}$ and $(\mathbf{I} - \mathbf{M})\mathbf{Z} = \mathbf{0}$ have probability zero.)

■

The class of error laws covered is large. If $\mathbf{Z} \sim N_n(\mathbf{0}, \mathbf{I})$ and $S > 0$ is any random scale independent of $\mathbf{Z}$, or not, then $\boldsymbol{\varepsilon} = S\mathbf{Z} = (S\|\mathbf{Z}\|)(\mathbf{Z}/\|\mathbf{Z}\|)$ qualifies. Taking S^{-2} proportional to a $\chi^2(\nu)$ variable gives the spherical multivariate t distribution. Its components are uncorrelated and heavy-tailed but *not* independent: one common scale inflates or shrinks them all. The F test is exactly valid for all such laws. What the proposition does not cover are independent non-normal errors, whose direction is not uniform. For them the null distribution is only approximately F .

Example. (Four error laws)

Keep the design of Example (Do regions matter?) and simulate 20000 data sets with the reduced model true, each with errors from one of four laws: independent normal; spherical t with three degrees of freedom; independent t with three degrees of freedom; independent centred exponential. The spherical errors are uncorrelated, but their sizes move together: the rank correlation between the absolute values of two components is 0.20. The rejection rates of the 5% test are

Errors	normal	spherical t_3	independent t_3	exponential
Rejection rate	0.0502	0.0502	0.0444	0.0474

with Monte Carlo standard error about 0.0015. The first two agree exactly, not just approximately: the spherical errors were made by rescaling the normal ones, and each rescaled data set gives the same F as its normal original. The two independent non-normal laws give rates slightly below 0.05. For this design, with 43 residual degrees of freedom and no extreme leverage, the independent t_3 errors make the test slightly conservative, the exponential rate is within Monte Carlo error of 0.05, and both are close to the nominal level.

```
import numpy as np
import statsmodels.api as sm
from scipy import stats

rng = np.random.default_rng(1103)

data = sm.datasets.statcrime.load_pandas().data
data.index = data.index.str.strip()
data = data.drop(index="District of Columbia")
codes = "SWWSWNSSSWMMMMSSNSNMMSMWNNWNSMMSWNNSSWNSWSMW"
region = np.array(list(codes))
n = len(data)
covariates = np.column_stack([data["poverty"], data["single"], data["urban"]])
X0 = np.column_stack([np.ones(n), covariates])
D = np.column_stack([(region == g).astype(float) for g in "NSW"])
X = np.column_stack([X0, D])

def proj(Z):
    Q, _ = np.linalg.qr(Z)
    return Q @ Q.T

M, M0 = proj(X), proj(X0)
q, nu = 3, n - 7
crit = stats.f.ppf(0.95, q, nu)

def F_stat(E):
    """F statistic for each row of E, used as a data vector under the reduced model.
    """
    num = np.sum((E @ (M - M0)) ** 2, axis=1) / q
    den = np.sum((E - E @ M) ** 2, axis=1) / nu
    return num / den

reps = 20000
Z = rng.standard_normal((reps, n))
scale = 1 / np.sqrt(rng.chisquare(3, size=(reps, 1)) / 3)  # one scale per data
set
errors = {
    "normal": Z,
    "spherical t3": Z * scale,  # dependent, heavy
    -tailed
    "independent t3": rng.standard_t(3, size=(reps, n)),
    "exponential": rng.exponential(size=(reps, n)) - 1.0,  # skewed
}
for name, E in errors.items():
    print(f"{name:15s} rejection rate {np.mean(F_stat(E) > crit):.4f}")
```

Remark (Only the null distribution). [Proposition 11.2.4](#) concerns the level of the test. Under an alternative, F depends on the error length R as well as the direction, and the

power under spherical t errors differs from the power under normal errors with the same covariance. Nor does the proposition say that the F test is a *good* test for heavy-tailed errors. Its power can be poor compared with robust methods. Those questions belong to Chapter 19 and Chapter 21.

11.2.6 Exercises

11.2.6.1 A. Check your understanding

Exercise A1.

Use [Proposition 11.2.1](#) to express $\cos^2 \theta$ in terms of F , q and $n - r$. Check the value 0.145 of [Example \(The region test as an angle\)](#) from $F = 2.440$, $q = 3$ and $n - r = 43$, and compute θ_α from $F_{0.05}(3, 43) = 2.822$.

11.2.6.2 B. Practice

Exercise B1.

Show that under H_0 the squared cosine $\cos^2 \theta$ of [Proposition 11.2.1](#) has the $\text{Beta}(q/2, (n-r)/2)$ distribution, and that it is independent of $\|(\mathbf{I} - \mathbf{M}_0)\mathbf{Y}\|^2$. What is its expectation?

Solution. By [Proposition 11.2.3](#), $\cos^2 \theta = U/(U + V)$ with $U = \|\mathbf{z}_1\|^2/\sigma^2 \sim \chi^2(q)$ and $V = \|\mathbf{z}_2\|^2/\sigma^2 \sim \chi^2(n - r)$ independent under H_0 , and $\|(\mathbf{I} - \mathbf{M}_0)\mathbf{Y}\|^2 = \sigma^2(U + V)$. [Exercise \(B4\)](#) gives both the Beta law and the independence of $U/(U + V)$ from $U + V$. The mean of $\text{Beta}(a, b)$ is $a/(a + b)$, here $q/(n - r_0)$: under the hypothesis the residual vector of the reduced model puts, on average, a share of its squared length in the test space equal to that space's share of the dimensions.

Exercise B2.

Prove that $\mathcal{C}(\mathbf{M} - \mathbf{M}_0) = \mathcal{C}((\mathbf{I} - \mathbf{M}_0)\mathbf{X})$, and that for $\mathbf{X} = [\mathbf{X}_0, \mathbf{X}_2]$ this equals $\mathcal{C}((\mathbf{I} - \mathbf{M}_0)\mathbf{X}_2)$. Show that $\dim \mathcal{C}((\mathbf{I} - \mathbf{M}_0)\mathbf{X}_2) = \text{rank}(\mathbf{X}) - \text{rank}(\mathbf{X}_0)$.

Solution. $\mathbf{M} - \mathbf{M}_0 = (\mathbf{I} - \mathbf{M}_0)\mathbf{M}$ by [Theorem 6.5.1\(a\)](#), so $\mathcal{C}(\mathbf{M} - \mathbf{M}_0) \subseteq \mathcal{C}((\mathbf{I} - \mathbf{M}_0)\mathbf{M}) = (\mathbf{I} - \mathbf{M}_0)\mathcal{C}(\mathbf{M}) = (\mathbf{I} - \mathbf{M}_0)\mathcal{C}(\mathbf{X}) = \mathcal{C}((\mathbf{I} - \mathbf{M}_0)\mathbf{X})$. Conversely each column $(\mathbf{I} - \mathbf{M}_0)\mathbf{x}_j = \mathbf{x}_j - \mathbf{M}_0\mathbf{x}_j$ lies in $\mathcal{C}(\mathbf{X})$ and is orthogonal to $\mathcal{C}(\mathbf{X}_0)$, so it lies in $\mathcal{C}(\mathbf{X}) \cap \mathcal{C}(\mathbf{X}_0)^\perp = \mathcal{C}(\mathbf{M} - \mathbf{M}_0)$. For $\mathbf{X} = [\mathbf{X}_0, \mathbf{X}_2]$, $(\mathbf{I} - \mathbf{M}_0)\mathbf{X} = [\mathbf{0}, (\mathbf{I} - \mathbf{M}_0)\mathbf{X}_2]$. The dimension is $\text{rank}(\mathbf{M} - \mathbf{M}_0) = r - r_0$ ([Theorem 6.5.1\(c\)](#)).

Exercise B3.

Which of the following error vectors satisfy the hypothesis of [Proposition 11.2.4](#)? (i) $\boldsymbol{\varepsilon} = \sigma\mathbf{Z}$ with σ fixed; (ii) $\boldsymbol{\varepsilon} = S\mathbf{Z}$ with S uniform on $(1, 2)$ and independent of $\mathbf{Z}$; (iii) $\boldsymbol{\varepsilon}$ with independent Laplace components; (iv) $\boldsymbol{\varepsilon} = \mathbf{D}\mathbf{Z}$ with $\mathbf{D}$ a fixed diagonal matrix with unequal entries; (v) $\boldsymbol{\varepsilon} = \mathbf{Z}/\|\mathbf{Z}\|^2$.

11.2.6.3 C. Going deeper

Exercise C1.

In the canonical form, let two data vectors $\mathbf{y}$ and $\mathbf{y}'$ with $\mathbf{z}_2, \mathbf{z}'_2 \neq \mathbf{0}$ have the same value of F . Construct $a > 0$, $\mathbf{v} \in \mathcal{C}(\mathbf{X}_0)$ and an orthogonal $\mathbf{H}$ of the kind allowed in [Proposition 11.2.2](#) such that $\mathbf{y}' = a\mathbf{H}\mathbf{y} + \mathbf{v}$. (So F is a *maximal invariant*: any statistic unchanged by these transformations is a function of F .)

Exercise C2.

Take a one-way layout with two groups of sizes $n_1 = 5$ and $n_2 = 20$, independent normal errors with standard deviations 3 in the small group and 1 in the large one, and equal means. Compute, by simulation or numerical integration, the true level of the nominal 5% F test of equal means. Explain the direction of the error using the angle picture: which way does the unequal variance tilt the residual vector?

11.3 Testable hypotheses

Scientific questions are usually phrased in terms of coefficients rather than subspaces: “the coefficients of poverty and single parenthood are equal”, “the four regions have the same intercept”, “the elasticity is one”. Such statements have the form

$$H : \mathbf{\Lambda}^\top \boldsymbol{\beta} = \mathbf{d},$$

with a known $p \times m$ matrix $\mathbf{\Lambda}$ and a known $\mathbf{d} \in \mathbb{R}^m$. This is the **general linear hypothesis**. This section decides when such a statement is a hypothesis about the distribution of $\mathbf{Y}$ at all, gives its F statistic in terms of the unrestricted fit alone, shows that the statistic agrees with the reduced-model statistic of [Section 11.1](#), and describes least squares under the hypothesis. Nothing here assumes that $\mathbf{X}$ has full column rank.

11.3.1 Which constraints are hypotheses

In the full model the distribution of $\mathbf{Y}$ depends on $\boldsymbol{\beta}$ only through $\mathbf{X}\boldsymbol{\beta}$. A statement about $\boldsymbol{\beta}$ that cannot be read off from $\mathbf{X}\boldsymbol{\beta}$ is therefore not a statement about the data-generating distribution, and no data can test it.

Definition 11.3.1 — Testable hypothesis

Let $\mathbf{\Lambda}$ be a $p \times m$ matrix and $\mathbf{d} \in \mathbb{R}^m$. The hypothesis $H : \mathbf{\Lambda}^\top \boldsymbol{\beta} = \mathbf{d}$ is **testable** if

- (i) $\mathbf{\Lambda}^\top \boldsymbol{\beta}$ is estimable, that is, $\mathcal{C}(\mathbf{\Lambda}) \subseteq \mathcal{C}(\mathbf{X}^\top)$ ([Definition 8.2.1](#) and [Theorem 8.3.1](#)), so that $\mathbf{\Lambda} = \mathbf{X}^\top \mathbf{T}$ for some $n \times m$ matrix $\mathbf{T}$; and
- (ii) the equations $\mathbf{\Lambda}^\top \mathbf{b} = \mathbf{d}$ are consistent, that is, $\mathbf{d} \in \mathcal{C}(\mathbf{\Lambda}^\top)$ ([Theorem 1.4.1](#)).

Its number of degrees of freedom is $q = \text{rank}(\mathbf{\Lambda})$. We assume $q \geq 1$.

Condition (ii) is automatic when $\mathbf{\Lambda}$ has full column rank, for then $\mathcal{C}(\mathbf{\Lambda}^\top) = \mathbb{R}^m$. It matters when the columns of $\mathbf{\Lambda}$ are redundant, as when all six pairwise differences of four group effects are set to zero. Redundant constraints are harmless if they agree with each other. If they do not, as in $\alpha_1 - \alpha_2 = 1$, $\alpha_2 - \alpha_3 = 1$, $\alpha_1 - \alpha_3 = 0$, the “hypothesis” is false for every $\boldsymbol{\beta}$, and no test is possible.

Condition (i) is the substantive one. Without it the constraint cannot be settled by any amount of data.

Proposition 11.3.1 — Nonestimable constraints cannot be tested

Let $\boldsymbol{\lambda} \notin \mathcal{C}(\mathbf{X}^\top)$ and $c \in \mathbb{R}$.

- (a) For every $\boldsymbol{\beta} \in \mathbb{R}^p$ there is $\boldsymbol{\beta}'$ with $\mathbf{X}\boldsymbol{\beta}' = \mathbf{X}\boldsymbol{\beta}$ and $\boldsymbol{\lambda}^\top \boldsymbol{\beta}' = c$.
- (b) Let ϕ be any test of $H : \boldsymbol{\lambda}^\top \boldsymbol{\beta} = c$ in the model Eq. 7.3.1, with rejection probability $\pi(\boldsymbol{\beta}, \sigma^2) = E_{\boldsymbol{\beta}, \sigma^2} \phi(\mathbf{Y})$. For every point $(\boldsymbol{\beta}, \sigma^2)$ of the alternative there is a point $(\boldsymbol{\beta}', \sigma^2)$ of the hypothesis with $\pi(\boldsymbol{\beta}', \sigma^2) = \pi(\boldsymbol{\beta}, \sigma^2)$. So no test has power exceeding its size anywhere.

Proof.

- (a) By Theorem 8.3.1(c), $\boldsymbol{\lambda} \notin \mathcal{C}(\mathbf{X}^\top)$ means that $\boldsymbol{\lambda}$ is not orthogonal to $\mathcal{N}(\mathbf{X})$: there is $\mathbf{v}$ with $\mathbf{X}\mathbf{v} = \mathbf{0}$ and $\boldsymbol{\lambda}^\top \mathbf{v} \neq 0$. Put $\boldsymbol{\beta}' = \boldsymbol{\beta} + t\mathbf{v}$ with $t = (c - \boldsymbol{\lambda}^\top \boldsymbol{\beta})/\boldsymbol{\lambda}^\top \mathbf{v}$. Then $\mathbf{X}\boldsymbol{\beta}' = \mathbf{X}\boldsymbol{\beta}$ and $\boldsymbol{\lambda}^\top \boldsymbol{\beta}' = c$. (b) The law of $\mathbf{Y}$, and hence π , depends on $(\boldsymbol{\beta}, \sigma^2)$ only through $(\mathbf{X}\boldsymbol{\beta}, \sigma^2)$. Apply (a).

■

For a matrix $\mathbf{\Lambda}$, Proposition 9.2.6 gives the general statement: the constraint $\mathbf{\Lambda}^\top \mathbf{b} = \mathbf{0}$ restricts the mean vector only through its estimable part $\mathcal{C}(\mathbf{\Lambda}) \cap \mathcal{C}(\mathbf{X}^\top)$, and the extra sum of squares has $\dim(\mathcal{C}(\mathbf{\Lambda}) \cap \mathcal{C}(\mathbf{X}^\top))$ degrees of freedom. In the one-way layout $E(Y_{ij}) = \mu + \alpha_i$, the constraint $\alpha_1 = 0$ is a *side condition* (Theorem 8.4.2). It selects one of the equivalent parameterizations and says nothing about the data. Programs that report a “test” of $\alpha_1 = 0$ are testing something else, usually a contrast such as $\alpha_1 - \alpha_g$ that the program’s coding makes estimable.

11.3.2 The F statistic of a testable hypothesis

The sum of squares of a testable hypothesis was computed in Theorem 9.2.4 for $\mathbf{\Lambda}$ of full column rank. The next theorem removes that restriction, identifies the reduced model, and supplies the distribution.

Theorem 11.3.2 — The F test of a testable hypothesis

Assume Eq. 7.3.1 with $r < n$, and let $H : \mathbf{\Lambda}^\top \boldsymbol{\beta} = \mathbf{d}$ be testable with $q = \text{rank}(\mathbf{\Lambda})$. Let $\mathbf{G}$ be any generalized inverse of $\mathbf{X}^\top \mathbf{X}$, $\hat{\boldsymbol{\beta}}$ any least squares estimate, $\mathbf{W} = \mathbf{\Lambda}^\top \mathbf{G} \mathbf{\Lambda}$ and $\mathbf{W}^-$ any generalized inverse of $\mathbf{W}$. Put

$$SS_H = (\mathbf{\Lambda}^\top \hat{\boldsymbol{\beta}} - \mathbf{d})^\top \mathbf{W}^- (\mathbf{\Lambda}^\top \hat{\boldsymbol{\beta}} - \mathbf{d}), \quad F_H = \frac{SS_H/q}{SSE/(n-r)}. \quad (11.3.1)$$

- (a) SS_H does not depend on the choices of $\hat{\boldsymbol{\beta}}$, $\mathbf{G}$ and $\mathbf{W}^-$.
 (b) Let $\mathbf{b}_0$ be any solution of $\mathbf{\Lambda}^\top \mathbf{b}_0 = \mathbf{d}$ and $\mathcal{S}_0 = \{\mathbf{Xb} : \mathbf{\Lambda}^\top \mathbf{b} = \mathbf{0}\}$. Then $\mathcal{S}_0$ is a subspace of $\mathcal{C}(\mathbf{X})$ of dimension $r - q$, the set of mean vectors allowed by H is $\mathbf{Xb}_0 + \mathcal{S}_0$, and

$$SS_H = \min_{\mathbf{\Lambda}^\top \mathbf{b} = \mathbf{d}} \|\mathbf{y} - \mathbf{Xb}\|^2 - SSE = \|(\mathbf{M} - \mathbf{P}_{\mathcal{S}_0})(\mathbf{y} - \mathbf{Xb}_0)\|^2.$$

- (c) $F_H \sim F(q, n - r, \gamma)$ with

$$\gamma = \frac{(\mathbf{\Lambda}^\top \boldsymbol{\beta} - \mathbf{d})^\top \mathbf{W}^- (\mathbf{\Lambda}^\top \boldsymbol{\beta} - \mathbf{d})}{\sigma^2}, \quad (11.3.2)$$

and $\gamma = 0$ iff $\mathbf{\Lambda}^\top \boldsymbol{\beta} = \mathbf{d}$. In particular F_H is the F statistic of [Theorem 11.1.1](#) for the data $\mathbf{y} - \mathbf{Xb}_0$ and the reduced model space $\mathcal{S}_0$.

Proof. Reduction to full column rank. Choose q linearly independent columns of $\mathbf{\Lambda}$, forming a $p \times q$ matrix $\mathbf{\Lambda}_1$. Every column of $\mathbf{\Lambda}$ is a combination of them, so $\mathbf{\Lambda} = \mathbf{\Lambda}_1 \mathbf{K}$ for a $q \times m$ matrix $\mathbf{K}$, which has rank q because $\text{rank } \mathbf{\Lambda} = q$. Hence $\mathbf{K}\mathbf{K}^\top$ is nonsingular. Put $\mathbf{d}_1 = \mathbf{\Lambda}_1^\top \mathbf{b}_0$, so that $\mathbf{d} = \mathbf{K}^\top \mathbf{d}_1$. Since $\mathbf{K}^\top$ has full column rank it is one-to-one, so $\mathbf{d}_1$ is determined by $\mathbf{d}$ and does not depend on the choice of $\mathbf{b}_0$. Moreover $\mathbf{\Lambda}^\top \mathbf{b} = \mathbf{d} \iff \mathbf{K}^\top (\mathbf{\Lambda}_1^\top \mathbf{b} - \mathbf{d}_1) = \mathbf{0} \iff \mathbf{\Lambda}_1^\top \mathbf{b} = \mathbf{d}_1$. The same argument with $\mathbf{d}$ replaced by $\mathbf{0}$ shows that $\mathcal{S}_0$ is also $\{\mathbf{Xb} : \mathbf{\Lambda}_1^\top \mathbf{b} = \mathbf{0}\}$. The columns of $\mathbf{\Lambda}_1$ are columns of $\mathbf{\Lambda}$, so $\mathbf{\Lambda}_1^\top \boldsymbol{\beta}$ is estimable.

(a). By [Theorem 9.2.4\(a\)](#), $\mathbf{u}_1 = \mathbf{\Lambda}_1^\top \hat{\boldsymbol{\beta}} - \mathbf{d}_1$ and the positive definite matrix $\mathbf{W}_1 = \mathbf{\Lambda}_1^\top \mathbf{G} \mathbf{\Lambda}_1$ do not depend on $\hat{\boldsymbol{\beta}}$ and $\mathbf{G}$. Now $\mathbf{u} = \mathbf{\Lambda}^\top \hat{\boldsymbol{\beta}} - \mathbf{d} = \mathbf{K}^\top \mathbf{u}_1$ and $\mathbf{W} = \mathbf{K}^\top \mathbf{W}_1 \mathbf{K}$. Let $\mathbf{v} = \mathbf{K}(\mathbf{K}\mathbf{K}^\top)^{-1} \mathbf{W}_1^{-1} \mathbf{u}_1$, so that $\mathbf{Kv} = \mathbf{W}_1^{-1} \mathbf{u}_1$. Then $\mathbf{Wv} = \mathbf{K}^\top \mathbf{W}_1 \mathbf{Kv} = \mathbf{K}^\top \mathbf{u}_1 = \mathbf{u}$, and for every generalized inverse,

$$\mathbf{u}^\top \mathbf{W}^- \mathbf{u} = \mathbf{v}^\top \mathbf{W} \mathbf{W}^- \mathbf{Wv} = \mathbf{v}^\top \mathbf{Wv} = (\mathbf{Kv})^\top \mathbf{W}_1 (\mathbf{Kv}) = \mathbf{u}_1^\top \mathbf{W}_1^{-1} \mathbf{u}_1.$$

The right side involves none of the choices.

(b). $\mathcal{S}_0$ is the image of the subspace $\mathcal{N}(\mathbf{\Lambda}^\top)$ under $\mathbf{b} \mapsto \mathbf{Xb}$, so it is a subspace of $\mathcal{C}(\mathbf{X})$. Its dimension is $r - \dim(\mathcal{C}(\mathbf{\Lambda}) \cap \mathcal{C}(\mathbf{X}^\top)) = r - q$ by [Proposition 9.2.6](#), because $\mathcal{C}(\mathbf{\Lambda}) \subseteq \mathcal{C}(\mathbf{X}^\top)$. The solutions of $\mathbf{\Lambda}^\top \mathbf{b} = \mathbf{d}$ are $\mathbf{b}_0 + \mathcal{N}(\mathbf{\Lambda}^\top)$, so the allowed means are $\mathbf{Xb}_0 + \mathcal{S}_0$. For the minimum, write $\mathbf{y}_* = \mathbf{y} - \mathbf{Xb}_0$. As $\mathbf{b}$ ranges over the constraint set, $\mathbf{Xb} - \mathbf{Xb}_0$ ranges over $\mathcal{S}_0$, so

$$\min_{\mathbf{\Lambda}^\top \mathbf{b} = \mathbf{d}} \|\mathbf{y} - \mathbf{Xb}\|^2 = \min_{\mathbf{m} \in \mathcal{S}_0} \|\mathbf{y}_* - \mathbf{m}\|^2 = \|(\mathbf{I} - \mathbf{P}_{\mathcal{S}_0})\mathbf{y}_*\|^2$$

by [Theorem 6.3.1](#). Also $SSE = \|(\mathbf{I} - \mathbf{M})\mathbf{y}\|^2 = \|(\mathbf{I} - \mathbf{M})\mathbf{y}_*\|^2$, since $\mathbf{Xb}_0 \in \mathcal{C}(\mathbf{X})$. Because $\mathcal{S}_0 \subseteq \mathcal{C}(\mathbf{X})$, [Eq. 6.5.2](#) turns the difference into $\|(\mathbf{M} - \mathbf{P}_{\mathcal{S}_0})\mathbf{y}_*\|^2$. By [Theorem 9.2.4\(c\)](#) applied to $\mathbf{\Lambda}_1$ and $\mathbf{d}_1$, the same difference equals $\mathbf{u}_1^\top \mathbf{W}_1^{-1} \mathbf{u}_1$, which is SS_H by (a).

(c). $\mathbf{Y} - \mathbf{Xb}_0 \sim N_n(\mathbf{X}(\boldsymbol{\beta} - \mathbf{b}_0), \sigma^2 \mathbf{I})$ is a normal linear model with mean in $\mathcal{C}(\mathbf{X})$, and by (b) F_H is its F statistic for the reduced model space $\mathcal{S}_0$, which has dimension $r - q$. [Theorem 11.1.1](#) gives $F_H \sim F(q, n - r, \gamma)$ with $\sigma^2 \gamma = \|(\mathbf{M} - \mathbf{P}_{\mathcal{S}_0})(\mathbf{X}\boldsymbol{\beta} - \mathbf{Xb}_0)\|^2$. To evaluate this, note that the identity in (b) holds for every data vector. Apply it to the

vector $\mathbf{X}\boldsymbol{\beta}$, for which $\boldsymbol{\beta}$ itself is a least squares estimate: $\sigma^2\gamma = (\boldsymbol{\Lambda}^\top \boldsymbol{\beta} - \mathbf{d})^\top \mathbf{W}^{-1}(\boldsymbol{\Lambda}^\top \boldsymbol{\beta} - \mathbf{d})$. Finally, by (a) this equals $\mathbf{u}_1(\boldsymbol{\beta})^\top \mathbf{W}_1^{-1} \mathbf{u}_1(\boldsymbol{\beta})$ with $\mathbf{u}_1(\boldsymbol{\beta}) = \boldsymbol{\Lambda}_1^\top \boldsymbol{\beta} - \mathbf{d}_1$. It is zero iff $\boldsymbol{\Lambda}_1^\top \boldsymbol{\beta} = \mathbf{d}_1$, that is, iff $\boldsymbol{\Lambda}^\top \boldsymbol{\beta} = \mathbf{d}$. ■

Formula Eq. 11.3.1 needs only the unrestricted fit. Its ingredients have a statistical meaning: $\boldsymbol{\Lambda}^\top \hat{\boldsymbol{\beta}} - \mathbf{d}$ is the estimated departure from the hypothesis, and $\sigma^2 \mathbf{W}$ is its covariance matrix (Theorem 7.5.1(a)). So SS_H/σ^2 is the squared Mahalanobis length of the estimated departure, and the noncentrality Eq. 11.3.2 is the same length for the true departure. Under H the quadratic form SS_H/σ^2 is $\chi^2(q)$, which is Corollary 7.5.3. The theorem adds the reduced model, the distribution under alternatives, and the fact that a redundant $\boldsymbol{\Lambda}$ can be used as it stands, with any generalized inverse of $\mathbf{W}$.

The test space of the hypothesis is the orthogonal complement of $\mathcal{S}_0$ within $\mathcal{C}(\mathbf{X})$. By Theorem 6.5.2 it is $\mathcal{C}(\mathbf{MT})$, where $\boldsymbol{\Lambda} = \mathbf{X}^\top \mathbf{T}$: the hypothesis constrains the mean vector to be orthogonal to the fitted values of the columns of $\mathbf{T}$.

11.3.3 Least squares under the hypothesis

The minimum in Theorem 11.3.2(b) is attained, and its minimizers have a closed form. The following extends Theorem 9.2.5, which assumed full rank, to every model matrix.

Proposition 11.3.3 — Least squares under a testable hypothesis

Let $H : \boldsymbol{\Lambda}^\top \boldsymbol{\beta} = \mathbf{d}$ be testable with $\boldsymbol{\Lambda}$ of full column rank q , and write $\boldsymbol{\Lambda} = \mathbf{X}^\top \mathbf{T}$. Let $\mathbf{G}$ be a generalized inverse of $\mathbf{X}^\top \mathbf{X}$, $\hat{\boldsymbol{\beta}} = \mathbf{GX}^\top \mathbf{y}$, $\mathbf{W} = \boldsymbol{\Lambda}^\top \mathbf{G} \boldsymbol{\Lambda}$ and $\mathbf{u} = \boldsymbol{\Lambda}^\top \hat{\boldsymbol{\beta}} - \mathbf{d}$. Then $\mathbf{W}$ is positive definite, and:

- (a) $\hat{\boldsymbol{\beta}}_H = \hat{\boldsymbol{\beta}} - \mathbf{G} \boldsymbol{\Lambda} \mathbf{W}^{-1} \mathbf{u}$ satisfies $\boldsymbol{\Lambda}^\top \hat{\boldsymbol{\beta}}_H = \mathbf{d}$ and minimizes $\|\mathbf{y} - \mathbf{X}\mathbf{b}\|^2$ subject to $\boldsymbol{\Lambda}^\top \mathbf{b} = \mathbf{d}$. The set of all minimizers is $\hat{\boldsymbol{\beta}}_H + (\mathcal{N}(\mathbf{X}) \cap \mathcal{N}(\boldsymbol{\Lambda}^\top))$.
- (b) The restricted fitted vector is unique: $\mathbf{X}\hat{\boldsymbol{\beta}}_H = \mathbf{M}\mathbf{y} - \mathbf{MTW}^{-1} \mathbf{u} = \mathbf{X}\mathbf{b}_0 + \mathbf{P}_{\mathcal{S}_0}(\mathbf{y} - \mathbf{X}\mathbf{b}_0)$, and $\|\mathbf{y} - \mathbf{X}\hat{\boldsymbol{\beta}}_H\|^2 = \text{SSE} + \text{SS}_H$, with $\text{SS}_H = \mathbf{u}^\top \mathbf{W}^{-1} \mathbf{u}$.
- (c) With $\hat{\boldsymbol{\theta}} = \mathbf{W}^{-1} \mathbf{u}$, the restricted estimate solves the constrained normal equations (compare Theorem 6.4.2)

$$\mathbf{X}^\top \mathbf{X} \hat{\boldsymbol{\beta}}_H + \boldsymbol{\Lambda} \hat{\boldsymbol{\theta}} = \mathbf{X}^\top \mathbf{y}, \quad \boldsymbol{\Lambda}^\top \hat{\boldsymbol{\beta}}_H = \mathbf{d},$$

$$\text{and } \text{SS}_H = \hat{\boldsymbol{\theta}}^\top \mathbf{W} \hat{\boldsymbol{\theta}}.$$

Proof. $\mathbf{W} = \mathbf{T}^\top \mathbf{X} \mathbf{G} \mathbf{X}^\top \mathbf{T} = \mathbf{T}^\top \mathbf{MT}$ by Theorem 6.3.7. It is symmetric, and positive definite by Theorem 9.2.4(a). Also $\mathbf{X} \mathbf{G} \boldsymbol{\Lambda} = \mathbf{X} \mathbf{G} \mathbf{X}^\top \mathbf{T} = \mathbf{MT}$.

- (a) $\boldsymbol{\Lambda}^\top \hat{\boldsymbol{\beta}}_H = \boldsymbol{\Lambda}^\top \hat{\boldsymbol{\beta}} - \mathbf{W} \mathbf{W}^{-1} \mathbf{u} = \mathbf{d}$. For any feasible $\mathbf{b}$, Eq. 6.3.1 with $\mathbf{u} = \mathbf{X}\mathbf{b}$ gives $\|\mathbf{y} - \mathbf{X}\mathbf{b}\|^2 = \text{SSE} + \|\mathbf{X}\hat{\boldsymbol{\beta}} - \mathbf{X}\mathbf{b}\|^2$. Write $\mathbf{b} = \hat{\boldsymbol{\beta}}_H + \mathbf{v}$ with $\boldsymbol{\Lambda}^\top \mathbf{v} = \mathbf{0}$. Then $\mathbf{X}\hat{\boldsymbol{\beta}} - \mathbf{X}\mathbf{b} = \mathbf{MTW}^{-1} \mathbf{u} - \mathbf{X}\mathbf{v}$, and the cross term vanishes:

$$(\mathbf{MTW}^{-1} \mathbf{u})^\top \mathbf{X}\mathbf{v} = \mathbf{u}^\top \mathbf{W}^{-1} \mathbf{T}^\top \mathbf{X}\mathbf{v} = \mathbf{u}^\top \mathbf{W}^{-1} \boldsymbol{\Lambda}^\top \mathbf{v} = 0,$$

using $\mathbf{MX} = \mathbf{X}$. Hence $\|\mathbf{y} - \mathbf{X}\mathbf{b}\|^2 = \text{SSE} + \|\mathbf{MTW}^{-1}\mathbf{u}\|^2 + \|\mathbf{X}\mathbf{v}\|^2$, which is smallest exactly when $\mathbf{X}\mathbf{v} = \mathbf{0}$. The minimizers are $\hat{\boldsymbol{\beta}}_H + \mathbf{v}$ with $\mathbf{v} \in \mathcal{N}(\mathbf{X}) \cap \mathcal{N}(\mathbf{\Lambda}^\top)$.

- (b) $\mathbf{X}\hat{\boldsymbol{\beta}}_H = \mathbf{X}\hat{\boldsymbol{\beta}} - \mathbf{XG}\mathbf{\Lambda}\mathbf{W}^{-1}\mathbf{u} = \mathbf{M}\mathbf{y} - \mathbf{MTW}^{-1}\mathbf{u}$, and both terms are unique by [Theorem 9.2.4\(a\)](#). The minimum value is $\text{SSE} + \mathbf{u}^\top \mathbf{W}^{-1} \mathbf{T}^\top \mathbf{MTW}^{-1} \mathbf{u} = \text{SSE} + \mathbf{u}^\top \mathbf{W}^{-1} \mathbf{u}$, and $\mathbf{u}^\top \mathbf{W}^{-1} \mathbf{u} = \text{SS}_H$ by [Theorem 11.3.2\(a\)](#). The fitted vector $\mathbf{X}\hat{\boldsymbol{\beta}}_H$ is the point of $\mathbf{X}\mathbf{b}_0 + \mathcal{S}_0$ nearest to $\mathbf{y}$, which is $\mathbf{X}\mathbf{b}_0 + \mathbf{P}_{\mathcal{S}_0}(\mathbf{y} - \mathbf{X}\mathbf{b}_0)$ by the argument in the proof of [Theorem 11.3.2\(b\)](#).
- (c) $\mathbf{X}^\top \mathbf{X}\hat{\boldsymbol{\beta}}_H = \mathbf{X}^\top \mathbf{X}\hat{\boldsymbol{\beta}} - \mathbf{X}^\top \mathbf{XG}\mathbf{\Lambda}\mathbf{W}^{-1}\mathbf{u} = \mathbf{X}^\top \mathbf{y} - \mathbf{X}^\top \mathbf{MT}\hat{\boldsymbol{\theta}} = \mathbf{X}^\top \mathbf{y} - \mathbf{\Lambda}\hat{\boldsymbol{\theta}}$, because $\mathbf{X}^\top \mathbf{X}\hat{\boldsymbol{\beta}} = \mathbf{X}^\top \mathbf{y}$ and $\mathbf{X}^\top \mathbf{M} = \mathbf{X}^\top$. The last identity is $\mathbf{u}^\top \mathbf{W}^{-1} \mathbf{u} = \hat{\boldsymbol{\theta}}^\top \mathbf{W}\hat{\boldsymbol{\theta}}$.

■

Part (c) identifies $\hat{\boldsymbol{\theta}}$ as a vector of Lagrange multipliers (compare [Proposition 1.11.3](#)). Each multiplier measures how hard the data push against its constraint, and the hypothesis sum of squares is their squared Mahalanobis length. This is the “Lagrange multiplier” form of the test met again in [Section 11.4](#).

11.3.4 Nonzero right-hand sides: offsets

When $\mathbf{d} \neq \mathbf{0}$ the hypothesis restricts the mean to the *affine* set $\mathbf{X}\mathbf{b}_0 + \mathcal{S}_0$, not to a subspace. [Theorem 11.3.2\(c\)](#) shows how to handle it: subtract a known point of the set from the data. The reduced model becomes

$$\mathbf{E}(\mathbf{Y} - \mathbf{X}\mathbf{b}_0) \in \mathcal{S}_0,$$

a linear model for the adjusted response $\mathbf{Y} - \mathbf{X}\mathbf{b}_0$ with a **known offset** $\mathbf{X}\mathbf{b}_0$. Any solution $\mathbf{b}_0$ of $\mathbf{\Lambda}^\top \mathbf{b}_0 = \mathbf{d}$ may be used. Two solutions differ by an element of $\mathcal{N}(\mathbf{\Lambda}^\top)$, so their offsets differ by an element of $\mathcal{S}_0$, which the reduced model absorbs. In practice one fits $\mathbf{y} - \mathbf{X}\mathbf{b}_0$ on a basis of $\mathcal{S}_0$, for example the columns of $\mathbf{X}\mathbf{U}$ where the columns of $\mathbf{U}$ span $\mathcal{N}(\mathbf{\Lambda}^\top)$, and takes the difference of residual sums of squares. The formula [Eq. 11.3.1](#) gives the same number with no second fit.

Example. (Regional hypotheses with an over-parameterized model)

Return to the state murder data, now with an intercept *and* an indicator for each of the four regions, so that $\boldsymbol{\beta} = (\mu, \alpha_N, \alpha_M, \alpha_S, \alpha_W, \beta_{\text{pov}}, \beta_{\text{sin}}, \beta_{\text{urb}})^\top$. The model matrix is 50×8 of rank 7. No region effect is estimable on its own, but every contrast among them is.

All four regions equal. Write the hypothesis with all six pairwise differences $\alpha_i - \alpha_j$ as the columns of $\mathbf{\Lambda}$. They are consistent and have rank $q = 3$. With the Moore–Penrose inverses of $\mathbf{X}^\top \mathbf{X}$ and of $\mathbf{W}$, [Eq. 11.3.1](#) gives $\text{SS}_H = 14.83$ and $F = 2.440$, exactly the reduced-model test of [Example \(Do regions matter?\)](#), as [Theorem 11.3.2\(b\)](#) requires. A second generalized inverse, made by deleting the row and column of α_W from $\mathbf{X}^\top \mathbf{X}$ before inverting, gives a different $\hat{\boldsymbol{\beta}}$ but the same $\mathbf{\Lambda}^\top \hat{\boldsymbol{\beta}}$, $\mathbf{W}$ and F .

Midwest, South and West equal. With $\mathbf{\Lambda}$ made of $\alpha_M - \alpha_S$ and $\alpha_S - \alpha_W$, the sum of squares is 3.881 on 2 degrees of freedom, $F = 0.958$, $p = 0.392$. The restricted fit of [Proposition 11.3.3](#) coincides with the fit of the model in which the three regions are merged. Under the constraint, the Northeast lies 1.379 below the common level of the

other three, and the coefficient of single parenthood moves from 0.428 to 0.441.

A nonzero right-hand side. The contrast $\alpha_N - (\alpha_M + \alpha_S + \alpha_W)/3$, the Northeast against the average of the other regions, is estimated as $\hat{\psi} = -1.426$ with standard error 0.595. To test whether it equals -1 , take the offset $\mathbf{b}_0 = (0, -1, 0, \dots, 0)^\top$ and fit $\mathbf{y} - \mathbf{X}\mathbf{b}_0$ on a basis of $\mathcal{S}_0$. The drop in residual sum of squares is 1.040, equal to $(\hat{\psi} + 1)^2 / \lambda^\top \mathbf{G} \lambda$ as Eq. 11.3.1 predicts, and $F = 0.513$ with $p = 0.478$. Any other solution of $\lambda^\top \mathbf{b}_0 = -1$ gives the same answer.

A constraint that is not a hypothesis. The constraint $\alpha_N = 0$ is not estimable. Minimizing the residual sum of squares subject to it gives exactly SSE: the constrained model spans the whole of $\mathcal{C}(\mathbf{X})$, in agreement with Proposition 11.3.1.

```
import numpy as np
import statsmodels.api as sm
from scipy import linalg, stats

data = sm.datasets.statcrime.load_pandas().data
data.index = data.index.str.strip()
data = data.drop(index="District of Columbia")
codes = "SWWSWNSSSWMMMMSSNSNMMSMWNNWNMSMMSWNMSWNSWSMW"
region = np.array(list(codes))
y = data["murder"].to_numpy()
n = len(y)
D = np.column_stack([(region == g).astype(float) for g in "NMSW"]) # all four
                                regions
X = np.column_stack([np.ones(n), D, data["poverty"], data["single"], data["urban"]])
r = np.linalg.matrix_rank(X) # 7, not 8

G = np.linalg.pinv(X.T @ X) # one generalized inverse of X'X
b = G @ X.T @ y # one least squares solution
s2 = np.sum((y - X @ b) ** 2) / (n - r)

def glh_test(Lam, d):
    """F test of the testable hypothesis Lam' beta = d, via any g-inverse of Lam' G
    Lam. """
    u = Lam.T @ b - d
    W = Lam.T @ G @ Lam
    q = np.linalg.matrix_rank(Lam)
    ss_h = u @ np.linalg.pinv(W) @ u
    F = ss_h / q / s2
    return ss_h, q, F, stats.f.sf(F, q, n - r)

def contrast(pairs):
    """Columns e_i - e_j on the region coefficients (positions 1..4 = N, M, S, W).
    """
    L = np.zeros((8, len(pairs)))
    for k, (i, j) in enumerate(pairs):
        L[1 + i, k], L[1 + j, k] = 1.0, -1.0
    return L

all_pairs = contrast([(0, 1), (0, 2), (0, 3), (1, 2), (1, 3), (2, 3)]) # rank 3,
                                redundant
three_equal = contrast([(1, 2), (2, 3)]) # M = S = W
for name, L in (("four regions equal", all_pairs), ("M = S = W", three_equal)):
    ss_h, q, F, p = glh_test(L, np.zeros(L.shape[1]))
    print(f"{name:20s} SS_H = {ss_h:.3f} q = {q} F = {F:.3f} p = {p:.4f}")
```

The offset computation fits the reduced model directly, as a check on the formula.

```
def sse(Z, target):
    coef, *_ = np.linalg.lstsq(Z, target, rcond=None)
    return np.sum((target - Z @ coef) ** 2)

sse_full = sse(X, y)
lam = np.array([0, 1, -1 / 3, -1 / 3, -1 / 3, 0, 0, 0]) # Northeast minus the
other three
d = -1.0 # hypothesized difference
b0 = np.array([0, -1.0, 0, 0, 0, 0, 0, 0]) # one solution of lam' b0
    = d
U = linalg.null_space(lam[None, :]) # beta - b0 ranges over C(
    U)
sse_H = sse(X @ U, y - X @ b0) # reduced model, offset X
    b0
est = lam @ b
se = np.sqrt(s2 * lam @ G @ lam)
print(f"estimate {est:.3f} (se {se:.3f}); SSE_H - SSE = {sse_H - sse_full:.3f}")
print(f"formula: {(est - d) ** 2 / (lam @ G @ lam):.3f}")
```

The restricted estimates, from two different generalized inverses:

```
keep = [0, 1, 2, 3, 5, 6, 7]
G2 = np.zeros((8, 8))
G2[np.ix_(keep, keep)] = np.linalg.inv((X.T @ X)[np.ix_(keep, keep)])

def restricted(Lam, d, Ginv):
    """Least squares under the testable constraint Lam' b = d (Lam of full column
    rank)."""
    bh = Ginv @ X.T @ y
    W = Lam.T @ Ginv @ Lam
    return bh - Ginv @ Lam @ np.linalg.solve(W, Lam.T @ bh - d)

bH = restricted(three_equal, np.zeros(2), G)
bH2 = restricted(three_equal, np.zeros(2), G2)
print("constraint holds:", np.allclose(three_equal.T @ bH, 0))
print("fitted values agree:", np.allclose(X @ bH, X @ bH2))
```

Warning (Hypotheses chosen after seeing the data). The same Northeast contrast, tested against zero, gives $t = -2.398$ and $p = 0.021$. That p-value would be valid had the contrast been specified before the data were examined. It was not: it was suggested by the estimates, after the joint test of all regional differences had failed to reject. Among the many contrasts one could have tested, the one that looks largest is, by construction, the one most likely to look significant by chance. [Chapter 13](#) gives procedures, notably Scheffé's, that remain valid when the contrast is chosen by looking at the data.

11.3.5 Exercises

11.3.5.1 A. Check your understanding

Exercise A1.

In the one-way layout $E(Y_{ij}) = \mu + \alpha_i$, $i = 1, 2, 3$, with all groups nonempty, decide which of the following are testable hypotheses: (i) $\alpha_1 = \alpha_2$; (ii) $\alpha_1 = 0$; (iii) $\alpha_1 + \alpha_2 + \alpha_3 = 0$; (iv) $\mu + \alpha_1 = 5$; (v) $\alpha_1 - \alpha_2 = 1$, $\alpha_2 - \alpha_3 = 2$, $\alpha_1 - \alpha_3 = 3$; (vi) $\alpha_1 - \alpha_2 = 1$, $\alpha_2 - \alpha_3 = 2$, $\alpha_1 - \alpha_3 = 4$. Give q for those that are.

Solution. The estimable functions are the combinations $c_0\mu + \sum_i c_i\alpha_i$ with $c_0 = \sum_i c_i$ (Example (Four answers, one fit) and Section 8.2). (i) is testable, $q = 1$. (ii) and (iii) are not estimable: their coefficient of μ is 0 while the α coefficients sum to 1 and 3. They are side conditions (Theorem 8.4.2). (iv) is testable, $q = 1$. (v) is testable with $q = 2$: the three differences are estimable, have rank 2, and the right-hand sides are consistent because $1 + 2 = 3$. (vi) is estimable but inconsistent, since $(\alpha_1 - \alpha_2) + (\alpha_2 - \alpha_3) = \alpha_1 - \alpha_3$ would need $1 + 2 = 4$. No parameter satisfies it.

11.3.5.2 B. Practice**Exercise B1.**

Let Λ be $p \times m$ and $\mathbf{A}$ a nonsingular $m \times m$ matrix. Show that the hypotheses $\Lambda^\top \boldsymbol{\beta} = \mathbf{d}$ and $(\Lambda\mathbf{A})^\top \boldsymbol{\beta} = \mathbf{A}^\top \mathbf{d}$ are the same, and that Eq. 11.3.1 gives the same SS_H for both. Conclude that the test does not depend on how a set of constraints is written.

Solution. $(\Lambda\mathbf{A})^\top \mathbf{b} = \mathbf{A}^\top \mathbf{d} \iff \mathbf{A}^\top (\Lambda^\top \mathbf{b} - \mathbf{d}) = \mathbf{0} \iff \Lambda^\top \mathbf{b} = \mathbf{d}$, since $\mathbf{A}^\top$ is nonsingular. So the constraint sets, and by Theorem 11.3.2(b) the sums of squares, agree. Directly: with $\mathbf{u} = \Lambda^\top \hat{\boldsymbol{\beta}} - \mathbf{d}$, the new departure is $\mathbf{A}^\top \mathbf{u}$ and the new middle matrix is $\mathbf{A}^\top \mathbf{W} \mathbf{A}$, and one generalized inverse of it is $\mathbf{A}^{-1} \mathbf{W}^{-1} \mathbf{A}^{-\top}$, which gives $\mathbf{u}^\top \mathbf{A} \mathbf{A}^{-1} \mathbf{W}^{-1} \mathbf{A}^{-\top} \mathbf{A}^\top \mathbf{u} = \mathbf{u}^\top \mathbf{W}^{-1} \mathbf{u}$. The choice of generalized inverse does not matter by Theorem 11.3.2(a).

Exercise B2.

In Proposition 11.3.3, show that $E(\mathbf{X}\hat{\boldsymbol{\beta}}_H) = \mathbf{X}\boldsymbol{\beta} - \mathbf{M}\mathbf{T}\mathbf{W}^{-1}(\Lambda^\top \boldsymbol{\beta} - \mathbf{d})$ and $\text{Cov}(\mathbf{X}\hat{\boldsymbol{\beta}}_H) = \sigma^2 \mathbf{P}_{S_0}$, where $\mathbf{P}_{S_0} = \mathbf{M} - \mathbf{M}\mathbf{T}\mathbf{W}^{-1}\mathbf{T}^\top \mathbf{M}$. Conclude that restricted fitted values are unbiased when H holds, have smaller variance than $\mathbf{M}\mathbf{Y}$ in every direction, and are biased when H fails.

Solution. $\mathbf{X}\hat{\boldsymbol{\beta}}_H = \mathbf{M}\mathbf{Y} - \mathbf{M}\mathbf{T}\mathbf{W}^{-1}(\mathbf{T}^\top \mathbf{M}\mathbf{Y} - \mathbf{d})$, using $\Lambda^\top \hat{\boldsymbol{\beta}} = \mathbf{T}^\top \mathbf{M}\mathbf{Y}$. Its mean is $\mathbf{X}\boldsymbol{\beta} - \mathbf{M}\mathbf{T}\mathbf{W}^{-1}(\mathbf{T}^\top \mathbf{X}\boldsymbol{\beta} - \mathbf{d})$, and $\mathbf{T}^\top \mathbf{X}\boldsymbol{\beta} = \Lambda^\top \boldsymbol{\beta}$. The random part is $(\mathbf{M} - \mathbf{M}\mathbf{T}\mathbf{W}^{-1}\mathbf{T}^\top \mathbf{M})\mathbf{Y}$. The matrix in parentheses is the projection onto $\mathcal{C}(\mathbf{X})$ minus the projection onto $\mathcal{C}(\mathbf{M}\mathbf{T})$ (Theorem 6.3.7, since $\mathbf{W} = \mathbf{T}^\top \mathbf{M}\mathbf{T}$), which is $\mathbf{P}_{S_0}$ by Theorem 6.5.2. It is symmetric idempotent, so the covariance is $\sigma^2 \mathbf{P}_{S_0}$. The difference $\sigma^2(\mathbf{M} - \mathbf{P}_{S_0})$ from $\text{Cov}(\mathbf{M}\mathbf{Y})$ is a projection times σ^2 , so it is nonnegative definite. The bias is $-\mathbf{M}\mathbf{T}\mathbf{W}^{-1}(\Lambda^\top \boldsymbol{\beta} - \mathbf{d})$, zero iff H holds, because $\mathbf{M}\mathbf{T}$ has full column rank.

Exercise B3.

In simple regression $E(Y_i) = \beta_0 + \beta_1 x_i$, derive the F statistic for $\beta_1 = 1$ in two ways:

from Eq. 11.3.1, and by regressing $y_i - x_i$ on an intercept and comparing with the full fit. Show that both give $F = (\hat{\beta}_1 - 1)^2 S_{xx}/s^2$.

Exercise B4.

Show that the hypothesis $\mathbf{X}\boldsymbol{\beta} = \mathbf{X}\boldsymbol{\beta}_0$, with $\boldsymbol{\beta}_0$ known, is testable, that $q = r$, and that its F statistic is $\{\|\mathbf{M}(\mathbf{y} - \mathbf{X}\boldsymbol{\beta}_0)\|^2/r\}/s^2$. What is the reduced model?

11.3.5.3 C. Going deeper

Exercise C1.

In the one-way layout with two groups of sizes 2 and 3, take $\boldsymbol{\Lambda} = (0, 1, 0)^\top$, so that the “hypothesis” is $\alpha_1 = 0$. Evaluate $(\boldsymbol{\Lambda}^\top \hat{\boldsymbol{\beta}})^2/(\boldsymbol{\Lambda}^\top \mathbf{G} \boldsymbol{\Lambda})$ for two different generalized inverses $\mathbf{G}$ of $\mathbf{X}^\top \mathbf{X}$ and show that the values differ. Explain which step of the proof of Theorem 11.3.2 fails.

Exercise C2.

Let $\mathbf{X}$ have full column rank and write $\boldsymbol{\Lambda}^\top = [\boldsymbol{\Lambda}_1^\top, \boldsymbol{\Lambda}_2^\top]$ with $\boldsymbol{\Lambda}_2^\top$ a nonsingular $q \times q$ block, after permuting the coefficients so that $\boldsymbol{\beta} = (\boldsymbol{\beta}_1^\top, \boldsymbol{\beta}_2^\top)^\top$ conformably. Show that $\boldsymbol{\Lambda}^\top \boldsymbol{\beta} = \mathbf{0}$ is equivalent to $\boldsymbol{\beta}_2 = -\boldsymbol{\Lambda}_2^{-\top} \boldsymbol{\Lambda}_1^\top \boldsymbol{\beta}_1$, that the reduced model matrix is $\mathbf{X}_1 - \mathbf{X}_2 \boldsymbol{\Lambda}_2^{-\top} \boldsymbol{\Lambda}_1^\top$, and that this matrix has full column rank $p - q$. This is how the reduced model is fitted when no offset or null-space basis is at hand.

11.4 The likelihood ratio test and its relatives

The F test was motivated in Section 11.1 by a heuristic: compare the extra sum of squares with the noise level. General statistical theory offers other routes to a test, chiefly the likelihood ratio, the Wald statistic and the score statistic. In the normal linear model all three routes lead back to F . They differ only when they are calibrated by their common large-sample χ^2 approximation instead of the exact distribution, and then they can disagree about the same data. This section derives all three, shows how different the approximate versions can be, and then proves that the F test is the best test among those that respect the symmetries of Section 11.2.

11.4.1 The likelihood ratio

For a hypothesis H_0 about the parameter of a family of densities $L(\boldsymbol{\theta}; \mathbf{y})$, the **likelihood ratio** is

$$\Lambda(\mathbf{y}) = \frac{\sup_{\boldsymbol{\theta} \in H_0} L(\boldsymbol{\theta}; \mathbf{y})}{\sup_{\boldsymbol{\theta}} L(\boldsymbol{\theta}; \mathbf{y})},$$

and the likelihood ratio test rejects when Λ is small, that is, when the best explanation the hypothesis can offer is much worse than the best explanation overall. In the linear model both suprema are available in closed form from [Theorem 7.3.1](#), and [Exercise \(A2\)](#) already carried out the computation for reduced models. The theorem records it for testable hypotheses, including the calibration.

Theorem 11.4.1 — The F test is the likelihood ratio test

Assume [Eq. 7.3.1](#) with $r < n$, and let $H : \mathbf{\Lambda}^\top \boldsymbol{\beta} = \mathbf{d}$ be testable with $q = \text{rank}(\mathbf{\Lambda})$. Let $\text{SSE}_H = \min_{\mathbf{\Lambda}^\top \mathbf{b} = \mathbf{d}} \|\mathbf{y} - \mathbf{X}\mathbf{b}\|^2$ and suppose $\text{SSE} > 0$, which holds with probability one. Then:

- (a) under H the likelihood is maximized at any minimizer $\hat{\boldsymbol{\beta}}_H$ of $\|\mathbf{y} - \mathbf{X}\mathbf{b}\|^2$ subject to $\mathbf{\Lambda}^\top \mathbf{b} = \mathbf{d}$ (for $\mathbf{\Lambda}$ of full column rank, the estimate of [Proposition 11.3.3](#); compare [Theorem 9.2.5](#)), with $\tilde{\sigma}_H^2 = \text{SSE}_H/n$;
- (b) the likelihood ratio is

$$\Lambda = \left(\frac{\text{SSE}}{\text{SSE}_H} \right)^{n/2} = \left(1 + \frac{q F_H}{n - r} \right)^{-n/2}, \quad (11.4.1)$$

a strictly decreasing function of the statistic F_H of [Eq. 11.3.1](#);

- (c) the likelihood ratio test of size α rejects exactly when $F_H > F_\alpha(q, n - r)$.

The same holds for a reduced model $E(\mathbf{Y}) \in \mathcal{C}(\mathbf{X}_0)$, with $q = r - r_0$ and the statistic of [Theorem 11.1.1](#).

Proof.

- (a) By [Theorem 11.3.2\(b\)](#), H holds iff $E(\mathbf{Y}) \in \mathbf{X}\mathbf{b}_0 + \mathcal{S}_0$. Maximizing the likelihood over H is therefore maximizing it in the linear model $\mathbf{Y} - \mathbf{X}\mathbf{b}_0 \sim N_n(\mathbf{m}, \sigma^2 \mathbf{I})$, $\mathbf{m} \in \mathcal{S}_0$. By [Theorem 7.3.1](#) applied to that model (with any matrix whose columns span $\mathcal{S}_0$), the maximizing mean is the least squares fit and the maximizing variance is its residual sum of squares divided by n . That fit is $\mathbf{P}_{\mathcal{S}_0}(\mathbf{y} - \mathbf{X}\mathbf{b}_0)$, so the maximizing mean of $\mathbf{Y}$ is $\mathbf{X}\mathbf{b}_0 + \mathbf{P}_{\mathcal{S}_0}(\mathbf{y} - \mathbf{X}\mathbf{b}_0)$, the point of $\mathbf{X}\mathbf{b}_0 + \mathcal{S}_0$ nearest to $\mathbf{y}$. As shown in the proof of [Theorem 11.3.2\(b\)](#), no rank condition on $\mathbf{\Lambda}$ is needed for this. That point is $\mathbf{X}\mathbf{b}$ exactly for the minimizers $\mathbf{b}$ of $\|\mathbf{y} - \mathbf{X}\mathbf{b}\|^2$ subject to $\mathbf{\Lambda}^\top \mathbf{b} = \mathbf{d}$, so it is $\mathbf{X}\hat{\boldsymbol{\beta}}_H$, and its residual sum of squares is SSE_H . This needs $\text{SSE}_H > 0$, which follows from $\text{SSE}_H \geq \text{SSE} > 0$.
- (b) By [Theorem 7.3.1](#) the two maxima are $(2\pi e \text{SSE}/n)^{-n/2}$ and $(2\pi e \text{SSE}_H/n)^{-n/2}$. Their ratio is $(\text{SSE}/\text{SSE}_H)^{n/2}$, and $\text{SSE}_H = \text{SSE} + \text{SS}_H = \text{SSE}\{1 + qF_H/(n - r)\}$ by [Theorem 11.3.2\(b\)](#).
- (c) By (b), $\Lambda \leq k$ iff $F_H \geq c$ for a constant c determined by k . By [Theorem 11.3.2\(c\)](#) the null distribution of F_H is $F(q, n - r)$ at every point of H , so size α requires $c = F_\alpha(q, n - r)$.

■

The large-sample theory of likelihood ratios (Wilks's theorem) would calibrate $-2\log \Lambda = n \log\{1 + qF_H/(n - r)\}$ by the $\chi^2(q)$ distribution. That calibration is only approximate. In the normal linear model the exact distribution is available through F_H and should be used.

11.4.2 Wald and score statistics

Two other large-sample recipes are in wide use. The **Wald statistic** measures the estimated departure $\mathbf{\Lambda}^\top \hat{\boldsymbol{\beta}} - \mathbf{d}$ against its estimated covariance, both computed at the unrestricted maximum likelihood estimate. The **score statistic**, also called the Lagrange multiplier statistic, measures the gradient of the log-likelihood at the *restricted* maximum likelihood estimate against the information there. In the linear model, with $\mathbf{\Lambda}$ of full column rank q and the maximum likelihood variance estimates $\hat{\sigma}^2 = \text{SSE}/n$ and $\tilde{\sigma}_H^2 = \text{SSE}_H/n$, they are

$$\text{Wald} = \frac{(\mathbf{\Lambda}^\top \hat{\boldsymbol{\beta}} - \mathbf{d})^\top (\mathbf{\Lambda}^\top \mathbf{G} \mathbf{\Lambda})^{-1} (\mathbf{\Lambda}^\top \hat{\boldsymbol{\beta}} - \mathbf{d})}{\hat{\sigma}^2}$$

and

$$\text{Score} = \mathbf{U}^\top \left(\frac{\mathbf{X}^\top \mathbf{X}}{\tilde{\sigma}_H^2} \right)^{-} \mathbf{U} = \tilde{\sigma}_H^2 \mathbf{U}^\top (\mathbf{X}^\top \mathbf{X})^{-} \mathbf{U},$$

where $\mathbf{U} = \mathbf{X}^\top (\mathbf{y} - \mathbf{X} \hat{\boldsymbol{\beta}}_H) / \tilde{\sigma}_H^2$. Here $\mathbf{U}$ is the gradient of the log-likelihood Eq. 7.3.1 with respect to $\boldsymbol{\beta}$ at $(\hat{\boldsymbol{\beta}}_H, \tilde{\sigma}_H^2)$, and $\mathbf{X}^\top \mathbf{X} / \tilde{\sigma}_H^2$ is the corresponding block of the Fisher information. The gradient with respect to σ^2 vanishes at the restricted maximum, and the information matrix is block diagonal, because $\mathbf{E}\{\mathbf{X}^\top (\mathbf{Y} - \mathbf{X} \boldsymbol{\beta})\} = \mathbf{0}$. So the σ^2 coordinates contribute nothing, and the formula is the full score statistic. The generalized inverse is harmless because $\mathbf{U} \in \mathcal{C}(\mathbf{X}^\top \mathbf{X})$.

Proposition 11.4.2 — Wald, likelihood ratio and score

In the setting above, let $u = \text{SS}_H / \text{SSE} = q F_H / (n - r)$. Then

$$\text{Wald} = n u, \quad -2 \log \Lambda = n \log(1 + u), \quad \text{Score} = \frac{n u}{1 + u}.$$

Each is a strictly increasing function of F_H , so the three tests, calibrated exactly, coincide with the F test. For every sample with $\text{SS}_H > 0$,

$$\text{Wald} > -2 \log \Lambda > \text{Score}.$$

If each is compared with the upper α point c of $\chi^2(q)$, the exact sizes are

$$\Pr\left\{F(q, n - r) > \frac{n - r}{q} u_c\right\}, \quad u_c = \frac{c}{n}, \quad e^{c/n} - 1, \quad \frac{c}{n - c} \quad (n > c)$$

for the Wald, likelihood ratio and score tests respectively.

Proof. With $\hat{\sigma}^2 = \text{SSE}/n$, $\text{Wald} = n \text{SS}_H / \text{SSE} = n u$ by Eq. 11.3.1. For the score, $\mathbf{X}^\top \mathbf{y} = \mathbf{X}^\top \mathbf{X} \hat{\boldsymbol{\beta}}$, so $\mathbf{X}^\top (\mathbf{y} - \mathbf{X} \hat{\boldsymbol{\beta}}_H) = \mathbf{X}^\top \mathbf{X} (\hat{\boldsymbol{\beta}} - \hat{\boldsymbol{\beta}}_H)$ and

$$\mathbf{U}^\top (\mathbf{X}^\top \mathbf{X})^{-} \mathbf{U} \cdot \tilde{\sigma}_H^2 = \frac{(\hat{\boldsymbol{\beta}} - \hat{\boldsymbol{\beta}}_H)^\top \mathbf{X}^\top \mathbf{X} (\mathbf{X}^\top \mathbf{X})^{-} \mathbf{X}^\top \mathbf{X} (\hat{\boldsymbol{\beta}} - \hat{\boldsymbol{\beta}}_H)}{\tilde{\sigma}_H^2} = \frac{\|\mathbf{X}(\hat{\boldsymbol{\beta}} - \hat{\boldsymbol{\beta}}_H)\|^2}{\tilde{\sigma}_H^2}.$$

By the proof of Proposition 11.3.3, $\|\mathbf{X}(\hat{\boldsymbol{\beta}} - \hat{\boldsymbol{\beta}}_H)\|^2 = \text{SS}_H$, so $\text{Score} = n \text{SS}_H / \text{SSE}_H = n u / (1 + u)$. The likelihood ratio is Eq. 11.4.1. All three are increasing in u , which is proportional to F_H . For $u > 0$, $\log(1 + u) = \int_0^u (1 + t)^{-1} dt$ lies strictly between $u / (1 + u)$ and u , since the integrand lies strictly between $(1 + u)^{-1}$ and 1 on $(0, u)$. Finally, each statistic exceeds c iff $u > u_c$, with u_c as stated (for the score, $n u / (1 + u) > c$ iff $u(n - c) > c$, and it never exceeds c when $n \leq c$), and $u > u_c$ iff $F_H > (n - r) u_c / q$.

The inequality $\text{Wald} \geq \text{LR} \geq \text{Score}$ is a general feature of linear restrictions in normal regression, and it has a practical consequence: with χ^2 critical values, the Wald test rejects most often and the score test least often, *on the same data*.

Example. (Three verdicts from one data set)

For the region test of [Example \(Do regions matter?\)](#), $n = 50$, $q = 3$ and $n - r = 43$. The three statistics are

Statistic	Value	p-value from $\chi^2(3)$
Wald	8.512	0.0365
Likelihood ratio, $-2 \log \Lambda$	7.861	0.0490
Score	7.274	0.0637
$F = 2.440$, exact		0.0773

With the 5% critical value 7.815 of $\chi^2(3)$, the Wald test rejects, the likelihood ratio test rejects barely, and the score test does not. The exact test, which all three approximate, does not reject. The disagreement is not a paradox. The approximate tests are different tests, and their true sizes differ from 0.05. With $q = 3$ and $r = 7$, the formulas of [Proposition 11.4.2](#) give

n	15	30	50	100	400
Wald	0.315	0.143	0.097	0.071	0.055
Likelihood ratio	0.221	0.106	0.079	0.063	0.053
Score	0.102	0.069	0.060	0.055	0.051

At the sample size of the example the nominal 5% Wald test has true size close to 10%. All three sizes approach 0.05 as n grows, but slowly when r is not small compared with n .

```
import numpy as np
import statsmodels.api as sm
from scipy import stats

data = sm.datasets.statcrime.load_pandas().data
data.index = data.index.str.strip()
data = data.drop(index="District of Columbia")
codes = "SWWSWWNSSSWMMMMSSNSNMMSMWNNWNSMMSWNSMSSWNSWSMW"
region = np.array(list(codes))
y = data["murder"].to_numpy()
n = len(y)
covariates = np.column_stack([data["poverty"], data["single"], data["urban"]])
X0 = np.column_stack([np.ones(n), covariates])
D = np.column_stack([(region == g).astype(float) for g in "NSW"])
X = np.column_stack([X0, D])
r, q = 7, 3
```

```

def fit(Z):
    b, *_ = np.linalg.lstsq(Z, y, rcond=None)
    return b, np.sum((y - Z @ b) ** 2)

b, sse = fit(X)
b0, sse0 = fit(X0)
F = (sse0 - sse) / q / (sse / (n - r))

u = (sse0 - sse) / sse          # the ratio behind every statistic
wald = n * u                  # n (SSE0 - SSE) / SSE
lr = n * np.log1p(u)           # n log(SSE0 / SSE)
score = n * u / (1 + u)        # n (SSE0 - SSE) / SSE0
crit = stats.chi2.ppf(0.95, q)
for name, stat in (("Wald", wald), ("LR", lr), ("score", score)):
    print(f"{name:5s} {stat:.3f}  chi2 p = {stats.chi2.sf(stat, q):.4f}")
print(f"F      {F:.3f}  exact p = {stats.f.sf(F, q, n - r):.4f}")

```

The lesson for other models is double. In the normal linear model none of the three approximations is needed, because the exact F test is available. In models without an exact theory, generalized linear models among them (Chapter 34 onwards), the three statistics are the standard tools. Their disagreement in small samples is a warning that the χ^2 approximation is poor. The linear model suggests a remedy: dividing by q and referring to an F distribution with finite denominator degrees of freedom often calibrates better.

11.4.3 Invariance and optimality

Is the F test a *good* test? For a single constraint and a one-sided alternative, the t test of Section 11.6 has strong optimality properties. For $q \geq 2$ no test is uniformly most powerful: the most powerful test against alternatives in one direction of the test space rejects for large values of the estimated departure *in that direction*, and it is a different test for each direction (Exercise C1). Some restriction on the class of tests is needed. The natural one is invariance. By Proposition 11.2.2, the problem does not change if the data are rescaled, shifted within the reduced model, or rotated within the test space and within the residual space. A test whose verdict changes under such operations would be favouring particular directions for no reason given by the problem. We show that among tests that do not, the F test is the most powerful, simultaneously against every alternative.

We need one property of the noncentral F family.

Lemma 11.4.3 — Monotone likelihood ratio of the noncentral F

For $q, \nu \geq 1$ let f_γ be the density of $F(q, \nu, \gamma)$. For every $\gamma > 0$, the ratio $f_\gamma(x)/f_0(x)$ is a strictly increasing function of $x > 0$.

Proof. Let $V \sim \chi^2(\nu)$, and for $k = 0, 1, \dots$ let h_k be the density of $(U_k/q)/(V/\nu)$, where $U_k \sim \chi^2(q + 2k)$ is independent of V . By Exercise B4, $B = U_k/(U_k + V) \sim \text{Beta}(a_k, b)$ with $a_k = q/2 + k$ and $b = \nu/2$. The map $B \mapsto x = \nu B/\{q(1 - B)\}$ is increasing, with inverse $B = t(x) := qx/(\nu + qx)$, and $1 - t(x) = \nu/(\nu + qx)$. Changing variables,

$$h_k(x) = \frac{t(x)^{a_k-1} \{1 - t(x)\}^{b-1}}{B(a_k, b)} \cdot \frac{q\nu}{(\nu + qx)^2}, \quad \text{so} \quad \frac{h_k(x)}{h_0(x)} = \frac{B(q/2, \nu/2)}{B(q/2 + k, \nu/2)} t(x)^k = c_k t(x)^k,$$

where B is the beta function and $c_k > 0$. By [Theorem 4.1.2\(d\)](#), $U \sim \chi^2(q, \gamma)$ has the law of U_K with $K \sim \text{Poisson}(\gamma/2)$ independent of the U_k and V . Conditioning on K , $F = (U/q)/(V/\nu)$ has density $f_\gamma(x) = \sum_{k \geq 0} p_k h_k(x)$ with $p_k = e^{-\gamma/2}(\gamma/2)^k/k!$, and $f_0 = h_0$. Hence

$$\frac{f_\gamma(x)}{f_0(x)} = \sum_{k \geq 0} p_k c_k t(x)^k.$$

Every coefficient is positive when $\gamma > 0$, so the series is strictly increasing in t , and $t(x)$ is strictly increasing in x . ■

A test is a measurable function ϕ of the data with values in $[0, 1]$, the probability of rejecting. It is **invariant** if $\phi(g(\mathbf{y})) = \phi(\mathbf{y})$ for all $\mathbf{y}$ and all g in the set $\mathcal{G}$ of maps

$$g(\mathbf{y}) = a\mathbf{H}\mathbf{y} + \mathbf{v}, \quad a > 0, \quad \mathbf{v} \in \mathcal{C}(\mathbf{X}_0), \quad \mathbf{H} \text{ orthogonal with } \mathbf{H}\mathcal{C}(\mathbf{X}_0) = \mathcal{C}(\mathbf{X}_0), \quad \mathbf{H}\mathcal{C}(\mathbf{X}) = \mathcal{C}(\mathbf{X}).$$

For a testable hypothesis with $\mathbf{d} \neq \mathbf{0}$, the same applies to the data $\mathbf{y} - \mathbf{X}\mathbf{b}_0$, with $\mathcal{C}(\mathbf{X}_0)$ replaced by $\mathcal{S}_0$.

Theorem 11.4.4 — The F test is uniformly most powerful invariant

Assume [Eq. 7.3.1](#) and the setting of [Definition 11.1.1](#). Let ϕ be an invariant test with $E\phi(\mathbf{Y}) \leq \alpha$ at every point of H_0 . Then at every point $(\boldsymbol{\beta}, \sigma^2)$ of the full model,

$$E_{\boldsymbol{\beta}, \sigma^2} \phi(\mathbf{Y}) \leq \Pr\{F(q, n-r, \gamma) > F_\alpha(q, n-r)\},$$

with γ given by [Eq. 11.1.2](#). The right side is the power of the F test, which is itself invariant and of size α .

Proof. Use the canonical coordinates $\mathbf{z}_k = \mathbf{Q}_k^\top \mathbf{y}$ of [Proposition 11.2.3](#), and let $\mathcal{Y} = \{\mathbf{y} : \mathbf{z}_2 \neq \mathbf{0}\}$, a set of probability one on which F is defined.

Step 1: an invariant test depends on $\mathbf{y}$ only through F . Let $\mathbf{y}, \mathbf{y}' \in \mathcal{Y}$ with $F(\mathbf{y}) = F(\mathbf{y}')$. Put $a = \|\mathbf{z}'_2\|/\|\mathbf{z}_2\| > 0$. Equality of the F values gives $\|\mathbf{z}'_1\| = a\|\mathbf{z}_1\|$. Choose orthogonal matrices $\mathbf{O}_1$ ($q \times q$) and $\mathbf{O}_2$ $((n-r) \times (n-r))$ with $\mathbf{O}_1(a\mathbf{z}_1) = \mathbf{z}'_1$ and $\mathbf{O}_2(a\mathbf{z}_2) = \mathbf{z}'_2$. This is possible because the vectors in each pair have equal lengths, and if $\mathbf{z}_1 = \mathbf{0}$ then $\mathbf{z}'_1 = \mathbf{0}$ and $\mathbf{O}_1 = \mathbf{I}$ will do. Put

$$\mathbf{H} = \mathbf{Q}_0\mathbf{Q}_0^\top + \mathbf{Q}_1\mathbf{O}_1\mathbf{Q}_1^\top + \mathbf{Q}_2\mathbf{O}_2\mathbf{Q}_2^\top, \quad \mathbf{v} = \mathbf{Q}_0(\mathbf{z}'_0 - a\mathbf{z}_0).$$

$\mathbf{H}$ is orthogonal, maps $\mathcal{C}(\mathbf{X}_0) = \mathcal{C}(\mathbf{Q}_0)$ and $\mathcal{C}(\mathbf{X}) = \mathcal{C}([\mathbf{Q}_0, \mathbf{Q}_1])$ onto themselves, and $\mathbf{v} \in \mathcal{C}(\mathbf{X}_0)$. The canonical coordinates of $a\mathbf{H}\mathbf{y} + \mathbf{v}$ are $a\mathbf{z}_0 + \mathbf{z}'_0 - a\mathbf{z}_0 = \mathbf{z}'_0$, $a\mathbf{O}_1\mathbf{z}_1 = \mathbf{z}'_1$ and $a\mathbf{O}_2\mathbf{z}_2 = \mathbf{z}'_2$, so $a\mathbf{H}\mathbf{y} + \mathbf{v} = \mathbf{y}'$ and $\phi(\mathbf{y}') = \phi(\mathbf{y})$. Now fix, for each $f \geq 0$, the point $\mathbf{y}_f = \sqrt{qf/(n-r)}\mathbf{Q}_1\mathbf{e}_1 + \mathbf{Q}_2\mathbf{e}_1$, where $\mathbf{e}_1$ denotes a first coordinate vector, and put $\psi(f) = \phi(\mathbf{y}_f)$. Then $F(\mathbf{y}_f) = f$, so $\phi(\mathbf{y}) = \psi(F(\mathbf{y}))$ for every $\mathbf{y} \in \mathcal{Y}$. The map $f \mapsto \mathbf{y}_f$ is continuous, so ψ is measurable.

Step 2: reduction to one dimension. By Step 1 and [Theorem 11.1.1\(b\)](#), $E\phi(\mathbf{Y}) = E\psi(F) = \int_0^\infty \psi(x)f_\gamma(x)dx$, with f_γ the density of $F(q, n-r, \gamma)$. The hypothesis is $\gamma = 0$ ([Theorem 11.1.1\(c\)](#)), so the size condition says $\int \psi f_0 \leq \alpha$.

Step 3: Neyman–Pearson. Fix $\gamma > 0$, let $c = F_\alpha(q, n-r)$, $\phi^*(x) = 1\{x > c\}$ and $k = f_\gamma(c)/f_0(c)$. By [Lemma 11.4.3](#), $f_\gamma(x) - kf_0(x)$ is positive for $x > c$ and negative

for $x < c$. Since $\phi^* - \psi \geq 0$ where $x > c$ and $\phi^* - \psi \leq 0$ where $x < c$, the product $(\phi^* - \psi)(f_\gamma - kf_0)$ is nonnegative everywhere. Integrating,

$$\int (\phi^* - \psi)f_\gamma \geq k \int (\phi^* - \psi)f_0 = k\left(\alpha - \int \psi f_0\right) \geq 0.$$

So $E\phi(\mathbf{Y}) = \int \psi f_\gamma \leq \int \phi^* f_\gamma$, which is the power of the F test. For $\gamma = 0$ the inequality is the size condition. The F test is invariant by [Proposition 11.2.2](#). ■

The theorem says more than “the F test is good”. It says that once one agrees that no direction in the test space and no scale deserves special treatment, the choice of test is forced: every invariant test is a function of F (Step 1), and among functions of F the upper tail is best against every alternative at once. The likelihood ratio principle arrives at the same test without any appeal to optimality, which is reassuring but not a proof. In other models the likelihood ratio test need not be optimal in any sense.

Remark (What invariance costs). Invariance is a restriction, and the F test is not the most powerful test against every alternative. If one contrast in the test space is of special interest, the t test for that contrast is more powerful against departures in its direction and less powerful against others ([Section 11.6](#)). A Bayesian analysis with a prior concentrated on some directions behaves similarly. Invariance expresses the absence of such preferences. It is appropriate for an omnibus question like “do the regions differ?”, and inappropriate when the scientific question singles out a direction.

11.4.4 Exercises

11.4.4.1 A. Check your understanding

Exercise A1.

From $F = 2.440$, $q = 3$, $n = 50$ and $r = 7$, compute the three statistics of [Proposition 11.4.2](#) and check them against [Example \(Three verdicts from one data set\)](#).

11.4.4.2 B. Practice

Exercise B1.

Fix q and r . Show that under H , as $n \rightarrow \infty$, qF_H converges in distribution to $\chi^2(q)$, and that the differences Wald $- qF_H$, $-2\log \Lambda - qF_H$ and Score $- qF_H$ converge to zero in probability. Deduce that the sizes in [Example \(Three verdicts from one data set\)](#) tend to 0.05.

Solution. Write $qF_H = (SS_H/\sigma^2)/(s^2/\sigma^2)$. The numerator is $\chi^2(q)$ for every n , and $s^2/\sigma^2 \rightarrow 1$ in probability ([Exercise \(B1\)](#)), so $qF_H \rightarrow \chi^2(q)$ in distribution. With $u = qF_H/(n-r)$, Wald $- qF_H = qF_H\{n/(n-r) - 1\} = qF_H r/(n-r) \rightarrow 0$ in probability. For the others, $nu = O_p(1)$, so $u \rightarrow 0$, and $n \log(1+u) - nu = nu\{\log(1+u)/u - 1\}$ and $nu/(1+u) - nu = -nu^2/(1+u)$ both tend to zero. Each statistic therefore converges in distribution to $\chi^2(q)$, whose upper 5% point is the critical value used, so the sizes

converge to 0.05.

Exercise B2.

Let $\hat{\epsilon}_H = \mathbf{y} - \mathbf{X}\hat{\beta}_H$ be the residual vector of the restricted fit. Show that the score statistic equals $n\|\mathbf{M}\hat{\epsilon}_H\|^2/\|\hat{\epsilon}_H\|^2$, which is n times the uncentred R^2 from regressing $\hat{\epsilon}_H$ on $\mathbf{X}$. Explain why this “ nR^2 ” recipe needs only the restricted fit.

Solution. $\mathbf{M}\hat{\epsilon}_H = \mathbf{M}\mathbf{y} - \mathbf{X}\hat{\beta}_H = \mathbf{X}(\hat{\beta} - \hat{\beta}_H)$, whose squared length is SS_H (proof of Proposition 11.4.2), and $\|\hat{\epsilon}_H\|^2 = \text{SSE}_H$. So $n\|\mathbf{M}\hat{\epsilon}_H\|^2/\|\hat{\epsilon}_H\|^2 = n\text{SS}_H/\text{SSE}_H = \text{Score}$. The uncentred R^2 of a regression of $\hat{\epsilon}_H$ on $\mathbf{X}$ is $\|\mathbf{M}\hat{\epsilon}_H\|^2/\|\hat{\epsilon}_H\|^2$. Computing it requires the restricted residuals and one auxiliary regression, but not $\hat{\beta}$ or the unrestricted model’s residual sum of squares as such. This is why score tests are popular when the full model is hard to fit, as in tests for omitted nonlinear terms or for heteroscedasticity (Chapter 21).

Exercise B3.

Suppose $\sigma^2 = \sigma_0^2$ is known. Show that the likelihood ratio test of a reduced model rejects for large $(\text{SSE}_0 - \text{SSE})/\sigma_0^2$, the statistic of Exercise (B1), and that $-2\log\Lambda$ is then exactly $\chi^2(q)$ under H_0 .

11.4.4.3 C. Going deeper

Exercise C1.

In the canonical form with σ^2 known and $q \geq 2$, consider testing $\eta_1 = \mathbf{0}$ against the single alternative $\eta_1 = \mathbf{a} \neq \mathbf{0}$. Show by the Neyman–Pearson lemma that the most powerful level- α test rejects when $\mathbf{a}^\top \mathbf{z}_1 > \sigma\|\mathbf{a}\|z_\alpha$, where z_α is the upper α point of $N(0, 1)$. Show that this test has power below α against $-\mathbf{a}$, and conclude that no uniformly most powerful test of $\eta_1 = \mathbf{0}$ exists.

Solution. The likelihood ratio of $N(\mathbf{a}, \sigma^2\mathbf{I})$ to $N(\mathbf{0}, \sigma^2\mathbf{I})$ at $\mathbf{z}_1$ is $\exp\{(\mathbf{a}^\top \mathbf{z}_1 - \|\mathbf{a}\|^2/2)/\sigma^2\}$, increasing in $\mathbf{a}^\top \mathbf{z}_1$. Under the null, $\mathbf{a}^\top \mathbf{z}_1 \sim N(0, \sigma^2\|\mathbf{a}\|^2)$, which gives the critical value, and the Neyman–Pearson argument of Step 3 of the proof of Theorem 11.4.4 shows that this test is most powerful. Against $-\mathbf{a}$, $\mathbf{a}^\top \mathbf{z}_1 \sim N(-\|\mathbf{a}\|^2, \sigma^2\|\mathbf{a}\|^2)$, so the power is $\Pr\{Z > z_\alpha + \|\mathbf{a}\|/\sigma\} < \alpha$. A uniformly most powerful test would have to be most powerful against both $\mathbf{a}$ and $-\mathbf{a}$. The most powerful tests against these two alternatives are essentially unique and different, so no single test can be both.

Exercise C2.

Extend Lemma 11.4.3: show that for $0 \leq \gamma_1 < \gamma_2$ the ratio $f_{\gamma_2}(x)/f_{\gamma_1}(x)$ is increasing in x . (Write both densities as power series in $t(x)$ with coefficients $a_k(\gamma)$, and show that $a_k(\gamma_2)/a_k(\gamma_1)$ increases with k .) Deduce that the F test is also uniformly most powerful invariant for testing $\gamma \leq \gamma_1$ against $\gamma > \gamma_1$, with a noncentral critical value. Such tests appear in equivalence testing.

11.5 Power and noncentrality

A test is judged by two probabilities. Its size, the probability of rejecting a true hypothesis, is fixed by the critical value. Its **power**, the probability of rejecting a false one, depends on how false the hypothesis is. For the F test the whole dependence runs through the single number γ , and that makes power calculations unusually transparent. This section states the power function, interprets the noncentrality, uses it to plan the size of a study, and asks how sensitive the region test of [Section 11.1](#) actually was. It ends with a popular calculation that should not be made.

11.5.1 The power function

Theorem 11.5.1 — Power of the F test

Assume [Eq. 7.3.1](#) with $r < n$, and let $H : \Lambda^\top \beta = \mathbf{d}$ be testable with $q = \text{rank}(\Lambda)$ (or let H be a reduced model with $q = r - r_0$). The probability that the level- α F test rejects is

$$\pi(\gamma) = \Pr\{F(q, n - r, \gamma) > F_\alpha(q, n - r)\}, \quad (11.5.1)$$

where γ is the noncentrality of [Eq. 11.3.2](#), or of [Eq. 11.1.2](#) for a reduced model. Moreover:

- (a) π depends on (β, σ^2) only through γ . It is continuous and strictly increasing, with $\pi(0) = \alpha$ and $\pi(\gamma) \rightarrow 1$ as $\gamma \rightarrow \infty$. In particular the test is *unbiased*: its power exceeds α at every alternative.
- (b) $\sigma^2\gamma$ is the squared distance from the mean vector $\mathbf{X}\beta$ to the set of mean vectors allowed by H :

$$\sigma^2\gamma = \min_{\Lambda^\top \mathbf{b} = \mathbf{d}} \|\mathbf{X}\beta - \mathbf{X}\mathbf{b}\|^2 = \min_{\Lambda^\top \mathbf{b} = \mathbf{d}} \sum_{i=1}^n (\mathbf{x}_{(i)}^\top \beta - \mathbf{x}_{(i)}^\top \mathbf{b})^2.$$

- (c) If further observations are added, with the same β, σ^2 and hypothesis, the noncentrality does not decrease. If the whole design is replicated k times, γ is multiplied by k .

Proof. The distribution of F_H is [Theorem 11.3.2\(c\)](#), which gives [Eq. 11.5.1](#). (a) is [Theorem 4.1.4](#) with $c = F_\alpha(q, n - r)$. For (b), [Theorem 11.3.2\(b\)](#) applied to the data vector $\mathbf{X}\beta$, for which β is a least squares estimate and the residual sum of squares is zero, shows that $\min_{\Lambda^\top \mathbf{b} = \mathbf{d}} \|\mathbf{X}\beta - \mathbf{X}\mathbf{b}\|^2 = (\Lambda^\top \beta - \mathbf{d})^\top \mathbf{W}^- (\Lambda^\top \beta - \mathbf{d}) = \sigma^2\gamma$. For (c), adding rows to $\mathbf{X}$ enlarges $\mathcal{C}(\mathbf{X}^\top)$, so H remains testable (the constraint set $\{\mathbf{b} : \Lambda^\top \mathbf{b} = \mathbf{d}\}$ is unchanged), and the sum in (b) acquires extra nonnegative terms for every $\mathbf{b}$. The minimum of a larger function over the same set is at least as large. Replicating the design k times multiplies every sum of squares in (b) by k , and hence the minimum. ■

Part (b) is the reading of γ to remember. The noncentrality is the smallest total squared

discrepancy, over all observations, between the true mean and a mean the hypothesis allows, in units of the error variance. It is large when the hypothesis is badly wrong, when σ^2 is small, and when there are many observations at which the discrepancy shows. Part (c) makes the last point precise. Note that adding observations also increases $n - r$, which helps as well, and that the sum in (b) does *not* depend on the parameterization.

Two consequences follow at once. First, for a fixed alternative, γ grows in proportion to the number of replicates, so the power tends to one: *every fixed departure from the hypothesis is detected with certainty in large enough samples*. A significant result from a very large sample may therefore reflect a departure too small to matter, and effect sizes, not p-values, answer the question “how much?”. Second, the ratio γ/n is a natural *effect size*, the average squared discrepancy per observation relative to σ^2 . It is closely related to the effect size f^2 of Cohen’s power tables, which are widely used in the behavioural sciences.

11.5.2 Planning a study

Before the data are collected, Eq. 11.5.1 turns the question “how many observations?” into arithmetic. One specifies the smallest departure worth detecting, a guess at σ , the level and the desired power, and solves for n . The difficulty is the first ingredient. A departure is a whole vector $\mathbf{A}^\top \boldsymbol{\beta} - \mathbf{d}$, and different vectors of the same apparent size give different γ . For the comparison of group means there is a clean way out.

Proposition 11.5.2 — Least and most favourable configurations

In the balanced one-way layout with $g \geq 2$ groups of m observations, group means $\mu_1, \dots, \mu_g$ and common variance σ^2 , the noncentrality of the F test of equal means is

$$\gamma = \frac{m}{\sigma^2} \sum_{i=1}^g (\mu_i - \bar{\mu})^2, \quad \bar{\mu} = \frac{1}{g} \sum_i \mu_i.$$

Among configurations whose range $\max_i \mu_i - \min_i \mu_i$ equals Δ ,

$$\frac{m\Delta^2}{2\sigma^2} \leq \gamma \leq \frac{mg\Delta^2}{4\sigma^2}.$$

The lower bound is attained exactly when one mean is at each extreme and all others are at the midpoint. For even g , the upper bound is attained exactly when half the means are at each extreme.

Proof. The full model has one indicator per group. The reduced model is $\mathcal{C}(\mathbf{1})$, so by Theorem 11.1.1(c), $\sigma^2\gamma = \|(\mathbf{I} - n^{-1}\mathbf{1}\mathbf{1}^\top)\boldsymbol{\mu}\|^2$, where $\boldsymbol{\mu}$ repeats μ_i on the m observations of group i . This is $m \sum_i (\mu_i - \bar{\mu})^2$. Let $\mu_{\max}$ and $\mu_{\min}$ be two groups at the extremes and put $a = \mu_{\max} - \bar{\mu} \geq 0$, $b = \bar{\mu} - \mu_{\min} \geq 0$, so $a + b = \Delta$. Dropping the other groups from the sum,

$$\sum_i (\mu_i - \bar{\mu})^2 \geq a^2 + b^2 \geq \frac{1}{2}(a + b)^2 = \frac{1}{2}\Delta^2,$$

with equality in the first step iff every other mean equals $\bar{\mu}$, and in the second iff $a = b$. If all other means equal $\bar{\mu}$, then $\bar{\mu}$ is also the average of the two extremes, so $a = b$ holds automatically, and the equality case is the stated configuration. For the upper bound, let c be the midpoint of the range. Since $\bar{\mu}$ minimizes $t \mapsto \sum_i (\mu_i - t)^2$, $\sum_i (\mu_i - \bar{\mu})^2 \leq$

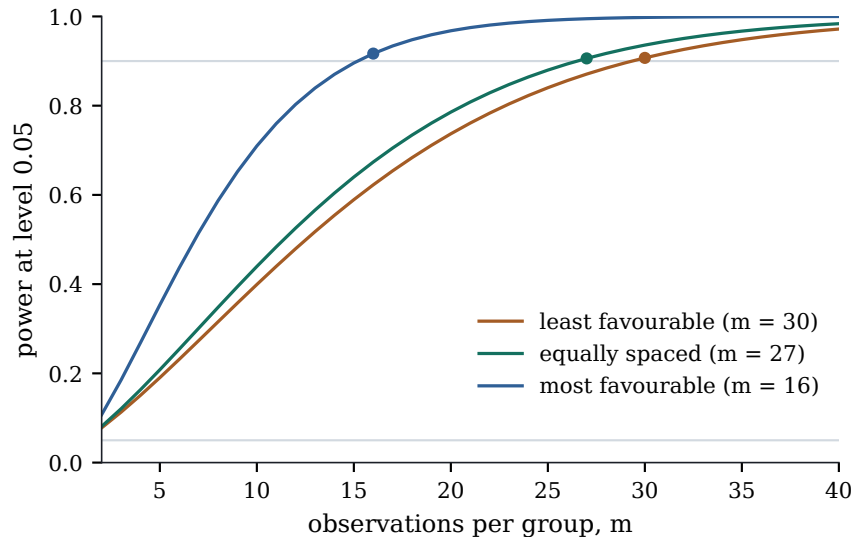


Figure 11.5.1: Power of the level-0.05 F test of equal means for four groups of m observations, when the group means span one error standard deviation. The three curves are the least favourable configuration of Proposition 11.5.2, equally spaced means, and the most favourable configuration. Dots mark the smallest m giving power 0.9.

$\sum_i (\mu_i - c)^2 \leq g\Delta^2/4$, because every $|\mu_i - c| \leq \Delta/2$. Equality requires every mean at an extreme and $\bar{\mu} = c$, which for even g means half at each end. ■

A design that has power 0.9 at the least favourable configuration has at least that power at every configuration with range Δ or more. That is the guarantee a planner usually wants.

Example. (How many observations per group?)

Four treatments are to be compared at level 0.05, and a spread of one standard deviation between the best and the worst treatment ($\Delta/\sigma = 1$) should be detected with probability at least 0.9. The noncentrality per observation per group is 0.500 for the least favourable configuration, 0.556 for equally spaced means, and 1.000 for the most favourable one (two means at each end). Searching over m with Eq. 11.5.1 gives

Configuration	γ/m	smallest m with power ≥ 0.9	power at that m
least favourable	0.500	30	0.907
equally spaced	0.556	27	0.906
most favourable	1.000	16	0.917

To be safe against every configuration with range at least σ , the study needs 30 observations per group. A simulation of 20000 data sets at the least favourable configuration with $m = 30$ rejects in a proportion 0.909 of them, against the computed 0.907. Figure 11.5.1 shows the whole power curves.

```
import numpy as np
from scipy import stats
```

```

def power(gamma, q, nu, alpha=0.05):
    """P{F(q, nu, gamma) > F_alpha(q, nu)}."""
    crit = stats.f.ppf(1 - alpha, q, nu)
    return stats.ncf.sf(crit, q, nu, gamma) if gamma > 0 else alpha

g, delta_over_sigma = 4, 1.0
patterns = {
    "least favourable": np.array([0, 0.5, 0.5, 1]),
    "equally spaced": np.array([0, 1 / 3, 2 / 3, 1]),
    "most favourable": np.array([0, 0, 1, 1]),
}

def gamma_oneway(m, means):
    """Noncentrality m * sum (mu_i - mu_bar)^2 / sigma^2 for m observations per
    group."""
    dev = (means - means.mean()) * delta_over_sigma
    return m * np.sum(dev ** 2)

for name, means in patterns.items():
    m = 2
    while power(gamma_oneway(m, means), g - 1, g * (m - 1)) < 0.9:
        m += 1
    print(f"{name:17s} m = {m} per group, power {power(gamma_oneway(m, means), g -
    1, g * (m - 1)):.3f}")

```

The same logic applies to any linear hypothesis. For a regression coefficient β_j , with the other regressors in the model, $\sigma^2\gamma = \beta_j^2 \|(\mathbf{I} - \mathbf{M}_{(j)})\mathbf{x}_j\|^2$, where $\mathbf{M}_{(j)}$ projects onto the other columns. The design enters through the length of the part of $\mathbf{x}_j$ that the other regressors cannot explain (Theorem 6.6.2). Spreading the values of x_j increases power, and collinearity with the other regressors destroys it. Adding a covariate that is unrelated to $\mathbf{x}_j$ but explains part of the response reduces σ^2 , and so increases γ , with little loss in the length of the residualized $\mathbf{x}_j$.

11.5.3 How sensitive was the region test?

Power calculations are also useful *after* a nonsignificant result, provided the alternatives are fixed by the scientific question and not by the data. For the region test of Example (Do regions matter?), consider the alternative in which the Northeast differs from the other three regions by δ murders per 100,000, with everything else as in the reduced model. The mean vector is then $\mathbf{X}_0\mathbf{c} + \delta\mathbf{d}_N$, where $\mathbf{d}_N$ is the Northeast indicator, and by Theorem 11.1.1(c),

$$\gamma = \frac{\delta^2 \|(\mathbf{I} - \mathbf{M}_0)\mathbf{d}_N\|^2}{\sigma^2}.$$

There are 9 Northeast states, but $\|(\mathbf{I} - \mathbf{M}_0)\mathbf{d}_N\|^2 = 5.76$ only: the covariates already account for part of the Northeast's distinctiveness, and only the rest counts as evidence about regions. Setting σ at $s = 1.423$ as a planning value, the power of the 5% test is

δ	0.5	1.0	1.5	2.0
power	0.091	0.241	0.506	0.778

and power 0.8 needs $\delta = 2.05$. A Northeast shortfall of 1.5 murders per 100,000, about the size estimated in Example (Regional hypotheses with an over-parameterized model) (1.379 below the

other three regions), would have been missed about half the time. The nonsignificant verdict of [Example \(Do regions matter?\)](#) therefore says little about whether regional differences of that size exist.

```
import statsmodels.api as sm

data = sm.datasets.statelcrime.load_pandas().data
data.index = data.index.str.strip()
data = data.drop(index="District of Columbia")
codes = "SWWSWNSSSSWMMMMSSNSNMMSMWNNWNSMMSWNSMSSWNSWSMW"
region = np.array(list(codes))
y = data["murder"].to_numpy()
n = len(y)
X0 = np.column_stack([np.ones(n), data["poverty"], data["single"], data["urban"]])
D = np.column_stack([(region == c).astype(float) for c in "NSW"])
X = np.column_stack([X0, D])

b0, *_ = np.linalg.lstsq(X0, D[:, 0], rcond=None)
k2 = np.sum((D[:, 0] - X0 @ b0) ** 2)      #  $|| (I - M_0) d_N ||^2$ :  $\gamma = \delta^2 k2$ 
    / sigma^2
bf, *_ = np.linalg.lstsq(X, y, rcond=None)
sigma = np.sqrt(np.sum((y - X @ bf) ** 2) / (n - 7))  # planning value for sigma
for delta in (0.5, 1.0, 1.5, 2.0):
    print(f"Northeast lower by {delta}:  $\gamma = \{\delta^2 * k2 / \sigma^2\}$ , "
          f"power = {power(delta^2 * k2 / sigma^2, 3, n - 7):.3f}")
```

The calculation uses the estimate s in place of σ , which is itself uncertain. A more careful version would repeat it for a range of plausible values of σ , or replace the whole exercise by a confidence region for the regional effects, which shows directly which differences the data rule out ([Chapter 12](#)).

11.5.4 Observed power

A common calculation takes the observed data, estimates the noncentrality from them, and reports the power “the test had” at that estimate. It is meant to show whether a nonsignificant result was due to low power. It cannot show that.

Proposition 11.5.3 — Observed power is a function of the p-value

Fix q , $n - r$ and α , and define the **observed power** as $\pi(\hat{\gamma})$, with π as in [Eq. 11.5.1](#) and $\hat{\gamma} = qF_{\text{obs}}$. Then the observed power is a strictly decreasing function of the p-value of the F test. The same holds for any estimate $\hat{\gamma} = h(F_{\text{obs}})$ with h strictly increasing and positive.

Proof. The p-value is $1 - \Phi_F(F_{\text{obs}})$, with Φ_F the continuous, strictly increasing distribution function of $F(q, n - r)$, so F_{obs} is a strictly decreasing function of the p-value. The map $F_{\text{obs}} \mapsto h(F_{\text{obs}})$ is strictly increasing, and π is strictly increasing by [Theorem 11.5.1\(a\)](#). The composition is strictly decreasing in the p-value. ■

So observed power adds nothing to the p-value. In the setting of the region test ($q = 3$,

$n - r = 43$), a p-value of 0.2 always corresponds to an observed power of 0.394, a p-value of exactly 0.05 to 0.637, and 0.01 to 0.830. “The test was not significant and its observed power was low” says the same thing twice. The informative calculations are the ones above: power at alternatives chosen for their scientific meaning, or a confidence region.

Warning (Reading a nonsignificant F). A nonsignificant F statistic is weak evidence for the reduced model unless the test had high power against every alternative that matters. Whether it had is a question about the design and about which alternatives matter, answered by [Eq. 11.5.1](#) at alternatives fixed in advance. It is not answered by the observed power, which merely restates the p-value, and it is not answered by the p-value being large. A p-value of 0.9 is no stronger evidence for H_0 than one of 0.3 when the test has low power.

11.5.5 Exercises

11.5.5.1 A. Check your understanding

Exercise A1.

For the test of $\beta_1 = 0$ in simple regression, show that $\gamma = \beta_1^2 S_{xx} / \sigma^2$, with $S_{xx} = \sum_i (x_i - \bar{x})^2$. With $n = 20$ design points in $[0, 1]$, compare S_{xx} for equal spacing and for ten points at each end. What does the second design give up?

Exercise A2.

For the one-way layout with group sizes $n_1, \dots, n_g$, show that the noncentrality of the test of equal means is $\gamma = \sum_i n_i (\mu_i - \bar{\mu}_w)^2 / \sigma^2$, with $\bar{\mu}_w = \sum_i n_i \mu_i / n$.

11.5.5.2 B. Practice

Exercise B1.

Extend [Proposition 11.5.2](#) to unequal group sizes: show that among configurations with range Δ ,

$$\gamma \geq \frac{\Delta^2}{\sigma^2} \frac{1}{1/n_{(1)} + 1/n_{(2)}},$$

where $n_{(1)} \leq n_{(2)}$ are the two smallest group sizes, and describe the configuration attaining it.

Solution. Let groups a and b carry the largest and smallest means. Dropping the other terms of [Exercise \(A2\)](#), $\sigma^2 \gamma \geq n_a (\mu_a - \bar{\mu}_w)^2 + n_b (\mu_b - \bar{\mu}_w)^2 \geq \min_t \{n_a (\mu_a - t)^2 + n_b (\mu_b - t)^2\}$. The minimum is at $t = (n_a \mu_a + n_b \mu_b) / (n_a + n_b)$ and equals $\Delta^2 n_a n_b / (n_a + n_b) = \Delta^2 / (1/n_a + 1/n_b)$. This is smallest when a and b are the two smallest groups. It is attained when those two groups carry the extremes and all other means equal $(n_a \mu_a + n_b \mu_b) / (n_a + n_b)$, which is then also $\bar{\mu}_w$. A departure is hardest to detect when it is confined to the groups with the fewest observations.

Exercise B2.

Take the straight-line model with design points 0, 1, 2, 3, one observation each, and the hypothesis $\beta_1 = 0$. Compute γ as a multiple of β_1^2/σ^2 and the power at $\beta_1/\sigma = 1$, level 0.05. Then compare (i) replicating the design twice, (ii) adding four observations at $x = 3$, and (iii) adding four observations at $x = 1.5$. Verify [Theorem 11.5.1\(c\)](#) in each case and explain the ranking.

Exercise B3.

For $q = 1$ and large $n - r$, the F test is approximately the test that rejects when $|Z| > z_{\alpha/2}$, with $Z \sim N(\delta, 1)$. Show that if the observed $|z|$ equals $z_{\alpha/2}$ exactly, the observed power $\Pr\{|Z| > z_{\alpha/2}\}$ at $\delta = z_{\alpha/2}$ is just above 1/2. Conclude that a result on the edge of significance always has observed power about one half, whatever the design.

Solution. At $\delta = z_{\alpha/2}$, $\Pr\{|Z| > z_{\alpha/2}\} = \Pr\{Z > z_{\alpha/2}\} + \Pr\{Z < -z_{\alpha/2}\} = \frac{1}{2} + \Pr\{Z - \delta < -2z_{\alpha/2}\}$, and the last probability is $\Phi(-2z_{\alpha/2})$, which is $\Phi(-3.92) \approx 4 \times 10^{-5}$ for $\alpha = 0.05$. The value depends only on α , not on n , σ or the design, as [Proposition 11.5.3](#) predicts.

11.5.5.3 C. Going deeper

Exercise C1.

In the sensitivity calculation, let n_N be the number of Northeast states and R_N^2 the coefficient of determination from regressing the Northeast indicator on the reduced model's columns (which include $\mathbf{1}$). Show that $\|(\mathbf{I} - \mathbf{M}_0)\mathbf{d}_N\|^2 = n_N(1 - n_N/n)(1 - R_N^2)$. Interpret each factor, and relate the formula to [Eq. 6.6.2](#).

11.6 Testing single coefficients and groups of coefficients

The tests most often printed by regression software are instances of the general theory: a t test for each coefficient, an F test that all slopes vanish, and on request an F test for a group of coefficients. This section derives each from the preceding sections, explains what each one tests, and studies how a group test and the individual tests inside it can disagree, in both directions.

11.6.1 The t test

When the hypothesis consists of a single estimable function, $q = 1$, the F statistic is the square of a statistic with a sign, and the sign allows one-sided alternatives.

Theorem 11.6.1 — The t test for an estimable function

Assume [Eq. 7.3.1](#) with $r < n$. Let $\lambda \in \mathcal{C}(\mathbf{X}^\top)$, $\lambda \neq \mathbf{0}$, let $d \in \mathbb{R}$, let $\mathbf{G}$ be a generalized inverse of $\mathbf{X}^\top \mathbf{X}$, and put

$$T = \frac{\lambda^\top \hat{\boldsymbol{\beta}} - d}{s \sqrt{\lambda^\top \mathbf{G} \lambda}}, \quad s^2 = \frac{\text{SSE}}{n - r}. \quad (11.6.1)$$

- (a) $T \sim t(n - r, \delta)$ with $\delta = (\lambda^\top \boldsymbol{\beta} - d) / (\sigma \sqrt{\lambda^\top \mathbf{G} \lambda})$.
- (b) T^2 equals the statistic F_H of [Eq. 11.3.1](#) for $H : \lambda^\top \boldsymbol{\beta} = d$, and $\gamma = \delta^2$. The two-sided test that rejects when $|T| > t_{n-r, \alpha/2}$ is the F test of H , with the same p-value.
- (c) The one-sided test that rejects when $T > t_{n-r, \alpha}$ has size exactly α for the hypothesis $\lambda^\top \boldsymbol{\beta} \leq d$: its rejection probability is at most α whenever $\lambda^\top \boldsymbol{\beta} \leq d$, with equality when $\lambda^\top \boldsymbol{\beta} = d$. Its power increases strictly with $\lambda^\top \boldsymbol{\beta}$.

Proof.

- (a) is [Corollary 7.5.2](#). (b) With $\mathbf{A} = \lambda$, $\mathbf{W} = \lambda^\top \mathbf{G} \lambda > 0$ is a scalar, so $F_H = (\lambda^\top \hat{\boldsymbol{\beta}} - d)^2 / (\lambda^\top \mathbf{G} \lambda s^2) = T^2$, and [Eq. 11.3.2](#) gives $\gamma = \delta^2$. Under H , $T \sim t(n - r)$ is symmetric about zero, so $\Pr\{|T| > t\} = \Pr\{T^2 > t^2\}$ for every $t > 0$. Hence $t_{n-r, \alpha/2}^2 = F_\alpha(1, n - r)$, the two rejection regions coincide, and the two-sided t p-value $\Pr\{|T| \geq |T_{\text{obs}}|\}$ equals the F p-value $\Pr\{T^2 \geq T_{\text{obs}}^2\}$. (c) By [Exercise \(A2\)](#), the probability that a $t(n - r, \delta)$ variable exceeds a fixed constant is strictly increasing in δ , and δ is an increasing function of $\lambda^\top \boldsymbol{\beta}$. At $\delta = 0$ the probability is α . ■

For a single coefficient β_j in a full-rank model, [Eq. 11.6.1](#) becomes the familiar $t_j = \hat{\beta}_j / \{s \sqrt{c_{jj}}\}$, with c_{jj} the j th diagonal entry of $(\mathbf{X}^\top \mathbf{X})^{-1}$. By [Corollary 9.2.3](#), t_j^2 is the F statistic for dropping column j from the model. The t test of $\beta_j = 0$ therefore compares the model with all regressors to the model with all *except* $\mathbf{x}_j$. It asks whether $\mathbf{x}_j$ adds anything *given the others*. By the Frisch–Waugh–Lovell theorem ([Theorem 6.6.2](#)), what it measures is the association of $\mathbf{y}$ with the part of $\mathbf{x}_j$ orthogonal to the other columns. The same coefficient can be “significant” in one model and not in another, and there is no contradiction: the two tests answer different questions.

The one-sided test is appropriate only when the direction was specified before seeing the data, for example when theory says a coefficient cannot be negative, or when only an increase would lead to action. Choosing the direction after seeing the sign of the estimate doubles the actual size.

11.6.2 Tests and intervals are one computation

The two-sided test of $\boldsymbol{\lambda}^\top \boldsymbol{\beta} = d$ at level α *accepts* exactly when $|\boldsymbol{\lambda}^\top \hat{\boldsymbol{\beta}} - d| \leq t_{n-r, \alpha/2} s \sqrt{\boldsymbol{\lambda}^\top \mathbf{G} \boldsymbol{\lambda}}$. Collecting the values of d that are accepted gives the interval

$$\boldsymbol{\lambda}^\top \hat{\boldsymbol{\beta}} \pm t_{n-r, \alpha/2} s \sqrt{\boldsymbol{\lambda}^\top \mathbf{G} \boldsymbol{\lambda}},$$

which is the $1 - \alpha$ confidence interval [Eq. 7.5.1](#). Every interval of this kind is the set of hypothesized values that a test does not reject, and every test can be read off an interval: reject $\boldsymbol{\lambda}^\top \boldsymbol{\beta} = d$ iff d lies outside. The interval reports more. It shows which values are ruled out, and so how precisely the data determine $\boldsymbol{\lambda}^\top \boldsymbol{\beta}$, which the verdict of a single test does not. For several functions at once, inverting the F test of [Theorem 11.3.2](#) gives an ellipsoid. [Chapter 12](#) develops both and the duality between them.

11.6.3 The overall F test

The most common reduced model is the one with no regressors at all. Let $\mathbf{X}$ contain the intercept column $\mathbf{1}$ and have rank r . The reduced model $E(\mathbf{Y}) = \beta_0 \mathbf{1}$ has $\text{SSE}_0 = \text{SST} = \sum_i (y_i - \bar{y})^2$, and $\text{SST} - \text{SSE} = \text{SSR}$ is the regression sum of squares. By [Theorem 11.1.1](#) and [Theorem 9.5.2](#),

$$F = \frac{\text{SSR}/(r-1)}{\text{SSE}/(n-r)} = \frac{R^2/(r-1)}{(1-R^2)/(n-r)} \sim F(r-1, n-r, \gamma), \quad \gamma = \frac{\|(\mathbf{M} - n^{-1} \mathbf{1} \mathbf{1}^\top) \mathbf{X} \boldsymbol{\beta}\|^2}{\sigma^2}.$$

The **overall F test** asks whether any linear combination of the regressors helps to predict the response. The answer is almost always yes in observational data with a sensible choice of regressors, and a significant overall F is correspondingly weak information. It does not show that any particular regressor matters, that the model fits well, or that it predicts usefully: with n large, a model with $R^2 = 0.02$ has a highly significant overall F . A nonsignificant overall F is more informative. It says that the data give no evidence that the regressors, taken together, relate to the response at all.

11.6.4 Groups of coefficients

Between one coefficient and all of them lies the test of a group: whether the terms of a factor, the powers of a polynomial beyond the linear, or a set of related regressors can be dropped together. The statistic is the extra sum of squares of the group, divided by its degrees of freedom and by s^2 ([Theorem 11.1.1](#)). How does it relate to the t statistics of its members?

Proposition 11.6.2 — The group statistic is the largest single statistic

Let $H : \boldsymbol{\Lambda}^\top \boldsymbol{\beta} = \mathbf{d}$ be testable with $\boldsymbol{\Lambda}$ of full column rank q , let F_H be its statistic [Eq. 11.3.1](#), and for $\mathbf{a} \in \mathbb{R}^q$, $\mathbf{a} \neq \mathbf{0}$, let $T(\mathbf{a})$ be the t statistic [Eq. 11.6.1](#) for the single function $\mathbf{a}^\top \boldsymbol{\Lambda}^\top \boldsymbol{\beta} = \mathbf{a}^\top \mathbf{d}$. Then

$$qF_H = \max_{\mathbf{a} \neq \mathbf{0}} T(\mathbf{a})^2,$$

and the maximum is attained at $\mathbf{a} \propto \mathbf{W}^{-1}(\mathbf{\Lambda}^\top \hat{\boldsymbol{\beta}} - \mathbf{d})$, where $\mathbf{W} = \mathbf{\Lambda}^\top \mathbf{G} \mathbf{\Lambda}$. In particular $t_j^2 \leq qF_H$ for each coefficient β_j in a tested group $\beta_j = 0$, $j \in J$, with $q = |J|$.

Proof. Write $\mathbf{u} = \mathbf{\Lambda}^\top \hat{\boldsymbol{\beta}} - \mathbf{d}$. The function $\mathbf{a}^\top \mathbf{\Lambda}^\top \boldsymbol{\beta}$ is estimable with $\boldsymbol{\lambda} = \mathbf{\Lambda} \mathbf{a} \neq \mathbf{0}$ (the columns of $\mathbf{\Lambda}$ are independent), and $\boldsymbol{\lambda}^\top \mathbf{G} \boldsymbol{\lambda} = \mathbf{a}^\top \mathbf{W} \mathbf{a}$, so $T(\mathbf{a})^2 = (\mathbf{a}^\top \mathbf{u})^2 / (s^2 \mathbf{a}^\top \mathbf{W} \mathbf{a})$. The matrix $\mathbf{W}$ is positive definite (Theorem 9.2.4(a)) and has a positive definite square root (Theorem 1.7.6). By the Cauchy–Schwarz inequality (Proposition 1.1.1) applied to $\mathbf{W}^{1/2} \mathbf{a}$ and $\mathbf{W}^{-1/2} \mathbf{u}$,

$$(\mathbf{a}^\top \mathbf{u})^2 \leq (\mathbf{a}^\top \mathbf{W} \mathbf{a})(\mathbf{u}^\top \mathbf{W}^{-1} \mathbf{u}),$$

with equality iff $\mathbf{W}^{1/2} \mathbf{a}$ is proportional to $\mathbf{W}^{-1/2} \mathbf{u}$, that is, $\mathbf{a} \propto \mathbf{W}^{-1} \mathbf{u}$ (or $\mathbf{u} = \mathbf{0}$). Dividing by $s^2 \mathbf{a}^\top \mathbf{W} \mathbf{a}$ gives $T(\mathbf{a})^2 \leq \mathbf{u}^\top \mathbf{W}^{-1} \mathbf{u} / s^2 = qF_H$, with equality at the stated $\mathbf{a}$. For the coefficient β_j , take $\mathbf{a}$ to be the corresponding coordinate vector. ■

The proposition is the geometric heart of Scheffé’s method for multiple comparisons, and it explains the two ways in which a group test and its members can disagree.

Dilution. A single large t_j need not make the group significant. The group test rejects when $qF_H > qF_\alpha(q, n-r)$, and $qF_\alpha(q, n-r)$ grows with q (Exercise (B2)). A t_j^2 just above $F_\alpha(1, n-r)$ can therefore leave qF_H below the group’s critical value, especially when the other members of the group contribute nothing. This is what happened in Example (Do regions matter?): the Northeast contrast had $|t| = 2.519$, but the three-degree-of-freedom test of all regional differences was not significant.

Collinearity. Conversely, the group can be highly significant while every member has a small t . The direction $\mathbf{a}$ that attains the maximum is generally not a coordinate direction. When the estimates are strongly correlated, the data can say clearly that *some* combination of the coefficients is nonzero without being able to attribute it to any single one.

Example. (Five regressors for the murder rate)

Regress the murder rate of the 50 states on poverty, high-school graduation, single-parent households, percentage white and urbanization, with an intercept ($p = 6$, $n - p = 44$).

Regressor	Estimate	Standard error	t	p-value
intercept	−10.0586	17.6785	−0.569	0.572
poverty	0.2378	0.1412	1.684	0.099
high-school graduates	−0.0047	0.1530	−0.030	0.976
single-parent households	0.4280	0.1372	3.120	0.003
white	0.0108	0.0270	0.398	0.692
urban	0.0046	0.0145	0.315	0.754

The overall test gives $R^2 = 0.643$ and $F = 15.87$ on 5 and 44 degrees of freedom, $p = 6.4 \times 10^{-9}$. Only single parenthood has a t statistic beyond the critical value 2.015. The group of three demographic and educational regressors, high-school graduation, percentage white and urbanization, has $F = 0.096$ on 3 and 44 degrees of freedom, $p = 0.962$: the data give no evidence that these three add anything once poverty and single parenthood are in the model. Every t_j^2 of the group is at most $3F = 0.29$, as [Proposition 11.6.2](#) requires.

This group was chosen here because its members have small t statistics. A group selected that way will tend to have a small F , and its large p-value makes dropping the group look safer than it is. Groups should be defined by the scientific question, before the individual results are seen.

```
import numpy as np
import statsmodels.api as sm
from scipy import stats

data = sm.datasets.statcrime.load_pandas().data
data.index = data.index.str.strip()
data = data.drop(index="District of Columbia")
names = ["poverty", "hs_grad", "single", "white", "urban"]
y = data["murder"].to_numpy()
n = len(y)
X = np.column_stack([np.ones(n), data[names]])
p = X.shape[1]

C = np.linalg.inv(X.T @ X)
b = C @ X.T @ y
sse = np.sum((y - X @ b) ** 2)
s2 = sse / (n - p)
se = np.sqrt(s2 * np.diag(C))
t = b / se
pv = 2 * stats.t.sf(np.abs(t), n - p)
for name, bj, sj, tj, pj in zip(["intercept"] + names, b, se, t, pv):
    print(f"{name:10s} {bj:9.4f} {sj:8.4f} {tj:7.3f} {pj:7.4f}")

sst = np.sum((y - y.mean()) ** 2)
R2 = 1 - sse / sst
F_all = (R2 / (p - 1)) / ((1 - R2) / (n - p))

def extra_F(keep):
    """F test that the coefficients not in `keep` (columns of X) are zero."""
    Xr = X[:, keep]
    br, *_ = np.linalg.lstsq(Xr, y, rcond=None)
    sse_r = np.sum((y - Xr @ br) ** 2)
    q = p - len(keep)
    F = ((sse_r - sse) / q) / s2
    return F, stats.f.sf(F, q, n - p)

F_group, p_group = extra_F([0, 1, 3]) # drop hs_grad, white, urban
print(f"overall: R2 = {R2:.3f}, F = {F_all:.2f} on ({p - 1}, {n - p})")
print(f"hs_grad, white, urban: F = {F_group:.3f}, p = {p_group:.3f}")
```

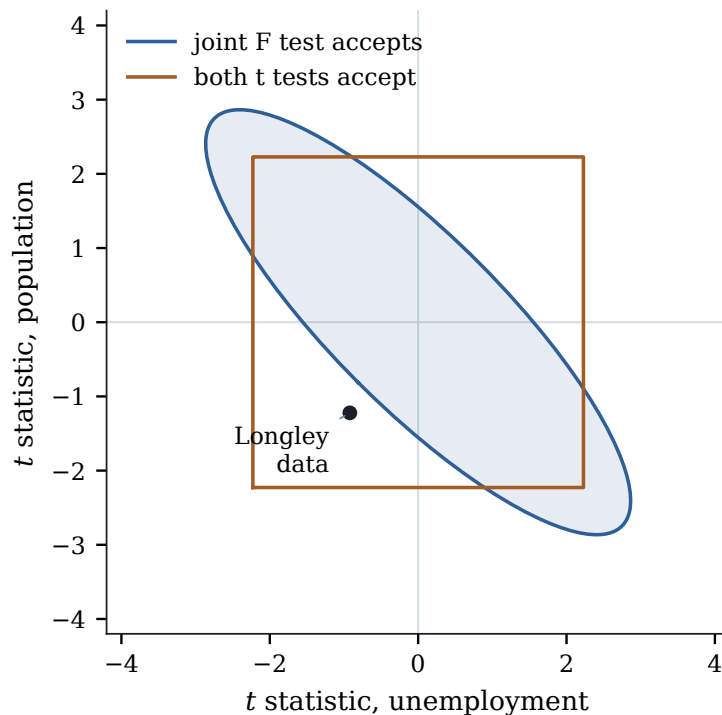


Figure 11.6.1: Acceptance regions for the hypothesis that the unemployment and population coefficients in Longley’s regression are both zero, drawn in the plane of the two t statistics. The square accepts when both individual t tests accept at level 0.05. The ellipse accepts when the joint F test does. The black point is the data.

Example. (Jointly significant, individually not)

Longley’s data record US employment for the 16 years 1947 to 1962, with five economic series: the GNP deflator, GNP, unemployment, the size of the armed forces and population. Regress employment on the five series and an intercept, leaving out the time trend, so that $n - p = 10$. For unemployment and population, the t statistics are -0.921 and -1.222 , with p -values 0.379 and 0.250. Neither comes near the critical value 2.228. The test that *both* coefficients are zero gives $F = 7.175$ on 2 and 10 degrees of freedom, $p = 0.0117$.

Figure 11.6.1 shows why. The two estimates have correlation -0.840 . Under the hypothesis, their t statistics are then likely to have opposite signs, and the F test’s acceptance region is an ellipse stretched along the line $t_1 = -t_2$. The observed pair has both statistics negative, a combination the ellipse excludes although each coordinate on its own is unremarkable. The data are clear that unemployment and population *together* matter, and unclear about how to divide the credit. With 16 consecutive years of trending series, the errors are also unlikely to be independent, a caution taken up in Chapter 21.

```
lg = sm.datasets.longley.load_pandas()
Z = sm.add_constant(lg.exog.drop(columns="YEAR"))
res = sm.OLS(lg.endog, Z).fit()
joint = res.f_test("UNEMP = 0, POP = 0")
print(res.tvalues[["UNEMP", "POP"]].round(3).to_dict(), res.pvalues[["UNEMP", "POP"]].round(3).to_dict())
print(f"joint F = {float(joint.fvalue):.3f}, p = {float(joint.pvalue):.4f}")
```

The figure also shows the opposite disagreement. Points inside the ellipse but outside the square have one significant t and a nonsignificant F . Neither verdict is wrong: the individual tests ask about the coordinate directions, and the joint test asks about all directions at once, paying for that breadth with a larger critical value.

Warning (Several t tests are not a test of the group). Rejecting the group hypothesis “all $\beta_j = 0, j \in J$ ” whenever at least one $|t_j|$ exceeds $t_{n-r, \alpha/2}$ is not a level- α test: when the k coefficient estimates are independent and $n - r$ is large, its size is about $1 - (1 - \alpha)^k$, about 0.23 for $k = 5$ at $\alpha = 0.05$. (The t statistics share the estimate s , so they are not exactly independent.) Reporting the smallest of several p-values as if it were the only one overstates the evidence in the same way. The F test of the group, or a multiple-comparison procedure designed for the family (Chapter 13), controls the familywise error rate. Choosing regressors by their t statistics and then testing the chosen model on the same data invalidates all the p-values of the final fit. Chapter 29 returns to this problem.

11.6.5 Exercises

11.6.5.1 A. Check your understanding

Exercise A1.

Using the table of [Example \(Five regressors for the murder rate\)](#), compute the F statistic for dropping poverty from the five-regressor model and its p-value, without refitting.

Exercise A2.

Show that if the direction of a one-sided t test is chosen after seeing the sign of T , and the test then rejects when $|T| > t_{n-r, \alpha}$, its size is 2α .

11.6.5.2 B. Practice

Exercise B1.

In a full-rank model, show that the t statistic for $\beta_j = \beta_k$ is

$$T = \frac{\hat{\beta}_j - \hat{\beta}_k}{s \sqrt{c_{jj} + c_{kk} - 2c_{jk}}},$$

where c_{jk} are the entries of $(\mathbf{X}^\top \mathbf{X})^{-1}$. Show that it equals the t statistic of the coefficient of $\mathbf{x}_j$ in the reparameterized model with regressors $\mathbf{x}_j$ and $\mathbf{x}_j + \mathbf{x}_k$ in place of $\mathbf{x}_j$ and $\mathbf{x}_k$.

Exercise B2.

Show that $qF_\alpha(q, \nu)$ is strictly increasing in q for fixed ν and α . Deduce that a single t statistic with t^2 just above $F_\alpha(1, \nu)$ does not force the rejection of a group containing it.

Solution. $qF(q, \nu)$ has the law of $U_q/(V/\nu)$ with $U_q \sim \chi^2(q)$ independent of $V \sim \chi^2(\nu)$. Taking $U_{q+1} = U_q + W$ with $W \sim \chi^2(1)$ independent of U_q and V , we have $U_{q+1}/(V/\nu) > U_q/(V/\nu)$ with probability one, so $\Pr\{U_{q+1}/(V/\nu) > c\} > \Pr\{U_q/(V/\nu) > c\}$ for every $c > 0$ (strictly, since $\Pr\{U_q/(V/\nu) \leq c < U_{q+1}/(V/\nu)\} > 0$). Taking $c = qF_\alpha(q, \nu)$, where the right side equals α , shows that the upper α point of $U_{q+1}/(V/\nu)$ exceeds $qF_\alpha(q, \nu)$. For the deduction, if t_j^2 lies between $F_\alpha(1, \nu)$ and $qF_\alpha(q, \nu)$ and the other members contribute nothing, then qF_H can equal $t_j^2 < qF_\alpha(q, \nu)$, and the group is not rejected.

Exercise B3.

For two coefficients with t statistics t_1, t_2 whose estimates have correlation ρ , show that the F statistic for both being zero is

$$F = \frac{t_1^2 + t_2^2 - 2\rho t_1 t_2}{2(1 - \rho^2)}.$$

Check the value 7.175 of [Example \(Jointly significant, individually not\)](#) from $t_1 = -0.921$, $t_2 = -1.222$ and $\rho = -0.840$, and find the value of F had the correlation been $+0.840$.

Solution. Let $\mathbf{D} = \text{diag}(\text{se}_1, \text{se}_2)$ and $\mathbf{R}$ the correlation matrix of the estimates, so that the estimated covariance is $\mathbf{D}\mathbf{R}\mathbf{D}$. With $\mathbf{A}$ selecting the two coefficients, $2F = \hat{\boldsymbol{\beta}}_J^\top (\mathbf{D}\mathbf{R}\mathbf{D})^{-1} \hat{\boldsymbol{\beta}}_J = \mathbf{t}^\top \mathbf{R}^{-1} \mathbf{t}$, where $\mathbf{t} = \mathbf{D}^{-1} \hat{\boldsymbol{\beta}}_J$, and $\mathbf{R}^{-1} = (1 - \rho^2)^{-1} \begin{pmatrix} 1 & -\rho \\ -\rho & 1 \end{pmatrix}$. Numerically, $t_1^2 + t_2^2 = 2.341$, $-2\rho t_1 t_2 = 1.890$ and $2(1 - \rho^2) = 0.589$, so $F = 7.175$, as in the example, up to the rounding of these three terms. With $\rho = +0.840$ the middle term changes sign and $F = 0.765$, about $(2.341 - 1.890)/0.589$: the same two t statistics would then be entirely unremarkable.

11.6.5.3 C. Going deeper

Exercise C1.

Show that $\max T(\boldsymbol{\lambda})^2$ over all nonzero estimable $\boldsymbol{\lambda}$, for the hypotheses $\boldsymbol{\lambda}^\top \boldsymbol{\beta} = 0$, equals rF_* , where F_* is the statistic for the reduced model $E(\mathbf{Y}) = \mathbf{0}$. Deduce that $\Pr\{T(\boldsymbol{\lambda})^2 \leq rF_\alpha(r, n-r) \text{ for all estimable } \boldsymbol{\lambda}\} = 1 - \alpha$ when $\boldsymbol{\beta} = \mathbf{0}$. What changes if $\boldsymbol{\beta} \neq \mathbf{0}$ and $T(\boldsymbol{\lambda})$ is centred at $\boldsymbol{\lambda}^\top \boldsymbol{\beta}$? ([Chapter 13](#) builds simultaneous inference on this.)

Exercise C2.

Find the likelihood ratio test of $\boldsymbol{\lambda}^\top \boldsymbol{\beta} \leq d$ against $\boldsymbol{\lambda}^\top \boldsymbol{\beta} > d$ for estimable $\boldsymbol{\lambda}$, and show that it rejects for large values of T in [Eq. 11.6.1](#). (Maximize the likelihood under the inequality constraint: if $\boldsymbol{\lambda}^\top \hat{\boldsymbol{\beta}} \leq d$ the unrestricted maximum is feasible; otherwise the maximum is on the boundary.)

Solution. If $\boldsymbol{\lambda}^\top \hat{\boldsymbol{\beta}} \leq d$, the unrestricted maximizer satisfies the constraint and $\Lambda = 1$. Otherwise the log-likelihood, maximized over σ^2 , is a decreasing function of $\|\mathbf{y} - \mathbf{X}\mathbf{b}\|^2$, which is a convex quadratic in $\mathbf{b}$ with minimum at $\hat{\boldsymbol{\beta}}$. Its minimum over the half-space $\boldsymbol{\lambda}^\top \mathbf{b} \leq d$ therefore lies on the boundary $\boldsymbol{\lambda}^\top \mathbf{b} = d$, where it equals $\text{SSE} + (\boldsymbol{\lambda}^\top \hat{\boldsymbol{\beta}} - d)^2 / \boldsymbol{\lambda}^\top \mathbf{G} \boldsymbol{\lambda}$

(Eq. 7.3.3). So $\Lambda = \{1 + T^2/(n - r)\}^{-n/2}$ when $T > 0$ and $\Lambda = 1$ when $T \leq 0$. This is a nonincreasing function of T , strictly decreasing for $T > 0$, so $\Lambda \leq k < 1$ iff $T \geq c$ for some $c > 0$, which is the one-sided t test.

11.7 Summary and notes

11.7.1 Summary

Idea.

1. A linear hypothesis says that the mean vector lies in a subspace $\mathcal{C}(\mathbf{X}_0)$ of the model space. Its F statistic compares the extra sum of squares $\|(\mathbf{M} - \mathbf{M}_0)\mathbf{y}\|^2$, per dimension of the test space, with s^2 . Under normal errors it is $F(r - r_0, n - r, \gamma)$, central under the hypothesis, with γ the squared distance from the mean to the reduced model in units of σ^2 (Definition 11.1.1, Theorem 11.1.1, Proposition 11.1.2).
2. Geometrically, F is a function of the angle between the reduced model's residual vector and the test space. The rejection region is a cone, the statistic is invariant under shifts within the reduced model, rescaling and rotations of the test and residual spaces, and in canonical coordinates every hypothesis is a test that a q -dimensional normal mean is zero. The null distribution holds for every spherically symmetric error law (Proposition 11.2.1, Proposition 11.2.2, Proposition 11.2.3, Proposition 11.2.4).
3. A constraint $\mathbf{\Lambda}^\top \boldsymbol{\beta} = \mathbf{d}$ is a hypothesis about the data only if it is estimable and consistent. Nonestimable constraints cannot be tested at all. For a testable hypothesis, $(\mathbf{\Lambda}^\top \hat{\boldsymbol{\beta}} - \mathbf{d})^\top [\mathbf{\Lambda}^\top (\mathbf{X}^\top \mathbf{X})^{-} \mathbf{\Lambda}]^{-} (\mathbf{\Lambda}^\top \hat{\boldsymbol{\beta}} - \mathbf{d})$ is the extra sum of squares of the reduced model with offset $\mathbf{X}\mathbf{b}_0$, whatever the rank of $\mathbf{X}$ or $\mathbf{\Lambda}$, and restricted least squares has a closed form through any generalized inverse (Definition 11.3.1, Proposition 11.3.1, Theorem 11.3.2, Proposition 11.3.3).
4. The F test is the likelihood ratio test. The Wald, likelihood ratio and score statistics are nu , $n \log(1 + u)$ and $nu/(1 + u)$ with $u = qF/(n - r)$, so they always satisfy Wald > LR > Score. Calibrated by χ^2 , they disagree and have inflated sizes in small samples (Theorem 11.4.1, Proposition 11.4.2).
5. The noncentral F family has monotone likelihood ratio, and every test invariant under the symmetries of the problem is a function of F . Hence the F test is uniformly most powerful among invariant tests (Lemma 11.4.3, Theorem 11.4.4).
6. Power depends only on γ , increases with it, and grows with replication. In the one-way layout the least favourable configuration for a given range puts two means at the extremes and the rest at the midpoint. Observed power is a function of the p-value and adds nothing to it (Theorem 11.5.1, Proposition 11.5.2, Proposition 11.5.3).
7. For one estimable function the F test is the two-sided t test, and the one-sided version has exact size for one-sided hypotheses. A group statistic qF is the largest squared t statistic over all combinations in the group, which explains why group and individual tests can disagree in both directions (Theorem 11.6.1, Proposition 11.6.2).

11.7.2 Notes and sources

Origins. The ratio of mean squares as a test statistic is Fisher's. It grew out of his analysis of variance in the early 1920s, and Fisher (1925) tabulated it on the logarithmic scale, as $z = \frac{1}{2} \log F$. Snedecor (1934) introduced the letter F in Fisher's honour and published tables of the ratio itself. The exact t distribution of a regression coefficient, with $n - p$ degrees of freedom, is also due to Fisher (1922). The geometry of [Section 11.2](#) follows the coordinate-free tradition described in the notes to [Chapter 6](#). Christensen (2020, chapter 3) develops testing entirely in that language, with test spaces, offsets and nonestimable constraints. Rencher and Schaalje (2008, chapters 8 and 12) and Seber and Lee (2003, chapter 4) give the matrix treatment for full-rank and rank-deficient models, including the canonical elimination of [Exercise \(C2\)](#). Sen and Srivastava (1990, chapter 3) and Agresti (2015, sections 3.1–3.2) emphasize applied examples. Note that Rencher and Schaalje and Christensen define the noncentrality as half of our γ ([Section 4.1](#)).

Testable hypotheses. The insistence that only estimable functions can be tested goes back at least to Searle (1971), whose treatment of “testable hypotheses” in models not of full rank is the standard reference; [Proposition 11.3.1](#) is a minimal version of his argument. Christensen (2020, section 3.3) shows how a nonestimable constraint reduces to its estimable part, which in our treatment is [Proposition 9.2.6](#).

Likelihood ratio, Wald and score. Neyman and Pearson (1933) introduced the most powerful test and the likelihood ratio principle for composite hypotheses. The large-sample χ^2 calibration of the likelihood ratio is due to Wilks (1938), the Wald statistic to Wald (1943), and the score statistic to Rao (1948). The inequality $\text{Wald} \geq \text{LR} \geq \text{score}$ for linear restrictions in normal regression, and the resulting possibility of conflicting verdicts, were pointed out by Berndt and Savin (1977) and discussed further by Breusch (1979). The “ nR^2 ” form of the score test in [Exercise \(B2\)](#) is widely used in econometrics.

Invariance. Lehmann and Romano (2005) develop the theory of invariant tests and prove that the F test is uniformly most powerful invariant; [Theorem 11.4.4](#) follows their approach in outline. Scheffé (1959) treats the power of F tests in the analysis of variance. The robustness of the null distribution to spherically symmetric errors ([Proposition 11.2.4](#)) is part of the theory of spherical and elliptical distributions surveyed by Fang, Kotz and Ng (1990).

Power. The noncentral F distribution was tabulated for power calculations by Tang (1938). The least favourable configuration of [Proposition 11.5.2](#) is the basis of the classical sample-size tables for the analysis of variance. Cohen (1988) popularized effect-size based power analysis. Hoenig and Heisey (2001) explain why “observed power” computed from the data is uninformative, the point of [Proposition 11.5.3](#). Wasserstein and Lazar (2016) discuss common misreadings of p-values. Equivalence tests, which can support a claim of “no important difference”, go back to Schuirmann (1987).

Other. The test of equality of two regressions in [Exercise \(B4\)](#) and [Exercise \(C1\)](#) is due to Chow (1960). Longley (1967) assembled his data to test the numerical accuracy of regression programs ([Chapter 10](#)). Monotonicity properties of the χ^2 , F and t distributions in their degrees of freedom, of the kind behind the dilution effect of [Exercise \(B2\)](#), are studied systematically by Ghosh (1973).

11.7.3 References

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Chapter 12

Confidence Intervals, Regions and Prediction Bands

A test asks whether one value of a parameter is compatible with the data; an interval reports all such values, and how precisely they are pinned down. Under normal errors the answers are exact for every sample size, and they come from three facts: the fitted vector is normal, the residual sum of squares is an independent chi-squared variable, and their ratio gives t and F pivots. One estimable function gives a t interval; several give an ellipsoid, the dual of the F test. A future observation gives a prediction interval, whose width never falls below the noise and whose validity rests on the normality of a single error. The whole regression surface gives the Working–Hotelling band, from the Cauchy–Schwarz inequality. Inverting a prediction band gives a calibration set, sometimes an interval, sometimes two rays, sometimes the whole line, and no honest procedure can avoid the last two. Finally, in the conjugate model of [Chapter 7](#), credible and predictive intervals have the same shapes, and under the flat prior they are the frequentist ones.

What you need. The sampling distributions of the normal linear model ([Chapter 7: Theorem 7.5.1](#), [Corollary 7.5.2](#) and [Corollary 7.5.3](#)) and the conjugate posterior ([Theorem 7.6.2](#), [Lemma 7.6.1](#), [Corollary 7.6.4](#)). Projections, leverage and the Frisch–Waugh–Lovell theorem ([Chapter 6: Theorem 6.3.7](#), [Proposition 6.8.1](#), [Theorem 6.6.2](#)). Estimability ([Chapter 8: Definition 8.2.1](#), [Theorem 8.3.1](#)) and the sum of squares of a linear hypothesis ([Chapter 9: Theorem 9.2.4](#)). Chi-squared and F distributions of quadratic forms ([Chapter 4: Theorem 4.3.2](#), [Definition 4.1.2](#), [Definition 4.1.3](#)). The tests of [Chapter 11](#) are the duals of the intervals here, but the proofs below do not depend on them.

12.0.1 Roadmap

- [Intervals for estimable functions](#)
- [Confidence ellipsoids](#)
- [Prediction intervals](#)
- [Simultaneous bands for the regression line and surface](#)
- [Inverse prediction and calibration](#)
- [Bayesian thread II: credible intervals and predictive distributions](#)
- [Summary and notes](#)

12.1 Intervals for estimable functions

A point estimate needs a statement of its precision. A confidence interval gives one exactly: a range computed from the data that contains the unknown quantity with a probability fixed in advance. This section builds the interval for one estimable function, relates it to the t test, and records two facts that are easy to overlook: an interval for a non-estimable function is useless, and the interval for σ^2 , unlike the one for a coefficient, depends delicately on normality.

Throughout the chapter we work in the normal linear model

$$\mathbf{Y} \sim N_n(\mathbf{X}\boldsymbol{\beta}, \sigma^2\mathbf{I}), \quad \boldsymbol{\beta} \in \mathbb{R}^p, \sigma^2 > 0,$$

of Eq. 7.3.1, with $r = \text{rank}(\mathbf{X}) < n$. We write $\mathbf{G}$ for any generalized inverse of $\mathbf{X}^\top \mathbf{X}$, $\hat{\boldsymbol{\beta}}$ for any least squares estimate, $\text{SSE} = \|(\mathbf{I} - \mathbf{M})\mathbf{Y}\|^2$ and $s^2 = \text{SSE}/(n - r)$. For $0 < \gamma < 1$, $t_{\nu, \gamma}$, $F_\gamma(q, \nu)$ and $\chi_\gamma^2(\nu)$ denote the upper γ points of $t(\nu)$, $F(q, \nu)$ and $\chi^2(\nu)$.

12.1.1 Confidence sets

Definition 12.1.1 — Confidence set

Let $\theta = g(\boldsymbol{\beta}, \sigma^2)$ be a quantity with values in a set Θ . A **confidence set** for θ with **confidence level** $1 - \alpha$ is a rule that assigns to each data vector $\mathbf{y}$ a subset $C(\mathbf{y}) \subseteq \Theta$, such that

$$\Pr_{\boldsymbol{\beta}, \sigma^2} \{\theta \in C(\mathbf{Y})\} \geq 1 - \alpha$$

for every $\boldsymbol{\beta} \in \mathbb{R}^p$ and $\sigma^2 > 0$. The left side is the **coverage probability** at $(\boldsymbol{\beta}, \sigma^2)$. The set is **exact** if the coverage equals $1 - \alpha$ at every parameter value. When $C(\mathbf{y})$ is an interval it is a **confidence interval**.

The probability is taken over $\mathbf{Y}$ with the parameters fixed: the set is random and the target is not. A realized interval either contains θ or it does not; $1 - \alpha$ describes the procedure.

The standard way to build an exact set is a **pivot**: a function $Q(\mathbf{Y}, \theta)$ whose distribution is the same at every parameter value. If $\Pr\{Q(\mathbf{Y}, \theta) \in A\} = 1 - \alpha$ for a fixed set A , then $C(\mathbf{y}) = \{\theta_0 : Q(\mathbf{y}, \theta_0) \in A\}$ is exact, because the events $\{\theta \in C(\mathbf{Y})\}$ and $\{Q(\mathbf{Y}, \theta) \in A\}$ are the same event. All the sets of this chapter are built this way, from pivots supplied by Chapter 7.

12.1.2 The t interval

Theorem 12.1.1 — The t interval for an estimable function

Let $\boldsymbol{\lambda} \in \mathcal{C}(\mathbf{X}^\top)$, $\boldsymbol{\lambda} \neq \mathbf{0}$, and put $\text{se}(\boldsymbol{\lambda}^\top \hat{\boldsymbol{\beta}}) = s\sqrt{\boldsymbol{\lambda}^\top \mathbf{G} \boldsymbol{\lambda}}$. Let

$$I(\mathbf{Y}) = [\boldsymbol{\lambda}^\top \hat{\boldsymbol{\beta}} - t_{n-r, \alpha/2} \text{se}(\boldsymbol{\lambda}^\top \hat{\boldsymbol{\beta}}), \quad \boldsymbol{\lambda}^\top \hat{\boldsymbol{\beta}} + t_{n-r, \alpha/2} \text{se}(\boldsymbol{\lambda}^\top \hat{\boldsymbol{\beta}})]. \quad (12.1.1)$$

- (a) $I(\mathbf{Y})$ does not depend on the choice of $\hat{\boldsymbol{\beta}}$ or of $\mathbf{G}$.
- (b) $I(\mathbf{Y})$ is an exact $1 - \alpha$ confidence interval for $\boldsymbol{\lambda}^\top \boldsymbol{\beta}$.

- (c) Its length $L = 2t_{n-r,\alpha/2} s \sqrt{\lambda^\top \mathbf{G} \lambda}$ has expectation $E L = 2t_{n-r,\alpha/2} \kappa_{n-r} \sigma \sqrt{\lambda^\top \mathbf{G} \lambda}$, where

$$\kappa_\nu = E \sqrt{V/\nu} = \sqrt{\frac{2}{\nu}} \frac{\Gamma((\nu+1)/2)}{\Gamma(\nu/2)} < 1, \quad V \sim \chi^2(\nu).$$

- (d) The one-sided intervals $(-\infty, \lambda^\top \hat{\beta} + t_{n-r,\alpha} \text{se}]$ and $[\lambda^\top \hat{\beta} - t_{n-r,\alpha} \text{se}, \infty)$ are exact $1 - \alpha$ confidence intervals for $\lambda^\top \beta$.

Proof.

- (a) Write $\lambda = \mathbf{X}^\top \rho$. Then $\lambda^\top \hat{\beta} = \rho^\top \mathbf{X} \hat{\beta} = \rho^\top \mathbf{M} \mathbf{Y}$ and $\lambda^\top \mathbf{G} \lambda = \rho^\top \mathbf{X} \mathbf{G} \mathbf{X}^\top \rho = \rho^\top \mathbf{M} \rho$ by Theorem 6.3.7. Neither involves the choices, and neither does s .
- (b) By Corollary 7.5.2, $T = (\lambda^\top \hat{\beta} - \lambda^\top \beta) / \text{se}(\lambda^\top \hat{\beta}) \sim t(n-r)$ at every (β, σ^2) . The event $\lambda^\top \beta \in I(\mathbf{Y})$ is the event $|T| \leq t_{n-r,\alpha/2}$, whose probability is $1 - \alpha$. So part (b) restates Eq. 7.5.1 as a confidence statement.
- (c) L is a constant multiple of $s = \sigma \sqrt{V/(n-r)}$ with $V = \text{SSE}/\sigma^2 \sim \chi^2(n-r)$ (Theorem 7.5.1(b)). With the $\chi^2(\nu)$ density of Eq. 4.1.1,

$$E \sqrt{V} = \int_0^\infty \frac{v^{1/2} v^{\nu/2-1} e^{-v/2}}{2^{\nu/2} \Gamma(\nu/2)} dv = \frac{2^{(\nu+1)/2} \Gamma((\nu+1)/2)}{2^{\nu/2} \Gamma(\nu/2)},$$

because the integrand is a multiple of the $\chi^2(\nu+1)$ density. Dividing by $\sqrt{\nu}$ gives κ_ν . By Jensen's inequality $E \sqrt{V/\nu} < \sqrt{E(V/\nu)} = 1$, with strict inequality since V is not constant.

- (d) The events are $\{T \geq -t_{n-r,\alpha}\}$ and $\{T \leq t_{n-r,\alpha}\}$, each of probability $1 - \alpha$.

■

Part (a) matters when $\mathbf{X}$ is rank deficient: the interval is a property of the model, not of the software's choice of generalized inverse. The degrees of freedom are $n - r$, not $n - p$.

Part (c) shows what the interval costs. Its expected length is proportional to $\sigma \sqrt{\lambda^\top \mathbf{G} \lambda}$, the standard deviation of the estimate. Estimating σ adds the factor $t_{n-r,\alpha/2} \kappa_{n-r}$ in place of the normal point $z_{\alpha/2}$ that would be used if σ were known. For $\alpha = 0.05$ that factor is 2.45 at $n - r = 5$ and 2.06 at $n - r = 20$, against 1.96. Beyond about twenty residual degrees of freedom, precision improves only through $\lambda^\top \mathbf{G} \lambda$, that is, through the design.

Example. (Murder rates: coefficients and a mean)

Return to the regression of the 2009 murder rate of the 50 states on the poverty rate, the percentage of single-parent households and the percentage urban, with an intercept (Example (Poverty and murder rates)). Here $n = 50$, $r = 4$, $s^2 = 2.217$, and $t_{46,0.025} = 2.013$. The 95% intervals for the coefficients are

Coefficient	Estimate	Standard error	95% interval
intercept	−9.0946	1.5555	(−12.226, −5.964)
poverty	0.2538	0.0934	(0.066, 0.442)
single parent	0.3980	0.0829	(0.231, 0.565)
urban	0.0046	0.0131	(−0.022, 0.031)

The data are consistent with anything from 0.066 to 0.442 murders per 100,000 for each point of poverty, with the other regressors held fixed in the sense of [Theorem 6.6.2](#). The urban interval straddles zero but is narrow, which says more than “not significant”: any such association is at most 0.031 per percentage point.

A mean response is an estimable function too. For a hypothetical state with poverty 18%, single-parent households 28% and 70% urban, $\mathbf{x}_0 = (1, 18, 28, 70)^\top$, the estimated mean murder rate is 6.937 with standard error 0.412, and the 95% interval is (6.107, 7.767). Here $\mathbf{x}_0^\top \mathbf{G} \mathbf{x}_0 = 0.0767$. [Section 12.3](#) interprets this number as the leverage of the new point.

```
import numpy as np
import statsmodels.api as sm
from scipy import stats

data = sm.datasets.statcrime.load_pandas().data
data = data.drop(index="District of Columbia")
y = data["murder"].to_numpy()
X = np.column_stack([np.ones(len(y)), data[["poverty", "single", "urban"]]])
n, p = X.shape
r = np.linalg.matrix_rank(X)
G = np.linalg.pinv(X.T @ X)                    # any generalized inverse will do
beta_hat = G @ X.T @ y
s2 = np.sum((y - X @ beta_hat) ** 2) / (n - r)
q = stats.t.ppf(0.975, n - r)                  # upper 2.5% point of t(n - r)

def t_interval(lam):
    """95% interval for the estimable function lam^T beta."""
    est = lam @ beta_hat
    se = np.sqrt(s2 * lam @ G @ lam)
    return est, se, est - q * se, est + q * se

for j, name in enumerate(["intercept", "poverty", "single", "urban"]):
    est, se, lo, hi = t_interval(np.eye(p)[j])
    print(f"{name:9s} {est:8.4f} se {se:.4f} 95% interval ({lo:.3f}, {hi:.3f})")
```

12.1.3 What the standard error measures

For a coefficient in a full-rank model with an intercept, [Eq. 6.6.2](#) gives

$$\text{se}(\hat{\beta}_j) = \frac{s}{\|(\mathbf{I} - \mathbf{M}_{(j)})\mathbf{x}_j\|} = \frac{s}{\sqrt{S_{jj}(1 - R_j^2)}},$$

where $\mathbf{M}_{(j)}$ projects onto the span of the other columns. The interval is short when $\mathbf{x}_j$ has a large component orthogonal to the other regressors; collinearity lengthens it through $1 - R_j^2$.

In general $\boldsymbol{\lambda}^\top \mathbf{G} \boldsymbol{\lambda} = \|\mathbf{M} \boldsymbol{\rho}\|^2$, with $\boldsymbol{\lambda} = \mathbf{X}^\top \boldsymbol{\rho}$, is the smallest squared length of a vector $\mathbf{a}$ with $E(\mathbf{a}^\top \mathbf{Y}) = \boldsymbol{\lambda}^\top \boldsymbol{\beta}$ ([Theorem 7.2.1](#)).

12.1.4 Duality with tests

An interval and a family of tests carry the same information, in any model.

Proposition 12.1.2 — Tests and confidence sets

For each $\theta_0 \in \Theta$, let $A(\theta_0)$ be the acceptance region of a test of $H : \theta = \theta_0$ with $\Pr_{\beta, \sigma^2}\{\mathbf{Y} \in A(\theta_0)\} \geq 1 - \alpha$ whenever $g(\beta, \sigma^2) = \theta_0$. Then

$$C(\mathbf{y}) = \{\theta_0 \in \Theta : \mathbf{y} \in A(\theta_0)\}$$

is a $1 - \alpha$ confidence set for θ , and it is exact if every test has size exactly α at every parameter value in its hypothesis. Conversely, if C is a $1 - \alpha$ confidence set, the rule “reject $H : \theta = \theta_0$ iff $\theta_0 \notin C(\mathbf{y})$ ” is a level- α test for every θ_0 .

Proof. For every parameter value, with $\theta = g(\beta, \sigma^2)$, the events $\{\theta \in C(\mathbf{Y})\}$ and $\{\mathbf{Y} \in A(\theta)\}$ coincide, so they have the same probability. Both directions follow. ■

For an estimable function, the two-sided t test of $\lambda^\top \beta = d$ accepts iff $|\lambda^\top \hat{\beta} - d| \leq t_{n-r, \alpha/2} \text{se}$, which is the condition $d \in I(\mathbf{y})$. So the interval Eq. 12.1.1 is exactly the set of values d that the t test of Theorem 11.6.1 does not reject. The p -value of the test of d is the smallest α for which the $1 - \alpha$ interval excludes d .

The interval is the more informative report. A test of $\beta_j = 0$ reports one bit, confounded with the sample size; the interval shows location and resolution. Exercise (B2) uses it to answer “is the effect negligible?”, which a test of $\beta_j = 0$ cannot answer at all.

12.1.5 Functions that are not estimable

What if $\lambda \notin \mathcal{C}(\mathbf{X}^\top)$? The recipe above fails because $\lambda^\top \hat{\beta}$ depends on the choice of $\hat{\beta}$. The failure is not a defect of the recipe. No procedure can do better.

Proposition 12.1.3 — No useful interval for a non-estimable function

Let $\lambda \notin \mathcal{C}(\mathbf{X}^\top)$ and let C be a $1 - \alpha$ confidence set for $\lambda^\top \beta$ with $\alpha < 1$. Then at every parameter value (β, σ^2) and for every real v , $\Pr_{\beta, \sigma^2}\{v \in C(\mathbf{Y})\} \geq 1 - \alpha$. If the sets $C(\mathbf{y})$ are such that $\{(\mathbf{y}, v) : v \in C(\mathbf{y})\}$ is measurable, the expected length of $C(\mathbf{Y})$ is infinite at every parameter value.

Proof. By Theorem 8.3.1 there is $\mathbf{w} \in \mathcal{N}(\mathbf{X})$ with $\lambda^\top \mathbf{w} \neq 0$. Fix (β, σ^2) and v , and put $\beta' = \beta + c\mathbf{w}$ with $c = (v - \lambda^\top \beta)/\lambda^\top \mathbf{w}$. Then $\mathbf{X}\beta' = \mathbf{X}\beta$, so $\mathbf{Y}$ has the same distribution under (β', σ^2) as under (β, σ^2) , while $\lambda^\top \beta' = v$. Hence $\Pr_{\beta, \sigma^2}\{v \in C(\mathbf{Y})\} =$

$\Pr_{\beta', \sigma^2} \{\lambda^\top \beta' \in C(\mathbf{Y})\} \geq 1 - \alpha$. For the length, Fubini's theorem gives, for every $a > 0$,

$$E[\text{length}(C(\mathbf{Y}) \cap [-a, a])] = \int_{-a}^a \Pr\{v \in C(\mathbf{Y})\} dv \geq 2a(1 - \alpha),$$

and the left side is at most the expected length of $C(\mathbf{Y})$. Let $a \rightarrow \infty$. ■

Thus the algebraic fact of [Chapter 8](#) becomes a statement about inference.

12.1.6 An interval for σ^2

Proposition 12.1.4 — Interval for the error variance

The interval

$$\left[\frac{\text{SSE}}{\chi_{\alpha/2}^2(n-r)}, \frac{\text{SSE}}{\chi_{1-\alpha/2}^2(n-r)} \right]$$

is an exact $1 - \alpha$ confidence interval for σ^2 , and the square roots of its endpoints give one for σ .

Proof. $\text{SSE}/\sigma^2 \sim \chi^2(n-r)$ at every parameter value ([Theorem 7.5.1\(b\)](#)), so it is a pivot. The event $\chi_{1-\alpha/2}^2(n-r) \leq \text{SSE}/\sigma^2 \leq \chi_{\alpha/2}^2(n-r)$ has probability $1 - \alpha$, and it is the event that σ^2 lies in the interval, which is the interval of [Exercise \(B3\)](#). Taking square roots is monotone. ■

For the murder-rate regression, $\text{SSE} = 46s^2$ with $s^2 = 2.217$, the chi-squared points are 29.160 and 66.617, and the 95% interval for σ^2 is (1.531, 3.497). The interval is not symmetric about s^2 , and it is not the shortest one with this coverage ([Exercise \(C2\)](#)).

The two pivots of this section react differently to non-normal errors. The t statistic needs normality of a weighted average of errors, which a central limit theorem restores, and $s \rightarrow \sigma$, which holds for any error law. The chi-squared pivot needs the whole error distribution, because the variance of SSE depends on the fourth moment ([Proposition 7.5.4](#)). Take the fitted murder-rate model as the truth, with $\sigma^2 = s^2$, and simulate 40000 data sets. With normal errors the two 95% intervals cover in proportions 0.9514 (poverty coefficient) and 0.9491 (σ^2). With errors from a $t(5)$ distribution rescaled to the same variance, the coefficient interval still covers 0.9498 of the time, but the interval for σ^2 covers only 0.8165. More data do not repair this ([Exercise \(B3\)](#)).

```
x0 = np.array([1.0, 18.0, 28.0, 70.0])           # a state with high poverty and
single_parenthood
est0, se0, lo0, hi0 = t_interval(x0)
print(f"mean murder rate at x0: {est0:.3f}, se {se0:.3f}, interval ({lo0:.3f}, {hi0:.3f})")

sse = s2 * (n - r)
lo_s2 = sse / stats.chi2.ppf(0.975, n - r)       # interval for sigma^2 from SSE/
sigma^2 ~ chi^2(n-r)
```



```
hi_s2 = sse / stats.chi2.ppf(0.025, n - r)
print(f"s^2 = {s2:.3f}; 95% interval for sigma^2 ({lo_s2:.3f}, {hi_s2:.3f})")
```

12.1.7 Exercises

12.1.7.1 A. Check your understanding

Exercise A1.

A design is replicated m times: $\mathbf{X}$ is replaced by m stacked copies of itself. For a fixed estimable $\boldsymbol{\lambda}^\top \boldsymbol{\beta}$, how does $\boldsymbol{\lambda}^\top \mathbf{G} \boldsymbol{\lambda}$ change? Roughly how many replicates are needed to halve the expected length of the 95% interval when $n - r$ is already large, and why is the answer slightly less than 4 when $n - r$ is small?

Solution. The stacked matrix has $\mathbf{X}^\top \mathbf{X}$ replaced by $m\mathbf{X}^\top \mathbf{X}$, and $\mathbf{G}/m$ is a generalized inverse of $m\mathbf{X}^\top \mathbf{X}$, so $\boldsymbol{\lambda}^\top \mathbf{G} \boldsymbol{\lambda}$ is divided by m . By [Theorem 12.1.1\(c\)](#) the expected length is proportional to $t_{\nu, \alpha/2} \kappa_\nu / \sqrt{m}$, with $\nu = mn - r$. For large ν the factor $t\kappa$ is nearly constant and $m = 4$ halves the length; for small ν , replication also lowers $t\kappa$, so slightly less suffices.

12.1.7.2 B. Practice

Exercise B1.

Show that $\kappa_\nu \rightarrow 1$ as $\nu \rightarrow \infty$, and that s/κ_{n-r} is an unbiased estimator of σ . Compute κ_ν for $\nu = 1$ and $\nu = 2$.

Solution. $V/\nu \rightarrow 1$ in probability by the law of large numbers, and $\sqrt{V/\nu}$ is bounded in mean square, hence uniformly integrable, so $E \sqrt{V/\nu} \rightarrow 1$. Unbiasedness: $E s = \sigma E \sqrt{V/(n-r)} = \sigma \kappa_{n-r}$. For $\nu = 1$, $\kappa_1 = \sqrt{2} \Gamma(1)/\Gamma(1/2) = \sqrt{2/\pi} \approx 0.798$. For $\nu = 2$, $\kappa_2 = \Gamma(3/2)/\Gamma(1) = \sqrt{\pi}/2 \approx 0.886$.

Exercise B2.

Let $\delta > 0$ be a margin below which $|\boldsymbol{\lambda}^\top \boldsymbol{\beta}|$ is considered negligible. The *two one-sided tests* procedure declares equivalence when both $H_1 : \boldsymbol{\lambda}^\top \boldsymbol{\beta} \leq -\delta$ and $H_2 : \boldsymbol{\lambda}^\top \boldsymbol{\beta} \geq \delta$ are rejected by one-sided t tests at level α . Show that this happens iff the $1 - 2\alpha$ interval [Eq. 12.1.1](#) lies inside $(-\delta, \delta)$. Show that the procedure has level α : when $|\boldsymbol{\lambda}^\top \boldsymbol{\beta}| \geq \delta$, the probability of declaring equivalence is at most α .

Solution. H_1 is rejected iff $(\boldsymbol{\lambda}^\top \hat{\boldsymbol{\beta}} + \delta)/\text{se} > t_{n-r, \alpha}$, that is iff the lower end of the $1 - 2\alpha$ interval, $\boldsymbol{\lambda}^\top \hat{\boldsymbol{\beta}} - t_{n-r, \alpha} \text{se}$, exceeds $-\delta$. Symmetrically H_2 is rejected iff the upper end is below δ . For the level: if $\boldsymbol{\lambda}^\top \boldsymbol{\beta} \leq -\delta$, declaring equivalence requires rejecting H_1 . Write $\tau = \sigma \sqrt{\boldsymbol{\lambda}^\top \mathbf{G} \boldsymbol{\lambda}}$, $Z = (\boldsymbol{\lambda}^\top \hat{\boldsymbol{\beta}} - \boldsymbol{\lambda}^\top \boldsymbol{\beta})/\tau \sim N(0, 1)$ and $V = \text{SSE}/\sigma^2 \sim \chi^2(\nu)$, $\nu = n - r$, independent of Z ([Theorem 7.5.1](#)). The test statistic is $(Z + \eta)/\sqrt{V/\nu}$ with $\eta = (\boldsymbol{\lambda}^\top \boldsymbol{\beta} + \delta)/\tau$. For every realization of (Z, V) it increases in η , so the probability that it exceeds $t_{\nu, \alpha}$ increases in η . At $\eta = 0$ that probability is α , so it is at most α whenever

$\lambda^\top \beta \leq -\delta$. The case $\lambda^\top \beta \geq \delta$ is symmetric.

Exercise B3.

Suppose $\mathbf{Y} \sim N_n(\mathbf{X}\beta, \sigma^2\mathbf{V})$ with $\mathbf{V}$ known and positive definite and $\mathbf{X}$ of full column rank. Using the generalized least squares estimator $\tilde{\beta} = (\mathbf{X}^\top \mathbf{V}^{-1} \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{V}^{-1} \mathbf{Y}$ and $\tilde{s}^2 = (\mathbf{Y} - \mathbf{X}\tilde{\beta})^\top \mathbf{V}^{-1} (\mathbf{Y} - \mathbf{X}\tilde{\beta}) / (n - p)$, derive an exact $1 - \alpha$ interval for $\lambda^\top \beta$. Write it out for a straight line $E Y_i = \beta_0 + \beta_1 x_i$ with $\mathbf{V} = \text{diag}(1/w_i)$.

Solution. Let $\mathbf{V} = \mathbf{L}\mathbf{L}^\top$ (Cholesky) and $\mathbf{Z} = \mathbf{L}^{-1}\mathbf{Y} \sim N_n(\mathbf{L}^{-1}\mathbf{X}\beta, \sigma^2\mathbf{I})$. The ordinary least squares estimate for $(\mathbf{Z}, \mathbf{L}^{-1}\mathbf{X})$ is $\tilde{\beta}$, and its residual sum of squares is $(n - p)\tilde{s}^2$. Applying [Theorem 12.1.1](#) to the transformed model gives $\lambda^\top \tilde{\beta} \pm t_{n-p, \alpha/2} \tilde{s} \sqrt{\lambda^\top (\mathbf{X}^\top \mathbf{V}^{-1} \mathbf{X})^{-1} \lambda}$. For the weighted straight line, with $\bar{x}_w = \sum w_i x_i / \sum w_i$, $\bar{y}_w$ defined likewise and $S_w = \sum w_i (x_i - \bar{x}_w)^2$, the slope estimate is $\tilde{\beta}_1 = \sum w_i (x_i - \bar{x}_w)(y_i - \bar{y}_w) / S_w$ and its interval is $\tilde{\beta}_1 \pm t_{n-2, \alpha/2} \tilde{s} / \sqrt{S_w}$.

Exercise B4.

In a one-way layout with a groups of sizes $n_1, \dots, n_a$ and the overparameterized model $E Y_{ij} = \mu + \alpha_i$, show that the interval for $\alpha_1 - \alpha_2$ is $\bar{y}_1 - \bar{y}_2 \pm t_{n-a, \alpha/2} s \sqrt{1/n_1 + 1/n_2}$, with $n = \sum n_i$. What does [Proposition 12.1.3](#) say about an interval for α_1 alone?

Solution. The rank is $r = a$. With $\lambda = \mathbf{e}_2 - \mathbf{e}_3$ (coordinates $\mu, \alpha_1, \dots, \alpha_a$), $\lambda^\top \beta = \alpha_1 - \alpha_2$ is estimable, and $\lambda^\top \hat{\beta} = \bar{y}_1 - \bar{y}_2$ for every least squares estimate. Its variance is $\sigma^2(1/n_1 + 1/n_2)$ since the group means are independent, so $\lambda^\top \mathbf{G}\lambda = 1/n_1 + 1/n_2$. Substitute in [Eq. 12.1.1](#) with $n - r = n - a$. The function α_1 is not estimable (adding c to every α_i and subtracting it from μ leaves $\mathbf{X}\beta$ unchanged), so every 95% confidence set for it covers every real number with probability at least 0.95.

12.1.7.3 C. Going deeper

Exercise C1.

Among intervals $[\lambda^\top \hat{\beta} - a \text{ se}, \lambda^\top \hat{\beta} + b \text{ se}]$ with $a, b \geq 0$ and coverage exactly $1 - \alpha$, show that the expected length is smallest when $a = b = t_{n-r, \alpha/2}$. (Use that the t density is symmetric and strictly decreasing in $|x|$.)

Solution. Coverage is $F(b) - F(-a)$, with F the $t(n - r)$ distribution function, and the expected length is proportional to $a + b$. If $a < b$, shifting $[-a, b]$ slightly left gains density $f(a)$ and loses $f(b) < f(a)$, raising the coverage, so a strictly shorter interval has coverage $1 - \alpha$. Hence $a = b$.

Exercise C2.

Intervals for σ^2 of the form $[\text{SSE}/c_2, \text{SSE}/c_1]$ with $\Pr(c_1 \leq \chi^2(\nu) \leq c_2) = 1 - \alpha$ all have exact coverage. Show that the one with the smallest expected length satisfies $c_1^2 f(c_1) = c_2^2 f(c_2)$, where f is the $\chi^2(\nu)$ density, and that the equal-tailed choice of

[Proposition 12.1.4](#) does not satisfy this condition.

12.2 Confidence ellipsoids

Two 95% intervals, one for each of two coefficients, do not make a 95% statement about the pair, and they ignore the correlation between the estimates. The natural joint region is an ellipsoid: the set of values the F test does not reject, shaped like the covariance matrix of the estimates, with shadows that are wider than the t intervals. This geometry explains an F test that rejects while every t test accepts, and the reverse.

12.2.1 The ellipsoid

Let $\mathbf{\Lambda}$ be a $p \times q$ matrix of rank q whose columns lie in $\mathcal{C}(\mathbf{X}^\top)$, so that every entry of $\boldsymbol{\phi} = \mathbf{\Lambda}^\top \boldsymbol{\beta}$ is estimable. Put

$$\hat{\boldsymbol{\phi}} = \mathbf{\Lambda}^\top \hat{\boldsymbol{\beta}}, \quad \mathbf{W} = \mathbf{\Lambda}^\top \mathbf{G} \mathbf{\Lambda}.$$

By [Theorem 9.2.4\(a\)](#), neither depends on the choices of $\hat{\boldsymbol{\beta}}$ and $\mathbf{G}$, and $\mathbf{W}$ is positive definite. By [Theorem 7.5.1\(a\)](#), $\hat{\boldsymbol{\phi}} \sim N_q(\boldsymbol{\phi}, \sigma^2 \mathbf{W})$.

Theorem 12.2.1 — Confidence ellipsoid for $\mathbf{\Lambda}^\top \boldsymbol{\beta}$

With the notation above, let $F_\alpha = F_\alpha(q, n - r)$ and

$$E(\mathbf{Y}) = \{\boldsymbol{\phi}_0 \in \mathbb{R}^q : (\hat{\boldsymbol{\phi}} - \boldsymbol{\phi}_0)^\top \mathbf{W}^{-1} (\hat{\boldsymbol{\phi}} - \boldsymbol{\phi}_0) \leq q s^2 F_\alpha\}. \quad (12.2.1)$$

- (a) $E(\mathbf{Y})$ is an exact $1 - \alpha$ confidence set for $\mathbf{\Lambda}^\top \boldsymbol{\beta}$.
- (b) $E(\mathbf{Y})$ is a solid ellipsoid centred at $\hat{\boldsymbol{\phi}}$. If $\mathbf{W} = \sum_{k=1}^q w_k \mathbf{u}_k \mathbf{u}_k^\top$ is a spectral decomposition, its principal axes point along $\mathbf{u}_1, \dots, \mathbf{u}_q$ and have half-lengths $\sqrt{q s^2 F_\alpha w_k}$. Its volume is

$$\frac{\pi^{q/2}}{\Gamma(q/2 + 1)} (q s^2 F_\alpha)^{q/2} \det(\mathbf{W})^{1/2}.$$

- (c) (*Duality with the F test.*) For $\mathbf{d} \in \mathbb{R}^q$, let $\text{SSE}_{\mathbf{d}}$ be the minimum of $\|\mathbf{Y} - \mathbf{X}\mathbf{b}\|^2$ over $\{\mathbf{b} : \mathbf{\Lambda}^\top \mathbf{b} = \mathbf{d}\}$ and let

$$F_{\mathbf{d}} = \frac{(\text{SSE}_{\mathbf{d}} - \text{SSE})/q}{s^2} = \frac{(\hat{\boldsymbol{\phi}} - \mathbf{d})^\top \mathbf{W}^{-1} (\hat{\boldsymbol{\phi}} - \mathbf{d})}{q s^2}.$$

Then $\mathbf{d} \in E(\mathbf{Y})$ iff $F_{\mathbf{d}} \leq F_\alpha$, that is, iff the level- α F test of $H : \mathbf{\Lambda}^\top \boldsymbol{\beta} = \mathbf{d}$ does not reject.

Proof.

- (a) Since $\mathbf{W}$ is invertible, $\mathbf{W}^{-1}$ is its only generalized inverse, and [Corollary 7.5.3](#) with

$\text{rank}(\mathbf{\Lambda}) = q$ gives

$$\frac{(\hat{\boldsymbol{\Phi}} - \boldsymbol{\Phi})^\top \mathbf{W}^{-1}(\hat{\boldsymbol{\Phi}} - \boldsymbol{\Phi})}{q s^2} \sim F(q, n - r)$$

at every parameter value. This is a pivot, and $\boldsymbol{\Phi} \in E(\mathbf{Y})$ is the event that it is at most F_α .

- (b) Put $c = q s^2 F_\alpha$ and write $\boldsymbol{\Phi}_0 - \hat{\boldsymbol{\Phi}} = \sum_k c_k \mathbf{u}_k$. Since $\mathbf{W}^{-1} = \sum_k w_k^{-1} \mathbf{u}_k \mathbf{u}_k^\top$, the defining inequality becomes $\sum_k c_k^2 / w_k \leq c$, the standard form of an ellipsoid with half-axes $\sqrt{c w_k}$ along $\mathbf{u}_k$. Equivalently $E(\mathbf{Y}) = \hat{\boldsymbol{\Phi}} + \mathbf{W}^{1/2} B$, where B is the ball of radius $\sqrt{c}$ in $\mathbb{R}^q$. A linear map multiplies volume by the absolute value of its determinant, so the volume is $\det(\mathbf{W})^{1/2} c^{q/2}$ times the volume $\pi^{q/2} / \Gamma(q/2 + 1)$ of the unit ball.
- (c) The two expressions for F_d agree by [Theorem 9.2.4\(c\)](#), and the equivalence is the definition of $E(\mathbf{Y})$. ■

The statistic F_d is the F statistic of the general linear hypothesis in both of its forms (see [Theorem 11.3.2](#)), and part (c) is [Proposition 12.1.2](#) in concrete form. In particular the test of $\mathbf{\Lambda}^\top \boldsymbol{\beta} = \mathbf{0}$ rejects iff the ellipsoid misses the origin.

In the full-rank case with $\mathbf{\Lambda} = \mathbf{I}_p$, the region for the whole coefficient vector is

$$\begin{aligned} \{\mathbf{b} : (\hat{\boldsymbol{\beta}} - \mathbf{b})^\top \mathbf{X}^\top \mathbf{X}(\hat{\boldsymbol{\beta}} - \mathbf{b}) &\leq p s^2 F_\alpha(p, n - p)\} \\ &= \{\mathbf{b} : \|\mathbf{X}\hat{\boldsymbol{\beta}} - \mathbf{X}\mathbf{b}\|^2 \leq p s^2 F_\alpha(p, n - p)\}. \end{aligned}$$

The second form says that $\mathbf{b}$ is in the region iff its fitted vector $\mathbf{X}\mathbf{b}$ is close to the least squares fit $\hat{\mathbf{Y}}$. This suggests looking at the mean vector directly.

12.2.2 A ball in the column space

Proposition 12.2.2 — Confidence ball for the mean vector

Let $\boldsymbol{\mu} = \mathbf{X}\boldsymbol{\beta}$ and $c_r = r s^2 F_\alpha(r, n - r)$. The set

$$B(\mathbf{Y}) = \{\boldsymbol{\mu}_0 \in \mathcal{C}(\mathbf{X}) : \|\hat{\mathbf{Y}} - \boldsymbol{\mu}_0\|^2 \leq c_r\}$$

is an exact $1 - \alpha$ confidence set for $\boldsymbol{\mu}$: a ball of radius $\sqrt{c_r}$ in the r -dimensional space $\mathcal{C}(\mathbf{X})$, centred at the fitted vector. Moreover, if $\mathbf{\Lambda} = \mathbf{X}^\top \mathbf{T}$ for an $n \times q$ matrix $\mathbf{T}$ (such a $\mathbf{T}$ exists by [Theorem 8.3.1](#)), the image of $B(\mathbf{Y})$ under $\boldsymbol{\mu}_0 \mapsto \mathbf{T}^\top \boldsymbol{\mu}_0$ is the set

$$\{\boldsymbol{\Phi}_0 : (\hat{\boldsymbol{\Phi}} - \boldsymbol{\Phi}_0)^\top \mathbf{W}^{-1}(\hat{\boldsymbol{\Phi}} - \boldsymbol{\Phi}_0) \leq c_r\},$$

which contains the ellipsoid [Eq. 12.2.1](#).

Proof. $\hat{\mathbf{Y}} - \boldsymbol{\mu} = \mathbf{M}(\mathbf{Y} - \mathbf{X}\boldsymbol{\beta})$, and $\mathbf{M}$ is symmetric idempotent of rank r , so $\|\hat{\mathbf{Y}} - \boldsymbol{\mu}\|^2 / \sigma^2 \sim \chi^2(r)$ by [Theorem 4.3.2\(a\)](#), independently of SSE ([Theorem 7.5.1\(c\)](#)). Dividing by $r s^2 / \sigma^2$ gives an $F(r, n - r)$ pivot, and the first claim follows as before.

For the image, put $\mathbf{A} = \mathbf{T}^\top \mathbf{M}$, a $q \times n$ matrix. For $\boldsymbol{\mu}_0 \in \mathcal{C}(\mathbf{X})$, $\mathbf{T}^\top \boldsymbol{\mu}_0 = \mathbf{A}\boldsymbol{\mu}_0$, and $\mathbf{A}\mathbf{A}^\top =$

$\mathbf{T}^\top \mathbf{M} \mathbf{T} = \mathbf{W}$ (Theorem 9.2.4(a)). Also $\hat{\boldsymbol{\phi}} = \mathbf{T}^\top \hat{\mathbf{Y}} = \mathbf{A} \hat{\mathbf{Y}}$. So the image consists of the points $\hat{\boldsymbol{\phi}} - \mathbf{A} \mathbf{u}$ with $\mathbf{u} \in \mathcal{C}(\mathbf{X})$ and $\|\mathbf{u}\|^2 \leq c_r$. If $\mathbf{z} = \mathbf{A} \mathbf{u}$ with $\|\mathbf{u}\|^2 \leq c_r$, then $\mathbf{z}^\top \mathbf{W}^{-1} \mathbf{z} = \mathbf{u}^\top \mathbf{A}^\top \mathbf{W}^{-1} \mathbf{A} \mathbf{u} \leq \|\mathbf{u}\|^2 \leq c_r$, because $\mathbf{A}^\top \mathbf{W}^{-1} \mathbf{A}$ is symmetric and idempotent, hence a projection, and a projection does not increase length (Proposition 6.3.4(d)). Conversely, if $\mathbf{z}^\top \mathbf{W}^{-1} \mathbf{z} \leq c_r$, then $\mathbf{u} = \mathbf{A}^\top \mathbf{W}^{-1} \mathbf{z}$ lies in $\mathcal{C}(\mathbf{M}) = \mathcal{C}(\mathbf{X})$, satisfies $\mathbf{A} \mathbf{u} = \mathbf{z}$, and has $\|\mathbf{u}\|^2 = \mathbf{z}^\top \mathbf{W}^{-1} \mathbf{z} \leq c_r$.

For the containment it remains to show $q F_\alpha(q, \nu) \leq r F_\alpha(r, \nu)$ when $q \leq r$. Let $U_q \sim \chi^2(q)$, $U' \sim \chi^2(r-q)$ and $V \sim \chi^2(\nu)$ be independent. Then $U_q + U' \sim \chi^2(r)$, so for every c , $\Pr\{(U_q + U')/(V/\nu) \leq c\} \leq \Pr\{U_q/(V/\nu) \leq c\}$. The upper α point of the variable on the left is $r F_\alpha(r, \nu)$, and that of the one on the right is $q F_\alpha(q, \nu)$.

■

So all confidence ellipsoids are views of one ball around $\hat{\mathbf{Y}}$ in the column space. Each linear summary $\mathbf{A}^\top \boldsymbol{\beta}$ maps the ball to an ellipsoid, and since the ball covers $\boldsymbol{\mu}$ with probability $1 - \alpha$, all the images cover their targets *simultaneously* with probability at least $1 - \alpha$, at the price of the constant $r F_\alpha(r, \nu)$ in place of $q F_\alpha(q, \nu)$. Section 12.4 takes $q = 1$ and all directions at once.

12.2.3 Shadows

One inequality gives the extent of an ellipsoid in every direction.

Lemma 12.2.3 — Extent of an ellipsoid

Let $\mathbf{W}$ be a $q \times q$ positive definite matrix and $c > 0$. For every $\mathbf{a} \in \mathbb{R}^q$,

$$\max\{\mathbf{a}^\top \mathbf{u} : \mathbf{u}^\top \mathbf{W}^{-1} \mathbf{u} \leq c\} = \sqrt{c \mathbf{a}^\top \mathbf{W} \mathbf{a}},$$

attained at $\mathbf{u} = \sqrt{c/(\mathbf{a}^\top \mathbf{W} \mathbf{a})} \mathbf{W} \mathbf{a}$ when $\mathbf{a} \neq \mathbf{0}$. Consequently $\mathbf{u}^\top \mathbf{W}^{-1} \mathbf{u} \leq c$ iff $|\mathbf{a}^\top \mathbf{u}| \leq \sqrt{c \mathbf{a}^\top \mathbf{W} \mathbf{a}}$ for every $\mathbf{a} \in \mathbb{R}^q$.

Proof. The bound is Corollary 1.7.8(c) with $\mathbf{B} = \mathbf{W}^{-1}$; directly, write $\mathbf{a}^\top \mathbf{u} = (\mathbf{W}^{1/2} \mathbf{a})^\top (\mathbf{W}^{-1/2} \mathbf{u})$. By the Cauchy–Schwarz inequality, $(\mathbf{a}^\top \mathbf{u})^2 \leq (\mathbf{a}^\top \mathbf{W} \mathbf{a})(\mathbf{u}^\top \mathbf{W}^{-1} \mathbf{u}) \leq c \mathbf{a}^\top \mathbf{W} \mathbf{a}$, with equality when $\mathbf{W}^{-1/2} \mathbf{u}$ is a nonnegative multiple of $\mathbf{W}^{1/2} \mathbf{a}$ and $\mathbf{u}^\top \mathbf{W}^{-1} \mathbf{u} = c$, which is the stated $\mathbf{u}$. For the last statement, “only if” follows from the maximum applied to $\mathbf{a}$ and $-\mathbf{a}$. For “if”, take $\mathbf{a} = \mathbf{W}^{-1} \mathbf{u}$: then $(\mathbf{u}^\top \mathbf{W}^{-1} \mathbf{u})^2 \leq c \mathbf{u}^\top \mathbf{W}^{-1} \mathbf{u}$, so $\mathbf{u}^\top \mathbf{W}^{-1} \mathbf{u} \leq c$.

■

Corollary 12.2.4 — Shadows of the confidence ellipsoid

For every $\mathbf{a} \in \mathbb{R}^q$, the projection of $E(\mathbf{Y})$ onto the line spanned by $\mathbf{a}$ is the interval

$$\mathbf{a}^\top \hat{\boldsymbol{\phi}} \pm \sqrt{q F_\alpha(q, n-r)} \text{se}(\mathbf{a}^\top \hat{\boldsymbol{\phi}}), \quad \text{se}(\mathbf{a}^\top \hat{\boldsymbol{\phi}}) = s \sqrt{\mathbf{a}^\top \mathbf{W} \mathbf{a}},$$

and $E(\mathbf{Y})$ is the intersection of the slabs $\{\boldsymbol{\phi}_0 : |\mathbf{a}^\top \hat{\boldsymbol{\phi}} - \mathbf{a}^\top \boldsymbol{\phi}_0| \leq \sqrt{q F_\alpha} \text{se}(\mathbf{a}^\top \hat{\boldsymbol{\phi}})\}$ over all $\mathbf{a}$. Hence, with probability exactly $1 - \alpha$, every one of these intervals contains $\mathbf{a}^\top \boldsymbol{\Lambda}^\top \boldsymbol{\beta}$.

Proof. Apply [Lemma 12.2.3](#) to $\mathbf{u} = \boldsymbol{\phi}_0 - \hat{\boldsymbol{\phi}}$ with $c = q s^2 F_\alpha$. The extreme values of $\mathbf{a}^\top \boldsymbol{\phi}_0$ over $E(\mathbf{Y})$ are $\mathbf{a}^\top \hat{\boldsymbol{\phi}} \pm \sqrt{c \mathbf{a}^\top \mathbf{W} \mathbf{a}}$, and the set of values in between is attained because $E(\mathbf{Y})$ is convex. The intersection statement is the last part of the lemma. The final claim holds because the event “all intervals cover” is the event $\boldsymbol{\phi} \in E(\mathbf{Y})$. ■

A shadow is a t interval with $t_{n-r, \alpha/2}$ replaced by $\sqrt{q F_\alpha(q, n-r)}$. The multipliers agree when $q = 1$ ([Exercise \(A1\)](#)) and the shadow is wider when $q > 1$, as it must be: the shadows hold simultaneously for all directions $\mathbf{a}$, including directions chosen after seeing the data. This is Scheffé’s method, developed in [Chapter 13](#) ([Theorem 13.2.2](#)).

The box of one-at-a-time t intervals undercovers, and it disagrees with the ellipsoid in both directions: a corner of the box can lie outside the ellipsoid, and the ellipsoid pokes out of the box along its long axis.

Example. (A joint region for two coefficients)

In the murder-rate regression of [Example \(Murder rates: coefficients and a mean\)](#), take Λ to pick out the coefficients of poverty and single parenthood, so $q = 2$, $\hat{\boldsymbol{\phi}} = (0.2538, 0.3980)^\top$, and the correlation of the two estimates is -0.639 . The negative correlation reflects the positive correlation of the regressors: the data can credit murder to either, but not to both at once. With $F_{0.05}(2, 46) = 3.200$, the 95% ellipse is drawn in [Figure 12.2.1](#). The F statistic for $\boldsymbol{\phi} = \mathbf{0}$ is 39.81, so the origin is far outside.

The shadows of the ellipse on the two axes are (0.018, 0.490) and (0.188, 0.608), wider than the t intervals (0.066, 0.442) and (0.231, 0.565) by the factor $\sqrt{2 \times 3.200}/2.013 = 1.257$. Two points show how the ellipse and the box disagree.

- $\mathbf{A} = (0.442, 0.565)$, the upper corner of the box. Neither t test rejects, but $\mathbf{A}$ raises both coefficients at once, against the correlation, and the joint test rejects: $F = 11.230$, $p = 0.0001$.
- $\mathbf{B} = (0.463, 0.224)$, on the long axis of the ellipse. Both t statistics exceed 2.013 (2.242 and 2.098), but $\mathbf{B}$ trades one coefficient for the other, which the data cannot resolve, and $F = 2.888 < 3.200$.

The ellipse and the box have similar areas (0.1197 and 0.1255) but different shapes. In 40000 simulated data sets from the fitted model the ellipse covered the true pair in a proportion 0.9500 and the box in only 0.9164.

```
import numpy as np
import statsmodels.api as sm
from scipy import stats

data = sm.datasets.statelcrime.load_pandas().data
data = data.drop(index="District of Columbia")
y = data["murder"].to_numpy()
X = np.column_stack([np.ones(len(y)), data[["poverty", "single", "urban"]]])
n, p = X.shape
C = np.linalg.inv(X.T @ X)
beta_hat = C @ X.T @ y
s2 = np.sum((y - X @ beta_hat) ** 2) / (n - p)
```

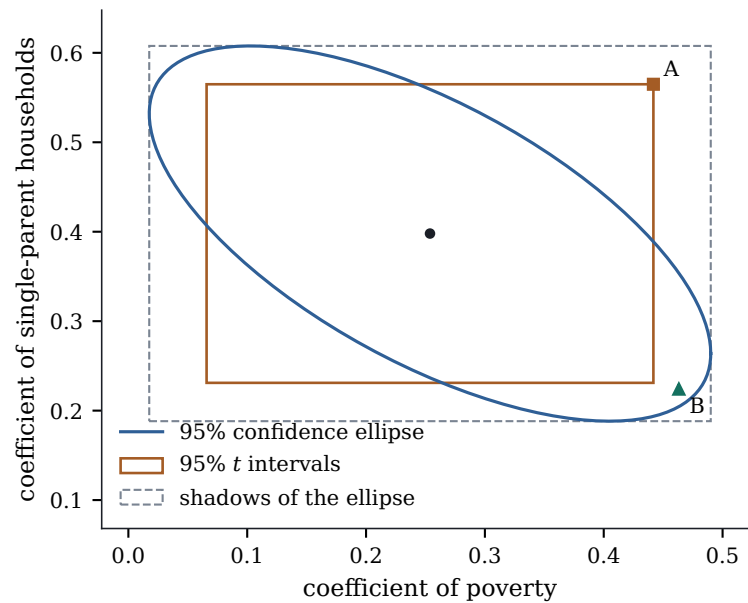


Figure 12.2.1: The 95% confidence ellipse for the poverty and single-parent coefficients, the box of 95% t intervals, and the ellipse's shadows (dashed). A is inside the box but outside the ellipse; B the reverse.

```

L = np.eye(p)[: , [1, 2]]                                # Lambda: the poverty and single
    coefficients
q = L.shape[1]
phi_hat = L.T @ beta_hat
W = L.T @ C @ L                                           # Cov(phi_hat) / sigma^2
Fq = stats.f.ppf(0.95, q, n - p)

def F_stat(d):
    """F statistic for the hypothesis  $\Lambda^T \beta = d$ ."""
    u = phi_hat - d
    return u @ np.linalg.solve(W, u) / (q * s2)

def in_ellipse(d):
    return F_stat(d) <= Fq

corr = W[0, 1] / np.sqrt(W[0, 0] * W[1, 1])
print(f"estimates {phi_hat}, correlation of the estimates {corr:.3f}")
print(f"F({q}, {n - p}) 95% point {Fq:.3f}; F statistic for (0, 0): {F_stat(np.zeros
(2)):.2f}")

tq = stats.t.ppf(0.975, n - p)
se = np.sqrt(s2 * np.diag(W))
t_box = np.column_stack([phi_hat - tq * se, phi_hat + tq * se])    # one-at-
    a-time t intervals
k = np.sqrt(q * Fq)
shadow = np.column_stack([phi_hat - k * se, phi_hat + k * se])    #
    projections of the ellipse
print("t intervals:\n", t_box.round(3))
print("shadows of the ellipse:\n", shadow.round(3), f"\nwider by the factor {k / tq
:.3f}")

```

12.2.4 Two cautions

The ellipsoid is for a question fixed in advance. If the coefficients are chosen because they look interesting, the ball of [Proposition 12.2.2](#) or the methods of [Chapter 13](#) are the honest tools.

Intervals after a test. If intervals are reported only when a preliminary test of $\mathbf{A}^\top \boldsymbol{\beta} = \mathbf{0}$ rejects, their coverage *given* the rejection is no longer $1 - \alpha$, and can be larger or smaller (Olshen 1973).

12.2.5 Exercises

12.2.5.1 A. Check your understanding

Exercise A1.

Show that for $q = 1$ the ellipsoid [Eq. 12.2.1](#) is the t interval [Eq. 12.1.1](#). Which fact about the $F(1, \nu)$ distribution does this use?

Solution. With $q = 1$, $\mathbf{W} = w$ is a positive number and the region is $(\hat{\phi} - \phi_0)^2 \leq s^2 w F_\alpha(1, n - r)$. If $T \sim t(\nu)$, then $T^2 \sim F(1, \nu)$ ([Definition 4.1.3](#) and the remark after it), so $\Pr\{T^2 \leq t_{\nu, \alpha/2}^2\} = \Pr\{|T| \leq t_{\nu, \alpha/2}\} = 1 - \alpha$ and $F_\alpha(1, \nu) = t_{\nu, \alpha/2}^2$. The region is $|\hat{\phi} - \phi_0| \leq t_{\nu, \alpha/2} s \sqrt{w}$.

Exercise A2.

For the straight line $E Y_i = \beta_0 + \beta_1 x_i$, write the 95% confidence ellipse for (β_0, β_1) in terms of n , $\bar{x}$ and $\sum x_i^2$. Show that in the centred parameterization $E Y_i = \gamma_0 + \beta_1(x_i - \bar{x})$ the axes of the ellipse for (γ_0, β_1) are parallel to the coordinate axes. When is it a circle?

Solution. $\mathbf{X}^\top \mathbf{X} = \begin{pmatrix} n & n\bar{x} \\ n\bar{x} & \sum x_i^2 \end{pmatrix}$, and the region is $n(\hat{\beta}_0 - b_0)^2 + 2n\bar{x}(\hat{\beta}_0 - b_0)(\hat{\beta}_1 - b_1) + \sum x_i^2(\hat{\beta}_1 - b_1)^2 \leq 2s^2 F_\alpha(2, n - 2)$. In the centred parameterization the cross-product matrix is $\text{diag}(n, S_{xx})$, so the quadratic form has no cross term, and the ellipse has axes along the coordinates with half-lengths $\sqrt{2s^2 F_\alpha/n}$ and $\sqrt{2s^2 F_\alpha/S_{xx}}$. It is a circle iff $S_{xx} = n$, that is, iff the x_i have mean square deviation one.

12.2.5.2 B. Practice

Exercise B1.

Take $q = 2$ and let ρ be the correlation of $\hat{\phi}_1$ and $\hat{\phi}_2$. Show that the area of the ellipse divided by the area of the box of the two t intervals is

$$\frac{\pi F_\alpha(2, n - r)}{2 t_{n-r, \alpha/2}^2} \sqrt{1 - \rho^2},$$

so that it depends on the design only through ρ . Check it against [Example \(A joint](#)

region for two coefficients), where the areas are 0.1197 and 0.1255 at $\rho = -0.639$. For which $|\rho|$ do the two regions have equal area there, and what happens to the ratio as the two regressors become collinear?

Solution. By [Theorem 12.2.1\(b\)](#) the ellipse has area $\pi \cdot 2F_\alpha s^2 \det(\mathbf{W})^{1/2}$, and writing ω_j for the diagonal entries of $\mathbf{W}$, $\det \mathbf{W} = \omega_1 \omega_2 (1 - \rho^2)$. The box has sides $2ts\sqrt{\omega_j}$, so area $4t^2 s^2 \sqrt{\omega_1 \omega_2}$; divide. In the example $F_{0.05}(2, 46) = 3.200$ and $t = 2.013$ give 1.240 at $\rho = 0$, and $1.240\sqrt{1 - \rho^2} = 0.954$ at $\rho = -0.639$, which is $0.1197/0.1255$. The areas agree when $\sqrt{1 - \rho^2} = 1/1.240$, that is $|\rho| = 0.592$. With uncorrelated estimates the ellipse is the larger region; as the regressors become collinear, $|\rho| \rightarrow 1$, the ratio tends to 0 and the ellipse becomes a thin needle across a large box, which the joint region mostly excludes.

Exercise B2.

In [Example \(A joint region for two coefficients\)](#), the hypothesis that one point of poverty and one point of single parenthood have the same coefficient is the line $\phi_1 = \phi_2$. Using [Lemma 12.2.3](#), show that this line meets the ellipse iff $|\hat{\phi}_1 - \hat{\phi}_2| \leq \sqrt{2F_\alpha(2, n - r)} \text{se}(\hat{\phi}_1 - \hat{\phi}_2)$. Compare this with the t test of $\phi_1 = \phi_2$. Which is appropriate if the hypothesis was formulated before seeing the data?

Solution. The line is $\{\boldsymbol{\phi}_0 : \mathbf{a}^\top \boldsymbol{\phi}_0 = 0\}$ with $\mathbf{a} = (1, -1)^\top$. By [Corollary 12.2.4](#), the values of $\mathbf{a}^\top \boldsymbol{\phi}_0$ over the ellipse fill the interval $\mathbf{a}^\top \hat{\boldsymbol{\phi}} \pm \sqrt{2F_\alpha} s \sqrt{\mathbf{a}^\top \mathbf{W} \mathbf{a}}$, so the line meets the ellipse iff this interval contains 0. The t test uses the smaller multiplier $t_{n-r, \alpha/2}$. It is the right test for a hypothesis fixed in advance. The ellipse criterion is the right one for a contrast chosen after looking at the estimates, because it holds simultaneously for all $\mathbf{a}$.

Exercise B3.

Let the columns of $\mathbf{\Lambda}$ lie in $\mathcal{C}(\mathbf{X}^\top)$ but be linearly dependent, with $\text{rank}(\mathbf{\Lambda}) = k < q$. Show that $\mathbf{W} = \mathbf{\Lambda}^\top \mathbf{G} \mathbf{\Lambda}$ is singular, that $\hat{\boldsymbol{\phi}} - \boldsymbol{\phi} \in \mathcal{C}(\mathbf{W})$ with probability one, and that

$$\left\{ \boldsymbol{\phi}_0 \in \hat{\boldsymbol{\phi}} + \mathcal{C}(\mathbf{W}) : (\hat{\boldsymbol{\phi}} - \boldsymbol{\phi}_0)^\top \mathbf{W}^{-} (\hat{\boldsymbol{\phi}} - \boldsymbol{\phi}_0) \leq k s^2 F_\alpha(k, n - r) \right\}$$

is an exact $1 - \alpha$ confidence set, a k -dimensional ellipsoid inside a k -dimensional affine subspace of $\mathbb{R}^q$.

Solution. $\mathbf{W} = (\mathbf{M}\mathbf{T})^\top (\mathbf{M}\mathbf{T})$ has rank $\text{rank}(\mathbf{M}\mathbf{T}) = \text{rank}(\mathbf{\Lambda}) = k$ (proof of [Corollary 7.5.3](#)). The vector $\hat{\boldsymbol{\phi}} - \boldsymbol{\phi}$ is $N_q(\mathbf{0}, \sigma^2 \mathbf{W})$, so it lies in $\mathcal{C}(\mathbf{W})$ almost surely ([Theorem 2.2.3](#)). By [Corollary 7.5.3](#) the quadratic form over $k s^2$ is $F(k, n - r)$ and does not depend on the generalized inverse. So the displayed set covers $\boldsymbol{\phi}$ with probability $1 - \alpha$. Restricted to the affine subspace, write $\mathbf{W} = \mathbf{U}\mathbf{D}\mathbf{U}^\top$ with $\mathbf{U}$ ($q \times k$) orthonormal and $\mathbf{D}$ positive diagonal. For $\boldsymbol{\phi}_0 - \hat{\boldsymbol{\phi}} = \mathbf{U}\mathbf{c}$, the form is $\mathbf{c}^\top \mathbf{D}^{-1} \mathbf{c}$, an ellipsoid in the coordinates $\mathbf{c}$.

12.2.5.3 C. Going deeper

Exercise C1.

Let $\mathbf{X}$ have full column rank. Show that the confidence ellipsoid for $\boldsymbol{\beta}$ is a region of the form $\{\mathbf{b} : \ell_p(\mathbf{b}) \geq \ell_p(\hat{\boldsymbol{\beta}}) - k\}$, where $\ell_p(\mathbf{b}) = \max_{\sigma^2} \ell(\mathbf{b}, \sigma^2)$ is the profile log-likelihood of Chapter 7, and find k in terms of $F_\alpha(p, n - p)$. How does k compare with the value $\frac{1}{2}\chi_\alpha^2(p)$ suggested by the asymptotic theory of likelihood ratios?

Solution. The profile log-likelihood is $\ell_p(\mathbf{b}) = \text{const} - \frac{n}{2} \log \|\mathbf{Y} - \mathbf{X}\mathbf{b}\|^2$, and $\|\mathbf{Y} - \mathbf{X}\mathbf{b}\|^2 = \text{SSE} + (\hat{\boldsymbol{\beta}} - \mathbf{b})^\top \mathbf{X}^\top \mathbf{X} (\hat{\boldsymbol{\beta}} - \mathbf{b})$ by Eq. 6.3.1. So $\ell_p(\hat{\boldsymbol{\beta}}) - \ell_p(\mathbf{b}) = \frac{n}{2} \log\{1 + pF_{\mathbf{b}}/(n - p)\}$, where $F_{\mathbf{b}}$ is the statistic of Theorem 12.2.1(c). This is increasing in $F_{\mathbf{b}}$, so the ellipsoid is the region with $k = \frac{n}{2} \log\{1 + pF_\alpha(p, n - p)/(n - p)\}$. As $n \rightarrow \infty$, $pF_\alpha(p, n - p) \rightarrow \chi_\alpha^2(p)$ and $k \rightarrow \frac{1}{2}\chi_\alpha^2(p)$. For finite n , k is larger, and the exact region is larger than the asymptotic likelihood region.

12.3 Prediction intervals

Often the quantity of interest is an observation not yet made: tomorrow's stock loss, or the reading of the next specimen. An interval for it must allow for the error in the estimated mean, which shrinks as data accumulate, and the error of the new observation itself, which does not. The first depends on where the new point lies relative to the data; the second makes prediction intervals depend on normality in a way that confidence intervals do not.

12.3.1 Predicting a new observation

Let the data follow the normal linear model of Section 12.1, and let

$$Y_0 = \mathbf{x}_0^\top \boldsymbol{\beta} + \varepsilon_0, \quad \varepsilon_0 \sim N(0, \sigma^2), \quad \varepsilon_0 \text{ independent of } \mathbf{Y}, \quad (12.3.1)$$

be a future observation at a regressor vector $\mathbf{x}_0 \in \mathcal{C}(\mathbf{X}^\top)$. The requirement $\mathbf{x}_0 \in \mathcal{C}(\mathbf{X}^\top)$ makes the mean $\mathbf{x}_0^\top \boldsymbol{\beta}$ estimable. Without it, Proposition 12.1.3 shows that nothing useful can be said. We call

$$h_0 = \mathbf{x}_0^\top \mathbf{G} \mathbf{x}_0$$

the **leverage** of the new point. When $\mathbf{x}_0$ is the i th row of $\mathbf{X}$, h_0 is the diagonal entry h_{ii} of $\mathbf{M}$ (Proposition 6.8.1), and $\sigma^2 h_0$ is always the variance of $\hat{Y}_0 = \mathbf{x}_0^\top \hat{\boldsymbol{\beta}}$ (Theorem 7.5.1(a)).

Theorem 12.3.1 — Prediction interval

Under Eq. 12.3.1, put $\hat{Y}_0 = \mathbf{x}_0^\top \hat{\boldsymbol{\beta}}$.

- (a) $Y_0 - \hat{Y}_0 \sim N(0, \sigma^2(1 + h_0))$, it is independent of s^2 , and

$$\frac{Y_0 - \hat{Y}_0}{s\sqrt{1 + h_0}} \sim t(n - r).$$

- (b) The interval $\hat{Y}_0 \pm t_{n-r, \alpha/2} s\sqrt{1 + h_0}$ contains Y_0 with probability exactly $1 - \alpha$, the

probability being computed over $\mathbf{Y}$ and Y_0 jointly, at every $(\boldsymbol{\beta}, \sigma^2)$.

- (c) More generally, if $\bar{Y}_0$ is the mean of m independent new observations at $\mathbf{x}_0$, each distributed as in Eq. 12.3.1, then $\hat{Y}_0 \pm t_{n-r, \alpha/2} s \sqrt{1/m + h_0}$ contains $\bar{Y}_0$ with probability exactly $1 - \alpha$. As $m \rightarrow \infty$ this interval becomes the confidence interval $\hat{Y}_0 \pm t_{n-r, \alpha/2} s \sqrt{h_0}$ for $\mathbf{x}_0^\top \boldsymbol{\beta}$.

Proof. We prove (c), of which (a) and (b) are the case $m = 1$. The new observations are independent of $\mathbf{Y}$, and $\bar{Y}_0 \sim N(\mathbf{x}_0^\top \boldsymbol{\beta}, \sigma^2/m)$. The vector $(\bar{Y}_0, \mathbf{Y}^\top)^\top$ is normal with independent blocks, so $\bar{Y}_0 - \hat{Y}_0$, a linear function of it, is normal (Theorem 3.3.1) with mean $\mathbf{x}_0^\top \boldsymbol{\beta} - \mathbf{x}_0^\top \boldsymbol{\beta} = 0$ and variance $\sigma^2/m + \sigma^2 h_0$, the two terms adding because $\bar{Y}_0$ and $\hat{Y}_0$ are independent. For independence from s^2 : $\hat{Y}_0$ is a function of $\mathbf{MY}$, and $\mathbf{MY}$ and SSE are independent (Theorem 7.5.1(c)). Since $\bar{Y}_0$ is independent of $\mathbf{Y}$, for any events,

$$\begin{aligned} \Pr(\bar{Y}_0 \in A, \mathbf{MY} \in B, \text{SSE} \in D) \\ = \Pr(\bar{Y}_0 \in A) \Pr(\mathbf{MY} \in B) \Pr(\text{SSE} \in D), \end{aligned}$$

so $(\bar{Y}_0, \mathbf{MY})$ is independent of SSE, and so is $\bar{Y}_0 - \hat{Y}_0$. Dividing the standardized normal variable by $\sqrt{(\text{SSE}/\sigma^2)/(n-r)}$ gives $t(n-r)$ (Definition 4.1.3), and the interval statement is $\Pr\{|T| \leq t_{n-r, \alpha/2}\} = 1 - \alpha$. The limit as $m \rightarrow \infty$ is immediate, and the limiting interval is Eq. 12.1.1 with $\boldsymbol{\lambda} = \mathbf{x}_0$. ■

Part (a) was sketched in Exercise (B2). A *prediction* interval is for a random variable, and its probability refers to the joint experiment: collect the data, compute the interval, observe Y_0 . Given the data, the probability that the interval contains Y_0 is random,

$$\begin{aligned} \Pr(Y_0 \in \text{interval} \mid \mathbf{Y}) &= \Phi\left(\frac{\mathbf{x}_0^\top (\hat{\boldsymbol{\beta}} - \boldsymbol{\beta}) + t s \sqrt{1 + h_0}}{\sigma}\right) \\ &\quad - \Phi\left(\frac{\mathbf{x}_0^\top (\hat{\boldsymbol{\beta}} - \boldsymbol{\beta}) - t s \sqrt{1 + h_0}}{\sigma}\right), \end{aligned}$$

with $t = t_{n-r, \alpha/2}$ and Φ the standard normal distribution function, and it is only its expectation that equals $1 - \alpha$.

In the variance $\sigma^2(1 + h_0)$, the leverage h_0 is the price of not knowing $\boldsymbol{\beta}$, of order r/n at a typical point (Proposition 6.8.1). The 1 is the new error, which no amount of data removes: the prediction interval never becomes shorter than about $2z_{\alpha/2}\sigma$, while the interval for the mean shrinks to a point.

Example. (Predicting a state's murder rate)

In the murder-rate regression, consider a state at the centroid of the regressors. Its leverage is $h_0 = 1/n = 0.0200$ (see Proposition 12.3.2), the fitted rate is 4.514, and the 95% intervals are

$$\text{mean response: } (4.090, 4.938), \quad \text{new state: } (1.487, 7.541).$$

The prediction interval is 7.14 times as long as the interval for the mean, and nearly all of its length, 6.053 per 100,000, comes from s .

```

import numpy as np
import statsmodels.api as sm
from scipy import stats

data = sm.datasets.statelcrime.load_pandas().data
data = data.drop(index="District of Columbia")
y = data["murder"].to_numpy()
X = np.column_stack([np.ones(len(y)), data[["poverty", "single", "urban"]]])
n, p = X.shape
C = np.linalg.inv(X.T @ X)
beta_hat = C @ X.T @ y
s = np.sqrt(np.sum((y - X @ beta_hat) ** 2) / (n - p))
q = stats.t.ppf(0.975, n - p)
h = np.einsum("ij,jk,ik->i", X, C, X) # leverages of the 50 states

def intervals(x0):
    """Leverage h0, 95% interval for the mean response, 95% prediction interval."""
    h0 = x0 @ C @ x0
    fit = x0 @ beta_hat
    ci = (fit - q * s * np.sqrt(h0), fit + q * s * np.sqrt(h0))
    pi = (fit - q * s * np.sqrt(1 + h0), fit + q * s * np.sqrt(1 + h0))
    return h0, fit, ci, pi

x_centre = X.mean(axis=0) # the average state
x_hidden = np.array([1.0, 20.0, 19.0, 60.0]) # each coordinate inside its
    observed range
for name, x0 in [("centre", x_centre), ("hidden", x_hidden)]:
    h0, fit, ci, pi = intervals(x0)
    print(f"{name}: h0 = {h0:.4f}, fit {fit:.3f}, mean ({ci[0]:.3f}, {ci[1]:.3f}), "
          f"new state ({pi[0]:.3f}, {pi[1]:.3f})")
print(f"largest leverage in the data: {h.max():.4f}")

```

12.3.2 Leverage and extrapolation

Proposition 12.3.2 — Leverage of a new point

Let $\mathbf{x}_0 = \mathbf{X}^\top \boldsymbol{\rho} \in \mathcal{C}(\mathbf{X}^\top)$ and $h_0 = \mathbf{x}_0^\top \mathbf{G} \mathbf{x}_0$.

- (a) $h_0 = \min\{\|\mathbf{a}\|^2 : \mathbf{X}^\top \mathbf{a} = \mathbf{x}_0\}$, and the minimum is attained only at $\mathbf{a} = \mathbf{M}\boldsymbol{\rho}$.
- (b) If $\mathbf{X} = [\mathbf{1}, \mathbf{X}_1]$ has full column rank, $\mathbf{x}_0 = (1, \mathbf{x}_{01}^\top)^\top$, $\bar{\mathbf{x}}$ is the vector of column means of $\mathbf{X}_1$ and $\tilde{\mathbf{X}}_1 = \mathbf{X}_1 - \mathbf{1}\bar{\mathbf{x}}^\top$, then

$$h_0 = \frac{1}{n} + (\mathbf{x}_{01} - \bar{\mathbf{x}})^\top (\tilde{\mathbf{X}}_1^\top \tilde{\mathbf{X}}_1)^{-1} (\mathbf{x}_{01} - \bar{\mathbf{x}}).$$

- (c) If further columns are added, $\mathbf{X}_+ = [\mathbf{X}, \mathbf{Z}]$, and $\mathbf{x}_{0+} = (\mathbf{x}_0^\top, \mathbf{z}_0^\top)^\top \in \mathcal{C}(\mathbf{X}_+^\top)$, then the leverage of $\mathbf{x}_{0+}$ in the enlarged model is at least h_0 .

Proof.

- (a) Let $\mathbf{X}^\top \mathbf{a} = \mathbf{x}_0$. Then $\mathbf{X}^\top \mathbf{M} \mathbf{a} = \mathbf{X}^\top \mathbf{a} = \mathbf{x}_0 = \mathbf{X}^\top \mathbf{M} \boldsymbol{\rho}$, so $\mathbf{M} \mathbf{a} - \mathbf{M} \boldsymbol{\rho}$ lies in $\mathcal{C}(\mathbf{X})$

and is orthogonal to $\mathcal{C}(\mathbf{X})$; hence $\mathbf{M}\mathbf{a} = \mathbf{M}\boldsymbol{\rho}$. By Pythagoras, $\|\mathbf{a}\|^2 = \|\mathbf{M}\boldsymbol{\rho}\|^2 + \|(\mathbf{I} - \mathbf{M})\mathbf{a}\|^2 \geq \|\mathbf{M}\boldsymbol{\rho}\|^2$, with equality iff $\mathbf{a} = \mathbf{M}\boldsymbol{\rho}$, and $\|\mathbf{M}\boldsymbol{\rho}\|^2 = \boldsymbol{\rho}^\top \mathbf{X}\mathbf{G}\mathbf{X}^\top \boldsymbol{\rho} = h_0$ by Theorem 6.3.7.

- (b) $\mathbf{X} = [\mathbf{1}, \tilde{\mathbf{X}}_1]\mathbf{A}$ with $\mathbf{A} = \begin{pmatrix} 1 & \bar{\mathbf{x}}^\top \\ \mathbf{0} & \mathbf{I} \end{pmatrix}$, which is invertible. Since $\mathbf{1}^\top \tilde{\mathbf{X}}_1 = \mathbf{0}^\top$, $\mathbf{X}^\top \mathbf{X} = \mathbf{A}^\top \text{diag}(n, \tilde{\mathbf{X}}_1^\top \tilde{\mathbf{X}}_1) \mathbf{A}$, and so $h_0 = \mathbf{w}^\top \text{diag}(1/n, (\tilde{\mathbf{X}}_1^\top \tilde{\mathbf{X}}_1)^{-1}) \mathbf{w}$ with $\mathbf{w} = \mathbf{A}^{-\top} \mathbf{x}_0$. Solving $\mathbf{A}^\top \mathbf{w} = \mathbf{x}_0$ gives $\mathbf{w} = (1, (\mathbf{x}_{01} - \bar{\mathbf{x}})^\top)^\top$.
- (c) Every $\mathbf{a}$ with $\mathbf{X}_+^\top \mathbf{a} = \mathbf{x}_{0+}$ also satisfies $\mathbf{X}^\top \mathbf{a} = \mathbf{x}_0$. A minimum over a smaller set is at least as large, so (a), applied to both models, gives the claim. ■

Part (a) is the Gauss–Markov theorem in disguise. Part (b) makes leverage a distance: the quadratic form is $(n - 1)^{-1}$ times the squared Mahalanobis distance of $\mathbf{x}_{01}$ from the centroid of the regressors, as for the observed points (Proposition 6.8.2). So the interval for the mean is shortest at the centroid, where $h_0 = 1/n$, and widens along ellipsoidal contours. Part (c) says a larger model never predicts more precisely at a given point, though it may predict with less bias; Chapter 29 weighs the two.

With several correlated regressors, a point can lie inside the range of every regressor and still be far from the data, because it breaks their correlation. This **hidden extrapolation** is detected by the leverage, not by the separate ranges.

Example. (Hidden extrapolation)

Poverty ranges from 8.5% to 21.9% over the 50 states, and single-parent households from 17.9% to 32.8%. A state with poverty 20%, single-parent households 19% and 60% urban is inside every range, but poor with few single parents, a combination absent from the data. Its leverage is $h_0 = 0.435$, far above the largest leverage among the actual states, 0.260 (Mississippi). Figure 12.3.1 shows the contours of h_0 over the two regressors, with urbanization held at its mean.

The fitted rate is 3.817. The 95% interval for the mean is (1.840, 5.794), longer than at the centroid by the factor $\sqrt{0.435/0.0200} = 4.66$. The prediction interval, (0.227, 7.407), is only 1.186 times as long as at the centroid, because it is dominated by s . The prediction interval barely registers the extrapolation, and the real danger, that the model may be wrong where there are no data to check it, is in neither formula. If urbanization is dropped from the model the leverage of this point falls to 0.287, as Proposition 12.3.2(c) requires.

12.3.3 When the errors are not normal

The estimate $\hat{Y}_0$ is a weighted sum of n errors, approximately normal by a central limit theorem when no weight dominates (Chapter 19). The new error ε_0 is a single draw, and its distribution enters the prediction interval unchanged, however large n is.

Proposition 12.3.3 — Prediction intervals without normality

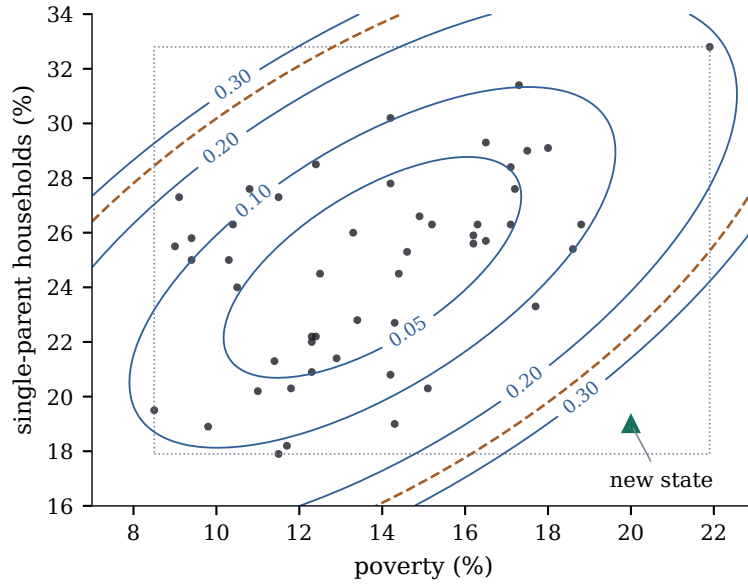


Figure 12.3.1: Contours of the leverage h_0 of a new state (urbanization at its mean), with the 50 states, the box of observed ranges (dotted) and the largest observed leverage (dashed). The marked state is inside the box but far from the data.

Consider a sequence of full-rank models $\mathbf{Y}_n = \mathbf{X}_n \boldsymbol{\beta} + \boldsymbol{\varepsilon}_n$ with p fixed and $n \rightarrow \infty$, and a fixed $\mathbf{x}_0$ whose leverage $h_{0,n} = \mathbf{x}_0^\top (\mathbf{X}_n^\top \mathbf{X}_n)^{-1} \mathbf{x}_0$ tends to zero. Let $\varepsilon_1, \dots, \varepsilon_n$ and ε_0 be independent and identically distributed with mean 0, variance σ^2 , finite fourth moment and a continuous distribution function, and let $Y_0 = \mathbf{x}_0^\top \boldsymbol{\beta} + \varepsilon_0$. Then the coverage of the nominal $1 - \alpha$ prediction interval $\hat{Y}_0 \pm t_{n-p, \alpha/2} s \sqrt{1 + h_{0,n}}$ satisfies

$$\begin{aligned} \Pr\{|Y_0 - \hat{Y}_0| \leq t_{n-p, \alpha/2} s \sqrt{1 + h_{0,n}}\} \\ \longrightarrow \Pr\{|\varepsilon_0| \leq z_{\alpha/2} \sigma\}, \end{aligned}$$

where $z_{\alpha/2}$ is the upper $\alpha/2$ point of $N(0, 1)$. Likewise the probabilities of missing above and below tend to $\Pr\{\varepsilon_0 > z_{\alpha/2} \sigma\}$ and $\Pr\{\varepsilon_0 < -z_{\alpha/2} \sigma\}$.

Proof. Write $T_n = (Y_0 - \hat{Y}_0)/(s \sqrt{1 + h_{0,n}})$, $t_n = t_{n-p, \alpha/2}$ and $E = \varepsilon_0/\sigma$. First, $Y_0 - \hat{Y}_0 = \varepsilon_0 - \mathbf{x}_0^\top (\hat{\boldsymbol{\beta}} - \boldsymbol{\beta})$, and $\mathbf{x}_0^\top (\hat{\boldsymbol{\beta}} - \boldsymbol{\beta})$ has mean zero and variance $\sigma^2 h_{0,n} \rightarrow 0$ (Theorem 5.5.1), so it tends to zero in probability by Chebyshev's inequality. Second, $s \rightarrow \sigma$ in probability (Corollary 5.6.4), and $t_n \rightarrow z_{\alpha/2}$. Hence $D_n = T_n - E \rightarrow 0$ in probability. Fix $\eta > 0$. On the event $\{|D_n| \leq \eta/2, |t_n - z_{\alpha/2}| \leq \eta/2\}$, whose probability tends to one,

$$|E| \leq z_{\alpha/2} - \eta \implies |T_n| \leq t_n \implies |E| \leq z_{\alpha/2} + \eta.$$

So $\Pr\{|E| \leq z_{\alpha/2} - \eta\} \leq \liminf \Pr\{|T_n| \leq t_n\} \leq \limsup \Pr\{|T_n| \leq t_n\} \leq \Pr\{|E| \leq z_{\alpha/2} + \eta\}$. The distribution function of E is continuous, so letting $\eta \rightarrow 0$ gives the limit. The one-sided statements are proved the same way. ■

The limit is the nominal $1 - \alpha$ only if $\Pr\{|\varepsilon_0| \leq z_{\alpha/2} \sigma\} = 1 - \alpha$, a property of the normal distribution, not of large samples. A simulation shows both effects: a straight line with n equally spaced design points on $[0, 1]$, prediction at $x_0 = 0.8$, and errors with mean zero and

variance one that are normal, centred exponential (skewed) or uniform (short-tailed). With 20000 data sets for each case, the coverages of the nominal 80% and 95% intervals are:

Errors	n	mean, 80%	mean, 95%	new obs., 80%	new obs., 95%
normal	20	0.799	0.949	0.797	0.949
normal	400	0.805	0.953	0.802	0.953
exponential	20	0.774	0.931	0.863	0.939
exponential	400	0.799	0.949	0.896	0.949
uniform	20	0.797	0.948	0.787	0.983
uniform	400	0.800	0.950	0.741	1.000

At $n = 400$ the intervals for the mean are near nominal for every law, while the prediction intervals approach the limits of [Proposition 12.3.3](#): 0.898 and 0.948 (exponential), 0.740 and 1.000 (uniform). The exponential 95% interval looks fine only by accident: at $n = 400$ it misses above in a proportion 0.051 of the samples and below in 0.0000, since a centred exponential error is never below -1 .

```

rng = np.random.default_rng(1203)
laws = {
    "normal": lambda size: rng.normal(size=size),
    "exponential": lambda size: rng.exponential(size=size) - 1.0,          #
    "uniform": lambda size: rng.uniform(-np.sqrt(3), np.sqrt(3), size=size), #
    "skewed": lambda size: rng.exponential(size=size) - 1.0,             #
    "short tails": lambda size: rng.uniform(-np.sqrt(3), np.sqrt(3), size=size), #
}
all: mean 0, variance 1
x0_sim = 0.8

def simulate(law, n_sim, reps=20_000):
    x = np.linspace(0.0, 1.0, n_sim)
    Xs = np.column_stack([np.ones(n_sim), x])
    Hs = np.linalg.solve(Xs.T @ Xs, Xs.T)                                # beta_hat = Hs y
    z0 = np.array([1.0, x0_sim])
    h0 = z0 @ np.linalg.solve(Xs.T @ Xs, z0)
    E = laws[law](reps, n_sim)                                           # true beta = 0,
    sigma = 1
    fit = E @ (Hs.T @ z0)
    s_sim = np.sqrt(np.sum((E - (E @ Hs.T) @ Xs.T) ** 2, axis=1) / (n_sim - 2))
    new = laws[law](reps)                                                # the new
    observation's error
    out = {}
    for level in (0.80, 0.95):
        t = stats.t.ppf(0.5 + level / 2, n_sim - 2)
        out[("mean", level)] = np.mean(np.abs(fit) <= t * s_sim * np.sqrt(h0))
        out[("new", level)] = np.mean(np.abs(new - fit) <= t * s_sim * np.sqrt(1 +
h0))
    return out

table = {(law, m): simulate(law, m) for law in laws for m in (20, 400)}
for (law, m), out in table.items():
    print(f"{law:12s} n = {m:3d}: " + ", ".join(f"{k[0]} {k[1]:.2f}: {v:.3f}" for k,
v in out.items()))

```

Remedies, transformation (Chapter 22) and resampling (Chapter 23), come later. Checking normality matters much more for prediction than for estimation.

Remark (Prediction is not intervention). A prediction interval describes the next observation from the *same process*. If the regressors are set by intervention, say a policy that changes a state's poverty rate, the observed association need not describe what happens (Chapter 25).

12.3.4 Exercises

12.3.4.1 A. Check your understanding

Exercise A1.

Show that the ratio of the length of the prediction interval to that of the confidence interval for the mean at $\mathbf{x}_0$ is $\sqrt{(1 + h_0)/h_0}$. What is it in [Example \(Predicting a state's murder rate\)](#), and what happens to it as $n \rightarrow \infty$ with the design growing regularly?

Solution. Both intervals use the multiplier $t_{n-r, \alpha/2} s$, with $\sqrt{1 + h_0}$ and $\sqrt{h_0}$ respectively. At the centroid, $h_0 = 0.0200$ and the ratio is $\sqrt{51} = 7.14$. As the design grows, $h_0 \rightarrow 0$ and the ratio tends to infinity: the confidence interval shrinks to a point while the prediction interval does not.

12.3.4.2 B. Practice

Exercise B1.

Two new observations will be made, at $\mathbf{x}_{01}$ and $\mathbf{x}_{02}$, with independent errors. Find an exact $1 - \alpha$ prediction interval for their difference $Y_{01} - Y_{02}$. Show that when $\mathbf{x}_{01} = \mathbf{x}_{02}$ it does not depend on the data except through s .

Solution. $Y_{01} - Y_{02} - (\mathbf{x}_{01} - \mathbf{x}_{02})^\top \hat{\boldsymbol{\beta}}$ is normal with mean zero and variance $\sigma^2\{2 + (\mathbf{x}_{01} - \mathbf{x}_{02})^\top \mathbf{G}(\mathbf{x}_{01} - \mathbf{x}_{02})\}$, independent of s^2 by the argument of [Theorem 12.3.1](#). The interval is $(\mathbf{x}_{01} - \mathbf{x}_{02})^\top \hat{\boldsymbol{\beta}} \pm t_{n-r, \alpha/2} s \sqrt{2 + (\mathbf{x}_{01} - \mathbf{x}_{02})^\top \mathbf{G}(\mathbf{x}_{01} - \mathbf{x}_{02})}$. When the two points coincide it is $\pm t_{n-r, \alpha/2} s \sqrt{2}$: the difference of two new errors, for which the fitted model contributes only its estimate of σ .

Exercise B2.

In [Proposition 12.3.2\(c\)](#), let $\mathbf{X}$ have full column rank and add a single column $\mathbf{z} \notin \mathcal{C}(\mathbf{X})$. Show that the new leverage is

$$h_{0+} = h_0 + \frac{(z_0 - \mathbf{k}^\top \mathbf{x}_0)^2}{\|(\mathbf{I} - \mathbf{M})\mathbf{z}\|^2}, \quad \mathbf{k} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{z}.$$

So the leverage is unchanged iff z_0 equals the value $\mathbf{k}^\top \mathbf{x}_0$ predicted for it by the other regressors.

Solution. Use [Proposition 12.3.2\(a\)](#) for the enlarged model. Write $\mathbf{x}_0 = \mathbf{X}^\top \boldsymbol{\rho}$ and split any feasible $\mathbf{a}$ as $\mathbf{a} = \mathbf{M}\mathbf{a} + \mathbf{b}$ with $\mathbf{b} \perp \mathcal{C}(\mathbf{X})$. The constraint $\mathbf{X}^\top \mathbf{a} = \mathbf{x}_0$ forces $\mathbf{M}\mathbf{a} = \mathbf{M}\boldsymbol{\rho}$, as in the proof of the proposition. The constraint $\mathbf{z}^\top \mathbf{a} = z_0$ then reads $\mathbf{z}^\top \mathbf{M}\boldsymbol{\rho} + \mathbf{z}^\top \mathbf{b} = z_0$, and $\mathbf{z}^\top \mathbf{M}\boldsymbol{\rho} = \mathbf{z}^\top \mathbf{X}(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \boldsymbol{\rho} = \mathbf{k}^\top \mathbf{x}_0$. Since $\mathbf{b} \perp \mathcal{C}(\mathbf{X})$, $\mathbf{z}^\top \mathbf{b} = ((\mathbf{I} - \mathbf{M})\mathbf{z})^\top \mathbf{b}$, and by the Cauchy–Schwarz inequality the shortest $\mathbf{b}$ with $((\mathbf{I} - \mathbf{M})\mathbf{z})^\top \mathbf{b} = z_0 - \mathbf{k}^\top \mathbf{x}_0$ is a multiple of $(\mathbf{I} - \mathbf{M})\mathbf{z}$, with $\|\mathbf{b}\|^2 = (z_0 - \mathbf{k}^\top \mathbf{x}_0)^2 / \|(\mathbf{I} - \mathbf{M})\mathbf{z}\|^2$. Adding $\|\mathbf{M}\boldsymbol{\rho}\|^2 = h_0$ gives the formula.

Exercise B3.

For the line through the origin, $E Y_i = \beta x_i$ with $\sum x_i^2 > 0$, write the $1 - \alpha$ confidence interval for the mean at x_0 and the prediction interval for Y_0 . Show that the confidence interval for the mean has length proportional to $|x_0|$, and is a point at $x_0 = 0$. Why is that reasonable, and what does it assume?

Solution. Here $r = 1$, $\hat{\beta} = \sum x_i y_i / \sum x_i^2$, $s^2 = \sum (y_i - \hat{\beta} x_i)^2 / (n - 1)$ and $h_0 = x_0^2 / \sum x_i^2$. The intervals are $\hat{\beta} x_0 \pm t_{n-1, \alpha/2} s |x_0| / \sqrt{\sum x_i^2}$ and $\hat{\beta} x_0 \pm t_{n-1, \alpha/2} s \sqrt{1 + x_0^2 / \sum x_i^2}$. At $x_0 = 0$ the model asserts that the mean is exactly zero, so there is nothing to estimate. The interval is only as good as that assertion: if the true line has a small intercept, the interval near $x_0 = 0$ is too short.

Exercise B4.

Suppose the new regressor vector $\mathbf{X}_0$ is random, independent of the pair $(\mathbf{Y}, \varepsilon_0)$, and always lies in $\mathcal{C}(\mathbf{X}^\top)$. Show that the prediction interval of [Theorem 12.3.1](#), computed at the realized $\mathbf{X}_0$, still covers Y_0 with probability exactly $1 - \alpha$.

Solution. Condition on $\mathbf{X}_0 = \mathbf{x}_0$. By independence, given this event $(\mathbf{Y}, \varepsilon_0)$ has its unconditional distribution, so [Theorem 12.3.1](#) applies and the conditional coverage is $1 - \alpha$ for every $\mathbf{x}_0$. Averaging over the distribution of $\mathbf{X}_0$ gives $1 - \alpha$.

12.3.4.3 C. Going deeper

Exercise C1.

Let k new observations be made at the rows of a $k \times p$ matrix $\mathbf{X}_0$ whose rows lie in $\mathcal{C}(\mathbf{X}^\top)$, with errors independent of each other and of $\mathbf{Y}$. Show that $\mathbf{D} = \mathbf{Y}_0 - \mathbf{X}_0 \hat{\boldsymbol{\beta}} \sim N_k(\mathbf{0}, \sigma^2(\mathbf{I}_k + \mathbf{X}_0 \mathbf{G} \mathbf{X}_0^\top))$, independent of s^2 , and deduce that the intervals

$$\mathbf{x}_{0j}^\top \hat{\boldsymbol{\beta}} \pm \sqrt{k F_\alpha(k, n - r)} s \sqrt{1 + h_{0j}}, \quad j = 1, \dots, k,$$

contain all k new observations simultaneously with probability at least $1 - \alpha$.

Solution. Normality, the mean and the covariance follow as in [Theorem 12.3.1](#), and $\mathbf{S} = \mathbf{I}_k + \mathbf{X}_0 \mathbf{G} \mathbf{X}_0^\top$ is positive definite. So $\mathbf{D}^\top \mathbf{S}^{-1} \mathbf{D} / (ks^2) \sim F(k, n - r)$ ([Corollary 4.3.3](#) and [Definition 4.1.2](#)). By [Lemma 12.2.3](#) with $\mathbf{a} = \mathbf{e}_j$, the event $\mathbf{D}^\top \mathbf{S}^{-1} \mathbf{D} \leq ks^2 F_\alpha$ implies $|D_j| \leq \sqrt{k F_\alpha} s \sqrt{S_{jj}}$ for every j , and $S_{jj} = 1 + h_{0j}$. [Chapter 13](#) compares this with Bonferroni's method, which is shorter when k is small.

Exercise C2.

A $(P, 1 - \alpha)$ **tolerance interval** at $\mathbf{x}_0$ is a random interval that, with probability at least $1 - \alpha$ over $\mathbf{Y}$, contains at least a proportion P of the distribution of Y_0 . Show that the prediction interval of [Theorem 12.3.1](#) is not a $(1 - \alpha, 1 - \alpha)$ tolerance interval, by showing that its conditional coverage given $\mathbf{Y}$ is below $1 - \alpha$ with probability bounded away from zero. Construct a conservative tolerance interval from a lower confidence bound for the conditional coverage, using a $1 - \alpha/2$ upper confidence bound for σ and a $1 - \alpha/2$ confidence interval for $\mathbf{x}_0^\top \boldsymbol{\beta}$.

12.4 Simultaneous bands for the regression line and surface

A fitted line is usually drawn inside a shaded region. If the region is made of pointwise intervals, each vertical slice covers the true mean with probability $1 - \alpha$, but the region does not cover the whole true line with that probability. A reader who asks “which lines are consistent with the data?” needs a simultaneous band, and for a linear model the Cauchy–Schwarz inequality gives one with exact coverage.

12.4.1 Pointwise is not simultaneous

Definition 12.4.1 — Simultaneous confidence band

Let $\mathcal{S} \subseteq \mathcal{C}(\mathbf{X}^\top)$ be a set of regressor vectors. Random functions $\ell(\mathbf{x}) = \ell(\mathbf{x}; \mathbf{Y})$ and $u(\mathbf{x}) = u(\mathbf{x}; \mathbf{Y})$ form a $1 - \alpha$ **simultaneous confidence band** for the regression function over $\mathcal{S}$ if

$$\Pr_{\boldsymbol{\beta}, \sigma^2} \{ \ell(\mathbf{x}) \leq \mathbf{x}^\top \boldsymbol{\beta} \leq u(\mathbf{x}) \text{ for every } \mathbf{x} \in \mathcal{S} \} \geq 1 - \alpha$$

for every $(\boldsymbol{\beta}, \sigma^2)$.

Pointwise t intervals fail this: for the stack loss line of [Example \(A band for the stack loss line\)](#), the pointwise 95% intervals cover the true line over the observed range in only 0.8705 of simulated samples.

12.4.2 The Working–Hotelling band

Theorem 12.4.1 — Working–Hotelling band

Let $\mathcal{L}$ be a linear subspace of $\mathcal{C}(\mathbf{X}^\top)$ of dimension $k \geq 1$, and let $w_k = \sqrt{k F_\alpha(k, n - r)}$. Then

$$\Pr \{ |\mathbf{x}^\top \hat{\boldsymbol{\beta}} - \mathbf{x}^\top \boldsymbol{\beta}| \leq w_k s \sqrt{\mathbf{x}^\top \mathbf{G} \mathbf{x}} \text{ for all } \mathbf{x} \in \mathcal{L} \} = 1 - \alpha. \quad (12.4.1)$$

In particular:

- (a) with $\mathcal{L} = \mathcal{C}(\mathbf{X}^\top)$ and $k = r$, the functions $\mathbf{x}^\top \hat{\boldsymbol{\beta}} \pm w_r s \sqrt{\mathbf{x}^\top \mathbf{G} \mathbf{x}}$ form an exact $1 - \alpha$ simultaneous band for the regression surface over all estimable $\mathbf{x}$;
- (b) for a straight line $E Y_i = \beta_0 + \beta_1 x_i$ with the x_i not all equal, the band

$$\hat{\beta}_0 + \hat{\beta}_1 x \pm \sqrt{2F_\alpha(2, n-2)} s \sqrt{\frac{1}{n} + \frac{(x - \bar{x})^2}{S_{xx}}}, \quad x \in \mathbb{R},$$

contains the true line $\beta_0 + \beta_1 x$ for every real x with probability exactly $1 - \alpha$.

Proof. Step 1: from $\mathcal{L}$ to a subspace of $\mathcal{C}(\mathbf{X})$. The map $\mathbf{u} \mapsto \mathbf{X}^\top \mathbf{u}$ is one to one on $\mathcal{C}(\mathbf{X})$, since $\mathbf{X}^\top \mathbf{u} = \mathbf{0}$ with $\mathbf{u} \in \mathcal{C}(\mathbf{X})$ forces $\mathbf{u} \in \mathcal{C}(\mathbf{X}) \cap \mathcal{C}(\mathbf{X})^\perp = \{\mathbf{0}\}$. It maps $\mathcal{C}(\mathbf{X})$ onto $\mathcal{C}(\mathbf{X}^\top)$, because $\mathbf{X}^\top \boldsymbol{\rho} = \mathbf{X}^\top \mathbf{M} \boldsymbol{\rho}$. So $\mathcal{U} = \{\mathbf{u} \in \mathcal{C}(\mathbf{X}) : \mathbf{X}^\top \mathbf{u} \in \mathcal{L}\}$ is a subspace of $\mathcal{C}(\mathbf{X})$ of dimension k , and every $\mathbf{x} \in \mathcal{L}$ is $\mathbf{X}^\top \mathbf{u}$ for exactly one $\mathbf{u} \in \mathcal{U}$. For such a pair,

$$\begin{aligned} \mathbf{x}^\top \hat{\boldsymbol{\beta}} - \mathbf{x}^\top \boldsymbol{\beta} &= \mathbf{u}^\top (\mathbf{X} \hat{\boldsymbol{\beta}} - \mathbf{X} \boldsymbol{\beta}) = \mathbf{u}^\top \mathbf{M} (\mathbf{Y} - \mathbf{X} \boldsymbol{\beta}) = \mathbf{u}^\top \mathbf{v}, \\ \mathbf{x}^\top \mathbf{G} \mathbf{x} &= \mathbf{u}^\top \mathbf{X} \mathbf{G} \mathbf{X}^\top \mathbf{u} = \mathbf{u}^\top \mathbf{M} \mathbf{u} = \|\mathbf{u}\|^2, \end{aligned}$$

where $\mathbf{v} = \mathbf{P}_\mathcal{U}(\mathbf{Y} - \mathbf{X} \boldsymbol{\beta})$. The middle equality uses $\mathbf{X} \hat{\boldsymbol{\beta}} = \mathbf{M} \mathbf{Y}$ and $\mathbf{M} \mathbf{X} = \mathbf{X}$. The last step of the first chain uses $\mathbf{u} = \mathbf{M} \mathbf{u} = \mathbf{P}_\mathcal{U} \mathbf{u}$, and the second chain uses [Theorem 6.3.7](#).

Step 2: the supremum. By the Cauchy–Schwarz inequality, for $\mathbf{u} \in \mathcal{U}$, $\mathbf{u} \neq \mathbf{0}$,

$$\frac{(\mathbf{x}^\top \hat{\boldsymbol{\beta}} - \mathbf{x}^\top \boldsymbol{\beta})^2}{\mathbf{x}^\top \mathbf{G} \mathbf{x}} = \frac{(\mathbf{u}^\top \mathbf{v})^2}{\|\mathbf{u}\|^2} \leq \|\mathbf{v}\|^2,$$

with equality at $\mathbf{u} = \mathbf{v}$ when $\mathbf{v} \neq \mathbf{0}$, and $\mathbf{v} \in \mathcal{U}$. So the supremum of the left side over $\mathbf{x} \in \mathcal{L} \setminus \{\mathbf{0}\}$ equals $\|\mathbf{v}\|^2$ and is attained.

Step 3: its distribution. $\mathbf{Y} - \mathbf{X} \boldsymbol{\beta} \sim N_n(\mathbf{0}, \sigma^2 \mathbf{I})$, and $\mathbf{P}_\mathcal{U}$ is symmetric idempotent of rank k , so $\|\mathbf{v}\|^2 / \sigma^2 \sim \chi^2(k)$ ([Theorem 4.3.2\(a\)](#)). Since $\mathcal{U} \subseteq \mathcal{C}(\mathbf{X})$, $\mathbf{P}_\mathcal{U} = \mathbf{P}_\mathcal{U} \mathbf{M}$, so $\mathbf{v}$ is a function of $\mathbf{M} \mathbf{Y}$ and is independent of SSE ([Theorem 7.5.1\(c\)](#)). Hence $\|\mathbf{v}\|^2 / (k s^2) \sim F(k, n-r)$ ([Definition 4.1.2](#)), and the event in [Eq. 12.4.1](#), which is $\|\mathbf{v}\|^2 \leq w_k^2 s^2$ by Step 2, has probability $1 - \alpha$. (At $\mathbf{x} = \mathbf{0}$ both sides of the inequality are zero.)

Part (a) is the case $\mathcal{L} = \mathcal{C}(\mathbf{X}^\top)$, $\mathcal{U} = \mathcal{C}(\mathbf{X})$.

Part (b). Here $r = k = 2$ and $\mathcal{L} = \mathbb{R}^2$, but the band uses only the vectors $\mathbf{x} = (1, x)^\top$. The ratio in Step 2 is unchanged when $\mathbf{x}$ is multiplied by a nonzero scalar, so its supremum over $\{(1, x)^\top : x \in \mathbb{R}\}$ equals its supremum over all $\mathbf{x} = (a, b)^\top$ with $a \neq 0$. By Step 2 the supremum over all of $\mathbb{R}^2$ is attained at $\mathbf{x}^* = \mathbf{X}^\top \mathbf{v}$, whose first coordinate is $\mathbf{1}^\top \mathbf{v}$. The random variable $\mathbf{1}^\top \mathbf{v}$ is normal with variance $\sigma^2 \|\mathbf{P}_\mathcal{U} \mathbf{1}\|^2 = \sigma^2 n > 0$, because $\mathbf{1} \in \mathcal{C}(\mathbf{X}) = \mathcal{U}$. So with probability one $\mathbf{x}^*$ has a nonzero first coordinate, the two suprema are equal, and the band event has the probability computed in Step 3. The formula for $\mathbf{x}^\top \mathbf{G} \mathbf{x}$ is [Eq. 5.5.2](#) divided by σ^2 . ■

The proof has one idea: every linear function of the fitted mean is an inner product with $\mathbf{v}$, and the worst case over all directions is $\|\mathbf{v}\|$. With $\mathcal{L} = \mathcal{C}(\mathbf{X}^\top)$ the band holds exactly when the confidence ball of [Proposition 12.2.2](#) contains $\mathbf{X} \boldsymbol{\beta}$: the band is that ball, viewed one linear

functional at a time. Applied to general families of estimable functions, the argument gives Scheffé's intervals, which [Chapter 13](#) develops ([Theorem 13.2.2](#)) and compares with Bonferroni and Tukey.

The straight-line band is hyperbolic, narrowest at $\bar{x}$. Although x varies along a line, the multiplier uses $k = 2$, because the band must hold for intercept and slope together.

12.4.3 How much wider?

The band is wider than the pointwise intervals by the factor $w_k/t_{n-r,\alpha/2}$, the same at every $\mathbf{x}$. The multipliers $w_k = \sqrt{kF_{0.05}(k, \nu)}$ for 95% coverage are:

k	$\nu = 5$	$\nu = 10$	$\nu = 20$	$\nu = 60$	$\nu = \infty$
1	2.571	2.228	2.086	2.000	1.960
2	3.402	2.865	2.643	2.510	2.448
3	4.028	3.335	3.049	2.876	2.795
5	5.025	4.078	3.682	3.441	3.327
10	6.881	5.457	4.845	4.464	4.279

The row $k = 1$ is the t multiplier; the column $\nu = \infty$ is $\sqrt{\chi_{0.05}^2(k)}$, for known σ . For a straight line the band is about a quarter wider than the pointwise intervals, for ten coefficients more than twice as wide: for large k simultaneity costs roughly a factor $\sqrt{k}$.

Example. (A band for the stack loss line)

Regress stack loss on air flow for the 21 days of Brownlee's plant data ([Example \(Stack loss\)](#)), ignoring the other two regressors. The fitted line is $-44.132 + 1.0203x$ with $s = 4.098$, $\bar{x} = 60.43$ and $S_{xx} = 1681.1$. The pointwise multiplier is $t_{19,0.025} = 2.093$, and the Working–Hotelling multiplier is $\sqrt{2F_{0.05}(2, 19)} = 2.654$, larger by the factor 1.268. [Figure 12.4.1](#) shows the data, the band, the pointwise intervals, and the 95% prediction intervals of [Section 12.3](#).

Near the centre, at $x = 60$, the fitted mean is 17.09, and the half-widths are 1.87 (pointwise), 2.38 (band) and 8.78 (prediction). At $x = 80$, the edge of the data, the fitted mean is 37.49 and the half-widths are 4.50, 5.71 and 9.69.

In 40000 data sets simulated from the fitted line, the band contained the true line over the whole real line in a proportion 0.9500 of the samples and over the observed range $50 \leq x \leq 80$ in 0.9537. The pointwise intervals covered at $x = 60$ in 0.9517 of the samples, but covered the line over the whole observed range in only 0.8705.

```
import numpy as np
import statsmodels.api as sm
from scipy import stats

data = sm.datasets.stackloss.load_pandas().data
x = data["AIRFLOW"].to_numpy()
y = data["STACKLOSS"].to_numpy()
n = len(y)
X = np.column_stack([np.ones(n), x])
```

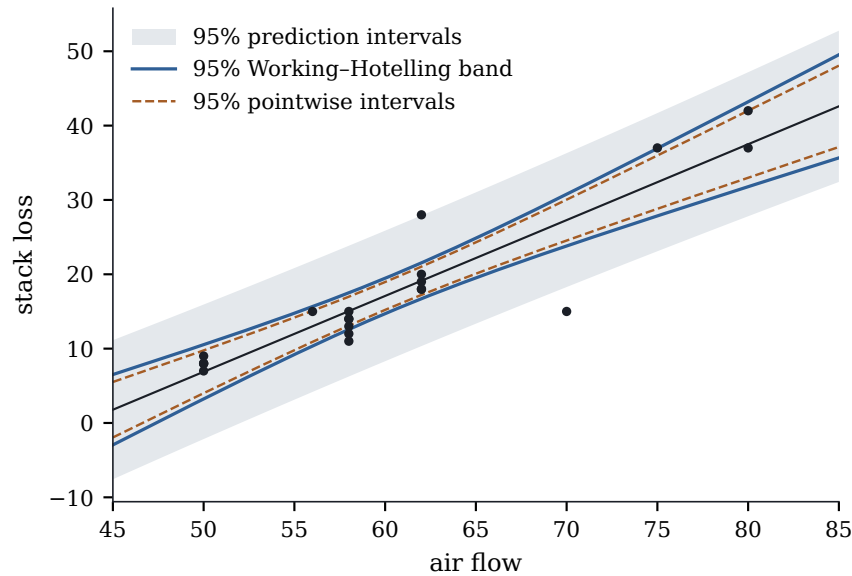


Figure 12.4.1: Stack loss against air flow: least squares line, 95% Working–Hotelling band (solid), pointwise 95% intervals for the mean (dashed) and pointwise 95% prediction intervals (shaded).

```
C = np.linalg.inv(X.T @ X)
beta_hat = C @ X.T @ y
s = np.sqrt(np.sum((y - X @ beta_hat) ** 2) / (n - 2))

grid = np.linspace(45, 85, 401)
Z = np.column_stack([np.ones_like(grid), grid])
fit = Z @ beta_hat
se_mean = s * np.sqrt(np.einsum("ij,jk,ik->i", Z, C, Z)) # s * sqrt(1/n + (x -
xbar)^2 / Sxx)
t_pt = stats.t.ppf(0.975, n - 2) # pointwise multiplier
w_wh = np.sqrt(2 * stats.f.ppf(0.95, 2, n - 2)) # Working-Hotelling
multiplier
pointwise = (fit - t_pt * se_mean, fit + t_pt * se_mean)
band = (fit - w_wh * se_mean, fit + w_wh * se_mean)
se_new = np.sqrt(s**2 + se_mean**2)
prediction = (fit - t_pt * se_new, fit + t_pt * se_new)
print(f"slope {beta_hat[1]:.4f}, s = {s:.3f}; multipliers: t {t_pt:.3f}, Working-
Hotelling {w_wh:.3f}")
```

12.4.4 Bands over part of the space, and other shapes

The band holds over the whole line. If only $x \in [a, b]$ matters it is conservative (the simulated 0.9537), and an exact band over $[a, b]$ uses a smaller multiplier (Wynn and Bloomfield 1971). The hyperbolic shape is also a choice: bands of constant width, or with straight edges, come from replacing the Euclidean norm in Step 2 by another norm (Gafarian 1964; Liu 2010). And if only some regressors vary, $\mathcal{L}$ is smaller, and so is k .

12.4.5 Exercises

12.4.5.1 A. Check your understanding

Exercise A1.

For a straight line with many residual degrees of freedom, compute the ratio of the Working–Hotelling multiplier to the pointwise one at $\alpha = 0.05$. How does the ratio change at $\alpha = 0.01$?

Solution. As $\nu \rightarrow \infty$, $2F_\alpha(2, \nu) \rightarrow \chi_\alpha^2(2) = -2 \log \alpha$, since $\chi^2(2)$ is exponential with mean 2. At $\alpha = 0.05$ the ratio is $\sqrt{2 \log 20} / z_{0.025} = 2.448 / 1.960 = 1.249$. At $\alpha = 0.01$ it is $\sqrt{2 \log 100} / z_{0.005} = 3.035 / 2.576 = 1.178$. The relative cost of simultaneity falls as the level becomes more demanding. As $\alpha \rightarrow 0$, both squared multipliers, $z_{\alpha/2}^2$ and $\chi_\alpha^2(2) = 2 \log(1/\alpha)$, grow like $2 \log(1/\alpha) + O(\log \log(1/\alpha))$, so the ratio of the multipliers tends to 1.

Exercise A2.

For the line through the origin $EY_i = \beta x_i$, show that a simultaneous band for βx over all real x is $\hat{\beta}x \pm t_{n-1, \alpha/2} s|x| / \sqrt{\sum x_i^2}$, with no widening at all. Explain this with the dimension k of [Theorem 12.4.1](#).

Solution. Here $\mathcal{C}(\mathbf{X}^\top) = \mathbb{R}$, so $k = 1$ and $w_1 = t_{n-1, \alpha/2}$. Directly: the band holds at every x iff it holds at $x = 1$, since both sides scale with $|x|$.

12.4.5.2 B. Practice

Exercise B1.

For the straight line, show that the Working–Hotelling band contains the line $b_0 + b_1x$ for every x iff (b_0, b_1) lies in the $1 - \alpha$ confidence ellipse for (β_0, β_1) of [Theorem 12.2.1](#). Deduce that the band is the envelope of all lines whose coefficients lie in the ellipse.

Solution. Put $\mathbf{u} = (\hat{\beta}_0 - b_0, \hat{\beta}_1 - b_1)^\top$ and $\mathbf{W} = (\mathbf{X}^\top \mathbf{X})^{-1}$. The line $b_0 + b_1x$ is inside the band at x iff $|(1, x)\mathbf{u}| \leq w_2 s \sqrt{(1, x)\mathbf{W}(1, x)^\top}$. By scale invariance this holds for all x iff it holds for all $\mathbf{a} = (a_1, a_2)^\top$ with $a_1 \neq 0$, and by continuity iff it holds for all $\mathbf{a}$. By [Lemma 12.2.3](#) that is $\mathbf{u}^\top \mathbf{W}^{-1} \mathbf{u} \leq w_2^2 s^2$, which is the ellipse. At each x , the upper edge of the band is the maximum of $b_0 + b_1x$ over the ellipse ([Corollary 12.2.4](#)), so the band is the envelope.

Exercise B2.

For quadratic regression $EY_i = \beta_0 + \beta_1 x_i + \beta_2 x_i^2$, the natural band over all x uses $\mathbf{x} = (1, x, x^2)^\top$, a curve in $\mathbb{R}^3$ whose directions do not fill $\mathbb{R}^3$. Show that the band with multiplier w_3 has coverage at least $1 - \alpha$ over all x , and argue that its coverage is strictly

larger than $1 - \alpha$. (Hint: when is the maximizing direction $\mathbf{X}^\top \mathbf{v}$ of Step 2 a multiple of some $(1, x, x^2)^\top$?)

Solution. The band event over all $\mathbf{x} \in \mathbb{R}^3$ has probability $1 - \alpha$ and is contained in the event over the curve, so the coverage is at least $1 - \alpha$. Write $S = \|\mathbf{v}\|^2$ and S_c for the supremum of the ratio of Step 2 over the curve. The ratio depends on $\mathbf{v}$ only through S and the direction $\mathbf{v}/\|\mathbf{v}\|$, so $S_c = \rho S$, where $\rho \leq 1$ is a continuous function of the direction alone. Equality $\rho = 1$ requires either that $\mathbf{X}^\top \mathbf{v}$ be a multiple of some $(1, x, x^2)^\top$, that is, its coordinates (c_0, c_1, c_2) satisfy $c_0 c_2 = c_1^2$, or that $\mathbf{X}^\top \mathbf{v}$ be a multiple of $(0, 0, 1)^\top$, the limit of $(1, x, x^2)^\top/x^2$ as $x \rightarrow \infty$, where the supremum is approached but not attained. Both are events of probability zero. For a spherical normal vector in $\mathcal{U}$, the length and the direction are independent, and both are independent of s^2 . So the coverage over the curve exceeds $1 - \alpha$ by $\Pr\{w_3^2 s^2 < S \leq w_3^2 s^2 / \rho\}$, which is positive because S/σ^2 has a positive density on $(0, \infty)$ and $\rho < 1$ almost surely.

12.4.5.3 C. Going deeper

Exercise C1.

A single new observation $Y_0 = \mathbf{x}_0^\top \boldsymbol{\beta} + \varepsilon_0$ will be taken at a regressor vector $\mathbf{x}_0 \in \mathcal{C}(\mathbf{X}^\top)$ that is not known in advance. Show that the band $\mathbf{x}^\top \hat{\boldsymbol{\beta}} \pm \sqrt{(r+1)F_\alpha(r+1, n-r)} s \sqrt{1 + \mathbf{x}^\top \mathbf{G} \mathbf{x}}$, $\mathbf{x} \in \mathcal{C}(\mathbf{X}^\top)$, contains Y_0 with probability at least $1 - \alpha$, whatever $\mathbf{x}_0$ turns out to be. (Hint: apply Step 2 of the proof to the $(r+1)$ -dimensional vector $(\mathbf{v}, \varepsilon_0)$.)

Solution. For $\mathbf{x} = \mathbf{X}^\top \mathbf{u}$ with $\mathbf{u} \in \mathcal{C}(\mathbf{X})$, $Y_0 - \mathbf{x}^\top \hat{\boldsymbol{\beta}} = \varepsilon_0 - \mathbf{u}^\top \mathbf{v}$, with $\mathbf{v} = \mathbf{M}(\mathbf{Y} - \mathbf{X}\boldsymbol{\beta})$, and $1 + \mathbf{x}^\top \mathbf{G} \mathbf{x} = 1 + \|\mathbf{u}\|^2$. By Cauchy–Schwarz in $\mathcal{C}(\mathbf{X}) \times \mathbb{R}$, $(\varepsilon_0 - \mathbf{u}^\top \mathbf{v})^2 \leq (1 + \|\mathbf{u}\|^2)(\varepsilon_0^2 + \|\mathbf{v}\|^2)$. The variable $(\varepsilon_0^2 + \|\mathbf{v}\|^2)/\sigma^2$ is $\chi^2(r+1)$, independent of SSE, so $(\varepsilon_0^2 + \|\mathbf{v}\|^2)/((r+1)s^2) \sim F(r+1, n-r)$, and on the event that it is at most $F_\alpha(r+1, n-r)$, which has probability $1 - \alpha$, the band contains Y_0 at every $\mathbf{x}$, in particular at $\mathbf{x}_0$.

Exercise C2.

Two straight lines are fitted in one model, with separate intercepts and slopes and a common σ^2 . Construct simultaneous bands for both lines with joint coverage at least $1 - \alpha$ in two ways: (i) Theorem 12.4.1 with $\mathcal{L} = \mathbb{R}^4$, (ii) a band for each line with $\mathcal{L} = \mathbb{R}^2$ at level $\alpha/2$, combined by the Bonferroni inequality. Compare the multipliers for $n - r = 20$ and explain which is better and why.

12.5 Inverse prediction and calibration

Prediction runs from x to y . Calibration runs the other way: an instrument is calibrated on specimens of known value, a new specimen is measured, and its value is wanted. The estimate reads the fitted line backwards; the interval reads the prediction band backwards, and it is not always an interval.

12.5.1 Ratios of estimable functions

Inverse prediction belongs to a wider problem: a confidence set for a *ratio* of linear functions of the coefficients, such as the zero crossing $-\beta_0/\beta_1$ of a line or the turning point $-\beta_1/(2\beta_2)$ of a quadratic. A delta-method interval ignores that the denominator is uncertain and may be near zero. Fieller's approach tests, for each candidate value, a hypothesis that is linear in the coefficients.

Proposition 12.5.1 — Fieller's confidence set for a ratio

Let $\lambda_1, \lambda_2 \in \mathcal{C}(\mathbf{X}^\top)$ be linearly independent, put $a = \lambda_1^\top \beta$, $b = \lambda_2^\top \beta$, their estimates $\hat{a} = \lambda_1^\top \hat{\beta}$, $\hat{b} = \lambda_2^\top \hat{\beta}$, and $v_{ij} = \lambda_i^\top \mathbf{G} \lambda_j$, so that $\mathbf{V} = (v_{ij})$ is positive definite. Suppose $b \neq 0$ and let $\theta = a/b$. With $t = t_{n-r, \alpha/2}$, let

$$\mathcal{F}(\mathbf{Y}) = \{\theta_0 \in \mathbb{R} : (\hat{a} - \theta_0 \hat{b})^2 \leq t^2 s^2 (v_{11} - 2\theta_0 v_{12} + \theta_0^2 v_{22})\}. \quad (12.5.1)$$

- (a) $\mathcal{F}(\mathbf{Y})$ is an exact $1 - \alpha$ confidence set for θ , at every parameter value with $b \neq 0$.
- (b) Let $Q = (\hat{a}, \hat{b})\mathbf{V}^{-1}(\hat{a}, \hat{b})^\top$. Outside an event of probability zero, $\mathcal{F}(\mathbf{Y})$ is
 - a bounded closed interval containing $\hat{\theta} = \hat{a}/\hat{b}$, if $\hat{b}^2 > t^2 s^2 v_{22}$;
 - the union of two disjoint closed rays, if $\hat{b}^2 < t^2 s^2 v_{22}$ and $Q > t^2 s^2$;
 - the whole real line, if $\hat{b}^2 < t^2 s^2 v_{22}$ and $Q < t^2 s^2$.

Proof.

- (a) Fix θ_0 . The vector $\lambda_1 - \theta_0 \lambda_2$ lies in $\mathcal{C}(\mathbf{X}^\top)$ and is not zero, so $(\lambda_1 - \theta_0 \lambda_2)^\top \beta = a - \theta_0 b$ is estimable, with estimate $\hat{a} - \theta_0 \hat{b}$ and $(\lambda_1 - \theta_0 \lambda_2)^\top \mathbf{G} (\lambda_1 - \theta_0 \lambda_2) = v_{11} - 2\theta_0 v_{12} + \theta_0^2 v_{22}$. The inequality in Eq. 12.5.1 says that the t interval of Theorem 12.1.1 for $a - \theta_0 b$ contains 0. At $\theta_0 = \theta$, $a - \theta b = 0$, so the probability of this event is exactly $1 - \alpha$.
- (b) The defining inequality is $q(\theta_0) = A\theta_0^2 - 2B\theta_0 + K \leq 0$, with

$$A = \hat{b}^2 - t^2 s^2 v_{22}, \quad B = \hat{a}\hat{b} - t^2 s^2 v_{12}, \quad K = \hat{a}^2 - t^2 s^2 v_{11}.$$

Expanding and using $(\hat{a}, \hat{b})\mathbf{V}^{-1}(\hat{a}, \hat{b})^\top = (v_{22}\hat{a}^2 - 2v_{12}\hat{a}\hat{b} + v_{11}\hat{b}^2)/\det \mathbf{V}$,

$$\begin{aligned} B^2 - AK &= t^2 s^2 (v_{22}\hat{a}^2 - 2v_{12}\hat{a}\hat{b} + v_{11}\hat{b}^2 - t^2 s^2 \det \mathbf{V}) \\ &= t^2 s^2 \det(\mathbf{V}) (Q - t^2 s^2). \end{aligned}$$

If $A > 0$, then $q(\hat{\theta}) = -t^2 s^2 (v_{11} - 2\hat{a}\hat{b} + \hat{b}^2 v_{22}) < 0$, so the upward parabola q has two real roots, and $\{q \leq 0\}$ is the closed interval between them, which contains $\hat{\theta}$. If $A < 0$, the parabola opens downward. When $B^2 - AK < 0$ it is negative everywhere and $\mathcal{F}(\mathbf{Y}) = \mathbb{R}$. When $B^2 - AK > 0$ it is nonpositive exactly outside the open interval between its roots. The events $A = 0$ and $Q = t^2 s^2$ have probability zero, because $\hat{b}$ and Q/s^2 have continuous distributions. ■

The first case says that the t test rejects $b = 0$: the denominator is away from zero and the

ratio is confined. Otherwise very large $|\theta|$ cannot be excluded, and $Q/(2s^2)$, the F statistic for $a = b = 0$, decides between two rays and the whole line. [Exercise \(C1\)](#) gives a picture: $\mathcal{F}(\mathbf{Y})$ is the set of slopes of lines through the origin that meet a certain confidence ellipse for (a, b) .

12.5.2 The calibration problem

A calibration experiment measures n standards with known values $x_1, \dots, x_n$, not all equal, and records readings

$$Y_i = \beta_0 + \beta_1 x_i + \varepsilon_i, \quad i = 1, \dots, n.$$

A new specimen with unknown value x_0 is then measured $m \geq 1$ times, $Y_{0j} = \beta_0 + \beta_1 x_0 + \varepsilon_{0j}$, and all $n + m$ errors are independent $N(0, \sigma^2)$. Let $\bar{Y}_0$ be the mean of the new readings, and let $\hat{\beta}_0$, $\hat{\beta}_1$ and s^2 (with $n - 2$ degrees of freedom) come from the standards alone. The **classical estimator** reads the fitted line backwards:

$$\hat{x}_0 = \frac{\bar{Y}_0 - \hat{\beta}_0}{\hat{\beta}_1} = \bar{x} + \frac{\bar{Y}_0 - \bar{Y}}{\hat{\beta}_1}.$$

It is the maximum likelihood estimator ([Exercise \(A2\)](#)) and, being a ratio, inherits a pathology of ratios.

Proposition 12.5.2 — The classical estimator has no mean

If $\Pr(\bar{Y}_0 \neq \bar{Y}) > 0$, which always holds here, then $E|\hat{x}_0| = \infty$.

Proof. $\bar{Y}$ and $\hat{\beta}_1$ are jointly normal and uncorrelated ([Corollary 5.5.4](#)), hence independent ([Theorem 3.4.1](#)), and $\bar{Y}_0$ is independent of both. So $g = \bar{Y}_0 - \bar{Y}$ is independent of $\hat{\beta}_1$ and $E|\hat{x}_0 - \bar{x}| = E|g| E|1/\hat{\beta}_1|$, with $E|g| > 0$. The density f of $\hat{\beta}_1$ is continuous and positive at 0, so $f \geq c > 0$ on some $[-\epsilon, \epsilon]$ and $E|1/\hat{\beta}_1| \geq c \int_{-\epsilon}^{\epsilon} |u|^{-1} du = \infty$. Finally $|\hat{x}_0| \geq |\hat{x}_0 - \bar{x}| - |\bar{x}|$.

■

So a delta-method “standard error” of $\hat{x}_0$ approximates a quantity that does not exist. That is harmless when $\hat{\beta}_1$ is many standard errors from zero, and harmful exactly when the slope is poorly determined.

Theorem 12.5.3 — Calibration set

In the calibration model, let $t = t_{n-2, \alpha/2}$ and

$$\begin{aligned} \mathcal{C} = \left\{ x \in \mathbb{R} : (\bar{Y}_0 - \hat{\beta}_0 - \hat{\beta}_1 x)^2 \right. \\ \left. \leq t^2 s^2 \left(\frac{1}{m} + \frac{1}{n} + \frac{(x - \bar{x})^2}{S_{xx}} \right) \right\}. \end{aligned} \quad (12.5.2)$$

- (a) $\mathcal{C}$ is an exact $1 - \alpha$ confidence set for x_0 , at every $(\beta_0, \beta_1, \sigma^2, x_0)$, including $\beta_1 = 0$.
- (b) $\mathcal{C}$ is the set of x at which $\bar{Y}_0$ lies inside the $1 - \alpha$ prediction interval of [Theorem 12.3.1\(c\)](#) for the mean of m new readings at x .
- (c) Let $T_1 = \hat{\beta}_1 \sqrt{S_{xx}}/s$ be the t statistic for the slope and $g = \bar{Y}_0 - \bar{Y}$. Outside an

event of probability zero, $\mathcal{C}$ is

- a bounded interval containing $\hat{x}_0$, if $T_1^2 > t^2$;
- the union of two disjoint closed rays, if $T_1^2 < t^2$ and $T_1^2 + g^2 / \{s^2(1/m + 1/n)\} > t^2$;
- the whole real line, if $T_1^2 + g^2 / \{s^2(1/m + 1/n)\} < t^2$.

Proof.

- (a) and (b). For fixed x , $\bar{Y}_0 - \hat{\beta}_0 - \hat{\beta}_1 x$ is the difference between the mean of the new readings and the estimated mean response at x . At $x = x_0$, [Theorem 12.3.1\(c\)](#), with $\mathbf{x}_0 = (1, x_0)^\top$ and $h_0 = 1/n + (x_0 - \bar{x})^2 / S_{xx}$ ([Eq. 5.5.2](#)), shows that

$$\frac{\bar{Y}_0 - \hat{\beta}_0 - \hat{\beta}_1 x_0}{s\sqrt{1/m + 1/n + (x_0 - \bar{x})^2 / S_{xx}}} \sim t(n-2),$$

whatever the parameters are. This is a pivot, and $x_0 \in \mathcal{C}$ is the event that its absolute value is at most t . Nothing in the argument requires $\beta_1 \neq 0$. Part (b) restates the definition.

- (b) Put $d = x - \bar{x}$ and $c = 1/m + 1/n$. Since $\bar{Y}_0 - \hat{\beta}_0 - \hat{\beta}_1 x = g - \hat{\beta}_1 d$, the defining inequality is $Ad^2 - 2\hat{\beta}_1 g d + (g^2 - t^2 s^2 c) \leq 0$ with $A = \hat{\beta}_1^2 - t^2 s^2 / S_{xx}$. Its discriminant is

$$\begin{aligned} \hat{\beta}_1^2 g^2 - A(g^2 - t^2 s^2 c) &= t^2 s^2 \left(\frac{g^2}{S_{xx}} + cA \right) \\ &= \frac{t^2 s^4 c}{S_{xx}} \left(T_1^2 + \frac{g^2}{s^2 c} - t^2 \right). \end{aligned}$$

The sign of A is that of $T_1^2 - t^2$. If $A > 0$, the quadratic is negative at $d = g/\hat{\beta}_1$, that is at $x = \hat{x}_0$, where it equals $-t^2 s^2 (c + d^2 / S_{xx})$; so it has two real roots and $\mathcal{C}$ is the interval between them. If $A < 0$, the argument of [Proposition 12.5.1\(b\)](#) applies: two rays if the discriminant is positive and the whole line if it is negative. Equalities have probability zero. ■

Part (b) is the practical recipe ([Figure 12.5.1](#)): draw the prediction band for the mean of m readings and the horizontal line at $\bar{Y}_0$, and read off where the line is inside the band. A steep calibration line crosses the band in a short interval. A flat one lets both hyperbolic edges pass the horizontal line, giving two rays or everything. The set is bounded iff the slope test rejects, as it does for any useful instrument.

Proposition 12.5.4 — Unbounded calibration sets cannot be avoided

Let $\mathcal{D}$ be any confidence set for x_0 with coverage at least $1 - \alpha > 0$ at every parameter value, including those with $\beta_1 = 0$. Then at every parameter value with $\beta_1 = 0$, $\Pr\{v \in \mathcal{D}\} \geq 1 - \alpha$ for every real v , and if $\mathcal{D}$ is measurable in the sense of [Proposition 12.1.3](#), its expected length is infinite.

Proof. When $\beta_1 = 0$, the joint distribution of the standards and the new readings does not involve x_0 . So for every v , the probability that $v \in \mathcal{D}$ is the same as it would be if x_0 were v , and that probability is at least $1 - \alpha$. The rest is the Fubini argument in the

proof of [Proposition 12.1.3](#).

■

By continuity the same happens approximately for small β_1 . A procedure that always reports a bounded interval, like the delta method, pays with coverage below $1 - \alpha$ somewhere. The Fieller set's unbounded answers are the honest answer to a question the data cannot settle.

Example. (Calibrating an assay)

In a simulated assay, nine standards of concentration $0, 1, \dots, 8$ are each measured three times ($n = 27$). The fitted line is $0.0338 + 0.1161x$, with $s = 0.0131$ and slope t statistic 118.7. A new specimen is read $m = 3$ times, with mean reading 0.6521. The classical estimate is $\hat{x}_0 = 5.327$, and the 95% calibration interval is $(5.184, 5.471)$ ($t_{25,0.025} = 2.060$). For an assay this precise the delta-method interval, $\hat{x}_0 \pm t(s/|\hat{\beta}_1|)\sqrt{1/m + 1/n + (\hat{x}_0 - \bar{x})^2/S_{xx}}$, agrees to three decimals: $(5.184, 5.471)$.

Now consider a weak assay with the same design, in which the fitted slope is 0.0021 and its t statistic is only 1.67. The mean reading of the standards is 0.0475. A specimen with mean reading 0.0525, close to that mean, gives the whole real line: nothing about its concentration can be concluded. A specimen with mean reading 0.0775 gives two rays, $(-\infty, -55.3] \cup [7.6, \infty)$: the middle of the scale is excluded, but not extreme concentrations of either sign. [Figure 12.5.1](#) shows both assays.

In 20000 simulated experiments at $x_0 = 5.3$, both intervals cover in 0.9517 for the precise assay. With the slope three standard errors from zero ($\beta_1 = 0.00335$) the Fieller set again covers in 0.9517, since its event involves only the errors, and is bounded in 0.817 of the experiments, close to the power 0.822 of the slope t test; the delta-method interval covers in 0.9827, wastefully. With the slope two standard errors from zero and $x_0 = 12$, beyond the standards, the Fieller set covers in 0.9528 and the delta-method interval in only 0.9337.

```
import numpy as np
from scipy import stats

rng = np.random.default_rng(1206)
x = np.repeat(np.arange(9.0), 3)          # standards, in triplicate
n = len(x)
y = 0.04 + 0.115 * x + 0.015 * rng.normal(size=n) # absorbance readings
m = 3
y0 = 0.04 + 0.115 * 5.3 + 0.015 * rng.normal(size=m) # the unknown sample, true x0
    = 5.3

def calibrate(x, y, y0bar, m, level=0.95):
    """Fieller set {x0 : |y0bar - b0 - b1 x0| <= t s sqrt(1/m + 1/n + (x0 - xbar)^2
    / Sxx)}. """
    n = len(x)
    xbar, Sxx = x.mean(), np.sum((x - x.mean()) ** 2)
    b1 = np.sum((x - xbar) * (y - y.mean())) / Sxx
    s2 = np.sum((y - y.mean() - b1 * (x - xbar)) ** 2) / (n - 2)
    t2 = stats.t.ppf(0.5 + level / 2, n - 2) ** 2
    g = y0bar - y.mean()
    # quadratic in d = x0 - xbar: A d^2 - 2 B d + K <= 0
```

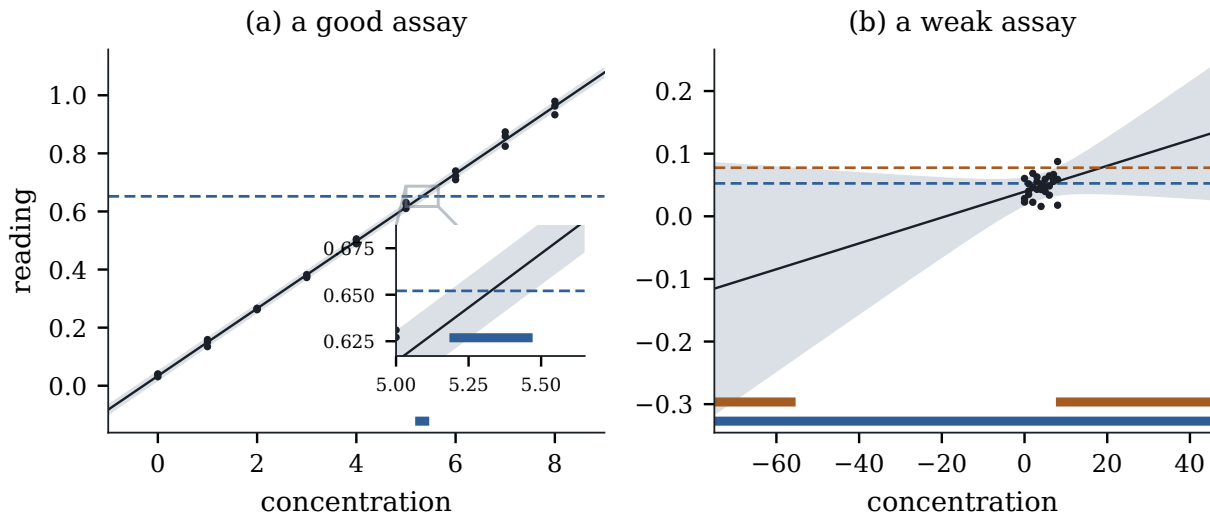


Figure 12.5.1: Calibration inverts the 95% prediction band for the mean of $m = 3$ readings (shaded); bars mark the calibration sets for the observed readings (horizontal lines). (a) A precise assay, magnified in the inset. (b) A weak assay: whole line (lower bar) and two rays (upper bar).

```

A = b1**2 - t2 * s2 / Sxx
B = b1 * g
K = g**2 - t2 * s2 * (1 / m + 1 / n)
disc = B**2 - A * K
if disc < 0:
    return "whole line", (-np.inf, np.inf), xbar + g / b1
r1, r2 = sorted([(B - np.sqrt(disc)) / A, (B + np.sqrt(disc)) / A])
if A > 0:
    return "interval", (xbar + r1, xbar + r2), xbar + g / b1
return "two rays", (xbar + r1, xbar + r2), xbar + g / b1 # (-inf, first] and [
second, inf)

shape, (lo, hi), x0_hat = calibrate(x, y, y0.mean(), m)
print(f"estimate {x0_hat:.3f}; 95% calibration set: {shape} ({lo:.3f}, {hi:.3f})")

```

Remark (Classical or inverse estimation?). The *inverse* estimator regresses the x_i on the readings and predicts from $\bar{Y}_0$: $\tilde{x}_0 = \bar{x} + (S_{xy}/S_{yy})(\bar{Y}_0 - \bar{Y})$, with $S_{xy} = \sum(x_i - \bar{x})(Y_i - \bar{Y})$ and $S_{yy} = \sum(Y_i - \bar{Y})^2$. Since $\bar{Y}_0 - \bar{Y} = \hat{\beta}_1(\hat{x}_0 - \bar{x})$ and $\hat{\beta}_1 = S_{xy}/S_{xx}$,

$$\tilde{x}_0 - \bar{x} = \frac{S_{xy}^2}{S_{xx}S_{yy}}(\hat{x}_0 - \bar{x}) = R^2(\hat{x}_0 - \bar{x}),$$

where R is the sample correlation of the calibration pairs. So $\tilde{x}_0$ shrinks $\hat{x}_0$ toward $\bar{x}$ by the factor $R^2 \in [0, 1]$. It has finite moments and can beat $\hat{x}_0$ in mean squared error near the centre of the standards (Krutchhoff 1967); away from the centre it does worse (Exercise (B3)). Hoadley (1970) showed that it is a Bayes estimator under a particular prior for x_0 .

12.5.3 Exercises

12.5.3.1 A. Check your understanding

Exercise A1.

For a straight line, write down Fieller's $1 - \alpha$ confidence set for the point $\xi = -\beta_0/\beta_1$ where the line crosses zero, and state the condition under which it is a bounded interval. Show that it is the calibration set of [Theorem 12.5.3](#) with $\bar{Y}_0$ replaced by 0 and $1/m$ replaced by 0.

Solution. With $\lambda_1 = (1, 0)^\top$ and $\lambda_2 = (0, 1)^\top$, $\xi = -a/b$. Apply [Proposition 12.5.1](#) to $\theta = a/b$ and change sign, or directly: ξ_0 is in the set iff the t interval for $\beta_0 + \beta_1\xi_0$ contains 0, that is iff $(\hat{\beta}_0 + \hat{\beta}_1\xi_0)^2 \leq t^2 s^2 \{1/n + (\xi_0 - \bar{x})^2/S_{xx}\}$. This is [Eq. 12.5.2](#) with $\bar{Y}_0 = 0$ and the $1/m$ term absent, because 0 is a known number rather than an average of new readings. It is bounded iff $T_1^2 > t^2$.

Exercise A2.

Show that $\hat{x}_0$, together with the least squares estimates from the standards, maximizes the joint likelihood of the $n + m$ readings over $(\beta_0, \beta_1, \sigma^2, x_0)$ when $\hat{\beta}_1 \neq 0$. What is the maximum likelihood estimate of σ^2 ?

Solution. For fixed σ^2 the likelihood is maximized by minimizing $\sum_i (Y_i - \beta_0 - \beta_1 x_i)^2 + \sum_j (Y_{0j} - \beta_0 - \beta_1 x_0)^2$. For any $\beta_1 \neq 0$ the second sum is minimized over x_0 by choosing $\beta_0 + \beta_1 x_0 = \bar{Y}_0$, leaving $\sum_j (Y_{0j} - \bar{Y}_0)^2$, which is free of the parameters. The first sum is then minimized by least squares on the standards, and $x_0 = (\bar{Y}_0 - \hat{\beta}_0)/\hat{\beta}_1 = \hat{x}_0$. The maximum likelihood estimate of σ^2 is $\{\text{SSE} + \sum_j (Y_{0j} - \bar{Y}_0)^2\}/(n + m)$.

12.5.3.2 B. Practice

Exercise B1.

When $m \geq 2$, the new readings carry their own information about σ^2 . Let $s_p^2 = \{\text{SSE} + \sum_j (Y_{0j} - \bar{Y}_0)^2\}/(n + m - 3)$. Show that replacing s by s_p and $t_{n-2, \alpha/2}$ by $t_{n+m-3, \alpha/2}$ in [Eq. 12.5.2](#) gives another exact $1 - \alpha$ set. Why is it shorter on average, and why is it more often bounded?

Solution. The within-specimen sum of squares is $\sigma^2 \chi^2(m - 1)$, independent of $\bar{Y}_0$ (the sample mean and sample variance of a normal sample are independent) and of the standards. So $\{\text{SSE} + \sum_j (Y_{0j} - \bar{Y}_0)^2\}/\sigma^2 \sim \chi^2(n + m - 3)$, independent of the numerator of the pivot, and the pivot with s_p is $t(n + m - 3)$. More degrees of freedom give a smaller multiplier and a less variable estimate of σ , so the set is shorter on average. The bounded case requires $\hat{\beta}_1^2 S_{xx} > t^2 s_p^2$, and a smaller t makes this easier.

Exercise B2.

In the quadratic regression $EY_i = \beta_0 + \beta_1 x_i + \beta_2 x_i^2$, use [Proposition 12.5.1](#) to give

a confidence set for the location $x^* = -\beta_1/(2\beta_2)$ of the turning point. When is it a bounded interval, and what does that condition mean for the curvature?

Solution. Take $\lambda_1 = (0, 1, 0)^\top$, $\lambda_2 = (0, 0, 2)^\top$ and $\theta = \beta_1/(2\beta_2)$, so $x^* = -\theta$. The set is $\{x_0 : (\hat{\beta}_1 + 2\hat{\beta}_2 x_0)^2 \leq t^2 s^2 (c_{11} + 4c_{12}x_0 + 4c_{22}x_0^2)\}$, where c_{ij} are the entries of $(\mathbf{X}^\top \mathbf{X})^{-1}$ belonging to β_1, β_2 . It is the set of x_0 at which the estimated derivative $\beta_1 + 2\beta_2 x_0$ is not significantly different from zero. It is bounded iff $\hat{\beta}_2^2 > t^2 s^2 c_{22}$, that is iff the curvature is significant: without significant curvature, the fitted curve may be nearly straight and its turning point anywhere.

Exercise B3.

Study the two estimators of the Remark “Classical or inverse estimation?” in an idealized large calibration experiment: the standards are so numerous that $\hat{\beta}_0, \hat{\beta}_1$ and R^2 may be replaced by constants $\beta_0, \beta_1 \neq 0$ and $\rho^2 \in (0, 1)$, while $\bar{Y}_0$ is still the mean of m new readings. Let $v = \sigma^2/(m\beta_1^2)$.

1. Show that $\hat{x}_0$ is unbiased with mean squared error v , and that $\tilde{x}_0 = \bar{x} + \rho^2(\hat{x}_0 - \bar{x})$ has mean squared error $\rho^4 v + (1 - \rho^2)^2(x_0 - \bar{x})^2$.
2. Show that $\tilde{x}_0$ has the smaller mean squared error iff $(x_0 - \bar{x})^2 < v(1 + \rho^2)/(1 - \rho^2)$.
3. Let $\tau^2 = S_{xx}/n$, so that in this limit $\rho^2 = \beta_1^2 \tau^2 / (\beta_1^2 \tau^2 + \sigma^2)$. Show that the condition of part 2 reads $(x_0 - \bar{x})^2 < v + 2\tau^2/m$. With $m = 1$ and a precise instrument, roughly how far from $\bar{x}$, in units of the spread τ of the standards, does the inverse estimator stay ahead?

Solution.

1. Here $\hat{x}_0 = x_0 + \bar{\varepsilon}_0/\beta_1$ with $\bar{\varepsilon}_0 \sim N(0, \sigma^2/m)$, so it is unbiased with variance v . Then $\tilde{x}_0 - x_0 = \rho^2(\hat{x}_0 - x_0) - (1 - \rho^2)(x_0 - \bar{x})$: variance $\rho^4 v$ plus squared bias $(1 - \rho^2)^2(x_0 - \bar{x})^2$.
2. The inequality $\rho^4 v + (1 - \rho^2)^2(x_0 - \bar{x})^2 < v$ is $(1 - \rho^2)^2(x_0 - \bar{x})^2 < (1 - \rho^2)(1 + \rho^2)v$; divide by $(1 - \rho^2)^2 > 0$.
3. $(1 + \rho^2)/(1 - \rho^2) = (2\beta_1^2 \tau^2 + \sigma^2)/\sigma^2$, so $v(1 + \rho^2)/(1 - \rho^2) = v + 2\tau^2/m$. When σ is small, $v \approx 0$ and the inverse estimator wins only for $|x_0 - \bar{x}| < \sqrt{2}\tau$, a little more than one spread from the centre; beyond that its bias dominates.

12.5.3.3 C. Going deeper

Exercise C1.

In the setting of [Proposition 12.5.1](#), let $E_t = \{\boldsymbol{\phi} \in \mathbb{R}^2 : ((\hat{a}, \hat{b})^\top - \boldsymbol{\phi})^\top \mathbf{V}^{-1}((\hat{a}, \hat{b})^\top - \boldsymbol{\phi}) \leq t^2 s^2\}$, an ellipse with the t constant in place of $2F_\alpha(2, n - r)$. Show that $\theta_0 \in \mathcal{F}(\mathbf{Y})$ iff the line $\{(\phi_1, \phi_2) : \phi_1 = \theta_0 \phi_2\}$ meets E_t . Use the picture to explain the three cases: the set is the whole line iff E_t contains the origin, and it is bounded iff E_t does not meet the axis $\phi_2 = 0$.

Solution. The line is $\{\boldsymbol{\phi} : \mathbf{c}^\top \boldsymbol{\phi} = 0\}$ with $\mathbf{c} = (1, -\theta_0)^\top$. By [Corollary 12.2.4](#) (with the constant $t^2 s^2$), the values of $\mathbf{c}^\top \boldsymbol{\phi}$ over E_t form the interval $\hat{a} - \theta_0 \hat{b} \pm ts\sqrt{\mathbf{c}^\top \mathbf{V} \mathbf{c}}$, and $\mathbf{c}^\top \mathbf{V} \mathbf{c} = v_{11} - 2\theta_0 v_{12} + \theta_0^2 v_{22}$. So the line meets E_t iff this interval contains 0, which is [Eq. 12.5.1](#). Every line through the origin meets E_t iff the origin is in E_t ,

which is $Q \leq t^2 s^2$. Lines of large slope approach the axis $\phi_2 = 0$ (the ϕ_1 -axis in these coordinates); the set is unbounded iff lines of arbitrarily large slope meet E_t , iff E_t meets that axis, which by [Corollary 12.2.4](#) with $\mathbf{c} = (0, 1)^\top$ is $\hat{b}^2 \leq t^2 s^2 v_{22}$.

Exercise C2.

Two straight lines $EY = \alpha_k + \beta_k x$, $k = 1, 2$, are fitted to two independent samples in one model with a common σ^2 , as in a two-phase regression. Use [Proposition 12.5.1](#) to construct a confidence set for the x -coordinate $(\alpha_1 - \alpha_2)/(\beta_2 - \beta_1)$ of their intersection. When is it bounded? Interpret the unbounded cases in terms of parallel lines.

12.6 Bayesian thread II: credible intervals and predictive distributions

[Section 7.6](#) derived the posterior distribution of $(\boldsymbol{\beta}, \sigma^2)$ under the conjugate normal-inverse-gamma prior, and showed that the flat prior $\pi(\boldsymbol{\beta}, \sigma^2) \propto 1/\sigma^2$ reproduces least squares. Here the posterior is summarized by credible regions and used to predict. In the conjugate model these have the shapes of the frequentist sets, and under the flat prior they coincide with them. We end by asking how often a credible interval covers in repeated sampling.

12.6.1 Credible sets

Definition 12.6.1 — Credible set

Let θ be a function of the parameters and let $\pi(\cdot | \mathbf{y})$ be its posterior distribution. A set $C(\mathbf{y})$ with $\Pr\{\theta \in C(\mathbf{y}) | \mathbf{y}\} \geq 1 - \alpha$ is a $1 - \alpha$ **credible set** for θ . If the posterior has a density $\pi(\theta | \mathbf{y})$, a set of the form $\{\theta : \pi(\theta | \mathbf{y}) \geq k\}$ with posterior probability $1 - \alpha$ is a **highest posterior density** (HPD) set.

Here the statement concerns the realized set. An HPD set has the smallest volume among sets with its posterior probability. We need three facts about the multivariate t distribution of [Definition 7.6.2](#).

Lemma 12.6.1 — Multivariate t

Let $\mathbf{u} \sim t_\nu(\mathbf{m}, \mathbf{S})$ in $\mathbb{R}^p$, with $\mathbf{S}$ positive definite.

- (a) If $\mathbf{A}$ is $q \times p$ of rank q , then $\mathbf{A}\mathbf{u} \sim t_\nu(\mathbf{A}\mathbf{m}, \mathbf{A}\mathbf{S}\mathbf{A}^\top)$.
- (b) $(\mathbf{u} - \mathbf{m})^\top \mathbf{S}^{-1}(\mathbf{u} - \mathbf{m})/p \sim F(p, \nu)$.
- (c) $\mathbf{u}$ has density proportional to $\{1 + (\mathbf{u} - \mathbf{m})^\top \mathbf{S}^{-1}(\mathbf{u} - \mathbf{m})/\nu\}^{-(\nu+p)/2}$.

Proof. By Definition 7.6.2, $\mathbf{u} = \mathbf{m} + \mathbf{Z}/\sqrt{W/\nu}$ with $\mathbf{Z} \sim N_p(\mathbf{0}, \mathbf{S})$ and $W \sim \chi^2(\nu)$ independent.

- (a) $\mathbf{A}\mathbf{u} = \mathbf{A}\mathbf{m} + \mathbf{A}\mathbf{Z}/\sqrt{W/\nu}$, and $\mathbf{A}\mathbf{Z} \sim N_q(\mathbf{0}, \mathbf{A}\mathbf{S}\mathbf{A}^\top)$ (Theorem 3.3.1) is independent of W , with $\mathbf{A}\mathbf{S}\mathbf{A}^\top$ positive definite.
- (b) $(\mathbf{u} - \mathbf{m})^\top \mathbf{S}^{-1}(\mathbf{u} - \mathbf{m}) = \mathbf{Z}^\top \mathbf{S}^{-1} \mathbf{Z}/(W/\nu)$, and $\mathbf{Z}^\top \mathbf{S}^{-1} \mathbf{Z} \sim \chi^2(p)$ independently of W (Corollary 4.3.3(a)). Divide by p and use Definition 4.1.2.
- (c) Given $W = w$, $\mathbf{u} \sim N_p(\mathbf{m}, (\nu/w)\mathbf{S})$. Write $D = (\mathbf{u} - \mathbf{m})^\top \mathbf{S}^{-1}(\mathbf{u} - \mathbf{m})$. Integrating the normal density (Theorem 3.2.2) against the $\chi^2(\nu)$ density of W , and dropping factors free of $\mathbf{u}$,

$$\begin{aligned} \int_0^\infty w^{p/2} e^{-wD/(2\nu)} w^{\nu/2-1} e^{-w/2} dw &= \int_0^\infty w^{(\nu+p)/2-1} e^{-w(1+D/\nu)/2} dw \\ &\propto (1 + D/\nu)^{-(\nu+p)/2}, \end{aligned}$$

by the substitution $w' = w(1 + D/\nu)$. ■

12.6.2 Credible sets and predictive distributions in the conjugate model

Theorem 12.6.2 — Credible sets and posterior predictive distribution

Let the prior be $\text{NIG}(\mathbf{m}_0, \mathbf{V}_0, a_0, b_0)$, so that by Theorem 7.6.2 the posterior is $\text{NIG}(\mathbf{m}_n, \mathbf{V}_n, a_n, b_n)$. Put $\nu = 2a_n$ and $\tau^2 = b_n/a_n$.

- (a) For $\boldsymbol{\lambda} \neq \mathbf{0}$, the interval

$$\boldsymbol{\lambda}^\top \mathbf{m}_n \pm t_{\nu, \alpha/2} \tau \sqrt{\boldsymbol{\lambda}^\top \mathbf{V}_n \boldsymbol{\lambda}}$$

is the $1 - \alpha$ HPD credible interval for $\boldsymbol{\lambda}^\top \boldsymbol{\beta}$; it is also equal-tailed.

- (b) For a $p \times q$ matrix $\boldsymbol{\Lambda}$ of rank q , the set

$$\begin{aligned} \{ \boldsymbol{\Phi}_0 : (\boldsymbol{\Phi}_0 - \boldsymbol{\Lambda}^\top \mathbf{m}_n)^\top (\boldsymbol{\Lambda}^\top \mathbf{V}_n \boldsymbol{\Lambda})^{-1} (\boldsymbol{\Phi}_0 - \boldsymbol{\Lambda}^\top \mathbf{m}_n) \\ \leq q \tau^2 F_\alpha(q, \nu) \} \end{aligned}$$

is the $1 - \alpha$ HPD credible region for $\boldsymbol{\Lambda}^\top \boldsymbol{\beta}$.

- (c) Let $\mathbf{Y}_0 = \mathbf{X}_0 \boldsymbol{\beta} + \boldsymbol{\varepsilon}_0$ be k future observations, with $\boldsymbol{\varepsilon}_0 \sim N_k(\mathbf{0}, \sigma^2 \mathbf{I}_k)$ independent of $\mathbf{Y}$ and of $\boldsymbol{\beta}$ given σ^2 . The posterior predictive distribution of $\mathbf{Y}_0$ is

$$\mathbf{Y}_0 \mid \mathbf{y} \sim t_\nu(\mathbf{X}_0 \mathbf{m}_n, \tau^2(\mathbf{I}_k + \mathbf{X}_0 \mathbf{V}_n \mathbf{X}_0^\top)).$$

In particular, for one future observation at $\mathbf{x}_0$, the interval $\mathbf{x}_0^\top \mathbf{m}_n \pm t_{\nu, \alpha/2} \tau \sqrt{1 + \mathbf{x}_0^\top \mathbf{V}_n \mathbf{x}_0}$ has posterior predictive probability $1 - \alpha$.

(d) $[2b_n/\chi_{\alpha/2}^2(\nu), 2b_n/\chi_{1-\alpha/2}^2(\nu)]$ is an equal-tailed $1 - \alpha$ credible interval for σ^2 .

Proof. By Theorem 7.6.2 and Lemma 7.6.1, $\boldsymbol{\beta} \mid \mathbf{y} \sim t_\nu(\mathbf{m}_n, \tau^2 \mathbf{V}_n)$ and $2b_n/\sigma^2 \mid \mathbf{y} \sim \chi^2(\nu)$.

- (a) By Lemma 12.6.1(a) with $\mathbf{A} = \boldsymbol{\lambda}^\top$, $\boldsymbol{\lambda}^\top \boldsymbol{\beta} = \boldsymbol{\lambda}^\top \mathbf{m}_n + \tau \sqrt{\boldsymbol{\lambda}^\top \mathbf{V}_n \boldsymbol{\lambda}} T$ with $T \sim t(\nu)$. The $t(\nu)$ density is symmetric and decreasing in $|x|$, so the symmetric interval is both equal-tailed and HPD.
- (b) By Lemma 12.6.1(a), $\boldsymbol{\Lambda}^\top \boldsymbol{\beta} \mid \mathbf{y} \sim t_\nu(\boldsymbol{\Lambda}^\top \mathbf{m}_n, \tau^2 \boldsymbol{\Lambda}^\top \mathbf{V}_n \boldsymbol{\Lambda})$. By (b) of the lemma the quadratic form divided by $q\tau^2$ is $F(q, \nu)$, so the set has posterior probability $1 - \alpha$. By (c) the density is a decreasing function of the quadratic form, so the set has the form $\{\text{density} \geq k\}$.
- (c) Given σ^2 and $\mathbf{y}$, $\boldsymbol{\beta} \sim N_p(\mathbf{m}_n, \sigma^2 \mathbf{V}_n)$ and $\boldsymbol{\varepsilon}_0 \sim N_k(\mathbf{0}, \sigma^2 \mathbf{I}_k)$ independently, so $\mathbf{Y}_0 \sim N_k(\mathbf{X}_0 \mathbf{m}_n, \sigma^2(\mathbf{X}_0 \mathbf{V}_n \mathbf{X}_0^\top + \mathbf{I}_k))$ (Theorem 3.3.1). Together with $\sigma^2 \mid \mathbf{y} \sim \text{IG}(a_n, b_n)$, this says that $(\mathbf{Y}_0, \sigma^2) \mid \mathbf{y}$ is NIG($\mathbf{X}_0 \mathbf{m}_n, \mathbf{I}_k + \mathbf{X}_0 \mathbf{V}_n \mathbf{X}_0^\top, a_n, b_n$) (Definition 7.6.1), the matrix being positive definite. Lemma 7.6.1 gives the t_ν law, and (a) of the present theorem, applied to it with $k = 1$, gives the interval.
- (d) $\Pr\{\chi_{1-\alpha/2}^2(\nu) \leq 2b_n/\sigma^2 \leq \chi_{\alpha/2}^2(\nu) \mid \mathbf{y}\} = 1 - \alpha$; invert the inequalities.

■

The formulas mirror the frequentist ones: $\hat{\boldsymbol{\beta}}$ becomes $\mathbf{m}_n$, $\mathbf{G}$ becomes $\mathbf{V}_n = (\mathbf{V}_0^{-1} + \mathbf{X}^\top \mathbf{X})^{-1}$, s^2 becomes τ^2 , and $n - r$ becomes $\nu = 2a_0 + n$, which counts every observation plus $2a_0$ “prior observations” about σ^2 . The 1 in $1 + \mathbf{x}_0^\top \mathbf{V}_n \mathbf{x}_0$ is again the new error.

Corollary 12.6.3 — The flat prior reproduces the frequentist sets

Let $\text{rank}(\mathbf{X}) = p < n$, $\text{SSE} > 0$, and let the prior be $\pi(\boldsymbol{\beta}, \sigma^2) \propto 1/\sigma^2$. Then the credible interval (a), the region (b), the predictive interval (c) and the interval for σ^2 (d) of Theorem 12.6.2 are, respectively, the t interval of Theorem 12.1.1, the confidence ellipsoid of Theorem 12.2.1, the prediction interval of Theorem 12.3.1 and the interval of Proposition 12.1.4.

Proof. By Corollary 7.6.4 the posterior is NIG($\hat{\boldsymbol{\beta}}, (\mathbf{X}^\top \mathbf{X})^{-1}, (n - p)/2, \text{SSE}/2$). The proof of Theorem 12.6.2 used only that the posterior is normal-inverse-gamma, so its conclusions hold with $\mathbf{m}_n = \hat{\boldsymbol{\beta}}$, $\mathbf{V}_n = (\mathbf{X}^\top \mathbf{X})^{-1} = \mathbf{G}$, $\nu = n - p = n - r$, $\tau^2 = \text{SSE}/(n - p) = s^2$ and $2b_n = \text{SSE}$. Substituting gives the four frequentist sets.

■

The same set then carries two readings: “this procedure covers the fixed truth in 95% of samples” and “given these data, the parameter lies here with probability 0.95”. The agreement is special; it fails for proper priors and for most other models.

12.6.3 A worked example

Example. (Predicting a day of stack loss)

Take the stack loss regression with the engineer's prior of [Example \(A prior for the stack loss plant\)](#), and consider a new day with air flow 65, water temperature 23 and acid concentration 87. The posterior has $a_n = 13.5$, so $\nu = 27$ and $t_{27,0.025} = 2.052$. By [Theorem 12.6.2](#), the 95% credible interval for the mean stack loss at these conditions is 23.12 ± 1.51 , and the 95% posterior predictive interval for the day's stack loss is 23.12 ± 6.11 . The least squares answers are 23.15 ± 1.77 for the mean and 23.15 ± 7.07 for the new day, with $t_{17,0.025} = 2.110$. The Bayesian intervals are shorter because the prior informs the slopes, shrinking $\mathbf{x}_0^\top \mathbf{V}_n \mathbf{x}_0$, and σ^2 , raising the degrees of freedom from 17 to 27.

The 2.5% and 97.5% points of 400000 posterior predictive draws are 17.00 and 29.21, matching the t interval within Monte Carlo error. The simulated posterior probability of the credible ellipse of (b) for the air-flow and water-temperature coefficients is 0.9503.

```
import numpy as np
import statsmodels.api as sm
from scipy import stats

data = sm.datasets.stackloss.load_pandas().data
y = data["STACKLOSS"].to_numpy()
X = np.column_stack([np.ones(len(y)), data[["AIRFLOW", "WATERTEMP", "ACIDCONC"]]])
n, p = X.shape
XtX, Xty = X.T @ X, X.T @ y

def nig_posterior(m0, P0, a0, b0, y=y):
    """Conjugate update; P0 is the prior precision V0^{-1} (per unit sigma^2)."""
    Vn = np.linalg.inv(P0 + XtX)
    mn = Vn @ (P0 @ m0 + X.T @ y)
    an = a0 + n / 2
    bn = b0 + (y @ y + m0 @ P0 @ m0 - mn @ (P0 + XtX) @ mn) / 2
    return mn, Vn, an, bn

m0 = np.array([0.0, 1.0, 1.0, 0.0]) # the prior of Chapter 7
P0 = np.linalg.inv(np.diag([100.0**2, 0.1**2, 0.2**2, 0.1**2]))
a0, b0 = 3.0, 18.0
mn, Vn, an, bn = nig_posterior(m0, P0, a0, b0)

x0 = np.array([1.0, 65.0, 23.0, 87.0]) # a new operating condition
qb = stats.t.ppf(0.975, 2 * an)
centre = x0 @ mn
half_mean = qb * np.sqrt(bn / an * (x0 @ Vn @ x0)) # credible interval for x0^T
beta
half_new = qb * np.sqrt(bn / an * (1 + x0 @ Vn @ x0)) # posterior predictive
interval for Y0

beta_hat = np.linalg.solve(XtX, Xty) # frequentist intervals, for
comparison
s2 = np.sum((y - X @ beta_hat) ** 2) / (n - p)
qf = stats.t.ppf(0.975, n - p)
h0 = x0 @ np.linalg.solve(XtX, x0)
print(f"Bayes: mean response {centre:.2f} +- {half_mean:.2f}; new day {centre:.2f}
```

```

+- {half_new:.2f}")
print(f"least squares: mean response {x0 @ beta_hat:.2f} +- {qf * np.sqrt(s2 * h0)
:.2f}; "
      f"new day +- {qf * np.sqrt(s2 * (1 + h0)):.2f}")

```

12.6.4 How often does a credible interval cover?

One coverage statement holds for every proper prior.

Proposition 12.6.4 — Coverage averaged over the prior

Let the prior π be proper, and let $C(\mathbf{y})$ be a $1 - \alpha$ credible set for θ for every $\mathbf{y}$. If $(\boldsymbol{\beta}, \sigma^2)$ is drawn from π and then $\mathbf{Y}$ from the model, $\Pr\{\theta \in C(\mathbf{Y})\} \geq 1 - \alpha$, with equality if every $C(\mathbf{y})$ has posterior probability exactly $1 - \alpha$. Equivalently, the frequentist coverage $\Pr_{\boldsymbol{\beta}, \sigma^2}\{\theta \in C(\mathbf{Y})\}$, averaged over π , is at least $1 - \alpha$.

Proof. Under the joint distribution of parameters and data, the conditional distribution of the parameters given $\mathbf{Y} = \mathbf{y}$ is the posterior. By the tower property, $\Pr\{\theta \in C(\mathbf{Y})\} = \mathbb{E}[\Pr\{\theta \in C(\mathbf{Y}) \mid \mathbf{Y}\}] \geq 1 - \alpha$. Conditioning instead on the parameters gives the second form. ■

The average hides variation: coverage is high where the prior is right and low where it is wrong. Consider the credible interval for the air-flow coefficient in [Example \(Predicting a day of stack loss\)](#), and simulate its frequentist coverage when the true air-flow coefficient is varied, the other coefficients are held at their least squares values, and $\sigma^2 = s^2$. [Figure 12.6.1](#) shows the result. With the prior of Chapter 7, the coverage is 0.959 when the truth is the prior mean 1, and 0.936 at 0.7 and at 1.5. It ranges from 0.877 to 0.965 over the plotted range. This prior is mild, and prior–data conflict inflates b_n , which widens the interval. A prior with slope standard deviations five times smaller behaves very differently: its coverage is 1.000 at the prior mean but 0.007 at 0.7, close to the least squares estimate, and 0.000 at 1.5. Averaged over the prior of Chapter 7, the simulated coverage is 0.9512 (40000 draws), as [Proposition 12.6.4](#) requires. The flat-prior interval is the t interval and covers exactly 95% at every parameter value.

An informative prior makes intervals shorter, and better when the prior is sound; when it is not, they are confidently wrong. The frequentist interval makes no bet and pays in length. Chapter 27 meets the same trade-off as bias against variance.

12.6.5 Exercises

12.6.5.1 A. Check your understanding

Exercise A1.

Is the equal-tailed interval of [Theorem 12.6.2\(d\)](#) the HPD interval for σ^2 ? Describe the HPD interval and explain the difference.

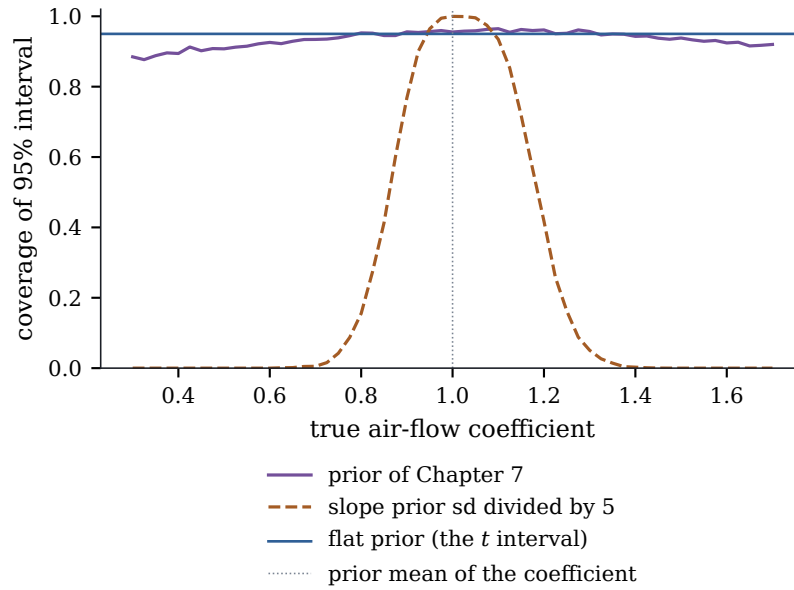


Figure 12.6.1: Frequentist coverage of the 95% credible interval for the air-flow coefficient against the true coefficient, for the prior of Chapter 7 and for a prior with slope standard deviations divided by five (4000 data sets per point). The flat-prior interval covers 0.95 everywhere.

Solution. No. The inverse gamma density is skewed to the right, so the HPD interval $[l, u]$, with $\pi(l | \mathbf{y}) = \pi(u | \mathbf{y})$, is shifted toward smaller values. It also depends on the parameterization: the HPD interval for $\log \sigma^2$ does not transform into the one for σ^2 , while equal-tailed intervals do.

Exercise A2.

For $\nu > 2$, find the posterior predictive mean and variance of a single future observation at $\mathbf{x}_0$. Why is the variance larger than $\tau^2(1 + \mathbf{x}_0^\top \mathbf{V}_n \mathbf{x}_0)$?

Solution. From Theorem 12.6.2(c), $Y_0 | \mathbf{y} = \mathbf{x}_0^\top \mathbf{m}_n + \tau \sqrt{1 + \mathbf{x}_0^\top \mathbf{V}_n \mathbf{x}_0} T$ with $T \sim t(\nu)$, so the mean is $\mathbf{x}_0^\top \mathbf{m}_n$ and the variance is $\tau^2(1 + \mathbf{x}_0^\top \mathbf{V}_n \mathbf{x}_0) \nu/(\nu - 2)$. The extra factor comes from the uncertainty about σ^2 : given σ^2 the predictive variance is $\sigma^2(1 + \mathbf{x}_0^\top \mathbf{V}_n \mathbf{x}_0)$, and averaging over the posterior gives $E(\sigma^2 | \mathbf{y}) = b_n/(a_n - 1)$, which exceeds $\tau^2 = b_n/a_n$ by the factor $a_n/(a_n - 1) = \nu/(\nu - 2)$.

12.6.5.2 B. Practice

Exercise B1.

Under the g -prior of Proposition 7.6.3(c) with $\mathbf{m}_0 = \mathbf{0}$, show that the credible interval for $\lambda^\top \boldsymbol{\beta}$ is centred at $\{g/(1+g)\} \lambda^\top \hat{\boldsymbol{\beta}}$ and has half-width $t_{\nu, \alpha/2} \tau \sqrt{g/(1+g)} \sqrt{\lambda^\top (\mathbf{X}^\top \mathbf{X})^{-1} \lambda}$. Compare it with the t interval as $g \rightarrow \infty$, paying attention to ν and τ^2 .

Solution. With $\mathbf{V}_0 = g(\mathbf{X}^\top \mathbf{X})^{-1}$, $\mathbf{V}_n = \{g/(1+g)\}(\mathbf{X}^\top \mathbf{X})^{-1}$ and $\mathbf{m}_n = \{g/(1+g)\} \hat{\boldsymbol{\beta}}$ (Proposition 7.6.3(c)). Substitute in Theorem 12.6.2(a). As $g \rightarrow \infty$ the centre and the matrix tend to those of least squares, but $\nu = 2a_0 + n$ stays larger than $n - p$ and

$\tau^2 = b_n/a_n$ tends to $(2b_0 + \text{SSE})/(2a_0 + n)$, not to s^2 . The interval does not become the t interval: that needs the flat prior on σ^2 as well, with $a_0 = -p/2$ and $b_0 = 0$ (Corollary 7.6.4).

Exercise B2.

Two future observations are taken at the same $\mathbf{x}_0$. Show that, under the posterior predictive distribution, they are correlated, and find the correlation. Why are they correlated when their errors are independent? Are they independent under the frequentist model?

Solution. By Theorem 12.6.2(c) with $\mathbf{X}_0$ having two equal rows $\mathbf{x}_0^\top$, the scale matrix is $\tau^2(\mathbf{I}_2 + h\mathbf{1}\mathbf{1}^\top)$ with $h = \mathbf{x}_0^\top \mathbf{V}_n \mathbf{x}_0$, so the correlation is $h/(1+h)$. Both observations share the unknown $\boldsymbol{\beta}$ (and σ^2), and uncertainty about the shared mean correlates them. Under the frequentist model with fixed parameters the two observations are independent, but the two prediction errors $Y_{0j} - \hat{Y}_0$ share $\hat{Y}_0$ and have the same correlation $h_0/(1+h_0)$ under the flat prior.

12.6.5.3 C. Going deeper

Exercise C1.

Let $\mathbf{X}$ have rank $r < p$ and take the flat prior $\pi(\boldsymbol{\beta}, \sigma^2) \propto 1/\sigma^2$. Show that the posterior of $\boldsymbol{\beta}$ is improper, but that for estimable $\boldsymbol{\lambda}^\top \boldsymbol{\beta}$ the posterior of $(\boldsymbol{\lambda}^\top \boldsymbol{\beta}, \sigma^2)$ can be defined by reparameterizing with $\boldsymbol{\gamma} = \mathbf{U}^\top \boldsymbol{\beta}$ and $\boldsymbol{\delta} = \mathbf{N}^\top \boldsymbol{\beta}$, where the columns of $\mathbf{U}$ and $\mathbf{N}$ are orthonormal bases of $\mathcal{C}(\mathbf{X}^\top)$ and $\mathcal{N}(\mathbf{X})$, and integrating over nothing but $\boldsymbol{\gamma}$. Show that the resulting credible interval is the t interval of Theorem 12.1.1 with $n-r$ degrees of freedom, provided the flat prior is taken as $1/\sigma^2$ in $(\boldsymbol{\gamma}, \sigma^2)$ alone.

Solution. The likelihood depends on $\boldsymbol{\beta}$ only through $\mathbf{X}\boldsymbol{\beta} = \mathbf{X}\mathbf{U}\boldsymbol{\gamma}$, since $\mathbf{X}\mathbf{N} = \mathbf{0}$. So the posterior is flat in $\boldsymbol{\delta}$, which is not integrable: the posterior of $\boldsymbol{\beta}$ is improper. In the model $\mathbf{Y} \sim N_n(\mathbf{X}\mathbf{U}\boldsymbol{\gamma}, \sigma^2 \mathbf{I})$ the matrix $\mathbf{X}\mathbf{U}$ has full column rank r , and Corollary 7.6.4 with p replaced by r gives a proper posterior with $t(n-r)$ marginals. An estimable $\boldsymbol{\lambda}$ lies in $\mathcal{C}(\mathbf{X}^\top) = \mathcal{C}(\mathbf{U})$, so $\boldsymbol{\lambda} = \mathbf{U}\mathbf{U}^\top \boldsymbol{\lambda}$ and $\boldsymbol{\lambda}^\top \boldsymbol{\beta} = (\mathbf{U}^\top \boldsymbol{\lambda})^\top \boldsymbol{\gamma}$ involves $\boldsymbol{\gamma}$ only. Its credible interval is centred at $(\mathbf{U}^\top \boldsymbol{\lambda})^\top \hat{\boldsymbol{\gamma}} = \boldsymbol{\lambda}^\top \hat{\boldsymbol{\beta}}$ with scale $s^2(\mathbf{U}^\top \boldsymbol{\lambda})^\top (\mathbf{U}^\top \mathbf{X}^\top \mathbf{X} \mathbf{U})^{-1} (\mathbf{U}^\top \boldsymbol{\lambda})$, and $\mathbf{U}(\mathbf{U}^\top \mathbf{X}^\top \mathbf{X} \mathbf{U})^{-1} \mathbf{U}^\top$ is a generalized inverse of $\mathbf{X}^\top \mathbf{X}$ (check $\mathbf{X}^\top \mathbf{X} \mathbf{U}(\mathbf{U}^\top \mathbf{X}^\top \mathbf{X} \mathbf{U})^{-1} \mathbf{U}^\top \mathbf{X}^\top \mathbf{X} = \mathbf{X}^\top \mathbf{X}$ using $\mathbf{X} = \mathbf{X}\mathbf{U}\mathbf{U}^\top$), so the scale is $s^2 \boldsymbol{\lambda}^\top \mathbf{G} \boldsymbol{\lambda}$.

Exercise C2.

In the calibration model of Section 12.5, give x_0 a normal prior independent of the regression parameters, and use the flat prior $1/\sigma^2$ for $(\beta_0, \beta_1, \sigma^2)$. Write down the posterior density of x_0 up to a constant, and show that it is proper even when the slope is not significant. Compare the behaviour of the resulting credible interval with the Fieller set in the weak assay of Example (Calibrating an assay).

12.7 Summary and Notes

12.7.1 Summary

Idea.

1. Confidence sets come from pivots, and are the sets of values that a family of level- α tests accepts (Definition 12.1.1, Proposition 12.1.2).
2. For estimable $\lambda^\top \beta$, $\lambda^\top \hat{\beta} \pm t_{n-r, \alpha/2} s \sqrt{\lambda^\top \mathbf{G} \lambda}$ is exact and free of the generalized inverse (Theorem 12.1.1). A non-estimable function has no confidence set of finite expected length (Proposition 12.1.3). The interval for σ^2 is exact under normality and fragile without it (Proposition 12.1.4).
3. The confidence ellipsoid for $\Lambda^\top \beta$ is the acceptance region of the F test, shaped by the covariance of the estimates (Theorem 12.2.1); all such ellipsoids are images of one ball around $\hat{Y}$ (Proposition 12.2.2). Its shadows are Scheffé-type intervals (Lemma 12.2.3, Corollary 12.2.4).
4. The prediction interval $\hat{Y}_0 \pm t_{n-r, \alpha/2} s \sqrt{1 + h_0}$ never shrinks below the noise (Theorem 12.3.1). The leverage h_0 is a Mahalanobis distance that reveals hidden extrapolation (Proposition 12.3.2). Without normality prediction intervals stay wrong as $n \rightarrow \infty$ (Proposition 12.3.3).
5. The Working–Hotelling band, with multiplier $\sqrt{k F_\alpha(k, n-r)}$, covers the regression surface over a k -dimensional family exactly; $k = 2$ for a straight line (Theorem 12.4.1).
6. Fieller sets for ratios, and calibration sets, are exact, and are intervals, two rays or the whole line (Proposition 12.5.1, Theorem 12.5.3). The classical calibration estimator has no mean (Proposition 12.5.2), and unbounded sets cannot be avoided (Proposition 12.5.4).
7. Conjugate credible sets and predictive intervals have the frequentist shapes (Lemma 12.6.1, Theorem 12.6.2) and equal them under the flat prior (Corollary 12.6.3). Their coverage is nominal on average over the prior, not pointwise (Proposition 12.6.4).

12.7.2 Notes and sources

Confidence intervals and duality. Confidence intervals, and their construction by inverting tests, are due to Neyman (1937). Lehmann and Romano (2005) develop the duality and its optimality consequences. The two one-sided tests procedure of Exercise (B2) is due to Schuirman (1987). The shortest interval for a normal variance (Exercise (C2)) was worked out by Tate and Klett (1959).

The source books. The chapter covers Seber and Lee (2003, chapter 5 and sections 6.1–6.5), Rencher and Schaalje (2008, sections 8.6–8.7) and Sen and Srivastava (1990, section 3.8). These books state the intervals for full-rank models; here they are stated for estimable functions in models of any rank, as in Christensen (2020). Rencher and Schaalje (2008, section 8.7) derive the F test as a likelihood ratio test, which Chapter 11 covers (Theorem 11.4.1). Hahn and Meeker (1991) is a practical guide to confidence, prediction and tolerance intervals.

Bands. The band for a straight line is due to Working and Hotelling (1929), and the general

projection argument to Scheffé (1953). Exact bands over finite intervals are due to Wynn and Bloomfield (1971) and constant-width bands to Gafarian (1964). Miller (1981) surveys the early literature and Liu (2010) gives a modern book-length treatment.

Normality and prediction. The warning about conditional coverage after a preliminary test is from Olshen (1973).

Ratios and calibration. Fieller’s method came from bioassay and is set out in general form in Fieller (1954). Krutchkoff (1967) proposed the inverse estimator and Hoadley (1970) gave its Bayesian interpretation. Osborne (1991) reviews calibration, and Brown (1993) treats the multivariate case. Gleser and Hwang (1987) proved the general form of [Proposition 12.5.4](#).

Bayesian intervals. Box and Tiao (1973) and Gelman et al. (2013) treat credible and predictive intervals for the normal linear model under conjugate and flat priors.

Data. The murder-rate data are the 2009 US state data of statsmodels; the stack loss data are from Brownlee (1965); the assay is simulated.

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Chapter 13

Multiplicity: Scheffé, Tukey, Bonferroni and the False Discovery Rate

A test with error rate 0.05 keeps that promise only when it is the only inference drawn from the data. A regression analysis rarely stops at one: it compares every pair of groups, examines each of many coefficients, and follows a significant F test with the contrast that looks most interesting. This chapter measures the error of a whole family of inferences and controls it in four ways. Scheffé's method covers every estimable function in a subspace, even functions chosen after seeing the data, and is exactly the F test read one direction at a time. Tukey's studentized range is exact for all pairwise differences of balanced means. Bonferroni's inequality, and Holm's sharper step-down version, need only the level of each test. The Benjamini–Hochberg procedure controls the expected proportion of false discoveries, which is what screening many hypotheses needs. The chapter ends with a guide to choosing among them.

What you need. The normal linear model and the sampling distributions of [Chapter 7](#) ([Theorem 7.5.1](#), [Corollary 7.5.2](#), [Corollary 7.5.3](#)), independence of projections from [Chapter 4](#) ([Theorem 4.4.4](#)), and orthogonal projections onto subspaces from [Chapter 6](#) ([Theorem 6.5.1](#), [Theorem 6.5.2](#)). Estimable functions and contrasts in the one-way layout come from [Chapter 8](#) ([Theorem 8.7.1](#), [Proposition 8.7.2](#)), and the sum of squares of a linear hypothesis from [Chapter 9](#) ([Theorem 9.2.4](#)). The F test of [Chapter 11](#) ([Theorem 11.1.1](#), [Theorem 11.3.2](#)) and the confidence intervals and ellipsoids of [Chapter 12](#) ([Theorem 12.1.1](#), [Theorem 12.2.1](#)) are the single-inference versions of what this chapter does for families.

13.0.1 Roadmap

- [The multiple testing problem](#)
- [Scheffé's projection method](#)
- [Tukey's studentized range](#)
- [Bonferroni and Holm](#)
- [False discovery rate control](#)
- [Choosing a procedure](#)
- [Summary and notes](#)

13.1 The multiple testing problem

Seven groups give 21 pairs, and a two-sided t test at level 0.05 for each pair is a correct test. But an analyst who runs all 21 and reports the significant ones is not using a procedure with error rate 0.05: even if all seven means are equal, about one pair is expected to be significant, and the reader cannot tell it from a real difference. This section defines the error rates of a family of inferences, measures by simulation how large they become without adjustment, and separates strong control from a weak form that protects only against the situation in which nothing is going on.

13.1.1 A family of hypotheses

Let $H_1, \dots, H_m$ be null hypotheses about the parameters of a model, tested on the same data. The model's true parameter makes some of them true and the others false. Write $I_0 \subseteq \{1, \dots, m\}$ for the set of true hypotheses and $m_0 = |I_0|$ for their number. A **multiple testing procedure** maps the data to a set $\mathcal{R} \subseteq \{1, \dots, m\}$ of rejected hypotheses. The outcome can be summarized by three counts:

	not rejected	rejected	total
true hypotheses	$m_0 - V$	V	m_0
false hypotheses	$m - m_0 - S$	S	$m - m_0$
total	$m - R$	R	m

Here $R = |\mathcal{R}|$ is the number of rejections, $V = |\mathcal{R} \cap I_0|$ the number of **false rejections** (Type I errors), and $S = R - V$ the number of correct ones. The counts R , V and S are random, and m_0 is unknown. The analyst sees R but not how it splits into V and S .

With one true hypothesis, a level- α test makes $\Pr(V = 1) \leq \alpha$. For a family there are several reasonable ways to summarize the distribution of V , and they lead to different procedures.

Definition 13.1.1 — Error rates of a family

For a multiple testing procedure and a given value of the parameters:

- (a) the **familywise error rate** is $\text{FWER} = \Pr(V \geq 1)$, the probability of at least one false rejection;
- (b) the **per-family error rate** is $\text{PFER} = \mathbb{E}(V)$, the expected number of false rejections, and the **per-comparison error rate** is $\text{PCER} = \mathbb{E}(V)/m$;
- (c) the **false discovery proportion** is $Q = V/\max(R, 1)$, which is V/R when something is rejected and 0 otherwise, and the **false discovery rate** is its expectation, $\text{FDR} = \mathbb{E}(Q)$.

A procedure **controls** one of these rates at level α if the rate is at most α for every value of the parameters.

The familywise error rate treats one false claim as a failure of the whole family; it suits conclusions that will each be acted on. The false discovery rate accepts some errors in a long list of

discoveries and bounds their expected share. The unadjusted procedure, testing each hypothesis at level α , controls only the per-comparison rate.

Proposition 13.1.1 — How the error rates compare

For every procedure and every value of the parameters,

$$\text{PCER} \leq \text{FWER} \leq \text{PFER}, \quad \text{FDR} \leq \text{FWER}.$$

If all the hypotheses are true ($m_0 = m$), then $\text{FDR} = \text{FWER}$. Consequently a procedure that controls the per-family error rate at α controls the familywise rate at α , and a procedure that controls the familywise rate controls the false discovery rate.

Proof. All four rates are expectations of functions of V and R , so it is enough to compare the functions pointwise. Since $0 \leq V \leq m$, we have $V/m \leq 1_{\{V \geq 1\}} \leq V$, which gives the first chain after taking expectations. Since $V \leq R$, the ratio $V/\max(R, 1)$ is at most 1, and it is 0 when $V = 0$; so $V/\max(R, 1) \leq 1_{\{V \geq 1\}}$, which gives $\text{FDR} \leq \text{FWER}$. If every hypothesis is true, every rejection is false, so $V = R$ and $V/\max(R, 1) = 1_{\{V \geq 1\}}$ exactly. ■

In particular, a procedure that controls the false discovery rate is a valid level- α test of the hypothesis that *all* the H_j are true, but not in general a procedure with familywise control (Section 13.5).

13.1.2 How often something is significant by chance

If the m tests are independent and each has exact level α , then when all hypotheses are true,

$$\text{FWER} = 1 - \Pr(\text{no rejection}) = 1 - (1 - \alpha)^m. \quad (13.1.1)$$

With $\alpha = 0.05$ this is 0.401 for ten tests and 0.994 for a hundred. In a linear model the tests are rarely independent. They share the estimate s of σ , and comparisons involving a common group share that group's mean. The next example measures the effect of that dependence for the most common family of all.

Example. (All pairs of equal means)

Take g groups of 5 observations each, with all group means equal, and compare every pair with an unadjusted two-sided t test at level 0.05, using the pooled variance estimate on $g(5 - 1)$ degrees of freedom. The listing simulates 20000 data sets for each g . For $g = 3, 5, 7, 10$ (so 3, 10, 21, 45 pairs), the proportion of data sets with at least one significant pair is

$$0.116, \quad 0.265, \quad 0.412, \quad 0.589.$$

Independent tests would give 0.143, 0.401, 0.659 and 0.901. The average number of significant pairs is 0.15, 0.50, 1.08 and 2.23, in each case 0.05 times the number of pairs up to simulation error. Figure 13.1.1 plots the familywise rate against the number of tests.

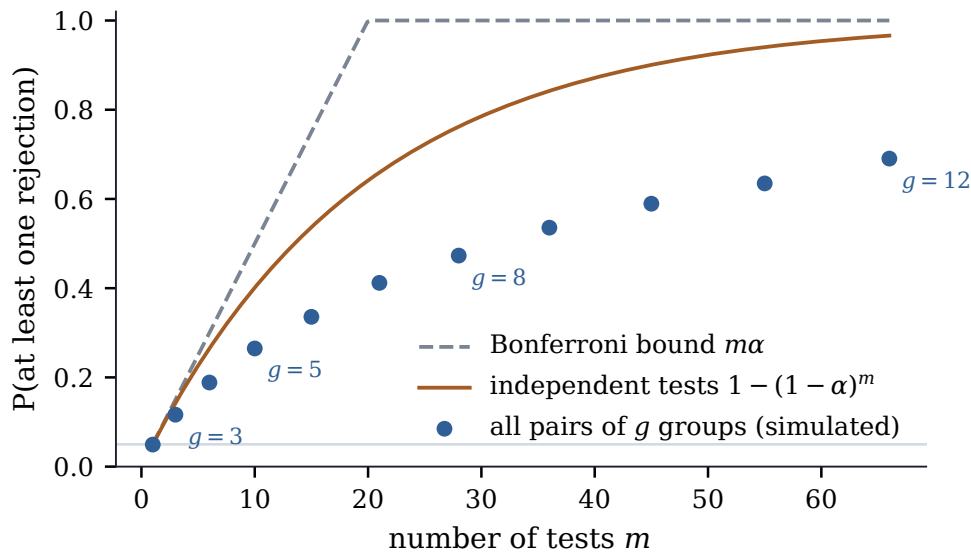


Figure 13.1.1: Probability of at least one false rejection among m unadjusted level-0.05 tests: all pairs of $g = 2, \dots, 12$ equal means (dots, simulated), independent tests (curve, [Eq. 13.1.1](#)) and the Bonferroni bound $m\alpha$ (dashed).

```
import itertools
import numpy as np
from scipy import stats

alpha, reps, m_rep = 0.05, 20_000, 5                                # 5 observations per group

def any_significant(g, m_rep, reps, alpha):
    """Share of data sets with at least one unadjusted pairwise rejection, and mean
    count."""
    rng = np.random.default_rng([20260919, g])                      # one fixed stream per g
    nu = g * (m_rep - 1)                                           # error degrees of freedom
    y = rng.normal(size=(reps, g, m_rep))                          # all group means equal
    means = y.mean(axis=2)
    s2 = ((y - means[:, :, None]) ** 2).sum(axis=(1, 2)) / nu
    se = np.sqrt(s2 * 2 / m_rep)                                    # standard error of a
    difference
    crit = stats.t.ppf(1 - alpha / 2, nu)
    count = np.zeros(reps)
    for k, l in itertools.combinations(range(g), 2):
        count += np.abs(means[:, k] - means[:, l]) / se > crit
    return np.mean(count > 0), np.mean(count)

for g in (3, 5, 10):
    pairs = g * (g - 1) // 2
    fwer, mean_count = any_significant(g, m_rep, reps, alpha)
    print(f"g = {g:2d}, {pairs:2d} pairs: P(some rejection) = {fwer:.3f}, "
          f"mean number = {mean_count:.2f} (alpha x pairs = {alpha * pairs:.2f}), "
          f"independent tests would give {1 - (1 - alpha) ** pairs:.3f}")
```

The expected number of false rejections is additive whatever the dependence, which is why the Bonferroni bound of [Section 13.4](#) needs no assumptions. The probability of at least one grows more slowly than for independent tests, because false rejections cluster when s is small or one mean is extreme. Still, with ten groups some pair is “significant” in well over half of all

experiments in which nothing differs.

13.1.3 Simultaneous intervals and the tests they induce

Let ψ_j , $j \in J$, be real parameters, where the index set J may be infinite. Intervals $[L_j, U_j]$ computed from the data are **simultaneous confidence intervals** at level $1 - \alpha$ if

$$\Pr(L_j \leq \psi_j \leq U_j \text{ for every } j \in J) \geq 1 - \alpha$$

for every value of the parameters. Scheffé's and Tukey's methods and the Bonferroni intervals all produce them, and they yield tests with the strongest guarantees.

Proposition 13.1.2 — Tests from simultaneous intervals

Let $[L_j, U_j]$, $j \in J$, be simultaneous confidence intervals at level $1 - \alpha$, and for given constants d_j reject $H_j : \psi_j = d_j$ exactly when $d_j \notin [L_j, U_j]$. Then:

- (a) the probability of rejecting at least one true H_j is at most α , for every value of the parameters;
- (b) if, whenever H_j is rejected, one also declares $\psi_j > d_j$ or $\psi_j < d_j$ according to the side of d_j on which the interval lies, the probability of at least one false rejection or wrong declaration of sign is at most α .

Proof. (a) If a true H_j is rejected, then $\psi_j = d_j \notin [L_j, U_j]$, so some interval fails to cover. That event has probability at most α . (b) A wrong declaration, say $\psi_j > d_j$ when in fact $\psi_j \leq d_j$, happens only if $L_j > d_j \geq \psi_j$; again an interval fails to cover. Both kinds of error are contained in the event that some interval misses, whose probability is at most α . ■

Part (b) matters because few parameters are exactly equal to their hypothesized values; the real question is which way each difference goes. Some good procedures, such as Holm's, give tests but no simple companion intervals.

13.1.4 Weak control is not enough

The error rates depend on which hypotheses are true. The strongest requirement bounds the rate in every configuration; a much weaker one only when every hypothesis is true.

Definition 13.1.2 — Strong and weak control

A procedure controls the familywise error rate **strongly** at level α if $\Pr(V \geq 1) \leq \alpha$ for every value of the parameters, and **weakly** if $\Pr(V \geq 1) \leq \alpha$ whenever all of $H_1, \dots, H_m$ are true.

Weak control is easy to obtain. Let $H_0 = \bigcap_j H_j$ be the **global null hypothesis** that every H_j holds.

Proposition 13.1.3 — A gate gives weak control

Let ϕ_0 be a level- α test of the global null H_0 . A procedure that rejects no H_j unless ϕ_0 rejects H_0 controls the familywise error rate weakly at level α .

Proof. When all H_j are true, $V \geq 1$ requires a rejection by ϕ_0 , which has probability at most α because H_0 is true. ■

The best-known gated procedure is Fisher's **protected least significant difference** (LSD): in a one-way layout, test equality of all g means with the F test at level α , and only if it rejects, compare each pair with an unadjusted t test at level α . Weak control is all it has.

Example. (The protected LSD without strong control)

With $g = 10$ groups of 5 observations and all means equal, the simulation below gives a familywise error rate of 0.049, as Proposition 13.1.3 promises. Now move one mean far from the other nine, which remain equal. The F test rejects in essentially every data set, the gate is always open, and the 36 pairwise comparisons among the nine equal means are made without adjustment. The familywise error rate rises to 0.539.

```
import itertools
import numpy as np
from scipy import stats

alpha, reps, m_rep = 0.05, 20_000, 5

def protected_lsd_fwer(mu, m_rep, reps, alpha, seed):
    """Share of data sets in which the protected LSD rejects some true equality."""
    rng = np.random.default_rng(seed)
    g = len(mu)
    nu = g * (m_rep - 1)
    y = np.asarray(mu)[None, :, None] + rng.normal(size=(reps, g, m_rep))
    means = y.mean(axis=2)
    s2 = ((y - means[:, :, None]) ** 2).sum(axis=(1, 2)) / nu
    between = m_rep * ((means - means.mean(axis=1, keepdims=True)) ** 2).sum(axis=1)
    gate = between / (g - 1) / s2 > stats.f.ppf(1 - alpha, g - 1, nu)  # the F test
    crit = stats.t.ppf(1 - alpha / 2, nu) * np.sqrt(2 * s2 / m_rep)
    false = np.zeros(reps, dtype=bool)
    for k, l in itertools.combinations(range(g), 2):
        if mu[k] == mu[l]:  # a true null hypothesis
            false |= gate & (np.abs(means[:, k] - means[:, l]) > crit)
    return false.mean()

all_equal = protected_lsd_fwer([0.0] * 10, m_rep, reps, alpha, seed=1)
one_apart = protected_lsd_fwer([0.0] * 9 + [10.0], m_rep, reps, alpha, seed=2)
print(f"10 groups, all means equal: FWER = {all_equal:.3f}")
print(f"10 groups, one mean far from rest: FWER = {one_apart:.3f}")
```

Weak control protects only against the one configuration in which there is nothing to find. Every procedure recommended in this chapter controls its error rate strongly.

13.1.5 Exercises

13.1.5.1 A. Check your understanding

Exercise A1.

Ten and twenty independent tests of true hypotheses are made, each at level 0.01. Find the familywise error rate in each case. What common level for twenty independent tests gives a familywise rate of exactly 0.05?

Solution. By [Eq. 13.1.1](#) the rates are $1 - 0.99^{10} = 0.0956$ and $1 - 0.99^{20} = 0.1821$. The level a with $1 - (1 - a)^{20} = 0.05$ is $a = 1 - 0.95^{1/20} = 0.00256$, slightly more than $0.05/20 = 0.00250$. This is the Šidák level of [Section 13.4](#).

Exercise A2.

Give a configuration of true and false hypotheses, and a procedure, for which the familywise error rate is close to one while the false discovery rate is close to zero.

13.1.5.2 B. Practice

Exercise B1.

Show that with $g = 3$ groups the protected LSD controls the familywise error rate strongly. *Hint:* consider separately the configurations with three, one and no true equalities among the three pairwise hypotheses.

Solution. The three hypotheses are $\mu_1 = \mu_2$, $\mu_1 = \mu_3$, $\mu_2 = \mu_3$. If two of them are true, the third is true as well, so the possible numbers of true hypotheses are 3, 1 and 0. With three, all means are equal and weak control ([Proposition 13.1.3](#)) applies. With none, no false rejection is possible. With exactly one, say $\mu_1 = \mu_2 \neq \mu_3$, a false rejection needs the unadjusted t test of $\mu_1 = \mu_2$ to reject, an event of probability α whether or not the gate opens. So the familywise rate is at most α in every configuration. With four or more groups, a configuration can hold two disjoint true equalities, and the argument fails.

Exercise B2.

Let m_0 true hypotheses have independent uniform p -values, and let m_1 false hypotheses have p -values equal to 0. The unadjusted procedure rejects when $p_j \leq \alpha$. Find its familywise error rate, show that its false discovery rate is at most $m_0\alpha/m_1$, and conclude that the false discovery rate tends to 0 as $m_1 \rightarrow \infty$ with m_0 fixed.

13.1.5.3 C. Going deeper

Exercise C1.

In a one-way layout with $g \geq 3$ groups of m observations and known σ , run the protected LSD with z tests: the gate is the level- α test that rejects when $\sum_k m(\bar{y}_k - \bar{y})^2/\sigma^2$ exceeds the upper α point of $\chi^2(g-1)$, and a pair is declared different when $|\bar{y}_k - \bar{y}_l| > z_{\alpha/2} \sigma \sqrt{2/m}$, where $z_{\alpha/2}$ is the upper $\alpha/2$ point of $N(0, 1)$. Let $\mu_1 = \dots = \mu_{g-1} = 0$ and $\mu_g = \Delta$. Show that as $\Delta \rightarrow \infty$ the familywise error rate tends to $\Pr(W_{g-1} > \sqrt{2} z_{\alpha/2})$, where W_{g-1} is the range of $g-1$ independent $N(0, 1)$ variables. Show that this limit is at least $1 - (1 - \alpha)^{\lfloor (g-1)/2 \rfloor}$ and that it tends to 1 as $g \rightarrow \infty$.

Solution. With $Z_k = \sqrt{m} \bar{y}_k / \sigma$, $k < g$, independent $N(0, 1)$ and free of Δ , some pair $k, l < g$ is declared different iff $W_{g-1} = \max_{k < g} Z_k - \min_{k < g} Z_k > \sqrt{2} z_{\alpha/2}$, so the rate is $\Pr(G \cap \{W_{g-1} > \sqrt{2} z_{\alpha/2}\})$, with G the event that the gate opens. The gate statistic is at least $m(\bar{y}_g - \bar{y})^2/\sigma^2$, where $\bar{y}_g - \bar{y}$ is $\frac{g-1}{g}\Delta$ plus a variable free of Δ ; so $\Pr(G) \rightarrow 1$, giving the limit. The disjoint pairs $(1, 2), (3, 4), \dots$ give $\lfloor (g-1)/2 \rfloor$ independent statistics $|Z_{2i-1} - Z_{2i}|/\sqrt{2}$, each exceeding $z_{\alpha/2}$ with probability α , and W_{g-1} is at least $\sqrt{2}$ times each of them; hence the lower bound, which tends to 1.

13.2 Scheffé's projection method

Scheffé's method answers the hardest version of the problem: the family is every estimable function in a subspace, and the analyst may choose which ones to look at after seeing the data. The intervals have simultaneous coverage exactly $1 - \alpha$, because the largest standardized error in the family is the length of a projection, and Scheffé's method *is* the F test, read one direction at a time.

Throughout this section the model is the normal linear model [Eq. 7.3.1](#), $\mathbf{Y} \sim N_n(\mathbf{X}\boldsymbol{\beta}, \sigma^2\mathbf{I})$, with $r = \text{rank}(\mathbf{X}) < n$. Write $\boldsymbol{\mu} = \mathbf{X}\boldsymbol{\beta}$, $\nu = n - r$ for the error degrees of freedom, $s^2 = \text{SSE}/\nu$, $\mathbf{G} = (\mathbf{X}^\top \mathbf{X})^-$ for any generalized inverse, and $F_\alpha(q, \nu)$ for the upper α point of $F(q, \nu)$. For an estimable $\boldsymbol{\lambda}^\top \boldsymbol{\beta}$ the standard error is $\text{se}(\boldsymbol{\lambda}^\top \hat{\boldsymbol{\beta}}) = s\sqrt{\boldsymbol{\lambda}^\top \mathbf{G} \boldsymbol{\lambda}}$ ([Corollary 7.5.2](#)).

13.2.1 Linear functions of the mean as vectors

An estimable function is a linear function of the mean vector, and a linear function of $\boldsymbol{\mu} \in \mathcal{C}(\mathbf{X})$ can be represented by a vector of $\mathcal{C}(\mathbf{X})$. If $\boldsymbol{\lambda}^\top \boldsymbol{\beta}$ is estimable, then $\boldsymbol{\lambda} = \mathbf{X}^\top \boldsymbol{\rho}$ for some vector $\boldsymbol{\rho} \in \mathbb{R}^n$ ([Theorem 8.3.1](#)), and with $\mathbf{v} = \mathbf{M}\boldsymbol{\rho}$,

$$\boldsymbol{\lambda}^\top \boldsymbol{\beta} = \boldsymbol{\rho}^\top \mathbf{X}\boldsymbol{\beta} = \mathbf{v}^\top \boldsymbol{\mu}, \quad \boldsymbol{\lambda}^\top \hat{\boldsymbol{\beta}} = \boldsymbol{\rho}^\top \mathbf{M}\mathbf{Y} = \mathbf{v}^\top \mathbf{Y}, \quad \boldsymbol{\lambda}^\top \mathbf{G} \boldsymbol{\lambda} = \boldsymbol{\rho}^\top \mathbf{X}\mathbf{G}\mathbf{X}^\top \boldsymbol{\rho} = \boldsymbol{\rho}^\top \mathbf{M}\boldsymbol{\rho} = \|\mathbf{v}\|^2. \quad (13.2.1)$$

The second identity is [Theorem 6.4.4](#), and the third uses $\mathbf{M} = \mathbf{X}\mathbf{G}\mathbf{X}^\top$ ([Theorem 6.3.7](#)). The vector $\mathbf{v}$ does not depend on the choice of $\boldsymbol{\rho}$, since two choices differ by some $\boldsymbol{\delta}$ with $\mathbf{X}^\top \boldsymbol{\delta} = \mathbf{0}$, hence $\mathbf{M}\boldsymbol{\delta} = \mathbf{0}$. So the estimate is the inner product of $\mathbf{v}$ with the data, the estimand its inner product with the mean, and the standard error is $s\|\mathbf{v}\|$.

A *family* of estimable functions closed under linear combination is then a subspace $\mathcal{V} \subseteq \mathcal{C}(\mathbf{X})$.

It is usually described in one of two ways. If the family is $\{\boldsymbol{\lambda}^\top \boldsymbol{\beta} : \boldsymbol{\lambda} \in \mathcal{C}(\boldsymbol{\Lambda})\}$ for a $p \times q$ matrix $\boldsymbol{\Lambda} = \mathbf{X}^\top \mathbf{T}$, then $\mathbf{v} = \mathbf{M}\mathbf{T}\mathbf{a}$ and $\mathcal{V} = \mathcal{C}(\mathbf{M}\mathbf{T})$, the test space of $\boldsymbol{\Lambda}^\top \boldsymbol{\beta} = \mathbf{0}$ (Theorem 6.5.2); if $\boldsymbol{\Lambda}$ has rank q , so does $\mathbf{M}\mathbf{T}$, because $\mathbf{X}^\top \mathbf{M}\mathbf{T} = \boldsymbol{\Lambda}$. If instead the family is the set of functions about which a reduced model $\mathcal{C}(\mathbf{X}_0) \subseteq \mathcal{C}(\mathbf{X})$ makes a claim, those vanishing on $\mathcal{C}(\mathbf{X}_0)$, then $\mathcal{V} = \mathcal{C}(\mathbf{M} - \mathbf{M}_0)$, of dimension $r - r_0$ (Theorem 6.5.1).

13.2.2 The theorem

Lemma 13.2.1 — The largest standardized component

Let $\mathcal{V}$ be a subspace of $\mathbb{R}^n$ with orthogonal projection $\mathbf{P}_{\mathcal{V}}$, and let $\mathbf{u} \in \mathbb{R}^n$. Then

$$\max_{\mathbf{v} \in \mathcal{V}, \mathbf{v} \neq \mathbf{0}} \frac{(\mathbf{v}^\top \mathbf{u})^2}{\|\mathbf{v}\|^2} = \|\mathbf{P}_{\mathcal{V}} \mathbf{u}\|^2,$$

and if $\mathbf{P}_{\mathcal{V}} \mathbf{u} \neq \mathbf{0}$ the maximum is attained exactly at the nonzero multiples of $\mathbf{P}_{\mathcal{V}} \mathbf{u}$.

Proof. For $\mathbf{v} \in \mathcal{V}$, $\mathbf{v} = \mathbf{P}_{\mathcal{V}} \mathbf{v}$, and since $\mathbf{P}_{\mathcal{V}}$ is symmetric, $\mathbf{v}^\top \mathbf{u} = \mathbf{v}^\top \mathbf{P}_{\mathcal{V}} \mathbf{u}$. By Cauchy–Schwarz (Proposition 1.1.1), $(\mathbf{v}^\top \mathbf{P}_{\mathcal{V}} \mathbf{u})^2 \leq \|\mathbf{v}\|^2 \|\mathbf{P}_{\mathcal{V}} \mathbf{u}\|^2$, with equality iff $\mathbf{v}$ is a multiple of $\mathbf{P}_{\mathcal{V}} \mathbf{u}$. That multiple lies in $\mathcal{V}$, so the bound is attained. If $\mathbf{P}_{\mathcal{V}} \mathbf{u} = \mathbf{0}$ both sides are zero. ■

The lemma is Exercise (C1) of Chapter 9, where it described the sum of squares of a row of an analysis of variance table as the largest one-degree-of-freedom sum of squares inside it. Applied to $\mathbf{u} = \mathbf{Y} - \boldsymbol{\mu}$, it describes the worst standardized error in a family.

Theorem 13.2.2 — Scheffé's simultaneous intervals

Assume Eq. 7.3.1 with $r < n$, and let $0 < \alpha < 1$.

(a) Let $\mathcal{V} \subseteq \mathcal{C}(\mathbf{X})$ be a subspace of dimension $q \geq 1$. Then

$$\max_{\mathbf{v} \in \mathcal{V}, \mathbf{v} \neq \mathbf{0}} \frac{(\mathbf{v}^\top \mathbf{Y} - \mathbf{v}^\top \boldsymbol{\mu})^2}{q s^2 \|\mathbf{v}\|^2} = \frac{\|\mathbf{P}_{\mathcal{V}}(\mathbf{Y} - \boldsymbol{\mu})\|^2 / q}{s^2} \sim F(q, \nu),$$

and consequently

$$\Pr\left(|\mathbf{v}^\top (\mathbf{Y} - \boldsymbol{\mu})| \leq \sqrt{q F_\alpha(q, \nu)} s \|\mathbf{v}\| \quad \forall \mathbf{v} \in \mathcal{V}\right) = 1 - \alpha.$$

(b) Let $\boldsymbol{\Lambda}$ be $p \times q$ of rank q with $\mathcal{C}(\boldsymbol{\Lambda}) \subseteq \mathcal{C}(\mathbf{X}^\top)$. With probability exactly $1 - \alpha$, for every $\boldsymbol{\lambda} \in \mathcal{C}(\boldsymbol{\Lambda})$ simultaneously,

$$\boldsymbol{\lambda}^\top \boldsymbol{\beta} \in \left[\boldsymbol{\lambda}^\top \hat{\boldsymbol{\beta}} - \sqrt{q F_\alpha(q, \nu)} \text{se}(\boldsymbol{\lambda}^\top \hat{\boldsymbol{\beta}}), \quad \boldsymbol{\lambda}^\top \hat{\boldsymbol{\beta}} + \sqrt{q F_\alpha(q, \nu)} \text{se}(\boldsymbol{\lambda}^\top \hat{\boldsymbol{\beta}}) \right]. \quad (13.2.2)$$

Moreover, with $\boldsymbol{\theta} = \boldsymbol{\Lambda}^\top \boldsymbol{\beta}$, $\hat{\boldsymbol{\theta}} = \boldsymbol{\Lambda}^\top \hat{\boldsymbol{\beta}}$ and $\mathbf{W} = \boldsymbol{\Lambda}^\top \mathbf{G} \boldsymbol{\Lambda}$,

$$\max_{\lambda \in \mathcal{C}(\Lambda), \lambda \neq \mathbf{0}} \frac{(\lambda^\top \hat{\beta} - \lambda^\top \beta)^2}{\text{se}(\lambda^\top \hat{\beta})^2} = \frac{(\hat{\theta} - \theta)^\top \mathbf{W}^{-1}(\hat{\theta} - \theta)}{s^2}, \quad (13.2.3)$$

attained at $\lambda = \Lambda \mathbf{W}^{-1}(\hat{\theta} - \theta)$.

Proof. (a) The first equality is Lemma 13.2.1 with $\mathbf{u} = \mathbf{Y} - \boldsymbol{\mu}$. The map $\mathbf{v} \mapsto \mathbf{X}^\top \mathbf{v}$ is one to one on $\mathcal{C}(\mathbf{X})$ (Step 1 of the proof of Theorem 12.4.1), so $\mathcal{L} = \{\mathbf{X}^\top \mathbf{v} : \mathbf{v} \in \mathcal{V}\}$ is a subspace of $\mathcal{C}(\mathbf{X}^\top)$ of dimension q . For $\mathbf{x} = \mathbf{X}^\top \mathbf{v}$, Eq. 13.2.1 with $\boldsymbol{\rho} = \mathbf{v}$ gives $\mathbf{x}^\top \hat{\beta} - \mathbf{x}^\top \beta = \mathbf{v}^\top (\mathbf{Y} - \boldsymbol{\mu})$ and $\mathbf{x}^\top \mathbf{G} \mathbf{x} = \|\mathbf{v}\|^2$. So the event in the display is the event Eq. 12.4.1 with $k = q$, and it has probability exactly $1 - \alpha$ by Theorem 12.4.1. That event is also the event that the maximum is at most $F_\alpha(q, \nu)$. Since this holds for every $\alpha \in (0, 1)$, the maximum has the distribution function of $F(q, \nu)$.

(b) Write $\Lambda = \mathbf{X}^\top \mathbf{T}$ and $\mathcal{V} = \mathcal{C}(\mathbf{MT})$, of dimension q . Every $\lambda \in \mathcal{C}(\Lambda)$ is $\Lambda \mathbf{a} = \mathbf{X}^\top (\mathbf{T} \mathbf{a})$, and by Eq. 13.2.1 $\lambda^\top \hat{\beta} - \lambda^\top \beta = \mathbf{v}^\top \mathbf{Y} - \mathbf{v}^\top \boldsymbol{\mu}$ and $\text{se}(\lambda^\top \hat{\beta}) = s \|\mathbf{v}\|$ with $\mathbf{v} = \mathbf{MT} \mathbf{a} \in \mathcal{V}$. As $\mathbf{a}$ ranges over $\mathbb{R}^q$, $\mathbf{v}$ ranges over all of $\mathcal{V}$. So the event in (b) is the event in (a). Finally, Eq. 13.2.3 is Proposition 11.6.2 for the hypothesis $\Lambda^\top \beta = \theta$, which is testable because θ is the true value: its statistic Eq. 11.3.1 is $(\hat{\theta} - \theta)^\top \mathbf{W}^{-1}(\hat{\theta} - \theta)/(q s^2)$, and each $T(\mathbf{a})^2$ is the ratio on the left of Eq. 13.2.3 at $\lambda = \Lambda \mathbf{a}$, so the maximum is attained at $\mathbf{a} \propto \mathbf{W}^{-1}(\hat{\theta} - \theta)$. ■

The coverage is exact, not a bound: some function in the family, depending on the data, always lies on the edge of the event. Only the subspace matters, so dependent columns of Λ may be dropped first. For $q = 1$ the multiplier is $t_{\nu, \alpha/2}$, since $F_\alpha(1, \nu) = t_{\nu, \alpha/2}^2$, and the interval is that of Theorem 12.1.1.

Part (b) is the observation-space form of a fact met twice already. Section 11.6 proved Eq. 13.2.3, with $\mathbf{d}$ in place of θ , as the statement that a group F statistic is the largest single t statistic in the group, and Section 12.2 showed that the intervals Eq. 13.2.2 are the *shadows* of the confidence ellipsoid for $\theta = \Lambda^\top \beta$ (Theorem 12.2.1). The ellipsoid covers θ iff every shadow covers the corresponding $\mathbf{a}^\top \theta$, which is why an infinite family costs no more than one ellipsoid. The coverage statement itself is the Working–Hotelling band of Theorem 12.4.1 for the subspace $\mathcal{L} = \mathcal{C}(\Lambda)$: a band for a regression surface and a family of intervals for estimable functions are one object, written in the coordinates of $\mathcal{C}(\mathbf{X}^\top)$ or of $\mathcal{C}(\mathbf{X})$. Figure 13.2.1 extends Figure 12.2.1, which drew the same ellipse for two slopes of the crime regression, by adding the Bonferroni rectangle and the tangent lines that bound the Scheffé interval for a combination $a_1 \beta_1 + a_2 \beta_2$.

Example. (Three slopes of a crime regression)

Return to the regression of the murder rate on poverty, single parenthood and urbanization for the 50 states of Example (Poverty and murder rates), with $\nu = 46$ error degrees of freedom. The poverty coefficient is 0.2538 with standard error 0.0934, so $t = 2.718$. Suppose the family of interest is every linear combination of the three slopes, $q = 3$. The 95% intervals for the poverty coefficient are:

method	multiplier	interval
unadjusted t	2.013	[0.066, 0.442]
Bonferroni, three slopes	2.485	[0.022, 0.486]

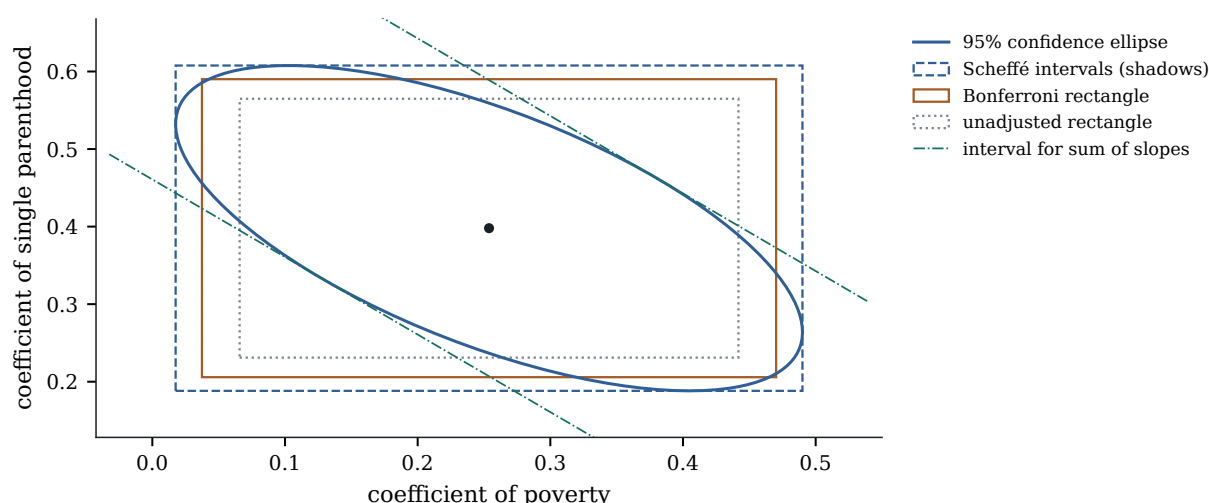


Figure 13.2.1: The coefficients of poverty and single parenthood in the crime regression. The Scheffé intervals (dashed box, multiplier 2.530) are the shadows of the 95% confidence ellipse; the Bonferroni rectangle (multiplier 2.317) covers only the two coefficients. The tangent lines bound the Scheffé interval for the sum of the slopes. The ellipse is that of Figure 12.2.1.

Scheffé, all combinations 2.902 $[-0.017, 0.525]$

The poverty coefficient is significant on its own and among three planned comparisons, but not once every combination of the slopes is admitted. The F statistic for the hypothesis that all three slopes are zero is 27.49, and the largest $|t|$ over all combinations is $9.081 = \sqrt{3 \times 27.49}$, in agreement with Eq. 13.2.3.

```
import numpy as np
import statsmodels.api as sm
from scipy import stats
data = sm.datasets.statecrime.load_pandas().data
data = data.drop(index="District of Columbia")

y = data["murder"].to_numpy()
X = np.column_stack([np.ones(len(y)), data[["poverty", "single", "urban"]]])
n, p = X.shape
nu = n - p                                # error degrees of freedom
XtX_inv = np.linalg.inv(X.T @ X)
beta_hat = XtX_inv @ X.T @ y
s2 = np.sum((y - X @ beta_hat) ** 2) / nu
se = np.sqrt(s2 * np.diag(XtX_inv))
```

```
alpha, slopes = 0.05, [1, 2, 3]           # poverty, single, urban
k = len(slopes)
mult = {
    "unadjusted": stats.t.ppf(1 - alpha / 2, nu),
    "Bonferroni": stats.t.ppf(1 - alpha / (2 * k), nu),
    "Scheffe": np.sqrt(k * stats.f.ppf(1 - alpha, k, nu)),    # q = 3 slopes
}
for name, c in mult.items():
    lo, hi = beta_hat[slopes] - c * se[slopes], beta_hat[slopes] + c * se[slopes]
    print(f"{name:10s} c = {c:.3f} poverty [{lo[0]:.3f}, {hi[0]:.3f}]")
```

```

L = np.eye(p)[: , slopes]                                # Lambda: the three slopes
W = L.T @ XtX_inv @ L
d = L.T @ beta_hat
F = d @ np.linalg.solve(W, d) / (k * s2)                # F test that all slopes are 0
a_star = np.linalg.solve(W, d)                           # the most significant
combination
t_star = (a_star @ d) / np.sqrt(s2 * a_star @ W @ a_star)
print(f"F = {F:.2f}; largest |t| over all combinations = {t_star:.3f} = sqrt(3F)")

```

13.2.3 Scheffé's method is the F test

Let $\mathbf{d} \in \mathbb{R}^q$ and consider the hypothesis $\mathbf{\Lambda}^\top \boldsymbol{\beta} = \mathbf{d}$, with $\mathbf{\Lambda}$ as in [Theorem 13.2.2\(b\)](#). Because $\mathbf{\Lambda}^\top$ has rank q , the hypothesis is consistent for every $\mathbf{d}$. By [Theorem 11.3.2](#) its F statistic is

$$F = \frac{(\hat{\boldsymbol{\theta}} - \mathbf{d})^\top \mathbf{W}^{-1}(\hat{\boldsymbol{\theta}} - \mathbf{d})}{q s^2}.$$

The hypothesis implies a single-degree-of-freedom statement for every $\boldsymbol{\lambda} = \mathbf{\Lambda} \mathbf{a}$ in the family, namely $\boldsymbol{\lambda}^\top \boldsymbol{\beta} = \mathbf{a}^\top \mathbf{d}$. Call such a statement **Scheffé-rejected** if its hypothesized value lies outside the Scheffé interval [Eq. 13.2.2](#).

Corollary 13.2.3 — The F test and Scheffé's intervals

In the setting of [Theorem 13.2.2\(b\)](#), for every $\mathbf{d} \in \mathbb{R}^q$:

- (a) $\max_{\mathbf{a} \neq \mathbf{0}} \frac{(\mathbf{a}^\top \hat{\boldsymbol{\theta}} - \mathbf{a}^\top \mathbf{d})^2}{s^2 \mathbf{a}^\top \mathbf{W} \mathbf{a}} = qF$, attained at $\mathbf{a} = \mathbf{W}^{-1}(\hat{\boldsymbol{\theta}} - \mathbf{d})$;
- (b) the level- α F test rejects $\mathbf{\Lambda}^\top \boldsymbol{\beta} = \mathbf{d}$ iff at least one statement $\boldsymbol{\lambda}^\top \boldsymbol{\beta} = \mathbf{a}^\top \mathbf{d}$, $\boldsymbol{\lambda} = \mathbf{\Lambda} \mathbf{a}$, is Scheffé-rejected;
- (c) if $\mathbf{\Lambda}^\top \boldsymbol{\beta} = \mathbf{d}$ is true, the probability that any of these statements is Scheffé-rejected is exactly α . For any $\boldsymbol{\beta}$, the probability of Scheffé-rejecting at least one *true* statement $\boldsymbol{\lambda}^\top \boldsymbol{\beta} = c$, $\boldsymbol{\lambda} \in \mathcal{C}(\mathbf{\Lambda})$, is at most α .

Proof. (a) For $\boldsymbol{\lambda} = \mathbf{\Lambda} \mathbf{a}$, the ratio is $(\mathbf{a}^\top \mathbf{w})^2 / (s^2 \mathbf{a}^\top \mathbf{W} \mathbf{a})$ with $\mathbf{w} = \hat{\boldsymbol{\theta}} - \mathbf{d}$, and its maximum is $\mathbf{w}^\top \mathbf{W}^{-1} \mathbf{w} / s^2 = qF$, attained at $\mathbf{a} = \mathbf{W}^{-1} \mathbf{w}$, by [Corollary 1.7.8\(c\)](#) with $\mathbf{W}$ positive definite ([Theorem 9.2.4\(a\)](#)); this is the maximum identity of [Section 11.6](#). (b) The statement for $\mathbf{a}$ is Scheffé-rejected iff its squared t statistic exceeds $qF_\alpha(q, \nu)$, so some statement is rejected iff the maximum in (a) exceeds $qF_\alpha(q, \nu)$, that is, iff $F > F_\alpha(q, \nu)$. (c) The first claim follows from (b) and the exact level of the F test ([Theorem 11.3.2](#)). The second is [Proposition 13.1.2\(a\)](#) applied to the simultaneous intervals of [Theorem 13.2.2](#). ■

When the F test rejects, *some* function in the family is Scheffé-significant, though perhaps not one of interest; conversely, no function is Scheffé-significant unless the F test rejects, so no separate protecting F test is needed. For a reduced model the same holds with $\mathcal{V} = \mathcal{C}(\mathbf{M} - \mathbf{M}_0)$: by [Lemma 13.2.1](#) the F statistic of [Theorem 11.1.1](#) is the maximum over $\mathbf{v} \in \mathcal{V}$ of $(\mathbf{v}^\top \mathbf{Y})^2 / ((r - r_0)s^2 \|\mathbf{v}\|^2)$ ([Exercise \(B1\)](#)). Taking $\mathcal{C}(\mathbf{\Lambda}) = \mathcal{C}(\mathbf{X}^\top)$, so that $q = r$ and the family contains

the mean response $\mathbf{x}^\top \boldsymbol{\beta}$ at every $\mathbf{x} \in \mathcal{C}(\mathbf{X}^\top)$, gives the band $\mathbf{x}^\top \hat{\boldsymbol{\beta}} \pm \sqrt{rF_\alpha(r, \nu)} s \sqrt{\mathbf{x}^\top \mathbf{G} \mathbf{x}}$, the Working–Hotelling band of [Theorem 12.4.1](#).

13.2.4 Smaller families give shorter intervals

The Scheffé multiplier depends on the family only through its dimension q , and $qF_\alpha(q, \nu)$ is strictly increasing in q for fixed ν and α , as a χ^2 coupling shows (an exercise of [Section 11.6](#)). So one should use the smallest subspace containing every function of interest, whose dimension can be smaller than the length of a list spanning it ([Exercise \(A2\)](#)). For a few planned functions, Bonferroni's method ([Section 13.4](#)) is usually shorter still, as in [Example \(Three slopes of a crime regression\)](#); Scheffé's multiplier is earned when functions are picked after the estimates are seen.

13.2.5 All contrasts in a one-way layout

In the one-way model $E(y_{kj}) = \mu + \alpha_k$ with g levels, $n_k \geq 1$ observations at level k and $n = \sum_k n_k$, the estimable functions of the effects alone are the contrasts $\psi = \sum_k c_k \mu_k$ with $\sum_k c_k = 0$ ([Theorem 8.7.1](#)). They form a space of dimension $g - 1$, the test space of the hypothesis of equal means. The estimate is $\hat{\psi} = \sum_k c_k \bar{y}_k$ with standard error $s(\sum_k c_k^2/n_k)^{1/2}$, and $\nu = n - g$. Scheffé's intervals are

$$\sum_k c_k \bar{y}_k \pm \sqrt{(g-1)F_\alpha(g-1, n-g)} s \left(\sum_k \frac{c_k^2}{n_k} \right)^{1/2} \quad \text{for every contrast } \mathbf{c}, \quad (13.2.4)$$

with simultaneous coverage exactly $1 - \alpha$, balanced or not. The most significant contrast has $c_k = n_k(\bar{y}_k - \bar{y})$ ([Exercise \(B1\)](#)), and its squared t statistic is $(g-1)F$, where F is the one-way F statistic.

Example. (Education and the perceived position of a candidate)

The 1996 American National Election Study asked respondents to place Bill Clinton on a seven-point scale from extremely liberal (1) to extremely conservative (7). Group the $n = 944$ respondents by education, coded from 1 (least) to 7 (most). The group sizes are 13, 52, 248, 187, 90, 227, 127, and the mean placements are

$$4.08, 3.58, 3.21, 2.95, 2.79, 2.66, 2.65.$$

More educated respondents place Clinton further to the left. The pooled standard deviation is $s = 1.354$ on $\nu = 937$ degrees of freedom, and the one-way F statistic is 7.88, with p -value 2.7×10^{-8} . The between-groups sum of squares is 86.72.

Having seen the means, one might ask how the two least educated groups compare with the two most educated, the contrast $\psi = \frac{1}{2}(\mu_1 + \mu_2) - \frac{1}{2}(\mu_6 + \mu_7)$. Its estimate is 1.176 with standard error 0.223, so $t = 5.27$. Because the contrast was suggested by the data, the unadjusted t interval is not legitimate. The Scheffé interval [Eq. 13.2.4](#), with multiplier $\sqrt{6F_{0.05}(6, 937)} = 3.557$, is $[0.383, 1.969]$, and it is legitimate however the contrast was found. The most significant contrast of all has $t = 6.88 = \sqrt{6 \times 7.88}$, and its sum of squares is the whole between-groups sum of squares.

```

import numpy as np
import statsmodels.api as sm
from scipy import stats
anes = sm.datasets.anes96.load_pandas().data

y = anes["ClinLR"].to_numpy()
level = anes["educ"].to_numpy().astype(int) - 1      # 0, ..., 6
g = 7
n_k = np.bincount(level).astype(float)
means = np.bincount(level, weights=y) / n_k
n = len(y)
nu = n - g                                           # error degrees of freedom
s2 = np.sum((y - means[level]) ** 2) / nu           # pooled variance estimate
s = np.sqrt(s2)
between = np.sum(n_k * (means - y.mean()) ** 2)
F = between / (g - 1) / s2
print("group sizes", n_k.astype(int))
print("means", means.round(3))
print(f"s = {s:.4f} on {nu} df; F = {F:.3f}, p = {stats.f.sf(F, g - 1, nu):.2g}")

alpha = 0.05
c_sch = np.sqrt((g - 1) * stats.f.ppf(1 - alpha, g - 1, nu)) # Scheffe multiplier,
q = g - 1

def contrast(c):
    """Estimate, standard error and sum of squares of the contrast sum_k c_k mu_k.
    """
    est = c @ means
    v = np.sum(c ** 2 / n_k)
    return est, s * np.sqrt(v), est ** 2 / v

c_max = n_k * (means - y.mean())                    # the most significant
contrast                                         contrast
c_lohi = np.array([0.5, 0.5, 0, 0, 0, -0.5, -0.5]) # levels 1-2 against levels
6-7
for name, c in (("maximal", c_max), ("1-2 vs 6-7", c_lohi)):
    est, se, ss = contrast(c)
    print(f"{name:11s} estimate {est:8.3f} t = {est / se:6.2f} SS = {ss:8.2f}"
          f" Scheffe interval [{est - c_sch * se:.3f}, {est + c_sch * se:.3f}]")
print(f"between-groups SS = {between:.2f}; Scheffe multiplier = {c_sch:.3f}")

```

13.2.6 Exercises

13.2.6.1 A. Check your understanding

Exercise A1.

Show that for $q = 1$ Scheffé's interval Eq. 13.2.2 is the t interval of Theorem 12.1.1, and explain why no multiplicity adjustment appears.

Exercise A2.

In [Example \(Education and the perceived position of a candidate\)](#), an analyst decides before seeing the data to examine the three contrasts $\frac{1}{2}(\mu_1 + \mu_2) - \frac{1}{2}(\mu_6 + \mu_7)$, $\mu_1 - \mu_7$ and $\mu_2 - \mu_6$. Show that they span a space of dimension 2, and state the Scheffé multiplier for this family.

13.2.6.2 B. Practice**Exercise B1.**

Let $\mathcal{C}(\mathbf{X}_0) \subseteq \mathcal{C}(\mathbf{X})$ with ranks $r_0 < r$, and $\mathcal{V} = \mathcal{C}(\mathbf{M} - \mathbf{M}_0)$. Show that a vector $\mathbf{v} \in \mathcal{C}(\mathbf{X})$ lies in $\mathcal{V}$ iff $\mathbf{v}^\top \boldsymbol{\mu} = 0$ for every $\boldsymbol{\mu} \in \mathcal{C}(\mathbf{X}_0)$. Show that the maximum over $\mathbf{v} \in \mathcal{V}$ of the squared t statistic for $\mathbf{v}^\top \boldsymbol{\mu} = 0$ equals $(r - r_0)F$, where F is the statistic of [Theorem 11.1.1](#).

13.2.6.3 C. Going deeper**Exercise C1.**

Show that Scheffé's multiplier cannot be reduced: if $c < \sqrt{qF_\alpha(q, \nu)}$, the intervals $\boldsymbol{\lambda}^\top \hat{\boldsymbol{\beta}} \pm c \text{se}(\boldsymbol{\lambda}^\top \hat{\boldsymbol{\beta}})$, $\boldsymbol{\lambda} \in \mathcal{C}(\boldsymbol{\Lambda})$, have simultaneous coverage less than $1 - \alpha$. So among intervals of the form estimate $\pm$ constant $\times$ standard error, valid for the whole family, Scheffé's are the shortest.

Solution. By [Eq. 13.2.3](#), all intervals cover iff the maximal squared t statistic, which has q times an $F(q, \nu)$ distribution, is at most c^2 . So the coverage is $\Pr\{F(q, \nu) \leq c^2/q\}$, which is less than $1 - \alpha$ when $c^2/q < F_\alpha(q, \nu)$, because the F distribution has a positive density.

13.3 Tukey's studentized range

For the family of all pairwise differences $\mu_k - \mu_l$, Tukey saw that the largest standardized error is the range of the estimated means divided by an estimate of their common standard deviation. When the means are estimated independently and with equal precision, as in a balanced one-way layout, this ratio is pivotal, and its upper α point gives intervals for all pairs with simultaneous coverage exactly $1 - \alpha$, shorter than Scheffé's or Bonferroni's.

13.3.1 The studentized range distribution

Definition 13.3.1 — Studentized range

Let $Z_1, \dots, Z_g$ be independent $N(0, 1)$ variables, $g \geq 2$, and let $V \sim \chi^2(\nu)$ be indepen-

dent of them. The law of

$$Q = \frac{\max_k Z_k - \min_k Z_k}{\sqrt{V/\nu}}$$

is the **studentized range distribution** with parameters g and ν . Its upper α point is written $q_\alpha(g, \nu)$.

The numerator is the range of a normal sample of size g . Dividing by $\sqrt{V/\nu}$ replaces the known standard deviation 1 by an independent estimate of it with ν degrees of freedom, just as a t variable is a studentized normal. For $g = 2$ the range is $|Z_1 - Z_2|$, a $N(0, 2)$ variable in absolute value, so $Q/\sqrt{2}$ is the absolute value of a $t(\nu)$ variable and

$$q_\alpha(2, \nu) = \sqrt{2} t_{\nu, \alpha/2}. \quad (13.3.1)$$

For larger g the distribution function is a double integral: conditioning on the smallest Z_k gives $\Pr(\max_k Z_k - \min_k Z_k \leq w) = g \int \varphi(z) [\Phi(z + w) - \Phi(z)]^{g-1} dz$, and one further integration over V gives the distribution of Q . SciPy evaluates it as `scipy.stats.studentized_range`. For $g = 5$ and $\nu = 20$ it gives $q_{0.05}(5, 20) = 4.232$, and 200000 simulated values of Q put the upper 5% point at 4.230.

```
import itertools
import numpy as np
from scipy import stats

alpha, reps = 0.05, 40_000

g, nu = 5, 20
rng = np.random.default_rng(20260919)
Z = rng.normal(size=(200_000, g))
V = rng.chisquare(nu, size=200_000)
Q = (Z.max(axis=1) - Z.min(axis=1)) / np.sqrt(V / nu)    # studentized range
q_sim = np.quantile(Q, 1 - alpha)
q_exact = stats.studentized_range.ppf(1 - alpha, g, nu)
print(f"upper 5% point of the studentized range, g = {g}, nu = {nu}:"
      f" simulated {q_sim:.3f}, scipy {q_exact:.3f}")
```

13.3.2 Tukey's intervals for a balanced layout

Consider the one-way model with g levels and the same number m of observations at every level, so $n = gm$, and assume normal errors. The level means $\bar{y}_1, \dots, \bar{y}_g$ are the least squares estimates of $\mu_1, \dots, \mu_g$, the pooled variance is $s^2 = \text{SSE}/\nu$ with $\nu = g(m - 1)$, and the standard error of every difference $\bar{y}_k - \bar{y}_l$ is $s\sqrt{2/m}$.

Lemma 13.3.1 — A contrast is bounded by the range

If $\sum_k c_k = 0$, then for all real $x_1, \dots, x_g$,

$$\left| \sum_k c_k x_k \right| \leq \frac{1}{2} \sum_k |c_k| \left(\max_k x_k - \min_k x_k \right),$$

with equality when $\mathbf{c}$ is a multiple of a pairwise difference $\mathbf{e}_k - \mathbf{e}_l$ and x_k, x_l are the

largest and smallest of the x 's.

Proof. Let $a = \frac{1}{2}(\max_k x_k + \min_k x_k)$. Because $\sum_k c_k = 0$, $\sum_k c_k x_k = \sum_k c_k (x_k - a)$, and $|x_k - a| \leq \frac{1}{2}(\max_k x_k - \min_k x_k)$ for every k . The triangle inequality gives the bound. For $\mathbf{c} = \mathbf{e}_k - \mathbf{e}_l$, the left side is $|x_k - x_l|$ and the right side is the range, which agree when x_k, x_l are the extremes. ■

Theorem 13.3.2 — Tukey's simultaneous intervals

In the balanced one-way model with normal errors, $g \geq 2$ and $\nu = g(m-1) \geq 1$, let $c_T = q_\alpha(g, \nu)$. Then:

(a) with probability exactly $1 - \alpha$,

$$\mu_k - \mu_l \in \left[\bar{y}_k - \bar{y}_l - c_T \frac{s}{\sqrt{m}}, \bar{y}_k - \bar{y}_l + c_T \frac{s}{\sqrt{m}} \right] \quad \text{for all } k \neq l; \quad (13.3.2)$$

(b) on the same event, for every contrast $\psi = \sum_k c_k \mu_k$,

$$\left| \sum_k c_k \bar{y}_k - \psi \right| \leq c_T \frac{s}{\sqrt{m}} \cdot \frac{1}{2} \sum_k |c_k|,$$

so these intervals for all contrasts also have simultaneous coverage exactly $1 - \alpha$.

Proof. By Theorem 7.5.1, the means $\bar{y}_k$ are independent $N(\mu_k, \sigma^2/m)$, and they are independent of s^2 , with $\nu s^2/\sigma^2 \sim \chi^2(\nu)$. Put $Z_k = \sqrt{m}(\bar{y}_k - \mu_k)/\sigma$ and $V = \nu s^2/\sigma^2$. These satisfy the assumptions of Definition 13.3.1, whatever the μ_k are. Now

$$\max_{k \neq l} |(\bar{y}_k - \bar{y}_l) - (\mu_k - \mu_l)| = \max_{k, l} [(\bar{y}_k - \mu_k) - (\bar{y}_l - \mu_l)] = \frac{\sigma}{\sqrt{m}} (\max_k Z_k - \min_k Z_k),$$

so all intervals Eq. 13.3.2 cover iff $(\max_k Z_k - \min_k Z_k)/\sqrt{V/\nu} \leq c_T$, an event of probability $1 - \alpha$, since the studentized range has a continuous distribution. For (b), apply Lemma 13.3.1 with $x_k = \bar{y}_k - \mu_k$: on that event the range of the x_k is at most $c_T s/\sqrt{m}$. The pairwise differences belong to the family of contrasts, so the coverage of the enlarged family cannot exceed that of the pairs, and it equals $1 - \alpha$. ■

Proposition 13.3.3 — Tukey's intervals are the shortest for all pairs

In the balanced one-way model, suppose intervals $\bar{y}_k - \bar{y}_l \pm c s \sqrt{2/m}$ with a common multiplier c have simultaneous coverage at least $1 - \alpha$ for all pairs. Then $c \geq q_\alpha(g, \nu)/\sqrt{2}$. In particular the Bonferroni, Šidák and Scheffé intervals for all pairs are at least as long as Tukey's.

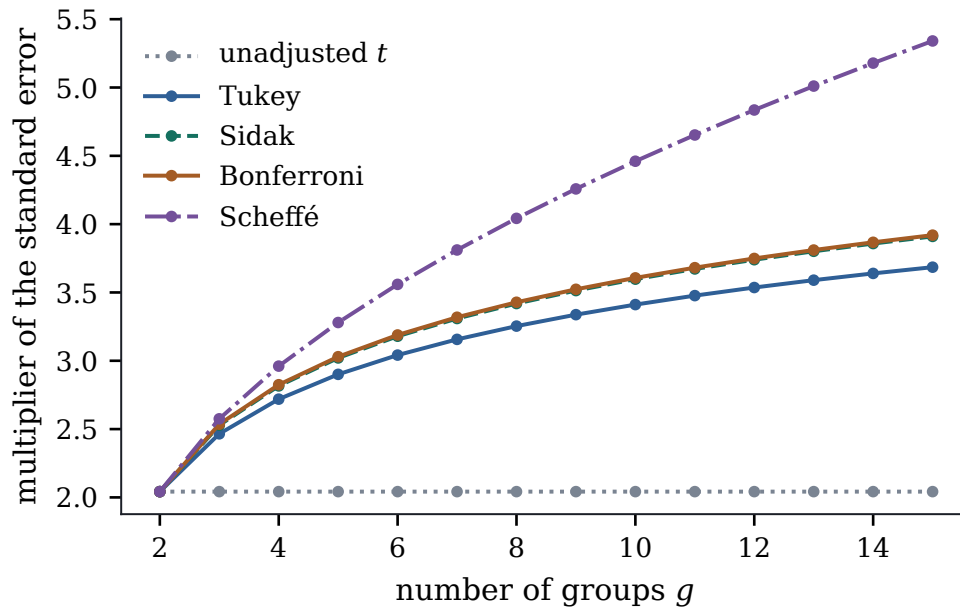


Figure 13.3.1: Multipliers of the standard error for 95% simultaneous intervals for all pairwise differences of g balanced means, $\nu = 30$ error degrees of freedom. Tukey's multiplier is $q_{0.05}(g, 30)/\sqrt{2}$. The Šidák and Bonferroni curves nearly coincide.

Proof. By the proof of [Theorem 13.3.2](#), the simultaneous coverage of such intervals is $\Pr(Q \leq c\sqrt{2})$, where Q has the studentized range distribution. This is at least $1 - \alpha$ only if $c\sqrt{2} \geq q_\alpha(g, \nu)$. The other three methods give intervals of this form with coverage at least $1 - \alpha$ ([Theorem 13.2.2](#) and [Section 13.4](#)).

■

[Figure 13.3.1](#) compares the multipliers of the standard error of a difference for $\nu = 30$. With $g = 5$ groups they are 2.901 for Tukey, 3.021 for Šidák, 3.030 for Bonferroni and 3.280 for Scheffé. With $g = 10$ they are 3.411, 3.597, 3.607 and 4.461. The unadjusted multiplier is 2.042 throughout. For $g = 2$ all methods coincide, by [Eq. 13.3.1](#). Scheffé's method falls further behind as g grows, because it covers a space of dimension $g - 1$ while the family of pairs is finite.

```
import numpy as np
from scipy import stats

alpha, nu = 0.05, 30

def pairwise_multipliers(g, nu, alpha=0.05):
    m = g * (g - 1) // 2                                # number of pairs
    return {
        "unadjusted": stats.t.ppf(1 - alpha / 2, nu),
        "Tukey": stats.studentized_range.ppf(1 - alpha, g, nu) / np.sqrt(2),
        "Sidak": stats.t.ppf((1 + (1 - alpha) ** (1 / m)) / 2, nu),
        "Bonferroni": stats.t.ppf(1 - alpha / (2 * m), nu),
        "Scheffe": np.sqrt((g - 1) * stats.f.ppf(1 - alpha, g - 1, nu)),
    }

for g in (2, 3, 5, 10):
    row = pairwise_multipliers(g, nu)
    print(f"g = {g:2d}: " + " ".join(f"{k} {v:.3f}" for k, v in row.items()))
```

For contrasts other than pairs the comparison can reverse. Part (b) of [Theorem 13.3.2](#) gives the half-width $q_\alpha(g, \nu) (s/\sqrt{m}) \frac{1}{2} \sum_k |c_k|$, while Scheffé's is $\sqrt{(g-1)F_\alpha(g-1, \nu)} (s/\sqrt{m}) (\sum_k c_k^2)^{1/2}$. With $g = 6$ and $\nu = 30$, in units of $s/\sqrt{m}$, a pairwise difference has half-width 4.301 by Tukey's method and 5.033 by Scheffé's; for $c = \frac{1}{3}(1, 1, 1, -1, -1, -1)$ the half-widths are 4.301 and 2.906. Tukey's bound is sharp for pairs and crude for contrasts spread over many means.

13.3.3 Unbalanced layouts: the Tukey–Kramer intervals

With unequal group sizes n_k the maximum standardized difference is no longer a studentized range. Tukey and Kramer proposed keeping the multiplier and using the standard error of each difference:

$$\bar{y}_k - \bar{y}_l \pm \frac{q_\alpha(g, \nu)}{\sqrt{2}} s \sqrt{\frac{1}{n_k} + \frac{1}{n_l}}, \quad \nu = n - g. \quad (13.3.3)$$

Tukey conjectured, and Hayter (1984) proved, that the simultaneous coverage of these intervals is never less than $1 - \alpha$, whatever the group sizes. The proof is beyond this chapter. How conservative the intervals are is a separate question; the simulation below estimates the coverage of 95% intervals in three designs from 40000 data sets each (standard error about 0.0011):

design	coverage
balanced, five groups of 5	0.9511
the education example (sizes 13 to 248)	0.9562
four groups of 2 and four of 40	0.9614

The balanced design is exact; the unbalanced ones overcover only modestly, even with very different group sizes.

```
import itertools
import numpy as np
from scipy import stats

alpha, reps = 0.05, 40_000

def tukey_kramer_coverage(n_k, reps, alpha, seed):
    """Share of data sets in which every Tukey-Kramer interval covers its difference
    ."""
    rng = np.random.default_rng(seed)
    n_k = np.asarray(n_k)
    g, nu = len(n_k), n_k.sum() - len(n_k)
    q = stats.studentized_range.ppf(1 - alpha, g, nu)
    # group means (true means 0) and an independent pooled variance, as in the model
    means = rng.normal(size=(reps, g)) / np.sqrt(n_k)
    s = np.sqrt(rng.chisquare(nu, size=reps) / nu)
    ok = np.ones(reps, dtype=bool)
    for k, l in itertools.combinations(range(g), 2):
        half = q / np.sqrt(2) * s * np.sqrt(1 / n_k[k] + 1 / n_k[l])
        ok &= np.abs(means[:, k] - means[:, l]) <= half
    return ok.mean()
```

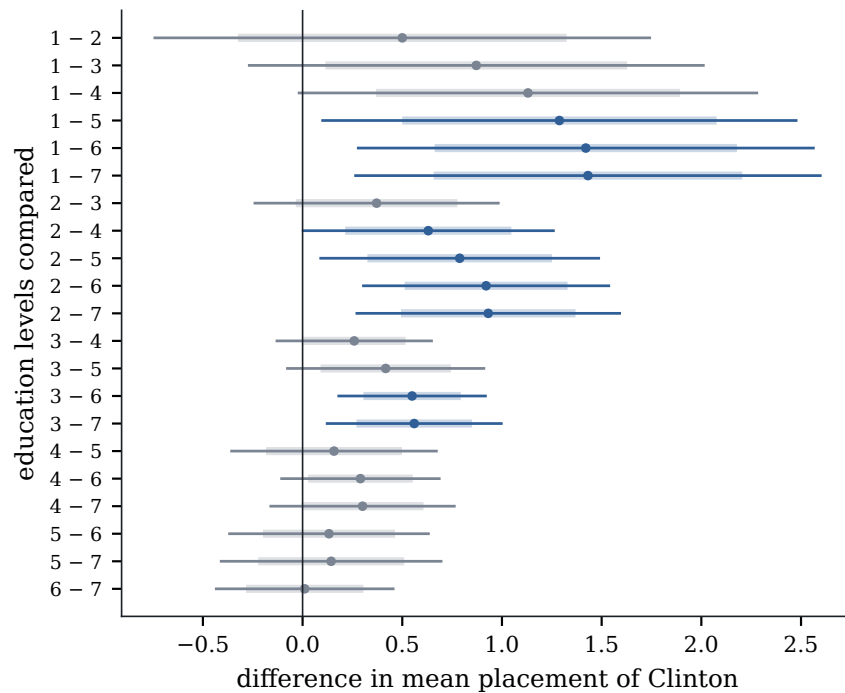


Figure 13.3.2: Differences in mean placement of Clinton between education levels, with 95% Tukey–Kramer simultaneous intervals (thin lines) and unadjusted t intervals (thick pale bands). Differences whose Tukey–Kramer interval excludes zero are drawn in blue.

```
balanced = tukey_kramer_coverage([5] * 5, reps, alpha, seed=1)
education = tukey_kramer_coverage([13, 52, 248, 187, 90, 227, 127], reps, alpha,
    seed=2)
lopsided = tukey_kramer_coverage([2, 2, 2, 2, 40, 40, 40, 40], reps, alpha, seed=3)
print(f"balanced (5 x 5):           {balanced:.4f}")
print(f"education design:          {education:.4f}")
print(f"four groups of 2, four of 40: {lopsided:.4f}")
```

Example. (All pairs of education levels)

In [Example \(Education and the perceived position of a candidate\)](#) there are $g = 7$ education levels, so 21 pairs, and $\nu = 937$. The Tukey–Kramer multiplier is $q_{0.05}(7, 937)/\sqrt{2} = 4.179/\sqrt{2} = 2.955$. It declares 9 of the 21 differences nonzero, against 14 for unadjusted t intervals ([Section 13.6](#) compares all the methods). [Figure 13.3.2](#) shows the intervals. The significant differences all separate a lower education level from a higher one, and the three most educated levels do not differ from one another. The smallest group, with only 13 respondents, has the highest mean, but its intervals are so wide that only its differences from levels 5, 6 and 7 are established.

```
import itertools
import numpy as np
import statsmodels.api as sm
from scipy import stats
anes = sm.datasets.anes96.load_pandas().data
y = anes["ClinLR"].to_numpy()
level = anes["educ"].to_numpy().astype(int) - 1      # 0, ..., 6
```

```

g = 7
n_k = np.bincount(level).astype(float)
means = np.bincount(level, weights=y) / n_k
n = len(y)
nu = n - g                                # error degrees of freedom
s2 = np.sum((y - means[level]) ** 2) / nu  # pooled variance estimate
s = np.sqrt(s2)
alpha = 0.05
c_sch = np.sqrt((g - 1) * stats.f.ppf(1 - alpha, g - 1, nu))  # Scheffe multiplier,
    q = g - 1

pairs = list(itertools.combinations(range(g), 2))
m = len(pairs)                            # 21 comparisons
diff = np.array([means[k] - means[l] for k, l in pairs])
se_d = np.array([s * np.sqrt(1 / n_k[k] + 1 / n_k[l]) for k, l in pairs])
t = diff / se_d
pval = 2 * stats.t.sf(np.abs(t), nu)

mult = {                                  # half-width = multiplier x se
    "unadjusted": stats.t.ppf(1 - alpha / 2, nu),
    "Bonferroni": stats.t.ppf(1 - alpha / (2 * m), nu),
    "Sidak": stats.t.ppf((1 + (1 - alpha) ** (1 / m)) / 2, nu),
    "Tukey-Kramer": stats.studentized_range.ppf(1 - alpha, g, nu) / np.sqrt(2),
    "Scheffe": c_sch,
}
for name, c in mult.items():
    print(f"{name:13s} multiplier {c:.3f}   pairs declared different: {np.sum(np.abs(
        t) > c)}")

```

13.3.4 Beyond the one-way layout

The proof of [Theorem 13.3.2](#) used only that the g estimates are independent, with a common variance $\sigma^2 c$ for a known c , and independent of s^2 . That also holds for the levels of one factor in a balanced multi-way design ([Exercise B1](#)), the subject of [Chapters 15](#) and [16](#). For correlated estimates, such as regression coefficients or adjusted means, neither form is exact; the exact critical value for any planned family then comes from a multivariate t distribution ([Exercise C1](#)), which standard software computes.

13.3.5 Exercises

13.3.5.1 A. Check your understanding

Exercise A1.

Prove [Eq. 13.3.1](#), and deduce that Tukey's intervals for $g = 2$ are the ordinary t intervals.

13.3.5.2 B. Practice

Exercise B1.

In a two-way additive model $E(y_{ijk}) = \mu + \alpha_i + \beta_j$ with a rows, b columns and N

observations in every cell, show that the column means $\bar{y}_{\cdot j}$ are independent, each with variance $\sigma^2/(aN)$, and independent of the residual mean square. Deduce Tukey intervals for all differences $\beta_j - \beta_{j'}$, and state their degrees of freedom.

Solution. Each $\bar{y}_{\cdot j}$ averages the aN observations of column j , and different columns use disjoint sets of independent observations, so the column means are independent with variance $\sigma^2/(aN)$. Their expectations are $\mu + \bar{\alpha} + \beta_j$, with $\bar{\alpha}$ the average row effect, so differences estimate $\beta_j - \beta_{j'}$. The column means are functions of the fitted values (the fitted value of cell (i, j) is $\bar{y}_{i\cdot} + \bar{y}_{\cdot j} - \bar{y}$, and averaging it over i gives $\bar{y}_{\cdot j}$), so they are independent of s^2 by [Theorem 7.5.1](#). The residual degrees of freedom are $\nu = abN - a - b + 1$, and the intervals are $\bar{y}_{\cdot j} - \bar{y}_{\cdot j'} \pm q_\alpha(b, \nu) s / \sqrt{aN}$, exact for all pairs of columns.

Exercise B2.

Treatments $1, \dots, g-1$ are to be compared with a control, level 0, in a balanced layout. Show that the exact simultaneous multiplier for the $g-1$ differences $\mu_k - \mu_0$ is the upper α point of $\max_{k \geq 1} |Z_k - Z_0| / \sqrt{V/\nu}$, and that it is smaller than $q_\alpha(g, \nu)$.

Solution. As in the proof of [Theorem 13.3.2](#), the largest standardized error among the differences with the control is $(\sigma/\sqrt{m}) \max_{k \geq 1} |Z_k - Z_0|$, and the statistic $D = \max_{k \geq 1} |Z_k - Z_0| / \sqrt{V/\nu}$ is pivotal. Let Q be the studentized range of $Z_0, \dots, Z_{g-1}$, so that $D \leq Q$. The event $\{D \leq q_\alpha(g, \nu) < Q\}$ has positive probability: it contains an open set of values of $(Z_0, \dots, Z_{g-1}, V)$, for instance those with Z_0 near the middle of the others and the range just above the critical value, and these variables have a positive joint density. Hence $\Pr(D > q_\alpha(g, \nu)) = \alpha - \Pr(D \leq q_\alpha(g, \nu) < Q) < \alpha$, and since D has a continuous distribution its upper α point is smaller than $q_\alpha(g, \nu)$. This is Dunnett's many-to-one procedure.

Exercise B3.

In the balanced one-way layout, the **studentized range test** rejects $\mu_1 = \dots = \mu_g$ when $Q_{\text{obs}} = (\max_k \bar{y}_k - \min_k \bar{y}_k) / (s/\sqrt{m}) > q_\alpha(g, \nu)$. Show that it has exact level α , and that it rejects iff some Tukey interval [Eq. 13.3.2](#) excludes 0. Show that the F test rejects iff $\sum_k (\bar{y}_k - \bar{y})^2 > (g-1)F_\alpha(g-1, \nu) s^2/m$, and describe, for $g=3$, configurations of the means that one test rejects and the other does not.

Solution. Under the hypothesis Q_{obs} has the studentized range distribution (proof of [Theorem 13.3.2](#) with all μ_k equal), so the level is exactly α , and $Q_{\text{obs}} > q_\alpha(g, \nu)$ says that the extreme pair's interval excludes 0; every other pair has a smaller difference. The F statistic is $m \sum_k (\bar{y}_k - \bar{y})^2 / \{(g-1)s^2\}$. For $g=3$ and a fixed range $2a$, the sum of squares is $2a^2$ for means $(-a, 0, a)$ and $8a^2/3$ for $(-a, -a, a)$. In the limit $\nu \rightarrow \infty$, in units of $s/\sqrt{m}$, the range test rejects when $2a > 3.314$ and the F test when the sum of squares exceeds 5.991: so $(-a, 0, a)$ with $1.657 < a < 1.731$ is rejected only by the range test, and $(-a, -a, a)$ with $1.499 < a < 1.657$ only by the F test. The range test is the natural gate for pairs, the F test for contrasts in general.

13.3.5.3 C. Going deeper

Exercise C1.

Let $\lambda_1, \dots, \lambda_k \in \mathcal{C}(\mathbf{X}^\top)$ be chosen in advance, with t statistics $T_j = (\lambda_j^\top \hat{\beta} - \lambda_j^\top \beta) / \text{se}(\lambda_j^\top \hat{\beta})$. Show that $(T_1, \dots, T_k) = \mathbf{Z} / (s/\sigma)$, where $\mathbf{Z} \sim N_k(\mathbf{0}, \mathbf{R})$ is independent of s , and $\mathbf{R}$ is the correlation matrix with entries $\lambda_i^\top \mathbf{G} \lambda_j / \sqrt{\lambda_i^\top \mathbf{G} \lambda_i \cdot \lambda_j^\top \mathbf{G} \lambda_j}$. Deduce that the constant c with $\Pr(\max_j |T_j| \leq c) = 1 - \alpha$ depends only on $\mathbf{R}$ and ν , and that it is at most the Bonferroni multiplier and at most Scheffé's multiplier for the span of the λ_j .

13.4 Bonferroni and Holm

Scheffé's and Tukey's methods need an exact distribution for the worst error in a family. The methods of this section need only the level of each separate test and a probability inequality, so they apply to any finite list of hypotheses in any model, whatever the dependence. Holm's step-down procedure recovers part of the resulting conservatism at no cost in generality.

13.4.1 The Bonferroni inequality

- Theorem 13.4.1 — Bonferroni** (a) For any events $A_1, \dots, A_m$, $\Pr(\bigcup_j A_j) \leq \sum_j \Pr(A_j)$.
- (b) Let $\alpha_1, \dots, \alpha_m > 0$ with $\sum_j \alpha_j = \alpha$. If $[L_j, U_j]$ is a confidence interval for ψ_j with coverage at least $1 - \alpha_j$, for each j , then the m intervals have simultaneous coverage at least $1 - \alpha$.
- (c) Let p_j be a p -value for H_j , valid in the sense that $\Pr(p_j \leq u) \leq u$ for $0 \leq u \leq 1$ whenever H_j is true. The procedure that rejects H_j when $p_j \leq \alpha_j$ has PFER $\leq \sum_{j \in I_0} \alpha_j \leq \alpha$, and hence controls the familywise error rate strongly at level α .

No assumption is made about the dependence among the events, intervals or p -values.

Proof. (a) Pointwise, the indicator of a union is at most the sum of the indicators: $1_{\bigcup_j A_j} \leq \sum_j 1_{A_j}$, since the right side is at least 1 wherever the left side is. Take expectations. (b) Apply (a) to the events $A_j = \{\psi_j \notin [L_j, U_j]\}$. (c) The number of false rejections is $V = \sum_{j \in I_0} 1_{\{p_j \leq \alpha_j\}}$, so $E(V) = \sum_{j \in I_0} \Pr(p_j \leq \alpha_j) \leq \sum_{j \in I_0} \alpha_j$. Strong control follows from [Proposition 13.1.1](#). ■

Part (c) bounds even the expected *number* of false rejections, which is why the procedure is conservative when false rejections tend to occur together.

In the linear model, the **Bonferroni intervals** for k estimable functions $\lambda_1^\top \beta, \dots, \lambda_k^\top \beta$ chosen in advance are

$$\lambda_j^\top \hat{\beta} \pm t_{\nu, \alpha/(2k)} \text{se}(\lambda_j^\top \hat{\beta}), \quad j = 1, \dots, k, \quad (13.4.1)$$

each the t interval of [Theorem 12.1.1](#) at level $1 - \alpha/k$. The equal split is conventional; any split

fixed before the data are seen is valid.

Warning. The Bonferroni guarantee is for a list of k hypotheses fixed in advance. If the analyst looks at the estimates and then chooses which k to test, the hypotheses actually tested are a data-dependent selection from a larger family, and the relevant k is the size of that larger family. For data-driven choices within a subspace, use Scheffé's method.

13.4.2 Independent tests: Šidák's improvement

If the tests of the true hypotheses are independent and each has exact level α , then by Eq. 13.1.1 the familywise error rate is $1 - (1 - \alpha)^{m_0}$. Choosing

$$\alpha = 1 - (1 - \alpha)^{1/m} \quad (13.4.2)$$

makes this at most $1 - (1 - \alpha)^m = \alpha$, with equality when all hypotheses are true. This **Šidák level** is slightly larger than α/m , because $(1 - \alpha/m)^m > 1 - \alpha$ for $m \geq 2$ (Bernoulli's inequality), so the Šidák procedure is slightly less conservative than Bonferroni's.

In a linear model, however, t statistics are never independent: they share s . The following result shows that the sharing works in the right direction.

Lemma 13.4.2 — Chebyshev's association inequality

Let U be a real random variable and $f_1, \dots, f_k$ nonnegative, nondecreasing functions with $E f_j(U)^k < \infty$. Then $E[\prod_j f_j(U)] \geq \prod_j E f_j(U)$.

Proof. For two functions f, h , let U' be an independent copy of U . Since f and h are both nondecreasing, $(f(U) - f(U'))(h(U) - h(U')) \geq 0$. Taking expectations gives $2E[f(U)h(U)] - 2E f(U) E h(U) \geq 0$. For k functions, argue by induction: the product $f_1 \cdots f_{k-1}$ of nonnegative nondecreasing functions is nonnegative and nondecreasing, so $E[f_1 \cdots f_k(U)] \geq E[f_1 \cdots f_{k-1}(U)] E f_k(U)$, and the induction hypothesis applies to the first factor. The moment condition makes every expectation finite, by Hölder's inequality. ■

Proposition 13.4.3 — Šidák intervals

Let $Z_1, \dots, Z_k$ be independent $N(0, 1)$ variables and $V \sim \chi^2(\nu)$ independent of them, and put $T_j = Z_j / \sqrt{V/\nu}$. Then for every $c > 0$,

$$\Pr(|T_j| \leq c \text{ for all } j) \geq \prod_{j=1}^k \Pr(|T_j| \leq c).$$

Consequently, if $\lambda_1^\top \hat{\beta}, \dots, \lambda_k^\top \hat{\beta}$ are uncorrelated, the intervals $\lambda_j^\top \hat{\beta} \pm c \text{se}(\lambda_j^\top \hat{\beta})$ with c the upper $\frac{1}{2}(1 - (1 - \alpha)^{1/k})$ point of $t(\nu)$ have simultaneous coverage at least $1 - \alpha$.

Proof. Conditionally on $V = v$, the events $\{|Z_j| \leq c\sqrt{v/\nu}\}$ are independent, each with probability $f(v) = 2\Phi(c\sqrt{v/\nu}) - 1$. So $\Pr(|T_j| \leq c \ \forall j) = E[f(V)^k]$, while $\Pr(|T_j| \leq c) = E f(V)$. The function f is nonnegative, bounded and nondecreasing, and Lemma 13.4.2 gives $E[f(V)^k] \geq (E f(V))^k$. For the intervals, the estimates are jointly normal (Theorem 7.5.1), so uncorrelated estimates are independent (Theorem 3.4.1), and they are independent of s . The standardized errors are therefore of the form T_j , and each $\Pr(|T_j| \leq c) = (1 - \alpha)^{1/k}$. ■

The proposition covers orthogonal contrasts (Proposition 8.7.2(c)) and regressions with orthogonal columns. Šidák (1967) proved the corresponding inequality $\Pr(|Z_j| \leq c_j \ \forall j) \geq \prod_j \Pr(|Z_j| \leq c_j)$ for *any* centred jointly normal $(Z_1, \dots, Z_k)$; with the conditioning argument above it makes Šidák's intervals valid for any finite family of estimable functions. We use that theorem without proof. The gain over Bonferroni is small: for $k = 10$ and $\nu = 29$ the multipliers are 3.0289 and 3.0380, and with independent numerators and a shared s the exact coverage of the nominal 95% intervals, computed by integrating over s , is 0.9520 and 0.9530.

13.4.3 The closure principle

Bonferroni's procedure keeps the threshold α/m even after rejections, although a rejected false hypothesis can no longer be falsely rejected. The closure principle exploits this. For each nonempty $I \subseteq \{1, \dots, m\}$ let $H_I = \bigcap_{i \in I} H_i$ be the **intersection hypothesis**, and let ϕ_I be a level- α **local test** of it.

Theorem 13.4.4 — Closed testing

The procedure that rejects H_j iff ϕ_I rejects H_I for every I containing j controls the familywise error rate strongly at level α .

Proof. Fix the parameters and suppose $I_0 \neq \emptyset$; otherwise there is nothing to prove. The intersection hypothesis H_{I_0} is true, so $\Pr(\phi_{I_0} \text{ rejects}) \leq \alpha$. If some true H_j is rejected, then $j \in I_0$, so by the definition of the procedure ϕ_{I_0} rejects. Hence $\Pr(V \geq 1) \leq \Pr(\phi_{I_0} \text{ rejects}) \leq \alpha$. ■

Rejecting one hypothesis requires rejecting every intersection containing it, in general 2^{m-1} tests. Holm's procedure is a case where this collapses to m steps.

13.4.4 Holm's step-down procedure

Order the p -values as $p_{(1)} \leq p_{(2)} \leq \dots \leq p_{(m)}$, breaking ties arbitrarily, and let $H_{(i)}$ be the hypothesis with the i th smallest p -value.

Holm's procedure at level α . For $i = 1, 2, \dots, m$ in turn: if $p_{(i)} \leq \alpha/(m - i + 1)$, reject $H_{(i)}$ and continue; otherwise stop, and reject none of $H_{(i)}, \dots, H_{(m)}$.

The first threshold is Bonferroni's, α/m . Each rejection relaxes the next threshold: to $\alpha/(m-1)$, then $\alpha/(m-2)$, and so on, up to α for the last hypothesis.

Theorem 13.4.5 — Holm

Let each p_j be valid for H_j , as in Theorem 13.4.1(c), with any dependence among them. Then:

- (a) Holm's procedure controls the familywise error rate strongly at level α ;
- (b) it rejects every hypothesis that Bonferroni's procedure (all $\alpha_j = \alpha/m$) rejects, and possibly more.

Proof. (a) Suppose $m_0 \geq 1$, and let j^* be the rank, in the ordering of all m p -values, of the smallest p -value belonging to a true hypothesis, so that $p_{(j^*)} = \min_{i \in I_0} p_i$. The hypotheses ranked before j^* are all false, and there are only $m - m_0$ false hypotheses, so $j^* \leq m - m_0 + 1$ and $\alpha/(m - j^* + 1) \leq \alpha/m_0$. Holm's procedure rejects an initial segment $H_{(1)}, \dots, H_{(i)}$ of the ordering. If that segment contains any true hypothesis, it contains $H_{(j^*)}$, so $p_{(j^*)} \leq \alpha/(m - j^* + 1) \leq \alpha/m_0$. Therefore

$$\Pr(V \geq 1) \leq \Pr\left(\min_{i \in I_0} p_i \leq \frac{\alpha}{m_0}\right) \leq \sum_{i \in I_0} \Pr\left(p_i \leq \frac{\alpha}{m_0}\right) \leq \alpha,$$

by Theorem 13.4.1(a) and the validity of the p -values. (b) If Bonferroni rejects $H_{(i)}$, then $p_{(j)} \leq p_{(i)} \leq \alpha/m \leq \alpha/(m - j + 1)$ for every $j \leq i$, so Holm's procedure does not stop before step i and rejects $H_{(i)}$. ■

Bonferroni's α/m pays for m potential errors when only m_0 are possible. Holm's procedure is closed testing with the Bonferroni local tests (Exercise (B3)). It gives no simple simultaneous intervals, since the threshold for a hypothesis depends on the other p -values; its output is a list of rejections, or **adjusted p -values**

$$\tilde{p}_{(i)} = \max_{j \leq i} \min\{1, (m - j + 1)p_{(j)}\}, \quad (13.4.3)$$

with $H_{(i)}$ rejected at level α iff $\tilde{p}_{(i)} \leq \alpha$ (Exercise (B1)).

Example. (Holm for the education pairs)

For the 21 pairwise comparisons of Example (All pairs of education levels) the smallest p -value is 1.1×10^{-5} . Bonferroni's threshold is $0.05/21 = 0.00238$, and 8 of the p -values fall below it. Holm's procedure starts at the same threshold but relaxes it after each rejection. It rejects 10 hypotheses: the tenth smallest p -value is 0.0037, below its threshold $0.05/12$, and the eleventh, 0.0126, exceeds $0.05/11 = 0.0045$, so the procedure stops.

```
import itertools
import numpy as np
import statsmodels.api as sm
```

```

from scipy import stats
anes = sm.datasets.anes96.load_pandas().data
y = anes["ClinLR"].to_numpy()
level = anes["educ"].to_numpy().astype(int) - 1      # 0, ..., 6
g = 7
n_k = np.bincount(level).astype(float)
means = np.bincount(level, weights=y) / n_k
n = len(y)
nu = n - g                                           # error degrees of freedom
s2 = np.sum((y - means[level]) ** 2) / nu           # pooled variance estimate
s = np.sqrt(s2)
alpha = 0.05
pairs = list(itertools.combinations(range(g), 2))
diff = np.array([means[k] - means[l] for k, l in pairs])
se_d = np.array([s * np.sqrt(1 / n_k[k] + 1 / n_k[l]) for k, l in pairs])
t = diff / se_d
pval = 2 * stats.t.sf(np.abs(t), nu)

def holm(p, alpha):
    """Holm's step-down procedure: reject the i smallest p-values, i maximal with
    p_(j) <= alpha / (m - j + 1) for all j <= i."""
    m = len(p)
    order = np.argsort(p)
    reject = np.zeros(m, dtype=bool)
    for j, idx in enumerate(order):                  # j = 0, 1, ...
        if p[idx] > alpha / (m - j):
            break
        reject[idx] = True
    return reject

def benjamini_hochberg(p, q):
    """Reject the k smallest p-values, k the largest index with p_(k) <= k q / m."""
    m = len(p)
    order = np.argsort(p)
    below = np.nonzero(p[order] <= q * np.arange(1, m + 1) / m)[0]
    reject = np.zeros(m, dtype=bool)
    if below.size:
        reject[order[: below[-1] + 1]] = True
    return reject

print("Holm rejects", holm(pval, alpha).sum(), " BH (q = 0.05) rejects",
      benjamini_hochberg(pval, alpha).sum())

```

13.4.5 How much does Holm gain?

If all hypotheses are true, Holm and Bonferroni make a false rejection in the same data sets. The more hypotheses are false and rejected early, the larger the later thresholds and the gain.

Example. (Ten regression coefficients)

A regression has an intercept and ten centred, mutually orthogonal regressors, with $n = 40$. The ten t statistics for the slopes are then $T_j = Z_j / \sqrt{V/29}$ with independent $Z_j \sim N(\delta_j, 1)$ and a shared $V \sim \chi^2(29)$ (Theorem 7.5.1). Let k of the slopes be nonzero, each with $\delta_j = 3.5$, and test all ten at familywise level 0.05. Figure 13.4.1

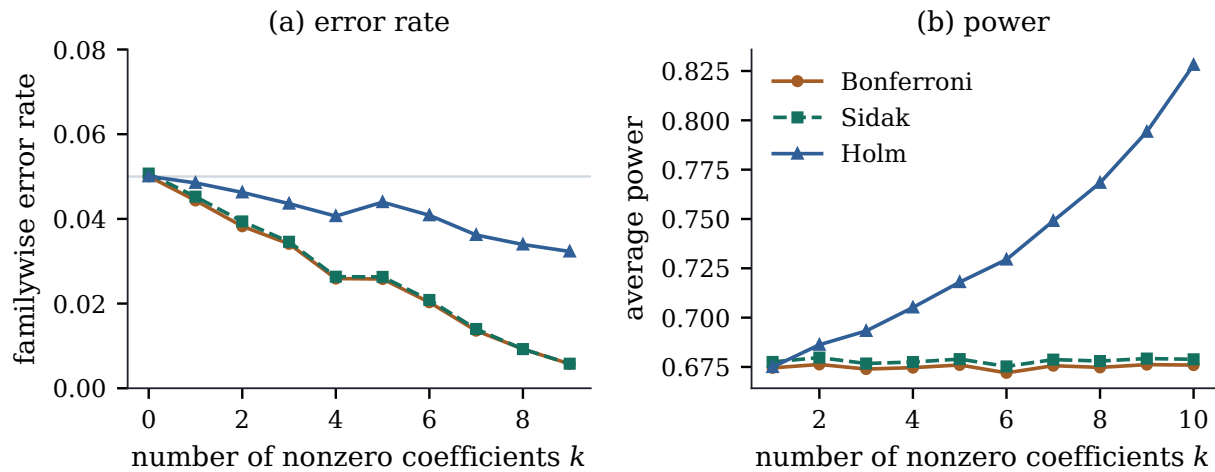


Figure 13.4.1: Ten orthogonal regression coefficients, k of them nonzero, tested at familywise level 0.05: (a) familywise error rate, (b) average power, the share of nonzero coefficients detected.

shows the result of 20000 simulated data sets for each k . All three procedures keep the familywise error rate at or below 0.05, up to simulation error. The average power of Bonferroni's procedure, the proportion of nonzero slopes it detects, hardly depends on k : it is 0.675 with one nonzero slope and 0.676 with five or nine. Šidák's is larger by a few thousandths. Holm's power is 0.675 with one nonzero slope, 0.718 with five and 0.794 with nine.

```
import numpy as np
from scipy import stats

alpha, m, nu, delta, reps = 0.05, 10, 29, 3.5, 20_000

def simulate(k, reps, seed):
    """FWER and average power of three procedures when k of m coefficients are
    nonzero."""
    rng = np.random.default_rng(seed)
    shift = np.r_[np.full(k, delta), np.zeros(m - k)]
    T = (rng.normal(size=(reps, m)) + shift) / np.sqrt(rng.chisquare(nu, (reps, 1))
    / nu)
    p = 2 * stats.t.sf(np.abs(T), nu)
    rules = {
        "Bonferroni": p <= alpha / m,
        "Sidak": p <= 1 - (1 - alpha) ** (1 / m),
    }
    # Holm: sort each row, compare p_(j) with alpha / (m - j + 1), stop at the first
    failure
    order = np.argsort(p, axis=1)
    ps = np.take_along_axis(p, order, axis=1)
    passed = np.cumprod(ps <= alpha / (m - np.arange(m)), axis=1).astype(bool)
    holm = np.zeros_like(passed)
    np.put_along_axis(holm, order, passed, axis=1)
    rules["Holm"] = holm
    out = {}
    for name, rej in rules.items():
        fwer = np.mean(rej[:, k:].any(axis=1)) if k < m else np.nan
        power = rej[:, :k].mean() if k > 0 else np.nan
        out[name] = (fwer, power, rej)
    return out
```

```
res = simulate(5, reps, seed=5)
for name, (fwer, power, _) in res.items():
    print(f"{name:10s} FWER = {fwer:.3f}    average power = {power:.3f}")
```

There is no reason to use Bonferroni's *tests* rather than Holm's, though Bonferroni's *intervals* remain useful. With independent p -values, Hochberg (1988) proposed the *step-up* version: reject $H_{(1)}, \dots, H_{(i)}$ for the largest i with $p_{(i)} \leq \alpha/(m - i + 1)$. It rejects everything Holm's procedure does ([Exercise \(C1\)](#)), but its validity rests on an inequality of Simes (1986) that holds under independence and some positive dependence, not in general. The step-up idea returns in the next section, controlling a different error rate.

13.4.6 Exercises

13.4.6.1 A. Check your understanding

Exercise A1.

Five hypotheses have p -values 0.001, 0.011, 0.012, 0.04, 0.2. At familywise level 0.05, which are rejected by Bonferroni's procedure and which by Holm's? Compute Holm's adjusted p -values [Eq. 13.4.3](#).

Solution. Bonferroni's threshold is 0.01, so only the first is rejected. Holm compares $0.001 \leq 0.05/5$, $0.011 \leq 0.05/4$, $0.012 \leq 0.05/3$, and then $0.04 > 0.05/2$, so it rejects the first three and stops. The adjusted p -values are $5 \times 0.001 = 0.005$, $\max(0.005, 4 \times 0.011) = 0.044$, $\max(0.044, 3 \times 0.012) = 0.044$, $\max(0.044, 2 \times 0.04) = 0.08$ and $\max(0.08, 0.2) = 0.2$. The three below 0.05 are the rejected hypotheses.

13.4.6.2 B. Practice

Exercise B1.

Show that Holm's procedure rejects $H_{(i)}$ iff the adjusted p -value [Eq. 13.4.3](#) satisfies $\tilde{p}_{(i)} \leq \alpha$, and that the adjusted p -values are nondecreasing in i .

Exercise B2.

All $m \geq 2$ hypotheses are true and the p -values are independent and uniform. Show that the per-family error rate of Bonferroni's procedure is exactly α , while that of Holm's procedure is larger than α . So Holm's procedure controls the familywise rate but not the per-family rate.

Exercise B3.

Show that Holm's procedure is closed testing ([Theorem 13.4.4](#)) with the Bonferroni local tests ϕ_I : reject H_I when $\min_{i \in I} p_i \leq \alpha/|I|$.

Solution. Suppose Holm rejects $H_{(1)}, \dots, H_{(i)}$ and stops at step $i + 1$. Take $j \leq i$ and any $I \ni (j)$, writing (j) for the index of $H_{(j)}$. If I contains some (l) with $l \leq j$, let l be the smallest such; then I contains no $(1), \dots, (l - 1)$, so $|I| \leq m - l + 1$ and $\min_I p \leq p_{(l)} \leq \alpha / (m - l + 1) \leq \alpha / |I|$: ϕ_I rejects. So closed testing rejects $H_{(j)}$. Conversely, the set $I = \{(i + 1), \dots, (m)\}$ has $m - i$ elements and minimum $p_{(i+1)} > \alpha / (m - i)$, so ϕ_I does not reject, and no hypothesis in I is rejected by closed testing. The two procedures therefore reject the same hypotheses.

13.4.6.3 C. Going deeper

Exercise C1.

Show that Hochberg's step-up procedure rejects every hypothesis rejected by Holm's step-down procedure with the same thresholds. Give an example with $m = 2$ where Hochberg rejects both hypotheses and Holm rejects neither.

13.5 False discovery rate control

With hundreds of hypotheses, as when a regression screens many candidate predictors, family-wise control becomes self-defeating: the threshold α/m shrinks with m , and power disappears exactly when there is most to find. The false discovery rate of [Definition 13.1.1](#) asks instead that, on average, only a small *proportion* of the rejections be false. This section proves that the procedure of Benjamini and Hochberg controls it for independent p -values, and describes what is known under dependence.

13.5.1 The Benjamini–Hochberg procedure

As before, $p_{(1)} \leq \dots \leq p_{(m)}$ are the ordered p -values, $H_{(i)}$ the corresponding hypotheses, and $q \in (0, 1)$ is the target false discovery rate.

Benjamini–Hochberg (BH) procedure at level q . Let

$$\hat{k} = \max \left\{ k \in \{1, \dots, m\} : p_{(k)} \leq \frac{kq}{m} \right\},$$

and reject $H_{(1)}, \dots, H_{(\hat{k})}$. If no k qualifies, reject nothing.

The thresholds rise linearly from Bonferroni's q/m to q . The procedure is a **step-up** procedure: it rejects everything up to the *last* p -value on or below the line, including smaller p -values that lie above the line at their own rank ([Figure 13.5.1](#)).

Why is the line the right one? Suppose k hypotheses are rejected, all with p -values at most $t = kq/m$. The m_0 true hypotheses have uniform p -values, so about $m_0 t \leq mt$ of them fall below t by chance, and the proportion of false ones among the k rejections is about $mt/k = q$ at most. The BH procedure chooses the largest threshold at which this estimate of the false discovery proportion is at most q . The theorem makes the heuristic exact.

Two simple descriptions of the rejected set are useful. Write $N(t) = \#\{j : p_j \leq t\}$ for the number of p -values at most t . Since $p_{(k)} \leq t$ iff $N(t) \geq k$,

$$\hat{k} = \max\left\{k : N(kq/m) \geq k\right\}. \quad (13.5.1)$$

And the rejected hypotheses are exactly those with $p_j \leq \hat{k}q/m$. Indeed, every rejected p -value is at most $p_{(\hat{k})} \leq \hat{k}q/m$. Conversely, if some p_j with j not rejected were at most $\hat{k}q/m$, then $N(\hat{k}q/m) \geq \hat{k} + 1$, hence $N((\hat{k} + 1)q/m) \geq \hat{k} + 1$, and $\hat{k} + 1$ would qualify, contradicting the maximality of $\hat{k}$.

13.5.2 Control under independence

Lemma 13.5.1 — Replacing one p -value by zero

Fix i , and let $\hat{k}^{(i)}$ be the number of rejections of the BH procedure applied to the p -values with p_i replaced by 0. Then $\hat{k}^{(i)}$ is a function of $(p_j)_{j \neq i}$ alone, and for each $k = 1, \dots, m$,

$$\{H_i \text{ is rejected and } \hat{k} = k\} = \{p_i \leq kq/m \text{ and } \hat{k}^{(i)} = k\}.$$

Proof. The first claim is clear. Let $N^{(i)}$ count the modified p -values. Then $N^{(i)}(t) = N(t)$ for $t \geq p_i$, because at such t the p -value p_i is counted in both. By the second description above, H_i is rejected and $\hat{k} = k$ iff $p_i \leq kq/m$ and $\hat{k} = k$. Assume $p_i \leq kq/m$. By Eq. 13.5.1, $\hat{k} = k$ iff $N(kq/m) \geq k$ and $N(k'q/m) < k'$ for every $k' > k$. These conditions involve N only at points $t \geq kq/m \geq p_i$, where N and $N^{(i)}$ agree, so they hold iff the same conditions hold for $N^{(i)}$, that is, iff $\hat{k}^{(i)} = k$. ■

Theorem 13.5.2 — Benjamini–Hochberg

Suppose that for each true hypothesis H_i , $i \in I_0$, the p -value p_i is independent of the other $m - 1$ p -values and satisfies $\Pr(p_i \leq u) \leq u$ for $0 \leq u \leq 1$. Then the BH procedure at level q has

$$\text{FDR} \leq \frac{m_0}{m} q \leq q,$$

with equality in the first inequality when every p_i , $i \in I_0$, is uniform on $(0, 1)$.

Proof. The false discovery proportion is $Q = \sum_{i \in I_0} 1_{\{H_i \text{ rejected}\}} / \max(\hat{k}, 1)$. A rejection forces $\hat{k} \geq 1$, so

$$\text{FDR} = \sum_{i \in I_0} \sum_{k=1}^m \frac{1}{k} \Pr(H_i \text{ rejected}, \hat{k} = k) = \sum_{i \in I_0} \sum_{k=1}^m \frac{1}{k} \Pr\left(p_i \leq \frac{kq}{m}\right) \Pr(\hat{k}^{(i)} = k),$$

by Lemma 13.5.1 and the independence of p_i from $\hat{k}^{(i)}$. Now $\Pr(p_i \leq kq/m) \leq kq/m$, so the inner sum is at most $(q/m) \sum_{k=1}^m \Pr(\hat{k}^{(i)} = k) = q/m$, because $\hat{k}^{(i)} \geq 1$ always (the replaced p -value 0 lies below the first threshold). Summing over the m_0 true hypotheses gives $\text{FDR} \leq m_0 q/m$. With uniform null p -values every inequality is an equality. ■

The false hypotheses' p -values may depend on each other in any way. With uniform null p -values the false discovery rate is exactly $\pi_0 q$, $\pi_0 = m_0/m$. With $m = 100$ independent z -tests, 90 of them null and the rest shifted by 3, 20000 simulated repetitions estimate it at 0.0439, against the exact 0.0450.

When all hypotheses are true the false discovery rate is the familywise error rate ([Proposition 13.1.1](#)), so BH is a level- q test of the global null. It rejects everything Bonferroni's and Holm's procedures at level q reject ([Exercise \(B2\)](#)). But it does *not* control the familywise error rate strongly: once many false hypotheses are rejected, the thresholds for the rest approach q .

13.5.3 Dependent tests

In a regression the t statistics share s , and their numerators are correlated unless $\mathbf{X}$ has orthogonal columns. Two results, stated here without proof, cover much of what is needed.

Positive dependence. Benjamini and Yekutieli (2001) showed that $\text{FDR} \leq m_0 q/m$ remains true if the p -values are *positively regression dependent on each one from the subset of true hypotheses* (PRDS): for every true H_i and every increasing set D of p -value vectors, $\Pr((p_1, \dots, p_m) \in D \mid p_i = u)$ is nondecreasing in u . One-sided tests based on jointly normal statistics with nonnegative correlations are an example; two-sided tests of correlated statistics need not be.

Arbitrary dependence. They also showed that under *any* dependence, BH at level q/c_m with $c_m = \sum_{i=1}^m 1/i \approx \log m + 0.577$ has false discovery rate at most $m_0 q/m$ ([Exercise \(C1\)](#)). For $m = 100$, $c_m = 5.187$, so a target of 0.05 becomes an effective level of 0.0096.

BH is routinely applied to regression t tests, and simulations, including the one below, find its false discovery rate below q . That is evidence, not a theorem; when a guarantee is essential, the Benjamini–Yekutieli version is the safe choice.

13.5.4 Adjusted p -values and q -values

As for Holm's procedure, the BH decisions for all levels at once are summarized by **adjusted p -values**

$$\tilde{p}_{(i)} = \min_{j \geq i} \min \left\{ 1, \frac{m p_{(j)}}{j} \right\}, \quad (13.5.2)$$

with $H_{(i)}$ rejected at level q iff $\tilde{p}_{(i)} \leq q$ ([Exercise \(B1\)](#)). Since BH controls the rate at $\pi_0 q$, it is conservative when many hypotheses are false. **Adaptive** procedures estimate π_0 , for instance by twice the proportion of p -values above one half, and run BH at level $q/\hat{\pi}_0$. Storey (2002) developed this approach and defined the **q -value** of a hypothesis, the smallest false discovery rate at which it would be rejected; with $\hat{\pi}_0 = 1$ the q -values are the adjusted values [Eq. 13.5.2](#). The gain is largest when π_0 is well below one.

13.5.5 Screening regression coefficients

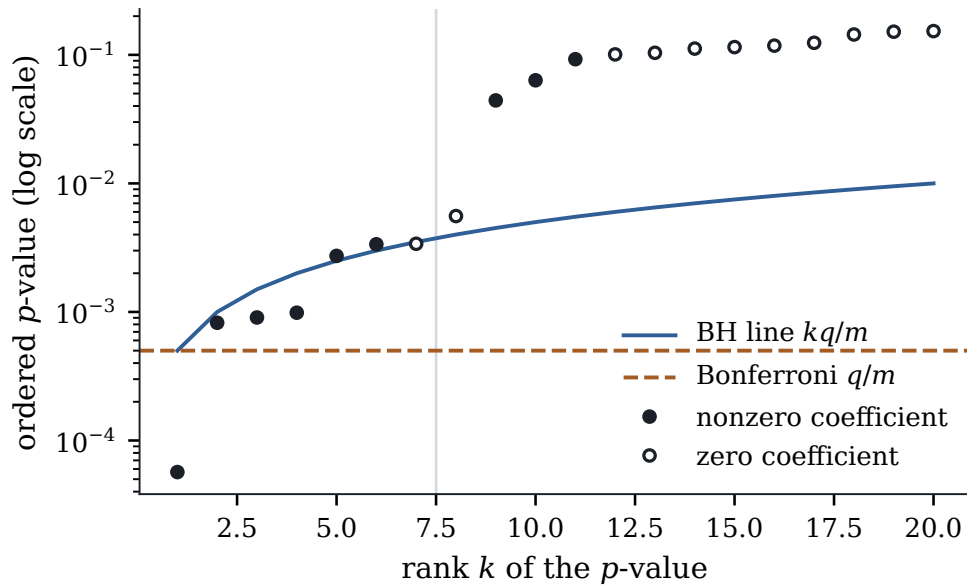


Figure 13.5.1: The twenty smallest of 100 ordered p -values (log scale), with the BH line kq/m and Bonferroni's threshold q/m , $q = 0.05$. Filled points are nonzero coefficients; BH rejects the seven left of the vertical line.

Example. (One hundred candidate regressors)

A simulated regression has $n = 250$ observations, an intercept and 100 regressors with independent standard normal entries, leaving $\nu = 149$ error degrees of freedom. The first 10 slopes are nonzero, each chosen so that the noncentrality of its t statistic is 3; the other 90 are zero. The 100 two-sided t tests are run on one data set at level 0.05:

procedure	rejections	of which false
unadjusted, $p \leq 0.05$	9	2
Bonferroni	1	0
Holm	1	0
BH, $q = 0.05$	7	1

Bonferroni's threshold is $0.05/100 = 0.0005$, and only one p -value is below it. The BH line reaches $7 \times 0.05/100 = 0.0035$ at rank 7, and the seventh smallest p -value is 0.0034 (Figure 13.5.1). Six of the seven BH discoveries are real.

```
import numpy as np
from scipy import stats

q = 0.05

rng = np.random.default_rng(20260919)
n, m = 250, 100
X = np.column_stack([np.ones(n), rng.normal(size=(n, m))])
Qx, Rx = np.linalg.qr(X)
nu = n - m - 1
c_diag = np.diag(np.linalg.inv(Rx.T @ Rx))[1:]
slopes
```

149 error degrees of freedom
[(X'X)^{-1}]_{jj} for the

```

def t_statistics(Y):
    """t statistics of the 100 slopes for each column of Y."""
    coef = np.linalg.solve(Rx, Qx.T @ Y)
    resid = Y - X @ coef
    s2 = np.sum(resid ** 2, axis=0) / nu
    return coef[1:] / np.sqrt(np.outer(c_diag, s2))

def bh(p, q):
    """Benjamini-Hochberg step-up at level q; returns a boolean rejection vector."""
    m = len(p)
    order = np.argsort(p)
    below = np.nonzero(p[order] <= q * np.arange(1, m + 1) / m)[0]
    reject = np.zeros(m, dtype=bool)
    if below.size:
        reject[order[: below[-1] + 1]] = True
    return reject

def holm(p, alpha):
    m = len(p)
    order = np.argsort(p)
    ok = np.cumprod(p[order] <= alpha / (m - np.arange(m))).astype(bool)
    reject = np.zeros(m, dtype=bool)
    reject[order] = ok
    return reject

procedures = {
    "unadjusted": lambda p: p <= q,
    "Bonferroni": lambda p: p <= q / m,
    "Holm": lambda p: holm(p, q),
    "BH": lambda p: bh(p, q),
}

```

```

m1 = 10                                # the first 10 slopes are
    nonzero
beta = np.zeros(m + 1)
beta[1:m1 + 1] = 3.0 * np.sqrt(c_diag[:m1])    # noncentrality 3 when sigma =
    1
y = X @ beta + rng.normal(size=n)
p = 2 * stats.t.sf(np.abs(t_statistics(y[:, None]))[:, 0]), nu)
for name, rule in procedures.items():
    rej = rule(p)
    print(f"{name:10s} rejects {rej.sum():3d}, of which {rej[m1:].sum()} are false
discoveries")

```

A simulation with the same design, $m_1 = 0, 5, 10, 25, 50$ nonzero slopes and 2000 responses for each, shows the average behaviour (Figure 13.5.2). With $m_1 = 10$, the false discovery rate is 0.327 without adjustment, 0.014 for Bonferroni and Holm and 0.046 for BH, and the average power is 0.845, 0.289, 0.291 and 0.476. With $m_1 = 50$, BH's power rises to 0.731 and its false discovery rate falls to 0.024, in line with $\pi_0 q$, while Bonferroni's and Holm's power stay near 0.3. BH pays with familywise errors: at least one of its rejections is false in 0.238 of the data sets with $m_1 = 10$ and 0.570 with $m_1 = 50$. With all slopes zero, its false discovery rate and familywise error rate coincide, at 0.050.

For the 21 education comparisons of Example (Holm for the education pairs), BH at $q = 0.05$ rejects 13 hypotheses: the thirteenth smallest p -value is $0.0303 \leq 13 \times 0.05/21 = 0.0310$, and no

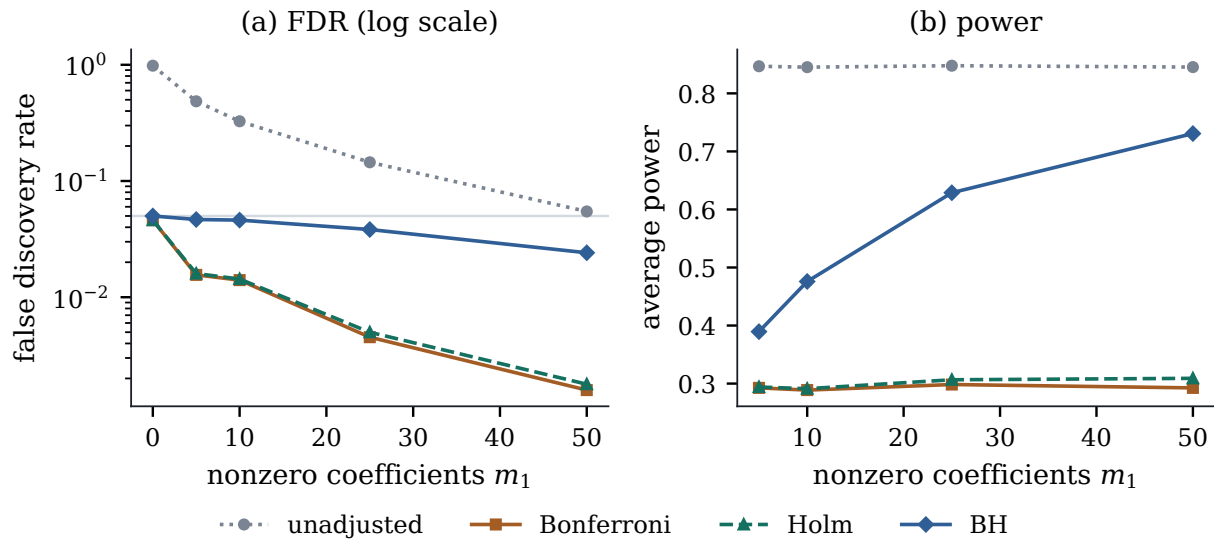


Figure 13.5.2: False discovery rate (a, log scale) and average power (b) of four procedures when m_1 of 100 regression slopes are nonzero, 2000 simulated data sets per point, nominal level 0.05. The pale line in (a) marks 0.05.

later one comes back below the line. The next section discusses how to choose between the two error rates.

13.5.6 Exercises

13.5.6.1 A. Check your understanding

Exercise A1.

Apply the BH procedure at $q = 0.05$ to the five p -values of [Exercise \(A1\)](#), and compute the adjusted p -values [Eq. 13.5.2](#).

Solution. The thresholds are 0.01, 0.02, 0.03, 0.04, 0.05. From the top: $0.2 > 0.05$, but $0.04 \leq 0.04$, so $\hat{k} = 4$ and the first four hypotheses are rejected, one more than Holm's procedure rejects. The adjusted values are $\min(5 \times 0.001, \dots) = 0.005$, then $\min(5 \times 0.011/2, 5 \times 0.012/3, 5 \times 0.04/4, 1) = 0.02, 0.02, 0.05$ and 0.2.

Exercise A2.

A report says: "Using BH at $q = 0.1$ we made 40 discoveries, so at most 4 of them are false." What is wrong with this statement? What would be a correct one?

13.5.6.2 B. Practice

Exercise B1.

Prove that BH at level q rejects $H_{(i)}$ iff the adjusted value [Eq. 13.5.2](#) satisfies $\tilde{p}_{(i)} \leq q$.

Exercise B2.

Show that the BH procedure at level q rejects every hypothesis that Holm's procedure at level q rejects. *Hint:* $q/(m - i + 1) \leq iq/m$ for $1 \leq i \leq m$.

Solution. The hint holds because $i(m - i + 1) \geq m$ for $1 \leq i \leq m$: the product of two positive integers summing to $m + 1$ is at least m . If Holm rejects $H_{(1)}, \dots, H_{(i)}$, then $p_{(i)} \leq q/(m - i + 1) \leq iq/m$, so $\hat{k} \geq i$.

Exercise B3.

With $m_0 = 2$ true hypotheses with independent uniform p -values and $m - 2$ false hypotheses whose p -values are 0, find the familywise error rate of BH at level q . Compare it with the false discovery rate given by [Theorem 13.5.2](#).

13.5.6.3 C. Going deeper**Exercise C1.**

Prove the Benjamini–Yekutieli bound: under arbitrary dependence, BH at level q/c_m has false discovery rate at most $m_0 q/m$. *Hint:* write the false discovery rate as $\sum_{i \in I_0} \sum_k \Pr(p_i \in (\frac{(j-1)q'}{m}, \frac{jq'}{m}], \hat{k} = k)/k$ summed over $j \leq k$, with $q' = q/c_m$, and exchange the order of summation.

13.6 Choosing a procedure

None of the procedures of this chapter is best in general, because they answer different questions. This section compares them on one data set, collects the comparison into a guide, and explains why several procedures common in software are not developed here.

13.6.1 Decide the family first

Every guarantee in this chapter is for a *family* of inferences, and is only as good as the honesty with which the family is defined. A short list of functions written down before the data arrive is a **planned** family. All $g(g - 1)/2$ pairwise differences form a family with its own exact method. A function chosen *because* the data make it look interesting belongs to the family of everything the analyst might have chosen, usually a whole subspace; this is **post hoc** inference, or data snooping. And a long list screened for leads worth following up calls for a different error rate altogether.

Warning. The family must be fixed before the data are examined, and reported. Choosing the adjustment after seeing the results, for example reporting Bonferroni's intervals for the three comparisons that turned out largest, reproduces the problem that adjustment was meant to solve. So does splitting one analysis into several “families” so that each is small.

13.6.2 One data set, five sets of intervals

The education example of [Example \(Education and the perceived position of a candidate\)](#) and [Example \(All pairs of education levels\)](#) makes the differences concrete. The table gives, for each method, the multiplier of the standard error for all 21 pairwise differences, the resulting half-width for two pairs with very different precision, and the number of pairs declared different at familywise level 0.05 (false discovery rate 0.05 for BH). The pair of levels 1 and 2 involves the two smallest groups, with standard error 0.420; the pair of levels 3 and 6 involves two large groups, with standard error 0.124.

method	multiplier	ratio to t	half-width, 1 vs 2	half-width, 3 vs 6	pairs declared different
unadjusted t	1.962	1.00	0.824	0.244	14
Tukey– Kramer	2.955	1.51	1.241	0.368	9
Šidák	3.039	1.55	1.276	0.378	8
Bonferroni	3.046	1.55	1.279	0.379	8
Scheffé	3.557	1.81	1.494	0.442	6
Holm					10
Benjamini– Hochberg					13

The ordering is the one the theory predicts: the method designed for pairs is shortest, the general finite-family methods are a few percent longer, and the method covering every contrast is much longer. Holm’s procedure rejects more by stepping down, and BH by controlling a weaker error rate. The Šidák row relies on Šidák’s inequality for correlated normal variables, stated without proof in [Section 13.4](#).

```
import itertools
import numpy as np
import statsmodels.api as sm
from scipy import stats
anes = sm.datasets.anes96.load_pandas().data
y = anes["ClinLR"].to_numpy()
level = anes["educ"].to_numpy().astype(int) - 1      # 0, ..., 6
g = 7
n_k = np.bincount(level).astype(float)
means = np.bincount(level, weights=y) / n_k
n = len(y)
nu = n - g                                           # error degrees of freedom
s2 = np.sum((y - means[level]) ** 2) / nu           # pooled variance estimate
s = np.sqrt(s2)
alpha = 0.05
c_sch = np.sqrt((g - 1) * stats.f.ppf(1 - alpha, g - 1, nu)) # Scheffe multiplier,
q = g - 1
pairs = list(itertools.combinations(range(g), 2))
m = len(pairs)                                     # 21 comparisons
diff = np.array([means[k] - means[l] for k, l in pairs])
se_d = np.array([s * np.sqrt(1 / n_k[k] + 1 / n_k[l]) for k, l in pairs])
t = diff / se_d
mult = {                                           # half-width = multiplier x se
    "unadjusted": stats.t.ppf(1 - alpha / 2, nu),
```

```

    "Bonferroni": stats.t.ppf(1 - alpha / (2 * m), nu),
    "Sidak": stats.t.ppf((1 + (1 - alpha) ** (1 / m)) / 2, nu),
    "Tukey-Kramer": stats.studentized_range.ppf(1 - alpha, g, nu) / np.sqrt(2),
    "Scheffe": c_sch,
}
for name, c in mult.items():
    print(f"{name:13s} multiplier {c:.3f}    pairs declared different: {np.sum(np.abs
(t) > c)}")

```

13.6.3 A decision guide

The choice can be put as a short sequence of questions.

- (1) *Is there one question?* Then use the ordinary t or F procedure of [Chapter 11](#) or [Chapter 12](#). No adjustment is needed.
- (2) *Will the functions be chosen after seeing the data?* If they lie in a known subspace, use Scheffé's intervals for that subspace, choosing the smallest subspace that contains every function one might pick ([Section 13.2](#)).
- (3) *Is the family all pairs of means?* Use Tukey's intervals if the design is balanced and the means are independent, Tukey–Kramer if the one-way layout is unbalanced, and the exact multivariate t computation otherwise.
- (4) *Is the family a short list fixed in advance?* Use Bonferroni's (or Šidák's) intervals, or Holm's tests if intervals are not needed, or the exact multivariate t value if software is at hand.
- (5) *Is the family long, and the goal a list of leads for further study?* Control the false discovery rate with the Benjamini–Hochberg procedure, or with the Benjamini–Yekutieli version if dependence among the tests is strong and of unknown form.

Where intervals are available they are preferable to tests alone, because they report the size of each difference as well as its sign ([Proposition 13.1.2](#)). Adjusted p -values, [Eq. 13.4.3](#) and [Eq. 13.5.2](#), let readers apply a different level. And a report should say how many comparisons were made in all, not only which were significant.

13.6.4 Planned versus post hoc

A post hoc comparison is chosen because it looks large, and its t statistic has the distribution of a maximum. In the education example the most extreme contrast has $t = 6.88$. Under equal means its square is not an $F(1, 937)$ variable but 6 times an $F(6, 937)$ variable ([Corollary 13.2.3](#)), and treating it as planned would overstate the evidence by exactly the amount Scheffé's multiplier corrects. The same happens at a larger scale when regressors are selected by their t statistics; Chapter 29 takes up inference after model selection.

13.6.5 Procedures not developed in this book

Several procedures widely available in software are deliberately left out. They control the familywise error rate only weakly, or not at all.

Fisher's protected LSD controls the familywise error rate only weakly, as [Example \(The protected LSD without strong control\)](#) and [Exercise \(C1\)](#) show, except with three groups ([Exercise \(B1\)](#)).

The Newman–Keuls procedure orders the means and tests the range of each set of w adjacent means against $q_\alpha(w, \nu) s / \sqrt{m}$, stepping down from $w = g$ and never testing inside a set whose range was not significant. Under the global null it rejects exactly when Tukey's test does, so its control is weak; with means in well-separated groups, each small group is tested nearly unprotected.

Example. (Newman–Keuls without strong control)

Take $g = 10$ groups of 5 observations, with the true means equal in pairs and the five pairs far apart, and $\nu = 40$. The larger ranges are always significant, so the procedure reaches every pair and tests it with $q_\alpha(2, \nu)$, the critical value of a single unadjusted comparison. In 10000 simulated data sets, the Newman–Keuls procedure declares some pair of equal means different in a proportion 0.219 of them. Five independent level-0.05 tests would err with probability 0.226; the shared s lowers the rate slightly, as [Proposition 13.4.3](#) implies. Tukey's intervals err in 0.011 of the data sets.

```
import numpy as np
from scipy import stats

alpha, g, m_rep, reps = 0.05, 10, 5, 10_000
nu = g * (m_rep - 1)
mu = np.repeat(np.arange(5) * 20.0, 2)           # pairs 0,0, 20,20, 40,40, ...

def newman_keuls(means, s, m_rep, q_crit):
    """Pairs (k, l) declared different by the Newman-Keuls step-down procedure;
    q_crit[w] is the upper-alpha point of the studentized range of w means."""
    g = len(means)
    order = np.argsort(means)
    ybar = means[order]
    se = s / np.sqrt(m_rep)
    kept = []                                     # windows found homogeneous
    different = set()
    for width in range(g, 1, -1):                # ranges of width, width-1,
        ..., 2
        crit = q_crit[width] * se
        for i in range(g - width + 1):
            j = i + width - 1
            if any(a <= i and j <= b for a, b in kept):
                continue                         # inside a homogeneous window
            if ybar[j] - ybar[i] > crit:
                different.add((min(order[i], order[j]), max(order[i], order[j])))
            else:
                kept.append((i, j))
    return different

q_crit = {w: stats.studentized_range.ppf(1 - alpha, w, nu) for w in range(2, g + 1)}
rng = np.random.default_rng(20260919)
q_tukey = stats.studentized_range.ppf(1 - alpha, g, nu)
nk_false = tukey_false = 0
for r in range(reps):
    y = mu[:, None] + rng.normal(size=(g, m_rep))
```

```

means = y.mean(axis=1)
s = np.sqrt(np.sum((y - means[:, None]) ** 2) / nu)
true_pairs = [(2 * i, 2 * i + 1) for i in range(5)]
nk = newman_keuls(means, s, m_rep, q_crit)
nk_false += any(pair in nk for pair in true_pairs)
tukey_false += any(abs(means[a] - means[b]) > q_tukey * s / np.sqrt(m_rep)
                    for a, b in true_pairs)
print(f"P(some false difference): Newman-Keuls {nk_false / reps:.3f}, "
      f"Tukey {tukey_false / reps:.3f}")

```

With $2J$ means in J separated pairs, the familywise error rate of Newman–Keuls approaches $1 - (1 - \alpha)^J$ as the degrees of freedom grow ([Exercise \(B2\)](#)). No bound below one holds uniformly in g .

Duncan’s multiple range test. Duncan’s procedure has the structure of Newman–Keuls but raises the level used for a set of w means to $1 - (1 - \alpha)^{w-1}$. Its first step, the range of all g means, is therefore a test at level $1 - (1 - \alpha)^{g-1}$, and if it rejects, at least the most extreme pair is declared different. Under the global null its familywise error rate is exactly $1 - (1 - \alpha)^{g-1}$, which is 0.370 for $g = 10$ and $\alpha = 0.05$. It does not control the rate even weakly.

These procedures buy power over Tukey’s by relaxing the error rate when some hypotheses are false. If that trade is wanted, it is better made openly, by controlling the false discovery rate, whose meaning is clear. Closed testing ([Theorem 13.4.4](#)) with range or F local tests gives step-down procedures that are more powerful than Tukey’s and still control the familywise error rate strongly; the books cited in the notes describe them.

A Bayesian response to multiplicity is a hierarchical model that treats the group means as draws from a common distribution and shrinks the estimated differences toward each other ([Exercise \(C1\)](#)). Chapter 32 develops random-effects models of this kind, and the conjugate analysis of [Theorem 7.6.2](#) is their simplest case.

13.6.6 Exercises

13.6.6.1 A. Check your understanding

Exercise A1.

Choose a procedure for each situation, and say what family its guarantee refers to: (a) a trial with three doses and a placebo, where the protocol specifies the three comparisons with placebo; (b) eight fertilizers compared in a balanced field trial, with interest in every pair; (c) 5000 genes screened for association with a trait, to select candidates for a second study; (d) a contrast suggested by a plot of the group means.

13.6.6.2 B. Practice

Exercise B1.

Show that under the global null the familywise error rate of Duncan’s procedure is $1 - (1 - \alpha)^{g-1}$ in a balanced one-way layout.

Exercise B2.

In [Example \(Newman–Keuls without strong control\)](#) with J pairs, let the separation between pairs tend to infinity. Show that the familywise error rate of Newman–Keuls converges to $1 - \Pr(|T_j| \leq q_\alpha(2, \nu)/\sqrt{2}, j = 1, \dots, J)$, where the T_j are t statistics with independent numerators and a common s . Using [Proposition 13.4.3](#), show that the limit is at most $1 - (1 - \alpha)^J$, and that it tends to this value as $\nu \rightarrow \infty$.

13.6.6.3 C. Going deeper**Exercise C1.**

In the balanced one-way layout with known σ^2 , give the group means the prior $\mu_k \stackrel{\text{iid}}{\sim} N(\mu_0, \tau^2)$, with μ_0 and τ^2 known. Find the posterior distribution of $\mu_k - \mu_l$ and compare the width and centre of its central 95% posterior interval with the unadjusted confidence interval. Explain in what sense the shrinkage addresses the multiplicity of comparisons, and in what sense it does not.

13.7 Summary and Notes

13.7.1 Summary

Idea.

1. The error rates of a family, familywise, per-family and false discovery, are ordered ([Definition 13.1.1](#), [Proposition 13.1.1](#)). Control should be strong; a gate such as the protected LSD gives only weak control ([Definition 13.1.2](#), [Proposition 13.1.3](#)). Simultaneous intervals give strong control, including of sign errors ([Proposition 13.1.2](#)).
2. Scheffé: a family of estimable functions closed under linear combination is a subspace of $\mathcal{C}(\mathbf{X})$, and its largest squared t statistic is q times an $F(q, n - r)$ variable. The intervals with multiplier $\sqrt{qF_\alpha(q, \nu)}$ are exact, even for functions chosen after seeing the data ([Theorem 13.2.2](#)); the F test rejects iff some Scheffé interval excludes its hypothesized value ([Corollary 13.2.3](#)); smaller subspaces give shorter intervals ([Section 13.2](#)).
3. Tukey: in a balanced one-way layout the intervals $\bar{y}_k - \bar{y}_l \pm q_\alpha(g, \nu)s/\sqrt{m}$ are exact for all pairs, extend to all contrasts, and are the shortest of their form ([Definition 13.3.1](#), [Theorem 13.3.2](#), [Proposition 13.3.3](#)). The Tukey–Kramer intervals [Eq. 13.3.3](#) are conservative.
4. Bonferroni needs no assumption on dependence and bounds the expected number of false rejections ([Theorem 13.4.1](#)); Šidák’s level is valid for independent estimates, and a shared s only helps ([Proposition 13.4.3](#)). Closed testing gives strong control ([Theorem 13.4.4](#)), and Holm’s step-down procedure, its Bonferroni case, rejects everything Bonferroni does and possibly more ([Theorem 13.4.5](#)).
5. The Benjamini–Hochberg procedure controls the false discovery rate at $\pi_0 q$ for independent null p -values ([Theorem 13.5.2](#)), and Benjamini and Yekutieli’s results

extend it to positively dependent (PRDS) and arbitrarily dependent p -values.

6. Fix the family first: planned lists call for Bonferroni, Holm or exact multivariate t values, all pairs for Tukey, data snooping for Scheffé, screening for the false discovery rate ([Section 13.6](#)).

13.7.2 Notes and sources

Sources for this chapter. Christensen (2020, chapter 5) treats multiple comparisons within the projection framework used here, describing Scheffé’s method through a test space and showing that the F test rejects exactly when some single-degree-of-freedom hypothesis in that space does. [Section 13.2](#) follows that view. His chapter also compares the least significant difference, Bonferroni, Tukey, Newman–Keuls and Duncan procedures and discusses the Fisherian and Neyman–Pearson views of multiple testing. Agresti (2015, section 3.5) gives a concise account of Bonferroni’s and Tukey’s methods, the Tukey–Kramer extension and the Benjamini–Hochberg procedure. The examples, simulations and the treatment of error rates, Šidák’s inequality and closed testing are this book’s own.

Scheffé and Tukey. Scheffé (1953) introduced the method for all contrasts, and his book (Scheffé 1959) develops it for general linear hypotheses and relates it to the F test and the confidence ellipsoid. Tukey’s 1953 manuscript circulated for decades before it appeared in his collected works (Tukey 1994). Kramer (1956) proposed the extension to unequal group sizes, and Hayter (1984) proved Tukey’s conjecture that these intervals are conservative. The case of estimates with general correlation, where neither form is exact, is handled by computing multivariate t probabilities; Hothorn, Bretz and Westfall (2008) describe a general framework and Bretz, Hothorn and Westfall (2010) a practical guide. Dunnett (1955) gave the exact procedure for comparisons with a control.

Bonferroni, Šidák and Holm. The name commemorates Bonferroni’s (1936) work on inequalities for probabilities of unions. Dunn (1961) popularized its use for simultaneous confidence intervals. Šidák (1967) proved the inequality for rectangular regions of the multivariate normal distribution that underlies [Proposition 13.4.3](#) for correlated estimates. Holm (1979) introduced the step-down procedure, and Marcus, Peritz and Gabriel (1976) the closure principle, of which it is a case. Hochberg (1988) proposed the step-up version, whose validity rests on the inequality of Simes (1986).

False discovery rate. Benjamini and Hochberg (1995) defined the false discovery rate and proved [Theorem 13.5.2](#). The leave-one-out proof given here is a standard later simplification. Benjamini and Yekutieli (2001) proved control under positive regression dependence and gave the version valid under arbitrary dependence. Storey (2002) introduced estimation of π_0 and the q -value, and Benjamini, Krieger and Yekutieli (2006) gave adaptive procedures with proven control.

Books. Miller (1981) is the classic account of simultaneous inference, and Hochberg and Tamhane (1987) and Hsu (1996) are comprehensive monographs. All three treat the procedures of Duncan (1955) and of Newman (1939) and Keuls (1952), and the closed step-down procedures that improve on Tukey’s. Gelman, Hill and Yajima (2012) give a Bayesian view of why hierarchical models reduce the need for multiplicity adjustments.

13.7.3 References

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Chapter 14

Correlation, Lack of Fit and Prediction Theory

So far the regressors have been fixed numbers. In most observational studies they are random, drawn along with the response. This chapter shows that the exact tests and intervals of Part III survive that change, because their conditional laws do not depend on the design, while the laws of the estimates, and the power of tests, do change. With random rows a regression is a prediction rule, and the chapter measures how well an estimated predictor predicts new cases and how much its in-sample error flatters it. It then studies the correlations that measure linear association: the multiple correlation, whose sample version R^2 is the overall F statistic in disguise; the simple correlation and Fisher's transformation; and the partial correlation, whose test is the t test of a coefficient. The last two sections test whether a mean function is adequate: pure error gives an exact test when rows are replicated, and several exact tests, none dominating the others, replace it when they are not.

What you need. The normal linear model and its sampling distributions from [Chapter 7](#) ([Theorem 7.5.1](#), [Corollary 7.5.2](#), [Corollary 7.5.3](#)), nested projections and the Frisch–Waugh–Lovell theorem from [Chapter 6](#) ([Theorem 6.5.1](#), [Theorem 6.6.2](#)), best linear prediction in the population ([Theorem 2.5.2](#), [Theorem 6.11.1](#), [Proposition 6.11.3](#)) and the multiple correlation ([Proposition 2.5.3](#)) from [Chapter 2](#) and [Chapter 6](#), normal conditional distributions and partial correlations from [Chapter 3](#) ([Theorem 3.5.2](#), [Definition 3.6.1](#)), the noncentral distributions of [Chapter 4](#) ([Theorem 4.4.4](#), [Theorem 4.1.4](#)), and R^2 and partial R^2 from [Chapter 9](#) ([Theorem 9.5.2](#), [Theorem 9.5.4](#)). The F test of a reduced model ([Chapter 11](#), [Theorem 11.1.1](#)) is the template for every lack-of-fit test.

14.0.1 Roadmap

- [Random regressors and conditional inference](#)
- [Best linear prediction](#)
- [Multiple correlation](#)
- [The correlation coefficient and Fisher's \$z\$](#)
- [Partial correlation](#)
- [Testing lack of fit with replicates](#)
- [Near replicates and nonparametric lack-of-fit tests](#)
- [Summary and notes](#)

14.1 Random regressors and conditional inference

The theory of Part III was built for a model matrix $\mathbf{X}$ of known constants. In observational data the regressors are drawn with the response, and a second sample would bring a different $\mathbf{X}$. [Section 5.1](#) already argued that the assumptions of the linear model should then be read *conditionally on $\mathbf{X}$* , and [Corollary 5.5.5](#) carried unbiasedness over to random regressors. This section does the same for the exact distribution theory. Tests and intervals stay exact; the estimates acquire heavier-tailed distributions, and power becomes a random quantity that depends on the design the data deliver.

14.1.1 The model with random regressors

Definition 14.1.1 — Normal linear model with random regressors

Let $\mathbf{X}$ be a random $n \times p$ matrix and $\mathbf{Y}$ a random n -vector. They follow the **normal linear model with random regressors** if $\text{rank}(\mathbf{X}) = p$ with probability one and, conditionally on $\mathbf{X}$,

$$\mathbf{Y} \mid \mathbf{X} \sim N_n(\mathbf{X}\boldsymbol{\beta}, \sigma^2 \mathbf{I}_n)$$

for constants $\boldsymbol{\beta} \in \mathbb{R}^p$ and $\sigma^2 > 0$. Equivalently, $\mathbf{Y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}$ with $\boldsymbol{\varepsilon} \sim N_n(\mathbf{0}, \sigma^2 \mathbf{I}_n)$ independent of $\mathbf{X}$.

The leading case has independent and identically distributed rows $(\mathbf{x}_{(i)}^\top, Y_i)$ with $Y_i = \mathbf{x}_{(i)}^\top \boldsymbol{\beta} + \varepsilon_i$, where $\varepsilon_i \sim N(0, \sigma^2)$ is independent of $\mathbf{x}_{(i)}$. Independence of the rows then makes the errors independent given the whole of $\mathbf{X}$. Jointly normal rows satisfy the definition ([Eq. 3.5.2](#)), but the regressors may equally be skewed, discrete or dependent on one another; only the errors must be normal, homoscedastic and independent of the regressors.

14.1.2 Conditioning on the design

Lemma 14.1.1 — Pivots given the design

Let S be a random vector and U any random element. Suppose there is a distribution Q such that, for almost every value u , the conditional distribution of S given $U = u$ is Q . Then $S \sim Q$, and S is independent of U .

Proof. For a Borel set B and an event $\{U \in A\}$, the tower rule gives

$$\begin{aligned} \Pr(S \in B, U \in A) &= E[\mathbf{1}\{U \in A\} \Pr(S \in B \mid U)] \\ &= E[\mathbf{1}\{U \in A\} Q(B)] = Q(B) \Pr(U \in A). \end{aligned}$$

Taking A to be the whole space gives $\Pr(S \in B) = Q(B)$, and then the display says that the joint probability factorizes, which is independence.

A statistic whose conditional law given $\mathbf{X}$ is the same for every design is a *pivot given the design*. The t and F statistics of Part III are of this kind.

Theorem 14.1.2 — Conditional inference is valid unconditionally

Assume the model of Definition 14.1.1, with $n > p$. Let $\hat{\boldsymbol{\beta}} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{Y}$ and $s^2 = \|\mathbf{Y} - \mathbf{X}\hat{\boldsymbol{\beta}}\|^2 / (n - p)$.

- (a) For every fixed $\boldsymbol{\lambda} \neq \mathbf{0}$, the statistic $T = (\boldsymbol{\lambda}^\top \hat{\boldsymbol{\beta}} - \boldsymbol{\lambda}^\top \boldsymbol{\beta}) / (s \sqrt{\boldsymbol{\lambda}^\top (\mathbf{X}^\top \mathbf{X})^{-1} \boldsymbol{\lambda}})$ has the $t(n - p)$ distribution and is independent of $\mathbf{X}$.
- (b) Let $\boldsymbol{\Lambda}$ be a fixed $p \times q$ matrix of rank q , let $\mathbf{d} \in \mathbb{R}^q$, and put

$$F = \frac{(\boldsymbol{\Lambda}^\top \hat{\boldsymbol{\beta}} - \mathbf{d})^\top [\boldsymbol{\Lambda}^\top (\mathbf{X}^\top \mathbf{X})^{-1} \boldsymbol{\Lambda}]^{-1} (\boldsymbol{\Lambda}^\top \hat{\boldsymbol{\beta}} - \mathbf{d})}{q s^2}.$$

Given $\mathbf{X}$, $F \sim F(q, n - p, \gamma(\mathbf{X}))$ with

$$\gamma(\mathbf{X}) = \frac{(\boldsymbol{\Lambda}^\top \boldsymbol{\beta} - \mathbf{d})^\top [\boldsymbol{\Lambda}^\top (\mathbf{X}^\top \mathbf{X})^{-1} \boldsymbol{\Lambda}]^{-1} (\boldsymbol{\Lambda}^\top \boldsymbol{\beta} - \mathbf{d})}{\sigma^2}. \quad (14.1.1)$$

If $\boldsymbol{\Lambda}^\top \boldsymbol{\beta} = \mathbf{d}$, then $F \sim F(q, n - p)$ unconditionally, independently of $\mathbf{X}$.

- (c) A confidence set whose coverage probability given $\mathbf{X}$ equals $1 - \alpha$ for almost every $\mathbf{X}$ has unconditional coverage $1 - \alpha$. A test whose rejection probability under the null hypothesis equals α given almost every $\mathbf{X}$ has unconditional size α .
- (d) The unconditional power of the level- α test that rejects when $F > F_\alpha(q, n - p)$ is $E[\pi(\gamma(\mathbf{X}))]$, where $\pi(\gamma) = \Pr\{F(q, n - p, \gamma) > F_\alpha(q, n - p)\}$.

Proof. Fix a value of $\mathbf{X}$ of rank p . Given this value, $\mathbf{Y}$ follows the normal linear model with a fixed model matrix, so the results of Chapter 7 apply to the conditional law given $\mathbf{X}$.

- (a) By Corollary 7.5.2 with $d = \boldsymbol{\lambda}^\top \boldsymbol{\beta}$, the conditional law of T is $t(n - p)$. It is the same for every design, so Lemma 14.1.1 gives the unconditional law and the independence.
- (b) Write $\mathbf{W} = \boldsymbol{\Lambda}^\top (\mathbf{X}^\top \mathbf{X})^{-1} \boldsymbol{\Lambda}$, which is positive definite because $\boldsymbol{\Lambda}$ has rank q . By Theorem 7.5.1(a), given $\mathbf{X}$, $\mathbf{D} = \boldsymbol{\Lambda}^\top \hat{\boldsymbol{\beta}} - \mathbf{d} \sim N_q(\boldsymbol{\Lambda}^\top \boldsymbol{\beta} - \mathbf{d}, \sigma^2 \mathbf{W})$, so by Theorem 4.3.2(b) with $\mathbf{A} = \mathbf{W}^{-1} / \sigma^2$ (for which $\mathbf{A} \cdot \sigma^2 \mathbf{W} = \mathbf{I}$ is idempotent) the quadratic form $\mathbf{D}^\top \mathbf{W}^{-1} \mathbf{D} / \sigma^2$ is $\chi^2(q, \gamma(\mathbf{X}))$. It is independent of $(n - p)s^2 / \sigma^2 \sim \chi^2(n - p)$ by Theorem 7.5.1(b) and (c). Hence the conditional law of F is $F(q, n - p, \gamma(\mathbf{X}))$ (Definition 4.1.2). Under $\boldsymbol{\Lambda}^\top \boldsymbol{\beta} = \mathbf{d}$ the noncentrality is zero for every design, and Lemma 14.1.1 applies.
- (c) Let A be the event that the set covers the true value, or that the test rejects. Then $\Pr(A) = E[\Pr(A | \mathbf{X})] = E(1 - \alpha) = 1 - \alpha$, and likewise for the size.
- (d) Given $\mathbf{X}$, the rejection probability is $\pi(\gamma(\mathbf{X}))$ by (b). Take expectations.

The statistic in (b) is the F statistic of the general linear hypothesis ([Theorem 11.3.2](#)) for a full-rank design. So every test of [Chapter 11](#) and every interval of [Chapter 12](#) that is exact for fixed $\mathbf{X}$ is exact for random $\mathbf{X}$ too, whatever the distribution of the regressors, even without finite moments.

Example. (Skewed regressors)

Take $n = 10$ cases with an intercept and two regressors drawn afresh for every sample, one exponential and one lognormal, and $E(Y) = 1 + 0 \cdot x_1 + 0.5x_2$. The t test of $\beta_1 = 0$ at level 0.05 uses the $t(7)$ quantile. Over 100000 samples with standard normal errors independent of the regressors, the test rejects in a proportion 0.0508 of them, as [Theorem 14.1.2](#) requires; panel (b) of [Figure 14.1.1](#) shows the simulated statistics against the $t(7)$ density. If instead the error of each case is multiplied by its own value of x_1 , so that the spread depends on the regressor, the same test rejects in a proportion 0.2798. The conditional model no longer holds, and nothing in the theorem protects the test.

```
import numpy as np
from scipy import stats
reps = 100_000

rng = np.random.default_rng(1402)
n2, p2 = 10, 3
beta = np.array([1.0, 0.0, 0.5])          # H0: beta_1 = 0 is true
tq = stats.t.ppf(0.975, n2 - p2)

def t_stats(errors_from):
    """t statistics for beta_1 = 0 over many samples with skewed random regressors.
    """
    out = np.empty(reps)
    for s in range(reps):
        X = np.column_stack([np.ones(n2), rng.exponential(size=n2), rng.lognormal(
            size=n2)])
        y = X @ beta + errors_from(X)
        XtX_inv = np.linalg.inv(X.T @ X)
        b = XtX_inv @ X.T @ y
        s2 = np.sum((y - X @ b) ** 2) / (n2 - p2)
        out[s] = b[1] / np.sqrt(s2 * XtX_inv[1, 1])
    return out

t_indep = t_stats(lambda X: rng.normal(size=n2))          # errors independent of X
t_hetero = t_stats(lambda X: X[:, 1] * rng.normal(size=n2)) # spread grows with x_1
for label, t in [("independent errors", t_indep), ("spread depends on X", t_hetero)]:
    print(f"{label:20s} rejection rate {np.mean(np.abs(t) > tq):.4f}")
```

Warning. The conditional model can fail in ways no amount of data repairs. If the error variance depends on the regressors, the t and F statistics lose their exact laws ([Chapter 21](#)). If the regressors are correlated with the errors, as with measurement error ([Chapter 24](#)), an omitted common cause ([Chapter 25](#)) or a lagged response, least squares estimates something other than β . And if the mean is not linear, the population projection of [Section 6.11](#) is still estimated consistently, but its errors depend on the regressors and the usual standard errors are wrong.

14.1.3 What changes: the law of the estimates

Pivots keep their laws, but estimates do not. Given the design, $\hat{\beta} \sim N_p(\beta, \sigma^2(\mathbf{X}^\top \mathbf{X})^{-1})$. Unconditionally $\hat{\beta}$ is a mixture of normal distributions over the random covariance $\sigma^2(\mathbf{X}^\top \mathbf{X})^{-1}$. Its mean is still β , and its covariance is $\sigma^2 E[(\mathbf{X}^\top \mathbf{X})^{-1}]$ by [Corollary 5.5.5](#), but it is not normal. For a single normal regressor the mixture can be written down exactly. We need one moment of the chi-squared distribution.

Lemma 14.1.3 — Mean of an inverse chi-squared variable

If $V \sim \chi^2(\nu)$ with $\nu > 2$, then $E(1/V) = 1/(\nu - 2)$, the fact used without proof for [Eq. 9.6.2](#). For $\nu \leq 2$ the mean is infinite.

Proof. With the density [Eq. 4.1.1](#),

$$E\left(\frac{1}{V}\right) = \int_0^\infty \frac{v^{\nu/2-2} e^{-v/2}}{2^{\nu/2} \Gamma(\nu/2)} dv = \frac{2^{\nu/2-1} \Gamma(\nu/2 - 1)}{2^{\nu/2} \Gamma(\nu/2)} = \frac{1}{2(\nu/2 - 1)} = \frac{1}{\nu - 2},$$

using $\Gamma(a + 1) = a\Gamma(a)$. The integral converges at zero iff $\nu/2 - 2 > -1$, that is, $\nu > 2$. ■

Proposition 14.1.4 — The slope under a random normal regressor

Let (X_i, Y_i) , $i = 1, \dots, n$ with $n \geq 3$, be independent draws from a bivariate normal distribution with $\text{Var}(X_i) = \sigma_X^2$, regression slope β_1 and conditional variance $\sigma^2 = \text{Var}(Y_i | X_i)$. Let $\hat{\beta}_1$ be the least squares slope. Then

$$\frac{\sigma_X \sqrt{n-1}}{\sigma} (\hat{\beta}_1 - \beta_1) \sim t(n-1),$$

so $\hat{\beta}_1$ has heavier tails than any normal distribution, and for $n > 3$

$$\text{Var}(\hat{\beta}_1) = \frac{\sigma^2}{\sigma_X^2 (n-3)}. \quad (14.1.2)$$

Proof. Given the X values, $\hat{\beta}_1 \sim N(\beta_1, \sigma^2/S_{xx})$ with $S_{xx} = \sum_i (X_i - \bar{X})^2$ ([Theorem 7.5.1\(a\)](#) and [Corollary 5.5.4](#)). So $Z = \sqrt{S_{xx}}(\hat{\beta}_1 - \beta_1)/\sigma$ is standard normal given every design, and by [Lemma 14.1.1](#) it is standard normal and independent of the X values, hence of $V = S_{xx}/\sigma_X^2$. The X_i are independent $N(\mu_X, \sigma_X^2)$, so $V \sim \chi^2(n-1)$ by [Corollary 4.4.5\(b\)](#). Therefore

$$\frac{\sigma_X \sqrt{n-1}}{\sigma} (\hat{\beta}_1 - \beta_1) = \frac{Z}{\sqrt{V/(n-1)}} \sim t(n-1)$$

by [Definition 4.1.3](#). For the variance, $E(Z^2/V) = E(Z^2) E(1/V) = 1/(n-3)$ by independence and [Lemma 14.1.3](#), so $\text{Var}(\hat{\beta}_1) = \sigma^2 \sigma_X^{-2}/(n-3)$. ■

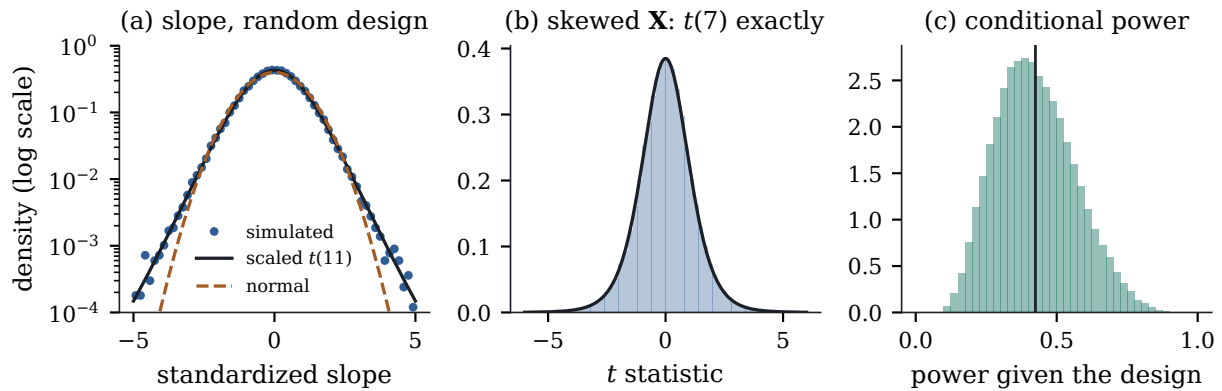


Figure 14.1.1: Random regressors. (a) The standardized slope with a random normal regressor ($n = 12$) follows the scaled $t(11)$ law of Proposition 14.1.4. (b) With skewed regressors the t statistic is exactly $t(7)$. (c) Conditional power varies with the design; the line marks the unconditional power.

The naive guess, the fixed-design variance σ^2/S_{xx} with S_{xx} replaced by its mean $(n-1)\sigma_X^2$, is too small by the factor $(n-3)/(n-1)$. This is Jensen's inequality at work, as in Example (One slope, two sampling schemes): designs that happen to bunch the regressor values together cost more precision than spread-out designs gain. Section 14.2 extends Eq. 14.1.2 to several regressors.

Example. (A heavy-tailed slope)

With $n = 12$, $\sigma_X = 2$, $\sigma = 1$ and slope 0.5, the variance of the slope over 100000 simulated samples, each with its own design, is 0.02751, against 0.02778 from Eq. 14.1.2 and 0.02273 from the naive guess. The excess kurtosis of the simulated slopes is 0.85, close to the value 0.86 of the $t(11)$ distribution. Panel (a) of Figure 14.1.1 shows the tails on a logarithmic scale. The scaled $t(11)$ density follows the simulated slopes, and the normal density with the same variance falls away too fast.

```
rng = np.random.default_rng(1401)
n, beta1, sigma, sigma_x = 12, 0.5, 1.0, 2.0
x = sigma_x * rng.normal(size=(reps, n))           # a new design for every sample
y = 1.0 + beta1 * x + sigma * rng.normal(size=(reps, n))
xc = x - x.mean(axis=1, keepdims=True)
Sxx = np.sum(xc ** 2, axis=1)
b1 = np.sum(xc * y, axis=1) / Sxx                  # least squares slope, one per
sample

scale = sigma / (sigma_x * np.sqrt(n - 1))          # b1 - beta1 = scale * t(n-1)
print(f"variance of the slope   {b1.var():.5f}")
print(f"sigma^2/(sigma_x^2 (n-3)) {sigma**2 / (sigma_x**2 * (n - 3)):.5f}")
print(f"sigma^2/E(Sxx)           {sigma**2 / (sigma_x**2 * (n - 1)):.5f}")
print("KS p-value against the scaled t(n-1):",
      round(stats.kstest((b1 - beta1) / scale, "t", args=(n - 1,)).pvalue, 3))
```

Which of the two distributions should a report describe? For the data in hand, the conditional one. The design is *ancillary*: its distribution does not involve (β, σ^2) , and it tells us how precise this particular sample is. A slope estimated from widely spread regressor values deserves a shorter interval than one estimated from bunched values, which the conditional standard error

$s/\sqrt{S_{xx}}$ delivers, and by [Theorem 14.1.2](#) the interval is also exact on average over designs. The unconditional law answers questions asked before the data arrive, such as the power of a planned study.

14.1.4 Power depends on the design

Part (d) of [Theorem 14.1.2](#) says that power, unlike size, is not free of the design. For a single normal regressor the noncentrality of the test of a zero slope is $\gamma(\mathbf{X}) = \beta_1^2 S_{xx} / \sigma^2$, which is proportional to a $\chi^2(n-1)$ variable.

Example. (How much power has this design?)

Take $n = 12$, $\sigma = \sigma_X = 1$ and slope 0.6. The test of a zero slope at level 0.05 has conditional power between 0.321 and 0.518 for the middle half of the designs, with median 0.414 (panel (c) of [Figure 14.1.1](#)). The unconditional power, the average over designs, is 0.425; a direct simulation of the whole procedure gives 0.426. Plugging in the average design, $S_{xx} = n - 1$, gives 0.436, which overstates the power a study should expect.

```
rng = np.random.default_rng(1403)
n3, b3, df3 = 12, 0.6, 10                                # slope 0.6, sigma = sigma_x =
1                                                         #
Sxx3 = stats.chi2.rvs(n3 - 1, size=reps, random_state=rng) # Sxx / sigma_x^2 ~ chi
~2(n-1)
gamma = b3 ** 2 * Sxx3                                    # noncentrality given the
design
fq = stats.f.ppf(0.95, 1, df3)
cond_power = stats.ncf.sf(fq, 1, df3, gamma)              # power of the F = t^2 test
given X
print(f"conditional power: quartiles {np.percentile(cond_power, [25, 50, 75]).round
(3)}")
print(f"unconditional power {cond_power.mean():.3f}")
print(f"power at the average design {stats.ncf.sf(fq, 1, df3, b3**2 * (n3 - 1)):.3f}
")
```

14.1.5 Exercises

14.1.5.1 A. Check your understanding

Exercise A1.

In the setting of [Proposition 14.1.4](#) take $n = 20$, $\sigma = 2$ and $\sigma_X = 1$. Compute the unconditional variance of $\hat{\beta}_1$ and compare it with $\sigma^2 / E(S_{xx})$. Which quantile would you use for a 95% interval for β_1 computed from one sample?

Solution. $\text{Var}(\hat{\beta}_1) = 4/17 = 0.235$, against $4/19 = 0.211$ for the naive value. For an interval from one sample, use the conditional standard error $s/\sqrt{S_{xx}}$ and the $t(18)$ quantile, as for a fixed design; by [Theorem 14.1.2](#) its coverage is exactly 95% over random designs too. The unconditional $t(19)$ law of the proposition involves the unknown σ/σ_X

and is not used for intervals.

Exercise A2.

Under [Definition 14.1.1](#), which of the following hold unconditionally: (i) $E(s^2) = \sigma^2$; (ii) $\text{Cov}(\hat{\boldsymbol{\beta}}) = \sigma^2(\mathbf{X}^\top \mathbf{X})^{-1}$; (iii) $(n-p)s^2/\sigma^2 \sim \chi^2(n-p)$, independent of $\mathbf{X}$; (iv) the 95% confidence ellipsoid for $\boldsymbol{\beta}$ has coverage 0.95?

14.1.5.2 B. Practice

Exercise B1.

In the setting of [Proposition 14.1.4](#), let x_0 be a fixed value and $\hat{\mu}_0 = \hat{\beta}_0 + \hat{\beta}_1 x_0$ the fitted mean there. Show that

$$\text{Var}(\hat{\mu}_0) = \sigma^2 \left[\frac{1}{n} + \frac{(x_0 - \mu_X)^2 + \sigma_X^2/n}{\sigma_X^2(n-3)} \right].$$

Hint: $\bar{X}$ and S_{xx} are independent.

Solution. Given the design, $\hat{\mu}_0$ is unbiased with variance $\sigma^2[1/n + (x_0 - \bar{X})^2/S_{xx}]$ ([Eq. 5.5.2](#)). By [Proposition 2.5.4](#), the unconditional variance is the expectation of this, since the conditional mean $\beta_0 + \beta_1 x_0$ is constant. By [Corollary 4.4.5](#), $\bar{X} \sim N(\mu_X, \sigma_X^2/n)$ is independent of $S_{xx} = \sigma_X^2 V$ with $V \sim \chi^2(n-1)$. Hence $E[(x_0 - \bar{X})^2/S_{xx}] = E(x_0 - \bar{X})^2 E(1/V)/\sigma_X^2 = [(x_0 - \mu_X)^2 + \sigma_X^2/n]/[\sigma_X^2(n-3)]$ by [Lemma 14.1.3](#).

Exercise B2.

Suppose a pilot sample is used to choose the regressor values of the main study, so that the distribution of $\mathbf{X}$ in the main study depends on $\boldsymbol{\beta}$. Assume that, given $\mathbf{X}$, the main-study responses follow the normal linear model with errors independent of the pilot. Show that the t test of [Theorem 14.1.2\(a\)](#), computed from the main study alone, still has exact size. Why is the design no longer ancillary, and what information about $\boldsymbol{\beta}$ does the t test ignore?

14.1.5.3 C. Going deeper

Exercise C1.

In the setting of [Proposition 14.1.4](#), with σ known, consider the interval $\hat{\beta}_1 \pm 1.96 \sigma / \sqrt{(n-1)\sigma_X^2}$, which uses the average design instead of the observed one. Show that its coverage given the design is $2\Phi(1.96\sqrt{V/(n-1)}) - 1$ with $V \sim \chi^2(n-1)$, and that its unconditional coverage is below 0.95. Which designs make it badly wrong, and in which direction?

14.2 Best linear prediction

With random rows, a regression is a rule for predicting the response of a new case from its regressors. The population theory is in place: the conditional mean $m(\mathbf{X}) = E(Y | \mathbf{X})$ is the best predictor ([Proposition 6.11.2](#)), the best *linear* predictor $L(Y | \mathbf{X}) = \alpha^* + \boldsymbol{\beta}^{*\top} \mathbf{X}$ is a projection ([Theorem 6.11.1](#), [Theorem 2.5.2](#)), the two coincide for jointly normal variables ([Proposition 3.5.3](#)), and least squares estimates L consistently ([Proposition 6.11.3](#)). This section asks the finite-sample questions: how large is the error of the estimated predictor on a new case, and how misleading is its error on the data used to fit it?

14.2.1 Characterizing the best predictor

The next result gives a test for whether a predictor is the best one, and a correlation that measures how well the best predictor can do.

Proposition 14.2.1 — The best predictor and the correlation ratio

Let $EY^2 < \infty$, $\text{Var}(Y) > 0$, and $m(\mathbf{X}) = E(Y | \mathbf{X})$.

- (a) A predictor $g(\mathbf{X})$ with $Eg(\mathbf{X})^2 < \infty$ equals $m(\mathbf{X})$ with probability one iff $Eg(\mathbf{X}) = EY$ and $\text{Cov}(h(\mathbf{X}), Y - g(\mathbf{X})) = 0$ for every square-integrable h .
- (b) For every such g with $\text{Var } g(\mathbf{X}) > 0$,

$$\text{corr}(Y, g(\mathbf{X}))^2 \leq \eta^2 = \frac{\text{Var } m(\mathbf{X})}{\text{Var } Y}, \quad (14.2.1)$$

with equality for $g = m$ when $\text{Var } m(\mathbf{X}) > 0$. The **correlation ratio** η^2 exceeds the squared multiple correlation $\rho_{Y \cdot X}^2$ of [Proposition 2.5.3](#) by $E(m(\mathbf{X}) - L(Y | \mathbf{X}))^2 / \text{Var } Y$, so $\eta^2 = \rho_{Y \cdot X}^2$ iff m is affine.

Proof.

- (a) If $g = m$, then $Em(\mathbf{X}) = EY$ by the tower rule, and $E[(Y - m)h] = 0$ by [Proposition 6.11.2\(a\)](#), so the covariance is zero. Conversely, let $d = g - m$. Then $Ed = 0$ and, taking $h = d$, $\text{Var } d = \text{Cov}(d, Y - m) - \text{Cov}(d, Y - g) = 0 - 0$, so $d = 0$ with probability one.
- (b) By (a) with $h = g$, $\text{Cov}(Y, g) = \text{Cov}(m, g)$. Cauchy–Schwarz for covariances gives $\text{Cov}(m, g)^2 \leq \text{Var } m \text{ Var } g$, hence $\text{corr}(Y, g)^2 = \text{Cov}(m, g)^2 / (\text{Var } Y \text{ Var } g) \leq \text{Var } m / \text{Var } Y$. For $g = m$ the covariance is $\text{Var } m$ and equality holds. For the last claim write $L = L(Y | \mathbf{X}) = L(m | \mathbf{X})$ ([Proposition 6.11.2\(b\)](#)). The error $m - L$ has mean zero and is uncorrelated with the affine function L , so $\text{Var } m = \text{Var } L + E(m - L)^2$. Divide by $\text{Var } Y$ and use $\text{Var } L / \text{Var } Y = \rho_{Y \cdot X}^2$ ([Eq. 6.11.1](#)). ■

Part (a) is the population basis of residual plots. If a fitted model reproduces the conditional mean, its residuals are uncorrelated with *every* function of the regressors, not only with the regressors themselves. A residual plot that shows a trend against x_j^2 , against a product $x_j x_k$ or

against the fitted values is evidence that the fit is not m . Sections 14.6 and 14.7 turn this into tests. Part (b) says that no transformation of the regressors predicts better, in the correlation sense, than the conditional mean. The gap $\eta^2 - \rho_{Y \cdot X}^2$ is the share of the variance of Y that a linear predictor leaves on the table.

14.2.2 The error of an estimated predictor

In practice α^* and β^* are replaced by least squares estimates from a training sample $(\mathbf{X}_i, Y_i)$, $i = 1, \dots, n$. By centring and Theorem 6.6.2 these are the plug-in estimates $\hat{\beta} = \mathbf{S}_{XX}^{-1} \mathbf{s}_{XY}$ and $\hat{\alpha} = \bar{Y} - \hat{\beta}^\top \bar{\mathbf{X}}$, built from the sample means and covariances. Their performance is judged on a new case $(\mathbf{X}_0, Y_0)$ drawn from the same distribution, independently of the training data.

Proposition 14.2.2 — Excess error of a fitted linear predictor

Let $(\mathbf{X}_0, Y_0)$ have finite second moments, $\text{Cov}(\mathbf{X}_0) = \Sigma_{XX}$, mean μ_X , and best linear predictor $L_0 = \alpha^* + \beta^{*\top} \mathbf{X}_0$ with error variance $\sigma_*^2 = E(Y_0 - L_0)^2$. Let $\hat{Y}_0 = a + \mathbf{b}^\top \mathbf{X}_0$, where $(a, \mathbf{b})$ depend only on training data independent of $(\mathbf{X}_0, Y_0)$. Then, given the training data,

$$E[(Y_0 - \hat{Y}_0)^2 \mid \text{training data}] = \sigma_*^2 + (\alpha^* - a + (\beta^* - \mathbf{b})^\top \mu_X)^2 + (\beta^* - \mathbf{b})^\top \Sigma_{XX} (\beta^* - \mathbf{b}).$$

Proof. Condition on the training data, so that a and $\mathbf{b}$ are constants. Write $Y_0 - \hat{Y}_0 = (Y_0 - L_0) + (L_0 - \hat{Y}_0)$. The second term is an affine function of $\mathbf{X}_0$, and by Theorem 6.11.1 the first has mean zero and is uncorrelated with every affine function of $\mathbf{X}_0$. So the cross term vanishes, and the mean square of the affine term is its squared mean plus its variance $(\beta^* - \mathbf{b})^\top \Sigma_{XX} (\beta^* - \mathbf{b})$. ■

The first term is the error of the best linear predictor, which no amount of data can remove. The other two are the price of estimation, and by Proposition 6.11.3 they vanish as $n \rightarrow \infty$ whatever the joint distribution. To see how large they are for moderate n , we need the distribution of the regressors. Under joint normality the calculation can be done exactly. It uses one new distribution.

Definition 14.2.1 — Wishart distribution

Let the rows of the $m \times k$ matrix $\mathbf{Z}$ be independent $N_k(\mathbf{0}, \Sigma)$ vectors. The law of $\mathbf{W} = \mathbf{Z}^\top \mathbf{Z} = \sum_{i=1}^m \mathbf{z}_i \mathbf{z}_i^\top$ is the **Wishart distribution** $W_k(m, \Sigma)$ with m degrees of freedom. For $k = 1$, $W_1(m, \sigma^2)$ is the law of $\sigma^2 \chi^2(m)$.

Lemma 14.2.3 — Centred cross-products

Let $\mathbf{X}_1, \dots, \mathbf{X}_n$ ($n \geq 2$) be independent $N_k(\mu, \Sigma)$ vectors with mean $\bar{\mathbf{X}}$, and let $\mathbf{W} = \sum_i (\mathbf{X}_i - \bar{\mathbf{X}})(\mathbf{X}_i - \bar{\mathbf{X}})^\top$. Then $\mathbf{W} \sim W_k(n-1, \Sigma)$, and $\mathbf{W}$ is independent of $\bar{\mathbf{X}}$.

Proof. Let $\mathbf{X}_1$ be the $n \times k$ matrix with rows $\mathbf{X}_i^\top$. Write $\mathbf{X}_1 = \mathbf{1}\boldsymbol{\mu}^\top + \mathbf{Z}_0\boldsymbol{\Sigma}^{1/2}$, where $\boldsymbol{\Sigma}^{1/2}$ is the symmetric square root (Theorem 1.7.6) and $\mathbf{Z}_0$ has independent standard normal entries. Let $\mathbf{O}$ be an orthogonal $n \times n$ matrix with first row $n^{-1/2}\mathbf{1}^\top$, for instance the Helmert matrix of Example (Helmert matrices), and let $\mathbf{K}$ consist of its last $n-1$ rows. From $\mathbf{O}^\top\mathbf{O} = \mathbf{I}$, $\mathbf{K}^\top\mathbf{K} = \mathbf{I} - n^{-1}\mathbf{1}\mathbf{1}^\top$, so $\mathbf{W} = \mathbf{X}_1^\top\mathbf{K}^\top\mathbf{K}\mathbf{X}_1 = (\mathbf{K}\mathbf{X}_1)^\top(\mathbf{K}\mathbf{X}_1)$, and $\mathbf{K}\mathbf{1} = \mathbf{0}$ gives $\mathbf{K}\mathbf{X}_1 = (\mathbf{K}\mathbf{Z}_0)\boldsymbol{\Sigma}^{1/2}$.

Each column of $\mathbf{O}\mathbf{Z}_0$ is $N_n(\mathbf{0}, \mathbf{O}\mathbf{O}^\top) = N_n(\mathbf{0}, \mathbf{I})$ by Theorem 3.3.1, and the columns are independent, so $\mathbf{O}\mathbf{Z}_0$ again has independent standard normal entries. Its first row is $\sqrt{n}(\bar{\mathbf{X}} - \boldsymbol{\mu})^\top\boldsymbol{\Sigma}^{-1/2}$, and its other rows form $\mathbf{K}\mathbf{Z}_0$. Hence the $n-1$ rows of $\mathbf{K}\mathbf{X}_1$ are independent $N_k(\mathbf{0}, \boldsymbol{\Sigma})$ vectors, independent of $\bar{\mathbf{X}}$, and $\mathbf{W} \sim W_k(n-1, \boldsymbol{\Sigma})$. ■

Lemma 14.2.4 — Mean of an inverse Wishart matrix

Let $\mathbf{W} \sim W_k(m, \boldsymbol{\Sigma})$ with $\boldsymbol{\Sigma}$ positive definite and $m \geq k+2$. Then $\mathbf{W}$ is nonsingular with probability one, and

$$\mathbb{E}(\mathbf{W}^{-1}) = \frac{\boldsymbol{\Sigma}^{-1}}{m - k - 1}.$$

Proof. Reduction. Write $\mathbf{Z} = \mathbf{Z}_0\boldsymbol{\Sigma}^{1/2}$ with $\mathbf{Z}_0$ as in the previous proof. Then $\mathbf{W} = \boldsymbol{\Sigma}^{1/2}\mathbf{W}_0\boldsymbol{\Sigma}^{1/2}$ with $\mathbf{W}_0 = \mathbf{Z}_0^\top\mathbf{Z}_0 \sim W_k(m, \mathbf{I})$, and $\mathbf{W}^{-1} = \boldsymbol{\Sigma}^{-1/2}\mathbf{W}_0^{-1}\boldsymbol{\Sigma}^{-1/2}$. It suffices to show $\mathbb{E}(\mathbf{W}_0^{-1}) = \mathbf{I}/(m-k-1)$.

Nonsingularity. $\mathbf{W}_0$ is singular iff some column $\mathbf{z}_j$ of $\mathbf{Z}_0$ lies in the span of the others. Given the other columns, that span has dimension at most $k-1 < m$, and $\mathbf{z}_j \sim N_m(\mathbf{0}, \mathbf{I})$, which has a density, falls in it with probability zero.

Diagonal entries. Partition $\mathbf{Z}_0 = [\mathbf{z}_1, \mathbf{Z}_2]$. By Theorem 1.6.4, the $(1,1)$ entry of $\mathbf{W}_0^{-1}$ is $1/(\mathbf{z}_1^\top\mathbf{z}_1 - \mathbf{z}_1^\top\mathbf{Z}_2(\mathbf{Z}_2^\top\mathbf{Z}_2)^{-1}\mathbf{Z}_2^\top\mathbf{z}_1) = 1/\mathbf{z}_1^\top(\mathbf{I} - \mathbf{P}_2)\mathbf{z}_1$, where $\mathbf{P}_2$ projects onto $\mathcal{C}(\mathbf{Z}_2)$, of dimension $k-1$. Given $\mathbf{Z}_2$, $\mathbf{z}_1$ is still $N_m(\mathbf{0}, \mathbf{I})$, so $\mathbf{z}_1^\top(\mathbf{I} - \mathbf{P}_2)\mathbf{z}_1 \sim \chi^2(m-k+1)$ by Theorem 4.3.2(a). This law does not depend on $\mathbf{Z}_2$, so it holds unconditionally (Lemma 14.1.1), and Lemma 14.1.3 gives mean $1/(m-k-1)$, finite because $m-k+1 > 2$. The same argument applies to every diagonal entry.

Off-diagonal entries. Let $\mathbf{D}_i$ be the identity with its i th diagonal entry replaced by -1 . Then $\mathbf{Z}_0\mathbf{D}_i$ has the same law as $\mathbf{Z}_0$, so $\mathbf{D}_i\mathbf{W}_0^{-1}\mathbf{D}_i = (\mathbf{D}_i\mathbf{W}_0\mathbf{D}_i)^{-1}$ has the same law as $\mathbf{W}_0^{-1}$. The mean of an off-diagonal entry (i,j) therefore equals its own negative, and is zero. It exists because $|w^{ij}| \leq \frac{1}{2}(w^{ii} + w^{jj})$ for the entries of the positive definite $\mathbf{W}_0^{-1}$. ■

With these two lemmas the random-design analogue of Theorem 5.5.1 follows at once.

Corollary 14.2.5 — Covariance of the slopes under normal regressors

Let the rows $(\mathbf{X}_i^\top, Y_i)$ be independent draws from a $(k+1)$ -variate normal distribution with $\boldsymbol{\Sigma}_{XX}$ positive definite and conditional variance $\sigma^2 = \text{Var}(Y | \mathbf{X})$, and let $\hat{\boldsymbol{\beta}}$ be the

least squares slopes from a fit with an intercept. If $n \geq k + 3$,

$$\mathbb{E}(\hat{\boldsymbol{\beta}}) = \boldsymbol{\beta}, \quad \text{Cov}(\hat{\boldsymbol{\beta}}) = \frac{\sigma^2}{n - k - 2} \boldsymbol{\Sigma}_{XX}^{-1}.$$

Proof. Given the regressors, the slopes are unbiased with covariance $\sigma^2 \mathbf{W}^{-1}$, where $\mathbf{W}$ is the centred cross-product matrix (Eq. 6.6.1 with $\mathbf{X}_2 = \mathbf{1}$). By Lemma 14.2.3 $\mathbf{W} \sim W_k(n - 1, \boldsymbol{\Sigma}_{XX})$, and by Proposition 2.5.4 and Lemma 14.2.4 the unconditional covariance is $\sigma^2 \mathbb{E}(\mathbf{W}^{-1}) = \sigma^2 \boldsymbol{\Sigma}_{XX}^{-1} / (n - k - 2)$. ■

For $k = 1$ this is Eq. 14.1.2. The factor $1/(n - k - 2)$, against $1/(n - 1)$ for a fixed design with the average spread, grows quickly with k .

Theorem 14.2.6 — Prediction error under normal regressors

In the setting of Corollary 14.2.5, let $(\mathbf{X}_0, Y_0)$ be a further independent draw and $\hat{Y}_0 = \hat{\alpha} + \hat{\boldsymbol{\beta}}^\top \mathbf{X}_0$ the fitted predictor. If $n \geq k + 3$, then

$$\mathbb{E}(Y_0 - \hat{Y}_0)^2 = \sigma^2 \left(1 + \frac{1}{n}\right) \frac{n - 2}{n - k - 2}. \quad (14.2.2)$$

Proof. Write $\hat{Y}_0 = \bar{Y} + \hat{\boldsymbol{\beta}}^\top (\mathbf{X}_0 - \bar{\mathbf{X}})$, and use the model $Y_i = \alpha + \boldsymbol{\beta}^\top \mathbf{X}_i + e_i$ of Eq. 3.5.2 for $i = 0, \dots, n$. Since $\bar{Y} = \alpha + \boldsymbol{\beta}^\top \bar{\mathbf{X}} + \bar{e}$,

$$Y_0 - \hat{Y}_0 = e_0 - \bar{e} - (\hat{\boldsymbol{\beta}} - \boldsymbol{\beta})^\top (\mathbf{X}_0 - \bar{\mathbf{X}}).$$

Condition on $\mathbf{X}_0, \dots, \mathbf{X}_n$. Then e_0 , $\bar{e}$ and $\hat{\boldsymbol{\beta}} - \boldsymbol{\beta} = \mathbf{W}^{-1} \sum_i (\mathbf{X}_i - \bar{\mathbf{X}}) e_i$ are normal with mean zero, and they are uncorrelated: e_0 is independent of the training errors, and $\text{Cov}(\bar{e}, \hat{\boldsymbol{\beta}}) = \sigma^2 n^{-1} \mathbf{W}^{-1} \sum_i (\mathbf{X}_i - \bar{\mathbf{X}}) = \mathbf{0}$. So the conditional mean square is

$$\sigma^2 \left[1 + \frac{1}{n} + (\mathbf{X}_0 - \bar{\mathbf{X}})^\top \mathbf{W}^{-1} (\mathbf{X}_0 - \bar{\mathbf{X}}) \right].$$

Now $\mathbf{u} = \mathbf{X}_0 - \bar{\mathbf{X}} \sim N_k(\mathbf{0}, (1 + 1/n) \boldsymbol{\Sigma}_{XX})$, and $\mathbf{u}$ is independent of $\mathbf{W}$, because $\mathbf{X}_0$ is independent of the training sample and $\bar{\mathbf{X}}$ is independent of $\mathbf{W}$ (Lemma 14.2.3). Hence $\mathbb{E}(\mathbf{u}^\top \mathbf{W}^{-1} \mathbf{u}) = \text{tr}[\mathbb{E}(\mathbf{W}^{-1}) \mathbb{E}(\mathbf{u} \mathbf{u}^\top)] = (1 + 1/n) k / (n - k - 2)$ by Lemma 14.2.4, and

$$\mathbb{E}(Y_0 - \hat{Y}_0)^2 = \sigma^2 \left(1 + \frac{1}{n}\right) \left(1 + \frac{k}{n - k - 2}\right) = \sigma^2 \left(1 + \frac{1}{n}\right) \frac{n - 2}{n - k - 2}. \quad \blacksquare$$

The conditional mean square in the proof, $\sigma^2 [1 + 1/n + (\mathbf{X}_0 - \bar{\mathbf{X}})^\top \mathbf{W}^{-1} (\mathbf{X}_0 - \bar{\mathbf{X}})]$, is the variance behind the prediction interval of Theorem 12.3.1, so that interval is exact for random regressors as well (Theorem 14.1.2). What the theorem adds is the average over new cases and over training designs.

14.2.3 The optimism of the in-sample error

The residual mean square SSE/n is the error the fitted predictor makes on the cases used to fit it. It is a biased guide to the error on new cases, in both the fixed and the random design.

Proposition 14.2.7 — In-sample and out-of-sample error (a) (*Fixed design.*) Let $E(\mathbf{Y}) = \mathbf{X}\boldsymbol{\beta}$ and $\text{Cov}(\mathbf{Y}) = \sigma^2 \mathbf{I}_n$ with $\text{rank}(\mathbf{X}) = r$, and let $\mathbf{Y}^*$ be a vector of new responses at the same design, uncorrelated with $\mathbf{Y}$, with the same mean and covariance. Then

$$E \frac{\|\mathbf{Y} - \mathbf{X}\hat{\boldsymbol{\beta}}\|^2}{n} = \sigma^2 \frac{n-r}{n}, \quad E \frac{\|\mathbf{Y}^* - \mathbf{X}\hat{\boldsymbol{\beta}}\|^2}{n} = \sigma^2 \frac{n+r}{n},$$

so the in-sample error understates the error on new responses by $2r\sigma^2/n$ on average.

(b) (*Normal random design.*) In the setting of [Theorem 14.2.6](#), the in-sample error has $E(\text{SSE}/n) = \sigma^2(n-k-1)/n$, and the error on a new case is [Eq. 14.2.2](#).

Proof.

- (a) The first identity is $E(\text{SSE}) = \sigma^2(n-r)$ ([Theorem 5.6.1](#), or [Eq. 2.3.1](#) with a mean in $\mathcal{C}(\mathbf{X})$). For the second, $\mathbf{Y}^* - \mathbf{X}\hat{\boldsymbol{\beta}} = (\mathbf{Y}^* - \mathbf{X}\boldsymbol{\beta}) - \mathbf{M}(\mathbf{Y} - \mathbf{X}\boldsymbol{\beta})$, and the two terms are uncorrelated, so the expected squared length is $n\sigma^2 + \sigma^2 \text{tr}(\mathbf{M}) = \sigma^2(n+r)$ by [Theorem 2.3.1](#).
- (b) Given the regressors, $E(\text{SSE} \mid \mathbf{X}) = \sigma^2(n-k-1)$, which does not depend on $\mathbf{X}$; take expectations. ■

A new random case brings its own, possibly outlying, regressor values, where the fitted plane is least reliable, so the random design is worse. [Figure 14.2.1](#) shows the three expected errors for $n = 30$. With $k = 10$ regressors the in-sample error is $0.633\sigma^2$, the error on new responses at the same design is $1.367\sigma^2$, and the error on a new random case is $1.607\sigma^2$: the in-sample figure flatters the fit by a factor of about 2.5. At $k = 20$ the three values are 0.300, 1.700 and 3.617, in units of σ^2 . The simulated averages over 20000 training samples per point, for instance 1.604 at $k = 10$ and 3.612 at $k = 20$, agree with [Eq. 14.2.2](#). Estimating the out-of-sample error honestly, by penalties such as Mallows' C_p or by cross-validation, is the subject of Chapter 29.

```
import numpy as np
rng = np.random.default_rng(1422)
n, sigma2, reps = 30, 1.0, 20_000

def simulate(k):
    """Average in-sample and out-of-sample squared error over many training samples.
    """
    ins, out = np.empty(reps), np.empty(reps)
    beta = np.full(k, 0.5)
    for s in range(reps):
        X = np.column_stack([np.ones(n), rng.normal(size=(n, k))])
        y = X @ np.r_[1.0, beta] + rng.normal(size=n)
        b, *_ = np.linalg.lstsq(X, y, rcond=None)
        ins[s] = np.mean((y - X @ b) ** 2)           # SSE / n
```

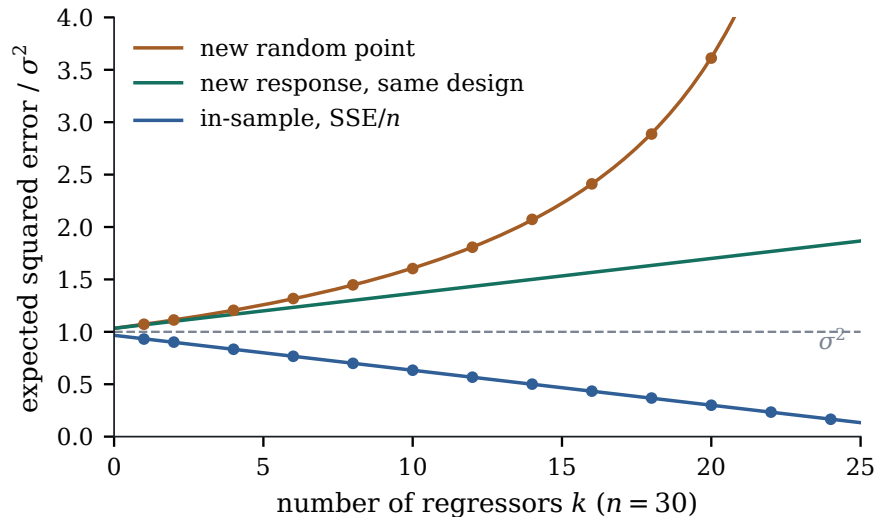


Figure 14.2.1: Expected squared error of a least squares fit, $n = 30$: in-sample (bottom), new responses at the same design (middle) and a new case with normal regressors (top, Eq. 14.2.2). Points: simulation.

```
# new point  $X_0 \sim N(0, I)$ ,  $Y_0 = 1 + \beta^T X_0 + e_0$ : exact conditional error
out[s] = sigma2 + (b[0] - 1.0) ** 2 + np.sum((b[1:] - beta) ** 2)
return ins.mean(), out.mean()

results = {}
for k in (1, 5, 10, 20):
    ins, out = simulate(k)
    results[k] = (ins, out)
    theory = sigma2 * (1 + 1 / n) * (n - 2) / (n - k - 2)
    print(f"k = {k:2d}: in-sample {ins:.3f} (theory {sigma2 * (n - k - 1) / n:.3f}),",
          "
          f" new point {out:.3f} (theory {theory:.3f})")
```

Example. (Predicting doctor visits)

The RAND Health Insurance Experiment records, for 20190 person-years, the number of outpatient visits and characteristics of the insurance plan and of the person. Treat these rows as the population, and predict $\log(1 + \text{visits})$ linearly from 6 of the characteristics (coinsurance rate, deductible and participation-incentive terms, maximum expenditure, physical limitation and a chronic-disease index). The population projection has mean squared error 0.6330 and explains a fraction 0.094 of the variance. Draw training samples of 40 rows, fit by least squares, and evaluate each fit exactly on the whole population. Over 4000 training samples the average in-sample error is 0.5182 and the average error on new cases is 0.8201, against normal-theory values 0.5223 and 0.7705 from Proposition 14.2.7(b). The normal-theory value understates the error on new cases: the regressors are far from normal (two are indicators, one skewed), the linear predictor need not be the conditional mean, and the error variance need not be constant; the formula assumes none of this. It is a first guide to the optimism, not a substitute for checking it.

```
import statsmodels.api as sm
data = sm.datasets.randhie.load_pandas().data
```

```

cols = ["lncoins", "idp", "lpi", "fmde", "physlm", "disea"]
Xpop = np.column_stack([np.ones(len(data)), data[cols].to_numpy()])
ypop = np.log1p(data["mdvis"].to_numpy()) # log(1 + number of visits)
b_pop, *_ = np.linalg.lstsq(Xpop, ypop, rcond=None) # the population projection
sigma2_pop = np.mean((ypop - Xpop @ b_pop) ** 2) # its mean squared error

rng = np.random.default_rng(1423)
n_train, draws = 40, 4000
k_rand = len(cols)
ins, out = np.empty(draws), np.empty(draws)
for s in range(draws):
    idx = rng.integers(0, len(ypop), n_train) # iid draws from the population
    b, *_ = np.linalg.lstsq(Xpop[idx], ypop[idx], rcond=None)
    ins[s] = np.mean((ypop[idx] - Xpop[idx] @ b) ** 2)
    out[s] = np.mean((ypop - Xpop @ b) ** 2) # exact error over the
    population
print(f"population projection error {sigma2_pop:.4f}")
print(f"in-sample {ins.mean():.4f}, new case {out.mean():.4f}")
print(f"normal-theory values: {sigma2_pop * (n_train - k_rand - 1) / n_train:.4f},"
      f" {sigma2_pop * (1 + 1 / n_train) * (n_train - 2) / (n_train - k_rand - 2):.4f}")

```

14.2.4 Best linear unbiased prediction

So far the new response was independent of the training data. When it is not, as for the next value of a time series or the yield at an unsampled site near sampled ones, the training residuals carry information about the new error, and a good predictor should use it.

Theorem 14.2.8 — Best linear unbiased predictor

Let $\mathbf{Y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}$ and $y_0 = \mathbf{x}_0^\top \boldsymbol{\beta} + e_0$, with $E(\boldsymbol{\varepsilon}) = \mathbf{0}$, $E(e_0) = 0$, $\text{Cov}(\boldsymbol{\varepsilon}) = \sigma^2 \mathbf{V}$ for a known positive definite $\mathbf{V}$, $\text{Cov}(\boldsymbol{\varepsilon}, e_0) = \sigma^2 \mathbf{v}_0$ and $\text{Var}(e_0) = \sigma^2 v_{00}$. Suppose $\mathbf{x}_0 \in \mathcal{C}(\mathbf{X}^\top)$. A predictor $\mathbf{a}^\top \mathbf{Y} + a_0$ is **unbiased** if $E(\mathbf{a}^\top \mathbf{Y} + a_0 - y_0) = 0$ for every $\boldsymbol{\beta}$. Among unbiased linear predictors, the mean squared error $E(y_0 - \mathbf{a}^\top \mathbf{Y} - a_0)^2$ is minimized by

$$\hat{y}_0 = \mathbf{x}_0^\top \tilde{\boldsymbol{\beta}} + \mathbf{v}_0^\top \mathbf{V}^{-1}(\mathbf{Y} - \mathbf{X}\tilde{\boldsymbol{\beta}}), \quad (14.2.3)$$

where $\tilde{\boldsymbol{\beta}}$ solves the generalized least squares equations $\mathbf{X}^\top \mathbf{V}^{-1} \mathbf{X} \mathbf{b} = \mathbf{X}^\top \mathbf{V}^{-1} \mathbf{Y}$. With $\mathbf{c} = \mathbf{x}_0 - \mathbf{X}^\top \mathbf{V}^{-1} \mathbf{v}_0$, its mean squared error is $\sigma^2 [v_{00} - \mathbf{v}_0^\top \mathbf{V}^{-1} \mathbf{v}_0 + \mathbf{c}^\top (\mathbf{X}^\top \mathbf{V}^{-1} \mathbf{X})^{-1} \mathbf{c}]$.

Proof. Put $\boldsymbol{\delta} = \mathbf{V}^{-1} \mathbf{v}_0$ and split the prediction error of any linear predictor:

$$y_0 - \mathbf{a}^\top \mathbf{Y} - a_0 = \underbrace{(e_0 - \boldsymbol{\delta}^\top \boldsymbol{\varepsilon})}_u + \underbrace{(\mathbf{x}_0^\top \boldsymbol{\beta} - \boldsymbol{\delta}^\top \mathbf{X} \boldsymbol{\beta} - (\mathbf{a} - \boldsymbol{\delta})^\top \mathbf{Y} - a_0)}_w.$$

The term u is uncorrelated with $\boldsymbol{\varepsilon}$, since $\text{Cov}(\boldsymbol{\varepsilon}, e_0 - \boldsymbol{\delta}^\top \boldsymbol{\varepsilon}) = \sigma^2 (\mathbf{v}_0 - \mathbf{V} \boldsymbol{\delta}) = \mathbf{0}$. The term w is a constant plus a linear function of $\mathbf{Y}$, hence of $\boldsymbol{\varepsilon}$, so $E(uw) = 0$ and $E(y_0 - \mathbf{a}^\top \mathbf{Y} - a_0)^2 = E u^2 + E w^2$. Only $E w^2$ depends on the predictor.

Unbiasedness means $\mathbf{a}^\top \mathbf{X} \boldsymbol{\beta} + a_0 = \mathbf{x}_0^\top \boldsymbol{\beta}$ for all $\boldsymbol{\beta}$, that is, $a_0 = 0$ and $\mathbf{X}^\top \mathbf{a} = \mathbf{x}_0$.

Equivalently, $(\mathbf{a} - \boldsymbol{\delta})^\top \mathbf{Y}$ is a linear unbiased estimator of $\mathbf{c}^\top \boldsymbol{\beta}$, which is estimable because $\mathbf{x}_0$ and $\mathbf{X}^\top \boldsymbol{\delta}$ lie in $\mathcal{C}(\mathbf{X}^\top)$. For such predictors $Ew = 0$, so Ew^2 is the variance of $(\mathbf{a} - \boldsymbol{\delta})^\top \mathbf{Y}$. By [Corollary 7.2.3](#) this variance is smallest when $(\mathbf{a} - \boldsymbol{\delta})^\top \mathbf{Y} = \mathbf{c}^\top \tilde{\boldsymbol{\beta}}$, the generalized least squares estimator, with variance $\sigma^2 \mathbf{c}^\top (\mathbf{X}^\top \mathbf{V}^{-1} \mathbf{X})^{-1} \mathbf{c}$. The predictor is then $\boldsymbol{\delta}^\top \mathbf{Y} + \mathbf{c}^\top \tilde{\boldsymbol{\beta}} = \mathbf{x}_0^\top \tilde{\boldsymbol{\beta}} + \boldsymbol{\delta}^\top (\mathbf{Y} - \mathbf{X} \tilde{\boldsymbol{\beta}})$, which is [Eq. 14.2.3](#). Finally $E u^2 = \sigma^2 (v_{00} - 2\boldsymbol{\delta}^\top \mathbf{v}_0 + \boldsymbol{\delta}^\top \mathbf{V} \boldsymbol{\delta}) = \sigma^2 (v_{00} - \mathbf{v}_0^\top \mathbf{V}^{-1} \mathbf{v}_0)$. ■

The predictor is the estimated mean at the new point plus a prediction of the new error e_0 from the residuals, by the best linear predictor of [Theorem 2.5.2](#). In the ordinary model, $\mathbf{V} = \mathbf{I}$ and $\mathbf{v}_0 = \mathbf{0}$, the correction vanishes and [Eq. 14.2.3](#) is the fitted value $\mathbf{x}_0^\top \hat{\boldsymbol{\beta}}$, with mean squared error $\sigma^2 [v_{00} + \mathbf{x}_0^\top (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{x}_0]$. The same theorem, with $\mathbf{v}_0$ built from a spatial covariance function, is the kriging predictor of geostatistics, and with random effects in place of e_0 it gives the predictors of mixed models (Chapter 32).

14.2.5 Exercises

14.2.5.1 A. Check your understanding

Exercise A1.

With normal regressors, $n = 50$ and $k = 10$, compute the expected in-sample error $E(\text{SSE}/n)$ and the expected error on a new case, in units of σ^2 . By what factor does the first understate the second?

14.2.5.2 B. Practice

Exercise B1.

For any predictor $g(\mathbf{X})$ with $0 < \text{Var } g < \infty$, let $\tilde{g} = EY + b(g - EY)$ with $b = \text{Cov}(Y, g) / \text{Var } g$. Show that $\tilde{g}$ predicts at least as well as g and that $E(Y - \tilde{g})^2 = \text{Var}(Y)(1 - \text{corr}(Y, g)^2)$. Conclude that among such *linearized* predictors, a higher squared correlation with Y means a smaller mean squared error, and that [Eq. 14.2.1](#) identifies the best of them.

Solution. $\tilde{g}$ is the best linear predictor of Y from the single variable g ([Theorem 6.11.1](#) with $k = 1$), and g itself is one such linear function (intercept 0, slope 1), so $E(Y - \tilde{g})^2 \leq E(Y - g)^2$. By [Eq. 6.11.1](#) the error of $\tilde{g}$ is $\text{Var } Y - \text{Cov}(Y, g)^2 / \text{Var } g = \text{Var } Y (1 - \text{corr}(Y, g)^2)$, which is decreasing in the squared correlation. The largest squared correlation is η^2 , attained by $g = m$, whose linearization is m itself.

Exercise B2.

Let the errors of a regression follow a stationary autoregression, $e_t = \phi e_{t-1} + u_t$, $|\phi| < 1$, with white noise u_t , so that $\text{Cov}(e_s, e_t) \propto \phi^{|s-t|}$. Observe $t = 1, \dots, n$ and predict $y_{n+1} = \mathbf{x}_{n+1}^\top \boldsymbol{\beta} + e_{n+1}$. Show that $\mathbf{V}^{-1} \mathbf{v}_0 = \phi \mathbf{e}_n$, the last standard basis vector scaled by ϕ , and hence that the BLUP is $\mathbf{x}_{n+1}^\top \tilde{\boldsymbol{\beta}} + \phi(y_n - \mathbf{x}_n^\top \tilde{\boldsymbol{\beta}})$.

Solution. With $\mathbf{V}_{st} = \phi^{|s-t|}$ (the common factor cancels in $\mathbf{V}^{-1}\mathbf{v}_0$), the covariance of e_s with e_{n+1} is $\phi^{n+1-s} = \phi \cdot \phi^{n-s} = \phi \mathbf{V}_{sn}$. So $\mathbf{v}_0 = \phi \mathbf{V} \mathbf{e}_n$ and $\mathbf{V}^{-1}\mathbf{v}_0 = \phi \mathbf{e}_n$. Substituting into Eq. 14.2.3, the correction term is ϕ times the last residual $y_n - \mathbf{x}_n^\top \hat{\boldsymbol{\beta}}$. Only the most recent residual matters, because given e_n the earlier errors carry no further linear information about e_{n+1} .

Exercise B3.

In the setting of Corollary 14.2.5, let $s^2 = \text{SSE}/(n - k - 1)$ and let $\mathbf{W}$ be the centred cross-product matrix. Show that $s^2 \mathbf{W}^{-1}$, the usual estimate of the covariance of the slopes, is an unbiased estimator of their *unconditional* covariance $\sigma^2 \boldsymbol{\Sigma}_{XX}^{-1}/(n - k - 2)$.

Solution. Given the regressors, $(n - k - 1)s^2/\sigma^2 \sim \chi^2(n - k - 1)$ (Theorem 7.5.1(b)), a law free of the design, so by Lemma 14.1.1 s^2 is independent of the regressors and hence of $\mathbf{W}$. Therefore $E(s^2 \mathbf{W}^{-1}) = E(s^2) E(\mathbf{W}^{-1}) = \sigma^2 \boldsymbol{\Sigma}_{XX}^{-1}/(n - k - 2)$ by Lemma 14.2.3 and Lemma 14.2.4.

14.2.5.3 C. Going deeper

Exercise C1.

Let the rows $(\mathbf{X}_i^\top, Y_i)$ be jointly normal with k regressors, and fit only the first $k_1 < k$ of them. Show that the expected error on a new case is $\sigma_1^2(1 + 1/n)(n - 2)/(n - k_1 - 2)$, where σ_1^2 is the error variance of the best linear predictor from the first k_1 regressors. *Hint:* under joint normality the smaller model is itself a correct normal model with random regressors. For $n = 30$, $k = 10$, $k_1 = 4$ and $\sigma_1^2 = 1.2\sigma^2$, is the smaller model better for prediction? What is the largest k_1 for which it is?

Solution. The pairs $(\mathbf{X}_{1i}, Y_i)$, with $\mathbf{X}_{1i}$ the first k_1 regressors, are jointly normal, so given $\mathbf{X}_1$ the response follows a normal linear model with k_1 regressors and error variance σ_1^2 (Eq. 3.5.2). The fitted predictor uses only these pairs, so Theorem 14.2.6 applies with k_1 and σ_1^2 in place of k and σ^2 . With the numbers given, the full model has expected error $1.607\sigma^2$ and the smaller one $1.2 \cdot (31/30) \cdot 28/24 \sigma^2 = 1.447\sigma^2$, so the smaller model predicts better despite its larger population error. Since the error of the smaller model increases with k_1 , it wins as long as $1.2 \cdot 28/(28 - k_1) < 28/18$, that is $k_1 < 6.4$: at $k_1 = 6$ its error is $1.578\sigma^2$, at $k_1 = 7$ it is $1.653\sigma^2$. The largest such k_1 is 6.

14.3 Multiple correlation

The population squared multiple correlation

$$\rho_{Y \cdot X}^2 = \frac{\boldsymbol{\sigma}_{XY}^\top \boldsymbol{\Sigma}_{XX}^{-1} \boldsymbol{\sigma}_{XY}}{\sigma_Y^2}$$

is the largest squared correlation between Y and a linear combination of $\mathbf{X}$ (Proposition 2.5.3). Section 9.5 studied its sample counterpart R^2 for a fixed design (Theorem 9.5.1, Theorem 9.5.2). With random rows, R^2 estimates $\rho_{Y \cdot X}^2$. This section finds its exact distribution under multivariate normality, and uses it to test $\rho_{Y \cdot X}^2 = 0$, to build an interval and to measure the bias.

Throughout, $(\mathbf{X}_i^\top, Y_i)$, $i = 1, \dots, n$, are independent draws from a $(k+1)$ -variate normal distribution with Σ_{XX} positive definite and $\sigma_Y^2 > 0$, and $n \geq k+2$. By [Eq. 3.5.2](#), $Y_i = \alpha + \boldsymbol{\beta}^\top \mathbf{X}_i + e_i$ with independent $e_i \sim N(0, \sigma^2)$ independent of the regressors, where

$$\sigma^2 = \sigma_Y^2(1 - \rho_{Y \cdot X}^2), \quad \boldsymbol{\beta}^\top \Sigma_{XX} \boldsymbol{\beta} = \sigma_Y^2 \rho_{Y \cdot X}^2, \quad \text{so} \quad \frac{\boldsymbol{\beta}^\top \Sigma_{XX} \boldsymbol{\beta}}{\sigma^2} = \frac{\rho_{Y \cdot X}^2}{1 - \rho_{Y \cdot X}^2}. \quad (14.3.1)$$

The ratio on the right is the population *signal-to-noise ratio*. We write ρ^2 for $\rho_{Y \cdot X}^2$ when no confusion can arise.

14.3.1 The sample multiple correlation as an estimator

Let $\mathbf{S}_{XX}$, $\mathbf{s}_{XY}$ and s_Y^2 be the sample covariances, with any common divisor. The fit with an intercept has $\text{SSR} = \hat{\boldsymbol{\beta}}^\top \mathbf{W} \hat{\boldsymbol{\beta}}$, where $\mathbf{W}$ is the centred cross-product matrix of the regressors and $\hat{\boldsymbol{\beta}} = \mathbf{W}^{-1} \sum_i (\mathbf{X}_i - \bar{\mathbf{X}})(Y_i - \bar{Y})$ ([Theorem 6.6.2](#) with $\mathbf{X}_2 = \mathbf{1}$). Dividing by SST gives

$$R^2 = \frac{\mathbf{s}_{XY}^\top \mathbf{S}_{XX}^{-1} \mathbf{s}_{XY}}{s_Y^2}, \quad (14.3.2)$$

the population formula with sample moments in place of population moments. Under normality it is more than a plug-in.

Proposition 14.3.1 — Maximum likelihood with normal rows

Suppose the sample covariance matrix of the rows is nonsingular. The maximum likelihood estimates of $(\alpha, \boldsymbol{\beta}, \sigma^2)$ are the least squares estimates and SSE/n , and the maximum likelihood estimate of $\rho_{Y \cdot X}^2$ is R^2 .

Proof. The joint density of a row factorizes as $f(\mathbf{x}; \boldsymbol{\mu}_X, \Sigma_{XX}) f(y | \mathbf{x}; \alpha, \boldsymbol{\beta}, \sigma^2)$, a normal density for the regressors times the conditional normal density of [Theorem 3.5.2](#). The map from $(\boldsymbol{\mu}, \Sigma)$ to $(\boldsymbol{\mu}_X, \Sigma_{XX}, \alpha, \boldsymbol{\beta}, \sigma^2)$ is one-to-one onto the set where Σ_{XX} is positive definite and $\sigma^2 > 0$, with inverse $\boldsymbol{\sigma}_{XY} = \Sigma_{XX} \boldsymbol{\beta}$, $\sigma_Y^2 = \sigma^2 + \boldsymbol{\beta}^\top \Sigma_{XX} \boldsymbol{\beta}$, $\mu_Y = \alpha + \boldsymbol{\beta}^\top \boldsymbol{\mu}_X$. So the log-likelihood is a sum of two terms with separate parameters, and each can be maximized on its own. The conditional term is the likelihood of a normal linear model with model matrix $[\mathbf{1}, \mathbf{X}_1]$, maximized by least squares and SSE/n ([Theorem 7.3.1](#)).

In the marginal term the quadratic form is minimized over $\boldsymbol{\mu}_X$ at $\bar{\mathbf{x}}$, leaving $-\frac{n}{2} [\log \det \Sigma_{XX} + \text{tr}(\Sigma_{XX}^{-1} \mathbf{S})]$ with $\mathbf{S} = \mathbf{W}/n$ positive definite. If $\lambda_1, \dots, \lambda_k > 0$ are the eigenvalues of $\mathbf{S}^{1/2} \Sigma_{XX}^{-1} \mathbf{S}^{1/2}$, then

$$\log \det \Sigma_{XX} + \text{tr}(\Sigma_{XX}^{-1} \mathbf{S}) - \log \det \mathbf{S} - k = \sum_{j=1}^k (\lambda_j - \log \lambda_j - 1) \geq 0,$$

with equality iff every $\lambda_j = 1$, that is, iff $\Sigma_{XX} = \mathbf{S}$. By invariance of maximum likelihood under reparameterization, the estimate of $\rho^2 = \boldsymbol{\beta}^\top \Sigma_{XX} \boldsymbol{\beta} / (\sigma^2 + \boldsymbol{\beta}^\top \Sigma_{XX} \boldsymbol{\beta})$ is $\hat{\boldsymbol{\beta}}^\top \mathbf{S} \hat{\boldsymbol{\beta}} / (\text{SSE}/n + \hat{\boldsymbol{\beta}}^\top \mathbf{S} \hat{\boldsymbol{\beta}}) = \text{SSR} / (\text{SSE} + \text{SSR}) = R^2$.

The rest of the section concerns the sampling distribution of this estimator. ■

14.3.2 The distribution of R-squared

Theorem 14.3.2 — Distribution of R^2 with normal rows

Under the assumptions above, let $F = \frac{n-k-1}{k} \frac{R^2}{1-R^2}$ and $\lambda = \rho^2/(1-\rho^2)$.

- (a) Given the regressors, $F \sim F(k, n-k-1, \gamma)$ with $\gamma = \boldsymbol{\beta}^\top \mathbf{W} \boldsymbol{\beta} / \sigma^2$, where $\mathbf{W}$ is the centred cross-product matrix of the regressors.
- (b) If $\rho^2 = 0$, then $F \sim F(k, n-k-1)$ and $R^2 \sim \text{Beta}(\frac{k}{2}, \frac{n-k-1}{2})$, independently of the regressors. The overall F test of the regression is an exact size- α test of $\rho^2 = 0$, which under joint normality is the hypothesis that Y is independent of $\mathbf{X}$.
- (c) If $\rho^2 > 0$, then $\gamma = \lambda V$ with $V \sim \chi^2(n-1)$, and for $0 < c < 1$

$$\Pr(R^2 \leq c) = \mathbb{E} \left[\Pr \left\{ F(k, n-k-1, \lambda V) \leq \frac{(n-k-1)c}{k(1-c)} \right\} \right]. \quad (14.3.3)$$

- (d) For each $c \in (0, 1)$, $\Pr(R^2 > c)$ is a strictly increasing function of $\rho^2 \in [0, 1)$.

Proof.

- (a) The centred regressor matrix has rank k with probability one, because $\mathbf{W} \sim W_k(n-1, \boldsymbol{\Sigma}_{XX})$ (Lemma 14.2.3) is nonsingular when $n-1 \geq k$ (the nonsingularity step in the proof of Lemma 14.2.4 needs only $m \geq k$). Given the regressors, the model is a normal linear model with $r = k+1$ and mean $\alpha \mathbf{1} + \mathbf{X}_1 \boldsymbol{\beta}$. By Theorem 9.5.2(a), $F \sim F(k, n-k-1, \gamma)$ with $\gamma = \|(\mathbf{M} - \mathbf{P}_1)(\alpha \mathbf{1} + \mathbf{X}_1 \boldsymbol{\beta})\|^2 / \sigma^2$. Since $(\mathbf{M} - \mathbf{P}_1)\mathbf{1} = \mathbf{0}$ and $(\mathbf{M} - \mathbf{P}_1)\mathbf{X}_1 \boldsymbol{\beta} = (\mathbf{I} - \mathbf{P}_1)\mathbf{X}_1 \boldsymbol{\beta}$, this is $\boldsymbol{\beta}^\top \mathbf{X}_1^\top (\mathbf{I} - \mathbf{P}_1) \mathbf{X}_1 \boldsymbol{\beta} / \sigma^2 = \boldsymbol{\beta}^\top \mathbf{W} \boldsymbol{\beta} / \sigma^2$.
- (b) $\rho^2 = 0$ means $\boldsymbol{\sigma}_{XY} = \mathbf{0}$, so $\boldsymbol{\beta} = \mathbf{0}$ and $\gamma = 0$ for every design. By Lemma 14.1.1, $F \sim F(k, n-k-1)$ independently of the regressors, and so is $R^2 = kF/(kF + n - k - 1)$, which is $\text{Beta}(k/2, (n-k-1)/2)$ by Theorem 9.5.2(b). For jointly normal variables, zero covariance is independence (Theorem 3.4.1).
- (c) By the proof of Lemma 14.2.3, $\mathbf{W} = (\mathbf{KX}_1)^\top (\mathbf{KX}_1)$, where the $n-1$ rows of $\mathbf{KX}_1$ are independent $N_k(\mathbf{0}, \boldsymbol{\Sigma}_{XX})$. So $\mathbf{KX}_1 \boldsymbol{\beta}$ has independent $N(0, \boldsymbol{\beta}^\top \boldsymbol{\Sigma}_{XX} \boldsymbol{\beta})$ entries, and $\boldsymbol{\beta}^\top \mathbf{W} \boldsymbol{\beta} = \|\mathbf{KX}_1 \boldsymbol{\beta}\|^2 = (\boldsymbol{\beta}^\top \boldsymbol{\Sigma}_{XX} \boldsymbol{\beta}) V$ with $V \sim \chi^2(n-1)$. By Eq. 14.3.1, $\gamma = \lambda V$. Since $R^2 \leq c$ iff $F \leq (n-k-1)c/(k(1-c))$, taking expectations of the conditional probability from (a) gives Eq. 14.3.3.
- (d) Fix $v > 0$ and $f > 0$. By Theorem 4.1.4, $\Pr\{F(k, n-k-1, \lambda v) > f\}$ is strictly increasing in λ , and $\lambda = \rho^2/(1-\rho^2)$ is strictly increasing in ρ^2 . If $\rho_1^2 < \rho_2^2$, the conditional tail probabilities therefore differ by a positive amount for every value of $V > 0$, and so do their expectations. ■

Part (c) is Fisher's (1928) distribution of the multiple correlation coefficient, reached here with-

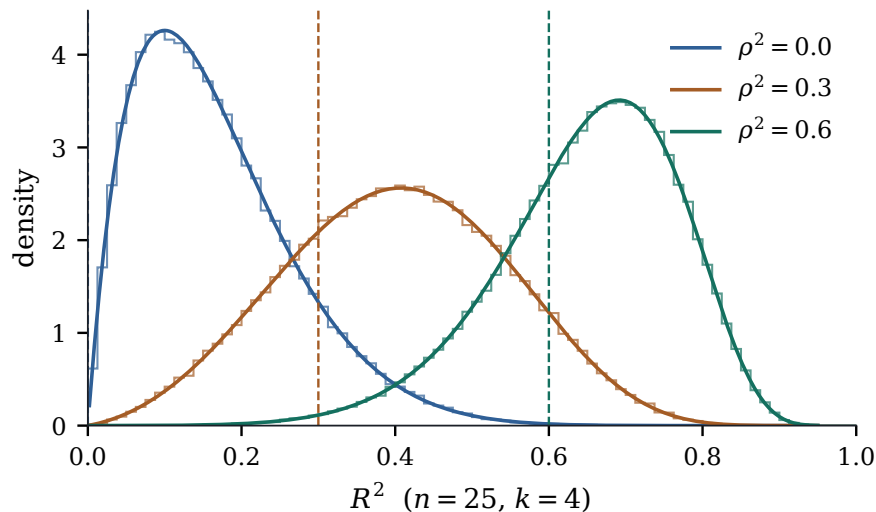


Figure 14.3.1: The law of R^2 with normal rows, $n = 25$, $k = 4$, for $\rho^2 = 0, 0.3$ and 0.6 (dashed). Step outlines: simulation. Curves: Eq. 14.3.3.

out any integration. Given the design, R^2 is a transformed noncentral F variable, and the design enters only through the single number $\beta^\top \mathbf{W} \beta$, which is a scaled chi-squared variable because the regressors are normal. Figure 14.3.1 shows the resulting densities for $n = 25$ and $k = 4$, computed from Eq. 14.3.3 by averaging over a grid of quantiles of V .

Example. (How biased is R-squared?)

With $n = 25$, $k = 4$ and 100000 simulated samples, the averages of R^2 are 0.1669, 0.4026 and 0.6500 for $\rho^2 = 0, 0.3$ and 0.6 , against exact values 0.1667, 0.4026 and 0.6498. The exact values come from the formula $E(R^2) = 1 - \frac{n-k-1}{n-1}(1-\rho^2) {}_2F_1(1, 1; \frac{n+1}{2}; \rho^2)$, which follows from Fisher's (1928) distribution of R^2 (see Olkin and Pratt 1958), where ${}_2F_1$ is the hypergeometric function; it can also be computed from Eq. 14.3.3. The bias shrinks as ρ^2 grows. The adjusted R^2 of Eq. 9.5.1 averages 0.0003, 0.2831 and 0.5800: unbiased under the null (Proposition 9.5.3(e)), but biased *downwards* when $\rho^2 > 0$. Neither is unbiased everywhere. The standard deviations of R^2 are 0.104, 0.146 and 0.118: with 25 cases, a single R^2 says little about ρ^2 .

```
import numpy as np
import statsmodels.api as sm
from scipy import optimize, special, stats

def r2_cdf(r, rho2, n, k, m=4000):
    """P(R^2 <= r) for iid normal rows: average the conditional noncentral F over V.
    """
    df2 = n - k - 1
    f = df2 * r / (k * (1 - r))
    if rho2 == 0:
        return stats.f.cdf(f, k, df2)
    V = stats.chi2.ppf((np.arange(m) + 0.5) / m, n - 1)  # equal-probability
    nodes
    return np.mean(stats.ncf.cdf(f, k, df2, rho2 / (1 - rho2) * V))
```

Part (d) says that the overall F test is unbiased and has power increasing in ρ^2 , as a test of $\rho^2 = 0$ against random-regressor alternatives. For $n = 25$ and $k = 4$ it rejects when $R^2 > 0.364$,

and its power is 0.182, 0.377, 0.599 and 0.922 at $\rho^2 = 0.1, 0.2, 0.3$ and 0.5. A population in which the regressors account for a fifth of the variance of the response is detected less than half the time.

14.3.3 An interval for the squared multiple correlation

Part (d) also makes an exact confidence interval possible. For observed $R^2 = r$, let ρ_L^2 be the value at which $\Pr(R^2 \leq r) = 0.975$ and ρ_U^2 the value at which it equals 0.025 (with $\rho_L^2 = 0$ if $\Pr(R^2 \leq r) < 0.975$ already at $\rho^2 = 0$). Because the probability decreases in ρ^2 , $[\rho_L^2, \rho_U^2]$ covers the true ρ^2 exactly when R^2 lies between the 2.5% and 97.5% points of its law at that ρ^2 , which has probability 0.95. This is the usual inversion of a family of tests ([Proposition 12.1.2](#)).

Example. (Violent crime in the states)

For the 50 states in 2009, regress the violent crime rate on the high-school graduation rate, the poverty rate, the percentage of single-parent households, the percentage of white residents and the urbanization rate. Then $R^2 = 0.4607$, the adjusted value is 0.3994, and $F = 7.52$ on 5 and 44 degrees of freedom, with p -value 3.6×10^{-5} . Read with random regressors, the 95% interval for ρ^2 is (0.175, 0.610). The interval is wide: five regressors and fifty cases leave much room for chance. It also treats the states as a random sample, which they are not; it describes hypothetical replications of a process generating rows like these, and here, as in [Example \(Poverty and graduation\)](#), no more can be claimed.

```
data = sm.datasets.statecrime.load_pandas().data.drop(index="District of Columbia")
regs = ["hs_grad", "poverty", "single", "white", "urban"]
y = data["violent"].to_numpy()
X = np.column_stack([np.ones(len(y))] + [data[c].to_numpy() for c in regs])
fit = sm.OLS(y, X).fit()
n_s, k_s = len(y), len(regs)
R2_obs = fit.rsquared
print(f"R^2 = {R2_obs:.4f}, adjusted {fit.rsquared_adj:.4f}, F = {fit.fvalue:.2f}, "
      f"p = {fit.f_pvalue:.2g}")

# 95% interval for rho^2: the values at which the observed R^2 is not extreme
def limit(prob):
    g = lambda rho2: r2_cdf(R2_obs, rho2, n_s, k_s) - prob
    return 0.0 if g(1e-12) < 0 else optimize.brentq(g, 1e-12, 0.999)

lower, upper = limit(0.975), limit(0.025)
print(f"95% interval for rho^2: ({lower:.3f}, {upper:.3f})")
```

Remark (Fixed or random?). Under a fixed design the expectation of R^2 depends on the chosen regressor values ([Exercise \(B1\)](#)). With random rows R^2 estimates $\rho_{Y.X}^2$, which still depends on the regressor distribution: a narrower range gives a smaller ρ^2 . The null law of the F test is the same in both readings.

14.3.4 Exercises

14.3.4.1 A. Check your understanding

Exercise A1.

A regression on $k = 3$ random regressors from $n = 40$ cases has $R^2 = 0.25$. Compute the overall F statistic and its degrees of freedom. Under the null hypothesis $\rho^2 = 0$, what are the mean and the distribution of R^2 ?

14.3.4.2 B. Practice

Exercise B1.

Let $\mathbf{R}$ be the sample correlation matrix of $(Y, X_1, \dots, X_k)$, assumed nonsingular, and let r^{YY} be the $(1, 1)$ entry of $\mathbf{R}^{-1}$. Show that $R^2 = 1 - 1/r^{YY}$, and that $1 - R^2 = \det \mathbf{R} / \det \mathbf{R}_{XX}$, where $\mathbf{R}_{XX}$ is the correlation matrix of the regressors.

Solution. Eq. 14.3.2 is unchanged when every variable is standardized, so $R^2 = \mathbf{r}_{XY}^\top \mathbf{R}_{XX}^{-1} \mathbf{r}_{XY}$. By Theorem 1.6.4, the leading entry of $\mathbf{R}^{-1}$ is the inverse of the Schur complement $1 - \mathbf{r}_{XY}^\top \mathbf{R}_{XX}^{-1} \mathbf{r}_{XY} = 1 - R^2$, so $r^{YY} = 1/(1 - R^2)$. By Theorem 1.6.3(b), $\det \mathbf{R} = \det \mathbf{R}_{XX} (1 - R^2)$.

Exercise B2.

Show that R^2 is unchanged if Y is replaced by $aY + b$ with $a \neq 0$, or the regressors by $\mathbf{B}^\top \mathbf{X} + \mathbf{c}$ with $\mathbf{B}$ nonsingular. Show that $\rho_{Y \cdot X}^2$ has the same invariance.

Exercise B3.

Show that $R^2 \rightarrow \rho_{Y \cdot X}^2$ with probability one as $n \rightarrow \infty$, without assuming normality, provided the rows are independent and identically distributed with finite second moments and Σ_{XX} positive definite.

14.3.4.3 C. Going deeper

Exercise C1.

With normal rows, show that the likelihood ratio statistic for $\rho_{Y \cdot X}^2 = 0$ against an unrestricted alternative is $(1 - R^2)^{n/2}$, so the likelihood ratio test is the overall F test. *Hint:* under the null the covariance matrix is block diagonal; use the proof of Proposition 14.3.1 and Exercise (B1).

Solution. With $\mathbf{S}$ the covariance matrix of the rows with divisor n , the maximized log-likelihood of a $(k + 1)$ -variate normal sample is $-\frac{n}{2}[\log \det \mathbf{S} + (k + 1)(1 + \log 2\pi)]$, by the argument in the proof of Proposition 14.3.1 applied to all $k + 1$ variables. Under the null, Y and $\mathbf{X}$ are independent, the likelihood factorizes, and the maximum is the same expression with $\log(s_Y^2 \det \mathbf{S}_{XX})$ in place of $\log \det \mathbf{S}$. The likelihood ratio is therefore

$(\det \mathbf{S}/(s_Y^2 \det \mathbf{S}_{XX}))^{n/2} = (1 - R^2)^{n/2}$, by [Theorem 1.6.3\(b\)](#) as in [Exercise \(B1\)](#). It is a decreasing function of R^2 , hence of F .

14.4 The correlation coefficient and Fisher's Z

With a single regressor, the multiple correlation reduces to the ordinary correlation ρ of the pair (X, Y) , and R^2 to the square of the sample correlation

$$r = \frac{\sum_i (X_i - \bar{X})(Y_i - \bar{Y})}{\sqrt{\sum_i (X_i - \bar{X})^2 \sum_i (Y_i - \bar{Y})^2}}.$$

Keeping the sign, this section develops two routes to inference about ρ : the exact distribution of r , and Fisher's transformation to a nearly normal variable whose variance does not depend on ρ . Throughout, (X_i, Y_i) , $i = 1, \dots, n$, are independent draws from a bivariate normal distribution with correlation $|\rho| < 1$, and $n \geq 3$.

14.4.1 The exact distribution of r

Proposition 14.4.1 — Exact law of the sample correlation

Let $T = r\sqrt{n-2}/\sqrt{1-r^2}$ and $\delta = \rho/\sqrt{1-\rho^2}$.

- (a) Given the X values, $T \sim t(n-2, \delta\sqrt{V})$, where $V = \sum_i (X_i - \bar{X})^2/\sigma_X^2$.
- (b) $V \sim \chi^2(n-1)$, and for $-1 < c < 1$,

$$\Pr(r \leq c) = \mathbb{E}[\Pr\{t(n-2, \delta\sqrt{V}) \leq c\sqrt{n-2}/\sqrt{1-c^2}\}]. \quad (14.4.1)$$

- (c) If $\rho = 0$, then $T \sim t(n-2)$ and $r^2 \sim \text{Beta}(\frac{1}{2}, \frac{n-2}{2})$, independently of the X values.
- (d) The law of r is stochastically increasing in ρ : for each $c \in (-1, 1)$, $\Pr(r \leq c)$ is a strictly decreasing function of ρ .

Proof.

- (a) Let $\hat{\beta}_1$ be the least squares slope of Y on X and t_1 its t statistic for the value zero. By [Theorem 9.5.4\(c\)](#) (see also [Exercise \(B3\)](#)), with only the intercept as the other column, $r^2 = t_1^2/(t_1^2 + n-2)$, and t_1 has the sign of $\hat{\beta}_1$, which is the sign of r . Solving, $t_1 = T$. Given the X values the pairs follow a normal simple regression with slope $\beta_1 = \rho\sigma_Y/\sigma_X$ and error variance $\sigma^2 = \sigma_Y^2(1-\rho^2)$ ([Eq. 3.5.2](#)), so by [Corollary 7.5.2](#) with $d = 0$, T is noncentral t with $n-2$ degrees of freedom and noncentrality $\beta_1\sqrt{S_{xx}}/\sigma = \rho\sqrt{S_{xx}}/(\sigma_X\sqrt{1-\rho^2}) = \delta\sqrt{V}$.
- (b) $V \sim \chi^2(n-1)$ by [Corollary 4.4.5](#). Since T is an increasing function of r , $r \leq c$ iff $T \leq c\sqrt{n-2}/\sqrt{1-c^2}$, and [Eq. 14.4.1](#) is the expectation of the conditional probability.
- (c) With $\delta = 0$ the conditional law is $t(n-2)$ for every design; apply [Lemma 14.1.1](#). Then $r^2 = T^2/(T^2 + n-2)$ is the $k = 1$ case of [Theorem 14.3.2\(b\)](#).

- (d) Write a $t(\nu, \theta)$ variable as $(Z + \theta)/\sqrt{W/\nu}$ (Definition 4.1.3). For $\theta_1 < \theta_2$ and any u , $\Pr\{t(\nu, \theta_1) \leq u\} - \Pr\{t(\nu, \theta_2) \leq u\} = \Pr\{u\sqrt{W/\nu} - \theta_2 < Z \leq u\sqrt{W/\nu} - \theta_1\} > 0$. Since $\delta = \rho/\sqrt{1-\rho^2}$ is strictly increasing in ρ , for every c and every $V > 0$ the conditional probability in Eq. 14.4.1 strictly decreases in ρ , and so does its expectation. ■

Fisher (1915) found the density of r as an infinite series; for computation the mixture Eq. 14.4.1, an average of noncentral t probabilities over quantiles of V , is enough. Part (c) is the familiar test of $\rho = 0$, the t test of a zero slope, symmetric in the two variables. Part (d) makes exact intervals possible, by the inversion used for ρ^2 in Section 14.3.

```
import numpy as np
import statsmodels.api as sm
from scipy import optimize, stats

def r_cdf(c, rho, n, m=4000):
    """Exact P(r <= c) for n iid bivariate normal pairs with correlation rho."""
    if c < 0:
        # law of -r is that for
        -rho
        return 1 - r_cdf(-c, -rho, n, m)
    t = c * np.sqrt(n - 2) / np.sqrt(1 - c ** 2)
    # increasing in c
    if rho == 0:
        return stats.t.cdf(t, n - 2)
    V = stats.chi2.ppf((np.arange(m) + 0.5) / m, n - 1)
    # equal-probability
    nodes
    delta = rho / np.sqrt(1 - rho ** 2)
    return np.mean(stats.nct.cdf(t, n - 2, delta * np.sqrt(V)))
```

When $\rho \neq 0$ the distribution of r is skewed towards zero, because r cannot pass ± 1 . For $\rho = 0.8$ and $n = 15$, 200000 simulated samples give a mean of 0.7886, a median of 0.811, a skewness of -1.40 and a variance of 0.01195 (panel (a) of Figure 14.4.1). The variance depends strongly on ρ : the standard deviation of r is 0.267 at $\rho = 0$ and 0.032 at $\rho = 0.95$ (panel (c)). An interval $r \pm \text{constant} \times \text{standard error}$ would inherit both problems.

14.4.2 Fisher's transformation

Fisher (1921) proposed to work instead with

$$z = \tanh^{-1} r = \frac{1}{2} \log \frac{1+r}{1-r}, \quad \zeta = \tanh^{-1} \rho. \quad (14.4.2)$$

To see why this helps we need the large-sample behaviour of r , and for that three standard tools from probability theory, used here without proof (see van der Vaart 1998, chapters 2 and 3).

Remark (Three limit theorems).

- (i) *Central limit theorem.* If $\mathbf{W}_1, \mathbf{W}_2, \dots$ are independent copies of a random vector with mean $\boldsymbol{\mu}$ and covariance matrix $\boldsymbol{\Gamma}$, then $\sqrt{n}(\bar{\mathbf{W}}_n - \boldsymbol{\mu}) \rightarrow N(\mathbf{0}, \boldsymbol{\Gamma})$ in distribution.
- (ii) *Slutsky's lemma.* If $\mathbf{U}_n \rightarrow \mathbf{U}$ in distribution and $\mathbf{V}_n \rightarrow \mathbf{c}$ in probability, then $\mathbf{U}_n + \mathbf{V}_n \rightarrow \mathbf{U} + \mathbf{c}$, and $V_n U_n \rightarrow cU$ for scalars.
- (iii) *Delta method.* If $\sqrt{n}(\mathbf{T}_n - \boldsymbol{\theta}) \rightarrow$

$N(\mathbf{0}, \mathbf{\Gamma})$ and g is differentiable at $\boldsymbol{\theta}$ with gradient ∇g , then $\sqrt{n}(g(\mathbf{T}_n) - g(\boldsymbol{\theta})) \rightarrow N(0, \nabla g^\top \mathbf{\Gamma} \nabla g)$.

The limiting variance of r needs fourth moments of the bivariate normal distribution.

Lemma 14.4.2 — Fourth moments of a standard bivariate normal pair

Let (X, Y) be bivariate normal with means 0, variances 1 and correlation ρ . Then $E X^4 = 3$, $E X^3 Y = 3\rho$ and $E X^2 Y^2 = 1 + 2\rho^2$. Consequently the covariance matrix of (X^2, Y^2, XY) is

$$\mathbf{\Gamma} = \begin{pmatrix} 2 & 2\rho^2 & 2\rho \\ 2\rho^2 & 2 & 2\rho \\ 2\rho & 2\rho & 1 + \rho^2 \end{pmatrix}.$$

Proof. Write $Y = \rho X + \sqrt{1 - \rho^2} U$ with $U \sim N(0, 1)$ independent of X ; the pair has the right means, variances and covariance, and it is normal by [Theorem 3.3.1](#). Using $E X^2 = 1$, $E X^4 = 3$ (the fourth moment of a standard normal) and the vanishing of odd moments, $E X^3 Y = \rho E X^4 = 3\rho$ and $E X^2 Y^2 = \rho^2 E X^4 + (1 - \rho^2) E X^2 E U^2 = 3\rho^2 + 1 - \rho^2 = 1 + 2\rho^2$. The covariances follow by subtracting products of means, $E X^2 = E Y^2 = 1$ and $E XY = \rho$: for instance $\text{Var}(XY) = 1 + 2\rho^2 - \rho^2$ and $\text{Cov}(X^2, XY) = E X^3 Y - \rho = 2\rho$. ■

Theorem 14.4.3 — Fisher's z transformation

As $n \rightarrow \infty$,

$$\sqrt{n}(r - \rho) \rightarrow N(0, (1 - \rho^2)^2) \quad \text{and} \quad \sqrt{n}(z - \zeta) \rightarrow N(0, 1)$$

in distribution. The same limits hold with $\sqrt{n-3}$ in place of $\sqrt{n}$. The limiting variance of z does not depend on ρ : $\tanh^{-1}$ is a variance-stabilizing transformation for r .

Proof. The correlation is unchanged by the maps $X \mapsto a + bX$, $Y \mapsto c + dY$ with $b, d > 0$, so we may assume $E X = E Y = 0$ and $\text{Var } X = \text{Var } Y = 1$. Let $\bar{\mathbf{W}}_n$ be the average of $\mathbf{W}_i = (X_i^2, Y_i^2, X_i Y_i)$, whose mean is $\boldsymbol{\theta} = (1, 1, \rho)$ and whose covariance is $\mathbf{\Gamma}$ of [Lemma 14.4.2](#). With divisor n , the sample variances and covariance are $\bar{\mathbf{W}}_n - (\bar{X}^2, \bar{Y}^2, \bar{X}\bar{Y})$. Since $\sqrt{n} \bar{X}$ converges in distribution and $\bar{X} \rightarrow 0$ in probability, $\sqrt{n} \bar{X}^2 \rightarrow 0$ in probability, and similarly for the other two products, so by Slutsky's lemma the centring does not affect the limit. The divisor cancels in $r = g(\text{sample moments})$, where $g(a, b, c) = c/\sqrt{ab}$, and $\nabla g(\boldsymbol{\theta}) = (-\rho/2, -\rho/2, 1)$. By the central limit theorem and the delta method, $\sqrt{n}(r - \rho)$ is asymptotically normal with variance

$$\nabla g^\top \mathbf{\Gamma} \nabla g = \frac{\rho^2}{4}(2+2+4\rho^2) - 2 \cdot \frac{\rho}{2}(2\rho+2\rho) + (1+\rho^2) = \rho^2 + \rho^4 - 4\rho^2 + 1 + \rho^2 = (1 - \rho^2)^2.$$

The derivative of $\tanh^{-1}$ at ρ is $1/(1 - \rho^2)$, so a second application of the delta method gives variance 1 for z . Finally $\sqrt{n-3}/\sqrt{n} \rightarrow 1$, and Slutsky's lemma allows the substitution.

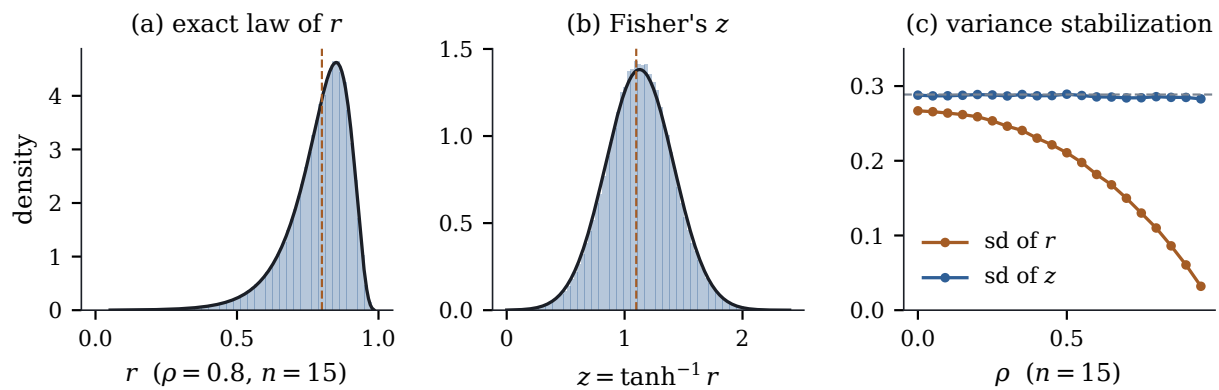


Figure 14.4.1: The sample correlation of $n = 15$ normal pairs. (a) r for $\rho = 0.8$, with the exact density, obtained by differentiating Eq. 14.4.1. (b) Fisher's z , with the normal approximation. (c) The standard deviations of r and z across ρ ; dashed: $1/\sqrt{n-3}$.

The transformation solves $g'(\rho) = 1/(1 - \rho^2)$, the condition for a limiting standard deviation of one (Exercise (B1)). Why the approximation is so good for small n is another matter: Fisher (1921) showed that the mean of z exceeds ζ by about $\rho/(2(n-1))$ and that its variance is close to $1/(n-3)$, and the distribution of z is nearly symmetric even when that of r is strongly skewed. In the simulation with $\rho = 0.8$ and $n = 15$, z has mean 1.1295 against $\zeta = 1.0986$ and $\zeta + \rho/(2(n-1)) = 1.1272$, variance 0.0811 against $1/(n-3) = 0.0833$, and skewness 0.01. Its standard deviation is 0.288 at $\rho = 0$ and 0.283 at $\rho = 0.95$ (panel (c)).

14.4.3 Intervals and tests for a correlation

The approximation $z \approx N(\zeta, 1/(n-3))$ gives the interval

$$\tanh\left(z - \frac{z_{\alpha/2}}{\sqrt{n-3}}\right) \leq \rho \leq \tanh\left(z + \frac{z_{\alpha/2}}{\sqrt{n-3}}\right), \quad (14.4.3)$$

with $z_{\alpha/2}$ the upper $\alpha/2$ point of $N(0,1)$. It is asymmetric about r , as it should be, and it never leaves $(-1,1)$. The exact interval inverts Eq. 14.4.1, using part (d) of Proposition 14.4.1 exactly as for ρ^2 : its limits are the values of ρ at which the observed r is the 97.5% and the 2.5% point.

How good is the approximation? For $n = 10$ pairs, the coverage of the 95% interval Eq. 14.4.3 over 200000 samples is 0.949, 0.949 and 0.952 at $\rho = 0, 0.5$ and 0.9 . The interval $r \pm 1.96(1 - r^2)/\sqrt{n}$, which uses the asymptotic standard error of Theorem 14.4.3 directly, covers only 0.839, 0.838 and 0.834.

```
def sample_r(rho, n, reps, rng):
    x = rng.normal(size=(reps, n))
    y = rho * x + np.sqrt(1 - rho ** 2) * rng.normal(size=(reps, n))
    xc = x - x.mean(axis=1, keepdims=True)
    yc = y - y.mean(axis=1, keepdims=True)
    return np.sum(xc * yc, axis=1) / np.sqrt(np.sum(xc ** 2, axis=1) * np.sum(yc **
    2, axis=1))

rng = np.random.default_rng(1443)
```

```

n_c, reps_c, zq = 10, 200_000, stats.norm.ppf(0.975)
for rho_c in (0.0, 0.5, 0.9):
    r_c = sample_r(rho_c, n_c, reps_c, rng)
    z_c = np.arctanh(r_c)
    fisher = np.abs(z_c - np.arctanh(rho_c)) <= zq / np.sqrt(n_c - 3)
    naive = np.abs(r_c - rho_c) <= zq * (1 - r_c ** 2) / np.sqrt(n_c)
    print(f"rho = {rho_c}: Fisher z covers {fisher.mean():.4f}, naive covers {naive.
          mean():.4f}")

```

Because z has an approximately known variance, it also compares correlations. With independent samples of n_1 and n_2 cases, the hypothesis $\rho_1 = \rho_2$ is tested by referring

$$\frac{z_1 - z_2}{\sqrt{1/(n_1 - 3) + 1/(n_2 - 3)}}$$

to the standard normal distribution. The t statistic of [Proposition 14.4.1\(c\)](#) remains the exact test of $\rho = 0$; for any other null value ρ_0 , either $(z - \tanh^{-1} \rho_0)\sqrt{n - 3}$ or the exact law [Eq. 14.4.1](#) is used.

Example. (Poverty and graduation)

Across the 50 states, the correlation between the poverty rate and the high-school graduation rate is $r = -0.7613$. Its t statistic for $\rho = 0$ is -8.13 on 48 degrees of freedom, far out in the tail. The Fisher interval [Eq. 14.4.3](#) is $(-0.858, -0.613)$ and the exact interval is $(-0.855, -0.609)$, almost the same, and both are noticeably asymmetric about r .

```

data = sm.datasets.statelcrime.load_pandas().data.drop(index="District of Columbia")
x, y = data["poverty"].to_numpy(), data["hs_grad"].to_numpy()
n_s = len(x)
r_s = np.corrcoef(x, y)[0, 1]
t_s = r_s * np.sqrt(n_s - 2) / np.sqrt(1 - r_s ** 2)
lo_z, hi_z = np.tanh(np.arctanh(r_s) + np.array([-1, 1]) * zq / np.sqrt(n_s - 3))
# exact interval: the rho at which the observed r sits at the 97.5% and 2.5% points
lo_x = optimize.brentq(lambda rh: r_cdf(r_s, rh, n_s) - 0.975, -0.99, 0.99)
hi_x = optimize.brentq(lambda rh: r_cdf(r_s, rh, n_s) - 0.025, -0.99, 0.99)
print(f"r = {r_s:.4f}, t = {t_s:.2f} on {n_s - 2} df")
print(f"Fisher z interval ({lo_z:.3f}, {hi_z:.3f}); exact interval ({lo_x:.3f}, {
    hi_x:.3f})")

```

Warning. Everything in this section uses bivariate normality, and not only through the conditional linear model. The asymptotic variance $(1 - \rho^2)^2/n$ comes from the fourth moments of [Lemma 14.4.2](#). For other distributions it can be quite different, and then neither z nor the exact interval is reliable ([Exercise \(B2\)](#)). The t test of $\rho = 0$ is more robust: by [Theorem 14.1.2](#) it needs only that Y given X follow a normal linear model with constant variance, and when X and Y are independent this holds as soon as Y is normal.

14.4.4 Exercises

14.4.4.1 A. Check your understanding

Exercise A1.

A sample of $n = 28$ pairs has $r = 0.62$. Compute the 95% Fisher interval for ρ .

Solution. $z = \tanh^{-1} 0.62 = 0.7250$ and $1.96/\sqrt{25} = 0.392$, so ζ lies in $(0.333, 1.117)$, and ρ in $(\tanh 0.333, \tanh 1.117) = (0.321, 0.806)$. The interval is longer below r than above.

Exercise A2.

Two independent samples give $r_1 = 0.45$ with $n_1 = 60$ and $r_2 = 0.20$ with $n_2 = 80$. Test $\rho_1 = \rho_2$ at level 0.05.

14.4.4.2 B. Practice**Exercise B1.**

Suppose $\sqrt{n}(T_n - \theta) \rightarrow N(0, s(\theta)^2)$ for every θ in an interval, with s positive and continuous. Show that if $g'(\theta) = 1/s(\theta)$, then $\sqrt{n}(g(T_n) - g(\theta)) \rightarrow N(0, 1)$. Solve $g'(\rho) = 1/(1 - \rho^2)$ and recover [Eq. 14.4.2](#).

Solution. By the delta method the limiting variance is $g'(\theta)^2 s(\theta)^2 = 1$. With partial fractions, $1/(1 - \rho^2) = \frac{1}{2}[1/(1 + \rho) + 1/(1 - \rho)]$, whose antiderivative vanishing at zero is $\frac{1}{2}[\log(1 + \rho) - \log(1 - \rho)] = \tanh^{-1} \rho$.

Exercise B2.

Let X and U be independent standard normal variables and $Y = XU$. Show that X and Y are uncorrelated but not independent, and that $\sqrt{n}r \rightarrow N(0, 3)$. What happens to the level of the test that rejects $\rho = 0$ when $|T| > t_{n-2, 0.025}$ in large samples? Which assumption of [Theorem 14.1.2](#) fails?

Solution. $EXY = EX^2EU = 0$, but $E(X^2Y^2) = EX^4EU^2 = 3 \neq EX^2EY^2 = 1$, so they are dependent. Repeat the proof of [Theorem 14.4.3](#) at $\rho = 0$: the gradient is $(0, 0, 1)$, so the limiting variance of $\sqrt{n}r$ is $\text{Var}(XY)/(\text{Var } X \text{Var } Y) = 3$. Since $T \approx \sqrt{n}r$ for large n , the test rejects with probability tending to $2\{1 - \Phi(1.96/\sqrt{3})\} = 0.258$, not 0.05. Given X , Y is normal with mean zero but variance X^2 : the errors are not homoscedastic, so the conditional model of [Definition 14.1.1](#) fails.

14.4.4.3 C. Going deeper**Exercise C1.**

For a general pair (X, Y) with finite fourth moments, standardized to means 0 and variances 1, show that $\sqrt{n}(r - \rho)$ is asymptotically normal with variance $(1 + \frac{\rho^2}{2})EX^2Y^2 + \frac{\rho^2}{4}(EX^4 + EY^4) - \rho(EX^3Y + EY^3X)$. Check that it reduces to $(1 - \rho^2)^2$ under normality, and explain why no transformation of r stabilizes the variance across all distributions.

14.5 Partial correlation

A partial correlation measures linear association after the linear effects of conditioning variables have been removed from both variables (Definition 3.6.1, Proposition 3.6.1). Section 3.6 computed the sample version and Theorem 9.5.4 identified its square with a partial R^2 ; the sampling distribution is still missing. This section shows that a partial correlation from n cases with q conditioning variables behaves exactly like an ordinary correlation from $n - q$ cases, and that its test is the t test of a regression coefficient.

14.5.1 The sample partial correlation

Definition 14.5.1 — Sample partial correlation

Let $\mathbf{y}, \mathbf{x} \in \mathbb{R}^n$, let $\mathbf{Z}$ be an $n \times q$ matrix, and let $\mathbf{D} = [\mathbf{1}, \mathbf{Z}]$ have rank $q + 1 < n$. With $\mathbf{M}_D$ the projection onto $\mathcal{C}(\mathbf{D})$, put $\tilde{\mathbf{y}} = (\mathbf{I} - \mathbf{M}_D)\mathbf{y}$ and $\tilde{\mathbf{x}} = (\mathbf{I} - \mathbf{M}_D)\mathbf{x}$, and suppose both are nonzero. The **sample partial correlation** of $\mathbf{y}$ and $\mathbf{x}$ given $\mathbf{Z}$ is

$$r_{yx \cdot Z} = \frac{\tilde{\mathbf{y}}^\top \tilde{\mathbf{x}}}{\|\tilde{\mathbf{y}}\| \|\tilde{\mathbf{x}}\|}.$$

Both residual vectors have mean zero, because $\mathbf{1} \in \mathcal{C}(\mathbf{D})$, so $r_{yx \cdot Z}$ is the ordinary sample correlation of the two residual vectors. With $q = 0$ it is the ordinary correlation of $\mathbf{y}$ and $\mathbf{x}$.

14.5.2 Its distribution

The key is that residualizing on $q + 1$ columns leaves $n - q - 1$ dimensions, exactly as centring alone leaves $n - 1$.

Theorem 14.5.1 — Distribution of a sample partial correlation

Let $\mathbf{Z}$ be a random $n \times q$ matrix with $\text{rank}[\mathbf{1}, \mathbf{Z}] = q + 1$ with probability one, and $n \geq q + 3$. Suppose that, given $\mathbf{Z}$, the rows (y_i, x_i) are independent bivariate normal vectors with means $\mathbf{a} + \mathbf{B}\mathbf{z}_{(i)}$, affine in the i th row $\mathbf{z}_{(i)}$ of $\mathbf{Z}$, and a common covariance matrix with correlation ρ . Then $r_{yx \cdot Z}$ has the same distribution as the sample correlation of $n - q$ independent bivariate normal pairs with correlation ρ , and it is independent of $\mathbf{Z}$.

In particular, if the rows $(y_i, x_i, \mathbf{z}_{(i)}^\top)$ are independent draws from a multivariate normal distribution with a nonsingular covariance matrix, the conclusion holds with $\rho = \rho_{yx \cdot Z}$, the population partial correlation.

Proof. Fix $\mathbf{Z}$ and write $\mathbf{D} = [\mathbf{1}, \mathbf{Z}]$. Let $\mathbf{K}$ be an $(n - q - 1) \times n$ matrix whose rows form an orthonormal basis of $\mathcal{C}(\mathbf{D})^\perp$, so that $\mathbf{KD} = \mathbf{0}$ and $\mathbf{K}^\top \mathbf{K} = \mathbf{I} - \mathbf{M}_D$. Then $\tilde{\mathbf{y}}^\top \tilde{\mathbf{x}} = \mathbf{y}^\top (\mathbf{I} - \mathbf{M}_D) \mathbf{x} = (\mathbf{Ky})^\top (\mathbf{Kx})$, and similarly for the squared lengths, so

$$r_{yx \cdot Z} = \frac{(\mathbf{Ky})^\top (\mathbf{Kx})}{\|\mathbf{Ky}\| \|\mathbf{Kx}\|}.$$

Stack the pairs as the $n \times 2$ matrix $[\mathbf{y}, \mathbf{x}]$. Given $\mathbf{Z}$, $[\mathbf{y}, \mathbf{x}] = \mathbf{D}\mathbf{A} + \mathbf{E}$ for a $(q+1) \times 2$ matrix of coefficients $\mathbf{A}$, where $\mathbf{E}$ has independent $N_2(\mathbf{0}, \mathbf{C})$ rows. Hence $\mathbf{K}[\mathbf{y}, \mathbf{x}] = \mathbf{K}\mathbf{E}$. Writing $\mathbf{E} = \mathbf{E}_0\mathbf{C}^{1/2}$ with independent standard normal entries in $\mathbf{E}_0$, each column of $\mathbf{K}\mathbf{E}_0$ is $N(\mathbf{0}, \mathbf{K}\mathbf{K}^\top) = N(\mathbf{0}, \mathbf{I}_{n-q-1})$ (Theorem 3.3.1), so the $n - q - 1$ rows of $\mathbf{K}\mathbf{E}$ are independent $N_2(\mathbf{0}, \mathbf{C})$ vectors. Thus, given $\mathbf{Z}$, $r_{yx \cdot Z}$ is the *uncentred* correlation of $m = n - q - 1$ independent $N_2(\mathbf{0}, \mathbf{C})$ pairs.

The same argument with $q = 0$ and $\mathbf{Z}$ absent shows that the ordinary sample correlation of $m + 1$ bivariate normal pairs is the uncentred correlation of m independent $N_2(\mathbf{0}, \mathbf{C})$ pairs. So the conditional law of $r_{yx \cdot Z}$ is that of an ordinary correlation from $m + 1 = n - q$ pairs, whatever $\mathbf{Z}$ is, and Lemma 14.1.1 gives the unconditional statement.

For the multivariate normal case, Theorem 3.5.2 says that given the conditioning variables the pair is bivariate normal with mean affine in them and covariance $\Sigma_{11 \cdot 2}$, whose correlation is $\rho_{yx \cdot Z}$ by Definition 3.6.1. ■

This is the result promised in Section 3.6: each conditioning variable costs one case. Everything in Section 14.4 transfers with n replaced by $n - q$: the exact law Eq. 14.4.1, the t test with $n - q - 2$ degrees of freedom, and Fisher's interval with $\sqrt{n - q - 3}$ (Fisher 1924). The conditioning variables need not be normal.

Example. (Twelve cases, two skewed conditioning variables)

Take $n = 12$, $q = 2$ conditioning variables drawn as squares of exponential variables, which are very skewed, a pair whose conditional means depend linearly on them, and conditional correlation $\rho = 0.6$. Over 100000 samples, the 10%, 50% and 90% points of the sample partial correlation are 0.257, 0.624 and 0.834. For ordinary correlations from $n - q = 10$ normal pairs they are 0.256, 0.625 and 0.835. A two-sample Kolmogorov–Smirnov test finds no difference between the two sets of simulated values (p -value 0.09).

```
import numpy as np
import statsmodels.api as sm
from scipy import optimize, stats

rng = np.random.default_rng(1451)
n, q, rho, reps = 12, 2, 0.6, 100_000
B = np.array([[1.0, -2.0], [0.5, 3.0]])          # dependence of the pair on Z
C = np.array([[1.0, rho], [rho, 1.0]])          # conditional covariance of the
pair
Lc = np.linalg.cholesky(C)

partial = np.empty(reps)
for s in range(reps):
    Zc = rng.exponential(size=(n, q)) ** 2      # skewed conditioning variables
    pair = Zc @ B.T + rng.normal(size=(n, 2)) @ Lc.T
    D = np.column_stack([np.ones(n), Zc])
    coef, *_ = np.linalg.lstsq(D, pair, rcond=None)
    e = pair - D @ coef                          # residuals of both variables
    partial[s] = e[:, 0] @ e[:, 1] / np.sqrt((e[:, 0] @ e[:, 0]) * (e[:, 1] @ e[:, 1]))

# ordinary correlations from n - q pairs with correlation rho
```

```

x = rng.normal(size=(reps, n - q))
y = rho * x + np.sqrt(1 - rho ** 2) * rng.normal(size=(reps, n - q))
ordinary = np.array([np.corrcoef(a, b)[0, 1] for a, b in zip(x, y)])
print("quantiles, partial :", np.quantile(partial, [0.1, 0.5, 0.9]).round(3))
print("quantiles, ordinary:", np.quantile(ordinary, [0.1, 0.5, 0.9]).round(3))
print("two-sample KS p-value:", round(stats.ks_2samp(partial, ordinary).pvalue, 3))

```

14.5.3 The test of a partial correlation is a coefficient test

In the population, a zero partial correlation and a zero regression coefficient are the same thing.

Proposition 14.5.2 — Partial correlation and the regression coefficient

Let $(Y, X, \mathbf{Z})$ have finite second moments, with $\text{Cov}((\mathbf{Z}^\top, X)^\top)$ positive definite, and let β_X be the coefficient of X in the best linear predictor of Y from $(\mathbf{Z}, X)$. Then

$$\beta_X = \frac{\sigma_{yx \cdot Z}}{\sigma_{xx \cdot Z}} = \rho_{yx \cdot Z} \sqrt{\frac{\sigma_{yy \cdot Z}}{\sigma_{xx \cdot Z}}},$$

so $\beta_X = 0$ iff $\rho_{yx \cdot Z} = 0$.

Proof. Let $e_X = X - L(X | \mathbf{Z})$ and $e_Y = Y - L(Y | \mathbf{Z})$ be the prediction errors of [Proposition 3.6.1\(a\)](#), with $\text{Var}(e_X) = \sigma_{xx \cdot Z} > 0$ and $\text{Cov}(e_Y, e_X) = \sigma_{yx \cdot Z}$. The affine span of $(\mathbf{Z}, X)$ is the orthogonal sum, in L^2 , of the affine span of $\mathbf{Z}$ and the line spanned by e_X , which is orthogonal to it. By the population version of [Theorem 6.3.8](#), projecting Y onto the sum projects it onto each piece: $L(Y | \mathbf{Z}, X) = L(Y | \mathbf{Z}) + (\text{Cov}(Y, e_X) / \text{Var}(e_X)) e_X$. Since $L(Y | \mathbf{Z})$ is orthogonal to e_X , $\text{Cov}(Y, e_X) = \text{Cov}(e_Y, e_X) = \sigma_{yx \cdot Z}$. The coefficient of X in this expression comes only from the last term, because $e_X = X -$ (affine function of $\mathbf{Z}$); it is $\sigma_{yx \cdot Z} / \sigma_{xx \cdot Z}$. ■

This is the Frisch–Waugh–Lovell theorem in the population, and the sample version is [Theorem 6.6.2](#) itself. Combining the two with [Theorem 14.5.1](#):

Theorem 14.5.3 — The partial correlation test is the coefficient t test

Regress $\mathbf{y}$ on the $p = q + 2$ columns $[\mathbf{1}, \mathbf{Z}, \mathbf{x}]$, assumed of full rank, and let t be the t statistic of the coefficient of $\mathbf{x}$ for the value zero. Then:

- (a) for every data set, $t = \sqrt{n - p} r_{yx \cdot Z} / \sqrt{1 - r_{yx \cdot Z}^2}$;
- (b) under the assumptions of [Theorem 14.5.1](#) with $\rho = 0$, $t \sim t(n - p)$, whatever the distribution of $\mathbf{Z}$;
- (c) when the rows are multivariate normal, the t test of the coefficient of $\mathbf{x}$ and the


```

def resid(v, Z):
    coef, *_ = np.linalg.lstsq(Z, v, rcond=None)
    return v - Z @ coef

ey, ex = resid(y, Z), resid(x, Z)
r_p = ey @ ex / np.sqrt((ey @ ey) * (ex @ ex))      # sample partial correlation
fit = sm.OLS(y, np.column_stack([Z, x])).fit()      # p = 4 columns
t = fit.tvalues[-1]
df = n_s - 4
print(f"partial r = {r_p:.4f}, t = {t:.3f}, sqrt(df) r/sqrt(1-r^2) = "
      f"{np.sqrt(df) * r_p / np.sqrt(1 - r_p**2):.3f}")
zq = stats.norm.ppf(0.975)
lo_z, hi_z = np.tanh(np.arctanh(r_p) + np.array([-1, 1]) * zq / np.sqrt(n_s - q_s -
3))
lo_x = optimize.brentq(lambda rh: r_cdf(r_p, rh, n_s - q_s) - 0.975, -0.99, 0.99)
hi_x = optimize.brentq(lambda rh: r_cdf(r_p, rh, n_s - q_s) - 0.025, -0.99, 0.99)
print(f"Fisher z interval ({lo_z:.3f}, {hi_z:.3f}); exact ({lo_x:.3f}, {hi_x:.3f})")

```

Remark (Which variables to condition on). Conditioning can shrink an association, as in the example, reverse its sign, as in [Example \(Employment and unemployment\)](#) and Simpson's paradox, or create one where none exists, when the conditioning variable is a common effect of the two ([Exercise \(C1\)](#)). The distribution theory is silent on the choice of conditioning set; Chapter 25 takes it up.

14.5.4 Exercises

14.5.4.1 A. Check your understanding

Exercise A1.

From $n = 40$ cases, the partial correlation of y and x given $q = 3$ other variables is -0.30 . Compute the t statistic, its degrees of freedom, and the 95% Fisher interval for the population partial correlation.

Exercise A2.

Explain, in terms of dimensions of subspaces, why the t test of a partial correlation given q variables has $n - q - 2$ degrees of freedom, and why this equals $n - p$ for the corresponding regression.

14.5.4.2 B. Practice

Exercise B1.

Let $\mathbf{X} = [\mathbf{X}_1, \mathbf{X}_2]$ with $\mathbf{1} \in \mathcal{C}(\mathbf{X}_1)$, $\text{rank } \mathbf{X}_1 = r_1$, $\text{rank } \mathbf{X} = r$ almost surely, and suppose that, given the regressors, $\mathbf{Y}$ follows the normal linear model with mean $\mathbf{X}_1 \boldsymbol{\beta}_1$. Show that the partial R^2 of [Definition 9.5.1](#) has the $\text{Beta}(\frac{r-r_1}{2}, \frac{n-r}{2})$ distribution, whatever the distribution of the regressors.

Solution. By [Theorem 9.5.4\(b\)](#), $R_{\mathbf{X}|\mathbf{X}_1}^2 = qF/(qF+n-r)$ with $q = r-r_1$, where F is the F ratio for the added block. Given the regressors, the null hypothesis holds, so $F \sim F(q, n-r)$ for every design ([Theorem 14.1.2\(b\)](#)), and by [Lemma 14.1.1](#) unconditionally. Writing $F = (U/q)/(V/(n-r))$ with independent chi-squared variables, $R_{\mathbf{X}|\mathbf{X}_1}^2 = U/(U+V)$, which is $\text{Beta}(q/2, (n-r)/2)$ by [Exercise \(B4\)](#).

Exercise B2.

The *semipartial* correlation of $\mathbf{y}$ and $\mathbf{x}$ given $\mathbf{Z}$ is the correlation between $\mathbf{y}$ itself and the residual $\tilde{\mathbf{x}} = (\mathbf{I} - \mathbf{M}_D)\mathbf{x}$. Show that its square equals $R_{\text{full}}^2 - R_{\text{reduced}}^2$, the increase in R^2 when $\mathbf{x}$ is added to $[\mathbf{1}, \mathbf{Z}]$. How does it relate to the squared partial correlation?

Solution. By [Lemma 9.2.1](#), the extra sum of squares of $\mathbf{x}$ is $(\tilde{\mathbf{x}}^\top \mathbf{y})^2 / \|\tilde{\mathbf{x}}\|^2$. Dividing by $\text{SST} = \|(\mathbf{I} - \mathbf{P}_1)\mathbf{y}\|^2$ gives $R_{\text{full}}^2 - R_{\text{reduced}}^2$. Since $\tilde{\mathbf{x}}$ has mean zero, $\tilde{\mathbf{x}}^\top \mathbf{y} = \tilde{\mathbf{x}}^\top (\mathbf{I} - \mathbf{P}_1)\mathbf{y}$, so the quotient is the squared correlation of $\mathbf{y}$ and $\tilde{\mathbf{x}}$. The squared partial correlation divides the same extra sum of squares by $\text{SSE}_{\text{reduced}}$ instead of SST, so it equals the squared semipartial correlation divided by $1 - R_{\text{reduced}}^2$, and is never smaller.

14.5.4.3 C. Going deeper

Exercise C1.

Let X and Y be independent $N(0, 1)$ and $C = X + Y + U$ with $U \sim N(0, 1)$ independent of both. Compute $\rho_{XY \cdot C}$. With $n = 200$ cases, what does the t test of the coefficient of X in the regression of Y on C and X tend to conclude? Is the conclusion about the relation between X and Y wrong, or is the question wrong?

14.6 Testing lack of fit with replicates

Every test and interval so far has assumed that the mean function is right, that $E(\mathbf{Y}) \in \mathcal{C}(\mathbf{X})$. When it is wrong, the residual mean square overestimates the error variance: [Example \(Residual sum of squares under a wrong mean\)](#) showed that $E(\text{SSE}) = \sigma^2(n-r) + \|(\mathbf{I} - \mathbf{M})\boldsymbol{\theta}\|^2$ for a mean vector $\boldsymbol{\theta}$ outside the model space. Separating noise from systematic error needs an estimate of σ^2 free of the form of the mean, and repeated rows provide one: observations at identical settings differ only by noise. The idea, and the test built on it, go back to Fisher (1922).

14.6.1 Replicates and the cell-means model

Suppose the model matrix has c distinct rows $\mathbf{x}_1^\top, \dots, \mathbf{x}_c^\top$, the i th occurring $n_i \geq 1$ times, with $n = \sum_i n_i$. Index the observations by the distinct row and the replicate, y_{ij} with $j = 1, \dots, n_i$, so that the model is

$$E(y_{ij}) = \mathbf{x}_i^\top \boldsymbol{\beta}, \quad i = 1, \dots, c, \quad j = 1, \dots, n_i.$$

Let $\mathbf{Z}$ be the $n \times c$ matrix of group indicators, the model matrix of the one-way layout in which each distinct row has its own mean μ_i . Call two matrices with n rows *alike in row structure* as

$\mathbf{X}$ if their rows are equal whenever the corresponding rows of $\mathbf{X}$ are equal.

Lemma 14.6.1 — The largest model with the same replicates (a) $\mathcal{C}(\mathbf{X}) \subseteq \mathcal{C}(\mathbf{Z})$. More generally, every matrix $\mathbf{W}$ with n rows whose rows are constant within the groups satisfies $\mathcal{C}(\mathbf{W}) \subseteq \mathcal{C}(\mathbf{Z})$.
 (b) $\mathbf{M}_Z \mathbf{y}$ replaces each observation by its group mean $\bar{y}_i$, and $\mathbf{M} \mathbf{y}$ is constant within groups.

Proof.

- (a) Let $\bar{\mathbf{W}}$ be the $c \times s$ matrix whose i th row is the common row of $\mathbf{W}$ in group i . Then $\mathbf{W} = \mathbf{Z} \bar{\mathbf{W}}$, so $\mathcal{C}(\mathbf{W}) \subseteq \mathcal{C}(\mathbf{Z})$. The matrix $\mathbf{X}$ is a special case.
 (b) The first claim is [Example \(The one-way layout\)](#). For the second, $\mathbf{M} \mathbf{y} \in \mathcal{C}(\mathbf{X}) \subseteq \mathcal{C}(\mathbf{Z})$, and the vectors of $\mathcal{C}(\mathbf{Z})$ are constant within groups. ■

So the cell-means model is the largest linear model that treats identical rows identically. Any mean function of the regressors at all, linear or not, produces a mean vector in $\mathcal{C}(\mathbf{Z})$.

Definition 14.6.1 — Pure error and lack of fit

With $\mathbf{M}_Z$ the projection onto $\mathcal{C}(\mathbf{Z})$, the **pure error** and **lack-of-fit** sums of squares are

$$\begin{aligned} \text{SSPE} &= \|(\mathbf{I} - \mathbf{M}_Z) \mathbf{y}\|^2 = \sum_{i=1}^c \sum_{j=1}^{n_i} (y_{ij} - \bar{y}_i)^2, \\ \text{SSLF} &= \|(\mathbf{M}_Z - \mathbf{M}) \mathbf{y}\|^2 = \sum_{i=1}^c n_i (\bar{y}_i - \hat{y}_i)^2, \end{aligned} \tag{14.6.1}$$

with $n - c$ and $c - r$ degrees of freedom, where $\hat{y}_i = \mathbf{x}_i^\top \hat{\boldsymbol{\beta}}$ is the common fitted value of group i and $r = \text{rank}(\mathbf{X})$.

The two expressions for each sum of squares agree by [Lemma 14.6.1\(b\)](#): $(\mathbf{I} - \mathbf{M}_Z) \mathbf{y}$ has entries $y_{ij} - \bar{y}_i$, and $(\mathbf{M}_Z - \mathbf{M}) \mathbf{y}$ has entries $\bar{y}_i - \hat{y}_i$, repeated n_i times.

14.6.2 The test

Theorem 14.6.2 — Pure-error lack-of-fit test

Let $\mathbf{Y} \sim N_n(\boldsymbol{\theta}, \sigma^2 \mathbf{I})$ with $\boldsymbol{\theta} \in \mathcal{C}(\mathbf{Z})$, that is, $E(y_{ij}) = \mu_i$ depends only on the row of $\mathbf{X}$; assume $r < c < n$. Then:

- (a) $\text{SSE} = \text{SSLF} + \text{SSPE}$, and SSLF and SSPE are independent;
 (b) $\text{SSPE}/\sigma^2 \sim \chi^2(n - c)$ for every $\mu_1, \dots, \mu_c$;

(c) $\text{SSLF}/\sigma^2 \sim \chi^2(c-r, \gamma)$ with

$$\gamma = \frac{\|(\mathbf{I} - \mathbf{M})\boldsymbol{\theta}\|^2}{\sigma^2} = \frac{1}{\sigma^2} \sum_{i=1}^c n_i (\mu_i - \tilde{\mu}_i)^2, \quad (14.6.2)$$

where $\tilde{\mu}_i$ is the i th group value of $\mathbf{M}\boldsymbol{\theta}$, the least squares fit to the true means;

(d) the ratio

$$F = \frac{\text{SSLF}/(c-r)}{\text{SSPE}/(n-c)} \sim F(c-r, n-c, \gamma),$$

and $\gamma = 0$ iff $\boldsymbol{\theta} \in \mathcal{C}(\mathbf{X})$, that is, iff the model is correct. The test that rejects the model when $F > F_\alpha(c-r, n-c)$ has size α , and its power increases strictly with γ .

Proof. By Lemma 14.6.1, $\mathcal{C}(\mathbf{X}) \subseteq \mathcal{C}(\mathbf{Z})$, so by Theorem 6.5.1 $\mathbf{M}_Z - \mathbf{M}$ is the projection onto the orthogonal complement of $\mathcal{C}(\mathbf{X})$ in $\mathcal{C}(\mathbf{Z})$, of rank $c-r$, and $\mathbf{I} - \mathbf{M} = (\mathbf{M}_Z - \mathbf{M}) + (\mathbf{I} - \mathbf{M}_Z)$ splits into two orthogonal projections. This gives the identity in (a). By Theorem 4.4.4, SSLF/σ^2 and SSPE/σ^2 are independent noncentral chi-squared variables with $c-r$ and $n-c$ degrees of freedom and noncentralities $\|(\mathbf{M}_Z - \mathbf{M})\boldsymbol{\theta}\|^2/\sigma^2$ and $\|(\mathbf{I} - \mathbf{M}_Z)\boldsymbol{\theta}\|^2/\sigma^2$. The second is zero because $\boldsymbol{\theta} \in \mathcal{C}(\mathbf{Z})$, which proves (b). In the first, $\mathbf{M}_Z\boldsymbol{\theta} = \boldsymbol{\theta}$, so $(\mathbf{M}_Z - \mathbf{M})\boldsymbol{\theta} = (\mathbf{I} - \mathbf{M})\boldsymbol{\theta}$, whose entries are $\mu_i - \tilde{\mu}_i$, repeated n_i times; this is (c). Part (d) follows from Definition 4.1.2, the fact that $(\mathbf{I} - \mathbf{M})\boldsymbol{\theta} = \mathbf{0}$ iff $\boldsymbol{\theta} \in \mathcal{C}(\mathbf{X})$, and Theorem 4.1.4. ■

The test is the F test of the reduced model $\mathcal{C}(\mathbf{X})$ inside the full model $\mathcal{C}(\mathbf{Z})$ (Theorem 11.1.1, Theorem 4.4.6). What makes it special is the full model, which is chosen not by the analyst but by the replication in the data, and which contains every mean function of the regressors. The residual sum of squares of the working model is split into two rows of an analysis of variance:

Source	Degrees of freedom	Sum of squares	Mean square
Lack of fit	$c-r$	$\sum_i n_i (\bar{y}_i - \hat{y}_i)^2$	$\text{SSLF}/(c-r)$
Pure error	$n-c$	$\sum_i \sum_j (y_{ij} - \bar{y}_i)^2$	$\text{SSPE}/(n-c)$
Residual	$n-r$	SSE	s^2

The noncentrality Eq. 14.6.2 is the weighted squared distance between the true group means and their best approximation by the model, in units of σ^2 . It grows with the replication, so replicates buy power twice: through γ and through the pure-error degrees of freedom.

Example. (Stack loss and air flow)

Brownlee's stack loss data record $n = 21$ days of operation of a plant that oxidizes ammonia. Air flow, the rate of operation, takes only $c = 7$ distinct values, so a straight-line regression of stack loss on air flow can be tested for lack of fit with 14 degrees of freedom for pure error. The line has intercept -44.13 and slope 1.020 , and $R^2 = 0.846$. Its residual sum of squares 319.12 splits into $\text{SSPE} = 96.53$ and $\text{SSLF} = 222.58$ on 5 degrees of freedom. The mean squares are 44.52 and 6.90 , so $F = 6.46$ with p -value 0.0026 ; the 5% point of $F(5, 14)$ is 2.96 . The residual mean square of the line, 16.80 , is more than twice the pure-error estimate of σ^2 .

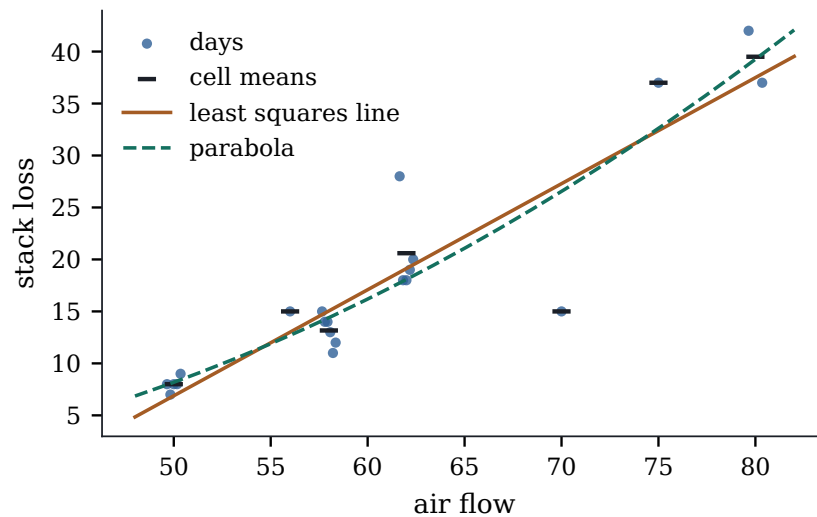


Figure 14.6.1: Stack loss against air flow on 21 days (replicates spread sideways), the seven group means (bars), the least squares line and parabola.

Figure 14.6.1 shows why. The group means do not lie on any smooth curve: the single day at air flow 70 has stack loss 15.0, below the mean 20.6 of the five days at air flow 62. A parabola does not help, with $F = 7.22$ on 4 and 14 degrees of freedom (p -value 0.0023). The mean of stack loss is not a function of air flow alone, which points to the other operating variables, taken up in [Example \(Is the three-variable model adequate?\)](#).

```
import numpy as np
import statsmodels.api as sm
from scipy import stats
data = sm.datasets.stackloss.load_pandas().data
y = data["STACKLOSS"].to_numpy()
x = data["AIRFLOW"].to_numpy()
n = len(y)
levels, groups = np.unique(x, return_inverse=True)  # c distinct rows of X
c = len(levels)
Z = np.eye(c)[groups]  # cell-means model matrix

def sse(A):
    coef, *_ = np.linalg.lstsq(A, y, rcond=None)
    return np.sum((y - A @ coef) ** 2)

ss_pe = sse(Z)  # pure error, n - c df
for name, X in [("line", np.column_stack([np.ones(n), x])),
                ("parabola", np.column_stack([np.ones(n), x, x ** 2]))]:
    r = np.linalg.matrix_rank(X)
    ss_lf = sse(X) - ss_pe  # lack of fit, c - r df
    F = (ss_lf / (c - r)) / (ss_pe / (n - c))
    print(f"{name:8s}: SSE {sse(X):7.2f} = SSLF {ss_lf:7.2f} ({c - r} df)"
          f" + SSPE {ss_pe:6.2f} ({n - c} df); F = {F:.2f}, p = {stats.f.sf(F, c - r, n - c):.4f}")
```

14.6.3 When pure error is not pure

The theorem assumes $\boldsymbol{\theta} \in \mathcal{C}(\mathbf{Z})$: the mean depends on the observation only through its row of $\mathbf{X}$. If a variable that is not in the model affects the response and varies within the groups of replicates, as water temperature does among the days with equal air flow in the stack loss data, then observations in a group do not share a mean. By [Theorem 9.6.1\(b\)](#), applied to the cell-means model,

$$E\left(\frac{\text{SSPE}}{n-c}\right) = \sigma^2 + \frac{\|(\mathbf{I} - \mathbf{M}_Z)\boldsymbol{\theta}\|^2}{n-c},$$

so the “pure” error estimate is inflated, and F becomes a ratio of two noncentral chi-squared variables, a *doubly noncentral* F , which rejects less often than it would with a pure estimate of σ^2 . Two further cautions: the pooled estimate assumes a common variance across groups ([Exercise \(C2\)](#)), and a non-significant result with few degrees of freedom is weak evidence that the model is adequate.

Remark (Designing for a lack-of-fit test). The test needs $c > r$ distinct rows and $n > c$ observations. The design that estimates a straight line most precisely puts half the observations at each end of the range; it has $c = r = 2$ and cannot detect any curvature. A few intermediate settings buy a check of the model at a small cost in precision. Replicates also cap what any model of the regressors can achieve: $R^2 \leq 1 - \text{SSPE}/\text{SST}$ ([Exercise \(B2\)](#)).

14.6.4 Exercises

14.6.4.1 A. Check your understanding

Exercise A1.

A straight line is fitted to $n = 24$ observations at $c = 6$ distinct values of x , each replicated four times. The residual sum of squares is 50.0 and the pure-error sum of squares is 36.0. Complete the analysis of variance table for lack of fit and compute F . Is there evidence of lack of fit at level 0.05, given that $F_{0.05}(4, 18) = 2.93$?

14.6.4.2 B. Practice

Exercise B1.

Show that the least squares estimate $\hat{\boldsymbol{\beta}}$ from all n observations is also the weighted least squares estimate from the c group means, with weights n_i , and that SSLF is the weighted residual sum of squares of that fit.

Solution. By [Theorem 6.5.1](#), $\mathbf{M}\mathbf{y} = \mathbf{M}\mathbf{M}_Z\mathbf{y}$: projecting $\mathbf{y}$ onto $\mathcal{C}(\mathbf{X})$ is the same as projecting the vector of group means $\mathbf{M}_Z\mathbf{y}$. For a vector constant within groups, $\|\mathbf{M}_Z\mathbf{y} - \mathbf{X}\mathbf{b}\|^2 = \sum_i n_i(\bar{y}_i - \mathbf{x}_i^\top \mathbf{b})^2$, so minimizing it over $\mathbf{b}$ is weighted least squares on the c means with weights n_i . Its minimum is $\|(\mathbf{M}_Z - \mathbf{M})\mathbf{y}\|^2 = \text{SSLF}$.

Exercise B2.

Suppose $\mathbf{1} \in \mathcal{C}(\mathbf{X})$. Show that every model whose model matrix has the row structure of $\mathbf{X}$ has $R^2 \leq 1 - \text{SSPE}/\text{SST}$, with equality for the cell-means model.

Solution. By Lemma 14.6.1(a) the column space $\mathcal{V}$ of such a model lies in $\mathcal{C}(\mathbf{Z})$, so its residual sum of squares $\|(\mathbf{I} - \mathbf{P}_{\mathcal{V}})\mathbf{y}\|^2$ is at least $\|(\mathbf{I} - \mathbf{M}_{\mathbf{Z}})\mathbf{y}\|^2 = \text{SSPE}$ (projection onto a smaller space leaves a longer residual). Hence $R^2 = 1 - \text{SSE}/\text{SST} \leq 1 - \text{SSPE}/\text{SST}$, with equality when $\mathcal{V} = \mathcal{C}(\mathbf{Z})$.

Exercise B3.

Show that rows i and j of $\mathbf{M}$ are equal iff rows i and j of $\mathbf{X}$ are equal. So replicates can be read off the hat matrix.

14.6.4.3 C. Going deeper**Exercise C1.**

A straight line is to be tested for lack of fit at the four values $x = 0, 1, 2, 3$, each replicated m times, when the true mean is $\beta_0 + \beta_1 x + \beta_2 x^2$ with $\beta_2/\sigma = 0.5$. Show that $\gamma = m\beta_2^2 \|(\mathbf{I} - \mathbf{M}_4)\mathbf{q}\|^2/\sigma^2$, where $\mathbf{q} = (0, 1, 4, 9)^\top$ and $\mathbf{M}_4$ is the projection for a line on the four points, and find the smallest m for which the test at level 0.05 has power at least 0.8.

Exercise C2.

Suppose the groups have different variances σ_i^2 . Compute $E(\text{SSPE})$ and $E(\text{SSLF})$ under a correct mean model, and show by an example with two large groups of very different sizes and variances that the F test can then reject far more or far less often than α .

14.7 Near replicates and nonparametric lack-of-fit tests

Exact replicates are a luxury of designed experiments. With continuous regressors, rows of the model matrix are rarely identical: stack loss with all three operating variables has 21 rows and a single repeated pair, one degree of freedom for pure error. The pure-error test survives in a more general form: any larger model chosen without looking at the response gives an exact test, and so do regressors built from the fitted values. What is lost is the guarantee that the larger model contains every alternative: each test looks in some directions and is blind in others.

14.7.1 A general recipe

Proposition 14.7.1 — Testing against any larger model

Let $\mathbf{Y} \sim N_n(\boldsymbol{\theta}, \sigma^2 \mathbf{I})$, let $\mathbf{X}$ have rank r , and let $\mathbf{Z}$ be a fixed matrix, chosen without reference to $\mathbf{Y}$, with $\mathcal{C}(\mathbf{X}) \subseteq \mathcal{C}(\mathbf{Z})$ and $\text{rank}(\mathbf{Z}) = s$, $r < s < n$. Let $\mathbf{M}$ and $\mathbf{M}_Z$ be the two projections and

$$F = \frac{[\text{SSE}(\mathbf{X}) - \text{SSE}(\mathbf{Z})]/(s - r)}{\text{SSE}(\mathbf{Z})/(n - s)}.$$

- (a) If $\boldsymbol{\theta} \in \mathcal{C}(\mathbf{X})$, then $F \sim F(s - r, n - s)$.
- (b) If $\boldsymbol{\theta} \in \mathcal{C}(\mathbf{Z})$, then $F \sim F(s - r, n - s, \|(\mathbf{I} - \mathbf{M})\boldsymbol{\theta}\|^2/\sigma^2)$.
- (c) In general $F = \frac{U/(s - r)}{V/(n - s)}$ with independent $U \sim \chi^2(s - r, \gamma_1)$ and $V \sim \chi^2(n - s, \gamma_2)$, where $\gamma_1 = \|(\mathbf{M}_Z - \mathbf{M})\boldsymbol{\theta}\|^2/\sigma^2$ and $\gamma_2 = \|(\mathbf{I} - \mathbf{M}_Z)\boldsymbol{\theta}\|^2/\sigma^2$. For every $c > 0$, $\Pr(F > c)$ is strictly increasing in γ_1 and strictly decreasing in γ_2 .

Proof. The decomposition $\mathbf{I} - \mathbf{M} = (\mathbf{M}_Z - \mathbf{M}) + (\mathbf{I} - \mathbf{M}_Z)$ into orthogonal projections of ranks $s - r$ and $n - s$ comes from [Theorem 6.5.1](#), and [Theorem 4.4.4](#) gives (c) apart from the monotonicity. Parts (a) and (b) are the cases $\gamma_1 = \gamma_2 = 0$ and $\gamma_2 = 0$. For the monotonicity, write $\Pr(F > c) = E[\Pr(U > c'V \mid V)]$ with $c' = c(s - r)/(n - s)$. For each value $V > 0$ the inner probability is strictly increasing in γ_1 by [Proposition 4.1.3](#). Similarly $\Pr(F > c) = E[\Pr(V < U/c' \mid U)]$, and $\Pr\{\chi^2(n - s, \gamma_2) < u\}$ is strictly decreasing in γ_2 for $u > 0$. ■

The size is exact however $\mathbf{Z}$ is chosen without using $\mathbf{Y}$; with random regressors, $\mathbf{Z}$ may be any function of $\mathbf{X}$ ([Theorem 14.1.2](#)). The pure-error test is the case $\mathbf{Z} = \text{cell means}$. Part (c) describes what happens under the alternatives that matter, where the true mean lies in neither model. Departures captured by the larger model raise γ_1 and help; departures outside it raise γ_2 , inflate the denominator and hurt. A departure orthogonal to $\mathcal{C}(\mathbf{Z})$ gives power *below* the level ([Exercise \(B3\)](#)).

14.7.2 Near replicates, partitions and the rainbow test

Without exact replicates, the natural substitute is to group rows that are *nearly* identical. A cluster analysis of the standardized rows of $\mathbf{X}$ forms g clusters of near replicates, with indicator matrix $\mathbf{D}$. Testing $\mathbf{X}$ against $\mathbf{Z} = [\mathbf{X}, \mathbf{D}]$ allows each cluster its own shift from the fitted plane. This is the near-replicate test studied by Christensen (1989); Shillington (1979) proposed a variant based on the cluster averages of the regressors, and Christensen (1991) characterized the kinds of lack of fit each detects. With exact replicates all reduce to the pure-error test.

A second family partitions the cases instead of the regressor space. If the rows are split into two groups, with model matrices $\mathbf{X}_1$ and $\mathbf{X}_2$, the model that fits a separate regression to each group has $\mathbf{Z} = \begin{pmatrix} \mathbf{X}_1 & \mathbf{0} \\ \mathbf{0} & \mathbf{X}_2 \end{pmatrix}$, whose column space contains $\mathcal{C}(\mathbf{X})$, and $\text{SSE}(\mathbf{Z})$ is the sum of the two residual sums of squares ([Exercise \(B1\)](#)). A straight line fitted to a curved mean is beaten by two lines, one for each half of the range. Utts (1982) proposed an extreme partition, the **rainbow test**: fit the model to a central subset of m cases, those with the smallest leverages,

and give each of the remaining $n - m$ cases a free mean of its own. Then $\text{SSE}(\mathbf{Z})$ is the residual sum of squares of the central fit, and the test asks whether the model fitted to the middle of the design extrapolates to its edges.

All of these are exact by [Proposition 14.7.1](#), and all require a decision made from $\mathbf{X}$ alone: how many clusters, where to split, how large a central subset. Choosing the clusters or the split after looking at the residuals destroys the exactness, because $\mathbf{Z}$ then depends on $\mathbf{Y}$.

14.7.3 Richer mean functions

The most direct alternative is a larger parametric model: add squares and cross-products of the regressors, or a regression spline. For a single regressor, a cubic spline with interior knots $\kappa_1, \dots, \kappa_m$ adds the columns x^2 , x^3 and $(x - \kappa_l)_+^3$, $l = 1, \dots, m$, and can follow any smooth curve if the knots are dense enough. These are reduced-model F tests ([Theorem 11.1.1](#)), exact by [Proposition 14.7.1\(a\)](#) provided the larger model is fixed before seeing the data. Their power is concentrated on departures that the added columns can represent. A quadratic term is excellent against smooth curvature and nearly useless against a departure confined to a small part of the range.

14.7.4 Regressors built from the fitted values

A classical alternative looks for departures along the fitted values: if the mean is a nonlinear function of $\mathbf{x}^\top \boldsymbol{\beta}$, then powers of $\hat{y}$ should help predict y . But $\hat{\mathbf{Y}} = \mathbf{M}\mathbf{Y}$ depends on the response, so [Proposition 14.7.1](#) does not apply. Remarkably, the F test is exact anyway.

Theorem 14.7.2 — Regressors that are functions of the fitted values

Let $\mathbf{Y} \sim N_n(\mathbf{X}\boldsymbol{\beta}, \sigma^2\mathbf{I})$ with $\text{rank}(\mathbf{X}) = r$. Let $\mathbf{W} = w(\mathbf{M}\mathbf{Y})$ be an $n \times q$ matrix computed from the fitted values alone, with $\text{rank}[\mathbf{X}, \mathbf{W}] = r + q < n$ with probability one. Then

$$F = \frac{[\text{SSE}(\mathbf{X}) - \text{SSE}([\mathbf{X}, \mathbf{W}])]/q}{\text{SSE}([\mathbf{X}, \mathbf{W}])/(n - r - q)} \sim F(q, n - r - q).$$

Proof. Let $\tilde{\mathbf{W}} = (\mathbf{I} - \mathbf{M})\mathbf{W}$, which has rank q , and let $\mathbf{P}$ project onto $\mathcal{C}(\tilde{\mathbf{W}})$. As in [Lemma 6.6.1](#), $\mathcal{C}([\mathbf{X}, \mathbf{W}]) = \mathcal{C}(\mathbf{X}) \oplus \mathcal{C}(\tilde{\mathbf{W}})$ with orthogonal summands, so the projection onto $\mathcal{C}([\mathbf{X}, \mathbf{W}])$ is $\mathbf{M} + \mathbf{P}$. With $\hat{\boldsymbol{\epsilon}} = (\mathbf{I} - \mathbf{M})\mathbf{Y}$, and since $\mathbf{P}\mathbf{M} = \mathbf{0}$,

$$\begin{aligned} \text{SSE}(\mathbf{X}) - \text{SSE}([\mathbf{X}, \mathbf{W}]) &= \|\mathbf{P}\mathbf{Y}\|^2 = \|\mathbf{P}\hat{\boldsymbol{\epsilon}}\|^2, \\ \text{SSE}([\mathbf{X}, \mathbf{W}]) &= \|\hat{\boldsymbol{\epsilon}}\|^2 - \|\mathbf{P}\hat{\boldsymbol{\epsilon}}\|^2. \end{aligned}$$

By [Theorem 4.4.4](#), $\mathbf{M}\mathbf{Y}$ and $\hat{\boldsymbol{\epsilon}}$ are independent, and $\hat{\boldsymbol{\epsilon}} \sim N_n(\mathbf{0}, \sigma^2(\mathbf{I} - \mathbf{M}))$. So, given $\mathbf{M}\mathbf{Y}$, the matrix $\mathbf{P}$ is fixed and $\hat{\boldsymbol{\epsilon}}$ keeps its distribution, which is that of $(\mathbf{I} - \mathbf{M})\boldsymbol{\epsilon}^*$ for $\boldsymbol{\epsilon}^* \sim N_n(\mathbf{0}, \sigma^2\mathbf{I})$. Because $\mathcal{C}(\mathbf{P}) \subseteq \mathcal{C}(\mathbf{X})^\perp$, $\mathbf{P}(\mathbf{I} - \mathbf{M}) = \mathbf{P}$, and the two sums of squares become $\|\mathbf{P}\boldsymbol{\epsilon}^*\|^2$ and $\|(\mathbf{I} - \mathbf{M} - \mathbf{P})\boldsymbol{\epsilon}^*\|^2$. These are quadratic forms in orthogonal projections of ranks q and $n - r - q$, independent $\sigma^2\chi^2(q)$ and $\sigma^2\chi^2(n - r - q)$ variables by [Theorem 4.4.4](#). The conditional law of F is therefore $F(q, n - r - q)$ whatever $\mathbf{M}\mathbf{Y}$ is, and by [Lemma 14.1.1](#) it is also the unconditional law.

The theorem is due to Milliken and Graybill (1970). Its best-known case is Ramsey's (1969) **RESET**, which adds the squares and cubes of the fitted values, entry by entry. Tukey's (1949) one degree of freedom for nonadditivity, which tests a two-way additive layout against a multiplicative interaction, can be written in the same form with the squared fitted values. With a single regressor the construction brings nothing new: when $\hat{\beta}_1 \neq 0$, the columns $\hat{\mathbf{y}}^2$ and $\hat{\mathbf{y}}^3$ span, together with $\mathbf{1}$ and $\mathbf{x}$, the same space as x^2 and x^3 , so RESET is the cubic polynomial test ([Exercise \(B4\)](#)). Its value is in many dimensions, where it spends two degrees of freedom on the one direction along which the fit varies. ■

14.7.5 Smoothing-based tests

Instead of committing to a parametric alternative, one can ask whether the residuals contain any smooth structure. Let $\mathbf{S}$ be a fixed smoother matrix, for instance kernel weights that average each residual with those of its neighbours in the regressor space, and consider

$$T = \frac{\|\mathbf{S}\hat{\boldsymbol{\varepsilon}}\|^2}{\|\hat{\boldsymbol{\varepsilon}}\|^2}. \quad (14.7.1)$$

If the model is right, the residuals are noise and smoothing them leaves little; if the mean has structure the model misses, the smoothed residuals keep it and T is large.

Proposition 14.7.3 — The smoother statistic is pivotal

If $\mathbf{Y} \sim N_n(\mathbf{X}\boldsymbol{\beta}, \sigma^2\mathbf{I})$, the distribution of T in [Eq. 14.7.1](#) depends only on $\mathbf{X}$ and $\mathbf{S}$, not on $\boldsymbol{\beta}$ or σ^2 . It equals the distribution of $\|\mathbf{S}(\mathbf{I} - \mathbf{M})\mathbf{u}\|^2 / \|(\mathbf{I} - \mathbf{M})\mathbf{u}\|^2$ with $\mathbf{u} \sim N_n(\mathbf{0}, \mathbf{I})$.

Proof. $\hat{\boldsymbol{\varepsilon}} = (\mathbf{I} - \mathbf{M})\mathbf{Y} = (\mathbf{I} - \mathbf{M})\boldsymbol{\varepsilon}$, and $\boldsymbol{\varepsilon} = \sigma\mathbf{u}$ with $\mathbf{u} \sim N_n(\mathbf{0}, \mathbf{I})$. The factor σ^2 cancels in the ratio. ■

So the null distribution of T can be simulated as precisely as desired, from the design alone, and the test has exact size apart from Monte Carlo error. Statistics of this kind were proposed by Azzalini and Bowman (1993), who compare the residual sums of squares of the parametric fit and of a nonparametric smoother, and they are developed at length by Hart (1997). The power depends on the amount of smoothing, and a bandwidth chosen after inspecting the data breaks the exact size. With several regressors, neighbourhoods become sparse, so in practice one smooths along a single direction, such as the fitted values. And exactness depends, as for every test in this section, on normal homoscedastic errors.

14.7.6 Stack loss with all three regressors

Example. (Is the three-variable model adequate?)

Return to the stack loss data with air flow, water temperature and acid concentration,

$n = 21$ and $r = 4$. Ward clustering of the standardized rows into 8 clusters and testing $\mathbf{X}$ against $[\mathbf{X}, \mathbf{D}]$ gives $F = 1.43$ on 7 and 10 degrees of freedom (p -value 0.292). Adding the squares of the three regressors gives $F = 1.92$ on 3 and 14 (p -value 0.172). The rainbow test with the 11 cases of smallest leverage as the central subset gives $F = 1.04$ on 10 and 7 (p -value 0.497), and RESET gives $F = 1.17$ on 2 and 15 (p -value 0.336). None of the tests finds lack of fit. The strong lack of fit of the line in air flow alone (Example (Stack loss and air flow)) was the omitted water temperature, which varies within the groups of equal air flow. With 21 cases none of these tests is powerful, and a few unusual days deserve the scrutiny of Chapter 20.

```
import numpy as np
import statsmodels.api as sm
from scipy import stats
from scipy.cluster.hierarchy import fcluster, linkage
from statsmodels.stats.diagnostic import linear_reset

def proj(A):
    """Orthogonal projection onto  $C(A)$ , via an orthonormal basis of the column space
    ."""
    U, s, _ = np.linalg.svd(A, full_matrices=False)
    U = U[:, s > s[0] * 1e-10]
    return U @ U.T

def f_test(y, X, Zbig):
    """F test of  $C(X)$  against a larger  $C(Zbig)$   $C(X)$ ; returns F, df1, df2, p."""
    n = len(y)
    M, MZ = proj(X), proj(Zbig)
    r, s = round(np.trace(M)), round(np.trace(MZ))
    sse, sse_z = y @ (y - M @ y), y @ (y - MZ @ y)
    F = ((sse - sse_z) / (s - r)) / (sse_z / (n - s))
    return F, s - r, n - s, stats.f.sf(F, s - r, n - s)

data = sm.datasets.stackloss.load_pandas().data
y = data["STACKLOSS"].to_numpy()
R = data[["AIRFLOW", "WATERTEMP", "ACIDCONC"]].to_numpy()
n = len(y)
X = np.column_stack([np.ones(n), R])
fit = sm.OLS(y, X).fit()

# exact replicates: rows of X that occur more than once
_, rows = np.unique(R, axis=0, return_inverse=True)
print("distinct rows:", rows.max() + 1, "of", n)

# near replicates: cluster the standardized regressor rows (Ward linkage), c
clusters
Rs = (R - R.mean(axis=0)) / R.std(axis=0)
clusters = fcluster(linkage(Rs, method="ward"), t=8, criterion="maxclust")
D = np.eye(clusters.max())[clusters - 1] # cluster indicators
tests = {"near replicates": f_test(y, X, np.column_stack([X, D])),
        "added squares": f_test(y, X, np.column_stack([X, R ** 2]))}

# rainbow: refit on the half of the cases with the smallest leverages
h = np.diag(proj(X))
central = np.argsort(h)[: n // 2 + 1]
E = np.eye(n)[:, np.setdiff1d(np.arange(n), central)] # one free mean per outer
case
tests["rainbow"] = f_test(y, X, np.column_stack([X, E]))
```

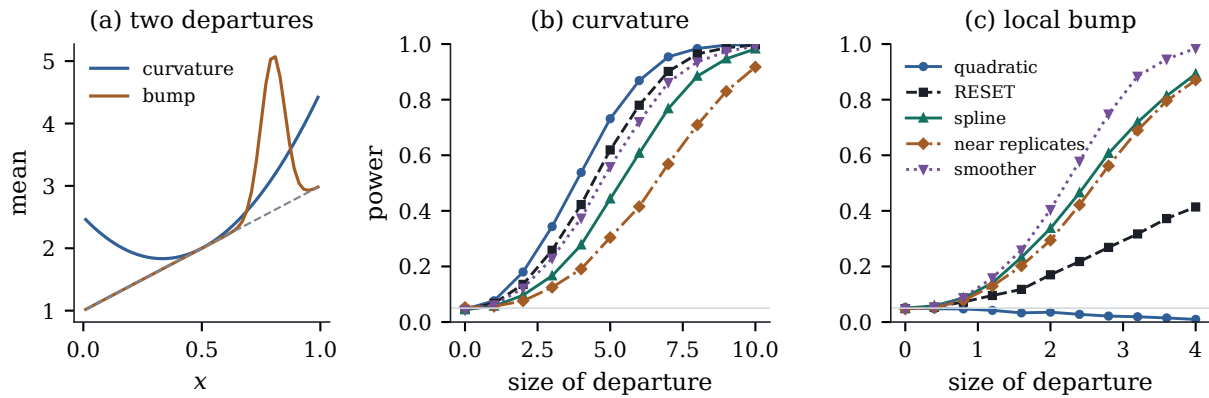


Figure 14.7.1: Power of five exact lack-of-fit tests of a straight line ($n = 50$, level 0.05). (a) The two departures. (b) Smooth curvature. (c) A local bump, against which the quadratic test has power below its level.

```
# RESET: add squares and cubes of the fitted values
reset = linear_reset(fit, power=3, test_type="fitted", use_f=True)
tests["RESET"] = (reset.fvalue, reset.df_num, reset.df_denom, reset.pvalue)
for name, (F, d1, d2, p) in tests.items():
    print(f"{name:16s} F = {F:6.2f} on ({d1:.0f}, {d2:.0f}) df, p = {p:.3f}")
```

14.7.7 Which test?

No lack-of-fit test is best against every alternative: by [Proposition 14.7.1\(c\)](#), a test gains power from departures its larger model can represent and loses power from the others.

Example. (Two departures, five tests)

Take $n = 50$ equally spaced values of x in $(0, 1)$, a straight-line working model, standard normal errors, and five tests at level 0.05: adding x^2 ; RESET with squared and cubed fitted values; a cubic spline with knots at 0.25, 0.5 and 0.75; near replicates formed by 10 clusters of 5 neighbouring values; and the smoother statistic [Eq. 14.7.1](#) with Gaussian kernel weights of bandwidth 0.06, its critical value simulated from 50000 null residual vectors. Over 4000 samples from the straight line, the rejection rates are 0.050, 0.054, 0.054, 0.052 and 0.054: all five tests are exact.

Against smooth curvature, a mean of $1 + 2x + 6(x - 0.5)^2$, the powers are 0.874 for the quadratic term, 0.786 for RESET (which here is the cubic polynomial test), 0.606 for the spline, 0.422 for near replicates and 0.723 for the smoother. Against a narrow bump of height 2.5 near $x = 0.8$, the order is reversed: the smoother has power 0.622, the spline 0.501, near replicates 0.459, RESET 0.227, and the quadratic term 0.029, *below* the level: the bump is nearly orthogonal to x^2 after the line is removed, so it inflates the denominator of that test without feeding its numerator. [Figure 14.7.1](#) shows the power curves.

```
rng = np.random.default_rng(1471)
n2, reps = 50, 4000
x = (np.arange(n2) + 0.5) / n2
```

fixed design on (0, 1)


```

X2 = np.column_stack([np.ones(n2), x])
M2 = proj(X2)
knots = (0.25, 0.5, 0.75)
spline = np.column_stack([x ** 2, x ** 3] + [np.clip(x - k, 0, None) ** 3 for k in
    knots])
cells = np.eye(10)[np.arange(n2) // 5]          # 10 clusters of 5
    neighbouring x
bigger = {"quadratic": np.column_stack([X2, x ** 2]),
    "spline": np.column_stack([X2, spline]),
    "near replicates": np.column_stack([X2, cells])}
proj_big = {k: proj(v) for k, v in bigger.items()}

# kernel smoother of the residuals;  $T = |S e|^2 / |e|^2$ , null law free of beta and
    sigma
bw = 0.06
K = np.exp(-0.5 * ((x[:, None] - x[None, :]) / bw) ** 2)
S = K / K.sum(axis=1, keepdims=True)
null_e = (np.eye(n2) - M2) @ rng.normal(size=(n2, 50_000))
T_null = np.sum((S @ null_e) ** 2, axis=0) / np.sum(null_e ** 2, axis=0)
T_crit = np.quantile(T_null, 0.95)

def powers(mean):
    """Rejection rates at level 0.05 of each test when  $E(Y) = \text{mean}$ ."""
    Y = mean[:, None] + rng.normal(size=(n2, reps))
    e = Y - M2 @ Y
    sse = np.sum(e ** 2, axis=0)
    out = {}
    for name, P in proj_big.items():
        s = round(np.trace(P))
        sse_z = np.sum((Y - P @ Y) ** 2, axis=0)
        F = ((sse - sse_z) / (s - 2)) / (sse_z / (n2 - s))
        out[name] = np.mean(F > stats.f.ppf(0.95, s - 2, n2 - s))
    Fr = np.empty(reps)
    for j in range(reps):          # RESET: regressors depend on
        y
        yh = M2 @ Y[:, j]
        W = np.column_stack([yh ** 2, yh ** 3])
        Wt = W - M2 @ W
        g, *_ = np.linalg.lstsq(Wt, e[:, j], rcond=None)
        ss_add = e[:, j] @ (Wt @ g)
        Fr[j] = (ss_add / 2) / ((sse[j] - ss_add) / (n2 - 4))
    out["RESET"] = np.mean(Fr > stats.f.ppf(0.95, 2, n2 - 4))
    T = np.sum((S @ e) ** 2, axis=0) / sse
    out["smoother"] = np.mean(T > T_crit)
    return out

curve = (x - 0.5) ** 2          # smooth curvature
bump = np.exp(-0.5 * ((x - 0.8) / 0.05) ** 2)          # a local bump
sizes = powers(1 + 2 * x)
print("sizes:", {k: round(float(v), 3) for k, v in sizes.items()})
reported = {}
for label, shape, a in [("curvature", curve, 6.0), ("bump", bump, 2.5)]:
    reported[label] = powers(1 + 2 * x + a * shape)
    print(f"{label:9s}:", {k: round(float(v), 3) for k, v in reported[label].items()
    })

```

The lesson is to decide which inadequacy matters before testing: a low-degree polynomial or

RESET for smooth curvature, a flexible alternative or a smoother for local departures. Only the pure-error test of [Theorem 14.6.2](#) has a larger model containing every mean function of the regressors. Residual plots, which look in all directions informally, remain indispensable (Chapter 20).

14.7.8 Exercises

14.7.8.1 A. Check your understanding

Exercise A1.

A regression with an intercept and two regressors has $n = 30$ cases, whose rows are grouped into 8 clusters of near replicates. Assuming no combination of cluster indicators lies in $\mathcal{C}(\mathbf{X})$ except the constant, give the degrees of freedom of the near-replicate test of $\mathbf{X}$ against $[\mathbf{X}, \mathbf{D}]$.

14.7.8.2 B. Practice

Exercise B1.

Let $\mathbf{X} = \begin{pmatrix} \mathbf{x}_1 \\ \mathbf{x}_2 \end{pmatrix}$ and $\mathbf{Z} = \begin{pmatrix} \mathbf{x}_1 & \mathbf{0} \\ \mathbf{0} & \mathbf{x}_2 \end{pmatrix}$. Show that $\mathcal{C}(\mathbf{X}) \subseteq \mathcal{C}(\mathbf{Z})$, that the projection onto $\mathcal{C}(\mathbf{Z})$ is $\begin{pmatrix} \mathbf{M}_1 & \mathbf{0} \\ \mathbf{0} & \mathbf{M}_2 \end{pmatrix}$, and that $\text{SSE}(\mathbf{Z})$ is the sum of the residual sums of squares of the two separate regressions.

Solution. $\mathbf{X} = \mathbf{Z} \begin{pmatrix} \mathbf{I} \\ \mathbf{0} \end{pmatrix}$, so $\mathcal{C}(\mathbf{X}) \subseteq \mathcal{C}(\mathbf{Z})$. The block-diagonal matrix of the two projections is symmetric and idempotent, and its column space is $\mathcal{C}(\mathbf{X}_1) \times \mathcal{C}(\mathbf{X}_2) = \mathcal{C}(\mathbf{Z})$, so it is the projection onto $\mathcal{C}(\mathbf{Z})$ ([Theorem 6.3.3](#)). Applying $\mathbf{I}$ minus it to $\mathbf{y} = (\mathbf{y}_1^\top, \mathbf{y}_2^\top)^\top$ gives the two residual vectors $(\mathbf{I} - \mathbf{M}_1)\mathbf{y}_1$ and $(\mathbf{I} - \mathbf{M}_2)\mathbf{y}_2$, whose squared lengths add.

Exercise B2.

In the rainbow test, order the cases so that the central subset of size m comes first, and let its model matrix $\mathbf{X}_c$ have rank r . Show that $\mathbf{Z} = \begin{pmatrix} \mathbf{x}_c & \mathbf{0} \\ \mathbf{0} & \mathbf{I}_{n-m} \end{pmatrix}$ has $\mathcal{C}(\mathbf{X}) \subseteq \mathcal{C}(\mathbf{Z})$, that $\text{SSE}(\mathbf{Z})$ is the residual sum of squares of the central fit, and that the test has $n - m$ and $m - r$ degrees of freedom.

Exercise B3.

In [Proposition 14.7.1](#), suppose $\boldsymbol{\theta} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\delta}$ with $\boldsymbol{\delta} \neq \mathbf{0}$ orthogonal to $\mathcal{C}(\mathbf{Z})$. Show that the power of the test is strictly less than its size.

Solution. Here $(\mathbf{M}_Z - \mathbf{M})\boldsymbol{\theta} = \mathbf{M}_Z\boldsymbol{\delta} - \mathbf{M}\boldsymbol{\delta} = \mathbf{0}$, since $\boldsymbol{\delta} \perp \mathcal{C}(\mathbf{Z}) \supseteq \mathcal{C}(\mathbf{X})$, so $\gamma_1 = 0$, while $\gamma_2 = \|\boldsymbol{\delta}\|^2/\sigma^2 > 0$. By part (c), $\Pr(F > c)$ is strictly smaller than its value at $\gamma_2 = 0$, which is the size by part (a).

Exercise B4.

With a single regressor and an intercept, show that if $\hat{\beta}_1 \neq 0$, then $\mathcal{C}([\mathbf{1}, \mathbf{x}, \hat{\mathbf{y}}^2, \hat{\mathbf{y}}^3]) = \mathcal{C}([\mathbf{1}, \mathbf{x}, \mathbf{x}^2, \mathbf{x}^3])$, powers taken entry by entry. Conclude that RESET is then the F test of the cubic polynomial.

14.7.8.3 C. Going deeper**Exercise C1.**

In a two-way layout with one observation per cell and the additive model $E(y_{ij}) = \mu + \alpha_i + \beta_j$, Tukey's statistic tests the added regressor $(\hat{\alpha}_i \hat{\beta}_j)$. Show that, after the additive model is projected out, this regressor spans the same space as the entrywise square of the fitted values, and deduce from [Theorem 14.7.2](#) that Tukey's test has an exact $F(1, (a-1)(b-1)-1)$ null distribution.

14.8 Summary and notes

14.8.1 Summary

Idea.

1. If the errors are normal, homoscedastic and independent of the regressors, every t and F statistic has the same law given every design, so tests and intervals stay exact with random regressors of any distribution. Power depends on the design drawn and is averaged over it ([Lemma 14.1.1](#), [Theorem 14.1.2](#)).
2. The estimates themselves change law. With normal regressors the slope of a simple regression is a scaled $t(n-1)$ variable with variance $\sigma^2/(\sigma_X^2(n-3))$, and with k regressors the slopes have covariance $\sigma^2 \Sigma_{XX}^{-1}/(n-k-2)$ ([Proposition 14.1.4](#), [Lemma 14.2.4](#), [Corollary 14.2.5](#)).
3. The conditional mean is the only predictor whose errors are uncorrelated with every function of the regressors, and the correlation ratio bounds the correlation any predictor can reach ([Proposition 14.2.1](#)).
4. A fitted linear predictor pays for estimation. With normal regressors its expected error on a new case is $\sigma^2(1+1/n)(n-2)/(n-k-2)$, while its in-sample error has mean $\sigma^2(n-k-1)/n$; with a fixed design the gap is $2r\sigma^2/n$ ([Proposition 14.2.2](#), [Theorem 14.2.6](#), [Proposition 14.2.7](#)).
5. When the new error is correlated with the data, the best linear unbiased predictor adds to the estimated mean the best linear prediction of that error from the generalized least squares residuals ([Theorem 14.2.8](#)).
6. R^2 is the maximum likelihood estimate of $\rho_{Y.X}^2$. Given the design it is a transformed noncentral F variable; with normal regressors its noncentrality is $\lambda\chi^2(n-1)$, $\lambda = \rho^2/(1-\rho^2)$, which gives its exact law, its bias, the power of the overall F test and an exact interval for ρ^2 ([Proposition 14.3.1](#), [Theorem 14.3.2](#)).
7. The sample correlation has an exact law expressed through noncentral t probabili-

ties, and it is stochastically increasing in ρ . Fisher's $z = \tanh^{-1} r$ is asymptotically $N(\zeta, 1/n)$ whatever ρ , and in practice $N(\zeta, 1/(n-3))$ ([Proposition 14.4.1](#), [Theorem 14.4.3](#)).

8. A sample partial correlation given q variables behaves exactly like an ordinary correlation from $n-q$ cases, and its test is the t test of the corresponding regression coefficient ([Theorem 14.5.1](#), [Proposition 14.5.2](#), [Theorem 14.5.3](#)).
9. With replicated rows, the residual sum of squares splits into pure error and lack of fit, and their ratio is an exact F test of the model against every mean function of the regressors ([Lemma 14.6.1](#), [Theorem 14.6.2](#)).
10. Without replicates, any larger model chosen from $\mathbf{X}$ alone, any regressors built from the fitted values, and any smoother of the residuals give exact tests. Each has power only against departures its alternative can represent, and some departures reduce its power below the level ([Proposition 14.7.1](#), [Theorem 14.7.2](#), [Proposition 14.7.3](#)).

14.8.2 Notes and sources

Source texts. The chapter covers Christensen (2020, sections 6.3 to 6.7) on prediction theory, multiple and partial correlation, best linear unbiased prediction and lack of fit; Rencher and Schaalje (2008, chapter 10) on regression with random regressors; and Agresti (2015, sections 3.3 and 3.4), whose worked example includes partial correlations and model checking. Christensen's near-replicate tests and his view of prediction as the goal of regression shaped [Section 14.2](#) and [Section 14.7](#). Rencher and Schaalje give the normal-theory formulas for R^2 , z and partial correlations. Agresti's intervals of section 3.3 belong to [Chapter 12](#).

Random regressors and prediction. Reporting precision conditionally on an ancillary design is Fisher's conditionality principle; Cox (1958) gave the standard discussion. The Wishart distribution is due to Wishart (1928), and Anderson (2003) treats it and the mean of its inverse through densities; the proof of [Lemma 14.2.4](#) here uses only conditioning and symmetry. Breiman and Freedman (1983) use the prediction error of regression with normal regressors to study how many variables to enter, a question Chapter 29 takes up. Best linear unbiased prediction is due to Goldberger (1962); Chapter 31 and Chapter 32 develop it for general covariance structures and mixed models.

Correlation coefficients. Fisher found the exact distribution of r (1915), introduced z with its approximate mean and variance (1921), showed that partial correlations follow the same law with the sample size reduced by the number of conditioning variables (1924), and derived the distribution of the multiple correlation (1928). Hotelling (1953) refined the moments of r and z . [Lemma 14.4.2](#) is a case of Isserlis's (1918) formula, and the mean of R^2 in [Example \(How biased is R-squared?\)](#) follows from Fisher's (1928) distribution; see Olkin and Pratt (1958), who also give the unbiased estimator of ρ^2 . For the limit theorems of [Section 14.4](#), see van der Vaart (1998).

Lack of fit. The pure-error test is Fisher's (1922). Near-replicate tests were proposed by Shillington (1979) and analysed by Christensen (1989, 1991); the rainbow test is due to Utts (1982). Ramsey (1969) introduced RESET, and Milliken and Graybill (1970) proved [Theorem 14.7.2](#), which also covers Tukey's (1949) one degree of freedom for nonadditivity. Smoothing-based tests were proposed by Azzalini and Bowman (1993) and are the subject of Hart (1997).

Data. The stack loss data are from Brownlee (1965). The RAND Health Insurance Experiment

is described by Manning et al. (1987); the extract and the 2009 state data are public-domain data distributed with statsmodels.

What is new here. The conditional derivations of the laws of R^2 , r and partial correlations (Theorem 14.3.2, Proposition 14.4.1, Theorem 14.5.1), the proof of Lemma 14.2.4, the organization of lack-of-fit tests around Proposition 14.7.1, and the examples and simulations are this book's.

14.8.3 References

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Chapter 15

Analysis of Variance as Subspace Comparison

An experimenter assigns treatments to units at random and measures a response. The simplest such experiment has one factor, and its analysis is the prototype for Part IV. The mean vector lies in the space of vectors constant within groups, that space splits orthogonally into the constants and a treatment space, and every question about the treatments concerns the component of the mean in the treatment space. This chapter turns that geometry into a working analysis. It states the one-way model and what its parameters mean, shows that single contrasts, families of contrasts and the overall F test are projections of the data onto pieces of one subspace, and derives the analysis of variance table and its expected mean squares. It splits the treatment sum of squares into single degrees of freedom along directions chosen for their meaning, including polynomial trends in a quantitative factor. Finally it asks what balance buys: plain orthogonality, the largest guaranteed power, and a test that tolerates unequal variances. When the design is unbalanced and the variances differ, the classical test can go badly wrong, and the chapter describes the heteroscedasticity-robust tests of Welch and of Brown and Forsythe.

What you need. The one-way projection and its generalized inverses from [Chapter 6](#) ([Example \(A rank-deficient layout\)](#), [Example \(The one-way layout\)](#), [Theorem 6.5.1](#)); estimability, side conditions and contrasts from [Chapter 8](#) ([Theorem 8.7.1](#), [Proposition 8.7.2](#), [Theorem 8.4.2](#)); orthogonal decompositions and expected mean squares from [Chapter 9](#) ([Theorem 9.6.1](#), [Theorem 9.4.4](#)); the F test and its power from [Chapter 11](#) ([Theorem 11.1.1](#), [Theorem 11.5.1](#)); the t interval from [Chapter 12](#) ([Theorem 12.1.1](#)); and the simultaneous methods of [Chapter 13](#) ([Theorem 13.2.2](#), [Theorem 13.3.2](#), [Theorem 13.4.1](#)). The pure-error lack-of-fit test of [Chapter 14](#) ([Theorem 14.6.2](#)) reappears as trend analysis.

15.0.1 Roadmap

- [The one-way layout as a linear model](#)
- [Estimable contrasts](#)
- [The analysis of variance table from projections](#)
- [Orthogonal contrasts](#)
- [Balanced and unbalanced one-way layouts](#)
- [Summary and notes](#)

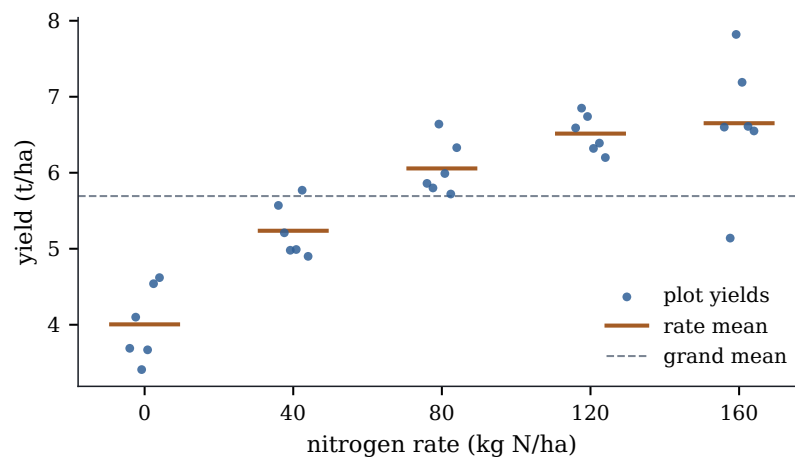


Figure 15.1.1: The nitrogen trial: yields of thirty plots at five nitrogen rates, six plots per rate (slightly spread horizontally), with the rate means and the grand mean.

15.1 The one-way layout as a linear model

Earlier chapters used the one-way layout as an example of rank deficiency ([Chapter 6](#)), estimability ([Chapter 8](#)), orthogonal decomposition ([Chapter 9](#)) and simultaneous inference ([Chapter 13](#)). Here it is the subject. This section fixes the model, recalls its projection, and settles what its parameters mean and where they come from.

15.1.1 A designed experiment

An agronomist wants to know how the yield of a wheat variety responds to nitrogen fertilizer. Thirty field plots of equal size are available. Five application rates are chosen, 0, 40, 80, 120 and 160 kilograms of nitrogen per hectare, and each rate is assigned to six plots chosen at random from the thirty. At harvest the grain yield of each plot is recorded in tonnes per hectare. This is a **completely randomized design** with one **factor** (nitrogen) at five **levels**, also called **treatments**, with six **replicates** of each. The data used throughout this chapter were simulated from a smooth response curve with plot-to-plot standard deviation 0.5 t/ha, which the analysis does not know about.

Rate (kg N/ha)	Yields (t/ha)	Mean	Standard deviation
0	3.69, 4.10, 3.41, 3.67, 4.54, 4.62	4.005	0.498
40	5.57, 5.21, 4.98, 4.99, 5.77, 4.90	5.237	0.357
80	5.86, 5.80, 6.64, 5.99, 5.72, 6.33	6.057	0.357
120	6.59, 6.85, 6.74, 6.32, 6.39, 6.20	6.515	0.254
160	6.60, 5.14, 7.82, 7.19, 6.61, 6.55	6.652	0.889

[Figure 15.1.1](#) shows the thirty yields. Mean yield rises with the rate, steeply at first and then more slowly, and the plots within a rate scatter by roughly half a tonne. The later sections compare particular treatments ([Section 15.2](#)), ask whether the rates differ at all ([Section 15.3](#)), and describe the shape of the response ([Section 15.4](#)).

15.1.2 The model in two forms

Index the groups by $k = 1, \dots, g$ and the observations within group k by $j = 1, \dots, n_k$. The observations may be listed in any order in the vector $\mathbf{Y}$; what matters is which group each belongs to.

Definition 15.1.1 — The one-way layout

A **one-way layout** has $g \geq 2$ groups with $n_k \geq 1$ observations in group k and $n = \sum_k n_k > g$ observations in all. Its **cell-means form** is

$$Y_{kj} = \mu_k + \varepsilon_{kj}, \quad \text{that is,} \quad \mathbf{Y} = \mathbf{Z}\boldsymbol{\mu} + \boldsymbol{\varepsilon},$$

where $\mathbf{Z}$ is the $n \times g$ matrix whose column k is the indicator $\mathbf{z}_k$ of group k and $\boldsymbol{\mu} = (\mu_1, \dots, \mu_g)^\top$. Its **effects form** is

$$Y_{kj} = \mu + \alpha_k + \varepsilon_{kj}, \quad \text{that is,} \quad \mathbf{Y} = [\mathbf{1}, \mathbf{Z}] \begin{pmatrix} \mu \\ \boldsymbol{\alpha} \end{pmatrix} + \boldsymbol{\varepsilon}.$$

In both, the errors have $E(\boldsymbol{\varepsilon}) = \mathbf{0}$ and $\text{Cov}(\boldsymbol{\varepsilon}) = \sigma^2 \mathbf{I}$. The **normal one-way model** adds $\boldsymbol{\varepsilon} \sim N_n(\mathbf{0}, \sigma^2 \mathbf{I})$. The layout is **balanced** if $n_1 = \dots = n_g = m$, say. The treatment parameters μ_k (or α_k) are **fixed effects**: unknown constants attached to the g treatments actually studied.

The two forms describe the same mean vectors: since $\mathbf{z}_1 + \dots + \mathbf{z}_g = \mathbf{1}$, the column space of $[\mathbf{1}, \mathbf{Z}]$ is $\mathcal{C}(\mathbf{Z})$, the vectors constant within each group (Example (A rank-deficient layout)). The effects matrix has $g + 1$ columns but rank g , so its parameters are not identified. Fitted values, residuals, sums of squares and tests depend only on $\mathcal{C}(\mathbf{Z})$, so the choice of form decides which parameters one talks about, never the results of the analysis.

A word on notation. Group means are written $\bar{y}_k$ (the literature also writes $\bar{y}_{k\cdot}$), the grand mean is $\bar{y} = n^{-1} \sum_{k,j} y_{kj}$, and $\bar{\mu} = n^{-1} \sum_k n_k \mu_k$ is the *weighted* average of the group means, which is $E(\bar{Y})$. The unweighted average $g^{-1} \sum_k \mu_k$ is a different quantity unless the layout is balanced.

15.1.3 The model space and its projection

The orthogonal projection onto $\mathcal{C}(\mathbf{Z})$ was computed in Example (The one-way layout):

$$\mathbf{M} = \sum_{k=1}^g \frac{\mathbf{z}_k \mathbf{z}_k^\top}{n_k}, \quad (\mathbf{M}\mathbf{y})_{kj} = \bar{y}_k, \quad (15.1.1)$$

so the fitted value of every observation is its group mean and the residual is its deviation from that mean. Whatever the parameterization and whatever generalized inverse is used to solve the normal equations, this is the fit (Theorem 6.3.7). In the cell-means form the least squares estimate is unique, $\hat{\boldsymbol{\mu}} = (\bar{y}_1, \dots, \bar{y}_g)^\top$, because $\mathbf{Z}^\top \mathbf{Z} = \text{diag}(n_1, \dots, n_g)$ is invertible.

The residual sum of squares pools the variation inside the groups:

$$\text{SSE} = \|(\mathbf{I} - \mathbf{M})\mathbf{y}\|^2 = \sum_{k=1}^g \sum_{j=1}^{n_k} (y_{kj} - \bar{y}_k)^2 = \sum_{k=1}^g (n_k - 1)s_k^2, \quad (15.1.2)$$

where s_k^2 is the sample variance of group k (taken as zero when $n_k = 1$). The error space $\mathcal{C}(\mathbf{Z})^\perp$ has dimension $n - g$, so the unbiased estimator of σ^2 ([Theorem 5.6.1](#)) is

$$s^2 = \frac{\text{SSE}}{n - g} = \sum_{k=1}^g \frac{n_k - 1}{n - g} s_k^2,$$

a weighted average of the group variances with weights proportional to their degrees of freedom. In the nitrogen trial $s = 0.5211$ t/ha on 25 degrees of freedom, and since the design is balanced s^2 is the plain average of the five group variances.

```
import numpy as np

rates = np.array([0, 40, 80, 120, 160])          # kg N per hectare
g, m = len(rates), 6                             # 5 rates, 6 plots per rate
true_means = 6.8 - 2.7 * np.exp(-0.018 * rates)  # unknown to the analyst
rng = np.random.default_rng(1515)
group = np.repeat(np.arange(g), m)               # plot i received rate group[i]
y = np.round(true_means[group] + rng.normal(0, 0.5, g * m), 2) # yield, t/ha
n = len(y)

Z = (group[:, None] == np.arange(g)).astype(float) # cell-means model matrix
n_k = Z.sum(axis=0)
means = Z.T @ y / n_k                             # least squares: the group means
fitted = Z @ means                                 # M y
resid = y - fitted                                 # (I - M) y
s2 = resid @ resid / (n - g)                       # pooled within-group variance
print("group means:", means.round(3))
print(f"s = {np.sqrt(s2):.4f} on {n - g} degrees of freedom")
```

15.1.4 What the parameters mean

The group means μ_k are estimable, and so is every linear function of them. The effects form adds one parameter that the data cannot determine. By [Theorem 8.7.1](#), a function $\lambda_0\mu + \sum_k \lambda_k\alpha_k$ is estimable iff $\lambda_0 = \sum_k \lambda_k$; in particular no single α_k and not μ itself is estimable, while every difference $\alpha_k - \alpha_l$ is.

Software nevertheless reports values of μ and α_k , by adding a side condition, one nonestimable linear restriction that picks a single solution of the normal equations ([Theorem 8.4.2](#)). Under the restriction, each parameter becomes a particular estimable function of the group means ([Proposition 8.4.3](#)), and it is worth knowing which:

Side condition	μ becomes	α_k becomes
$\sum_k \alpha_k = 0$	$g^{-1} \sum_l \mu_l$, the unweighted average	$\mu_k - g^{-1} \sum_l \mu_l$
$\sum_k n_k \alpha_k = 0$	$\bar{\mu} = n^{-1} \sum_l n_l \mu_l$, the weighted average	$\mu_k - \bar{\mu}$
$\alpha_1 = 0$	μ_1 , the reference group	$\mu_k - \mu_1$

Each row solves the g equations $\mu + \alpha_k = \mu_k$ together with the side condition, and the estimates put $\bar{y}_k$ for μ_k (Exercise (B1)). The second gives the classical $\hat{\mu} = \bar{y}$, $\hat{\alpha}_k = \bar{y}_k - \bar{y}$. The first two agree in a balanced layout; in an unbalanced one they define different “overall means”, and a reported $\hat{\alpha}_k$ means different things under each (Example (Five side conditions, one fit)). Contrasts are unaffected by the choice.

Warning. A side condition is a convention for printing, not a statement about the treatments. “The effect of treatment 3 is 0.4” means nothing until the side condition is named. Report group means and contrasts.

15.1.5 Where the model comes from

In a randomized experiment much of the one-way model can be derived rather than assumed. Suppose each plot i has a yield u_i it would give without nitrogen, reflecting its soil and position, and suppose rate k adds a fixed amount τ_k , the same on every plot: this is **unit–treatment additivity**. Random assignment then makes the set of plots receiving rate k a simple random sample of the thirty. The observed yields in group k are τ_k plus a sample of the u_i , so each has expectation $\tau_k + \bar{u}$, with $\bar{u}$ the average of all thirty plot values. The expectations therefore have exactly the cell-means structure, with $\mu_k = \tau_k + \bar{u}$, whatever the pattern of fertility in the field. Had the six plots given the highest rate been chosen for convenience, say all at one end of the field, a fertility gradient would be confounded with the rate, and no analysis could separate the two.

The derived errors are not exactly those of Definition 15.1.1: sampling without replacement makes them slightly negatively correlated (Exercise (B3)), and nothing makes them normal. But under the hypothesis that no treatment has any effect, every rearrangement of the yields among the treatments was equally likely, so the **randomization distribution** of the F statistic of Section 15.3 can be computed exactly by permuting the data, and for moderate designs the normal-theory F distribution approximates it closely. Chapter 23 develops permutation tests. Here the normal model is the working model, and randomization is the reason for trusting it.

15.1.6 Fixed and random effects

The five nitrogen rates were chosen deliberately, and the conclusions are about them: the treatment parameters are fixed constants, the setting of this chapter. In other studies the levels are themselves a sample. A laboratory measuring an enzyme activity in batches of reagent from a production line wants to know how much batches vary, not how batch 3 compares with batch 5. The natural model then takes the effects α_k to be random, uncorrelated with mean zero and variance σ_a^2 , and independent of the errors. The mean vector is $\mu\mathbf{1}$ and the covariance matrix is $\sigma^2\mathbf{I} + \sigma_a^2\mathbf{Z}\mathbf{Z}^\top$, so the model is no longer a linear model with covariance $\sigma^2\mathbf{I}$.

The projections and sums of squares are the same in both models, but their expectations differ (Section 15.3), and so does the target: contrasts among the treatments studied, or the variance components σ_a^2 and σ^2 (Exercise (B1)). Chapter 32 develops random and mixed effects.

15.1.7 Exercises

15.1.7.1 A. Check your understanding

Exercise A1.

For a one-way layout with group sizes 2, 1, 3, write down $\mathbf{Z}$, $\mathbf{M}$ and $\mathbf{I} - \mathbf{M}$ with the observations listed group by group. What is the residual of the observation in group 2, and why? How many degrees of freedom does the error space have, and which groups contribute to them?

Exercise A2.

Show that $\mu_1 = \cdots = \mu_g$ iff $\sum_k c_k \mu_k = 0$ for every contrast $\mathbf{c}$ ($\sum_k c_k = 0$), and iff $\mu_k - \mu_g = 0$ for $k = 1, \dots, g-1$.

Solution. If all means are equal to μ , then $\sum_k c_k \mu_k = \mu \sum_k c_k = 0$. Conversely, the differences $\mu_k - \mu_g$ are contrasts, and if they vanish the means are equal. So the three conditions are equivalent. The differences $\mu_k - \mu_g$ span the space of contrasts, which has dimension $g-1$.

15.1.7.2 B. Practice

Exercise B1.

Derive the three rows of the table in “What the parameters mean”. Then show that under $\sum_k n_k \alpha_k = 0$ the least squares estimates are $\hat{\mu} = \bar{y}$ and $\hat{\alpha}_k = \bar{y}_k - \bar{y}$, and that $\sum_k n_k \hat{\alpha}_k^2 = \sum_k n_k (\bar{y}_k - \bar{y})^2$.

Solution. A solution of the normal equations ([Theorem 6.4.2](#)) reproduces the fitted values, so $\hat{\mu} + \hat{\alpha}_k = \bar{y}_k$ for every k . With $\sum_k \alpha_k = 0$, summing over k gives $g\hat{\mu} = \sum_k \bar{y}_k$. With $\sum_k n_k \alpha_k = 0$, multiplying by n_k and summing gives $n\hat{\mu} = \sum_k n_k \bar{y}_k = n\bar{y}$. With $\alpha_1 = 0$, the first equation gives $\hat{\mu} = \bar{y}_1$. In each case $\hat{\alpha}_k = \bar{y}_k - \hat{\mu}$. The same algebra with μ_k in place of $\bar{y}_k$ gives the parameter column. The last identity is immediate from $\hat{\alpha}_k = \bar{y}_k - \bar{y}$.

Exercise B2.

For $g = 2$, show that the pooled variance of the one-way model is the pooled variance of the two-sample t test, and that the t statistic for $\mu_1 - \mu_2$ is $(\bar{y}_1 - \bar{y}_2)/(s\sqrt{1/n_1 + 1/n_2})$. ([Section 15.3](#) shows that its square is the one-way F statistic.)

Exercise B3.

In the setting of “Where the model comes from”, suppose N units with values $u_1, \dots, u_N$ are divided at random into groups of fixed sizes $n_1, \dots, n_g$, and unit i in group k yields $u_i + \tau_k$. Show that $E(\bar{Y}_k) = \tau_k + \bar{u}$ and that $\text{Cov}(Y_{kj}, Y_{k'j'}) = -S_u^2/N$ for two different units, where $S_u^2 = \sum_i (u_i - \bar{u})^2/(N-1)$. *Hint:* the sum of the N deviations $u_i - \bar{u}$ is

zero.

Solution. Under random assignment the unit placed in position (k, j) is equally likely to be any of the N , so $E(Y_{kj}) = \tau_k + \bar{u}$, and so is the average $E(\bar{Y}_k)$. For two different positions, the pair of units is equally likely to be any ordered pair of distinct units, so $\text{Cov}(Y_{kj}, Y_{k'j'}) = E\{(u_I - \bar{u})(u_J - \bar{u})\}$ with (I, J) a random ordered pair of distinct indices. Since $\sum_i (u_i - \bar{u}) = 0$,

$$\sum_{i \neq l} (u_i - \bar{u})(u_l - \bar{u}) = \left(\sum_i (u_i - \bar{u}) \right)^2 - \sum_i (u_i - \bar{u})^2 = -(N-1)S_u^2,$$

and dividing by the $N(N-1)$ ordered pairs gives $-S_u^2/N$. The variance of a single observation is $(N-1)S_u^2/N$, so the correlation between two observations is $-1/(N-1)$, small when N is large.

15.1.7.3 C. Going deeper

Exercise C1.

Let $\mathcal{G}$ be the set of $n \times n$ permutation matrices that permute observations only within groups. Show that

$$\mathbf{M} = \frac{1}{|\mathcal{G}|} \sum_{\mathbf{\Pi} \in \mathcal{G}} \mathbf{\Pi},$$

so the one-way projection is the average over the symmetries of the design. Deduce that $\mathcal{C}(\mathbf{Z})$ is exactly the set of vectors fixed by every $\mathbf{\Pi} \in \mathcal{G}$.

Solution. Take observations a and b . If they are in different groups, no $\mathbf{\Pi} \in \mathcal{G}$ sends b to a , so entry (a, b) of the average is $0 = m_{ab}$. If both are in group k , the permutations of group k that send b to a are a fraction $1/n_k$ of all permutations of group k , and the other groups are permuted freely, so entry (a, b) is $1/n_k = m_{ab}$. A vector fixed by every $\mathbf{\Pi}$ is constant within groups, because a transposition of two members of a group exchanges their entries. Conversely, a vector constant within groups is fixed by each $\mathbf{\Pi}$, hence by their average $\mathbf{M}$, so it lies in $\mathcal{C}(\mathbf{M}) = \mathcal{C}(\mathbf{Z})$.

Exercise C2.

In the random-effects model of “Fixed and random effects”, show that $\text{Var}(\bar{Y}) = \sigma^2/n + \sigma_a^2 \sum_k n_k/n^2$. For fixed n and g , which group sizes make the grand mean most precise? Contrast this with the fixed-effects model, where $\text{Var}(\bar{Y}) = \sigma^2/n$ whatever the group sizes.

15.2 Estimable contrasts

Most questions about the treatments of a one-way experiment are contrasts. Does fertilizer help at all? Is the highest rate worth its cost? Which pairs of treatments differ? Earlier chapters supplied the theory: estimability (Chapter 8), the t test and interval (Chapter 11, Chapter 12) and simultaneous methods (Chapter 13). This section collects them for the one-way analysis,

adds a geometric fact about power, and applies them to the nitrogen trial.

15.2.1 Contrasts are the treatment space

Recall that a contrast is a linear function $\psi = \sum_k c_k \mu_k$ with $\sum_k c_k = 0$ and $\mathbf{c} \neq \mathbf{0}$ (Definition 8.7.1). In the effects form it equals $\sum_k c_k \alpha_k$, and the contrasts are exactly the estimable functions of the effects alone (Theorem 8.7.1). Their estimates are $\hat{\psi} = \sum_k c_k \bar{y}_k$, whatever side condition or coding was used to fit the model.

Behind this lies a simple geometry. Let

$$\mathcal{T} = \mathcal{C}(\mathbf{Z}) \cap \mathbf{1}^\perp$$

be the **treatment space**: the vectors that are constant within groups and sum to zero. Its dimension is $g - 1$ (Theorem 6.5.1). A vector $\mathbf{u} \in \mathcal{T}$ takes some value v_k on every observation of group k , with $\sum_k n_k v_k = 0$. Put $c_k = n_k v_k$. Then $\mathbf{c}$ is a contrast and $\mathbf{u}$ is the vector $\mathbf{u}_{\mathbf{c}} = \mathbf{Z}\mathbf{D}^{-1}\mathbf{c}$ of Eq. 8.7.2, with $\mathbf{D} = \text{diag}(n_1, \dots, n_g)$. Conversely every $\mathbf{u}_{\mathbf{c}}$ lies in $\mathcal{T}$ (Proposition 8.7.2). So contrasts and nonzero vectors of the treatment space correspond one to one, and $\hat{\psi} = \mathbf{u}_{\mathbf{c}}^\top \mathbf{y}$. A contrast is a direction in the treatment space, and its estimate is the component of the data along that direction.

Equal treatment means say that the mean vector $\boldsymbol{\theta} = \mathbf{E}(\mathbf{Y})$ has no component in $\mathcal{T}$, that is, that every contrast vanishes (Exercise (A2)). The overall test of Section 15.3 looks at all of $\mathcal{T}$; a single contrast looks along one line of it.

15.2.2 The test and interval for one contrast

For a contrast $\mathbf{c}$ write

$$v_{\mathbf{c}} = \sum_{k=1}^g \frac{c_k^2}{n_k},$$

so that $\text{Var}(\hat{\psi}) = \sigma^2 v_{\mathbf{c}}$ (Eq. 8.7.1).

Proposition 15.2.1 — Sum of squares and test for a contrast

In the normal one-way model, let $\psi = \sum_k c_k \mu_k$ be a contrast, $\hat{\psi} = \sum_k c_k \bar{Y}_k$, and $\psi_0 \in \mathbb{R}$. Then:

- (a) $\hat{\psi} \sim N(\psi, \sigma^2 v_{\mathbf{c}})$, and $\hat{\psi}$ is independent of s^2 ;
- (b) the statistic

$$T = \frac{\hat{\psi} - \psi_0}{s\sqrt{v_{\mathbf{c}}}} \sim t(n - g, \delta), \quad \delta = \frac{\psi - \psi_0}{\sigma\sqrt{v_{\mathbf{c}}}},$$

so the test of $\psi = \psi_0$ rejects when $|T| > t_{n-g, \alpha/2}$, and $\hat{\psi} \pm t_{n-g, \alpha/2} s\sqrt{v_{\mathbf{c}}}$ is an exact $1 - \alpha$ confidence interval for ψ ;

- (c) the **sum of squares of the contrast**

$$\text{SS}(\mathbf{c}) = \frac{\hat{\psi}^2}{v_{\mathbf{c}}} = \|\mathbf{P}_{\mathbf{c}} \mathbf{Y}\|^2,$$

where $\mathbf{P}_{\mathbf{c}}$ projects onto the line spanned by $\mathbf{u}_{\mathbf{c}}$, has one degree of freedom, and $SS(\mathbf{c})/s^2 = T^2$ for $\psi_0 = 0$ has the $F(1, n - g, \delta^2)$ distribution. Replacing $\mathbf{c}$ by $a\mathbf{c}$ with $a \neq 0$ changes neither $SS(\mathbf{c})$ nor the test;

(d) $SS(\mathbf{c}) \leq \sum_k n_k (\bar{Y}_k - \bar{Y})^2$, with equality when $c_k = n_k(\bar{Y}_k - \bar{Y})$ for every k .

Proof. Use the cell-means form, with model matrix $\mathbf{Z}$ of full rank g , so that $(\mathbf{Z}^\top \mathbf{Z})^{-1} = \mathbf{D}^{-1}$ and $\mathbf{c}^\top \mathbf{D}^{-1} \mathbf{c} = v_{\mathbf{c}}$. The function $\mathbf{c}^\top \boldsymbol{\mu}$ is estimable with estimate $\mathbf{c}^\top \hat{\boldsymbol{\mu}} = \hat{\psi}$. (a) and (b) are [Theorem 11.6.1\(a\)](#) and [Corollary 7.5.2](#) with $\boldsymbol{\lambda} = \mathbf{c}$, $r = g$, and the interval is [Theorem 12.1.1](#). The independence of $\hat{\psi}$ and s^2 is part of [Theorem 7.5.1](#). (c) The identity $\hat{\psi}^2/v_{\mathbf{c}} = \|\mathbf{P}_{\mathbf{c}} \mathbf{Y}\|^2$ is [Proposition 8.7.2\(b\)](#), and T^2 is the F statistic of the hypothesis $\psi = 0$ by [Theorem 11.6.1\(b\)](#), with noncentrality δ^2 . Scaling $\mathbf{c}$ by a multiplies $\hat{\psi}^2$ and $v_{\mathbf{c}}$ by a^2 , and leaves the line spanned by $\mathbf{u}_{\mathbf{c}}$ unchanged. (d) is [Exercise \(B1\)](#). ■

By part (c) a contrast's sum of squares is a projection onto a line of $\mathcal{T}$, and by part (d) no line captures more than the whole space. The line that captures all of it is chosen by the data, which is why its t statistic needs Scheffé's multiplier ([Theorem 13.2.2](#)).

15.2.3 Planned contrasts in the nitrogen trial

Before the trial the agronomist wrote down two questions. First, does nitrogen raise yield at all? The natural contrast compares the average of the four fertilized rates with the control, $\psi_1 = \frac{1}{4}(\mu_2 + \mu_3 + \mu_4 + \mu_5) - \mu_1$. Second, is the highest rate better than the moderate one? That is $\psi_2 = \mu_5 - \mu_3$, the difference between 160 and 80 kg N/ha.

Example. (Two planned contrasts)

The pooled standard deviation is $s = 0.5211$ on 25 degrees of freedom. For ψ_1 , $v_{\mathbf{c}} = \frac{1}{6}(1 + 4 \cdot \frac{1}{16})$, the estimate is 2.110 t/ha with standard error 0.238, so $t = 8.87$ and the 95% interval is [1.620, 2.600]. Fertilizer raises yield by roughly two tonnes per hectare. Its sum of squares is 21.370.

For ψ_2 the estimate is 0.595 t/ha with standard error 0.301, $t = 1.98$, two-sided $p = 0.059$, and the interval [−0.025, 1.215]. The data are consistent with anything from no gain to a gain of more than a tonne: with six plots per rate, half a tonne is only about 1.7 standard errors of a pairwise difference.

```
import numpy as np
from scipy import stats

rates = np.array([0, 40, 80, 120, 160])           # kg N per hectare
g, m = len(rates), 6                             # 5 rates, 6 plots per rate
true_means = 6.8 - 2.7 * np.exp(-0.018 * rates)  # unknown to the analyst
rng = np.random.default_rng(1515)
group = np.repeat(np.arange(g), m)                # plot i received rate group[i]
y = np.round(true_means[group] + rng.normal(0, 0.5, g * m), 2) # yield, t/ha
n = len(y)

n_k = np.bincount(group).astype(float)
```

```

means = np.bincount(group, weights=y) / n_k
nu = n - g
s = np.sqrt(np.sum((y - means[group]) ** 2) / nu)

def contrast(c, psi0=0.0, mult=None):
    """Estimate, standard error, t statistic, SS and interval for sum_k c_k mu_k."""
    est = c @ means
    v = np.sum(c ** 2 / n_k) # Var(estimate) / sigma^2
    se = s * np.sqrt(v)
    mult = stats.t.ppf(0.975, nu) if mult is None else mult
    return est, se, (est - psi0) / se, est ** 2 / v, (est - mult * se, est + mult * se)

c_fert = np.array([-1, 0.25, 0.25, 0.25, 0.25]) # fertilized (average) minus control
c_top = np.array([0, 0, -1, 0, 1]) # 160 against 80 kg N/ha
for name, c in (("fertilized - control", c_fert), ("160 - 80", c_top)):
    est, se, t, ss, (lo, hi) = contrast(c)
    print(f"{name:21s} {est:6.3f} se {se:.3f} t = {t:5.2f} SS = {ss:6.3f}"
          f" 95% interval [{lo:.3f}, {hi:.3f}]")

```

Each was a single planned question, so each gets an unadjusted t interval; joint coverage for the pair would use Bonferroni's $t_{25,0.0125}$. Writing the questions down in advance fixes a small family (Section 13.6).

15.2.4 Families of comparisons

Three families come up in almost every one-way analysis, and Chapter 13 supplies a method for each. The multipliers below multiply the standard error of a difference of two means, $s\sqrt{2/m} = 0.3009$ here.

Comparisons with a control. The four differences $\mu_k - \mu_1$, $k = 2, \dots, 5$, form a planned family of size four. Bonferroni's method (Theorem 13.4.1) uses the multiplier $t_{25,0.05/8} = 2.692$ in place of the unadjusted 2.060:

Rate against control	Difference	Bonferroni 95% interval
40	1.232	[0.422, 2.041]
80	2.052	[1.242, 2.861]
120	2.510	[1.700, 3.320]
160	2.647	[1.837, 3.456]

Every fertilized rate beats the control. The four estimates share $\bar{y}_1$, so they are positively correlated, with correlation $\frac{1}{2}$ in a balanced design (Exercise (B4)). Dunnett (1955) computed the exact joint distribution of the four t statistics under this correlation; his multiplier here is 2.606 (Exercise (B2) characterizes it), a few percent shorter than Bonferroni's. We do not derive its tables.

All pairs. Tukey's method (Theorem 13.3.2) covers all ten differences $\mu_k - \mu_l$ with exact joint coverage in a balanced design. With $q_{0.05}(5, 25) = 4.153$ the half-width is $q s / \sqrt{m} = 0.884$, and six of the ten pairs are declared different. The four that are not are 40 and 80, 80 and 120, 80 and 160, and 120 and 160. The conclusions are not transitive: 40 is not distinguished from

80, nor 80 from 160, yet 40 and 160 are declared different, since $6.652 - 5.237 = 1.415$ exceeds 0.884. “Not declared different” means “not enough evidence of a difference”, which is not an equivalence relation, so displays that sort treatments into groups of “equal” means should be read with care, especially in unbalanced designs.

Contrasts suggested by the data. Looking at Figure 15.1.1, one might ask whether the response has levelled off, by comparing the average of the two highest rates with 80 kg N/ha, $\frac{1}{2}(\mu_4 + \mu_5) - \mu_3$. The estimate is 0.527 with standard error 0.261. The unadjusted interval $[-0.010, 1.063]$ is not legitimate, because the contrast was chosen after seeing the means. Scheffé’s interval (Eq. 13.2.4), with multiplier $\sqrt{4F_{0.05}(4, 25)} = 3.322$, is $[-0.339, 1.392]$. It is valid however the contrast was found, and it says the question is open.

```
import itertools

k_ctrl = g - 1                                # four comparisons with the
control                                       control
bonf = stats.t.ppf(1 - 0.05 / (2 * k_ctrl), nu) # Bonferroni multiplier
se_pair = s * np.sqrt(2 / m)
for k in range(1, g):
    d = means[k] - means[0]
    print(f"{rates[k]:3d} - 0: {d:6.3f} Bonferroni interval"
          f" [{d - bonf * se_pair:.3f}, {d + bonf * se_pair:.3f}]")

q = stats.studentized_range.ppf(0.95, g, nu)   # q_{0.05}(g, nu)
hsd = q * s / np.sqrt(m)                      # Tukey half-width, all pairs
pairs = list(itertools.combinations(range(g), 2))
different = [(int(rates[k]), int(rates[l])) for k, l in pairs if abs(means[k] -
    means[l]) > hsd]
print(f"Tukey half-width {hsd:.3f}; pairs declared different: {len(different)} of {
    len(pairs)}")
print("not declared different:",
      [(int(rates[k]), int(rates[l])) for k, l in pairs if abs(means[k] - means[l])
    <= hsd])
```

15.2.5 Why planned contrasts are powerful

A single contrast is tested with one numerator degree of freedom, the overall test with $g - 1$. When the scientific question is sharp, the contrast is the better test, and the reason is geometric. Write $\mathbf{P}_{\mathcal{T}}$ for the projection onto the treatment space and

$$\gamma_{\mathcal{T}} = \frac{\|\mathbf{P}_{\mathcal{T}}\boldsymbol{\theta}\|^2}{\sigma^2} = \frac{1}{\sigma^2} \sum_{k=1}^g n_k (\mu_k - \bar{\mu})^2,$$

the squared length of the treatment component of the mean in units of σ^2 . (Section 15.3 shows that it is the noncentrality of the overall F test.)

Proposition 15.2.2 — The noncentrality of a contrast

Let $\mathbf{c}$ be a contrast with vector $\mathbf{u}_{\mathbf{c}} \in \mathcal{T}$, and suppose $\mathbf{P}_{\mathcal{T}}\boldsymbol{\theta} \neq \mathbf{0}$. The noncentrality of the test of $\psi = 0$ in Proposition 15.2.1 is

$$\delta^2 = \frac{\psi^2}{\sigma^2 v_{\mathbf{c}}} = \gamma_{\mathcal{T}} \cos^2 \varphi,$$

where φ is the angle between $\mathbf{u}_c$ and $\mathbf{P}_T\boldsymbol{\theta}$. In particular $\delta^2 \leq \gamma_T$, with equality iff $\mathbf{u}_c$ is parallel to $\mathbf{P}_T\boldsymbol{\theta}$, that is, iff $c_k \propto n_k(\mu_k - \bar{\mu})$.

Proof. By Proposition 8.7.2(a) applied to the mean vector, $\psi = \mathbf{u}_c^\top \boldsymbol{\theta}$, and $v_c = \|\mathbf{u}_c\|^2$ by the computation in the proof of that result. Since $\mathbf{u}_c \in \mathcal{T}$, $\mathbf{u}_c^\top \boldsymbol{\theta} = \mathbf{u}_c^\top \mathbf{P}_T \boldsymbol{\theta}$. So

$$\delta^2 = \frac{(\mathbf{u}_c^\top \mathbf{P}_T \boldsymbol{\theta})^2}{\sigma^2 \|\mathbf{u}_c\|^2} = \frac{\|\mathbf{P}_T \boldsymbol{\theta}\|^2}{\sigma^2} \cdot \frac{(\mathbf{u}_c^\top \mathbf{P}_T \boldsymbol{\theta})^2}{\|\mathbf{u}_c\|^2 \|\mathbf{P}_T \boldsymbol{\theta}\|^2} = \gamma_T \cos^2 \varphi$$

by Eq. 6.2.2. Equality in $\cos^2 \varphi \leq 1$ holds iff the two vectors are parallel. The vector $\mathbf{P}_T \boldsymbol{\theta}$ has entry $\mu_k - \bar{\mu}$ in group k (Theorem 6.5.1), so it is $\mathbf{u}_c$ for $c_k = n_k(\mu_k - \bar{\mu})$. ■

If the departure from equal means lies along the planned contrast, the contrast test gets all of the noncentrality on one degree of freedom, while the overall test spreads the same noncentrality over $g - 1$. For the nitrogen design, with 25 error degrees of freedom, the powers at level 0.05 are:

γ_T , all along the contrast	4	8	12
contrast test, $F(1, 25)$	0.485	0.776	0.914
overall test, $F(4, 25)$	0.269	0.516	0.714

The price is paid when the question is wrong: a contrast orthogonal to $\mathbf{P}_T \boldsymbol{\theta}$ has power α however large the treatment differences, and a direction chosen at random captures on average only $1/(g - 1)$ of γ_T (Exercise (C1)).

```
def power_f(gamma, q, nu, alpha=0.05):
    return stats.ncf.sf(stats.f.ppf(1 - alpha, q, nu), q, nu, gamma)

for gamma in (4.0, 8.0, 12.0):
    # all of it along the planned contrast
    print(f"gamma = {gamma:4.1f}: contrast test {power_f(gamma, 1, nu):.3f},"
          f" omnibus F test {power_f(gamma, g - 1, nu):.3f}")
```

15.2.6 Exercises

15.2.6.1 A. Check your understanding

Exercise A1.

In the nitrogen trial, compute by hand the estimate and standard error of $\frac{1}{2}(\mu_4 + \mu_5) - \frac{1}{2}(\mu_1 + \mu_2)$, the average of the two highest rates minus the average of the two lowest, from the rate means of Section 15.1 and $s = 0.5211$.

Solution. The estimate is $\frac{1}{2}(6.515 + 6.652) - \frac{1}{2}(4.005 + 5.237) = 6.5835 - 4.621 = 1.9625$. With $m = 6$, $v_{\mathbf{c}} = 4 \cdot \frac{1}{4}/6 = 1/6$, so the standard error is $0.5211/\sqrt{6} \approx 0.213$.

Exercise A2.

Show directly that the t statistic of Proposition 15.2.1(b) with $\psi_0 = 0$ is unchanged when $\mathbf{c}$ is multiplied by $a > 0$ and changes sign when $a < 0$. What happens to the confidence interval?

15.2.6.2 B. Practice

Exercise B1.

In a balanced layout with m observations per group, show that the noncentrality of the test of a contrast is $\delta^2 = m\psi^2/(\sigma^2 \sum_k c_k^2)$. How many observations per group are needed for the test of $\mu_5 - \mu_3$ in the nitrogen trial to have power 0.8 at level 0.05 when the true difference is 0.5 t/ha and $\sigma = 0.5$? (Use software for the noncentral F , or the normal approximation.)

Solution. Here $v_{\mathbf{c}} = \sum_k c_k^2/m$, so $\delta^2 = \psi^2/(\sigma^2 v_{\mathbf{c}}) = m\psi^2/(\sigma^2 \sum_k c_k^2)$. For $\mu_5 - \mu_3$, $\sum_k c_k^2 = 2$, $\psi/\sigma = 1$, and $\delta^2 = m/2$. The normal approximation asks for $\delta \approx z_{0.025} + z_{0.2} = 1.96 + 0.84 = 2.80$, so $m \approx 2 \cdot 2.80^2 \approx 15.7$. The exact noncentral F with $5(m-1)$ error degrees of freedom gives $m = 17$, since the t quantile exceeds the normal one. The trial's $m = 6$ is far short.

Exercise B2.

A single contrast $\mathbf{c}$ matters above all. For a fixed total n , show that $v_{\mathbf{c}} = \sum_k c_k^2/n_k$ is minimized, over positive real n_k with $\sum_k n_k = n$, by $n_k \propto |c_k|$, with minimum $(\sum_k |c_k|)^2/n$. What does this say about how to allocate plots between two treatments, and between a control and the average of four treatments?

Solution. By the Cauchy–Schwarz inequality, $(\sum_k |c_k|)^2 = (\sum_k (|c_k|/\sqrt{n_k})\sqrt{n_k})^2 \leq \sum_k (c_k^2/n_k) \sum_k n_k = nv_{\mathbf{c}}$, with equality iff $|c_k|/\sqrt{n_k} \propto \sqrt{n_k}$, that is, $n_k \propto |c_k|$. Groups with $c_k = 0$ get no observations. For $\mu_1 - \mu_2$, equal allocation is best. For $\mu_1 - \frac{1}{4}(\mu_2 + \dots + \mu_5)$, the control should get as many plots as the four treatments together, $n_1 = n/2$ and $n_k = n/8$ for the others.

Exercise B3.

In a comparison of $g - 1$ treatments with a control, suppose the control gets n_0 observations and each treatment gets n_t , with $n_0 + (g - 1)n_t = n$ fixed. Show that $\sum_k \text{Var}(\bar{Y}_k - \bar{Y}_0)$ is minimized when $n_0 = \sqrt{g-1} n_t$. For $g = 5$ and $n = 30$, what allocation does this suggest?

Solution. The sum is $\sigma^2(g-1)(1/n_0 + 1/n_t)$. Minimizing $1/n_0 + 1/n_t$ subject to $n_0 + (g-1)n_t = n$ with a Lagrange multiplier gives $1/n_0^2 = \lambda$ and $1/n_t^2 = (g-1)\lambda$, so $n_0/n_t = \sqrt{g-1}$. For $g = 5$ the control should get twice as many observations as each

treatment: $n_0 = 10$ and $n_t = 5$. Contrast [Exercise \(B2\)](#): for the average of the four comparisons the optimal control share is $n/2$, that is $n_0 = 4n_t$. The best allocation depends on whether the four comparisons or their average is the target.

Exercise B4.

With n_0 observations on the control and n_t on each treatment, show that the estimates $\bar{Y}_k - \bar{Y}_0$ and $\bar{Y}_l - \bar{Y}_0$ of two treatment-versus-control differences have correlation $n_t/(n_t + n_0)$. In a balanced design it is $\frac{1}{2}$.

Solution. The covariance is $\text{Var}(\bar{Y}_0) = \sigma^2/n_0$, since the treatment means are independent of each other and of $\bar{Y}_0$. Each variance is $\sigma^2(1/n_t + 1/n_0)$. The ratio is $(1/n_0)/(1/n_t + 1/n_0) = n_t/(n_t + n_0)$.

15.2.6.3 C. Going deeper

Exercise C1.

Let $q = g - 1 \geq 2$ and let $\mathbf{u}$ be uniformly distributed on the unit sphere of the q -dimensional space $\mathcal{T}$ (for example $\mathbf{u} = \mathbf{w}/\|\mathbf{w}\|$ with $\mathbf{w} \sim N$ on $\mathcal{T}$ with covariance $\mathbf{P}_{\mathcal{T}}$). Show that $E \cos^2 \varphi = 1/q$, where φ is the angle between $\mathbf{u}$ and a fixed nonzero $\mathbf{v} \in \mathcal{T}$. Interpret this with [Proposition 15.2.2](#).

Solution. Choose an orthonormal basis of $\mathcal{T}$ whose first vector is $\mathbf{v}/\|\mathbf{v}\|$. In these coordinates $\cos^2 \varphi = u_1^2$. By symmetry the u_i^2 have a common mean, and they sum to 1, so $E u_1^2 = 1/q$. By [Proposition 15.2.2](#), a contrast whose direction is unrelated to the truth captures on average a fraction $1/q$ of $\gamma_{\mathcal{T}}$ on its one degree of freedom, the same average share per degree of freedom as the overall test. Knowledge, not the count of degrees of freedom, is what makes a planned contrast more powerful.

Exercise C2.

The Scheffé multiplier for a family of contrasts depends on the dimension of the space the family spans ([Theorem 13.2.2](#)). Show that the $g - 1$ comparisons with a control span $\mathcal{T}$, and that the family of the four “successive differences” $\mu_{k+1} - \mu_k$ does too. Why, then, are Bonferroni or Dunnett intervals shorter than Scheffé’s for these families?

15.3 The analysis of variance table from projections

Whether the treatments differ at all is answered by the F test, with the analysis of variance table as its bookkeeping. Both come from one orthogonal decomposition of the observation space into the constants, the treatment space and the error space, and every entry of the table is a property of one piece.

15.3.1 Three subspaces

Write $\mathbf{P}_0 = n^{-1}\mathbf{1}\mathbf{1}^\top$ for the projection onto the constants and $\mathbf{M}$ for the projection onto $\mathcal{C}(\mathbf{Z})$ (Eq. 15.1.1). Since $\mathbf{1} \in \mathcal{C}(\mathbf{Z})$, Theorem 6.5.1 applies to the pair $\text{span}(\mathbf{1}) \subseteq \mathcal{C}(\mathbf{Z})$ and gives the orthogonal decomposition

$$\mathbb{R}^n = \underbrace{\text{span}(\mathbf{1})}_{\dim 1} \oplus \underbrace{\mathcal{T}}_{\dim g-1} \oplus \underbrace{\mathcal{C}(\mathbf{Z})^\perp}_{\dim n-g}, \quad \mathbf{I} = \mathbf{P}_0 + (\mathbf{M} - \mathbf{P}_0) + (\mathbf{I} - \mathbf{M}), \quad (15.3.1)$$

with the treatment space $\mathcal{T} = \mathcal{C}(\mathbf{Z}) \cap \mathbf{1}^\perp$ of Section 15.2 and $\mathbf{P}_{\mathcal{T}} = \mathbf{M} - \mathbf{P}_0$. The three projections act on the data in the familiar way:

$$(\mathbf{P}_0 \mathbf{y})_{kj} = \bar{y}, \quad (\mathbf{P}_{\mathcal{T}} \mathbf{y})_{kj} = \bar{y}_k - \bar{y}, \quad ((\mathbf{I} - \mathbf{M}) \mathbf{y})_{kj} = y_{kj} - \bar{y}_k.$$

So every observation is split as $y_{kj} = \bar{y} + (\bar{y}_k - \bar{y}) + (y_{kj} - \bar{y}_k)$, grand mean plus treatment deviation plus residual, and the three vectors so formed are mutually orthogonal. Pythagoras gives

$$\sum_{k,j} y_{kj}^2 = n\bar{y}^2 + \sum_k n_k(\bar{y}_k - \bar{y})^2 + \sum_{k,j} (y_{kj} - \bar{y}_k)^2. \quad (15.3.2)$$

The middle term is the **between-groups sum of squares** $\text{SSB} = \|\mathbf{P}_{\mathcal{T}} \mathbf{y}\|^2$, and the last is the **within-groups** or residual sum of squares $\text{SSE} = \|(\mathbf{I} - \mathbf{M}) \mathbf{y}\|^2$ of Eq. 15.1.2. Their sum $\sum_{k,j} (y_{kj} - \bar{y})^2$ is the corrected total sum of squares. The same decomposition, with the distributions attached, was the first application of Cochran's theorem (Example (The one-way layout)).

15.3.2 The F test

Theorem 15.3.1 — The one-way F test

In the normal one-way model of Definition 15.1.1, with mean vector $\boldsymbol{\theta} = \mathbf{Z}\boldsymbol{\mu}$, put

$$\text{MSB} = \frac{\text{SSB}}{g-1}, \quad s^2 = \text{MSE} = \frac{\text{SSE}}{n-g}, \quad F = \frac{\text{MSB}}{\text{MSE}}.$$

- (a) $\bar{Y}$, $\text{SSB} = \sum_k n_k(\bar{Y}_k - \bar{Y})^2$ and $\text{SSE} = \sum_{k,j} (Y_{kj} - \bar{Y}_k)^2$ are mutually independent.
- (b) $\text{SSE}/\sigma^2 \sim \chi^2(n-g)$, and $\text{SSB}/\sigma^2 \sim \chi^2(g-1, \gamma)$ with

$$\gamma = \frac{\|\mathbf{P}_{\mathcal{T}} \boldsymbol{\theta}\|^2}{\sigma^2} = \frac{1}{\sigma^2} \sum_{k=1}^g n_k(\mu_k - \bar{\mu})^2, \quad \bar{\mu} = \frac{1}{n} \sum_{k=1}^g n_k \mu_k. \quad (15.3.3)$$

- (c) $F \sim F(g-1, n-g, \gamma)$, and $\gamma = 0$ iff $\mu_1 = \dots = \mu_g$. In particular, when the means are equal, $F \sim F(g-1, n-g)$ whatever their common value and σ^2 .
- (d) F is the statistic of the F test of the reduced model $\mathbf{E}(\mathbf{Y}) \in \text{span}(\mathbf{1})$ against $\mathcal{C}(\mathbf{Z})$, and of the testable hypothesis $\alpha_1 = \dots = \alpha_g$ in the effects form. The test that rejects when $F > F_\alpha(g-1, n-g)$ is the likelihood ratio test of equal means. Its

power is a continuous, strictly increasing function of γ , equal to α at $\gamma = 0$.

Proof. By Eq. 15.3.1 the matrices $\mathbf{P}_0$, $\mathbf{P}_T$ and $\mathbf{I} - \mathbf{M}$ are symmetric, idempotent and mutually orthogonal, with ranks 1, $g - 1$ and $n - g$. Theorem 4.4.4 gives (a), since $\bar{Y}$ is a function of $\mathbf{P}_0\mathbf{Y}$, and it gives the chi-squared laws of (b) with noncentralities $\|\mathbf{P}_T\boldsymbol{\theta}\|^2/\sigma^2$ and $\|(\mathbf{I} - \mathbf{M})\boldsymbol{\theta}\|^2/\sigma^2$. The second is zero because $\boldsymbol{\theta} \in \mathcal{C}(\mathbf{Z})$. The vector $\boldsymbol{\theta}$ has entry μ_k in group k , so $\mathbf{M}\boldsymbol{\theta} = \boldsymbol{\theta}$ and $\mathbf{P}_0\boldsymbol{\theta} = \bar{\mu}\mathbf{1}$, and $\mathbf{P}_T\boldsymbol{\theta}$ has entry $\mu_k - \bar{\mu}$ in group k . Its squared length is $\sum_k n_k(\mu_k - \bar{\mu})^2$, which proves Eq. 15.3.3.

For (c): by (a) and (b), F is a ratio of independent chi-squared variables divided by their degrees of freedom, the numerator noncentral, which is the noncentral F law (Definition 4.1.2). Since every $n_k \geq 1$, $\gamma = 0$ iff $\mu_k = \bar{\mu}$ for every k .

For (d), take $\mathbf{X}_0 = \mathbf{1}$ inside $\mathbf{X} = \mathbf{Z}$. Then $\text{SSE}_0 - \text{SSE} = \|(\mathbf{M} - \mathbf{P}_0)\mathbf{Y}\|^2 = \text{SSB}$ by Eq. 6.5.2, with $r - r_0 = g - 1$, so F is the statistic of Theorem 11.1.1. In the effects form the hypothesis is $\alpha_k - \alpha_g = 0$, $k < g$: $g - 1$ estimable, linearly independent contrasts, so it is testable (Definition 11.3.1). The mean vectors it allows are the constant vectors, so by Theorem 11.3.2(c) its F statistic is the same one. The likelihood ratio property is Theorem 11.4.1, and the power statement is Theorem 11.5.1(a). ■

The noncentrality Eq. 15.3.3 is a weighted sum of squared deviations of the group means from their *weighted* mean. The weights are the group sizes, because a group of n_k observations carries n_k copies of its mean in $\boldsymbol{\theta}$. In a balanced layout $\gamma = m \sum_k (\mu_k - \bar{\mu})^2 / \sigma^2$, the case studied for planning in Proposition 11.5.2 and Example (How many observations per group?). The dependence on the group sizes in unbalanced layouts is taken up in Section 15.5.

15.3.3 Expected mean squares

The distributions in Theorem 15.3.1 need normality. The expectations of the mean squares do not, and they explain why the ratio of two mean squares is the right statistic.

Theorem 15.3.2 — Expected mean squares in the one-way layout

In the one-way model of Definition 15.1.1 (uncorrelated errors with common variance σ^2 , not necessarily normal):

- (a) $E(\text{MSE}) = \sigma^2$;
- (b) $E(\text{MSB}) = \sigma^2 + \frac{1}{g-1} \sum_{k=1}^g n_k(\mu_k - \bar{\mu})^2$;
- (c) in the effects form, $\sum_k n_k(\mu_k - \bar{\mu})^2 = \sum_k n_k(\alpha_k - \bar{\alpha})^2$ with $\bar{\alpha} = n^{-1} \sum_k n_k \alpha_k$, which is $\sum_k n_k \alpha_k^2$ under the side condition $\sum_k n_k \alpha_k = 0$. In a balanced layout it is $m \sum_k (\mu_k - \bar{\mu})^2$ with $\bar{\mu} = g^{-1} \sum_k \mu_k$, which is $m \sum_k \alpha_k^2$ under $\sum_k \alpha_k = 0$;
- (d) $E(\text{MSB}) \geq E(\text{MSE})$, with equality iff $\mu_1 = \dots = \mu_g$.

Proof. The spaces of Eq. 15.3.1 form an analysis of variance decomposition of $\mathcal{C}(\mathbf{Z})$ in the sense of Theorem 9.6.1, which gives $E(\text{SSE}) = \sigma^2(n - g) + \|(\mathbf{I} - \mathbf{M})\boldsymbol{\theta}\|^2 = \sigma^2(n - g)$

and $E(\text{SSB}) = \sigma^2(g-1) + \|\mathbf{P}_{\mathcal{T}}\boldsymbol{\theta}\|^2$. The squared length was computed in the proof of [Theorem 15.3.1](#). Dividing by the degrees of freedom gives (a) and (b). For (c), $\mu_k = \mu + \alpha_k$ and $\bar{\mu} = \mu + \bar{\alpha}$, so $\mu_k - \bar{\mu} = \alpha_k - \bar{\alpha}$. In a balanced layout $\bar{\mu} = \tilde{\mu}$ and $\bar{\alpha} = g^{-1} \sum_k \alpha_k$. (d) follows from (a) and (b). Part (b) is also [Exercise \(A2\)](#), which computes it directly from the group means. ■

Both mean squares estimate σ^2 when the treatments do not differ; otherwise MSB estimates σ^2 plus the treatment variation per degree of freedom. Every F ratio in an analysis of variance follows this logic: the denominator's expectation equals the numerator's exactly when the hypothesis holds ([Section 9.6](#)).

The same projections serve the random-effects model of [Section 15.1](#), but the expectations change. With $\text{Cov}(\mathbf{Y}) = \sigma^2 \mathbf{I} + \sigma_a^2 \mathbf{Z}\mathbf{Z}^\top$ and a constant mean, $E(\text{MSE}) = \sigma^2$ still, while $E(\text{MSB}) = \sigma^2 + n_0 \sigma_a^2$ with $n_0 = (n - \sum_k n_k^2/n)/(g-1)$, which is m in a balanced layout ([Exercise \(C1\)](#)). The ratio F then tests $\sigma_a^2 = 0$, and the difference of the two mean squares divided by n_0 estimates σ_a^2 . Chapter 32 develops this.

15.3.4 The table, derived

Collecting the pieces gives the one-way analysis of variance table. Each row is a subspace, its degrees of freedom are the dimension, its sum of squares is the squared length of the projection of $\mathbf{y}$, and its expected mean square is σ^2 plus the squared length of the projection of $\boldsymbol{\theta}$ per dimension.

Source	Subspace	df	Sum of squares	Mean square	F	Expected mean square
treatments	$\mathcal{T}$	$g-1$	$\sum_k n_k (\bar{y}_k - \bar{y})^2$	MSB	MSB/MSE	$\sigma^2 + \frac{\sum_k n_k (\mu_k - \bar{\mu})^2}{g-1}$
error	$\mathcal{C}(\mathbf{Z})^\perp$	$n-g$	$\sum_{k,j} (y_{kj} - \bar{y}_k)^2$	MSE		σ^2
corrected total	$\mathbf{1}^\perp$	$n-1$	$\sum_{k,j} (y_{kj} - \bar{y})^2$			

Some tables add a row for the mean, with 1 degree of freedom and sum of squares $n\bar{y}^2 = \|\mathbf{P}_0 \mathbf{y}\|^2$, and an uncorrected total $\sum y_{kj}^2$; its F ratio would test $\bar{\mu} = 0$, rarely of interest.

Hand computation used the group totals T_k and grand total T : $\|\mathbf{M}\mathbf{y}\|^2 = \sum_k T_k^2/n_k$ and $\|\mathbf{P}_0 \mathbf{y}\|^2 = T^2/n$, the “correction factor”, so $\text{SSB} = \sum_k T_k^2/n_k - T^2/n$. Such differences of nearly equal numbers lose accuracy in floating point ([Chapter 10](#)); computing deviations first, as the projections do, avoids this.

Example. (The analysis of variance for the nitrogen trial)

For the thirty yields of [Section 15.1](#), with grand mean $\bar{y} = 5.6930$ t/ha:

Source	df	Sum of squares	Mean square	F
nitrogen	4	28.7074	7.1768	26.43
rates				
plots within	25	6.7887	0.2715	
rates				
corrected	29	35.4960		
total				

The mean row would add $n\bar{y}^2 = 972.307$ on one degree of freedom. The critical value is $F_{0.05}(4, 25) = 2.759$, and the p -value is 1.2×10^{-8} . The table says only that the treatment component of the mean is not zero; the contrasts of [Section 15.2](#) and [Section 15.4](#) say what it is. Since the data were simulated, the true noncentrality is known: $\gamma = 103.4$, far more than needed for power near one.

```
import numpy as np
from scipy import stats

rates = np.array([0, 40, 80, 120, 160])          # kg N per hectare
g, m = len(rates), 6                             # 5 rates, 6 plots per rate
true_means = 6.8 - 2.7 * np.exp(-0.018 * rates)  # unknown to the analyst
rng = np.random.default_rng(1515)
group = np.repeat(np.arange(g), m)               # plot i received rate group[i]
y = np.round(true_means[group] + rng.normal(0, 0.5, g * m), 2) # yield, t/ha
n = len(y)
Z = (group[:, None] == np.arange(g)).astype(float) # cell-means model matrix
n_k = Z.sum(axis=0)

M = Z @ np.diag(1 / n_k) @ Z.T
P0 = np.full((n, n), 1 / n)                     # projection onto span(1)
ss_mean = y @ P0 @ y                           # n * ybar^2
ss_between = y @ (M - P0) @ y                   # sum_k n_k (ybar_k - ybar)^2
ss_within = y @ (np.eye(n) - M) @ y             # sum_kj (y_kj - ybar_k)^2
df_between, df_within = g - 1, n - g
F = (ss_between / df_between) / (ss_within / df_within)
p_value = stats.f.sf(F, df_between, df_within)
print(f"{'source':10s} {'df':>3s} {'SS':>9s} {'MS':>8s}")
print(f"{'rates':10s} {'df_between':3d} {'ss_between':9.4f} {'ss_between / df_between':8.4f}")
print(f"{'plots':10s} {'df_within':3d} {'ss_within':9.4f} {'ss_within / df_within':8.4f}")
print(f"{'corrected':10s} {'n - 1':3d} {'ss_between + ss_within':9.4f}")
print(f"F = {F:.2f} on ({df_between}, {df_within}) df, p = {p_value:.2g}")
```

15.3.5 Three ways to read F

As a proportion of variation. Let $\eta^2 = \text{SSB}/(\text{SSB} + \text{SSE})$, the fraction of the corrected total sum of squares that lies in the treatment space. It is the R^2 of the regression of $\mathbf{y}$ on the group indicators, and the sample version of the correlation ratio of [Eq. 14.2.1](#). By [Theorem 9.5.2](#),

$$F = \frac{\eta^2/(g-1)}{(1-\eta^2)/(n-g)},$$

and in the normal model, under equal means, η^2 has the Beta law of that theorem, with $E(\eta^2) = (g-1)/(n-1)$, not small with many groups and few observations. In the nitrogen trial $\eta^2 = 0.809$: four fifths of the variation among plots is variation among rates.

As the largest contrast. By Proposition 15.2.1(d), every contrast has $SS(c) \leq SSB$, with equality for $c_k \propto n_k(\bar{y}_k - \bar{y})$. Dividing by s^2 , the largest squared t statistic over all contrasts is $(g-1)F$. Hence the F test rejects iff some contrast, perhaps not an interesting one, is declared nonzero by Scheffé's method (Corollary 13.2.3).

As a two-sample test, for $g = 2$. Then $\mathcal{T}$ is a line, the only contrast is $\mu_1 - \mu_2$ up to scale, and $F = T^2$ for the pooled two-sample t statistic of Exercise (B2), whose sensitivity to unequal variances the one-way test inherits (Section 15.5).

15.3.6 Exercises

15.3.6.1 A. Check your understanding

Exercise A1.

A one-way experiment with four treatments and six observations per treatment gives a treatment mean square of 12.0 and a corrected total sum of squares of 96.0. Complete the analysis of variance table, compute F and η^2 , and state the reference distribution of F under equal means.

Solution. The degrees of freedom are 3, 20 and 23. The treatment sum of squares is $3 \times 12 = 36$, so the error sum of squares is $96 - 36 = 60$ and $MSE = 3$. Then $F = 12/3 = 4$, $\eta^2 = 36/96 = 0.375$, and under equal means $F \sim F(3, 20)$.

Exercise A2.

Show from Eq. 15.3.2 that $SSB = \sum_k T_k^2/n_k - T^2/n$ and $SSE = \sum_{k,j} y_{kj}^2 - \sum_k T_k^2/n_k$.

15.3.6.2 B. Practice

Exercise B1.

For $g = 2$ with $n_1 + n_2 = n$ fixed, show that $\gamma = (\mu_1 - \mu_2)^2 n_1 n_2 / (n\sigma^2)$, and that it is largest for $n_1 = n_2$. By how much does the noncentrality fall for the split $n_1 = n/4$, $n_2 = 3n/4$?

Solution. With $\bar{\mu} = (n_1\mu_1 + n_2\mu_2)/n$, $\mu_1 - \bar{\mu} = n_2(\mu_1 - \mu_2)/n$ and $\mu_2 - \bar{\mu} = -n_1(\mu_1 - \mu_2)/n$. So $\sigma^2\gamma = (\mu_1 - \mu_2)^2(n_1n_2^2 + n_2n_1^2)/n^2 = (\mu_1 - \mu_2)^2 n_1 n_2 / n$. The product $n_1 n_2$ with fixed sum is largest at $n_1 = n_2 = n/2$, where it is $n^2/4$. For the split $(n/4, 3n/4)$ it is $3n^2/16$, so the noncentrality falls by a quarter.

Exercise B2.

In the normal model with equal means $E(\eta^2) = (g-1)/(n-1)$, which is not zero, so η^2

overstates the treatment variation in small experiments. Show that the numerator of

$$\hat{\omega}^2 = \frac{\text{SSB} - (g-1)\text{MSE}}{\text{SSB} + \text{SSE}}$$

has expectation $\sum_k n_k(\mu_k - \bar{\mu})^2$, whatever the means, so that it removes the bias from the numerator. When is $\hat{\omega}^2$ negative?

Solution. By [Theorem 15.3.2](#), $E(\text{SSB}) = (g-1)\sigma^2 + \sum_k n_k(\mu_k - \bar{\mu})^2$ and $E\{(g-1)\text{MSE}\} = (g-1)\sigma^2$, so the difference has the stated expectation. The numerator is negative iff $\text{MSB} < \text{MSE}$, that is, iff $F < 1$, which often has null probability above one half: 0.57 for the nitrogen design's $F(4, 25)$ in the normal model. The ratio itself is not unbiased, since its denominator is random.

Exercise B3.

In the nitrogen trial, compute the Scheffé-maximal contrast $c_k = n_k(\bar{y}_k - \bar{y})$ from the rate means, and verify that its squared t statistic is $4F$. Describe in words what it compares.

15.3.6.3 C. Going deeper

Exercise C1.

In the random-effects model $\text{Cov}(\mathbf{Y}) = \sigma^2\mathbf{I} + \sigma_a^2\mathbf{Z}\mathbf{Z}^\top$, $E(\mathbf{Y}) = \mu\mathbf{1}$, use [Theorem 2.3.1](#) to show that $E(\text{SSB}) = (g-1)\sigma^2 + (n - \sum_k n_k^2/n)\sigma_a^2$ and $E(\text{SSE}) = (n-g)\sigma^2$. Show that $n_0 = (n - \sum_k n_k^2/n)/(g-1) \leq n/g$, with equality iff the layout is balanced.

Solution. By [Theorem 2.3.1](#), $E(\mathbf{Y}^\top\mathbf{A}\mathbf{Y}) = \text{tr}(\mathbf{A}\Sigma) + \boldsymbol{\theta}^\top\mathbf{A}\boldsymbol{\theta}$, and $\boldsymbol{\theta} = \mu\mathbf{1}$ is annihilated by both $\mathbf{P}_\mathcal{T}$ and $\mathbf{I} - \mathbf{M}$. Now $\text{tr}(\mathbf{P}_\mathcal{T}) = g-1$ and $\text{tr}(\mathbf{P}_\mathcal{T}\mathbf{Z}\mathbf{Z}^\top) = \text{tr}(\mathbf{Z}^\top\mathbf{M}\mathbf{Z}) - \text{tr}(\mathbf{Z}^\top\mathbf{P}_0\mathbf{Z})$. Since $\mathbf{M}\mathbf{Z} = \mathbf{Z}$, the first term is $\text{tr}(\mathbf{Z}^\top\mathbf{Z}) = n$. The second is $n^{-1}\|\mathbf{Z}^\top\mathbf{1}\|^2 = \sum_k n_k^2/n$. For the error, $(\mathbf{I} - \mathbf{M})\mathbf{Z} = \mathbf{0}$, so only $\sigma^2\text{tr}(\mathbf{I} - \mathbf{M}) = (n-g)\sigma^2$ remains. Finally $\sum_k n_k^2 \geq n^2/g$ by the Cauchy-Schwarz inequality, with equality iff all $n_k = n/g$, so $(g-1)n_0 \leq n - n/g = (g-1)n/g$.

Exercise C2.

Show that F is unchanged when $\mathbf{y}$ is replaced by $a\mathbf{y} + b\mathbf{1}$ with $a \neq 0$, and when the observations are permuted within groups. Show that it changes when a constant is added to the observations of one group only, and explain why it should.

15.4 Orthogonal contrasts

Once the F test of [Section 15.3](#) says that the treatment component of the mean is not zero, the next step is to split that component into interpretable pieces. A complete set of orthogonal contrasts chooses an orthogonal basis of the treatment space, one direction per question, and splits the between-groups sum of squares along it. [Chapter 8](#) proved the arithmetic ([Propo-](#)

sition 8.7.2) and Chapter 9 placed it in the theory of single-degree-of-freedom decompositions (Theorem 9.4.4). This section adds the distribution theory, builds orthogonal sets, and analyses the shape of the nitrogen response.

15.4.1 The decomposition

Contrasts $\mathbf{c}$ and $\mathbf{d}$ are orthogonal for the design if $\sum_k c_k d_k / n_k = 0$ (Definition 8.7.1), which says exactly that $\mathbf{u}_c \perp \mathbf{u}_d$ (Proposition 8.7.2(c)). Since $\dim \mathcal{T} = g - 1$, a set of $g - 1$ pairwise orthogonal contrasts is as large as possible; we call it **complete**.

Theorem 15.4.1 — Orthogonal contrasts decompose the treatment sum of squares

In the one-way model, let $\mathbf{c}_1, \dots, \mathbf{c}_{g-1}$ be contrasts that are pairwise orthogonal for the design,

$$\sum_{k=1}^g \frac{c_{ik} c_{lk}}{n_k} = 0 \quad (i \neq l), \quad (15.4.1)$$

and write $\psi_i = \sum_k c_{ik} \mu_k$ and $F_i = \text{SS}(\mathbf{c}_i) / s^2$.

- (a) The vectors $\mathbf{u}_{\mathbf{c}_1}, \dots, \mathbf{u}_{\mathbf{c}_{g-1}}$ form an orthogonal basis of the treatment space $\mathcal{T}$, and

$$\text{SSB} = \sum_{i=1}^{g-1} \text{SS}(\mathbf{c}_i).$$

- (b) In the normal one-way model, $\text{SS}(\mathbf{c}_1), \dots, \text{SS}(\mathbf{c}_{g-1})$ and SSE are mutually independent, and $\text{SS}(\mathbf{c}_i) / \sigma^2 \sim \chi^2(1, \gamma_i)$ with $\gamma_i = \psi_i^2 / (\sigma^2 v_{\mathbf{c}_i})$. The noncentralities add up to that of the F test: $\sum_i \gamma_i = \gamma$ of Eq. 15.3.3.
- (c) The F statistic is the average of the single-degree-of-freedom statistics: $F = (g - 1)^{-1} \sum_i F_i$.
- (d) In a balanced layout, Eq. 15.4.1 reduces to $\sum_k c_{ik} c_{lk} = 0$.
- (e) Conversely, every orthogonal basis $\mathbf{u}_1, \dots, \mathbf{u}_{g-1}$ of $\mathcal{T}$ arises in this way: if $\mathbf{u}_i$ takes the value v_{ik} on group k , then $c_{ik} = n_k v_{ik}$ defines a complete set of orthogonal contrasts with $\mathbf{u}_{\mathbf{c}_i} = \mathbf{u}_i$.

Proof. Part (a) is Proposition 8.7.2(d), whose proof shows that the $\mathbf{u}_{\mathbf{c}_i}$ are nonzero, pairwise orthogonal vectors of $\mathcal{T}$, hence a basis, and that the projections $\mathbf{P}_i$ onto the lines they span sum to $\mathbf{P}_{\mathcal{T}}$. (b) The $\mathbf{P}_i$ are mutually orthogonal projections of rank one, and each is orthogonal to $\mathbf{I} - \mathbf{M}$ because its range lies in $\mathcal{C}(\mathbf{Z})$. By Theorem 4.4.4 the sums of squares $\|\mathbf{P}_i \mathbf{Y}\|^2 = \text{SS}(\mathbf{c}_i)$ (Proposition 15.2.1(c)) and SSE are independent, with noncentralities $\|\mathbf{P}_i \boldsymbol{\theta}\|^2 / \sigma^2$. By the computation in the proof of Proposition 15.2.2, $\|\mathbf{P}_i \boldsymbol{\theta}\|^2 = (\mathbf{u}_{\mathbf{c}_i}^\top \boldsymbol{\theta})^2 / \|\mathbf{u}_{\mathbf{c}_i}\|^2 = \psi_i^2 / v_{\mathbf{c}_i}$. Since $\sum_i \mathbf{P}_i = \mathbf{P}_{\mathcal{T}}$ and the $\mathbf{P}_i \boldsymbol{\theta}$ are orthogonal, $\sum_i \|\mathbf{P}_i \boldsymbol{\theta}\|^2 = \|\mathbf{P}_{\mathcal{T}} \boldsymbol{\theta}\|^2$, which is $\sigma^2 \gamma$ by Theorem 15.3.1. (c) Divide (a) by $(g - 1)s^2$. (d) Put $n_k = m$. (e) In Section 15.2 we saw that each $\mathbf{u} \in \mathcal{T}$ equals $\mathbf{u}_c$ for the contrast

$c_k = n_k v_k$, and by Proposition 8.7.2(c), $\sum_k c_{ik} c_{lk} / n_k = \mathbf{u}_{\mathbf{c}_i}^\top \mathbf{u}_{\mathbf{c}_l}$, which is zero for $i \neq l$. ■

By part (e) there are infinitely many complete orthogonal sets, one for each orthogonal basis of $\mathcal{T}$, and by part (a) all of them split the same total. The individual pieces depend on the basis (Exercise (C1)): choosing the basis is choosing the questions, and the design of the study, not the mathematics, should make that choice.

Warning. Orthogonality is a convenience, not a requirement. Planned contrasts that are not orthogonal are valid, each with the test and interval of Proposition 15.2.1. What is lost is additivity: their sums of squares need not total SSB, and their estimates are correlated. No contrast should be replaced by a less relevant one merely to achieve orthogonality.

15.4.2 Building orthogonal sets

Three constructions cover most designed experiments.

Splitting the treatments in stages. Many questions form a hierarchy: control against treatments, then old treatments against new, then one new formulation against another. Each splits a set of levels into two parts, and the comparisons are orthogonal for any group sizes provided each part is summarized by the mean of all its observations.

Proposition 15.4.2 — Contrasts from successive splits

For a set S of levels split into disjoint nonempty parts A and B , let $N_A = \sum_{k \in A} n_k$ and $N_B = \sum_{k \in B} n_k$, and define the contrast

$$c_k^{(S)} = \begin{cases} n_k / N_A, & k \in A, \\ -n_k / N_B, & k \in B, \\ 0, & k \notin S. \end{cases}$$

Its estimate is $\bar{y}_A - \bar{y}_B$, the difference between the means of all observations in A and in B , and

$$\text{SS}(\mathbf{c}^{(S)}) = \frac{N_A N_B}{N_A + N_B} (\bar{y}_A - \bar{y}_B)^2.$$

- (a) Every contrast $\mathbf{d}$ that is zero outside A (or outside B) is orthogonal to $\mathbf{c}^{(S)}$ for the design.
- (b) Split the set of all g levels into two parts, split each part with more than one level into two, and continue until every part is a single level. The $g - 1$ contrasts of the splits form a complete set of orthogonal contrasts, whatever the group sizes.

Proof. The estimate is $\sum_{k \in A} n_k \bar{y}_k / N_A - \sum_{k \in B} n_k \bar{y}_k / N_B = \bar{y}_A - \bar{y}_B$, and $v_{\mathbf{c}} = \sum_{k \in A} n_k / N_A^2 + \sum_{k \in B} n_k / N_B^2 = 1/N_A + 1/N_B$, which gives the sum of squares. (a) If $\mathbf{d}$ is zero outside A , then $\sum_k c_k^{(S)} d_k / n_k = \sum_{k \in A} (n_k / N_A) d_k / n_k = N_A^{-1} \sum_{k \in A} d_k = 0$, since $\mathbf{d}$ is a contrast; the case of B is the same. (b) A sequence of splits ending in single levels is a binary tree with g leaves, so it has $g - 1$ internal nodes, one split each. Take two different splits,

of sets S and S' . By construction either S and S' are disjoint, in which case the two contrasts have disjoint supports and are orthogonal, or one of them, say S' , lies inside one part of the split of S , in which case (a) applies. So the $g - 1$ contrasts are pairwise orthogonal, and they form a complete set. ■

The weighting matters: comparing plain averages of the levels in A and B gives orthogonal contrasts, in general, only in a balanced layout, as [Example \(Splitting the education sum of squares\)](#) showed numerically. The Helmert contrasts of [Exercise \(B3\)](#), which compare each level with the average of those before it, are the special case of the proposition that splits off one level at a time; with the weights n_k/N_A they stay orthogonal for unequal group sizes.

Factorial treatments. Four treatments may be the combinations of two fertilizers, each present or absent: none, P only, K only, and both. The contrasts

$$\begin{aligned} \text{effect of } P &: (-1, 1, -1, 1), & \text{effect of } K &: (-1, -1, 1, 1), \\ \text{interaction} &: (1, -1, -1, 1) \end{aligned}$$

are orthogonal in a balanced layout. The third asks whether the effect of P depends on K . A one-way analysis with these contrasts is a two-factor analysis in disguise; [Chapter 16](#) treats such layouts directly and defines interaction in general ([Definition 16.2.1](#)).

Polynomial trends. When the levels are values of a quantitative variable, such as a dose or a fertilizer rate, the orthogonal polynomial contrasts describe the shape of the response one degree at a time. For equally spaced levels and equal group sizes they are the integer vectors obtained from Gram–Schmidt applied to $1, s, s^2, \dots$ in the level scores ([Section 8.7](#), [Section 9.4](#)). For five levels they are

$$\begin{aligned} \text{linear} & (-2, -1, 0, 1, 2), & \text{quadratic} & (2, -1, -2, -1, 2), \\ \text{cubic} & (-1, 2, 0, -2, 1), & \text{quartic} & (1, -4, 6, -4, 1). \end{aligned}$$

The degree- d contrast is orthogonal to every polynomial of lower degree in the score, so it vanishes whenever the means lie on a polynomial of degree less than d . With unequal spacing the Gram–Schmidt step uses the actual level values ([Exercise \(B2\)](#)), and with unequal group sizes it uses the weighted inner product $\sum_k n_k p_k q_k$ and the contrasts $c_k = n_k p_k$ ([Example \(Trend in ideology across party identification\)](#)).

15.4.3 Trend analysis of the nitrogen trial

Example. (The shape of the nitrogen response)

The five rates are equally spaced and the design is balanced, so the integer contrasts above apply. The four single-degree-of-freedom sums of squares, each tested against $s^2 = 0.2715$ on 25 degrees of freedom, are:

Component	Sum of squares	F	p	Share of SSB
linear	25.9121	95.42	5.1×10^{-10}	90.3%
quadratic	2.7904	10.28	0.0037	9.7%
cubic	0.0049	0.02	0.89	0.0%
quartic	8.6×10^{-6}	3.2×10^{-5}	0.996	0.0%
rates (total)	28.7074	26.43		

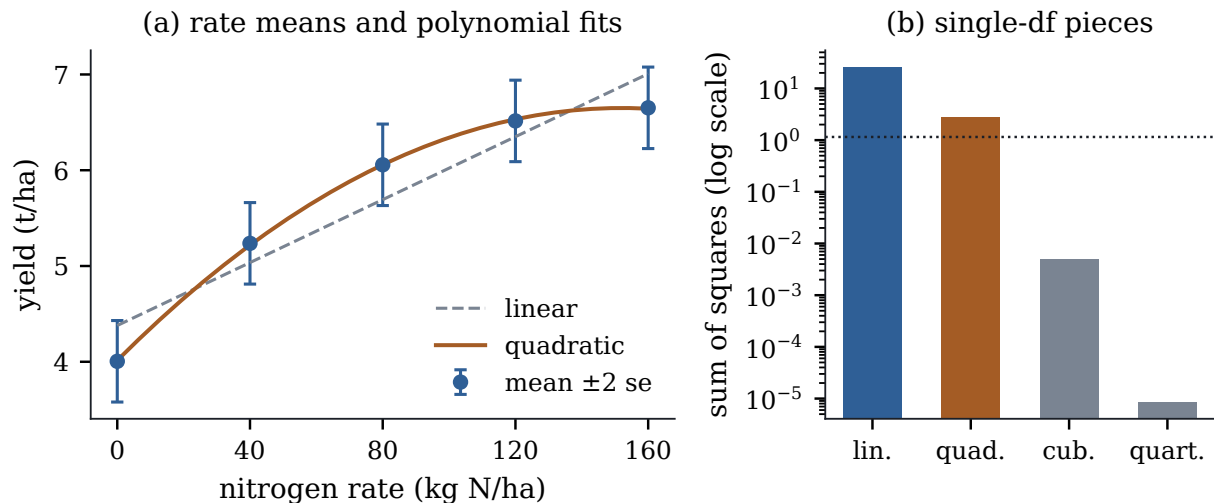


Figure 15.4.1: Trend analysis of the nitrogen trial. (a) Rate means with bars of two standard errors, and the least squares linear and quadratic curves in the rate fitted to all thirty plots. (b) The four orthogonal polynomial sums of squares (logarithmic scale); the dotted line is the value a single degree of freedom needs to be significant at level 0.05, $s^2 F_{0.05}(1, 25)$.

The pieces add to the between-rates sum of squares of [Example \(The analysis of variance for the nitrogen trial\)](#), and F is the average of the four single-degree-of-freedom F values. Yield rises with the rate and the rise flattens: the quadratic component is clearly significant. The cubic and quartic components together test whether a quadratic in the rate describes the five means; their sum of squares 0.0049 on two degrees of freedom gives $F = 0.009$ and $p = 0.99$ ([Figure 15.4.1](#)).

```
import numpy as np
from scipy import stats

rates = np.array([0, 40, 80, 120, 160])           # kg N per hectare
g, m = len(rates), 6                             # 5 rates, 6 plots per rate
true_means = 6.8 - 2.7 * np.exp(-0.018 * rates)   # unknown to the analyst
rng = np.random.default_rng(1515)
group = np.repeat(np.arange(g), m)                # plot i received rate group[i]
y = np.round(true_means[group] + rng.normal(0, 0.5, g * m), 2) # yield, t/ha
n = len(y)

means = np.array([y[group == k].mean() for k in range(g)])
nu = n - g
s2 = np.sum((y - means[group]) ** 2) / nu
ss_between = m * np.sum((means - means.mean()) ** 2)

C_poly = np.array([[-2, -1, 0, 1, 2],            # linear
                   [2, -1, -2, -1, 2],          # quadratic
                   [-1, 2, 0, -2, 1],            # cubic
                   [1, -4, 6, -4, 1]], float)     # quartic

def contrast_ss(c):
    return (c @ means) ** 2 / (c @ c / m)         # balanced: sum c_k^2 / n_k = c'
    c / m

ss_poly = np.array([contrast_ss(c) for c in C_poly])
F_poly = ss_poly / s2
```

```

p_poly = stats.f.sf(F_poly, 1, nu)
for name, ss, F, p in zip(["linear", "quadratic", "cubic", "quartic"], ss_poly,
    F_poly, p_poly):
    print(f"{name:9s} SS = {ss:8.4f} F = {F:7.2f} p = {p:.2g}")
print(f"sum = {ss_poly.sum():.4f}    between rates = {ss_between:.4f}")

```

The test of the cubic and quartic components is not a new test. Every model in which the mean depends on the rate alone has its mean space inside $\mathcal{C}(\mathbf{Z})$ (Lemma 14.6.1), so the one-way residual sum of squares is the pure-error sum of squares of Section 14.6. The quadratic regression on the rate misses exactly the cubic and quartic directions of $\mathcal{C}(\mathbf{Z})$, so their sum of squares is its lack-of-fit sum of squares, and the trend test is the pure-error test of Theorem 14.6.2 (Exercise (C2)). The listing checks the identity: the quadratic regression has residual sum of squares 6.7935, the lack of fit plus the within-rates sum of squares.

```

ss_lof = ss_poly[2] + ss_poly[3]                                # cubic + quartic, 2 df
F_lof = (ss_lof / 2) / s2
X2 = np.column_stack([np.ones(n), rates[group], rates[group] ** 2])
coef, *_ = np.linalg.lstsq(X2, y, rcond=None)                  # quadratic regression on the
    rate
sse_quad = np.sum((y - X2 @ coef) ** 2)
print(f"lack of fit of the quadratic: SS = {ss_lof:.4f}, F = {F_lof:.3f},"
    f" p = {stats.f.sf(F_lof, 2, nu):.2f}")
print(f"quadratic regression SSE = {sse_quad:.4f} = lack of fit + within = {ss_lof +
    s2 * nu:.4f}")

```

The fitted quadratic is $4.014 + 0.03466 N - 0.0001139 N^2$ t/ha at N kg N/ha. It has a maximum at $N = 0.03466 / (2 \times 0.0001139) \approx 152$, inside the range of rates, and it declines after that. That decline is a property of the parabola, not of the crop: the data were generated from a response that rises at every rate, and five noisy means cannot distinguish a parabola that turns over from a curve that levels off. A trend analysis describes the means at the rates used and should not be extrapolated.

15.4.4 One data set, two decompositions

A different set of questions gives a different complete set. Suppose the agronomist had planned to ask first whether fertilizer helps at all, and then how the response behaves among the fertilized rates. The contrast of the control against the four fertilized rates is a split of the kind in Proposition 15.4.2, with coefficients proportional to $(-4, 1, 1, 1, 1)$. The linear, quadratic and cubic polynomial contrasts among the four fertilized rates, $(0, -3, -1, 1, 3)$, $(0, 1, -1, -1, 1)$ and $(0, -1, 3, -3, 1)$, are zero at the control, so they are orthogonal to it by part (a) of that proposition and to each other by balance. Their sums of squares are 21.3701 ($F = 78.70$), 6.6364 ($F = 24.44$), 0.7004 ($F = 2.58$, $p = 0.12$) and 0.0005, and they add to the same 28.7074.

```

C_ctrl = np.array([[-4, 1, 1, 1, 1],                          # control against the four
    fertilized
                    [0, -3, -1, 1, 3],                          # linear among 40, ..., 160
                    [0, 1, -1, -1, 1],                          # quadratic among them
                    [0, -1, 3, -3, 1]], float)                  # cubic among them
ss_ctrl = np.array([contrast_ss(c) for c in C_ctrl])
print("second set:", ss_ctrl.round(4), " sum =", round(ss_ctrl.sum(), 4))

```

The two decompositions tell different but compatible stories. In the first, curvature is significant;

in the second, the curvature among the fertilized rates is not, because most of the bend is the jump from no nitrogen to 40 kg, which the second set assigns to the control contrast. Computing several decompositions and reporting the one whose pieces look best is the data snooping of [Section 13.6](#).

15.4.5 Exercises

15.4.5.1 A. Check your understanding

Exercise A1.

Check that the four polynomial contrasts for five levels are pairwise orthogonal. Show that the quadratic contrast vanishes when $\mu_k = a + bk$, and the cubic contrast vanishes when $\mu_k = a + bk + ck^2$.

Exercise A2.

For the four fertilizer combinations of “Factorial treatments”, verify that the three contrasts are orthogonal when every group has the same size, and that the P and K contrasts are not orthogonal for group sizes 3, 3, 3, 5 (in the order none, P , K , both). What does the lack of orthogonality mean for the estimates?

Solution. With equal sizes, orthogonality is the plain inner product: $(-1, 1, -1, 1) \cdot (-1, -1, 1, 1) = 1 - 1 - 1 + 1 = 0$, and similarly for the other two pairs. With sizes 3, 3, 3, 5, $\sum_k c_k d_k / n_k = \frac{1}{3} - \frac{1}{3} - \frac{1}{3} + \frac{1}{5} = -\frac{2}{15} \neq 0$. By [Proposition 8.7.2\(c\)](#) the two estimates are then correlated, with covariance $-\frac{2}{15}\sigma^2$, and their sums of squares do not add to the part of SSB that the two contrasts span.

15.4.5.2 B. Practice

Exercise B1.

Show that the two planned contrasts of [Example \(Two planned contrasts\)](#), $(-1, \frac{1}{4}, \frac{1}{4}, \frac{1}{4}, \frac{1}{4})$ and $(0, 0, -1, 0, 1)$, are orthogonal. Find two further contrasts that complete them to a complete orthogonal set, and say what they compare.

Solution. The design is balanced, and $-1 \cdot 0 + \frac{1}{4}(0 - 1 + 0 + 1) = 0$. A contrast $\mathbf{e}$ orthogonal to the first satisfies $-e_1 + \frac{1}{4}(e_2 + e_3 + e_4 + e_5) = 0$, and since $e_2 + e_3 + e_4 + e_5 = -e_1$ this forces $e_1 = 0$. Orthogonality to the second forces $e_3 = e_5$. So $\mathbf{e} = (0, a, b, c, b)$ with $a + 2b + c = 0$. Two orthogonal choices are $(0, 1, 0, -1, 0)$, comparing 40 with 120, and $(0, 1, -1, 1, -1)$, comparing 40 and 120 together with 80 and 160 together. Their inner product is $1 - 1 = 0$. The completion is valid but not very natural: completing a planned set is optional.

Exercise B2.

Five plots at each of four rates 0, 1, 2 and 4 (in units of 50 kg N/ha) are to be analysed for trend. Show that the linear and quadratic orthogonal polynomial contrasts are

proportional to $(-7, -3, 1, 9)$ and $(7, -4, -8, 5)$. Check that the quadratic contrast vanishes when the means are linear in the rate.

Solution. The rates have mean $7/4$, so the linear contrast is proportional to $4(s_k - 7/4) = (-7, -3, 1, 9)$. For the quadratic, centre $s_k^2 = (0, 1, 4, 16)$ at its mean $21/4$ and multiply by four: $(-21, -17, -5, 43)$. Its inner product with $(-7, -3, 1, 9)$ is 580, and the linear vector has squared length 140, so subtract $\frac{580}{140}(-7, -3, 1, 9)$ to get $(8, -\frac{32}{7}, -\frac{64}{7}, \frac{40}{7})$, which is proportional to $(7, -4, -8, 5)$. Its entries sum to zero and its inner product with the linear contrast is $-49 + 12 - 8 + 45 = 0$. For means $a + bs_k$ the quadratic contrast gives $b(0 - 4 - 16 + 20) = 0$.

Exercise B3.

Four equally spaced levels have group sizes 2, 4, 4, 2. Show that the classical linear contrast $(-3, -1, 1, 3)$ is orthogonal for this design to the quadratic $(1, -1, -1, 1)$ but not to the cubic $(-1, 3, -3, 1)$. Explain the first fact by symmetry.

Solution. For linear and quadratic, $\sum_k c_k d_k / n_k = -\frac{3}{2} + \frac{1}{4} - \frac{1}{4} + \frac{3}{2} = 0$. For linear and cubic, it is $\frac{3}{2} - \frac{3}{4} - \frac{3}{4} + \frac{3}{2} = \frac{3}{2} \neq 0$. The sizes are symmetric about the middle of the scale. The linear contrast is odd under the reflection $k \mapsto 5 - k$, the quadratic is even, and the weights $1/n_k$ are even, so the weighted inner product of an odd and an even vector vanishes. Linear and cubic are both odd, and nothing forces their weighted inner product to vanish.

Exercise B4.

Use the formula of [Proposition 15.4.2](#) to recompute the sum of squares of the control against the fertilized rates in the nitrogen trial from the rate means, and compare it with the value 21.3701 in the text.

Solution. Here $N_A = 24$, $N_B = 6$, $\bar{y}_A = \frac{1}{4}(5.237 + 6.057 + 6.515 + 6.652) = 6.11525$ and $\bar{y}_B = 4.005$. The sum of squares is $\frac{24 \cdot 6}{30}(6.11525 - 4.005)^2 = 4.8 \times 2.11025^2 \approx 21.375$. The small difference from 21.3701 comes from using the rounded rate means.

15.4.5.3 C. Going deeper

Exercise C1.

Let $\mathbf{U} = [\mathbf{u}_1, \dots, \mathbf{u}_{g-1}]$ and $\mathbf{V} = [\mathbf{v}_1, \dots, \mathbf{v}_{g-1}]$ be two orthonormal bases of the treatment space. Show that $\mathbf{V} = \mathbf{U}\mathbf{Q}$ for an orthogonal $(g-1) \times (g-1)$ matrix $\mathbf{Q}$, that the vectors of squared coordinates of $\mathbf{y}$ in the two bases have the same sum SSB, and that the individual squared coordinates generally differ. More generally, show that for any symmetric idempotent $\mathbf{A}$ of rank q and $\mathbf{Y} \sim N_n(\boldsymbol{\theta}, \sigma^2 \mathbf{I})$, $\mathbf{Y}^\top \mathbf{A} \mathbf{Y}$ splits into q independent scaled noncentral $\chi^2(1)$ variables, one for each vector of an orthonormal basis of $\mathcal{C}(\mathbf{A})$.

Solution. Both $\mathbf{U}\mathbf{U}^\top$ and $\mathbf{V}\mathbf{V}^\top$ equal $\mathbf{P}_\mathcal{T}$, the unique projection onto $\mathcal{T}$. Put $\mathbf{Q} = \mathbf{U}^\top \mathbf{V}$. Then $\mathbf{U}\mathbf{Q} = \mathbf{P}_\mathcal{T} \mathbf{V} = \mathbf{V}$ and $\mathbf{Q}^\top \mathbf{Q} = \mathbf{V}^\top \mathbf{P}_\mathcal{T} \mathbf{V} = \mathbf{V}^\top \mathbf{V} = \mathbf{I}$. The coordinate vectors satisfy $\mathbf{V}^\top \mathbf{y} = \mathbf{Q}^\top \mathbf{U}^\top \mathbf{y}$, and an orthogonal matrix preserves length, so both

have squared length $\mathbf{y}^\top \mathbf{P}_\tau \mathbf{y} = \text{SSB}$. The individual coordinates are mixed by $\mathbf{Q}$ and generally change. For the general statement, write $\mathbf{A} = \sum_{i=1}^q \mathbf{w}_i \mathbf{w}_i^\top$ with $\mathbf{w}_1, \dots, \mathbf{w}_q$ an orthonormal basis of $\mathcal{C}(\mathbf{A})$ (Proposition 6.3.2). The rank-one projections $\mathbf{w}_i \mathbf{w}_i^\top$ are mutually orthogonal, so by Theorem 4.4.4 the terms $(\mathbf{w}_i^\top \mathbf{Y})^2$ are independent, and $(\mathbf{w}_i^\top \mathbf{Y})^2 / \sigma^2 \sim \chi^2(1, (\mathbf{w}_i^\top \boldsymbol{\theta})^2 / \sigma^2)$. Eigenvectors of $\mathbf{A}$ for the eigenvalue one are one such basis.

Exercise C2.

For g equally spaced levels with equal group sizes, show that the sum of the sums of squares of the polynomial contrasts of degrees $d+1, \dots, g-1$ is the lack-of-fit sum of squares of the polynomial regression of degree $d < g-1$ on the level score, and that the sum for degrees $1, \dots, d$ is that regression's sum of squares after the mean.

Solution. Let $\mathcal{P}_d \subseteq \mathcal{C}(\mathbf{Z})$ be the mean space of the degree- d regression, spanned by the vectors whose entry in group k is s_k^i , $i = 0, \dots, d$ (Lemma 14.6.1). The contrast vector of the degree- i orthogonal polynomial p_i has entry $p_i(s_k)/m$ in group k , and in a balanced layout orthogonality of these vectors is orthogonality of the p_i in the plain inner product. By construction $p_0, \dots, p_d$ span the polynomials of degree at most d on the scores, so the vectors of degrees $1, \dots, d$ span $\mathcal{P}_d \cap \mathbf{1}^\perp$, and those of degrees $d+1, \dots, g-1$ span $\mathcal{C}(\mathbf{Z}) \cap \mathcal{P}_d^\perp$. The projection onto the first space gives the regression sum of squares after the mean (Theorem 6.5.1), and the projection onto the second gives $\|(\mathbf{M} - \mathbf{P}_{\mathcal{P}_d})\mathbf{y}\|^2$, the lack-of-fit sum of squares of Theorem 14.6.2. Each projection is the sum of the projections onto the lines, by Theorem 6.3.8.

15.5 Balanced and unbalanced one-way layouts

Every result so far holds for any group sizes, yet experimenters aim for balance. This section says what it buys: plain orthogonality, the largest guaranteed power for a fixed number of observations, and a test of equal means that tolerates unequal variances. The third matters most in practice, because unbalanced data with unequal variances are common and the classical F test can then go badly wrong.

15.5.1 Orthogonality and simplicity

In a balanced layout, orthogonality of contrasts is plain orthogonality $\sum_k c_k d_k = 0$ (Theorem 15.4.1(d)), so tabulated sets such as the integer polynomial contrasts can be used as they stand. The side conditions $\sum_k \alpha_k = 0$ and $\sum_k n_k \alpha_k = 0$ define the same parameters, the pooled variance is the plain average of the group variances, and every pairwise difference has the standard error $s\sqrt{2/m}$. Tukey's intervals have exact simultaneous coverage (Theorem 13.3.2), whereas with unequal sizes the Tukey–Kramer intervals (Eq. 13.3.3) are conservative by an unknown amount.

15.5.2 Balance maximizes guaranteed power

The noncentrality [Eq. 15.3.3](#) depends on the group sizes as well as the means. No allocation of a fixed total is best for every configuration of means, and the planner does not know the means, so the relevant criterion is the power *guaranteed* against every configuration whose means span at least a range Δ , as in [Proposition 11.5.2](#). For unequal group sizes, [Exercise \(B1\)](#) already identifies the least favourable configuration; what is new here is the comparison of allocations.

Proposition 15.5.1 — Balance maximizes the guaranteed noncentrality

In a one-way layout with group sizes $n_1, \dots, n_g$, $g \geq 2$, let

$$h = \min_{k \neq l} \frac{n_k n_l}{n_k + n_l} = \min_{k \neq l} \left(\frac{1}{n_k} + \frac{1}{n_l} \right)^{-1}.$$

- (a) For all means with $\max_k \mu_k - \min_k \mu_k \geq \Delta > 0$, the noncentrality of the F test satisfies $\gamma \geq h\Delta^2/\sigma^2$, and there is a configuration with range exactly Δ at which equality holds.
- (b) For fixed $n = \sum_k n_k$, $h \leq n/(2g)$, with equality iff the layout is balanced.

Proof. Part (a) is [Exercise \(B1\)](#), rewritten with h : its bound is $\Delta^2/(\sigma^2(1/n_{(1)} + 1/n_{(2)})) = h\Delta^2/\sigma^2$, since $1/n_k + 1/n_l$ is largest for the two smallest groups. Equality holds when those two groups carry means Δ apart and every other mean equals their weighted average.

For (b): as just noted, $1/n_k + 1/n_l$ is largest for the two smallest groups. Call their sizes $a \leq b$. The other $g-2$ groups have at least b observations each, so $n \geq a + (g-1)b$, and

$$\left(\frac{1}{a} + \frac{1}{b} \right) n \geq \left(\frac{1}{a} + \frac{1}{b} \right) (a + (g-1)b) = g + (g-1)t + \frac{1}{t},$$

where $t = b/a \geq 1$. Now $(g-1)t + 1/t - g = ((g-1)t - 1)(t-1)/t \geq 0$, because both factors are nonnegative when $t \geq 1$ and $g \geq 2$. Hence $1/a + 1/b \geq 2g/n$, that is, $h \leq n/(2g)$. Equality requires $t = 1$ and $n = a + (g-1)b$, so all groups have the same size. Conversely, a balanced layout has $h = n/2 = n/(2g)$. ■

A departure is hardest to detect when it sits in the least informative pair of groups. The balanced value $h = n/2$ is the bound of [Proposition 11.5.2](#).

Example. (Forty observations in four groups)

With $n = 40$ and $g = 4$, the guaranteed noncentrality per unit of $(\Delta/\sigma)^2$ is 5.000 for the balanced allocation 10, 10, 10, 10, 3.938 for the mildly unbalanced 7, 9, 11, 13, and 2.400 for 4, 6, 10, 20. At $\Delta = \sigma$ the guaranteed powers of the level-0.05 test are 0.399, 0.320 and 0.205. To guarantee power 0.8, the range of the means must be at least 1.56σ , 1.76σ and 2.25σ respectively ([Figure 15.5.1](#)). The mild imbalance gives the same

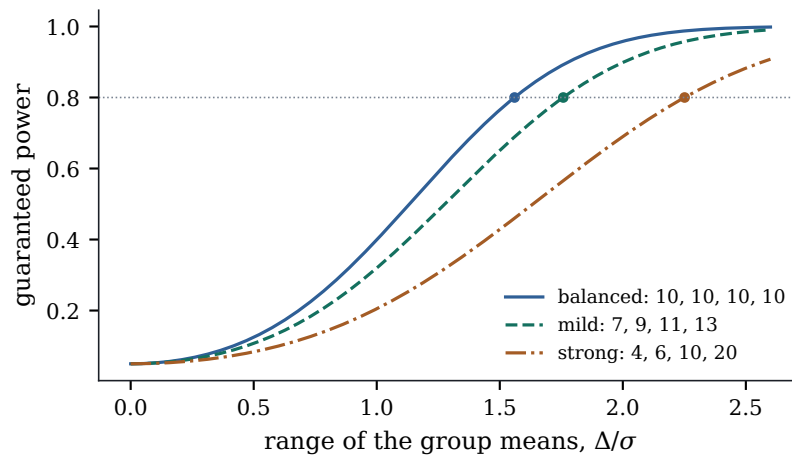


Figure 15.5.1: Guaranteed power of the level-0.05 F test with 40 observations in four groups, against the range of the group means in units of σ , for three allocations. The curves are the power at the least favourable configuration of Proposition 15.5.1; dots mark where each reaches 0.8.

guarantee as a balanced design with $8 \times 3.938 \approx 31.5$ observations, so it wastes a fifth of the experiment.

```
import itertools

import numpy as np
from scipy import stats

def guaranteed_gamma(sizes, delta_over_sigma):
    """Smallest noncentrality over all mean configurations with range at least Delta
    """
    h = min(a * b / (a + b) for a, b in itertools.combinations(sizes, 2))
    return delta_over_sigma ** 2 * h

def power_f(gamma, q, nu, alpha=0.05):
    return stats.ncf.sf(stats.f.ppf(1 - alpha, q, nu), q, nu, gamma)

allocations = {"balanced": [10, 10, 10, 10], "mild": [7, 9, 11, 13], "strong": [4,
    6, 10, 20]}
for name, sizes in allocations.items():
    gam = guaranteed_gamma(sizes, 1.0)
    print(f"{name:9s} {sizes}: guaranteed gamma = {gam:.3f},"
        f" guaranteed power at Delta = sigma: {power_f(gam, 3, 36):.3f}")
```

Balance is optimal for the overall test and for the family of all pairwise comparisons. It is not optimal for every purpose. A single planned contrast is estimated most precisely with $n_k \propto |c_k|$ (Exercise (B2)), and comparisons with a control favour a larger control group (Exercise (B3)). The allocation should follow the questions.

15.5.3 Unequal variances

The one-way model assumes a common variance. Suppose instead that the observations in group k have variance σ_k^2 , with everything else as before. What do the two mean squares then

estimate?

Theorem 15.5.2 — Expected mean squares under unequal variances

Let the Y_{kj} be uncorrelated with $E(Y_{kj}) = \mu_k$ and $\text{Var}(Y_{kj}) = \sigma_k^2$, and write $\bar{n} = n/g$. Then:

- (a) $E(\text{SSE}) = \sum_k (n_k - 1)\sigma_k^2$, so $E(\text{MSE})$ is the average of the σ_k^2 weighted by $n_k - 1$;
- (b) $E(\text{SSB}) = \sum_k (1 - n_k/n)\sigma_k^2 + \sum_k n_k(\mu_k - \bar{\mu})^2$;
- (c) when the means are equal,

$$E(\text{MSB}) - E(\text{MSE}) = \frac{g(n-1)}{n(g-1)(n-g)} \sum_{k=1}^g (\bar{n} - n_k) \sigma_k^2. \quad (15.5.1)$$

In a balanced layout this is zero, and both mean squares estimate $g^{-1} \sum_k \sigma_k^2$. In an unbalanced layout it is positive when the larger variances belong to the smaller groups and negative when they belong to the larger groups.

Proof. The argument of Theorem 9.6.1 applies with a diagonal covariance. Now $\text{Cov}(\mathbf{Y}) = \mathbf{\Sigma}$, diagonal with σ_k^2 on the observations of group k , and $E(\mathbf{Y}) = \boldsymbol{\theta}$. By Theorem 2.3.1, $E(\mathbf{Y}^\top \mathbf{A} \mathbf{Y}) = \text{tr}(\mathbf{A} \mathbf{\Sigma}) + \boldsymbol{\theta}^\top \mathbf{A} \boldsymbol{\theta}$. (a) With $\mathbf{A} = \mathbf{I} - \mathbf{M}$, whose diagonal entries in group k are $1 - 1/n_k$, $\text{tr}(\mathbf{A} \mathbf{\Sigma}) = \sum_k n_k(1 - 1/n_k)\sigma_k^2$, and $\mathbf{A} \boldsymbol{\theta} = \mathbf{0}$. (b) With $\mathbf{A} = \mathbf{M} - \mathbf{P}_0$, the diagonal entries in group k are $1/n_k - 1/n$, so $\text{tr}(\mathbf{A} \mathbf{\Sigma}) = \sum_k (1 - n_k/n)\sigma_k^2$, and $\boldsymbol{\theta}^\top \mathbf{A} \boldsymbol{\theta} = \sum_k n_k(\mu_k - \bar{\mu})^2$ as in Theorem 15.3.1. (c) With equal means, multiply the difference of the two expectations by $(g-1)(n-g)$:

$$\begin{aligned} & (n-g) \sum_k (1 - n_k/n) \sigma_k^2 - (g-1) \sum_k (n_k - 1) \sigma_k^2 \\ &= \sum_k (n-1 - g(n-1)n_k/n) \sigma_k^2 = \frac{g(n-1)}{n} \sum_k (\bar{n} - n_k) \sigma_k^2, \end{aligned}$$

where the middle step collects the coefficient of σ_k^2 : $(n-g) - (n-g)n_k/n - (g-1)n_k + (g-1) = n-1 - gn_k + gn_k/n$. Since $\sum_k (\bar{n} - n_k) = 0$, the sum equals $-\sum_k (n_k - \bar{n})(\sigma_k^2 - \bar{\sigma}^2)$ for any constant $\bar{\sigma}^2$, which is negative when n_k and σ_k^2 move together and positive when they move in opposite directions. In a balanced layout every $\bar{n} - n_k$ is zero, and then $E(\text{MSE}) = \sum_k (m-1)\sigma_k^2/(g(m-1)) = g^{-1} \sum_k \sigma_k^2$. ■

The theorem predicts the behaviour of the F test. In a balanced layout both mean squares estimate the same average variance under the hypothesis, and although the ratio is no longer exactly F -distributed, Box (1954) showed that the effect on the level is modest. In an unbalanced layout MSE leans towards the variances of the large groups and MSB towards those of the small groups. When the small groups are the variable ones the test rejects too often; when the large groups are, it rejects too rarely and loses power.

Example. (How far the level moves)

Four normal groups have equal means and standard deviations increasing geometrically

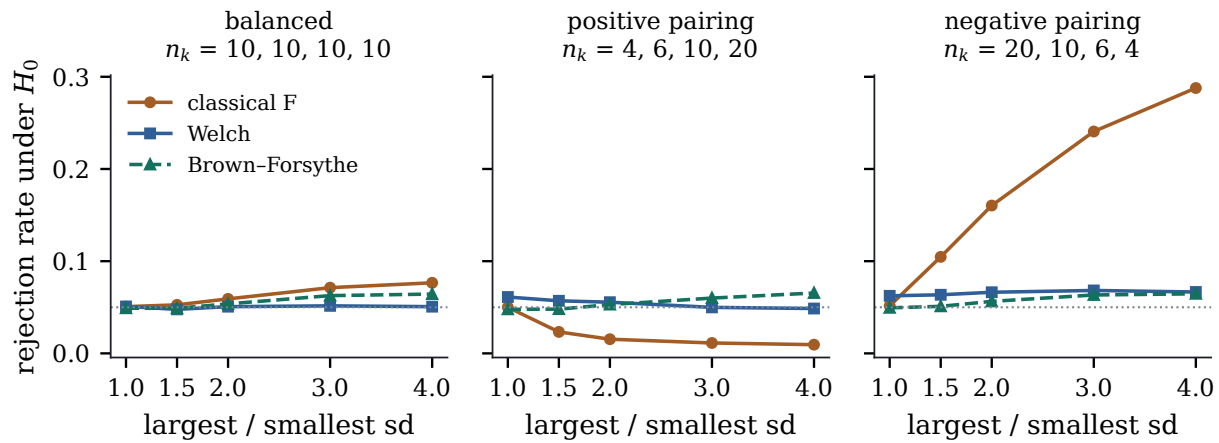


Figure 15.5.2: Rejection rates under equal means of three tests at nominal level 0.05 (dotted), for four normal groups whose standard deviations increase from 1 to the ratio on the horizontal axis. Each point is based on 40000 simulated data sets.

from 1 to R . Three designs of 40 observations are compared: balanced (10 per group), *positive pairing* (sizes 4, 6, 10, 20, the largest group most variable) and *negative pairing* (sizes 20, 10, 6, 4, the smallest group most variable). For each design and each R , 40000 data sets were simulated and three level-0.05 tests applied: the classical F test and the tests of Welch and of Brown and Forsythe described below (Figure 15.5.2).

At $R = 3$, Eq. 15.5.1 gives expected mean square ratios $E(\text{MSB})/E(\text{MSE})$ of 1.000, 0.559 and 1.923 for the three designs. The simulated levels of the classical test follow: 0.071 when balanced, 0.011 with positive pairing, and 0.241 with negative pairing, almost five times the nominal level. At $R = 4$ the negative pairing gives 0.288. Welch's test stays between 0.048 and 0.068 throughout, and the Brown–Forsythe test between 0.048 and 0.065.

The simulation supports the common rule of thumb that a ratio of about two between the largest and smallest standard deviations does little harm, but only with balance: at $R = 2$ the balanced level is 0.059, while the two unbalanced designs give 0.015 and 0.160.

15.5.4 Tests that allow unequal variances

If the variances σ_k^2 were known, weighted least squares (Section 6.9) would compare the group means with their precision-weighted average $\tilde{\mu} = \sum_k w_k \bar{Y}_k / \sum_k w_k$, $w_k = n_k / \sigma_k^2$, and $\sum_k w_k (\bar{Y}_k - \tilde{\mu})^2$ would have an exact $\chi^2(g-1)$ law under equal means. In practice the σ_k^2 are replaced by the s_k^2 and the reference distribution is approximated. We describe the two approximations in common use without deriving them; Chapter 21 treats heteroscedasticity in general.

Welch's test. Welch (1951) replaced σ_k^2 by s_k^2 in the weights, $w_k = n_k / s_k^2$, and divided the weighted mean square by a correction factor:

$$F_W = \frac{\sum_k w_k (\bar{y}_k - \tilde{\mu})^2 / (g-1)}{1 + \frac{2(g-2)}{g^2-1} \sum_k h_k}, \quad h_k = \frac{(1 - w_k / \sum_l w_l)^2}{n_k - 1},$$

referred to $F(g-1, \nu_W)$ with $\nu_W = (g^2-1)/(3 \sum_k h_k)$. The correction and the degrees of freedom come from an approximation to the statistic's null distribution that is accurate to first

order in the quantities $1/(n_k - 1)$. For two groups F_W is the square of Welch's two-sample t statistic with Satterthwaite's degrees of freedom ([Exercise \(B1\)](#)).

The Brown–Forsythe test. Brown and Forsythe (1974a) kept the unweighted numerator and changed the denominator. By [Theorem 15.5.2\(b\)](#), under equal means SSB has expectation $\sum_k (1 - n_k/n) \sigma_k^2$, whatever the variances. Replacing σ_k^2 by s_k^2 gives an unbiased estimate of that expectation, and the statistic

$$F_{BF} = \frac{\text{SSB}}{\sum_k (1 - n_k/n) s_k^2}$$

compares SSB with an estimate of what it should be under the hypothesis. It is referred to $F(g - 1, f)$, where $1/f = \sum_k c_k^2 / (n_k - 1)$ with $c_k = (1 - n_k/n) s_k^2 / \sum_l (1 - n_l/n) s_l^2$, Satterthwaite's approximation to the degrees of freedom of a linear combination of independent variance estimates. In a balanced layout the denominator is $(g - 1) s^2$, so $F_{BF} = F$ and only the degrees of freedom change.

Individual contrasts need the same care, since $s\sqrt{v_c}$ estimates the wrong variance when the σ_k^2 differ, even in a balanced layout ([Exercise \(C2\)](#)). The **Welch–Satterthwaite** interval uses

$$\widehat{\text{Var}}(\hat{\psi}) = \sum_k \frac{c_k^2 s_k^2}{n_k}, \quad \nu = \frac{(\sum_k c_k^2 s_k^2 / n_k)^2}{\sum_k (c_k^2 s_k^2 / n_k)^2 / (n_k - 1)},$$

and the multiplier $t_{\nu, \alpha/2}$ ([Exercise \(B3\)](#)), adjusted as in [Chapter 13](#) for simultaneous inference.

Example. (An unbalanced trial with unequal variances)

A repair mortar was made with four admixture formulations, and its setting time was measured in minutes. The standard formulation A was tested most often, and the newer formulations least (simulated data, rounded to 0.1 minute):

Formulation	n_k	Setting times (min)	Mean	s_k
A	12	18.9, 20.2, 20.6, 21.2, 20.3, 19.0, 20.2, 19.0, 20.5, 20.6, 20.1, 20.3	20.075	0.73
B	9	22.6, 21.3, 22.2, 22.8, 19.3, 20.7, 21.5, 20.8, 18.8	21.111	1.38
C	6	20.7, 18.2, 18.6, 23.6, 21.3, 22.6	20.833	2.14
D	4	20.1, 22.6, 28.3, 25.3	24.075	3.53

The standard deviations grow as the group sizes shrink, the negative pairing of [Example \(How far the level moves\)](#). The classical analysis gives $F = 5.34$ on $(3, 27)$ degrees of freedom and $p = 0.0051$, apparently strong evidence, driven by the high mean of D. But $\text{MSE} = 3.015$ gives the variance of D weight $3/27$, while the null expectation of MSB in [Theorem 15.5.2\(b\)](#) gives it weight $(1 - 4/31)/3 \approx 0.29$; estimated with the s_k^2 , that expectation is 5.407, far above MSE. Welch's test gives $F_W = 2.64$ on $(3, 8.3)$ degrees of freedom and $p = 0.119$, and the Brown–Forsythe test gives $F_{BF} = 2.98$ on $(3, 6.2)$ degrees of freedom and $p = 0.115$. The difference between D and A is 4.00 minutes. With the pooled standard error 1.003 its 95% interval is $[1.94, 6.06]$; with the Welch–Satterthwaite standard error 1.776 on 3.1 degrees of freedom it is $[-1.56, 9.56]$. Four observations with a standard deviation of 3.5 minutes cannot establish a difference of four minutes.

The weighted and unweighted averages of the group means are 21.039 and 21.524, so the side conditions $\sum_k n_k \alpha_k = 0$ and $\sum_k \alpha_k = 0$ of [Section 15.1](#) ([Theorem 8.4.2](#)) define overall means that differ by half a minute here.

```
import numpy as np
from scipy import stats
from statsmodels.stats.oneway import anova_oneway

labels = ["A", "B", "C", "D"]
n_k = np.array([12, 9, 6, 4])
rng = np.random.default_rng(1520)
samples = [np.round(rng.normal(mu, sd, k), 1)
            for mu, sd, k in zip([20.0, 20.6, 21.0, 22.2], [0.8, 1.2, 2.0, 3.0], n_k)
          ]

g, n = len(samples), n_k.sum()
means = np.array([x.mean() for x in samples])
s2_k = np.array([x.var(ddof=1) for x in samples])      # group variances
s2 = np.sum((n_k - 1) * s2_k) / (n - g)              # pooled within-group
               variance
grand = means @ n_k / n
ss_b = np.sum(n_k * (means - grand) ** 2)
F = ss_b / (g - 1) / s2
print("means:", means.round(3), " sds:", np.sqrt(s2_k).round(2))
print(f"classical F = {F:.2f}, p = {stats.f.sf(F, g - 1, n - g):.4f}")
# null expectation of MS_between if the variances really are s2_k (the theorem of
  this section)
print(f"MS_within = {s2:.3f}; estimated null E(MS_between) = "
      f"{np.sum((1 - n_k / n) * s2_k) / (g - 1):.3f}")

welch = anova_oneway(samples, use_var="unequal")
bf = anova_oneway(samples, use_var="bf")
print(f"Welch: F = {welch.statistic:.2f} on ({g - 1}, {welch.df_denom:.1f}) df, p = "
      f"{welch.pvalue:.3f}")
print(f"Brown-Forsythe: F = {bf.statistic:.2f} on ({g - 1}, {bf.df2[1]:.1f}) df, p = "
      f"{bf.pvalue2:.3f}")
lev = stats.levene(*samples, center="median")          # dispersion test on |y -
               median|
print(f"variance test on absolute deviations from medians: p = {lev.pvalue:.4f}")
```

(By default `statsmodels` refines the numerator degrees of freedom of the Brown–Forsythe test; `pvalue2` is the original version.)

The simulation of [Example \(How far the level moves\)](#) used the same formulas, applied to group means and variances drawn directly from their sampling distributions:

```
import numpy as np
from scipy import stats

def three_tests(ybar, s2, n_k):
    """p-values of the classical F, Welch and Brown-Forsythe tests; arrays of shape
    (reps, g)."""
    g, n = n_k.size, n_k.sum()
    grand = ybar @ n_k / n
    ss_b = ((ybar - grand[:, None]) ** 2) @ n_k
    ss_w = s2 @ (n_k - 1)
    p_f = stats.f.sf(ss_b / (g - 1) / (ss_w / (n - g)), g - 1, n - g)
    # Welch (1951): weights n_k / s_k^2 and a Satterthwaite-type denominator df
    w = n_k / s2
    mu_w = np.sum(w * ybar, axis=1) / w.sum(axis=1)
    h = (1 - w / w.sum(axis=1)[:, None]) ** 2 / (n_k - 1)
    a = np.sum(w * (ybar - mu_w[:, None]) ** 2, axis=1) / (g - 1)
    b = 1 + 2 * (g - 2) / (g ** 2 - 1) * h.sum(axis=1)
    p_w = stats.f.sf(a / b, g - 1, (g ** 2 - 1) / (3 * h.sum(axis=1)))
    # Brown-Forsythe (1974): SS_between over its null expectation under unequal
    # variances
    d = (1 - n_k / n) * s2
    c = d / d.sum(axis=1)[:, None]
    p_bf = stats.f.sf(ss_b / d.sum(axis=1), g - 1, 1 / np.sum(c ** 2 / (n_k - 1),
    axis=1))
    return p_f, p_w, p_bf

def size(n_k, sds, reps=40000, seed=0, alpha=0.05):
    """Rejection rates under equal means, from group means and variances drawn
    exactly."""
    rng = np.random.default_rng(seed)
    n_k, sds = np.asarray(n_k, float), np.asarray(sds, float)
    ybar = rng.normal(0, sds / np.sqrt(n_k), (reps, n_k.size))
    s2 = sds ** 2 * rng.chisquare(n_k - 1, (reps, n_k.size)) / (n_k - 1)
    return [np.mean(p < alpha) for p in three_tests(ybar, s2, n_k)]

designs = {"balanced": [10, 10, 10, 10],
          "positive pairing": [4, 6, 10, 20],          # big groups get the big
          "negative pairing": [20, 10, 6, 4]}          # small groups get the big
          variances
          variances
ratios = [1, 1.5, 2, 3, 4]
res = {}
for name, sizes in designs.items():
    # sds grow geometrically from 1
    # to R
    res[name] = np.array([size(sizes, R ** (np.arange(4) / 3), seed=int(10 * R)) for
    R in ratios])
for R, row in zip(ratios, res[name]):
    print(f"{name:17s} R = {R:3.1f}: F, Welch, BF sizes = {np.round(row, 3)}")
```

Robustness has a small price. With equal variances and a group of only four observations, Welch's test rejects slightly too often, 0.061 and 0.062 in the two unbalanced designs. The practical advice is simple: with unequal group sizes, use a heteroscedasticity-robust test unless there is good reason to believe the variances are equal; with equal sizes the classical test is acceptable, although individual contrasts still need care.

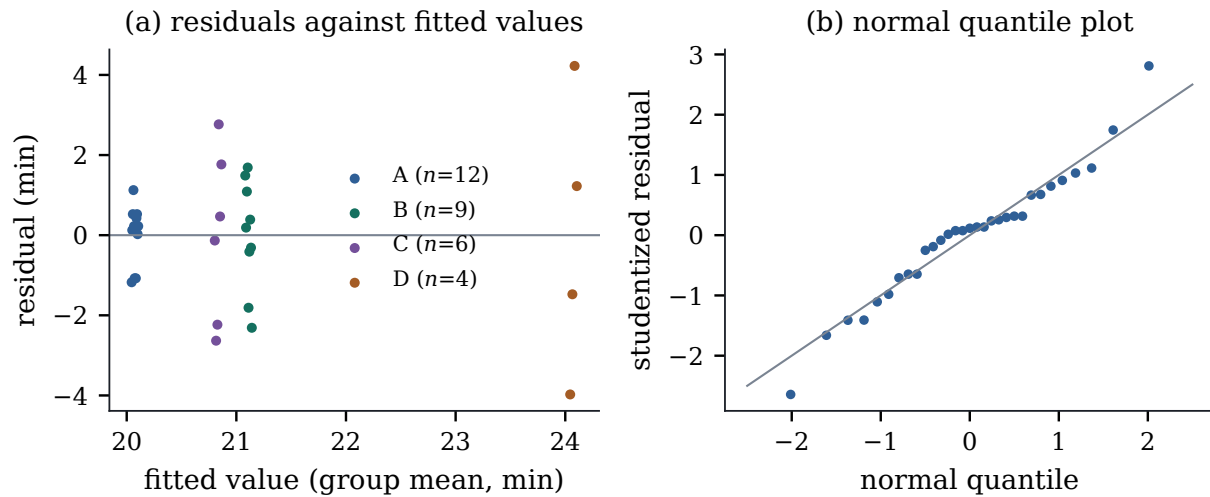


Figure 15.5.3: Residual plots for the mortar trial. (a) Residuals against fitted values: the spread grows from formulation A to formulation D. (b) Normal quantile plot of the studentized residuals: the curvature at both ends reflects the mixture of groups with different variances, not a heavy-tailed error distribution.

15.5.5 Checking the assumptions

Independence is secured by the design, above all by randomization, and cannot be checked well from the data. Correlation within groups, which arises when the observations in a group share a batch, an operator or a day, inflates the F statistic badly ([Exercise \(C1\)](#)). A common variance and normality are checked with the residuals $e_{kj} = y_{kj} - \bar{y}_k$.

The leverage of an observation in group k is $1/n_k$ ([Proposition 6.8.1](#)), so $\text{Var}(e_{kj}) = \sigma^2(1 - 1/n_k)$. Residuals from small groups are shrunk towards zero, so they are compared on the common scale $r_{kj} = e_{kj}/(s\sqrt{1 - 1/n_k})$. A plot of residuals against the group means shows whether the spread differs between groups, and a spread growing with the mean suggests a transformation (Chapter 22). A normal quantile plot of the studentized residuals shows skewness and heavy tails. Chapter 20 develops residual diagnostics in general.

[Figure 15.5.3](#) shows the two plots for the mortar trial. The fan shape in panel (a) is unmistakable. The classical tests of equal variances, which compare the logarithms of the s_k^2 , are so sensitive to nonnormality that a rejection says as much about the tails as about the variances. Levene (1960) proposed instead to apply the one-way F test to the absolute deviations $|y_{kj} - \bar{y}_k|$, which are larger on average in a more variable group; Brown and Forsythe (1974b) used deviations from the group *medians*, which behave better for skewed data. For the mortar trial that version gives $p = 0.0019$. Using such a test as a gate for the classical F test is not recommended, since it has little power in small groups, exactly where unequal variances do the most harm. The analysis should be chosen in advance, from the design and from what is known about the response.

Nonnormality is usually the least serious problem for tests about means: the level of the F test is fairly robust to it, more so in balanced designs (Chapter 19), though heavy tails cost power by inflating MSE. Transformations (Chapter 22) and resampling (Chapter 23), for example the Kruskal–Wallis permutation test on ranks, are the usual remedies.

15.5.6 Exercises

15.5.6.1 A. Check your understanding

Exercise A1.

For the balanced design with $m = 10$ and variances 1, 2, 4, 8, compute $E(\text{MSB})$ under equal means and $E(\text{MSE})$ from [Theorem 15.5.2](#). Repeat for the sizes 4, 6, 10, 20 and for 20, 10, 6, 4.

Solution. Balanced: both equal $(1 + 2 + 4 + 8)/4 = 3.75$. Sizes 4, 6, 10, 20 ($n = 40$): $E(\text{SSB}) = 0.9 + 0.85 \cdot 2 + 0.75 \cdot 4 + 0.5 \cdot 8 = 9.6$, so $E(\text{MSB}) = 3.2$; $E(\text{SSE}) = 3 + 10 + 36 + 152 = 201$, so $E(\text{MSE}) = 201/36 \approx 5.58$. The test is conservative. Sizes 20, 10, 6, 4: $E(\text{SSB}) = 0.5 + 0.75 \cdot 2 + 0.85 \cdot 4 + 0.9 \cdot 8 = 12.6$, so $E(\text{MSB}) = 4.2$; $E(\text{SSE}) = 19 + 18 + 20 + 24 = 81$, so $E(\text{MSE}) = 2.25$. The test is liberal.

Exercise A2.

Show that the hat matrix of the one-way model has diagonal entries $1/n_k$ and that $\text{Var}(e_{kj}) = \sigma^2(1 - 1/n_k)$. What is the residual of an observation that is alone in its group, and why is it useless for checking the model?

Solution. The hat matrix is $\mathbf{M}$ of [Eq. 15.1.1](#), whose diagonal entry for an observation of group k is $1/n_k$. By [Proposition 5.5.3](#), $\text{Cov}(\hat{\boldsymbol{\epsilon}}) = \sigma^2(\mathbf{I} - \mathbf{M})$, so $\text{Var}(e_{kj}) = \sigma^2(1 - 1/n_k)$. An observation alone in its group has leverage one, its fitted value is itself, and its residual is identically zero: it carries no information about the variance or about the model.

15.5.6.2 B. Practice

Exercise B1.

Show that for $g = 2$ the Welch statistic F_W equals t_W^2 , where $t_W = (\bar{y}_1 - \bar{y}_2)/\sqrt{s_1^2/n_1 + s_2^2/n_2}$, and that ν_W is Satterthwaite's $(s_1^2/n_1 + s_2^2/n_2)^2 / [(s_1^2/n_1)^2/(n_1 - 1) + (s_2^2/n_2)^2/(n_2 - 1)]$.

Solution. With $g = 2$ the correction factor is 1. Write $v_k = s_k^2/n_k = 1/w_k$ and $W = w_1 + w_2$. The weighted sum of squares of two numbers about their weighted mean is $w_1 w_2 (\bar{y}_1 - \bar{y}_2)^2 / W = (\bar{y}_1 - \bar{y}_2)^2 / (v_1 + v_2)$, which is t_W^2 . Next $1 - w_1/W = w_2/W = v_1/(v_1 + v_2)$, so $h_1 = v_1^2 / \{(v_1 + v_2)^2(n_1 - 1)\}$ and similarly for h_2 . Then $\nu_W = 3 / (3 \sum_k h_k) = (v_1 + v_2)^2 / \{v_1^2/(n_1 - 1) + v_2^2/(n_2 - 1)\}$.

Exercise B2.

Show that for $g = 2$ the Brown–Forsythe statistic also equals t_W^2 , and that its degrees of freedom f equal ν_W . So the two robust tests coincide for two groups and differ only for three or more.

Solution. With two groups, $SSB = n_1 n_2 (\bar{y}_1 - \bar{y}_2)^2 / n$, and the denominator is $(n_2 s_1^2 + n_1 s_2^2) / n$. Their ratio is $(\bar{y}_1 - \bar{y}_2)^2 / (s_1^2 / n_1 + s_2^2 / n_2) = t_W^2$. The weights are $c_1 = (n_2 s_1^2 / n) / \{(n_2 s_1^2 + n_1 s_2^2) / n\} = v_1 / (v_1 + v_2)$ and $c_2 = v_2 / (v_1 + v_2)$, so $1/f = \sum_k c_k^2 / (n_k - 1) = 1/\nu_W$ by the previous exercise.

Exercise B3.

Let $\hat{V} = \sum_k c_k^2 s_k^2 / n_k$ with independent $s_k^2 \sim \sigma_k^2 \chi^2(n_k - 1) / (n_k - 1)$ under normality. Show that $E \hat{V} = V = \text{Var}(\hat{\psi})$ and $\text{Var} \hat{V} = 2 \sum_k (c_k^2 \sigma_k^2 / n_k)^2 / (n_k - 1)$. Choose ν so that $\nu \hat{V} / V$ has the mean and variance of a $\chi^2(\nu)$ variable, and show that replacing σ_k^2 by s_k^2 in the answer gives the degrees of freedom of the Welch–Satterthwaite interval.

Solution. Since $E s_k^2 = \sigma_k^2$, $E \hat{V} = \sum_k c_k^2 \sigma_k^2 / n_k = V$, which is $\text{Var}(\hat{\psi})$ because the group means are independent with variances σ_k^2 / n_k . $\text{Var}(s_k^2) = 2\sigma_k^4 / (n_k - 1)$, and the terms are independent, which gives $\text{Var} \hat{V}$. A $\chi^2(\nu)$ variable has mean ν and variance 2ν . The mean of $\nu \hat{V} / V$ is ν for every ν , and its variance is $\nu^2 \text{Var}(\hat{V}) / V^2$. Setting this equal to 2ν gives $\nu = 2V^2 / \text{Var} \hat{V} = V^2 / \sum_k (c_k^2 \sigma_k^2 / n_k)^2 / (n_k - 1)$. With s_k^2 for σ_k^2 this is the formula in the text.

Exercise B4.

For $n = 30$ observations in five groups, compare the quantity h of [Proposition 15.5.1](#) for the balanced allocation and for the allocation 10, 5, 5, 5, 5 suggested by [Exercise \(B3\)](#) for comparisons with a control. What does each allocation optimize?

Solution. Balanced: $h = 6/2 = 3$. For 10, 5, 5, 5, 5, the least informative pair is two treatments of size 5, with $h = 25/10 = 2.5$. The guaranteed noncentrality of the overall test is $3(\Delta/\sigma)^2$ against $2.5(\Delta/\sigma)^2$. But each comparison with the control has variance $\sigma^2(1/10 + 1/5) = 0.3\sigma^2$ instead of $\sigma^2/3$. Balance protects the overall test and the comparisons among treatments; the unbalanced allocation sharpens the comparisons with the control.

15.5.6.3 C. Going deeper

Exercise C1.

In a balanced layout, suppose that under equal means the errors within a group are equicorrelated, $\text{Cov}(\varepsilon_{kj}, \varepsilon_{kj'}) = \rho\sigma^2$ for $j \neq j'$ with $\rho \geq 0$, and independent between groups. Show that $E(\text{MSE}) = \sigma^2(1 - \rho)$ and $E(\text{MSB}) = \sigma^2(1 - \rho + m\rho)$. What happens to the level of the F test with $m = 10$ and $\rho = 0.1$?

Solution. The covariance matrix is $\sigma^2\{(1 - \rho)\mathbf{I} + \rho\mathbf{Z}\mathbf{Z}^\top\}$, the random-effects form with $\sigma_a^2 = \rho\sigma^2$ and error variance $(1 - \rho)\sigma^2$. By [Exercise \(C1\)](#) with $n_0 = m$, $E(\text{MSE}) = (1 - \rho)\sigma^2$ and $E(\text{MSB}) = (1 - \rho)\sigma^2 + m\rho\sigma^2$. Under normality $\text{MSB}/E(\text{MSB})$ and $\text{MSE}/E(\text{MSE})$ are independent scaled chi-squared variables, so F is $1 + m\rho/(1 - \rho)$ times an $F(g - 1, n - g)$ variable. With $m = 10$ and $\rho = 0.1$ the factor is $1 + 1/0.9 \approx 2.1$, and a nominal 5% test rejects far more often: with $g = 4$ groups its level is 0.271. A small intraclass correlation does great damage when groups are large.

Exercise C2.

In a balanced layout with variances σ_k^2 , show that the pooled variance estimate of the pairwise difference $\bar{Y}_k - \bar{Y}_l$, $2s^2/m$, has expectation $2\bar{\sigma}^2/m$ with $\bar{\sigma}^2 = g^{-1} \sum_i \sigma_i^2$, while the true variance is $(\sigma_k^2 + \sigma_l^2)/m$. Conclude that balance protects the overall test but not individual comparisons, and find the ratio of true to estimated variance for the two most variable groups when the variances are 1, 2, 4, 8.

Solution. By [Theorem 15.5.2\(c\)](#), in a balanced layout $E(s^2) = \bar{\sigma}^2$, so $E(2s^2/m) = 2\bar{\sigma}^2/m$, and the true variance is $\sigma_k^2/m + \sigma_l^2/m$ because the means are independent. For the groups with variances 4 and 8, the true variance is $12/m$ and the pooled estimate targets $2 \cdot 3.75/m = 7.5/m$, a ratio of 1.6. Intervals for that difference built from s^2 are too short by a factor of about $\sqrt{1.6} \approx 1.26$, and intervals for the difference of the two least variable groups are too long.

15.6 Summary and Notes

15.6.1 Summary

Idea.

1. The one-way layout has mean space $\mathcal{C}(\mathbf{Z})$, the vectors constant within groups, in both its forms ([Definition 15.1.1](#)). The fit is the group means, and s^2 pools the group variances. Different side conditions define different “overall means” when the layout is unbalanced ([Section 15.1](#)).
2. Contrasts are the directions of the treatment space $\mathcal{T} = \mathcal{C}(\mathbf{Z}) \cap \mathbf{1}^\perp$, and a contrast’s test, interval and sum of squares come from the projection onto its line ([Proposition 15.2.1](#)). Families use Bonferroni or Dunnett for comparisons with a control, Tukey for all pairs and Scheffé for contrasts suggested by the data ([Section 15.2](#)).
3. A contrast test captures the fraction $\cos^2 \varphi$ of the overall noncentrality, φ being its angle with the true treatment component ([Proposition 15.2.2](#)): a well-chosen planned contrast beats the overall test, a badly chosen one has no power.
4. The decomposition $\mathbf{I} = \mathbf{P}_0 + (\mathbf{M} - \mathbf{P}_0) + (\mathbf{I} - \mathbf{M})$ gives the analysis of variance table. The F statistic has the $F(g-1, n-g, \gamma)$ law with $\gamma = \sum_k n_k (\mu_k - \bar{\mu})^2 / \sigma^2$, and it is the likelihood ratio test of equal means ([Theorem 15.3.1](#)). Without normality, $E(\text{MSE}) = \sigma^2$ and $E(\text{MSB}) = \sigma^2 + \sum_k n_k (\mu_k - \bar{\mu})^2 / (g-1)$ ([Theorem 15.3.2](#)). The quantity $(g-1)F$ is the largest squared contrast t statistic.
5. A complete set of orthogonal contrasts, orthogonal in the sense $\sum_k c_k d_k / n_k = 0$, splits SSB into $g-1$ independent single-degree-of-freedom pieces whose noncentralities add up to γ , and F is the average of their F statistics ([Theorem 15.4.1](#)). Successive splits of the treatments give such sets for any group sizes ([Proposition 15.4.2](#)), and polynomial contrasts give a trend analysis, whose higher-degree pieces are the pure-error lack-of-fit test of a low-degree curve ([Example \(The shape of the nitrogen response\)](#)).
6. Balance makes design orthogonality plain orthogonality and maximizes the noncentrality guaranteed against every configuration of means with a given range

([Proposition 15.5.1](#)).

7. With unequal variances, MSB and MSE estimate the same average variance under equal means if the layout is balanced, but not otherwise. The classical test is liberal when the small groups are the variable ones and conservative when the large groups are ([Theorem 15.5.2](#)). The tests of Welch and of Brown and Forsythe hold their level approximately, and Welch–Satterthwaite intervals do the same for contrasts ([Section 15.5](#)).
8. Residuals have variance $\sigma^2(1 - 1/n_k)$; plotted against the group means and on a normal quantile plot after studentizing, they check the variances and normality. Within-group correlation, which the data cannot reveal, is the most damaging failure.

15.6.2 Notes and sources

Origins. The analysis of variance is Fisher’s. It appears in *Statistical Methods for Research Workers* (Fisher 1925), developed for the field experiments at Rothamsted, and *The Design of Experiments* (Fisher 1935) made randomization the logical basis of the analysis. The derivation of the one-way model from unit–treatment additivity and random assignment, sketched in [Section 15.1](#), is developed in detail by Kempthorne (1952). Scheffé (1959) remains the classic treatment of the fixed-effects theory, including the effects of departures from the assumptions.

The textbook treatments. Christensen (2020, chapter 4) derives the one-way analysis entirely from the projections $\mathbf{M}$ and $\mathbf{M} - \mathbf{P}_0$, including the characterization of contrasts through the space onto which $\mathbf{M} - \mathbf{P}_0$ projects; this chapter follows his geometric approach, with the treatment space as the organizing object. Rencher and Schaalje (2008, chapter 13) give the balanced case through full-and-reduced models and through the general linear hypothesis, derive the expected mean squares both ways, and treat orthogonal and orthogonal polynomial contrasts, including the result that the overall F statistic averages the single-degree-of-freedom statistics of a complete orthogonal set. Their section 12.7 covers the testable hypotheses of non-full-rank models used in [Theorem 15.3.1](#). Seber and Lee (2003, section 8.2) discuss simultaneous intervals and the assumptions, including unequal variances. Note that Rencher and Schaalje and Christensen define noncentrality as half of our γ .

Contrasts and multiple comparisons. Tables of orthogonal polynomials for equally spaced levels go back at least to Fisher and Yates (1938). The exact method for comparisons with a control is due to Dunnett (1955), and the conservativeness of the Tukey–Kramer intervals for unequal group sizes was proved by Hayter (1984); [Chapter 13 \(Exercise \(B2\)\)](#) characterizes the many-to-one multiplier and gives further references. [Proposition 15.2.2](#) and [Proposition 15.4.2](#) are elementary consequences of the projection picture. Part (a) of [Proposition 15.5.1](#) is [Exercise \(B1\)](#) of Chapter 11; part (b), the optimality of balance, is a short calculation.

Unequal variances. Box (1954) computed the effect of unequal variances on the one-way F test, showing that it is small when the groups are of equal size and can be large otherwise, in the directions described by [Theorem 15.5.2](#). The test of Welch (1951) weights the groups by their estimated precisions; the test of Brown and Forsythe (1974a) keeps the unweighted numerator and estimates its null expectation. Welch obtained his correction factor and degrees of freedom from a series expansion of the statistic’s distribution to first order in $1/(n_k - 1)$; the Brown–Forsythe degrees of freedom, like those of the contrast intervals, are Satterthwaite’s (1946) moment match. The robust test of equal variances by analysis of variance of absolute deviations is due to Levene (1960), and the version based on deviations from medians to Brown

and Forsythe (1974b). The simulation of [Example \(How far the level moves\)](#) is ours. Chapter 21 returns to heteroscedasticity with weighted least squares and heteroscedasticity-consistent standard errors, and Chapter 32 to random effects.

15.6.3 References

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Chapter 16

Multi-Way Layouts and Interaction

An experiment with several factors asks more than one question. It asks whether each factor matters, and also whether the effect of one factor depends on the level of another. In a balanced layout, where every combination of levels is observed equally often, these questions correspond to mutually orthogonal subspaces of observation space. Each subspace has a projection that is a Kronecker product of averaging and centring matrices, and the analysis of variance is Pythagoras applied to that decomposition. This chapter builds the decomposition for two factors, with and without replication, for any number of crossed factors, and for factors nested inside other factors. The algebra is the easy part. The harder part is interpretation: what an interaction is, when a transformation removes it, what a main effect means when interaction is present, and when it is safe to pool one term into another.

What you need. Kronecker products ([Section 1.10](#)); independence of projections of a normal vector and the noncentral F distribution ([Chapter 4](#)); nested projections and the two-factor projection identity of [Example \(Additive two-factor model, one observation per cell\)](#) ([Section 6.5](#)); estimability with two factors ([Section 8.6](#)); analysis of variance decompositions, orthogonal designs and expected mean squares ([Chapter 9](#)); F tests of reduced models ([Chapter 11](#)); simultaneous inference ([Chapter 13](#)); the lack-of-fit test and regressors built from fitted values ([Sections 14.6 and 14.7](#)); and the one-way analysis of variance ([Chapter 15](#)).

16.0.1 Roadmap

- [Two-way additive models](#)
- [Interaction and its interpretation](#)
- [Balanced two-way layouts via orthogonal projections](#)
- [Higher-way layouts](#)
- [Nested factors](#)
- [Summary and notes](#)

16.1 Two-way additive models

Two factors are observed together. Factor A has levels $i = 1, \dots, a$ and factor B has levels $j = 1, \dots, b$, with $a, b \geq 2$. Every one of the ab combinations, or **cells**, is observed the same

number $m \geq 1$ of times. We write y_{ijk} for the k th observation in cell (i, j) , so that $n = abm$.

Example (Additive two-factor model, one observation per cell) found the projections of the additive model for $m = 1$ by a direct argument. This section redoes it with the tools used throughout the chapter, and adds replication, estimable functions, tests, and a test of additivity with one observation per cell.

16.1.1 Averaging and centring matrices

Stack the observations with the replicate index k running fastest, then j , then i :

$$\mathbf{y} = (y_{111}, \dots, y_{11m}, y_{121}, \dots, y_{12m}, \dots, y_{ab1}, \dots, y_{abm})^\top.$$

With this order a matrix of the form $\mathbf{E}_a \otimes \mathbf{E}_b \otimes \mathbf{E}_m$ acts on each index separately, as in **Example (The balanced two-way layout)**. For $k \geq 1$ write

$$\bar{\mathbf{J}}_k = \frac{1}{k} \mathbf{1}_k \mathbf{1}_k^\top, \quad \mathbf{C}_k = \mathbf{I}_k - \bar{\mathbf{J}}_k.$$

The **averaging matrix** $\bar{\mathbf{J}}_k$ replaces each entry of a k -vector by the mean of the entries. The **centring matrix** $\mathbf{C}_k$ subtracts that mean. With the usual dot notation for means, the following products replace y_{ijk} by

$$\begin{aligned} (\mathbf{I}_a \otimes \mathbf{I}_b \otimes \bar{\mathbf{J}}_m) \mathbf{y} &: \bar{y}_{ij\cdot}, & (\mathbf{I}_a \otimes \bar{\mathbf{J}}_b \otimes \bar{\mathbf{J}}_m) \mathbf{y} &: \bar{y}_{i\cdot\cdot}, \\ (\bar{\mathbf{J}}_a \otimes \mathbf{I}_b \otimes \bar{\mathbf{J}}_m) \mathbf{y} &: \bar{y}_{\cdot j\cdot}, & (\bar{\mathbf{J}}_a \otimes \bar{\mathbf{J}}_b \otimes \bar{\mathbf{J}}_m) \mathbf{y} &: \bar{y}_{\cdot\cdot\cdot}, \end{aligned}$$

the cell, row, column and grand means. Each operator averages over the indices where it has a $\bar{\mathbf{J}}$ and leaves the others alone. Every projection in this chapter is a Kronecker product of factors $\mathbf{I}$, $\bar{\mathbf{J}}$ and $\mathbf{C}$, and the following lemma is all the algebra they need.

Lemma 16.1.1 — Kronecker products of averaging and centring matrices

Let $l_1, \dots, l_t \geq 1$.

- (a) $\bar{\mathbf{J}}_k$ and $\mathbf{C}_k$ are symmetric and idempotent, with ranks 1 and $k - 1$, and $\bar{\mathbf{J}}_k \mathbf{C}_k = \mathbf{0}$, $\bar{\mathbf{J}}_k + \mathbf{C}_k = \mathbf{I}_k$.
- (b) If each $\mathbf{E}_s$ is one of $\mathbf{I}_{l_s}$, $\bar{\mathbf{J}}_{l_s}$, $\mathbf{C}_{l_s}$, then $\mathbf{E}_1 \otimes \dots \otimes \mathbf{E}_t$ is an orthogonal projection whose rank is the product of the ranks of the $\mathbf{E}_s$.
- (c) If each $\mathbf{E}_s$ and each $\mathbf{F}_s$ is one of $\bar{\mathbf{J}}_{l_s}$, $\mathbf{C}_{l_s}$, and $\mathbf{E}_s \neq \mathbf{F}_s$ for at least one s , then $(\mathbf{E}_1 \otimes \dots \otimes \mathbf{E}_t)(\mathbf{F}_1 \otimes \dots \otimes \mathbf{F}_t) = \mathbf{0}$.
- (d) The 2^t products $\mathbf{E}_1 \otimes \dots \otimes \mathbf{E}_t$ with each $\mathbf{E}_s \in \{\bar{\mathbf{J}}_{l_s}, \mathbf{C}_{l_s}\}$ sum to $\mathbf{I}$.

Proof.

- (a) $\bar{\mathbf{J}}_k$ is symmetric and $\bar{\mathbf{J}}_k^2 = k^{-2} \mathbf{1}(\mathbf{1}^\top \mathbf{1}) \mathbf{1}^\top = \bar{\mathbf{J}}_k$. Its trace is 1. Then $\mathbf{C}_k = \mathbf{I} - \bar{\mathbf{J}}_k$ is the complementary projection, with trace $k - 1$, and $\bar{\mathbf{J}}_k \mathbf{C}_k = \bar{\mathbf{J}}_k - \bar{\mathbf{J}}_k^2 = \mathbf{0}$. Ranks equal traces by **Proposition 6.3.4**. (b) By **Proposition 1.10.1(e)** a Kronecker product of symmetric idempotent matrices is symmetric and idempotent, and by **Proposition 1.10.1(c)** its rank is the product of the ranks. (c) By the mixed-product rule the product equals $\mathbf{E}_1 \mathbf{F}_1 \otimes \dots \otimes \mathbf{E}_t \mathbf{F}_t$, and the factor with $\mathbf{E}_s \neq \mathbf{F}_s$ is $\bar{\mathbf{J}} \mathbf{C} = \mathbf{0}$ or $\mathbf{C} \bar{\mathbf{J}} = \mathbf{0}$. (d) Expand $\mathbf{I} = (\bar{\mathbf{J}}_{l_1} + \mathbf{C}_{l_1}) \otimes \dots \otimes (\bar{\mathbf{J}}_{l_t} + \mathbf{C}_{l_t})$ by bilinearity of the Kronecker product.

The rank $k - 1$ of $\mathbf{C}_k$ is the source of every “levels minus one” in the analysis of variance, and part (d) is the source of every analysis of variance table in this chapter. Part (b) allows $k = 1$: then $\bar{\mathbf{J}}_1 = \mathbf{1}$ and $\mathbf{C}_1 = \mathbf{0}$.

```
import itertools

import numpy as np

def Jbar(k):
    return np.full((k, k), 1.0 / k)          # averaging: replaces a k-vector by its
    mean

def Cen(k):
    return np.eye(k) - Jbar(k)              # centring: subtracts the mean

def kron_all(mats):
    out = np.ones((1, 1))
    for A in mats:
        out = np.kron(out, A)
    return out

def factorial_projections(levels, m):
    """P_S for every subset S of the factors (a 0/1 tuple), and the error projection
    ."""
    P = {}
    for S in itertools.product([0, 1], repeat=len(levels)):
        P[S] = kron_all([Cen(l) if s else Jbar(l) for l, s in zip(levels, S)] + [
            Jbar(m)])
    P["error"] = kron_all([np.eye(l) for l in levels] + [Cen(m)])
    return P

def rank(A):
    return int(round(np.trace(A)))           # rank = trace for a projection
```

16.1.2 The additive model

The **additive two-way model** is

$$E(y_{ijk}) = \mu + \alpha_i + \beta_j, \quad \text{Cov}(\mathbf{Y}) = \sigma^2 \mathbf{I}, \quad (16.1.1)$$

with model matrix $\mathbf{X} = [\mathbf{1}_n, \mathbf{Z}_A, \mathbf{Z}_B]$, where $\mathbf{Z}_A = \mathbf{I}_a \otimes \mathbf{1}_b \otimes \mathbf{1}_m$ and $\mathbf{Z}_B = \mathbf{1}_a \otimes \mathbf{I}_b \otimes \mathbf{1}_m$ are the indicator matrices of the two factors. It says that the difference between two levels of A is the same in every column and, symmetrically, that the difference between two levels of B is the same in every row. The model has $1 + a + b$ parameters and, by [Theorem 8.6.1\(a\)](#), rank $a + b - 1$. Define

$$\begin{aligned}
\mathbf{P}_0 &= \bar{\mathbf{J}}_a \otimes \bar{\mathbf{J}}_b \otimes \bar{\mathbf{J}}_m, \\
\mathbf{P}_A &= \mathbf{C}_a \otimes \bar{\mathbf{J}}_b \otimes \bar{\mathbf{J}}_m, \\
\mathbf{P}_B &= \bar{\mathbf{J}}_a \otimes \mathbf{C}_b \otimes \bar{\mathbf{J}}_m,
\end{aligned} \tag{16.1.2}$$

and $\mathbf{R} = \mathbf{I} - \mathbf{P}_0 - \mathbf{P}_A - \mathbf{P}_B$. By the table of means above, $\mathbf{P}_A \mathbf{y}$ has entry $\bar{y}_{i..} - \bar{y}_{...}$ in every position of row i , and $\mathbf{P}_B \mathbf{y}$ has entry $\bar{y}_{.j.} - \bar{y}_{...}$ in every position of column j .

16.1.3 Additivity with one observation per cell

Many two-factor data sets have one observation per cell, such as methods each run once on several test problems. By [Example \(Additive two-factor model, one observation per cell\)](#) the residual space of the additive model then has dimension $(a-1)(b-1)$, and the residuals are the interaction terms $y_{ij} - \bar{y}_{i.} - \bar{y}_{.j} + \bar{y}_{...}$. The residual sum of squares then serves two purposes. It estimates σ^2 if the model is additive, and it is the only place where nonadditivity could show. There is no model-free estimate of σ^2 , and no test of additivity against a general alternative.

A test is still possible against a *specific* kind of departure. Suppose the cell means are additive on some other scale, $\mu_{ij} = f(\nu + a_i + b_j)$, for a smooth increasing function f and small effects a_i, b_j with $\sum a_i = \sum b_j = 0$. A second-order Taylor expansion about ν gives

$$\mu_{ij} \approx f(\nu) + f'(\nu)(a_i + b_j) + \frac{1}{2}f''(\nu)(a_i^2 + b_j^2) + f''(\nu)a_i b_j.$$

The terms in a_i^2 and b_j^2 are additive. Only the product term is not. To first order the row and column effects of μ_{ij} are $\alpha_i \approx f'(\nu)a_i$ and $\beta_j \approx f'(\nu)b_j$, so the interaction is approximately

$$\mu_{ij} - \bar{\mu}_{i.} - \bar{\mu}_{.j} + \bar{\mu}_{...} \approx \theta \alpha_i \beta_j, \quad \theta = \frac{f''(\nu)}{f'(\nu)^2}. \tag{16.1.3}$$

Tukey (1949) proposed to look for interaction of exactly this shape. The row and column effects are estimated by $r_i = \bar{y}_{i.} - \bar{y}_{...}$ and $c_j = \bar{y}_{.j} - \bar{y}_{...}$ (with $m = 1$ we drop the third subscript). The residuals are then regressed on the products $r_i c_j$. The table $(r_i c_j)$ has zero row and column sums, so it lies in the residual space. The regression through the origin of the residual table on it has slope and sum of squares

$$\hat{\theta} = \frac{\sum_{i,j} r_i c_j y_{ij}}{\sum_i r_i^2 \sum_j c_j^2}, \quad \text{SS}_N = \frac{(\sum_{i,j} r_i c_j y_{ij})^2}{\sum_i r_i^2 \sum_j c_j^2}. \tag{16.1.4}$$

The regressor is computed from the same data, so [Proposition 14.7.1](#) does not apply. But a regressor that is a function of the fitted values alone leaves the F test exact ([Theorem 14.7.2](#)), and part (f) of the next theorem shows that Tukey's regressor is of this kind.

16.1.4 Estimation and tests

The following theorem collects what the additive model delivers in a balanced layout, with and without replication.

Theorem 16.1.2 — The additive two-way model

Assume Eq. 16.1.1 with $m \geq 1$ observations in every cell.

- (a) $\mathbf{P}_0, \mathbf{P}_A, \mathbf{P}_B$ and $\mathbf{R}$ are mutually orthogonal projections summing to $\mathbf{I}$, with ranks 1, $a-1$, $b-1$ and $n-a-b+1$. The projection onto $\mathcal{C}(\mathbf{X})$ is $\mathbf{M} = \mathbf{P}_0 + \mathbf{P}_A + \mathbf{P}_B$, and the fitted values are $\hat{y}_{ijk} = \bar{y}_{i..} + \bar{y}_{.j.} - \bar{y}_{...}$.
- (b) $\lambda_0\mu + \sum_i \lambda_i \alpha_i + \sum_j \kappa_j \beta_j$ is estimable iff $\lambda_0 = \sum_i \lambda_i = \sum_j \kappa_j$. The BLUE of a contrast $\sum_i c_i \alpha_i$ is $\sum_i c_i \bar{y}_{i..}$, with variance $\sigma^2 \sum_i c_i^2 / (bm)$. These are the estimate and variance of the one-way layout in A that ignores B . The estimate of a contrast in the α_i is uncorrelated with the estimate of any contrast in the β_j .
- (c) Under normal errors, $SS_A = \|\mathbf{P}_A \mathbf{y}\|^2 = bm \sum_i (\bar{y}_{i..} - \bar{y}_{...})^2$, $SS_B = \|\mathbf{P}_B \mathbf{y}\|^2 = am \sum_j (\bar{y}_{.j.} - \bar{y}_{...})^2$ and $SSE = \|\mathbf{R} \mathbf{y}\|^2$ are independent, $SSE/\sigma^2 \sim \chi^2(n-a-b+1)$, and

$$F_A = \frac{SS_A/(a-1)}{SSE/(n-a-b+1)} \sim F(a-1, n-a-b+1, \gamma_A), \quad \gamma_A = \frac{bm}{\sigma^2} \sum_i (\alpha_i - \bar{\alpha})^2,$$

with $\bar{\alpha} = a^{-1} \sum_i \alpha_i$. So F_A is an exact test of $\alpha_1 = \cdots = \alpha_a$. Its numerator is that of the one-way F test in A that ignores B (Theorem 15.3.1); only the error term differs. The same holds for B , with $\gamma_B = (am/\sigma^2) \sum_j (\beta_j - \bar{\beta})^2$.

- (d) If $m \geq 2$, then $\mathbf{R} = \mathbf{P}_{AB} + \mathbf{P}_E$ with $\mathbf{P}_{AB} = \mathbf{C}_a \otimes \mathbf{C}_b \otimes \bar{\mathbf{J}}_m$ and $\mathbf{P}_E = \mathbf{I}_a \otimes \mathbf{I}_b \otimes \mathbf{C}_m$, of ranks $(a-1)(b-1)$ and $ab(m-1)$. The squared lengths $\|\mathbf{P}_{AB} \mathbf{y}\|^2$ and $\|\mathbf{P}_E \mathbf{y}\|^2$ are the lack-of-fit and pure-error sums of squares of the additive model (Definition 14.6.1). If the cell means μ_{ij} are arbitrary, the lack-of-fit F test of additivity has noncentrality

$$\gamma_{AB} = \frac{m}{\sigma^2} \sum_{i,j} (\mu_{ij} - \bar{\mu}_{i.} - \bar{\mu}_{.j} + \bar{\mu}_{..})^2.$$

- (e) If $m = 1$, then $\mathbf{R} = \mathbf{C}_a \otimes \mathbf{C}_b$, and the residual space consists of the $a \times b$ tables whose rows and columns all sum to zero. There is no pure error.
- (f) (*Tukey's test.*) If $m = 1$, the errors are normal and $(a-1)(b-1) \geq 2$, then with SS_N as in Eq. 16.1.4,

$$F_N = \frac{SS_N}{(SSE - SS_N)/((a-1)(b-1)-1)} \sim F(1, (a-1)(b-1)-1).$$

Proof.

- (a) By Lemma 16.1.1(c) the three projections $\mathbf{P}_0, \mathbf{P}_A, \mathbf{P}_B$ are mutually orthogonal, and their ranks are 1, $a-1$, $b-1$ by Lemma 16.1.1(b). Moreover $\mathbf{P}_0 + \mathbf{P}_A = \mathbf{I}_a \otimes \bar{\mathbf{J}}_b \otimes \bar{\mathbf{J}}_m$ replaces each observation by its row mean. It is the projection onto $\mathcal{C}(\mathbf{Z}_A)$, the vectors that are constant on rows (Example (The one-way layout)). Likewise $\mathbf{P}_0 + \mathbf{P}_B$ projects onto $\mathcal{C}(\mathbf{Z}_B)$. Hence

$$\mathcal{C}(\mathbf{X}) = \mathcal{C}(\mathbf{Z}_A) + \mathcal{C}(\mathbf{Z}_B) = (\mathcal{C}(\mathbf{P}_0) \oplus \mathcal{C}(\mathbf{P}_A)) + (\mathcal{C}(\mathbf{P}_0) \oplus \mathcal{C}(\mathbf{P}_B)) = \mathcal{C}(\mathbf{P}_0) \oplus \mathcal{C}(\mathbf{P}_A) \oplus \mathcal{C}(\mathbf{P}_B),$$

a sum of mutually orthogonal subspaces. By Theorem 6.3.8 its projection is $\mathbf{P}_0 + \mathbf{P}_A + \mathbf{P}_B$. Then $\mathbf{R} = \mathbf{I} - \mathbf{M}$ is the complementary projection, orthogonal to the other three, with rank $n - (a+b-1)$. The fitted values are $\mathbf{M} \mathbf{y} = (\mathbf{P}_0 + \mathbf{P}_A) \mathbf{y} +$

$$(\mathbf{P}_0 + \mathbf{P}_B)\mathbf{y} - \mathbf{P}_0\mathbf{y}.$$

- (b) The design is complete, hence connected, and the estimability condition is [Exercise \(C1\)](#) with a single component. For the estimator, $E(\bar{y}_{i..}) = \mu + \alpha_i + \bar{\beta}$, so $\sum_i c_i \bar{y}_{i..}$ is unbiased for $\sum_i c_i \alpha_i$ when $\sum_i c_i = 0$. It equals $\boldsymbol{\rho}^\top \mathbf{y}$ with $\boldsymbol{\rho} = (bm)^{-1} \mathbf{c} \otimes \mathbf{1}_b \otimes \mathbf{1}_m$, and $\boldsymbol{\rho} = \mathbf{P}_A \boldsymbol{\rho} \in \mathcal{C}(\mathbf{X})$ because $\mathbf{c} \perp \mathbf{1}_a$. So $(\mathbf{I} - \mathbf{M})\boldsymbol{\rho} = \mathbf{0}$, and by [Eq. 7.2.1](#) its variance equals that of the BLUE. By [Theorem 7.2.1\(c\)](#) it is the BLUE. Its variance is $\sigma^2 \|\boldsymbol{\rho}\|^2 = \sigma^2 \sum_i c_i^2 / (bm)$. The one-way layout in A has bm observations per level, and [Theorem 8.7.1](#) gives the same estimator and variance. A contrast in the β_j has coefficient vector $\boldsymbol{\rho}_B \in \mathcal{C}(\mathbf{P}_B)$, and $\text{Cov}(\boldsymbol{\rho}^\top \mathbf{Y}, \boldsymbol{\rho}_B^\top \mathbf{Y}) = \sigma^2 \boldsymbol{\rho}^\top \boldsymbol{\rho}_B = 0$.
- (c) The spaces $\mathcal{C}(\mathbf{P}_0), \mathcal{C}(\mathbf{P}_A), \mathcal{C}(\mathbf{P}_B)$ form an analysis of variance decomposition of $\mathcal{C}(\mathbf{X})$ ([Definition 9.1.1](#)), so [Corollary 9.6.2](#) applies. The mean vector has entries $\mu + \alpha_i + \beta_j$. Its row means minus its grand mean are $\alpha_i - \bar{\alpha}$, so $\|\mathbf{P}_A \boldsymbol{\mu}\|^2 = bm \sum_i (\alpha_i - \bar{\alpha})^2$. This is zero iff all α_i are equal. The formula for SS_A is the same computation applied to $\mathbf{y}$.
- (d) By [Lemma 16.1.1\(d\)](#) with $t = 3$, $\mathbf{I}$ is the sum of the eight products $\mathbf{E}_a \otimes \mathbf{E}_b \otimes \mathbf{E}_m$. The four with $\mathbf{E}_m = \bar{\mathbf{J}}_m$ are $\mathbf{P}_0, \mathbf{P}_A, \mathbf{P}_B, \mathbf{P}_{AB}$. The four with $\mathbf{E}_m = \mathbf{C}_m$ add up to $\mathbf{I}_a \otimes \mathbf{I}_b \otimes \mathbf{C}_m = \mathbf{P}_E$. So $\mathbf{R} = \mathbf{P}_{AB} + \mathbf{P}_E$, with ranks from [Lemma 16.1.1\(b\)](#). The matrix $\mathbf{I} - \mathbf{P}_E = \mathbf{I}_a \otimes \mathbf{I}_b \otimes \bar{\mathbf{J}}_m$ replaces each observation by its cell mean. It is the projection $\mathbf{M}_Z$ onto the cell-indicator space, the $\mathbf{Z}$ of [Definition 14.6.1](#). Hence $\mathbf{P}_E = \mathbf{I} - \mathbf{M}_Z$ and $\mathbf{P}_{AB} = \mathbf{M}_Z - \mathbf{M}$. By [Theorem 14.6.2](#) the noncentrality is $\|(\mathbf{I} - \mathbf{M})\boldsymbol{\theta}\|^2 / \sigma^2$, where $\boldsymbol{\theta}$ has entries μ_{ij} . Here $(\mathbf{I} - \mathbf{M})\boldsymbol{\theta} = \mathbf{P}_{AB}\boldsymbol{\theta}$, because $\mathbf{P}_E\boldsymbol{\theta} = \mathbf{0}$. Its entries are the displayed interaction terms, each repeated m times.
- (e) With $m = 1$, $\bar{\mathbf{J}}_1 = \mathbf{1}$ and $\mathbf{C}_1 = \mathbf{0}$, so $\mathbf{P}_E = \mathbf{0}$ and $\mathbf{R} = \mathbf{P}_{AB} = \mathbf{C}_a \otimes \mathbf{C}_b$. A vector $\mathbf{v}$, read as an $a \times b$ table $\mathbf{V}$, satisfies $(\mathbf{C}_a \otimes \mathbf{C}_b)\mathbf{v} = \mathbf{v}$ iff $\mathbf{C}_a \mathbf{V} \mathbf{C}_b = \mathbf{V}$, which holds iff $\mathbf{V} \mathbf{1}_b = \mathbf{0}$ and $\mathbf{1}_a^\top \mathbf{V} = \mathbf{0}^\top$.
- (f) Write $\hat{y}_{ij} = \bar{y}_{..} + r_i + c_j$, let $\mathbf{r} = (r_i)$ and $\mathbf{c} = (c_j)$, and put $\mathbf{U} = \mathbf{r} \otimes \mathbf{c}$, the table $(r_i c_j)$. Expanding,

$$\hat{y}_{ij}^2 = (\bar{y}_{..}^2 + r_i^2 + 2\bar{y}_{..}r_i) + (c_j^2 + 2\bar{y}_{..}c_j) + 2r_i c_j.$$

The first bracket depends on i alone and the second on j alone, so both lie in $\mathcal{C}(\mathbf{X})$. Since $\mathbf{1}_a^\top \mathbf{r} = \mathbf{1}_b^\top \mathbf{c} = 0$, $\mathbf{R}\mathbf{U} = \mathbf{C}_a \mathbf{r} \otimes \mathbf{C}_b \mathbf{c} = \mathbf{U}$ by (e). Hence the column $\mathbf{W} = (\hat{y}_{ij}^2)$, a function of $\mathbf{M}\mathbf{Y}$, satisfies $(\mathbf{I} - \mathbf{M})\mathbf{W} = 2\mathbf{U}$. By [Theorem 9.2.2](#) the extra sum of squares from adding $\mathbf{W}$ to the additive model is $(\mathbf{U}^\top \mathbf{y})^2 / \mathbf{U}^\top \mathbf{U}$, and $\mathbf{U}^\top \mathbf{U} = \sum_i r_i^2 \sum_j c_j^2$, so it is SS_N , and $\text{SSE} - \text{SS}_N$ is the residual sum of squares of the augmented model. The rank condition of [Theorem 14.7.2](#) holds because $\mathbf{U} \neq \mathbf{0}$ with probability one: $\mathbf{U} = \mathbf{0}$ only if $\mathbf{P}_A \mathbf{Y} = \mathbf{0}$ or $\mathbf{P}_B \mathbf{Y} = \mathbf{0}$, and each event has probability zero, because $\|\mathbf{P}_A \mathbf{Y}\|^2 / \sigma^2$ has a continuous (noncentral χ^2) distribution on $a - 1 \geq 1$ degrees of freedom, and likewise for B . So $\text{rank}[\mathbf{X}, \mathbf{W}] = a + b$, which is less than $n = ab$ because $(a - 1)(b - 1) \geq 2$. [Theorem 14.7.2](#) with $q = 1$ and $n - r - q = (a - 1)(b - 1) - 1$ gives the distribution of F_N .

■

Part (b) is the practical content of balance: rows are compared exactly as if the columns did not exist. The columns still matter, because they remove their own variation from the error.

The listing builds the five projections of part (d) for $a = 3$, $b = 4$, $m = 2$ and checks their ranks, orthogonality and sum. The script also checks each identification in the proof against projections computed from indicator columns.

```

a, b, m = 3, 4, 2
P = factorial_projections([a, b], m)
names = {(0, 0): "mean", (1, 0): "A", (0, 1): "B", (1, 1): "AB", "error": "error"}
for key, name in names.items():
    print(f"{name:6s} rank {rank(P[key])}")
keys = list(names)
worst = max(np.abs(P[s] @ P[t]).max() for s, t in itertools.combinations(keys, 2))
print("largest entry of any product P_s P_t, s != t:", f"{worst:.1e}")
print("sum is the identity:", np.allclose(sum(P[k] for k in keys), np.eye(a * b * m)
))

```

16.1.5 Tukey's test in practice

Tukey's test ([Theorem 16.1.2\(f\)](#)) is exact against every additive model, whatever the values of μ , α_i , β_j . It has power against interactions roughly proportional to $\alpha_i\beta_j$, the kind produced by measuring on the wrong scale, and little against anything else. The estimated slope says which scale. If f is the inverse of a power transformation, so that y^p is additive, then $f(t) = t^{1/p}$, and [Eq. 16.1.3](#) gives $\theta = (1 - p)/f(\nu)$ ([Exercise \(B3\)](#)). Replacing $f(\nu)$ by the grand mean suggests the power

$$\hat{p} = 1 - \hat{\theta} \bar{y}_{..}, \quad (16.1.5)$$

with $\hat{p} = 0$ read as the logarithm. The rule is rough, but it turns a vague complaint into a specific suggestion.

Example. (Solvers on benchmark problems)

Six solvers were each run once on eight benchmark problems, and the running time was recorded in seconds. The data are synthetic, generated by the script `tukey_runtime.py` so that the logarithm of time is additive in solver and problem, with a little noise:

	P1	P2	P3	P4	P5	P6	P7	P8
S1	1.7	3.4	5.6	14.1	28.1	50.2	82.9	249.4
S2	2.1	5.7	9.0	14.7	36.9	76.0	116.7	292.0
S3	1.3	3.1	5.1	10.5	18.8	37.6	82.2	148.3
S4	3.3	9.1	12.7	19.1	52.8	110.0	183.1	371.8
S5	2.0	4.7	7.2	13.5	31.4	65.2	111.1	289.3
S6	1.5	3.4	5.4	10.4	26.0	43.2	106.3	172.0

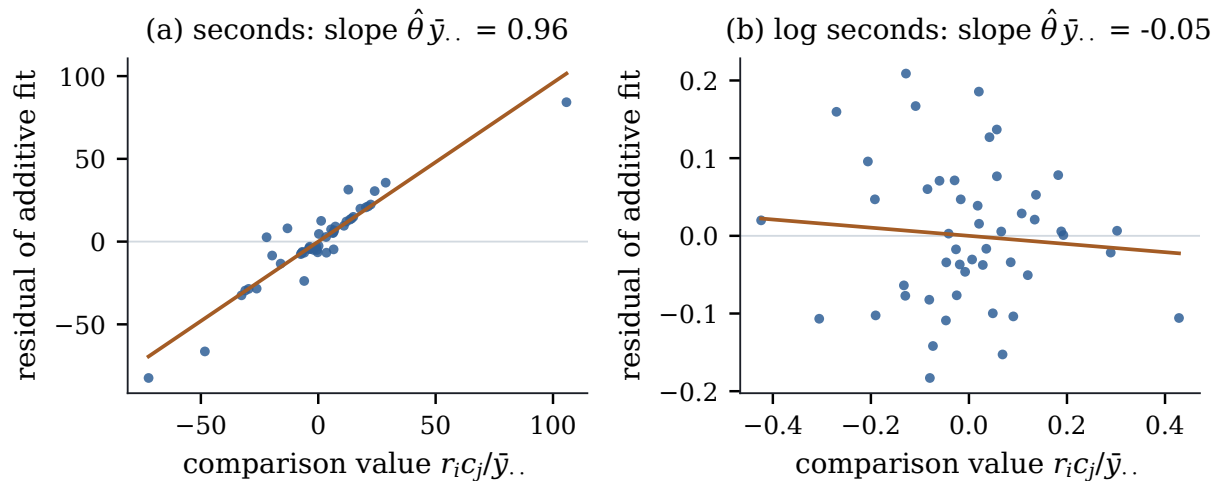


Figure 16.1.1: Tukey's diagnostic plot for the solver data: residuals of the additive fit against the comparison values $r_i c_j / \bar{y}_{..}$, with the fitted line of slope $\hat{\theta}\bar{y}_{..} = 1 - \hat{p}$. (a) In seconds the residuals lie along a line of slope near one, which points to the logarithm. (b) In log seconds there is no trend.

The additive model in seconds leaves a residual sum of squares of 29396 on 35 degrees of freedom. Tukey's single direction takes $SS_N = 26126$ of it, a fraction 0.889, and $F_N = 271.7$ on 1 and 34 degrees of freedom ($p = 9 \times 10^{-18}$). The interaction has exactly the shape of Eq. 16.1.3: fast problems separate the solvers by fractions of a second, slow problems by minutes. The slope is $\hat{\theta} = 0.01563$ and the grand mean is 61.46, so Eq. 16.1.5 suggests the power 0.04, essentially the logarithm. Figure 16.1.1 shows the diagnostic plot. On the log scale, $F_N = 0.27$ ($p = 0.60$), and the residuals show no trend against $r_i c_j / \bar{y}_{..}$.

The scale changes the conclusion about the solvers. In seconds, $F_A = 3.88$ ($p = 0.007$), because the nonadditivity inflates the error term. In log seconds, $F_A = 77.7$. The residual standard deviation is $s = 0.105$, so differences between solvers are measured in proportions of about ten per cent. By Exercise (B1) the solver means of log time are independent with common variance $\sigma^2/8$ and independent of s^2 . Tukey's method therefore gives simultaneous intervals for all 15 ratios of solver speeds, with multiplier $q_{0.05}(6, 35) = 4.261$. Solver 4, the slowest, takes 2.50 times as long as solver 3, the fastest, with simultaneous 95% interval [2.14, 2.94].

```
import numpy as np
from scipy import stats

gen_rng = np.random.default_rng(16_01)
a, b = 6, 8
solver_log = np.array([0.0, 0.25, -0.35, 0.55, 0.1, -0.2]) # log speed
factors
problem_log = np.log([2.0, 4.5, 7.0, 12.0, 25.0, 60.0, 110.0, 240.0]) # typical
seconds
noise = gen_rng.normal(scale=0.12, size=(a, b))
times = np.round(np.exp(solver_log[:, None] + problem_log[None, :] + noise), 1)

def additive_fit(Y):
    """Row effects, column effects and residuals of the additive fit to an a x b
    table."""
    g = Y.mean()
```

```

r = Y.mean(axis=1) - g                                # ybar_i. - ybar..
c = Y.mean(axis=0) - g                                # ybar_.j - ybar..
resid = Y - g - r[:, None] - c[None, :]
return g, r, c, resid

def tukey_test(Y):
    """Tukey's one-degree-of-freedom test for nonadditivity in an a x b table."""
    a, b = Y.shape
    g, r, c, resid = additive_fit(Y)
    u = np.outer(r, c)                                # lies in the interaction space
    ss_n = (u * Y).sum() ** 2 / (u * u).sum()
    sse = (resid ** 2).sum()
    df = (a - 1) * (b - 1) - 1
    F = ss_n / ((sse - ss_n) / df)
    theta = (u * Y).sum() / (u * u).sum()             # coefficient of a_i b_j
    return F, stats.f.sf(F, 1, df), theta, g, sse, ss_n

for label, Y in [("seconds", times), ("log seconds", np.log(times))]:
    F, p, theta, g, sse, ss_n = tukey_test(Y)
    print(f"{label:12s} F = {F:9.2f} p = {p:.2g} theta = {theta:.4f}"
          f" suggested power 1 - theta*ybar = {1 - theta * g:.2f}")

```

The listing below checks [Theorem 16.1.2\(f\)](#) by simulation. It draws 20000 tables from an additive normal model with $a = 6$ and $b = 8$. The 5% test rejects in a fraction 0.0508 of them, and a Kolmogorov–Smirnov comparison of the simulated statistics with $F(1, 34)$ gives $p = 0.98$.

```

rng = np.random.default_rng(20260919)
reps = 20_000
Fs = np.empty(reps)
for t in range(reps):
    Y = 3.0 + np.arange(a)[:, None] * 0.5 + np.arange(b)[None, :] + rng.normal(size=(a, b))
    Fs[t] = tukey_test(Y)[0]
size = np.mean(Fs > stats.f.ppf(0.95, 1, (a - 1) * (b - 1) - 1))
ks = stats.kstest(Fs, stats.f(1, (a - 1) * (b - 1) - 1).cdf)
print(f"simulated size of the 5% test: {size:.4f} Kolmogorov-Smirnov p-value: {ks.pvalue:.2f}")

```

Warning. A nonsignificant Tukey test does not establish additivity. It rules out only interaction of the form $\theta\alpha_i\beta_j$. A single aberrant cell or row can leave F_N small and still distort every conclusion, so look at the residual table and an interaction plot ([Section 16.2](#)) as well.

16.1.6 Exercises

16.1.6.1 A. Check your understanding

Exercise A1.

For $a = b = 2$ and $m = 1$, write out the 4×4 matrices $\mathbf{P}_0$, $\mathbf{P}_A$, $\mathbf{P}_B$ and $\mathbf{R}$ of [Eq. 16.1.2](#). Show that $\mathbf{R}\mathbf{y} = \frac{1}{4}(y_{11} - y_{12} - y_{21} + y_{22})(1, -1, -1, 1)^\top$, and explain why Tukey's test

is not available here.

Exercise A2.

In the additive model with m observations per cell, show that the sum of squares for testing a single contrast $\sum_i c_i \alpha_i = 0$ is $(\sum_i c_i \bar{y}_{i..})^2 / (\sum_i c_i^2 / (bm))$, the same as in the one-way layout in A ([Proposition 15.2.1](#)). What differs between the two F tests of this contrast, the one here and the one-way test that ignores B ?

16.1.6.2 B. Practice

Exercise B1.

In the additive model, suppose $\alpha_1 = \delta$ and $\alpha_2 = \dots = \alpha_a = 0$. Find γ_A . With $a = 4$, $b = 5$ and $\delta = \sigma$, compare the power of the 5% test for $m = 1$ and $m = 2$ using [Theorem 11.5.1](#), and explain why doubling m more than doubles the error degrees of freedom.

Solution. Here $\bar{\alpha} = \delta/a$, and $\sum_i (\alpha_i - \bar{\alpha})^2 = \delta^2(1 - 1/a)^2 + (a - 1)\delta^2/a^2 = \delta^2(a - 1)/a$. So $\gamma_A = bm\delta^2(a - 1)/(a\sigma^2)$. With $a = 4$, $b = 5$, $\delta = \sigma$ this is $3.75m$. The error degrees of freedom are $n - a - b + 1 = 20m - 8$, which is 12 for $m = 1$ and 32 for $m = 2$. The power is $\Pr\{F(3, 20m - 8, 3.75m) > F_{0.05}(3, 20m - 8)\}$, about 0.25 for $m = 1$ and 0.56 for $m = 2$ (computed by the script `exercise_checks.py`). The noncentrality doubles, and the error degrees of freedom more than double because the $a + b - 1$ degrees of freedom spent on the mean structure are a fixed cost. Both effects raise the power.

Exercise B2.

Show that Tukey's statistic F_N is unchanged if $\mathbf{y}$ is replaced by $\lambda\mathbf{y} + \mathbf{t}$, where $\lambda \neq 0$ and $\mathbf{t}$ is any additive table $t_{ij} = u_i + v_j$. Conclude that F_N has the same distribution under every additive model with the same σ^2 .

Exercise B3.

Suppose y^p is additive for some $p \neq 0$, so that $\mu_{ij} = f(\nu + a_i + b_j)$ with $f(t) = t^{1/p}$. Verify that $\theta = f''(\nu)/f'(\nu)^2 = (1 - p)/f(\nu)$, and derive [Eq. 16.1.5](#). What does $\hat{\theta}\bar{y}_{..} \approx 1$ suggest, and why is $p = 0$ the right reading?

Solution. $f'(t) = p^{-1}t^{1/p-1}$ and $f''(t) = p^{-1}(p^{-1} - 1)t^{1/p-2}$. So

$$\frac{f''(\nu)}{f'(\nu)^2} = \frac{p^{-1}(p^{-1} - 1)\nu^{1/p-2}}{p^{-2}\nu^{2/p-2}} = p\left(\frac{1}{p} - 1\right)\nu^{-1/p} = \frac{1 - p}{f(\nu)}.$$

Solving $\theta = (1 - p)/f(\nu)$ for p , and estimating θ by $\hat{\theta}$ and $f(\nu)$, the typical size of a cell mean, by $\bar{y}_{..}$, gives $\hat{p} = 1 - \hat{\theta}\bar{y}_{..}$. A value $\hat{\theta}\bar{y}_{..} \approx 1$ gives $\hat{p} \approx 0$. For $f(t) = e^t$, which makes $\log y$ additive, $f''/f'^2 = e^{-\nu} = 1/f(\nu)$, which is the limit $p \rightarrow 0$ of the power formula. This matches the convention that $(y^p - 1)/p \rightarrow \log y$.

Exercise B4.

With $m = 1$ and normal errors, suppose the cell means are $\mu_{ij} = \mu + \alpha_i + \beta_j + \theta\alpha_i\beta_j$ with $\sum_i \alpha_i = \sum_j \beta_j = 0$, exactly the shape of [Eq. 16.1.3](#). Show that $r_i = \alpha_i + \bar{\varepsilon}_{i\cdot} - \bar{\varepsilon}_{\cdot\cdot}$, and that, given $\mathbf{P}_A \mathbf{Y}$ and $\mathbf{P}_B \mathbf{Y}$,

$$E(\hat{\theta} \mid \mathbf{P}_A \mathbf{Y}, \mathbf{P}_B \mathbf{Y}) = \theta \frac{\sum_i r_i \alpha_i}{\sum_i r_i^2} \cdot \frac{\sum_j c_j \beta_j}{\sum_j c_j^2}.$$

Explain why $\hat{\theta}$ is nearly unbiased when the row and column effects are large compared with σ , and why it tends to be pulled towards zero when they are not. What does this imply for the power rule [Eq. 16.1.5](#)?

Solution. The interaction terms $\theta\alpha_i\beta_j$ have zero row and column means, so $\bar{\mu}_{i\cdot} - \bar{\mu}_{\cdot\cdot} = \alpha_i$ and $r_i = \alpha_i + \bar{\varepsilon}_{i\cdot} - \bar{\varepsilon}_{\cdot\cdot}$; likewise for c_j . With $\mathbf{U} = \mathbf{r} \otimes \mathbf{c}$ as in the proof of [Theorem 16.1.2\(f\)](#), $\hat{\theta} = \mathbf{U}^\top \mathbf{R} \mathbf{Y} / \mathbf{U}^\top \mathbf{U}$. By [Theorem 4.4.4](#), $\mathbf{R} \mathbf{Y}$ is independent of $(\mathbf{P}_A \mathbf{Y}, \mathbf{P}_B \mathbf{Y})$, whatever the mean, so given those two vectors $\mathbf{U}$ is fixed and $E(\mathbf{R} \mathbf{Y}) = \mathbf{R} \boldsymbol{\mu}$, the table $(\theta\alpha_i\beta_j)$. Hence the conditional mean is $\theta \sum_{i,j} r_i c_j \alpha_i \beta_j / (\sum_i r_i^2 \sum_j c_j^2)$, which factorizes as displayed. If the effects dominate the noise, $\mathbf{r} \approx \boldsymbol{\alpha}$ and both factors are near one. If not, $\sum_i r_i^2$ carries the extra noise $\sigma^2(a-1)/b$ on average while $\sum_i r_i \alpha_i$ does not: the attenuation of a regression on a noisy regressor. The power rule then suggests a $\hat{p}$ too close to one, too weak a transformation.

16.1.6.3 C. Going deeper**Exercise C1.**

Extend Tukey's test to $m \geq 2$ observations per cell. Replace y_{ij} by the cell mean $\bar{y}_{ij}$ in [Eq. 16.1.4](#) and multiply the result by m , and compare it with the pure-error mean square. Show that under the additive model the resulting statistic has the $F(1, ab(m-1))$ distribution. Show also that $SS_{AB} - SS_N$ gives a test of the remaining nonadditivity.

16.2 Interaction and its interpretation

The additive model says that the difference between two levels of A is the same at every level of B . *Interaction* is the failure of that statement. This section defines interaction through the cell means, so that its meaning does not depend on a parameterization. It then shows how to see interaction in a plot and when a change of scale removes it, and explains why a “main effect” in the presence of interaction is a weighted average whose weights someone has to choose.

16.2.1 Cell means and the definition of interaction

Write μ_{ij} for the mean response in cell (i, j) . The **cell-means model** $E(y_{ijk}) = \mu_{ij}$ places no restriction on the ab means. By [Proposition 8.6.2\(a\)](#) it has the same mean space as the overparameterized model $\mu + \alpha_i + \beta_j + \gamma_{ij}$, and every cell mean of an occupied cell is estimable. Averages with equal weights split the table of means into four parts:

$$\mu_{ij} = \bar{\mu}_{..} + (\bar{\mu}_{i.} - \bar{\mu}_{..}) + (\bar{\mu}_{.j} - \bar{\mu}_{..}) + (\mu_{ij} - \bar{\mu}_{i.} - \bar{\mu}_{.j} + \bar{\mu}_{..}). \quad (16.2.1)$$

This is an algebraic identity. The four parts are the images of the vector of cell means under $\bar{\mathbf{J}}_a \otimes \bar{\mathbf{J}}_b$, $\mathbf{C}_a \otimes \bar{\mathbf{J}}_b$, $\bar{\mathbf{J}}_a \otimes \mathbf{C}_b$ and $\mathbf{C}_a \otimes \mathbf{C}_b$, which are mutually orthogonal projections (Lemma 16.1.1).

Definition 16.2.1 — Interaction

Let μ_{ij} be the cell means of a two-way layout.

- (a) An **interaction contrast** is a linear function $\sum_{i,j} d_{ij} \mu_{ij}$ whose coefficient table (d_{ij}) is not zero and has every row sum and every column sum equal to zero. The simplest is the **tetrad difference** $\mu_{ij} - \mu_{il} - \mu_{kj} + \mu_{kl}$ of rows $i \neq k$ and columns $j \neq l$. It is the difference between columns j and l in row i , minus the same difference in row k .
- (b) The factors **interact** if some interaction contrast is nonzero. Otherwise the layout is **additive**.
- (c) In the *effects form* $\mu_{ij} = \mu + \alpha_i + \beta_j + \gamma_{ij}$ of Eq. 16.2.1, the **main effects** are $\alpha_i = \bar{\mu}_{i.} - \bar{\mu}_{..}$ and $\beta_j = \bar{\mu}_{.j} - \bar{\mu}_{..}$, and the **interaction effects** are $\gamma_{ij} = \mu_{ij} - \bar{\mu}_{i.} - \bar{\mu}_{.j} + \bar{\mu}_{..}$. They satisfy $\sum_i \alpha_i = \sum_j \beta_j = 0$, and every row and column of (γ_{ij}) sums to zero.

The cell means are what the data estimate. The effects in (c) are one coordinate system for them, fixed by the choice of equal weights. The next result shows that whether interaction is present does not depend on that choice.

Proposition 16.2.1 — Equivalent forms of additivity

For a table of cell means (μ_{ij}) , the following are equivalent:

- (i) every tetrad difference is zero;
- (ii) for all rows i, k , the difference $\mu_{ij} - \mu_{kj}$ does not depend on j ;
- (iii) $\mu_{ij} = u_i + v_j$ for some numbers $u_1, \dots, u_a$ and $v_1, \dots, v_b$;
- (iv) every interaction contrast is zero;
- (v) for some pair of weight vectors $\mathbf{s} \in \mathbb{R}^a$ and $\mathbf{w} \in \mathbb{R}^b$ with $\sum_k s_k = \sum_l w_l = 1$,

$$\mu_{ij} - \sum_l w_l \mu_{il} - \sum_k s_k \mu_{kj} + \sum_{k,l} s_k w_l \mu_{kl} = 0 \quad \text{for all } i, j;$$

- (vi) the condition in (v) holds for every such pair of weight vectors, in particular $\gamma_{ij} = 0$ for all i, j .

Moreover, the coefficient tables of interaction contrasts, together with the zero table, form a subspace of dimension $(a-1)(b-1)$, and $\sum_{i,j} d_{ij} \mu_{ij} = \sum_{i,j} d_{ij} \gamma_{ij}$ for each of them.

Proof. (i) $\Rightarrow$ (ii): a zero tetrad difference says $\mu_{ij} - \mu_{kj} = \mu_{il} - \mu_{kl}$. (ii) $\Rightarrow$ (iii): put $u_i = \mu_{i1} - \mu_{11}$ and $v_j = \mu_{1j}$. By (ii), $\mu_{ij} - \mu_{1j} = \mu_{i1} - \mu_{11}$, that is, $\mu_{ij} = u_i + v_j$. (iii) $\Rightarrow$ (iv): $\sum_{i,j} d_{ij}(u_i + v_j) = \sum_i u_i \sum_j d_{ij} + \sum_j v_j \sum_i d_{ij} = 0$. (iv) $\Rightarrow$ (i): a tetrad difference is an interaction contrast. (iii) $\Rightarrow$ (vi): substituting $u_i + v_j$, the four terms are

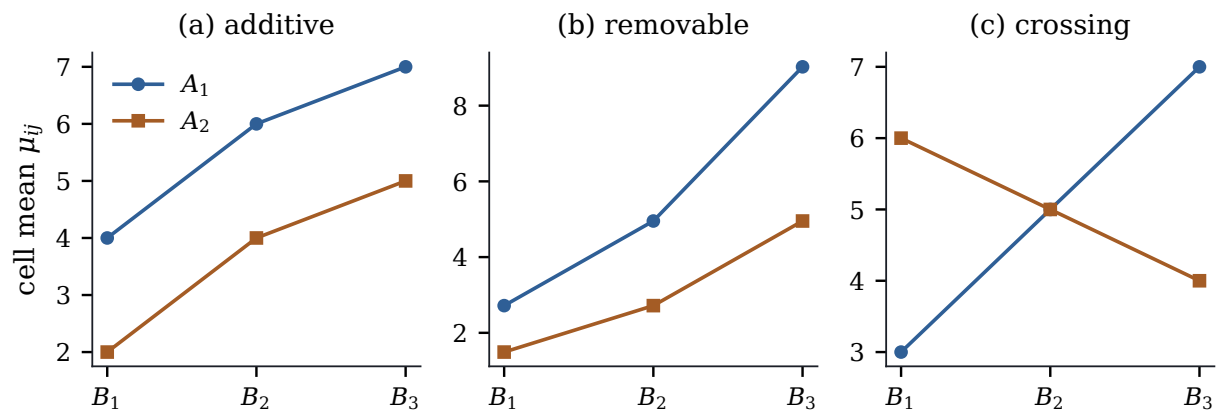


Figure 16.2.1: Three tables of cell means for a 2×3 layout. (a) Additive: parallel profiles. (b) Interaction that the logarithm removes: the means are exp of an additive table. (c) Crossing interaction: the order of the rows reverses between B_1 and B_3 , and no increasing transformation can make the profiles parallel.

$u_i + v_j$, $u_i + \sum_l w_l v_l$, $\sum_k s_k u_k + v_j$ and $\sum_k s_k u_k + \sum_l w_l v_l$, and the signed sum is zero. (vi) $\Rightarrow$ (v) is trivial. (v) $\Rightarrow$ (iii): (v) writes μ_{ij} as $\sum_l w_l \mu_{il}$, which depends only on i , plus $\sum_k s_k \mu_{kj} - \sum_{k,l} s_k w_l \mu_{kl}$, which depends only on j .

For the last statement, read tables as vectors in $\mathbb{R}^{ab}$ with j running fastest. A table has zero row and column sums iff it is fixed by $\mathbf{C}_a \otimes \mathbf{C}_b$, as in the proof of Theorem 16.1.2(e). So the coefficient tables form $\mathcal{C}(\mathbf{C}_a \otimes \mathbf{C}_b)$, of dimension $(a-1)(b-1)$ by Lemma 16.1.1(b). Such a table is orthogonal to the first three parts of Eq. 16.2.1, which leaves $\sum d_{ij} \gamma_{ij}$.

■

So interaction is a property of the table of cell means, with $(a-1)(b-1)$ dimensions. In the overparameterized model $\mu + \alpha_i + \beta_j + \gamma_{ij}$ without side conditions, $\alpha_i - \alpha_k$ is not estimable (Proposition 8.6.2). The function $\alpha_i - \alpha_k + \bar{\gamma}_i - \bar{\gamma}_k$ is estimable, because it equals $\bar{\mu}_{i\cdot} - \bar{\mu}_{k\cdot}$. Software that prints a “main effect” in such a model has chosen side conditions (Theorem 8.4.2) and hence a set of weights, and the printed number depends on that choice. The last part of this section explains why.

16.2.2 Interaction plots

An **interaction plot** draws the estimated cell means $\bar{y}_{ij}$ against the levels of one factor, with one line (a *profile*) for each level of the other. By Proposition 16.2.1(ii), the layout is additive iff the profiles of the true means are parallel, that is, vertical translates of each other. Figure 16.2.1 shows three patterns of true means. In (a) the profiles are parallel. In (b) they fan out, but the row that is higher in one column is higher in every column. In (c) they cross.

Estimated profiles are never exactly parallel. Whether the departures exceed the noise is a question for the F test of Section 16.3. The plot shows *where* the interaction is, which is usually more useful.

Example. (Flour and proofing temperature)

A bakery trial measured the volume (ml) of loaves made from three flours (white, whole-

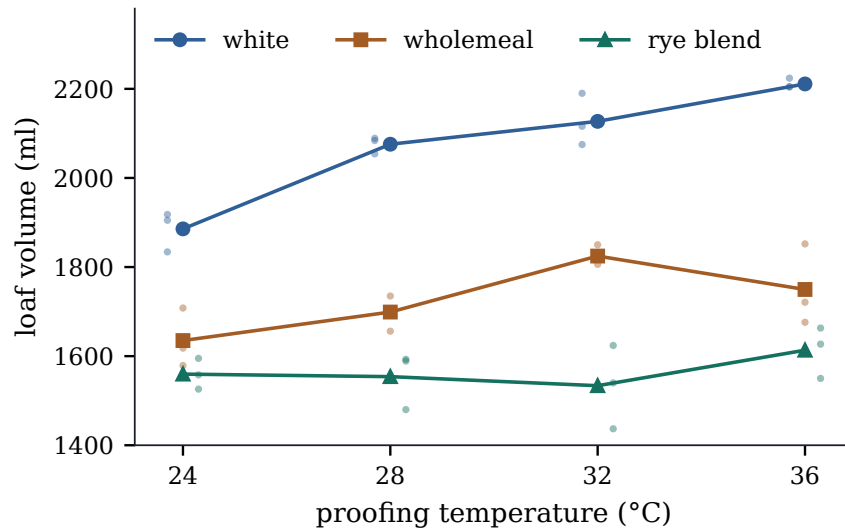


Figure 16.2.2: Interaction plot for the bread trial: cell means of loaf volume against proofing temperature, one profile per flour, with the individual loaves as faint points.

meal and a rye blend), with the dough proofed at four temperatures (24, 28, 32 and 36 °C). Three loaves were baked for each of the 12 combinations, 36 in all, in random order. The data are synthetic, generated by the script `bread_factorial.py`. The cell means are

	24 °C	28 °C	32 °C	36 °C	mean
white	1885.7	2075.7	2127.0	2211.0	2074.8
wholemeal	1635.0	1699.0	1824.7	1749.7	1727.1
rye blend	1559.7	1554.0	1533.7	1613.3	1565.2
mean	1693.4	1776.2	1828.4	1858.0	1789.0

Figure 16.2.2 is the interaction plot. White flour gains volume steadily with temperature. Wholemeal gains up to 32 °C and then loses. The rye blend hardly responds. The profiles are far from parallel. The within-cell standard deviation, estimated in Section 16.3, is $s = 56.4$ ml, so a cell mean of three loaves has standard error $56.4/\sqrt{3} \approx 32.6$ ml, much smaller than the departures from parallelism. Yet the flours never change order: at every temperature white beats wholemeal, which beats the rye blend. The interaction changes the *size* of the flour differences but not their *direction*.

16.2.3 Removable and nonremovable interaction

Panel (b) of Figure 16.2.1 shows the pattern of the solver data of Example (Solvers on benchmark problems): the effect of A is a constant *ratio*, and on the logarithmic scale the table is additive. Such interaction is **removable**: some strictly increasing g makes the table $g(\mu_{ij})$ additive. When can no transformation remove an interaction?

Proposition 16.2.2 — An order condition for removable interaction

If g is strictly increasing and the table $g(\mu_{ij})$ is additive, then the rows are ordered in

the same way in every column, and the columns in the same way in every row. That is, for all i, k the sign of $\mu_{ij} - \mu_{kj}$ does not depend on j , and for all j, l the sign of $\mu_{ij} - \mu_{il}$ does not depend on i . In particular, if $\mu_{ij} > \mu_{kj}$ and $\mu_{il} < \mu_{kl}$ for some i, k, j, l , no strictly increasing transformation removes the interaction.

Proof. Write $g(\mu_{ij}) = u_i + v_j$ (Proposition 16.2.1(iii)). Since g is strictly increasing, $\mu_{ij} > \mu_{kj}$ iff $g(\mu_{ij}) > g(\mu_{kj})$ iff $u_i > u_k$, a condition that does not involve j . The same holds with $>$ replaced by $=$ or $<$. The statement for columns is symmetric. ■

This justifies a distinction that is often drawn informally. An interaction is **quantitative** (or *ordinal*) if the orders are consistent in the sense of the proposition, and **qualitative** (or *crossing, disordinal*) if the order of some pair of levels reverses. Qualitative interaction is a fact about the phenomenon on every monotone scale. Quantitative interaction may be a fact about the scale of measurement. Whether an interaction is qualitative can depend on which factor one looks along. In the bread trial the flours keep their order at every temperature, but the temperatures do not keep their order for every flour. White flour gives more volume at 36 °C than at 32 °C, and wholemeal gives less. If the true means follow the same pattern as the estimates, the proposition says that no rescaling of volume makes the table additive. The condition is necessary but not sufficient. With three or more levels of each factor, a table can have consistent orders and still admit no additive rescaling (Exercise (C1)).

The listing checks the three tables of Figure 16.2.1. It also shows that the logarithm *creates* interaction in the additive table (a). Additivity is a property of a table on a given scale, and so is its absence.

```
import itertools

import numpy as np

additive = np.array([[4.0, 6.0, 7.0], [2.0, 4.0, 5.0]])           # rows: levels of
A
removable = np.exp(np.array([[1.0, 1.6, 2.2], [0.4, 1.0, 1.6]])) # exp of an
additive table
crossing = np.array([[3.0, 5.0, 7.0], [6.0, 5.0, 4.0]])

def interaction(mu):
    """mu_ij - mu_i. - mu_.j + mu_. (equal weights)."""
    return mu - mu.mean(1, keepdims=True) - mu.mean(0, keepdims=True) + mu.mean()

def same_orders(mu):
    """Necessary condition for removability: every column orders the rows alike, and
    every row the columns."""
    a, b = mu.shape
    rows_ok = all(len(set(np.sign(mu[i] - mu[k]))) == 1 for i, k in itertools.
combinations(range(a), 2))
    cols_ok = all(len(set(np.sign(mu[:, j] - mu[:, l]))) == 1 for j, l in itertools.
combinations(range(b), 2))
    return rows_ok and cols_ok

for name, mu in [("additive", additive), ("removable", removable), ("crossing",
```

```
crossing)]:
print(f"{name:9s} max |interaction| raw {np.abs(interaction(mu)).max():.3f}"
      f"   log {np.abs(interaction(np.log(mu))).max():.3f}   same orders: {
same_orders(mu)}")
```

Two cautions apply to transforming data rather than means. Since $E g(Y) \neq g(E Y)$ for nonlinear g , additivity of $g(y)$ and of $g(\mu_{ij})$ agree only approximately, for small errors. And a transformation also changes the shape and variance of the errors. In [Example \(Solvers on benchmark problems\)](#) the logarithm happened to fix both, because the noise was multiplicative. The choice of scale is the subject of Chapter 22.

16.2.4 Main effects in the presence of interaction

Without interaction, “the effect of A ” is the common difference $\mu_{ij} - \mu_{kj}$. With interaction, the difference between rows i and k is a list of **simple effects** $\delta_j = \mu_{ij} - \mu_{kj}$, and a single summary must average them with weights. For a weight vector $\mathbf{w}$ with $w_j \geq 0$ and $\sum_j w_j = 1$, define the **w-weighted main effect** difference of rows i and k as $\sum_j w_j \delta_j = \sum_j w_j \mu_{ij} - \sum_j w_j \mu_{kj}$.

Proposition 16.2.3 — Main effects depend on the weights

Fix rows $i \neq k$ and let $\delta_j = \mu_{ij} - \mu_{kj}$.

- (a) If the δ_j are all equal, every weighted main effect difference equals their common value.
- (b) If they are not all equal, the values of $\sum_j w_j \delta_j$ over weight vectors with every $w_j > 0$ fill the open interval $(\min_j \delta_j, \max_j \delta_j)$. In particular, if the simple effects change sign, some weighting with all weights positive makes the main effect of A for these two rows exactly zero, and other weightings give it either sign.
- (c) If all $\delta_j > 0$, every weighted main effect difference is positive.

Proof.

- (a) and (c) are immediate, and a weighted average with positive weights of numbers that are not all equal lies strictly between their minimum and maximum. Conversely, let $\min_j \delta_j < t < \max_j \delta_j$, attained at columns j_0 and j_1 , and let $\bar{\delta} = b^{-1} \sum_j \delta_j$. For $0 < \varepsilon < 1$ put $t_\varepsilon = (t - \varepsilon \bar{\delta}) / (1 - \varepsilon)$. Since $t_\varepsilon \rightarrow t$ as $\varepsilon \rightarrow 0$, there is an ε with $\delta_{j_0} < t_\varepsilon < \delta_{j_1}$. Then $t_\varepsilon = \lambda \delta_{j_0} + (1 - \lambda) \delta_{j_1}$ for some $\lambda \in (0, 1)$. The weights ε/b on every column, plus $(1 - \varepsilon)\lambda$ on j_0 and $(1 - \varepsilon)(1 - \lambda)$ on j_1 , are all positive and give $\varepsilon \bar{\delta} + (1 - \varepsilon) t_\varepsilon = t$.

■

In table (c) of [Figure 16.2.1](#) the simple effects of A_1 against A_2 are $-3, 0$ and 3 . With equal weights the main effect of A is exactly zero, although A matters in two of the three columns. With weights $(\frac{1}{2}, \frac{1}{2}, 0)$ it is -1.5 , and with $(0, \frac{1}{2}, \frac{1}{2})$ it is $+1.5$.

The balanced analysis of [Section 16.3](#) uses equal weights: its sum of squares for A tests $\bar{\mu}_{1.} = \dots = \bar{\mu}_{a.}$. In unbalanced data the various sums of squares use different weights, some determined by the cell counts ([Theorem 9.3.2](#)), and [Chapter 17](#) sorts them out ([Theorem 17.2.2](#)). Equal

weights are a convention, natural when the levels of B were chosen as a representative spread.

For the bread trial the simple effects of white against wholemeal flour are 250.7, 376.7, 302.3 and 461.3 ml at the four temperatures. The equal-weight main effect is their average, 347.7 ml. A bakery that proofs mostly warm might weight the temperatures in the proportions 1 : 1 : 2 : 4, and then the relevant difference is 384.7 ml. Both figures answer different questions correctly. Because all four simple effects are positive, part (c) guarantees that white flour gives larger loaves than wholemeal under every weighting, a statement that needs no choice of weights.

Idea. Report main effects when the interaction is small or quantitative, and say what they average over. When the interaction is qualitative, report simple effects. A main effect that averages a positive and a negative effect answers a question nobody asked.

16.2.5 Exercises

16.2.5.1 A. Check your understanding

Exercise A1.

For the crossing table $\mu = \begin{pmatrix} 3 & 5 & 7 \\ 6 & 5 & 4 \end{pmatrix}$, compute $\bar{\mu}_{..}$, α_i , β_j and γ_{ij} of Definition 16.2.1(c). Check that the tetrad difference of columns 1 and 3 equals $\gamma_{11} - \gamma_{13} - \gamma_{21} + \gamma_{23}$.

Solution. $\bar{\mu}_{..} = 5$, row means 5, 5, so $\alpha_1 = \alpha_2 = 0$. Column means 4.5, 5, 5.5, so $\beta = (-0.5, 0, 0.5)$. Then $\gamma_{1j} = \mu_{1j} - 5 - \beta_j = (-1.5, 0, 1.5)$ and $\gamma_{2j} = (1.5, 0, -1.5)$. The tetrad difference is $3 - 7 - 6 + 4 = -6$, and $\gamma_{11} - \gamma_{13} - \gamma_{21} + \gamma_{23} = -1.5 - 1.5 - 1.5 - 1.5 = -6$.

Exercise A2.

From the table of cell means in Example (Flour and proofing temperature), compute the simple effects of wholemeal against the rye blend at the four temperatures. Is their interaction quantitative or qualitative? Which pair of temperatures, if any, reverses its order for some flour?

16.2.5.2 B. Practice

Exercise B1.

Show that the $(a-1)(b-1)$ tetrad differences $\mu_{ij} - \mu_{ib} - \mu_{aj} + \mu_{ab}$, $i < a$, $j < b$, have linearly independent coefficient tables. Conclude that they form a basis of the interaction contrasts, so that additivity is equivalent to $(a-1)(b-1)$ linear restrictions on the cell means.

Solution. The coefficient table of the (i, j) tetrad has entry 1 at (i, j) and its other nonzero entries in row a or column b . In a linear combination $\sum_{i < a, j < b} c_{ij}(\text{table}_{ij})$, the entry at position (i, j) with $i < a$, $j < b$ is c_{ij} , because no other table in the list has a nonzero entry there. So the combination vanishes only if all $c_{ij} = 0$. There are $(a-1)(b-1)$ independent tables in a space of that dimension (Proposition 16.2.1), so

they form a basis. Additivity means all interaction contrasts vanish, which is equivalent to the basis contrasts vanishing.

Exercise B2.

Let $\mu_{ij} = u_i + v_j > 0$ be an additive table. Show that $\log \mu_{ij}$ is additive only if all u_i are equal or all v_j are equal. So on a positive scale a transformation can create interaction as well as remove it.

Solution. By Proposition 16.2.1(i), $\log \mu_{ij}$ is additive iff $\mu_{ij}\mu_{kl} = \mu_{il}\mu_{kj}$ for all i, k, j, l . Expanding $(u_i + v_j)(u_k + v_l) - (u_i + v_l)(u_k + v_j) = u_i v_l + u_k v_j - u_i v_j - u_k v_l = -(u_i - u_k)(v_j - v_l)$. This vanishes for all index choices iff $u_i = u_k$ for all i, k or $v_j = v_l$ for all j, l , since if some $u_i \neq u_k$ and some $v_j \neq v_l$, that choice of indices gives a nonzero product.

Exercise B3.

Show that the converse of Proposition 16.2.2 holds for 2×2 tables. If the rows are ordered alike in both columns and the columns alike in both rows, then some strictly increasing function g makes $g(\mu_{ij})$ additive.

16.2.5.3 C. Going deeper

Exercise C1.

Consider the 3×3 table

$$\mu = \begin{pmatrix} 1 & 3 & 6 \\ 4 & 7 & 9 \\ 5 & 10 & 11 \end{pmatrix}.$$

Check that every row and every column is increasing, so that the order condition of Proposition 16.2.2 holds. Show that nevertheless no strictly increasing g makes $g(\mu_{ij})$ additive. (*Hint:* compare μ_{21} with μ_{12} , μ_{32} with μ_{23} , and μ_{31} with μ_{13} .) The condition violated here is the *double cancellation* axiom of conjoint measurement.

Solution. The rows $(1, 3, 6)$, $(4, 7, 9)$, $(5, 10, 11)$ and the columns $(1, 4, 5)$, $(3, 7, 10)$, $(6, 9, 11)$ are increasing. Suppose $g(\mu_{ij}) = u_i + v_j$ with g strictly increasing. From $\mu_{21} = 4 > 3 = \mu_{12}$ we get $u_2 + v_1 > u_1 + v_2$. From $\mu_{32} = 10 > 9 = \mu_{23}$ we get $u_3 + v_2 > u_2 + v_3$. Adding, $u_3 + v_1 > u_1 + v_3$, so $\mu_{31} > \mu_{13}$. But $\mu_{31} = 5 < 6 = \mu_{13}$, a contradiction.

Exercise C2.

For a weight vector $\mathbf{w}$, let $H_A^{\mathbf{w}}$ be the hypothesis that $\sum_j w_j \mu_{ij}$ is the same for all i . Show that $H_A^{\mathbf{w}}$ and $H_A^{\mathbf{w}'}$ define the same set of cell-mean tables only if $\mathbf{w} = \mathbf{w}'$. Show also that on the set of additive tables all of them coincide with $\alpha_1 = \cdots = \alpha_a$.

16.3 Balanced two-way layouts via orthogonal projections

With $m \geq 2$ observations in every cell, the replicates supply an estimate of σ^2 that is valid whatever the cell means are, and interaction can be tested without any assumption about its form. This section gives the complete analysis of the balanced two-way layout: the orthogonal decomposition, exact tests, contrasts, and simultaneous inference. [Exercise \(C1\)](#) asked for the five projections, and [Theorem 16.1.2\(d\)](#) has already met four of them.

16.3.1 The decomposition

The model is the cell-means model

$$E(y_{ijk}) = \mu_{ij}, \quad \text{Cov}(\mathbf{Y}) = \sigma^2 \mathbf{I}, \quad i \leq a, j \leq b, k \leq m, \quad (16.3.1)$$

written in the effects form of [Definition 16.2.1](#) as $\mu_{ij} = \mu + \alpha_i + \beta_j + \gamma_{ij}$, with equal-weight effects. Its mean space $\mathcal{C}(\mathbf{X})$ is the set of vectors that are constant within cells, of dimension ab ([Proposition 8.6.2](#)). With the ordering and notation of [Section 16.1](#), define

$$\begin{aligned} \mathbf{P}_0 &= \bar{\mathbf{J}}_a \otimes \bar{\mathbf{J}}_b \otimes \bar{\mathbf{J}}_m, & \mathbf{P}_A &= \mathbf{C}_a \otimes \bar{\mathbf{J}}_b \otimes \bar{\mathbf{J}}_m, & \mathbf{P}_B &= \bar{\mathbf{J}}_a \otimes \mathbf{C}_b \otimes \bar{\mathbf{J}}_m, \\ \mathbf{P}_{AB} &= \mathbf{C}_a \otimes \mathbf{C}_b \otimes \bar{\mathbf{J}}_m, & \mathbf{P}_E &= \mathbf{I}_a \otimes \mathbf{I}_b \otimes \mathbf{C}_m. \end{aligned} \quad (16.3.2)$$

Theorem 16.3.1 — The balanced two-way layout

Assume [Eq. 16.3.1](#) with $a, b \geq 2$ and $m \geq 2$, and let $\alpha_i, \beta_j, \gamma_{ij}$ be the equal-weight effects of [Definition 16.2.1](#).

- (a) The five matrices [Eq. 16.3.2](#) are mutually orthogonal projections that sum to $\mathbf{I}$, with ranks 1, $a - 1$, $b - 1$, $(a - 1)(b - 1)$ and $ab(m - 1)$. The projection onto $\mathcal{C}(\mathbf{X})$ is $\mathbf{M} = \mathbf{P}_0 + \mathbf{P}_A + \mathbf{P}_B + \mathbf{P}_{AB}$, and that onto the additive model space is $\mathbf{P}_0 + \mathbf{P}_A + \mathbf{P}_B$. So $\mathcal{C}(\mathbf{P}_{AB})$ is the orthogonal complement of the additive model within the cell-means model.
- (b) The projections act by

$$\begin{aligned} (\mathbf{P}_A \mathbf{y})_{ijk} &= \bar{y}_{i..} - \bar{y}_{...}, \\ (\mathbf{P}_B \mathbf{y})_{ijk} &= \bar{y}_{.j.} - \bar{y}_{i..} - \bar{y}_{.j.} + \bar{y}_{...}, \\ (\mathbf{P}_E \mathbf{y})_{ijk} &= y_{ijk} - \bar{y}_{ij.}, \end{aligned}$$

and symmetrically for $\mathbf{P}_B$. The sums of squares are

$$\begin{aligned} \text{SS}_A &= bm \sum_i (\bar{y}_{i..} - \bar{y}_{...})^2, \\ \text{SS}_B &= am \sum_j (\bar{y}_{.j.} - \bar{y}_{...})^2, \\ \text{SS}_{AB} &= m \sum_{i,j} (\bar{y}_{ij.} - \bar{y}_{i..} - \bar{y}_{.j.} + \bar{y}_{...})^2, \\ \text{SSE} &= \sum_{i,j,k} (y_{ijk} - \bar{y}_{ij.})^2. \end{aligned}$$

(c) The mean squares have expectations

$$\begin{aligned} \text{E MS}_A &= \sigma^2 + \frac{bm}{a-1} \sum_i \alpha_i^2, \\ \text{E MS}_B &= \sigma^2 + \frac{am}{b-1} \sum_j \beta_j^2, \\ \text{E MS}_{AB} &= \sigma^2 + \frac{m}{(a-1)(b-1)} \sum_{i,j} \gamma_{ij}^2, \\ \text{E MSE} &= \sigma^2. \end{aligned}$$

(d) Under normal errors the four sums of squares are independent, $\text{SSE}/\sigma^2 \sim \chi^2(ab(m-1))$, and

$$\begin{aligned} F_A &= \frac{\text{MS}_A}{\text{MSE}} \sim F\left(a-1, ab(m-1), \frac{bm}{\sigma^2} \sum_i \alpha_i^2\right), \\ F_{AB} &= \frac{\text{MS}_{AB}}{\text{MSE}} \sim F\left((a-1)(b-1), ab(m-1), \frac{m}{\sigma^2} \sum_{i,j} \gamma_{ij}^2\right), \end{aligned}$$

and likewise for F_B . So F_{AB} is an exact test of additivity, F_A of $\bar{\mu}_{1.} = \cdots = \bar{\mu}_{a.}$, and F_B of $\bar{\mu}_{.1} = \cdots = \bar{\mu}_{.b}$.

(e) SS_A is the sum of squares of the hypothesis $\alpha_1 = \cdots = \alpha_a = 0$ in the cell-means model. It is also the sequential sum of squares of A whether A is entered before or after B , and it equals the Type II and Type III sums of squares of A . The same holds for B , and SS_{AB} is the sequential, Type II and Type III sum of squares of the interaction.

Proof.

- (a) Orthogonality, ranks and the sum $\mathbf{I}$ follow from [Lemma 16.1.1](#), as in the proof of [Theorem 16.1.2\(d\)](#). The sum of the first four is $(\bar{\mathbf{J}}_a + \mathbf{C}_a) \otimes (\bar{\mathbf{J}}_b + \mathbf{C}_b) \otimes \bar{\mathbf{J}}_m = \mathbf{I}_a \otimes \mathbf{I}_b \otimes \bar{\mathbf{J}}_m$. This matrix replaces each observation by its cell mean, so it is the projection onto the vectors that are constant within cells, which is $\mathcal{C}(\mathbf{X})$. The additive projection is [Theorem 16.1.2\(a\)](#).
- (b) Expand each Kronecker product with $\mathbf{C} = \mathbf{I} - \bar{\mathbf{J}}$ and read off the means from the table in [Section 16.1](#). For example,

$$\begin{aligned} \mathbf{P}_{AB} &= \mathbf{I}_a \otimes \mathbf{I}_b \otimes \bar{\mathbf{J}}_m - \mathbf{I}_a \otimes \bar{\mathbf{J}}_b \otimes \bar{\mathbf{J}}_m \\ &\quad - \bar{\mathbf{J}}_a \otimes \mathbf{I}_b \otimes \bar{\mathbf{J}}_m + \bar{\mathbf{J}}_a \otimes \bar{\mathbf{J}}_b \otimes \bar{\mathbf{J}}_m. \end{aligned}$$

Each sum of squares is a sum over the n positions of a quantity that is constant over m , bm or am of them.

- (c) Apply (b) to the mean vector $\boldsymbol{\mu}$, whose entries are μ_{ij} . Then $\mathbf{P}_A \boldsymbol{\mu}$ has entries $\bar{\mu}_{i.} - \bar{\mu}_{..} = \alpha_i$, so $\|\mathbf{P}_A \boldsymbol{\mu}\|^2 = bm \sum_i \alpha_i^2$. Likewise $\|\mathbf{P}_{AB} \boldsymbol{\mu}\|^2 = m \sum_{i,j} \gamma_{ij}^2$, and $\mathbf{P}_E \boldsymbol{\mu} = \mathbf{0}$. The four subspaces $\mathcal{C}(\mathbf{P}_0), \dots, \mathcal{C}(\mathbf{P}_{AB})$ form an analysis of variance decomposition of $\mathcal{C}(\mathbf{X})$, so [Theorem 9.6.1](#) gives the expectations.
- (d) This is [Corollary 9.6.2](#) with the noncentralities of (c). The noncentrality of F_A vanishes iff every $\alpha_i = 0$, that is, iff the row means $\bar{\mu}_{i.}$ are equal. The noncentrality of F_{AB} vanishes iff every $\gamma_{ij} = 0$, which is additivity by [Proposition 16.2.1](#)(vi).
- (e) Each α_i is a linear function of the cell means, so the hypothesis is testable. The mean vectors it allows are the $\boldsymbol{\mu} \in \mathcal{C}(\mathbf{X})$ with $\mathbf{P}_A \boldsymbol{\mu} = \mathbf{0}$. Since $\boldsymbol{\mu} = \mathbf{P}_0 \boldsymbol{\mu} + \mathbf{P}_A \boldsymbol{\mu} + \mathbf{P}_B \boldsymbol{\mu} + \mathbf{P}_{AB} \boldsymbol{\mu}$, these form $\mathcal{S}_0 = \mathcal{C}(\mathbf{P}_0) \oplus \mathcal{C}(\mathbf{P}_B) \oplus \mathcal{C}(\mathbf{P}_{AB})$. By [Theorem 11.3.2](#)(b) the sum of squares of the hypothesis is $\|(\mathbf{M} - \mathbf{P}_{\mathcal{S}_0}) \mathbf{y}\|^2 = \|\mathbf{P}_A \mathbf{y}\|^2$. The rest is Chapter 9: equal counts are proportional ([Theorem 9.4.3](#)), so [Theorem 9.4.1](#) applies, and [Exercise \(B1\)](#) shows that the sequential sums of squares of A in either order and its Type II and Type III sums of squares all equal $\|\mathbf{P}_A \mathbf{y}\|^2$. The interaction enters last in every order, so all its sums of squares are $\|(\mathbf{M} - \mathbf{P}_0 - \mathbf{P}_A - \mathbf{P}_B) \mathbf{y}\|^2 = \|\mathbf{P}_{AB} \mathbf{y}\|^2$.

■

The theorem is summarized in the analysis of variance table:

Source	df	Sum of squares	E(MS)
A	$a - 1$	$\ \mathbf{P}_A \mathbf{y}\ ^2$	$\sigma^2 + bm \sum_i \alpha_i^2 / (a - 1)$
B	$b - 1$	$\ \mathbf{P}_B \mathbf{y}\ ^2$	$\sigma^2 + am \sum_j \beta_j^2 / (b - 1)$
AB	$(a - 1)(b - 1)$	$\ \mathbf{P}_{AB} \mathbf{y}\ ^2$	$\sigma^2 + m \sum_{i,j} \gamma_{ij}^2 / ((a - 1)(b - 1))$
error	$ab(m - 1)$	$\ \mathbf{P}_E \mathbf{y}\ ^2$	σ^2
corrected total	$abm - 1$	$\ (\mathbf{I} - \mathbf{P}_0) \mathbf{y}\ ^2$	

The test for A is valid with or without interaction, because $\mathcal{C}(\mathbf{P}_{AB}) \perp \mathcal{C}(\mathbf{P}_A)$. What the interaction changes is the *meaning* of H_A : equality of equal-weight row means is one hypothesis among many when the simple effects differ ([Section 16.2](#)). Part (e) is special to balance. With unequal cell counts the sequential, Type II and Type III sums of squares of A test hypotheses with different weights ([Theorem 9.3.2](#)), the subject of [Chapter 17](#).

16.3.2 Contrasts and their sums of squares

The F tests say whether a term matters. Contrasts say how. A **main-effect contrast** in A is $\psi = \sum_i c_i \bar{\mu}_{i.} = \sum_i c_i \alpha_i$ with $\sum_i c_i = 0$. An **interaction contrast** is $\sum_{i,j} d_{ij} \mu_{ij} = \sum_{i,j} d_{ij} \gamma_{ij}$, with (d_{ij}) having zero row and column sums ([Proposition 16.2.1](#)). Their least squares estimators replace means by sample means. Since $\bar{y}_{i.}$ averages bm independent observations and $\bar{y}_{ij.}$ averages m ,

$$\text{Var}\left(\sum_i c_i \bar{y}_{i.}\right) = \frac{\sigma^2}{bm} \sum_i c_i^2, \quad \text{Var}\left(\sum_{i,j} d_{ij} \bar{y}_{ij.}\right) = \frac{\sigma^2}{m} \sum_{i,j} d_{ij}^2.$$

The sums of squares of the corresponding single-degree-of-freedom hypotheses are the squared estimates divided by these variances without the σ^2 ([Theorem 9.4.4](#)), as for a one-way contrast

(Proposition 15.2.1):

$$SS(\mathbf{c}) = \frac{(\sum_i c_i \bar{y}_{i\cdot})^2}{\sum_i c_i^2 / (bm)}, \quad SS(\mathbf{d}) = \frac{(\sum_{i,j} d_{ij} \bar{y}_{ij\cdot})^2}{\sum_{i,j} d_{ij}^2 / m}. \quad (16.3.3)$$

For main effects, a complete set of $a - 1$ mutually orthogonal contrasts ($\sum_i c_i c'_i = 0$) splits SS_A into $a - 1$ independent pieces. This is the one-way result [Theorem 15.4.1](#) applied to the row means, which are one-way means with bm observations each. For interaction, orthogonal contrasts are most easily built as products.

Proposition 16.3.2 — Product contrasts

Let $\mathbf{c}^{(1)}, \dots, \mathbf{c}^{(a-1)}$ be mutually orthogonal contrast vectors in $\mathbb{R}^a$, and $\mathbf{e}^{(1)}, \dots, \mathbf{e}^{(b-1)}$ mutually orthogonal contrast vectors in $\mathbb{R}^b$. Then the $(a - 1)(b - 1)$ tables $\mathbf{d}^{(rs)} = \mathbf{c}^{(r)} \mathbf{e}^{(s)\top}$, with entries $c_i^{(r)} e_j^{(s)}$, are coefficient tables of mutually orthogonal interaction contrasts, and

$$\sum_{r=1}^{a-1} \sum_{s=1}^{b-1} SS(\mathbf{d}^{(rs)}) = SS_{AB}.$$

Under normality the $(a - 1)(b - 1)$ sums of squares are independent of each other and of SSE.

Proof. The row sums of $\mathbf{d}^{(rs)}$ are $c_i^{(r)} \sum_j e_j^{(s)} = 0$ and the column sums vanish likewise. Let $\mathbf{v}_{rs} = \mathbf{c}^{(r)} \otimes \mathbf{e}^{(s)} \otimes \mathbf{1}_m \in \mathbb{R}^n$. Since $\mathbf{C}_a \mathbf{c}^{(r)} = \mathbf{c}^{(r)}$ and $\mathbf{C}_b \mathbf{e}^{(s)} = \mathbf{e}^{(s)}$, the mixed-product rule gives $\mathbf{P}_{AB} \mathbf{v}_{rs} = \mathbf{v}_{rs}$, so $\mathbf{v}_{rs} \in \mathcal{C}(\mathbf{P}_{AB})$. Also $\mathbf{v}_{rs}^\top \mathbf{v}_{r's'} = m (\mathbf{c}^{(r)\top} \mathbf{c}^{(r')})(\mathbf{e}^{(s)\top} \mathbf{e}^{(s')})$, which is zero unless $r = r'$ and $s = s'$. So the $\mathbf{v}_{rs}$ are $(a - 1)(b - 1)$ nonzero orthogonal vectors in a space of that dimension, an orthogonal basis of $\mathcal{C}(\mathbf{P}_{AB})$. Hence $\|\mathbf{P}_{AB} \mathbf{y}\|^2 = \sum_{r,s} (\mathbf{v}_{rs}^\top \mathbf{y})^2 / \|\mathbf{v}_{rs}\|^2$. Finally $\mathbf{v}_{rs}^\top \mathbf{y} = m \sum_{i,j} d_{ij}^{(rs)} \bar{y}_{ij\cdot}$ and $\|\mathbf{v}_{rs}\|^2 = m \sum_{i,j} (d_{ij}^{(rs)})^2$, so each term is $SS(\mathbf{d}^{(rs)})$ of [Eq. 16.3.3](#). Independence follows from [Theorem 4.4.4](#) applied to the projections onto the lines $\text{span}(\mathbf{v}_{rs})$ and $\mathbf{P}_E$. ■

Product contrasts suit factors with structure. If $\mathbf{c}$ compares a control with the treatments and $\mathbf{e}$ is a linear trend in a quantitative factor, then $\mathbf{c}\mathbf{e}^\top$ asks whether the trend differs between them. The cell-means model of a balanced layout with quantitative levels w_i and z_j is the same as the polynomial model with all terms $w^r z^s$, $r < a$, $s < b$ ([Exercise \(B3\)](#)), and the product contrasts of orthogonal polynomials give its sequential sums of squares.

16.3.3 Simultaneous inference

The tests and contrasts above come in families, and the methods of [Chapter 13](#) apply directly.

- **Tukey for the levels of one factor.** The row means $\bar{y}_{i\cdot}$ are independent, each with variance $\sigma^2 / (bm)$. They are functions of $(\mathbf{I} - \mathbf{P}_E) \mathbf{Y}$, hence independent of MSE, with or without interaction ([Exercise \(B2\)](#)). So [Theorem 13.3.2](#) applies with $g = a$, multiplier $q_\alpha(a, ab(m - 1))$ and standard error $s / \sqrt{bm}$.

- **Scheffé for interaction contrasts.** Taking $\mathcal{V} = \mathcal{C}(\mathbf{P}_{AB})$ in [Theorem 13.2.2](#) gives intervals for all interaction contrasts at once, with multiplier $\sqrt{(a-1)(b-1)F_\alpha((a-1)(b-1), ab(m-1))}$. By [Corollary 13.2.3](#), some interaction contrast is Scheffé-significant iff the F_{AB} test rejects. This is the honest way to report a contrast suggested by the interaction plot. The most extreme contrast always accounts for the whole interaction sum of squares ([Exercise \(C1\)](#)); the question is whether it is interpretable.
- **Bonferroni or Holm for planned families** ([Theorem 13.4.1](#), [Theorem 13.4.5](#)), such as the comparison of two levels of A separately at each level of B .

Example. (Analysis of the bread trial)

For the data of [Example \(Flour and proofing temperature\)](#), the decomposition of [Theorem 16.3.1](#) gives

Source	df	Sum of squares	Mean square	F	p
flour	2	1627629	813814	255.74	6.6×10^{-17}
temperature	3	140499	46833	14.72	1.2×10^{-5}
interaction	6	98947	16491	5.18	1.5×10^{-3}
error	24	76373	3182		

so $s = 56.4$ ml, and the interaction is clearly present. The script confirms that statsmodels reports the same sums of squares for Types I, II and III, as part (e) says.

Where is the interaction? Take the flour contrasts $(2, -1, -1)$ (white against the other two) and $(0, 1, -1)$ (wholemeal against rye), and the orthogonal polynomial contrasts $(-3, -1, 1, 3)$, $(1, -1, -1, 1)$, $(-1, 3, -3, 1)$ for the equally spaced temperatures. The six products split SS_{AB} as in [Proposition 16.3.2](#):

	linear	quadratic	cubic
white vs others	52152 ($F = 16.39$)	3134 ($F = 0.98$)	6011 ($F = 1.89$)
wholemeal vs rye	8118 ($F = 2.55$)	18872 ($F = 5.93$)	10660 ($F = 3.35$)

The six add up to 98947. The largest piece, a fraction 0.527 of the interaction, says that white flour responds to temperature more steeply than the others. The second is the wholemeal peak at 32 °C. Does the white-by-linear component survive the most generous correction? Its estimate is 1444.3. The temperature contrast has $\sum_j e_j t_j = -72 - 28 + 32 + 108 = 40$, and the flour contrast is twice “white minus the average of the others”, so the estimate divided by 80 is the difference between white flour’s slope and the average slope of the other two: 18.05 ml per °C. The Scheffé multiplier for interaction contrasts is 3.879, and the simultaneous interval for this slope difference is $[0.75, 35.35]$ ml per °C. It excludes zero even though Scheffé’s correction allows for every contrast the plot might have suggested. The temperature main effect is almost entirely linear, with sequential sums of squares 134098, 6373 and 28 for the linear, quadratic and cubic components.

Comparing the flours. The equal-weight flour means are 2074.8, 1727.1 and 1565.2 ml. Tukey’s multiplier $q_{0.05}(3, 24) = 3.532$ with standard error $s/\sqrt{12}$ gives half-width 57.5 ml. White minus wholemeal is 347.7, with interval $[290.2, 405.3]$. White minus rye blend is 509.7 $[452.2, 567.2]$, and wholemeal minus rye blend is 161.9 $[104.4, 219.4]$. Because of the interaction, these average over the four temperatures the trial happened to use. The simple effects of white against wholemeal, with Bonferroni intervals for the four temperatures (multiplier $t_{24, 0.05/8} = 2.700$, half-width 124.4), are 250.7 $[126.3, 375.0]$, 376.7 $[252.3, 501.0]$, 302.3 $[178.0, 426.7]$ and 461.3 $[337.0, 585.7]$. All four lie above zero, so with simultaneous confidence 95% white flour is better at every temperature, and hence ([Proposition 16.2.3\(c\)](#)) under any weighting of the temperatures.

```
import itertools

import numpy as np
from scipy import stats

flours = ["white", "wholemeal", "rye blend"]
temps = [24, 28, 32, 36]
true_means = np.array([[1880, 2010, 2130, 2190],
                        [1660, 1760, 1810, 1750],
                        [1500, 1540, 1580, 1590]], dtype=float)

a, b, m = 3, 4, 3
data_rng = np.random.default_rng(16_03)
volume = np.round(true_means[:, :, None] + data_rng.normal(scale=55, size=(a, b, m))
                  )

def Jbar(k):
    return np.full((k, k), 1.0 / k)
```

```

def Cen(k):
    return np.eye(k) - Jbar(k)

y = volume.ravel()                                # replicate fastest, then
                                                temperature, then flour
P = {"flour": np.kron(np.kron(Cen(a), Jbar(b)), Jbar(m)),
     "temperature": np.kron(np.kron(Jbar(a), Cen(b)), Jbar(m)),
     "interaction": np.kron(np.kron(Cen(a), Cen(b)), Jbar(m)),
     "error": np.kron(np.kron(np.eye(a), np.eye(b)), Cen(m))}
ss = {k: y @ Pk @ y for k, Pk in P.items()}
df = {k: int(round(np.trace(Pk))) for k, Pk in P.items()}
mse = ss["error"] / df["error"]
for k in ["flour", "temperature", "interaction", "error"]:
    F = ss[k] / df[k] / mse
    tail = f"F = {F:7.2f}  p = {stats.f.sf(F, df[k], df['error']):.2g}" if k != "
    error" else ""
    print(f"{k:12s} df {df[k]:2d}  SS {ss[k]:10.0f}  MS {ss[k] / df[k]:9.0f}  {tail}"
          ")

```

```

cell = volume.mean(axis=2)
flour_c = {"white vs others": np.array([2.0, -1, -1]), "wholemeal vs rye": np.array
          ([0.0, 1, -1])}
temp_c = {"linear": np.array([-3.0, -1, 1, 3]), "quadratic": np.array([1.0, -1, -1,
          1]),
          "cubic": np.array([-1.0, 3, -3, 1])}
rows = []
for (fn, c), (tn, d) in itertools.product(flour_c.items(), temp_c.items()):
    D = np.outer(c, d)                                # an interaction contrast in
    the cell means
    est = (D * cell).sum()
    ss_c = est ** 2 / ((D ** 2).sum() / m)              # its one-degree-of-freedom
    sum of squares
    rows.append((fn, tn, est, ss_c, ss_c / mse))
    print(f"{fn:17s} x {tn:9s}  estimate {est:8.1f}  SS {ss_c:8.0f}  F {ss_c / mse
    :6.2f}")
print(f"sum of the six: {sum(r[3] for r in rows):.0f}  interaction SS: {ss['
interaction']:.0f}")

```

```

fmean, tmean, grand = cell.mean(1), cell.mean(0), cell.mean()
s = np.sqrt(mse)
nu = df["error"]
q = stats.studentized_range.ppf(0.95, a, nu)
half_tukey = q * s / np.sqrt(b * m)                  # flour means average b*m
loaves
for i, k in itertools.combinations(range(a), 2):
    d = fmean[i] - fmean[k]
    print(f"{flours[i]:9s} - {flours[k]:9s}: {d:7.1f}  Tukey [{d - half_tukey:7.1f},
    {d + half_tukey:7.1f}]")
t_bonf = stats.t.ppf(1 - 0.05 / (2 * b), nu)          # white - wholemeal at each
temperature
half_simple = t_bonf * s * np.sqrt(2 / m)
for j in range(b):
    d = cell[0, j] - cell[1, j]
    print(f"white - wholemeal at {temps[j]} C: {d:6.1f}  Bonferroni [{d -

```



```
half_simple:6.1f}, {d + half_simple:6.1f}])")
c_scheffe = np.sqrt(df["interaction"] * stats.f.ppf(0.95, df["interaction"], nu))
```

Warning. It is tempting to drop a nonsignificant interaction and “pool” its sum of squares into the error. With $ab(m-1)$ error degrees of freedom already available this gains little. If the interaction is real, the pooled mean square overestimates σ^2 , and if the decision to pool depends on the data, the nominal levels no longer hold (Section 16.4).

16.3.4 Exercises

16.3.4.1 A. Check your understanding

Exercise A1.

Let $SS_{\text{cells}} = m \sum_{i,j} (\bar{y}_{ij} - \bar{y}_{...})^2$, the between-cells sum of squares of the one-way layout with ab groups. Show that $SS_{\text{cells}} = SS_A + SS_B + SS_{AB}$, and identify each term with a projection.

16.3.4.2 B. Practice

Exercise B1.

Suppose all cell means are equal except μ_{11} , which is larger by δ . Find α_i , β_j and γ_{ij} , and show that the noncentralities of F_A , F_B and F_{AB} are

$$\frac{m\delta^2}{\sigma^2} \cdot \frac{a-1}{ab}, \quad \frac{m\delta^2}{\sigma^2} \cdot \frac{b-1}{ab}, \quad \frac{m\delta^2}{\sigma^2} \cdot \frac{(a-1)(b-1)}{ab}.$$

Which test is most likely to detect a single deviant cell when a and b are large?

Solution. The row means are $\bar{\mu}_{1.} = \mu + \delta/b$ and $\bar{\mu}_{i.} = \mu$ otherwise, and the grand mean is $\mu + \delta/(ab)$. So $\alpha_1 = \delta(a-1)/(ab)$ and $\alpha_i = -\delta/(ab)$ for $i \geq 2$, and

$$\sum_i \alpha_i^2 = \frac{\delta^2}{a^2 b^2} ((a-1)^2 + (a-1)) = \frac{\delta^2(a-1)}{ab^2},$$

so $bm \sum_i \alpha_i^2 / \sigma^2 = m\delta^2(a-1)/(ab\sigma^2)$. By symmetry the noncentrality for B is $m\delta^2(b-1)/(ab\sigma^2)$. For the interaction, $\gamma_{ij} = \delta(g_i - 1/a)(h_j - 1/b)$, where $g_1 = h_1 = 1$ and $g_i = h_j = 0$ for $i, j \geq 2$, so $\sum \gamma_{ij}^2 = \delta^2 \cdot \frac{a-1}{a} \cdot \frac{b-1}{b}$, and the noncentrality is $m\delta^2(a-1)(b-1)/(ab\sigma^2)$. For large a, b the interaction noncentrality approaches $m\delta^2/\sigma^2$, while the main-effect noncentralities approach $m\delta^2/(b\sigma^2)$ and $m\delta^2/(a\sigma^2)$. Almost all of a single deviant cell shows up as interaction. It is, however, spread over $(a-1)(b-1)$ degrees of freedom, and the Scheffé-type contrast aimed at that cell is the one to look at.

Exercise B2.

In the cell-means model Eq. 16.3.1 with normal errors, show that $\bar{y}_{1..}, \dots, \bar{y}_{a..}$ are in-

dependent, each with variance $\sigma^2/(bm)$, and jointly independent of SSE, whatever the μ_{ij} . Deduce Tukey intervals for all $\bar{\mu}_{i.} - \bar{\mu}_{k.}$ with exact simultaneous coverage.

Solution. Different rows use disjoint sets of the independent observations, so the row means are independent, and each averages bm observations. The vector of row means is a linear function of $(\mathbf{I}_a \otimes \bar{\mathbf{J}}_b \otimes \bar{\mathbf{J}}_m)\mathbf{Y}$, whose range lies in $\mathcal{C}(\mathbf{P}_0 + \mathbf{P}_A) \perp \mathcal{C}(\mathbf{P}_E)$. By [Theorem 4.4.4](#), $(\mathbf{P}_0 + \mathbf{P}_A)\mathbf{Y}$ and $\mathbf{P}_E\mathbf{Y}$ are independent, and $\text{SSE} = \|\mathbf{P}_E\mathbf{Y}\|^2$. The row means have expectations $\bar{\mu}_{i.}$. These are exactly the conditions used in the proof of [Theorem 13.3.2](#), with $g = a$, common variance $\sigma^2/(bm)$ and $\nu = ab(m-1)$. So the intervals $\bar{y}_{i.} - \bar{y}_{k.} \pm q_\alpha(a, ab(m-1)) s/\sqrt{bm}$ have simultaneous coverage exactly $1 - \alpha$.

Exercise B3.

Let the levels of A and B be distinct numbers $w_1, \dots, w_a$ and $z_1, \dots, z_b$. Show that the vectors with entries $w_i^r z_j^s$ (repeated over replicates), $0 \leq r < a$, $0 \leq s < b$, span the cell-means space $\mathcal{C}(\mathbf{X})$. Show that the additive model corresponds to the terms with $r = 0$ or $s = 0$. What does the coefficient of wz measure?

16.3.4.3 C. Going deeper

Exercise C1.

Let $\hat{\gamma}_{ij} = \bar{y}_{ij.} - \bar{y}_{i.} - \bar{y}_{.j} + \bar{y}_{...}$. Show that among all interaction contrasts, $\text{SS}(\mathbf{d})$ of [Eq. 16.3.3](#) is largest for $d_{ij} \propto \hat{\gamma}_{ij}$, and that the maximum is SS_{AB} . Relate this to [Corollary 13.2.3\(a\)](#). Why does it mean that the F statistic of a contrast chosen after inspecting the interaction plot cannot be compared with $F(1, ab(m-1))$?

Solution. Since (d_{ij}) has zero margins, $\sum d_{ij} \bar{y}_{ij.} = \sum d_{ij} \hat{\gamma}_{ij}$ ([Proposition 16.2.1](#) applied to the table of cell means). By the Cauchy–Schwarz inequality, $(\sum d_{ij} \hat{\gamma}_{ij})^2 \leq \sum d_{ij}^2 \sum \hat{\gamma}_{ij}^2$, with equality iff $\mathbf{d} \propto \hat{\boldsymbol{\gamma}}$. Hence $\text{SS}(\mathbf{d}) \leq m \sum \hat{\gamma}_{ij}^2 = \text{SS}_{AB}$, with equality at $\mathbf{d} = \hat{\boldsymbol{\gamma}}$, which is itself a zero-margin table. In the notation of [Corollary 13.2.3\(a\)](#) this says that the largest single-contrast F ratio is $(a-1)(b-1)F_{AB}$. A contrast chosen for its large value is, in the extreme, this maximizing contrast. Its null distribution is that of $(a-1)(b-1)$ times an $F((a-1)(b-1), ab(m-1))$ variable, not $F(1, ab(m-1))$.

Exercise C2.

For each column j , let $\text{SS}_{A|j} = m \sum_i (\bar{y}_{ij.} - \bar{y}_{.j})^2$ be the sum of squares for A computed within column j alone. Show that $\sum_j \text{SS}_{A|j} = \text{SS}_A + \text{SS}_{AB}$, and find the projection whose squared length is the left side. What hypothesis does the corresponding F test with $b(a-1)$ numerator degrees of freedom address?

16.4 Higher-way layouts

Nothing in the two-way analysis depended on there being two factors. With t crossed factors, all combinations observed equally often, the observation space splits into 2^t mutually orthogonal

pieces plus error, one for each set of factors. This section proves that decomposition, interprets the interaction of three or more factors, specializes to two-level factorial designs, and ends with a warning about pooling “negligible” terms into error.

16.4.1 The general balanced factorial

Let factors $F_1, \dots, F_t$ have $l_1, \dots, l_t \geq 2$ levels. Suppose all $N = l_1 \cdots l_t$ combinations are observed $m \geq 1$ times each, so $n = Nm$. Stack the observations with the replicate index fastest, then the level of F_t , and so on, with F_1 slowest. For a subset $S \subseteq \{1, \dots, t\}$, a *term*, define

$$\begin{aligned} \mathbf{P}_S &= \mathbf{E}_1 \otimes \cdots \otimes \mathbf{E}_t \otimes \bar{\mathbf{J}}_m, & \mathbf{E}_s &= \begin{cases} \mathbf{C}_{l_s}, & s \in S, \\ \bar{\mathbf{J}}_{l_s}, & s \notin S, \end{cases} \\ \mathbf{P}_E &= \mathbf{I}_{l_1} \otimes \cdots \otimes \mathbf{I}_{l_t} \otimes \mathbf{C}_m, \end{aligned} \quad (16.4.1)$$

and the **marginal averaging operator**

$$\mathbf{A}_S = \mathbf{G}_1 \otimes \cdots \otimes \mathbf{G}_t \otimes \bar{\mathbf{J}}_m, \quad \mathbf{G}_s = \begin{cases} \mathbf{I}_{l_s}, & s \in S, \\ \bar{\mathbf{J}}_{l_s}, & s \notin S. \end{cases}$$

The operator $\mathbf{A}_S$ replaces each observation by the mean of all observations that share its levels of the factors in S . For example, with three factors A, B, C , $\mathbf{A}_{\{A,C\}}\mathbf{y}$ has entries $\bar{y}_{i.k.}$. Let $\mathcal{V}_S$ be the set of vectors whose entries depend only on the levels of the factors in S . It is the column space of the indicator matrix of the combinations of those factors. So $\mathcal{V}_\emptyset = \text{span}(\mathbf{1})$, and $\mathcal{V}_{\{1, \dots, t\}}$ is the space of the full cell-means model.

Theorem 16.4.1 — The balanced factorial decomposition

In the setting above:

- (a) The 2^t matrices $\mathbf{P}_S$ and $\mathbf{P}_E$ are mutually orthogonal projections that sum to $\mathbf{I}$. Their ranks are $r_S = \prod_{s \in S} (l_s - 1)$ (with $r_\emptyset = 1$) and $N(m - 1)$.
- (b) $\mathbf{A}_S$ is the orthogonal projection onto $\mathcal{V}_S$, and $\mathbf{A}_S = \sum_{T \subseteq S} \mathbf{P}_T$. So $\mathcal{V}_S = \bigoplus_{T \subseteq S} \mathcal{C}(\mathbf{P}_T)$.
- (c) $\mathbf{P}_S = \sum_{T \subseteq S} (-1)^{|S|-|T|} \mathbf{A}_T$. Thus $\mathbf{P}_S \mathbf{y}$ is the alternating sum of the marginal means over the subsets of S .
- (d) Let $\mathcal{H}$ be a *hierarchical* family of terms: $\emptyset \in \mathcal{H}$, and $T \subseteq S \in \mathcal{H}$ implies $T \in \mathcal{H}$. The model $\mathbf{E}(\mathbf{Y}) \in \sum_{S \in \mathcal{H}} \mathcal{V}_S$, which contains the terms in $\mathcal{H}$ and nothing else, has projection $\sum_{S \in \mathcal{H}} \mathbf{P}_S$. If terms are entered one at a time so that each term follows all of its subsets, the sequential sum of squares of term S is $\|\mathbf{P}_S \mathbf{y}\|^2$, whatever the order.
- (e) If $\mathbf{E}(\mathbf{Y}) = \boldsymbol{\mu} \in \mathcal{V}_{\{1, \dots, t\}}$ and $\text{Cov}(\mathbf{Y}) = \sigma^2 \mathbf{I}$, then $\mathbf{E}\|\mathbf{P}_S \mathbf{Y}\|^2 = \sigma^2 r_S + \|\mathbf{P}_S \boldsymbol{\mu}\|^2$. Under normality the $\|\mathbf{P}_S \mathbf{Y}\|^2$ and $\text{SSE} = \|\mathbf{P}_E \mathbf{Y}\|^2$ are independent. When $m \geq 2$,

$$F_S = \frac{\|\mathbf{P}_S \mathbf{Y}\|^2 / r_S}{\text{SSE} / (N(m - 1))} \sim F(r_S, N(m - 1), \|\mathbf{P}_S \boldsymbol{\mu}\|^2 / \sigma^2).$$

Proof.

- (a) By Lemma 16.1.1(d) with $t + 1$ factors, $\mathbf{I}$ is the sum of the 2^{t+1} products with $\bar{\mathbf{J}}$ or

- $\mathbf{C}$ in each position. Those with $\bar{\mathbf{J}}_m$ last are the $\mathbf{P}_S$. Those with $\mathbf{C}_m$ last add up to $\mathbf{I} \otimes \cdots \otimes \mathbf{I} \otimes \mathbf{C}_m = \mathbf{P}_E$. Orthogonality and ranks come from [Lemma 16.1.1](#)(b) and (c).
- (b) In each position $s \in S$ write $\mathbf{I}_{l_s} = \bar{\mathbf{J}}_{l_s} + \mathbf{C}_{l_s}$ and expand. This gives $\mathbf{A}_S = \sum_{T \subseteq S} \mathbf{P}_T$, a sum of orthogonal projections, hence a projection ([Theorem 6.3.8](#)) onto $\bigoplus_{T \subseteq S} \mathcal{C}(\mathbf{P}_T)$. The entries of $\mathbf{A}_S \mathbf{y}$ are means over sets of observations with the same levels of the factors in S , so $\mathcal{C}(\mathbf{A}_S) \subseteq \mathcal{V}_S$. Conversely, averaging a vector in $\mathcal{V}_S$ over such a set leaves it unchanged, so $\mathcal{V}_S \subseteq \mathcal{C}(\mathbf{A}_S)$.
- (c) In each position $s \in S$ write $\mathbf{C}_{l_s} = \mathbf{I}_{l_s} - \bar{\mathbf{J}}_{l_s}$ and expand. Choosing $\mathbf{I}$ in the positions of $T \subseteq S$ and $-\bar{\mathbf{J}}$ in the rest of S gives $(-1)^{|S|-|T|} \mathbf{A}_T$.
- (d) By (b), $\sum_{S \in \mathcal{H}} \mathcal{V}_S = \sum_{S \in \mathcal{H}} \bigoplus_{T \subseteq S} \mathcal{C}(\mathbf{P}_T)$. Every T appearing belongs to $\mathcal{H}$, and every $T \in \mathcal{H}$ appears (take $S = T$). So the model space is $\bigoplus_{T \in \mathcal{H}} \mathcal{C}(\mathbf{P}_T)$, with projection $\sum_{T \in \mathcal{H}} \mathbf{P}_T$ ([Theorem 6.3.8](#)). If the terms enter in an order in which each follows its subsets, every intermediate model is hierarchical. Two consecutive models differ by one term S , so their projections differ by $\mathbf{P}_S$ ([Definition 9.3.1](#)).
- (e) The subspaces $\mathcal{C}(\mathbf{P}_S)$ form an analysis of variance decomposition of $\mathcal{V}_{\{1, \dots, t\}}$ by (a) and (b). Apply [Theorem 9.6.1](#) and [Corollary 9.6.2](#).

■

With $t = 2$ the theorem is [Theorem 16.3.1](#), since $\mathbf{P}_{\{1\}} = \mathbf{P}_A$, $\mathbf{P}_{\{2\}} = \mathbf{P}_B$ and $\mathbf{P}_{\{1,2\}} = \mathbf{P}_{AB}$. Part (c) gives the familiar formulas. For three factors A, B, C with levels indexed by i, j, k and replicates by r , the three-factor term has entries

$$(\mathbf{P}_{ABC} \mathbf{y})_{ijk r} = \bar{y}_{ijk \cdot} - \bar{y}_{ij \cdot \cdot} - \bar{y}_{i \cdot k \cdot} - \bar{y}_{\cdot j k \cdot} + \bar{y}_{i \cdot \cdot} + \bar{y}_{\cdot j \cdot \cdot} + \bar{y}_{\cdot \cdot k \cdot} - \bar{y}_{\cdot \cdot \cdot \cdot}.$$

Part (d) explains why balanced factorials have one analysis of variance table and not several. The qualification “each term follows its subsets” matters: entering AB before A gives A nothing, because $\mathcal{V}_A \subseteq \mathcal{V}_{AB}$ ([Exercise \(B1\)](#)). The listing builds the decomposition for a $2 \times 3 \times 4$ layout with $m = 2$ and checks the ranks of part (a). The script also checks parts (b), (c) and (d).

```
import itertools

import numpy as np

def Jbar(k):
    return np.full((k, k), 1.0 / k)          # averaging: replaces a k-vector by its
    mean

def Cen(k):
    return np.eye(k) - Jbar(k)              # centring: subtracts the mean

def kron_all(mats):
    out = np.ones((1, 1))
    for A in mats:
        out = np.kron(out, A)
    return out

def factorial_projections(levels, m):
    """P_S for every subset S of the factors (a 0/1 tuple), and the error projection
    ."""
```

```

P = {}
for S in itertools.product([0, 1], repeat=len(levels)):
    P[S] = kron_all([Cen(1) if s else Jbar(1) for l, s in zip(levels, S)] + [
Jbar(m)])
P["error"] = kron_all([np.eye(1) for l in levels] + [Cen(m)])
return P

def rank(A):
    return int(round(np.trace(A)))          # rank = trace for a projection

levels, m3 = [2, 3, 4], 2
P3 = factorial_projections(levels, m3)
for S in P3:
    if S == "error":
        continue
    label = "".join(f for f, s in zip("ABC", S) if s) or "mean"
    print(f"{label:5s} rank {rank(P3[S]):2d}    predicted {int(np.prod([1 - 1 for l,
s in zip(levels, S) if s]))}")
print("error rank", rank(P3["error"]), "    n =", int(np.prod(levels)) * m3)

```

16.4.2 Three-factor interaction

In a three-factor layout with cell means μ_{ijk} , the two-factor interaction of A and B can be computed separately at each level k of C :

$$\gamma_{ij}^{(k)} = \mu_{ijk} - \bar{\mu}_{i..k} - \bar{\mu}_{.jk} + \bar{\mu}_{...k}.$$

The AB interaction of the analysis of variance, the entries of $\mathbf{P}_{AB}\boldsymbol{\mu}$, is the average of these over k . A three-factor interaction is present when they are not all the same.

Proposition 16.4.2 — Three-factor interaction

For the cell means μ_{ijk} of a three-factor layout, the following are equivalent:

- (i) $\mathbf{P}_{ABC}\boldsymbol{\mu} = \mathbf{0}$;
- (ii) the AB interaction effects $\gamma_{ij}^{(k)}$ at level k of C do not depend on k ;
- (iii) every AB interaction contrast $\sum_{i,j} d_{ij}\mu_{ijk}$ has the same value at every level k of C ;
- (iv) statements (ii) and (iii) with the roles of the three factors permuted in any way;
- (v) $\mu_{ijk} = f_{ij} + g_{ik} + h_{jk}$ for some numbers f_{ij}, g_{ik}, h_{jk} .

Proof. It suffices to consider $m = 1$, since the replicate factor $\bar{\mathbf{J}}_m$ only repeats entries. Then $\mathbf{P}_{ABC} = \mathbf{C}_a \otimes \mathbf{C}_b \otimes \mathbf{C}_c = \mathbf{C}_a \otimes \mathbf{C}_b \otimes \mathbf{I}_c - \mathbf{C}_a \otimes \mathbf{C}_b \otimes \bar{\mathbf{J}}_c$. Applied to $\boldsymbol{\mu}$, the first term produces $\gamma_{ij}^{(k)}$ in position (i, j, k) , because it centres over i and over j within each k . The second produces the average of $\gamma_{ij}^{(k)}$ over k . So (i) holds iff $\gamma_{ij}^{(k)}$ equals its average over k for all i, j, k , which is (ii). By Proposition 16.2.1, $\sum_{i,j} d_{ij}\mu_{ijk} = \sum_{i,j} d_{ij}\gamma_{ij}^{(k)}$ for every zero-margin table (d_{ij}) . Since the tables $\gamma^{(k)}$ themselves have zero margins, $\gamma^{(k)} = \gamma^{(k')}$ iff their inner products with every zero-margin $\mathbf{d}$ agree (take $\mathbf{d} = \gamma^{(k)} - \gamma^{(k')}$). This proves (ii) $\Leftrightarrow$ (iii). Statement (i) is symmetric in the three factors, which gives (iv).

Finally, the vectors of the form in (v) make up $\mathcal{V}_{AB} + \mathcal{V}_{AC} + \mathcal{V}_{BC}$, the model space of the hierarchical family of all terms except ABC . By [Theorem 16.4.1\(d\)](#) this is the orthogonal complement of $\mathcal{C}(\mathbf{P}_{ABC})$ in the cell-means space, so $\boldsymbol{\mu}$ lies in it iff $\mathbf{P}_{ABC}\boldsymbol{\mu} = \mathbf{0}$. ■

So the three-factor interaction is an “interaction of interactions”. It measures how much the pattern of AB interaction changes from one level of C to another, and by (iv) it measures the same thing whichever factor plays the role of C . When it is present, the AB interaction in the table is an equal-weight average over the levels of C , with all the caveats of [Section 16.2](#). Interpretation proceeds from the top down: examine the highest-order interactions clearly present, then read lower-order terms as averages or look at simple effects. Models are usually required to be hierarchical, because a model with AB but without A constrains the equal-weight average $\bar{\mu}_{i..}$, and so depends on an arbitrary choice of weights.

16.4.3 Two-level factorial designs

When every factor has two levels, each $\mathbf{C}_2$ has rank one, and every term of [Theorem 16.4.1](#) has a single degree of freedom. Code the levels of each factor as -1 (“low”) and $+1$ (“high”). For a nonempty term S , let $\mathbf{x}_S$ be the column whose entry for an observation is the product of the codes of the factors in S . For example, $\mathbf{x}_{AC}$ is $+1$ when A and C are at the same level and -1 otherwise. The **effect** of S is

$$\hat{E}_S = \bar{y}(\mathbf{x}_S = +1) - \bar{y}(\mathbf{x}_S = -1),$$

the mean response where $\mathbf{x}_S$ is $+1$ minus the mean where it is -1 . For a main effect this is the average change in response when the factor moves from low to high. For AC it is half the difference between the effect of C at high A and at low A ([Exercise \(B2\)](#)).

Proposition 16.4.3 — Effects in a 2^t factorial

In a 2^t factorial with m replicates, $n = 2^t m$, and a nonempty term S :

- (a) $\mathbf{P}_S = \mathbf{x}_S \mathbf{x}_S^\top / n$. The columns $\mathbf{x}_S$ are mutually orthogonal and orthogonal to $\mathbf{1}$.
- (b) $\hat{E}_S = 2\mathbf{x}_S^\top \mathbf{y} / n$, and $\text{Var}(\hat{E}_S) = 4\sigma^2 / n$. Estimated effects of different terms are uncorrelated.
- (c) The sum of squares of S is $\|\mathbf{P}_S \mathbf{y}\|^2 = n\hat{E}_S^2 / 4$.
- (d) In the regression of $\mathbf{y}$ on $\mathbf{1}$ and all $2^t - 1$ columns $\mathbf{x}_S$, the coefficient of $\mathbf{x}_S$ is $\hat{E}_S / 2$, and it is the same in every submodel that contains $\mathbf{x}_S$.

Proof.

- (a) With $\mathbf{u} = (-1, 1)^\top$, $\mathbf{C}_2 = \frac{1}{2}\mathbf{u}\mathbf{u}^\top$ and $\bar{\mathbf{J}}_2 = \frac{1}{2}\mathbf{1}_2\mathbf{1}_2^\top$. By the mixed-product rule $\mathbf{P}_S = (2^t m)^{-1} \mathbf{x}\mathbf{x}^\top$, where $\mathbf{x}$ is the Kronecker product with $\mathbf{u}$ in the positions of S , $\mathbf{1}_2$ elsewhere, and $\mathbf{1}_m$ last. Its entries are the products of the codes of the factors in S , so $\mathbf{x} = \mathbf{x}_S$. Orthogonality follows from [Theorem 16.4.1\(a\)](#), since $\mathbf{x}_S \in \mathcal{C}(\mathbf{P}_S)$ and $\mathbf{1} \in \mathcal{C}(\mathbf{P}_\emptyset)$.
- (b) Since $\mathbf{x}_S \perp \mathbf{1}$, exactly $n/2$ entries are $+1$, so $\mathbf{x}_S^\top \mathbf{y} = \frac{n}{2}(\bar{y}(+) - \bar{y}(-))$. Then $\text{Var}(\hat{E}_S) = (4/n^2)\sigma^2\|\mathbf{x}_S\|^2 = 4\sigma^2/n$. Covariances are $(4/n^2)\sigma^2\mathbf{x}_S^\top \mathbf{x}_T = 0$.

- (c) $\|\mathbf{P}_S \mathbf{y}\|^2 = (\mathbf{x}_S^\top \mathbf{y})^2 / n = n \hat{E}_S^2 / 4$.
- (d) With orthogonal columns each coefficient is $\mathbf{x}_S^\top \mathbf{y} / \|\mathbf{x}_S\|^2 = \mathbf{x}_S^\top \mathbf{y} / n$ and does not depend on the other columns (Proposition 7.2.4).

■

Example. (Heat seals: an unreplicated 2^4 design)

A packaging line makes heat seals on plastic film. Four factors were each set at two levels: A , jaw temperature; B , jaw pressure; C , dwell time; D , film gauge. One seal was made at each of the 16 combinations, in random order, and its peel strength was measured in newtons. The data are synthetic, generated by the script `two_level.py`. In standard order (A changing fastest), the strengths are

run	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
A		+		+		+		+		+		+		+		+
B			+	+			+	+			+	+			+	+
C					+	+	+	+					+	+	+	+
D									+	+	+	+	+	+	+	+
y	28.5	32.1	26.7	31.1	29.0	37.4	27.2	37.4	25.8	26.6	24.2	28.8	24.9	34.3	25.9	36.7

The mean is 29.79. The largest estimated effects are A (6.52), C (3.62), AC (3.17) and D (−2.78). They are followed by BD (1.08) and AB (0.98), and none of the other nine exceeds 0.53 in absolute value. There is no pure error. The experimenters had decided *before the trial* to treat the five interactions of three or four factors as negligible and to use their sums of squares as error. That gives $s^2 = 0.514$ on 5 degrees of freedom. Against the critical value $F_{0.05}(1, 5) = 6.61$, the tests reject for A , C and AC ($F = 331.0, 102.2, 78.4$, each $p < 0.001$), for D ($F = 59.9, p = 0.001$), and, nominally, for BD ($p = 0.030$) and AB ($p = 0.042$). The last two are among ten tests, however. Holm's procedure at familywise level 0.05 (Theorem 13.4.5) accepts A , C , AC and D and stops at BD , whose p -value exceeds 0.05/6. The data were in fact generated with exactly those four effects.

The interaction AC matters in practice. The effect of dwell time is $\hat{E}_C + \hat{E}_{AC} = 3.62 + 3.17 = 6.79$ N at high temperature and $\hat{E}_C - \hat{E}_{AC} = 0.45$ N at low temperature (Exercise B2)). Longer dwell helps only when the jaws are hot. Figure 16.4.1 shows the half-normal plot of Daniel (1959): absolute effects against half-normal quantiles. Noise falls near a line through the origin and real effects stand out above it, with no estimate of σ needed.

```
import itertools

import numpy as np
from scipy import stats

k = 4
runs = np.array(list(itertools.product([-1, 1], repeat=k))[:, :-1]) # standard
                        order: A changes fastest
names = []
columns = []
```

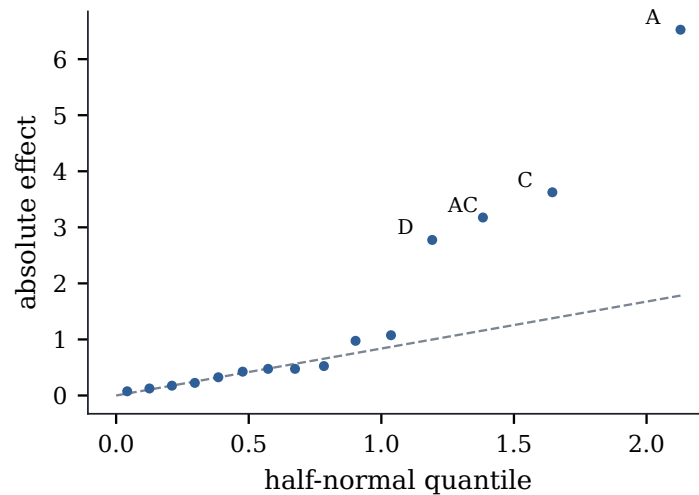


Figure 16.4.1: Half-normal plot of the 15 absolute effects of the heat-seal experiment. The dashed line through the origin is fitted to the ten smallest. *A*, *C*, *AC* and *D* stand clear of it.

```
for size in range(1, k + 1):
    for S in itertools.combinations(range(k), size):
        names.append("".join("ABCD"[i] for i in S))
        columns.append(np.prod(runs[:, list(S)], axis=1))
Xc = np.column_stack(columns)           # 16 x 15 matrix of +-1 contrast
columns
print("columns orthogonal:", np.allclose(Xc.T @ Xc, 16 * np.eye(15)))
```

```
true = 30 + 3.0 * runs[:, 0] + 2.0 * runs[:, 2] + 1.5 * runs[:, 0] * runs[:, 2] -
    1.25 * runs[:, 3]
rng = np.random.default_rng(16_04)
y = np.round(true + rng.normal(scale=1.0, size=16), 1)
n = len(y)
effects = Xc.T @ y / (n / 2)           # mean at + minus mean at -
ss = n * effects ** 2 / 4              # one degree of freedom each
order = np.argsort(-np.abs(effects))
for i in order[:6]:
    print(f"{names[i]:4s} effect {effects[i]:6.2f}    SS {ss[i]:7.2f}")
high = [i for i, nm in enumerate(names) if len(nm) >= 3]  # ABC, ABD, ACD, BCD,
ABCD
ss_pooled = ss[high].sum()
s2 = ss_pooled / len(high)
F = ss / s2
p = stats.f.sf(F, 1, len(high))
for i in order[:6]:
    print(f"{names[i]:4s} F = {F[i]:7.2f}    p = {p[i]:.3f}")
```

Two-level designs are economical: every main effect is estimated with the precision $4\sigma^2/n$ of the whole experiment, as if each factor had been studied alone with all n runs. With many factors, *fractional* factorials use a chosen subset of the 2^t runs and deliberately confound some effects with others. Box, Hunter and Hunter (2005) and Wu and Hamada (2009) develop that theory.

16.4.4 Pooling and its dangers

In [Example \(Heat seals: an unreplicated \$2^4\$ design\)](#) the error was built from the sums of squares of terms assumed to be zero. This is **pooling**: replacing SSE by $\text{SSE} + \sum_{T \in \mathcal{Q}} \|\mathbf{P}_T \mathbf{y}\|^2$, with $\nu_E + \sum_{T \in \mathcal{Q}} r_T$ degrees of freedom, for a set $\mathcal{Q}$ of terms. Without replication it is the only source of an error term. The theory is simple when $\mathcal{Q}$ is fixed in advance.

Proposition 16.4.4 — Pooling a fixed set of terms

In the setting of [Theorem 16.4.1\(e\)](#) with normal errors, let $\mathcal{Q}$ be a set of terms fixed before the data are seen. Let ν be the pooled degrees of freedom and MS_{pool} the pooled mean square. Let $S \notin \mathcal{Q}$ and $F_S^{\text{pool}} = (\|\mathbf{P}_S \mathbf{Y}\|^2 / r_S) / \text{MS}_{\text{pool}}$.

- (a) $\text{E MS}_{\text{pool}} = \sigma^2 + \nu^{-1} \sum_{T \in \mathcal{Q}} \|\mathbf{P}_T \boldsymbol{\mu}\|^2$.
- (b) If $\mathbf{P}_T \boldsymbol{\mu} = \mathbf{0}$ for every $T \in \mathcal{Q}$, then $F_S^{\text{pool}} \sim F(r_S, \nu, \|\mathbf{P}_S \boldsymbol{\mu}\|^2 / \sigma^2)$. The pooled test is exact and has more denominator degrees of freedom.
- (c) If $\mathbf{P}_S \boldsymbol{\mu} = \mathbf{0}$ but some pooled term is not null, then $\Pr\{F_S^{\text{pool}} > F_\alpha(r_S, \nu)\} < \alpha$. The test is conservative, and its power is reduced for the same reason.

Proof. The matrix $\mathbf{P}_{\text{pool}} = \mathbf{P}_E + \sum_{T \in \mathcal{Q}} \mathbf{P}_T$ is a projection of rank ν , orthogonal to $\mathbf{P}_S$ ([Theorem 16.4.1\(a\)](#)). By [Theorem 4.4.4](#), $U = \|\mathbf{P}_S \mathbf{Y}\|^2 / \sigma^2$ and $V = \|\mathbf{P}_{\text{pool}} \mathbf{Y}\|^2 / \sigma^2$ are independent, with $V \sim \chi^2(\nu, \delta)$ and $\delta = \sum_{T \in \mathcal{Q}} \|\mathbf{P}_T \boldsymbol{\mu}\|^2 / \sigma^2$, because $\mathbf{P}_E \boldsymbol{\mu} = \mathbf{0}$. Part (a) is the mean $\nu + \delta$ of V , times σ^2 / ν . If $\delta = 0$, (b) is the definition of the F distribution. For (c), $U \sim \chi^2(r_S)$, and with $c = F_\alpha(r_S, \nu)$,

$$\Pr\{F_S^{\text{pool}} > c\} = \text{E} \left[\Pr \left\{ V < \frac{\nu U}{r_S c} \mid U \right\} \right].$$

By [Proposition 4.1.3](#), $\Pr\{\chi^2(\nu, \delta) < x\}$ is strictly smaller for $\delta > 0$ than for $\delta = 0$, for every $x > 0$. Taking expectations gives a probability smaller than its value at $\delta = 0$, which is α . ■

The dangers come when the choice of $\mathcal{Q}$ depends on the data. There are two common ways that happens.

Pooling after a preliminary test. A common rule pools an interaction into error if its own F test is not significant at a lenient level such as 0.25. Consider a $2 \times 3 \times 4$ layout with $m = 2$, so $\nu_E = 24$ and $r_{ABC} = 6$, and test a null main effect A at level 0.05 after this preliminary test of ABC . By [Theorem 16.4.1\(e\)](#), SS_A , SS_{ABC} and SSE are independent, with distributions $\sigma^2 \chi^2(1)$, $\sigma^2 \chi^2(6, \gamma_{ABC})$ and $\sigma^2 \chi^2(24)$. So the rule can be simulated exactly by drawing these three variables ([Figure 16.4.2](#)). When ABC is absent, the rule pools in 75% of samples, precisely those with small MS_{ABC} . The pooled denominator is biased downwards, and the test of A rejects a true null with probability 0.0551 instead of 0.05. When ABC is large the rule rarely pools, and the level returns to 0.05 (0.0504 at $\gamma_{ABC} = 30$). Always pooling is very conservative there (0.0064), as [Proposition 16.4.4\(c\)](#) predicts. Here the damage is modest, but a preliminary test buys degrees of freedom with a size that is no longer what it claims to be (Bancroft 1944).

```
import numpy as np
from scipy import stats
```

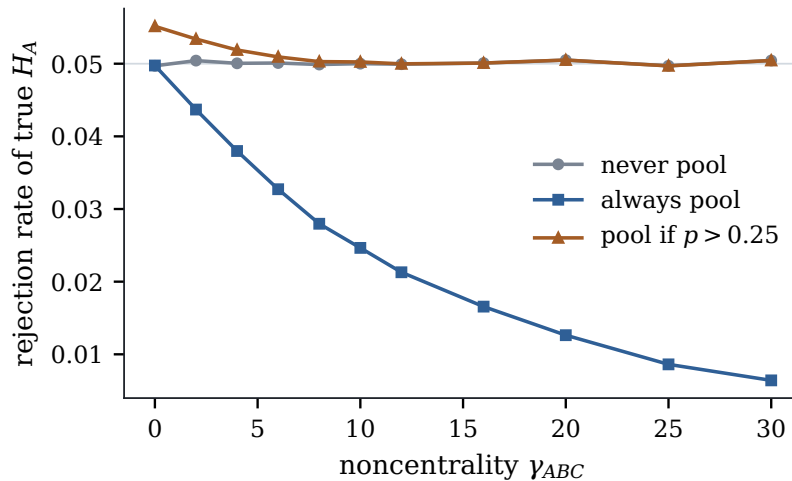


Figure 16.4.2: Rejection rate of the level-0.05 test of a true null main effect in a $2 \times 3 \times 4$ layout with two replicates, as the three-factor noncentrality γ_{ABC} grows. Never pooling is exact. Always pooling is conservative once ABC is present. Pooling after a preliminary test at level 0.25 is anticonservative near $\gamma_{ABC} = 0$. Each point is based on 400000 simulated samples.

```

alpha, alpha_pre = 0.05, 0.25
df_A, df_ABC, df_E = 1, 6, 24

def rejection_rates(gamma, reps, rng):
    """Size of the level-0.05 test of A under three pooling rules, when A has no
    effect."""
    ssA = rng.chisquare(df_A, reps)
    ssABC = rng.noncentral_chisquare(df_ABC, gamma, reps) if gamma > 0 else rng.
    chisquare(df_ABC, reps)
    ssE = rng.chisquare(df_E, reps)
    never = ssA / (ssE / df_E) > stats.f.ppf(1 - alpha, df_A, df_E)
    pooled_F = ssA / ((ssE + ssABC) / (df_E + df_ABC))
    always = pooled_F > stats.f.ppf(1 - alpha, df_A, df_E + df_ABC)
    pool = ssABC / df_ABC / (ssE / df_E) <= stats.f.ppf(1 - alpha_pre, df_ABC, df_E)
    pretest = np.where(pool, always, never)
    return never.mean(), always.mean(), pretest.mean(), pool.mean()

rng = np.random.default_rng(20260919)
gammas = np.array([0, 2, 4, 6, 8, 10, 12, 16, 20, 25, 30])
rates = np.array([rejection_rates(g, 400_000, rng) for g in gammas])
for g, (nv, al, pt, pl) in zip(gammas, rates):
    print(f"gamma {g:3d}: never {nv:.4f}  always {al:.4f}  pre-test {pt:.4f} (
    pooled in {pl:.2f})")

```

Pooling the smallest effects. The second temptation is much worse. In an unreplicated 2^t design one may look at the estimated effects, declare the smallest few to be “error”, and test the rest against them. If no factor does anything, the 15 estimated effects of a 2^4 design are independent normal variables with a common variance (Proposition 16.4.3(b)). Pool the five smallest in absolute value and test the largest against them with the $F(1, 5)$ critical value. It is declared significant in a fraction 0.999 of null data sets (and so, of course, at least one effect is declared significant). The five smallest of fifteen normal variables are the bottom third of the noise, and a mean square built from them underestimates the variance by a factor of 11.5 on average. This

is why the error terms in [Example \(Heat seals: an unreplicated \$2^4\$ design\)](#) were chosen before the data were seen, and why the half-normal plot and Lenth's (1989) robust standard error ([Exercise \(C1\)](#)) are the standard tools for unreplicated designs.

```
rng2 = np.random.default_rng(16_0404)
reps = 200_000
effects = rng2.normal(size=(reps, 15))           # 15 null effect
           estimates, common variance
sq = np.sort(effects ** 2, axis=1)
F_largest = sq[:, -1] / sq[:, :5].mean(axis=1)    # largest against the
           five smallest
size_largest = np.mean(F_largest > stats.f.ppf(1 - alpha, 1, 5))
pooled_ms = sq[:, :5].mean(axis=1)               # the "error" mean
           square; true variance 1
underestimate = 1 / pooled_ms.mean()             # factor by which it
           falls short on average
print(f"largest effect declared significant: {size_largest:.3f};"
      f" pooled mean square averages {pooled_ms.mean():.3f}, an underestimate by a
      factor {underestimate:.1f}")
```

Idea. Pooling is safe when the pooled terms are chosen before the data are seen and are genuinely negligible. It is conservative when they are chosen in advance but are not negligible. It is invalid when they are chosen because they look small.

16.4.5 Exercises

16.4.5.1 A. Check your understanding

Exercise A1.

List the terms and their degrees of freedom for a balanced $3 \times 2 \times 4$ layout with $m = 2$, and check that they add up to $n - 1$.

Exercise A2.

For t factors each with l levels, show that the number of terms involving exactly q factors is $\binom{t}{q}$, each with $(l - 1)^q$ degrees of freedom, and that these add up to $l^t - 1$.

16.4.5.2 B. Practice

Exercise B1.

In a balanced two-way layout, show that if AB (the cell indicators) is entered before A , the sequential sum of squares of A is zero. What does that say about software output for models written in a non-hierarchical order?

Exercise B2.

In a 2^t factorial, show that the effect of C computed only from the runs with A at its

high level is $\hat{E}_C + \hat{E}_{AC}$, and at the low level it is $\hat{E}_C - \hat{E}_{AC}$. Conclude that $\hat{E}_{AC}$ is half the difference between the two simple effects of C .

Solution. Let $\bar{y}_{ac}$ be the mean of the runs with codes $a, c \in \{-1, 1\}$ for A and C , each averaging $n/4$ runs. Then $\hat{E}_C = \frac{1}{2}(\bar{y}_{1,1} + \bar{y}_{-1,1}) - \frac{1}{2}(\bar{y}_{1,-1} + \bar{y}_{-1,-1})$, and, since $\mathbf{x}_{AC} = +1$ when the codes agree, $\hat{E}_{AC} = \frac{1}{2}(\bar{y}_{1,1} + \bar{y}_{-1,-1}) - \frac{1}{2}(\bar{y}_{1,-1} + \bar{y}_{-1,1})$. Adding, $\hat{E}_C + \hat{E}_{AC} = \bar{y}_{1,1} - \bar{y}_{-1,-1}$, the effect of C at high A . Subtracting gives $\bar{y}_{-1,1} - \bar{y}_{1,-1}$. Half the difference of the two is $\hat{E}_{AC}$.

Exercise B3.

Use [Theorem 16.4.1\(c\)](#) to derive the displayed formula for $(\mathbf{P}_{ABC}\mathbf{y})_{ijkl}$. Write the analogous formula for $(\mathbf{P}_{AB}\mathbf{y})_{ijkl}$ in a three-factor layout, and check that it does not involve k .

16.4.5.3 C. Going deeper

Exercise C1.

In an unreplicated 2^t design, let $\hat{E}_1, \dots, \hat{E}_K$ be the $K = 2^t - 1$ estimated effects. Lenth (1989) proposed $s_0 = 1.5 \text{median}|\hat{E}_S|$ and the *pseudo standard error* $\text{PSE} = 1.5 \text{median}\{|\hat{E}_S| : |\hat{E}_S| < 2.5s_0\}$. Show that if all effects are null and $\tau^2 = 4\sigma^2/n$, then $\text{median}|\hat{E}_S| \rightarrow 0.6745\tau$ as $K \rightarrow \infty$, so that s_0 estimates 1.01τ . Explain why the trimming at $2.5s_0$ makes PSE insensitive to a few large active effects, and why that is the property pooling lacks.

Solution. Under the null hypothesis the $\hat{E}_S$ are independent $N(0, \tau^2)$ variables ([Proposition 16.4.3\(b\)](#)), so $|\hat{E}_S|$ has the half-normal distribution with median $\tau \Phi^{-1}(0.75) = 0.6745\tau$. The sample median converges to it, and $1.5 \times 0.6745 \approx 1.01$, so $s_0 \approx \tau$. The median is unaffected by a few large values. The trimmed median in PSE additionally removes effects that are large compared with the first estimate before re-estimating, so active effects enter neither estimate as long as they are few (*effect sparsity*). Pooling the smallest effects uses the bottom of the distribution only and so is biased downwards. The median of all effects is instead a consistent estimate of the scale when most effects are null.

16.5 Nested factors

So far every level of one factor has appeared with every level of the other. Many studies are built differently. Pupils are taught in classrooms, and each classroom follows one curriculum. Specimens are cut from batches, and each batch comes from one supplier. Classroom 3 under one curriculum has nothing to do with classroom 3 under another: the factors are *nested*, not crossed. This section gives the decomposition for balanced nested layouts, explains why a nested factor has no main effect, and shows why the right denominator for the outer factor depends on what the inner levels represent.

16.5.1 Nested and crossed factors

Definition 16.5.1 — Nested factors

Factor B is **nested** in factor A if every level of B occurs together with exactly one level of A . Factors A and B are **crossed** if every level of B occurs with every level of A . A **balanced nested layout** has a levels of A , b levels of B within each level of A , and m observations at each level of B . We write $B(A)$ for the nested factor, index its levels within level i of A by $j = 1, \dots, b$, and write y_{ijk} , $k = 1, \dots, m$.

The labels j are local: (i, j) names a level of B , but j alone names nothing. The model is

$$E(y_{ijk}) = \mu + \alpha_i + \beta_{j(i)}, \quad \text{Cov}(\mathbf{Y}) = \sigma^2 \mathbf{I}, \quad (16.5.1)$$

with $1 + a + ab$ parameters. Every level of B has its own mean $\mu_{ij} = \mu + \alpha_i + \beta_{j(i)}$, so the mean space is the set of vectors constant on each level of B , of dimension ab . This is the mean space of the crossed cell-means model [Eq. 16.3.1](#) on an array of the same shape. Nesting changes not the space of possible means but which questions about them make sense. In a crossed layout, averaging over i at fixed j is meaningful. In a nested layout it is not, so there is no main effect of B and no AB interaction, only variation among the levels of B within each level of A .

16.5.2 The nested decomposition

With the observations stacked as in [Section 16.1](#) (k fastest, then j , then i), define

$$\begin{aligned} \mathbf{P}_0 &= \bar{\mathbf{J}}_a \otimes \bar{\mathbf{J}}_b \otimes \bar{\mathbf{J}}_m, & \mathbf{P}_A &= \mathbf{C}_a \otimes \bar{\mathbf{J}}_b \otimes \bar{\mathbf{J}}_m, \\ \mathbf{P}_{B(A)} &= \mathbf{I}_a \otimes \mathbf{C}_b \otimes \bar{\mathbf{J}}_m, & \mathbf{P}_E &= \mathbf{I}_a \otimes \mathbf{I}_b \otimes \mathbf{C}_m. \end{aligned} \quad (16.5.2)$$

Theorem 16.5.1 — The balanced nested layout

Assume [Eq. 16.5.1](#) for a balanced nested layout with $a \geq 2$, $b \geq 2$, $m \geq 2$. Write μ_{ij} for the mean of level (i, j) of B and $\bar{\mu}_i = b^{-1} \sum_j \mu_{ij}$.

- (a) The four matrices [Eq. 16.5.2](#) are mutually orthogonal projections summing to $\mathbf{I}$, with ranks 1, $a - 1$, $a(b - 1)$ and $ab(m - 1)$. The first three sum to the projection onto the mean space. Moreover $\mathbf{P}_{B(A)} = \mathbf{P}_B + \mathbf{P}_{AB}$, where $\mathbf{P}_B$ and $\mathbf{P}_{AB}$ are the projections [Eq. 16.3.2](#) of the crossed layout of the same shape.
- (b) The sums of squares are

$$SS_A = bm \sum_i (\bar{y}_{i..} - \bar{y}_{...})^2, \quad SS_{B(A)} = m \sum_i \sum_j (\bar{y}_{ij.} - \bar{y}_{i..})^2, \quad SSE = \sum_{i,j,k} (y_{ijk} - \bar{y}_{ij.})^2.$$

So $SS_{B(A)}$ is the sum over i of the between-levels sums of squares of B computed separately within each level of A .

- (c) The function $\lambda_0 \mu + \sum_i \lambda_i \alpha_i + \sum_{i,j} \kappa_{ij} \beta_{j(i)}$ is estimable iff $\lambda_i = \sum_j \kappa_{ij}$ for every i and $\lambda_0 = \sum_i \lambda_i$. In particular $\beta_{j(i)} - \beta_{l(i)}$ is estimable, but $\beta_{j(i)} - \beta_{l(k)}$ with $i \neq k$

is not, and $\alpha_i - \alpha_k$ is not, while $\alpha_i - \alpha_k + \bar{\beta}_{\cdot(i)} - \bar{\beta}_{\cdot(k)} = \bar{\mu}_{i\cdot} - \bar{\mu}_{k\cdot}$ is.

(d) The expected mean squares are

$$E \text{ MS}_A = \sigma^2 + \frac{bm}{a-1} \sum_i (\bar{\mu}_{i\cdot} - \bar{\mu}_{\cdot\cdot})^2, \quad E \text{ MS}_{B(A)} = \sigma^2 + \frac{m}{a(b-1)} \sum_{i,j} (\mu_{ij} - \bar{\mu}_{i\cdot})^2, \quad E \text{ MSE} = \sigma^2.$$

Under normal errors the three sums of squares are independent, and $F_A = \text{MS}_A/\text{MSE}$ and $F_{B(A)} = \text{MS}_{B(A)}/\text{MSE}$ have noncentral F distributions with $(a-1, ab(m-1))$ and $(a(b-1), ab(m-1))$ degrees of freedom and noncentralities equal to σ^{-2} times the sums in the numerators above. So F_A is an exact test of $\bar{\mu}_{1\cdot} = \cdots = \bar{\mu}_{a\cdot}$, and $F_{B(A)}$ of $\mu_{ij} = \bar{\mu}_{i\cdot}$ for all i, j .

Proof.

- (a) Ranks and symmetric idempotence follow from [Lemma 16.1.1\(b\)](#). The mixed-product rule gives $\mathbf{P}_0 \mathbf{P}_A = \mathbf{0}$ (from $\bar{\mathbf{J}}_a \mathbf{C}_a$), $\mathbf{P}_0 \mathbf{P}_{B(A)} = \mathbf{P}_A \mathbf{P}_{B(A)} = \mathbf{0}$ (from $\bar{\mathbf{J}}_b \mathbf{C}_b$), and each of the first three is orthogonal to $\mathbf{P}_E$ (from $\bar{\mathbf{J}}_m \mathbf{C}_m$). Next, $\mathbf{P}_0 + \mathbf{P}_A = \mathbf{I}_a \otimes \bar{\mathbf{J}}_b \otimes \bar{\mathbf{J}}_m$, and adding $\mathbf{P}_{B(A)}$ gives $\mathbf{I}_a \otimes \mathbf{I}_b \otimes \bar{\mathbf{J}}_m$, the projection onto the vectors constant on each level of B , which is the mean space. Adding $\mathbf{P}_E$ gives $\mathbf{I}$. Finally, $\mathbf{I}_a \otimes \mathbf{C}_b \otimes \bar{\mathbf{J}}_m = (\bar{\mathbf{J}}_a + \mathbf{C}_a) \otimes \mathbf{C}_b \otimes \bar{\mathbf{J}}_m = \mathbf{P}_B + \mathbf{P}_{AB}$.
- (b) $\mathbf{P}_{B(A)} = \mathbf{I}_a \otimes \mathbf{I}_b \otimes \bar{\mathbf{J}}_m - \mathbf{I}_a \otimes \bar{\mathbf{J}}_b \otimes \bar{\mathbf{J}}_m$ maps y_{ijk} to $\bar{y}_{ij\cdot} - \bar{y}_{i\cdot\cdot}$. The other two are as in [Theorem 16.3.1\(b\)](#).
- (c) By [Theorem 8.2.2](#) the estimable functions are the linear combinations $\sum_{i,j} w_{ij}(\mu + \alpha_i + \beta_{j(i)})$ of the expected responses. Such a combination has $\kappa_{ij} = w_{ij}$, $\lambda_i = \sum_j w_{ij}$ and $\lambda_0 = \sum_{i,j} w_{ij}$, which gives the conditions. Conversely, if they hold, $w_{ij} = \kappa_{ij}$ reproduces the function. The three examples follow by checking the conditions. For $\alpha_i - \alpha_k$ all κ are zero but $\lambda_i = 1 \neq 0$.
- (d) Apply [Theorem 9.6.1](#) and [Corollary 9.6.2](#) to the decomposition in (a). The vector $\mathbf{P}_A \boldsymbol{\mu}$ has entries $\bar{\mu}_{i\cdot} - \bar{\mu}_{\cdot\cdot}$, repeated bm times, and $\mathbf{P}_{B(A)} \boldsymbol{\mu}$ has entries $\mu_{ij} - \bar{\mu}_{i\cdot}$, repeated m times.

■

Part (a) is the precise sense in which a nested factor “absorbs” the interaction. If nested data are analysed as though crossed, with the local labels j treated as real levels, the sums of squares for B and AB add up to $\text{SS}_{B(A)}$, but each depends on the arbitrary labelling ([Exercise \(B1\)](#)). Part (c) is the nested analogue of [Proposition 8.6.2](#). The contrasts among the outer levels that can be estimated are contrasts among averages over their own inner levels.

More stages work the same way. With A, B nested in A , and C nested in B , the projections are $\mathbf{C} \otimes \bar{\mathbf{J}} \otimes \bar{\mathbf{J}} \otimes \bar{\mathbf{J}}, \mathbf{I} \otimes \mathbf{C} \otimes \bar{\mathbf{J}} \otimes \bar{\mathbf{J}}, \mathbf{I} \otimes \mathbf{I} \otimes \mathbf{C} \otimes \bar{\mathbf{J}}$ and the error $\mathbf{I} \otimes \mathbf{I} \otimes \mathbf{I} \otimes \mathbf{C}$ ([Exercise \(B3\)](#)). Each nested term has an identity matrix in the positions of the factors it is nested in and a centring matrix in its own position. The listing checks the identity of part (a) and the ranks for $a = 3, b = 4, m = 2$.

```
import itertools

import numpy as np

def Jbar(k):
    return np.full((k, k), 1.0 / k)           # averaging: replaces a k-vector by its
```

```

mean

def Cen(k):
    return np.eye(k) - Jbar(k)          # centring: subtracts the mean

def kron_all(mats):
    out = np.ones((1, 1))
    for A in mats:
        out = np.kron(out, A)
    return out

def factorial_projections(levels, m):
    """P_S for every subset S of the factors (a 0/1 tuple), and the error projection
    . """
    P = {}
    for S in itertools.product([0, 1], repeat=len(levels)):
        P[S] = kron_all([Cen(l) if s else Jbar(l) for l, s in zip(levels, S)] + [
            Jbar(m)])
    P["error"] = kron_all([np.eye(l) for l in levels] + [Cen(m)])
    return P

def rank(A):
    return int(round(np.trace(A)))        # rank = trace for a projection

a, b, m = 3, 4, 2
PB_in_A = kron_all([np.eye(a), Cen(b), Jbar(m)])          # B within A
P2 = factorial_projections([a, b], m)
print("B(A) = B + AB of the crossed layout:", np.allclose(PB_in_A, P2[(0, 1)] + P2
    [(1, 1)]))
print("rank of B(A):", rank(PB_in_A), " = a(b-1) =", a * (b - 1))

```

16.5.3 What the inner levels represent

The tests of [Theorem 16.5.1\(d\)](#) treat the levels of B as fixed. Then H_A says that the averages over *these particular* inner levels are equal, and the error is the variation within inner levels. That is right when the inner levels matter in themselves, such as three specific machines in each of two factories.

Often, though, the inner levels are a sample. The classrooms in a curriculum study stand for all classrooms that might follow that curriculum. Then the question about A concerns populations of inner levels, and the variation *between* inner levels is part of the noise against which A must be judged. A preview of Chapters 32 and 33 makes this precise. If the $\beta_{j(i)}$ are independent $N(0, \sigma_B^2)$ variables, independent of normal errors, then $E MS_{B(A)} = \sigma^2 + m\sigma_B^2$, and $E MS_A$ has the same expression plus a term that vanishes under H_A ([Exercise \(C1\)](#)). The ratio $MS_A/MS_{B(A)}$ then has an exact $F(a-1, a(b-1))$ null distribution. The ratio MS_A/MSE does not, because its numerator is inflated by $m\sigma_B^2$ even when H_A is true. With $a(b-1)$ denominator degrees of freedom instead of $ab(m-1)$, the correct test is based, in effect, on the number of inner levels rather than the number of observations ([Exercise \(B2\)](#)).

Example. (Curricula and classrooms)

Seventy-two pupils took the same reading test after a year in one of three reading curricula. Each curriculum was followed in four classrooms, with six pupils tested per classroom. The data are synthetic, generated by the script `nested_classrooms.py` with random classroom effects (standard deviation 4 points), pupil-level noise (standard deviation 8), and curriculum means differing by a few points. The classroom means are

curriculum	classroom 1	classroom 2	classroom 3	classroom 4	mean
phonics	77.0	67.8	68.7	60.8	68.6
whole language	52.7	64.8	64.3	60.8	60.7
mixed	61.8	61.8	63.8	58.5	61.5

and the nested analysis of variance is

Source	df	Sum of squares	Mean square
curriculum	2	908.3	454.2
classroom(curriculum)	9	1445.8	160.6
pupils (error)	60	3063.3	51.1

Classrooms within curricula differ: $F_{B(A)} = 3.15$ on 9 and 60 degrees of freedom, $p = 0.004$. For curricula, the fixed-effects test of [Theorem 16.5.1\(d\)](#) gives $F = 8.90$ on 2 and 60 degrees of freedom ($p = 0.0004$). It says, correctly, that these twelve classrooms differ between curricula on average. Against the classroom mean square the ratio is $F = 2.83$ on 2 and 9 degrees of freedom ($p = 0.111$). For classrooms in general, no difference has been shown. [Figure 16.5.1](#) shows why: with only four classrooms per curriculum, the gap rests on a handful of classroom means. One phonics classroom (77.0) contributes much of it, and the lowest phonics classroom (60.8) sits among those of the other curricula.

The listing checks by simulation which test to trust. It generates 20000 data sets from the same design with random classroom effects and *no* curriculum effect. The test against pupils rejects at the nominal 5% level in a fraction 0.289 of them, and the test against classrooms in a fraction 0.051. Under the null hypothesis the ratio of the two expected mean squares is $(8^2 + 6 \cdot 4^2)/8^2 = 2.5$.

```
import numpy as np
from scipy import stats

a, b, m = 3, 4, 6
data_rng = np.random.default_rng(16_05)
curriculum_effect = np.array([3.0, -2.0, 0.0])
classroom_effect = data_rng.normal(scale=4.0, size=(a, b))
scores = np.round(62 + curriculum_effect[:, None, None] + classroom_effect[:, :,
    None]
    + data_rng.normal(scale=8.0, size=(a, b, m)))

def Jbar(k):
    return np.full((k, k), 1.0 / k)
```

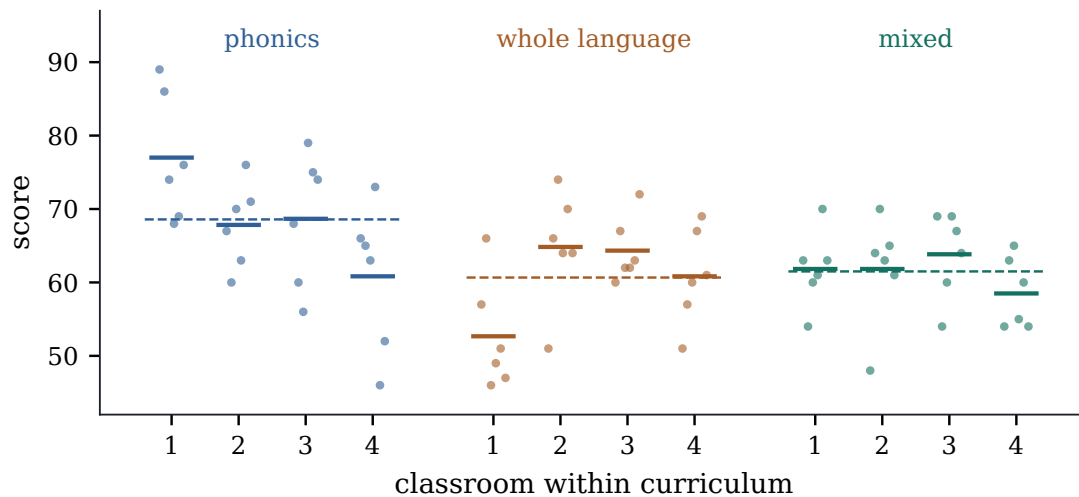



Figure 16.5.1: Reading scores of the 72 pupils by classroom, grouped by curriculum. Solid bars mark classroom means and dashed lines curriculum means. Classrooms vary visibly within each curriculum.

```
def Cen(k):
    return np.eye(k) - Jbar(k)

y = scores.ravel()  # pupils fastest, then
                    # classroom, then curriculum
P = {"curriculum": np.kron(np.kron(Cen(a), Jbar(b)), Jbar(m)),
     "classroom(curriculum)": np.kron(np.kron(np.eye(a), Cen(b)), Jbar(m)),
     "error": np.kron(np.kron(np.eye(a), np.eye(b)), Cen(m))}
ss = {k: y @ Pk @ y for k, Pk in P.items()}
df = {k: int(round(np.trace(Pk))) for k, Pk in P.items()}
ms = {k: ss[k] / df[k] for k in P}
for k in P:
    print(f"{k:22s} df {df[k]:2d} SS {ss[k]:8.1f} MS {ms[k]:7.1f}")
F_cls = ms["classroom(curriculum)"] / ms["error"]
F_cur_fixed = ms["curriculum"] / ms["error"]
F_cur_random = ms["curriculum"] / ms["classroom(curriculum)"]
print(f"classrooms within curricula: F = {F_cls:.2f}, p = {stats.f.sf(F_cls, df['classroom(curriculum)'], df['error']):.3f}")
print(f"curricula against pupils: F = {F_cur_fixed:.2f}, p = {stats.f.sf(F_cur_fixed, 2, df['error']):.3f}")
print(f"curricula against classrooms: F = {F_cur_random:.2f}, p = {stats.f.sf(F_cur_random, 2, df['classroom(curriculum)']):.3f}")
```

```
rng = np.random.default_rng(20260919)
reps = 20_000
Pc, Pb, Pe = P["curriculum"], P["classroom(curriculum)"], P["error"]
rej_fixed = rej_random = 0
for _ in range(reps):
    sim = (rng.normal(scale=4.0, size=(a, b))[:, :, None] + rng.normal(scale=8.0,
    size=(a, b, m))).ravel()
    msc, msb, mse = sim @ Pc @ sim / 2, sim @ Pb @ sim / 9, sim @ Pe @ sim / 60
    rej_fixed += msc / mse > stats.f.ppf(0.95, 2, 60)
    rej_random += msc / msb > stats.f.ppf(0.95, 2, 9)
```

```
print(f"no curriculum effect, random classrooms: rejection rate {rej_fixed / reps:.3f} "
      f"against pupils, {rej_random / reps:.3f} against classrooms")
```

Warning. The mistake of testing an outer factor against the within-level error, when the inner levels are a sample, is common enough to have a name, *pseudoreplication* (Hurlbert 1984). The pupils are not independent replicates of a curriculum. The classrooms are. With a fixed number of pupils, adding classrooms improves the curriculum comparison much more than adding pupils per classroom.

Nested and crossed structure are often combined, for example when every classroom is tested both before and after the year. The balanced decomposition is again a sum of Kronecker products, one for each meaningful term ([Exercise \(C2\)](#)). Unbalanced nested data lose the Kronecker structure, and random inner levels lead to the mixed models of Chapters 32 and 33.

16.5.4 Exercises

16.5.4.1 A. Check your understanding

Exercise A1.

Say whether each pair of factors is crossed or nested, and if nested, which is inside which. (a) Four fertilizers applied to plots of each of three wheat varieties. (b) Two suppliers, each delivering five batches, with several specimens tested from each batch. (c) Hospitals within regions, with patients within hospitals. (d) Three laboratories, each measuring portions of the same six reference samples.

16.5.4.2 B. Practice

Exercise B1.

In a balanced nested layout, permute the local labels j of B within one level i of A , leaving the data unchanged. Show that $SS_{B(A)}$ does not change, but that SS_B and SS_{AB} of the crossed analysis generally do. Conclude that only their sum has a meaning for nested data.

Solution. $SS_{B(A)} = m \sum_i \sum_j (\bar{y}_{ij.} - \bar{y}_{i..})^2$ sums over j within each i , so the order of the j within a level of A does not matter. $SS_B = am \sum_j (\bar{y}_{.j.} - \bar{y}_{...})^2$ depends on which inner levels share a label j across different i . For example, with $a = b = 2$, $m = 1$ and inner means $(1, 3)$ and $(1, 3)$, the column means are 1 and 3 and $SS_B = 4$. Swapping the labels in the second row gives inner means $(1, 3)$ and $(3, 1)$, column means 2 and 2, and $SS_B = 0$. By [Theorem 16.5.1\(a\)](#), $SS_B + SS_{AB} = SS_{B(A)}$ is unchanged, so the interaction absorbs the difference.

Exercise B2.

Show that $MS_A/MS_{B(A)}$ equals the one-way F statistic for A computed from the ab inner-level means $\bar{y}_{ij.}$, treated as b observations at each of the a levels. Interpret this as

an analysis in which the inner level is the unit.

Solution. Let $z_{ij} = \bar{y}_{ij..}$. The one-way analysis of the z_{ij} has between sum of squares $b \sum_i (\bar{z}_{i.} - \bar{z}_{..})^2$ on $a - 1$ degrees of freedom and within sum of squares $\sum_{i,j} (z_{ij} - \bar{z}_{i.})^2$ on $a(b - 1)$. Since $\bar{z}_{i.} = \bar{y}_{i..}$, these are SS_A/m and $SS_{B(A)}/m$ by [Theorem 16.5.1\(b\)](#). The factor m cancels in the ratio of mean squares. So the test of A against $B(A)$ is exactly the one-way analysis of variance of the inner-level means. The individual observations enter only through those means.

Exercise B3.

For a three-stage nested layout (a levels of A , b levels of B in each, c levels of C in each level of B , m observations at each level of C), verify that the four projections given in the text and $\mathbf{P}_0$ are mutually orthogonal and sum to $\mathbf{I}$, and find their ranks. Give the sum of squares for $C(B)$ in terms of means.

16.5.4.3 C. Going deeper

Exercise C1.

In the balanced nested layout of [Theorem 16.5.1](#), let $y_{ijk} = \mu + \alpha_i + b_{ij} + \varepsilon_{ijk}$, where the b_{ij} and ε_{ijk} are uncorrelated random variables with means zero and variances σ_B^2 and σ^2 . Show that $\text{Cov}(\mathbf{Y}) = \sigma^2 \mathbf{I} + m\sigma_B^2(\mathbf{I}_a \otimes \mathbf{I}_b \otimes \bar{\mathbf{J}}_m)$, and use [Theorem 2.3.1](#) to show that

$$E \text{MS}_A = \sigma^2 + m\sigma_B^2 + \frac{bm}{a-1} \sum_i (\alpha_i - \bar{\alpha})^2, \quad E \text{MS}_{B(A)} = \sigma^2 + m\sigma_B^2.$$

Compare [Exercise \(B1\)](#).

Solution. The vector of b_{ij} , each repeated m times, is $(\mathbf{I}_a \otimes \mathbf{I}_b \otimes \mathbf{1}_m)\mathbf{b}$, with covariance $\sigma_B^2(\mathbf{I}_a \otimes \mathbf{I}_b \otimes \mathbf{1}_m \mathbf{1}_m^\top) = m\sigma_B^2(\mathbf{I}_a \otimes \mathbf{I}_b \otimes \bar{\mathbf{J}}_m)$. Adding the error covariance gives $\Sigma = \text{Cov}(\mathbf{Y})$. By [Theorem 2.3.1](#), $E(\mathbf{Y}^\top \mathbf{P} \mathbf{Y}) = \text{tr}(\mathbf{P} \Sigma) + \boldsymbol{\mu}^\top \mathbf{P} \boldsymbol{\mu}$. For $\mathbf{P} = \mathbf{P}_A$: $\mathbf{P}_A(\mathbf{I} \otimes \mathbf{I} \otimes \bar{\mathbf{J}}_m) = \mathbf{P}_A$, so $\text{tr}(\mathbf{P}_A \Sigma) = (\sigma^2 + m\sigma_B^2)(a - 1)$, and $\boldsymbol{\mu}^\top \mathbf{P}_A \boldsymbol{\mu} = bm \sum_i (\alpha_i - \bar{\alpha})^2$. For $\mathbf{P}_{B(A)}$ likewise $\text{tr} = (\sigma^2 + m\sigma_B^2)a(b - 1)$, and $\mathbf{P}_{B(A)} \boldsymbol{\mu} = \mathbf{0}$ because the mean $\mu + \alpha_i$ does not vary within levels of A . Dividing by the degrees of freedom gives the result. Under H_A both mean squares have expectation $\sigma^2 + m\sigma_B^2$, which is why the ratio $\text{MS}_A/\text{MS}_{B(A)}$ is the natural test statistic.

Exercise C2.

Suppose each classroom of [Example \(Curricula and classrooms\)](#) is tested on two occasions (factor C , crossed with classroom), with m pupils per classroom and occasion. Order the observations by curriculum, classroom, occasion and replicate. Write the Kronecker projections for the terms A , $B(A)$, C , AC , $C \times B(A)$ and error, check that they are mutually orthogonal and sum to $\mathbf{I}$, and find their ranks.

16.6 Summary and Notes

16.6.1 Summary

Idea.

1. In a balanced layout every projection of interest is a Kronecker product of averaging matrices $\bar{\mathbf{J}}$, centring matrices $\mathbf{C}$ and identities. Products that differ by $\bar{\mathbf{J}}$ against $\mathbf{C}$ in some position are orthogonal, and ranks multiply ([Lemma 16.1.1](#)).
2. The additive two-way model has projection $\mathbf{P}_0 + \mathbf{P}_A + \mathbf{P}_B$. Row contrasts are estimated and judged as in a one-way layout that ignores the columns. With replication, the residual splits into interaction (lack of fit) and pure error. With one observation per cell, Tukey's one-degree-of-freedom test is an exact test against the kind of nonadditivity a wrong scale produces, and its slope suggests a power transformation ([Theorem 16.1.2](#)).
3. Interaction is a property of the table of cell means. It is present iff some interaction contrast (a zero-margin combination of cell means) is nonzero, and it has $(a - 1)(b - 1)$ dimensions whatever parameterization is used ([Definition 16.2.1](#), [Proposition 16.2.1](#)).
4. An increasing transformation can remove interaction only if the orders of the levels are the same everywhere. Qualitative (crossing) interaction cannot be removed. Main effects in the presence of interaction are weighted averages of simple effects, and their values, even their signs, depend on the weights ([Proposition 16.2.2](#), [Proposition 16.2.3](#)).
5. The balanced two-way layout with replication splits into grand mean, A , B , AB and error, with ranks 1, $a - 1$, $b - 1$, $(a - 1)(b - 1)$, $ab(m - 1)$. The F tests are exact, and the sequential, Type II and Type III sums of squares all coincide ([Theorem 16.3.1](#)).
6. Products of orthogonal contrasts split the interaction sum of squares into single degrees of freedom. Tukey's method covers the levels of one factor, Scheffé's all interaction contrasts, and Bonferroni or Holm planned families ([Proposition 16.3.2](#)).
7. A balanced factorial with t factors has one term for each subset of factors, with projection $\mathbf{P}_S$ built from centring matrices in the positions of S . Hierarchical models have order-free sums of squares, and $\mathbf{P}_S \mathbf{y}$ is an alternating sum of marginal means ([Theorem 16.4.1](#)). The three-factor interaction measures how a two-factor interaction changes across the third factor ([Proposition 16.4.2](#)).
8. In a 2^t factorial every term has one degree of freedom, and its effect is a difference of two means with variance $4\sigma^2/n$ ([Proposition 16.4.3](#)). Pooling terms into error is exact if they are null and chosen in advance, and conservative if they are chosen in advance but not null. It is invalid if they are chosen because they look small ([Proposition 16.4.4](#)).
9. A nested factor has no main effect. Its sum of squares is the crossed B plus AB , the variation among inner levels within each outer level ([Definition 16.5.1](#), [Theorem 16.5.1](#)). When the inner levels are a sample, the outer factor must be tested against the inner mean square, not against the within-level error.

16.6.2 Notes and sources

Sources. The coverage of this chapter was checked against Rencher and Schaalje (2008, chapter 14), Christensen (2020, chapter 7) and Seber and Lee (2003, sections 8.4–8.6). The projection approach follows the vector-space treatment of Christensen (2020), which develops interaction contrasts and their product construction ([Proposition 16.3.2](#)) with particular care. Rencher and Schaalje compute the expected mean squares directly, and projections replace that computation here. Scheffé (1959) remains the classical account of fixed-effects multiway layouts, including main effects and interactions defined with general weights (compare [Proposition 16.2.1\(v\)](#)).

Factorial experiments. Fisher (1935) argued for factorial experiments over one-factor-at-a-time trials: every observation contributes to every main effect, and interactions can be studied at all. Yates (1937) developed the analysis of 2^t and 3^t designs, including the algorithm for 2^t effects that carries his name. Box, Hunter and Hunter (2005) and Wu and Hamada (2009) are standard modern texts.

Tukey's test. The one-degree-of-freedom test is due to Tukey (1949). Its exact distribution follows from the theorem of Milliken and Graybill (1970) on regressors computed from the fitted values ([Theorem 14.7.2](#)), because Tukey's regressor is, after the additive model is projected out, the table of squared fitted values; [Exercise \(C1\)](#) made the same connection. The power rule [Eq. 16.1.5](#) is a heuristic, and Chapter 22 treats transformations systematically.

Interaction and scale. The dependence of interaction on the scale of measurement, and the order condition of [Proposition 16.2.2](#), belong to the theory of conjoint measurement of Luce and Tukey (1964). The counterexample of [Exercise \(C1\)](#) is an instance of their double cancellation condition. In clinical trials a treatment effect that changes sign across patient subgroups is called a *qualitative interaction*, and Gail and Simon (1985) gave a likelihood ratio test for it.

Unreplicated factorials and pooling. The half-normal plot is due to Daniel (1959), and the pseudo standard error of [Exercise \(C1\)](#) to Lenth (1989). The effect of preliminary tests on subsequent inference was first studied by Bancroft (1944), and the simulation of [Figure 16.4.2](#) illustrates his conclusions in a simple case.

Nested designs. Hurlbert (1984) named *pseudoreplication*, treating observations within an experimental unit as independent replicates of the unit's treatment, and documented how widespread it was in ecology. Random inner levels are taken up with variance components in Chapters 32 and 33. Unbalanced multiway data are the subject of [Chapter 17](#), and randomized blocks and Latin squares, additive layouts whose structure comes from randomization, of [Chapter 18](#).

16.6.3 References

- Bancroft, T. A. (1944). On Biases in Estimation Due to the Use of Preliminary Tests of Significance. *The Annals of Mathematical Statistics* 15(2), 190–204.
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Chapter 17

Unbalanced Data, Cell Means and Empty Cells

Balanced layouts give orthogonal decompositions, and then every sum of squares answers one clear question. Real experiments lose plots and patients, and observational data are unbalanced from the start. With unequal cell counts the same data produce several different “sums of squares for A ”, each testing its own hypothesis. This chapter starts from the parameterization in which nothing is ambiguous, a mean for every cell, and writes every hypothesis as linear constraints on the cell means. That shows exactly what the Type I, II and III sums of squares test, why Type III needs contrast codings, how the additive model is a constrained cell-means model, and which questions survive when some cells are empty.

What you need. Projections and nested models from [Chapter 6](#) ([Theorem 6.5.1](#), [Theorem 6.5.2](#)). Estimability, factor codings and connectedness from [Chapter 8](#) ([Theorem 8.3.1](#), [Definition 8.5.1](#), [Theorem 8.6.1](#), [Proposition 8.6.2](#)). The definitions of the Type I, II and III sums of squares and the first cell-mean results in [Chapter 9](#) ([Definition 9.3.1](#), [Definition 9.3.2](#), [Theorem 9.3.2](#), [Theorem 9.2.4](#)). Testable hypotheses and restricted least squares from [Chapter 11](#) ([Theorem 11.3.2](#), [Proposition 11.3.3](#)). The one-way layout of [Chapter 15](#) and the balanced two-way layout of [Chapter 16](#).

17.0.1 Roadmap

- [The cell-means model](#)
- [Types of sums of squares and what they test](#)
- [Constrained models](#)
- [Empty cells and estimability](#)
- [Summary and notes](#)

17.1 The cell-means model

With several factors the observations fall into *cells*, one for each combination of levels. The effects model of [Chapter 8](#) writes the mean of a cell as $\mu + \alpha_i + \beta_j + \gamma_{ij}$ and pays for it with side conditions and codings. The cell-means model gives each cell its own parameter and nothing

else. It fits the same values as the effects model with all interactions, but every question about the factors becomes a statement about a table of means, and the counts enter only through the precision with which that table is estimated.

17.1.1 The model

Definition 17.1.1 — Cell-means model

Let factors $F_1, \dots, F_k$ have $a_1, \dots, a_k$ levels. A **cell** is a combination $\omega = (i_1, \dots, i_k)$ of levels, and there are $p = a_1 a_2 \cdots a_k$ cells. Cell ω contains $n_\omega \geq 0$ observations $y_{\omega 1}, \dots, y_{\omega n_\omega}$. It is **occupied** if $n_\omega \geq 1$ and **empty** otherwise. Write O for the set of occupied cells, $m = |O|$ and $n = \sum_\omega n_\omega$. The **cell-means model** is

$$y_{\omega l} = \mu_\omega + \varepsilon_{\omega l}, \quad l = 1, \dots, n_\omega,$$

with errors that are uncorrelated with mean zero and variance σ^2 , and independent $N(0, \sigma^2)$ when we test. In matrix form $E(\mathbf{Y}) = \mathbf{W}\boldsymbol{\mu}$, where $\boldsymbol{\mu} \in \mathbb{R}^p$ collects the cell means and $\mathbf{W}$ is the $n \times p$ matrix whose column ω indicates the observations in cell ω . We write $\mathbf{D} = \mathbf{W}^\top \mathbf{W} = \text{diag}(n_\omega)$, and $\bar{y}_\omega$ for the average of an occupied cell.

Empty cells keep their entries in $\boldsymbol{\mu}$: their means do not enter the distribution of the data, but they are properties of the population (Section 17.4). In two-factor notation, cells are pairs (i, j) , counts are n_{ij} , and we order the cells row by row: $\boldsymbol{\mu} = (\mu_{11}, \dots, \mu_{1b}, \mu_{21}, \dots, \mu_{ab})^\top$. With this ordering, a Kronecker product $\mathbf{P} \otimes \mathbf{Q}$ acts on $\boldsymbol{\mu}$ by transforming the $a \times b$ table (μ_{ij}) into $\mathbf{P}(\mu_{ij})\mathbf{Q}^\top$ (Proposition 1.10.1(f) applied to the transposed table).

By Proposition 8.6.2(a), the overparameterized two-way model with interaction has column space $\mathcal{C}(\mathbf{W})$, so the cell-means model is its identifiable parameterization. It is also a one-way layout (Definition 15.1.1) whose groups are the m occupied cells. The factorial structure plays no part in fitting.

Proposition 17.1.1 — Inference with cell means

In the model of Definition 17.1.1, let $\mathbf{D}^+$ be the diagonal matrix with entries $1/n_\omega$ for occupied cells and 0 for empty ones, and let $\bar{\mathbf{y}} \in \mathbb{R}^p$ hold $\bar{y}_\omega$ for occupied cells and 0 for empty ones. Then:

- (a) $\text{rank}(\mathbf{W}) = m$, and $\mathbf{M} = \mathbf{W}\mathbf{D}^+\mathbf{W}^\top$ replaces each observation by the average of its cell;
- (b) $\boldsymbol{\lambda}^\top \boldsymbol{\mu}$ is estimable iff $\lambda_\omega = 0$ for every empty cell. Its best linear unbiased estimator is $\boldsymbol{\lambda}^\top \bar{\mathbf{y}} = \sum_{\omega \in O} \lambda_\omega \bar{y}_\omega$, with variance $\sigma^2 \sum_{\omega \in O} \lambda_\omega^2 / n_\omega$;
- (c) $\text{SSE} = \sum_{\omega \in O} \sum_l (y_{\omega l} - \bar{y}_\omega)^2$ has $n - m$ degrees of freedom, and $s^2 = \text{SSE} / (n - m)$, the pooled within-cell variance, is unbiased for σ^2 ;
- (d) if $\mathbf{L}$ is a $q \times p$ matrix of rank q whose rows vanish on the empty cells, $\mathbf{d} \in \mathbb{R}^q$, and $n > m$, the F statistic for $H : \mathbf{L}\boldsymbol{\mu} = \mathbf{d}$ is

$$F = \frac{(\mathbf{L}\bar{\mathbf{y}} - \mathbf{d})^\top (\mathbf{L}\mathbf{D}^+\mathbf{L}^\top)^{-1} (\mathbf{L}\bar{\mathbf{y}} - \mathbf{d})}{q s^2} \sim F(q, n - m, \gamma_H), \quad (17.1.1)$$

where $\gamma_H = (\mathbf{L}\boldsymbol{\mu} - \mathbf{d})^\top (\mathbf{LD}^+\mathbf{L}^\top)^{-1} (\mathbf{L}\boldsymbol{\mu} - \mathbf{d}) / \sigma^2$.

Proof.

- (a) The columns of $\mathbf{W}$ for occupied cells are nonzero indicators with disjoint supports, and those for empty cells are zero. So $\text{rank}(\mathbf{W}) = m$. Since $\mathbf{D}\mathbf{D}^+\mathbf{D} = \mathbf{D}$, $\mathbf{D}^+$ is a generalized inverse of $\mathbf{W}^\top\mathbf{W}$, and $\mathbf{M} = \mathbf{W}\mathbf{D}^+\mathbf{W}^\top$ by Theorem 6.3.7. The entry of $\mathbf{W}\mathbf{D}^+\mathbf{W}^\top\mathbf{y}$ for an observation in cell ω is $n_\omega^{-1} \sum_l y_{\omega l} = \bar{y}_\omega$.
- (b) The rows of $\mathbf{W}$ are the coordinate vectors $\mathbf{e}_\omega^\top$ of the occupied cells, so $\mathcal{C}(\mathbf{W}^\top) = \text{span}\{\mathbf{e}_\omega : \omega \in O\}$, and Theorem 8.3.1 gives the criterion. The vector $\mathbf{D}^+\mathbf{W}^\top\mathbf{y} = \bar{\mathbf{y}}$ solves the normal equations (Theorem 6.4.2), so $\boldsymbol{\lambda}^\top \bar{\mathbf{y}}$ is the least squares estimator of an estimable $\boldsymbol{\lambda}^\top \boldsymbol{\mu}$, and it is best linear unbiased by the Gauss–Markov theorem (Theorem 7.2.1). The cell averages are uncorrelated with variances σ^2/n_ω , which gives the variance.
- (c) $\text{SSE} = \|(\mathbf{I} - \mathbf{M})\mathbf{y}\|^2$ is the displayed sum by (a), and $E(\text{SSE}) = \sigma^2 \text{tr}(\mathbf{I} - \mathbf{M}) = \sigma^2(n - m)$ by Theorem 2.3.1, since $(\mathbf{I} - \mathbf{M})\mathbf{W}\boldsymbol{\mu} = \mathbf{0}$.
- (d) Apply Theorem 11.3.2 with $\mathbf{G} = \mathbf{D}^+$, $\hat{\boldsymbol{\beta}} = \bar{\mathbf{y}}$ and $\boldsymbol{\Lambda}^\top = \mathbf{L}$, which is testable by (b). The matrix $\mathbf{LD}^+\mathbf{L}^\top$ is nonsingular: for $\mathbf{x} \neq \mathbf{0}$, $\mathbf{x}^\top \mathbf{LD}^+\mathbf{L}^\top \mathbf{x} = \sum_{\omega \in O} (\mathbf{L}^\top \mathbf{x})_\omega^2 / n_\omega$. If this were zero, $\mathbf{L}^\top \mathbf{x}$ would vanish on O , and hence everywhere, because the rows of $\mathbf{L}$ vanish off O . That contradicts $\text{rank}(\mathbf{L}) = q$.

■

The counts enter only through $\mathbf{D}^+$, the variances of the cell averages. An unbalanced cell-means analysis has one fitting step, averaging within cells, and one inferential tool, Eq. 17.1.1. The work lies in choosing $\mathbf{L}$.

17.1.2 Standard hypotheses as Kronecker products

Let every cell of a two-way layout be occupied. Let $\mathbf{C}_a$ be an $a \times (a - 1)$ matrix whose columns form a basis of the contrasts $\{\mathbf{u} \in \mathbb{R}^a : \mathbf{1}^\top \mathbf{u} = 0\}$, and define $\mathbf{C}_b$ in the same way. These are coding matrices in the sense of Definition 8.5.1. The $a \times a$ centring matrix of Chapter 16 has the same column space, but its columns are linearly dependent, so we keep the thinner matrix here and count restrictions by its columns. The three classical hypotheses are

$$H_A : (\mathbf{C}_a^\top \otimes b^{-1} \mathbf{1}_b^\top) \boldsymbol{\mu} = \mathbf{0}, \quad H_B : (a^{-1} \mathbf{1}_a^\top \otimes \mathbf{C}_b^\top) \boldsymbol{\mu} = \mathbf{0}, \quad H_{AB} : (\mathbf{C}_a^\top \otimes \mathbf{C}_b^\top) \boldsymbol{\mu} = \mathbf{0}. \quad (17.1.2)$$

By the remark on Kronecker products above, H_A says that $\mathbf{C}_a^\top$ annihilates the vector of **unweighted row means** $\bar{\mu}_{i.} = b^{-1} \sum_j \mu_{ij}$, that is, that these means are all equal. H_B says the same of the unweighted column means. H_{AB} says that $\mathbf{C}_a^\top (\mu_{ij}) \mathbf{C}_b = \mathbf{0}$. Because $\mathcal{C}(\mathbf{C}_a) = \mathbf{1}_a^\perp$ and $\mathcal{C}(\mathbf{C}_b) = \mathbf{1}_b^\perp$, this holds iff the doubly centred table $\mu_{ij} - \bar{\mu}_{i.} - \bar{\mu}_{.j} + \bar{\mu}_{..}$ vanishes, that is, iff the cell means are additive (Definition 16.2.1). The ranks are $a - 1$, $b - 1$ and $(a - 1)(b - 1)$ (Proposition 1.10.1(c)), and the hypotheses do not depend on the basis of contrasts (Exercise (B2)). They are the hypotheses tested in the balanced layout by Theorem 16.3.1, where every type of sum of squares tests them.

17.1.3 Weighted and unweighted marginal means

A row of the table can be summarized by any average of its cells. Two choices dominate practice. The **weighted row mean** $\bar{\mu}_i^w = n_{i\cdot}^{-1} \sum_j n_{ij} \mu_{ij}$ weights each cell by its count. Its natural estimator is the raw average $\bar{y}_{i\cdot}$ of all observations in row i . The **unweighted row mean** $\bar{\mu}_i$ gives each column the same weight. Its estimator is the average of the cell averages, $\bar{y}_i = b^{-1} \sum_j \bar{y}_{ij}$. More generally, one may fix weights $w_1, \dots, w_b$ summing to one, such as the shares of the levels of B in a target population, and estimate $\sum_j w_j \mu_{ij}$.

Proposition 17.1.2 — Precision of marginal means

Fix a row i with all $n_{ij} \geq 1$, and weights $w_1, \dots, w_b$ with $\sum_j w_j = 1$. The estimator $\sum_j w_j \bar{y}_{ij}$ of $\sum_j w_j \mu_{ij}$ is unbiased with variance $\sigma^2 \sum_j w_j^2 / n_{ij}$. Among all such weights the variance is smallest, and equal to $\sigma^2 / n_{i\cdot}$, exactly for the count weights $w_j = n_{ij} / n_{i\cdot}$. In particular

$$\text{Var}(\bar{y}_i) = \frac{\sigma^2}{b^2} \sum_j \frac{1}{n_{ij}} \geq \frac{\sigma^2}{n_{i\cdot}} = \text{Var}(\bar{y}_{i\cdot}),$$

with equality iff the counts $n_{i1}, \dots, n_{ib}$ are equal.

Proof. Unbiasedness and the variance come from Proposition 17.1.1(b). By the Cauchy–Schwarz inequality,

$$1 = \left(\sum_j \frac{w_j}{\sqrt{n_{ij}}} \sqrt{n_{ij}} \right)^2 \leq \left(\sum_j \frac{w_j^2}{n_{ij}} \right) n_{i\cdot},$$

with equality iff $w_j / \sqrt{n_{ij}}$ is proportional to $\sqrt{n_{ij}}$, that is, $w_j \propto n_{ij}$. The unweighted mean is the case $w_j = 1/b$, which is proportional to n_{ij} iff the counts are equal. ■

The count weights are the most *precise*, but precision is the wrong criterion for choosing a target. The weighted mean $\bar{\mu}_i^w$ is the mean of row i with the levels of B mixed as they happened to be mixed in this study. If the counts are an accident of the design, such as lost plots, that mixture describes nothing. If they are proportional to a real population, as in a random sample, $\bar{\mu}_i^w$ may be exactly the target. When the rows have different mixes, comparing weighted means also compares mixes of B , and that is what the Type I sum of squares of a factor entered first tests (Section 17.2). Choose the weights to describe a population, and accept the variance that follows.

17.1.4 An unbalanced field trial

Example. (Three varieties at three sites)

A (synthetic) trial compared three wheat varieties V1, V2 and V3 at three sites S1, S2 and S3, with yield in tonnes per hectare. Each site grew mostly the variety it already knew. There are 37 plots in the $m = 9$ cells:

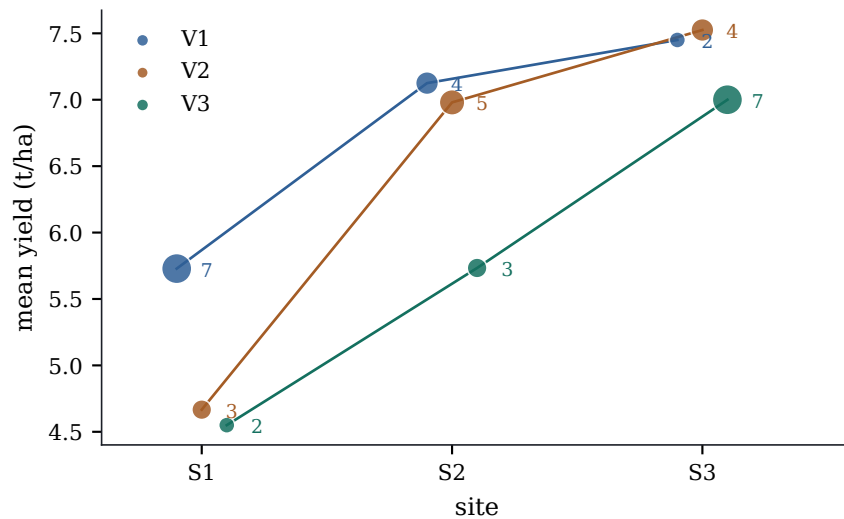


Figure 17.1.1: Cell means of the variety trial; point areas and labels give the cell counts.

	S1	S2	S3	all sites
V1	7 plots, 5.73	4, 7.12	2, 7.45	13, 6.423
V2	3, 4.67	5, 6.98	4, 7.53	12, 6.583
V3	2, 4.55	3, 5.73	7, 7.00	12, 6.275

The last column gives each variety's raw average, the estimate of its weighted mean $\bar{\mu}_i^w$. The pooled within-cell variance is $s^2 = 0.3136$ on $n - m = 28$ degrees of freedom, so the standard errors of the cell averages range from 0.212 (seven plots) to 0.396 (two plots).

Figure 17.1.1 shows the table. V1 is the best variety at S1 and S2 and nearly ties with V2 at S3, yet its raw average, 6.423, is below that of V2, 6.583, because V1 was grown mostly at the poorest site. The unweighted means 6.768, 6.391 and 5.761 compare the varieties over the same three sites, and V1 comes first. Their standard errors, 0.176, 0.165 and 0.184, exceed those of the raw averages (0.155, 0.162 and 0.162), as Proposition 17.1.2 requires. For V1 the ratio depends only on the counts: $\sqrt{13(1/7 + 1/4 + 1/2)}/3 = \sqrt{11.607}/3 \approx 1.136$.

The tests Eq. 17.1.2, computed with Eq. 17.1.1, give $F = 7.90$ on 2 and 28 degrees of freedom for equal unweighted variety means ($p = 0.0019$), $F = 44.06$ for equal unweighted site means, and $F = 1.71$ on 4 and 28 degrees of freedom for additivity ($p = 0.1755$). The difference between V1 and V2, averaged over sites without weights, is estimated as 0.377 with standard error 0.242. With $t_{28,0.025} = 2.048$, the 95% interval is from -0.12 to 0.87 . The raw averages give a difference of -0.160 , with the opposite sign.

```
import numpy as np
import pandas as pd
from scipy import stats
counts = np.array([[7, 4, 2],          # rows: varieties V1-V3; columns: sites S1-S3
                  [3, 5, 4],
                  [2, 3, 7]])
true_means = np.array([[5.8, 7.2, 7.6],
                      [5.0, 6.5, 7.5],
                      [4.4, 5.9, 7.2]])
```

```
rng = np.random.default_rng(20172)
rows = [(f"V{i + 1}", f"S{j + 1}", round(true_means[i, j] + rng.normal(0, 0.5), 1))
         for i in range(3) for j in range(3) for _ in range(counts[i, j])]
trial = pd.DataFrame(rows, columns=["variety", "site", "y"])
```

```
cells = trial.groupby(["variety", "site"])["y"]
ybar = cells.mean().unstack().to_numpy()           # 3 x 3 table of cell means
n_ij = cells.size().unstack().to_numpy()
n, m = n_ij.sum(), (n_ij > 0).sum()
sse = ((trial["y"] - cells.transform("mean")) ** 2).sum()
s2 = sse / (n - m)                                # pooled within-cell variance
se_cell = np.sqrt(s2 / n_ij)
weighted_rows = (n_ij * ybar).sum(axis=1) / n_ij.sum(axis=1) # = raw variety
                        averages
unweighted_rows = ybar.mean(axis=1)
se_weighted = np.sqrt(s2 / n_ij.sum(axis=1))
se_unweighted = np.sqrt(s2 * (1 / n_ij).sum(axis=1)) / 3
print(np.round(ybar, 3))
print("weighted :", weighted_rows.round(3), " se", se_weighted.round(3))
print("unweighted:", unweighted_rows.round(3), " se", se_unweighted.round(3))
```

```
def hypothesis_test(L, ybar, n_ij, s2, dfe):
    """F test of  $L\mu = 0$  in the cell-means model, all cells occupied (cells in row-
    major order)."""
    u = L @ ybar.ravel()
    ss = u @ np.linalg.solve(L @ np.diag(1 / n_ij.ravel()) @ L.T, u)
    q = np.linalg.matrix_rank(L)
    F = ss / q / s2
    return ss, q, F, stats.f.sf(F, q, dfe)

C3 = np.array([[1.0, 0.0], [0.0, 1.0], [-1.0, -1.0]]) # any 3 x 2 matrix of
                        contrasts will do
avg3 = np.ones((1, 3)) / 3
L_A = np.kron(C3.T, avg3)           # unweighted variety means equal
L_B = np.kron(avg3, C3.T)           # unweighted site means equal
L_AB = np.kron(C3.T, C3.T)          # all interaction contrasts zero
for name, L in [("variety", L_A), ("site", L_B), ("interaction", L_AB)]:
    ss, q, F, p = hypothesis_test(L, ybar, n_ij, s2, n - m)
    print(f"{name:12s} SS = {ss:6.3f} df = {q} F = {F:6.2f} p = {p:.4f}")
```

The script also checks that the variety sum of squares 4.958 is the increase in residual sum of squares when the cell means are fitted subject to H_A (Theorem 11.3.2(b)), and that a Helmert-type contrast matrix gives the same test. With one factor, counts only decide which contrasts split the between-groups sum of squares orthogonally (Theorem 15.4.1). With several they decide which hypotheses the printed sums of squares address.

17.1.5 Exercises

17.1.5.1 A. Check your understanding

Exercise A1.

A 2×3 layout has counts $n_{11} = 4$, $n_{12} = 0$, $n_{13} = 2$, $n_{21} = 3$, $n_{22} = 5$, $n_{23} = 1$. Give m ,

n , the rank of $\mathbf{W}$ and the error degrees of freedom of the cell-means model. Which of $\mu_{11} - \mu_{21}$, $\bar{\mu}_{1\cdot} - \bar{\mu}_{2\cdot}$ and $(\mu_{11} + \mu_{13})/2 - (\mu_{21} + \mu_{23})/2$ are estimable?

Exercise A2.

In [Example \(Three varieties at three sites\)](#), compute from the counts alone the ratio of the standard errors of the unweighted and weighted means of V3, and check it against the quoted standard errors.

17.1.5.2 B. Practice

Exercise B1.

Show that the F statistic of [Eq. 17.1.1](#) for a single contrast $\boldsymbol{\lambda}^\top \boldsymbol{\mu}$ with all cells occupied is the one-way contrast statistic of [Proposition 15.2.1](#), applied to the one-way layout whose groups are the cells. Which t statistic is it the square of, and what is its distribution under the hypothesis?

Solution. With $q = 1$, $\mathbf{L} = \boldsymbol{\lambda}^\top$ and $\mathbf{d} = 0$, [Eq. 17.1.1](#) is $F = (\sum_{\omega} \lambda_{\omega} \bar{y}_{\omega})^2 / (s^2 \sum_{\omega} \lambda_{\omega}^2 / n_{\omega})$. This is the contrast sum of squares $(\sum \lambda_{\omega} \bar{y}_{\omega})^2 / \sum (\lambda_{\omega}^2 / n_{\omega})$ of the one-way layout with groups ω , divided by the one-way error mean square, which is the pooled within-cell variance. Its signed root is $t = \boldsymbol{\lambda}^\top \bar{\mathbf{y}} / (s \sqrt{\sum \lambda_{\omega}^2 / n_{\omega}})$, which has a $t(n - m)$ distribution under the hypothesis. The factorial labels of the cells play no role.

Exercise B2.

Let $\mathbf{C}_a$ and $\mathbf{C}'_a$ be two $a \times (a - 1)$ matrices whose columns are bases of $\mathbf{1}_a^\perp$. Show that $\mathbf{C}'_a = \mathbf{C}_a \mathbf{T}$ for a nonsingular $\mathbf{T}$, and deduce that the hypotheses H_A and H_{AB} of [Eq. 17.1.2](#), and their F statistics, are the same for both.

Solution. Both column spaces equal $\mathbf{1}_a^\perp$, so each column of $\mathbf{C}'_a$ is a combination of the columns of $\mathbf{C}_a$: $\mathbf{C}'_a = \mathbf{C}_a \mathbf{T}$. The rank of $\mathbf{C}'_a$ is $a - 1$, so $\mathbf{T}$ is nonsingular. By [Proposition 1.10.1\(a\)](#), $(\mathbf{C}'_a)^\top \otimes \mathbf{v}^\top = (\mathbf{T}^\top \otimes \mathbf{1})(\mathbf{C}_a^\top \otimes \mathbf{v}^\top)$ for any row vector $\mathbf{v}^\top$, and similarly for $(\mathbf{C}'_a)^\top \otimes \mathbf{C}_b^\top$. So the new hypothesis matrix is $\mathbf{S}\mathbf{L}$ with $\mathbf{S}$ nonsingular. Then $\mathbf{S}\mathbf{L}\boldsymbol{\mu} = \mathbf{0}$ iff $\mathbf{L}\boldsymbol{\mu} = \mathbf{0}$. In [Eq. 17.1.1](#),

$$(\mathbf{S}\mathbf{L}\bar{\mathbf{y}})^\top (\mathbf{S}\mathbf{L}\mathbf{D}^+ \mathbf{L}^\top \mathbf{S}^\top)^{-1} \mathbf{S}\mathbf{L}\bar{\mathbf{y}} = (\mathbf{L}\bar{\mathbf{y}})^\top (\mathbf{L}\mathbf{D}^+ \mathbf{L}^\top)^{-1} \mathbf{L}\bar{\mathbf{y}}.$$

Exercise B3.

In [Example \(Three varieties at three sites\)](#), suppose that sites like S1, S2 and S3 make up 50%, 30% and 20% of the region where the varieties would be grown. Write down the estimate of the region-weighted mean yield of V1 and its standard error, and do the same for the region-weighted difference between V1 and V3. Compute them with the listing. Which of the three kinds of marginal mean in this section would you report to a grower in the region?

17.1.5.3 C. Going deeper

Exercise C1.

A row of a two-way layout will receive N observations, to be allocated as $n_{i1}, \dots, n_{ib}$, with the aim of estimating $\sum_j w_j \mu_{ij}$ for fixed weights $w_j > 0$ summing to one. Treating the n_{ij} as real numbers, show that the variance of $\sum_j w_j \bar{y}_{ij}$ is minimized by $n_{ij} = N w_j$, with minimum σ^2/N . What does this say about how the counts should relate to the target of inference?

Solution. Minimize $\sum_j w_j^2/n_{ij}$ subject to $\sum_j n_{ij} = N$. By Cauchy–Schwarz, $1 = (\sum_j w_j)^2 = (\sum_j (w_j/\sqrt{n_{ij}})\sqrt{n_{ij}})^2 \leq N \sum_j w_j^2/n_{ij}$, with equality iff $n_{ij} \propto w_j$, that is, $n_{ij} = N w_j$. So the variance is at least σ^2/N , with equality exactly there. When the counts are proportional to the weights of the target, the target is estimated by the raw row average, the weighted and target means coincide, and no precision is lost. A design should allocate observations in proportion to the target weights, which for an unweighted target means equal replication.

Exercise C2.

In an $a \times b \times d$ layout, where C has d levels and all cells are occupied, order the cells lexicographically. Write, as Kronecker products, hypothesis matrices for: (a) equality of the unweighted A means $\bar{\mu}_{i..}$; (b) absence of the AB interaction in the table $\bar{\mu}_{ij.}$ of means averaged over C ; (c) absence of the three-factor interaction (Proposition 16.4.2). Give their ranks.

17.2 Types of sums of squares and what they test

Section 9.3 defined the Type I, II and III sums of squares by the subspaces they project onto (Definition 9.3.1, Definition 9.3.2). It also found the cell-mean hypotheses of three of them in the two-way layout: equal weighted row means for a factor entered first, equal unweighted row means for Type III, and a single column of the table for last-in sums of squares under reference coding (Theorem 9.3.2). This section completes the list with one lemma, and then shows that the hypothesis of a main effect's last-in sum of squares is decided by the coding of the *other* factor.

17.2.1 One lemma for every sum of squares

Every term of a factorial model has columns that are constant within cells, so every model matrix we meet has the form $\mathbf{W}\mathbf{A}$ for a matrix $\mathbf{A}$ with p rows, one for each cell. On the cell scale, projections become weighted by the counts: write

$$\mathbf{H}_\mathbf{A} = \mathbf{A}(\mathbf{A}^\top \mathbf{D} \mathbf{A})^{-1} \mathbf{A}^\top \mathbf{D},$$

the operator that fits a table of cell means by the columns of $\mathbf{A}$ with weights n_ω .

Lemma 17.2.1 — The hypothesis of an extra sum of squares

Let $\mathbf{X}_1 = \mathbf{W}\mathbf{A}_1$ and $\mathbf{X}_2 = \mathbf{W}\mathbf{A}_2$, let $\mathbf{P}_1$ project onto $\mathcal{C}(\mathbf{X}_1)$, and let $\mathbf{Q}$ be the projection onto $\mathcal{C}([\mathbf{X}_1, \mathbf{X}_2])$ minus $\mathbf{P}_1$, so that $\text{SS}(\mathbf{X}_2 \mid \mathbf{X}_1) = \mathbf{y}^\top \mathbf{Q}\mathbf{y}$. For every table of cell means $\boldsymbol{\mu}$:

- (a) $\mathbf{P}_1 \mathbf{W}\boldsymbol{\mu} = \mathbf{W}\mathbf{H}_{\mathbf{A}_1} \boldsymbol{\mu}$;
- (b) $\mathbf{Q}\mathbf{W}\boldsymbol{\mu} = \mathbf{0}$ iff $\mathbf{A}_2^\top \mathbf{D}(\mathbf{I} - \mathbf{H}_{\mathbf{A}_1})\boldsymbol{\mu} = \mathbf{0}$;
- (c) in the cell-means model, $\text{SS}(\mathbf{X}_2 \mid \mathbf{X}_1)$ is the sum of squares of the hypothesis in (b), and its F ratio with the cell-means error mean square is the F statistic Eq. 17.1.1 of that hypothesis.

Proof.

- (a) $\mathbf{X}_1^\top \mathbf{X}_1 = \mathbf{A}_1^\top \mathbf{D}\mathbf{A}_1$, so by Theorem 6.3.7 $\mathbf{P}_1 \mathbf{W}\boldsymbol{\mu} = \mathbf{W}\mathbf{A}_1(\mathbf{A}_1^\top \mathbf{D}\mathbf{A}_1)^{-1} \mathbf{A}_1^\top \mathbf{W}^\top \mathbf{W}\boldsymbol{\mu} = \mathbf{W}\mathbf{H}_{\mathbf{A}_1} \boldsymbol{\mu}$.
- (b) By Lemma 9.2.1, $\mathbf{Q}$ projects onto $\mathcal{C}(\mathbf{T})$ with $\mathbf{T} = (\mathbf{I} - \mathbf{P}_1)\mathbf{X}_2$. A projection annihilates a vector iff the vector is orthogonal to its range, so $\mathbf{Q}\mathbf{W}\boldsymbol{\mu} = \mathbf{0}$ iff $\mathbf{T}^\top \mathbf{W}\boldsymbol{\mu} = \mathbf{0}$. By (a),

$$\mathbf{T}^\top \mathbf{W}\boldsymbol{\mu} = \mathbf{X}_2^\top (\mathbf{I} - \mathbf{P}_1) \mathbf{W}\boldsymbol{\mu} = \mathbf{A}_2^\top \mathbf{W}^\top \mathbf{W}(\mathbf{I} - \mathbf{H}_{\mathbf{A}_1})\boldsymbol{\mu} = \mathbf{A}_2^\top \mathbf{D}(\mathbf{I} - \mathbf{H}_{\mathbf{A}_1})\boldsymbol{\mu}.$$

- (c) Let $\mathcal{S}_0 = \{\mathbf{W}\boldsymbol{\mu} : \mathbf{Q}\mathbf{W}\boldsymbol{\mu} = \mathbf{0}\}$, the mean vectors allowed by the hypothesis. Since $\mathcal{C}(\mathbf{Q}) \subseteq \mathcal{C}(\mathbf{W})$, the orthogonal complement of $\mathcal{S}_0$ within $\mathcal{C}(\mathbf{W})$ is $\mathcal{C}(\mathbf{Q})$, and $\mathbf{y}^\top \mathbf{Q}\mathbf{y}$ is the drop in residual sum of squares from $\mathcal{S}_0$ to $\mathcal{C}(\mathbf{W})$ (Eq. 6.5.2). That is the sum of squares of the hypothesis (Theorem 11.3.2(b)), whose F statistic is Eq. 17.1.1 by Proposition 17.1.1(d). ■

Part (b) says that the extra sum of squares of $\mathbf{X}_2$ tests whether the residual of the true cell means, after the terms already fitted and weighted by the counts, is orthogonal to the new term. Unless $\mathbf{D}$ is a multiple of the identity, the counts enter the hypothesis. That is the root of every difference between the types.

17.2.2 The complete list for two factors

Let the $a \times b$ layout have all $n_{ij} \geq 1$, with row totals $n_{i.}$, column totals $n_{.j}$ and $n = \sum_{ij} n_{ij}$. Besides the weighted row means $\bar{\mu}_i^w$ of Section 17.1, write

$$\bar{\mu}_{.j}^w = \frac{1}{n_{.j}} \sum_i n_{ij} \mu_{ij}, \quad \bar{\mu}^w = \frac{1}{n} \sum_{i,j} n_{ij} \mu_{ij}$$

for the weighted column means and the weighted grand mean. As in Chapter 9, $\text{SS}(B \mid \mu, A)$ denotes the extra sum of squares of the columns of B given the intercept and A .

Theorem 17.2.2 — The cell-mean hypotheses of the types of sums of squares

In the two-way layout with all cells occupied:

- (a) $SS(A \mid \mu)$, the Type I sum of squares of A entered first, addresses $\bar{\mu}_1^w = \cdots = \bar{\mu}_a^w$;
- (b) $SS(B \mid \mu, A)$, which is both the Type I sum of squares of B entered second and the Type II sum of squares of B , addresses

$$\sum_i n_{ij} (\mu_{ij} - \bar{\mu}_i^w) = 0 \quad \text{for } j = 1, \dots, b; \quad (17.2.1)$$

- (c) $SS(A \mid \mu, B)$, the Type II sum of squares of A , addresses

$$\sum_j n_{ij} (\mu_{ij} - \bar{\mu}_{.j}^w) = 0 \quad \text{for } i = 1, \dots, a; \quad (17.2.2)$$

- (d) the Type III sum of squares of A addresses $\bar{\mu}_{1.} = \cdots = \bar{\mu}_{a.}$;
- (e) $SS(AB \mid \mu, A, B)$, the interaction entry of every type, addresses additivity: $\mu_{ij} - \mu_{il} - \mu_{kj} + \mu_{kl} = 0$ for all i, k, j, l .

The hypotheses have $a - 1$, $b - 1$, $a - 1$, $a - 1$ and $(a - 1)(b - 1)$ degrees of freedom, and each sum of squares is the sum of squares [Eq. 17.1.1](#) of its hypothesis. The statements for B follow by exchanging the factors.

Proof. Parts (a) and (d) are [Theorem 9.3.2\(a\)](#) and (b). By [Lemma 17.2.1\(c\)](#), each sum of squares is the sum of squares of the hypothesis given by [Lemma 17.2.1\(b\)](#), and it remains to identify that hypothesis and count its degrees of freedom.

- (b) Take $\mathbf{X}_1 = [\mathbf{1}, \mathbf{Z}_A]$ and $\mathbf{X}_2 = \mathbf{Z}_B$. The columns of $\mathbf{X}_1$ span the vectors that are constant within rows, so $\mathbf{P}_1$ replaces each entry of a vector by the average of its row. Applied to $\mathbf{W}\boldsymbol{\mu}$, this gives $\bar{\mu}_i^w$ in row i . The entry j of $\mathbf{Z}_B^\top (\mathbf{I} - \mathbf{P}_1) \mathbf{W}\boldsymbol{\mu}$ sums $\mu_{ij} - \bar{\mu}_i^w$ over the observations in column j , which is the left side of [Eq. 17.2.1](#). Part (c) is the same argument with the factors exchanged.

For part (e), the projection is $\mathbf{Q} = \mathbf{M} - \mathbf{P}_{A+B}$, where $\mathbf{P}_{A+B}$ projects onto the additive model space, so $\mathbf{QW}\boldsymbol{\mu} = \mathbf{0}$ iff $\mathbf{W}\boldsymbol{\mu}$ lies in that space, that is, iff $\mu_{ij} = \alpha_i + \beta_j$ for some α, β . Additive tables have vanishing tetrads. Conversely, if all tetrads vanish, then $\mu_{ij} = \mu_{i1} + \mu_{1j} - \mu_{11}$, which is additive.

Degrees of freedom. A complete layout is connected, so the additive model matrix has rank $a + b - 1$ ([Theorem 8.6.1\(a\)](#)). The degrees of freedom of the extra sums of squares are therefore $a - 1$ for (a), $(a + b - 1) - a = b - 1$ for (b), $a - 1$ for (c) and $ab - (a + b - 1)$ for (e). In each case this is $\text{rank}(\mathbf{Q})$, the number of independent linear restrictions that the hypothesis places on $\boldsymbol{\mu}$. ■

In the Type II hypothesis [Eq. 17.2.2](#), row i is compared cell by cell with the weighted means $\bar{\mu}_{.j}^w$ of its columns, with the row's own counts as weights. The a equations sum to zero ([Exercise \(B1\)](#)). With two rows the hypothesis is $\sum_j h_j (\mu_{1j} - \mu_{2j}) = 0$ with $h_j = n_{1j}n_{2j}/n_{.j}$ ([Exercise \(B2\)](#)), which favours columns where *both* rows are well replicated. That is the most precise comparison when the row difference is the same in every column. When it is not, the weights are set by the design rather than by anyone's question.

Under additivity, $\mu_{ij} = \alpha_i + \beta_j$, the hypotheses separate cleanly. The Type II and Type III

Type I, variety first				Type II				Type III			
V1	+0.35	+0.20	+0.10	V1	+0.22	+0.21	+0.13	V1	+0.22	+0.22	+0.22
V2	-0.08	-0.14	-0.11	V2	-0.13	-0.13	-0.05	V2	-0.11	-0.11	-0.11
V3	-0.05	-0.08	-0.19	V3	-0.09	-0.08	-0.08	V3	-0.11	-0.11	-0.11
	S1	S2	S3		S1	S2	S3		S1	S2	S3

Figure 17.2.1: Coefficients on the cell means of the first equation of each variety hypothesis in [Example \(Three varieties at three sites\)](#) (Type II divided by n_1).

hypotheses both reduce to $\alpha_1 = \dots = \alpha_a$: Type III because unweighted row means are $\alpha_i + \bar{\beta}$, and Type II by [Theorem 9.3.2\(d\)](#). The Type I hypothesis becomes

$$\alpha_i + \sum_j \frac{n_{ij}}{n_{i.}} \beta_j \text{ the same for all } i,$$

which involves the column effects unless the counts are proportional ([Theorem 9.4.3](#)). A factor entered first is credited with the effects of the factors it is confounded with.

In [Figure 17.2.1](#) the Type I coefficients (V1's weighted mean minus the weighted grand mean) do not sum to zero within columns, so that hypothesis compares sites as well as varieties. The Type II and Type III tables have zero column sums and compare varieties within sites. Type II gives site j the weight $n_{1j}(n_{.j} - n_{1j})/n_{.j}$, which is large only where V1 and the other varieties are both well replicated. Type III weights the sites equally.

17.2.3 Why Type III needs contrast codings

[Definition 9.3.2](#) defines Type III through sum-to-zero coding, and [Theorem 9.3.2\(c\)](#) showed that reference coding gives a different hypothesis. The next result settles which codings give Type III and why.

Proposition 17.2.3 — Which coding a last-in sum of squares depends on

In the two-way layout with all cells occupied, code A by an $a \times (a - 1)$ coding matrix $\mathbf{C}_A$ and B by a $b \times (b - 1)$ coding matrix $\mathbf{C}_B$, with $[\mathbf{1}_a, \mathbf{C}_A]$ and $[\mathbf{1}_b, \mathbf{C}_B]$ nonsingular ([Definition 8.5.1](#)). Let the interaction columns be the products of the coded columns of A and B , and let $\mathbf{w} \neq \mathbf{0}$ span the line $\mathcal{C}(\mathbf{C}_B)^\perp \subseteq \mathbb{R}^b$. Then the partial sum of squares of the columns of A , with the intercept, B and the interaction in the model, addresses

$$\sum_j w_j \mu_{ij} \text{ is the same for all } i. \quad (17.2.3)$$

The hypothesis depends neither on $\mathbf{C}_A$ nor on the counts. It is the Type III hypothesis

$\bar{\mu}_{1\cdot} = \cdots = \bar{\mu}_{a\cdot}$ iff every column of $\mathbf{C}_B$ sums to zero.

Proof. Put $\mathbf{K}_A = [\mathbf{1}_a, \mathbf{C}_A]$ and $\mathbf{K}_B = [\mathbf{1}_b, \mathbf{C}_B]$. For an observation in cell (i, j) , the product of the k th coded column of A and the l th coded column of B takes the value $(\mathbf{C}_A)_{ik}(\mathbf{C}_B)_{jl}$, the entry in row (i, j) of the corresponding column of $\mathbf{C}_A \otimes \mathbf{C}_B$. So the full model matrix is $\mathbf{W}$ times the columns of $\mathbf{K}_A \otimes \mathbf{K}_B$, namely $\mathbf{1}_a \otimes \mathbf{1}_b$, $\mathbf{1}_a \otimes \mathbf{C}_B$, $\mathbf{C}_A \otimes \mathbf{1}_b$ and $\mathbf{C}_A \otimes \mathbf{C}_B$. The matrix $\mathbf{K}_A \otimes \mathbf{K}_B$ is nonsingular (Proposition 1.10.1(b)), so the full model space is $\mathcal{C}(\mathbf{W})$ and its ab columns are linearly independent. Deleting the $a - 1$ columns $\mathbf{C}_A \otimes \mathbf{1}_b$ leaves the reduced model $\mathbf{WF}$ with

$$\mathbf{F} = [\mathbf{1}_a \otimes \mathbf{1}_b, \mathbf{1}_a \otimes \mathbf{C}_B, \mathbf{C}_A \otimes \mathbf{C}_B], \quad \text{rank}(\mathbf{F}) = ab - (a - 1).$$

Let $\mathcal{V} = \{\boldsymbol{\mu} : (\mathbf{u} \otimes \mathbf{w})^\top \boldsymbol{\mu} = 0 \text{ for all } \mathbf{u} \in \mathbf{1}_a^\perp\}$. Each column of $\mathbf{F}$ lies in $\mathcal{V}$: by the mixed-product rule, $(\mathbf{u}^\top \otimes \mathbf{w}^\top)(\mathbf{1}_a \otimes \mathbf{v}) = (\mathbf{u}^\top \mathbf{1}_a)(\mathbf{w}^\top \mathbf{v}) = 0$ for any $\mathbf{v}$, and $(\mathbf{u}^\top \otimes \mathbf{w}^\top)(\mathbf{C}_A \otimes \mathbf{C}_B) = (\mathbf{u}^\top \mathbf{C}_A) \otimes (\mathbf{w}^\top \mathbf{C}_B) = \mathbf{0}$ because $\mathbf{w} \perp \mathcal{C}(\mathbf{C}_B)$. The functionals $\mathbf{u} \otimes \mathbf{w}$, $\mathbf{u} \in \mathbf{1}_a^\perp$, form a space of dimension $a - 1$, since $\mathbf{u} \mapsto \mathbf{u} \otimes \mathbf{w}$ is injective. So $\dim \mathcal{V} = ab - (a - 1) = \text{rank}(\mathbf{F})$, and $\mathcal{C}(\mathbf{F}) = \mathcal{V}$. The partial sum of squares is the drop in residual sum of squares from $\mathcal{C}(\mathbf{WF})$ to $\mathcal{C}(\mathbf{W})$, so it addresses the hypothesis $\boldsymbol{\mu} \in \mathcal{V}$. Now $(\mathbf{u} \otimes \mathbf{w})^\top \boldsymbol{\mu} = \sum_i u_i \sum_j w_j \mu_{ij}$ vanishes for every contrast $\mathbf{u}$ iff $\sum_j w_j \mu_{ij}$ does not depend on i , which is Eq. 17.2.3. Finally, $\mathbf{w} \propto \mathbf{1}_b$ iff $\mathbf{1}_b \perp \mathcal{C}(\mathbf{C}_B)$, that is, iff $\mathbf{1}_b^\top \mathbf{C}_B = \mathbf{0}$. ■

Sum-to-zero, Helmert and orthogonal polynomial codings of B all give the Type III sum of squares for A , whatever coding A receives. Reference coding of B at level r gives $\mathbf{w} = \mathbf{e}_r$ and the simple effect of Theorem 9.3.2(c). With more factors (Exercise (C2)), the last-in sum of squares of a main effect is the Type III one when every *other* factor is coded by contrasts. In statsmodels, write `C(x, Sum)` for every factor.

17.2.4 The variety trial, five ways

Example. (Five tables for one data set)

The table has the layout of Example (Party, education and political self-placement) in Chapter 9, and the listing computes its entries the same way, as squared lengths of projections. For the trial of Example (Three varieties at three sites):

Term	df	Type I, variety first	Type I, site first	Type II	Type III	reference coding
variety	2	0.571	6.184	6.184	4.958	3.601
site	2	31.175	25.562	31.175	27.634	7.456
interaction	4	2.146	2.146	2.146	2.146	2.146
residual	28	8.781	8.781	8.781	8.781	8.781

Only the Type I columns add up to the model sum of squares 33.892; the Type II and Type III entries add up to 39.506 and 34.738. The verdicts on varieties differ. Entered first, $F = 0.91$ and $p = 0.414$: the raw averages 6.423, 6.583 and 6.275 mix each variety with the sites it was grown at. After sites (Type II), $F = 9.86$ and $p = 0.001$. Type III gives $F = 7.90$ and $p = 0.002$, the test of [Example \(Three varieties at three sites\)](#). The last column compares the varieties at S1 only ($F = 5.74$, $p = 0.008$).

The script checks every entry against statsmodels' `anova_lm` (types 1, 2 and 3, with sum-to-zero and Helmert coding for type 3), and recovers each entry from the matrix **L** of its part of [Theorem 17.2.2](#). It also confirms [Proposition 17.2.3](#): reference coding of the varieties with contrast coding of the sites reproduces 4.958, and a random coding of the sites gives the sum of squares of [Eq. 17.2.3](#).

```
import numpy as np
import pandas as pd
counts = np.array([[7, 4, 2],          # rows: varieties V1-V3; columns: sites S1-S3
                  [3, 5, 4],
                  [2, 3, 7]])
true_means = np.array([[5.8, 7.2, 7.6],
                      [5.0, 6.5, 7.5],
                      [4.4, 5.9, 7.2]])
rng = np.random.default_rng(20172)
rows = [(f"V{i + 1}", f"S{j + 1}", round(true_means[i, j] + rng.normal(0, 0.5), 1))
        for i in range(3) for j in range(3) for _ in range(counts[i, j])]
trial = pd.DataFrame(rows, columns=["variety", "site", "y"])

def proj(*blocks):
    """Orthogonal projection onto the span of the given column blocks (any rank)."""
    Z = np.column_stack(blocks)
    U, s, _ = np.linalg.svd(Z, full_matrices=False)
    U = U[:, s > 1e-10 * s[0]]
    return U @ U.T

y = trial["y"].to_numpy()
n = len(y)
one = np.ones((n, 1))
ZA = pd.get_dummies(trial["variety"]).to_numpy(float)      # variety indicators
ZB = pd.get_dummies(trial["site"]).to_numpy(float)          # site indicators
W = np.column_stack([ZA[:, i] * ZB for i in range(3)])      # the nine cell indicators

def ss(P):
    return y @ P @ y

P0, PA, PB, PAB, M = proj(one), proj(one, ZA), proj(one, ZB), proj(one, ZA, ZB),
proj(W)
typeI_AB = {"A": ss(PA - P0), "B": ss(PAB - PA), "AB": ss(M - PAB)} # variety
first
typeI_BA = {"B": ss(PB - P0), "A": ss(PAB - PB), "AB": ss(M - PAB)} # site first
typeII = {"A": ss(PAB - PB), "B": ss(PAB - PA), "AB": ss(M - PAB)}

def coding(labels, kind):
    """Columns coding a factor: sum-to-zero, or reference level (treatment)."""
    D = pd.get_dummies(labels).to_numpy(float)
    g = D.shape[1]
    C = np.vstack([np.eye(g - 1), -np.ones(g - 1)]) if kind == "sum" else np.eye(g)
   [:, 1:]
```

```

return D @ C

def last_in(CA, CB):
    """Partial (last-in) sums of squares of A and B with interaction columns =
    products."""
    CAB = np.column_stack([CA[:, [i]] * CB for i in range(CA.shape[1])])
    full = proj(one, CA, CB, CAB)
    assert np.allclose(full, M) # every coding spans the cell-means
    space
    return {"A": ss(full - proj(one, CB, CAB)), "B": ss(full - proj(one, CA, CAB))}

typeIII = last_in(coding(trial["variety"], "sum"), coding(trial["site"], "sum"))
reference = last_in(coding(trial["variety"], "treatment"), coding(trial["site"], "
    treatment"))
sse = ss(np.eye(n) - M)
for name, t in [("Type I, variety first", typeI_AB), ("Type I, site first", typeI_BA
    ),
                ("Type II", typeII), ("Type III (sum)", typeIII), ("reference coding
    ", reference)]:
    print(f"{name:22s}", {k: round(float(v), 3) for k, v in t.items()})

```

The next cell builds the Type II hypothesis [Eq. 17.2.2](#) and the Type I hypothesis as matrices on the nine cell means, and [Eq. 17.1.1](#) recovers their entries in the table.

```

cells = trial.groupby(["variety", "site"])["y"]
ybar = cells.mean().to_numpy() # nine cell means, row-major
N = cells.size().to_numpy().reshape(3, 3) # counts n_ij
npi, npj = N.sum(axis=1), N.sum(axis=0)

def hypothesis_ss(L):
    """Sum of squares of L mu = 0 in the cell-means model (redundant rows allowed).
    """
    u = L @ ybar
    return u @ np.linalg.pinv(L @ np.diag(1 / N.ravel()) @ L.T) @ u

# Type II for variety: row i has coefficient n_ij (1[i = k] - n_kj / n_.j) on cell (
    k, j)
L2A = np.array([[N[i, j] * ((i == k) - N[k, j] / npj[j]) for k in range(3) for j in
    range(3)]
                for i in range(3)])
# Type I, variety first: the count-weighted variety means are equal
wrow = np.array([[ (i == k) * N[k, j] / npi[k] for k in range(3) for j in range(3)]
    for i in range(3)])
L1 = wrow[[0]] - wrow[1:]
print(f"Type II hypothesis: SS = {hypothesis_ss(L2A):.3f}; Type I hypothesis: SS =
    {hypothesis_ss(L1):.3f}")

```

17.2.5 Choosing a sum of squares

The advice of [Section 9.3](#) stands: decide the hypothesis first, use Type I only when the order of entry is part of the question, and without interaction prefer Type II. The cell-mean hypotheses sharpen it in three ways.

1. *Test the interaction first.* It is the same in every type and tests additivity ([Theorem 17.2.2\(e\)](#)).

Under additivity Type II and Type III test the same hypothesis, and Type II is never less powerful ([Exercise \(C1\)](#)).

2. *If the factors interact*, the weights of a main-effect hypothesis decide the answer. Choose weights that describe a population (equal weights give Type III), or report simple effects within each column with multiplicity-adjusted intervals ([Chapter 13](#)).
3. *Report the hypothesis, not the type*. Anyone can read $\mathbf{L}\boldsymbol{\mu} = \mathbf{0}$, and [Eq. 17.1.1](#) tests it. With empty cells this is the only safe course ([Section 17.4](#)).

17.2.6 Exercises

17.2.6.1 A. Check your understanding

Exercise A1.

In a 2×2 layout with counts $n_{11} = 6$, $n_{12} = 2$, $n_{21} = 1$, $n_{22} = 3$, write the hypothesis of the Type I sum of squares of A entered first as one linear equation in the four cell means. Show that for the additive table $\mu_{ij} = \beta_j$, in which A has no effect, the hypothesis holds only if $\beta_1 = \beta_2$. In particular it fails for $\beta_1 = 0$, $\beta_2 = 1$.

17.2.6.2 B. Practice

Exercise B1.

Show that the left sides of the a equations [Eq. 17.2.2](#) add up to zero for every table $\boldsymbol{\mu}$. Show also that, for a connected design, the equations have rank exactly $a - 1$. *Hint:* count degrees of freedom.

Solution. Summing over i gives

$$\sum_{ij} n_{ij} \mu_{ij} - \sum_j n_{.j} \bar{\mu}_{.j}^w = \sum_{ij} n_{ij} \mu_{ij} - \sum_j \sum_i n_{ij} \mu_{ij} = 0.$$

So at most $a - 1$ of the equations are independent. By [Lemma 17.2.1\(b\)](#) the equations are $\mathbf{Z}_A^\top (\mathbf{I} - \mathbf{P}_B) \mathbf{W} \boldsymbol{\mu} = \mathbf{0}$, where $\mathbf{P}_B$ projects onto $\mathcal{C}([\mathbf{1}, \mathbf{Z}_B])$. The number of independent restrictions they impose on $\boldsymbol{\mu}$ is the rank of the projection onto $\mathcal{C}((\mathbf{I} - \mathbf{P}_B) \mathbf{Z}_A)$, which is $\text{rank}[\mathbf{1}, \mathbf{Z}_A, \mathbf{Z}_B] - \text{rank}[\mathbf{1}, \mathbf{Z}_B] = (a + b - 1) - b = a - 1$ for a connected design ([Theorem 8.6.1\(a\)](#)).

Exercise B2.

Show that when $a = 2$, the Type II hypothesis [Eq. 17.2.2](#) for A is the single equation $\sum_j h_j (\mu_{1j} - \mu_{2j}) = 0$ with $h_j = n_{1j} n_{2j} / (n_{1j} + n_{2j})$. Show that σ^2 / h_j is the variance of $\bar{y}_{1j} - \bar{y}_{2j}$, and conclude that under additivity the Type II hypothesis compares the rows with inverse-variance weights.

Solution. The first equation is

$$\sum_j n_{1j} (\mu_{1j} - (n_{1j} \mu_{1j} + n_{2j} \mu_{2j}) / n_{.j}) = \sum_j (n_{1j} n_{2j} / n_{.j}) (\mu_{1j} - \mu_{2j}),$$

and the second is its negative ([Exercise B1](#)). The variance of $\bar{y}_{1j} - \bar{y}_{2j}$ is $\sigma^2(1/n_{1j} + 1/n_{2j}) = \sigma^2 n_{\cdot j} / (n_{1j} n_{2j}) = \sigma^2 / h_j$. If the difference $\delta = \mu_{1j} - \mu_{2j}$ is the same in every column, the most precise combination of the column differences weights them in proportion to h_j , and the hypothesis says that this combination is zero.

Exercise B3.

Suppose the counts are proportional, $n_{ij} = n_{i\cdot} n_{\cdot j} / n$. Show that the hypotheses of [Theorem 17.2.2](#)(a) and (c) coincide, and that both say that the row means weighted by the column shares $n_{\cdot j} / n$ are equal. Why does this agree with [Theorem 9.4.3](#)?

Solution. With proportional counts, $n_{ij} / n_{i\cdot} = n_{\cdot j} / n =: \pi_j$ for every i , so $\bar{\mu}_i^w = \sum_j \pi_j \mu_{ij}$. The left side of [Eq. 17.2.2](#) is $n_{i\cdot} \sum_j \pi_j (\mu_{ij} - \bar{\mu}_{\cdot j}^w) = n_{i\cdot} (\bar{\mu}_i^w - \sum_j \pi_j \bar{\mu}_{\cdot j}^w)$, and $\sum_j \pi_j \bar{\mu}_{\cdot j}^w = \bar{\mu}^w$. So (c) says $\bar{\mu}_i^w = \bar{\mu}^w$ for all i , which is (a). In [Theorem 9.4.3](#) the centred factor spaces are orthogonal, so $\text{SS}(A \mid \mu, B) = \text{SS}(A \mid \mu)$ for every data set. Two sums of squares that agree for every $\mathbf{y}$ must address the same hypothesis.

Exercise B4.

Use [Proposition 17.2.3](#) to find the hypothesis of the last-in sum of squares of A when B (with $b = 3$) is coded by (a) the Helmert columns $(-1, 1, 0)^\top, (-1, -1, 2)^\top$; (b) the orthogonal polynomial columns $(-1, 0, 1)^\top, (1, -2, 1)^\top$; (c) the columns $(1, 0, 0)^\top, (0, 0, 1)^\top$, reference coding with level 2 as reference; (d) the columns $(1, 0, 0)^\top, (0, 1, 0)^\top$ plus $\frac{1}{2}(1, 1, 1)^\top$ added to both.

17.2.6.3 C. Going deeper

Exercise C1.

Suppose the cell means are additive. Let $\mathbf{P}_{\text{II}} = \mathbf{P}_{A+B} - \mathbf{P}_B$ and let $\mathbf{P}_{\text{III}}$ be the projection of the Type III sum of squares of A . Both have rank $a-1$, and both are used with the cell-means error mean square. Show that the noncentrality parameters satisfy $\|\mathbf{P}_{\text{II}} \mathbf{W}\boldsymbol{\mu}\|^2 \geq \|\mathbf{P}_{\text{III}} \mathbf{W}\boldsymbol{\mu}\|^2$, so that the Type II test is at least as powerful. *Hint:* both are squared distances from $\mathbf{W}\boldsymbol{\mu}$ to a subspace.

Solution. Write $\mathbf{v} = \mathbf{W}\boldsymbol{\mu}$, which lies in the additive space $\mathcal{C}(\mathbf{P}_{A+B})$. Then $\mathbf{P}_{\text{II}} \mathbf{v} = \mathbf{v} - \mathbf{P}_B \mathbf{v}$, so $\|\mathbf{P}_{\text{II}} \mathbf{v}\|^2$ is the squared distance from $\mathbf{v}$ to $\mathcal{C}([\mathbf{1}, \mathbf{Z}_B])$. Also $\mathbf{P}_{\text{III}} = \mathbf{M} - \mathbf{P}_R$, where $\mathbf{P}_R$ projects onto the reduced space $\mathcal{C}([\mathbf{1}, \mathbf{C}_B\text{-columns, interaction columns}])$, and $\mathbf{M}\mathbf{v} = \mathbf{v}$. So $\|\mathbf{P}_{\text{III}} \mathbf{v}\|^2$ is the squared distance from $\mathbf{v}$ to $\mathcal{C}(\mathbf{P}_R)$. This space contains $\mathcal{C}([\mathbf{1}, \mathbf{Z}_B])$, and the distance to a larger subspace is smaller. The two F statistics have the same degrees of freedom, and the power of an F test increases with its noncentrality ([Theorem 11.5.1](#)).

Exercise C2.

In an $a \times b \times d$ layout, where C has d levels and all cells are occupied, code the factors by $\mathbf{C}_A, \mathbf{C}_B, \mathbf{C}_C$, and let the model contain all main effects and all two- and three-factor interactions, formed as products of coded columns. Show that the last-in sum of squares

of the columns of A addresses equality over i of $\sum_{j,k} v_j w_k \mu_{ijk}$, where $\mathbf{v}$ spans $\mathcal{C}(\mathbf{C}_B)^\perp$ and $\mathbf{w}$ spans $\mathcal{C}(\mathbf{C}_C)^\perp$. When is this the unweighted hypothesis $\bar{\mu}_{1..} = \cdots = \bar{\mu}_{a..}$?

17.3 Constrained models

The cell-means model assumes nothing about the table of means. Smaller models, such as the additive model or a three-way model without the three-factor interaction, are the cell-means model with linear constraints $\mathbf{R}\boldsymbol{\mu} = \mathbf{0}$. This section fits and tests them with the restricted least squares of Section 11.3, and shows that main-effect hypotheses can merge once the constraints hold.

17.3.1 Estimation and tests under constraints

Theorem 17.3.1 — Constrained cell-means models

In the cell-means model of Definition 17.1.1, let $\mathbf{R}$ be a $g \times p$ matrix of rank g , and let the columns of the $p \times (p - g)$ matrix $\mathbf{K}$ form a basis of $\mathcal{N}(\mathbf{R})$. Assume that $\mathbf{WK}$ has full column rank $p - g$. The **constrained model** is $E(\mathbf{Y}) = \mathbf{W}\boldsymbol{\mu}$ with $\mathbf{R}\boldsymbol{\mu} = \mathbf{0}$.

- (a) The constrained model is the full-rank linear model $E(\mathbf{Y}) = \mathbf{WK}\boldsymbol{\theta}$, with $\boldsymbol{\mu} = \mathbf{K}\boldsymbol{\theta}$. Every cell mean, including the means of empty cells, is estimable in it. The least squares estimate is

$$\hat{\boldsymbol{\mu}}_{\mathbf{R}} = \mathbf{V}_{\mathbf{R}}\mathbf{D}\bar{\mathbf{y}}, \quad \mathbf{V}_{\mathbf{R}} = \mathbf{K}(\mathbf{K}^\top \mathbf{D}\mathbf{K})^{-1}\mathbf{K}^\top, \quad \text{Cov}(\hat{\boldsymbol{\mu}}_{\mathbf{R}}) = \sigma^2 \mathbf{V}_{\mathbf{R}}, \quad (17.3.1)$$

and its residual sum of squares $\text{SSE}_{\mathbf{R}}$ has $n - p + g$ degrees of freedom.

- (b) If every cell is occupied, then $\mathbf{V}_{\mathbf{R}} = \mathbf{D}^{-1} - \mathbf{D}^{-1}\mathbf{R}^\top(\mathbf{R}\mathbf{D}^{-1}\mathbf{R}^\top)^{-1}\mathbf{R}\mathbf{D}^{-1}$, so $\hat{\boldsymbol{\mu}}_{\mathbf{R}} = \bar{\mathbf{y}} - \mathbf{D}^{-1}\mathbf{R}^\top(\mathbf{R}\mathbf{D}^{-1}\mathbf{R}^\top)^{-1}\mathbf{R}\bar{\mathbf{y}}$, and $\text{SSE}_{\mathbf{R}} - \text{SSE} = (\mathbf{R}\bar{\mathbf{y}})^\top(\mathbf{R}\mathbf{D}^{-1}\mathbf{R}^\top)^{-1}\mathbf{R}\bar{\mathbf{y}}$ is the sum of squares, on g degrees of freedom, of the hypothesis $\mathbf{R}\boldsymbol{\mu} = \mathbf{0}$ in the unconstrained model.
- (c) Under normality, for an $\ell \times p$ matrix $\mathbf{L}$ with $q = \text{rank}(\mathbf{L}\mathbf{K}) \geq 1$, the F statistic for $H : \mathbf{L}\boldsymbol{\mu} = \mathbf{0}$ within the constrained model is

$$F = \frac{(\mathbf{L}\hat{\boldsymbol{\mu}}_{\mathbf{R}})^\top(\mathbf{L}\mathbf{V}_{\mathbf{R}}\mathbf{L}^\top)^{-1}\mathbf{L}\hat{\boldsymbol{\mu}}_{\mathbf{R}}}{q s_{\mathbf{R}}^2}, \quad s_{\mathbf{R}}^2 = \frac{\text{SSE}_{\mathbf{R}}}{n - p + g}, \quad (17.3.2)$$

for any generalized inverse. It has the $F(q, n - p + g, \gamma)$ distribution with $\gamma = (\mathbf{L}\boldsymbol{\mu})^\top(\mathbf{L}\mathbf{V}_{\mathbf{R}}\mathbf{L}^\top)^{-1}\mathbf{L}\boldsymbol{\mu}/\sigma^2$.

- (d) Within the constrained model, $\mathbf{L}_1\boldsymbol{\mu} = \mathbf{0}$ and $\mathbf{L}_2\boldsymbol{\mu} = \mathbf{0}$ are the same hypothesis iff $\mathbf{L}_1\mathbf{K}$ and $\mathbf{L}_2\mathbf{K}$ have the same row space, and then their F statistics coincide.

Proof.

- (a) A table satisfies $\mathbf{R}\boldsymbol{\mu} = \mathbf{0}$ iff $\boldsymbol{\mu} = \mathbf{K}\boldsymbol{\theta}$ for a unique $\boldsymbol{\theta}$, so the constrained model is the linear model with model matrix $\mathbf{WK}$, of full column rank by assumption. Every coordinate of $\boldsymbol{\theta}$ is then estimable, and so is every linear function of $\boldsymbol{\mu} = \mathbf{K}\boldsymbol{\theta}$.

Since $(\mathbf{W}\mathbf{K})^\top \mathbf{W}\mathbf{K} = \mathbf{K}^\top \mathbf{D}\mathbf{K}$ and $\mathbf{W}^\top \mathbf{y} = \mathbf{D}\bar{\mathbf{y}}$, the least squares estimate is $\hat{\boldsymbol{\theta}} = (\mathbf{K}^\top \mathbf{D}\mathbf{K})^{-1} \mathbf{K}^\top \mathbf{D}\bar{\mathbf{y}}$, with covariance $\sigma^2(\mathbf{K}^\top \mathbf{D}\mathbf{K})^{-1}$ (Theorem 5.5.1). Multiplying by $\mathbf{K}$ gives Eq. 17.3.1. The model matrix has rank $p - g$, which gives the degrees of freedom.

- (b) With every cell occupied, $\mathbf{W}$ has full column rank, $\mathbf{D}^{-1}$ is the inverse of $\mathbf{W}^\top \mathbf{W}$, and $\bar{\mathbf{y}}$ is the unrestricted estimate. The constrained model is the reduced model of the testable hypothesis $\mathbf{R}\boldsymbol{\mu} = \mathbf{0}$ in the unconstrained one. So Proposition 11.3.3(a) and (b), with $\boldsymbol{\Lambda}^\top = \mathbf{R}$ and $\mathbf{G} = \mathbf{D}^{-1}$, give the displayed formula for $\hat{\boldsymbol{\mu}}_{\mathbf{R}}$ and $\text{SSE}_{\mathbf{R}} = \text{SSE} + (\mathbf{R}\bar{\mathbf{y}})^\top (\mathbf{R}\mathbf{D}^{-1}\mathbf{R}^\top)^{-1} \mathbf{R}\bar{\mathbf{y}}$, the sum of squares Eq. 17.1.1 of the hypothesis. For the covariance, let $\mathbf{V}'$ be the right side of the claimed identity. On vectors $\mathbf{D}\mathbf{K}\mathbf{a}$ both matrices give $\mathbf{K}\mathbf{a}$, because $\mathbf{R}\mathbf{K} = \mathbf{0}$. On vectors $\mathbf{R}^\top \mathbf{b}$ both give $\mathbf{0}$, because $\mathbf{K}^\top \mathbf{R}^\top = \mathbf{0}$ and $\mathbf{V}'\mathbf{R}^\top \mathbf{b} = \mathbf{D}^{-1}\mathbf{R}^\top \mathbf{b} - \mathbf{D}^{-1}\mathbf{R}^\top \mathbf{b}$. These vectors span $\mathbb{R}^p$: the spaces $\mathcal{C}(\mathbf{D}\mathbf{K})$ and $\mathcal{C}(\mathbf{R}^\top)$ have dimensions $p - g$ and g , and if $\mathbf{D}\mathbf{K}\mathbf{a} = \mathbf{R}^\top \mathbf{b}$, then $\mathbf{a}^\top \mathbf{K}^\top \mathbf{D}\mathbf{K}\mathbf{a} = (\mathbf{R}\mathbf{K}\mathbf{a})^\top \mathbf{b} = 0$, so $\mathbf{K}\mathbf{a} = \mathbf{0}$ because $\mathbf{D}$ is positive definite. Two matrices that agree on a spanning set are equal.
- (c) In the full-rank model $\mathbf{W}\mathbf{K}\boldsymbol{\theta}$ the hypothesis is $\mathbf{L}\mathbf{K}\boldsymbol{\theta} = \mathbf{0}$, which is testable. Theorem 11.3.2 with $\boldsymbol{\Lambda}^\top = \mathbf{L}\mathbf{K}$ and $\mathbf{G} = (\mathbf{K}^\top \mathbf{D}\mathbf{K})^{-1}$ gives the statistic, since $\mathbf{L}\mathbf{K}\hat{\boldsymbol{\theta}} = \mathbf{L}\hat{\boldsymbol{\mu}}_{\mathbf{R}}$ and $\mathbf{L}\mathbf{K}(\mathbf{K}^\top \mathbf{D}\mathbf{K})^{-1} \mathbf{K}^\top \mathbf{L}^\top = \mathbf{L}\mathbf{V}_{\mathbf{R}}\mathbf{L}^\top$.
- (d) Within the model, $\mathbf{L}\boldsymbol{\mu} = \mathbf{0}$ iff $\boldsymbol{\theta} \in \mathcal{N}(\mathbf{L}\mathbf{K})$, and two matrices have the same null space iff they have the same row space. The hypotheses then have the same reduced model, and by Theorem 11.3.2(b) the same F statistic.

■

Part (d) is the new phenomenon: hypotheses that differ in the cell-means model can coincide once the constraints hold. Part (b) pulls the cell averages onto the constraint set along the directions $\mathbf{D}^{-1}\mathbf{R}^\top$. The rank assumption on $\mathbf{W}\mathbf{K}$ holds whenever every cell is occupied; with empty cells it is a real condition (Exercise (C2)).

17.3.2 The additive model

Take $\mathbf{R} = \mathbf{C}_a^\top \otimes \mathbf{C}_b^\top$, the interaction hypothesis matrix of Eq. 17.1.2, with $g = (a - 1)(b - 1)$. Its null space is the space of additive tables $\mu_{ij} = \alpha_i + \beta_j$ (Section 17.1), so the constrained model is the additive model Eq. 8.6.1 written in cell means.

Corollary 17.3.2 — Main effects in the additive model

In the two-way layout with all cells occupied, let $\mathbf{R} = \mathbf{C}_a^\top \otimes \mathbf{C}_b^\top$. Then:

- (a) $\hat{\boldsymbol{\mu}}_{\mathbf{R}}$ is the table of fitted cell means of the additive model, and $\text{SSE}_{\mathbf{R}} - \text{SSE} = \text{SS}(AB \mid \mu, A, B)$;
- (b) for any weights $\mathbf{w} \in \mathbb{R}^b$ with $\sum_j w_j \neq 0$, the hypothesis that $\sum_j w_j \mu_{ij}$ is the same for all i is, within the additive model, the hypothesis $\alpha_1 = \cdots = \alpha_a$. Its sum of squares is $\text{SS}(A \mid \mu, B)$, whatever $\mathbf{w}$;
- (c) weights that differ from row to row do not have this property: the hypothesis that $\sum_j w_{ij} \mu_{ij}$ is the same for all i depends on the column effects unless $\sum_j w_{ij} \beta_j$ does not depend on i .

Proof.

- (a) The constrained mean vectors $\mathbf{W}\boldsymbol{\mu}$ with $\boldsymbol{\mu}$ additive form $\mathcal{C}([\mathbf{1}, \mathbf{Z}_A, \mathbf{Z}_B])$, so the constrained fit is the additive fit. The difference of residual sums of squares is then $SS(AB \mid \mu, A, B)$ by definition. (b) For an additive table, $\sum_j w_j \mu_{ij} = (\sum_j w_j) \alpha_i + \sum_j w_j \beta_j$, which is the same for all i iff the α_i are equal. So all these hypotheses define the same subspace of the constrained model, and by [Theorem 17.3.1\(d\)](#) the same test. Taking $\mathbf{w} = \mathbf{e}_1$ shows that the subspace is $\{\mu_{ij} = \beta_j\}$, the model with B only, and the drop in residual sum of squares from it to the additive model is $SS(A \mid \mu, B)$. (c) Now $\sum_j w_{ij} \mu_{ij} = (\sum_j w_{ij}) \alpha_i + \sum_j w_{ij} \beta_j$, and the second term varies with i in general.

■

So once additivity is accepted, the argument about types disappears for the main effects: any common weighting, and even a single column, gives the comparison $\alpha_i - \alpha_k$, tested by the Type II sum of squares (and by Type III, which coincides with it there). Only the count-weighted means of Type I stay contaminated, by part (c). The usual summaries after an additive fit are the **least squares means** (estimated marginal means), the unweighted row means of $\hat{\boldsymbol{\mu}}_{\mathbf{R}}$, with standard errors from $\mathbf{V}_{\mathbf{R}}$.

Example. (The variety trial under additivity)

In [Example \(Five tables for one data set\)](#) the interaction sum of squares 2.146 gave $F = 1.71$ on 4 and 28 degrees of freedom, $p = 0.176$. Suppose we accept additivity. The constrained fit [Eq. 17.3.1](#) gives the table of fitted means

	S1	S2	S3
V1	5.56	7.17	7.95
V2	5.11	6.73	7.51
V3	4.48	6.09	6.87

with $SSE_{\mathbf{R}} = 10.927$ on $37 - 9 + 4 = 32$ degrees of freedom and $s_{\mathbf{R}}^2 = 0.3415$. The fitted mean of V1 at S3 is 7.95, although its two plots averaged 7.45: additivity carries V1's advantage at S1 and S2 over to S3. The test of equal unweighted variety means, [Eq. 17.3.2](#), gives 6.184, the Type II sum of squares, with $F = 9.06$ on 2 and 32 degrees of freedom, $p = 0.00076$. The script checks that site weights 0.5, 0.3, 0.2, and the column S1 alone, give the same 6.184 ([Corollary 17.3.2\(b\)](#)), while each variety's own count weights give 0.571, the Type I value. The least squares means are 6.894, 6.449 and 5.811 (standard errors 0.169, 0.170, 0.178), and V1 exceeds V2 by 0.445 (standard error 0.243).

The generating means had a small interaction: V1's advantage over V2 is 0.8 at S1, 0.7 at S2 and 0.1 at S3, and the test on 4 and 28 degrees of freedom has little power to detect it. A non-significant interaction test is not evidence of additivity.

```
import numpy as np
import pandas as pd
from scipy import stats
from scipy.linalg import null_space
```

```

counts = np.array([[7, 4, 2],          # rows: varieties V1-V3; columns: sites S1-S3
                  [3, 5, 4],
                  [2, 3, 7]])
true_means = np.array([[5.8, 7.2, 7.6],
                      [5.0, 6.5, 7.5],
                      [4.4, 5.9, 7.2]])
rng = np.random.default_rng(20172)
rows = [(f"V{i + 1}", f"S{j + 1}", round(true_means[i, j] + rng.normal(0, 0.5), 1))
        for i in range(3) for j in range(3) for _ in range(counts[i, j])]
trial = pd.DataFrame(rows, columns=["variety", "site", "y"])

cells = trial.groupby(["variety", "site"])["y"]
ybar = cells.mean().to_numpy()          # 9 cell means, row-major
n_c = cells.size().to_numpy()
Dinv = np.diag(1 / n_c)
C3 = np.array([[1.0, 0.0], [0.0, 1.0], [-1.0, -1.0]])
G = np.kron(C3.T, C3.T)                # 4 interaction contrasts: G
mu = 0
K = null_space(G)                      # 9 x 5 basis of additive
                                     tables

V_G = K @ np.linalg.inv(K.T @ np.diag(n_c) @ K) @ K.T          # Cov(mu_hat_G) /
sigma^2
mu_G = V_G @ np.diag(n_c) @ ybar          # constrained
estimate
mu_G_direct = ybar - Dinv @ G.T @ np.linalg.solve(G @ Dinv @ G.T, G @ ybar)
print("additive fitted cell means:\n", mu_G.reshape(3, 3).round(3))

n, m, g = n_c.sum(), 9, G.shape[0]
sse = ((trial["y"] - cells.transform("mean")) ** 2).sum()
Gy = G @ ybar
ss_constraint = Gy @ np.linalg.solve(G @ Dinv @ G.T, Gy)        # test of additivity
sse_G = sse + ss_constraint
s2_G = sse_G / (n - m + g)          # residual mean square of the
additive model

def constrained_test(L):
    """F test of L mu = 0 within the constrained model G mu = 0."""
    u = L @ mu_G
    V = L @ V_G @ L.T
    ss = u @ np.linalg.pinv(V) @ u
    q = np.linalg.matrix_rank(L @ K)
    F = ss / q / s2_G
    return ss, q, F, stats.f.sf(F, q, n - m + g)

avg3 = np.ones((1, 3)) / 3
print("variety, unweighted:", np.round(constrained_test(np.kron(C3.T, avg3)), 4))
print("site, unweighted:", np.round(constrained_test(np.kron(avg3, C3.T)), 4))

lsmeans = (mu_G.reshape(3, 3)).mean(axis=1)          # estimated unweighted variety
means
L_rows = np.kron(np.eye(3), avg3)
se_ls = np.sqrt(np.diag(L_rows @ V_G @ L_rows.T) * s2_G)
d = np.array([1.0, -1.0, 0.0]) @ L_rows
diff12, se_diff12 = d @ mu_G, np.sqrt(d @ V_G @ d * s2_G)
print("least squares means:", lsmeans.round(3), " se", se_ls.round(3))
print(f"V1 - V2: {diff12:.3f} (se {se_diff12:.3f})")

```

In the listing the constraint matrix is called **G**. The script checks the fit, the identity of [Theorem 17.3.1\(b\)](#) and the tests against statsmodels.

17.3.3 Several factors: choosing among constrained models

With three or more factors the useful constrained models are the hierarchical ones, named by their highest interactions. In an $a \times b \times d$ layout, where C has d levels, $[AB][C]$ contains the AB interaction, the main effect of C , and everything below them. The model $[AB][AC][BC]$ sets the $(a-1)(b-1)(d-1)$ contrasts $\mathbf{C}_a^\top \otimes \mathbf{C}_b^\top \otimes \mathbf{C}_d^\top$ to zero. Unbalanced data make these models non-orthogonal, so we remove one term at a time, judging each step by its drop in residual sum of squares over the error mean square of the full model.

Example. (Simplifying a three-factor curing study)

A (synthetic) study of a resin coating measured hardness after curing at three temperatures (T), with two hardeners (H) and resin from three suppliers (S). The 18 cells received one to three panels each, 37 in all, so the full model $[THS]$ leaves 19 error degrees of freedom, with mean square 0.408. The backward path:

Step	From	To	Term removed	df	SS	F	p
1	$[THS]$	$[TH][TS][HS]$	THS	4	1.137	0.70	0.6036
2	$[TH][TS][HS]$	$[TH][HS]$	TS	4	1.845	1.13	0.3719
3	$[TH][HS]$	$[TH][S]$	HS	2	4.132	5.06	0.0173
4	$[TH][HS]$	$[T][HS]$	TH	2	13.716	16.81	0.0001
5	$[TH][HS]$	$[HS]$	T and TH	4	19.100	11.70	0.0001

Steps 1 and 2 cost little. Step 1 is the test of the constraints $\mathbf{G}_3\boldsymbol{\mu} = \mathbf{0}$ with $\mathbf{G}_3 = \mathbf{C}_3^\top \otimes \mathbf{C}_2^\top \otimes \mathbf{C}_3^\top$, and the script recovers its sum of squares from the cell means through [Theorem 17.3.1\(b\)](#). Steps 3 and 4 show that neither interaction with the hardener can go, so the path stops at $[TH][HS]$. In that model the effect of temperature under a given hardener is the same for all three suppliers, so whether temperature matters at all is a single question inside the model, answered by step 5.

```
import statsmodels.formula.api as smf
rng3 = np.random.default_rng(31)
levels = [(i, j, k) for i in range(3) for j in range(2) for k in range(3)]
n3 = rng3.integers(1, 4, size=len(levels))  # 1 to 3 runs per cell

def true_hardness(i, j, k):
    """Temperature i, hardener j, supplier k: hardener interacts with both, nothing else does."""
    return (40 + [0, 1.0, 2.0][i] + 1.5 * j + [0, 0.5, -0.5][k]
            + j * [0, -1.0, -2.0][i] + j * [0, 1.2, -0.8][k])

cure = pd.DataFrame([(f"t{i + 1}", f"h{j + 1}", f"s{k + 1}", true_hardness(i, j, k)
                      + rng3.normal(0, 0.7))
```

```

        for (i, j, k), r in zip(levels, n3) for _ in range(r)],
        columns=["temp", "hardener", "supplier", "y"])

def fit(formula):
    f = smf.ols(formula, cure).fit()
    return f.ssr, int(f.df_resid)

sse_full, df_full = fit("y ~ temp * hardener * supplier")
mse_full = sse_full / df_full
path = [("THS", "[TH] [TS] [HS]", "y ~ temp * hardener * supplier", "y ~ (temp +
    hardener + supplier) ** 2"),
        ("TH [TS] [HS]", "[TH] [HS]", "y ~ (temp + hardener + supplier) ** 2", "y ~
    temp * hardener + hardener * supplier"),
        ("TH [HS]", "[TH] [S]", "y ~ temp * hardener + hardener * supplier", "y ~
    temp * hardener + supplier"),
        ("TH [HS]", "[T] [HS]", "y ~ temp * hardener + hardener * supplier", "y ~
    temp + hardener * supplier"),
        ("TH [HS]", "[HS]", "y ~ temp * hardener + hardener * supplier", "y ~
    hardener * supplier")]
steps = []
for big, small, f_big, f_small in path:
    (s1, d1), (s0, d0) = fit(f_big), fit(f_small)
    F = (s0 - s1) / (d0 - d1) / mse_full
    steps.append((big, small, d0 - d1, s0 - s1, F, stats.f.sf(F, d0 - d1, df_full)))
print(f"{big:13s} -> {small:13s} df = {d0 - d1}  SS = {s0 - s1:6.3f}  F = {F:5.2f}  p = {steps[-1][5]:.4f}")

```

Such comparisons can look contradictory: a factor may matter alone but not after another. That is the unbalanced analogue of collinearity, and it should be reported as such, not resolved by choosing a type of sum of squares.

17.3.4 Exercises

17.3.4.1 A. Check your understanding

Exercise A1.

In an $a \times b \times d$ layout with every cell occupied and n observations, how many constraints define the model $[AB][AC][BC]$, and how many error degrees of freedom does it have? Check your answer against step 1 of [Example \(Simplifying a three-factor curing study\)](#).

17.3.4.2 B. Practice

Exercise B1.

For a 2×2 layout with all cells occupied, take $\mathbf{R} = (1, -1, -1, 1)$. Use [Theorem 17.3.1\(b\)](#) to show that the additive fit is $\hat{\mu}_{ij} = \bar{y}_{ij} - (-1)^{i+j} \hat{\delta} / (n_{ij}H)$, where $\hat{\delta} = \bar{y}_{11} - \bar{y}_{12} - \bar{y}_{21} + \bar{y}_{22}$ and $H = \sum_{ij} 1/n_{ij}$. Which cells move most?

Solution. Here $\mathbf{R}\bar{\mathbf{y}} = \hat{\delta}$ and $\mathbf{R}\mathbf{D}^{-1}\mathbf{R}^\top = H$, so $\hat{\boldsymbol{\mu}}_{\mathbf{R}} = \bar{\mathbf{y}} - \mathbf{D}^{-1}\mathbf{R}^\top \hat{\delta} / H$. The (i, j) entry of $\mathbf{D}^{-1}\mathbf{R}^\top$ is $(-1)^{i+j} / n_{ij}$. Each cell average moves by $\hat{\delta} / (n_{ij}H)$ in the direction that removes the interaction, so the least replicated cells move most. The four moves remove

$\hat{\delta}$ exactly: $\sum_{ij} (-1)^{i+j} \cdot (-1)^{i+j} \hat{\delta} / (n_{ij} H) = \hat{\delta}$.

Exercise B2.

In a 2×3 layout with every cell occupied, impose only $\mu_{11} - \mu_{12} - \mu_{21} + \mu_{22} = 0$. Describe the constrained model in words. How many degrees of freedom does its residual have, and what does the F test of the remaining interaction contrast within it test?

Exercise B3.

In the additive model with all cells occupied, show that the least squares means satisfy $\hat{\mu}_{i\cdot} - \hat{\mu}_{k\cdot} = \hat{\alpha}_i - \hat{\alpha}_k$ for any solution of the normal equations, and that the same difference is obtained if the fitted table is averaged over j with any fixed weights summing to one. Why does the choice of weights still matter for the least squares means themselves?

Solution. The fitted table is $\hat{\mu}_{ij} = \hat{m} + \hat{\alpha}_i + \hat{\beta}_j$, unique as a table (Theorem 6.4.2). Averaging over j with weights w_j summing to one gives $\hat{m} + \hat{\alpha}_i + \sum_j w_j \hat{\beta}_j$, and the last term does not depend on i . So the differences are $\hat{\alpha}_i - \hat{\alpha}_k$, an estimable function whose value does not depend on the solution chosen. The means themselves contain $\sum_j w_j \hat{\beta}_j$, which does depend on the weights. They are levels, not comparisons, and a level is a mean over some population of columns.

17.3.4.3 C. Going deeper

Exercise C1.

The *method of unweighted means* estimates $\alpha_1 - \alpha_2$ in a $2 \times b$ additive layout by $b^{-1} \sum_j (\bar{y}_{1j} - \bar{y}_{2j})$, as if every cell had the same count. Show that the least squares estimate is $\sum_j h_j (\bar{y}_{1j} - \bar{y}_{2j}) / \sum_j h_j$ with $h_j = n_{1j} n_{2j} / n_{\cdot j}$. Deduce that both are unbiased under additivity, and that the least squares estimate has the smaller variance unless all h_j are equal.

Solution. Under additivity each $\bar{y}_{1j} - \bar{y}_{2j}$ is unbiased for $\delta = \alpha_1 - \alpha_2$, with variance σ^2/h_j (Exercise (B2)), and the b differences are independent. Any weighted average with weights summing to one is unbiased. Its variance $\sigma^2 \sum_j v_j^2 / h_j$ is minimized, by the Cauchy-Schwarz argument of Proposition 17.1.2, at $v_j \propto h_j$. The minimum variance is $\sigma^2 / \sum_j h_j$. The least squares estimate is the best linear unbiased estimator (Theorem 7.2.1), and it is a linear function of the cell averages, which are sufficient here. So it is the h -weighted average. The unweighted average has $v_j = 1/b$, and it is optimal only if all h_j are equal.

Exercise C2.

Allow empty cells. Show that **WK** in Theorem 17.3.1 has full column rank iff no nonzero table satisfying $\mathbf{R}\boldsymbol{\mu} = \mathbf{0}$ vanishes on every occupied cell. Deduce that for the additive constraints this holds iff the design graph of Theorem 8.6.1 is connected.

Solution. $\mathbf{WK}\boldsymbol{\theta} = \mathbf{0}$ iff the table $\mathbf{K}\boldsymbol{\theta}$ vanishes on every occupied cell, since the rows of $\mathbf{W}$ pick out the occupied cells. The columns of $\mathbf{K}$ are independent, so $\mathbf{WK}$ has a nonzero null vector iff some nonzero table satisfying the constraints vanishes on O . For additive tables $\alpha_i + \beta_j$, vanishing on O means $\alpha_i = -\beta_j$ for every occupied (i, j) . As in the proof of [Theorem 8.6.1](#), this forces $\alpha_i = t_C$ and $\beta_j = -t_C$ on each component C of the design graph. The table is nonzero iff the t_C are not all equal, which is possible iff there are at least two components.

17.4 Empty cells and estimability

Some combinations of levels are never observed, or are impossible. An empty cell's mean does not appear in the distribution of the data, so no unconstrained analysis can estimate it ([Proposition 17.1.1\(b\)](#)). This section works out what survives, building on connectedness ([Theorem 8.6.1](#)) and the estimability of interaction contrasts ([Proposition 8.6.2](#)).

Throughout, the two-way layout has occupied cells O , with $m = |O|$, every row and every column contains at least one occupied cell, and c is the number of connected components of the design graph of [Theorem 8.6.1](#). A row is **complete** if none of its cells is empty.

17.4.1 What remains estimable

Theorem 17.4.1 — Estimability and tests with empty cells

In the two-way layout described above:

- (a) In the cell-means model, $\sum_i u_i \sum_j w_j \mu_{ij}$ is estimable iff $u_i w_j = 0$ for every empty cell (i, j) . In particular the unweighted mean $\bar{\mu}_i$ is estimable iff row i is complete.
- (b) The estimable interaction contrasts, the tables (d_{ij}) with zero row and column sums and $d_{ij} = 0$ on every empty cell ([Proposition 8.6.2\(c\)](#)), form a space of dimension $m - a - b + c$. The interaction sum of squares $\text{SS}(AB \mid \mu, A, B)$ has $m - a - b + c$ degrees of freedom. It addresses the hypothesis that all these contrasts vanish, which holds iff the occupied cell means are additive: $\mu_{ij} = \alpha_i + \beta_j$ for all $(i, j) \in O$ and some α, β .
- (c) If the design is connected, the additive model estimates every cell mean, including those of empty cells, and under additivity $\text{SS}(A \mid \mu, B)$ addresses $\alpha_1 = \cdots = \alpha_a$, with $a - 1$ degrees of freedom.
- (d) Code both factors by contrasts and form the interaction columns as products. If at least one row is incomplete and k rows are complete, the last-in sum of squares of A addresses equality of the unweighted means of the complete rows, and it has $\max(k - 1, 0)$ degrees of freedom.

Proof.

- (a) The coefficient of μ_{ij} is $u_i w_j$, and [Proposition 17.1.1\(b\)](#) applies. For $\bar{\mu}_i$ the coefficients are $1/b$ on row i and zero elsewhere.
- (b) Let $T : \mathbb{R}^O \rightarrow \mathbb{R}^a \times \mathbb{R}^b$ send a table $(d_{ij})_{(i,j) \in O}$ to its row sums and column sums.

The estimable interaction contrasts form $\mathcal{N}(T)$. The transpose sends $(\mathbf{u}, \mathbf{v})$ to the table $(u_i + v_j)_{(i,j) \in O}$. As in the proof of [Theorem 8.6.1](#), $u_i + v_j = 0$ on O forces $u_i = t_C$ and $v_j = -t_C$ on each component C , with arbitrary t_C . So $\mathcal{N}(T^\top)$ has dimension c , $\text{rank}(T) = a + b - c$, and $\dim \mathcal{N}(T) = m - a - b + c$. The tables $(u_i + v_j)_{(i,j) \in O}$ are exactly the occupied parts of additive tables. They form $\mathcal{C}(T^\top)$, whose orthogonal complement in $\mathbb{R}^O$ is $\mathcal{N}(T)$. So the occupied cell means are additive iff they are orthogonal to every estimable interaction contrast, that is, iff all those contrasts vanish. The interaction sum of squares is $\mathbf{y}^\top (\mathbf{M} - \mathbf{P}_{A+B}) \mathbf{y}$, which addresses the hypothesis that $\mathbf{W}\boldsymbol{\mu}$ lies in the additive model space, that is, that the occupied cell means are additive. Its degrees of freedom are $m - (a + b - c)$ by [Proposition 17.1.1\(a\)](#) and [Theorem 8.6.1\(a\)](#).

- (c) The first claim is [Theorem 8.6.1\(c\)](#). For the second, let $\mu_{ij} = \alpha_i + \beta_j$. Then $(\mathbf{P}_{A+B} - \mathbf{P}_B)\mathbf{W}\boldsymbol{\mu} = \mathbf{0}$ iff $\mathbf{W}\boldsymbol{\mu} \in \mathcal{C}([\mathbf{1}, \mathbf{Z}_B])$, that is, iff $\alpha_i + \beta_j = \beta'_j$ on O for some β' . Then $\alpha_i + (\beta_j - \beta'_j) = 0$ on O , and in a connected design this forces all α_i to be equal, as in (b). Conversely, equal α_i give such a β' . The degrees of freedom are $\text{rank}[\mathbf{1}, \mathbf{Z}_A, \mathbf{Z}_B] - \text{rank}[\mathbf{1}, \mathbf{Z}_B] = (a + b - 1) - b$.
- (d) As in the proof of [Proposition 17.2.3](#), the full model matrix is $\mathbf{W}$ times the ab independent columns of $\mathbf{K}_A \otimes \mathbf{K}_B$, so its column space is $\mathcal{C}(\mathbf{W})$. The reduced model is $\mathbf{W}\mathbf{F}$. The argument in that proof does not use the counts, and with $\mathbf{w} = \mathbf{1}$ it shows that $\mathcal{C}(\mathbf{F})$ is the space $\mathcal{V}$ of tables with equal unweighted row means. Hence the reduced mean vectors are $\{\mathbf{W}\boldsymbol{\mu} : \boldsymbol{\mu} \in \mathcal{V}\}$: the tables of occupied cell means that can be completed, by choosing values in the empty cells, to a table with equal unweighted row means. Suppose the complete rows share a common unweighted mean t , with t arbitrary if there are none. Each incomplete row can then be given the mean t by choosing the value in one of its empty cells, so the occupied means can be completed. Conversely, a completion cannot change the means of complete rows. So the hypothesis is equality of the unweighted means of the complete rows. It consists of the $k - 1$ restrictions $\bar{\mu}_i = \bar{\mu}_{i_0}$, for the complete rows $i \neq i_0$, which are independent because they involve disjoint sets of cells, and of none if $k \leq 1$. The last-in sum of squares is the drop in residual sum of squares from the reduced model to $\mathcal{C}(\mathbf{W})$, so it addresses this hypothesis ([Eq. 6.5.2](#)), and its degrees of freedom are the number of independent restrictions. ■

In part (b), a table with zero row and column sums on the edges of the design graph is a *circulation*. The simplest ones run around cycles with alternating signs. A four-cycle is the familiar tetrad $\mu_{ij} - \mu_{il} - \mu_{kj} + \mu_{kl}$; longer cycles give contrasts on six or more cells that survive when every tetrad through them is broken. The number $m - a - b + c$ counts the independent cycles ([Exercise \(C1\)](#)).

Part (d) is the result to remember about Type III. With empty cells it ignores every incomplete row, and it may test nothing at all: an empty cell can hold whatever value makes its row's unweighted mean agree, and such means are not estimable, by (a).

17.4.2 An example with two combinations never grown

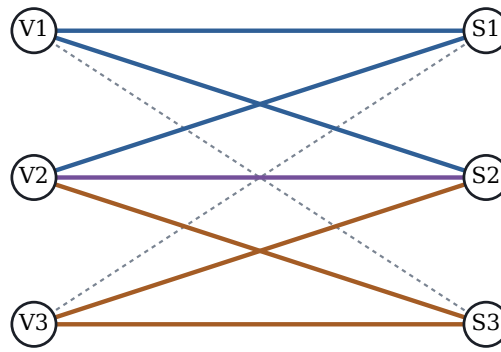


Figure 17.4.1: Design graph of [Example \(The variety trial with two empty cells\)](#) (dashed: empty cells). The blue and orange tetrads share the purple edge V2–S2.

Example. (The variety trial with two empty cells)

Suppose that in the trial of [Example \(Three varieties at three sites\)](#) V1 had never been grown at S3, nor V3 at S1. That leaves $n = 33$ plots in $m = 7$ cells, with 26 error degrees of freedom and error mean square 0.3096. The design graph ([Figure 17.4.1](#)) is connected, so the estimable interaction contrasts have dimension $7 - 3 - 3 + 1 = 2$, spanned by two tetrads. The interaction sum of squares is 1.218, $F = 1.97$, $p = 0.160$, and the script checks that the tetrads give the same value through [Eq. 17.1.1](#). The six-cell contrast around V1–S1–V2–S3–V3–S2–V1 is the difference of the tetrads ([Exercise \(B3\)](#)).

The unweighted means of V1 and V3 are no longer estimable. V1 and V2 share S1 and S2, and the difference of their averages there is 0.603 (standard error 0.268). The additive model is still identified. Its Type II sum of squares for varieties is 5.766 on 2 degrees of freedom, and it fills the empty cells with 8.32 for V1 at S3 and 4.20 for V3 at S1. The first exceeds every observed cell mean: V1's advantage at S1 and S2 is carried to a site where V1 was never grown, although in the generating means it had nearly vanished there (the true mean is 7.6).

The “Type III” sum of squares for varieties is 0 on 0 degrees of freedom: V2 is the only complete row ([Theorem 17.4.1\(d\)](#)). If only V1 at S3 is removed, it is 2.027 on 1 degree of freedom, the sum of squares of $\bar{\mu}_{2.} = \bar{\mu}_{3.}$, for every ordering of the levels.

```
import numpy as np
import pandas as pd
counts = np.array([[7, 4, 2],          # rows: varieties V1-V3; columns: sites S1-S3
                  [3, 5, 4],
                  [2, 3, 7]])
true_means = np.array([[5.8, 7.2, 7.6],
                      [5.0, 6.5, 7.5],
                      [4.4, 5.9, 7.2]])
rng = np.random.default_rng(20172)
rows = [(f"V{i + 1}", f"S{j + 1}", round(true_means[i, j] + rng.normal(0, 0.5), 1))
        for i in range(3) for j in range(3) for _ in range(counts[i, j])]
trial = pd.DataFrame(rows, columns=["variety", "site", "y"])
```



```

def proj(*blocks):
    """Orthogonal projection onto the span of the given column blocks (any rank)."""
    Z = np.column_stack(blocks)
    U, s, _ = np.linalg.svd(Z, full_matrices=False)
    U = U[:, s > 1e-10 * s[0]]
    return U @ U.T

def rank(*blocks):
    return np.linalg.matrix_rank(np.column_stack(blocks))

missing = ((trial["variety"] == "V1") & (trial["site"] == "S3")) | \
           ((trial["variety"] == "V3") & (trial["site"] == "S1"))
sub = trial[~missing].reset_index(drop=True)          # V1 never grown at S3, V3
                                                    # never at S1
y = sub["y"].to_numpy()
n = len(y)
one = np.ones((n, 1))
ZA = pd.get_dummies(sub["variety"]).to_numpy(float)
ZB = pd.get_dummies(sub["site"]).to_numpy(float)
W = np.column_stack([ZA[:, [i]] * ZB for i in range(3)])
m = rank(W)                                           # occupied cells
M, PAB, PA, PB = proj(W), proj(one, ZA, ZB), proj(one, ZA), proj(one, ZB)
sse = y @ (np.eye(n) - M) @ y
df_inter = m - rank(one, ZA, ZB)                     # m - a - b + 1 for a
                                                    # connected design
ss_inter = y @ (M - PAB) @ y
F_inter = ss_inter / df_inter / (sse / (n - m))
print(f"occupied cells {m}, interaction df {df_inter}, SS {ss_inter:.3f}, F {F_inter
:.2f}")
ss_typeII_A = y @ (PAB - PB) @ y                    # well defined, a - 1 = 2 df

def sum_coding(labels):
    D = pd.get_dummies(labels).to_numpy(float)
    return D[:, :-1] - D[:, [-1]]

CA, CB = sum_coding(sub["variety"]), sum_coding(sub["site"])
CAB = np.column_stack([CA[:, [i]] * CB for i in range(2)])
df_III_A = m - rank(one, CB, CAB)
ss_III_A = y @ (M - proj(one, CB, CAB)) @ y
print(f"last-in SS for variety with sum coding: {abs(ss_III_A):.3f} on {df_III_A} df
")

```

17.4.3 How software behaves

A program fitting the rank-deficient effects model picks one coefficient vector and computes its sums of squares from it. The script fits the data of [Example \(The variety trial with two empty cells\)](#) with statsmodels 0.15 under three namings of the levels. It warns that the design is rank-deficient and prints tables anyway. The Type I interaction appears on 4 degrees of freedom instead of 2, with sums of squares 3.96, 2.15 and 2.05 instead of 1.218. The Type II interaction entries are 23.46, 53.21 and 3.17, and the Type III variety entries 9.43, 41.10 and 34.25 on 2 degrees of freedom, where there is nothing to test. The projections give the same answers under every naming, as they must. SAS's "Type IV" sums of squares for empty cells build hypotheses from the occupied cells in a way that depends on the order of the levels (Goodnight 1980). Whatever the program, check that the printed degrees of freedom equal the rank of the

projection, and write down the hypothesis.

17.4.4 Recommended practice

1. *Tabulate the counts first*, draw the design graph and count its components. Rows in different components cannot be compared, even under additivity ([Theorem 8.6.1\(b\)](#)).
2. *Ask why the cells are empty*. Cells lost at random fit the theory above. Impossible combinations have no mean, and a main effect averaged over them is fiction. Combinations avoided *because* they were expected to do badly bias every additive fill-in, and no analysis of the observed cells can detect it.
3. *Test the interaction on its $m - a - b + c$ degrees of freedom*. If there are few cycles, additivity is barely checked.
4. *If additivity is credible*, use the additive model and $SS(A \mid \mu, B)$, and flag least squares means that involve empty cells as extrapolations.
5. *If not*, ask estimable questions of the cell means: simple effects, comparisons over shared columns, or a complete sub-table, and state which cells each comparison uses. Do not report a Type III or Type IV table without identifying its hypotheses.

One can also impose just enough interaction constraints to identify the empty cells ([Section 17.3](#)). The resulting estimates are then set by constraints the data cannot test ([Exercise \(B4\)](#)), and should be presented as assumptions.

17.4.5 Exercises

17.4.5.1 A. Check your understanding

Exercise A1.

A 3×4 layout has every cell occupied except $(1, 4)$, $(2, 1)$ and $(3, 1)$. Find m , the number of components of the design graph, the degrees of freedom of the interaction, and the tetrads that are estimable.

Exercise A2.

In [Example \(The variety trial with two empty cells\)](#), which comparisons between V1 and V3 are estimable in the cell-means model? Why is the V1–V2 comparison over S1 and S2 estimable when neither unweighted mean is?

17.4.5.2 B. Practice

Exercise B1.

In a 2×2 layout with cell $(2, 2)$ empty, show that the additive model estimates μ_{22} by $\bar{y}_{21} + \bar{y}_{12} - \bar{y}_{11}$, and find its variance. Show that the last-in sum of squares of A with sum-to-zero coding is identically zero. Compare with [Exercise \(B4\)](#).

Solution. With three occupied cells and a connected graph, the additive model has $1 + 2 + 2 - 1 = 3 = m$ free parameters. Its fitted values are the three cell averages, since $\text{SS}(AB \mid \mu, A, B)$ has $3 - 2 - 2 + 1 = 0$ degrees of freedom. Additivity gives $\mu_{22} = \mu_{21} + \mu_{12} - \mu_{11}$, estimated by $\bar{y}_{21} + \bar{y}_{12} - \bar{y}_{11}$ with variance $\sigma^2(1/n_{21} + 1/n_{12} + 1/n_{11})$. Row 1 is the only complete row, so [Theorem 17.4.1\(d\)](#) with $k = 1$ gives zero degrees of freedom, and the sum of squares vanishes for every data set. Directly, the reduced model with columns **1**, the B column and the product column already has rank 3 on the three occupied cells.

Exercise B2.

Take the 4×5 design of [Example \(Two islands\)](#), with occupied cells $(1, 1), (1, 2), (2, 2), (2, 3), (3, 4), (4, 4), (4, 5)$. Find the degrees of freedom of the interaction and the Type II sum of squares of A . Which comparisons of rows can the additive model make?

Exercise B3.

In [Example \(The variety trial with two empty cells\)](#), show that the six-cell contrast $\mu_{11} - \mu_{12} - \mu_{21} + \mu_{23} + \mu_{32} - \mu_{33}$ has zero row and column sums and vanishes on the empty cells. Write it as a combination of the two tetrads, and trace the cycle it corresponds to in [Figure 17.4.1](#).

Solution. Row sums: $1 - 1 = 0$, $-1 + 1 = 0$, $1 - 1 = 0$. Column sums: $1 - 1 = 0$ for S1, $-1 + 1 = 0$ for S2, $1 - 1 = 0$ for S3. It has no coefficient on $(1, 3)$ or $(3, 1)$. With $T_1 = \mu_{11} - \mu_{12} - \mu_{21} + \mu_{22}$ and $T_2 = \mu_{22} - \mu_{23} - \mu_{32} + \mu_{33}$, the contrast is $T_1 - T_2$, because μ_{22} cancels. The cycle is V1–S1–V2–S3–V3–S2–V1. It uses every occupied cell except V2–S2, the edge the two tetrads share.

Exercise B4.

In a 2×3 layout with only cell $(2, 3)$ empty, impose the single constraint $\mu_{12} - \mu_{13} - \mu_{22} + \mu_{23} = 0$. Show that every cell mean becomes estimable and find the estimate of μ_{23} . Can the constraint be tested?

Solution. The constraint gives $\mu_{23} = \mu_{22} + \mu_{13} - \mu_{12}$, so any table satisfying it that vanishes on the occupied cells vanishes everywhere, and [Exercise \(C2\)](#) gives identifiability. The five occupied cell means are unrestricted, so their estimates remain the cell averages, and $\hat{\mu}_{23} = \bar{y}_{22} + \bar{y}_{13} - \bar{y}_{12}$. The constraint involves the empty cell with a nonzero coefficient, so it is not estimable in the cell-means model and cannot be tested ([Proposition 11.3.1](#)). The fit is the same whether it holds or not. It only decides what we say about μ_{23} .

17.4.5.3 C. Going deeper

Exercise C1.

Let F be a spanning forest of the design graph, with $a + b - c$ edges. Show that each occupied cell e not in F closes a unique cycle in $F \cup \{e\}$, and that the alternating-

sign contrast $d^{(e)}$ of that cycle is an estimable interaction contrast. Show that the $m - a - b + c$ contrasts $d^{(e)}$ are linearly independent and hence form a basis of the space in [Theorem 17.4.1\(b\)](#).

Solution. Adding an edge $e = (i, j)$ to a forest joins two vertices of the same tree (the graph and the forest have the same components), so it closes exactly one cycle, the tree path from i to j followed by e . A cycle in a bipartite graph has even length and alternates between row and column vertices. Giving its edges alternating signs ± 1 makes the two edges at each vertex cancel, so the row and column sums vanish, and the table is supported on occupied cells. The contrast $d^{(e)}$ is the only one of the family with a nonzero entry at e , because the other cycles use only forest edges and their own extra edge. So in any vanishing linear combination the coefficient of $d^{(e)}$ is zero, for each e . There are $m - (a + b - c)$ of them, the dimension of the space, so they form a basis.

Exercise C2.

In the trial with only V1 at S3 removed, what does the “Type III” sum of squares for sites test, and on how many degrees of freedom? Verify your answer by modifying the listing.

17.5 Summary and Notes

17.5.1 Summary

Idea.

1. The cell-means model is the one-way layout on the occupied cells. Any hypothesis $\mathbf{L}\boldsymbol{\mu} = \mathbf{d}$ on occupied cells is tested by [Eq. 17.1.1](#) ([Definition 17.1.1](#), [Proposition 17.1.1](#)).
2. Count-weighted marginal means are the most precise, but they estimate the mixture the design happened to produce. Choose the target before the weights ([Proposition 17.1.2](#)).
3. An extra sum of squares tests whether the count-weighted residual of the true cell means, after the terms already fitted, is orthogonal to the new term ([Lemma 17.2.1](#)). This gives the hypotheses of all the types ([Theorem 17.2.2](#)).
4. The last-in sum of squares of a main effect depends on the coding of the *other* factors. It is the Type III sum of squares exactly when they are coded by contrasts ([Proposition 17.2.3](#)).
5. Constrained models are cell-means models with $\mathbf{R}\boldsymbol{\mu} = \mathbf{0}$, fitted and tested by restricted least squares ([Theorem 17.3.1](#)). Under additivity every common weighting of the row means gives the Type II test ([Corollary 17.3.2](#)).
6. With empty cells, the interaction has $m - a - b + c$ degrees of freedom, one per independent cycle of the design graph, and the “Type III” sum of squares compares only the complete rows ([Theorem 17.4.1](#)).

17.5.2 Notes and sources

Cell means. Analysing unbalanced data through cell means goes back to Yates (1934), whose weighted squares of means test hypotheses about unweighted means. Speed, Hocking and Hackney (1978) identified the cell-mean hypotheses of the sums of squares computed by the programs of their day, and Hocking (1985) and Searle (1987) give book-length treatments. Rencher and Schaalje (2008, chapter 15) presents unconstrained and constrained cell-means models and empty cells, and was the main source for this chapter’s coverage. Christensen (2020, sections 7.5–7.6) treats unbalanced layouts as a problem of comparing models, an emphasis reflected in [Section 17.3](#).

Types of sums of squares. The Type I, II and III terminology comes from statistical software. Nelder (1977) argued from the marginality of model terms that main effects should not be tested in the presence of their interactions, and Nelder and Lane (1995) restated the case. Langsrud (2003) argued for Type II on grounds of power, which [Exercise \(C1\)](#) makes precise. The hypothesis formulas of [Theorem 17.2.2\(b\)](#) and (c) are classical. Christensen (2020, section 7.5) states them for the two-way layout with interaction, and [Lemma 17.2.1](#) is our route to them. The characterization of the coding dependence in [Proposition 17.2.3](#), stated through the orthogonal complement of the other factor’s coding columns, is this book’s formulation of the familiar advice to use sum-to-zero contrasts.

Empty cells. Goodnight (1980) described the construction of the hypotheses behind SAS’s Type IV sums of squares for designs with empty cells, and noted that they are not unique. Milliken and Johnson (2009) discuss the analysis of two-way layouts with missing cells at length, with an emphasis on estimable questions and on identifying the hypotheses that software tests. The reading of estimable interaction contrasts as the cycle space of the design graph ([Theorem 17.4.1\(b\)](#)) uses the standard fact that a graph with m edges, v vertices and c components has $m - v + c$ independent cycles. The description of the Type III hypothesis with empty cells in [Theorem 17.4.1\(d\)](#) and the software comparison of [Section 17.4](#) are this book’s.

Data. The variety trial and the curing study are synthetic, generated by the scripts in `code/ch17/` with fixed seeds, so that the true cell means are known.

17.5.3 References

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Chapter 18

Analysis of Covariance and Designed Experiments

An experiment is designed before any response is measured. The experimenter decides which units receive which treatment, which nuisance variation to remove by grouping units into blocks, and which extra measurements to take on each unit before treatment begins. Each decision shows up in the linear model: blocks as additional factors, extra measurements as covariates, and the randomization as the reason the error assumptions are believable. This chapter works through the standard designed structures with the projection tools of Parts II and III. Analysis of covariance turns out to be the Frisch–Waugh–Lovell theorem applied to a factor model. Adjusted means are estimable functions with a correction term. The test of parallel slopes compares nested subspaces. Randomized blocks, Latin squares and balanced incomplete blocks are layouts whose balance makes the projections simple. A lost observation is handled exactly by least squares on the observations that remain, and the classical ways of filling in the gap are shortcuts to the same answer.

What you need. [Chapter 6](#) (nested projections, [Theorem 6.5.1](#), and the Frisch–Waugh–Lovell theorem, [Theorem 6.6.2](#)), [Chapter 8](#) (estimability, and the parallel and separate lines models of [Section 8.6](#)), [Chapter 9](#) (extra sums of squares and orthogonal designs), [Chapter 11](#) and [Chapter 13](#) (F tests and simultaneous inference), [Chapter 12](#) (t intervals, prediction intervals and the Working–Hotelling band), and, from this part, [Chapter 15](#) and [Chapter 16](#) (the one-way and additive two-way layouts) and [Chapter 17](#) (cell means and least squares means in unbalanced layouts).

18.0.1 Roadmap

- [Analysis of covariance as a projection problem](#)
- [Adjusted treatment means](#)
- [Heterogeneous slopes](#)
- [Randomized complete blocks](#)
- [Latin squares and incomplete blocks](#)
- [Missing observations in designed experiments](#)
- [Summary and notes](#)

18.1 Analysis of covariance as a projection problem

In many experiments something about each unit is known before the treatments are applied: a pretest score, the weight of an animal at the start of a feeding trial, the yield of a plot in the previous season, a patient's blood pressure at enrolment. Such a measurement is a **covariate**. If it predicts the response, much of the variation that the one-way analysis (Chapter 15) calls error is not error at all. Part of it is the covariate at work, and it can be removed.

The **analysis of covariance** (ANCOVA) adds covariates to a factor model. Section 8.6 treated one factor and one covariate (Proposition 8.6.3). This section states the result for any design part, rank deficient or not, and any number of covariates, and then asks what a covariate does in a randomized experiment.

18.1.1 The model

Write the mean of the response as the sum of a *design part* and a *covariate part*,

$$E(\mathbf{Y}) = \mathbf{X}\boldsymbol{\beta} + \mathbf{Z}\boldsymbol{\gamma}, \quad \text{Cov}(\mathbf{Y}) = \sigma^2 \mathbf{I}_n. \quad (18.1.1)$$

Here $\mathbf{X}$ is $n \times p$ with rank r , built from the indicator columns of the treatment factor and of any blocking factors. It is usually overparameterized. The $n \times q$ matrix $\mathbf{Z}$ holds the covariate values, and $\boldsymbol{\gamma} \in \mathbb{R}^q$ their coefficients. Let $\mathbf{M}$ be the orthogonal projection onto $\mathcal{C}(\mathbf{X})$. Everything below is written with the three matrices

$$\mathbf{E}_{zz} = \mathbf{Z}^\top (\mathbf{I} - \mathbf{M}) \mathbf{Z}, \quad \mathbf{E}_{zy} = \mathbf{Z}^\top (\mathbf{I} - \mathbf{M}) \mathbf{y}, \quad E_{yy} = \mathbf{y}^\top (\mathbf{I} - \mathbf{M}) \mathbf{y}. \quad (18.1.2)$$

The letter E stands for *error* or *within*: in the one-way layout $(\mathbf{I} - \mathbf{M})\mathbf{y}$ holds the deviations from the group means (Example (The one-way layout)), so these are within-group sums of squares and products. We assume throughout that

$$\text{rank}[\mathbf{X}, \mathbf{Z}] = \text{rank}(\mathbf{X}) + q, \quad \text{equivalently} \quad (\mathbf{I} - \mathbf{M})\mathbf{Z} \text{ has full column rank } q. \quad (18.1.3)$$

The two forms are equivalent by Lemma 9.2.1, since $\mathcal{C}([\mathbf{X}, \mathbf{Z}]) = \mathcal{C}(\mathbf{X}) \oplus \mathcal{C}((\mathbf{I} - \mathbf{M})\mathbf{Z})$. The condition fails exactly when some combination of the covariates lies in the design space. With one factor and one covariate, it fails when the covariate is constant within every group. Then the covariate cannot be separated from the group effects (Exercise (B1)).

18.1.2 The main theorem

Theorem 18.1.1 — Analysis of covariance

Assume Eq. 18.1.1 and Eq. 18.1.3. Let $\mathbf{M}_W$ be the orthogonal projection onto $\mathcal{C}([\mathbf{X}, \mathbf{Z}])$.

- (a) $\mathbf{M}_W = \mathbf{M} + (\mathbf{I} - \mathbf{M})\mathbf{Z}\mathbf{E}_{zz}^{-1}\mathbf{Z}^\top(\mathbf{I} - \mathbf{M})$, and the two terms are orthogonal projections

with orthogonal ranges.

- (b) $\boldsymbol{\gamma}$ is estimable, with the unique least squares estimate $\hat{\boldsymbol{\gamma}} = \mathbf{E}_{zz}^{-1}\mathbf{E}_{zy}$.
 (c) A function $\boldsymbol{\lambda}^\top \boldsymbol{\beta}$ is estimable in Eq. 18.1.1 iff it is estimable in the model $E(\mathbf{Y}) = \mathbf{X}\boldsymbol{\beta}$ without covariates. If $\boldsymbol{\lambda} = \mathbf{X}^\top \boldsymbol{\rho}$, its least squares estimate is

$$\boldsymbol{\lambda}^\top \hat{\boldsymbol{\beta}} = \boldsymbol{\rho}^\top \mathbf{M}(\mathbf{y} - \mathbf{Z}\hat{\boldsymbol{\gamma}}), \quad (18.1.4)$$

so the fitted design part is $\mathbf{X}\hat{\boldsymbol{\beta}} = \mathbf{M}(\mathbf{y} - \mathbf{Z}\hat{\boldsymbol{\gamma}})$: the covariance-adjusted response $\mathbf{y} - \mathbf{Z}\hat{\boldsymbol{\gamma}}$ analysed by the formulas of the model without covariates.

- (d) The residual sum of squares is

$$\text{SSE} = E_{yy} - \mathbf{E}_{zy}^\top \mathbf{E}_{zz}^{-1} \mathbf{E}_{zy}, \quad (18.1.5)$$

on $n - r - q$ degrees of freedom.

- (e) $\text{Cov}(\hat{\boldsymbol{\gamma}}) = \sigma^2 \mathbf{E}_{zz}^{-1}$, $\text{Cov}(\boldsymbol{\lambda}^\top \hat{\boldsymbol{\beta}}, \hat{\boldsymbol{\gamma}}) = -\sigma^2 \boldsymbol{\rho}^\top \mathbf{M} \mathbf{Z} \mathbf{E}_{zz}^{-1}$, and

$$\text{Var}(\boldsymbol{\lambda}^\top \hat{\boldsymbol{\beta}}) = \sigma^2 (\boldsymbol{\rho}^\top \mathbf{M} \boldsymbol{\rho} + \boldsymbol{\rho}^\top \mathbf{M} \mathbf{Z} \mathbf{E}_{zz}^{-1} \mathbf{Z}^\top \mathbf{M} \boldsymbol{\rho}). \quad (18.1.6)$$

- (f) Let $\mathcal{C}(\mathbf{X}_0) \subseteq \mathcal{C}(\mathbf{X})$ with rank $r_0 < r$ and projection $\mathbf{M}_0$, and define $\mathbf{T}_{zz}, \mathbf{T}_{zy}, T_{yy}$ as in Eq. 18.1.2 with $\mathbf{M}_0$ in place of $\mathbf{M}$. Then Eq. 18.1.3 holds for $\mathbf{X}_0$ as well, and the hypothesis that the design part of the mean lies in $\mathcal{C}(\mathbf{X}_0)$, with $\boldsymbol{\gamma}$ unrestricted, has the **adjusted sum of squares**

$$\text{SS}_H = (T_{yy} - \mathbf{T}_{zy}^\top \mathbf{T}_{zz}^{-1} \mathbf{T}_{zy}) - (E_{yy} - \mathbf{E}_{zy}^\top \mathbf{E}_{zz}^{-1} \mathbf{E}_{zy}) \quad (18.1.7)$$

on $r - r_0$ degrees of freedom. If $\mathbf{Y}$ is normal,

$$F = \frac{\text{SS}_H / (r - r_0)}{\text{SSE} / (n - r - q)} \sim F(r - r_0, n - r - q, \gamma),$$

where the noncentrality γ , a scalar not to be confused with the coefficient vector $\boldsymbol{\gamma}$, is given by $\sigma^2 \gamma$ equal to the squared distance from $\mathbf{X}\boldsymbol{\beta} + \mathbf{Z}\boldsymbol{\gamma}$ to $\mathcal{C}([\mathbf{X}_0, \mathbf{Z}])$.

Proof.

- (a) By Lemma 9.2.1, $\mathcal{C}([\mathbf{X}, \mathbf{Z}]) = \mathcal{C}(\mathbf{X}) \oplus \mathcal{C}(\tilde{\mathbf{Z}})$ with $\tilde{\mathbf{Z}} = (\mathbf{I} - \mathbf{M})\mathbf{Z}$, and the summands are orthogonal. $\tilde{\mathbf{Z}}$ has full column rank by Eq. 18.1.3, so its projection is $\tilde{\mathbf{Z}}(\tilde{\mathbf{Z}}^\top \tilde{\mathbf{Z}})^{-1} \tilde{\mathbf{Z}}^\top$ (Theorem 6.3.7), and $\tilde{\mathbf{Z}}^\top \tilde{\mathbf{Z}} = \mathbf{E}_{zz}$ because $\mathbf{I} - \mathbf{M}$ is symmetric and idempotent. Theorem 6.3.8 adds the two projections.
 (b) and (c). Put $\hat{\boldsymbol{\gamma}} = \mathbf{E}_{zz}^{-1} \mathbf{E}_{zy}$. By (a),

$$\mathbf{M}_W \mathbf{y} = \mathbf{M} \mathbf{y} + (\mathbf{I} - \mathbf{M}) \mathbf{Z} \hat{\boldsymbol{\gamma}} = \mathbf{M}(\mathbf{y} - \mathbf{Z} \hat{\boldsymbol{\gamma}}) + \mathbf{Z} \hat{\boldsymbol{\gamma}}.$$

The first term lies in $\mathcal{C}(\mathbf{X})$, so it equals $\mathbf{X}\mathbf{b}$ for some $\mathbf{b}$, and $(\mathbf{b}, \hat{\boldsymbol{\gamma}})$ is a least squares estimate (Theorem 6.4.1). If $(\mathbf{b}', \boldsymbol{\gamma}')$ is another one, then $\mathbf{X}(\mathbf{b} - \mathbf{b}') = \mathbf{Z}(\boldsymbol{\gamma}' - \hat{\boldsymbol{\gamma}})$. Applying $\mathbf{I} - \mathbf{M}$ gives $\tilde{\mathbf{Z}}(\boldsymbol{\gamma}' - \hat{\boldsymbol{\gamma}}) = \mathbf{0}$, and so $\boldsymbol{\gamma}' = \hat{\boldsymbol{\gamma}}$. The coefficient vector of the covariates is therefore the same in every least squares estimate, and so $\boldsymbol{\gamma}$ is estimable (Theorem 8.3.1(g)). The same argument shows that $\mathbf{X}\hat{\boldsymbol{\beta}} = \mathbf{M}(\mathbf{y} - \mathbf{Z}\hat{\boldsymbol{\gamma}})$ for every least squares estimate.

For estimability of $\boldsymbol{\lambda}^\top \boldsymbol{\beta}$: in the covariance model this function has coefficient vector $(\boldsymbol{\lambda}^\top, \mathbf{0}^\top)$, which is estimable iff $\mathbf{a}^\top [\mathbf{X}, \mathbf{Z}] = (\boldsymbol{\lambda}^\top, \mathbf{0}^\top)$ for some $\mathbf{a}$ (Theorem 8.2.2). Such

an $\mathbf{a}$ gives $\boldsymbol{\lambda} \in \mathcal{C}(\mathbf{X}^\top)$. Conversely, if $\boldsymbol{\lambda} = \mathbf{X}^\top \boldsymbol{\rho}$, put

$$\mathbf{a} = \mathbf{M}\boldsymbol{\rho} - \tilde{\mathbf{Z}}\mathbf{E}_{zz}^{-1}\tilde{\mathbf{Z}}^\top\mathbf{M}\boldsymbol{\rho}.$$

Then $\mathbf{a}^\top\mathbf{X} = \boldsymbol{\rho}^\top\mathbf{M}\mathbf{X} - \boldsymbol{\rho}^\top\mathbf{M}\mathbf{Z}\mathbf{E}_{zz}^{-1}\tilde{\mathbf{Z}}^\top\mathbf{X} = \boldsymbol{\lambda}^\top$, because $\tilde{\mathbf{Z}}^\top\mathbf{X} = \mathbf{Z}^\top(\mathbf{I} - \mathbf{M})\mathbf{X} = \mathbf{0}$, and $\mathbf{a}^\top\mathbf{Z} = \boldsymbol{\rho}^\top\mathbf{M}\mathbf{Z} - \boldsymbol{\rho}^\top\mathbf{M}\mathbf{Z}\mathbf{E}_{zz}^{-1}\tilde{\mathbf{Z}}^\top\mathbf{Z} = \mathbf{0}^\top$, because $\tilde{\mathbf{Z}}^\top\mathbf{Z} = \mathbf{E}_{zz}$. So $\boldsymbol{\lambda}^\top\boldsymbol{\beta}$ is estimable. Since $\mathbf{a}$ lies in $\mathcal{C}([\mathbf{X}, \mathbf{Z}])$, its least squares estimate is $\mathbf{a}^\top\mathbf{M}_W\mathbf{y} = \mathbf{a}^\top\mathbf{y}$ (Proposition 8.3.2(a)):

$$\mathbf{a}^\top\mathbf{y} = \boldsymbol{\rho}^\top\mathbf{M}\mathbf{y} - \boldsymbol{\rho}^\top\mathbf{M}\mathbf{Z}\mathbf{E}_{zz}^{-1}\mathbf{E}_{zy} = \boldsymbol{\rho}^\top\mathbf{M}(\mathbf{y} - \mathbf{Z}\hat{\boldsymbol{\gamma}}).$$

- (d) $\text{SSE} = \|(\mathbf{I} - \mathbf{M}_W)\mathbf{y}\|^2 = \mathbf{y}^\top(\mathbf{I} - \mathbf{M})\mathbf{y} - \mathbf{y}^\top\tilde{\mathbf{Z}}\mathbf{E}_{zz}^{-1}\tilde{\mathbf{Z}}^\top\mathbf{y}$ by (a), and $\tilde{\mathbf{Z}}^\top\mathbf{y} = \mathbf{E}_{zy}$. The rank of $\mathbf{M}_W$ is $r + q$.
- (e) $\hat{\boldsymbol{\gamma}} = \mathbf{E}_{zz}^{-1}\tilde{\mathbf{Z}}^\top\mathbf{Y}$ has covariance $\sigma^2\mathbf{E}_{zz}^{-1}\tilde{\mathbf{Z}}^\top\tilde{\mathbf{Z}}\mathbf{E}_{zz}^{-1} = \sigma^2\mathbf{E}_{zz}^{-1}$. The estimate of $\boldsymbol{\lambda}^\top\boldsymbol{\beta}$ is $\mathbf{a}^\top\mathbf{Y}$, and $\mathbf{M}\boldsymbol{\rho}$ is orthogonal to $\tilde{\mathbf{Z}}$. So $\text{Var}(\mathbf{a}^\top\mathbf{Y}) = \sigma^2\|\mathbf{a}\|^2$ is the sum of the two squared lengths in Eq. 18.1.6, and $\text{Cov}(\mathbf{a}^\top\mathbf{Y}, \hat{\boldsymbol{\gamma}}) = \sigma^2\mathbf{a}^\top\tilde{\mathbf{Z}}\mathbf{E}_{zz}^{-1} = -\sigma^2\boldsymbol{\rho}^\top\mathbf{M}\mathbf{Z}\mathbf{E}_{zz}^{-1}$.
- (f) If $(\mathbf{I} - \mathbf{M}_0)\mathbf{Z}\mathbf{c} = \mathbf{0}$, then $\mathbf{Z}\mathbf{c} \in \mathcal{C}(\mathbf{X}_0) \subseteq \mathcal{C}(\mathbf{X})$, so $(\mathbf{I} - \mathbf{M})\mathbf{Z}\mathbf{c} = \mathbf{0}$ and $\mathbf{c} = \mathbf{0}$. So Eq. 18.1.3 holds for $\mathbf{X}_0$, and (d) applied to $[\mathbf{X}_0, \mathbf{Z}]$ gives the first bracket in Eq. 18.1.7 as the residual sum of squares of the reduced model. The rank of the reduced model is $r_0 + q$, and $\mathcal{C}([\mathbf{X}_0, \mathbf{Z}]) \subseteq \mathcal{C}([\mathbf{X}, \mathbf{Z}])$. The distribution is Theorem 11.1.1. ■

The proof is the Frisch–Waugh–Lovell theorem (Theorem 6.6.2) with the covariates as the residualized columns, stated through $\mathbf{M}$ so that the design part may be rank deficient.

Idea. Analysis of covariance is two ordinary analyses joined by one projection. The within-group regression of $\mathbf{y}$ on the covariates gives $\hat{\boldsymbol{\gamma}}$, and the analysis of variance of the adjusted response $\mathbf{y} - \mathbf{Z}\hat{\boldsymbol{\gamma}}$ gives the treatment estimates. Their variances Eq. 18.1.6 carry an extra term for the uncertainty in $\hat{\boldsymbol{\gamma}}$. Tests must compare models, as in Eq. 18.1.7: the between-groups sum of squares of the adjusted response is not the numerator of any F statistic, because the reduced model must be allowed to re-estimate $\boldsymbol{\gamma}$.

18.1.3 The one-way layout with one covariate

For a single factor with g levels and a single covariate x , the matrices of Eq. 18.1.2 are scalars:

$$E_{xx} = \sum_k \sum_{i \in k} (x_i - \bar{x}_k)^2, \quad E_{xy} = \sum_k \sum_{i \in k} (x_i - \bar{x}_k)(y_i - \bar{y}_k),$$

and E_{yy} is the within-groups sum of squares of y . With $\mathbf{X}_0 = \mathbf{1}$, the T quantities are the same sums around the grand means. Theorem 18.1.1 gives:

- (i) slope $\hat{\beta} = E_{xy}/E_{xx}$;
- (ii) $\text{SSE} = E_{yy} - E_{xy}^2/E_{xx}$ on $n - g - 1$ degrees of freedom;
- (iii) adjusted treatment sum of squares $\text{SS}_{\text{trt}} = (T_{yy} - T_{xy}^2/T_{xx}) - (E_{yy} - E_{xy}^2/E_{xx})$ on $g - 1$ degrees of freedom;
- (iv) level means adjusted to a common covariate value, $\bar{y}_k - \hat{\beta}(\bar{x}_k - x_0)$, which differ by $(\bar{y}_k - \bar{y}_l) - \hat{\beta}(\bar{x}_k - \bar{x}_l)$, as in Proposition 8.6.3;
- (v) the test of $\beta = 0$, with sum of squares E_{xy}^2/E_{xx} on one degree of freedom (Corollary 9.2.3).

Example. (Four study formats with a pretest)

In a synthetic randomized experiment, $n = 40$ students are assigned at random, ten to each of four study formats for an introductory statistics unit. A pretest score x is recorded before the assignment, and the response y is the score on the final examination. The data were generated from a parallel lines model. Randomization balanced the pretest only on average: the four groups have pretest means 62.1, 57.2, 59.9 and 60.5, and final means 72.86, 75.15, 69.09 and 78.09. Figure 18.1.1 shows the data, and the sums of squares and products are:

	xx	xy	yy	df
between formats	124.9	-31.4	432.94	3
within formats	3643.9	3159.2	4146.41	36
total	3768.8	3127.8	4579.35	39

Without the covariate, the one-way analysis has error mean square $4146.41/36 = 115.2$ and treatment $F = 1.25$ on 3 and 36 degrees of freedom ($p = 0.305$). There is no evidence that the formats differ.

With the covariate, the within-groups slope is $\hat{\beta} = 3159.2/3643.9 = 0.8670$ points on the final per point on the pretest. The residual sum of squares falls to

$$\text{SSE} = 4146.41 - 2738.94 = 1407.47$$

on 35 degrees of freedom, so the error mean square falls from 115.2 to 40.2. The residual standard deviation falls from 10.73 to 6.34. The reduced model, one line for all forty students, has residual sum of squares $4579.35 - 2595.83 = 1983.52$. The adjusted treatment sum of squares is therefore $1983.52 - 1407.47 = 576.05$, and

$$F = \frac{576.05/3}{1407.47/35} = 4.77$$

on 3 and 35 degrees of freedom, with $p = 0.0068$: clear differences between the formats. The test of $\beta = 0$ has $F = 68.1$. The adjusted treatment sum of squares, 576.05, is even *larger* than the unadjusted 432.94: format 2 had the weakest students on entry and still finished second, and adjustment gives it credit for that.

```
import numpy as np
from scipy import stats

rng = np.random.default_rng(84)
g, m = 4, 10                                # four formats, ten students
    each
n = g * m
group = rng.permutation(np.repeat(np.arange(g), m)) # the randomization
x = np.round(rng.normal(60, 10, n))             # pretest, measured before
    assignment
mu_true = np.array([70.0, 74.0, 71.0, 77.0])
y = np.round(mu_true[group] + 0.8 * (x - 60) + rng.normal(0, 6, n), 1)

Z = np.eye(g)[group]                         # indicator matrix of the
    formats
n_k = Z.sum(axis=0)
```

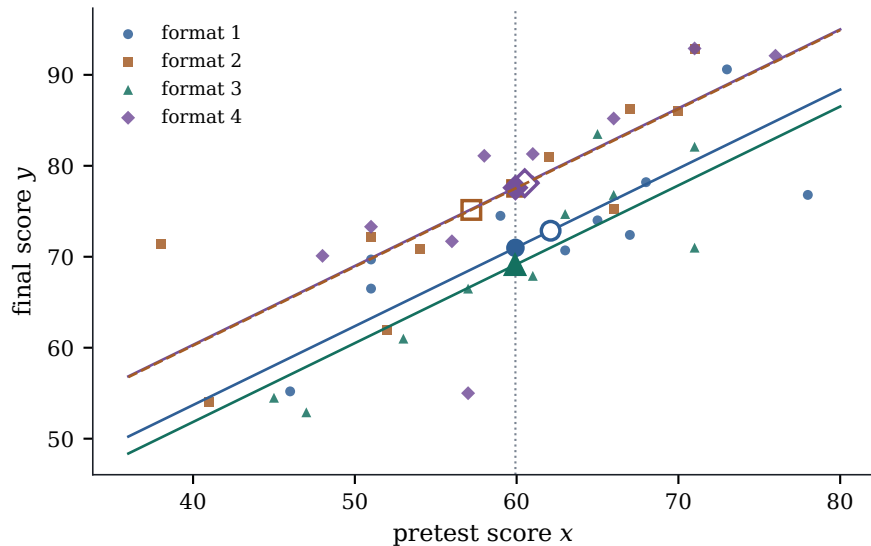


Figure 18.1.1: The tutoring experiment: final score against pretest score for the four formats, with the fitted parallel lines of the covariance model. Open symbols mark the raw group means $(\bar{x}_k, \bar{y}_k)$, and filled symbols on the dotted vertical line mark the adjusted means at the overall pretest mean $\bar{x}$. The lines of formats 2 (dashed, drawn on top) and 4 almost coincide.

```
x_bar, y_bar = Z.T @ x / n_k, Z.T @ y / n_k           # group means

def within(v):
    """(I - M) v: deviations from the group means (M projects onto C(Z))."""
    return v - (Z.T @ v / n_k)[group]

E_xx, E_xy, E_yy = within(x) @ within(x), within(x) @ within(y), within(y) @ within(
    y)
beta = E_xy / E_xx                                     # slope from within-group
                variation only
sse = E_yy - E_xy ** 2 / E_xx                         # ANCOVA residual sum of
                squares
nu = n - g - 1

T_xx = np.sum((x - x.mean()) ** 2)                   # the same with I - MO, MO =
                mean only
T_xy = np.sum((x - x.mean()) * (y - y.mean()))
T_yy = np.sum((y - y.mean()) ** 2)
sse0 = T_yy - T_xy ** 2 / T_xx                       # reduced model: one line for
                everyone
ss_trt = sse0 - sse                                   # adjusted treatment sum of
                squares
F_adj = (ss_trt / (g - 1)) / (sse / nu)
F_unadj = ((T_yy - E_yy) / (g - 1)) / (E_yy / (n - g))
print(f"slope {beta:.4f}; SSE: ANOVA {E_yy:.2f} on {n - g} df, ANCOVA {sse:.2f} on
      {nu} df")
print(f"treatments: unadjusted F = {F_unadj:.2f} (p = {stats.f.sf(F_unadj, g - 1, n
      - g):.3f}),"
      f" adjusted F = {F_adj:.2f} (p = {stats.f.sf(F_adj, g - 1, nu):.4f})")
```

18.1.4 What a covariate does in a randomized experiment

In a randomized experiment the groups differ on the covariate only by chance. Adjustment still helps, for two reasons.

Precision. The error variance of the covariance model is the variance of y given x , which is smaller whenever the two are correlated. The price is the estimated slope, which adds the second term in Eq. 18.1.6. The next result balances gain against price when the covariate is random, as it is in practice.

Proposition 18.1.2 — The precision of adjustment

Consider a one-way layout with g groups of m units, $N = gm$. Suppose the covariate is measured before randomization, so that the x_i are independent $N(\xi, \tau^2)$ whatever their group, and that given all the x 's the y_i are independent and normal, with $E(y_i | x_i) = \mu_k + \beta x_i$ for unit i in group k and $\text{Var}(y_i | x_i) = \sigma^2$. Let ρ be the within-group correlation of x and y . For two groups $k \neq l$, the unadjusted estimate $\bar{y}_k - \bar{y}_l$ and the adjusted estimate $(\bar{y}_k - \bar{y}_l) - \hat{\beta}(\bar{x}_k - \bar{x}_l)$ are both unbiased for $\mu_k - \mu_l$. If $N - g > 2$,

$$\frac{\text{Var}(\text{adjusted})}{\text{Var}(\text{unadjusted})} = (1 - \rho^2) \left(1 + \frac{1}{N - g - 2} \right). \quad (18.1.8)$$

Proof. Write $\bar{y}_k = \mu_k + \beta \bar{x}_k + \bar{\varepsilon}_k$, where $\varepsilon_i = y_i - \mu_k - \beta x_i$ has mean zero and variance σ^2 and is independent of all the x 's. Unconditionally $\bar{y}_k - \bar{y}_l$ has mean $\mu_k - \mu_l$, because $E \bar{x}_k = E \bar{x}_l$, and variance $(2/m)(\beta^2 \tau^2 + \sigma^2)$. The within-group variance of y is $\beta^2 \tau^2 + \sigma^2$, and $\sigma^2 = (1 - \rho^2)(\beta^2 \tau^2 + \sigma^2)$.

Given all the x 's, the covariance model holds with fixed covariates. By Theorem 18.1.1(e), with $\mathbf{p}$ the difference of the two scaled group indicators (so $\mathbf{M}\mathbf{p} = \mathbf{p}$ and $\mathbf{p}^\top \mathbf{Z} = \bar{x}_k - \bar{x}_l$), the adjusted estimate is conditionally unbiased with conditional variance $\sigma^2 \{2/m + (\bar{x}_k - \bar{x}_l)^2 / E_{xx}\}$. Its unconditional mean is therefore $\mu_k - \mu_l$, and its unconditional variance is the mean of the conditional variance. The group means of x are independent of the within-group sum of squares E_{xx} (Corollary 4.4.5, applied group by group), with $\bar{x}_k - \bar{x}_l \sim N(0, 2\tau^2/m)$ and $E_{xx}/\tau^2 \sim \chi^2(N - g)$. For $V \sim \chi^2(\nu)$ with $\nu > 2$, $E(1/V) = 1/(\nu - 2)$ (Exercise (B1)). Hence

$$E \frac{(\bar{x}_k - \bar{x}_l)^2}{E_{xx}} = \frac{2\tau^2}{m} \cdot \frac{1}{\tau^2(N - g - 2)}, \quad \text{Var}(\text{adjusted}) = \frac{2\sigma^2}{m} \left(1 + \frac{1}{N - g - 2} \right).$$

Dividing by the unadjusted variance gives Eq. 18.1.8. ■

The factor $1 - \rho^2$ is what adjustment would achieve if the slope were known. The factor $1 + 1/(N - g - 2)$ is the cost of estimating it. For the design of Example (Four study formats with a pretest) the cost is 1.0294. Adjustment pays whenever $\rho^2 > 1/(N - g - 1)$, that is, when $|\rho|$ exceeds 0.169. At $\rho = 0.8$, the correlation used to generate the tutoring data, the adjusted comparison has 0.371 of the variance of the unadjusted one (simulation: 0.372; Figure 18.1.2). Without the covariate the experiment would need almost three times as many students.

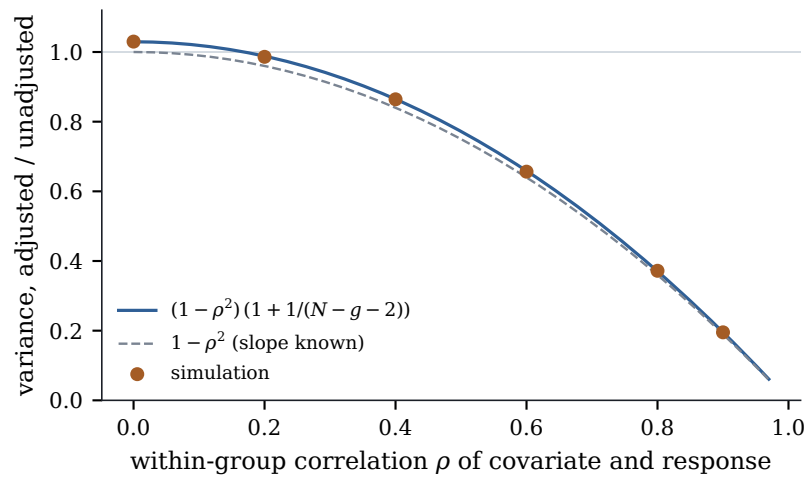


Figure 18.1.2: Variance of the adjusted difference of two group means relative to the unadjusted difference, for four groups of ten, as a function of the within-group correlation between covariate and response. The curve is Eq. 18.1.8, and the points are simulated from 100000 experiments each. The dashed curve is the ratio when the slope is known.

Chance imbalance. Randomization makes $\bar{y}_k - \bar{y}_l$ unbiased *over repetitions of the experiment*. In the experiment actually run, the proof above shows that, given the covariates,

$$E(\bar{y}_k - \bar{y}_l \mid x\text{'s}) = \mu_k - \mu_l + \beta(\bar{x}_k - \bar{x}_l),$$

which is *not* $\mu_k - \mu_l$ unless the covariate means agree. Once the pretests are known, the raw difference has an error of predictable sign. In [Example \(Four study formats with a pretest\)](#), format 1 started 4.9 points ahead of format 2 on the pretest, which predicts $0.8670 \times 4.9 \approx 4.25$ points of the final difference. The raw difference between formats 1 and 2 is -2.29 , and the conditionally unbiased adjusted difference is $-2.29 - 4.25 = -6.54$.

All this assumes the covariate was chosen in advance. Choosing, after seeing the data, the covariates that make the effect look best is a form of multiple testing, which is why trial protocols name their covariates beforehand.

Remark (Random covariates). The theorem ([Theorem 18.1.1](#)) treats $\mathbf{Z}$ as fixed. If, given random covariates, the model holds with normal errors, every F test and t interval of the theorem keeps its level unconditionally ([Theorem 14.1.2](#), applied after reparameterizing the design part to full rank, which changes neither the tests nor the estimable functions). Only the power depends on the covariate values that occur.

18.1.5 Adjusting for a covariate that the treatment affects

Warning. A covariate must not be affected by the treatment. If it is measured after assignment and the treatment changes it, adjusting for it removes the part of the treatment effect that acts through the covariate, and the adjusted comparison no longer estimates the effect of assigning the treatment.

If hours of study were used as a covariate, a format that works *by* getting students to study more would be compared with the others at equal hours of study, which removes the very thing

it does. In a synthetic randomized experiment with two arms of 100, let the treatment raise a post-assignment variable w by 2 units and let $E(y) = \theta T + 0.75w$ with direct effect $\theta = 1$. The effect of assigning the treatment is $1 + 0.75 \times 2 = 2.50$. Over 20000 replications, the unadjusted difference of means averages 2.500, and the coefficient of the treatment after adjustment for w averages 1.000, the direct effect, which is not what a decision about the treatment needs.

```
import numpy as np

rng = np.random.default_rng(2025)
n, reps = 200, 20_000
theta, delta, gamma = 1.0, 2.0, 0.75          # direct effect; T -> w; w ->
y                                              # effect of assigning
total = theta + gamma * delta
treatment
est_unadj, est_adj = np.empty(reps), np.empty(reps)
for r in range(reps):
    T = rng.permutation(np.repeat([0.0, 1.0], n // 2)) # randomized assignment
    w = rng.normal(size=n) + delta * T                 # measured after treatment
    y = theta * T + gamma * w + rng.normal(size=n)
    X = np.column_stack([np.ones(n), T, w])
    est_unadj[r] = y[T == 1].mean() - y[T == 0].mean()
    est_adj[r] = np.linalg.lstsq(X, y, rcond=None)[0][1]
print(f"total effect {total:.2f}: unadjusted mean {est_unadj.mean():.3f}, "
      f" adjusted for w mean {est_adj.mean():.3f}")
```

Chapter 25 develops the language for this in observational data: mediators, on the causal path from treatment to response, must not be adjusted for when the total effect is wanted. In a randomized experiment the safe rule is simple: *adjust only for variables fixed before randomization*. By [Proposition 18.1.2](#) this costs little even when the covariate turns out to be useless. A test of whether the groups are “balanced” on such a covariate is pointless, since its null hypothesis is true by construction. What matters is whether the covariate predicts the response.

18.1.6 Exercises

18.1.6.1 A. Check your understanding

Exercise A1.

In the one-way layout with one covariate, suppose x is constant within each group. Show that [Eq. 18.1.3](#) fails, and describe the set of least squares estimates of β . Is the adjusted difference of two group means estimable?

Solution. If $x_i = c_k$ for every unit in group k , then $\mathbf{x}$ is a combination of the group indicators, so $\mathbf{x} \in \mathcal{C}(\mathbf{X})$, $(\mathbf{I} - \mathbf{M})\mathbf{x} = \mathbf{0}$, and [Eq. 18.1.3](#) fails. The mean of group k is $\theta_k = \mu_k + \beta c_k$, and only the θ_k are estimable. Every value of β belongs to some least squares estimate: for any b , the group parameters $\bar{y}_k - bc_k$ together with b reproduce the fitted group means. The adjusted difference $\mu_k - \mu_l = (\theta_k - \theta_l) - \beta(c_k - c_l)$ is therefore estimable iff $c_k = c_l$. The data cannot separate a covariate that varies only between groups from the group effects.

Exercise A2.

From the numbers in [Example \(Four study formats with a pretest\)](#), compute the F statistic for $\beta = 0$ and check that its square root is the t statistic $\hat{\beta}/(s/\sqrt{E_{xx}})$.

18.1.6.2 B. Practice**Exercise B1.**

Let $V \sim \chi^2(\nu)$ with $\nu > 2$. Show that $E(1/V) = 1/(\nu - 2)$, and that $E(1/V) = \infty$ for $\nu \leq 2$.

Solution. The density of V is $f_\nu(v) = v^{\nu/2-1}e^{-v/2}/\{2^{\nu/2}\Gamma(\nu/2)\}$. Then

$$E(1/V) = \int_0^\infty \frac{v^{\nu/2-2}e^{-v/2}}{2^{\nu/2}\Gamma(\nu/2)} dv = \frac{2^{\nu/2-1}\Gamma(\nu/2-1)}{2^{\nu/2}\Gamma(\nu/2)} = \frac{1}{2(\nu/2-1)} = \frac{1}{\nu-2},$$

using $\Gamma(s+1) = s\Gamma(s)$. The integral converges at 0 iff $\nu/2 - 2 > -1$, that is, iff $\nu > 2$.

Exercise B2.

In the one-way layout with one covariate, write $B_{xx} = T_{xx} - E_{xx}$, and similarly B_{xy} and B_{yy} , for the between-groups sums of squares and products. Show that the adjusted treatment sum of squares equals

$$B_{yy} - \left(\frac{T_{xy}^2}{T_{xx}} - \frac{E_{xy}^2}{E_{xx}} \right),$$

and deduce that it exceeds B_{yy} iff $E_{xy}^2/E_{xx} > T_{xy}^2/T_{xx}$. Show that this happens in [Example \(Four study formats with a pretest\)](#), and explain it in terms of B_{xy} .

Solution. The formula is [Eq. 18.1.7](#) with $T_{yy} - E_{yy} = B_{yy}$. In the example $E_{xy}^2/E_{xx} = 2738.94$ and $T_{xy}^2/T_{xx} = 2595.83$, so the adjusted sum of squares is larger, by 143.11. The total regression uses both the within-group and the between-group variation of x . Here the between-group cross-product $B_{xy} = -31.4$ is negative: across the four formats, higher pretest means go with *lower* final means, the opposite of the within-group relationship. The single line for all students therefore explains less than the within-group lines do, and the difference is attributed to the formats.

Exercise B3.

In the one-way layout with q covariates, use [Eq. 18.1.6](#) to show that the adjusted difference $(\bar{y}_k - \bar{y}_l) - (\bar{\mathbf{z}}_k - \bar{\mathbf{z}}_l)^\top \hat{\boldsymbol{\gamma}}$ has variance $\sigma^2\{1/n_k + 1/n_l + (\bar{\mathbf{z}}_k - \bar{\mathbf{z}}_l)^\top \mathbf{E}_{zz}^{-1}(\bar{\mathbf{z}}_k - \bar{\mathbf{z}}_l)\}$.

18.1.6.3 C. Going deeper**Exercise C1.**

In the one-way covariance model, show that replacing x by $a + cx$ with $c \neq 0$ leaves the

fitted values, SSE, the adjusted treatment sum of squares and the adjusted differences unchanged. Then replace x by $x + d_k$ on group k , with group-specific constants d_k . Show that $\hat{\beta}$ and SSE are unchanged but the adjusted differences and the adjusted treatment sum of squares change. Explain why a covariate recorded with a group-specific calibration error is dangerous.

18.2 Adjusted treatment means

A covariance analysis is usually reported as a table of treatment means “adjusted” for the covariate. The raw mean $\bar{y}_k$ estimates the mean response of treatment k at whatever covariate values its units happened to have; the adjusted mean estimates it at a covariate value common to all treatments. This section defines adjusted means, gives their standard errors, and shows how to compare them.

18.2.1 Definition

Take the one-way covariance model with g groups of sizes $n_1, \dots, n_g$ and q covariates,

$$E(y_i) = \mu_k + \mathbf{z}_{(i)}^\top \boldsymbol{\gamma} \quad \text{for unit } i \text{ in group } k, \quad (18.2.1)$$

where $\mathbf{z}_{(i)} \in \mathbb{R}^q$ is the covariate vector of unit i , and write $\bar{\mathbf{z}}_k$ for the mean covariate vector of group k and $\bar{\mathbf{z}}$ for the overall mean. We assume [Eq. 18.1.3](#), and write $\mathbf{E}_{zz}$ and $\hat{\boldsymbol{\gamma}}$ as in [Theorem 18.1.1](#).

Definition 18.2.1 — Adjusted treatment means

For a fixed covariate vector $\mathbf{z}_0 \in \mathbb{R}^q$, the **adjusted mean** of treatment k at $\mathbf{z}_0$ is $\mu_k + \mathbf{z}_0^\top \boldsymbol{\gamma}$, the mean response of a unit with covariate $\mathbf{z}_0$ under treatment k . Its estimate is

$$\hat{\mu}_k(\mathbf{z}_0) = \bar{y}_k - (\bar{\mathbf{z}}_k - \mathbf{z}_0)^\top \hat{\boldsymbol{\gamma}}. \quad (18.2.2)$$

When no value is named, $\mathbf{z}_0 = \bar{\mathbf{z}}$, the overall mean of the covariates, and $\hat{\mu}_k(\bar{\mathbf{z}})$ is called simply the adjusted mean of treatment k .

The fitted line (or plane) of treatment k passes through $(\bar{\mathbf{z}}_k, \bar{y}_k)$ with slope $\hat{\boldsymbol{\gamma}}$; the adjusted mean is its height at $\mathbf{z}_0$. In [Figure 18.1.1](#) the filled symbols on the dashed line are the adjusted means.

The adjusted mean is estimable: $\mu_k + \mathbf{z}_0^\top \boldsymbol{\gamma}$ is the sum of $\mu_k + \bar{\mathbf{z}}_k^\top \boldsymbol{\gamma}$, the mean of the fitted values of group k , and $(\mathbf{z}_0 - \bar{\mathbf{z}}_k)^\top \boldsymbol{\gamma}$, a multiple of the estimable $\boldsymbol{\gamma}$ ([Theorem 18.1.1\(b\)](#)). By [Theorem 18.1.1\(c\)](#) with $\boldsymbol{\rho}$ the scaled indicator of group k , the least squares estimate of μ_k is $\bar{y}_k - \bar{\mathbf{z}}_k^\top \hat{\boldsymbol{\gamma}}$, which gives [Eq. 18.2.2](#).

18.2.2 Standard errors

Proposition 18.2.1 — Moments of adjusted means

In the model Eq. 18.2.1 with $\text{Cov}(\mathbf{Y}) = \sigma^2 \mathbf{I}$, and for a fixed $\mathbf{z}_0$:

- (a) $\bar{y}_1, \dots, \bar{y}_g$ are uncorrelated with $\hat{\gamma}$, and

$$\text{Var}(\hat{\mu}_k(\mathbf{z}_0)) = \sigma^2 \left(\frac{1}{n_k} + (\bar{\mathbf{z}}_k - \mathbf{z}_0)^\top \mathbf{E}_{zz}^{-1} (\bar{\mathbf{z}}_k - \mathbf{z}_0) \right); \quad (18.2.3)$$

- (b) for $k \neq l$, $\text{Cov}(\hat{\mu}_k(\mathbf{z}_0), \hat{\mu}_l(\mathbf{z}_0)) = \sigma^2 (\bar{\mathbf{z}}_k - \mathbf{z}_0)^\top \mathbf{E}_{zz}^{-1} (\bar{\mathbf{z}}_l - \mathbf{z}_0)$;

- (c) for a contrast $\psi = \sum_k c_k \mu_k$, the estimate $\hat{\psi} = \sum_k c_k \hat{\mu}_k(\mathbf{z}_0) = \sum_k c_k \bar{y}_k - \bar{\mathbf{z}}_c^\top \hat{\gamma}$, with $\bar{\mathbf{z}}_c = \sum_k c_k \bar{\mathbf{z}}_k$, does not depend on $\mathbf{z}_0$, and

$$\text{Var}(\hat{\psi}) = \sigma^2 \left(\sum_k \frac{c_k^2}{n_k} + \bar{\mathbf{z}}_c^\top \mathbf{E}_{zz}^{-1} \bar{\mathbf{z}}_c \right); \quad (18.2.4)$$

- (d) with $\mathbf{z}_0 = \bar{\mathbf{z}}$, the weighted average of the adjusted means is the grand mean: $\sum_k n_k \hat{\mu}_k(\bar{\mathbf{z}}) = \sum_i y_i$.

Proof. Let $\mathbf{M}$ project onto the span of the group indicators. The group mean $\bar{y}_k$ is $\mathbf{u}_k^\top \mathbf{Y}$ with $\mathbf{u}_k$ the indicator of group k divided by n_k , a vector in $\mathcal{C}(\mathbf{M})$. The estimate $\hat{\gamma} = \mathbf{E}_{zz}^{-1} \mathbf{Z}^\top (\mathbf{I} - \mathbf{M}) \mathbf{Y}$ is a linear function of $(\mathbf{I} - \mathbf{M}) \mathbf{Y}$. Hence $\text{Cov}(\bar{y}_k, \hat{\gamma}) = \sigma^2 \mathbf{u}_k^\top (\mathbf{I} - \mathbf{M}) \mathbf{Z} \mathbf{E}_{zz}^{-1} = \mathbf{0}$, since $(\mathbf{I} - \mathbf{M}) \mathbf{u}_k = \mathbf{0}$. Then (a) and (b) follow from $\text{Var}(\bar{y}_k) = \sigma^2/n_k$, $\text{Cov}(\bar{y}_k, \bar{y}_l) = 0$ and $\text{Cov}(\hat{\gamma}) = \sigma^2 \mathbf{E}_{zz}^{-1}$ (Theorem 18.1.1(e)). For (c), $\sum_k c_k = 0$ removes $\mathbf{z}_0$, and the variance follows in the same way. For (d), $\sum_k n_k \hat{\mu}_k(\bar{\mathbf{z}}) = \sum_k n_k \bar{y}_k - (\sum_k n_k \bar{\mathbf{z}}_k - n \bar{\mathbf{z}})^\top \hat{\gamma}$, and the bracket is zero. ■

Adjusted means are correlated, because their errors share $\hat{\gamma}$. Their variances grow with the distance of $\bar{\mathbf{z}}_k$ from $\mathbf{z}_0$: an adjusted mean is a prediction from the treatment's line, and for a group whose covariate values barely overlap the others' it rests on extrapolation, as in Section 12.3. Differences do not depend on $\mathbf{z}_0$; only the common level and the individual standard errors do. The overall mean $\bar{\mathbf{z}}$ keeps the adjusted means on the scale of the data (d) and minimizes $\sum_k n_k \text{Var}(\hat{\mu}_k(\mathbf{z}_0))$ (Exercise (B1)), but any stated value, such as a clinically meaningful baseline, is legitimate.

18.2.3 Comparing adjusted means

Pairwise differences are contrasts, so Eq. 18.2.4 gives

$$\text{se}(\hat{\mu}_k - \hat{\mu}_l) = s \sqrt{\frac{1}{n_k} + \frac{1}{n_l} + (\bar{\mathbf{z}}_k - \bar{\mathbf{z}}_l)^\top \mathbf{E}_{zz}^{-1} (\bar{\mathbf{z}}_k - \bar{\mathbf{z}}_l)}, \quad s^2 = \frac{\text{SSE}}{n - g - q}, \quad (18.2.5)$$

and a single comparison uses the t interval of Theorem 12.1.1 with $n - g - q$ degrees of freedom. For simultaneous comparisons the correlations matter.

- (i) *Scheffé*. The adjusted treatment F test of Theorem 18.1.1(f) is a test of $\mu_1 = \dots = \mu_g$, a $(g - 1)$ -dimensional space of estimable contrasts. By Theorem 13.2.2, the intervals

- $\hat{\psi} \pm \sqrt{(g-1)F_\alpha(g-1, n-g-q)} \text{se}(\hat{\psi})$ have simultaneous coverage exactly $1 - \alpha$ over *all* contrasts ψ . Some interval excludes zero iff the adjusted F test rejects ([Corollary 13.2.3](#)).
- (ii) *Bonferroni*. For a planned list of m comparisons, t intervals at level α/m have coverage at least $1 - \alpha$ whatever the correlations ([Theorem 13.4.1](#)).
 - (iii) *Tukey-type intervals*. The half-width $(q_\alpha(g, n-g-q)/\sqrt{2}) \text{se}(\hat{\mu}_k - \hat{\mu}_l)$ copies the Tukey–Kramer rule [Eq. 13.3.3](#). It is exact when the covariate means are equal and the groups balanced ([Exercise \(C1\)](#)). Otherwise its coverage is only approximately $1 - \alpha$: Hayter’s theorem covers unequal group sizes in the one-way layout, but not the correlations that the covariate introduces. When the covariate means are close, as they usually are after randomization, the departure is small ([Exercise \(C2\)](#) asks you to measure it). When they are far apart, use Scheffé’s intervals, or compute the exact multiplier for the given correlation matrix by simulation, as [Section 13.3](#) did for Tukey–Kramer.

Example. (Adjusted means for the four formats)

In [Example \(Four study formats with a pretest\)](#), the overall pretest mean is $\bar{x} = 59.92$. Format 2 had pretest mean 57.2, so its adjusted mean is

$$75.15 - 0.8670(57.2 - 59.92) = 75.15 + 2.36 = 77.51.$$

The four adjusted means are 70.97, 77.51, 69.11 and 77.59, averaging to the grand mean 73.80, with standard errors 2.02, 2.03, 2.01 and 2.01. These barely differ, because the pretest means are close to $\bar{x}$ compared with the within-group standard deviation of the pretest, 10.06.

The six pairwise comparisons, with $s = 6.34$ on 35 degrees of freedom, are as follows. The Tukey-type multiplier is $q_{0.05}(4, 35)/\sqrt{2} = 2.697$, and Scheffé’s is $\sqrt{3F_{0.05}(3, 35)} = 2.936$.

formats	raw difference	adjusted difference	se	Tukey-type half-width	Scheffé half-width
1 2	−2.29	−6.54	2.88	7.77	8.46
1 3	3.77	1.86	2.85	7.67	8.36
1 4	−5.23	−6.62	2.84	7.66	8.34
2 3	6.06	8.40	2.85	7.69	8.37
2 4	−2.94	−0.08	2.86	7.71	8.39
3 4	−9.00	−8.48	2.84	7.65	8.33

Both methods declare formats 2 and 4 better than format 3 and resolve nothing else, although the 2 3 comparison clears Scheffé’s bound by only 0.03. Without the covariate the standard error of a difference would be $10.73\sqrt{2/10} = 4.80$. The raw difference between formats 3 and 4, −9.00, would then have had a Tukey half-width near 13, and nothing would have been resolved.

Adjustment moved comparisons in both directions: format 2’s advantage over format 1 grew from 2.29 to 6.54, while format 4’s lead over format 3 shrank from 9.00 to 8.48.

```
import itertools
import numpy as np
from scipy import stats
```

```
rng = np.random.default_rng(84)
```

```

g, m = 4, 10                                # four formats, ten students
    each
n = g * m
group = rng.permutation(np.repeat(np.arange(g), m)) # the randomization
x = np.round(rng.normal(60, 10, n))            # pretest, measured before
    assignment
mu_true = np.array([70.0, 74.0, 71.0, 77.0])
y = np.round(mu_true[group] + 0.8 * (x - 60) + rng.normal(0, 6, n), 1)
Z = np.eye(g)[group]                          # indicator matrix of the
    formats
n_k = Z.sum(axis=0)
x_bar, y_bar = Z.T @ x / n_k, Z.T @ y / n_k    # group means

def within(v):
    """(I - M) v: deviations from the group means (M projects onto C(Z))."""
    return v - (Z.T @ v / n_k)[group]

E_xx, E_xy, E_yy = within(x) @ within(x), within(x) @ within(y), within(y) @ within(
    y)
beta = E_xy / E_xx                            # slope from within-group
    variation only
sse = E_yy - E_xy ** 2 / E_xx                  # ANCOVA residual sum of
    squares
nu = n - g - 1

s2 = sse / nu
x0 = x.mean()                                # the covariate value for
    adjustment
adj = y_bar - beta * (x_bar - x0)              # adjusted means
se_adj = np.sqrt(s2 * (1 / n_k + (x_bar - x0) ** 2 / E_xx))
pairs = list(itertools.combinations(range(g), 2))
q = stats.studentized_range.ppf(0.95, g, nu) / np.sqrt(2) # Tukey-Kramer
    multiplier
c_s = np.sqrt((g - 1) * stats.f.ppf(0.95, g - 1, nu))    # Scheffe multiplier
for k, l in pairs:
    d = adj[k] - adj[l]
    se_d = np.sqrt(s2 * (1 / n_k[k] + 1 / n_k[l] + (x_bar[k] - x_bar[l]) ** 2 / E_xx))
    print(f"{k + 1}-{l + 1}: raw {y_bar[k] - y_bar[l]:.2f} adjusted {d:.2f} se {
        se_d:.2f}")
        f" Tukey-type +-{q * se_d:.2f} Scheffe +-{c_s * se_d:.2f}")
print("adjusted means", adj.round(2), " standard errors", se_adj.round(2))

```

18.2.4 Adjusted means in other designs

For any design part, the adjusted version at $\mathbf{z}_0$ of an estimable $\lambda^\top \beta$, such as a treatment mean averaged over blocks, is $\lambda^\top \beta + \mathbf{z}_0^\top \gamma$, estimated by $\rho^\top \mathbf{M}(\mathbf{y} - \mathbf{Z}\hat{\gamma}) + \mathbf{z}_0^\top \hat{\gamma}$ with variance from [Theorem 18.1.1\(e\)](#). Software calls these **least squares means**. In unbalanced multi-way data the choice of λ , equal or count-proportional weights over cells, matters as much as $\mathbf{z}_0$ ([Chapter 17](#), [Theorem 17.2.2](#)); it is safest to write the intended function of the cell means ([Definition 17.1.1](#)) explicitly.

Remark (The adjusted mean is not a prediction interval). The standard error [Eq. 18.2.3](#) describes the uncertainty about the *mean* response of treatment k at $\mathbf{z}_0$. A new unit

with covariate $\mathbf{z}_0$ given treatment k will vary around that mean with variance σ^2 as well. Its prediction interval uses $s\sqrt{1 + 1/n_k + (\bar{\mathbf{z}}_k - \mathbf{z}_0)^\top \mathbf{E}_{zz}^{-1}(\bar{\mathbf{z}}_k - \mathbf{z}_0)}$, as in [Theorem 12.3.1](#).

18.2.5 Exercises

18.2.5.1 A. Check your understanding

Exercise A1.

From the numbers in [Example \(Four study formats with a pretest\)](#), compute the adjusted means of formats 1 and 3 by hand and check them against the values in [Example \(Adjusted means for the four formats\)](#).

Exercise A2.

Software that fits the model $E(y) = \mu_k + \beta x$ reports $\hat{\mu}_k$, the adjusted mean at $x_0 = 0$. For a pretest scored from 0 to 100, what does this number describe? Show that its standard error can be far larger than that of the adjusted mean at $\bar{x}$, although both give the same differences.

18.2.5.2 B. Practice

Exercise B1.

Show that $\sum_k n_k \text{Var}(\hat{\mu}_k(\mathbf{z}_0))$ is minimized over $\mathbf{z}_0$ by $\mathbf{z}_0 = \bar{\mathbf{z}}$.

Solution. By [Eq. 18.2.3](#), the sum is $\sigma^2(g + \sum_k n_k (\bar{\mathbf{z}}_k - \mathbf{z}_0)^\top \mathbf{E}_{zz}^{-1}(\bar{\mathbf{z}}_k - \mathbf{z}_0))$. Writing $\bar{\mathbf{z}}_k - \mathbf{z}_0 = (\bar{\mathbf{z}}_k - \bar{\mathbf{z}}) + (\bar{\mathbf{z}} - \mathbf{z}_0)$ and using $\sum_k n_k (\bar{\mathbf{z}}_k - \bar{\mathbf{z}}) = \mathbf{0}$, the cross terms vanish and the sum becomes

$$\sigma^2 \left(g + \sum_k n_k (\bar{\mathbf{z}}_k - \bar{\mathbf{z}})^\top \mathbf{E}_{zz}^{-1}(\bar{\mathbf{z}}_k - \bar{\mathbf{z}}) + n(\bar{\mathbf{z}} - \mathbf{z}_0)^\top \mathbf{E}_{zz}^{-1}(\bar{\mathbf{z}} - \mathbf{z}_0) \right).$$

The last term is nonnegative, and zero iff $\mathbf{z}_0 = \bar{\mathbf{z}}$, because $\mathbf{E}_{zz}$ is positive definite.

Exercise B2.

A common alternative to covariance adjustment analyses the *change* $y - x$. Show that this is the covariance model with β fixed at 1. Using [Proposition 18.1.2](#)'s setting with τ^2 the variance of x , find the variance of the difference of mean changes between two groups, and show that it exceeds the variance of the adjusted difference (ignoring the $1/(N - g - 2)$ term) unless $\beta = 1$.

Solution. The mean change in group k is $\bar{y}_k - \bar{x}_k = \mu_k + (\beta - 1)\bar{x}_k + \bar{\varepsilon}_k$. The difference of two groups has variance $(2/m)\{(\beta - 1)^2\tau^2 + \sigma^2\}$. The adjusted difference, with the slope known, has variance $2\sigma^2/m$. The change-score analysis is therefore worse by $(2/m)(\beta - 1)^2\tau^2$, which is zero only when $\beta = 1$. It beats the unadjusted analysis, which

has $(\beta - 1)^2$ replaced by β^2 , only if $\beta > 1/2$. Covariance adjustment estimates the slope rather than assuming one.

18.2.5.3 C. Going deeper

Exercise C1.

Suppose all groups have size m and the covariate means $\bar{\mathbf{z}}_1, \dots, \bar{\mathbf{z}}_g$ are equal. Show that the adjusted differences $\hat{\mu}_k - \hat{\mu}_l$ are the raw differences $\bar{y}_k - \bar{y}_l$, that these are independent of s^2 from the covariance model, and that the Tukey intervals built from s and $q_\alpha(g, n - g - q)$ have simultaneous coverage exactly $1 - \alpha$.

Solution. With equal covariate means, $\bar{\mathbf{z}}_c = \sum_k c_k \bar{\mathbf{z}}_k = \mathbf{0}$ for every contrast, so the adjusted contrast equals the raw one and has variance $\sigma^2 \sum_k c_k^2 / m$ by Eq. 18.2.4. The group means are functions of $\mathbf{M}\mathbf{Y}$. The residual vector $(\mathbf{I} - \mathbf{M}_W)\mathbf{Y}$ is orthogonal to $\mathcal{C}(\mathbf{M}_W) \supseteq \mathcal{C}(\mathbf{M})$, so under normality s^2 is independent of $\bar{y}_1, \dots, \bar{y}_g$ (Theorem 4.4.4), and $(n - g - q)s^2/\sigma^2 \sim \chi^2(n - g - q)$. The means are independent normal with common variance σ^2/m . The maximum of $|(\bar{y}_k - \bar{y}_l) - (\mu_k - \mu_l)|/(s/\sqrt{m})$ therefore has the studentized range distribution with parameters g and $n - g - q$, and the proof of Theorem 13.3.2 applies unchanged.

Exercise C2.

For four groups of ten and one covariate, simulate the simultaneous coverage of the Tukey-type intervals of Example (Adjusted means for the four formats) when the covariate means are held at (62, 57, 60, 61) and when they are held at (70, 50, 60, 60), with the within-group covariate variation as in the example. How far from 0.95 is the coverage in each case?

18.3 Heterogeneous slopes

The covariance model of Section 18.1 assumes a common slope, so each treatment comparison is a vertical shift between parallel lines, one number. If the slopes differ, the treatment effect depends on the covariate, and “the” adjusted difference no longer exists. This section tests parallelism and says what to report when it fails.

18.3.1 The separate lines model and the test of parallelism

Keep one factor with g levels and one covariate, and allow each level its own line:

$$E(y_i) = \mu_k + \beta_k x_i \quad \text{for unit } i \text{ in group } k.$$

This is the separate lines model Eq. 8.6.4 of Section 8.6. The parallel lines model is the special case $\beta_1 = \dots = \beta_g$. For group k write

$$E_{xx,k} = \sum_{i \in k} (x_i - \bar{x}_k)^2, \quad E_{xy,k} = \sum_{i \in k} (x_i - \bar{x}_k)(y_i - \bar{y}_k),$$

so that the pooled within-group quantities of Section 18.1 are $E_{xx} = \sum_k E_{xx,k}$ and $E_{xy} = \sum_k E_{xy,k}$.

Theorem 18.3.1 — The test of parallel slopes

Assume x varies within every group, so that every $E_{xx,k} > 0$, and $n > 2g$.

- (a) In the separate lines model, $\hat{\beta}_k = E_{xy,k}/E_{xx,k}$, these estimates are uncorrelated with variances $\sigma^2/E_{xx,k}$, and the residual sum of squares is

$$\text{SSE}_{\text{sep}} = \sum_k \left(E_{yy,k} - \frac{E_{xy,k}^2}{E_{xx,k}} \right)$$

on $n - 2g$ degrees of freedom.

- (b) The pooled slope of the parallel lines model is the weighted mean $\hat{\beta} = \sum_k E_{xx,k} \hat{\beta}_k / \sum_k E_{xx,k}$, and the extra sum of squares for the separate slopes is

$$\text{SS}_{\text{slopes}} = \sum_k E_{xx,k} (\hat{\beta}_k - \hat{\beta})^2 = \sum_k \frac{E_{xy,k}^2}{E_{xx,k}} - \frac{E_{xy}^2}{E_{xx}} \quad (18.3.1)$$

on $g - 1$ degrees of freedom.

- (c) If $\mathbf{Y}$ is normal,

$$F = \frac{\text{SS}_{\text{slopes}}/(g-1)}{\text{SSE}_{\text{sep}}/(n-2g)} \sim F(g-1, n-2g, \gamma),$$

with

$$\gamma = \frac{1}{\sigma^2} \sum_k E_{xx,k} (\beta_k - \bar{\beta})^2,$$

where $\bar{\beta} = \sum_k E_{xx,k} \beta_k / \sum_k E_{xx,k}$. The test that rejects for large F is the F test of parallelism.

Proof. Let $\mathbf{M}$ project onto the span of the group indicators, and let $\mathbf{w}_k$ be the vector that equals $x_i - \bar{x}_k$ for units of group k and zero elsewhere. The k th slope column of the separate lines model is $\mathbf{x}$ restricted to group k , and its residual after projection onto $\mathcal{C}(\mathbf{M})$ is $\mathbf{w}_k$. By Lemma 9.2.1 the separate lines model space is $\mathcal{C}(\mathbf{M}) \oplus \text{span}\{\mathbf{w}_1, \dots, \mathbf{w}_g\}$, and the parallel lines space is $\mathcal{C}(\mathbf{M}) \oplus \text{span}\{\mathbf{w}\}$ with $\mathbf{w} = \sum_k \mathbf{w}_k = (\mathbf{I} - \mathbf{M})\mathbf{x}$. The vectors $\mathbf{w}_k$ have disjoint supports, so they are mutually orthogonal, with $\|\mathbf{w}_k\|^2 = E_{xx,k}$ and $\mathbf{w}_k^\top \mathbf{y} = E_{xy,k}$.

- (a) The projection of $\mathbf{y}$ onto $\text{span}\{\mathbf{w}_k\}$ is $\sum_k (E_{xy,k}/E_{xx,k})\mathbf{w}_k$. Reading off the coefficient of the k th slope column gives $\hat{\beta}_k$, which is $\mathbf{w}_k^\top \mathbf{y} / \|\mathbf{w}_k\|^2$. Hence $\text{Cov}(\hat{\beta}_k, \hat{\beta}_l) = \sigma^2 \mathbf{w}_k^\top \mathbf{w}_l / (E_{xx,k} E_{xx,l})$, which is $\sigma^2/E_{xx,k}$ for $k = l$ and zero otherwise. The residual sum of squares is $\|(\mathbf{I} - \mathbf{M})\mathbf{y}\|^2 - \sum_k E_{xy,k}^2/E_{xx,k}$, and $\|(\mathbf{I} - \mathbf{M})\mathbf{y}\|^2 = \sum_k E_{yy,k}$. The model has rank $2g$.

- (b) The pooled slope is $\mathbf{w}^\top \mathbf{y} / \|\mathbf{w}\|^2 = E_{xy} / E_{xx}$, and $E_{xy} = \sum_k E_{xx,k} \hat{\beta}_k$. By [Theorem 6.5.1](#) the extra sum of squares is the squared length of the projection onto the orthogonal complement of $\mathbf{w}$ in $\text{span}\{\mathbf{w}_k\}$, which equals $\sum_k E_{xy,k}^2 / E_{xx,k} - E_{xy}^2 / E_{xx}$. Writing $E_{xy,k} = E_{xx,k} \hat{\beta}_k$ and expanding $\sum_k E_{xx,k} (\hat{\beta}_k - \hat{\beta})^2 = \sum_k E_{xx,k} \hat{\beta}_k^2 - \hat{\beta}^2 E_{xx}$ gives the first form.
- (c) By [Theorem 11.1.1](#), with $\sigma^2 \gamma$ equal to the extra sum of squares evaluated at the mean vector. The mean vector has $\mathbf{w}_k^\top \boldsymbol{\mu} = \beta_k E_{xx,k}$, because $\mathbf{w}_k$ is orthogonal to the group indicators. Substituting into (b) with $\hat{\beta}_k$ replaced by β_k gives γ .

■

The test is a weighted test of homogeneity of g independent slope estimates, with weights equal to their precisions $E_{xx,k} / \sigma^2$.

Example. (Parallelism in the tutoring experiment)

For the four formats of [Example \(Four study formats with a pretest\)](#), the separate slopes are 0.703, 0.823, 1.035 and 0.992, and their weighted mean is the pooled slope 0.8670. The separate lines model has residual sum of squares 1347.47 on 32 degrees of freedom, against 1407.47 on 35 for parallel lines. The test gives $F = 0.47$ on 3 and 32 degrees of freedom, $p = 0.70$, and the covariance analysis stands.

```
import numpy as np
from scipy import stats

rng = np.random.default_rng(84)
g, m = 4, 10                                # four formats, ten students
    each
n = g * m
group = rng.permutation(np.repeat(np.arange(g), m)) # the randomization
x = np.round(rng.normal(60, 10, n))            # pretest, measured before
    assignment
mu_true = np.array([70.0, 74.0, 71.0, 77.0])
y = np.round(mu_true[group] + 0.8 * (x - 60) + rng.normal(0, 6, n), 1)
Z = np.eye(g)[group]                          # indicator matrix of the
    formats
n_k = Z.sum(axis=0)

def within(v):
    """(I - M) v: deviations from the group means (M projects onto C(Z))."""
    return v - (Z.T @ v / n_k)[group]

E_xx, E_xy, E_yy = within(x) @ within(x), within(x) @ within(y), within(y) @ within(
    y)
sse = E_yy - E_xy ** 2 / E_xx                  # ANCOVA residual sum of
    squares
X_sep = np.column_stack([Z, Z * x[:, None]])   # a separate line for each
    format
E_xx_k = np.array([within(x)[group == k] @ within(x)[group == k] for k in range(g)])
E_xy_k = np.array([within(x)[group == k] @ within(y)[group == k] for k in range(g)])
slopes = E_xy_k / E_xx_k
sse_sep = sse - (np.sum(E_xy_k ** 2 / E_xx_k) - E_xy ** 2 / E_xx)
F_par = ((sse - sse_sep) / (g - 1)) / (sse_sep / (n - 2 * g))
print("separate slopes", slopes.round(3),
```



```
f" F = {F_par:.2f}, p = {stats.f.sf(F_par, g - 1, n - 2 * g):.3f}"
```

With ten students per group the test has little power, so the case for parallel lines rests mostly on the science. When a treatment is meant to help some units more than others, such as a remedial programme aimed at weak students, the separate lines model should be the starting point rather than a check.

18.3.2 When the slopes differ

If the slopes differ, the difference between treatments k and l at covariate value x is the line

$$\delta(x) = (\mu_k - \mu_l) + (\beta_k - \beta_l)x,$$

estimated by $\hat{\delta}(x) = (\hat{\mu}_k - \hat{\mu}_l) + (\hat{\beta}_k - \hat{\beta}_l)x$. The two lines are independent simple regressions, so by [Corollary 5.5.4](#)

$$\text{Var}(\hat{\delta}(x)) = \sigma^2 v(x), \quad v(x) = \frac{1}{n_k} + \frac{(x - \bar{x}_k)^2}{E_{xx,k}} + \frac{1}{n_l} + \frac{(x - \bar{x}_l)^2}{E_{xx,l}}. \quad (18.3.2)$$

Honest reports include:

- (i) Report $\hat{\delta}(x)$ with its standard error at a few covariate values chosen for their meaning, for instance the quartiles of x , or a clinical threshold. Each is a t interval ([Theorem 12.1.1](#)) with $n - 2g$ degrees of freedom.
- (ii) Draw $\hat{\delta}(x)$ as a function with a confidence band. The pointwise band uses $t_{n-2g, \alpha/2}$. A band that covers the whole line $\delta(\cdot)$ with probability exactly $1 - \alpha$ replaces this multiplier by $\sqrt{2F_\alpha(2, n - 2g)}$, because $\{\delta(x) : x \in \mathbb{R}\}$ lies in a two-dimensional space of estimable functions. This is the argument of [Theorem 13.2.2](#), as used for the Working–Hotelling band ([Theorem 12.4.1](#)).
- (iii) Give the set of covariate values at which the treatments differ significantly. This is the **Johnson–Neyman region** ([Proposition 18.3.2](#) below).

Reporting $\hat{\delta}(\bar{x})$ alone as *the* treatment effect hides the slope, which is the whole point.

Remark (The average effect in a randomized experiment). There is one exception. In a randomized experiment, δ at the population mean of the covariate is the *average* treatment effect, because δ is linear. Its estimate $\hat{\delta}(\bar{x})$ is the treatment coefficient in the regression with a treatment-by-covariate interaction and the covariate centred at $\bar{x}$ ([Exercise \(C1\)](#)). This “interacted” adjustment is asymptotically never less precise than the unadjusted difference, even when the model is wrong. Chapter 25 returns to it.

18.3.3 The Johnson–Neyman region

Proposition 18.3.2 — Johnson–Neyman region

In the separate lines model with normal errors, let $s^2 = \text{SSE}_{\text{sep}}/(n - 2g)$ and fix a multiplier $c > 0$. Write $\Delta_\mu = \hat{\mu}_k - \hat{\mu}_l$ and $\Delta_\beta = \hat{\beta}_k - \hat{\beta}_l$. The set of x with $|\hat{\delta}(x)| >$

$cs\sqrt{v(x)}$ is $\{x : Ax^2 + Bx + C > 0\}$, where

$$A = \Delta_\beta^2 - c^2 s^2 \left(\frac{1}{E_{xx,k}} + \frac{1}{E_{xx,l}} \right), \quad B = 2\Delta_\mu \Delta_\beta + 2c^2 s^2 \left(\frac{\bar{x}_k}{E_{xx,k}} + \frac{\bar{x}_l}{E_{xx,l}} \right),$$

$$C = \Delta_\mu^2 - c^2 s^2 \left(\frac{1}{n_k} + \frac{1}{n_l} + \frac{\bar{x}_k^2}{E_{xx,k}} + \frac{\bar{x}_l^2}{E_{xx,l}} \right).$$

If $A > 0$, which happens iff the t test of $\beta_k = \beta_l$ with multiplier c rejects, the region is the whole line with at most an interval removed: δ is significantly positive on one side and significantly negative on the other. If $A < 0$, the region is a bounded interval or empty. With $c = t_{n-2g, \alpha/2}$ each point of the region is significant at level α . With $c = \sqrt{2F_\alpha(2, n-2g)}$ the statement “ $\delta(x) \neq 0$ for every x in the region” holds with probability at least $1 - \alpha$.

Proof. Square both sides of $|\hat{\delta}(x)| > cs\sqrt{v(x)}$ and expand $\hat{\delta}(x)^2 = \Delta_\mu^2 + 2\Delta_\mu \Delta_\beta x + \Delta_\beta^2 x^2$ and $v(x)$ from Eq. 18.3.2 as polynomials in x . Collecting terms gives $Ax^2 + Bx + C$. The coefficient A is positive iff $\Delta_\beta^2 / \{s^2(1/E_{xx,k} + 1/E_{xx,l})\} > c^2$, which is the squared t statistic for $\beta_k - \beta_l$ exceeding c^2 . A quadratic with $A > 0$ is positive outside the interval between its real roots (everywhere if it has none), and with $A < 0$ it is positive between its real roots (nowhere if it has none). The last statement follows from the simultaneous band of item (ii) above: on the event that the band covers $\delta(\cdot)$ everywhere, $\delta(x)$ cannot be zero at any point where the band excludes zero. ■

Example. (Two versions of a reading programme)

In a second synthetic randomized experiment, 48 children with baseline reading fluency between 42 and 118 words per minute are assigned, 24 each, to the current and a new version of a reading programme. The response is the gain in fluency. The new version was designed for weak readers, and the fitted lines of Figure 18.3.1(a) show why that matters. The slope is 0.0156 for the current version and -0.1615 for the new one. The test of parallelism has $F = 11.2$ on 1 and 44 degrees of freedom, $p = 0.0017$. The pooled slope of the parallel lines model would be -0.0638 , a compromise that describes neither group.

The estimated lines cross at $x = 107.7$. At a baseline of 60 words per minute the new version adds 8.46 words per minute (standard error 1.56). At 100 it adds 1.37 (standard error 1.46), which is not significant. With $c = t_{44, 0.025}$, the Johnson–Neyman region (Proposition 18.3.2) is $x < 93.5$ together with $x > 152.0$. The second part lies far beyond the largest baseline in the data, and it is an artefact of extending two fitted lines. The useful conclusion is that the new version is better for children who start below about 93 words per minute, which covers 33 of the 48 children in the study. With the simultaneous multiplier $\sqrt{2F_{0.05}(2, 44)}$, the region shrinks slightly, to $x < 90.8$ (and $x > 195.6$).

```
import numpy as np
from scipy import stats
```

```
rng = np.random.default_rng(31)
n_per = 24
```

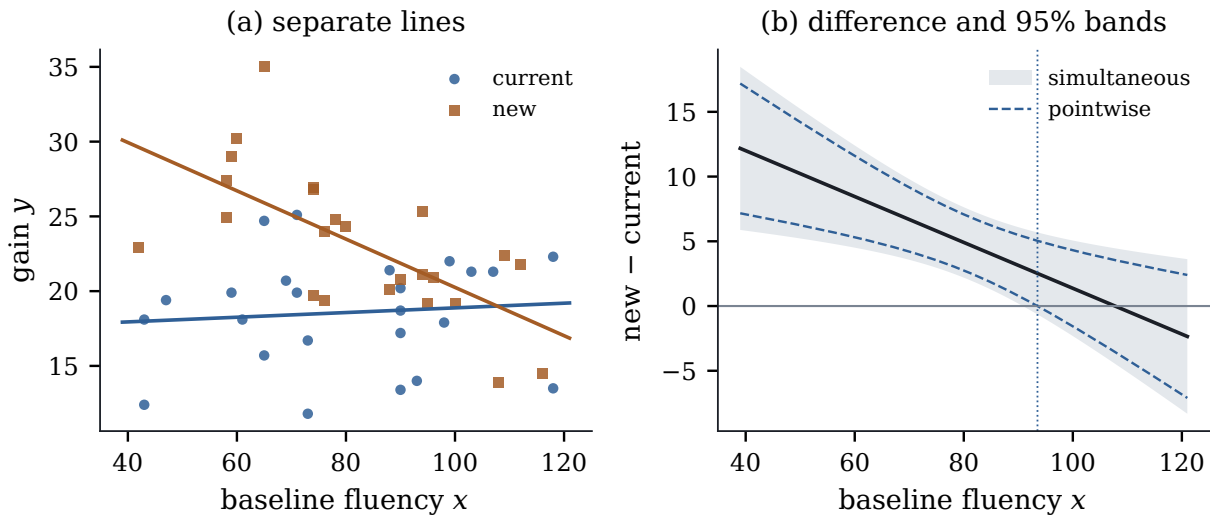


Figure 18.3.1: The reading programme. (a) Gain against baseline fluency with the separate least squares lines. (b) The estimated difference $\hat{\delta}(x)$ (new minus current) with a pointwise 95% band (dashed) and a simultaneous 95% band (shaded). The dotted line marks the lower end of the Johnson–Neyman region, $x = 93.5$; to its left the difference is significantly positive.

```

arm = rng.permutation(np.repeat([0, 1], n_per))      # 0 = current version, 1 = new
version
x = np.round(rng.uniform(40, 120, 2 * n_per))      # baseline fluency
y = np.where(arm == 0, 14 + 0.06 * x, 36 - 0.16 * x) + rng.normal(0, 4, 2 * n_per)
y = np.round(y, 1)
n = len(y)

def line(k):
    """Least squares line within arm k, and the arm's summaries."""
    xs, ys = x[arm == k], y[arm == k]
    Sxx = np.sum((xs - xs.mean()) ** 2)
    b = np.sum((xs - xs.mean()) * (ys - ys.mean())) / Sxx
    a = ys.mean() - b * xs.mean()
    sse = np.sum((ys - a - b * xs) ** 2)
    return a, b, xs.mean(), Sxx, len(xs), sse

(a0, b0, xb0, S0, n0, e0), (a1, b1, xb1, S1, n1, e1) = line(0), line(1)
nu = n - 4
s2 = (e0 + e1) / nu                                # separate-lines model

def significant_region(crit2):
    """Roots of  $d(x)^2 - \text{crit2} * s2 * v(x) = 0$ , a quadratic in  $x$ ."""
    A = (b1 - b0) ** 2 - crit2 * s2 * (1 / S0 + 1 / S1)
    B = 2 * (a1 - a0) * (b1 - b0) + 2 * crit2 * s2 * (xb0 / S0 + xb1 / S1)
    C = (a1 - a0) ** 2 - crit2 * s2 * (1 / n0 + 1 / n1 + xb0 ** 2 / S0 + xb1 ** 2 / S1)
    return np.sort(np.roots([A, B, C]).real)

t2 = stats.t.ppf(0.975, nu) ** 2                    # pointwise, 5% level
w2 = 2 * stats.f.ppf(0.95, 2, nu)                  # simultaneous over all x (
    Scheffe,  $q = 2$ )
x_cross = -(a1 - a0) / (b1 - b0)
print(f"lines cross at x = {x_cross:.1f}")
print("pointwise 5% region: difference significant outside", significant_region(t2).
    round(1))

```

```
print("simultaneous region: difference significant outside", significant_region(w2).
      round(1))
```

The region is only as good as the linearity of each line, it extrapolates outside the data, and it marks significance, not practical importance: with enough data every value except the crossing point becomes significant.

18.3.4 Several covariates, several factors

With q covariates, the test of a common γ against $\gamma_1, \dots, \gamma_g$ has $(g-1)q$ and $n - g(q+1)$ degrees of freedom. With two factors, slope terms can be built term by term like the means model of [Chapter 16](#). Each such test compares nested column spaces ([Theorem 11.1.1](#)).

18.3.5 Exercises

18.3.5.1 A. Check your understanding

Exercise A1.

For $g = 2$, show that [Eq. 18.3.1](#) reduces to $SS_{\text{slopes}} = \frac{E_{xx,1}E_{xx,2}}{E_{xx,1} + E_{xx,2}} (\hat{\beta}_1 - \hat{\beta}_2)^2$, and that the F statistic is the square of the two-sample t statistic for the difference of slopes.

Exercise A2.

In [Example \(Two versions of a reading programme\)](#), the estimated lines cross at $x = 107.7$. Explain why the Johnson–Neyman region never contains the crossing point, whatever the multiplier.

18.3.5.2 B. Practice

Exercise B1.

Give a numerical example of two groups for which $A < 0$ and the Johnson–Neyman region of [Proposition 18.3.2](#) is a bounded interval. Interpret the region.

Exercise B2.

In the tutoring experiment, suppose the true slopes were 0.8, 0.8, 0.8, 1.2 and $\sigma = 6$. Using the $E_{xx,k}$ of the example (their sum is 3643.9; take them equal for the calculation), compute the noncentrality of the test of parallelism and its power at level 0.05 ([Theorem 11.5.1](#)).

Exercise B3.

Prove the claim in item (ii): with $c = \sqrt{2F_\alpha(2, n-2g)}$, the band $\hat{\delta}(x) \pm cs\sqrt{v(x)}$

contains $\delta(x)$ for all x simultaneously with probability at least $1 - \alpha$. Is the probability exactly $1 - \alpha$?

Solution. The functions $\delta(x) = \lambda_x^\top \beta$ for $x \in \mathbb{R}$ have λ_x in a two-dimensional space $\mathcal{L}$ spanned by the coefficient vectors of $\mu_k - \mu_l$ and $\beta_k - \beta_l$, both estimable. By [Theorem 13.2.2\(b\)](#) with $q = 2$, with probability exactly $1 - \alpha$ every $\lambda \in \mathcal{L}$ satisfies $|\lambda^\top \hat{\beta} - \lambda^\top \beta| \leq c \text{se}(\lambda^\top \hat{\beta})$. The functions λ_x form a subset of $\mathcal{L}$, so the probability for them is at least $1 - \alpha$. It is in fact exactly $1 - \alpha$. In the coordinates $(\mu_k - \mu_l, \beta_k - \beta_l)$ we have $\lambda_x = (1, x)$, and the studentized ratio $(\lambda^\top \hat{\beta} - \lambda^\top \beta)^2 / \text{se}(\lambda^\top \hat{\beta})^2$ is unchanged when λ is rescaled and is continuous on $\mathcal{L} \setminus \{0\}$. Every direction of $\mathcal{L}$ except that of $(0, 1)$ is the direction of some λ_x , and $\lambda_x / \|\lambda_x\| \rightarrow \pm(0, 1)$ as $x \rightarrow \pm\infty$. So the supremum of the ratio over $x \in \mathbb{R}$ equals its supremum over $\mathcal{L} \setminus \{0\}$, and the two events coincide. This is the argument of [Theorem 12.4.1\(b\)](#) for a single line.

18.3.5.3 C. Going deeper

Exercise C1.

In a two-group randomized experiment, show that the treatment coefficient in the least squares fit of y on $1, T, x - \bar{x}$ and $T(x - \bar{x})$, where $\bar{x}$ is the overall mean, equals $(\bar{y}_1 - \hat{\beta}_1(\bar{x}_1 - \bar{x})) - (\bar{y}_0 - \hat{\beta}_0(\bar{x}_0 - \bar{x}))$, with $\hat{\beta}_k$ the within-group slopes. Compare with the adjusted difference of the parallel lines model.

Solution. The model with an interaction is the separate lines model, so its fitted lines are the two within-group least squares lines, $\bar{y}_k + \hat{\beta}_k(x - \bar{x}_k)$. In the given parameterization the line for group k is $(\beta_0 + \beta_T T) + (\beta_x + \beta_{Tx} T)(x - \bar{x})$, so the coefficient of T is the difference of the two lines at $x = \bar{x}$, which is the stated expression. The parallel lines model replaces both $\hat{\beta}_k$ by the pooled slope. The two estimates agree when the slopes are equal or when $\bar{x}_0 = \bar{x}_1 = \bar{x}$.

18.4 Randomized complete blocks

A covariate removes nuisance variation by regression. **Blocking** removes it by design: similar units (plots in one part of a field, animals from one litter, strips cut from one board) are grouped into blocks, and treatments are compared *within* blocks. This section analyses the randomized complete block design, explains how randomization justifies its model, and measures what blocking gains.

18.4.1 The design and its model

Definition 18.4.1 — Randomized complete block design

There are t treatments and b blocks of t units each. In every block, the t treatments are assigned to the t units by a random permutation, chosen uniformly and independently for different blocks. The response of the unit in block j that received treatment i is

written y_{ij} .

The usual model is the additive two-way model with one observation per cell,

$$y_{ij} = \mu + \tau_i + \beta_j + \varepsilon_{ij}, \quad \varepsilon_{ij} \stackrel{\text{iid}}{\sim} N(0, \sigma^2), \quad (18.4.1)$$

with treatment effects τ_i and block effects β_j . Its geometry is [Example \(Additive two-factor model, one observation per cell\)](#) and its analysis of variance is that of [Theorem 16.1.2](#); we restate it in the form the design needs.

Theorem 18.4.1 — Analysis of the randomized complete block design

Assume [Eq. 18.4.1](#) with $t, b \geq 2$. Let $\bar{y}_{i\cdot}$, $\bar{y}_{\cdot j}$ and $\bar{y}_{\cdot\cdot}$ be the treatment, block and grand means.

- (a) The sums of squares

$$\begin{aligned} SS_T &= b \sum_i (\bar{y}_{i\cdot} - \bar{y}_{\cdot\cdot})^2, & SS_B &= t \sum_j (\bar{y}_{\cdot j} - \bar{y}_{\cdot\cdot})^2, \\ SSE &= \sum_{i,j} (y_{ij} - \bar{y}_{i\cdot} - \bar{y}_{\cdot j} + \bar{y}_{\cdot\cdot})^2 \end{aligned}$$

are squared lengths of projections onto mutually orthogonal subspaces of dimensions $t-1$, $b-1$ and $(t-1)(b-1)$, and with the correction for the mean they add up to $\sum_{i,j} y_{ij}^2$.

- (b) A function $\sum_i c_i \tau_i$ is estimable iff $\sum_i c_i = 0$. Its least squares estimate is $\sum_i c_i \bar{y}_{i\cdot}$, and

$$\sum_i c_i \bar{y}_{i\cdot} = \sum_i c_i \tau_i + \sum_i c_i \bar{\varepsilon}_i.$$

identically, whatever the block effects. Its variance is $\sigma^2 \sum_i c_i^2 / b$.

- (c) The treatment F statistic satisfies

$$F_T = \frac{SS_T / (t-1)}{SSE / ((t-1)(b-1))} \sim F(t-1, (t-1)(b-1), \gamma),$$

with

$$\gamma = \frac{b}{\sigma^2} \sum_i (\tau_i - \bar{\tau})^2.$$

- (d) $E(SS_B / (b-1)) = \sigma^2 + t \sum_j (\beta_j - \bar{\beta})^2 / (b-1)$, and $E(SSE / ((t-1)(b-1))) = \sigma^2$.

Proof. Model [Eq. 18.4.1](#) is the additive two-way model of [Theorem 16.1.2](#) with factor A the treatments, factor B the blocks, $a = t$ and $m = 1$, so that $n - a - b + 1 = (t-1)(b-1)$. Parts (a), (b) and (c) are [Theorem 16.1.2](#)(a), (b) and (c), with (e) for the form of the residual space. The only new statement is the identity in (b): averaging [Eq. 18.4.1](#) over j gives $\bar{y}_{i\cdot} = \mu + \tau_i + \bar{\beta} + \bar{\varepsilon}_i$, and $\sum_i c_i (\mu + \bar{\beta}) = 0$ for a contrast. Part (d) is [Theorem 9.6.1](#) applied to the decomposition in (a); the block projection sends the mean vector to the

vector with entries $\beta_j - \bar{\beta}$, whose squared length is $t \sum_j (\beta_j - \bar{\beta})^2$.

■

Part (b) is the reason to block. However large the block effects, they cancel exactly from every treatment comparison and from the error mean square, which estimates the variation *within* blocks. The between-block variation moves into its own line, SS_B . A test of blocks is rarely of interest, since blocks were chosen because they differ; a large MS_B measures what blocking achieved ([Proposition 18.4.4](#) below).

Any block-by-treatment interaction is absorbed into the error, and without replication within cells it cannot be separated from it. Tukey's one-degree-of-freedom test ([Theorem 16.1.2\(f\)](#)) checks for its most common form. If treatment differences genuinely vary between blocks, the comparisons refer to the average over the blocks used.

18.4.2 Randomization as the justification of the model

The units of an experiment are not a sample from a normal population; they are the boards or patients at hand. What the experimenter controls is the randomization, and it alone supports much of the analysis. Give each unit u a fixed value ξ_u , and assume **unit–treatment additivity**: unit u responds to treatment i with $\xi_u + \tau_i$. The only probability below is that of the randomization.

Proposition 18.4.2 — Randomization moments of the block analysis

Let the units of block j have values $\xi_{j1}, \dots, \xi_{jt}$, with mean $\bar{\xi}_j$ and $W_j = \sum_s (\xi_{js} - \bar{\xi}_j)^2$, and put

$$S_W^2 = \frac{1}{b(t-1)} \sum_j W_j,$$

the average within-block variance of the unit values. Under unit–treatment additivity and the randomization of [Definition 18.4.1](#):

- (a) every contrast estimate $\sum_i c_i \bar{y}_i$ has randomization mean $\sum_i c_i \tau_i$ and randomization variance $S_W^2 \sum_i c_i^2 / b$;
- (b) $E(\text{SSE} / ((t-1)(b-1))) = S_W^2$;
- (c) $E(\text{SS}_T / (t-1)) = S_W^2 + b \sum_i (\tau_i - \bar{\tau})^2 / (t-1)$.

So the error mean square is unbiased for the variance that the model attributes to σ^2 , and when the treatments have no effect the treatment and error mean squares have the same expectation.

Proof. Let $\pi_j(i)$ be the unit of block j that receives treatment i , and $w_{ij} = \xi_{j\pi_j(i)} - \bar{\xi}_j$. Then $y_{ij} = \bar{\xi}_j + \tau_i + w_{ij}$. For fixed j , the vector $(w_{1j}, \dots, w_{tj})$ is a uniformly random permutation of the deviations $\xi_{js} - \bar{\xi}_j$, which sum to zero. Hence $E w_{ij} = 0$, $E w_{ij}^2 = W_j / t$, and, since $\sum_k w_{kj} = 0$, $E w_{ij} w_{kj} = -W_j / \{t(t-1)\}$ for $i \neq k$. Different blocks are independent.

- (a) $\sum_i c_i \bar{y}_i = \sum_i c_i \tau_i + b^{-1} \sum_j \sum_i c_i w_{ij}$, because $\sum_i c_i = 0$ removes the $\bar{\xi}_j$. The mean follows. For the variance,

$$\text{Var}\left(\sum_i c_i w_{ij}\right) = \frac{W_j}{t} \sum_i c_i^2 - \frac{W_j}{t(t-1)} \sum_{i \neq k} c_i c_k = \frac{W_j}{t-1} \sum_i c_i^2,$$

since $\sum_{i \neq k} c_i c_k = (\sum_i c_i)^2 - \sum_i c_i^2 = -\sum_i c_i^2$. Summing over the independent blocks and dividing by b^2 gives $S_W^2 \sum_i c_i^2 / b$.

- (b) The block means are $\bar{y}_{.j} = \bar{\xi}_j + \bar{\tau}$, and the treatment means are $\bar{y}_i = \bar{\xi} + \tau_i + \bar{w}_{i.}$, with $\bar{\xi} = b^{-1} \sum_j \bar{\xi}_j$. So the residual is $w_{ij} - \bar{w}_{i.}$, and

$$\text{SSE} = \sum_{i,j} w_{ij}^2 - b \sum_i \bar{w}_{i.}^2.$$

The first sum is $\sum_j W_j$ for every randomization. By independence across blocks, $E \bar{w}_{i.}^2 = b^{-2} \sum_j E w_{ij}^2 = b^{-2} \sum_j W_j / t$, so $E(b \sum_i \bar{w}_{i.}^2) = b^{-1} \sum_j W_j$. Hence $E(\text{SSE}) = (1 - 1/b) \sum_j W_j = (t-1)(b-1)S_W^2$.

- (c) $\text{SS}_T = b \sum_i (\tau_i - \bar{\tau} + \bar{w}_{i.})^2$, using $\sum_i \bar{w}_{i.} = 0$. The cross terms have mean zero, and $E(b \sum_i \bar{w}_{i.}^2) = (t-1)S_W^2$ by the computation in (b).

■

Independent measurement errors of variance σ_e^2 add σ_e^2 to every expectation. The proposition does not make F_T exactly F -distributed, but it shows that the estimates and their estimated variances are unbiased because of the randomization, whatever made the units differ. A systematic assignment could line treatments up with the ξ 's, and then neither statement would hold.

Randomization tests. Under the *sharp* null hypothesis that every unit would respond the same under every treatment, the responses are fixed and only the labels are random. The proportion of the $(t!)^b$ equally likely randomizations giving an F_T at least as large as the observed one is an exact p -value, estimated by sampling randomizations when there are too many. Chapter 23 develops permutation tests for linear models.

18.4.3 An example

Example. (Five wood adhesives on six boards)

In a synthetic experiment, five adhesives are compared for the shear strength of glued joints. Timber varies from board to board, so each of six boards is cut into five strips, assigned to the adhesives by a random permutation. The strengths (MPa) are:

adhesive	board 1	board 2	board 3	board 4	board 5	board 6
1	7.89	8.42	8.55	8.30	9.26	9.15
2	8.68	8.69	8.73	8.62	9.72	9.51
3	9.13	9.28	9.81	9.43	10.30	9.41
4	8.48	8.60	8.46	8.95	9.42	9.02
5	8.90	9.17	9.79	9.57	10.24	9.09

The analysis of variance is:

source	df	sum of squares	mean square	F
adhesives	4	4.106	1.0266	14.88
boards	5	4.068	0.8136	11.79
error	20	1.380	0.0690	
corrected total	29	9.555		

The adhesives differ clearly ($p = 8.6 \times 10^{-6}$). The adhesive means are 8.595, 8.992, 9.560, 8.822 and 9.460, and every difference has standard error $\sqrt{2 \times 0.0690/6} = 0.152$. Tukey's honestly significant difference at level 0.05, $q_{0.05}(5, 20)\sqrt{0.0690/6}$, is 0.454 ([Theorem 13.3.2](#) applies, with the error mean square of the block analysis, as in [Exercise \(B1\)](#)). It separates adhesives 3 and 5 from adhesives 1, 2 and 4: the significant pairs are 1–3, 1–5, 2–3, 2–5, 3–4 and 4–5. Tukey's test for nonadditivity gives $F = 0.07$ ($p = 0.80$), so there is no sign that the adhesive differences depend on the strength of the board.

Of 20000 randomizations sampled from the 2.99×10^{12} possible plans, 1 gave an adhesive F at least as large as 14.88: the randomization p -value is $(1 + 1)/(20000 + 1) = 0.00010$, in line with the normal theory.

```
import numpy as np
from scipy import stats

rng = np.random.default_rng(606)
t, b = 5, 6                                # adhesives, boards
plan = np.array([rng.permutation(t) for _ in range(b)]) # which strip of each
board gets which adhesive
tau = np.array([0.0, 0.40, 0.65, 0.15, 0.80]) # adhesive effects (unknown in
practice)
board = rng.normal(0, 0.75, b)              # board effects
y = np.round(9 + tau[:, None] + board[None, :] + rng.normal(0, 0.3, (t, b)), 2)

grand = y.mean()
trt_mean, blk_mean = y.mean(axis=1), y.mean(axis=0)
resid = y - trt_mean[:, None] - blk_mean[None, :] + grand
ss_trt = b * np.sum((trt_mean - grand) ** 2)
ss_blk = t * np.sum((blk_mean - grand) ** 2)
ss_err = np.sum(resid ** 2)
df_err = (t - 1) * (b - 1)
ms_trt, ms_blk, ms_err = ss_trt / (t - 1), ss_blk / (b - 1), ss_err / df_err
F_trt = ms_trt / ms_err
print(f"treatments SS {ss_trt:.2f}, blocks SS {ss_blk:.2f}, error SS {ss_err:.2f} on
{df_err} df")
print(f"F for adhesives = {F_trt:.2f}, p = {stats.f.sf(F_trt, t - 1, df_err):.2g}")
print("adhesive means", trt_mean.round(2), "se of a difference", np.sqrt(2 * ms_err
/ b).round(3))
```

```
def f_stat(yy):
    tm, bm, gm = yy.mean(axis=1), yy.mean(axis=0), yy.mean()
    sst = b * np.sum((tm - gm) ** 2)
    sse = np.sum((yy - tm[:, None] - bm[None, :] + gm) ** 2)
    return (sst / (t - 1)) / (sse / df_err)
```

```

rng_perm = np.random.default_rng(7)
draws = 20_000
F_perm = np.empty(draws)
for r in range(draws):
    # re-randomize: permute the adhesive labels independently within every board
    perm = np.array([rng_perm.permutation(t) for _ in range(b)]).T
    F_perm[r] = f_stat(np.take_along_axis(y, perm, axis=0))
p_rand = (1 + np.sum(F_perm >= F_trt)) / (draws + 1)
print(f"randomization p-value {p_rand:.4f} (F-test p-value {stats.f.sf(F_trt, t -
    1, df_err):.2g})")

```

18.4.4 How much did blocking gain?

Had the adhesives been assigned to the thirty strips completely at random, the completely randomized design (CRD) would have had the board-to-board variation in its error. The block experiment itself can estimate how large that error would have been, using a fact about sampling from a finite population.

Lemma 18.4.3 — Sample variance without replacement

Let $u_1, \dots, u_N$ be fixed numbers with mean $\bar{u}$ and $S^2 = \sum_a (u_a - \bar{u})^2 / (N - 1)$. If $\mathcal{S}$ is a simple random sample of $m \geq 2$ of the indices, drawn without replacement, then

$$E \sum_{a \in \mathcal{S}} (u_a - \bar{u}_{\mathcal{S}})^2 = (m - 1)S^2,$$

where $\bar{u}_{\mathcal{S}}$ is the sample mean.

Proof. For any m numbers, $\sum_a (u_a - \bar{u}_{\mathcal{S}})^2 = \frac{1}{2m} \sum_{a,c} (u_a - u_c)^2$, the sum being over ordered pairs of sample members. The $m(m - 1)$ ordered pairs with $a \neq c$ are each a uniformly random ordered pair of distinct population members. For such a pair, $E(u_a - u_c)^2 = \frac{1}{N(N-1)} \sum_{a \neq c} (u_a - u_c)^2 = \frac{2N \sum_a (u_a - \bar{u})^2}{N(N-1)} = 2S^2$. So the expectation is $\frac{1}{2m} m(m - 1) 2S^2 = (m - 1)S^2$. ■

Proposition 18.4.4 — What a completely randomized design would have given

Let the $N = bt$ units follow [Eq. 18.4.1](#), with fixed block effects β_j , and write $S_{\beta}^2 = t \sum_j (\beta_j - \bar{\beta})^2 / (N - 1)$ for the variance of the block effects over the N units.

- (a) If instead the treatments were assigned to the same units completely at random, b units each, the within-treatment mean square of the resulting one-way analysis would have expectation $\sigma^2 + S_{\beta}^2$, over the errors and the randomization.
- (b) In the block design,

$$\hat{\sigma}_{\text{CR}}^2 = \frac{(b - 1)\text{MS}_B + b(t - 1)\text{MSE}}{bt - 1} \quad (18.4.2)$$

is unbiased for $\sigma^2 + S_\beta^2$.

Proof.

- (a) The b units given treatment i form a simple random sample of the N units. Their responses are $\mu + \tau_i + \beta_{j(u)} + \varepsilon_u$, so the within-treatment sum of squares of treatment i is that of the numbers $\beta_{j(u)} + \varepsilon_u$ over the sample. Given the assignment, the errors contribute $(b-1)\sigma^2$ in expectation, and the cross terms have mean zero. By [Lemma 18.4.3](#) the block effects contribute $(b-1)S_\beta^2$ in expectation over the assignment. Summing over the t treatments and dividing by $t(b-1)$ gives $\sigma^2 + S_\beta^2$.
- (b) By [Theorem 18.4.1](#)(d), $E[(b-1)MS_B] = (b-1)\sigma^2 + t \sum_j (\beta_j - \bar{\beta})^2$ and $E[b(t-1)MSE] = b(t-1)\sigma^2$. The sum has expectation $(bt-1)\sigma^2 + (N-1)S_\beta^2$. ■

The **relative efficiency** $\hat{\sigma}_{CR}^2 / \text{MSE}$ estimates how many units a CRD would need per unit of the block design for the same precision. Some authors multiply it by Fisher's factor $(\nu_B + 1)(\nu_C + 3) / \{(\nu_B + 3)(\nu_C + 1)\}$, with $\nu_B = (t-1)(b-1)$ and $\nu_C = t(b-1)$ the two error degrees of freedom, to allow for estimating σ^2 with fewer degrees of freedom; this is a convention, not a theorem.

For the adhesives,

$$\hat{\sigma}_{CR}^2 = \frac{5 \times 0.8136 + 24 \times 0.0690}{29} = 0.1974,$$

and the relative efficiency is $0.1974/0.0690 = 2.86$ (2.81 with Fisher's factor). Blocking was worth almost three times as many strips; without it the adhesive F would have been about $1.0266/0.1974 = 5.20$.

```
s2_crd = ((b - 1) * ms_blk + b * (t - 1) * ms_err) / (b * t - 1)  # estimated CRD
                                error variance
re = s2_crd / ms_err
nu_b, nu_c = df_err, b * t - t                                     # error df: blocked,
                                completely randomized
re_fisher = re * (nu_b + 1) * (nu_c + 3) / ((nu_b + 3) * (nu_c + 1))
F_crd = ms_trt / s2_crd                                           # what the treatment F would
                                roughly have been
print(f"estimated CRD error variance {s2_crd:.2f} against {ms_err:.2f}:"
      f" relative efficiency {re:.2f} ({re_fisher:.2f} with Fisher's df factor)")
```

Remark (Fixed or random blocks). Treating the boards as a random sample of boards changes nothing about treatment comparisons in a complete block design, since block effects cancel from every contrast ([Theorem 18.4.1](#)(b)). It matters for incomplete blocks ([Section 18.5](#)) and for predicting a new block; Chapter 32 treats such mixed models.

18.4.5 Exercises

18.4.5.1 A. Check your understanding

Exercise A1.

An experiment has $t = 4$ treatments in $b = 8$ blocks. Give the degrees of freedom of the block design's error, and of the error of a completely randomized design with the same 32 units. How large must the relative efficiency (without Fisher's factor) be for blocking to be worthwhile?

Exercise A2.

Verify from the table in [Example \(Five wood adhesives on six boards\)](#) that the treatment mean of adhesive 3 is 9.560, and compute the difference between adhesives 3 and 5 with its standard error. Is it significant by Tukey's criterion?

18.4.5.2 B. Practice**Exercise B1.**

The strips cut from a board differ systematically: strips from the outer edge are stronger. How would you modify the design and analysis to remove this source of variation as well? Which design of the next section does your answer lead to?

Exercise B2.

In [Proposition 18.4.2](#), show that $\text{Cov}(\sum_i c_i \bar{y}_i, \sum_i d_i \bar{y}_i) = S_W^2 \sum_i c_i d_i / b$ for two contrasts. Conclude that orthogonal contrasts are uncorrelated over the randomization, as they are under the normal model.

Solution. As in the proof, $\text{Cov}(\sum_i c_i w_{ij}, \sum_k d_k w_{kj}) = \frac{W_j}{t} \sum_i c_i d_i - \frac{W_j}{t(t-1)} \sum_{i \neq k} c_i d_k$, and $\sum_{i \neq k} c_i d_k = (\sum_i c_i)(\sum_k d_k) - \sum_i c_i d_i = -\sum_i c_i d_i$. So the covariance is $W_j \sum_i c_i d_i / (t-1)$. Summing over blocks and dividing by b^2 gives $S_W^2 \sum_i c_i d_i / b$, which is zero for orthogonal contrasts.

Exercise B3.

Suppose the block effects are random, $\beta_j \stackrel{\text{iid}}{\sim} N(0, \sigma_\beta^2)$, independent of the errors. Show that $E(\hat{\sigma}_{\text{CR}}^2) = \sigma^2 + \sigma_\beta^2 t(b-1)/(bt-1)$, and interpret the factor $t(b-1)/(bt-1)$.

18.4.5.3 C. Going deeper**Exercise C1.**

In [Eq. 18.4.1](#) let the block effects be random, $\beta_j \stackrel{\text{iid}}{\sim} N(0, \sigma_\beta^2)$, independent of the errors. Show that the treatment means $\bar{y}_i$ are equicorrelated, with variance $(\sigma^2 + \sigma_\beta^2)/b$ and covariance σ_β^2/b , while every contrast estimate keeps the distribution given by [Theorem 18.4.1\(b\)](#). Show that $E(\text{MS}_B) = \sigma^2 + t\sigma_\beta^2$, and that $(\text{MS}_B + (t-1)\text{MSE})/(tb)$

is unbiased for $\text{Var}(\bar{y}_{i.})$. Why is $s/\sqrt{b}$ the right scale for comparing treatments but the wrong standard error for a single treatment mean?

Solution. Averaging over blocks, $\bar{y}_{i.} = \mu + \tau_i + \bar{\beta} + \bar{\varepsilon}_{i.}$, where $\bar{\beta} \sim N(0, \sigma_\beta^2/b)$ is common to all treatments and the $\bar{\varepsilon}_{i.}$ are independent $N(0, \sigma^2/b)$ and independent of $\bar{\beta}$. Hence $\text{Var}(\bar{y}_{i.}) = (\sigma^2 + \sigma_\beta^2)/b$ and $\text{Cov}(\bar{y}_{i.}, \bar{y}_{k.}) = \text{Var}(\bar{\beta}) = \sigma_\beta^2/b$ for $i \neq k$. In a contrast $\bar{\beta}$ is multiplied by $\sum_i c_i = 0$, so $\sum_i c_i \bar{y}_{i.} = \sum_i c_i \tau_i + \sum_i c_i \bar{\varepsilon}_{i.}$ exactly as with fixed blocks. The block means are $\bar{y}_{.j} = \mu + \bar{\tau} + \beta_j + \bar{\varepsilon}_{.j}$, and the $\beta_j + \bar{\varepsilon}_{.j}$ are independent with variance $\sigma_\beta^2 + \sigma^2/t$, so $E(\text{SS}_B) = t(b-1)(\sigma_\beta^2 + \sigma^2/t)$ and $E(\text{MS}_B) = \sigma^2 + t\sigma_\beta^2$. The error sum of squares is unchanged, because the residual projection annihilates every vector that is constant within blocks, so $E(\text{MSE}) = \sigma^2$. Then

$$E\left(\frac{\text{MS}_B + (t-1)\text{MSE}}{tb}\right) = \frac{t\sigma^2 + t\sigma_\beta^2}{tb} = \frac{\sigma^2 + \sigma_\beta^2}{b}.$$

A comparison of treatments uses only within-block information, in which $\bar{\beta}$ cancels, so $s/\sqrt{b}$ is its scale and the Tukey intervals of [Exercise \(B1\)](#) remain exact. A single treatment mean is a statement about the population of blocks, and its uncertainty includes the sampling of the b blocks: s^2/b misses the term σ_β^2/b . The estimate above mixes two mean squares with different degrees of freedom, so an interval built on it is only approximately t ; the proper treatment is the subject of Chapter 32.

18.5 Latin squares and incomplete blocks

A **Latin square** removes two crossed sources of nuisance variation at once. A **balanced incomplete block design** allows blocks smaller than the number of treatments, at a computable price in precision. The Latin square is orthogonal; the incomplete block design is not, and its treatment comparisons are adjusted for blocks exactly as they were adjusted for a covariate in [Section 18.1](#).

18.5.1 Latin squares

Definition 18.5.1 — Latin square design

An $a \times a$ **Latin square** is an arrangement of a symbols in an $a \times a$ array such that each symbol occurs exactly once in each row and exactly once in each column. A **Latin square design** compares a treatments on a^2 units classified by two crossed blocking factors with a levels each (rows and columns). The treatment applied to the unit in row i and column j is the symbol in cell (i, j) of a Latin square chosen at random.

The cyclic square $k(i, j) = i + j$ modulo a exists for every a ; the usual randomization permutes its rows, columns and symbols at random, which makes every unit equally likely to receive every treatment. With $k(i, j)$ the treatment in cell (i, j) , the model is additive in all three classifications:

$$y_{ij} = \mu + \rho_i + \kappa_j + \tau_{k(i,j)} + \varepsilon_{ij}, \quad \varepsilon_{ij} \stackrel{\text{iid}}{\sim} N(0, \sigma^2). \quad (18.5.1)$$

Theorem 18.5.1 — Analysis of the Latin square

Assume Eq. 18.5.1 with $a \geq 3$. Write $\bar{y}_{i\cdot}$, $\bar{y}_{\cdot j}$ and $\bar{y}^{(k)}$ for the means of row i , column j and treatment k , each an average of a observations, and $\bar{y}$ for the grand mean.

- (a) The centred row, column and treatment spaces are mutually orthogonal. So $\mathbf{I}$ is the sum of five mutually orthogonal projections: onto the mean, rows, columns, treatments (ranks 1, $a - 1$, $a - 1$, $a - 1$) and error (rank $(a - 1)(a - 2)$). The corresponding sums of squares are

$$\begin{aligned} SS_R &= a \sum_i (\bar{y}_{i\cdot} - \bar{y})^2, & SS_C &= a \sum_j (\bar{y}_{\cdot j} - \bar{y})^2, \\ SS_T &= a \sum_k (\bar{y}^{(k)} - \bar{y})^2, \end{aligned}$$

$$\text{and } SSE = \sum_{i,j} (y_{ij} - \bar{y}_{i\cdot} - \bar{y}_{\cdot j} - \bar{y}^{(k(i,j))} + 2\bar{y})^2.$$

- (b) Every contrast $\sum_k c_k \tau_k$ is estimable, with least squares estimate $\sum_k c_k \bar{y}^{(k)}$, free of the row and column effects, and variance $\sigma^2 \sum_k c_k^2 / a$.
- (c) The treatment F statistic satisfies

$$F_T = \frac{SS_T / (a - 1)}{SSE / ((a - 1)(a - 2))} \sim F(a - 1, (a - 1)(a - 2), \gamma),$$

with

$$\gamma = \frac{a}{\sigma^2} \sum_k (\tau_k - \bar{\tau})^2.$$

Proof.

- (a) Take any two of the three factors, say rows and treatments. Each treatment occurs exactly once in each row, so the two-way table of counts has every entry 1, with all row and column totals a , and it satisfies the proportional frequencies condition Eq. 9.4.1. By Theorem 9.4.3 the two centred spaces are orthogonal. The same argument applies to rows and columns (each cell once) and to columns and treatments. By Theorem 6.3.8 the projection onto the model space is $\mathbf{P}_0 + (\mathbf{P}_R - \mathbf{P}_0) + (\mathbf{P}_C - \mathbf{P}_0) + (\mathbf{P}_T - \mathbf{P}_0)$, of rank $1 + 3(a - 1)$. The error projection is $\mathbf{I}$ minus this, with rank $a^2 - 3a + 2 = (a - 1)(a - 2)$. Applying the projections to $\mathbf{y}$ gives the stated sums of squares and the residual.
- (b) Treatment k occurs once in each row and once in each column, so averaging Eq. 18.5.1 over its a cells gives $\bar{y}^{(k)} = \mu + \bar{\rho} + \bar{\kappa} + \tau_k + \bar{\varepsilon}^{(k)}$. For a contrast the constant cancels. Estimability follows as in Theorem 18.4.1(b), and by (a) and Theorem 9.4.1 the least squares estimate is the one computed from the treatment means alone. (c) is Corollary 9.6.2.

■

The price is the error: two degrees of freedom for $a = 3$ and six for $a = 4$, so small squares

are often replicated. The error consists of pieces of the row-by-column, row-by-treatment and column-by-treatment interactions, which cannot be told apart ([Exercise \(B2\)](#)). An interaction, such as a treatment working better in some rows, distorts the treatment comparisons: the design is only as good as its additivity assumption.

Example. (Keyboard layouts, typists and sessions)

In a synthetic experiment, five typists each type a standard passage in five sessions, comparing five keyboard layouts. Typists differ greatly, and speed rises with practice. A randomized 5×5 Latin square gives each layout once to each typist and once in each session:

	S1	S2	S3	S4	S5
typist 1	2	3	1	4	5
typist 2	3	5	4	2	1
typist 3	5	1	2	3	4
typist 4	4	2	5	1	3
typist 5	1	4	3	5	2

The analysis of variance of the speeds (words per minute) is:

source	df	sum of squares	mean square	F
typists	4	689.2	172.3	82.7
sessions	4	89.8	22.44	10.8
layouts	4	73.6	18.40	8.83
error	12	24.99	2.083	
corrected total	24	877.6		

The layouts differ ($p = 0.0015$). Their means are 47.06, 50.10, 45.92, 50.30 and 48.94 words per minute, and a difference has standard error $\sqrt{2 \times 2.083/5} = 0.913$.

Had the sessions been ignored, with layouts randomized within typists, the error variance would have been about $(MS_C + (a - 1)MSE)/a$ ([Exercise \(B1\)](#)). Relative to the Latin square this is

$$\frac{22.44 + 4 \times 2.083}{5 \times 2.083} = 2.95.$$

Ignoring the typists instead would have cost a factor of 17.35, and no blocking at all a factor of 16.25.

```
import numpy as np
from scipy import stats

rng = np.random.default_rng(62)
a = 5
cyclic = (np.arange(a)[: , None] + np.arange(a)[None, :]) % a      # a standard square
square = cyclic[rng.permutation(a)][: , rng.permutation(a)]      # permute rows and
    columns
square = rng.permutation(a)[square]                                # and relabel the
    layouts
typist = rng.normal(0, 8, a)                                       # typists differ a lot
session = np.array([-3.0, -1.0, 0.0, 1.5, 2.5])                  # practice effect over
    sessions
layout = np.array([0.0, 2.0, -1.0, 3.0, 1.0])
y = np.round(55 + typist[: , None] + session[None, :] + layout[square])
```

```

        + rng.normal(0, 1.5, (a, a)), 1)

grand = y.mean()
row_m, col_m = y.mean(axis=1), y.mean(axis=0)
trt_m = np.array([y[square == k].mean() for k in range(a)])
ss_row = a * np.sum((row_m - grand) ** 2)
ss_col = a * np.sum((col_m - grand) ** 2)
ss_trt = a * np.sum((trt_m - grand) ** 2)
ss_tot = np.sum((y - grand) ** 2)
ss_err = ss_tot - ss_row - ss_col - ss_trt
df_err = (a - 1) * (a - 2)
ms_err = ss_err / df_err
F_trt = ss_trt / (a - 1) / ms_err
print(f"SS rows {ss_row:.1f}, columns {ss_col:.1f}, layouts {ss_trt:.1f}, error {
    ss_err:.2f} on {df_err} df")
print(f"F for layouts = {F_trt:.2f}, p = {stats.f.sf(F_trt, a - 1, df_err):.2g}")
print("layout means", trt_m.round(2), " se of a difference", np.sqrt(2 * ms_err / a)
    .round(3))

```

Because the sessions are ordered in time, a layout may affect speed on the next one. Such **carryover** effects are not in [Eq. 18.5.1](#); *crossover designs*, such as Latin squares balanced for the preceding treatment, are built to estimate them.

18.5.2 Balanced incomplete block designs

Often a block cannot hold every treatment: a taster can judge only a few samples, a batch suffices for only a few runs. With blocks of size $k < t$, treatment comparisons are no longer free of block effects, and the fairest arrangement treats every pair alike.

Definition 18.5.2 — Balanced incomplete block design

A **balanced incomplete block design** (BIBD) with parameters (t, b, r, k, λ) arranges t treatments in b blocks of size $k < t$, so that each block contains k distinct treatments, each treatment appears in r blocks, and each pair of distinct treatments appears together in exactly λ blocks. Its **incidence matrix** is the $t \times b$ matrix $\mathbf{N}$ with $n_{ij} = 1$ if treatment i is in block j and 0 otherwise. Treatments are assigned to the units within each block at random, and the labels of treatments and blocks are randomized as well.

Lemma 18.5.2 — Identities for balanced incomplete block designs

For a BIBD:

- (a) $bk = tr$ and $\lambda(t - 1) = r(k - 1)$;
- (b) $\mathbf{N}\mathbf{N}^\top = (r - \lambda)\mathbf{I}_t + \lambda\mathbf{J}_t$, where $\mathbf{J}_t = \mathbf{1}\mathbf{1}^\top$;
- (c) $r > \lambda$, and $b \geq t$ (**Fisher's inequality**).

Proof.

- (a) Both sides of the first identity count the units. For the second, fix a treatment i .

The r blocks containing it hold $r(k-1)$ other units, and each of the other $t-1$ treatments occurs in exactly λ of these blocks.

- (b) The (i, i') entry of $\mathbf{NN}^\top$ counts the blocks containing both i and i' . It is r on the diagonal and λ off it.
- (c) $k < t$ gives $k-1 < t-1$, so $\lambda = r(k-1)/(t-1) < r$. The matrix in (b) has eigenvalue $r - \lambda + \lambda t$ on $\mathbf{1}$ and $r - \lambda$ on $\mathbf{1}^\perp$, all positive. So $\mathbf{NN}^\top$ is nonsingular, and $t = \text{rank}(\mathbf{NN}^\top) \leq \text{rank}(\mathbf{N}) \leq b$.

■

The model is the additive block model, observed only on the occupied cells:

$$y_{ij} = \mu + \tau_i + \beta_j + \varepsilon_{ij} \quad \text{for the } bk \text{ pairs with } n_{ij} = 1. \quad (18.5.2)$$

Let T_i be the total of the observations on treatment i , B_j the total of block j , and define the **adjusted treatment totals**

$$Q_i = T_i - \frac{1}{k} \sum_j n_{ij} B_j, \quad (18.5.3)$$

treatment i 's total minus the average level of the blocks it happened to be in. Then $\sum_i Q_i = 0$.

Theorem 18.5.3 — Intra-block analysis of a balanced incomplete block design

Assume Eq. 18.5.2 for a BIBD, with $\text{Cov}(\mathbf{Y}) = \sigma^2 \mathbf{I}$.

- (a) $\sum_i c_i \tau_i$ is estimable iff $\sum_i c_i = 0$. Its least squares estimate and variance are

$$\sum_i c_i \hat{\tau}_i = \frac{k}{\lambda t} \sum_i c_i Q_i, \quad \text{Var}\left(\sum_i c_i \hat{\tau}_i\right) = \frac{k}{\lambda t} \sigma^2 \sum_i c_i^2. \quad (18.5.4)$$

- (b) The sum of squares for treatments adjusted for blocks is $\text{SS}_{T|B} = \frac{k}{\lambda t} \sum_i Q_i^2$ on $t-1$ degrees of freedom. The residual sum of squares is $\text{SSE} = \text{SS}_{\text{tot}} - \text{SS}_B - \text{SS}_{T|B}$, where $\text{SS}_B = \sum_j B_j^2/k - (\sum_j B_j)^2/(bk)$ is the unadjusted block sum of squares, on $bk - b - t + 1$ degrees of freedom.
- (c) Under normality,

$$F = \frac{\text{SS}_{T|B}/(t-1)}{\text{SSE}/(bk - b - t + 1)} \sim F(t-1, bk - b - t + 1, \gamma),$$

with

$$\gamma = \frac{\lambda t}{k \sigma^2} \sum_i (\tau_i - \bar{\tau})^2.$$

- (d) A randomized complete block design with r blocks estimates each contrast with variance $\sigma^2 \sum_i c_i^2/r$. The ratio of that variance to Eq. 18.5.4 is the **efficiency**

factor

$$E = \frac{\lambda t}{rk} = \frac{t(k-1)}{(t-1)k} < 1.$$

Proof. Let $\mathbf{X}_B$ be the $n \times b$ indicator matrix of blocks (with the intercept in its span) and $\mathbf{Z}$ the $n \times t$ indicator matrix of treatments, with $n = bk$. Let $\mathbf{M}_B$ project onto $\mathcal{C}(\mathbf{X}_B)$, which replaces every observation by its block mean, and put $\tilde{\mathbf{Z}} = (\mathbf{I} - \mathbf{M}_B)\mathbf{Z}$. By Lemma 9.2.1, $\mathcal{C}([\mathbf{X}_B, \mathbf{Z}]) = \mathcal{C}(\mathbf{X}_B) \oplus \mathcal{C}(\tilde{\mathbf{Z}})$, orthogonally. This is the structure of Theorem 18.1.1 with the treatment indicators in the role of covariates, except that $\tilde{\mathbf{Z}}$ is not of full rank: $\mathbf{Z}\mathbf{1} = \mathbf{1}$ lies in $\mathcal{C}(\mathbf{X}_B)$, so $\tilde{\mathbf{Z}}\mathbf{1} = \mathbf{0}$.

The matrix $\mathbf{C} = \tilde{\mathbf{Z}}^\top \tilde{\mathbf{Z}} = \mathbf{Z}^\top \mathbf{Z} - \mathbf{Z}^\top \mathbf{M}_B \mathbf{Z}$ is the *information matrix* of the design. Here $\mathbf{Z}^\top \mathbf{Z} = r\mathbf{I}$, and $\mathbf{Z}^\top \mathbf{M}_B \mathbf{Z} = \mathbf{N}\mathbf{N}^\top/k$, because $\mathbf{Z}^\top \mathbf{X}_B = \mathbf{N}$ and $\mathbf{X}_B^\top \mathbf{X}_B = k\mathbf{I}$. By Lemma 18.5.2(b) and (a),

$$\mathbf{C} = r\mathbf{I} - \frac{(r-\lambda)\mathbf{I} + \lambda\mathbf{J}}{k} = \frac{r(k-1) + \lambda}{k}\mathbf{I} - \frac{\lambda}{k}\mathbf{J} = \frac{\lambda t}{k}\left(\mathbf{I} - \frac{1}{t}\mathbf{J}\right).$$

Also $\tilde{\mathbf{Z}}^\top \mathbf{y} = \mathbf{Z}^\top \mathbf{y} - \mathbf{Z}^\top \mathbf{M}_B \mathbf{y} = \mathbf{Q}$, since $\mathbf{Z}^\top \mathbf{y}$ is the vector of treatment totals and $\mathbf{Z}^\top \mathbf{M}_B \mathbf{y}$ has entries $\sum_j n_{ij} B_j/k$.

- (a) A vector $(\mathbf{0}, \mathbf{c})$, zero on the block and intercept parameters, is estimable iff $\mathbf{c}^\top = \mathbf{a}^\top \mathbf{Z}$ for some $\mathbf{a}$ with $\mathbf{a}^\top \mathbf{X}_B = \mathbf{0}$ (Theorem 8.2.2). Such an $\mathbf{a}$ satisfies $\mathbf{a} = (\mathbf{I} - \mathbf{M}_B)\mathbf{a}$, so $\mathbf{c} \in \mathcal{C}(\tilde{\mathbf{Z}}^\top) = \mathcal{C}(\mathbf{C})$, which is the space of contrasts because $\mathbf{C}$ is a positive multiple of the centring projection. Conversely, for a contrast $\mathbf{c} = \mathbf{C}\mathbf{h}$, the vector $\mathbf{a} = \tilde{\mathbf{Z}}\mathbf{h}$ works. Its least squares estimate is $\mathbf{a}^\top \mathbf{y} = \mathbf{h}^\top \mathbf{Q}$. With the generalized inverse $\mathbf{C}^- = \frac{k}{\lambda t}\mathbf{I}$, which satisfies $\mathbf{C}\mathbf{C}^-\mathbf{C} = \mathbf{C}$ because $\mathbf{I} - \mathbf{J}/t$ is idempotent, we have $\mathbf{C}\mathbf{C}^-\mathbf{c} = \mathbf{c}$ for a contrast, so we can take $\mathbf{h} = \mathbf{C}^-\mathbf{c}$, and the estimate is $\frac{k}{\lambda t}\mathbf{c}^\top \mathbf{Q}$. Its variance is $\sigma^2 \|\tilde{\mathbf{Z}}\mathbf{h}\|^2 = \sigma^2 \mathbf{h}^\top \mathbf{C}\mathbf{h} = \sigma^2 \mathbf{c}^\top \mathbf{C}^-\mathbf{c} = \frac{k}{\lambda t}\sigma^2 \|\mathbf{c}\|^2$.
- (b) The extra sum of squares for treatments after blocks is the squared length of the projection onto $\mathcal{C}(\tilde{\mathbf{Z}})$, $\mathbf{y}^\top \tilde{\mathbf{Z}}\mathbf{C}^-\tilde{\mathbf{Z}}^\top \mathbf{y} = \mathbf{Q}^\top \mathbf{C}^-\mathbf{Q} = \frac{k}{\lambda t} \sum_i Q_i^2$ (Lemma 9.2.1). The block sum of squares is $\|(\mathbf{M}_B - \mathbf{P}_0)\mathbf{y}\|^2$, and the block, adjusted treatment and residual projections are mutually orthogonal and sum to $\mathbf{I} - \mathbf{P}_0$ (Theorem 6.3.8). The model has rank $b + t - 1$.
- (c) is Theorem 11.1.1. The noncentrality is the same quadratic form evaluated at the mean, where $\tilde{\mathbf{Z}}^\top \boldsymbol{\mu} = \mathbf{C}\boldsymbol{\tau}$, giving $\boldsymbol{\tau}^\top \mathbf{C}\mathbf{C}^-\mathbf{C}\boldsymbol{\tau}/\sigma^2 = \frac{\lambda t}{k} \sum_i (\tau_i - \bar{\tau})^2/\sigma^2$.
- (d) The ratio is $(1/r)/(k/(\lambda t))$, and $\lambda = r(k-1)/(t-1)$ by Lemma 18.5.2(a). Since $k < t$, $(k-1)/k < (t-1)/t$.

■

The adjusted total Q_i compares treatment i with the blocks it appeared in, and $E Q_i = \frac{\lambda t}{k}(\tau_i - \bar{\tau})$ with no β 's left (Exercise B3): this is the **intra-block analysis**. The efficiency factor is the fraction of information that survives the elimination of blocks: about one half for $k = 2$, close to one for $k = t - 1$. Smaller blocks are often more homogeneous, and the smaller σ^2 can more than make up for E . Every pair sharing a block makes the design connected (Theorem 8.6.1), and balance gives every $\tau_i - \tau_{i'}$ the same variance $2k\sigma^2/(\lambda t)$. Unbalanced connected designs are analysed the same way with a different information matrix $\mathbf{C}$.

Example. (Seven recipes, fourteen tasters)

In a synthetic tasting experiment, seven recipes of a sports drink are rated on a 0–10 scale by 14 tasters, each of whom can judge only three. The blocks are the lines of two Fano planes, the cyclic sets $\{0, 1, 3\} + s$ and $\{0, 1, 5\} + s$ modulo 7 for $s = 0, \dots, 6$ (recipes numbered $0, \dots, 6$ in this description and $1, \dots, 7$ below). This is a BIBD with $t = 7$, $b = 14$, $k = 3$, $r = 6$ and $\lambda = 2$. There are 42 ratings and 22 error degrees of freedom.

Tasters use the scale very differently: the unadjusted block sum of squares is 69.72 out of a corrected total of 80.71. The raw recipe means, 6.15, 6.10, 5.40, 6.60, 6.30, 6.85 and 6.02, therefore mix recipe quality with the generosity of the tasters each recipe met. The adjusted totals are $Q = (-1.30, 0.90, -3.43, 3.13, 0.43, -1.50, 1.77)$, and the adjusted means $\bar{y} + \frac{k}{\lambda t} Q_i$ are 5.92, 6.40, 5.47, 6.87, 6.30, 5.88 and 6.58. Recipe 6 had the highest raw mean. Its adjusted mean is

$$6.202 + \frac{3}{14} \times (-1.50) = 5.88,$$

among the lowest: recipe 6 happened to meet generous tasters, and the adjustment removes 0.97 from its mean.

The adjusted recipe sum of squares is 6.36, and the residual sum of squares is 4.64, so

$$F = \frac{6.36/6}{4.64/22} = 5.03$$

on 6 and 22 degrees of freedom ($p = 0.0022$). A difference of two recipes has standard error $\sqrt{2 \times 3 \times 0.2108/14} = 0.301$. A complete block design with six tasters tasting all seven recipes, with the same σ^2 , would give 0.265; the variance ratio is the efficiency factor $7 \times 2/(6 \times 3) = 0.7778$. A taster judging seven drinks would probably not rate as consistently, though.

```
import numpy as np
from scipy import stats

t, k = 7, 3
blocks = [sorted((d + s) % t for d in base) for base in ([0, 1, 3], [0, 1, 5]) for s
           in range(t)]
b = len(blocks)                                # 14 tasters
N = np.zeros((t, b), dtype=int)                # incidence: recipe i tasted
                                           by taster j
for j, blk in enumerate(blocks):
    N[blk, j] = 1
r = N.sum(axis=1)[0]
lam = (N @ N.T)[0, 1]

rng = np.random.default_rng(77)
tau = np.array([0.0, 0.4, -0.3, 0.8, 0.1, -0.6, 0.5]) # recipe effects
taster = rng.normal(0, 1.2, b)                    # tasters use the scale
                                           differently
rows = [(i, j) for j, blk in enumerate(blocks) for i in blk]
recipe = np.array([i for i, j in rows])
block = np.array([j for i, j in rows])
y = np.round(6 + tau[recipe] + taster[block] + rng.normal(0, 0.5, len(rows)), 1)
n = len(y)
```

```

T_tot = np.bincount(recipe, weights=y, minlength=t)           # recipe totals
B_tot = np.bincount(block, weights=y, minlength=b)           # taster totals
Q = T_tot - N @ B_tot / k                                     # adjusted recipe
    totals
tau_hat = k * Q / (lam * t)                                    # intra-block estimates
    (sum to zero)
ss_trt_adj = k / (lam * t) * np.sum(Q ** 2)
ss_blk = np.sum(B_tot ** 2) / k - y.sum() ** 2 / n           # blocks, ignoring
    recipes
ss_tot = np.sum((y - y.mean()) ** 2)
ss_err = ss_tot - ss_blk - ss_trt_adj
df_err = n - b - t + 1
ms_err = ss_err / df_err
F = ss_trt_adj / (t - 1) / ms_err
se_diff = np.sqrt(2 * k * ms_err / (lam * t))
eff = lam * t / (r * k)                                       # efficiency factor
print("raw recipe means      ", (T_tot / r).round(2))
print("adjusted recipe means", (y.mean() + tau_hat).round(2))
print(f"adjusted recipe SS {ss_trt_adj:.3f}, error SS {ss_err:.3f} on {df_err} df,"
      f" F = {F:.2f}, p = {stats.f.sf(F, t - 1, df_err):.2g}")
print(f"se of a difference {se_diff:.3f}; efficiency factor {eff:.4f}")

```

Remark (Recovering inter-block information). The block totals, with $E B_j = k\mu + \sum_i n_{ij}\tau_i + k\beta_j$, also carry treatment information. With random block effects they give an independent *inter-block* estimate of every contrast, whose variance grows with σ_β^2 . Combining the two estimates by inverse-variance weights, as Yates proposed, is generalized least squares in a mixed model (Chapter 32). With blocks as different as these tasters, the inter-block estimate adds little.

18.5.3 Exercises

18.5.3.1 A. Check your understanding

Exercise A1.

Show that the cyclic array $k(i, j) = i + j \bmod a$ is a Latin square, and that permuting its rows, its columns, or its symbols gives a Latin square. For $a = 3$, how many distinct Latin squares are there, and does the randomization described above reach all of them?

Exercise A2.

Check the identities of [Lemma 18.5.2](#) for the design of [Example \(Seven recipes, fourteen tasters\)](#). Is there a BIBD with $t = 7$, $k = 3$ and $b = 7$? With $t = 6$, $k = 3$, $b = 6$?

Exercise A3.

From the numbers in [Example \(Seven recipes, fourteen tasters\)](#), compute the estimate

of $\tau_4 - \tau_3$ and its standard error, and decide whether recipe 4 is better than recipe 3 at level 0.05, considered alone.

18.5.3.2 B. Practice

Exercise B1.

In a Latin square, suppose the columns had not been used as a blocking factor, so that the treatments were randomized within rows only. Following [Proposition 18.4.4](#), show that the expected error mean square of that design is $\sigma^2 + \sum_j (\kappa_j - \bar{\kappa})^2 / (a - 1)$, and that $(\text{MS}_C + (a - 1)\text{MSE})/a$ is unbiased for it in the Latin square.

Solution. Randomizing within rows, the units of row i are assigned to the treatments by a random permutation. Within a row, the column effects $\kappa_1, \dots, \kappa_a$ play the role of the unit values, with within-row variance $S_\kappa^2 = \sum_j (\kappa_j - \bar{\kappa})^2 / (a - 1)$, the same for every row. By [Proposition 18.4.2\(b\)](#), applied to the unit values $\kappa_j + \varepsilon_{ij}$ and averaged over the errors, the expected error mean square of the row-blocked design is $\sigma^2 + S_\kappa^2$. In the Latin square, $\text{E MS}_C = \sigma^2 + a S_\kappa^2$ and $\text{E MSE} = \sigma^2$ ([Theorem 9.6.1](#)), so $\text{E}\{\text{MS}_C + (a - 1)\text{MSE}\}/a = \sigma^2 + S_\kappa^2$.

Exercise B2.

In a 4×4 Latin square, the row-by-column interaction space of the 16 cells has dimension 9. Show that the treatment space (centred) lies inside it, and that the error space is its orthogonal complement within it, of dimension 6. What does this imply about the effect of a row-by-column interaction on the treatment comparisons?

Solution. The row-by-column interaction space is the orthogonal complement, within $\mathbb{R}^{16}$, of the span of the mean, row and column indicators; it has dimension $16 - 1 - 3 - 3 = 9$. By [Theorem 18.5.1\(a\)](#) the centred treatment space is orthogonal to the mean, row and column spaces, so it lies in this complement, with dimension 3. The error space is the orthogonal complement of the mean, row, column and treatment spaces, so it is the rest of the interaction space, of dimension 6. A row-by-column interaction in the mean vector is a vector of the 9-dimensional space, and its component in the treatment space is indistinguishable from a treatment effect. Unless the interaction happens to be orthogonal to the treatment space, it biases the treatment comparisons.

Exercise B3.

Show that $\text{E } Q_i = \frac{\lambda t}{k}(\tau_i - \bar{\tau})$ in a BIBD, directly from [Eq. 18.5.2](#) and [Lemma 18.5.2](#).

Solution. $\text{E } T_i = r\mu + r\tau_i + \sum_j n_{ij}\beta_j$ and $\text{E } B_j = k\mu + \sum_{i'} n_{i'j}\tau_{i'} + k\beta_j$. So

$$\text{E } Q_i = r\tau_i - \frac{1}{k} \sum_j \sum_{i'} n_{ij} n_{i'j} \tau_{i'} = r\tau_i - \frac{1}{k} (r\tau_i + \lambda \sum_{i' \neq i} \tau_{i'}),$$

the μ 's and β 's cancelling. Writing $\sum_{i' \neq i} \tau_{i'} = t\bar{\tau} - \tau_i$ gives $\text{E } Q_i = \tau_i(r - (r - \lambda)/k) - \lambda t\bar{\tau}/k = \frac{\lambda t}{k}(\tau_i - \bar{\tau})$, using $r(k - 1) + \lambda = \lambda t$.

Exercise B4.

A **Graeco-Latin square** superimposes two Latin squares of the same order so that every ordered pair of symbols occurs exactly once. Write down one of order 3, state the model with rows, columns and two treatment factors, and give the analysis of variance table with its degrees of freedom. How many error degrees of freedom remain for $a = 3, 4$ and 5?

18.5.3.3 C. Going deeper**Exercise C1.**

For a general incomplete block design with incidence matrix $\mathbf{N}$, replication r_i and block sizes k_j , show that the information matrix is $\mathbf{C} = \text{diag}(r_i) - \mathbf{N} \text{diag}(1/k_j) \mathbf{N}^\top$, that $\mathbf{C}\mathbf{1} = \mathbf{0}$, and that every treatment contrast is estimable iff $\text{rank}(\mathbf{C}) = t - 1$. Relate this to the connectedness of [Theorem 8.6.1](#).

18.6 Missing observations in designed experiments

Designed experiments lose observations: a plot floods, a sample is dropped, a strip of timber splits. The balanced formulas of the last two sections then no longer apply. The correct response is simple: the model still holds for the remaining observations, so fit it to them and test by comparing models ([Chapter 11](#)). Before computers, experimenters instead *filled in the missing value so that the balanced formulas give the exact least squares answer*. The trick shows exactly what the balanced formulas get right and wrong with a filled-in value, and it is the simplest instance of the EM algorithm.

18.6.1 Filling in, exactly

Let $\mathbf{X}$ be the $n \times p$ model matrix of the complete design, of rank r , and suppose the observations in a set of m rows are missing. Reorder so that $\mathbf{X} = \begin{pmatrix} \mathbf{X}_o \\ \mathbf{X}_m \end{pmatrix}$ and $\mathbf{y} = \begin{pmatrix} \mathbf{y}_o \\ \cdot \end{pmatrix}$, with $\mathbf{y}_o$ the $n - m$ observed values. Let $\mathbf{M}$ be the projection onto $\mathcal{C}(\mathbf{X})$ for the *complete* design, partitioned conformably into blocks $\mathbf{M}_{oo}, \mathbf{M}_{om}, \mathbf{M}_{mo}, \mathbf{M}_{mm}$, and let $\mathbf{D} = \begin{pmatrix} \mathbf{0} \\ \mathbf{I}_m \end{pmatrix}$ be the $n \times m$ matrix of indicators of the missing rows. The key assumption is that the loss does not destroy estimability:

$$\text{rank}(\mathbf{X}_o) = \text{rank}(\mathbf{X}) = r. \quad (18.6.1)$$

Since $\mathcal{N}(\mathbf{X}) \subseteq \mathcal{N}(\mathbf{X}_o)$ always, [Eq. 18.6.1](#) says that $\mathcal{N}(\mathbf{X}_o) = \mathcal{N}(\mathbf{X})$: every function estimable in the complete design remains estimable. A single missing observation in a randomized block design with $t, b \geq 2$, or in a Latin square with $a \geq 3$, never violates it. Losing every observation on some treatment does, and so does losing the only observation of a cell in an unreplicated factorial fitted with all its interactions.

Theorem 18.6.1 — Missing observations

Assume Eq. 18.6.1, and let $\hat{\beta}_o$ be any least squares estimate from the observed data alone, that is, for the model $E(\mathbf{Y}_o) = \mathbf{X}_o\beta$. Put $\mathbf{z}^* = \mathbf{X}_m\hat{\beta}_o$, the fitted values at the missing positions, and for $\mathbf{z} \in \mathbb{R}^m$ let $\mathbf{y}(\mathbf{z}) = \begin{pmatrix} \mathbf{y}_o \\ \mathbf{z} \end{pmatrix}$ be the data completed with $\mathbf{z}$.

- (a) (*Filling in.*) $\hat{\beta}_o$ is a least squares estimate for the complete data $\mathbf{y}(\mathbf{z}^*)$, and the residual sum of squares of the complete-data analysis of $\mathbf{y}(\mathbf{z}^*)$ equals SSE_o , the residual sum of squares of the observed data.
- (b) (*Characterization.*) The eigenvalues of $\mathbf{M}_{mm}$ lie in $[0, 1)$, and $\mathbf{z}^*$ is the unique solution of

$$\mathbf{z} = \mathbf{M}_{mo}\mathbf{y}_o + \mathbf{M}_{mm}\mathbf{z}, \quad (18.6.2)$$

that is, the unique $\mathbf{z}$ that equals its own complete-data fitted value.

- (c) (*Iteration.*) For any starting value $\mathbf{z}_0$, the sequence $\mathbf{z}_{s+1} = \mathbf{M}_{mo}\mathbf{y}_o + \mathbf{M}_{mm}\mathbf{z}_s$, which fills in the current values, fits the complete-data model and replaces the missing values by their fitted values, converges to $\mathbf{z}^*$. The error $\mathbf{z}_s - \mathbf{z}^*$ equals $\mathbf{M}_{mm}^s(\mathbf{z}_0 - \mathbf{z}^*)$.
- (d) (*Covariate method.*) For any $\mathbf{w} \in \mathbb{R}^m$, the model $E \mathbf{y}(\mathbf{w}) = \mathbf{X}\beta + \mathbf{D}\gamma$ has rank $r + m$. Its least squares fit has $\mathbf{X}_o\hat{\beta} = \mathbf{X}_o\hat{\beta}_o$, $\hat{\gamma} = \mathbf{w} - \mathbf{z}^*$, and residual sum of squares SSE_o , on $n - m - r$ degrees of freedom.
- (e) (*Tests.*) Let $\mathcal{C}(\mathbf{X}_0) \subseteq \mathcal{C}(\mathbf{X})$ with $\text{rank}(\mathbf{X}_{0,o}) = \text{rank}(\mathbf{X}_0)$. The correct sum of squares for the reduced model is $\text{SSE}_{0,o} - \text{SSE}_o$, the difference of the residual sums of squares of the two models fitted to the observed data, on $r - r_0$ degrees of freedom, with error degrees of freedom $n - m - r$. The sum of squares computed from the complete-data formulas applied to $\mathbf{y}(\mathbf{z}^*)$ is at least as large.

Proof.

- (a) For any $\mathbf{b}$, the complete-data criterion splits as

$$\begin{aligned} \|\mathbf{y}(\mathbf{z}^*) - \mathbf{X}\mathbf{b}\|^2 &= \|\mathbf{y}_o - \mathbf{X}_o\mathbf{b}\|^2 + \|\mathbf{z}^* - \mathbf{X}_m\mathbf{b}\|^2 \\ &\geq \|\mathbf{y}_o - \mathbf{X}_o\mathbf{b}\|^2 \geq \text{SSE}_o. \end{aligned}$$

At $\mathbf{b} = \hat{\beta}_o$ both inequalities are equalities, because $\mathbf{z}^* = \mathbf{X}_m\hat{\beta}_o$ and $\hat{\beta}_o$ minimizes the observed-data criterion. So $\hat{\beta}_o$ minimizes the complete-data criterion, and the minimum is SSE_o .

- (b) By (a), the complete-data fitted vector of $\mathbf{y}(\mathbf{z}^*)$ is $\mathbf{X}\hat{\beta}_o$, whose missing part is $\mathbf{z}^*$. The fitted vector is $\mathbf{M}\mathbf{y}(\mathbf{z}^*)$, whose missing part is $\mathbf{M}_{mo}\mathbf{y}_o + \mathbf{M}_{mm}\mathbf{z}^*$. So $\mathbf{z}^*$ solves Eq. 18.6.2. For uniqueness, note that $\mathbf{M}_{mm} = \mathbf{D}^\top \mathbf{M} \mathbf{D}$ is symmetric and nonnegative definite, and for $\mathbf{u} \in \mathbb{R}^m$,

$$\mathbf{u}^\top \mathbf{M}_{mm} \mathbf{u} = \|\mathbf{M} \mathbf{D} \mathbf{u}\|^2 \leq \|\mathbf{D} \mathbf{u}\|^2 = \|\mathbf{u}\|^2,$$

with equality iff $\mathbf{D} \mathbf{u} \in \mathcal{C}(\mathbf{X})$, that is, iff $\mathbf{D} \mathbf{u} = \mathbf{X} \mathbf{b}$ for some $\mathbf{b}$. Such a $\mathbf{b}$ has $\mathbf{X}_o \mathbf{b} = \mathbf{0}$ and $\mathbf{X}_m \mathbf{b} = \mathbf{u}$. Under Eq. 18.6.1, $\mathbf{X}_o \mathbf{b} = \mathbf{0}$ implies $\mathbf{X} \mathbf{b} = \mathbf{0}$, so $\mathbf{u} = \mathbf{0}$. Hence $\mathbf{u}^\top \mathbf{M}_{mm} \mathbf{u} < \|\mathbf{u}\|^2$ for $\mathbf{u} \neq \mathbf{0}$, every eigenvalue of $\mathbf{M}_{mm}$ is less than 1, and

$\mathbf{I} - \mathbf{M}_{mm}$ is nonsingular. Equation Eq. 18.6.2 is $(\mathbf{I} - \mathbf{M}_{mm})\mathbf{z} = \mathbf{M}_{mo}\mathbf{y}_o$, which has exactly one solution.

- (c) Subtracting Eq. 18.6.2 for $\mathbf{z}^*$ from the recursion gives $\mathbf{z}_{s+1} - \mathbf{z}^* = \mathbf{M}_{mm}(\mathbf{z}_s - \mathbf{z}^*)$. The spectral radius of the symmetric matrix $\mathbf{M}_{mm}$ is its largest eigenvalue, which is less than 1 by (b), so $\mathbf{M}_{mm}^s \rightarrow \mathbf{0}$.
- (d) The argument of (b) shows that $\mathcal{C}(\mathbf{X}) \cap \mathcal{C}(\mathbf{D}) = \{\mathbf{0}\}$, so $\text{rank}[\mathbf{X}, \mathbf{D}] = r + m$. For any $\mathbf{b}$ and $\mathbf{g}$,

$$\|\mathbf{y}(\mathbf{w}) - \mathbf{X}\mathbf{b} - \mathbf{D}\mathbf{g}\|^2 = \|\mathbf{y}_o - \mathbf{X}_o\mathbf{b}\|^2 + \|\mathbf{w} - \mathbf{X}_m\mathbf{b} - \mathbf{g}\|^2.$$

For each $\mathbf{b}$ the second term is minimized, at zero, by $\mathbf{g} = \mathbf{w} - \mathbf{X}_m\mathbf{b}$. What remains is the observed-data criterion, so the minimizing $\mathbf{b}$ are the observed-data least squares estimates, $\mathbf{X}_o\hat{\boldsymbol{\beta}} = \mathbf{X}_o\hat{\boldsymbol{\beta}}_o$, and the minimum is SSE_o . Since $\mathbf{X}_m\mathbf{b}$ is the same for every least squares estimate (by Eq. 18.6.1, every row of $\mathbf{X}$ is an estimable function in the observed-data model), $\hat{\boldsymbol{\gamma}} = \mathbf{w} - \mathbf{X}_m\hat{\boldsymbol{\beta}}_o = \mathbf{w} - \mathbf{z}^*$.

- (e) The first statement is Theorem 11.1.1 for the observed-data model, of rank r with $n - m$ observations. For the second, the complete-data reduced-model fit of $\mathbf{y}(\mathbf{z}^*)$ has residual sum of squares $\min_{\mathbf{b}} (\|\mathbf{y}_o - \mathbf{X}_{0,o}\mathbf{b}\|^2 + \|\mathbf{z}^* - \mathbf{X}_{0,m}\mathbf{b}\|^2) \geq \text{SSE}_{0,o}$, while by (a) the complete-data full-model fit has residual sum of squares exactly SSE_o . Their difference is therefore at least $\text{SSE}_{0,o} - \text{SSE}_o$.

■

Part (a) is the classical claim. Part (e) gives the two classical corrections. The residual sum of squares is right, but it has $n - m - r$ degrees of freedom, not the $n - r$ that the complete-data formulas assign, since the filled-in values fit perfectly. And the treatment sum of squares is biased upward, because the filled-in value suits the full model and the reduced model cannot choose its own. Part (d) is Theorem 18.1.1 with indicator covariates: each gap gets its own parameter, the model fits it exactly and learns nothing from it, and the indicator's coefficient measures how far the number in the gap was from the least squares value.

Idea. A missing observation can be filled in with the value that the model, fitted to the remaining data, predicts for it. Then the balanced analysis of the completed data gives the exact least squares estimates and residual sum of squares. Subtract one error degree of freedom for each filled-in value, and compute tests by fitting the reduced model to the observed data.

18.6.2 The randomized block design with one missing value

For the randomized complete block design, Eq. 18.6.2 can be solved by hand. The projection of the complete design is $\mathbf{M} = \mathbf{P}_T + \mathbf{P}_B - \mathbf{P}_0$ (Example (Additive two-factor model, one observation per cell)), so the entry of $\mathbf{M}$ for two units is

$$m_{uv} = \frac{[\text{same treatment}]}{b} + \frac{[\text{same block}]}{t} - \frac{1}{bt},$$

where $[\cdot]$ is 1 if the condition holds and 0 otherwise.

Corollary 18.6.2 — Yates's missing value formula

In a randomized complete block design with t treatments and b blocks, suppose the observation on treatment i in block j is missing. Let T' and B' be the totals of the observed values of treatment i and of block j , and G' the total of all observed values. The least squares estimate of the missing value is

$$z^* = \frac{tT' + bB' - G'}{(t-1)(b-1)}. \quad (18.6.3)$$

The iteration of [Theorem 18.6.1\(c\)](#) reduces the error by the factor $1/b + 1/t - 1/(bt)$ at every step.

Proof. Here $m = 1$, and $\mathbf{M}_{mm} = 1/b + 1/t - 1/(bt)$, so $1 - \mathbf{M}_{mm} = (b-1)(t-1)/(bt)$. The row of $\mathbf{M}$ belonging to the missing unit, restricted to the observed units, sums the observed values of treatment i with weight $1/b - 1/(bt)$, those of block j with weight $1/t - 1/(bt)$, and all others with weight $-1/(bt)$. No observed unit shares both the treatment and the block. So $\mathbf{M}_{mo}\mathbf{y}_o = T'/b + B'/t - G'/(bt)$. Solving [Eq. 18.6.2](#),

$$z^* = \frac{bt}{(b-1)(t-1)} \left(\frac{T'}{b} + \frac{B'}{t} - \frac{G'}{bt} \right) = \frac{tT' + bB' - G'}{(t-1)(b-1)}.$$

The rate is $\mathbf{M}_{mm}$ by [Theorem 18.6.1\(c\)](#). ■

Example. (A strip that split)

Suppose that in [Example \(Five wood adhesives on six boards\)](#) the strip from board 4 for adhesive 2 had split while being cut, so its strength (in fact 8.62) was never measured. The observed totals are $T' = 45.33$ for adhesive 2, $B' = 36.25$ for board 4, and $G' = 263.95$ overall. Yates's formula gives

$$z^* = \frac{5 \times 45.33 + 6 \times 36.25 - 263.95}{4 \times 5} = 9.010.$$

The iteration started from the mean of the observed values, 9.102, gives 9.041, 9.020, 9.013, The distance to 9.010 shrinks by the factor $1/6 + 1/5 - 1/30 = 1/3$ at each step. Filling the gap with 0 and adding an indicator column gives $\hat{\gamma} = -9.010$, so the covariate method recovers the same value.

The completed table gives residual sum of squares 1.2788, exactly that of the 29 observed values, but on 19 degrees of freedom: the error mean square is $1.2788/19 = 0.0673$, not 0.0639. Its adhesive sum of squares, 4.0534, exceeds the correct 4.0512 by 0.0022. The correct F is 15.05 on 4 and 19 degrees of freedom; the naive analysis would report 15.85, mostly because of the degrees of freedom.

```
import numpy as np
from scipy import stats

rng = np.random.default_rng(606)
```

```

t, b = 5, 6                                # adhesives, boards
plan = np.array([rng.permutation(t) for _ in range(b)]) # which strip of each
board gets which adhesive
tau = np.array([0.0, 0.40, 0.65, 0.15, 0.80])    # adhesive effects (unknown in
practice)
board = rng.normal(0, 0.75, b)                  # board effects
y = np.round(9 + tau[:, None] + board[None, :] + rng.normal(0, 0.3, (t, b)), 2)

n = t * b
trt = np.repeat(np.arange(t), b)                # y.ravel() lists adhesive 1
first
blk = np.tile(np.arange(b), t)
X = np.column_stack([np.ones(n), np.eye(t)[trt], np.eye(b)[blk]]) # complete-data
design
miss = np.array([1 * b + 3])                    # adhesive 2, board 4
obs = np.setdiff1d(np.arange(n), miss)
yv = y.ravel().copy()
yv[miss] = np.nan                               # the value is lost

def fit(Xm, yy):
    coef = np.linalg.lstsq(Xm, yy, rcond=None)[0]
    return coef, np.sum((yy - Xm @ coef) ** 2)

coef_obs, sse_obs = fit(X[obs], yv[obs])          # exact least squares, 29
observations
df_obs = len(obs) - np.linalg.matrix_rank(X[obs]) # 29 - 10 = 19
X_red = np.column_stack([np.ones(n), np.eye(b)[blk]]) # blocks only
_, sse_red = fit(X_red[obs], yv[obs])
F_exact = ((sse_red - sse_obs) / (t - 1)) / (sse_obs / df_obs)
print(f"exact: SSE {sse_obs:.3f} on {df_obs} df, adhesive SS {sse_red - sse_obs:.3f}
",
      f" F = {F_exact:.2f}; fitted value for the lost cell {X[miss] @ coef_obs}")

```

```

Y_obs = np.where(np.isnan(yv), 0.0, yv).reshape(t, b)
T_i = Y_obs[1].sum()                          # observed total of adhesive 2
B_j = Y_obs[:, 3].sum()                       # observed total of board 4
G = Y_obs.sum()
z_yates = (t * T_i + b * B_j - G) / ((t - 1) * (b - 1))
print(f"Yates's value {z_yates:.3f}")

d = np.zeros(n)
d[miss] = 1.0                                # one indicator column per
missing cell
y_zero = np.where(np.isnan(yv), 0.0, yv)       # any number will do in the
gap
coef_cov, sse_cov = fit(np.column_stack([X, d]), y_zero)
print(f"covariate method: gamma-hat = {coef_cov[-1]:.3f}, so the filled value is {-
coef_cov[-1]:.3f};"
      f" SSE {sse_cov:.3f}")

z = np.array([np.nanmean(yv)])                # start from the grand mean
history = [z[0]]
for step in range(30):
    y_fill = yv.copy()
    y_fill[miss] = z
    coef_fill, _ = fit(X, y_fill)              # complete-data analysis
    z = X[miss] @ coef_fill                    # replace the gap by its
fitted value

```

```
history.append(z[0])
print("iterates", np.round(history[:6], 3))
```

With several missing values, Eq. 18.6.2 is an $m \times m$ system, and the iteration converges at the rate of the largest eigenvalue of $\mathbf{M}_{mm}$. In the adhesive experiment with a second value also lost (adhesive 5 on board 1), that rate is 0.367, a little slower than $1/3$, because the two gaps are coupled through the grand mean (Exercise (B2)).

18.6.3 Why the method works, and when it does not

The iteration of Theorem 18.6.1(c) is the EM algorithm for the normal linear model with missing responses: the E-step replaces each missing value by its current fitted value, and the M-step refits. Part (b) proves convergence directly.

Everything assumes that the loss is unrelated to what the value would have been. A strip that split before gluing qualifies; a joint that failed in handling because its glue was weak does not, and then both filling in and dropping make the adhesive look stronger than it is. The observed data cannot reveal this. Chapter 41 treats missing data in general, including the kinds of missingness and multiple imputation.

Warning. Never analyse filled-in values as if they were data without the corrections of Theorem 18.6.1(e). The residual degrees of freedom must be reduced by one for each filled-in value, and tests must be computed by comparing models fitted to the observed data. Standard errors computed from the completed data with the balanced formulas are also too small, because they treat $\mathbf{z}^*$ as observed.

18.6.4 Exercises

18.6.4.1 A. Check your understanding

Exercise A1.

Verify the arithmetic of Yates's formula in Example (A strip that split), and check that the filled-in value 9.010 makes the residual of that cell in the completed table exactly zero.

Exercise A2.

In a randomized block design with $t = 3$ and $b = 2$, suppose both observations on treatment 1 are lost. Show that Eq. 18.6.1 fails, and that $\mathbf{M}_{mm}$ has an eigenvalue equal to 1. What happens to the iteration?

Exercise A3.

Derive the analogue of Yates's formula for one missing value in an $a \times a$ Latin square: $z^* = \{a(R' + C' + T') - 2G'\} / \{(a-1)(a-2)\}$, where R' , C' and T' are the observed

totals of the row, column and treatment of the missing cell.

18.6.4.2 B. Practice

Exercise B1.

In the randomized block design with one missing value, show that the treatment sum of squares computed from the completed table exceeds the correct one by

$$\frac{\{B' - (t-1)z^*\}^2}{t(t-1)}.$$

Check the formula against the numbers in [Example \(A strip that split\)](#).

Solution. The correct treatment sum of squares is $\text{SSE}_{0,o} - \text{SSE}_o$, where $\text{SSE}_{0,o}$ is the residual sum of squares of the blocks-only model on the observed data. The complete-data sum of squares is $\text{SSE}_0(\mathbf{y}(z^*)) - \text{SSE}_o$ by [Theorem 18.6.1\(a\)](#). So the bias is $\text{SSE}_0(\mathbf{y}(z^*)) - \text{SSE}_{0,o}$. In the blocks-only model, the missing cell's fitted value from the observed data is the mean of the observed values of its block, $B'/(t-1)$, and by [Theorem 18.6.1\(a\)](#) applied to that model, filling the gap with $B'/(t-1)$ would give $\text{SSE}_{0,o}$ exactly. Only the within-block sum of squares of block j depends on the filled value z . It equals a constant plus $z^2 - (B' + z)^2/t$, a quadratic in z with leading coefficient $1 - 1/t$ and minimum at $z = B'/(t-1)$. Hence $\text{SSE}_0(\mathbf{y}(z)) - \text{SSE}_{0,o} = (1 - 1/t)\{z - B'/(t-1)\}^2 = \{(t-1)z - B'\}^2/(t(t-1))$. With $B' = 36.25$, $t = 5$ and $z^* = 9.010$: $(36.25 - 36.04)^2/20 = 0.0022$, as in the example.

Exercise B2.

Suppose two values are missing in a randomized block design, in different treatments and different blocks. Write $\mathbf{M}_{mm}$ explicitly and find its eigenvalues. Check the rate 0.367 quoted for the adhesive experiment.

Solution. With the two missing cells in different treatments and different blocks, they share neither, so $\mathbf{M}_{mm}$ has diagonal entries $d = 1/b + 1/t - 1/(bt)$ and off-diagonal entries $-1/(bt)$. Its eigenvalues are $d \pm 1/(bt)$, and the larger is $1/b + 1/t = 1/6 + 1/5 = 0.367$ for $t = 5$, $b = 6$.

18.6.4.3 C. Going deeper

Exercise C1.

Show that the covariance method of [Theorem 18.6.1\(d\)](#) gives correct standard errors for estimable functions of $\boldsymbol{\beta}$, without any correction, when the software's estimated covariance matrix of $(\hat{\boldsymbol{\beta}}, \hat{\boldsymbol{\gamma}})$ is used. Why does this not contradict the warning at the end of the section?

18.7 Summary and Notes

18.7.1 Summary

Idea.

1. Analysis of covariance is a projection problem. With design part $\mathbf{X}$ (any rank) and covariates $\mathbf{Z}$, the model space splits orthogonally as $\mathcal{C}(\mathbf{X}) \oplus \mathcal{C}((\mathbf{I} - \mathbf{M})\mathbf{Z})$. The covariate coefficients are estimated from within-group variation, $\hat{\boldsymbol{\gamma}} = \mathbf{E}_{zz}^{-1} \mathbf{E}_{zy}$, and the design part is fitted to the adjusted response $\mathbf{y} - \mathbf{Z}\hat{\boldsymbol{\gamma}}$. Tests compare residual sums of squares of nested models ([Theorem 18.1.1](#)).
2. In a randomized experiment a covariate measured before randomization reduces the variance of treatment comparisons by the factor $(1 - \rho^2)\{1 + 1/(N - g - 2)\}$ ([Proposition 18.1.2](#)), and it removes the conditional bias caused by chance imbalance ([Section 18.1](#)). A covariate affected by the treatment must not be adjusted for (same section).
3. Adjusted means are estimable functions $\mu_k + \mathbf{z}_0^\top \boldsymbol{\gamma}$, estimated by $\bar{y}_k - (\bar{\mathbf{z}}_k - \mathbf{z}_0)^\top \hat{\boldsymbol{\gamma}}$. They are correlated, their variances grow with the distance of $\bar{\mathbf{z}}_k$ from $\mathbf{z}_0$, and their differences do not depend on $\mathbf{z}_0$ ([Definition 18.2.1](#) and [Proposition 18.2.1](#)). Scheffé's intervals are exact for all their contrasts; Tukey-type intervals are approximate.
4. The test of parallel slopes is a weighted test of homogeneity of the within-group slopes ([Theorem 18.3.1](#)). When slopes differ, the treatment difference is a line in the covariate, to be reported with a band or a Johnson–Neyman region ([Proposition 18.3.2](#)).
5. In a randomized complete block design, block effects cancel from every treatment contrast. The analysis is the additive two-way analysis ([Theorem 18.4.1](#)). Randomization alone makes the contrast estimates unbiased and the error mean square unbiased for their variance ([Proposition 18.4.2](#)). The block experiment estimates the error variance a completely randomized design would have had ([Proposition 18.4.4](#)).
6. A Latin square blocks in two crossed directions with mutually orthogonal row, column and treatment spaces. Its error consists of confounded interaction pieces ([Theorem 18.5.1](#)).
7. A balanced incomplete block design has information matrix $(\lambda t/k)(\mathbf{I} - \mathbf{J}/t)$. Contrasts are estimated from the adjusted totals Q_i with variance $(k/\lambda t)\sigma^2 \sum c_i^2$, and the efficiency factor relative to complete blocks is $\lambda t/(rk)$ ([Theorem 18.5.3](#)).
8. A lost observation is handled exactly by least squares on the remaining data. Filling it with its fitted value, found by Yates's formula, by iteration or by the covariate method, reproduces the exact estimates and residual sum of squares. The error degrees of freedom must be reduced, and tests must be computed by comparing models ([Theorem 18.6.1](#) and [Corollary 18.6.2](#)).

18.7.2 Notes and sources

Analysis of covariance. Fisher introduced the analysis of covariance in the 1930s, in later editions of *Statistical Methods for Research Workers* (Fisher 1925 and later editions), as a

way of using a concomitant measurement to increase the precision of agricultural experiments. Cochran's review (Cochran 1957) remains an excellent account of its uses, including the dangers of adjusting for a covariate affected by treatment and of adjusting non-randomized groups. Rencher and Schaalje (2008, chapter 16) treat one-way, two-way and multi-covariate layouts and the test of homogeneous slopes. Christensen (2020, chapter 9) derives the analysis through the projection $(\mathbf{I} - \mathbf{M})\mathbf{Z}$, which is the route taken in [Theorem 18.1.1](#), and uses the same device for missing data and for incomplete block designs. The case against testing baseline balance in randomized trials was made forcefully by Senn (1994). The modern debate on regression adjustment in randomized experiments, which asks whether adjustment can hurt when the model is wrong, was opened by Freedman (2008) and answered, for the interacted estimator of [Exercise \(C1\)](#), by Lin (2013). Chapter 25 returns to it.

Heterogeneous slopes. The region of significance for non-parallel lines is due to Johnson and Neyman (1936), who developed it for comparisons of teaching methods with an ability covariate. Simultaneous versions use the Scheffé argument of [Chapter 13](#).

Block designs and randomization. Randomization, blocking and the Latin square as a design are Fisher's, set out in *The Design of Experiments* (Fisher 1935). The randomization moments of [Proposition 18.4.2](#) and the view of the linear model as an approximation justified by randomization are developed in Kempthorne (1952) and Hinkelmann and Kempthorne (2008). Cox (1958) remains the best short account of the ideas behind designed experiments, and Bailey (2008) gives a modern treatment of block designs built on the same orthogonal decompositions as this chapter. Relative efficiencies such as [Eq. 18.4.2](#) are standard in the design literature, for instance Cochran and Cox (1957). Christensen (2020, chapter 8) presents the randomized complete block and Latin square models as additive multi-way layouts and discusses their assumptions.

Incomplete blocks. Balanced incomplete block designs were introduced by Yates (1936), who also showed how to recover inter-block information (Yates 1940). Fisher (1940) proved the inequality $b \geq t$ of [Lemma 18.5.2](#). Christensen (2020, section 9.4) derives the intra-block analysis as an analysis of covariance with the treatment indicators as covariates, as in the proof of [Theorem 18.5.3](#).

Missing values. Yates (1933) derived the missing value formula [Eq. 18.6.3](#) and the rule of subtracting a degree of freedom for each estimated value. Bartlett (1937) proposed the covariance method with indicator covariates, and Healy and Westmacott (1956) the iterative method of [Theorem 18.6.1\(c\)](#), convenient for early computers. Dempster, Laird and Rubin (1977) placed the iteration within the EM algorithm. Christensen (2020, section 9.3) treats missing observations through the covariance model.

18.7.3 References

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